

Research continuation and complete source proofs

Split-Zero Cohomology and Arithmetic Weight Control

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This reader integrates the Split-Zero carrier, finite arithmetic operator comparisons, the historical packet ghost, chain retractions, arithmetic moments, Gaussian resolvents, full kernel layers, symmetric tensor frontiers, measured support branches, the original theta norm-volume identities, quartet derivative costs, gamma-reference determinants, and the complete two-Stieltjes-space and spectral-sum comparisons, the full relative connection, strict normal energy, the conormal derivative with its original residue maps, nested relative-frame transitions and density, the full limiting derivative graph, cyclic arithmetic metric comparisons, every retained conormal depth, exterior trace amplification, the exact equality and coupling maps, positive exterior formulas for arithmetic interpolation volumes, the original source and relation Toda flows with all four loss terms, exact endpoint comparisons, certified theta-seed input with explicit infinite tails, gamma convolution and its full arithmetic coefficient maps, the certified seed coefficients, finite source-to-quotient metric transfer, and the complete complex amplitude in the sum fibres. This edition also proves the original arithmetic monic-norm estimate, the balanced endpoint bound, exact products and their sharp scalar optimization, full confluent transfer, finite Hermitian window transport, the arithmetic central-volume upper comparison, and reflected norm/conormal kernel transport. Every original source note and the complete independent calculations supporting these statements accompany the proof chapters. The displayed upper and lower volume estimates retain their explicitly calculated gap.

The continuation also constructs the compact-support image and its dual, logarithmic specialization, exact invariant-subspace coupling, polynomial-exponential cohomology, its complete period determinant, and the translation maps carrying the original arithmetic operator and metric through this family. It retains the original arithmetic functions, measures, scalar actions, support fibres, and complete jets. Part I gives the local continuation with proofs. Part II preserves the complete selected source notes, their exact revisions, and their chronology.

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Part I

Integrated mathematical continuation

1 Exact Split-Zero carriers and the local spectral bridge

Throughout this section, semiring homomorphisms preserve additive zero and multiplicative unit. Let R be a nonzero commutative ring and $\mathbb{B} = (\{0, 1\}, \vee, \wedge)$. Define

$$G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset G(R).$$

Addition and multiplication of supported elements are those of R , including when their result is zero. The additional element satisfies

$$\tau + r = r + \tau = r, \quad \tau + \tau = \tau, \quad \tau r = r\tau = \tau, \quad \tau^2 = \tau.$$

Thus the semiring zero is τ , and $e \neq \tau$.

1.1 Scalar support and independent channel amplitudes

Proposition 1.1 (Pair carrier and scalar reflection). *There is a semiring isomorphism*

$$\eta_R : G(R) \longrightarrow \{(0, 0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}, \quad \tau \mapsto (0, 0), \quad r \mapsto (r, 1).$$

The coordinate maps

$$\begin{aligned} p_R : G(R) &\rightarrow R, & p_R(\tau) &= 0, & p_R(r) &= r, \\ \chi_R : G(R) &\rightarrow \mathbb{B}, & \chi_R(\tau) &= 0, & \chi_R(r) &= 1 \end{aligned}$$

are semiring homomorphisms. The map p_R is universal among semiring homomorphisms from $G(R)$ to a ring.

Proof. In the product semiring, two supported pairs have sum $(r + s, 1)$ and product $(rs, 1)$. These identities retain support when the amplitude vanishes. The pair $(0, 0)$ is an additive identity and an absorbing multiplicative zero. This proves closure and all operation identities, and the displayed map is bijective. Its coordinate projections give p_R, χ_R . If $f : G(R) \rightarrow A$ has ring target, then $e + e = e$ gives $f(e) + f(e) = f(e)$, hence $f(e) = 0_A$. Its restriction to the supported copy of R is therefore a unital ring homomorphism $h : R \rightarrow A$. The equality $f = hp_R$ holds on R and on τ , and determines h . Conversely every such h gives hp_R . $\square$

The larger carrier $R \times \mathbb{B}$ contains $(1, 0)$; its exact relation to $G(R)$ is the inclusion above. The pair presentation requires $R \times \{1\}$.

For a nontrivial bounded distributive lattice L , define

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), & (r, \lambda)(s, \mu) &= (rs, \lambda \wedge \mu). \end{aligned}$$

Write $z_\lambda = (0, \lambda)$, $\tau_L = z_{0_L}$, and $e_L = z_{1_L}$.

Proposition 1.2 (Lattice reflection with its original coefficient ring). *The displayed carrier is a commutative semiring. The projections*

$$p_L : G_L(R) \rightarrow R, \quad \chi_L : G_L(R) \rightarrow L$$

are respectively the universal ring reflection and the support homomorphism. In particular the amplitude codomain of $G_{\mathbb{B}^2}(\mathbb{C})$ is $\mathbb{C}$.

Proof. A summand with nonzero amplitude has top support, so its sum has top support. A factor with lower support has zero amplitude, so a product of lower support has zero amplitude. These facts prove closure. All laws restrict from the product semiring; distributivity uses that of L . Both projections preserve operations, zero and unit. If $f : G_L(R) \rightarrow A$ has ring target, $e_L + z_\lambda = e_L$ gives $f(z_\lambda) = 0$ by cancellation in A . Restriction to the supported copy of R is a ring homomorphism h and $f = hp_L$. Conversely each h gives this factorization. Surjectivity of p_L gives uniqueness. $\square$

Let $j : \mathbb{C} \rightarrow G(\mathbb{C})$ include the supported copy. It preserves addition, multiplication and unit; it takes ordinary zero to e , whereas the semiring zero is τ . Put

$$M = G(\mathbb{C})^2, \quad p^{(2)} = (p_{\mathbb{C}}, p_{\mathbb{C}}) : M \rightarrow \mathbb{C}^2, \quad \chi^{(2)} = (\chi_{\mathbb{C}}, \chi_{\mathbb{C}}) : M \rightarrow \mathbb{B}^2.$$

For $\epsilon \in \mathbb{B}$ let $u_0 = \tau, u_1 = e$.

Proposition 1.3 (Exact scalar-lattice and two-amplitude embeddings). *There are injective semiring homomorphisms*

$$b : G_{\mathbb{B}^2}(\mathbb{C}) \hookrightarrow M, \quad B : G_{\mathbb{B}^2}(\mathbb{C}^2) \hookrightarrow M$$

defined by

$$\begin{aligned} b(c, (1, 1)) &= (j(c), j(c)), \\ b(0, (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}), \\ B((a, b), (1, 1)) &= (j(a), j(b)), \\ B((0, 0), (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}). \end{aligned}$$

Their amplitude and support identities are

$$p^{(2)}b = \Delta_{\mathbb{C}}p_{\mathbb{B}^2}, \quad \chi^{(2)}b = \chi_{\mathbb{B}^2}, \quad p^{(2)}B = p_{\mathbb{B}^2}, \quad \chi^{(2)}B = \chi_{\mathbb{B}^2},$$

where $\Delta_{\mathbb{C}}(c) = (c, c)$.

Proof. The pair presentation identifies M with

$$\{((a, b), (\lambda_1, \lambda_2)) \in \mathbb{C}^2 \times \mathbb{B}^2 : \lambda_1 = 0 \Rightarrow a = 0, \quad \lambda_2 = 0 \Rightarrow b = 0\},$$

with componentwise operations. The map b restricts the product semiring map $(c, \lambda) \mapsto ((c, c), \lambda)$. Lower support in its source forces $c = 0$, so it lands in the displayed subset. It retains the amplitude and both support bits and is injective. The map B restricts the identity coordinate map on $\mathbb{C}^2 \times \mathbb{B}^2$; lower support in its source forces both amplitudes to vanish, so it too lands in M and is injective. The prescriptions agree at the top-support zero. Both maps preserve zero and unit. Projection proves all four identities. $\square$

The element $(j(1), \tau) \in M$ is outside both images. There is no function $r : M \rightarrow G_{\mathbb{B}^2}(\mathbb{C}^2)$ satisfying both $p_{\mathbb{B}^2}r = p^{(2)}$ and $\chi_{\mathbb{B}^2}r = \chi^{(2)}$: it would send that element to $((1, 0), (1, 0))$, outside its target carrier. This identifies the precise obstruction to a reverse map preserving both projections.

The synchronized carrier has the exact map

$$G(\mathbb{C}^2) \cong G(\mathbb{C}) \times_{\mathbb{B}} G(\mathbb{C}) \hookrightarrow M.$$

It takes τ to (τ, τ) and a supported (a, b) to $(j(a), j(b))$. Equality of the two support characters leaves exactly those cases, proving bijectivity onto the fibre product. Operations agree componentwise. Its image has masks $(0, 0), (1, 1)$, while the other masks in M carry independent mixed amplitudes.

The map $p^{(2)}$ is the universal ring reflection of M . Indeed, (e, e) is additively idempotent, so every ring-target map kills it. Adding (e, e) to $x \in M$ gives the fully supported element of amplitude $p^{(2)}x$. The map is consequently determined by a ring homomorphism from the supported copy of $\mathbb{C}^2$, proving the universal property. A semiring isomorphism $G_{\mathbb{B}^2}(\mathbb{C}) \cong M$ would induce a ring isomorphism $\mathbb{C} \cong \mathbb{C}^2$. This is impossible because $(1, 0)$ is a nontrivial idempotent of $\mathbb{C}^2$, and $\mathbb{C}$ has only 0, 1. The positive relation between the original carriers is b , with diagonal amplitude projection.

1.2 Tesserine sectors and a supported critical fixed locus

Reserve τ for split absence and use κ for conjugation. Let

$$\sigma(s) = 1 - s, \quad \kappa(s) = \bar{s}, \quad K_4 = \{1, \sigma, \kappa, \sigma\kappa\}.$$

The real tesserine algebra and its complexification are

$$A_{\mathbb{R}} = \mathbb{R}[K_4] = \mathbb{R}[\sigma, \kappa]/(\sigma^2 - 1, \kappa^2 - 1), \quad A_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} A_{\mathbb{R}}.$$

The polynomial algebra in this presentation is commutative.

Proposition 1.4 (Character algebra and quartet coordinates). *For $\epsilon, \delta \in \{1, -1\}$ let*

$$E_{\epsilon\delta} = \frac{1}{4}(1 + \epsilon\sigma)(1 + \delta\kappa).$$

Character evaluation gives an algebra isomorphism

$$\mathcal{F} : A_{\mathbb{C}} \rightarrow \mathbb{C}^4, \quad x \mapsto (\text{ev}_{\epsilon,\delta}(x))_{\epsilon,\delta},$$

with inverse $(d_{\epsilon\delta}) \mapsto \sum d_{\epsilon\delta} E_{\epsilon\delta}$. For the exact quartet element

$$Q(s) = s \cdot 1 + (1 - s)\sigma + \bar{s}\kappa + (1 - \bar{s})\sigma\kappa, \quad s = \beta + i\gamma,$$

the four character values are

$$c_{++} = 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1), \quad c_{--} = 4i\gamma.$$

Projection to the last two specified characters is a unital algebra map $\pi : A_{\mathbb{C}} \rightarrow \mathbb{C}^2$. The composite πQ is a real affine bijection

$$\mathbb{C} \rightarrow \mathbb{R} \times i\mathbb{R}, \quad s \mapsto (a, b) = (c_{-+}(s), c_{--}(s)), \quad s = \frac{1}{2} + \frac{1}{4}(a + b).$$

Proof. Evaluation of $E_{\epsilon\delta}$ at (ϵ', δ') is $\frac{1}{4}(1 + \epsilon'\sigma)(1 + \delta'\kappa)$. It is one when both signs agree and zero otherwise. This proves the two inverse formulas and all orthogonal idempotent identities, retaining the factors of four. Character evaluation preserves the algebra operations. Directly,

$$\begin{aligned} c_{++} &= s + (1 - s) + \bar{s} + (1 - \bar{s}) = 2, \\ c_{+-} &= s + (1 - s) - \bar{s} - (1 - \bar{s}) = 0, \\ c_{-+} &= s - (1 - s) + \bar{s} - (1 - \bar{s}) = 2(s + \bar{s}) - 2 = 4\beta - 2, \\ c_{--} &= s - (1 - s) - \bar{s} + (1 - \bar{s}) = 2(s - \bar{s}) = 4i\gamma. \end{aligned}$$

The proposed inverse recovers $\beta + i\gamma$. Conversely every real a and purely imaginary b is recovered by substituting that inverse into the two character expressions. $\square$

The quartet image has real affine dimension two. Its real linear span has dimension three, spanned by E_{++}, E_{-+}, iE_{--} , retaining the constant character coordinate. The real tesserine algebra has dimension four. The maps $Q, \mathcal{F}, \pi$ and the displayed affine inverse specify the exact relationship of these objects.

Proposition 1.5 (Supported sector lift and fixed locus). *Define*

$$\Psi : \mathbb{C} \rightarrow M, \quad \Psi(s) = (j(c_{-+}(s)), j(c_{--}(s))).$$

It has $\chi^{(2)}\Psi(s) = (1, 1)$ for all s . The $G(\mathbb{C})$ -linear automorphisms

$$T_{\sigma}(x, y) = ((-1)x, (-1)y), \quad T_{\kappa}(x, y) = (x, (-1)y)$$

of the free semimodule M give

$$\Psi(\sigma s) = T_{\sigma}\Psi(s), \quad \Psi(\kappa s) = T_{\kappa}\Psi(s).$$

Here $-1 = j(-1)$ is a supported unit. Their product satisfies

$$\text{Fix}(T_{\sigma}T_{\kappa}) = \{\tau, e\} \times G(\mathbb{C}), \quad \Psi^{-1} \text{Fix}(T_{\sigma}T_{\kappa}) = \{s : \Re s = \frac{1}{2}\}.$$

Proof. The sector formulas give $(a, b)(1 - s) = (-a, -b)$ and $(a, b)(\bar{s}) = (a, -b)$. Multiplication by $j(-1)$ negates supported amplitudes and fixes τ . It preserves addition, commutes with scalar multiplication and squares to identity. This proves the automorphism and equivariance statements. Its fixed points are τ and supported $j(c)$ with $c = -c$. Since $2 \neq 0$ in $\mathbb{C}$, the latter is precisely e . The product action is $((-1)x, y)$, proving its fixed locus. On the image of Ψ the first coordinate is supported and is fixed exactly when $4\beta - 2 = 0$. $\square$

The scalar-lattice channel atoms have images $b(z_{10}) = (e, \tau)$ and $b(z_{01}) = (\tau, e)$. The separate support swap has fixed masks $(0, 0), (1, 1)$. Since $\chi^{(2)}\Psi$ is constantly $(1, 1)$, it cannot detect the critical line by that swap. The support action induced by $T_\sigma T_\kappa$ is identity; its amplitude action is $(-a, b)$. The preceding proposition supplies the exact fixed-locus test with the first-channel supported zero retained.

1.3 Exact sector, holonomy and positive-moment maps

Fix a prime p , $\ell_p = \log p > 0$, and $u \in S^1$. For an arithmetic packet, take $u = \psi(\text{Frob}_p)$ with $L/\mathbb{Q}$ finite abelian, p unramified, and $\psi : \text{Gal}(L/\mathbb{Q}) \rightarrow S^1$ a character. All powers use $p^w = \exp(w\ell_p)$ with the positive real logarithm.

Proposition 1.6 (Sector-to-holonomy map with constants retained). *The analytic function*

$$H_{p,u} : \mathbb{C}^2 \rightarrow \mathbb{C}^\times, \quad H_{p,u}(a, b) = u \exp\left(\frac{\ell_p}{4}(a + b)\right)$$

gives a function $H_{p,u}p^{(2)}$ on M . On the quartet image,

$$H_{p,u}(c_{-+}(s), c_{--}(s)) = up^{s-1/2} =: z_{p,u}(s),$$

$$\frac{\log |z_{p,u}(s)|}{\ell_p} = \frac{c_{-+}(s)}{4} = \Re s - \frac{1}{2}, \quad |z_{p,u}(s)|^2 - 1 = \exp\left(\frac{\ell_p}{2}c_{-+}(s)\right) - 1.$$

Proof. The exact inverse for the sector coordinates gives $s - \frac{1}{2} = (a + b)/4$. Substitution proves the exponential identity. On that image a is real and b is purely imaginary. Since $|u| = 1$, taking absolute values yields $\exp(\ell_p a/4)$. Taking its real logarithm and, separately, its square proves the two remaining identities. $\square$

The radial exponent is $c_{-+}/4$; the exponent $c_{-+}/2$ belongs to the squared modulus. The map $H_{p,u}$ is used with its analytic-function type, without any assertion that it preserves semiring addition.

Proposition 1.7 (Rank-one packet). *For $z \in \mathbb{C}$ define*

$$v_2(z) = \begin{pmatrix} 1 \\ z \end{pmatrix}, \quad T_2(z) = v_2(z)v_2(z)^* = \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix}, \quad P_z(w) = w - \bar{z}.$$

Then $T_2(z)$ is Hermitian positive semidefinite of rank one and

$$\ker T_2(z) = \mathbb{C}(-\bar{z}, 1)^\top.$$

It is Toeplitz exactly when $|z| = 1$, equivalently when the unique root of P_z lies on S^1 . For $z = z_{p,u}(s)$ this is equivalent to $\Re s = \frac{1}{2}$.

Proof. For $x \in \mathbb{C}^2$, $x^*T_2(z)x = |v_2(z)^*x|^2 \geq 0$. The vector $v_2(z)$ is nonzero and hence its outer product has rank one. The adjoint kernel equation is $x_0 + \bar{z}x_1 = 0$, giving the displayed space. A two-by-two matrix is Toeplitz exactly when its diagonal entries agree, here $1 = |z|^2$. The root $\bar{z}$ has modulus $|z|$. The final equivalence follows from the preceding proposition and $\ell_p > 0$. $\square$

Theorem 1.8 (Positive finite packets and their exact diagonal defect). *Let $r \geq 1$, $\alpha_j > 0$, and $z_j \in \mathbb{C}$. For $n \geq 1$ put*

$$v_n(z) = (1, z, \dots, z^{n-1})^\top, \quad M_n = \sum_{j=1}^r \alpha_j v_n(z_j) v_n(z_j)^*.$$

The matrix M_n is Hermitian positive semidefinite. For $n \geq 3$, it is Toeplitz exactly when every $|z_j| = 1$. With matrix indices starting at zero,

$$\mathcal{D}(M_n) := (M_n)_{22} - 2(M_n)_{11} + (M_n)_{00} = \sum_{j=1}^r \alpha_j (|z_j|^2 - 1)^2.$$

For these packets and $n \geq 3$, vanishing of $\mathcal{D}$ already characterizes Toeplitzness. If the z_j are pairwise distinct and $r \leq n$, the rank is r . If $r < n$, coefficient vectors identify the kernel with

$$\left(\prod_{j=1}^r (w - \bar{z}_j) \right) \mathbb{C}[w]_{\leq n-1-r} \subset \mathbb{C}[w]_{\leq n-1}.$$

In particular $r = n - 1$ gives corank one and exactly the conjugated roots $\bar{z}_1, \dots, \bar{z}_{n-1}$.

Proof. The identity

$$x^* M_n x = \sum_j \alpha_j |v_n(z_j)^* x|^2$$

proves positivity and shows that its kernel is the simultaneous kernel of all $v_n(z_j)^*$. The first three diagonal entries are $\sum \alpha_j$, $\sum \alpha_j |z_j|^2$, and $\sum \alpha_j |z_j|^4$. Their displayed linear combination gives the defect without rescaling any weight. Toeplitzness equates these entries, forcing zero defect. Every weight is positive and every square nonnegative, hence all $|z_j| = 1$. Conversely that condition gives $(M_n)_{k\ell} = \sum_j \alpha_j z_j^{k-\ell}$, a Toeplitz matrix. The same argument proves the zero-defect characterization.

Let V have columns $v_n(z_j)$. Its first r rows have Vandermonde determinant $\prod_{i < j} (z_j - z_i) \neq 0$, so V has rank r . The established kernel identity is $\ker M_n = \ker V^*$, proving the rank. For $P_x(w) = \sum_{k=0}^{n-1} x_k w^k$, the kernel equations say $P_x(\bar{z}_j) = 0$. Distinct linear factors divide P_x exactly when these equations hold: division by one factor preserves the other root equations since that factor is nonzero at those roots. This proves the product and degree bound. At $r = n - 1$, the remaining factor is a scalar; for a nonzero kernel vector it is nonzero, so there are no further roots. $\square$

The size bound is sharp for general positive packets. At $n = 2$, the data $(\alpha_1, \alpha_2) = (4, 1)$ and $(z_1, z_2) = (1/2, 2)$ give

$$M_2 = 4 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This is positive definite and Toeplitz, while neither atom is on S^1 . At size three the diagonal is $(5, 5, 65/4)$ and the defect is $45/4$. One atom retains the two-by-two criterion. Zero-weight atoms are invisible; repeated atoms combine their weights before applying the distinct-column rank count.

The exact sector defect is therefore

$$\mathcal{D}(M_n) = \sum_{j=1}^r \alpha_j \left[\exp \left(\frac{\log p_j}{2} c_{-+}(s_j) \right) - 1 \right]^2.$$

The function chain

$$M \xrightarrow{p^{(2)}} \mathbb{C}^2 \xrightarrow{H_{p,u}} \mathbb{C}^\times \xrightarrow{z \mapsto v_n(z) v_n(z)^*} \text{Herm}_{\geq 0}(n)$$

has explicit maps at every step. Positive finite summation supplies the preceding theorem.

1.4 The theta-packet form and its scalar comparison

Proposition 1.9 (Eigenpair form quotient and sector channel). *For every complex d -by- d matrix A , Hermitian matrix G , and nonzero eigenvector v with $Av = \rho v$, let $W = A^*G + GA - G$. Then*

$$v^*Wv = (2\Re\rho - 1)v^*Gv.$$

*On the domain of these data where $v^*Gv \neq 0$, the quotient*

$$\omega(A, G, v, \rho) = \frac{v^*Wv}{v^*Gv} = 2\Re\rho - 1 = \frac{c_{-+}(\rho)}{2}$$

is real. Its associated holonomy satisfies

$$|z_{p,u}(\rho)|^2 = \exp(\ell_p\omega), \quad z_{p,u}(\rho) = u \exp\left(\frac{\ell_p}{4}(c_{-+}(\rho) + c_{--}(\rho))\right).$$

Proof. Since $(Av)^* = \bar{\rho}v^*$,

$$v^*(A^*G + GA - G)v = (Av)^*Gv + v^*G(Av) - v^*Gv = (\bar{\rho} + \rho - 1)v^*Gv.$$

This proves the unquotiented identity even when the right factor is zero. Division on the stated domain gives ω , and $c_{-+}(\rho) = 4\Re\rho - 2$ gives the factor $1/2$. The holonomy identities follow from the exact sector exponential above. $\square$

This is the typed map from an eigenpair of the form packet to its Split-Zero sector and local moment matrix. The moment matrix is positive for every parameter, so that positivity cannot supply a further zero-defect conclusion. For $\rho = 1, u = 1$,

$$z = \sqrt{p}, \quad T_2(z) = \begin{pmatrix} 1 & \sqrt{p} \\ \sqrt{p} & p \end{pmatrix} \geq 0,$$

and its diagonal entries differ. On the form side $A = (1), G = (1), v = (1)$ give $W = (1)$ and $\omega = 1$; the exact maps take these data to the same off-critical packet. Any arithmetic identity forcing the actual family of W to vanish on its eigenvectors must therefore be established from that family's construction. These maps retain that requirement rather than replacing it by Gram positivity.

1.5 Projective zeta at all zeros and at its pole

A pair in $G(\mathbb{C})^2$ is unimodular exactly when at least one amplitude is nonzero. Such a coordinate is a supported unit. If both amplitudes are zero, every linear combination has zero amplitude and cannot be the unit. Let $\mathbb{P}_{\text{um}}^1(G(\mathbb{C}))$ denote unimodular pairs modulo common unit multiplication.

Proposition 1.10 (Projective amplitude section and reciprocal). *The map*

$$p_* : \mathbb{P}_{\text{um}}^1(G(\mathbb{C})) \rightarrow \mathbb{P}^1(\mathbb{C}), \quad [x : y] \mapsto [p(x) : p(y)]$$

has an injective section

$$\iota : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad [a : b] \mapsto [j(a) : j(b)],$$

whose image is the full-support component. With

$$\mathcal{R} : G(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad x \mapsto [1 : x], \quad r : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad c \mapsto [1 : c],$$

one has $p_\mathcal{R} = rp_{\mathbb{C}}$ and $\mathcal{R}j = \iota r$.*

Proof. Unimodularity ensures the projected pair is nonzero. A common supported unit projects to a nonzero complex scalar, so p_* is well defined. A nonzero ordinary pair becomes a unimodular supported pair; changing its representative lifts to common supported unit multiplication. Thus ι is well defined, and $p_*\iota$ is identity. Every full-support pair is the supported lift of its amplitude pair. The pair $(1, x)$ is always unimodular, and projection of its coordinates proves both remaining identities. $\square$

In particular $\mathcal{R}(e) = [1 : e] = \infty_e$ and $\mathcal{R}(\tau) = [1 : \tau] = \infty_\tau$. Both project to $[1 : 0]$, while their Boolean projective supports are $[1 : 1]$ and $[1 : 0]$. Similarly $[e : 1]$ and $[\tau : 1]$ project to the ordinary zero while retaining different supports.

Proposition 1.11 (Two exact charts for the supported zeta map). *For the meromorphic zeta function with its simple pole of residue one at 1, the full-support lift is*

$$Z_{\text{sp}} : \mathbb{C} \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad Z_{\text{sp}}(s) = \begin{cases} [j(\zeta(s)) : 1], & s \neq 1, \\ [1 : e], & s = 1. \end{cases}$$

It equals $\iota\hat{\zeta}$. At every zeta zero ρ its value is $[e : 1]$. Near the pole it has the chart

$$Z_{\text{sp}}(s) = [1 : j(h(s))], \quad h(s) = 1/\zeta(s), \quad h(1) = 0, \quad h'(1) = 1.$$

The reciprocal is a finite-amplitude coordinate away from the zeta zeros, including its extension at 1. At a zero of order m , it has a pole of order m .

Proof. Put $t = s - 1$ and write $\zeta(1+t) = t^{-1} + g(t)$ with g holomorphic near zero. Then $h(1+t) = t/(1+tg(t))$. Its denominator has value one at zero, so this is holomorphic there with value zero and derivative one. On the overlap where $\zeta \neq 0$, common multiplication by $j(\zeta(s))$ gives $[1 : j(h(s))] = [j(\zeta(s)) : 1]$. This proves the transition and pole value. The first chart directly gives $[e : 1]$ at each zero. At a zero of order m , write $\zeta(\rho+t) = t^m u(t)$ with $u(0) \neq 0$. Its reciprocal is $t^{-m} u(t)^{-1}$, proving its pole order; the first chart retains the original order m . $\square$

The swap $J_*[x : y] = [y : x]$ is induced by an invertible permutation matrix and commutes with ι, p_* . Hence J_*Z_{sp} is the global reciprocal projective map, sending zeta zeros to ∞_e and the pole to $[e : 1]$. Analytic statements use the complex structure transported to the full-support component by ι ; no topology on the extra support components is needed.

1.6 Matrix packets and their full relation module

Theorem 1.12 (Block packet with explicit hypotheses). *Let V be a complex Hilbert space of finite dimension $d > 0$ with fixed positive-definite Hermitian inner product. Let $g \in \text{GL}(V)$ and $m \geq 1$. Use that metric and its orthogonal direct sums for all adjoints. Define*

$$W_m(g) : V \rightarrow V^{m+1}, \quad v \mapsto (v, gv, \dots, g^m v), \quad \mathbb{T}_m(g) = W_m(g)W_m(g)^*,$$

$$D_m(g) : V^m \rightarrow V^{m+1}, \quad \begin{cases} (D_m(g)y)_0 = -g^* y_1, \\ (D_m(g)y)_k = y_k - g^* y_{k+1} \quad (1 \leq k \leq m-1), \\ (D_m(g)y)_m = y_m. \end{cases}$$

The matrix $\mathbb{T}_m(g)$ is positive semidefinite and

$$\ker \mathbb{T}_m(g) = \text{im } D_m(g), \quad \text{rank } \mathbb{T}_m(g) = d, \quad \text{corank } \mathbb{T}_m(g) = md.$$

It is block Toeplitz exactly when g is unitary for the fixed metric.

Proof. The Gram factorization gives positivity and $\ker \mathbb{T}_m = \ker W_m^*$. The first component of $W_m v$ is v , so W_m is injective and the kernel dimension is $(m+1)d - d = md$. The adjoint is

$$W_m^*(v_0, \dots, v_m) = \sum_{k=0}^m (g^*)^k v_k.$$

Substitution of $D_m(y)$ gives

$$-g^* y_1 + \sum_{k=1}^{m-1} ((g^*)^k y_k - (g^*)^{k+1} y_{k+1}) + (g^*)^m y_m = 0.$$

Thus its image lies in the kernel. If $D_m(y) = 0$, the last component gives $y_m = 0$, and backward substitution gives each preceding $y_k = 0$. Its image dimension is md , proving equality with the kernel. The block of indices j, k is $g^j (g^*)^k$. For unitary g this is g^{j-k} , hence block Toeplitz. Conversely block Toeplitzness equates blocks $(0, 0), (1, 1)$, giving $I = gg^*$. Invertibility yields $g^* = g^{-1}$ and unitarity. $\square$

At $m = 0$ the only block is I , so the bound $m \geq 1$ is essential. At $m = 1$, the full kernel is $\{(-g^* y, y) : y \in V\}$; its vector polynomials are $(wI - g^*)y$. Taking determinant gives the exact scalar map

$$wI - g^* \mapsto \det(wI - g^*) = \overline{\det(\bar{w}I - g)}.$$

The equality follows by conjugating the determinant expansion and transposing. It retains conjugate eigenvalues and multiplicities. For the standard metric and $J_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ with $t > 0$, the polynomial is $(w - 1)^2$ but

$$J_t J_t^* = \begin{pmatrix} 1 + t^2 & t \\ t & 1 \end{pmatrix} \neq I.$$

Its block packet consequently fails Toeplitzness, although the determinant roots lie on the unit circle. The kernel remains $\{(-J_t^* y, y) : y \in \mathbb{C}^2\}$, retaining the matrix information that determinant forgets.

2 The historical packet ghost and its full arithmetic jet quotient

The source of the packet construction in this section is the response dated 24 June 2026, 11:13:04 AM in the SSD export *Globalization Semiring Analysis*, whose exact path, hash and response-line locators are recorded in `logbook/historical_split_transcripts_20260912.md`. The source proves the permutation-packet calculation. We give that calculation completely, then construct its comparison with the full local algebras of the arithmetic packet. In particular, we retain the nilpotent jet operators and distinguish the source's complex-linear permutation from arithmetic conjugation by explicit maps.

Let Z be a finite set of complex numbers stable under $c(\rho) = \bar{\rho}$ and $\sigma(\rho) = 1 - \rho$. Assign a positive integer m_ρ constant on each orbit. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A[f] = [sf].$$

We use the original coordinate s , without replacing its multiplicative operator by its semisimple part. Put

$$L_Z = \bigoplus_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad e_{\rho,k} = t_\rho^k \quad (0 \leq k < m_\rho)$$

in the indicated direct summand. The coefficient inner product makes all $e_{\rho,k}$ orthonormal; it is an explicitly chosen reference inner product, not the theta-source metric.

Proposition 2.1 (Original-coordinate identification with all jets). *The map*

$$T : E_h \longrightarrow L_Z, \quad T[f]_\rho = \sum_{k=0}^{m_\rho-1} \frac{f^{(k)}(\rho)}{k!} t_\rho^k$$

is a complex-algebra isomorphism. In these coordinates

$$TAT^{-1} = A_{ss} + N, \quad A_{ss}e_{\rho,k} = \rho e_{\rho,k}, \quad Ne_{\rho,k} = \begin{cases} e_{\rho,k+1} & k+1 < m_\rho, \\ 0 & k+1 = m_\rho. \end{cases}$$

Proof. Taylor multiplication modulo $t_\rho^{m_\rho}$ proves multiplicativity. Its kernel consists exactly of polynomials divisible by every $(s - \rho)^{m_\rho}$ and hence by their product h , since the factors are pairwise coprime. For completeness an inverse is explicit. Set $h_\rho = h/(s - \rho)^{m_\rho}$, and let a_ρ be the Taylor polynomial of $1/h_\rho(s)$ at ρ through degree $m_\rho - 1$. Then $E_\rho = [a_\rho h_\rho]$ has residue 1 at the ρ factor and 0 at every other factor. Thus

$$T^{-1}e_{\rho,k} = E_\rho[(s - \rho)^k]$$

gives the inverse. Multiplication by $s = \rho + t_\rho$ gives the displayed operator, including its last nilpotent column. $\square$

In every subsequent operator formula on L_Z , the symbol A denotes the transported operator $TAT^{-1} : L_Z \rightarrow L_Z$ of Proposition 2.1. The polynomial-coordinate quotient at the end is equipped with the original multiplication-by- s operator.

Define complex-linear permutations C_0, S_0 and a complex-linear parity operator H on this complete basis by

$$C_0e_{\rho,k} = e_{\bar{\rho},k}, \quad S_0e_{\rho,k} = e_{1-\rho,k}, \quad He_{\rho,k} = (-1)^k e_{\rho,k}.$$

Also let K be the conjugate-linear map $K(\sum a_{\rho,k}e_{\rho,k}) = \sum \bar{a}_{\rho,k}e_{\rho,k}$. All four maps commute, square to the identity, and preserve products in L_Z , with K conjugate-linear on scalars.

Proposition 2.2 (Typed arithmetic involutions). *Under T , ordinary conjugation of polynomial coefficients is the conjugate-linear map $\mathcal{C} = C_0K$, and pullback by $s \mapsto 1 - s$ is the complex-linear map $\mathcal{S} = S_0H$. In particular*

$$C_0 = \mathcal{C}K, \quad S_0 = \mathcal{S}H, \quad \mathcal{C}A = A\mathcal{C}, \quad \mathcal{S}AS = I - A.$$

Here the equality involving $\mathcal{C}$ is an equality of conjugate-linear maps. Moreover $C_0N = NC_0$, $S_0N = NS_0$, and $HN = -NH$.

Proof. If $f^\#(s) = \overline{f(\bar{s})}$, then $(f^\#)^{(k)}(\rho)/k! = \overline{f^{(k)}(\bar{\rho})/k!}$, giving $\mathcal{C} = C_0K$. The k th derivative of $f(1 - s)$ at ρ is $(-1)^k f^{(k)}(1 - \rho)$, giving $\mathcal{S} = S_0H$. The ideal (h) is preserved by both operations: conjugation fixes the monic polynomial h , and $h(1 - s) = (-1)^{\deg h} h(s)$. Conjugating coefficients fixes the indeterminate s , whereas substituting $1 - s$ replaces multiplication by s by multiplication by $1 - s$. This proves the two identities for A . The assertions about N follow on each basis vector, including the vector of largest jet degree, which both sides send to zero. $\square$

Theorem 2.3 (Historical ghost with complete multiplicities). *Set*

$$R = C_0S_0, \quad \Gamma = C_0 - S_0, \quad P_- = \frac{I - R}{2}, \quad P_+ = \frac{I + R}{2}.$$

These satisfy

$$\Gamma^* = \Gamma, \quad \Gamma^2 = 4P_-, \quad \Gamma e_{\rho,k} = 0 \iff \Re \rho = \frac{1}{2}.$$

On a nonreal off-axis quartet of common multiplicity m , Γ has eigenvalues -2 and 2 each with multiplicity m , and 0 with multiplicity $2m$. Furthermore

$$\text{rank } P_- = 2m, \quad \text{Tr}(\Gamma^*\Gamma) = 8m.$$

On a critical-line orbit $\Gamma = P_- = 0$ at every jet degree.

Proof. The permutations C_0, S_0 are commuting selfadjoint involutions in the declared reference inner product. Consequently $(C_0 - S_0)^2 = 2I - 2C_0S_0$. The equality $C_0e_{\rho,k} = S_0e_{\rho,k}$ says exactly that the two labelled basis vectors have the same root, namely $\bar{\rho} = 1 - \rho$, which is equivalent to $2\Re\rho = 1$.

On an off-axis nonreal quartet order the roots as $(\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho})$. At every fixed jet degree the four joint (C_0, S_0) eigenspaces have characters $(1, 1), (1, -1), (-1, 1), (-1, -1)$. Explicit joint eigenvectors are

$$e_{\rho,k} + \epsilon e_{\bar{\rho},k} + \delta e_{1-\rho,k} + \epsilon \delta e_{1-\bar{\rho},k}, \quad \epsilon, \delta \in \{1, -1\}.$$

They form a basis: their pairwise inner products are zero and each has squared norm 4. The eigenvalue of Γ is $\epsilon - \delta$. Thus its squared eigenvalues are 0, 4, 4, 0, giving the rank and trace with every one of the m jet degrees included. On a critical orbit the two permutations agree on every basis vector. $\square$

For real off-axis orbits the same proof gives rank m and squared trace $4m$, since the orbit has two elements. Actual nontrivial zeta zeros in $0 < \Re s < 1$ are nonreal: for real $0 < s < 1$ the alternating series $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is strictly positive by grouping successive pairs, and $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$. For completeness the paired terms satisfy

$$(2n-1)^{-s} - (2n)^{-s} = s \int_{2n-1}^{2n} x^{-s-1} dx.$$

They converge uniformly on each compact subset of $\Re s > 0$, because on such a subset $|s| \leq M$, $\Re s \geq \delta > 0$, and the absolute value is at most $M(2n-1)^{-1-\delta}$. Thus η is analytic there. The product identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ follows first in $\Re s > 1$ by splitting even terms and then by analytic continuation. Division is used here only on the real interval $0 < s < 1$, where the denominator is strictly negative. Hence the quartet formula applies to off-axis arithmetic packets.

Proposition 2.4 (The exact group-algebra quotient). *For $W = \langle c, \sigma \mid c^2 = \sigma^2 = 1, c\sigma = \sigma c \rangle$ the homomorphism*

$$q : \mathbb{C}[W] \longrightarrow \mathbb{C}[u]/(u^2 - 1), \quad c \longmapsto u, \quad \sigma \longmapsto u$$

is surjective, with

$$I_q := \ker q = (c - \sigma)\mathbb{C}[W] = \mathbb{C}(c - \sigma) \oplus \mathbb{C}(1 - c\sigma).$$

Under the representation $c \mapsto C_0, \sigma \mapsto S_0, I_q L_Z = P_- L_Z$. Thus $L_Z/P_- L_Z$ is the largest quotient of the W -representation on which $C_0 = S_0$.

Proof. In the basis $(1, c, \sigma, c\sigma)$, q sends $a + bc + d\sigma + ec\sigma$ to $(a + e) + (b + d)u$. Its kernel is therefore spanned by $c - \sigma$ and $1 - c\sigma$. Multiplication by c sends $c - \sigma$ to $1 - c\sigma$, and multiplication by any generator stays in their span. This proves the ideal formula. Since $P_- = \Gamma^2/4$ and $\Gamma = \Gamma P_-$, their images agree and equal $I_q L_Z$. A quotient kills I_q precisely when it kills this image; this proves its universal property without any invariance assumption about the arithmetic generator. $\square$

The generator acts on the complete local rings. The next calculation constructs exactly the extra kernel needed to descend that action.

Theorem 2.5 (Arithmetic saturation of the historical ghost quotient). *Put*

$$Be_{\rho,k} = (\Re\rho - \tfrac{1}{2})e_{\rho,k}, \quad Z_{\text{crit}} = \{\rho \in Z : \Re\rho = \tfrac{1}{2}\}, \quad L_{\text{off}} = \bigoplus_{\rho \notin Z_{\text{crit}}} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}).$$

The smallest A -invariant subspace containing $I_q L_Z$ is exactly

$$I_q L_Z + A(I_q L_Z) = L_{\text{off}}.$$

On each off-axis orbit this sum is direct. The largest quotient on which both the original generator A descends and $C_0 = S_0$ is therefore the exact algebra and W -module quotient

$$L_Z \longrightarrow L_{Z_{\text{crit}}}, \quad (f_\rho)_\rho \longmapsto (f_\rho)_{\rho \in Z_{\text{crit}}}.$$

Its kernel is L_{off} , including every local multiplicity. In polynomial coordinates this is the quotient

$$\mathbb{C}[s]/(h) \longrightarrow \mathbb{C}[s]/(h_{\text{crit}}), \quad [f]_h \longmapsto [f]_{h_{\text{crit}}}, \quad h_{\text{crit}} = \prod_{\rho \in Z_{\text{crit}}} (s - \rho)^{m_\rho},$$

whose kernel is $(h_{\text{crit}})/(h)$; the target is zero when $h_{\text{crit}} = 1$.

Proof. The permutation R sends ρ to $1 - \bar{\rho}$, preserving its imaginary part and reversing $\Re \rho - 1/2$. It preserves the jet index and commutes with N . Accordingly $RB = -BR$ and

$$A = \frac{1}{2}I + B + iY + N, \quad Ye_{\rho,k} = \Im \rho e_{\rho,k}, \quad RY = YR, \quad RN = NR.$$

This is an exact identity for A , with N retained. If $v \in P_-L_Z$, then $(\frac{1}{2}I + iY + N)v \in P_-L_Z$ whereas $Bv \in P_+L_Z$. Therefore

$$P_+Av = Bv \quad (v \in P_-L_Z).$$

On each off-axis orbit all diagonal entries of B are nonzero, so $B : P_-L_Z \rightarrow P_+L_Z$ is an isomorphism on that orbit, its inverse being the restriction of the original diagonal operator B^{-1} . For $w = w_- + w_+$ on that orbit, take $v = B^{-1}w_+$; then $P_+Av = w_+$ and $w - Av \in P_-L_Z$. Thus every w lies in $P_-L_Z + A(P_-L_Z)$. If $Av \in P_-L_Z$ with $v \in P_-L_Z$, then $Bv = P_+Av = 0$ and $v = 0$, proving directness.

On a critical orbit $P_- = 0$. Both A and all permutations preserve the direct sum of off-axis factors, so the subspace just calculated is invariant and is precisely the smallest A -invariant subspace containing P_-L_Z . If a linear map F intertwines A with an operator on its target and satisfies $FC_0 = FS_0$, its kernel contains I_qL_Z and is A -invariant. It consequently contains L_{off} . Conversely the displayed coordinate projection intertwines A , preserves products and has $C_0 = S_0$ on its target. This proves the universal quotient assertion, also for targets carrying all the stated structures. The polynomial quotient and its kernel follow from $h_{\text{crit}} \mid h$ and Proposition 2.1. $\square$

Proposition 2.6 (Exact displacement and nilpotent contribution). *In the declared reference coefficient metric, let $W_{\text{ref}} = A^* + A - I$. Then*

$$W_{\text{ref}} = 2B + N + N^*, \quad \frac{W_{\text{ref}} - RW_{\text{ref}}R}{2} = 2B, \quad \frac{W_{\text{ref}} + RW_{\text{ref}}R}{2} = N + N^*.$$

For the original tessarine coordinate $c_{-+}(\rho) = 4(\Re \rho - 1/2)$, the first displayed component equals multiplication by $c_{-+}/2$.

Proof. The adjoint of A_{ss} is diagonal with entries $\bar{\rho}$, so $A_{\text{ss}}^* + A_{\text{ss}} - I = 2B$. Because R is a self-adjoint permutation commuting with N , it also commutes with N^* . Conjugation by R reverses B . Adding and subtracting the two resulting identities proves both formulas, including the entire nilpotent term. The tessarine identification uses $c_{-+} = 2(2\Re \rho - 1) = 4B$ with its original factor 4. $\square$

The precise map from this reference calculation to a theta-source metric retains the full metric matrix. Let G_E be that positive Hermitian matrix in the original polynomial basis of E_h ; then in the declared coefficient basis of L_Z it is

$$G_L = (T^{-1})^* G_E T^{-1}.$$

Writing the original polynomial-coordinate generator as A_E and the transported one as $A_L = TA_ET^{-1}$ for this formula, one obtains

$$\begin{aligned} W_E &= A_E^* G_E + G_E A_E - G_E, \\ W_L &= A_L^* G_L + G_L A_L - G_L = (T^{-1})^* W_E T^{-1}, \\ W_L &= G_L W_{\text{ref}} + [A_L^*, G_L], \quad [A_L^*, G_L] = A_L^* G_L - G_L A_L^*. \end{aligned}$$

The first two identities follow by inserting $A_L = TA_ET^{-1}$ and cancelling adjacent factors $T^{-1}T$ and $T^*(T^{-1})^*$. The last identity follows by expanding $G_L(A_L^* + A_L - I) + (A_L^* G_L - G_L A_L^*)$. Thus both nilpotent and metric-commutator contributions are present in the exact arithmetic weight; no zero-location estimate is supplied by discarding either term.

Finally form the split-zero semiring $G(\text{End}_{\mathbb{C}} L_Z)$, using the associative endomorphism ring. The original ghost has supported lift $\Gamma^\bullet$; when the comparison cancels it is $e = 0^\bullet$, whereas the absent comparison is τ . Conjugation by T extends to a semiring isomorphism

$$G(\text{End}_{\mathbb{C}} E_h) \longrightarrow G(\text{End}_{\mathbb{C}} L_Z), \quad X^\bullet \longmapsto (T X T^{-1})^\bullet, \quad \tau \longmapsto \tau.$$

It preserves addition, composition, the unit, e , and τ directly. Thus the complete jet comparison preserves supported cancellation. Theorem 2.5 states exactly which arithmetic factors are removed if that cancellation is imposed as a generator-compatible quotient. It does not assert that those factors are absent from the original arithmetic cohomology.

The quotient itself induces an endomorphism map on an exact specified domain. Define

$$\mathcal{D} = \{X \in \text{End}_{\mathbb{C}} L_Z : X(L_{\text{off}}) \subseteq L_{\text{off}}\}.$$

It is a unital subalgebra, and the induced quotient action is a surjective unital homomorphism

$$d : \mathcal{D} \longrightarrow \text{End}_{\mathbb{C}} L_{Z_{\text{crit}}}, \quad \ker d = \text{Hom}_{\mathbb{C}}(L_Z, L_{\text{off}}).$$

Indeed invariance of L_{off} is exactly what makes the quotient action well defined; the kernel condition says exactly that every value of X lies in L_{off} . Any endomorphism of the critical direct summand lifts by making it zero on the off-axis direct summand, proving surjectivity. Thus

$$G(d) : G(\mathcal{D}) \longrightarrow G(\text{End}_{\mathbb{C}} L_{Z_{\text{crit}}})$$

is the explicit split-zero descent map. Both A and Γ belong to $\mathcal{D}$; it sends $A^\bullet$ to the supported critical generator, $\Gamma^\bullet$ to the supported zero e , and τ to τ . This statement includes the zero target algebra when the critical summand is absent.

3 Historical Hurwitz reconstruction and the complete Laurent jet bridge

This section reconstructs a compact probability measure from the complete local Hurwitz zero cluster at any nontrivial zeta zero. It also proves the complete finite-jet relation with the marked Laurent period, including the involution coefficients and the exact collision quotient. Two distinct prime periods recover the entire finite spectral algebra with every multiplicity retained.

3.1 Source receipts and exact locators

1. The original *Split-Zero CUE Hurwitz phase space* source has SHA256 95CCA3275F2012E32F9E506EC0211273D5E7DA1FD72AD90F7167AA37094E94CC. Its exact excerpts accompany this edition in `sources/historical/hurwitz_laurent/`: `compact_hurwitz_and_residues.tex`

reproduces source lines 722–856; `zero_motion_and_bernstein.tex` reproduces lines 1110–1257; and `all_moments_claim.tex` reproduces lines 1907–1912. The statements below retain the original coordinate s , displacement t , support coordinate y , support bound L , and moments $m_n(\mu)$.

2. The *Mathematical Functor Construction* transcript has SHA256 8F94D3A8172F8585837A4E4F6060B5F8DAC38CF627C9689335F13792B7710D85. Its complete mathematical proof block, source lines 14142–15894, is retained as `sources/historical/hurwitz_laurent/marked_period_laurent_conormal_proofs.md`. Theorems 3, 7, 11 and 13 give the marked Laurent augmentation and period maps; Theorems 15–18 give the packet and Laurent involution; Theorems 20–25 give the marked conormal chart. Within that source, lines 15143–15227 prove $J_u^* = J_{-u}$, lines 15397–15502 give the $d \log |a_\infty|$ chart, and lines 15504–15631 give the augmentation-conormal diagram.
3. The corresponding standalone *Terzo-CC-CVS-JT deformation intertwiner* witness has SHA256 BAF7DB6506710CA9AA5B7705C7D5687B4E66536C08A9FE77549ABBC041BC9044. Its Laurent deformation appears at source lines 490–542 and its conormal calculation at lines 638–760. The complete proofs needed here are reproduced below and in the accompanying transcript excerpt.

The historical source gives the forward zero-motion recursion and the marked conormal map. The proofs below give the inverse moment recursion, the complete multiple-zero version, the full nilpotent Laurent isomorphism, and its collision quotient. Source line numbers and hashes identify the particular source editions; these comparisons make no claim of global mathematical priority.

3.2 Every compact Hurwitz input is determined by one simple-zero germ

Keep the original family

$$Z_{\mu,t}(s) = \int_{[0,L]} \zeta(s, 1+ty) d\mu(y), \quad m_n(\mu) = \int_{[0,L]} y^n d\mu(y), \quad m_0(\mu) = 1.$$

Here μ is a Borel probability measure, L is any finite support bound, and t is the original displacement; no rescaling of either variable is made. For $|t|L < 1$, put

$$b_n(s) = \frac{(-1)^n}{n!} (s)_n \zeta(s+n), \quad Z_{\mu,t}(s) = \sum_{n \geq 0} m_n(\mu) b_n(s) t^n. \quad (\text{H1})$$

The expansion is the source’s compact Taylor formula. To verify its analytic scope directly, for $\text{Re } s > 1$ expand each $(k+ty)^{-s}$ in its binomial Taylor series. On compact s -sets and $|t|L \leq r < 1$, the ratios $|ty|/k \leq r$ give normally convergent double sums. Integration is termwise valid. The identity extends meromorphically in s ; near a nontrivial zero ρ all displayed coefficients are holomorphic. Joint holomorphicity and normal convergence in the strip can be checked directly: for a compact set K in $0 < \text{Re } s < 1$, the product

$$\left| \frac{(s)_n}{n!} \right| = \prod_{j=1}^n \left| 1 + \frac{s-1}{j} \right| \leq \prod_{j=1}^n \left(1 + \frac{C_K}{j} \right) \leq e^{C_K} (n+1)^{C_K}$$

is polynomially bounded uniformly in $s \in K$. The factor $\zeta(s+n)$ is uniformly bounded for $n \geq 1$ by the absolutely convergent Dirichlet series at $1 + \min_{s \in K} \text{Re } s$. Since $m_n \leq L^n$, the terms of (H1) are bounded by a constant times $(n+1)^{C_K} (L|t|)^n$. Thus the series is normally convergent when $L|t| < 1$, including on a surrounding compact annulus for any contour used below. Cauchy estimates give the same local derivative control. For $n \geq 1$, $\text{Re}(\rho+n) > 1$ and none of $\rho, \rho+1, \dots, \rho+n-1$ is zero, so

$$b_n(\rho) \neq 0. \quad (\text{H2})$$

The zeta factor in (H2) is nonzero because on $\operatorname{Re} z > 1$ the logarithmic Euler-product series

$$\sum_{p \text{ prime}} \sum_{k \geq 1} \frac{p^{-kz}}{k}$$

converges absolutely, and its exponential is $\zeta(z)$ there. This conclusion uses the actual zero's strip $0 < \operatorname{Re} \rho < 1$; it does not assume RH.

Fix a simple nontrivial zero ρ , so $\zeta(\rho) = 0$ and $\zeta'(\rho) \neq 0$. The analytic implicit function theorem produces the unique local holomorphic germ

$$\rho_\mu(t) = \rho + w_\mu(t), \quad w_\mu(t) = \sum_{j \geq 1} c_j(\mu) t^j, \quad Z_{\mu,t}(\rho_\mu(t)) = 0. \quad (\text{H3})$$

This is the existing source theorem. Its simplicity hypothesis is retained explicitly, rather than silently extending a scalar branch through a multiple zero.

The inverse recurrence. For each integer $N \geq 1$, the input moment is recovered by the exact finite formula

$$m_N(\mu) = -\frac{1}{b_N(\rho)} \sum_{n=0}^{N-1} m_n(\mu) [t^{N-n}] b_n \left(\rho + \sum_{j=1}^N c_j(\mu) t^j \right). \quad (\text{H4})$$

Thus $(c_1, \dots, c_N)$ determines $(m_1, \dots, m_N)$ and conversely. In particular, two compact probability inputs have the same zero-trajectory jet through order N exactly when their first N moments agree.

Proof. Substitute (H3) in (H1). Its coefficient of t^N is

$$0 = \sum_{n=0}^N m_n(\mu) [t^{N-n}] b_n(\rho + w_\mu(t)).$$

The $n = N$ summand is exactly $m_N(\mu) b_N(\rho)$. It contains no positive-order coefficient of w . Every other summand uses only $m_0, \dots, m_{N-1}$ and $c_1, \dots, c_N$. Division is legitimate by (H2), proving (H4). In the same coefficient equation the only occurrence of c_N is $\zeta'(\rho) c_N$, from the $n = 0$ summand, because every other Taylor term has either a positive power of t already or at least two positive-order factors of w . This proves the forward triangular recurrence and both directions of the finite-jet assertion by induction. $\square$

The first two inverse formulas, with every sign retained, are

$$m_1 = \frac{\zeta'(\rho) c_1}{\rho \zeta(\rho + 1)}, \quad (\text{H5})$$

$$m_2 = -\frac{2}{\rho(\rho + 1)\zeta(\rho + 2)} \left[\zeta'(\rho) c_2 + \frac{\zeta''(\rho)}{2} c_1^2 - m_1(\zeta(\rho + 1) + \rho \zeta'(\rho + 1)) c_1 \right]. \quad (\text{H6})$$

These follow by $b_1(s) = -s\zeta(s + 1)$ and $b_2(s) = s(s + 1)\zeta(s + 2)/2$, not by changing the Hurwitz convention.

Exact measure reconstruction on the original interval. Assume $L > 0$. For every integer $M \geq 1$ and $0 \leq j \leq M$, set

$$\lambda_{M,j} = \binom{M}{j} \sum_{k=0}^{M-j} (-1)^k \binom{M-j}{k} \frac{m_{j+k}(\mu)}{L^{j+k}}, \quad \mu_M = \sum_{j=0}^M \lambda_{M,j} \delta_{Lj/M}. \quad (\text{H7})$$

Then every $\lambda_{M,j} \geq 0$, their sum is one, and μ_M converges weakly to μ . Thus (H4) and (H7) give an explicit reconstruction of the entire input from the zero germ, with the original support coordinate and bound present in every formula.

Proof. Expanding $(1 - y/L)^{M-j}$ proves

$$\lambda_{M,j} = \int_{[0,L]} \binom{M}{j} (y/L)^j (1 - y/L)^{M-j} d\mu(y).$$

This proves positivity, and the binomial theorem proves the sum is one. At fixed original y , let J have binomial law with M trials and success probability y/L . Its original-coordinate variable LJ/M has mean y and variance

$$\mathbb{E}[(LJ/M - y)^2] = \frac{y(L - y)}{M} \leq \frac{L^2}{4M}. \quad (\text{H8})$$

For continuous f on $[0, L]$ and any $\delta > 0$, split the expectation according to $|LJ/M - y| \leq \delta$. Uniform continuity and Chebyshev's inequality give

$$|\mathbb{E}f(LJ/M) - f(y)| \leq \omega_f(\delta) + 2\|f\|_\infty \frac{L^2}{4M\delta^2},$$

uniformly in y . First take $M \rightarrow \infty$, then $\delta \rightarrow 0$, and integrate against μ . This is weak convergence. For Lipschitz f , Cauchy-Schwarz gives the explicit bound $\text{Lip}(f)L/(2\sqrt{M})$. If $L = 0$, the input is exactly δ_0 , all positive moments vanish, and the original zero branch is constant; this exceptional fibre is retained separately. $\square$

Injectivity without a prespecified common support bound. If two compactly supported probability measures have the same zero germ at the chosen simple ρ , (H4) gives equal moments. Choose any common finite support bound for the two measures. Their measures (H7) are identical at every M , and their weak limits are therefore identical. Hence the map

$$\left\{ \begin{array}{c} \text{compact probability measures} \\ \text{on } [0, \infty) \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{holomorphic zero germs} \\ \text{at the fixed simple } \rho \end{array} \right\},$$

$$\mu \longmapsto \rho_\mu$$

is injective onto its actual image. This statement asserts no surjectivity onto arbitrary holomorphic germs.

The exact support endpoint is also recovered:

$$\sup \text{supp } \mu = \lim_{n \rightarrow \infty} m_n(\mu)^{1/n}. \quad (\text{H9})$$

Indeed, if the endpoint is L_μ , then $m_n \leq L_\mu^n$. For any $0 < a < L_\mu$, positive mass lies in $(a, L_\mu]$, so $m_n \geq a^n \mu((a, L_\mu])$. Taking n th roots and then $a \uparrow L_\mu$ proves the claim. The zero endpoint gives δ_0 and both sides zero.

The completed residue comparison has its own exact factorization. Define

$$\Lambda_{\mu,t}(s) = \pi^{-s/2} \Gamma(s/2) Z_{\mu,t}(s).$$

Then, for the same displacement domain $|t|L < 1$,

$$\text{Res}_{s=1} \Lambda_{\mu,t} = 1, \quad \text{Res}_{s=0} \Lambda_{\mu,t} = -1 - 2tm_1(\mu), \quad (\text{HRes1})$$

and consequently

$$\mathfrak{B}(\mu, t) := \text{Res}_{s=0} \Lambda_{\mu,t} + \text{Res}_{s=1} \Lambda_{\mu,t} = -2tm_1(\mu). \quad (\text{HRes2})$$

Proof. For $\operatorname{Re} \mathfrak{a} > 0$ and $\operatorname{Re} s > 1$, termwise integration of the absolutely convergent geometric series gives

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \int_0^\infty x^{s-1} \frac{e^{-\mathfrak{a}x}}{1 - e^{-x}} dx.$$

At zero the integrand's second factor is $x^{-1} + (1/2 - \mathfrak{a}) + O(x)$, uniformly on compact subsets of the indicated parameter half-plane. Subtract these two terms on $(0, 1)$. The remaining integrals define a function $H(s, \mathfrak{a})$ holomorphic for $\operatorname{Re} s > -1$, and

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \frac{1}{s-1} + \frac{1/2 - \mathfrak{a}}{s} + H(s, \mathfrak{a}).$$

The tail integral is holomorphic there because of exponential decay. Since $\Gamma(1) = 1$ and $\Gamma(s) = s^{-1} + O(1)$, the formula proves $\operatorname{Res}_{s=1} \zeta(s, \mathfrak{a}) = 1$ and $\zeta(0, \mathfrak{a}) = 1/2 - \mathfrak{a}$. For $\mathfrak{a} = 1 + ty$, the parameters stay in a compact subset of that half-plane as y ranges over the original support. Uniformity therefore permits integration of both coefficients against the probability measure. It gives residue one at $s = 1$ and $Z_{\mu,t}(0) = -1/2 - tm_1(\mu)$. Finally $\pi^{-1/2}\Gamma(1/2) = 1$, $\Gamma(s/2) = 2/s + O(1)$, and $\pi^{-s/2} = 1 + O(s)$ give (HRes1) and their sum (HRes2). $\square$

Thus the residue map factors through the explicitly recovered first moment:

$$\rho_\mu \mapsto \frac{\zeta'(\rho)c_1(\mu)}{\rho\zeta(\rho+1)} = m_1(\mu) \mapsto -2tm_1(\mu).$$

The full germ retains every higher moment through (H4), with measure reconstruction given by (H7).

3.3 The complete zero cluster removes the simplicity restriction

The preceding source theorem used a simple scalar branch. The full cluster gives an inverse for **every** nontrivial zero, with its original multiplicity retained. Fix a nontrivial zero ρ of exact order $r \geq 1$. Choose a closed disc about ρ entirely inside $0 < \operatorname{Re} s < 1$, whose boundary Γ contains no zero and whose interior contains no zero of ζ other than ρ . Thus the entire disc also excludes the pole at 1. Shrink the complex t -disc so $Z_{\mu,t}$ is nonzero on Γ and has exactly r zeros inside, with multiplicity. The normal-convergence estimate following (H1) supplies joint holomorphicity and uniform convergence to ζ on Γ ; Rouché's theorem then proves the exact zero count.

For $1 \leq j \leq r$ define the holomorphic power-sum germs

$$q_j(t) = \frac{1}{2\pi i} \oint_\Gamma (s - \rho)^j \frac{\partial_s Z_{\mu,t}(s)}{Z_{\mu,t}(s)} ds.$$

They are exactly the sums of the j th powers of the displacements of the r zeros, counted with multiplicity, by the residue theorem. Set $e_0(t) = 1$ and recursively

$$je_j(t) = \sum_{i=1}^j (-1)^{i-1} e_{j-i}(t) q_i(t).$$

Newton's identities show that the polynomial

$$P_\mu(s, t) = \sum_{j=0}^r (-1)^j e_j(t) (s - \rho)^{r-j} = (s - \rho)^r + \sum_{N \geq 1} P_N(s) t^N \quad (\text{H10})$$

has precisely the full zero cluster as its divisor, and every $P_N(s)$ has degree at most $r - 1$ in the original displacement $s - \rho$. The original point ρ and monic leading coefficient one are both retained. The polynomial is independent of any enumeration or Puiseux branch choices.

There is a jointly holomorphic unit $U_\mu(s, t)$ on a smaller disc such that

$$Z_{\mu,t}(s) = U_\mu(s, t)P_\mu(s, t), \quad U_\mu(s, t) = \sum_{N \geq 0} U_N(s)t^N, \quad U_0(s) = \frac{\zeta(s)}{(s - \rho)^r}. \quad (\text{H11})$$

For completeness, joint holomorphicity does not need to be assumed from the root parametrization. Choose an inner circle still enclosing all zeros for small t ; on that circle the quotient Z/P has nonzero denominator. Its Cauchy integral against $1/(\zeta - s)$ defines a jointly holomorphic function of (s, t) inside, since the integrand is jointly holomorphic on the fixed contour. For each fixed t , the zeros of Z and P agree with their exact multiplicities, so the quotient has removable singularities at the roots and the Cauchy integral equals Z/P . The quotient has no zero, because the two divisors already coincide. This proves (H11). The contour variable ζ here denotes a complex integration variable, not the Riemann zeta function.

Now the input moments have the exact recursive recovery

$$m_N(\mu) = \frac{1}{b_N(\rho)} \sum_{k=1}^N U_{N-k}(\rho)P_k(\rho), \quad (\text{H12})$$

followed by the exact analytic division

$$U_N(s) = \frac{m_N(\mu)b_N(s) - \sum_{k=1}^N U_{N-k}(s)P_k(s)}{(s - \rho)^r}. \quad (\text{H13})$$

In (H13) the numerator vanishes to order at least r ; its holomorphic extension at ρ has value equal to its r th derivative there divided by $r!$. Thus the formula retains every derivative needed by a multiple zero, rather than dividing a nonvanishing germ by a zero.

Proof. Comparing the coefficient of t^N in (H1) and (H11) gives

$$m_N(\mu)b_N(s) = U_N(s)(s - \rho)^r + \sum_{k=1}^N U_{N-k}(s)P_k(s).$$

Evaluation at $s = \rho$ and the nonvanishing (H2) prove (H12). Rearranging the same analytic equality proves (H13), including the divisibility and the value at the centre. Starting from the completely known U_0 , the pair (H12)–(H13) is triangular and recovers every moment from the polynomial-cluster germ.

Two measures with the same polynomial-cluster jet through order N therefore have the same moments through N , by induction. Conversely, equal moments through N give $Z_{\mu,t} - Z_{\nu,t} = O(t^{N+1})$ as holomorphic functions on the fixed circle. Since both denominators remain nonzero there, their logarithmic derivatives, contour power sums, Newton coefficients and hence polynomials (H10) agree through order N . This proves the converse without choosing scalar zero branches. Finally (H7) reconstructs the measure, so the cluster germ at any nontrivial zero determines the entire compact probability input. $\square$

For $r = 1$, $P_N(s) = -c_N$, so (H12)–(H13) reduce to the inverse branch recurrence. For $r > 1$, the contour construction keeps all r zeros and their multiplicities even when individual zeros have fractional-power trajectories. This is a fully proved extension of the historical source's simple-zero statement.

3.4 The historical period map preserves every finite spectral jet

The transcript's marked Frobenius character is

$$z_{p,\omega}(s) = \omega p^{s-1/2}, \quad p > 1, \quad \omega \in S^1, \quad a = \log p > 0. \quad (\text{J1})$$

For the original prime-circle construction, p is a rational prime and $\omega = \psi(\text{Frob}_p)$; the proof below only uses the displayed positive a and unit-modulus ω . It retains the mark and phase.

Fix $\rho \in \mathbb{C}$ and an integer multiplicity $m \geq 1$, and write

$$z = \omega p^{\rho-1/2}, \quad E_{\rho,m} = \mathbb{C}[X]/(X - \rho)^m, \quad L_{z,m} = \mathbb{C}[U, U^{-1}]/(U - z)^m.$$

The original spectral coordinate remains X , and the original Laurent period generator remains U .

Exact jet isomorphism. There is a complex-algebra isomorphism

$$\Phi_{\rho,m} : L_{z,m} \xrightarrow{\sim} E_{\rho,m}, \quad [U] \mapsto z \sum_{k=0}^{m-1} \frac{a^k (X - \rho)^k}{k!}. \quad (\text{J2})$$

Its inverse is

$$[X] \mapsto \rho + \frac{1}{a} \sum_{k=1}^{m-1} \frac{(-1)^{k+1}}{k} \left(\frac{U - z}{z} \right)^k. \quad (\text{J3})$$

For every $0 \leq k < m$, (J2) induces on the k th associated conormal grade the exact multiplier

$$[(U - z)^k] \mapsto (za)^k [(X - \rho)^k]. \quad (\text{J4})$$

Proof. The target of (J2) is a unit: it is z times a polynomial with constant term one and nilpotent remaining part. Its difference from z lies in $(X - \rho)$, so its m th power vanishes. The Laurent universal property and the quotient therefore define (J2). In $\mathbb{C}[t]/t^m$, truncated $\exp(t)$ and $\log(1 + t)$ are inverse operations. This identity can be proved without convergence: formal differentiation shows $(\log(\exp t))' = 1$ and its constant term is zero; likewise $\exp(\log(1 + t))/(1 + t)$ has derivative zero and constant term one. Reducing the resulting formal identities modulo t^m gives (J3) and both composites. Finally

$$\Phi(U - z) = za(X - \rho) + (X - \rho)^2 \left(z \sum_{j=2}^{m-1} \frac{a^j (X - \rho)^{j-2}}{j!} \right),$$

so its k th power modulo $(X - \rho)^{k+1}$ is exactly $(za)^k (X - \rho)^k$. For $m = 1$, the sums in (J3) are empty and both rings are the specified scalar fibres, as required. $\square$

For a finite primary spectral block $A = \rho I + N$ with $N^m = 0$, this means that its actual period operator is exactly

$$\omega \exp(a(A - \tfrac{1}{2}I)) = z \sum_{k=0}^{m-1} \frac{a^k N^k}{k!}. \quad (\text{J5})$$

There is no deletion of nilpotent multiplicity in this map. Indeed, $\exp(aN) - I = N(aI + NB(N))$; its second factor is an invertible polynomial in N , so its powers have exactly the same kernels as the corresponding powers of N . This proves preservation of all Jordan block lengths locally.

3.5 Exact conjugate-reciprocal involution on the jets and conormal

Define the original spectral involution

$$\rho^\dagger = 1 - \bar{\rho}, \quad z^\dagger = \omega p^{\rho^\dagger - 1/2} = 1/\bar{z}. \quad (\text{J6})$$

On polynomial classes define the conjugate-linear algebra isomorphism

$$\mathcal{S}_E : E_{\rho,m} \longrightarrow E_{\rho^\dagger,m}, \quad [f(X)] \longmapsto [\bar{f}(1 - X)]. \quad (\text{J7})$$

On Laurent classes define

$$\mathcal{S}_L : L_{z,m} \longrightarrow L_{z^\dagger,m}, \quad \left[\sum c_n U^n \right] \longmapsto \left[\sum \bar{c}_n U^{-n} \right]. \quad (\text{J8})$$

Both have square the identity after returning to the original labelled block, and the full finite-jet diagram commutes:

$$\boxed{\mathcal{S}_E \Phi_{\rho,m} = \Phi_{\rho^\dagger,m} \mathcal{S}_L.} \quad (\text{J9})$$

Proof. The polynomial generator transforms by

$$X - \rho \longmapsto 1 - X - \bar{\rho} = -(X - \rho^\dagger).$$

The Laurent generator transforms by the exact identity

$$U - z \longmapsto U^{-1} - \bar{z} = -\bar{z} U^{-1} (U - z^\dagger). \quad (\text{J10})$$

The factor $-\bar{z}U^{-1}$ is a unit, so the respective m th-power ideals map to one another and both maps are defined. Applying each twice fixes constants and the original generator. To check (J9), it suffices to check the Laurent generator and constants. Its left image is

$$\bar{z} \sum_{k=0}^{m-1} \frac{(-a)^k (X - \rho^\dagger)^k}{k!},$$

which is the inverse of $z^\dagger \exp(a(X - \rho^\dagger))$ modulo $(X - \rho^\dagger)^m$ and hence the right image. $\square$

For clarity, the entire nonlinear mixing of Laurent Taylor coefficients under (J8) is explicit. With $t' = U - z^\dagger$, for $1 \leq k < m$ one has

$$\boxed{\mathcal{S}_L((U - z)^k) = \sum_{\ell=k}^{m-1} (-1)^\ell \binom{\ell-1}{\ell-k} \bar{z}^{k+\ell} (t')^\ell.} \quad (\text{J11})$$

Proof. By (J10), the expression is

$$(-\bar{z})^k U^{-k} (t')^k = (-1)^k \bar{z}^{2k} (t')^k (1 + \bar{z}t')^{-k}.$$

The formal binomial expansion $(1 + v)^{-k} = \sum_{j \geq 0} (-1)^j \binom{k+j-1}{j} v^j$, truncated when the total degree reaches m , gives (J11). The constant basis vector maps to one separately. $\square$

In particular, the conormal maps are

$$[X - \rho] \longmapsto -[X - \rho^\dagger], \quad [U - z] \longmapsto -\bar{z}^2 [U - z^\dagger], \quad (\text{J12})$$

and their compatibility with (J4) is an exact equality of coefficients:

$$-\bar{z}^2 (z^\dagger a) = -\bar{z}a = \bar{z}\bar{a}(-1).$$

The higher grade k has factors $(-1)^k$ and $(-1)^k \bar{z}^{2k}$ respectively, with the corresponding $(za)^k$ comparison. This supplies the all-order anti-involution/conormal morphism requested by the historical construction; it does not erase the higher terms in (J11).

On the real analytic radial chart $s = \beta + i\gamma$, the same exact map gives

$$\nu = \log |z| = (\beta - \tfrac{1}{2}) \log p, \quad d\nu = (\log p) d\beta, \quad c_{-+} = 4(\beta - \tfrac{1}{2}), \quad \nu = \frac{\log p}{4} c_{-+}. \quad (\text{J13})$$

The source's marked transverse chart $j_S(\nu) = [((e^\nu, 1, \dots, 1), 0)]$ pulls its displayed form $\alpha = d \log |a_\infty|$ back to $d\nu$, since $\log(e^\nu) = \nu$. Thus the composite character→radial chart has pullback $\alpha = (\log p) d\beta$, with the original prime and tessarine factor retained. The local identity requires no conjecture about zeros. A global identification with any separately defined Weil quadratic form is not asserted by this local chart calculation; the exact algebraic and chart maps proved here are the morphisms actually available from this source block.

3.6 Period collisions: the exact unlabelled image and its missing fibres

Let Z be a finite set of distinct spectral points ρ , with original multiplicities $m_\rho \geq 1$, and

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[X]/h(X) \cong \prod_{\rho \in Z} E_{\rho, m_\rho}.$$

The last map is the CRT map $f \mapsto (f \bmod (X - \rho)^{m_\rho})$; pairwise coprimality gives its inverse by the Bezout identity. No multiplicity is discarded.

Set $z_\rho = \omega p^{\rho-1/2}$. The equality of two period values is exactly

$$z_\rho = z_\sigma \iff \rho - \sigma \in \frac{2\pi i}{\log p} \mathbb{Z}. \quad (\text{J14})$$

The proof is the exact kernel $2\pi i \mathbb{Z}$ of the complex exponential. In particular, a fixed prime's character cannot automatically be treated as globally injective on a zero packet. No noncollision claim for actual zeta zeros is inserted here.

For each distinct period value z , put $M_z = \max_{\rho: z_\rho=z} m_\rho$ and

$$P(U) = \prod_z (U - z)^{M_z}, \quad L_P = \mathbb{C}[U, U^{-1}]/P(U).$$

There is an injective algebra map

$$\Phi_Z : L_P \hookrightarrow E_h \quad (\text{J15})$$

whose ρ th component is Laurent reduction modulo $(U - z_\rho)^{m_\rho}$ followed by (J2). Its image consists exactly of the tuples which, after (J3) on each labelled primary block, have the same Laurent Taylor coefficients at a common period value through every order visible in both blocks. More explicitly, if $a_\rho \in E_{\rho, m_\rho}$, write

$$\Phi_{\rho, m_\rho}^{-1}(a_\rho) = \sum_{k=0}^{m_\rho-1} b_{\rho, k} (U - z_\rho)^k.$$

Then (a_ρ) lies in the image exactly when

$$b_{\rho,k} = b_{\sigma,k} \quad \text{whenever } z_\rho = z_\sigma \text{ and } 0 \leq k < \min(m_\rho, m_\sigma). \quad (\text{J16})$$

The exact dimension and cokernel dimension as complex vector spaces are

$$\dim \operatorname{im} \Phi_Z = \sum_z M_z, \quad \dim \operatorname{coker} \Phi_Z = \sum_\rho m_\rho - \sum_z M_z. \quad (\text{J17})$$

Proof. Use CRT also on P . For a fixed common value z , a Laurent polynomial vanishes on every corresponding spectral block exactly when it is divisible by $(U - z)^{m_\rho}$ for every such ρ , which is equivalent to divisibility by $(U - z)^{M_z}$. Therefore the kernel of the original Laurent evaluation map is exactly (P) , giving (J15). Reduction of one degree- $< M_z$ polynomial to each smaller truncation gives (J16). Conversely choose any one block of maximal multiplicity M_z ; its Laurent coefficients define the common polynomial, and (J16) forces all other components to be its truncations. These constructions prove the image equality, while the dimensions follow from the explicit polynomial bases. $\square$

Keeping the labels restores a full exact isomorphism:

$$\prod_{\rho \in Z} L_{z_\rho, m_\rho} \xrightarrow{\Pi \Phi_{\rho, m_\rho}} E_h. \quad (\text{J18})$$

Thus the historical period map and the current finite arithmetic jet object have a proved relation at both the labelled and unlabelled levels. The collision compatibility space is exactly the image (J16). An explicit quotient records every lost independent coefficient: choose one maximal block ρ_z over each z , with $m_{\rho_z} = M_z$, and define the complex-linear map

$$\Delta_Z : E_h \longrightarrow \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho}, \quad (a_\rho) \longmapsto (b_{\rho,k} - b_{\rho_z,k})_{\substack{z, \rho \neq \rho_z \\ 0 \leq k < m_\rho}}. \quad (\text{J19})$$

It is surjective: set all maximal-block coefficients to zero and prescribe each remaining block's coefficients to be the desired output. Its kernel is exactly (J16). Consequently

$$0 \longrightarrow L_P \xrightarrow{\Phi_Z} E_h \xrightarrow{\Delta_Z} \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho} \longrightarrow 0 \quad (\text{J20})$$

is an exact sequence of complex vector spaces. The first map is an algebra homomorphism; the last map is explicitly linear and is not incorrectly promoted to a ring homomorphism. Its target has the cokernel dimension in (J17).

If Z is stable under $\rho \mapsto 1 - \bar{\rho}$ with the same multiplicities, the maps (J7) and (J8), together with permutation of the labelled factors, give conjugate-linear involutions on both sides of (J18). Formula (J9) proves that (J18) intertwines them. The unlabelled construction (J15) is also stable: $z \mapsto 1/\bar{z}$ permutes the common period values and preserves M_z ; (J10) carries the defining ideal of one value to that of the paired value.

3.7 Two prime periods recover the complete finite spectral algebra

Retain a finite nonempty set Z of distinct spectral points with the original orders $m_\rho \geq 1$, and $E_h = \mathbb{C}[X]/h(X)$, where $h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}$. Choose distinct rational primes p, q and fixed marks $\omega_p, \omega_q \in S^1$. Put

$$a = \log p, \quad b = \log q, \quad u_\rho = \omega_p e^{a(\rho-1/2)}, \quad v_\rho = \omega_q e^{b(\rho-1/2)}.$$

Define the units $\mathfrak{A}_p, \mathfrak{A}_q \in E_h$ by their complete original spectral CRT components:

$$\begin{aligned} (\mathfrak{A}_p)_\rho &= u_\rho \sum_{k=0}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \\ (\mathfrak{A}_q)_\rho &= v_\rho \sum_{k=0}^{m_\rho-1} \frac{b^k (X - \rho)^k}{k!}. \end{aligned} \tag{P1}$$

The period-algebra map therefore has the exact type

$$\mathcal{P}_{p,q} : \mathbb{C}[U^{\pm 1}, V^{\pm 1}] \longrightarrow E_h, \quad U \longmapsto \mathfrak{A}_p, \quad V \longmapsto \mathfrak{A}_q. \tag{P2}$$

This map uses the actual periods of the original generator; the notation $\mathcal{P}_{p,q}$ leaves the theta map Θ unchanged.

Theorem 3.1 (Complete two-prime Laurent presentation). *The map (P2) is surjective, with every local order retained. More precisely, in $\mathcal{L} = \mathbb{C}[U^{\pm 1}, V^{\pm 1}]$, put*

$$\begin{aligned} B_\rho(U) &= v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j, \\ I_\rho &= ((U - u_\rho)^{m_\rho}, V - B_\rho(U)). \end{aligned} \tag{P3}$$

Here $\binom{b/a}{0} = 1$, and for $j \geq 1$ the coefficient is $(b/a)(b/a - 1) \cdots (b/a - j + 1)/j!$. The ideals I_ρ are pairwise comaximal, and

$$\begin{aligned} \ker \mathcal{P}_{p,q} &= \bigcap_{\rho \in Z} I_\rho = \prod_{\rho \in Z} I_\rho, \\ \mathcal{L} / \bigcap_{\rho \in Z} I_\rho &\xrightarrow{\sim} E_h. \end{aligned} \tag{P4}$$

The map on each primary quotient is the original finite exponential map (J2), with the second generator determined by (P3).

Proof. First the two marked periods separate distinct complex points. If their values at ρ, σ agree, there exist integers k, l with

$$a(\rho - \sigma) = 2\pi i k, \quad b(\rho - \sigma) = 2\pi i l.$$

Eliminating $\rho - \sigma$ gives $k \log q = l \log p$, hence $q^k = p^l$. Prime valuations in the positive rationals give $k = l = 0$, including negative integer exponents. Thus $\rho = \sigma$. Each fixed phase cancels only within its own prime character; neither mark was changed.

For a fixed ρ , set $y = (U - u_\rho)/u_\rho$. The quotient $\mathcal{L}/I_\rho$ is exactly $\mathbb{C}[y]/y^{m_\rho}$: its generators satisfy

$$U = u_\rho(1 + y), \quad V = v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} y^j.$$

Both expressions are units, since their constant terms are nonzero and their remaining parts are nilpotent. The Laurent inverses therefore impose no extra relation. These expressions give a map from the Laurent quotient to $\mathbb{C}[y]/y^{m_\rho}$, whose inverse sends y to $(U - u_\rho)/u_\rho$; their composites fix the displayed generators.

The map to E_{ρ, m_ρ} is

$$y \longmapsto \sum_{k=1}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \quad X \longmapsto \rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} y^j \tag{P5}$$

for it and its inverse, respectively. The finite logarithm and exponential identities proved in (J2)–(J3) verify both composites. The identity $(1+y)^{b/a} = \exp((b/a) \log(1+y))$, reduced modulo y^{m_ρ} , verifies the prescribed V -image. For completeness its binomial coefficients are fixed by the formal differential equation $(1+y)F' = (b/a)F$, $F(0) = 1$: $(j+1)[y^{j+1}]F = (b/a-j)[y^j]F$. This recurrence gives exactly the coefficients in (P3). Thus I_ρ is precisely the kernel of the ρ -component of (P2), including order one.

For two different labels, either $u_\rho \neq u_\sigma$, or their v -values differ by the separation already proved. In the first case the powers of the two distinct linear U -factors have a polynomial Bezout identity, proving $I_\rho + I_\sigma = \mathcal{L}$. In the second case, $(V - v_\rho)^{m_\rho} \in I_\rho$: modulo I_ρ , $V - v_\rho = B_\rho(U) - v_\rho$ is a multiple of $U - u_\rho$. The corresponding powers of the two distinct linear V -factors again give a Bezout identity. Hence all primary kernels are pairwise comaximal.

The Chinese-remainder map now gives

$$\mathcal{L} / \bigcap_{\rho} I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} \mathcal{L} / I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} E_{\rho, m_{\rho}} \xleftarrow{\sim} E_h. \quad (\text{P6})$$

For clarity, the first map is onto because Bezout identities give classes ϵ_ρ equal to one in factor ρ and zero in every other factor; summing ϵ_ρ times prescribed representatives recovers any tuple. Its kernel is the intersection by definition. Pairwise comaximality also gives intersection equal to product, first for two ideals by multiplying a Bezout identity by an element of the intersection, and then by induction. These arguments prove every assertion in (P4).

The original coordinate itself has the explicit inverse image

$$[X] \mapsto \sum_{\rho \in Z} \epsilon_\rho \left(\rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j \right). \quad (\text{P7})$$

Each term is exactly (P5) in its own primary factor; the other factors vanish through ϵ_ρ . Thus the inverse retains the original generator and all of its nilpotent jets. $\square$

This presentation is an exact statement for each finite packet. The single-prime collision quotient (J20) and the two-prime map (P6) specify their respective information. The factorwise split lifts below retain their individual support labels.

3.8 Split-Zero support is preserved by the exact maps

For any ring R , keep the original $G(R) = \{\tau\} \sqcup R^\bullet$, global zero τ , and supported zero $e_R = 0_R^\bullet$. Every ring map $f : R \rightarrow S$ gives

$$G(f)(\tau) = \tau, \quad G(f)(r^\bullet) = f(r)^\bullet. \quad (\text{S1})$$

This preserves addition and multiplication: if one input is τ , use its defining identity/absorber laws, and if both are supported, use the ring homomorphism identities with the supported tag retained. In particular $G(f)(e_R) = e_S \neq \tau$, even if f has a nontrivial kernel.

Apply (S1) separately to each map (J2), and then take the product of these lifted semiring isomorphisms. The resulting map is

$$\prod_{\rho \in Z} G(L_{z_\rho, m_\rho}) \xrightarrow{\sim} \prod_{\rho \in Z} G(E_{\rho, m_\rho}). \quad (\text{S2})$$

Every individual supported-zero coordinate is retained. The support map on both sides is the full vector of presence indicators in $\{0, 1\}^Z$; under (S2) it is unchanged.

For a finite nonempty index set $\mathcal{I}$, the comparison with applying G once to a product ring is the map

$$\begin{aligned} \iota_R : G \left(\prod_{i \in \mathcal{I}} R_i \right) &\longrightarrow \prod_{i \in \mathcal{I}} G(R_i), \\ \tau &\longmapsto (\tau)_i, \quad (r_i)_i^\bullet \longmapsto (r_i^\bullet)_i. \end{aligned} \quad (\text{S4})$$

It is an injective unital semiring homomorphism. For two supported inputs, the operations agree coordinate by coordinate by the ring laws. When a τ occurs, its additive-identity and multiplicative-absorber laws also agree coordinate by coordinate. The two sorts of image are disjoint because $\mathcal{I}$ is nonempty; equality of supported images recovers every r_i . Its support image is exactly the empty and full support vectors. For ring maps $f_i : R_i \rightarrow S_i$, direct evaluation proves

$$\left(\prod_i G(f_i) \right) \iota_R = \iota_S G \left(\prod_i f_i \right). \quad (\text{S5})$$

This is the exact natural comparison with the factorwise map (S2). If $\mathcal{I}$ is empty, the same comparison has the two-element semiring $G(0)$ as source and the one-element empty product as target, so it is not injective; its unique map identifies e and τ . The product map (S2) itself remains the identity between empty products in that case.

For the unlabelled map, a source factor $G(L_{z,M_z})$ maps to all spectral factors over z , using (J2) after the quotient $L_{z,M_z} \twoheadrightarrow L_{z,m_\rho}$. This is injective because one target block has multiplicity M_z . Its support image consists exactly of the vectors which are constant on every collision class $q^{-1}(z)$, where $q : Z \rightarrow \{z_\rho\}$ is $\rho \mapsto z_\rho$. A source τ gives τ in every such factor, while a supported element gives a supported element in every factor, including factors in which its amplitude becomes zero. The support embedding is exactly

$$q^{-1} : \mathcal{P}(\{z_\rho\}) \hookrightarrow \mathcal{P}(Z), \quad (\text{S3})$$

which preserves empty set, unions and intersections. The direct-image map $q_* : \mathcal{P}(Z) \rightarrow \mathcal{P}(\{z_\rho\})$ preserves unions and has $q_* q^{-1} = \text{id}$, but need not preserve intersections. For two distinct labels ρ, σ in the same collision class, $\{\rho\} \cap \{\sigma\} = \emptyset$ while $q_* \{\rho\} \cap q_* \{\sigma\} = \{z\}$. Thus (S3), rather than an unsupported lattice quotient isomorphism, is the exact support relation; the amplitude compatibility is (J16).

For a same-centre truncation $m' \leq m$, the quotient

$$L_{z,m} \twoheadrightarrow L_{z,m'}, \quad E_{\rho,m} \twoheadrightarrow E_{\rho,m'}$$

commutes with (J2) because the finite power series simply reduce to the corresponding lower nilpotence order. Its lifted map sends the supported nonzero class $(U - z)^{m'}$ to e when $m' < m$, and never to τ . This supplies a concrete example of the original distinction between a vanished jet amplitude and absent support, inside the current arithmetic finite-jet packet.

3.9 All-order Hurwitz motion transported to the period and tesserine coordinates

For the actual zero branch (H3), the exact period germ is

$$z_\mu(t) = \omega p^{\rho_\mu(t)-1/2} = z_\rho \exp \left((\log p) \sum_{n \geq 1} c_n(\mu) t^n \right). \quad (\text{C1})$$

Its N th coefficient is

$$[t^N]z_\mu(t) = z_\rho \sum_{k=1}^N \frac{(\log p)^k}{k!} \sum_{\substack{n_1+\dots+n_k=N \\ n_j \geq 1}} \prod_{j=1}^k c_{n_j}(\mu) \quad (N \geq 1). \quad (\text{C2})$$

The empty coefficient is z_ρ . Conversely, using the unique logarithm germ vanishing at $z_\mu/z_\rho = 1$,

$$c_N(\mu) = \frac{1}{\log p} [t^N] \log(z_\mu(t)/z_\rho). \quad (\text{C3})$$

Equations (C2)–(C3) are inverse because the same formal exponential/logarithm calculation used in (J2)–(J3) applies at every finite order. Combining (C3) with (H4) proves that the marked local period germ also determines the entire compact Hurwitz input.

For real t in the germ's domain,

$$\begin{aligned} \log |z_\mu(t)| &= (\log p) \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right), \\ c_{-+}(\rho_\mu(t)) &= 4 \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right). \end{aligned} \quad (\text{C4})$$

All coefficients, including tangential ones, are retained in (C1)–(C3). The real projection in (C4) is explicitly the radial/tessierine observable; its loss of imaginary coefficients is a specified map. For $t \neq 0$, the Hurwitz family generally lacks the original functional equation, so $1 - \rho_\mu(t)$ is not asserted to be another zero of that family. The tessierine coordinate formula is the proved coordinate map evaluated at the actual branch, with this spectral distinction preserved.

3.10 Mathematical scope

The maps (J2)–(J20) identify the full finite arithmetic algebra with its labelled Laurent jets and calculate the exact quotient of unlabelled single-prime observations. Equations (H12)–(H13) recover all moments from the actual zero cluster, and (H7) reconstructs the entire measure. These statements retain the original multiplicities, support coordinates and marked periods.

The constructions give exact algebraic and analytic comparison maps. They do not supply the uniform estimate on the arithmetic moment matrices required by the weight-control calculation elsewhere in this reader.

The accompanying source records retain the historical locators and independent proof reviews. The proofs in this section specify their own hypotheses and maps in full.

4 Historical support branches, ideal completion, and measured observations

This chapter gives the complete branch, ideal-completion, atomic-kernel, and operator-algebra proofs extracted and continued from the historical Split-Zero support program. Its mathematical report and independent audit are, respectively, `logbook/historical_next_responses_20260912.md` and `logbook/frame_branch_independent_audit_20260912.md`. The equation identifiers B, I, M, A and O are local to this chapter.

Source locators and comparison. Exact complete response bodies are staged at `sources/historical/next_responses/preboolean_prime_branches.md` (source lines 23137–23956; response

heading 23104; exported timestamp 7/6/2026, 12:23:35 PM), sources/historical/next_response/preboolean_arithmetic_branch_maps.md (source lines 25162–25715; heading 25058; timestamp 7/6/2026, 7:58:41 PM), and sources/historical/next_responses/pointfree_frame_arithmetic_claim.md (source lines 25754–26221; heading 25721; timestamp 7/6/2026, 10:01:27 PM). Their shared original source has SHA-256 8d1780a24df3bb31db981b5e437558de31e6d94bf21bb54103a0a336d6a73091. The respective excerpt SHA-256 values are f5f11dd6b26a3a9d6a4c77cf2f7b86898759a19fa740c993d5749c8d47c02618, e84873955876b2be5f7738f746b4e133fc479cfe726bcb85d477f9aec7d6d10e, and 57c6662732b723e451a495f0224507f61e3a969956178018af7633ae06797ac0. For each excerpt, its first line is the first indicated original source line, so every original locator has the exact excerpt locator obtained by subtracting that first source line and adding one.

The core corpus comparison is split_zero_projective_monads_surcomplex/split_support_geometry_arithmetic_curve_v11.tex, SHA-256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d. Its lines 1772–2054 already prove the ordinary prime-filter branches and their arithmetic maps; lines 2001–2018 explicitly impose arbitrary-join preservation for locale points. The support and measured realizations were also compared at lines 5006–5096 and 5346–5378. Accordingly the ordinary branch classification is a surviving historical construction, while the ideal quotient, explicit descent witnesses, atomic kernel and full L^∞ correspondence below supply the additional proved maps.

The final historical response, source lines 25982–26020 (excerpt lines 229–267), equates ordinary amplitude-fixed semiring branches with frame points without imposing arbitrary-join preservation. The exact correction is the classification (B1)–(B5), the frame quotient (I2), and the point-descent criterion (I3)–(I10). The measure-algebra construction then exhibits the kernel pairs that ordinary branches can distinguish. No external KMS classification or Morita equivalence is used as a theorem of this chapter.

4.1 Exact ordinary and continuous Split-Zero branches

The results below are stated in classical set theory. The ultrafilter construction explicitly uses Zorn’s lemma. No claim about existence or nonexistence of points in a constructive foundation is inferred from that use of choice; such an internal comparison would also have to specify its truth-value object and its available joins.

Let R be a nonzero commutative ring and let L be a nontrivial bounded distributive lattice. Retain the source carrier and all its exceptional coordinates:

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subseteq R \times L,$$

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu),$$

$$\tau_L = (0, 0_L), \quad e_L = (0, 1_L), \quad 1 = (1_R, 1_L).$$

Write $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $e = 0_R^\bullet$ distinct from global zero τ . The indicated subset is the carrier throughout.

Here $\mathbb{B} = \{0, 1\}$ has join and meet as its semiring operations. The supported copy in $G(R)$ has $r^\bullet + s^\bullet = (r + s)^\bullet$ and $r^\bullet s^\bullet = (rs)^\bullet$; its global zero satisfies $\tau + x = x + \tau = x$ and $\tau x = x\tau = \tau$. The map $G_{\mathbb{B}}(R) \rightarrow G(R)$ defined by $(0, 0) \mapsto \tau$, $(r, 1) \mapsto r^\bullet$ is a bijection and preserves these operations in each support case, hence is a unital semiring isomorphism. To verify the general carrier, the displayed operations are the restrictions of the product semiring $R \times L$. This subset is closed: a sum of nonzero amplitude has top support because at least one input has nonzero amplitude; a product of nonzero amplitude has two nonzero input amplitudes and hence top support. Every zero-amplitude

output belongs to the zero fibre. The zero and unit belong to the subset. Thus associativity, commutativity and distributivity restrict from the product, including all supported cancellations. All homomorphisms below preserve zero and unit. Write $z_\lambda = (0, \lambda)$.

Theorem B1 — all ordinary amplitude-fixed branches. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \xrightarrow{\sim} \{\beta : G_L(R) \rightarrow G(R) \text{ unital semiring map} : \beta(r, 1_L) = r^\bullet\}. \quad (\text{B1})$$

It sends ϵ to

$$\beta_\epsilon(r, 1_L) = r^\bullet, \quad \beta_\epsilon(0, \lambda) = \begin{cases} \tau & \epsilon(\lambda) = 0, \\ e & \epsilon(\lambda) = 1. \end{cases} \quad (\text{B2})$$

The map is surjective as a semiring map; its exact kernel congruence is equality of amplitudes together with equality of selected support values:

$$(r, \lambda) \sim_\epsilon (s, \mu) \iff r = s \text{ and } \epsilon(\lambda) = \epsilon(\mu). \quad (\text{B3})$$

Proof. The source formulas agree at $(0, 1_L)$ since $\epsilon(1_L) = 1$. They preserve zero and unit. On the zero fibre, preservation of addition and multiplication is precisely preservation of join and meet. If one argument has nonzero amplitude, its lattice coordinate is 1_L ; adding a zero-fibre argument leaves it unchanged and multiplying by that argument gives $(0, \lambda)$, matching respectively addition by τ or e , and multiplication by τ or e in $G(R)$. With both arguments at full support, even if their sum or product cancels to amplitude zero, both outputs are supported and ring identities prove preservation. This verifies all exceptional products and cancellations.

Conversely, $(0, \lambda)$ is additively idempotent. The only additive idempotents of $G(R)$ are τ and e : a supported $r^\bullet + r^\bullet = r^\bullet$ implies $2r = r$ in the ring, hence $r = 0$. Therefore every amplitude-fixed β sends the zero fibre into $\{\tau, e\}$. Identifying this two-element subsemiring with $\mathbb{B}$ gives a map ϵ preserving bottom, top, meet and join. Formula (B2) reconstructs β ; both constructions are inverse. Every amplitude and τ occurs in the image, and inspecting (B2) gives exactly (B3). $\square$

Now let L be a frame: all joins exist and finite meets distribute over arbitrary joins. Its zero fibre has specified infinitary joins

$$\bigvee_{i \in I}^{\text{fib}} (0, \lambda_i) = (0, \bigvee_{i \in I} \lambda_i).$$

Theorem B2 — exactly the continuous branches. Under (B1), the subset of branches preserving all these zero-fibre joins is exactly

$$\text{Hom}_{\text{Frm}}(L, \mathbb{B}). \quad (\text{B4})$$

Indeed the corresponding equality for (B2) says exactly $\epsilon(\bigvee_i \lambda_i) = \bigvee_i \epsilon(\lambda_i)$, and finite meets are already preserved. This is the precise additional structure required by the transcript's final frame formulation. Its ordinary algebraic maps retain a larger, explicitly identified category.

The classification extends to every map between original lattice-split carriers. For commutative unital rings R, A and nontrivial bounded distributive lattices L, M ,

$$\text{Hom}_{\text{Semi}}(G_L(R), G_M(A)) \cong \text{Hom}_{\text{Ring}}(R, A) \times \text{Hom}_{\text{DLat}_{0,1}}(L, M). \quad (\text{B5})$$

The pair (h, f) gives $(r, \lambda) \mapsto (h(r), f(\lambda))$. Conversely an arbitrary semiring map sends each additively idempotent z_λ to amplitude zero, say $z_{f(\lambda)}$; this preserves finite joins, meets and bottom. The identity $e_L = (1_R, 1_L) + (-1_R, 1_L)$ makes its image have top support, because one summand is the target unit; additive idempotence makes its amplitude zero. Thus $e_L \mapsto e_M$ and f preserves

top. Now $(r, 1_L) + e_L = (r, 1_L)$ forces its image to have top support. Write that image $(h(r), 1_M)$. The supported operations give a unital ring map h , and the formulas recover every source element, consistently at e_L . Coordinate composition proves naturality and both inverse constructions. In particular the entire two-factor datum is recoverable from the semiring map; scalar amplitude and support are not freely interchanged.

4.2 The exact frame comparison retaining all ordinary branches

For the underlying bounded distributive lattice of a frame L , let $\text{Idl}(L)$ be the set of lattice ideals. An ideal is a nonempty lower set closed under finite joins. Its bottom is $\{0_L\}$, its top is L , its finite meet is intersection, and the join of a family of ideals is

$$\bigvee_{\alpha} I_{\alpha} = \{x \in L : x \leq y_1 \vee \cdots \vee y_n \text{ for some finite } n \geq 0, y_j \in \bigcup_{\alpha} I_{\alpha}\}. \quad (\text{I1})$$

The empty finite join is 0_L ; this retains the bottom ideal in every formula.

Theorem I1 — frame quotient and principal-ideal section. The maps

$$q : \text{Idl}(L) \rightarrow L, \quad I \mapsto \bigvee I, \quad \eta : L \rightarrow \text{Idl}(L), \quad \lambda \mapsto \downarrow \lambda \quad (\text{I2})$$

have the following exact properties:

1. $\text{Idl}(L)$ is a frame and q is a surjective frame homomorphism.
2. η is an injective bounded-lattice homomorphism, $q\eta = \text{id}_L$, and $q(I) \leq \lambda \iff I \subseteq \eta(\lambda)$.
3. The kernel congruence of q is exactly $I \sim J \iff \bigvee I = \bigvee J$. No claim that η preserves arbitrary joins is made.

Proof. Formula (I1) is an ideal and is the least ideal containing every I_{α} ; intersections give infima. To prove frame distributivity, let $x \in I \cap \bigvee_{\alpha} J_{\alpha}$. Choose $x \leq y_1 \vee \cdots \vee y_n$ with $y_j \in J_{\alpha_j}$. In the distributive lattice,

$$x = x \wedge (y_1 \vee \cdots \vee y_n) = (x \wedge y_1) \vee \cdots \vee (x \wedge y_n).$$

Each $x \wedge y_j$ belongs to $I \cap J_{\alpha_j}$, proving $I \cap \bigvee_{\alpha} J_{\alpha} \subseteq \bigvee_{\alpha} (I \cap J_{\alpha})$; the reverse inclusion follows from containment. Thus $\text{Idl}(L)$ is a frame.

The supremum of the right side of (I1) equals $\bigvee_{\alpha} \bigvee I_{\alpha}$, proving arbitrary-join preservation by q , including the empty family. Since L is a frame,

$$(\bigvee I) \wedge (\bigvee J) = \bigvee_{i \in I, j \in J} (i \wedge j) = \bigvee (I \cap J).$$

The last equality holds because each $i \wedge j$ is in the intersection, and each $x \in I \cap J$ is the term $x \wedge x$. Hence q preserves finite meets and the top. Principal ideals satisfy $\downarrow \lambda \cap \downarrow \mu = \downarrow (\lambda \wedge \mu)$ and $\downarrow \lambda \vee \downarrow \mu = \downarrow (\lambda \vee \mu)$; they also preserve bottom/top. Their suprema are the original elements, proving $q\eta = \text{id}_L$, injectivity and surjectivity. The adjunction equivalence and the kernel description follow immediately from the definition of supremum. $\square$

Theorem I2 — all algebraic branches become frame points before the quotient. There are mutually inverse bijections

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \cong \text{Hom}_{\text{Frm}}(\text{Idl}(L), \mathbb{B}), \quad (\text{I3})$$

given by

$$\widehat{\epsilon}(I) = \bigvee_{\lambda \in I} \epsilon(\lambda) \quad \text{and} \quad \epsilon = \widehat{\epsilon}\eta. \quad (I4)$$

A frame point $\widehat{\epsilon}$ in (I3) factors through q if and only if its original ϵ is a frame point of L . Thus the exact inclusion of point sets is

$$q^* : \text{Pt}(L) \hookrightarrow \text{Pt}(\text{Idl}(L)), \quad \epsilon \mapsto \epsilon q, \quad (I5)$$

and its image is the points constant on the kernel congruence in Theorem I1.

Proof. In (I4), the value is 1 precisely when the ideal meets the prime filter $\epsilon^{-1}(1)$. For a join of ideals, a value 1 is witnessed by an element below a finite join of source elements; finite primeness then gives an element of value 1 in one of those source ideals. The reverse implication follows by containment, proving arbitrary-join preservation. For intersections, if I and J each have a value 1 element i, j , their meet $i \wedge j \in I \cap J$ has value 1; the reverse implication is immediate. Top and bottom follow from $\epsilon(1) = 1, \epsilon(0) = 0$. Therefore $\widehat{\epsilon}$ is a frame homomorphism.

Every ideal is the join in $\text{Idl}(L)$ of its principal ideals, so an arbitrary frame point is completely determined by its restriction along η , giving both inverse identities. If ϵ preserves arbitrary joins, $\widehat{\epsilon}(I) = \epsilon(\bigvee I) = (\epsilon q)(I)$. Conversely if $\widehat{\epsilon} = \chi q$ for a frame map χ , then $\epsilon = \widehat{\epsilon}\eta = \chi q\eta = \chi$, so ϵ is a frame map. A set map constant on the fibres of surjective q factors uniquely; if that map is a frame map, the induced factor preserves arbitrary joins and finite meets by lifting families through q . This proves the final image claim. $\square$

Theorem I3 — original Split-Zero comparison and exact congruence. Every bounded-lattice map $f : L \rightarrow M$ induces

$$G_f(R) : G_L(R) \rightarrow G_M(R), \quad (r, \lambda) \mapsto (r, f(\lambda)). \quad (I6)$$

For the nontrivial frames in Theorem I1 this gives a split pair of unital semiring maps

$$G_L(R) \xrightarrow{G_\eta(R)} G_{\text{Idl}(L)}(R) \xrightarrow{G_q(R)} G_L(R), \quad G_q G_\eta = \text{id}. \quad (I7)$$

They preserve τ and supported e as distinct elements. The exact kernel of $G_q(R)$ consists of equal amplitudes and equal suprema of zero-fibre ideals:

$$(r, I) \sim (s, J) \iff r = s \text{ and } \bigvee I = \bigvee J. \quad (I8)$$

Proof. A nonzero amplitude in $G_L(R)$ has full support and maps to full support because $f(1_L) = 1_M$; hence (I6) lands in the exact target carrier. Finite join and meet preservation establish the semiring identities coordinate by coordinate, including zero-amplitude cancellation. Formula (I7) follows from $q\eta = \text{id}_L$, and (I8) is equality of the two coordinates of the image. The bottom and top of the nontrivial source and target lattices differ; their images under these bounded maps are again bottom and top, proving the exact zero statements. $\square$

Thus an ordinary branch which fails zero-fibre join continuity is a fully explicit frame point **upstairs**, and the arithmetic map fails to descend through a proved frame quotient **downstairs**. The original support object and all its algebraic branches remain related by the maps (I2)–(I8).

The two available lifts have an exact comparison, which prevents a split algebra map from being mistaken for a split frame map. Given an ordinary $\epsilon : L \rightarrow \mathbb{B}$, both

$$\epsilon q : \text{Idl}(L) \rightarrow \mathbb{B}, \quad \widehat{\epsilon} : \text{Idl}(L) \rightarrow \mathbb{B} \quad (I9)$$

are bounded-lattice maps and agree on every principal ideal. The first descends through q by definition; the second preserves arbitrary joins by Theorem I2. They coincide on all ideals exactly when

ϵ preserves arbitrary joins. The forward implication follows because $\epsilon q = \widehat{\epsilon}$ and surjectivity of the frame map q force its factor ϵ to preserve joins; the converse is (I4). This also yields two amplitude-fixed Split-Zero branch maps out of $G_{\text{Idl}(L)}(R)$, with exactly the same criterion for equality.

There is a further exact localic structure. The map $j = \eta q$ is a nucleus on $\text{Idl}(L)$: it is monotone, $I \subseteq j(I)$ because each $i \in I$ lies below $\bigvee I$, it satisfies $j^2 = j$ because $q\eta = \text{id}_L$, and it preserves finite meets because both q and η do. Its fixed ideals are exactly the principal ideals. Their inherited meet and nucleus-applied join

$$\bigvee_{\alpha}^{\text{fix}} \downarrow \lambda_{\alpha} = j \left(\bigvee_{\alpha}^{\text{Idl}(L)} \downarrow \lambda_{\alpha} \right) = \downarrow \left(\bigvee_{\alpha}^L \lambda_{\alpha} \right) \quad (\text{I10})$$

identify the fixed-point frame exactly with the original L . This proves the quotient/sub-locale correspondence at the level of operations, while retaining the precise failure of η to preserve the ambient ideal-frame joins.

4.3 A fully explicit nontrivial frame with algebraic branches and no frame points

Let L be the Lebesgue probability measure algebra of $[0, 1]$: measurable subsets modulo null symmetric difference. Write $[A]$ for the class of A , keep the original measure $\mu(A)$, and use union, intersection and complement for finite Boolean operations. These operations are well defined modulo null sets, since changing their inputs changes any output only within the union of those null differences. The exact order is $[A] \leq [C]$ if and only if $\mu(A \setminus C) = 0$.

Lemma M1 — completeness and atomlessness. This L is a complete Boolean algebra, hence a frame, and has no atoms.

Proof. For an arbitrary family $([A_i])_{i \in I}$, let

$$b = \sup \{ \mu(A_{i_1} \cup \dots \cup A_{i_n}) : n \geq 0 \}.$$

Choose finite unions F_n with measures tending to b , and put $B = \bigcup_n F_n$. Replacing F_n by the union of the first n such finite unions makes them increasing without changing their defining kind, so $\mu(B) = b$. For each original A_i , the finite-union supremum bounds $\mu(F_n \cup A_i) \leq b$, and continuity from below gives $\mu(B \cup A_i) \leq b = \mu(B)$. Thus $A_i \setminus B$ is null. If $[C]$ is another upper bound, every one of the countably many constituent sets in B is contained in C modulo a null set; hence $B \setminus C$ is null. This proves that $[B]$ is the supremum. Infima follow by complements.

In a complete Boolean algebra finite meets distribute over arbitrary joins: if $v = \bigvee_i (a \wedge b_i)$, then every $b_i \leq \neg a \vee v$; taking joins and intersecting with a gives $a \wedge \bigvee_i b_i \leq v$. The reverse inequality is immediate. Thus L is a frame.

For a nonzero $[A]$, the function $F(t) = \mu(A \cap [0, t])$ is continuous because $|F(t) - F(u)| \leq |t - u|$. It takes the values 0 and $\mu(A) > 0$ at the endpoints. There is therefore t with $F(t) = \mu(A)/2$. The class of $A \cap [0, t]$ is strictly between 0 and $[A]$, so no nonzero class is an atom. $\square$

Theorem M2 — exact frame points of a complete Boolean algebra. If B is any complete Boolean algebra, its frame points correspond bijectively to its atoms:

$$\text{Hom}_{\text{Frm}}(B, \mathbb{B}) \cong \text{At}(B), \quad a \mapsto \epsilon_a(b) = \mathbf{1}_{\{a \leq b\}}. \quad (\text{M1})$$

Proof. An atom below $\bigvee_i b_i$ lies below some b_i : otherwise every $a \wedge b_i = 0$, while frame distributivity gives $a = a \wedge \bigvee_i b_i = \bigvee_i (a \wedge b_i) = 0$, a contradiction. Thus ϵ_a preserves arbitrary joins. It preserves finite meets, top and bottom directly.

Conversely, a frame map $\epsilon : B \rightarrow \mathbb{B}$ into the two-element Boolean algebra preserves complements, since complements are characterized by their finite meet 0 and join 1. By De Morgan it therefore also

preserves arbitrary meets. Put $a = \bigwedge \{b : \epsilon(b) = 1\}$. Its image is 1, so $a \neq 0$. If $0 < c \leq a$ and $\epsilon(c) = 0$, then $\epsilon(\neg c) = 1$, forcing $a \leq \neg c$ and hence $c = 0$, a contradiction. Consequently $\epsilon(c) = 1$, which by definition of a implies $a \leq c$. Thus a is an atom. Every element of the filter contains a ; conversely if $a \leq b$, then $\epsilon(b) = 1$. This proves the inverse formulas and their uniqueness. $\square$

In particular the measure algebra of Lemma M1 has **no frame points**.

It nevertheless has ordinary lattice characters. Here is an explicit witnessed failure of the required continuity. Let $D_n = [0, 2^{-n}]$ as original measurable subsets of $[0, 1]$; their classes are nonzero and any finite intersection is the smallest selected D_n . The filter generated by these classes is proper. Order proper filters containing it by inclusion; the union of every chain is proper, so Zorn's lemma gives a maximal proper filter F . In a Boolean algebra, maximal proper filters contain exactly one of $b, \neg b$: if neither lay in F , adjoining b would produce 0, so some $f \in F$ satisfies $f \wedge b = 0$ and hence $f \leq \neg b$, which already puts $\neg b \in F$. Its characteristic function is therefore a bounded-lattice homomorphism $\epsilon_F : L \rightarrow \mathbb{B}$.

Every $\epsilon_F([D_n]) = 1$, while every $\epsilon_F([D_n^c]) = 0$. But

$$\bigvee_{n \geq 1} [D_n^c] = [[0, 1] \setminus \{0\}] = 1_L, \quad \epsilon_F(1_L) = 1 \neq \bigvee_{n \geq 1} 0 = 0. \quad (\text{M2})$$

Thus (B2) gives an actual surjective semiring branch $\beta_{\epsilon_F} : G_L(R) \rightarrow G(R)$ preserving amplitudes, even though the frame has no point. Formula (M2) gives its exact failed zero-fibre join equation. The point $\hat{\epsilon}_F$ of $\text{Idl}(L)$ does not descend through q . More explicitly, set

$$J = \bigvee_{n \geq 1}^{\text{Idl}(L)} \downarrow [D_n^c].$$

Then $q(J) = 1_L = q(L)$, but $\hat{\epsilon}_F(J) = 0$ and $\hat{\epsilon}_F(L) = 1$. This is an explicit pair in the kernel congruence of q distinguished by the upstairs point; it proves the obstruction to descent at an exact source/target pair.

In fact **every** ordinary branch on this measure algebra fails already countable-join preservation. For its ultrafilter F , start at $a_0 = 1$. Split a representative of a_n into two equal-measure pieces by the continuous cumulative-measure construction in Lemma M1. Exactly one of their classes belongs to F ; call it a_{n+1} . The resulting decreasing sequence satisfies $a_n \in F$ and $\mu(a_n) = 2^{-n}$. Its infimum is 0 by continuity of measure. Consequently $b_n = \neg a_n$ increases to 1 and each $\epsilon_F(b_n) = 0$, again contradicting countable-join preservation. The ideal $I = \bigcup_n \downarrow b_n$ then satisfies $q(I) = q(L) = 1$, $\hat{\epsilon}_F(I) = 0$, and $\hat{\epsilon}_F(L) = 1$. This gives a kernel-congruence witness for each individual algebraic branch, not merely for the displayed dyadic choice.

4.4 Atomic observation and the exact support data it removes

For a complete Boolean algebra B , put

$$b_{\text{at}} = \bigvee_{a \in \text{At}(B)} a, \quad b_{\text{na}} = \neg b_{\text{at}}.$$

The frame-point evaluation map is

$$\text{ev} : B \rightarrow \mathcal{P}(\text{At}(B)), \quad x \mapsto \{a : a \leq x\}. \quad (\text{A1})$$

Theorem A1 — exact atomic quotient. This is a surjective complete Boolean-algebra homomorphism. Its kernel congruence is

$$\text{ev}(x) = \text{ev}(y) \iff x \wedge b_{\text{at}} = y \wedge b_{\text{at}}, \quad (\text{A2})$$

and its inverse image of the empty set is the full principal ideal $\downarrow b_{\text{na}}$. The complete decomposition retaining that ideal is

$$B \xrightarrow{\sim} (\downarrow b_{\text{at}}) \times (\downarrow b_{\text{na}}), \quad x \mapsto (x \wedge b_{\text{at}}, x \wedge b_{\text{na}}), \quad (\text{A3})$$

with inverse $(u, v) \mapsto u \vee v$ and respective local units $b_{\text{at}}, b_{\text{na}}$.

Proof. Each coordinate of (A1) is the frame point in (M1), so it preserves arbitrary joins and finite meets. Complements are preserved because an atom lies below exactly one of $x, \neg x$. For any subset S of atoms, the element $\bigvee_{a \in S} a$ maps exactly to S , by pairwise disjointness of distinct atoms and the arbitrary-join argument in (M1). Thus evaluation is surjective. Distributivity gives $x \wedge b_{\text{at}} = \bigvee_{a \in \text{At}(B)} (x \wedge a) = \bigvee_{a \leq x} a$, which proves (A2) and the empty-fibre claim. The two complementary components in (A3) reconstruct x by distributivity, and disjointness of $b_{\text{at}}, b_{\text{na}}$ proves both inverse identities and preservation of complete Boolean operations. $\square$

The complement on either local Boolean factor $\downarrow b$ in (A3) is explicitly $u \mapsto b \wedge \neg u$; its local top is b , not the global top of B unless $b = 1$.

This supplies an exact relation to the source's adelic measure-algebra support. In a finite measure-algebra realization, frame-point evaluation detects precisely its measure atoms; (A3) retains the entire nonatomic component rather than treating lack of points as lack of support. For a discrete countable measure with positive measure at each singleton, the singleton classes are the atoms and $b_{\text{at}} = 1$; evaluation is an isomorphism to the power set of the original places. For the Lebesgue example, $b_{\text{at}} = 0, b_{\text{na}} = 1$, and every frame-point evaluation is absent while the entire nontrivial Boolean carrier persists.

When applying $G_{\text{ev}}(R)$, retain the exceptional case: if there are no atoms, $\mathcal{P}(\text{At}(B))$ is the one-element lattice, so the exact carrier $G_{\mathcal{P}(\emptyset)}(R)$ is isomorphic to R , and this quotient identifies τ_B and e_B with ordinary zero. It is the explicit ring-reflection map on this observation, not evidence that $\tau_B = e_B$ in the original carrier. The split comparison (I7) has nontrivial lattices on both sides and preserves them as distinct; it remains available when the point-observation quotient collapses them.

Concretely, $\Psi : G_{\{\ast\}}(R) \rightarrow R, (r, \ast) \mapsto r$, is bijective and preserves both operations because the unique support coordinate is fixed. In the atomless case its composition is the exact identity

$$\Psi \circ G_{\text{ev}}(R) = p_B : G_B(R) \rightarrow R, \quad p_B(r, \lambda) = r. \quad (\text{A4})$$

4.5 The exact operator-algebraic branch map

The core manuscript's L^∞ realization permits an additional precise bridge to state spaces. Work with the same finite probability measure space, complete measure algebra B , and the complex algebra $M = L^\infty(\Omega, \mu; \mathbb{C})$ equipped with essential-supremum norm and complex conjugation. The original measure and every atomic mass are retained.

Theorem O1 — all Boolean branches extend exactly to multiplicative states. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(B, \mathbb{B}) \cong \{\chi : M \rightarrow \mathbb{C} : \chi \text{ is a unital complex-linear } \ast\text{-homomorphism}\}. \quad (\text{O1})$$

The correspondence is specified on every projection by

$$\chi \epsilon(\mathbf{1}_A) = \epsilon([A]), \quad (\text{O2})$$

and on arbitrary $f \in M$ by the uniform extension of this finite formula:

$$\chi_\epsilon \left(\sum_{j=1}^n c_j \mathbf{1}_{A_j} \right) = \sum_{j=1}^n c_j \epsilon([A_j]), \quad (O3)$$

where the A_j form the specified measurable partition of Ω modulo null sets. Every such χ is positive and has norm 1.

Proof. Exactly one member of a finite partition has ϵ -value 1: finite-join preservation gives at least one, and pairwise disjointness with meet preservation gives at most one. Thus (O3) selects one of the original values c_j . Refining two partitions by all intersections shows the formula is independent of the simple-function presentation. The unique selected intersection gives additivity and multiplicativity for two simple functions, while conjugating their values gives star preservation. The selected value has modulus at most the essential supremum of the simple function, so the map is contractive.

Every bounded measurable complex function has simple functions tending to it in essential-supremum norm: partition a square containing its essential range into finitely many squares of diameter at most 2^{-n} and use one original value in each nonempty square. The resulting evaluations are Cauchy by contractivity. Their limit defines a unique continuous extension independent of the approximating sequence. Addition, multiplication and conjugation commute with these uniform limits, proving the required homomorphism identities. Positivity follows from $f = g^*g$ with $g = \sqrt{f}$ for every nonnegative measurable f ; hence $\chi(f) = |\chi(g)|^2 \geq 0$. Contractivity and $\chi(1) = 1$ prove norm 1.

Conversely a unital star homomorphism sends projections to real idempotents 0 or 1. Since $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B$ and $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A \cap B}$, its projection values form the required Boolean lattice map. It is contractive because $\|f\|_\infty^2 \mathbf{1} - f^*f$ is the square of a bounded measurable function, so positivity gives $|\chi(f)|^2 \leq \|f\|_\infty^2$. It agrees with (O3) on simple functions, and continuity forces equality on all of M . This proves both inverse constructions. $\square$

Theorem O2 — normal branches, exact atomic masses, and failure of normality. For the correspondence (O1), the following are equivalent:

1. χ preserves suprema of bounded increasing nets of nonnegative elements of M (normality).
2. ϵ is a frame point of the complete measure algebra.
3. There is a measure atom a such that

$$\chi(f) = \frac{1}{\mu(a)} \int_a f d\mu \quad (f \in M). \quad (O4)$$

In particular the atomless Lebesgue example has multiplicative states for all of its ordinary branches, and none is normal; its exact countable failure is exhibited by (M2).

All bounded increasing nets used here have a supremum in the specified M . An explicit proof for nonnegative nets retains the finite measure: put $b = \sup_i \int_\Omega f_i d\mu$. Choose indices with integrals tending to b , and use directedness to replace them by an increasing dominating sequence. Choose representatives outside one common null set for the countably many sequence comparisons, so this selected sequence is pointwise increasing there. Its pointwise supremum g has integral b . For every fixed original index i , each finite maximum with a sequence member lies below a net member, so $\int_\Omega \max(g, f_i) d\mu \leq b = \int_\Omega g d\mu$ by monotone convergence. Thus $f_i \leq g$ almost everywhere. Every upper bound of the original net bounds the selected sequence and hence g , proving that g is the required least upper bound.

Proof. The supremum in M of the finite-union projections from an arbitrary family $([A_i])$ is the projection $\mathbf{1}_{\bigvee_i [A_i]}$: Lemma M1's countable essential union is an upper bound, and its increasing finite selected unions show it is the least upper bound also among arbitrary measurable nonnegative functions. Normality applied to this increasing net therefore gives preservation of its full projection

join. Together with (O2), this is precisely frame continuity of ϵ , proving $1 \Rightarrow 2$. By Theorem M2, a frame point corresponds to a unique atom a .

Every bounded measurable f is essentially constant on a . To see this, use the finite measurable partitions of its essential range from the proof of Theorem O1. Each partition has exactly one cell whose intersection with a is nonzero, and by atomhood that intersection equals a modulo null sets. Its diameter tends to 0, so the simple approximants restrict to constants c_n on a , with $|c_n - c_m|$ bounded by their uniform difference. Their limit c is the almost-everywhere value of f on a . Formula (O3) selects exactly these constants and gives $\chi(f) = c$; integration over the original atom gives precisely (O4), with the factor $1/\mu(a)$ retained. Thus $2 \Rightarrow 3$.

Finally let a bounded increasing net $f_i \geq 0$ have supremum f in M , and let c_i be its respective constants on a . These constants increase; put $c = \sup_i c_i$. Every f_i is bounded above by the function equal to c on a and to f off a ; hence the least-upper-bound property of f forces its constant on a to be at most c . Conversely each $f_i \leq f$ forces $c_i \leq f|_a$, and taking their supremum gives the reverse inequality. Thus $\chi(f) = \sup_i \chi(f_i)$. This proves $3 \Rightarrow 1$, including every original coefficient and atomic mass. $\square$

Equations (B1),(O1),(O2) form the promised typed connection between algebraic Split-Zero branches and operator-algebraic states. They identify exactly which branches can be normal observations of the measured support.

5 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R -modules V_l , with linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f) , where $\alpha : L \rightarrow L'$ preserves bottom and joins with R -linear fibre maps

$$f_l : V_l \rightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l. \quad (\text{D1})$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)} f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 5.1 (Full support-index reconstruction). *The diagram category just defined is equivalent to the category of $G(R)$ -semimodules and their $G(R)$ -linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx). \quad (\text{D2})$$

Its global zero is $(\perp, 0)$. On morphisms it is $\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$.

The inverse sends a semimodule M to

$$\begin{aligned} L_M &= eM, \quad l \vee k = l + k, \quad \perp = 0_M, \\ V_l &= \{m \in M : em = l\}, \quad \rho_{lk}(m) = m + k \quad (l + k = k). \end{aligned} \quad (\text{D3})$$

The natural comparison maps are

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \mapsto m, \quad m \mapsto (em, m), \quad (\text{D4})$$

and the label-and-fibre isomorphism

$$L \longrightarrow e\mathcal{T}(L, V), \quad l \mapsto (l, 0), \quad x \in V_l \mapsto (l, x). \quad (\text{D5})$$

Proof. For (D2), threefold addition transports both parenthesizations to $l \vee k \vee j$. Linearity and composition of the ρ 's make both amplitudes $\rho_l x + \rho_k y + \rho_j z$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l . Their addition law at a vector in fibre l is the ordinary R -module addition in V_l ; in particular $0_R^\bullet$ gives $(l, 0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha, f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2 = e$, $e + e = e$, and $1 + e = 1$ in $G(R)$. Thus every $l \in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m \in V_l$,

$$m + l = (1 + e)m = m, \quad m + (-1_R)^\bullet m = em = l.$$

The inherited addition in V_l , with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r \in R$, $e(r^\bullet m) = em = l$, since $er^\bullet = e$. Its scalar action satisfies every R -module law inherited from the semimodule, and $0_R m = em = l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M .

If $l + k = k$, then $e(m + k) = l + k = k$, so ρ_{lk} has the correct codomain. Its zero law is $l + k = k$; its additivity uses $k + k = k$. Also $r^\bullet k = k$, proving R -linearity. Identity follows from $m + l = m$, and composition from $k + j = j$. A split-linear map $F : M \rightarrow N$ takes eM to eN , with $\alpha(l) = F(l)$. It preserves joins and bottom, restricts to R -linear maps on each fibre, and $F(m + k) = F(m) + F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

because $e(m + n) = l \vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ . It is a split-linear isomorphism. In the reconstructed total, $e(l, x) = (l, 0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l, x) + (k, 0) = (k, \rho_{lk}x)$, recovering the given ρ . For a morphism (α, f) , (D5) takes $(l, 0)$ to $(\alpha(l), 0)$; for F , (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. $\square$

5.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \rightarrow \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \quad (\text{D6})$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N , let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \rightarrow D$ of R -module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \text{im } d_C^{\text{prev}}$ inside Z_C , and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \quad [z] \mapsto [z] \quad (\text{D7})$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \rightarrow \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C ; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \rightarrow N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \quad (\text{D8})$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear map satisfying the respective condition has its unique factorization through the displayed subsemimodule. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

6 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R , while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m , including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

6.1 Source spaces and fixed arithmetic factors

Let

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \end{aligned}$$

Use the original Fourier, theta and Mellin transforms

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{A1})$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 6.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \rightarrow \mathcal{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \operatorname{Re} s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{A2})$$

For the exact original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \quad H_{P(D)\phi_*} = P, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP. \quad (\text{A3})$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\widehat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) \phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k) |\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\operatorname{Re} s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2} \{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. $\square$

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Let $Q = \mathcal{B}/\Theta V$ and $q : \mathcal{B} \rightarrow Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \quad (\text{A4})$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q . This follows from injectivity of theta and $\operatorname{im} d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-} , with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

6.2 The exact operator range, including its inverse

Proposition 6.2 (Source and target Euler inverses). *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Then

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)S_\rho = 1, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{A5})$$

The operator $D - \rho$ is injective. For $0 < \operatorname{Re} \rho < 1$, the same formula, extended evenly, gives

$$(D - \rho)V = \ker(H(\cdot)(\rho)). \quad (\text{A6})$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \operatorname{Re} \rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N - \sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D - \rho)S_\rho \phi = \phi$ on $\mathbb{R}$ gives $(1 - \rho) \int S_\rho \phi = \int \phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V . $\square$

For any polynomial $h(t) = \prod_i (t - \rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H), \quad E_h = \mathbb{C}[t]/(h). \quad (\text{A7})$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho_i y}/k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \quad (\text{A8})$$

Theorem 6.3 (Finite arithmetic kernel, quotient and packet section). *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{A9})$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu , and set

$$R_{\text{ref}}u = S_h^{\mathcal{B}}\Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M}F, \quad \sigma = qR_{\text{ref}}.$$

Then

$$\mathcal{M}R_{\text{ref}}u = (g/h)R_Z(u), \quad J_Z R_{\text{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \quad (\text{A10})$$

The projector is $\sigma \bar{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h \Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\text{ref}} = j_h v \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ . Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. $\square$

6.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E , define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$DR_{\text{ref}} - R_{\text{ref}}A = f_0 \ell, \quad f_0 = \Theta\phi_*. \quad (\text{A11})$$

Indeed $tR_Z(u) - R_Z(tu)$ is divisible by the monic h , and its degree is at most d ; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathcal{B}_Z = \Theta V + R_{\text{ref}}E$ fits into the actual extension

$$0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J_Z} E \rightarrow 0. \quad (\text{A12})$$

It splits complex-linearly through R_{ref} ; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_* \ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathcal{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\sim} [V \rightarrow \mathcal{B}_Z] \longrightarrow [V \rightarrow \mathcal{B}]. \quad (\text{A13})$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

6.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below

uses these specified metrics. Put $\phi_j = D^j \phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j \phi_j$. For $F_m = \Theta \Phi_m : \mathbb{C}^{m+1} \rightarrow \mathcal{H}$, define

$$\begin{aligned} H_m &= F_m^* F_m, & C_m &= F_m^* R_{\text{ref}}, & B_m &= -H_m^{-1} C_m, \\ \Pi_m &= F_m H_m^{-1} F_m^*, & R_m &= R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{aligned} \quad (\text{A14})$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \rightarrow \mathbb{C}^{m+1}$,

$$\begin{aligned} (R_{\text{ref}} + F_m B)^* (R_{\text{ref}} + F_m B) &= G_m + (B - B_m)^* H_m (B - B_m), \\ G_m &= R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{aligned} \quad (\text{A15})$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0 \ell + S_m B_m - I_m B_m A, \quad D R_m - R_m A = F_{m+1} K_m. \quad (\text{A16})$$

Integration by parts in dx , with the source boundary term $[x \bar{F} H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \quad (\text{A17})$$

In particular rank $W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \bar{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy-Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \quad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \quad (\text{A18})$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{A19})$$

The next sections compute the actual arithmetic moment problem behind G_m , sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

7 Certified moments from the original arithmetic theta source

The moment matrices in the arithmetic kernel formulas can be calculated directly from the specified theta source. This section gives an exact incomplete-gamma series, proves an explicit bound for the entire omitted double tail, and applies directed interval arithmetic to the retained terms. The construction retains the original factor $g = 2\xi$, the measure dx , the Euler operator $D = -x\partial_x$, and every polynomial coefficient. No information about the location of a zeta zero enters this calculation.

7.1 Source polynomials and the exact reflection map

Keep

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_j = D^j \Theta \phi_*, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx). \quad (\text{AM.1})$$

Introduce polynomials with their original integer coefficients by

$$P_0(z) = 4z^2 - 6z, \quad P_{j+1}(z) = 2z(P_j(z) - P'_j(z)), \quad Q_j(z) = \sum_{k=0}^j (-1)^k \binom{j}{k} P_k(z). \quad (\text{AM.2})$$

Write p_{ja} and q_{ja} for the coefficients of z^a . Both polynomials have degree $d_j = j + 2$, with leading coefficients 2^{j+2} and $(-1)^j 2^{j+2}$, respectively, and both have zero constant coefficient. The formula defining Q_j is an explicit change of polynomial coordinates; neither family is discarded.

Define the linear involution on the actual Hilbert observation by

$$\mathcal{T} : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}_{>0}, dx), \quad (\mathcal{T}F)(x) = x^{-1}F(1/x). \quad (\text{AM.3})$$

It is unitary and self-adjoint, satisfies $\mathcal{T}^2 = 1$, and, on the strong Schwartz space $\mathcal{B}$,

$$D\mathcal{T} = \mathcal{T}(1 - D), \quad \mathcal{T}f_0 = f_0, \quad \mathcal{T}f_j = (1 - D)^j f_0. \quad (\text{AM.4})$$

Consequently, the complete source series and its reflected series are

$$f_j(x) = 2 \sum_{n \geq 1} P_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \tilde{f}_j(x) := (1 - D)^j f_0(x) = 2 \sum_{n \geq 1} Q_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad (\text{AM.5})$$

and $f_j(1/x) = x \tilde{f}_j(x)$.

Proof. For $z = \pi x^2$, applying $-x\partial_x$ to $P(z)e^{-z}$ gives $2z(P - P')e^{-z}$. This proves the recurrence and (AM.5). The leading coefficients and zero constant coefficients follow by induction. Under the Fourier convention $\hat{\phi}(\xi) = \int \phi(x)e^{-2\pi i x \xi} dx$, differentiating the Fourier transform of the Gaussian gives

$$\widehat{x^2 e^{-\pi x^2}}(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}}(\xi) = \left(\xi^4 - \frac{3\xi^2}{\pi} + \frac{3}{4\pi^2} \right) e^{-\pi \xi^2}.$$

Multiplication by the exact coefficients $4\pi^2, -6\pi$ therefore gives $\hat{\phi}_* = \phi_*$. Its value at zero is zero, and its integral is $\hat{\phi}_*(0) = 0$. The theta-Fourier identity already proved for the source V then gives $\mathcal{T}f_0 = f_0$.

Substitution $u = 1/x$ gives $\int |x^{-1}F(1/x)|^2 dx = \int |F(u)|^2 du$, so $\mathcal{T}$ is unitary. It is an involution and hence self-adjoint. Direct differentiation proves $D\mathcal{T} = \mathcal{T}(1 - D)$, including the constant 1 arising from the original factor x^{-1} . Induction proves the other identities in (AM.4). Local uniform convergence of each differentiated Gaussian series follows on $x \geq \delta > 0$ by a polynomial times Gaussian majorant. The series gives decay of all Euler derivatives at infinity, and its exact reflected expression gives the corresponding decay at zero. Thus these operations apply to the original functions in $\mathcal{B}$. $\square$

7.2 An absolutely convergent double series for every entry

For integers $i, j \geq 0$, retain the raw moment entry

$$H_{ij} = \langle D^i f_0, D^j f_0 \rangle_{L^2(dx)} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j |2\xi(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{AM.6})$$

The equality is the original Mellin Plancherel identity. In particular H_{ij} is real because the source functions are real. For $L \geq 0$ and $\lambda > 0$, set

$$I_L(\lambda) = \int_1^\infty x^{2L} e^{-\lambda x^2} dx = \frac{\Gamma(L + 1/2, \lambda)}{2\lambda^{L+1/2}}, \quad \Gamma(a, \lambda) = \int_\lambda^\infty u^{a-1} e^{-u} du. \quad (\text{AM.7})$$

Theorem 7.1 (Exact original arithmetic moment series). *With $\lambda_{nm} = \pi(n^2 + m^2)$, the following double series converges absolutely, including after the finite coefficient sums are replaced by their absolute values:*

$$H_{ij} = 2 \sum_{n,m \geq 1} \sum_{a=1}^{d_i} \sum_{b=1}^{d_j} (p_{ia} p_{jb} + q_{ia} q_{jb}) \pi^{a+b} n^{2a} m^{2b} \cdot \frac{\Gamma(a+b+1/2, \lambda_{nm})}{\lambda_{nm}^{a+b+1/2}}. \quad (\text{AM.8})$$

Equivalently, it is the Gram integral on the positive half-domain

$$H_{ij} = \int_1^\infty (f_i(x) f_j(x) + \tilde{f}_i(x) \tilde{f}_j(x)) dx. \quad (\text{AM.9})$$

Thus the precise map to the positive Gram presentation is

$$\mathbb{C}[s] \longrightarrow L^2([1, \infty), dx)^{\oplus 2}, \quad P \longmapsto (P(D)f_0, P(1-D)f_0)|_{[1, \infty)}. \quad (\text{AM.10})$$

It preserves the original polynomial inner product (AM.6) and is injective. The signs in (AM.8) are retained; the positivity is the Gram positivity of this specified map.

Proof. Split the integral in (AM.6) at $x = 1$. On $(0, 1)$, set $x = 1/u$. The two factors u supplied by (AM.5) cancel the Jacobian u^{-2} exactly. This gives (AM.9). For absolute convergence, write

$$C_j = \sum_{a=1}^{d_j} |p_{ja}| \pi^a, \quad E_j = \sum_{a=1}^{d_j} |q_{ja}| \pi^a. \quad (\text{AM.11})$$

For $n, x \geq 1$, the polynomial bounds are

$$|P_j(\pi n^2 x^2)| \leq C_j n^{2d_j} x^{2d_j}, \quad |Q_j(\pi n^2 x^2)| \leq E_j n^{2d_j} x^{2d_j}.$$

Since $n^2 + m^2 \geq 2$ and $x^2 \geq 1$,

$$(n^2 + m^2)x^2 \geq \frac{1}{2}(n^2 + m^2) + x^2.$$

The absolute integrand series is therefore bounded by

$$4(C_i C_j + E_i E_j) x^{2(d_i + d_j)} e^{-\pi x^2} \left(\sum_{n \geq 1} n^{2d_i} e^{-\pi n^2/2} \right) \left(\sum_{m \geq 1} m^{2d_j} e^{-\pi m^2/2} \right),$$

which is integrable on $[1, \infty)$. This proves all required sum-integral interchanges. Substitution $u = \lambda x^2$ gives (AM.7). The two theta factors contribute 4, and (AM.7) contributes $1/2$, leaving exactly the factor 2 in (AM.8).

For complex polynomial coefficients, conjugate the coefficients of the first factor in the Gram integral. Equation (AM.9) then proves the inner-product assertion in (AM.10). If a nonzero polynomial P were in its kernel, its first component would vanish on $[1, \infty)$. Put $U(z) = \sum_j P_j(z) c_j$ when $P(s) = \sum_j c_j s^j$. The distinct degrees and nonzero leading coefficients of the P_j imply $U \neq 0$. Multiplying the vanishing theta series by $e^{\pi x^2}/2$ gives

$$U(\pi x^2) = - \sum_{n \geq 2} U(\pi n^2 x^2) e^{-\pi(n^2-1)x^2}.$$

The right side decays exponentially times a fixed polynomial as $x \rightarrow \infty$, by the same Gaussian bound. A nonzero polynomial on the left cannot have that decay. This contradiction proves injectivity. $\square$

7.3 A complete finite truncation certificate

Let $H_{ij}^{[N]}$ be (AM.8) with both theta indices restricted to $1 \leq n, m \leq N$, keeping all coefficients p_{ia}, q_{ia} . For the calculation of this entry choose an integer $N \geq \max(d_i, d_j)$. Define, for $k = i, j$,

$$\begin{aligned} r_{k,N} &= \left(\frac{N+2}{N+1} \right)^{2d_k} e^{-\pi(2N+3)}, & T_{k,N} &= \frac{(N+1)^{2d_k}}{1 - r_{k,N}}, \\ A_{k,N} &= \sum_{n=1}^N n^{2d_k} e^{-\pi(n^2-1)} + T_{k,N} e^{-\pi((N+1)^2-1)}. \end{aligned} \quad (\text{AM.12})$$

All these numbers are specified positive real constants. The choice of N ensures $r_{k,N} < 1$, since

$$\log r_{k,N} < \frac{2d_k}{N+1} - \pi(2N+3) < 2 - 5\pi < 0.$$

Theorem 7.2 (Rigorous complete double-tail bound). *For every $i, j \geq 0$ and $N \geq \max(i, j) + 2$,*

$$\boxed{|H_{ij} - H_{ij}^{[N]}| \leq \mathcal{E}_{ij,N} := 4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) \cdot I_{d_i+d_j}(\pi((N+1)^2 + 1))}. \quad (\text{AM.13})$$

In particular the bound tends to zero as $N \rightarrow \infty$, for each fixed pair i, j , and provides an explicit finite certificate for every raw arithmetic moment.

Proof. For $x \geq 1$, the ratio of successive terms of $n^{2d_k} e^{-\pi n^2 x^2}$ is at most

$$\left(\frac{n+1}{n} \right)^{2d_k} e^{-\pi(2n+1)}.$$

The logarithm of this expression has derivative $-2d_k/(n(n+1)) - 2\pi < 0$ for positive real n . For $n \geq N+1$ every ratio is consequently at most $r_{k,N}$. Summing the geometric majorant yields

$$\sum_{n \geq N} n^{2d_k} e^{-\pi n^2 x^2} \leq T_{k,N} e^{-\pi(N+1)^2 x^2}, \quad \sum_{n \geq 1} n^{2d_k} e^{-\pi n^2 x^2} \leq A_{k,N} e^{-\pi x^2}. \quad (\text{AM.14})$$

For the second inequality, divide by $e^{-\pi x^2}$, bound each retained factor $e^{-\pi(n^2-1)x^2}$ at $x = 1$, and apply the first inequality to the tail.

Write $F_{k,N}$ for the finite P_k -series and $\tilde{F}_{k,N}$ for the finite Q_k -series. Equations (AM.11) and (AM.14) give

$$\begin{aligned} |f_k(x)|, |F_{k,N}(x)| &\leq 2C_k x^{2d_k} A_{k,N} e^{-\pi x^2}, \\ |f_k(x) - F_{k,N}(x)| &\leq 2C_k x^{2d_k} T_{k,N} e^{-\pi(N+1)^2 x^2}. \end{aligned}$$

The reflected inequalities have exactly E_k in place of C_k . In each product of (AM.9), use the exact decomposition $ab - a_N b_N = (a - a_N)b + a_N(b - b_N)$. The resulting bound on the absolute difference of integrands is

$$4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) x^{2(d_i+d_j)} e^{-\pi((N+1)^2+1)x^2}.$$

Its integral is exactly the right side of (AM.13). Finally, the $A_{k,N}$ remain bounded for fixed k , the $T_{k,N}$ grow at most polynomially, and the integral decays exponentially in $(N+1)^2$. For completeness, with $L = d_i + d_j$ fixed and $\lambda \geq 2L$, the inequality $\log x \leq (x^2 - 1)/2$ for $x \geq 1$ gives

$$I_L(\lambda) \leq e^{-\lambda} \int_1^\infty e^{-(\lambda-L)(x^2-1)} dx \leq \frac{e^{-\lambda}}{2(\lambda-L)}.$$

Here the last step uses $x^2 - 1 \geq 2(x-1)$. This proves the asserted limit with an additional elementary upper bound when desired. $\square$

The finite matrix $(H_{ij}^{[N]})_{0 \leq i, j \leq m}$ is itself positive definite for every $N \geq 1$. Indeed it is the Gram matrix of $(F_{j,N}, \tilde{F}_{j,N})$ on the same half-domain; the injectivity proof in (AM.10) applies verbatim to the finite sum. This fact does not assign an ordering to its difference from the full matrix. The relation to the full matrix is the entrywise enclosure (AM.13), whose signs and error bounds are explicit.

7.4 Exact adjoint, parity, and finite-boundary checks

On $\mathcal{B}$ in $L^2(dx)$, integration by parts gives $\langle DF, H \rangle = \langle F, (1 - D)H \rangle$. Therefore the complete moments satisfy

$$H_{i+1,j} + H_{i,j+1} = H_{ij}, \quad H_{ij} = \sum_{k=0}^i (-1)^k \binom{i}{k} H_{0,j+k}, \quad H_{01} = \frac{1}{2} H_{00}. \quad (\text{AM.15})$$

All boundary terms vanish because the original functions have rapid decay at both ends. More explicitly,

$$(Df_i)f_j + f_i(Df_j) - f_i f_j = -\partial_x(x f_i f_j),$$

whose integral on $(0, \infty)$ is zero. Moving each of the i copies of D in the first argument to the second gives the middle identity; symmetry gives the last one.

The finite sums retain an exact boundary correction at $x = 1$:

$$H_{i+1,j}^{[N]} + H_{i,j+1}^{[N]} - H_{ij}^{[N]} = F_{i,N}(1)F_{j,N}(1) - \tilde{F}_{i,N}(1)\tilde{F}_{j,N}(1). \quad (\text{AM.16})$$

To prove this, use $DF_{j,N} = F_{j+1,N}$ for the first component and $D\tilde{F}_{j,N} = \tilde{F}_{j,N} - \tilde{F}_{j+1,N}$ for the second. Applying the preceding integration-by-parts calculation on $[1, \infty)$ gives the first displayed endpoint product for the first component and its negative for the second. The endpoints at infinity vanish term by term. In particular the exact finite identity $H_{01}^{[N]} = H_{00}^{[N]}/2$ follows even before the omitted tail is enclosed, since $F_{0,N} = \tilde{F}_{0,N}$.

One may also retain the exact triangular coordinate map

$$B_{ja} = \begin{cases} \binom{j}{a} (-1/2)^{j-a}, & a \leq j, \\ 0, & a > j, \end{cases} \quad C = BHB^T. \quad (\text{AM.17})$$

This is the Gram matrix of $u_j = (D - 1/2)^j f_0$, with inverse coordinate matrix $(B^{-1})_{ja} = \binom{j}{a} (1/2)^{j-a}$. No original moment is lost. Since $\mathcal{T}u_j = (-1)^j u_j$, unitarity proves $C_{ij} = 0$ for odd $i + j$. The operator $D - 1/2$ is skew-symmetric on this displayed strong Schwartz domain, so for $i + j = 2q$,

$$C_{ij} = (-1)^{i+q} \|u_q\|^2, \quad \|u_q\|^2 = \int_{\mathbb{R}} t^{2q} |2\xi(1/2 + it)|^2 \frac{dt}{2\pi} > 0. \quad (\text{AM.18})$$

For the first equality move all i powers to the second argument, and compare with moving the q powers in $\langle u_q, u_q \rangle$. Mellin Plancherel gives the second equality. Strict positivity follows from the nonzero original function and the polynomial injectivity already proved.

7.5 Reproducible certified finite arithmetic data

The companion program `scripts/check_arithmetic_moments.py` implements the exact integer recurrence (AM.2), evaluates every retained term of (AM.8) by Arb directed ball arithmetic, and enlarges its radius by an upward-rounded value of (AM.13). Thus the final balls enclose the infinite-source entries, including rounding error and both omitted theta tails. The default run uses $0 \leq i, j \leq 5, N = 7$, and 110 decimal working digits; its machine-readable output is `checks/arithmetic_moments.json`.

Let $\Delta_k = \det(H_{ij})_{0 \leq i,j < k}$ and $\Delta_0 = 1$. The squared norm h_k of the monic degree- k polynomial for the original measure μ is

$$h_k = \frac{\Delta_{k+1}}{\Delta_k}. \quad (\text{AM.19})$$

Indeed the coefficients below the leading term solve the Gram system making that monic polynomial orthogonal to degrees below k . Substitution of that unique solution into its squared norm gives the Schur complement of the leading k -by- k Gram block. The block determinant identity gives (AM.19). The interval calculation certifies strict positivity of all six leading principal determinants recorded in the companion certificate and all six corresponding h_k ; those are the actual arithmetic norms used by the orthogonal recurrence, through the finite degrees stated.

Here are explicit certified intervals from that run. Each row means the entire interval $[c - r, c + r]$, with the printed decimal centre c and radius r interpreted exactly. Rounding the stored Arb balls to the displayed decimals was performed outwards.

raw moment	c	r
H_{00}	1.279007247846485140480	$4.67 \cdot 10^{-22}$
H_{01}	0.6395036239232425702398	$3.33 \cdot 10^{-23}$
H_{02}	-12.73579749060893715027	$2.81 \cdot 10^{-21}$
H_{11}	13.37530111453217972051	$2.57 \cdot 10^{-21}$
H_{12}	6.687650557266089860256	$2.85 \cdot 10^{-22}$
H_{22}	389.8820543216399270934	$1.65 \cdot 10^{-20}$

Symmetry supplies the other entries of this original three-by-three moment block. The six original monic squared norms are enclosed by

k	c for h_k	r
0	1.279007247846485140480	$4.67 \cdot 10^{-22}$
1	13.05554930257055843539	$2.69 \cdot 10^{-21}$
2	250.0089768190499145175	$4.88 \cdot 10^{-20}$
3	6694.666685525627506773	$4.76 \cdot 10^{-19}$
4	224298.1953314520556463	$9.95 \cdot 10^{-18}$
5	9383436.471794727053469	$2.54 \cdot 10^{-16}$

The analytic double-tail bound is less than $2.961369 \cdot 10^{-63}$ for every entry of the six-by-six matrix, and is less than $1.150030 \cdot 10^{-83}$ for H_{00} . These bounds account for omitted source terms; the displayed intervals additionally include finite-term arithmetic and decimal conversion errors.

The checker also evaluates (AM.15) and the independent exact finite-boundary identity (AM.16), and detects removal of either the reflected half-domain, the theta factor 2, or the adjoint constant 1. Its checks execute identically under ordinary Python and Python's optimization flag. Finite interval certificates do not assert an infinite-degree weight estimate; (AM.8)–(AM.14) give the exact arithmetic input and its complete error control for calculating further degrees.

8 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

8.1 The fixed arithmetic maps and their boundary

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx). \end{aligned}$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta \phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ , and retain

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \dots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$\begin{aligned} R_{\text{ref}} u &= S_{h_Z}^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad JF = j_h \mathcal{M}F, \quad \sigma_Z = q R_{\text{ref}}, \\ JR_{\text{ref}} &= 1_E, \quad J\Theta = 0, \quad DR_{\text{ref}} - R_{\text{ref}} A = f_0 \ell_Z, \quad \ell_Z(u) = [t^{d-1}] R_Z(u). \end{aligned} \quad (\text{TC1})$$

Here $S_\rho^{\mathcal{B}} F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S_{h_Z}^{\mathcal{B}}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^L c_j \phi_j, \quad F_L = \Theta \Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathcal{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \quad C : E \longrightarrow \mathbb{C}^{L+1}. \quad (\text{TC2})$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \rightarrow \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1} . Directly,

$$DR - RA = \Theta K_R, \quad K_R = \phi_* \ell_Z + \Phi_{L+1}(S_L C - I_L C A) : E \longrightarrow V. \quad (\text{TC3})$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^* R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

because $[x \overline{F(x)} H(x)]_0^\infty = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^* G_R + G_R A - G_R = -(R^* \Theta K_R + (\Theta K_R)^* R). \quad (\text{TC4})$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

8.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \quad (\mathcal{U}F)(y) = e^{y/2} F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\operatorname{Im} z| < \pi/4$. On every closed narrower strip, $\operatorname{Re}(e^{2z}) \geq c_a e^{2\operatorname{Re} z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\operatorname{Re} z \rightarrow +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1} f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2} f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\operatorname{Im} z = a$ sign t has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \quad (\text{TC5})$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \bar{v} d\mu$ has a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2 + it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g . It follows that

$$\overline{\bigcup_{L \geq 0} F_L \mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{TC6})$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

8.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \quad \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m. \quad (\text{TC7})$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \quad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \quad (\text{TC8})$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \rightarrow 1$ strongly. Since E is finite dimensional, $R_m \rightarrow 0$ in $\operatorname{Hom}(E, \mathcal{H})$, and $G_m \rightarrow 0$ in matrix norm.

Exact fixed-level image. At every specified level m , the entire image of the source-Gram map is

$$\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \rightarrow \mathbb{C}^{m+1}\} = \{G_m + \Delta : \Delta \succeq 0, \operatorname{rank} \Delta \leq m+1\}. \quad (\text{TC8a})$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \quad (\text{TC8b})$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m + 1$ has a factorization $\Delta = T^*T$ with $T : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ , retain its nonnegative eigenvalues, and insert their square roots into $m + 1$ rows. Taking $C = B_m + H_m^{-1/2}T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E , there exist a finite L and a coefficient map $C : E \rightarrow \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \quad JR = 1, \quad qR = \sigma_Z, \quad R^*R = G. \quad (\text{TC9})$$

The construction is as follows. Choose $m \geq d - 1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1} . Put

$$\Delta = G - G_m > 0, \quad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \quad (\text{TC10})$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E . The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \quad R_m^* F_m = 0,$$

so its Gram is

$$R^*R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ . This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\text{Ext}_{\mathbb{C}[t]}^1(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E . The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \quad [p(t)] \longmapsto [p(D)\phi_*]. \quad (\text{TC10a})$$

It is well-defined and $\mathbb{C}[t]$ -linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \quad [\psi] \longmapsto j_h \left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*} \right). \quad (\text{TC10b})$$

The denominator is

$$\mathcal{M}\phi_*(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\text{Re } s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z ; this proves well-definedness of κ_h and $\kappa_h \iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z . For the first factor $t - \rho$, its source inverse on $x > 0$ is

$$S_\rho \psi(x) = x^{-\rho} \int_x^\infty \psi(y) y^{\rho-1} dy = - \int_0^1 \psi(tx) t^{\rho-1} dt.$$

The second expression and $0 < \text{Re } \rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\text{Re } \rho - 1}$. The first expression gives Schwartz decay at infinity. Integration of $(D - \rho)S_\rho \psi = \psi$ on the full real line gives $(1 - \rho) \int S_\rho \psi = \int \psi = 0$, so $S_\rho \psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s - \rho)$, so the remaining zero jets permit iteration through every factor of h_Z . Consequently $\ker(j_h \mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$,

where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h\kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D -equivariant section: an intertwiner $T : E \rightarrow \mathcal{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathcal{B}$. To see this injectivity, $(D - \rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathcal{B}$ force $c = 0$, and one iterates over the factors of h_Z . The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

8.4 Sharp control with all nilpotent jets present

For a positive definite G , set

$$W_A(G) = A^*G + GA - G, \quad \epsilon_A(G) = \sup_{u \neq 0} \frac{|u^*W_A(G)u|}{u^*Gu}, \quad \eta_Z = \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|. \quad (\text{TC11})$$

For every eigenvector $Au = \rho u$,

$$u^*W_A(G)u = (2 \operatorname{Re} \rho - 1)u^*Gu. \quad (\text{TC12})$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G . We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t}e^{tA^*}G_0e^{tA}, \\ G_\epsilon &= \int_{-\infty}^{\infty} e^{-\epsilon|t|}F(t)dt, \\ W_A(G_\epsilon) &= \epsilon \int_{-\infty}^{\infty} \operatorname{sign}(t)e^{-\epsilon|t|}F(t)dt. \end{aligned} \quad (\text{TC13})$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A - \rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho. \quad (\text{TC14})$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\bar{\rho} + \sigma - 1$ has absolute value at most η_Z . This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^*F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \quad (\text{TC15})$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \quad (\text{TC16})$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at} dt = k! a^{-k-1}$ show that

$$G_\epsilon = \sum_{\rho, \sigma \in Z} \sum_{i=0}^{m_\rho-1} \sum_{j=0}^{m_\sigma-1} \frac{I_{i+j}(\bar{\rho} + \sigma - 1, \epsilon)}{i!j!} \cdot (N_\rho^i P_\rho)^* G_0(N_\sigma^j P_\sigma). \quad (\text{TC17})$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \rightarrow \mathbb{C}^{L+1}}} \epsilon_A((R_{\text{ref}} + F_L C)^*(R_{\text{ref}} + F_L C)) = \eta_Z. \quad (\text{TC18})$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA} u\|_G \leq e^{(1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA} u\|_G \leq e^{(-1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t , hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z , and use (TC13) on that summand. Its control is strictly smaller than η_Z . The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

8.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^n I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \quad W_{A_n, n}(G) = A_n^* G + G A_n - nG. \quad (\text{TC19})$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \cdots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \operatorname{Spec} A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \quad (\text{TC20})$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \dots, \rho)$ for a zero attaining η_Z . Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\inf_{G>0} \sup_{v \neq 0} \frac{|v^* W_{A,n}(G) v|}{v^* G v} = n\eta_Z. \quad (\text{TC21})$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

8.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \bar{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \rightarrow \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \rightarrow E.$$

The source boundary of R_m lies in the span of $f_0, \dots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \text{rank } W_m \leq 2. \quad (\text{TC22})$$

Reflection stability gives

$$\text{Tr}(G_m^{-1} W_m) = 2 \text{Re Tr } A - d = 0. \quad (\text{TC23})$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m -self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \max(2 \text{Re } \rho - 1, 0) = \sum_{\rho \in Z} m_\rho |\text{Re } \rho - \tfrac{1}{2}|. \quad (\text{TC24})$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A -invariant flag, keeping all generalized eigenvectors. Choose a G_m -orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \rightarrow E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ , repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal entries $\delta_j = 2 \text{Re } \rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m . Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \leq P \leq I$ and $H \leq H_+$,

$$\sum_{\delta_j > 0} \delta_j = \text{Tr}(PH) \leq \text{Tr}(PH_+) \leq \text{Tr}(H_+) = \epsilon_m. \quad (\text{TC25})$$

Each trace inequality follows by inserting the positive square root of P or H_+ , so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+ , the same argument gives $\sum_j \max(\delta_j, 0) \leq r_+ \epsilon_A(G)$. In the present source construction $r_+ \leq 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\operatorname{Im} c_m)^2. \quad (\text{TC26})$$

Indeed the two nonzero eigenvalues are those of $-\begin{pmatrix} c_m & v_m \\ u_m & c_m \end{pmatrix}$, and (TC23) gives $\operatorname{Re} c_m = 0$. If $a_{m+1} = (1 - \Pi_m)f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \quad u_m = \eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right). \quad (\text{TC27})$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right) v_m - (\operatorname{Im} c_m)^2 \geq \left(\sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}| \right)^2. \quad (\text{TC28})$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

8.7 Original supported maps and the precise result delivered

The passage from a finite source correction C to its representative, Gram, and boundary is the explicit chain of maps

$$C \mapsto R_{\text{ref}} + \Theta \Phi_L C \mapsto (qR, JR, R^*R, K_R) = (\sigma_Z, 1_E, G_R, K_R). \quad (\text{TC29})$$

For every active support label λ , the linear maps lift as $(\lambda, u) \mapsto (\lambda, Ru)$ and $(\lambda, u) \mapsto (\lambda, K_R u)$. The quotient of the boundary is $(\lambda, \Theta K_R u) \mapsto (\lambda, 0_Q)$; external absence remains the external zero τ . For the two-leg differential $d(\phi, \psi) = \Theta(\phi - \widehat{\psi})$, the two primitives are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$. Their difference $(K_R u, \widehat{K_R u})$ is the retained degree-zero cycle. Thus all source changes in the optimization have actual typed cochain lifts with the same support synchronization as the input programme.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

9 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

9.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x.$$

Let $\mathcal{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b |D^k F(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j \phi_*$ and $f_j = D^j f_0 = \Theta\phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \quad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2} F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity} . Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathcal{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g , including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \dots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \quad \varepsilon = (j_h v)^{-1} \in E^\times, \quad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h , including the factor 2 in $g = 2\xi$. It commutes with A . For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu . Write $R_{\text{ref}} : E \rightarrow \mathcal{B}$ for the source note's fixed finite section R_0 . This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m . The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}} = I_E, \quad J_Z \Theta = 0. \quad (\text{TB.1})$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad d\nu_Z(t) = |v(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{TB.2})$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

9.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\text{ref}} u + \sum_{j=0}^m f_j(Bu)_j, \quad B : E \longrightarrow \mathbb{C}^{m+1}. \quad (\text{TB.3})$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N , with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \quad C_N P = j_h P. \quad (\text{TB.4})$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j d\nu_Z(t), \quad 0 \leq i, j \leq N. \quad (\text{TB.5})$$

It is positive definite by (TB.2). Define

$$K_N = C_N (H_N^\nu)^{-1} C_N^*. \quad (\text{TB.6})$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 9.1 (The actual source minimum). *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad \mathcal{M} R_m u = v p_{N,u}, \quad (\text{TB.7})$$

$$G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon. \quad (\text{TB.8})$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \quad (\text{TB.9})$$

has degree at most m . Its coefficients give exactly the map $B_m : E \rightarrow \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M} R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^m (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \quad (\text{TB.10})$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h , and the degree bound makes its quotient a polynomial of degree at most m . Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p + z)^* H_N^\nu (p + z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \quad (\text{TB.11})$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $q R_m = \sigma_Z$ and $J_Z R_m = I_E$. $\square$

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

9.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z , with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E , no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 9.2 (The retained vertical-line recurrence). *There are $b_n \in \mathbb{C}$, with $b_n + \bar{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \quad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \geq 1). \quad (\text{TB.12})$$

Proof. On the original line $s = 1/2 + it$, one has $\bar{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \leq n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1 - s)q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \bar{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. $\square$

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \quad (\text{TB.13})$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^N \frac{v_n v_n^*}{\kappa_n}. \quad (\text{TB.14})$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z, w) = \sum_{n=0}^N \frac{q_n(z) \overline{q_n(w)}}{\kappa_n}. \quad (\text{TB.15})$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0 \leq k < m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 9.3 (Confluent kernel boundary formula). *For every $N \geq d - 1$,*

$$\boxed{AK_N + K_N A^* - K_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N}}. \quad (\text{TB.16})$$

Before taking any jets the corresponding polynomial identity is

$$(z + \bar{w} - 1)\mathcal{K}_N(z, w) = \frac{q_{N+1}(z) \overline{q_N(w)} + q_N(z) \overline{q_{N+1}(w)}}{\kappa_N}. \quad (\text{TB.17})$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n , because $j_h(sq_n) = A(j_hq_n)$. Substitute these relations into

$$AK_N + K_N A^* - K_N = \sum_{n=0}^N \frac{(Av_n)v_n^* + v_n(Av_n)^* - v_n v_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n + \bar{b}_n = 1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncanceled term is the $n = N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z, w . This is a polynomial identity; no division by $z + \bar{w} - 1$ is performed at its zero set, and differentiating it proves the full confluent version. $\square$

9.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m = A^* G_m + G_m A - G_m = -\{R_m^*(DR_m - R_m A) + (DR_m - R_m A)^* R_m\}. \quad (\text{TB.18})$$

The boundary term $[x\bar{F}H]_0^\infty$ vanishes for $F, H \in \mathcal{B}$. Put

$$L_N = AK_N + K_N A^* - K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m = U_\varepsilon^* K_N^{-1} L_N K_N^{-1} U_\varepsilon.} \quad (\text{TB.19})$$

For clarity, multiplication gives

$$K_N^{-1} L_N K_N^{-1} = K_N^{-1} A + A^* K_N^{-1} - K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N = v_{N+1}^* K_N^{-1} v_{N+1}, \quad \beta_N = v_N^* K_N^{-1} v_N, \quad \gamma_N = v_N^* K_N^{-1} v_{N+1}. \quad (\text{TB.20})$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\boxed{\epsilon_m = \frac{|\operatorname{Re} \gamma_N| + \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.} \quad (\text{TB.21})$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1} U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y / (y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N} \begin{pmatrix} \gamma_N & \beta_N \\ \alpha_N & \bar{\gamma}_N \end{pmatrix}.$$

Its trace and determinant give

$$\lambda_{\pm} = \frac{\operatorname{Re} \gamma_N \pm \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.$$

Cauchy-Schwarz in K_N^{-1} gives $|\gamma_N|^2 \leq \alpha_N \beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re} \gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \bar{\rho}$, with the multiplicities retained, $2 \operatorname{Re} \operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1} L_N) = \frac{2 \operatorname{Re} \gamma_N}{\kappa_N}. \quad (\text{TB.22})$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{TB.23})$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \geq \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

9.5 Exact change when one more theta direction is admitted

The step $m \mapsto m + 1$ means $N \mapsto N + 1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \geq 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{w w^*}{k}. \quad (\text{TB.24})$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1} w w^* K_N^{-1}}{k + \theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1} w w^* K_N^{-1} U_\varepsilon}{k + \theta}, \quad (\text{TB.25})$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k + \theta} > 0. \quad (\text{TB.26})$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \neq 0$, the determinant formula follows by extending $K_N^{-1} w$ to a basis, in which the rank-one map $K_N^{-1} w w^* / k$ has one possible nonzero eigenvalue θ/k . Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h , as in (TB.9). This gives a source polynomial P_u of degree at most $m + 1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1} , where $W_{m+1} = \operatorname{span}(\phi_*, D\phi_*, \dots, D^{m+1}\phi_*)$. The quotient map $\mathcal{B}/\Theta W_m \rightarrow \mathcal{B}/\Theta W_{m+1}$ has kernel $\Theta W_{m+1}/\Theta W_m$; injectivity of Θ identifies this kernel with the original one-dimensional space W_{m+1}/W_m . The scalar e sends a class to the zero in its same support fibre, while τ sends it to external absence. None of (TB.24)–(TB.26) changes those maps.

9.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make this measure invariant under $s \mapsto 1 - s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1 - s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$s Q_n = Q_{n+1} + \frac{1}{2} Q_n - \frac{h_n}{h_{n-1}} Q_{n-1} \quad (n \geq 1). \quad (\text{TB.27})$$

For $n = 0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \quad d_n u = \frac{1}{h_n} \int_{\mathbb{R}} \overline{Q_n(1/2 + it)} r(u)(1/2 + it) d\mu(t). \quad (\text{TB.28})$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g \sum_{n>m} Q_n d_n u, \quad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n.} \quad (\text{TB.29})$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n , and use the adjoint relation $s^* = 1 - s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n} d_{n+1} = d_{n-1} + d_n \left(\frac{1}{2} I - A \right) - \delta_{n0} \ell_Z.} \quad (\text{TB.30})$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1} d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1} (d_{m+1}^* d_m + d_m^* d_{m+1}).} \quad (\text{TB.31})$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 9.4 (The full source defect and its observation of the packet). *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{DR_m - R_m A = -\frac{h_{m+1}}{h_m} Q_m(D) f_0 d_{m+1} - Q_{m+1}(D) f_0 d_m.} \quad (\text{TB.35})$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \quad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \quad (\text{TB.36})$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^m Q_j d_j$ and compute

$$sr_m - r_m A = \ell_Z - \sum_{j=0}^m (s Q_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j . For $0 \leq j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z . The coefficient at Q_m is $-h_{m+1} d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed

primitive belongs to V , since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D)f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \geq 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k . Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \geq 0$. In particular every term of the positive series in (TB.29) vanishes on u , so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. $\square$

9.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1 - \bar{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \rightarrow E_{Z^\dagger}$ defined in Section 60. The original arithmetic residue matrix S , satisfying $S^* = -S$ and $A^* S + S A = S$, is still related to its dual metric by

$$G_m^D = S^* G_m^{-1} S, \quad W_m^D = -S^* G_m^{-1} W_m G_m^{-1} S. \quad (\text{TB.32})$$

The original arithmetic Weil form, with $J_g = M_{j_h g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \quad (\text{TB.33})$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \rightarrow 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \rightarrow +\infty, \quad \prod_{m=m_0}^M \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \rightarrow 0. \quad (\text{TB.34})$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \rightarrow 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z -moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, <https://dlmf.nist.gov/18.2>. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

10 The full jet kernel, finite resolvents, and the actual analytic tail

Retain the arithmetic measure, coordinates and maps of Section 9:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = |g(s)/h(s)|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

In particular, every selected zero has its complete multiplicity in h , and c is the coordinate column of the unit in the retained monomial basis of E . Write q_k for the monic orthogonal polynomials of ν_Z , $\kappa_k = \|q_k\|_{\nu_Z}^2$, and $a_k = \kappa_k/\kappa_{k-1} > 0$ for $k \geq 1$. The exact recurrence is

$$sq_k = q_{k+1} + b_k q_k - a_k q_{k-1}, \quad b_k + \bar{b}_k = 1, \quad (\text{KR.1})$$

where the last term is omitted at $k = 0$.

10.1 A finite compression in the original Mellin coordinate

Fix $n \geq d$, so $N = n - 1$ is the degree bound in the preceding kernel construction. On the coefficient space with basis $q_0, \dots, q_{n-1}$, put

$$H_n = \text{diag}(\kappa_0, \dots, \kappa_{n-1}), \quad (J_n)_{k,k} = b_k, \quad (J_n)_{k+1,k} = 1, \quad (J_n)_{k,k+1} = -a_{k+1}. \quad (\text{KR.2})$$

All unspecified entries are zero, and indices start at zero. Multiplication of the displayed matrices proves

$$J_n^* H_n + H_n J_n = H_n. \quad (\text{KR.3})$$

The isometry from this coefficient space to its polynomial image is $u \mapsto \sum_k q_k u_k$. For a standard coefficient Hermitian form, the corresponding isometry is

$$\mathcal{I}_n u = \sum_{k=0}^{n-1} \frac{q_k}{\sqrt{\kappa_k}} u_k, \quad B_n = H_n^{1/2} J_n H_n^{-1/2}. \quad (\text{KR.4})$$

This explicitly stated auxiliary coordinate map leaves A , the packet coordinates, q_k , and κ_k unchanged. The matrix B_n has lower off-diagonal entry $\sqrt{a_{k+1}}$, upper off-diagonal entry $-\sqrt{a_{k+1}}$, and satisfies $B_n^* + B_n = I$.

Lemma 10.1 (Nodes and endpoint resolvents). *The characteristic polynomial of J_n , and hence of B_n , is $q_n(z)$. Its roots $\lambda_{n,1}, \dots, \lambda_{n,n}$ are distinct and lie on $\text{Re } z = 1/2$. There are positive numbers $\omega_{n,j}$ and $\sigma_{n,j}$ such that*

$$\begin{aligned} m_n(z) &= \kappa_0 e_0^* (zI - B_n)^{-1} e_0 = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \omega_{n,j} &= \kappa_0, \\ F_n(z) &= e_{n-1}^* (zI - B_n)^{-1} e_{n-1} = \frac{q_{n-1}(z)}{q_n(z)} = \sum_{j=1}^n \frac{\sigma_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \sigma_{n,j} &= 1. \end{aligned} \quad (\text{KR.5})$$

Here m_n and F_n are respectively the initial and terminal observations of the same finite compression.

Proof. Expanding the tridiagonal determinant along its last row gives $D_{k+1}(z) = (z - b_k)D_k(z) + a_k D_{k-1}(z)$, with $D_0 = 1$, $D_1 = z - b_0$. This is (KR.1), proving $D_n = q_n$. The matrix $(B_n - \frac{1}{2}I)/i$ is Hermitian. If an eigenvector has first coordinate zero, its first row equation forces its second coordinate to be zero; recurrence through the rows forces every coordinate to vanish. Thus each eigenspace has dimension at most one. Hermitian diagonalization gives a complete eigenbasis, hence all eigenvalues are simple. The same argument beginning at the last row shows every eigenvector has nonzero last coordinate. If P_j are its rank-one orthogonal eigenprojections, set $\omega_{n,j} = \kappa_0 e_0^* P_j e_0 > 0$ and $\sigma_{n,j} = e_{n-1}^* P_j e_{n-1} > 0$. Their sums and the two spectral expansions follow from $\sum_j P_j = I$. The last diagonal cofactor of $zI - B_n$ is $q_{n-1}(z)$, proving the ratio in (KR.5). $\square$

Define $\delta(z) = |\operatorname{Re} z - 1/2|$. For $\delta(z) > 0$, (KR.3)–(KR.4) give $\|(zI - B_n)^{-1}\| \leq \delta(z)^{-1}$: for a vector u , the real part of $u^*(zI - B_n)u$ is $(\operatorname{Re} z - 1/2)\|u\|^2$, and Cauchy–Schwarz then gives the lower singular-value bound. In particular $|F_n(z)| \leq \delta(z)^{-1}$, and

$$\operatorname{Re} F_n(z) = (\operatorname{Re} z - \tfrac{1}{2}) \sum_j \frac{\sigma_{n,j}}{|z - \lambda_{n,j}|^2}. \quad (\text{KR.6})$$

10.2 Exact quadrature and the full confluent kernel

Theorem 10.2 (A polynomial formula valid at every packet centre). *Put*

$$L_{n,j}(z) = \frac{q_n(z)}{(z - \lambda_{n,j})q'_n(\lambda_{n,j})}. \quad (\text{KR.7})$$

This denotes the polynomial obtained by cancelling the displayed linear factor. For every polynomial P of degree at most $2n - 1$,

$$\int P(s) d\nu_Z = \sum_j \omega_{n,j} P(\lambda_{n,j}). \quad (\text{KR.8})$$

Consequently the complete arithmetic kernel is

$$K_{n-1} = \sum_{j=1}^n \frac{L_{n,j}(A) c c^* L_{n,j}(A)^*}{\omega_{n,j}}. \quad (\text{KR.9})$$

Formula (KR.9) includes all Taylor derivatives at all selected zeros, even when a selected critical zero equals a quadrature node.

Proof. For degree below n , the vector $P(J_n)e_0$ records exactly the coefficients of P in the monic orthogonal basis: in the successive multiplications no term of degree n occurs. Its initial coefficient is $\kappa_0^{-1} \int P d\nu_Z$. Thus the spectral expansion in (KR.5) proves (KR.8) in that degree range. For a general P in the asserted range, polynomial division gives $P = q_n U + R$, where $\deg U \leq n - 1$ and $\deg R < n$. On the line, the conjugate of $U(s)$ is again a polynomial in s , namely $\overline{U(1 - \bar{s})}$. Orthogonality therefore gives $\int q_n U d\nu_Z = 0$. The same multiple vanishes at all the nodes, proving (KR.8).

The polynomials $L_{n,j}$ form the interpolation basis of $\mathcal{P}_{n-1}$. Apply (KR.8) to $\overline{L_{n,j}(s)} L_{n,k}(s)$, interpreted as a polynomial on the line; its degree is at most $2n - 2$. Their Gram matrix is $\operatorname{diag}(\omega_{n,j})$. In this basis the residue map has column $j_h L_{n,j} = L_{n,j}(A)c$. The definition $K = CH^{-1}C^*$ now proves (KR.9). This is a basis change in the polynomial source only. Polynomial evaluation at A retains its nilpotent powers, so the formula gives the complete jets without any distinct-centre or simple-zero assumption. $\square$

To identify the quadrature endpoints exactly, one has

$$\sigma_{n,j} = \frac{\omega_{n,j} |q_{n-1}(\lambda_{n,j})|^2}{\kappa_{n-1}} = \frac{\kappa_{n-1}}{\omega_{n,j} |q'_n(\lambda_{n,j})|^2}. \quad (\text{KR.10})$$

Indeed an eigenvector of B_n at λ has coordinates proportional to $\overline{q_k(\lambda)}/\sqrt{\kappa_k}$: substitution into the recurrence proves this assertion, including the endpoint row. The squared ratio of its last and first coordinates proves the first equality. The preceding polynomial kernel formula gives $\mathcal{K}_{n-1}(\lambda_{n,j}, \lambda_{n,j}) = 1/\omega_{n,j}$. In the original Christoffel–Darboux polynomial identity, set $w = \lambda_{n,j}$, divide its common polynomial factor $z - \lambda_{n,j}$, then evaluate at that point. This gives

$$\frac{1}{\omega_{n,j}} = \frac{q'_n(\lambda_{n,j}) \overline{q_{n-1}(\lambda_{n,j})}}{\kappa_{n-1}}.$$

Its squared modulus gives the second equality in (KR.10).

Theorem 10.3 (The exact resolvent metric for the packet). *Suppose, for this formula, that no selected centre equals one of the specified n quadrature nodes. Define maps on the original space E by*

$$Q_n = q_n(A), \quad X_{n,j} = (A - \lambda_{n,j}I)^{-1}c, \quad S_n = \sum_j \sigma_{n,j} X_{n,j} X_{n,j}^*, \quad r_n = F_n(A)c. \quad (\text{KR.11})$$

Then Q_n is invertible, $S_n > 0$, and

$$\boxed{K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*}. \quad (\text{KR.12})$$

In particular, the original relative-control quantity at $m = n - 1 - d$ is exactly

$$\epsilon_m = \sup_{y \neq 0} \frac{|y^*(cr_n^* + r_n c^*)y|}{y^* S_n y}. \quad (\text{KR.13})$$

The identity is valid whenever that index belongs to the original source sequence; $m = -1$ is its already specified fixed-section extension. Every off-line packet obeys the nonintersection condition for every n . At an intersecting critical centre, (KR.9) gives the exact polynomial continuation, with no deletion of that centre or its jets.

Proof. The finite algebra E has an inverse for a polynomial precisely when the polynomial is nonzero at every centre: the Euclidean algorithm gives an inverse modulo h , or, equivalently, the finite Taylor inverse at each centre gives the inverse under the complete Chinese-remainder map. This proves the asserted invertibilities. Substitution of (KR.7) and (KR.10) into (KR.9) proves the first identity in (KR.12). It proves $S_n > 0$, since the residue map is onto at $n \geq d$, hence $K_{n-1} > 0$. The identity $AX_{n,j} = \lambda_{n,j} X_{n,j} + c$, together with $\lambda_{n,j} + \bar{\lambda}_{n,j} = 1$, proves

$$AS_n + S_n A^* - S_n = c \left(\sum_j \sigma_{n,j} X_{n,j} \right)^* + \left(\sum_j \sigma_{n,j} X_{n,j} \right) c^*.$$

The partial fractions in (KR.5), evaluated in the finite algebra, identify the sum with $F_n(A)c$. Because Q_n commutes with A , both kernel and its weight form are congruent, with the same factor κ_{n-1}^{-1} , to S_n and its displayed weight form. The invertible change $y = Q_n^* x$ proves equality of their Rayleigh suprema. The previous exact map from K_{n-1} to G_m then proves (KR.13). All of these maps are stated in the original quotient coordinates; the Taylor expansion of each rational factor includes every power through $m_\rho - 1$. $\square$

The scalar form of (KR.13), with every inverse taken on the same retained packet coordinates, is

$$\alpha_n = r_n^* S_n^{-1} r_n, \quad \beta_n = c^* S_n^{-1} c, \quad \gamma_n = c^* S_n^{-1} r_n,$$

$$\boxed{\epsilon_m = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}}. \quad (\text{KR.13a})$$

Indeed the nonzero eigenvalues of $S_n^{-1/2}(cr_n^* + r_n c^*)S_n^{-1/2}$ coincide, with multiplicities, with the nonzero eigenvalues of $\begin{pmatrix} \gamma_n & \beta_n \\ \alpha_n & \bar{\gamma}_n \end{pmatrix}$: factor the former through its two displayed vectors and interchange the two rectangular factors. Its characteristic polynomial gives $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Cauchy-Schwarz gives $|\gamma_n|^2 \leq \alpha_n \beta_n$, so their greatest absolute value is (KR.13a), including zero or one-dimensional rank. The full congruence (KR.12) specifies exactly how the original q_n , κ_{n-1} and local unit enter the metric whose relative form is evaluated here.

10.3 The infinite multiplication operator and its genuine tail

Let $\mathcal{L} = L^2(\nu_Z)$, and let M be multiplication by s , with domain $\{f : sf \in \mathcal{L}\}$. It is a closed operator and $M^* = I - M$. These statements follow directly from the multiplication formula; its resolvent off the line is multiplication by $(z - s)^{-1}$, of norm at most $\delta(z)^{-1}$. The auxiliary vectors $\psi_k = q_k / \sqrt{\kappa_k}$ form a complete orthonormal basis. Completeness follows from the positive exponential moment of the actual arithmetic measure by the Fourier-annihilator proof already given for the theta measures.

Let P_n project onto $\psi_0, \dots, \psi_{n-1}$, and put $Q_n^{\text{tail}} = I - P_n$; this projection is distinct from the polynomial map $Q_n = q_n(A)$ in (KR.11). Define

$$M_n^{\text{tail}} = Q_n^{\text{tail}} M Q_n^{\text{tail}} \quad \text{on } Q_n^{\text{tail}} \text{Dom } M, \quad \zeta_n(z) = \langle \psi_n, (z - M_n^{\text{tail}})^{-1} \psi_n \rangle. \quad (\text{KR.14})$$

This tail resolvent exists at every off-line z , with norm at most $\delta(z)^{-1}$. Here is the domain justification. Every polynomial belongs to $\text{Dom } M$, so P_n preserves that domain and $Q_n^{\text{tail}} \text{Dom } M = \text{Dom } M \cap Q_n^{\text{tail}} \mathcal{L}$. The operator $(M - \frac{1}{2}I)/i$ is self-adjoint multiplication by t . Its off-diagonal blocks for $P_n \oplus Q_n^{\text{tail}}$ extend to a bounded, finite-rank self-adjoint operator, by (KR.1). Subtracting this operator preserves self-adjointness: for a self-adjoint T and bounded self-adjoint C , the definition of the adjoint gives $(T - C)^* = T^* - C$ on exactly $\text{Dom } T^*$. The resulting diagonal operator reduces the two projections. Its tail block is therefore self-adjoint on the displayed tail domain. Multiplication by i and addition of $1/2$ gives the claimed tail resolvent.

Theorem 10.4 (Exact finite-section, terminal, and tail comparison). *For $\delta(z) > 0$, let*

$$m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t).$$

Then

$$H_n^{\text{obs}}(z) = \frac{\zeta_n(z)}{1 + a_n F_n(z) \zeta_n(z)}, \quad (\text{KR.15})$$

$$m(z) - m_n(z) = \frac{(-1)^n \kappa_n \zeta_n(z)}{q_n(z)^2 [1 + a_n F_n(z) \zeta_n(z)]} = \frac{(-1)^n}{q_n(z)^2} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t). \quad (\text{KR.16})$$

Every denominator in these formulas is nonzero off the original Mellin line. The terminal finite observations additionally obey the exact Schur recurrence

$$F_1(z) = \frac{1}{z - b_0}, \quad F_{n+1}(z) = \frac{1}{z - b_n + a_n F_n(z)}. \quad (\text{KR.17})$$

Proof. Under $\mathcal{I}_n \oplus I$, the block operator M is

$$\begin{pmatrix} B_n & -\sqrt{a_n} e_{n-1} \langle \psi_n, \cdot \rangle \\ \sqrt{a_n} \psi_n e_{n-1}^* & M_n^{\text{tail}} \end{pmatrix}. \quad (\text{KR.18})$$

The signs follow directly from (KR.1); both coupling maps are bounded on the displayed summands. Eliminate the tail component in $z - M$. The finite Schur complement is $zI - B_n + a_n \zeta_n(z) e_{n-1} e_{n-1}^*$. For $\text{Re } z > 1/2$, the real part of its quadratic form is at least $\delta(z) \|u\|^2$, because $\text{Re } \zeta_n(z) > 0$; on the other side all signs reverse. The sign of this scalar real part follows either from the self-adjoint tail resolvent or from the resolvent identity $\text{Re} \langle v, (z - M_n^{\text{tail}})^{-1} v \rangle = (\text{Re } z - 1/2) \|(z - M_n^{\text{tail}})^{-1} v\|^2$. Thus the Schur complement is invertible and $1 + a_n \zeta_n F_n \neq 0$. Direct multiplication verifies the rank-one inverse formula

$$R_n - \frac{a_n \zeta_n}{1 + a_n \zeta_n F_n} R_n e_{n-1} e_{n-1}^* R_n, \quad R_n = (zI - B_n)^{-1}. \quad (\text{KR.19})$$

The analogous bottom Schur calculation, paired against ψ_n , gives (KR.15), since $\langle \psi_n, (z - M)^{-1} \psi_n \rangle = H_n^{\text{obs}}(z)$.

The two endpoint cofactors of $zI - B_n$ give

$$e_0^* R_n e_{n-1} = \frac{(-1)^{n-1} \sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}, \quad e_{n-1}^* R_n e_0 = \frac{\sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}. \quad (\text{KR.20})$$

These cofactors are the products of the successive upper and lower entries, respectively; the formula also covers $n = 1$. Pair (KR.19) against $\sqrt{\kappa_0} e_0$, the coefficient vector of the unchanged function 1. Substitute (KR.20) and $a_n \kappa_{n-1} = \kappa_n$. This proves the first equality in (KR.16); (KR.15) proves the second. Finally (KR.17) is (KR.1) divided by q_{n+1} , equivalently the last-coordinate finite Schur complement. $\square$

There is also a direct polynomial check on every factor in (KR.16). The polynomial

$$p_n(z) = \int \frac{q_n(z) - q_n(s)}{z - s} d\nu_Z(t)$$

has degree below n . At a quadrature node its ratio residue is $p_n(\lambda_{n,j})/q'_n(\lambda_{n,j}) = \int L_{n,j} d\nu_Z = \omega_{n,j}$, so $m_n = p_n/q_n$. Orthogonality to the polynomial $(q_n(z) - q_n(s))/(z - s)$ gives

$$m - m_n = \frac{1}{q_n(z)^2} \int \frac{q_n(s)^2}{z - s} d\nu_Z.$$

The polynomial $(-1)^n \overline{q_n(1 - \bar{s})}$ is monic and has the same orthogonality conditions as q_n ; uniqueness makes it equal to q_n . On the line this says $q_n(s)^2 = (-1)^n |q_n(s)|^2$, giving (KR.16) again. In particular the explicit error inequality is

$$|m(z) - m_n(z)| \leq \frac{\kappa_n}{\delta(z) |q_n(z)|^2}. \quad (\text{KR.21})$$

The initial and terminal observations have now been related by the actual block maps (KR.18)–(KR.20), the tail (KR.14), and the exact remainder (KR.16). The relative form uses the terminal observation through (KR.11)–(KR.13). An estimate for the initial observation can be propagated through these equalities while retaining the factors q_n , κ_n , a_n , and the tail denominator. None of those arithmetic quantities is discarded in passing from one observation to the other.

11 Quantitative convergence of the arithmetic Gaussian resolvents

Retain the actual measure and every coordinate of Section 10:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

The polynomial h contains the complete multiplicities of its selected actual zeros. The quotient g/h is therefore entire, including at the selected critical zeros. Use the initial Gaussian observation of (KR.5) and (KR.8),

$$m_n(z) = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \quad m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad \mathcal{E}_n(z) = m(z) - m_n(z), \quad \delta(z) = |\operatorname{Re} z - \tfrac{1}{2}|. \quad (\text{GC.1})$$

The Gaussian weights are positive and the distinct nodes lie on the original Mellin line, as proved in Section 10. Its exact quadrature (KR.8) and exact error (KR.16) are the inputs below. No asymptotic for q_n or κ_n is assumed.

For each fixed $0 < a < \pi/2$, the number

$$E_a = \max \left\{ \int e^{at} d\nu_Z(t), \int e^{-at} d\nu_Z(t) \right\} \quad (\text{GC.2})$$

is finite. Here is the connection to the proved arithmetic decay. The actual theta calculation in Section 59 gives $|g(1/2 + it)| \leq C_b e^{-b|t|}$ for every $0 < b < \pi/4$. Choose $a/2 < b < \pi/4$. Outside a sufficiently large compact interval, the monic polynomial h of degree d satisfies $|h(1/2 + it)| \geq c_h (1 + |t|)^d$. Thus the density multiplied by $e^{a|t|}$ is bounded there by a constant times $(1 + |t|)^{-2d} e^{-(2b-a)|t|}$. On the compact interval the entire quotient is bounded. Both parts are integrable, proving (GC.2) with the original factor $dt/(2\pi)$ retained.

11.1 The Gaussian rule inherits the original exponential moments

Lemma 11.1 (Exponential moment domination). *For every real v with $|v| \leq a$, write $\lambda_{n,j} = 1/2 + it_{n,j}$. Then*

$$\sum_j \omega_{n,j} e^{vt_{n,j}} \leq \int e^{vt} d\nu_Z(t) \leq E_a. \quad (\text{GC.3})$$

Proof. Let H be the real Hermite interpolation polynomial of degree at most $2n - 1$ whose values and first derivatives at the distinct real nodes $t_{n,j}$ equal those of $f(t) = e^{vt}$. Its existence and uniqueness follow from the invertibility of the interpolation map: a polynomial in its kernel has a double zero at each of the n nodes, hence is zero in degree below $2n$; domain and target both have dimension $2n$. For a real t outside the nodes the interpolation remainder is

$$f(t) - H(t) = \frac{f^{(2n)}(\xi)}{(2n)!} \prod_j (t - t_{n,j})^2 \geq 0 \quad (\text{GC.4})$$

for some real ξ between t and the nodes. To prove the remainder, subtract $C \prod_j (x - t_{n,j})^2$ from $f(x) - H(x)$, choosing C to produce an additional zero at $x = t$. The resulting function has the n double zeros and the additional zero. Repeated Rolle's theorem gives a zero of its $2n$ -th derivative; solving at that zero gives (GC.4). The last sign follows from $f^{(2n)}(x) = v^{2n} e^{vx} \geq 0$. At a node the remainder is zero.

Integrate the pointwise inequality. The exponential and every polynomial are integrable by (GC.2). Since H is a polynomial in $t = (s - 1/2)/i$ of degree at most $2n - 1$, (KR.8) yields

$$\int H(t) d\nu_Z(t) = \sum_j \omega_{n,j} H(t_{n,j}) = \sum_j \omega_{n,j} e^{vt_{n,j}}.$$

This proves the first inequality in (GC.3). Pointwise convexity of e^{vt} in $v \in [-a, a]$, followed by integration, proves the second. No bound on the largest quadrature node has been inserted. $\square$

11.2 Exact moments give a quantitative analytic-strip estimate

Define

$$\Phi(u) = \int e^{iut} d\nu_Z(t), \quad \Phi_n(u) = \sum_j \omega_{n,j} e^{iut_{n,j}}, \quad \Delta_n(u) = \Phi(u) - \Phi_n(u). \quad (\text{GC.5})$$

These functions are holomorphic on $|\operatorname{Im} u| < a$. Indeed, on each smaller closed strip, every differentiated integrand is dominated by a constant times $e^{at} + e^{-at}$, because $|t|^k e^{b|t|} \leq C_{k,b,a} (e^{at} + e^{-at})$ when $b < a$. Dominated convergence proves all the derivatives of Φ ; the finite sum is entire. Equation (GC.3), applied at $v = -\operatorname{Im} u$, gives

$$|\Delta_n(u)| \leq 2E_a \quad (|\operatorname{Im} u| < a), \quad \Delta_n^{(k)}(0) = 0 \quad (0 \leq k < 2n). \quad (\text{GC.6})$$

The derivative identities are exactly the moments in (KR.8).

Lemma 11.2 (The complete strip bound). *For $|\operatorname{Im} u| < a$,*

$$|\Delta_n(u)| \leq 2E_a \left| \tanh\left(\frac{\pi u}{4a}\right) \right|^{2n}. \quad (\text{GC.7})$$

Proof. The explicit holomorphic coordinate maps

$$w = \tanh\left(\frac{\pi u}{4a}\right), \quad u = \frac{2a}{\pi} \operatorname{Log} \frac{1+w}{1-w}$$

are inverse bijections between the strip and the unit disk. For $|w| < 1$, the fraction lies in the right half-plane, so its principal logarithm has imaginary part in $(-\pi/2, \pi/2)$, which proves the asserted strip range. Compose $\Delta_n/(2E_a)$ with this inverse. By (GC.6), the resulting disk function is bounded by one and has a zero of order at least $2n$ at zero. Divide by w^{2n} , filling in its removable value at zero. On each circle $|w| = r < 1$, its modulus is at most r^{-2n} ; the maximum principle gives this bound inside the circle. Letting r increase to one proves the required bound by $|w|^{2n}$, hence (GC.7). $\square$

Theorem 11.3 (Direct off-line convergence with an explicit rate). *For every $0 < a < \pi/2$, every $n \geq 1$, and every z off the original Mellin line,*

$$|m(z) - m_n(z)| \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\delta(z)}{\pi}\right) (4n)^{-2a\delta(z)/\pi}. \quad (\text{GC.8})$$

On a compact set K disjoint from that line the same bound holds uniformly after replacing $\delta(z)$ everywhere by $\delta_K = \min_{z \in K} \delta(z) > 0$.

Proof. Put $w = z - 1/2$. On $\operatorname{Re} w > 0$, the elementary Laplace identity and Fubini's theorem give

$$\mathcal{E}_n(z) = \int_0^\infty e^{-wu} \Delta_n(u) du. \quad (\text{GC.9})$$

For example the absolute integral for the original measure before subtraction is at most $\nu_Z(\mathbb{R})/\operatorname{Re} w$, so the interchange is justified. On $\operatorname{Re} w < 0$, the exact orientation is

$$\mathcal{E}_n(z) = - \int_0^\infty e^{wu} \Delta_n(-u) du. \quad (\text{GC.10})$$

These follow respectively from integrating $e^{-(w-it)u}$ and $-e^{(w-it)u}$. Set $b = \pi/(4a)$. Equation (GC.7) gives

$$|\mathcal{E}_n(z)| \leq 2E_a I_n(\delta(z)), \quad I_n(\delta) = \int_0^\infty e^{-\delta u} \tanh(bu)^{2n} du. \quad (\text{GC.11})$$

For $0 < x < 1$, direct integration gives

$$\log \frac{1-x}{1+x} = -2 \int_0^x \frac{dv}{1-v^2} \leq -2x.$$

Taking $x = e^{-2bu}$ yields $\tanh(bu)^{2n} \leq e^{-4ne^{-2bu}}$. With $\beta = \delta/(2b) = 2a\delta/\pi$, the same variable change gives

$$\begin{aligned} I_n(\delta) &\leq \frac{1}{2b} \int_0^1 x^{\beta-1} e^{-4nx} dx \\ &\leq \frac{1}{2b} (4n)^{-\beta} \Gamma(\beta), \quad \Gamma(\beta) = \int_0^\infty y^{\beta-1} e^{-y} dy. \end{aligned} \quad (\text{GC.12})$$

Substitution in (GC.11) proves (GC.8). On the compact set one uses $e^{-\delta(z)u} \leq e^{-\delta_K u}$ in (GC.11) and then (GC.12) at δ_K ; no monotonicity of the gamma factor is assumed. This proves locally uniform convergence on both components of the off-line domain. $\square$

11.3 Every confluent derivative and the original finite algebra

Write $F^{[k]} = F^{(k)}/k!$. The exact error (KR.16), with its vertical-line sign, gives for all $k \geq 0$

$$\mathcal{E}_n^{[k]}(z) = (-1)^n \sum_{j=0}^k (q_n^{-2})^{[k-j]}(z) (-1)^j \int \frac{|q_n(s)|^2}{(z-s)^{j+1}} d\nu_Z(t). \quad (\text{GC.13})$$

Every derivative under this integral is justified on a compact off-line neighborhood by its positive distance from the line and the finite measure $|q_n|^2 \nu_Z$. The product rule for Taylor coefficients has no binomial factors; those are precisely cancelled by the convention $F^{[k]} = F^{(k)}/k!$. For completeness its polynomial factor is explicitly

$$(q_n^{-2})^{[r]}(z) = \frac{(-1)^r}{q_n(z)^2} \sum_{r_1+\dots+r_n=r} \prod_{j=1}^n \frac{r_j+1}{(z-\lambda_{n,j})^{r_j}}. \quad (\text{GC.14})$$

This follows by multiplying the Taylor series of the n original factors $(z - \lambda_{n,j})^{-2}$, retaining every cross term.

Differentiating (GC.9)–(GC.10) gives the independent estimate

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{2E_a}{k!} \int_0^\infty u^k e^{-\delta(z)u} \tanh(bu)^{2n} du. \quad (\text{GC.15})$$

The integrable majorant $u^k e^{-\delta_K u}$ on a smaller off-line compact neighborhood justifies this differentiation. If $0 < \sigma < \delta$, the exponential series gives $u^k e^{-(\delta-\sigma)u} \leq k!/(\delta-\sigma)^k$. Equations (GC.11)–(GC.12) prove

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{4aE_a}{\pi(\delta-\sigma)^k} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi}, \quad 0 < \sigma < \delta \leq \delta(z). \quad (\text{GC.16})$$

Taking $\delta = \delta_K$ makes this a simultaneous compact-uniform bound on every fixed finite list of derivatives.

Here is the complete algebra transport, including the part of the original packet outside this resolvent domain. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{d_\rho}, \quad H_\rho(s) = \frac{h(s)}{(s - \rho)^{d_\rho}}, \quad U_\rho(s) = \sum_{k=0}^{d_\rho-1} (H_\rho^{-1})^{[k]}(\rho) (s - \rho)^k,$$

and define the actual polynomial remainder

$$e_\rho = j_h(H_\rho U_\rho) \in E. \quad (\text{GC.17})$$

Its remainder is one modulo $(s - \rho)^{d_\rho}$ and zero modulo every other factor of h . The distinct factors are coprime, and a polynomial divisible by each is divisible by their product: successive use of the Bezout identity proves this last statement. It follows directly that

$$e_\rho e_\eta = 0 \ (\rho \neq \eta), \quad e_\rho^2 = e_\rho, \quad \sum_{\rho \in Z} e_\rho = 1. \quad (\text{GC.18})$$

Restriction of a polynomial to its truncated Taylor coefficients therefore gives the exact algebra isomorphism

$$E \longrightarrow \bigoplus_{\rho \in Z} \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{d_\rho}), \quad j_h P \longmapsto \left(\sum_{k < d_\rho} P^{[k]}(\rho) \epsilon_\rho^k \right)_\rho. \quad (\text{GC.19})$$

Its inverse sends the coordinate ϵ_ρ^k to $e_\rho(s-\rho)^k$; composition is the identity by the just proved congruences. Multiplication is preserved because the Taylor coefficient of a product is the exact convolution of the two coefficient lists, truncated only by the displayed ideal.

Let $Z_{\text{off}} = \{\rho \in Z : \text{Re } \rho \neq 1/2\}$ and $Z_{\text{line}} = Z \setminus Z_{\text{off}}$. Define

$$e_{\text{off}} = \sum_{\rho \in Z_{\text{off}}} e_\rho, \quad e_{\text{line}} = 1 - e_{\text{off}}, \quad E_{\text{off}} = e_{\text{off}}E, \quad E_{\text{line}} = e_{\text{line}}E. \quad (\text{GC.20})$$

Then $E = E_{\text{off}} \oplus E_{\text{line}}$ as algebras with their respective idempotent units. The projection $u \mapsto e_{\text{off}}u$ has exactly kernel E_{line} and commutes with the original multiplication operator A . Under (GC.19), its restriction at every retained centre is $A_\rho = \rho I + N_\rho$, where multiplication by ϵ_ρ gives $N_\rho^{d_\rho} = 0$ and retains all lower powers. This constructs the precise map from the full packet to the off-line resolvent domain, together with its full kernel and the operator on that kernel. An empty off-line set gives the zero summand, with the full original packet in the kernel.

For a function f holomorphic near Z_{off} , define

$$f(A_{\text{off}}) = \sum_{\rho \in Z_{\text{off}}} \sum_{k=0}^{d_\rho-1} f^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \quad (\text{GC.21})$$

Here $e_\rho(A)$ means multiplication by the explicitly constructed remainder (GC.17). Equation (GC.19) proves that (GC.21) is a unital algebra homomorphism on the off-line summand. No diagonalizability of A has entered this construction.

In particular, finite geometric expansion in each nilpotent block proves

$$\begin{aligned} m(A_{\text{off}}) &= \int (A_{\text{off}} - sI)^{-1} d\nu_Z(t), \\ m_n(A_{\text{off}}) &= \sum_j \omega_{n,j}(A_{\text{off}} - \lambda_{n,j}I)^{-1}, \\ \mathcal{E}_n(A_{\text{off}}) &= \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \mathcal{E}_n^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \end{aligned} \quad (\text{GC.22})$$

For the first equality the expansion is

$$(A - sI)^{-1} e_\rho(A) = \sum_{k < d_\rho} \frac{(-1)^k (A - \rho I)^k e_\rho(A)}{(\rho - s)^{k+1}}.$$

Each coefficient is integrable since $\delta(\rho) > 0$, and its integral is exactly $m^{[k]}(\rho)$. This proves the integral identity, rather than assigning a formal meaning to it.

For a concrete bound keep any fixed positive Hermitian form G_* on the original coefficient space and restrict it to E_{off} . When that summand is nonzero, put

$$\delta_* = \min_{\rho \in Z_{\text{off}}} \delta(\rho) > 0, \quad C_{\rho,k} = \left\| (A - \rho I)^k e_\rho(A)|_{E_{\text{off}}} \right\|_{G_*}.$$

The triangle inequality and (GC.16) give, for every $0 < \sigma < \delta_*$,

$$\|m(A_{\text{off}}) - m_n(A_{\text{off}})\|_{G_*} \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi} \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \frac{C_{\rho,k}}{(\delta_* - \sigma)^k}. \quad (\text{GC.23})$$

Thus the full initial resolvent converges in the original finite coefficient space, with every nilpotent contribution controlled. Applying the displayed operators to the retained vector $e_{\text{off}}c$ gives the corresponding vector statement with the additional factor $\|e_{\text{off}}c\|_{G_*}$.

The polynomial kernel (KR.9) continues to act on all of E , including E_{line} and collisions with quadrature nodes. The off-line estimates above concern the initial observation m_n . Its exact map to the terminal observation F_n is the block and tail comparison (KR.14)–(KR.20), with the factors q_n, κ_n, a_n and the tail denominator still present. The estimate (GC.23) may be inserted into those identities; it does not remove their arithmetic factors or the critical-line summand of (GC.20).

12 The exact terminal step and its retained boundary energy

All measures, packet coordinates and maps in this section are the original arithmetic ones. Thus $s = 1/2 + it$, $d\nu_Z = |g(s)/h(s)|^2 dt/(2\pi)$, $E = \mathbb{C}[s]/(h)$, $A = M_s$, and $c = j_h 1$. Every selected zero occurs with its complete multiplicity in h . Retain the monic q_n , their actual norms κ_n , and

$$a_n = \frac{\kappa_n}{\kappa_{n-1}} > 0, \quad q_{n+1}(s) = (s - b_n)q_n(s) + a_n q_{n-1}(s), \quad b_n + \bar{b}_n = 1. \quad (\text{KT.1})$$

No recurrence coefficient is replaced by a limiting value.

12.1 A step written entirely in the original polynomial packet

For $n \geq d = \deg h$, put

$$v_j = q_j(A)c, \quad K = K_{n-1} = \sum_{j=0}^{n-1} \frac{v_j v_j^*}{\kappa_j} > 0, \quad M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}. \quad (\text{KT.2})$$

Use the positive form $\langle x, y \rangle_n = x^* M_n y$, conjugate-linear in its first argument, and define

$$u_n = v_n, \quad w_n = v_{n-1}, \quad \alpha_n = \langle w_n, w_n \rangle_n, \quad \beta_n = \langle u_n, u_n \rangle_n, \quad \gamma_n = \langle u_n, w_n \rangle_n. \quad (\text{KT.3})$$

The arithmetic relative allowance at source level $m = n - d - 1$ will be denoted by $\epsilon^{(n)}$. The fixed-section extension at $m = -1$ has already been specified; the original corrected sequence has $n \geq d + 1$. The exact kernel boundary identity gives

$$\epsilon^{(n)} = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}, \quad \operatorname{Re} \gamma_n = \operatorname{Re} \operatorname{tr} A - \frac{d}{2} =: g_Z. \quad (\text{KT.4})$$

Indeed the kernel weight form is $(u_n w_n^* + w_n u_n^*)/\kappa_{n-1}$. Its possible nonzero generalized eigenvalues are obtained by reversing the two rectangular factors, giving $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Taking the trace after multiplication by K^{-1} gives the second identity. Cauchy–Schwarz proves the nonnegativity of the radicand and justifies the maximum absolute value in (KT.4).

Theorem 12.1 (Universal scalar and metric step). *Write $a = a_n$, and retain the exact next vector*

$$t_n = (A - b_n I)u_n + a_n w_n = v_{n+1}. \quad (\text{KT.5})$$

Then

$$M_{n+1} = \frac{1}{a} \left(M_n - \frac{M_n u_n u_n^* M_n}{a + \beta_n} \right), \quad (\text{KT.6})$$

$$\begin{aligned} \alpha_{n+1} &= \frac{\beta_n}{a + \beta_n}, \\ \gamma_{n+1} &= \frac{\langle t_n, u_n \rangle_n}{a + \beta_n}, \\ \beta_{n+1} &= \frac{1}{a} \left(\langle t_n, t_n \rangle_n - \frac{|\langle u_n, t_n \rangle_n|^2}{a + \beta_n} \right). \end{aligned} \quad (\text{KT.7})$$

Every denominator is strictly positive. These identities hold also when any selected critical centre equals a quadrature node, and when $q_n(A)$ is singular.

The exact determinant changes of the kernel and original theta metric are

$$\boxed{\frac{\det K_n}{\det K_{n-1}} = 1 + \frac{\beta_n}{a_n}, \quad \frac{\det G_{m+1}}{\det G_m} = \frac{a_n}{a_n + \beta_n} = 1 - \alpha_{n+1}.} \quad (\text{KT.8})$$

Proof. The unchanged kernel update is $K_n = K_{n-1} + u_n u_n^* / \kappa_n$. Its inverse is verified by multiplication:

$$K_n^{-1} = K_{n-1}^{-1} - \frac{K_{n-1}^{-1} u_n u_n^* K_{n-1}^{-1}}{\kappa_n + u_n^* K_{n-1}^{-1} u_n}.$$

Insert $\kappa_n = a \kappa_{n-1}$ and the definition of M_n ; this gives (KT.6). At the next step the ordered vectors in (KT.3) are $u_{n+1} = t_n$, $w_{n+1} = u_n$. Pair each ordered pair with (KT.6). The three resulting expressions are precisely (KT.7), including their conjugations. The denominator is positive because $a > 0$, $\beta_n \geq 0$. The determinant of the rank-one update is $1 + u_n^* K_{n-1}^{-1} u_n / \kappa_n = 1 + \beta_n / a$; this follows directly by expanding the determinant multilinearly in its columns. The original metric is $G_m = U_\varepsilon^* K_{n-1}^{-1} U_\varepsilon$, with the same fixed invertible local unit at adjacent levels. Its determinant ratio is therefore the inverse kernel ratio. Equation (KT.7) gives the last equality in (KT.8). All calculations use polynomial residues at A ; no inverse of $q_n(A)$ has been used. $\square$

12.2 The complete boundary-energy increment

Suppose first that $\beta_n > 0$. The exact orthogonal projection for $\langle \cdot, \cdot \rangle_n$ is

$$\begin{aligned} P_n x &= x - u_n \frac{\langle u_n, x \rangle_n}{\beta_n}, & p_n &= P_n w_n, & z_n &= P_n A u_n, \\ \Delta_n &= \alpha_n \beta_n - |\gamma_n|^2 = \beta_n \|p_n\|_n^2, & \Omega_n &= \beta_n \|z_n\|_n^2. \end{aligned} \quad (\text{KT.9})$$

The projection records a specified map inside the original packet; $A u_n$, its removed coefficient, and its remaining vector z_n are all retained in (KT.9).

Theorem 12.2 (Exact transverse energy law). *At every such step,*

$$\boxed{\Delta_{n+1} = \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}}. \quad (\text{KT.10})$$

Consequently

$$\boxed{\Delta_{n+1} - \Delta_n = \frac{\beta_n}{a_n(a_n + \beta_n)} (\|z_n\|_n^2 + 2a_n \operatorname{Re} \langle z_n, p_n \rangle_n - a_n \beta_n \|p_n\|_n^2)}. \quad (\text{KT.11})$$

The exact energy law gives the following two-sided bounds by the sharp triangle inequalities:

$$\frac{|a_n \sqrt{\Delta_n} - \sqrt{\Omega_n}|}{\sqrt{a_n(a_n + \beta_n)}} \leq \sqrt{\Delta_{n+1}} \leq \frac{a_n \sqrt{\Delta_n} + \sqrt{\Omega_n}}{\sqrt{a_n(a_n + \beta_n)}}. \quad (\text{KT.12})$$

The actual relative allowance is recovered exactly by

$$\boxed{\epsilon^{(n)} = |g_Z| + \sqrt{g_Z^2 + \Delta_n}}. \quad (\text{KT.13})$$

For a reflection-stable packet with its multiplicities, $g_Z = 0$, so the energy law calculates the square of the actual allowance without a trace term.

Proof. Let $T = \langle t_n, t_n \rangle_n$ and $v = \langle u_n, t_n \rangle_n$. Substitution of (KT.7) gives

$$\alpha_{n+1}\beta_{n+1} - |\gamma_{n+1}|^2 = \frac{\beta_n T - |v|^2}{a_n(a_n + \beta_n)} = \frac{\beta_n \|P_n t_n\|_n^2}{a_n(a_n + \beta_n)}.$$

Since $P_n u_n = 0$, the exact recurrence (KT.5) gives $P_n t_n = P_n A u_n + a_n P_n w_n = z_n + a_n p_n$. This proves (KT.10). Expand the squared norm and subtract $\Delta_n = \beta_n \|p_n\|_n^2$ to prove (KT.11). The triangle inequality and its reverse, applied to these two specific vectors, prove (KT.12). Finally (KT.4) gives (KT.13). Its real trace term is unchanged with n ; equivalently the same trace computation can be performed at each adjacent positive kernel. Reflection permutes all roots and multiplicities, so $2 \operatorname{Re} \operatorname{tr} A = d$ in that case. $\square$

If $\beta_n = 0$, positivity of M_n forces $u_n = v_n = 0$. Then $\gamma_n = 0$, the current kernel weight is zero, $K_n = K_{n-1}$, and (KT.5)–(KT.7) give the entire exceptional step:

$$t_n = a_n w_n, \quad \alpha_{n+1} = \gamma_{n+1} = 0, \quad \beta_{n+1} = a_n \alpha_n, \quad \Delta_n = \Delta_{n+1} = 0, \quad \epsilon^{(n)} = \epsilon^{(n+1)} = 0. \quad (\text{KT.14})$$

Moreover $v_n = 0$ is exactly the divisibility $h \mid q_n$. Since every root of q_n is simple and lies on the Mellin line, this particular event forces the selected packet to consist of simple line roots. It does not remove the resulting packet or the next vectors in (KT.14).

12.3 The exact rational terminal step and its polynomial continuation

Where $q_n(A)$ and $q_{n+1}(A)$ are invertible, use the actual terminal maps and Gram from Section 10:

$$T_n = F_n(A), \quad r_n = T_n c, \quad Q_n = q_n(A), \quad K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*.$$

The map Q_n is the exact isometry

$$Q_n : (E, S_n^{-1}) \longrightarrow (E, M_n), \quad Q_n c = u_n, \quad Q_n r_n = w_n. \quad (\text{KT.15})$$

Indeed $Q_n^* M_n Q_n = S_n^{-1}$ follows by inverting the displayed Gram congruence, and the vector identity uses $F_n = q_{n-1}/q_n$. This proves the precise relation between the rational and polynomial forms of (KT.3)–(KT.13).

The terminal recurrence gives

$$T_{n+1} = (A - b_n I + a_n T_n)^{-1}, \quad \boxed{S_{n+1} = T_{n+1} (c c^* + a_n S_n) T_{n+1}^*}. \quad (\text{KT.16})$$

For a direct full-jet proof, the scalar kernel of a terminal observation is

$$H_{F_n}(z, w) = \frac{F_n(z) + \overline{F_n(w)}}{z + \overline{w} - 1}.$$

The quotient denotes its removable polynomial/rational continuation where numerator and denominator both vanish. It is identically the positive resolvent Gram $\sum_j \sigma_{n,j} / [(z - \lambda_{n,j})(\overline{w} - \overline{\lambda}_{n,j})]$. Substitution of $F_{n+1} = (z - b_n + a_n F_n)^{-1}$ gives

$$H_{F_{n+1}}(z, w) = F_{n+1}(z) \overline{F_{n+1}(w)} [1 + a_n H_{F_n}(z, w)]. \quad (\text{KT.17})$$

This identity holds first away from its removable sets and hence through them by continuation. Taking the complete residue on both variables proves (KT.16). It also follows from the original kernel update and $Q_{n+1}^{-1} Q_n = T_{n+1}$. Thus a quadrature collision is handled by the polynomial formulas (KT.2)–(KT.14), which retain all coordinates; (KT.15) states exactly the isometry on the open set where the rational presentation is defined.

In this presentation the determinant still retains the entire polynomial factor:

$$\frac{\det S_{n+1}}{\det S_n} = \frac{a_n^{d-1}(a_n + \beta_n)}{|\det(A - b_n I + a_n T_n)|^2}, \quad \det(A - b_n I + a_n T_n) = \prod_{\rho \in Z} \left(\frac{q_{n+1}(\rho)}{q_n(\rho)} \right)^{m_\rho}. \quad (\text{KT.18})$$

The determinant of $a_n S_n + cc^*$ gives the first formula. The second follows by evaluating the polynomial ratio on each full triangular Taylor block $\rho I + N_\rho$. The determinant has the stated multiplicity even when the block is not semisimple.

12.4 The extra jets required to recover the metric from the tail

There is an exact map from the scalar tail identity to the matrix energy, with a specified number of Taylor coefficients. For (KT.19)–(KT.22), take centres at which q_n is nonzero, so F_n is holomorphic there. Every off-line centre meets this condition. At a critical quadrature intersection the universal polynomial formulas (KT.2)–(KT.14) retain the complete packet. The terminal function satisfies $\overline{F_n(1 - \bar{z})} = -F_n(z)$, by its positive spectral sum on the line. Thus

$$H_{F_n}(z, w) = \frac{F_n(z) - F_n(1 - \bar{w})}{z - (1 - \bar{w})}. \quad (\text{KT.19})$$

For centres ρ, σ , and Taylor orders $0 \leq k < m_\rho, 0 \leq l < m_\sigma$, let $S_n^{\text{jet}}(\rho, k; \sigma, l)$ denote the coefficient of $(z - \rho)^k (\bar{w} - \bar{\sigma})^l$ in this kernel. In particular, at a reflection pair,

$$\boxed{S_n^{\text{jet}}(\rho, k; \sigma, l) = (-1)^l F_n^{[k+l+1]}(\rho) \quad \text{if } \rho = 1 - \bar{\sigma}.} \quad (\text{KT.20})$$

Here $F^{[j]} = F^{(j)}/j!$. To prove the formula, put $z = \rho + x, 1 - \bar{w} = \rho - y$ in (KT.19), and expand

$$\frac{x^r - (-y)^r}{x + y} = \sum_{j=0}^{r-1} x^{r-1-j} (-y)^j.$$

The coefficient of $x^k y^l$ is supplied exactly by $r = k + l + 1, j = l$, proving (KT.20) without dividing by zero.

For $D_{\rho\sigma} = \rho + \bar{\sigma} - 1 \neq 0$, the corresponding complete coefficient is

$$\begin{aligned} S_n^{\text{jet}}(\rho, k; \sigma, l) &= \sum_{i=0}^k F_n^{[i]}(\rho) \frac{(-1)^{k-i+l} \binom{k-i+l}{k-i}}{D_{\rho\sigma}^{k-i+l+1}} \\ &\quad + \sum_{j=0}^l \overline{F_n^{[j]}(\sigma)} \frac{(-1)^{k+l-j} \binom{k+l-j}{k}}{D_{\rho\sigma}^{k+l-j+1}}. \end{aligned} \quad (\text{KT.21})$$

This is the full product of the numerator Taylor series and the geometric expansion of $(D_{\rho\sigma} + x + y)^{-1}$. Both constant terms in the numerator are retained. If $V_Z : E \rightarrow \bigoplus_\rho \mathbb{C}^{m_\rho}$ is the fixed complete Taylor-coordinate map, then

$$S_n^{\text{jet}} = V_Z S_n V_Z^*, \quad S_n = V_Z^{-1} S_n^{\text{jet}} (V_Z^{-1})^*. \quad (\text{KT.22})$$

This follows by applying V_Z to each resolvent column in the positive Gram representation of S_n .

To specify the domain of the analytic extraction, let $h_{\text{off}} = \prod_{\text{Re } \rho \neq 1/2} (s - \rho)^{m_\rho}$. The full original algebra projects onto $E_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}})$; the kernel is the retained critical-line CRT summand, with its original action of A . Above the off-line summand use the doubled algebra $\widehat{E}_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}}^2)$. Its exact reduction sequence of $\mathbb{C}[s]$ -modules is

$$0 \longrightarrow E_{\text{off}} \xrightarrow{[p] \mapsto [h_{\text{off}} p]} \widehat{E}_{\text{off}} \xrightarrow{[p] \mapsto [p]} E_{\text{off}} \longrightarrow 0. \quad (\text{KT.23})$$

The first map is well-defined and injective because $h_{\text{off}}^2 \mid h_{\text{off}}p$ holds exactly when $h_{\text{off}} \mid p$; its image is precisely the kernel of the reduction. All maps commute with multiplication by s . The reduction is also a unital algebra map. The image of the injection is the square-zero ideal $(h_{\text{off}})/(h_{\text{off}}^2)$: the product of the images of $[p]$ and $[q]$ is $[h_{\text{off}}^2 pq] = 0$. This records its complete inherited multiplication as well as its module isomorphism with E_{off} . The nilpotent lengths in the middle term are exactly $2m_\rho$, and reduction retains the lower m_ρ coefficients. At reflected actual zeros, the full multiplicities are equal, by the functional equation and conjugation of g . Thus (KT.20) uses at most order $2m_\rho - 1$, exactly a coefficient of this doubled algebra. Formula (KT.21) uses only the stated lower coefficients. Equations (KT.20)–(KT.23) prove the precise passage from terminal Taylor data to the full metric.

Write $\mathcal{E}_n = m - m_n$ for the initial observation error, and retain the genuine tail ζ_n from (KR.14). For every off-line point its real part has the sign of $\text{Re } z - 1/2$, so $\zeta_n \neq 0$. The exact identity (KR.16) also gives

$$\mathcal{E}_n(z) = \frac{(-1)^n \kappa_n}{q_n(z)^2} H_n^{\text{obs}}(z), \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z.$$

The real part of H_n^{obs} has that same strict sign: its defining measure is positive of total mass one. Hence $\mathcal{E}_n(z) \neq 0$ off the line as well. Solving the actual tail equation, before taking any residues, therefore gives the exact extraction

$$F_n(z) = \frac{1}{a_n} \left(\frac{(-1)^n \kappa_n}{q_n(z)^2 \mathcal{E}_n(z)} - \frac{1}{\zeta_n(z)} \right). \quad (\text{KT.24})$$

It holds in the complete doubled algebra at the off-line centres. Every reciprocal there is explicit: if $f(\rho + x) = \sum_{j=0}^{2m_\rho-1} f_j x^j$ modulo x^{2m_ρ} , with $f_0 \neq 0$, the inverse coefficients are

$$b_0 = f_0^{-1}, \quad b_k = -f_0^{-1} \sum_{j=1}^k f_j b_{k-j} \quad (1 \leq k < 2m_\rho). \quad (\text{KT.25})$$

Multiplication of the two finite series proves all coefficients. The already proved bounds (GC.13)–(GC.16) apply to every one of the error coefficients required here. Equations (KT.24)–(KT.25), then (KT.20)–(KT.22), give a complete finite extraction of the terminal metric on E_{off} from the initial error and genuine tail data. In the retained monomial quotient coordinates, set $A_{\text{off}} = M_s$, $\pi_{\text{off}}([p]_h) = [p]_{h_{\text{off}}}$, and $c_{\text{off}} = [1] = \pi_{\text{off}}c$. Explicitly, retain the same ν_Z , q_n , nodes and terminal weights, and put

$$S_n^{\text{off}} = \sum_j \sigma_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1} c_{\text{off}} c_{\text{off}}^* (A_{\text{off}}^* - \bar{\lambda}_{n,j} I)^{-1}. \quad (\text{KT.26})$$

For the original quotient map $\pi_{\text{off}} : E \rightarrow E_{\text{off}}$, the resolvent intertwining identity gives $S_n^{\text{off}} = \pi_{\text{off}} S_n \pi_{\text{off}}^*$ whenever the full rational Gram is defined. The right side of (KT.26) remains defined at every off-line centre even when the full packet has a critical quadrature collision. No replacement of the measure by $|g/h_{\text{off}}|^2 dt / (2\pi)$ occurs in this projection. The full polynomial quadrature formula, applied to these projected residue columns, proves even at such a collision that

$$\pi_{\text{off}} K_{n-1} \pi_{\text{off}}^* = \kappa_{n-1}^{-1} q_n (A_{\text{off}}) S_n^{\text{off}} q_n (A_{\text{off}})^*. \quad (\text{KT.27})$$

These extraction formulas retain the inverse error, the inverse tail, q_n , κ_n , and a_n , including their cancellations. The initial error bound alone is not used to replace any of those terms. The polynomial metric step (KT.6) continues to include the critical-line summand, where the off-line resolvent domain ends.

12.5 Actual quartet packets close the energy in two volume ratios

Assume here that the selected actual packet is closed under both $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \bar{\rho}$, with the complete multiplicities. These are exact invariances of the specified divisor; no value of $\text{Re } \rho$ has been imposed.

Theorem 12.3 (Two consecutive arithmetic volume contractions). *Let G_{-1} be the fixed-section metric already identified with $U_\varepsilon^* K_{d-1}^{-1} U_\varepsilon$, and define the actual positive determinant ratios*

$$r_m = \frac{\det G_m}{\det G_{m-1}} \quad (m \geq 0).$$

Then $0 < r_m \leq 1$, and for every $m \geq 0$, with $n = d + m + 1$,

$$\epsilon_m^2 = \frac{\kappa_{d+m+1}}{\kappa_{d+m}} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{KT.28})$$

This is the original relative allowance and the original theta Gram sequence, including their local unit and all multiplicities.

Proof. Conjugation invariance of the divisor makes every coefficient of the monic h real. The entire function g obeys $g(\bar{s}) = \overline{g(s)}$, so the unchanged density $|g(1/2 + it)/h(1/2 + it)|^2$ is even in t . For a monic orthogonal polynomial q , form $q^c(s) = \overline{q(\bar{s})}$. Substitution $t \mapsto -t$ and conjugation of each orthogonality equation show that q^c is monic with the same degree and the same lower-degree orthogonality. Positivity of the polynomial Gram proves uniqueness, so $q^c = q$. Thus every q_n has real coefficients in the original variable s . The coefficient of q_n in sq_n is therefore real as well; (KT.1) proves its exact value $b_n = 1/2$. This is a consequence of the given packet symmetries, with every other recurrence coefficient retained.

The original companion matrix A , vector c , residue columns v_j , and Gram matrices K_{n-1}, M_n are real. Therefore the scalar γ_n in (KT.3) is real. The second specified symmetry gives $g_Z = 0$ in (KT.4), so $\gamma_n = 0$ exactly. Hence $\epsilon_m^2 = \alpha_n \beta_n$. At the preceding and following steps (KT.7)–(KT.8) give

$$\alpha_n = 1 - r_m, \quad \alpha_{n+1} = 1 - r_{m+1}, \quad \beta_n = \frac{a_n \alpha_{n+1}}{1 - \alpha_{n+1}}.$$

All these denominators are positive: $G_m > 0$ and $a_n/(a_n + \beta_n) > 0$. Substitution proves (KT.28), including the exceptional zero-column steps. The inequality $0 < r_m \leq 1$ follows directly from the same determinant formula. At $m = 0$ the preceding metric is precisely the stated G_{-1} , so no earlier undefined source level is used. $\square$

12.6 What the recurrence proves about boundary energy

The unchanged arithmetic source has supplied the explicit energy increment (KT.11). Its three terms are the squared transverse A -image, its real pairing with the retained terminal vector, and the specified negative term $-a_n \beta_n \|p_n\|_n^2$. Thus the source recurrence produces an exactly computable boundary energy, including its sign-determining cross term. No sign for their sum has been assumed. Every term is a polynomial residue, an actual moment Gram inverse, or a retained recurrence coefficient. The rational route to those same quantities is (KT.15)–(KT.25). The original theta representative and its boundary are recovered from the preceding kernel-minimum formula and its explicit division by h . Consequently all updates are maps between the original source representatives, with the same quotient class and supported zero. For each support label the displayed linear maps lift by $(\lambda, x) \mapsto (\lambda, Tx)$; external absence remains τ , and a zero vector in that fibre remains its supported zero throughout.

12.7 An exact positive-moment counterexample to consecutive energy monotonicity

This section concerns the moment-to-packet construction for the explicitly specified positive measure below. It proves that positivity, the vertical orthogonal-polynomial recurrence, reflection stability of the packet, and the terminal rank-two identity together allow an increase of the relative weight

energy at consecutive degrees. The measure is specified by its nine atoms; no assertion identifies it with the arithmetic measure $d\nu_Z(t) = |g(\frac{1}{2} + it)/h(\frac{1}{2} + it)|^2 dt/(2\pi)$. The exact correspondence with that construction will be the same polynomial residue map, multiplication operator, kernel sum and Hermitian defect, with the measure retained as part of the data throughout.

12.7.1 The measure and every moment needed below

Keep the original coordinate $s = \frac{1}{2} + it$. Define a probability measure ν on this line by the following table. In a row containing two signs, the displayed weight is assigned to each of the two atoms.

t	weight at each atom	
0	115/288	(TEC.1)
$\pm 1/4$	29/120	
$\pm 1/2$	13/240	
$\pm 3/4$	11/2520	
± 1	1/6720	

All nine weights are strictly positive, and

$$\frac{115}{288} + \frac{29}{60} + \frac{13}{120} + \frac{11}{1260} + \frac{1}{3360} = 1. \quad (\text{TEC.2})$$

Write $\mu_k = \int t^k d\nu$ and $v = 1/16$. The paired atoms give $\mu_k = 0$ for every odd integer $k \geq 1$, including $k = 1, 3, 5, 7, 9$. For positive integers r , the even-moment formula, with all original atom weights retained, is

$$\mu_{2r} = 4^{-2r} \left(\frac{29}{60} 1^{2r} + \frac{13}{120} 2^{2r} + \frac{11}{1260} 3^{2r} + \frac{1}{3360} 4^{2r} \right). \quad (\text{TEC.3})$$

Here the factor 4^{-2r} comes from the explicitly displayed atom locations in (TEC.1). The four contributions and their sums are

r	$\frac{29}{60} 1^{2r}$	$\frac{13}{120} 2^{2r}$	$\frac{11}{1260} 3^{2r}$	$\frac{1}{3360} 4^{2r}$	sum	
1	29/60	13/30	11/140	1/210	1	(TEC.4)
2	29/60	26/15	99/140	8/105	3	
3	29/60	104/15	891/140	128/105	15	
4	29/60	416/15	8019/140	2048/105	105	
5	29/60	1664/15	72171/140	32768/105	939	

Consequently

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8, \mu_{10}) = (1, v, 3v^2, 15v^3, 105v^4, 939v^5). \quad (\text{TEC.5})$$

Equations (TEC.2)–(TEC.5), together with cancellation of each odd moment between the two atoms, verify all moments through degree ten.

12.7.2 The actual monic orthogonal polynomials of this measure

Use the inner product

$$\langle P, Q \rangle_\nu = \int \overline{P(\frac{1}{2} + it)} Q(\frac{1}{2} + it) d\nu(t), \quad x = s - \frac{1}{2}. \quad (\text{TEC.6})$$

The following table retains the monic polynomials in the original s -coordinate and gives their actual squared norms.

k	$q_k(s)$	$\kappa_k = \langle q_k, q_k \rangle_\nu$
0	1	1
1	x	v
2	$x^2 + v$	$2v^2$
3	$x^3 + 3vx$	$6v^3$
4	$x^4 + 6vx^2 + 3v^2$	$24v^4$
5	$x^5 + 10vx^3 + 15v^2x$	$114v^5$

(TEC.7)

Proof. For an explicit verification, define real polynomials

$$\begin{aligned}
 P_0(t) &= 1, & P_1(t) &= t, & P_2(t) &= t^2 - v, \\
 P_3(t) &= t^3 - 3vt, & P_4(t) &= t^4 - 6vt^2 + 3v^2, \\
 P_5(t) &= t^5 - 10vt^3 + 15v^2t.
 \end{aligned}
 \tag{TEC.8}$$

Direct substitution gives $q_k(\frac{1}{2} + it) = i^k P_k(t)$. Thus

$$\langle q_j, q_k \rangle_\nu = (-i)^j i^k \int P_j(t) P_k(t) d\nu(t). \tag{TEC.9}$$

If j, k have different parity, the integrand on the right is odd and its integral is zero. The remaining six off-diagonal integrals, using (TEC.5), are precisely

$$\begin{aligned}
 \int P_0 P_2 d\nu &= v(1 - 1) = 0, \\
 \int P_0 P_4 d\nu &= v^2(3 - 6 + 3) = 0, \\
 \int P_2 P_4 d\nu &= v^3(15 - 21 + 9 - 3) = 0, \\
 \int P_1 P_3 d\nu &= v^2(3 - 3) = 0, \\
 \int P_1 P_5 d\nu &= v^3(15 - 30 + 15) = 0, \\
 \int P_3 P_5 d\nu &= v^4(105 - 195 + 135 - 45) = 0.
 \end{aligned}
 \tag{TEC.10}$$

For the diagonal integrals, direct polynomial multiplication gives

$$\begin{aligned}
 \int P_0^2 d\nu &= 1, & \int P_1^2 d\nu &= v, \\
 \int P_2^2 d\nu &= v^2(3 - 2 + 1) = 2v^2, \\
 \int P_3^2 d\nu &= v^3(15 - 18 + 9) = 6v^3, \\
 \int P_4^2 d\nu &= v^4(105 - 180 + 126 - 36 + 9) = 24v^4, \\
 \int P_5^2 d\nu &= v^5(939 - 2100 + 1950 - 900 + 225) = 114v^5.
 \end{aligned}
 \tag{TEC.11}$$

In (TEC.9), the diagonal phase is $(-i)^k i^k = 1$. These equations therefore prove every entry of the Gram matrix asserted in (TEC.7). Each polynomial in that table is monic. A nonzero polynomial of degree at most five has at most five distinct roots, so it cannot vanish at all nine atoms in (TEC.1). Positivity of the weights makes the inner product positive definite on polynomials of degree at most five.

If two monic polynomials of the same degree were both orthogonal to every lower degree polynomial, their difference would have lower degree and zero squared norm. Positive definiteness forces that difference to vanish. Thus (TEC.7) consists of the actual monic orthogonal polynomials and norms of the stated measure, including the fifth norm. $\square$

Polynomial multiplication in (TEC.7) also proves

$$sq_k = q_{k+1} + \frac{1}{2}q_k - kvq_{k-1} \quad (0 \leq k \leq 4), \quad (\text{TEC.12})$$

where the last term is omitted for $k = 0$. For $1 \leq k \leq 4$, the displayed coefficient is exactly $a_k = \kappa_k/\kappa_{k-1} = kv = k/16 > 0$. Thus (TEC.12) has the required vertical sign and $b_k + \overline{b_k} = 1$, with $b_k = 1/2$. The fifth norm in (TEC.7) is $\kappa_5 = 114/16^5 = 57/524288$; its value was derived from the actual tenth moment $939/16^5$.

12.7.3 The retained packet and its exact evaluation isomorphism

Set

$$h(s) = (s - \frac{3}{4})(s - \frac{1}{4}) = s^2 - s + \frac{3}{16}, \quad E = \mathbb{C}[s]/(h), \quad j_h : \mathbb{C}[s] \longrightarrow E. \quad (\text{TEC.13})$$

Both packet centres lie in $0 < \operatorname{Re} s < 1$, and the reflection $\rho \mapsto 1 - \overline{\rho}$ exchanges them. They are distinct, and each has its complete multiplicity one. In the retained monomial basis $(1, s)$ of E , the unit and multiplication operator are

$$c = j_h 1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = M_s = \begin{pmatrix} 0 & -3/16 \\ 1 & 1 \end{pmatrix}. \quad (\text{TEC.14})$$

Indeed $s^2 = s - 3/16$ in this quotient, which proves both columns of the multiplication matrix.

The complete evaluation map is

$$\begin{aligned} T : E &\longrightarrow \mathbb{C} \oplus \mathbb{C}, & [p] &\longmapsto (p(3/4), p(1/4)), \\ T &= \begin{pmatrix} 1 & 3/4 \\ 1 & 1/4 \end{pmatrix}, & T^{-1} &= \begin{pmatrix} -1/2 & 3/2 \\ 2 & -2 \end{pmatrix}. \end{aligned} \quad (\text{TEC.15})$$

Both evaluations vanish on the ideal (h) , so the map is well defined. Evaluations preserve addition, multiplication and the unit. For any pair (u_+, u_-) , the polynomial

$$2u_+(s - \frac{1}{4}) + 2u_-(\frac{3}{4} - s) \quad (\text{TEC.16})$$

evaluates to that pair. It has the coefficients given by T^{-1} , and multiplication of the two matrices in (TEC.15) gives the identity. This proves the typed algebra isomorphism, including its inverse. Evaluation of sp also proves

$$TA = \tilde{A}T, \quad \tilde{A} = \operatorname{diag}(3/4, 1/4), \quad Tc = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (\text{TEC.17})$$

In the original quotient E one has $x^2 = v$. Reducing the explicit polynomials of (TEC.7), with no deletion of their supported residue classes, gives

k	$v_k := j_h q_k$ in the monomial basis $(1, s)$	
0	$(1, 0)^T$	
1	$(-1/2, 1)^T$	
2	$(1/8, 0)^T$	
3	$(-1/8, 1/4)^T$	
4	$(5/128, 0)^T$	
5	$(-13/256, 13/128)^T$	(TEC.18)

For example $q_3 = 4vx = x/4$, $q_4 = 10v^2 = 5/128$, and $q_5 = 26v^2x = (13/128)x$ in E . These calculations establish every entry of (TEC.18) directly from the original ideal relation.

12.7.4 The original kernels, terminal defects and exact energies

For $1 \leq N \leq 4$, define

$$\begin{aligned} K_N &= \sum_{k=0}^N \frac{v_k v_k^*}{\kappa_k}, \\ L_N &= AK_N + K_N A^* - K_N, \\ \epsilon_N &= \left\| K_N^{-1/2} L_N K_N^{-1/2} \right\|. \end{aligned} \tag{TEC.19}$$

All adjoints in (TEC.19) are conjugate transposes in the explicitly retained monomial coordinates. The columns v_0, v_1 are linearly independent. If a nonzero column u had $u^* K_N u = 0$, the positive sum in (TEC.19) would force both $v_0^* u = 0$ and $v_1^* u = 0$, a contradiction. Thus each K_N is positive definite, so its positive inverse square root and the displayed Hermitian operator norm are well defined.

Using (TEC.7) and (TEC.18), the successive kernels are

$$\begin{aligned} K_1 &= \begin{pmatrix} 5 & -8 \\ -8 & 16 \end{pmatrix}, & K_2 &= \begin{pmatrix} 7 & -8 \\ -8 & 16 \end{pmatrix}, \\ K_3 &= \begin{pmatrix} 53/3 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}, & K_4 &= \begin{pmatrix} 131/6 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}. \end{aligned} \tag{TEC.20}$$

In particular the last step is the exact positive rank-one update

$$K_4 - K_3 = \frac{v_4 v_4^*}{\kappa_4} = \frac{25}{6} cc^*. \tag{TEC.21}$$

Multiplication by the matrix in (TEC.14) gives

$$L_3 = \begin{pmatrix} -20/3 & 20/3 \\ 20/3 & 0 \end{pmatrix}, \quad L_4 = \begin{pmatrix} -65/6 & 65/6 \\ 65/6 & 0 \end{pmatrix}. \tag{TEC.22}$$

These defects satisfy the original terminal identity, with its exact coefficients and signs:

$$L_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N} \quad (N = 3, 4). \tag{TEC.23}$$

For $N = 3$, the numerator calculated from (TEC.18) is

$$\begin{pmatrix} -5/512 & 5/512 \\ 5/512 & 0 \end{pmatrix}, \quad \kappa_3 = 3/2048.$$

For $N = 4$, it is

$$\begin{pmatrix} -65/16384 & 65/16384 \\ 65/16384 & 0 \end{pmatrix}, \quad \kappa_4 = 3/8192.$$

Division gives exactly the two matrices in (TEC.22), proving (TEC.23) without an appeal to a numerical rank calculation.

Theorem 12.4 (Strict increase at two consecutive permitted levels). *For the positive measure (TEC.1) and the complete reflection-stable packet (TEC.13), the energies in (TEC.19) satisfy*

$$\epsilon_3^2 = \frac{25}{99}, \quad \epsilon_4^2 = \frac{4225}{15136}, \quad \epsilon_4^2 - \epsilon_3^2 = \frac{39875}{1498464} > 0. \tag{TEC.24}$$

In the indexing $N = d + m$, with $d = \dim_{\mathbb{C}} E = 2$, these are the consecutive levels $m = 1$ and $m = 2$, both with $m \geq 0$.

Proof. Transport the two forms by the proved isomorphism (TEC.15):

$$\begin{aligned}\tilde{K}_3 &= TK_3T^* = \begin{pmatrix} 20/3 & -2/3 \\ -2/3 & 20/3 \end{pmatrix}, \\ \tilde{K}_4 &= TK_4T^* = \begin{pmatrix} 65/6 & 7/2 \\ 7/2 & 65/6 \end{pmatrix}.\end{aligned}\tag{TEC.25}$$

By (TEC.17), the transported defect is

$$\tilde{L}_N = TL_NT^* = \tilde{A}\tilde{K}_N + \tilde{K}_N\tilde{A}^* - \tilde{K}_N.\tag{TEC.26}$$

For either matrix $\tilde{K} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ in (TEC.25), one has

$$\tilde{L} = \text{diag}(a/2, -a/2), \quad \det(\tilde{L} - \lambda\tilde{K}) = \lambda^2(a^2 - b^2) - a^2/4.\tag{TEC.27}$$

To check that this calculation gives the norm defined in the original coordinates, set $H = K^{-1/2}LK^{-1/2}$. Then

$$\det(H - \lambda I) = \det(K)^{-1} \det(L - \lambda K), \quad \det(T(L - \lambda K)T^*) = |\det T|^2 \det(L - \lambda K).\tag{TEC.28}$$

The two determinant factors are nonzero and independent of λ . Consequently the eigenvalues of the Hermitian matrix H equal the roots of (TEC.27), with their multiplicities. The roots are the opposite real numbers $\pm a/(2\sqrt{a^2 - b^2})$. A Hermitian matrix has operator norm equal to the greatest absolute value of its eigenvalues, so

$$\epsilon^2 = \frac{a^2}{4(a^2 - b^2)}.\tag{TEC.29}$$

For $N = 3$, substitution of $a = 20/3, b = -2/3$ gives $\epsilon_3^2 = (400/9)/(4 \cdot 396/9) = 25/99$. For $N = 4$, substitution of $a = 65/6, b = 7/2$ gives $\epsilon_4^2 = (4225/36)/(4 \cdot 3784/36) = 4225/15136$. Their difference, retaining the two denominators, is

$$\frac{4225 \cdot 99 - 25 \cdot 15136}{15136 \cdot 99} = \frac{418275 - 378400}{1498464} = \frac{39875}{1498464} > 0.\tag{TEC.30}$$

Both energies are nonnegative, and their squared values have strictly positive difference, so $\epsilon_4 > \epsilon_3$. The original coordinate characteristic polynomials provide the same calculation:

$$\begin{aligned}\det(L_3 - \lambda K_3) &= 176\lambda^2 - 400/9, \\ \det(L_4 - \lambda K_4) &= (3784/9)\lambda^2 - 4225/36.\end{aligned}\tag{TEC.31}$$

Finally $N = 3 = 2 + 1$ and $N = 4 = 2 + 2$, proving the stated degree range without using the lower degree $N = 1$. $\square$

12.7.5 The exact scope of the obstruction

For this measure the construction is the composite of the polynomial source with the original residue quotient

$$\mathcal{P}_N \hookrightarrow \mathbb{C}[s] \xrightarrow{j_h} E, \quad E \xrightarrow{T} \mathbb{C} \oplus \mathbb{C},\tag{TEC.32}$$

and multiplication is intertwined exactly by (TEC.17). Its source inner product is (TEC.6); its monic source Gram matrix is $\text{diag}(\kappa_0, \dots, \kappa_N)$; its kernel is the full sum (TEC.19). Thus it has precisely the vertical recurrence, positive kernel update, complete packet quotient and terminal defect maps appearing in the arithmetic moment construction. Equations (TEC.24) and (TEC.30) prove that these data alone do not imply nonincreasing consecutive energy. An argument for that property for ν_Z must therefore establish and use additional identities or estimates of its stated arithmetic density.

The present example establishes the indicated obstruction to a universal recurrence-based argument; it asserts no consecutive increase for ν_Z .

The companion exact checker verifies 88 rational and complex-rational identities in both ordinary Python and optimized Python. It checks the atom weights and relevant moments, source Gram matrix, packet isomorphism, residue columns, kernel updates, terminal identities and generalized characteristic polynomials. Those finite checks supplement the complete computations above.

13 The exact metric departure and the cost of complete jets

Retain the original packet $E = \mathbb{C}[s]/h$, its polynomial coordinates and $A = M_s$. Write its eigenvalues with their algebraic multiplicities as $\rho_1, \dots, \rho_d$, and put $\delta_j = \operatorname{Re} \rho_j - 1/2$. No eigenvalue or repeated factor is discarded. For a positive Hermitian matrix G in these coordinates define

$$\begin{aligned} W_G &= A^*G + GA - G, \quad S_G = G^{-1/2}W_GG^{-1/2}, \quad \epsilon_G = \|S_G\|_{\text{op}}, \\ \mathfrak{d}_G(A) &= \operatorname{Tr}(G^{-1}A^*GA) - \sum_{j=1}^d |\rho_j|^2. \end{aligned} \tag{MD1}$$

The positive square root is used only for the explicit isometry $x \mapsto G^{1/2}x$ from the specified metric space (E, G) to the Euclidean coordinate space. The original A, G , their entries and their maps remain those in (MD1).

Theorem 13.1 (Exact departure identity). *The number $\mathfrak{d}_G(A)$ is real and nonnegative, and*

$$\operatorname{Tr} S_G = 2 \sum_j \delta_j, \quad \operatorname{Tr}(S_G^2) = 4 \sum_j \delta_j^2 + 2\mathfrak{d}_G(A). \tag{MD2}$$

*It vanishes precisely when A is normal for the specified metric, that is, when $A(G^{-1}A^*G) = (G^{-1}A^*G)A$. For the original orthogonal representatives R_m , their matrices G_m and rank-two forms W_m , one has exactly*

$$\epsilon_m = \left| \sum_j \delta_j \right| + \sqrt{2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j \right)^2}. \tag{MD3}$$

For a reflection-stable packet this becomes

$$\boxed{\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A)}. \tag{MD4}$$

Proof. Let $C = G^{1/2}AG^{-1/2}$. Then $S_G = C + C^* - I$ and $\operatorname{Tr}(G^{-1}A^*GA) = \operatorname{Tr}(C^*C)$. For completeness, choose a unit eigenvector of C , extend it to an orthonormal basis, and repeat on the lower-right compression. Induction gives a unitary matrix U with upper-triangular $T = U^*CU$ whose diagonal lists the same eigenvalues with their algebraic multiplicities. Unitary invariance of the trace gives

$$\mathfrak{d}_G(A) = \sum_{i < j} |T_{ij}|^2 \geq 0. \tag{MD5}$$

If it is zero, T is diagonal and C is normal. Conversely, an upper-triangular normal matrix is diagonal: its first row norm equals its first column norm, eliminating the strict first-row entries, and induction eliminates every remaining such entry. Thus normality of C is equivalent to the stated equality for A , through the specified isometry.

Expanding $(C + C^* - I)^2$ under the trace gives

$$\mathrm{Tr}(S_G^2) = 2 \mathrm{Tr}(C^* C) + 2 \mathrm{Re} \mathrm{Tr}(C^2) - 4 \mathrm{Re} \mathrm{Tr} C + d.$$

The traces of C and C^2 are respectively $\sum_j \rho_j$ and $\sum_j \rho_j^2$, even in the presence of nilpotents. Since $2|\rho|^2 + 2 \mathrm{Re} \rho^2 - 4 \mathrm{Re} \rho + 1 = 4(\mathrm{Re} \rho - 1/2)^2$, this proves (MD2). The trace formula follows directly from $S_G = C + C^* - I$.

For G_m , Section 6 proved $\mathrm{rank} S_{G_m} \leq 2$. List its two possibly nonzero real eigenvalues a, b , appending zero if necessary (also when $d = 1$). Equations (MD2) give

$$a + b = 2 \sum_j \delta_j, \quad (a - b)^2/4 = 2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j\right)^2.$$

Thus the radicand is nonnegative, and $\max(|a|, |b|) = |(a + b)/2| + |a - b|/2$, proving (MD3). Reflection-stability pairs δ with $-\delta$, with the same multiplicities, proving (MD4). $\square$

The actual arithmetic kernel of Section 9 also gives this quantity without changing the local unit. Indeed $G_m = U_\varepsilon^* K_N^{-1} U_\varepsilon$, $N = d + m$, and $U_\varepsilon A = A U_\varepsilon$. Cyclicity of trace gives

$$\mathfrak{d}_{G_m}(A) = \mathrm{Tr}(K_N A^* K_N^{-1} A) - \sum_j |\rho_j|^2. \quad (\text{MD6})$$

To verify the cancellation, substitute $G_m^{-1} = U_\varepsilon^{-1} K_N (U_\varepsilon^*)^{-1}$ in (MD1), commute A with U_ε and A^* with U_ε^* , and cancel the outer similar factors inside the trace. This calculation removes no source representative: their exact recovery remains (TB.7)–(TB.9).

13.1 A quantitative constraint on every retained nilpotent chain

For a selected root ρ of full order r , let

$$I_\rho : \mathbb{C}[z]/z^r \longrightarrow E$$

be the original Chinese-remainder inclusion: the ρ -component is the given Taylor polynomial in $z = s - \rho$, and every other component is zero. Its Gram matrix is $G_\rho = I_\rho^* G I_\rho$, in the basis $1, z, \dots, z^{r-1}$. Put $N_\rho z^k = z^{k+1}$, with $z^r = 0$. The exact intertwining relation is $A I_\rho = I_\rho(\rho I + N_\rho)$.

Theorem 13.2 (Full-jet metric inequalities). *For every positive original metric G , write $\delta = \mathrm{Re} \rho - 1/2$. Then $\epsilon_G \geq 2|\delta|$, and for all $0 \leq k < k + \ell < r$,*

$$\frac{(G_\rho)_{k+\ell, k+\ell}}{(G_\rho)_{k, k}} \leq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^\ell. \quad (\text{MD7})$$

In particular, for $r \geq 2$,

$$\kappa_2(G_\rho) \geq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^{-(r-1)}. \quad (\text{MD8})$$

The bracket is strictly positive when $r \geq 2$. Here κ_2 is the ratio of the largest and smallest eigenvalues in the specified Taylor coordinates.

Proof. The definition of ϵ_G is the two-sided form inequality $-\epsilon_G G \leq W_G \leq \epsilon_G G$. On an eigenvector it gives $\epsilon_G \geq 2|\delta|$. Pull this same inequality back by I_ρ . Let $L = G_\rho^{1/2} N_\rho G_\rho^{-1/2}$ and $T = 2\delta I + L + L^*$. The pulled-back inequality gives $\|T\| \leq \epsilon_G$. Nilpotence retains $\mathrm{Tr} L = \mathrm{Tr}(L^2) = 0$, whence

$$\mathrm{Tr} T = 2r\delta, \quad \|L\|_F^2 = \frac{1}{2} \mathrm{Tr}(L + L^*)^2 = \frac{1}{2} \mathrm{Tr} T^2 - 2r\delta^2 \leq \frac{r}{2} (\epsilon_G^2 - 4\delta^2). \quad (\text{MD9})$$

The first equality for the Frobenius norm follows by expanding the square and using the two zero traces. The final inequality uses the r real eigenvalues of T , each bounded in modulus by ϵ_G . For the

original metric norm, $\|N_\rho\|_{G_\rho} = \|L\| \leq \|L\|_F$. Apply its ℓ -th power to the specified basis vector z^k . Since $N_\rho^\ell z^k = z^{k+\ell}$, the resulting squared norm inequality is exactly (MD7).

When $r \geq 2$, $N_\rho \neq 0$, so (MD9) makes the bracket strictly positive. Put $k = 0, \ell = r - 1$. The Rayleigh inequalities give $\kappa_2(G_\rho) \geq (G_\rho)_{0,0}/(G_\rho)_{r-1,r-1}$; combine with (MD7) to obtain (MD8). $\square$

All entries in (MD7) for the original sequence are explicitly $I_\rho^* U_\varepsilon^* K_N^{-1} U_\varepsilon I_\rho$. Thus this theorem supplies constraints on the actual moment calculation, with its original Taylor factors. At a critical root of order $r \geq 2$, a sequence with relative control approaching zero necessarily has the stated diverging Taylor-metric condition number. The surviving jets have not been erased; the inequalities calculate their relative metric scales. This is compatible with the complete graph and weighted-jet maps in Section 59, which retain those same coordinates.

14 Coherent packets, joint tensor layers, and their exact spectral cost

This section integrates the complete arithmetic interpolation construction of the 12 September coherent-interpolation delivery, whose source revision is 3bff1b2bd87dfa2592f65c5ccc231f6e e9541dfc, and proves its comparison with the preceding kernel and departure calculations. All zeros in a packet retain their complete multiplicities. A reflected packet is required only in statements explicitly using its reflection. The original spaces $V, \mathcal{B}$, map Θ , generator $D = -x\partial_x$, and Hilbert observation $L^2(dx)$ remain fixed.

14.1 The arithmetic unit and the coherent one-variable maps

For $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, write

$$d = \deg h, \quad E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1}, \quad g = 2\xi. \quad (\text{CT.1})$$

Complete multiplicities make v_h invertible. Let $F_h \in \mathcal{B}$ be the original inverse satisfying $h(D)F_h = f_0$ and $\mathcal{M}F_h = g/h$, and put $\mathcal{T}_h P = P(D)F_h$. The integral construction of F_h , its estimates at both ends, and injectivity of the Mellin transform are proved in Section 6. They give

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (\text{CT.2})$$

Let $C_N : \mathcal{P}_N \rightarrow E_h$ be the full remainder map in the original monomial coordinates, let $M_{h,N}$ be the Gram matrix of $\mathcal{P}_N$ for $d\nu_h = |g(1/2 + it)/h(1/2 + it)|^2 dt/(2\pi)$, and use U_a for multiplication by a on E_h . The two kernel matrices occurring in the constructions have the exact types

$$K_N = C_N M_{h,N}^{-1} C_N^*, \quad J_{h,N} = U_{v_h} C_N, \quad \mathcal{C}_{h,N} = U_{v_h} K_N U_{v_h}^*. \quad (\text{CT.3})$$

Consequently their actual representative and metric coincide:

$$\begin{aligned} M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1} &= M_{h,N}^{-1} C_N^* K_N^{-1} U_{\varepsilon_h}, \\ G_{h,n} &= \mathcal{C}_{h,n+d}^{-1} = U_{\varepsilon_h}^* K_{n+d}^{-1} U_{\varepsilon_h}. \end{aligned} \quad (\text{CT.4})$$

Indeed invert the congruence in (CT.3) and cancel the adjacent inverse factors. The first line is exactly the polynomial numerator in (TB.7)–(TB.9). Its difference from $\text{rem}_h(\varepsilon_h u)$ is divisible by h and has quotient of degree at most n . Applying (CT.2) gives the same original theta correction. Thus the arithmetic unit remains in every coefficient map and every source representative.

For completeness, the minimum assertion follows directly without an existence assumption. For any positive Gram matrix M and surjective J , set $G = (JM^{-1}J^*)^{-1}$ and $L = M^{-1}J^*G$. Then $JL = I$, $L^*ML = G$, and for $Jz = 0$, $z^*ML = z^*J^*G = 0$. Every feasible vector is $Lu + z$, so

$$(Lu + z)^* M (Lu + z) = u^* G u + z^* M z. \quad (\text{CT.5})$$

Positivity makes this the unique minimum. In (CT.3), positivity holds because a nonzero polynomial cannot vanish almost everywhere on the original integration line; surjectivity follows from complete polynomial remainders. The kernel is exactly $h\mathcal{P}_n$ when $N = n + d$. Hence $R_{h,n} = (I - P_{L_n})s_h$, where $s_h = \mathcal{T}_h \text{rem}_h(\varepsilon_h \cdot)$ and $L_n = \Theta \text{span}(\phi_*, \dots, D^n \phi_*)$.

If $h \mid H$ are full packets, put $m = H/h$. The factors h, m are coprime. The original CRT injection $\iota : E_h \rightarrow E_H$ inserts zero jets at the additional centres, and satisfies $A_H \iota = \iota A_h$. Multiplication $P \mapsto mP$ is an isometry of the declared polynomial images, since $(g/H)mP = (g/h)P$; it satisfies

$$\mathcal{T}_H(mP) = \mathcal{T}_h P, \quad J_H \mathcal{T}_H(mP) = \iota J_h \mathcal{T}_h P. \quad (\text{CT.6})$$

At the added centres the second identity follows from the full zeros of g/h ; at the old centres it is the identity of entire functions. Both numerators for $s_H \iota = \mathcal{T}_H(m \text{rem}_h(\varepsilon_h \cdot))$ have degree below $\deg H$ and the same complete jets, proving equality. Since L_n is independent of the packet,

$$R_{H,n} \iota = R_{h,n}, \quad \iota^* G_{H,n} \iota = G_{h,n}, \quad \iota^* W_{H,n} \iota = W_{h,n}. \quad (\text{CT.7})$$

These are congruences, with no interchange of inversion and compression. Their split maps at a retained support λ are $(\lambda, u) \mapsto (\lambda, \iota u)$; all added zero jets are represented zeros at that support.

14.2 The original reflection and the exact diagonal coefficient

The original arithmetic density $|g(1/2 + it)|^2$ is even in t , since the real Taylor coefficients of g give $g(1/2 - it) = \overline{g(1/2 + it)}$. If Q_j is the monic orthogonal polynomial for that density in the variable s , the polynomial $(-1)^j Q_j(1 - s)$ is monic and orthogonal to all lower degrees, by the change $t \mapsto -t$. Uniqueness therefore gives

$$Q_j(1 - s) = (-1)^j Q_j(s), \quad \int t |Q_j(1/2 + it)|^2 |g(1/2 + it)|^2 \frac{dt}{2\pi} = 0. \quad (\text{CT.8})$$

The diagonal recurrence coefficient is thus exactly $\langle Q_j, sQ_j \rangle / \|Q_j\|^2 = 1/2$. For lower degrees the identity $D^* + D = I$ eliminates every coefficient except the adjacent one and gives its sign and actual norm ratio. More explicitly, sQ_j is orthogonal to Q_i for $i \leq j - 2$, because its inner product is the inner product of Q_j with $(1 - s)Q_i$. For $i = j - 1$ the leading term gives coefficient $-\|Q_j\|^2 / \|Q_{j-1}\|^2$. This proves, with the original vectors and norms,

$$Du_j = u_{j+1} + \frac{1}{2}u_j - \frac{\mathfrak{h}_j}{\mathfrak{h}_{j-1}}u_{j-1}, \quad u_j = Q_j(D)f_0, \quad \mathfrak{h}_j = \|u_j\|^2. \quad (\text{CT.9})$$

The final term is omitted at $j = 0$. The even density in (CT.8) is the additional reason the imaginary part of the diagonal coefficient vanishes; the adjoint identity by itself fixes only its real part.

For a packet stable under $\rho \mapsto 1 - \bar{\rho}$, including the same complete orders, the original maps are

$$\mathfrak{j}_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}, \quad \mathfrak{j}_{\mathcal{B}} F(x) = x^{-1} \overline{F(1/\bar{x})}.$$

The second is antiunitary for dx , and $D\mathfrak{j}_{\mathcal{B}} = \mathfrak{j}_{\mathcal{B}}(I - D)$. The identities $h^\dagger = (-1)^d h$ and $g^\dagger = g$ imply $(g/h)^\dagger = (-1)^d g/h$; they give $\mathfrak{j}_{\mathcal{B}} s_h = s_h \mathfrak{j}_h$ by the same complete jets and degree bounds as above. Equation (CT.9), or invariance of the polynomial space under $s \mapsto 1 - s$, shows $\mathfrak{j}_{\mathcal{B}} L_n = L_n$. The unique minimum (CT.5) then commutes with these maps. If $\mathfrak{j}_h u = C_h \bar{u}$,

$$C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}. \quad (\text{CT.10})$$

Substitution into a quadratic form proves that an upper bound and its reflected lower bound use this same $G_{h,n}$.

14.3 The joint total-degree source and its relation layer

Fix $k \geq 1$, $M \geq k(d-1)$, and set

$$E^{(k)} = E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{P}_M^{(k)} = \mathbb{C}[s_1, \dots, s_k]_{\leq M}, \quad \mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}.$$

Here $\leq M$ means total degree. The full jet map is reduction by each $h(s_j)$, followed by multiplication by $v_h^{\otimes k}$, and its kernel is

$$\mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)). \quad (\text{CT.11})$$

Successive monic division in the declared order $s_1, \dots, s_k$ gives $P = \sum_j h(s_j) P_j$ in this kernel, with $\deg P_j \leq M - d$. Division lowers total degree at every replacement, so this bound holds at every division step. In the original complex $[V \rightarrow \mathcal{B}]$, placed in degrees 0, 1, the primitive for the j -th term uses $(-1)^{j-1} \phi_*$ in the j -th factor and F_h in the others, with the polynomial $P_j(D_1, \dots, D_k)$ applied to the whole tensor. The preceding $j-1$ factors have degree one, so the tensor differential supplies the further factor $(-1)^{j-1}$. Its boundary is exactly $+\mathcal{T}_h^{(k)}(h(s_j) P_j)$.

Write $\mathcal{B}_M = \mathcal{T}_h^{(k)} \mathcal{I}_{h,k,M}$ inside $\mathcal{H}_k = L^2(d^k x)$. The arithmetic moment matrix in the retained monomials is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s^{\alpha_j}} s^{\beta_j} d\nu_h, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (\text{CT.12})$$

Product integration proves the first formula. A polynomial vanishing almost everywhere for this product measure is zero, by successive one-variable uniqueness, proving positivity. The complete remainder basis has total degree at most $k(d-1)$, proving surjectivity. Thus (CT.5) proves

$$R_M = (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, \quad J^{(k)} R_M = I, \quad q^{(k)} R_M = \sigma_h^{\otimes k}. \quad (\text{CT.13})$$

The kernel dimension is $\binom{M+k}{k} - d^k$. Since $\mathcal{T}_h^{(k)}$ is injective by its entire Mellin multiplier, the orthogonal relation layer $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ has exactly

$$\dim \mathcal{E}_M = \binom{M+k}{k-1}. \quad (\text{CT.14})$$

Define all of the following maps on the original $E^{(k)}$:

$$B_M = D^{(k)} R_M - R_M A^{(k)}, \quad Y_M = P_{\mathcal{E}_M} R_M, \quad C_M = (I - P_{\mathcal{B}_M}) B_M. \quad (\text{CT.15})$$

The complete jet of B_M is zero and its numerator has degree at most $M+1$, so $B_M(E^{(k)}) \subseteq \mathcal{B}_{M+1}$. Both Y_M, C_M therefore have target $\mathcal{E}_M$. Integration by parts on these rapidly decreasing functions gives $(D^{(k)})^* + D^{(k)} = kI$. Substitution into $R_M^*(D^{(k)} + (D^{(k)})^*) R_M = kG_M$, followed by orthogonality to $\mathcal{B}_M$, proves

$$\begin{aligned} W_M &= (A^{(k)})^* G_M + G_M A^{(k)} - kG_M = -(Y_M^* C_M + C_M^* Y_M), \\ \Delta_M &= G_M - G_{M+1} = Y_M^* Y_M \succeq 0. \end{aligned} \quad (\text{CT.16})$$

For the second identity use the orthogonal decomposition of $\mathcal{B}_{M+1}$ and $R_{M+1} = R_M - Y_M$. The actual quotient map $[z] \mapsto (I - P_{\mathcal{B}_M})z$ identifies $\mathcal{B}_{M+1}/\mathcal{B}_M$ with $\mathcal{E}_M$, with inverse the inclusion followed by quotient. A transported boundary is zero at its next support and remains represented there; this map sends no supported input to external τ .

For a reflection-stable packet, put $C_{h,k} = C_h^{\otimes k}$. The original product reflection acts on a numerator by $(-1)^{kd} \overline{P(1 - \overline{s_1}, \dots, 1 - \overline{s_k})}$. This retains total degree and preserves the ideal in (CT.11), with

its displayed factor. Its action on $\mathcal{H}_k$ is antiunitary. The unique minimum (CT.5) therefore commutes with reflection and gives

$$C_{h,k}^* G_M C_{h,k} = \overline{G_M}, \quad A^{(k)} C_{h,k} = C_{h,k} (kI - \overline{A^{(k)}}), \quad C_{h,k}^* W_M C_{h,k} = -\overline{W_M}. \quad (\text{CT.16a})$$

If $W_M u = \lambda G_M u$, conjugation followed by $(C_{h,k}^*)^{-1}$ in the last identity gives $W_M C_{h,k} \bar{u} = -\lambda G_M C_{h,k} \bar{u}$. All generalized eigenvalues are real because $G_M > 0$. This invertible antilinear map pairs their full multiplicities. The source-layer maps themselves retain the reflection types. If $j_E = j_h^{\otimes k}$ and $j_{\mathcal{E}}$ is the restriction of the original function reflection to $\mathcal{E}_M$, then

$$j_{\mathcal{E}} Y_M = Y_M j_E, \quad j_{\mathcal{E}} C_M = -C_M j_E. \quad (\text{CT.16b})$$

Both boundary projections commute with reflection, proving the first identity. For the second use $D^{(k)} j_{\mathcal{H}} = j_{\mathcal{H}} (kI - D^{(k)})$, $A^{(k)} j_E = j_E (kI - A^{(k)})$, and $j_{\mathcal{H}} R_M = R_M j_E$ in (CT.15). Expanding its two terms gives $j_{\mathcal{H}} B_M = -B_M j_E$; the projection then proves (CT.16b).

14.4 Nested packets on the joint source: the additional projection

Let $h \mid H$, $q = \deg(H/h)$, $N = M + kq$, and $\iota_k = \iota_{hH}^{\otimes k}$. Multiplication of a numerator by $\prod_{j=1}^k (H/h)(s_j)$ gives a degree-preserving-with-shift map $\mathcal{P}_M^{(k)} \rightarrow \mathcal{P}_N^{(k)}$, with

$$\mathcal{T}_H^{(k)} (\prod_j (H/h)(s_j) P) = \mathcal{T}_h^{(k)} P, \quad J_H^{(k)} \mathcal{T}_h^{(k)} P = \iota_k J_h^{(k)} \mathcal{T}_h^{(k)} P. \quad (\text{CT.17})$$

The second equality follows on pure monomials from (CT.6) and then by linearity. In particular it maps the smaller relation space into the larger one in the same $\mathcal{H}_k$. Let

$$\mathcal{F} = \mathcal{B}_{H,k,N} \cap \mathcal{B}_{h,k,M}^{\perp}, \quad Z = P_{\mathcal{F}} R_{h,k,M}.$$

Since $s_H^{\otimes k} \iota_k = s_h^{\otimes k}$, the exact comparison is

$$\begin{aligned} R_{H,k,N} \iota_k &= R_{h,k,M} - Z, \\ \iota_k^* G_{H,k,N} \iota_k &= G_{h,k,M} - Z^* Z, \\ \iota_k^* W_{H,k,N} \iota_k &= W_{h,k,M} - ((A_h^{(k)})^* Z^* Z + Z^* Z A_h^{(k)} - k Z^* Z). \end{aligned} \quad (\text{CT.18})$$

Indeed nested orthogonal projections commute, and their difference is $P_{\mathcal{F}}$; apply them to the common section to prove the first line. Pythagoras proves the second; intertwining of $A^{(k)}$ and direct expansion prove the third. Also

$$\dim \mathcal{F} = \binom{N+k}{k} - (\deg H)^k - \binom{M+k}{k} + d^k. \quad (\text{CT.19})$$

This is the difference of the exact kernel dimensions. For $k = 1$, $M = n + d$, both boundary spaces are the original L_n , so $\mathcal{F} = 0$ and (CT.18) recovers (CT.7). For joint tensors (CT.18) retains every additional allowed boundary and calculates its effect on the control form.

14.5 Positive layer rank and the complete tensor spectral departure

List $\rho_1, \dots, \rho_d$ with all algebraic multiplicities, put $\delta_i = \text{Re } \rho_i - 1/2$, $\sigma_1 = \sum_i \delta_i$, $\sigma_2 = \sum_i \delta_i^2$, and define

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2}, \quad \epsilon_M = \|S_M\|_{\text{op}}, \quad r_M = \text{rank } \Delta_M, \\ \mathfrak{d}_M^{(k)} &= \text{Tr}(G_M^{-1} (A^{(k)})^* G_M A^{(k)}) - \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \rho_{i_j} \right|^2, \\ \mathcal{D}_k &= \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \delta_{i_j} \right|. \end{aligned} \quad (\text{CT.20})$$

All sums are ordered and retain repeated entries. The isometry $x \mapsto G_M^{1/2}x$ in this calculation leaves the original jet coordinates, source functions, metric and layer maps unchanged. For a map into $\mathcal{E}_M$, the Frobenius symbol below denotes its Hilbert–Schmidt norm for the layer’s inherited $L^2(d^kx)$ inner product.

Theorem 14.1 (Exact tensor departure and relative layer cost). *One has $\mathfrak{d}_M^{(k)} \geq 0$, and exactly*

$$\begin{aligned} \text{Tr } S_M &= 2kd^{k-1}\sigma_1, \\ \text{Tr}(S_M^2) &= 4\{kd^{k-1}\sigma_2 + k(k-1)d^{k-2}\sigma_1^2\} + 2\mathfrak{d}_M^{(k)}. \end{aligned} \quad (\text{CT.21})$$

The positive and negative inertias of W_M each obey

$$n_+(W_M), n_-(W_M) \leq \min(\text{rank } Y_M, \text{rank } C_M), \quad \text{rank } Y_M = r_M \leq \binom{M+k}{k-1}. \quad (\text{CT.22})$$

The complete ordered spectral cost satisfies

$$\boxed{\mathcal{D}_k \leq \frac{1}{2}\|S_M\|_1 \leq \|Y_M G_M^{-1/2}\|_F \|C_M G_M^{-1/2}\|_F}, \quad \mathcal{D}_k \leq r_M \epsilon_M. \quad (\text{CT.23})$$

In particular, the actual source matrices satisfy

$$\mathcal{D}_k^2 \leq \text{Tr}(G_M^{-1} \Delta_M) \text{Tr}(G_M^{-1} C_M^* C_M). \quad (\text{CT.24})$$

If the packet is reflection-stable, put $p_M = n_+(W_M) = n_-(W_M)$. Then

$$p_M \leq \min\{r_M, \text{rank } C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k \leq p_M \epsilon_M. \quad (\text{CT.24a})$$

In this case (CT.21) becomes $\text{Tr}(S_M^2) = 4kd^{k-1}\sigma_2 + 2\mathfrak{d}_M^{(k)}$. When $p_M > 0$, it implies

$$\epsilon_M^2 \geq \frac{2kd^{k-1}\sigma_2 + \mathfrak{d}_M^{(k)}}{p_M}. \quad (\text{CT.25})$$

Proof. Triangularize A_h over $\mathbb{C}$. The tensor product of this basis makes $A^{(k)}$ triangular, with diagonal entries $\sum_j \rho_{ij}$. Thus these are all its eigenvalues with algebraic multiplicity; its nilpotent entries have not been discarded. Now apply the unitary triangularization argument of (MD5) to $T = G_M^{1/2} A^{(k)} G_M^{-1/2}$. The sum of squared absolute values of its strictly upper triangular entries is exactly $\mathfrak{d}_M^{(k)}$, proving nonnegativity. Since $S_M = T + T^* - kI$, expansion of its squared trace gives

$$\text{Tr}(S_M^2) = 4 \sum_{i_1, \dots, i_k} \left(\sum_j \delta_{i_j} \right)^2 + 2\mathfrak{d}_M^{(k)}.$$

The k diagonal terms in this ordered sum contribute $kd^{k-1}\sigma_2$; the $k(k-1)$ ordered cross terms contribute $k(k-1)d^{k-2}\sigma_1^2$. The linear trace is computed in the same way. This proves (CT.21), including $k = 1$.

Write $\hat{Y} = Y_M G_M^{-1/2}$ and $\hat{C} = C_M G_M^{-1/2}$. Then $S_M = -(\hat{Y}^* \hat{C} + \hat{C}^* \hat{Y})$. Its form is zero on $\ker \hat{Y}$. A positive definite subspace of dimension greater than $\text{rank } \hat{Y}$ intersects this kernel nontrivially, a contradiction; the same argument holds for a negative definite subspace and with $\hat{C}$ in place of $\hat{Y}$. Sylvester inertia under the displayed invertible congruence proves the first part of (CT.22). The identity $\Delta_M = Y_M^* Y_M$ gives equality of their kernels, hence their ranks, proving the rest.

In the unitary triangular basis of T , the diagonal of S_M consists of $2 \sum_j \delta_{i_j}$. For any Hermitian matrix S , its spectral decomposition gives $|S_{ii}| \leq \sum_a |\lambda_a| |U_{ia}|^2$. Sum over i to obtain $\sum_i |S_{ii}| \leq \sum_a |\lambda_a| = \|S\|_1$. This proves the first inequality in (CT.23). For finite matrices B, C with a common row space, $\|B^* C\|_1 \leq \|B\|_F \|C\|_F$: choose a singular-value decomposition of $B^* C$, use its left and

right singular vectors to express its trace norm as $|\text{Tr}(UB^*C)|$ for a unitary U , and apply Cauchy-Schwarz to the entries of BU^* and C . The triangle inequality now proves the second inequality of (CT.23). Finally (CT.22) bounds the number of nonzero eigenvalues of S_M by $2r_M$, so $\|S_M\|_1 \leq 2r_M\epsilon_M$. Squaring the middle bound and writing its two Frobenius norms as traces proves (CT.24).

For a reflected packet $\sigma_1 = 0$, with every multiplicity paired. The exact reflection (CT.16a) pairs the nonzero eigenvalues of S_M , giving equal positive and negative counts p_M . Equation (CT.22) proves the bounds on p_M , and there are exactly $2p_M$ nonzero eigenvalues of S_M , each of square at most ϵ_M^2 . Their trace norm is at most $2p_M\epsilon_M$, proving (CT.24a); their squared trace proves (CT.25). If $p_M = 0$, the pairing implies $W_M = 0$ directly, and (CT.21) states that all the nonnegative terms vanish; no division by p_M is made. $\square$

Every quantity in (CT.24) is attached to the same arithmetic moment matrix and its next original relation layer. In particular, $0 \preceq \Delta_M \prec G_M$, because $G_{M+1} > 0$, so all eigenvalues of $G_M^{-1/2}\Delta_M G_M^{-1/2}$ lie in $[0, 1)$, and its trace is strictly smaller than r_M when $r_M > 0$. Its actual value is retained in (CT.24). Equations (CT.21)–(CT.25) calculate necessary relative layer costs and the full nilpotent departure for the constructed tensor family. They assert no unproved upper bound on these arithmetic quantities.

There is an exact determinant refinement of the same source loss. For $r_M > 0$, put

$$\pi_M = \frac{\det G_{M+1}}{\det G_M}, \quad \tau_M = \text{Tr}(G_M^{-1}\Delta_M), \quad \chi_M = \text{Tr}(G_M^{-1}C_M^*C_M).$$

Then

$$1 - \pi_M \leq \tau_M \leq r_M(1 - \pi_M^{1/r_M}) < r_M, \quad \boxed{\mathcal{D}_k^2 \leq \tau_M \chi_M \leq r_M(1 - \pi_M^{1/r_M})\chi_M.} \quad (\text{CT.26})$$

To prove every factor, let $t_1, \dots, t_{r_M} \in (0, 1)$ be the positive eigenvalues of $G_M^{-1/2}\Delta_M G_M^{-1/2}$. The other eigenvalues are zero, so $\tau_M = \sum_i t_i$ and $\pi_M = \prod_i (1 - t_i)$. Induction on the number of factors proves $\prod_i (1 - t_i) \geq 1 - \sum_i t_i$. For positive numbers with a fixed sum, their product is at most the product with all entries equal: a maximum exists on the closed nonnegative simplex, lies in its interior when the sum is positive, and any unequal pair can be replaced by its average to increase the product, since $((a+b)/2)^2 - ab = (a-b)^2/4$. Apply this fact to $1 - t_i$ to obtain $\pi_M \leq (1 - \tau_M/r_M)^{r_M}$, proving both bounds. The last two inequalities in (CT.26) now follow from (CT.24). For $r_M = 1$, $\tau_M = 1 - \pi_M$ exactly; for $r_M = 0$, $\Delta_M = 0$, $\pi_M = 1$, and $W_M = 0$, so this case requires no exponent $1/r_M$.

15 Density of the original joint boundaries and determinant dissipation

Retain the complete packet h , its degree $d \geq 1$, the arithmetic unit v_h , and every original map in Section 14. In particular

$$s_j = \frac{1}{2} + it_j, \quad v_h = g/h, \quad g = 2\xi, \quad d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad E^{(k)} = E_h^{\otimes k}, \quad \mathcal{H}_k = L^2(d^k x). \quad (\text{JD.1})$$

The one-variable cyclic-density argument in the coherent-interpolation source has a full joint-source extension. We prove it here, including the exact relation between the disappearing Hilbert representatives and their unchanged finite jets.

15.1 Polynomial density with the original weighted measures

Lemma 15.1. *Let μ be a finite positive Borel measure on $\mathbb{R}^k$ such that $\int e^{a \sum_j |t_j|} d\mu < \infty$ for some $a > 0$. The complex polynomials in $t_1, \dots, t_k$ are dense in $L^2(\mu)$.*

Proof. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial. Define the finite complex measure $d\eta = \bar{f} d\mu$. For $0 < b < a/2$, Cauchy–Schwarz gives

$$\int e^{b \sum_j |t_j|} d|\eta| \leq \|f\|_{L^2(\mu)} \left(\int e^{2b \sum_j |t_j|} d\mu \right)^{1/2} < \infty. \quad (\text{JD.2})$$

Thus $\Phi(z) = \int e^{i \sum_j z_j t_j} d\eta(t)$ is holomorphic in the connected tube $|\text{Im } z_j| < b$. On every smaller closed tube, each differentiated integrand is dominated by the integrable function in (JD.2), multiplied by a constant; the strict exponent margin absorbs each monomial. This justifies every derivative and the convergent Taylor series on small polydiscs. All derivatives at zero vanish by polynomial orthogonality. The Taylor series therefore vanishes near zero, and successive overlapping polydiscs along any path in the tube show $\Phi = 0$ throughout it. In particular the Fourier transform of η vanishes on $\mathbb{R}^k$.

For clarity, uniqueness here follows from an explicit approximate identity. Put

$$\gamma_\epsilon(x) = (4\pi\epsilon)^{-k/2} e^{-|x|^2/(4\epsilon)} = (2\pi)^{-k} \int e^{-\epsilon|u|^2} e^{iu \cdot x} du. \quad (\text{JD.3})$$

The second equality is the product Gaussian integral, obtained in one variable by differentiating its Fourier integral in x , integrating by parts in u , and evaluating its value at zero. The Gaussian integrability and finiteness of $|\eta|$ justify Fubini. Consequently $\eta * \gamma_\epsilon = 0$. For any continuous compactly supported φ , integration and Fubini give $\int (\varphi * \gamma_\epsilon) d\eta = 0$. These convolutions converge uniformly to φ : split the integral at a fixed small radius, use uniform continuity within it, and use the Gaussian tail and boundedness outside it. Hence $\int \varphi d\eta = 0$. Uniqueness of finite Borel measures from their continuous compactly supported integrals gives $\eta = 0$, and therefore $f = 0$ almost everywhere. The orthogonal complement of the polynomial closure is zero, proving the lemma. $\square$

The actual measure ν_h has every exponential moment of order less than $\pi/2$, with its full factor $dt/(2\pi)$, as proved in (GC.2). Multiplying it by any fixed polynomial in t preserves the same open range of exponential moments: choose a strictly larger exponent still below $\pi/2$ and absorb the polynomial into the difference. Product integration therefore verifies the lemma for both measures

$$\nu_h^{\otimes k}, \quad d\mu_{h,k}(t) = |h(s_1)|^2 \prod_{j=1}^k d\nu_h(t_j). \quad (\text{JD.4})$$

No factor h has been discarded. Its finitely many zeros on the first integration line have measure zero. The multiplication map

$$U_h : L^2(\mu_{h,k}) \longrightarrow L^2(\nu_h^{\otimes k}), \quad U_h P = h(s_1)P \quad (\text{JD.5})$$

is a unitary surjection: its norm identity follows directly from (JD.4), and its inverse is division by $h(s_1)$ away from that null set, with an arbitrary value on the null set. Polynomials in $s_j = 1/2 + it_j$ are precisely polynomials in t_j , since these coordinate changes are invertible. Thus (JD.5) and the lemma prove density of $h(s_1)\mathbb{C}[s_1, \dots, s_k]$ in $L^2(\nu_h^{\otimes k})$. This dense space is contained in the original joint ideal $(h(s_1), \dots, h(s_k))$.

15.2 Strong limits of the actual source representatives

The Mellin Plancherel map, with its original sign and measure, is

$$(\mathcal{M}_k F)(t) = \int_{(0,\infty)^k} F(x) \prod_{j=1}^k x_j^{-1/2+it_j} d^k x, \quad \mathcal{M}_k : \mathcal{H}_k \longrightarrow L^2 \left(\frac{d^k t}{(2\pi)^k} \right). \quad (\text{JD.6})$$

Indeed the substitutions $x_j = e^{u_j}$ send F to $e^{\sum_j u_j/2} F(e^{u_1}, \dots, e^{u_k})$ isometrically in $L^2(d^k u)$, and (JD.6) becomes its Fourier transform with kernel $e^{it \cdot u}$. The inverse has kernel $e^{-it \cdot u}$ and factor $(2\pi)^{-k}$. Multiplication by $\prod_j v_h(s_j)$ is a unitary surjection from $L^2(\nu_h^{\otimes k})$ to the target in (JD.6). For surjectivity divide by this product almost everywhere: the entire nonzero function v_h has only isolated zeros on the line, and the product zero set is null. Composing with the inverse of (JD.6) gives the unitary extension

$$\mathcal{U}_{h,k} : L^2(\nu_h^{\otimes k}) \longrightarrow \mathcal{H}_k, \quad \mathcal{U}_{h,k} P = \mathcal{T}_h^{(k)} P \quad \text{for every polynomial } P. \quad (\text{JD.7})$$

The polynomial identity follows from the actual Mellin transform of F_h and D_j , so this is an extension of the original source map, not a change of representative.

Theorem 15.2 (Joint boundary density and exact finite-jet graph closure). *For each $k \geq 1$, the original total-degree boundary spaces $\mathcal{B}_M$ satisfy*

$$\overline{\bigcup_{M \geq k(d-1)} \mathcal{B}_M} = \mathcal{H}_k, \quad P_{\mathcal{B}_M} \longrightarrow I \quad \text{strongly.} \quad (\text{JD.8})$$

For the actual coherent minima of (CT.13), in the original fixed finite monomial tensor coordinates,

$$\|R_M\|_{E^{(k)} \rightarrow \mathcal{H}_k} \longrightarrow 0, \quad G_M \downarrow 0, \quad \det G_M \longrightarrow 0. \quad (\text{JD.9})$$

Here $G_M > 0$ at every finite M , and the middle limit is both decreasing in the positive-semidefinite order and convergent in matrix operator norm. Every finite jet is still retained: $J^{(k)} R_M = I$ for every M .

Define J_{pol} on the dense original source $\mathcal{D}_{h,k} = \mathcal{T}_h^{(k)} \mathbb{C}[s_1, \dots, s_k]$ by its complete arithmetic jets, including multiplication by $v_h^{\otimes k}$. Its graph has exactly the closure

$$\overline{\text{Graph}(J_{\text{pol}})} = \mathcal{H}_k \oplus E^{(k)}. \quad (\text{JD.10})$$

As a closed linear relation this has the split exact sequence

$$0 \longrightarrow E^{(k)} \xrightarrow{u \mapsto (0,u)} \overline{\text{Graph}(J_{\text{pol}})} \xrightarrow{(F,u) \mapsto F} \mathcal{H}_k \longrightarrow 0, \quad F \longmapsto (F, 0). \quad (\text{JD.11})$$

Proof. Every polynomial in $h(s_1) \mathbb{C}[s_1, \dots, s_k]$ belongs to one finite total-degree ideal $\mathcal{I}_{h,k,M}$. Its density proved in (JD.5), followed by the unitary map (JD.7), proves the first assertion in (JD.8). Given $F \in \mathcal{H}_k$ and $\epsilon > 0$, choose $b \in \mathcal{B}_N$ with $\|F - b\| < \epsilon$. For every $M \geq N$, the distance from F to $\mathcal{B}_M$ is at most that norm, so $\|(I - P_{\mathcal{B}_M})F\| < \epsilon$. This proves strong convergence.

Apply it to each of the d^k images of the fixed monomial basis under $s_h^{\otimes k}$. The identity $R_M = (I - P_{\mathcal{B}_M})s_h^{\otimes k}$ makes each column tend to zero. For a vector with coordinate norm one, Cauchy-Schwarz bounds its image by the square root of the sum of the squared column norms, proving the first limit in (JD.9). Since $G_M = R_M^* R_M$ and $G_M - G_{M+1} = Y_M^* Y_M$, the norm limit and monotonicity follow. The determinant limit follows by continuity of the determinant in these same finite coordinates. Positivity and $J^{(k)} R_M = I$ are the proved finite interpolation identities; none of these finite algebraic equations is lost in taking the Hilbert norm limit.

The kernel of J_{pol} is exactly the union of the $\mathcal{B}_M$, so its graph closure contains $\mathcal{H}_k \oplus \{0\}$ by (JD.8). For any $u \in E^{(k)}$, the vectors $(R_M u, u)$ belong to its graph and converge to $(0, u)$ by (JD.9). The closure is a linear subspace, and therefore contains every (F, u) ; the reverse inclusion is its declared ambient space. This proves (JD.10), and the maps, exactness and section in (JD.11) follow immediately by evaluating their two coordinates. In particular the closure is a relation with vertical fibre $E^{(k)}$, and it cannot be the graph of a single-valued operator because $d^k > 0$. $\square$

There are corresponding exact limits for the original weight form and both interpolation kernels. Write $U = U_{v_h^{\otimes k}}$, let K_M be the remainder-only polynomial kernel on the joint source, and let $\mathcal{C}_M = G_M^{-1} = UK_M U^*$ be the complete arithmetic interpolation matrix. In the same fixed coordinates,

$$\begin{aligned} \|W_M\| &\leq (2\|A^{(k)}\| + k)\|G_M\| \longrightarrow 0, \\ \lambda_{\min}(\mathcal{C}_M) &= \|G_M\|^{-1} \longrightarrow \infty, \\ \lambda_{\min}(K_M) &\geq \frac{1}{\|U\|^2\|G_M\|} \longrightarrow \infty. \end{aligned} \tag{JD.9a}$$

The first line follows by the triangle inequality applied to the definition of W_M . Positivity of G_M proves the second. For the third, use $K_M = U^{-1}\mathcal{C}_M U^{-*}$ and $\|U^{-*}z\| \geq \|z\|/\|U\|$. Thus these coercive limits include every retained local jet, and the arithmetic unit remains in the comparison. The relative weight is related to its absolute form by the exact congruence

$$W_M = G_M^{1/2} S_M G_M^{1/2}, \quad \lambda_{\min}(G_M)\epsilon_M \leq \|W_M\| \leq \|G_M\|\epsilon_M. \tag{JD.9b}$$

The upper bound is submultiplicativity; applying it instead to $S_M = G_M^{-1/2} W_M G_M^{-1/2}$ gives the lower bound. Equations (JD.9a)–(JD.9b) retain the metric factors needed to pass from the proved absolute limits to the relative quantity.

The complementary quotient of the same closed relation is also explicit:

$$\frac{\overline{\text{Graph}(J_{\text{pol}})}}{\mathcal{H}_k \oplus \{0\}} \xrightarrow{\sim} E^{(k)}, \quad [(F, u)] \mapsto u, \quad u \mapsto [(0, u)]. \tag{JD.11a}$$

The kernel, well-definedness and both inverse identities follow by subtracting the two displayed pairs. With the product Hilbert norm the quotient norm is exactly $\|u\|$, since $\inf_{H \in \mathcal{H}_k} (\|F - H\|^2 + \|u\|^2)^{1/2} = \|u\|$, attained at $H = F$. Thus every completed-boundary class retains its original finite jet through an explicit isometric quotient.

This limit has an exact Split-Zero morphism. For a complex vector space X , use its one-support lift $\hat{X} = \{\tau_X\} \sqcup X^\bullet$, with τ_X the additive identity, vector addition within $X^\bullet$, and the original $G(\mathbb{C})$ -scalar action. In particular $0_X^\bullet \neq \tau_X$. Any linear map $T : X \rightarrow Y$ lifts to $\hat{T}(\tau_X) = \tau_Y$ and $\hat{T}(u^\bullet) = (Tu)^\bullet$; direct substitution checks addition and every scalar, including the supported scalar zero and external scalar zero. Give the supported fibre its norm topology and τ_X its isolated topology. Equation (JD.9) then says exactly

$$\hat{R}_M \longrightarrow \hat{0}, \quad \hat{0}(\tau_E) = \tau_{\mathcal{H}}, \quad \hat{0}(u^\bullet) = 0_{\mathcal{H}}^\bullet \quad (u \in E^{(k)}). \tag{JD.12}$$

Thus the limiting morphism retains the support of every input. Equation (JD.11) identifies all finite jets in the vertical fibre of the closed relation; it supplies the full typed relation between these supported Hilbert limits and the unchanged algebraic jets.

15.3 The accumulated determinant cost of the same boundary sequence

Continue to use the original ordered spectral sum and layer energy

$$\mathcal{D}_k = \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k (\text{Re } \rho_{i_j} - 1/2) \right|, \quad \chi_M = \text{Tr}(G_M^{-1} C_M^* C_M), \quad \tau_M = \text{Tr}(G_M^{-1} (G_M - G_{M+1})). \tag{JD.13}$$

The list $\rho_1, \dots, \rho_d$ includes every algebraic multiplicity. Define $\pi_M = \det G_{M+1} / \det G_M$. For integers $k(d-1) \leq a < b$, the following exact finite inequality holds whenever $\mathcal{D}_k > 0$:

$$\boxed{\mathcal{D}_k^2 \sum_{M=a}^{b-1} \frac{1}{\chi_M} \leq \sum_{M=a}^{b-1} \tau_M \leq \log \frac{\det G_a}{\det G_b}.} \tag{JD.14}$$

All denominators in this statement are proved positive. Indeed (CT.24) gives $\mathcal{D}_k^2 \leq \tau_M \chi_M$, with both factors nonnegative; strict positivity of the left side forces strict positivity of each factor. Let $t_{M,j} \in [0, 1)$ be the full eigenvalue list of $G_M^{-1/2}(G_M - G_{M+1})G_M^{-1/2}$, including zeros. The finite original matrices give exactly

$$\tau_M = \sum_j t_{M,j}, \quad \pi_M = \prod_j (1 - t_{M,j}), \quad -\log \pi_M - \tau_M = \sum_j \int_0^{t_{M,j}} \frac{u}{1-u} du \geq 0. \quad (\text{JD.15})$$

The first two identities are congruence and determinant identities; the last follows by integrating the derivative of $-\log(1-t) - t$. Sum (CT.24) divided by χ_M , then sum (JD.15). The product of the determinant ratios telescopes without a change of coordinates, proving (JD.14).

If $\mathcal{D}_k = 0$, every repeated tuple $(i, \dots, i)$ has zero summand, hence $\text{Re } \rho_i = 1/2$ for every retained zero. Conversely these equalities give $\mathcal{D}_k = 0$. The second inequality in (JD.14) and the exact remainder (JD.15) hold also in this case; no quotient $0/0$ is introduced. Equation (JD.9) proves that the right-hand side tends to infinity as $b \rightarrow \infty$ for each fixed a . Equations (JD.13)–(JD.15) retain the actual rate through χ_M , the complete relative Gram losses and their explicit logarithmic remainder. They impose no unproved upper estimate on that rate.

The same sequence supplies two further exact energy identities:

$$G_a = \sum_{M=a}^{\infty} Y_M^* Y_M, \quad \sum_{M=a}^{\infty} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M=a}^{\infty} \tau_M = \infty. \quad (\text{JD.16})$$

The first series converges in matrix operator norm, by telescoping $G_M - G_{M+1} = Y_M^* Y_M$ and the proved limit $G_M \rightarrow 0$. Taking traces of its finite partial sums proves the second identity; all summands are nonnegative. To prove the third, suppose instead that $\sum_M \tau_M$ were finite. Then beyond some index all $\tau_M \leq 1/2$, so every $0 \leq t_{M,j} \leq 1/2$. Integrating $(1-u)^{-1} \leq 2$ over $[0, t_{M,j}]$ gives $-\log(1-t_{M,j}) \leq 2t_{M,j}$. Hence the tail sum of $-\log \pi_M$ would be bounded by $2 \sum_M \tau_M < \infty$. The finite initial terms are finite because $G_M > 0$. Telescoping would therefore bound $\log(\det G_a / \det G_b)$ uniformly in b , contradicting $\det G_b \rightarrow 0$. This proves the divergence, including packets entirely on the critical line and every retained nilpotent chain. The equality $\tau_M = \|Y_M G_M^{-1/2}\|_{\text{HS}}^2$ identifies the relative energy in this divergence with the same layer map appearing in the convergent absolute energy series.

16 The actual kernel layer, its derivative image, and its quotient

This section integrates the complete kernel-layer delivery of 12 September 2026. Its archive hash and member hashes are recorded in `sources/web_kernel_layer_delivery/LOCAL_STAGING_PROVENANCE.json`. The source note's formulas (9)–(28) identify the interpolation kernel with an explicit basis of the next original theta relation layer. We first prove those maps in the retained coordinates and then calculate the exact derivative image and the entire quotient of that layer. The new quotient calculation retains the part of the layer not reached by the summed Euler derivative.

16.1 The original arithmetic source and its constrained minimum

Keep $g = 2\xi$, a nonempty full packet $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, and $d = \deg h$. Fix $k \geq 1$, and set

$$E = E_h^{\otimes k}, \quad E_h = \mathbb{C}[s]/(h), \quad A = \sum_{j=1}^k A_{h,j}, \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times. \quad (\text{KL.1})$$

All tensor coordinates use the original monomial remainders. The function F_h , with Mellin transform g/h , is the original test function satisfying $h(D)F_h = f_0$. The exact source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{M}_k \mathcal{T}(P) = P(s_1, \dots, s_k) \prod_{j=1}^k v_h(s_j), \quad s_j = \frac{1}{2} + it_j. \quad (\text{KL.2})$$

The Mellin multiplier proves injectivity on polynomials. Its Hilbert norm is the polynomial norm for the unchanged product measure

$$d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad \nu_h^{\otimes k}. \quad (\text{KL.3})$$

Let $\mathcal{P}_N$ be the polynomials of total degree at most N , and let $\mathcal{J}_N : \mathcal{P}_N \rightarrow E$ be full reduction by $h(s_1), \dots, h(s_k)$, followed by multiplication by the original unit $v_h^{\otimes k}$. For $N \geq k(d-1)$ this is onto, since the monomial remainder basis has at most that total degree. Fix an integer $N \geq k(d-1)$ for the remainder of this section; all subsequent levels $N+1, N+2$ satisfy the same threshold. Its kernel and its original theta image are

$$\mathcal{I}_N = \mathcal{P}_N \cap (h(s_1), \dots, h(s_k)), \quad \mathcal{L}_N = \mathcal{T}\mathcal{I}_N. \quad (\text{KL.4})$$

Let $p_j(s)$ be the actual monic orthogonal polynomial of ν_h , and retain $\kappa_j = \|p_j\|_{\nu_h}^2 > 0$. All moments exist by the proved arithmetic decay. The Gram is strictly positive because the density is positive except at a discrete set; this proves existence and uniqueness by finite orthogonal projection. For multi-indices set

$$p_\alpha = \prod_j p_{\alpha_j}, \quad \kappa_\alpha = \prod_j \kappa_{\alpha_j}, \quad a_\alpha = \bigotimes_j (v_h[p_{\alpha_j}]_h). \quad (\text{KL.5})$$

The monic triangular change preserves the full space of degree at most N , and product integration makes its Gram diagonal. It follows, by direct constrained minimization, that the original canonical representative and its metric are

$$\mathcal{K}_N = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha} > 0, \quad G_N = \mathcal{K}_N^{-1}, \quad R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (\text{KL.6})$$

Indeed the full jet of this vector is $\mathcal{K}_N G_N u = u$. If a coefficient vector has zero full jet, its inner product with the displayed coefficient vector is zero. Every other representative of that same jet is its sum with such a vector; its squared norm is therefore $u^* G_N u$ plus the squared norm of the added relation. This proves the unique minimum, its metric, and $R_N \perp \mathcal{L}_N$. These are the original maps $J^{(k)} R_N = I$ and $q^{(k)} R_N = \sigma$. The notation $\mathcal{K}_N$ here includes the local unit. To specify the comparison in the full tensor dimension, put

$$r_\alpha = \bigotimes_j [p_{\alpha_j}]_h, \quad K_N^{\text{rem}} = \sum_{|\alpha| \leq N} \frac{r_\alpha r_\alpha^*}{\kappa_\alpha}.$$

Then $\mathcal{K}_N = U_{v_h^{\otimes k}} K_N^{\text{rem}} U_{v_h^{\otimes k}}^*$, as an equality of operators on $E_h^{\otimes k}$. When $k = 1$, K_N^{rem} is exactly the earlier remainder-only kernel K_N .

16.2 The complete next-layer frame and its theta primitives

Write $\Sigma_n = \{\alpha \in \mathbb{N}^k : |\alpha| = n\}$ and $\mathbf{H}_n = \mathbb{C}^{\Sigma_n}$. The coordinate identification sends $x \in \mathbf{H}_n$ to the homogeneous polynomial $\sum_\alpha x_\alpha z^\alpha$; these z_j are coefficient variables and are not substituted for the original Mellin variables s_j . For $\beta \in \Sigma_{N+1}$, define the actual functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \mathcal{E}_N x = \sum_\beta x_\beta e_\beta, \quad \mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp. \quad (\text{KL.7})$$

The full jet of e_β is zero, hence it is in $\mathcal{L}_{N+1}$. The new-degree polynomial is orthogonal to all old polynomials and $R_N a_\beta \perp \mathcal{L}_N$, so $e_\beta \in \mathcal{E}_N$. The leading total-degree terms of the e_β are the distinct monomials s^β , proving independence by the injectivity of $\mathcal{T}$. For any element of $\mathcal{E}_N$, subtract its full degree- $N+1$ combination of the e_β . The remainder has zero jet and degree at most N , hence is in $\mathcal{L}_N$. It is also orthogonal to that space, so it is zero. This proves the exact isomorphism

$$\mathcal{E}_N : \mathcal{H}_{N+1} \xrightarrow{\sim} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1}/\mathcal{L}_N. \quad (\text{KL.8})$$

The inverse of the second arrow is $[f] \mapsto (I - P_{\mathcal{L}_N})f$.

Every vector in (KL.7) has a primitive in the original tensor complex. Its numerator polynomial is

$$p_\beta - \sum_{|\alpha| \leq N} p_\alpha \frac{a_\alpha^* G_N a_\beta}{\kappa_\alpha}. \quad (\text{KL.9})$$

Successive monic division in the ordered variables $s_1, \dots, s_k$ expresses this zero-jet polynomial as $\sum_i h(s_i) Q_{i,\beta}$, where $\deg Q_{i,\beta} \leq N+1-d$. At each replacement the leading power of the divided variable decreases and total degree does not increase, proving the bound. After the last division, the remainder has degree below d in every variable; its full jet is zero, so uniqueness of the complete remainder forces it to vanish. In the i -th tensor factor replace F_h by $(-1)^{i-1} \phi_*$, and apply $Q_{i,\beta}(D_1, \dots, D_k)$ to the whole tensor. The preceding $i-1$ degree-one factors give the differential sign $(-1)^{i-1}$. Together with $h(D)F_h = \Theta \phi_*$, the boundary of this primitive is exactly $+\mathcal{T}(h(s_i)Q_{i,\beta})$. Summing proves the asserted primitive with all ordered signs retained.

Put

$$\mathbf{U}_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad \mathbf{D}_{N+1} = \text{diag}(\kappa_\beta), \quad \mathbf{H}_N^{\text{lay}} = \mathcal{E}_N^* \mathcal{E}_N.$$

Expanding (KL.7) and using old/new polynomial orthogonality proves

$$\mathbf{H}_N^{\text{lay}} = \mathbf{D}_{N+1} + \mathbf{U}_N^* G_N \mathbf{U}_N > 0, \quad \mathcal{E}_N^* R_N = -\mathbf{U}_N^* G_N. \quad (\text{KL.10})$$

Define the projection loss $Y_N = P_{\mathcal{E}_N} R_N$. In particular its exact formula is $Y_N = -\mathcal{E}_N (\mathbf{H}_N^{\text{lay}})^{-1} \mathbf{U}_N^* G_N$. Its rank is exactly $\text{rank } \mathbf{U}_N$, because every other displayed factor is injective or invertible on its declared space.

16.3 The exact incidence map and the derivative-image rank

Define the actual old top-degree columns and their original norms by

$$\mathbf{A}_N = (a_\alpha)_{\alpha \in \Sigma_N} : \mathcal{H}_N \rightarrow E, \quad \mathbf{D}_N = \text{diag}(\kappa_\alpha)_{\Sigma_N}. \quad (\text{KL.11})$$

Define the coordinate incidence map by

$$\mathbf{T}_n : \mathcal{H}_n \rightarrow \mathcal{H}_{n+1}, \quad (\mathbf{T}_n x)_\beta = \sum_{i: \beta_i > 0} x_{\beta - \mathbf{e}_i}. \quad (\text{KL.12})$$

On the specified homogeneous coefficient polynomials this is multiplication by $\ell(z) = z_1 + \dots + z_k$. A product of two nonzero polynomials over $\mathbb{C}$ is nonzero: choose a monomial order compatible with products and multiply the two leading terms. Since $\ell \neq 0$, $\mathbf{T}_n$ is injective.

The full unprojected derivative defect and its projected layer map are

$$B_N = D^{(k)} R_N - R_N A, \quad C_N = (I - P_{\mathcal{L}_N}) B_N : E \rightarrow \mathcal{E}_N.$$

They have the exact coordinate factorization

$$\boxed{C_N = \mathcal{E}_N \mathbf{T}_N \mathbf{D}_N^{-1} \mathbf{A}_N^* G_N.} \quad (\text{KL.13})$$

To prove it, multiply the finite numerator in (KL.6) by $s_1 + \cdots + s_k$. Monicity makes its coefficient at the new top polynomial p_β equal to $\sum_{i:\beta_i>0} a_{\beta-\mathbf{e}_i}^* G_{Nu}/\kappa_{\beta-\mathbf{e}_i}$, precisely the row on the right side of (KL.13). The numerator of $R_N Au$ has degree at most N . Subtract the right side of (KL.13) from $B_N u$: its numerator has degree at most N and its full jet is zero, so it is an original relation in $\mathcal{L}_N$. Orthogonal projection proves (KL.13), including the original quotient statement $[B_N u] = [C_N u]$ in $\mathcal{B}^{(k)}/\mathcal{L}_N$.

Theorem 16.1 (Exact derivative rank and its full kernel). *With every original jet column retained,*

$$\boxed{\text{rank } C_N = \text{rank } A_N, \quad \ker C_N = \mathcal{K}_N \ker A_N^*} \quad (\text{KL.14})$$

Define the original weight form $W_N = A^* G_N + G_N A - k G_N$. It obeys the refined bound

$$\text{rank } W_N \leq 2 \min\{\text{rank } A_N, \text{rank } U_N\}. \quad (\text{KL.15})$$

Proof. Both $\mathcal{E}_N$ and $\mathbf{T}_N$ are injective; D_N and G_N are invertible. Thus (KL.13) has rank $\text{rank } A_N^*$, which equals the rank of its conjugate transpose. Its kernel is $G_N^{-1} \ker A_N^* = \mathcal{K}_N \ker A_N^*$, with the displayed invertible coordinate map retained. The original Green identity and orthogonality give $W_N = -(Y_N^* C_N + C_N^* Y_N)$. Each summand has rank at most the lesser of the two map ranks; the rank of a sum is at most the sum of its ranks. Use (KL.10) and (KL.14) to obtain (KL.15). No nonzero homogeneous-leading-term assertion has been substituted for an actual column a_α . $\square$

16.4 The whole quotient of the next layer by the derivative image

For $k \geq 2$, let H'_n be the homogeneous polynomials of degree n in $z_1, \dots, z_{k-1}$. There is a split exact sequence of vector spaces

$$0 \longrightarrow H_n \xrightarrow{\mathbf{T}_n} H_{n+1} \xrightarrow{\rho_n} H'_{n+1} \longrightarrow 0, \quad \rho_n f = f(z_1, \dots, z_{k-1}, -\sum_{i < k} z_i). \quad (\text{KL.16})$$

For a complete proof and inverse, divide f by the monic linear polynomial $\ell = z_k + \sum_{i < k} z_i$ in the last variable. Every replacement retains total degree, giving uniquely

$$f = \ell q + \iota r, \quad q \in H_n, \quad r = \rho_n f \in H'_{n+1}, \quad (\text{KL.17})$$

where ι includes a polynomial independent of z_k . For uniqueness set $z_k = -\sum_{i < k} z_i$ first to obtain $r = 0$, then use injectivity of multiplication by ℓ . This also proves surjectivity, exactness and the specified section ι . The ordered last coordinate and the signs in the substitution remain part of the map.

Theorem 16.2 (Exact unused-layer quotient). *Set $Z_N = \text{im}(D_N^{-1} A_N^*) \subseteq H_N$. Then the actual quotient is*

$$\boxed{\mathcal{E}_N / \text{im } C_N \simeq (H_N / Z_N) \oplus H'_{N+1}} \quad (\text{KL.18})$$

For an original layer representative $e = \mathcal{E}_N f$, divide $f = \ell q + \iota r$ by (KL.17); the map sends $[e]$ to $([q], r)$. Its inverse sends $([q], r)$ to $[\mathcal{E}_N(\ell q + \iota r)]$.

Proof. The invertibility of G_N in (KL.13) gives $\text{im } C_N = \mathcal{E}_N \mathbf{T}_N Z_N$. Changing e by such an element changes exactly q by an element of Z_N , while leaving r unchanged. Thus the first map is well-defined. Changing the representative of $[q]$ has the same effect on the inverse, so the inverse is well-defined. The unique division (KL.17) proves both inverse identities. For $k = 1$, $\mathbf{T}_N$ is an isomorphism between the one-dimensional shell spaces and the same statement holds with $H'_{N+1} = 0$. $\square$

In particular $\dim H'_{N+1} = \binom{N+k-1}{k-2}$ for $k \geq 2$, by counting monomials; the first summand has dimension $\binom{N+k-1}{k-1} - \text{rank } A_N$. Their sum is exactly the original layer dimension minus $\text{rank } C_N$. The quotient therefore retains both the full-jet rank loss and the explicitly complementary relation directions. For every coordinate vector, (KL.9) supplies its original theta primitive. The quotient is of actual test-function relations, with those primitives, rather than only a quotient of dimensions.

16.5 The derivative between consecutive relation quotients

The original Euler sum has $D^{(k)}\mathcal{L}_N \subseteq \mathcal{L}_{N+1}$, because it multiplies the numerator polynomial by $s_1 + \cdots + s_k$ and preserves the ideal in (KL.4). Hence it induces the well-typed map

$$\partial_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [D^{(k)}f]. \quad (\text{KL.19})$$

Under the exact layer coordinates of (KL.8), its matrix is

$$\boxed{\partial_N = \mathcal{E}_{N+1} \mathsf{T}_{N+1} \mathcal{E}_N^{-1}}, \quad (\text{KL.20})$$

where each $\mathcal{E}$ is followed or preceded by its quotient isomorphism as in (KL.8). Write $e_\beta^{(N)}$ for (KL.7) at level N , and $e_{\beta+\mathbf{e}_i}^{(N+1)}$ for the next layer. The actual recurrence $sp_j = p_{j+1} + b_j p_j - (\kappa_j/\kappa_{j-1})p_{j-1}$ gives the complete retained difference

$$\begin{aligned} D^{(k)}e_\beta^{(N)} - \sum_i e_{\beta+\mathbf{e}_i}^{(N+1)} = \mathcal{T} \left[\left(\sum_i b_{\beta_i} \right) p_\beta - \sum_{i:\beta_i > 0} \frac{\kappa_{\beta_i}}{\kappa_{\beta_i-1}} p_{\beta-\mathbf{e}_i} \right] \\ - D^{(k)}R_N a_\beta + R_{N+1} \sum_i a_{\beta+\mathbf{e}_i}. \end{aligned}$$

The final term in the one-variable recurrence is omitted at $j = 0$. Every term on the right has numerator degree at most $N + 1$. Each term on the left has zero full jet, so their difference is in $\mathcal{L}_{N+1}$. This proves (KL.20) with all lower polynomial coefficients retained. Applying the explicit ordered division in (KL.9) to this displayed numerator supplies its original theta primitive.

It follows that ∂_N is injective and its cokernel is exactly H'_{N+2} , with maps (KL.16)–(KL.17) at index $N + 1$. Thus the original derivative connects consecutive relation quotient types by a proved injection; it has not been made an endomorphism of a fixed quotient. The ordinary relation-admission map has the same declared source and target:

$$\text{adm}_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [f] = 0.$$

Equation (KL.19) instead sends the representative to its next derivative class. These are explicitly different typed maps of the same original representative, related by (KL.20) and the inclusion of their actual numerator relations.

All maps here have their original supported lifts. At a fixed support label λ , a linear map T acts by $(\lambda, x) \mapsto (\lambda, Tx)$. The relation-admission map therefore sends a killed class to the receiving fibre's supported zero. The external zero τ remains external absence. Each displayed kernel and quotient records its full supported vector space and each derivative retains its consecutive source and target; no passage to the Hilbert closure is used.

16.6 The exact original-theta norm and volume bridge

Specialize to $k = 1$. We now distinguish the degree-indexed G_N of (KL.6) from the correction-indexed original metric by writing

$$\mathbf{G}_m = G_{d+m} \quad (m \geq -1), \quad \mathbf{R}_m = R_{d+m}, \quad r_m = \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{m-1}} \quad (m \geq 0). \quad (\text{KL.21})$$

At $m = -1$, $\mathcal{P}_{d-1}$ contains no nonzero multiple of h . Thus the original representative is exactly the fixed section

$$s_h u = \mathbf{R}_{-1} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u), \quad \varepsilon_h = v_h^{-1}, \quad \mathbf{G}_{-1} = s_h^* s_h.$$

This fixes the initial determinant in (KL.21), without introducing a negative-degree theta polynomial. Put $L_{-1} = 0$ and

$$f_j = D^j f_0, \quad L_j = \text{span}(f_0, \dots, f_j), \quad u_j = (I - P_{L_{j-1}})f_j, \quad \mathfrak{h}_j = \|u_j\|^2 > 0 \quad (j \geq 0). \quad (\text{KL.22})$$

These $u_j = Q_j(D)f_0$ are the monic orthogonal theta vectors for the original measure $|g(1/2 + it)|^2 dt/(2\pi)$. Their definition does not depend on h . The monic p_j and their norms κ_j continue to use the original packet-dependent measure $|g/h|^2 dt/(2\pi)$. Multiplication by the unchanged monic polynomial h is the exact isometry connecting their source polynomial spaces:

$$P \mapsto hP, \quad \|hP\|_{\nu_h}^2 = \|P\|_{|g|^2 dt/(2\pi)}^2, \quad \mathcal{T}_h(hP) = P(D)f_0. \quad (\text{KL.23})$$

This follows by cancellation of the pointwise factor $|h|^2$ in the integral; the identities extend across the discrete zeros by their entire expressions. It is an isometry onto the specified subspace of multiples of h , with its degree shifted by the full degree $d = \sum_\rho m_\rho$.

Theorem 16.3 (Original norm as an exact relative Gram volume). *For every $j \geq 0$, including the initial theta vector,*

$$\boxed{\kappa_{d+j} = \mathfrak{h}_j r_j}. \quad (\text{KL.24})$$

Consequently, for $m \geq 0$,

$$\boxed{\frac{\kappa_{d+m+1}}{\kappa_{d+m}} = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} \frac{r_{m+1}}{r_m}, \quad \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{-1}} = \prod_{j=0}^m \frac{\kappa_{d+j}}{\mathfrak{h}_j}}. \quad (\text{KL.25})$$

All ratios have positive denominators and all packet multiplicities are retained.

Proof. At the degree level $N = d + j - 1$, define $e_{d+j} = \mathcal{T}_h p_{d+j} - \mathbf{R}_{j-1} a_{d+j}$. Its numerator has zero full jet, is monic of degree $d + j$, and is divisible by the original monic h . By ordinary monic division its quotient is monic of degree j . Under (KL.23) this is f_j plus a vector in L_{j-1} . By (KL.7) it is orthogonal to that space. The defining orthogonal projection in (KL.22) therefore proves

$$e_{d+j} = u_j, \quad \mathfrak{h}_j = \kappa_{d+j} + a_{d+j}^* \mathbf{G}_{j-1} a_{d+j}. \quad (\text{KL.26})$$

When $j = 0$, the quotient of the numerator by h is the constant 1, so the same argument explicitly gives $e_d = f_0 = u_0$; no previous theta layer is required.

For the full unit cancellation put $v_l = [p_l]_h = p_l(A_h)[1]_h$, and $K_{l-1} = \sum_{i=0}^{l-1} v_i v_i^* / \kappa_i$, using the original monomial remainder coordinates. At $l = d + j$,

$$a_l = U_{v_h} v_l, \quad \mathbf{G}_{j-1} = U_{\varepsilon_h}^* K_{l-1}^{-1} U_{\varepsilon_h}, \quad a_l^* \mathbf{G}_{j-1} a_l = v_l^* K_{l-1}^{-1} v_l. \quad (\text{KL.27})$$

Both cancellations use $U_{\varepsilon_h} U_{v_h} = I$; hence they preserve every derivative of the original local unit at every full jet. The degree update $K_l = K_{l-1} + v_l v_l^* / \kappa_l$ has determinant ratio $1 + v_l^* K_{l-1}^{-1} v_l / \kappa_l$. To verify this scalar identity directly, expand $\det(I + xy^*)$ multilinearly in its columns: all terms with two updated columns vanish since those columns are multiples of x , leaving $1 + y^* x$. The fixed-unit congruences in (KL.27) then give

$$r_j = \frac{\kappa_l}{\kappa_l + v_l^* K_{l-1}^{-1} v_l} = \frac{\kappa_{d+j}}{\mathfrak{h}_j}.$$

This also proves $0 < r_j \leq 1$, including the case $v_l = 0$. Divide two consecutive identities and multiply the first $m + 1$ identities to obtain (KL.25). $\square$

For two full packets h, H , their original norm sequences are therefore connected at the same theta degree j by the exact scalar equality

$$\frac{\kappa_{\deg h+j}^{(h)}}{r_j^{(h)}} = \mathfrak{h}_j = \frac{\kappa_{\deg H+j}^{(H)}}{r_j^{(H)}}. \quad (\text{KL.28})$$

When $h \mid H$, the corresponding source isometry is $P \mapsto (H/h)P$, since $(g/H)(H/h)P = (g/h)P$. At the old centres this equality preserves every jet, and at each added centre g/h has the complete zero order, so it realizes exactly the CRT insertion of zero jets. The reference numerator has degree below $\deg H$ and is its unique full remainder. Both corrected representatives then subtract the same projection onto the packet-independent L_j , proving the coherent equality $\mathbf{R}_{H,j} \iota_{hH} = \mathbf{R}_{h,j}$, including $j = -1$. Thus (KL.28) connects precisely these original coherent source maps and their fixed-section metrics.

16.7 The complete scalar dictionary for the boundary energy

For $m \geq 0$, set $n = d + m + 1$ and retain the terminal notation

$$M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}, \quad \alpha_n = v_{n-1}^* M_n v_{n-1}, \quad \beta_n = v_n^* M_n v_n, \quad \gamma_n = v_n^* M_n v_{n-1}.$$

Define the original theta projection rows by

$$d_j = \mathfrak{h}_j^{-1} u_j^* s_h : E_h \rightarrow \mathbb{C}, \quad \mathbf{R}_m = s_h - \sum_{j=0}^m u_j d_j, \quad c_m = d_m \mathbf{G}_m^{-1} d_{m+1}^*. \quad (\text{KL.29})$$

Here an expression $u_j^* F$ denotes the Hilbert inner product $\langle u_j, F \rangle$, conjugate-linear in its first argument. The row d_j therefore retains the original unnormalized vector and the exact factor $\mathfrak{h}_j^{-1}$.

The complete row and scalar comparison is

$$\boxed{\begin{aligned} d_m &= -\frac{v_{n-1}^* K_{n-1}^{-1} U_{\varepsilon_h}}{\kappa_{n-1}}, \\ d_{m+1} &= -\frac{v_n^* K_{n-1}^{-1} U_{\varepsilon_h}}{\mathfrak{h}_{m+1}}, \\ d_m \mathbf{G}_m^{-1} d_m^* &= \frac{\alpha_n}{\kappa_{n-1}}, \\ d_{m+1} \mathbf{G}_m^{-1} d_{m+1}^* &= \frac{\kappa_{n-1} \beta_n}{\mathfrak{h}_{m+1}^2}, \\ \mathfrak{h}_{m+1} c_m &= \overline{\gamma_n}. \end{aligned}} \quad (\text{KL.30})$$

To prove the first row, the polynomial numerator of s_h has degree below d . In (KL.29), each u_j has numerator hQ_j , monic of degree $d+j$. Thus the coefficient of degree $d+m = n-1$ in the numerator of $\mathbf{R}_m$ is $-d_m$. The finite expansion (KL.6) gives the same coefficient as $a_{n-1}^* \mathbf{G}_m / \kappa_{n-1}$. Cancelling the same units as in (KL.27) proves the first row, also at $m = 0$. For the second row use $u_{m+1} = e_n$, so (KL.10) gives $u_{m+1}^* \mathbf{R}_m = -a_n^* \mathbf{G}_m$. The orthogonality of u_{m+1} to L_m also gives $u_{m+1}^* \mathbf{R}_m = u_{m+1}^* s_h = \mathfrak{h}_{m+1} d_{m+1}$. This proves its formula. Finally $\mathbf{G}_m^{-1} = U_{v_h} K_{n-1} U_{v_h}^*$; insert both rows, cancel both unit pairs and the adjacent kernel inverse. The three remaining products are exactly those in (KL.30), including the conjugation in the last line.

Theorem 16.4 (Exact original-norm formula for the reflected packet). *Suppose the original full packet is stable under $\rho \mapsto 1 - \bar{\rho}$. For its actual relative allowance ϵ_m ,*

$$\boxed{\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2.} \quad (\text{KL.31})$$

If the packet is also stable under ordinary conjugation, then $c_m = 0$, so its complete formula becomes

$$\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{KL.32})$$

Proof. The terminal identity (KT.4), with the original full multiplicities, gives $\text{Re } \gamma_n = \text{Re tr } A_h - d/2 = 0$: reflected centres contribute complementary real parts with equal orders. Its two possible nonzero generalized eigenvalues then yield $\epsilon_m^2 = \alpha_n \beta_n - |\gamma_n|^2$. The preceding and succeeding rank-one kernel updates, also proved directly in (KL.27), give

$$\alpha_n = 1 - r_m, \quad \beta_n = \frac{\kappa_n}{\kappa_{n-1}} \frac{1 - r_{m+1}}{r_{m+1}}.$$

Insert (KL.25) into the product $\alpha_n \beta_n$ and the last line of (KL.30) into $|\gamma_n|^2$. Every positive ratio is retained, and this is (KL.31).

For the final assertion, conjugation stability makes h real in the original monomial coordinates. The real Taylor coefficients of g give $(g/h)(1/2 - it) = \overline{(g/h)(1/2 + it)}$, so the actual ν_h is even in t . Its monomial Gram matrix is real: the integrand for entry $s^i s^j$ at $-t$ is the complex conjugate of its integrand at t . Therefore the unique solution of the real Gram linear equations for the monic p_l has real coefficients. Polynomial division by the real monic h makes each v_l real, and the finite sum for K_{n-1} is real. Its inverse and M_n are real as well. Consequently γ_n is real. The already proved reflection trace identity makes its real part zero, hence $\gamma_n = 0$. Equation (KL.30), with $\mathfrak{h}_{m+1} > 0$, proves $c_m = 0$. No division by a jet column, an off-line displacement, or a boundary energy has been used; all zero-column cases remain included. $\square$

The source PR15 denotes the relative determinant at correction m by $\delta_m = \det \mathbf{G}_{m+1} / \det \mathbf{G}_m$. Its exact index map is therefore $\delta_m = r_{m+1}$, with $\delta_{-1} = r_0$. Its cross-term row product is the c_m of (KL.29). Equations (KL.24)–(KL.31) prove the full map from its original-theta scalar formula to the terminal-kernel formula, including both monic norm families, the unit congruences, the conjugation, the fixed initial section, and the complete orders. These equalities calculate the same actual boundary energy at every finite stage. They do not supply an estimate tending to zero; that requires further information about these now explicitly related arithmetic quantities.

The packet-independent norms in (KL.24)–(KL.32) also have the complete positive-half-line formulas (SP.18)–(SP.21) and the gamma-reference determinant formulas (TG.16)–(TG.19). Those sections prove the original affine and quadratic coordinate maps, their inverses, both measures and every phase. Consequently these formulas can be substituted directly into the identities here; the actual unit, packet determinant ratios, full zero orders and source degree indices remain unchanged.

17 Original orthogonal shells and the exact projective-ratio algebra

The source's degree shell consists of its original orthogonal product polynomials. This section proves its full comparison with the homogeneous monomial shell, including every lower-degree coefficient. It also calculates the algebra generated by the ratio of the two original tensor coordinates. The arithmetic local unit and all multiplicities remain in the resulting linear maps.

17.1 Two sections of the same polynomial filtration

Retain a complete nonempty arithmetic packet

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(g/h), \quad g = 2\xi. \quad (\text{KPC.1})$$

The complete orders make v_h a unit. For $k \geq 1$, write $E^{(k)} = E_h^{\otimes k}$ in its original monomial tensor coordinates and use the original full-jet map

$$J(P) = U_{v_h^{\otimes k}}[P]_{(h(s_1), \dots, h(s_k))}, \quad J : \mathbb{C}[s_1, \dots, s_k] \longrightarrow E^{(k)}. \quad (\text{KPC.2})$$

Here U_a is multiplication by the stated element a . The integration variables remain $s_j = 1/2 + it_j$, with

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{KPC.3})$$

Let $p_j(s) = \sum_{r=0}^j p_{j,r} s^r$, with $p_{j,j} = 1$, be the original monic orthogonal polynomials for this measure. Retain $\kappa_j = \int |p_j(s)|^2 d\nu_h > 0$. For a multi-index α , put

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad a_\alpha = J(p_\alpha). \quad (\text{KPC.4})$$

All moment integrals in these definitions use the original density.

For $N \geq 0$, let $\mathcal{P}_{\leq N}$ be the polynomials of total degree at most N , let $\mathcal{P}_{\leq -1} = 0$, and let $\mathcal{H}_N$ be the homogeneous polynomials of degree N . The highest-degree map $\ell_N : \mathcal{P}_{\leq N} \rightarrow \mathcal{H}_N$ has kernel $\mathcal{P}_{\leq N-1}$. It therefore identifies $\mathcal{H}_N$ with the associated-graded space $\mathcal{P}_{\leq N}/\mathcal{P}_{\leq N-1}$. There are two specified sections:

$$\begin{aligned} i_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha s^\alpha, \\ O_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha p_\alpha. \end{aligned} \quad (\text{KPC.5})$$

Theorem 17.1 (The exact lower-degree correction). *Both maps in (KPC.5) are linear sections of ℓ_N . Their difference $L_N = O_N - i_N$ has target $\mathcal{P}_{\leq N-1}$ and is exactly*

$$L_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) = \sum_{|\gamma| < N} \left(\sum_{\substack{|\alpha|=N \\ \gamma_j \leq \alpha_j \ (1 \leq j \leq k)}} c_\alpha \prod_{j=1}^k p_{\alpha_j, \gamma_j} \right) s^\gamma. \quad (\text{KPC.6})$$

The image of O_N is the orthogonal complement of $\mathcal{P}_{\leq N-1}$ in $\mathcal{P}_{\leq N}$ for the original product measure, and its basis in (KPC.5) has Gram $\text{diag}(\kappa_\alpha)_{|\alpha|=N}$. Define the two full residue maps on the same $\mathcal{H}_N$ by

$$A_N^{\text{hom}} = J i_N, \quad A_N^{\text{orth}} = J O_N.$$

Then, with $J_{<N} = J|_{\mathcal{P}_{\leq N-1}}$,

$$A_N^{\text{orth}} = A_N^{\text{hom}} + J_{<N} L_N. \quad (\text{KPC.7})$$

In particular the original orthogonal-shell columns of A_N^{orth} are exactly a_α .

Proof. Expand each factor in p_α . Its coefficient of s^γ is $\prod_j p_{\alpha_j, \gamma_j}$ when $\gamma_j \leq \alpha_j$ for all j , and is zero otherwise. If $|\alpha| = |\gamma| = N$, these inequalities force $\gamma = \alpha$, whose coefficient is one. This proves both section identities and the complete expression (KPC.6). For a monomial s^γ of degree less than $N = |\alpha|$, at least one $\gamma_j < \alpha_j$. In the product integral for $\langle s^\gamma, p_\alpha \rangle$, that factor is zero by one-variable orthogonality. All product integrals are justified by their finite moments and Cauchy-Schwarz. Distinct product polynomials are mutually orthogonal, and their norms multiply to κ_α . Their leading monomials make them independent. Their number is $\dim \mathcal{P}_{\leq N} - \dim \mathcal{P}_{\leq N-1}$, so they exhaust the indicated orthogonal complement. Applying the unchanged linear map J to $O_N = i_N + L_N$ proves (KPC.7). This application includes the fixed multiplication by $v_h^{\otimes k}$. For $N = 0$ the sum in (KPC.6) is empty and $L_0 = 0$, so the same proof includes the initial shell. $\square$

The associated-graded target is explicit as well. Put $F_N E^{(k)} = J(\mathcal{P}_{\leq N})$ and $F_{-1} E^{(k)} = 0$. Projection onto $F_N E^{(k)} / F_{N-1} E^{(k)}$, composed with either map in (KPC.7), gives the same surjection

$$\overline{A}_N : \mathcal{H}_N \twoheadrightarrow F_N E^{(k)} / F_{N-1} E^{(k)}. \quad (\text{KPC.8})$$

Indeed their difference lands in $F_{N-1} E^{(k)}$, and $\mathcal{P}_{\leq N} = i_N \mathcal{H}_N + \mathcal{P}_{\leq N-1}$ proves surjectivity. If $Z_N = \ker \overline{A}_N$ and $K_N^a = \ker A_N^a$ for $a \in \{\text{hom}, \text{orth}\}$, then $K_N^a \subseteq Z_N$ and the exact sequence is

$$0 \longrightarrow Z_N / K_N^a \longrightarrow \mathcal{H}_N / K_N^a \longrightarrow \mathcal{H}_N / Z_N \longrightarrow 0. \quad (\text{KPC.9})$$

The middle space is identified with $\text{im } A_N^a$ by $[c] \mapsto A_N^a c$, the last space with the target of (KPC.8), and the first space with $\text{im } A_N^a \cap F_{N-1} E^{(k)}$. These identifications follow directly from their kernels and are bijective: for the first image, membership in F_{N-1} is exactly the condition $c \in Z_N$. Thus equality of the graded maps retains, rather than removes, the two potentially different lower-filtration image spaces. In particular, once $N-1 \geq k(d-1)$, complete monomial remainders show $F_{N-1} E^{(k)} = E^{(k)}$, so (KPC.8) has zero target; the full maps (KPC.7) still have their stated images and kernels.

17.2 The full correspondence between the two image spaces

Theorem 17.2 (Common quotient and exact graph correspondence). *Write $I_h = \text{im } A_N^{\text{hom}}$, $I_o = \text{im } A_N^{\text{orth}}$, $K_h = \ker A_N^{\text{hom}}$, and $K_o = \ker A_N^{\text{orth}}$. Define*

$$Q_N = \mathcal{H}_N / (K_h + K_o), \quad \Gamma_N = \{(A_N^{\text{hom}} c, A_N^{\text{orth}} c) : c \in \mathcal{H}_N\}. \quad (\text{KPC.10})$$

The maps $q_h : I_h \rightarrow Q_N$ and $q_o : I_o \rightarrow Q_N$, given respectively by $A_N^{\text{hom}} c \mapsto [c]$ and $A_N^{\text{orth}} c \mapsto [c]$, are well-defined surjections, with exact sequences

$$\begin{aligned} 0 \longrightarrow A_N^{\text{hom}}(K_o) \longrightarrow I_h \xrightarrow{q_h} Q_N \longrightarrow 0, \\ 0 \longrightarrow A_N^{\text{orth}}(K_h) \longrightarrow I_o \xrightarrow{q_o} Q_N \longrightarrow 0. \end{aligned} \quad (\text{KPC.11})$$

One has the exact vector-space identifications

$$\boxed{\Gamma_N = I_h \times_{Q_N} I_o \cong \mathcal{H}_N / (K_h \cap K_o)}. \quad (\text{KPC.12})$$

The projection $\Gamma_N \rightarrow I_h$ has kernel $\{0\} \oplus A_N^{\text{orth}}(K_h)$; the projection onto I_o has kernel $A_N^{\text{hom}}(K_o) \oplus \{0\}$.

Proof. Changing a representative of an element of I_h changes c by K_h , hence leaves its class modulo $K_h + K_o$ unchanged. The same argument applies to I_o . The formulas give surjectivity. If $q_h(A_N^{\text{hom}} c) = 0$, write $c = c_h + c_o$ with $c_h \in K_h, c_o \in K_o$. Then $A_N^{\text{hom}} c = A_N^{\text{hom}} c_o$. Conversely every such vector maps to zero. This proves the first kernel formula in (KPC.11); interchanging the two maps proves the second.

The paired map from $\mathcal{H}_N$ onto Γ_N has kernel exactly $K_h \cap K_o$, proving the last identification in (KPC.12). Every paired image belongs to the fibre product. Conversely, suppose $(A_N^{\text{hom}} c, A_N^{\text{orth}} b)$ belongs to the fibre product. Then $c - b = c_h + c_o$ for some $c_h \in K_h, c_o \in K_o$. Put $e = c - c_h = b + c_o$. It has the two desired images, so the pair belongs to Γ_N . This proves equality with the full fibre product, not merely an inclusion. Taking the kernel of each coordinate projection gives the asserted remaining formulas. $\square$

The quotient in (KPC.10) also has its precise universal property. If linear maps $f_h : I_h \rightarrow Z$ and $f_o : I_o \rightarrow Z$ satisfy $f_h A_N^{\text{hom}} = f_o A_N^{\text{orth}}$, their common composite on $\mathcal{H}_N$ vanishes on both K_h and K_o , hence on their sum. It therefore factors uniquely through Q_N , and that factor gives both original maps after the surjections in (KPC.11). Likewise, a linear map $T : I_h \rightarrow I_o$ satisfying $T A_N^{\text{hom}} = A_N^{\text{orth}}$ exists exactly when $K_h \subseteq K_o$. Necessity follows by evaluating on K_h ; under that inclusion the formula $T(A_N^{\text{hom}} c) = A_N^{\text{orth}} c$ is well-defined and proves sufficiency. It is an isomorphism exactly

when $K_h = K_o$. Thus (KPC.11)–(KPC.12) give the full typed correspondence even when no such isomorphism exists.

There is also an explicit polynomial deformation of the sections. For a complex parameter z , define, without division by z ,

$$p_j^{(z)}(s) = \sum_{r=0}^j p_{j,r} z^{j-r} s^r, \quad O_{N,z}(c) = \sum_{|\alpha|=N} c_\alpha \prod_j p_{\alpha_j}^{(z)}(s_j). \quad (\text{KPC.13})$$

For $|\gamma| < N$, its coefficient of s^γ is exactly the coefficient in (KPC.6) multiplied by $z^{N-|\gamma|}$; at degree N it is c_γ . Hence $O_{N,0} = i_N$, $O_{N,1} = O_N$, and $\ell_N O_{N,z} = I$ for every z . The maps $JO_{N,z}$ are matrices polynomial in the parameter, and (KPC.8) is unchanged throughout. A rank condition $\text{rank } JO_{N,z} \leq r$ is exactly the vanishing of every $(r+1)$ -minor. No constancy of their rank follows from this deformation. In particular (KPC.13) retains every original coefficient at $z = 1$; it does not replace the original measure or its orthogonal minimum.

17.3 The algebra generated by the original coordinate ratio

Now take $k = 2$ and retain $h(0) \neq 0$. Every original nontrivial zero in the arithmetic packet has $0 < \text{Re } \rho < 1$, so this domain includes that packet. Work in the original algebra

$$E^{(2)} = \mathbb{C}[s_1, s_2]/(h(s_1), h(s_2)), \quad \zeta = s_1 s_2^{-1}. \quad (\text{KPC.14})$$

The inverse is defined because s and $h(s)$ are coprime; the Euclidean identity for these polynomials supplies an inverse modulo h . All ordered pairs of centres and their multiplicities are retained.

Theorem 17.3 (Complete ratio algebra and homogeneous-shell rank). *For every distinct ratio $r = \rho/\sigma$, put*

$$M_r = \max_{\rho/\sigma=r} (m_\rho + m_\sigma - 1), \quad b = \sum_r M_r, \quad \mu_\zeta(Z) = \prod_r (Z - r)^{M_r}. \quad (\text{KPC.15})$$

Then evaluation at ζ has exactly kernel (μ_ζ) , and gives the unital algebra embedding

$$\mathbb{C}[Z]/(\mu_\zeta) \hookrightarrow E^{(2)}, \quad [P] \mapsto P(\zeta). \quad (\text{KPC.16})$$

Its image is the subalgebra $\mathbb{C}[\zeta]$. For every $N \geq 0$, the full arithmetic homogeneous-shell map is

$$\sum_{j=0}^N c_j s_1^j s_2^{N-j} \mapsto v_h^{\otimes 2} s_2^N \sum_{j=0}^N c_j \zeta^j. \quad (\text{KPC.17})$$

Its kernel consists exactly of the coefficient polynomials $\sum_{j=0}^N c_j Z^j$ divisible by μ_ζ , and

$$\boxed{\text{rank } A_N^{\text{hom}} = \min\{N+1, b\}.} \quad (\text{KPC.18})$$

The arithmetic unit and s_2^N in (KPC.17) are retained invertible multiplication maps. Formula (KPC.18) does not replace the orthogonal-shell map (KPC.7).

Proof. The factors $(s - \rho)^{m_\rho}$ of h are pairwise coprime. For each factor, a Euclidean identity with the product of the others gives a polynomial equal to one modulo that factor and zero modulo all others. Their residue classes are mutually orthogonal idempotents and sum to one: each of these assertions holds modulo every factor, hence modulo their product. They give the full Chinese-remainder isomorphism, with inverse obtained by multiplying each component representative by its idempotent and adding. Tensoring these explicit maps gives

$$E^{(2)} \cong \prod_{(\rho, \sigma)} \mathbb{C}[x, y]/(x^{m_\rho}, y^{m_\sigma}), \quad x = s_1 - \rho, \quad y = s_2 - \sigma. \quad (\text{KPC.19})$$

In each component the basis is the full set $x^a y^c$, $0 \leq a < m_\rho, 0 \leq c < m_\sigma$. The inverse of $\sigma + y$ is explicitly $\sum_{j=0}^{m_\sigma-1} (-1)^j y^j / \sigma^{j+1}$, as multiplication and cancellation of consecutive powers verify.

Write $m = m_\rho, n = m_\sigma$. In this component the ratio retains the exact identity

$$\zeta - \frac{\rho}{\sigma} = \frac{\sigma x - \rho y}{\sigma(\sigma + y)}. \quad (\text{KPC.20})$$

Every monomial of total degree $m + n - 1$ vanishes, since one exponent must reach m or n . Thus the $(m + n - 1)$ -st power of (KPC.20) is zero. Its preceding power has the exact value

$$\left(\zeta - \frac{\rho}{\sigma}\right)^{m+n-2} = \binom{m+n-2}{m-1} \frac{\sigma^{m-1}(-\rho)^{n-1}}{\sigma^{2(m+n-2)}} x^{m-1} y^{n-1} \neq 0. \quad (\text{KPC.21})$$

Indeed only the displayed monomial survives in the numerator power; in the inverse-denominator power only its constant coefficient can multiply that monomial without making it zero. The displayed coefficient is nonzero over $\mathbb{C}$ because $\rho, \sigma \neq 0$. For $m = n = 1$ the preceding power is the zeroth power one, and the first power is zero, so this case is included. The local nilpotent index is therefore exactly $m + n - 1$.

For any nonzero polynomial $P(Z)$, factor at the local ratio r as $P(Z) = (Z - r)^e B(Z)$ with $B(r) \neq 0$. The element $B(\zeta)$ is invertible: after division by its nonzero constant $B(r)$, it is one plus a nilpotent polynomial in $\zeta - r$, whose finite geometric series is an inverse. Hence $P(\zeta) = 0$ in that component exactly when $e \geq m + n - 1$. Across all ordered components, the condition at each coincident ratio is its maximum exponent, and conditions at distinct ratios are coprime. This proves that the global evaluation kernel is precisely (μ_ζ) , including the zero polynomial. It proves (KPC.16) and its stated image.

Each homogeneous monomial satisfies $s_1^j s_2^{N-j} = s_2^N \zeta^j$ inside the original algebra. Applying the original arithmetic unit proves (KPC.17). Both multiplication factors there are invertible, so its kernel is exactly the asserted restricted polynomial kernel. For $N < b$, no nonzero polynomial of degree at most N is divisible by the monic μ_ζ . For $N \geq b$, all such multiples are $\mu_\zeta \mathbb{C}[Z]_{\leq N-b}$, which has dimension $N - b + 1$, since multiplication by the nonzero polynomial is injective. Subtracting this dimension from $N + 1$ gives (KPC.18) in both cases. $\square$

In the retained monomial tensor coordinates, set $S_1 = A_h \otimes I, S_2 = I \otimes A_h, c = [1] \otimes [1]$, and $Z_{\text{mat}} = S_1 S_2^{-1}$. Then (KPC.17) is exactly

$$(c_0, \dots, c_N) \mapsto U_{v_h^{\otimes 2}} S_2^N \sum_{j=0}^N c_j Z_{\text{mat}}^j c. \quad (\text{KPC.22})$$

This retains each complete Taylor block under (KPC.19) rather than replacing either coordinate by its eigenvalue. Multiplication by $v_h^{\otimes 2} s_2^N$ is a linear isomorphism from $\mathbb{C}[\zeta]$ to its displayed image, with inverse multiplication by $s_2^{-N} (v_h^{\otimes 2})^{-1}$ on that image. The image after this multiplication is not asserted to be a unital subalgebra.

The original summed generator also has an exact map between adjacent shells. Write $A^{(2)} = M_{s_1+s_2}$ on the retained algebra and $U = v_h^{\otimes 2}$. Then

$$A^{(2)}(U s_2^N f(\zeta)) = U s_2^{N+1} (1 + \zeta) f(\zeta). \quad (\text{KPC.22a})$$

This follows from $s_1 + s_2 = s_2(1 + \zeta)$ and commutativity of the original multiplication maps. Multiplication by $1 + Z$ sends degree at most N to degree at most $N + 1$, and sends a multiple of μ_ζ to another such multiple. Therefore (KPC.22a) is well-defined on the complete source quotients in (KPC.17). It is an adjacent-shell action; it does not assert that $\mathbb{C}[\zeta]$ alone is invariant under $A^{(2)}$. For example, if the packet has two distinct centres ρ, σ , the diagonal pairs (ρ, ρ) and (σ, σ) both have ratio one, while the constant terms of their summed generators are 2ρ and 2σ . A polynomial in the ratio has the same constant term at both pairs, so it cannot equal that summed coordinate. Formula (KPC.22a) retains the missing second-coordinate multiplication explicitly.

Theorem 17.4 (Exact algebra-retraction criterion). *In the local algebra $R_{\rho,\sigma} = \mathbb{C}[x, y]/(x^m, y^n)$, with $\rho, \sigma \neq 0$, the ratio-subalgebra inclusion $\mathbb{C}[\zeta] \hookrightarrow R_{\rho,\sigma}$ admits a unital complex-algebra retraction exactly when $\min(m, n) = 1$. In the full original tensor algebra $E_h^{\otimes 2}$, the inclusion (KPC.16) admits a unital complex-algebra retraction exactly when h is squarefree.*

Proof. First suppose $m = 1$, so $x = 0$ and $\zeta = \rho/(\sigma + y)$. This element is a unit, and $y = \rho\zeta^{-1} - \sigma$. Its inverse belongs to the ratio subalgebra: $\zeta = r + w$, with $r = \rho/\sigma \neq 0$ and $w^n = 0$, has inverse $\sum_{j=0}^{n-1} (-1)^j w^j / r^{j+1}$. Thus the ratio subalgebra contains y and is the whole local algebra. If $n = 1$, then $y = 0$ and $x = \sigma\zeta - \rho$, giving the same conclusion. In both cases the inclusion is an isomorphism and its inverse is the asserted retraction.

Now suppose $m, n \geq 2$, and put $L = m + n - 1$. The exact local index proved above identifies the ratio subalgebra with $B = \mathbb{C}[t]/(t^L)$, by $t \mapsto \zeta - \rho/\sigma$. In any putative retraction $f : R_{\rho,\sigma} \rightarrow B$, the images of x and y have zero constant term because their powers vanish. If the coefficient of t in $f(x)$ were a nonzero number a , the coefficient of t^m in $f(x)^m$ would be $a^m \neq 0$. Since $m < L$, that term survives in B , contradicting $x^m = 0$. The same argument using $n < L$ shows that $f(y)$ also has zero linear coefficient. The exact expression (KPC.20) then makes $f(\zeta - \rho/\sigma)$ have zero linear coefficient: its numerator lies in (t^2) and its denominator is a unit. A retraction would send that element to t . This contradiction proves the local converse.

If h is squarefree, (KPC.19) identifies the full tensor algebra with $\mathbb{C}^{\mathbb{Z} \times \mathbb{Z}}$. The ratio subalgebra is the algebra of functions constant on every fibre of $(\rho, \sigma) \mapsto \rho/\sigma$. Indeed (KPC.15) has exponent one at each ratio, and ordinary polynomial interpolation gives every assignment of values on the distinct ratios. Choose one ordered pair in each such fibre. Restriction of a function to those chosen pairs, then assigning that value to the entire corresponding fibre, is a unital algebra map back to this subalgebra. It is the identity on the subalgebra, proving existence of a retraction.

Conversely, let $M = \max_{\rho} m_{\rho} \geq 2$. The ratio-one component in (KPC.15) is

$$B_1 = \mathbb{C}[t]/(t^{2M-1}), \quad \zeta \mapsto 1 + t. \quad (\text{KPC.22b})$$

This is because $\rho/\sigma = 1$ for nonzero centres exactly when $\rho = \sigma$, and the maximum of $2m_{\rho} - 1$ is $2M - 1$. Compose a proposed global retraction with projection onto B_1 . A unital map from the product (KPC.19) into B_1 selects exactly one local factor. To justify that statement, an idempotent of B_1 has constant term zero or one. If it is zero, the idempotent lies in the nilpotent ideal (t) , and an idempotent which is nilpotent is zero, since all its positive powers equal itself. If the constant term is one, apply this argument to one minus the idempotent. Hence the only idempotents are zero and one. The mutually orthogonal source-factor idempotents sum to one, so exactly one maps to one and all others map to zero, proving the selection statement.

Suppose the selected factor is indexed by (ρ, σ) . Images of its nilpotent coordinates have zero constant term, so the constant term of its ratio image is ρ/σ . Equation (KPC.22b) forces this ratio to be one; thus $\rho = \sigma$, with common multiplicity $m = m_{\rho} \leq M < 2M - 1$. The relation $x^m = 0$ again forces the coefficient of t in the image of x to vanish, since a nonzero coefficient would produce a nonzero t^m term in the target. The same holds for y . Their ratio minus one therefore has zero linear coefficient by (KPC.20), contradicting its required image t in (KPC.22b). This proves the global converse while retaining all source-factor idempotents and the original ratio embedding. $\square$

17.4 A complete finite witness for unequal shell ranks

The following positive polynomial model tests the proposed passage from homogeneous rank to the original orthogonal rank. It is not a claim that $1/2$ is an arithmetic zero. Use the specified measure

$$\nu_{\text{ex}} = \delta_{1/2-i} + \delta_{1/2} + \delta_{1/2+i} \quad \text{on the variable } s, \quad h_{\text{ex}}(s) = s - \frac{1}{2}, \quad v_{\text{ex}} = 1. \quad (\text{KPC.23})$$

All three masses are exactly one. Through degree two the monic orthogonal polynomials and norms are

$$\begin{array}{c|c|c} j & p_j(s) & \kappa_j \\ \hline 0 & 1 & 3 \\ 1 & s - \frac{1}{2} & 2 \\ 2 & (s - \frac{1}{2})^2 + \frac{2}{3} & \frac{2}{3}. \end{array} \quad (\text{KPC.24})$$

To verify the complete Gram, their three value columns are

$$(1, 1, 1)^T, \quad (-i, 0, i)^T, \quad (-1/3, 2/3, -1/3)^T.$$

Their pairwise Hermitian inner products are zero, and their squared norms are respectively 3, $1 + 0 + 1 = 2$, and $1/9 + 4/9 + 1/9 = 2/3$. Thus the moment form through degree two is positive definite, as required for the source and next layer used below.

Take $k = 2$, $N = 1$. Here $d = 1$ and the threshold $N \geq k(d - 1) = 0$ holds. The original tensor residue algebra is $E_{\text{ex}}^{(2)} = \mathbb{C}$, with $s_1 = s_2 = 1/2$, and its full jet map is evaluation at that ordered pair. For $c = c_{10}s_1 + c_{01}s_2$, the two section and image maps are exactly

$$\begin{aligned} i_1(c) &= c_{10}s_1 + c_{01}s_2, \\ O_1(c) &= c_{10}(s_1 - \frac{1}{2}) + c_{01}(s_2 - \frac{1}{2}), \\ L_1(c) &= -\frac{1}{2}(c_{10} + c_{01}), \\ A_1^{\text{hom}}(c) &= \frac{1}{2}(c_{10} + c_{01}), \quad A_1^{\text{orth}}(c) = 0. \end{aligned} \quad (\text{KPC.25})$$

Thus the lower coefficient in (KPC.7) cancels the entire homogeneous image. The ranks are one and zero. The kernels are $K_h = \{(c_{10}, c_{01}) : c_{10} + c_{01} = 0\}$ and $K_o = \mathbb{C}^2$, so the common quotient in (KPC.10) is zero and the full correspondence is $\mathbb{C} \oplus \{0\}$. The ratio is $\zeta = 1$, its minimal polynomial is $Z - 1$, and (KPC.18) gives the correct homogeneous rank $\min(2, 1) = 1$. Equation (KPC.25) proves explicitly why it gives no equality with the orthogonal rank.

The actual finite minimum and next layer in this model can also be retained. In the source with product measure $\nu_{\text{ex}}^{\otimes 2}$, the degree-at-most-one orthogonal basis is p_0p_0, p_1p_0, p_0p_1 , with norms 9, 6, 6 and jets 1, 0, 0. Hence the full interpolation kernel is $K_1 = 1/9$, the metric is $G_1 = 9$, and its minimum is the constant polynomial $R_1u = u$. Multiplication by the original sum $s_1 + s_2$ acts on the packet by $A = 1$. Consequently

$$B_1u = (s_1 + s_2 - 1)u \in \mathcal{L}_1, \quad C_1 = (I - P_{\mathcal{L}_1})B_1 = 0, \quad W_1 = A^*G_1 + G_1A - 2G_1 = 0. \quad (\text{KPC.26})$$

Indeed $\mathcal{L}_1$ is spanned by $s_1 - 1/2, s_2 - 1/2$, the complete degree-one relation kernel. Thus the derivative rank zero is verified directly without presupposing the shell-rank formula.

At the next shell, ordered as $(2, 0), (1, 1), (0, 2)$, the original jets and norm diagonal are

$$U = \begin{pmatrix} 2/3 & 0 & 2/3 \end{pmatrix}, \quad D_{\text{next}} = \text{diag}(2, 4, 2).$$

The actual layer vectors are $e_{20} = (s_1 - 1/2)^2$, $e_{11} = (s_1 - 1/2)(s_2 - 1/2)$, and $e_{02} = (s_2 - 1/2)^2$. Each has zero jet and is orthogonal to $\mathcal{L}_1$, either by its product moments or by the layer construction. Their Gram is exactly

$$H = D_{\text{next}} + U^*G_1U = \begin{pmatrix} 6 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 6 \end{pmatrix}. \quad (\text{KPC.27})$$

For example the diagonal squared norm of e_{20} is $(1 + 0 + 1)(1 + 1 + 1) = 6$, and its pairing with e_{02} is $(-1 + 0 - 1)^2 = 4$; the odd cross terms vanish. The eigenvalues of the outer two-coordinate block are 10, 2, and the middle eigenvalue is 4, proving positivity. The next full kernel and metric therefore are

$$K_2 = \frac{1}{9} + \frac{(2/3)^2}{2} + \frac{(2/3)^2}{2} = \frac{5}{9}, \quad G_2 = \frac{9}{5}. \quad (\text{KPC.28})$$

The representative itself is the explicit polynomial

$$R_2 u = \left[1 + \frac{3}{5} \left((s_1 - \frac{1}{2})^2 + (s_2 - \frac{1}{2})^2 \right) \right] u. \quad (\text{KPC.28a})$$

For a direct verification, it has jet u . Its values for u equal to one are $-1/5$ at the four product nodes where both coordinates differ from $1/2$, $2/5$ at the four nodes where exactly one differs, and one at the central node. Its squared norm is therefore $4/25 + 16/25 + 1 = 9/5$. It is orthogonal to the five relation basis vectors $s_1 - 1/2, s_2 - 1/2, e_{20}, e_{11}, e_{02}$. The first two and e_{11} have an odd coordinate, so their pairings vanish by the explicit sign symmetry of the nine-node measure. For e_{20} , only the two noncentral first-coordinate values contribute, and for each of them the sum of the three values of $R_2(1)$ is $-1/5 + 2/5 - 1/5 = 0$. Its pairing is thus zero; the same computation with the coordinates interchanged treats e_{02} . These five independent zero-jet polynomials span the kernel in the six-dimensional degree-at-most-two source, whose jet map is onto the one-dimensional target. Every other feasible polynomial differs by a vector in this kernel. Pythagoras and positive definiteness therefore prove the unique-minimum assertion directly, including its actual norm. Thus the nonzero original relation layer and its positive metric loss are retained alongside the zero derivative layer. This complete finite calculation rejects only the proposed equality between homogeneous and orthogonal-shell ranks. The original arithmetic shell continues to be given by (KPC.4)–(KPC.7), and the exact graph and quotient maps relating its two presentations are (KPC.9)–(KPC.12).

Every linear map $T : V \rightarrow W$ in these comparisons has a support-preserving Split-Zero lift on the following specified carriers at a fixed support λ :

$$\widehat{V}_\lambda = \{\tau_V\} \sqcup (\{\lambda\} \times V), \quad \widehat{W}_\lambda = \{\tau_W\} \sqcup (\{\lambda\} \times W).$$

Addition has external τ as identity and uses vector addition inside the supported fibre. The scalar action is $r^\bullet(\lambda, v) = (\lambda, rv)$, while the external scalar zero sends every input to external absence and every scalar sends external absence to external absence. The lift maps τ_V to τ_W and (λ, v) to (λ, Tv) . Its addition identity follows from $T(v + w) = Tv + Tw$; its supported-scalar identity follows from $T(rv) = rTv$, including $r = 0$. The external cases follow from the definitions. Thus every displayed kernel vector maps to the represented zero at λ , and the graph, quotient and lower-degree maps retain this support even when the two ranks differ.

18 The original frontier energy, determinant growth, and the entire border fibre

The symmetric-frontier delivery of 12 September computes coordinates on the actual next relation layer. We retain these coordinates and prove their exact relation to the accumulated source energy of Section 15. The calculation also identifies the entire zero fibre of the delivery's smaller border operator, including its nilpotent extension.

18.1 Original moment coordinates and the relation graph

Keep a nonempty full-order packet h , its degree $d, g = 2\xi, E = \mathbb{C}[s]/h, v = j_h(g/h)$, and $A = M_s$. The positive measure remains

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}.$$

Let p_n be the monic orthogonal polynomial obtained from $1, s, \dots, s^n$ without dividing by its norm, and let $\omega_n = \|p_n\|_{\nu_h}^2 > 0$. The original generator has the exact recurrence

$$sp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad a_n = \omega_n / \omega_{n-1} \ (n \geq 1), \quad a_0 = 0, \quad b_n + \bar{b}_n = 1. \quad (\text{FE.1})$$

Here $p_{-1} = 0$. To verify the coefficients, multiplication by s has adjoint multiplication by $1 - s$ on the original line. Its pairing against degrees below $n - 1$ vanishes, the leading coefficient is one, and the pairing against p_{n-1} is $-\omega_n$. Its diagonal is $\int s|p_n|^2 d\nu_h/\omega_n$, which proves the last identity and retains its possibly nonzero imaginary part.

Fix $k \geq 1$ and $M \geq k(d - 1)$, and use $A_k = \sum_j A_j$ on the unchanged $E_k = E^{\otimes k}$. For $\alpha \in \mathbb{N}^k$, put

$$p_\alpha = \prod_j p_{\alpha_j}(s_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = v^{\otimes k}[p_\alpha] \in E_k.$$

All remainder coordinates, including every nilpotent jet and the entire local arithmetic unit, are retained in z_α . The original source map $\mathcal{T}^{(k)}$ is the map in (CT.11)–(CT.13). Its orthogonal polynomial images have norms ω_α . The constrained minimum gives exactly

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}, \quad R_M u = \sum_{|\alpha| \leq M} \mathcal{T}^{(k)} p_\alpha \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (\text{FE.2})$$

This is (CT.5) in the explicitly declared orthogonal source basis; no moment mass or polynomial norm has been changed.

Order the indices β of total degree $M + 1$, and put

$$r = \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\omega_\beta), \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \quad T_+ = [\mathcal{T}^{(k)} p_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k.$$

The two source maps have the same column ordering. Since the image of T_+ is orthogonal to the entire degree- M image,

$$b = T_+ - R_M F : \mathbb{C}^r \rightarrow \mathcal{E}_M, \quad H = b^* b = \Omega + F^* G_M F, \quad b^* R_M = -F^* G_M. \quad (\text{FE.3})$$

Here $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ is the original relation layer. The complete jets of b vanish; both of its terms are orthogonal to $\mathcal{B}_M$. Taking the highest orthogonal degree of bu recovers $T_+ u$, proving injectivity. For any boundary of degree $M + 1$, subtract the image under b of its highest coefficient. The difference lies in $\mathcal{B}_M$. This gives the inverse on the orthogonal layer and proves surjectivity. It also gives an original primitive: divide the numerator of each column of b in the fixed order by $h(s_1), \dots, h(s_k)$. The primitive in factor j has coefficient $(-1)^{j-1}$; its tensor differential contributes the same sign, giving the original positive boundary.

Define the second frontier matrix by

$$E_+ = [w_\beta], \quad w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}. \quad (\text{FE.4})$$

The notation E_+ denotes a matrix $\mathbb{C}^r \rightarrow E_k$. From (FE.2), the highest coefficient of $D^{(k)} R_M - R_M A_k$ is $\Omega^{-1} E_+^* G_M$. Projection onto the graph (FE.3) therefore proves, with the original types and signs,

$$Y_M = -b H^{-1} F^* G_M, \quad C_M = b \Omega^{-1} E_+^* G_M, \quad \Delta_M = Y_M^* Y_M = G_M F H^{-1} F^* G_M. \quad (\text{FE.5})$$

The notation C_M is the original derivative-layer map in (CT.15), not the inverse interpolation matrix $\mathcal{C}_M$.

Applying (FE.1) to (FE.2) cancels every internal neighbouring pair, because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonals give exactly $k C_M$. The remaining pairs have top degree $M + 1$, proving

$$\mathcal{Z}_M = A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{FE.6})$$

Alternatively (FE.5) inserted into $W_M = -(Y_M^* C_M + C_M^* Y_M)$ gives exactly (FE.6). Thus both the polynomial cancellation and its actual boundary primitive yield the same original form.

18.2 Exact relative loss and the source derivative energy

For the next formulas only, use the displayed finite isometries from the specified metric coordinate spaces and define

$$U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}, \quad P = U U^*, \quad K_U = U^* U. \quad (\text{FE.7})$$

The original F, E_+, Ω, G_M are still given by (FE.1)–(FE.5), and can be recovered by the inverse coordinate maps in (FE.7). In particular no polynomial is assigned norm one and no arithmetic coefficient is suppressed.

Theorem 18.1 (Frontier energy and every relative Gram eigenvalue). *In these explicitly related coordinates,*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = U V^* + V U^*, \\ T_M &= G_M^{-1/2} \Delta_M G_M^{-1/2} = U (I + K_U)^{-1} U^* = P (I + P)^{-1}, \\ G_M^{-1/2} G_{M+1} G_M^{-1/2} &= (I + P)^{-1}. \end{aligned} \quad (\text{FE.8})$$

If $\lambda_1, \dots, \lambda_q > 0$ are all positive eigenvalues of P , including multiplicity, then

$$\begin{aligned} q &= \text{rank } F = \text{rank } Y_M = \text{rank } \Delta_M, \\ \tau_M &= \text{Tr}(G_M^{-1} \Delta_M) = \sum_{i=1}^q \frac{\lambda_i}{1 + \lambda_i}, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M} = \prod_{i=1}^q (1 + \lambda_i)^{-1}, \\ \chi_M &= \text{Tr}(G_M^{-1} C_M^* C_M) = \|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{FE.9})$$

The empty product equals one if $q = 0$. In addition $\text{rank } C_M = \text{rank } E_+$.

Proof. Equation (FE.6) proves the first line of (FE.8). The identity $H = \Omega^{1/2}(I + U^* U) \Omega^{1/2}$ substituted into (FE.5) proves the second line's first expression. The equality $(I + U U^*) U = U (I + U^* U)$, multiplied by the indicated invertible factors, identifies it with $P (I + P)^{-1}$. Subtract it from I to get the third line. The positive matrices $I + K_U$ and $I + P$ are invertible directly from their quadratic forms.

The graph map b is injective, and H, Ω, G_M are invertible, so (FE.5) proves both asserted ranks. Equivalently, the formula for T_M identifies its positive eigenvalues with $\lambda_i / (1 + \lambda_i)$, retaining zeros on $\ker P$. The trace and determinant formulas now follow from the finite spectral decomposition in these same coordinates. Finally

$$C_M G_M^{-1/2} = b \Omega^{-1/2} V^*, \quad (b \Omega^{-1/2})^* (b \Omega^{-1/2}) = I + U^* U.$$

Its squared Hilbert–Schmidt norm is $\text{Tr}(V(I + U^* U)V^*)$, which expands to the last line of (FE.9). This is the energy of the original source-layer map for its inherited $L^2(d^k x)$ inner product. $\square$

The exact remainder in the accumulated determinant calculation is therefore also a frontier formula:

$$-\log \pi_M - \tau_M = \sum_{i=1}^q \left(\log(1 + \lambda_i) - \frac{\lambda_i}{1 + \lambda_i} \right) = \sum_{i=1}^q \int_0^{\lambda_i} \frac{x}{(1+x)^2} dx. \quad (\text{FE.10})$$

Differentiate each displayed scalar expression and evaluate at zero to prove the equality, including every eigenvalue and its sign. Thus (JD.14) uses the exact original derivative cost $\|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2$, and (JD.16) is precisely the pair of identities

$$\sum_{M \geq a} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M \geq a} \sum_i \frac{\lambda_{M,i}}{1 + \lambda_{M,i}} = \infty. \quad (\text{FE.11})$$

Both use the already proved actual boundary density; the second retains its full inverse-interpolation cost.

18.3 The entire small border operator, including its zero extension

Let $n = d^k$, and define

$$B = [U, V] : \mathbb{C}^{2r} \rightarrow \mathbb{C}^n, \quad K = B^*B, \quad J_0 = \begin{pmatrix} 0 & I_r \\ I_r & 0 \end{pmatrix}, \quad A_0 = J_0 K, \quad H_0 = K^{1/2} J_0 K^{1/2}. \quad (\text{FE.12})$$

The symbol A_0 here is a finite border operator, not the arithmetic generator A or A_k . This distinction is accompanied by the exact maps below. The source's unscaled border operator is

$$J[F, E_+]^* G_M[F, E_+], \quad J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

With $D_\Omega = \text{diag}(\Omega^{1/2}, \Omega^{1/2})$, it is exactly $D_\Omega^{-1} A_0 D_\Omega$: substitution of (FE.7) gives

$$[F, E_+]^* G_M[F, E_+] = D_\Omega K D_\Omega.$$

This states the full coordinate similarity, including zero.

Put $N = \ker K = \ker B$ and $\mathcal{R} = \text{ran } K$. They are orthogonal complements since K is Hermitian, and $K^{1/2}$ restricts to an invertible positive map on $\mathcal{R}$. There is an exact sequence of spaces with their specified operators:

$$0 \longrightarrow (N, 0) \longrightarrow (\mathbb{C}^{2r}, A_0) \xrightarrow{K^{1/2}} (\mathcal{R}, H_0|_{\mathcal{R}}) \longrightarrow 0. \quad (\text{FE.13})$$

Indeed $K^{1/2} A_0 = H_0 K^{1/2}$, its kernel is N , and its image is $\mathcal{R}$. On N , $A_0 = 0$. The map

$$Q|_{\mathcal{R}} = B(K^{1/2}|_{\mathcal{R}})^{-1}, \quad Q|_N = 0 \quad (\text{FE.14})$$

is a partial isometry onto $\text{ran } B$: for $x, y \in \mathcal{R}$, substitution of $B^*B = K$ gives $\langle Qx, Qy \rangle = \langle x, y \rangle$. It satisfies $B = QK^{1/2}$ and

$$S_M = B J_0 B^* = Q H_0 Q^*, \quad S_M|_{(\text{ran } B)^\perp} = 0. \quad (\text{FE.15})$$

Thus the restriction of H_0 to $\mathcal{R}$ and that of S_M to $\text{ran } B$ are unitarily intertwined by an explicitly proved map; both remaining orthogonal complements and their zero actions are specified.

Theorem 18.2 (Exact zero fibre of the border comparison). *Let $Z_0 = \ker(H_0|_{\mathcal{R}})$ and $q_0 = \dim Z_0$. The complete generalized zero eigenspace of A_0 is*

$$\mathcal{N}_0 = (K^{1/2})^{-1}(Z_0), \quad A_0^2|_{\mathcal{N}_0} = 0. \quad (\text{FE.16})$$

Define $L = (K^{1/2}|_{\mathcal{R}})^{-1} : \mathcal{R} \rightarrow \mathcal{R}$. For its restriction to $z \in Z_0$, set

$$Lz = (K^{1/2}|_{\mathcal{R}})^{-1}z, \quad N_0 z = J_0 K^{1/2} z \in N. \quad (\text{FE.17})$$

The map $N_0 : Z_0 \rightarrow N$ is injective, and the exact vector space decomposition $\mathcal{N}_0 = N \oplus LZ_0$ gives

$$A_0(n + Lz) = N_0 z. \quad (\text{FE.18})$$

Consequently its zero Jordan data consist of exactly q_0 blocks of size two and $\dim N - q_0$ blocks of size one. Its nonzero Jordan blocks have size one and coincide, with full multiplicities, with the nonzero eigenvalues of S_M .

Proof. If $A_0^m x = 0$, (FE.13) gives $H_0^m K^{1/2} x = 0$. Hermitian spectral decomposition of $H_0|_{\mathcal{R}}$ identifies its generalized zero space with Z_0 , proving that x belongs to the preimage in (FE.16). Conversely, if $K^{1/2} x \in Z_0$, then $K^{1/2} A_0 x = 0$, so $A_0 x \in N$ and $A_0^2 x = 0$. This proves (FE.16) exactly.

For $z \in Z_0$, one has $K^{1/2}J_0K^{1/2}z = H_0z = 0$, proving $N_0z \in N$. Since J_0 is invertible and $K^{1/2}$ is injective on $\mathcal{R}$, N_0 is injective. Equation $K^{1/2}Lz = z$ gives the direct decomposition, and $A_0Lz = J_0KLz = J_0K^{1/2}z$ proves (FE.18). Choose a basis of Z_0 , use its injective image as part of a basis of N , and extend it in N . Each pair Lz, N_0z is a size-two chain, and the remaining basis vectors of N are size-one chains. This also proves $q_0 \leq \dim N$ without a rank assumption.

For a nonzero eigenvalue λ , take an eigenvector $z \in \mathcal{R}$ of H_0 . If $x = Lz$, then $n = A_0x - \lambda x \in N$, so $x + n/\lambda$ is an eigenvector of A_0 lifting z . The lift is unique because A_0 vanishes on N and $\lambda \neq 0$. This supplies every eigenvector of the Hermitian quotient and leaves no nontrivial Jordan chain at a nonzero eigenvalue: applying (FE.13) to such a chain would give one for the Hermitian quotient, and a residual chain contained in N is impossible at nonzero λ . Finally (FE.15) identifies those eigenvalues with the actual relative form. All claims include their algebraic multiplicities. $\square$

In particular,

$$\dim \ker S_M = n - \text{rank } B + q_0, \quad \dim \ker A_0 = 2r - \text{rank } B, \quad \text{mult}_0(A_0) = 2r - \text{rank } B + q_0. \quad (\text{FE.19})$$

The first equality is (FE.15), the second uses the invertibility of J_0 , and the third counts (FE.18). Thus replacing the source's border calculation by a Hermitian one has an exact kernel, quotient, complement and zero extension, all explicitly retained here. At any fixed retained support λ , each displayed linear map has its Split-Zero lift $(\lambda, x) \mapsto (\lambda, Tx)$, with external $\tau \mapsto \tau$. A size-two chain in (FE.18) is therefore the exact supported chain

$$(\lambda, Lz) \mapsto (\lambda, N_0z) \mapsto (\lambda, 0). \quad (\text{FE.19a})$$

The last value is the represented zero in that fibre. For $z \neq 0$, injectivity of N_0 proves that the intermediate value has nonzero amplitude; neither map sends it to external absence.

18.4 A quantified link between the frontier bound and determinant cost

Let $\lambda_M = \|UU^*\|_{\text{op}}$, and retain the original positive recurrence quantity

$$\Gamma_M = \max_{|\alpha|=M} \sum_{j=1}^k \{a_{\alpha_j+1} + (k-1)a_{\alpha_j}\} > 0. \quad (\text{FE.20})$$

The strict inequality follows from $a_{\alpha_j+1} > 0$. For clarity, its upper estimate uses an exact matrix comparison. Let $Z = [z_\alpha]_{|\alpha|=M}$, $\Omega_M = \text{diag}(\omega_\alpha)$, and let $L_{\alpha\beta} = a_{\beta_j}$ for $\alpha = \beta - e_j$, summed over those j . Then $E_+ = ZL$, and $\Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is the displayed sum in (FE.20): $\omega_\alpha a_{\alpha_j+1} / \omega_{\alpha+e_j} = 1$ for each outgoing edge, and the other predecessors contribute $\sum_{l \neq j} a_{\alpha_l}$. For a matrix with nonnegative entries, an eigenvector component of largest absolute value bounds the eigenvalue's absolute value by its largest row sum. The matrix is similar to the positive Hermitian matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$, so

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_M Z \Omega_M^{-1} Z^* \preceq \Gamma_M \mathcal{C}_M.$$

It follows that $VV^* \preceq \Gamma_M I$. For any vector x , Cauchy-Schwarz applied to (FE.8) gives $|x^* S_M x| \leq 2\|U^* x\| \|V^* x\|$. Thus

$$\epsilon_M \leq 2\sqrt{\Gamma_M \lambda_M}. \quad (\text{FE.21})$$

Every actual eigenline survives in the full tensor family. If $\delta_* = \max_{\rho \in Z} |\text{Re } \rho - 1/2|$, its repeated tensor gives $\epsilon_M \geq 2k\delta_*$. Combining this with (FE.9) and (FE.21) proves, at every finite degree,

$$\lambda_M \geq \frac{k^2 \delta_*^2}{\Gamma_M}, \quad \frac{\det G_M}{\det G_{M+1}} \geq 1 + \frac{k^2 \delta_*^2}{\Gamma_M}. \quad (\text{FE.22})$$

Indeed the determinant inverse is $\prod_i (1 + \lambda_i)$, which is at least $1 + \lambda_M$. For integers $k(d-1) \leq a < b$, multiplication, followed by the increasing logarithm, gives the explicit source constraint

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{k^2 \delta_*^2}{\Gamma_M} \right). \quad (\text{FE.23})$$

When $\delta_* = 0$, this remains a valid inequality with zero right side and no division by a zero spectral quantity. It is an exact necessary growth relation for the constructed arithmetic matrices, not an assumed upper estimate on that growth.

The original derivative energy supplies a second exact growth bound without the recurrence maximum. By (FE.8), $\|Y_M G_M^{-1/2}\|_{\text{op}}^2 = \lambda_M / (1 + \lambda_M)$. By (FE.9), $\|C_M G_M^{-1/2}\|_{\text{op}}^2 \leq \chi_M$. The original cross-pairing formula, the operator triangle inequality, and submultiplicativity therefore give

$$\epsilon_M^2 \leq 4 \frac{\lambda_M}{1 + \lambda_M} \chi_M. \quad (\text{FE.24})$$

If $\delta_* > 0$, combine it with the retained eigenline $\epsilon_M \geq 2k\delta_*$. The finite number $\lambda_M / (1 + \lambda_M)$ is strictly below one, so

$$\chi_M > k^2 \delta_*^2, \quad \lambda_M \geq \frac{k^2 \delta_*^2}{\chi_M - k^2 \delta_*^2}, \quad \log \frac{\det G_M}{\det G_{M+1}} \geq -\log \left(1 - \frac{k^2 \delta_*^2}{\chi_M} \right). \quad (\text{FE.25})$$

Every denominator and logarithm in (FE.25) has just been proved valid. The middle inequality follows by rearranging (FE.24); the last uses $\det G_M / \det G_{M+1} \geq 1 + \lambda_M$. Summing it over any $k(d-1) \leq a < b$ gives the corresponding finite accumulated determinant inequality. For $\delta_* = 0$, (FE.24) remains valid, and (FE.25) is left un-divided; no 0/0 value is assigned to an original zero-energy layer.

For the actual conjugation-and-reflection packets, the further exact original-layer calculation in Section 21 proves $\epsilon_M^2 \leq [\lambda_M / (1 + \lambda_M)] \zeta_M \leq [\lambda_M / (1 + \lambda_M)] \chi_M$. Here ζ_M is the operator-norm square of the same derivative map whose trace cost is χ_M . Its proof uses the two actual reflection parities in that layer. For $k = 1$, it proves the exact equality $\epsilon_M^2 = (1 - \pi_M) \chi_M$, retaining both zero frontier cases. Thus (FE.24)–(FE.25) retain their general hypotheses, while (QD.8)–(QD.15) give the stronger arithmetic conclusions and all their finite accumulated determinant bounds.

19 The signed symmetric summand and its complete spectral cost

This section proves the cochain, coordinate, and spectral comparisons for the symmetric summand in the arithmetic frontier delivery. The original packet is $E = \mathbb{C}[t]/h$, of dimension $d \geq 1$, and its operator is $A = M_t$. The tensor degree is $k \geq 1$. All orders in $h = \prod_\rho (t - \rho)^{m_\rho}$ remain present. The original joint representative, metric, and layer maps of (CT.12)–(CT.16) are denoted by R_M, G_M, Y_M, C_M , with $M \geq k(d-1)$. The source's symmetric construction is used through the maps proved below; its orbit sums keep their literal factorials.

19.1 The chain projector and its completed and finite targets

The completion in this subsection is specifically the completed projective tensor product $\widehat{\otimes}_\pi$ of the strong-Schwartz locally convex spaces. The global-retraction source, equations (16), (19), and (24), constructs continuous maps

$$\Lambda : \mathcal{B} \longrightarrow V, \quad \Lambda \Theta = 1, \quad s : Q \longrightarrow \mathcal{B}, \quad s[F] = (1 - \Theta \Lambda) F, \quad qs = 1. \quad (\text{SS.1})$$

Its specified cutoffs and moment projection occur in $\Lambda = P_V (1 - A_\alpha B_\beta)^{-1} Y_{\alpha, \beta}$. The convergence and continuity of this formula are proved in that source; no Hilbert-space replacement of these topologies

is used here. Put $C = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $P = sq$ as a cochain map with degree-zero part zero, and $H^1 = \Lambda$. Then in the two degrees respectively, $\Lambda\Theta = 1$ and $\Theta\Lambda = 1 - sq$ prove

$$dH + Hd = 1 - P. \quad (\text{SS.2})$$

For homogeneous inputs define the chain permutation by

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i||x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (\text{SS.3})$$

Here the degrees in the exponent belong to the original inputs. An adjacent interchange contributes $(-1)^{ab}$ for degrees a, b . Interchanging twice contributes $(-1)^{2ab} = 1$; disjoint interchanges commute; and the two three-interchange braid words both contribute $(-1)^{ab+ac+bc}$. Thus (SS.3) is a representation of S_k . On two adjacent factors, $d(x \otimes y) = dx \otimes y + (-1)^a x \otimes dy$. After interchange the coefficients of $dy \otimes x$ and $y \otimes dx$ are respectively $(-1)^{ab}$ and $(-1)^{ab+b}$, on either side of the cochain equation. This proves $dT_\pi = T_\pi d$, first for generators and then for every permutation. Each map is continuous and therefore extends to $C^{\widehat{\otimes}_\pi k}$.

The signed average and its complementary map are

$$S_k = \frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi, \quad 1 - S_k. \quad (\text{SS.4})$$

For each γ , exactly $k!$ pairs (π, σ) satisfy $\pi\sigma = \gamma$, and their signed coefficient is $\text{sgn}(\gamma)/(k!)^2$. Hence $S_k^2 = S_k$. In top degree every factor has degree one, so $T_\pi = \text{sgn}(\pi)P_\pi$ and S_k is the ordinary symmetric projection there. The unsigned chain average has top part $k!^{-1} \sum_{\pi} \text{sgn}(\pi)P_\pi$; this exact equation specifies its alternating target.

To verify the completed cohomology directly, extend the finite sum

$$H_k = \sum_{j=1}^k P^{\widehat{\otimes}_\pi(j-1)} \widehat{\otimes}_\pi H \widehat{\otimes}_\pi 1^{\widehat{\otimes}_\pi(k-j)}. \quad (\text{SS.5})$$

The degree -1 factor contributes the Koszul sign $(-1)^{\sum_{i < j} |x_i|}$. Expanding $dH_k + H_k d$, the terms in which the differential and homotopy occupy different slots cancel in pairs. The term in slot j is $P^{\widehat{\otimes}(j-1)} \otimes (1 - P) \otimes 1^{\widehat{\otimes}(k-j)}$ by (SS.2). Summing telescopes to $1 - P^{\widehat{\otimes} k}$. Continuity extends this identity from the dense algebraic tensor product. Replace H_k by the explicit equivariant homotopy

$$\overline{H}_k = \frac{1}{k!} \sum_{\pi} T_\pi H_k T_\pi^{-1}. \quad (\text{SS.6})$$

The operator $P^{\widehat{\otimes} k}$ commutes with every permutation, so (SS.6) has the same homotopy identity and commutes with S_k . The maps $q^{\widehat{\otimes}_\pi k}$ and $s^{\widehat{\otimes}_\pi k}$, restricted to the images of (SS.4), now give continuous inverse cohomology maps. Thus

$$H^r(\text{im } S_k) = \begin{cases} (Q^{\widehat{\otimes}_\pi k})^{S_k}, & r = k, \\ 0, & r \neq k. \end{cases} \quad (\text{SS.7})$$

The complementary cohomology is the image of $1 - \frac{1}{k!} \sum_{\pi} P_\pi$ on the same completed Q -tensor. The full cochain isomorphism and its inverse are $x \mapsto (S_k x, (1 - S_k)x)$ and $(u, v) \mapsto u + v$. Equations (SS.5)–(SS.7) prove the completion statement using the supplied continuous contraction; they require no exactness assertion for arbitrary completed tensor products.

Here is the finite-rank cochain map that supports the finite traces. The full packet jet $J_h : \mathcal{B} \rightarrow E$ annihilates ΘV , and descends to a continuous $j_h : Q \rightarrow E$. The original packet inclusion $\sigma_h : E \rightarrow Q$ satisfies $j_h \sigma_h = 1$, as $J_h R_M = 1$ and $q R_M = \sigma_h$ in one variable. Set $\iota_h = s \sigma_h : E \rightarrow \mathcal{B}$ and define

$$\Pi_h^0 = 0, \quad \Pi_h^1 = \iota_h J_h, \quad J_h \iota_h = 1. \quad (\text{SS.8})$$

This is a continuous finite-rank cochain projector: $\Pi_h^1 \Theta = 0$ and $(\Pi_h^1)^2 = \Pi_h^1$. It commutes with permutations after taking identical tensor powers. Consequently $\Pi_h^{\otimes k} S_k$ and $\Pi_h^{\otimes k} (1 - S_k)$ are the finite cochain projectors with ranks $\binom{k+d-1}{d-1}$ and $d^k - \binom{k+d-1}{d-1}$. The projector S_k on the full infinite complex itself has not been assigned either finite rank. The finite inclusion and retraction on cohomology are precisely $\sigma_h^{\otimes k}$ and $j_h^{\otimes k}$, restricted to the respective permutation summands. Their composite on the packet is the identity.

For $U(a) = e^{(\log a)A}$, $a > 0$, the arithmetic action obeys $J_h U_{\mathcal{B}}(a) = U(a) J_h$. Thus $J_h U_{\mathcal{B}}(a) \iota_h = U(a)$. If $L : F \rightarrow X$, $R : X \rightarrow F$, and F is finite dimensional, the trace of the finite-rank map LR is $\text{Tr}_F(RL)$: expanding LRx in a basis of F expresses both as the sum of the same coefficient functionals evaluated on the same image vectors. Apply this identity to (SS.8) and its tensor powers. Expanding a permutation trace in a tensor basis contracts one matrix index around each cycle, proving

$$\text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|}). \quad (\text{SS.9})$$

This proof uses all entries of $U(a)$ and applies to its nonsemisimple action. The cochain supertrace carries $(-1)^k$. The complementary finite supertrace is exactly $(-1)^k$ times $(\text{Tr } U(a))^k$ minus the trace in (SS.9). No infinite trace is assigned by these equations.

At a fixed permutation-invariant support fibre, every displayed linear map lifts by $(\lambda, x) \mapsto (\lambda, Tx)$. For arbitrary masks the support of (SS.4) is the specified orbit join $c(\lambda) = \bigvee_{\pi} \pi \lambda$, and the amplitude maps include the original transports into that fibre. The two projection maps then decompose the transported vector in the fibre over $c(\lambda)$; their sum is that transport. Their sum has not been declared the identity on a different original fibre. The upstream kernel is exactly the kernel of its specified transport $V_{\lambda} \rightarrow V_{c(\lambda)}$. At invariant masks the transport is the identity and the decomposition above applies. Every vanishing projected amplitude remains the supported zero in its target fibre; external absence is sent to external absence.

19.2 Orbit coordinates, their Gram, and all local nilpotents

Let $e_1, \dots, e_d$ denote the original power basis with indices shifted by one. For $n \in \mathbb{N}^d$, $|n| = k$, let b_n be the sum of the distinct basis tensors having occupancy n . Define

$$N = \binom{k+d-1}{d-1}, \quad I : \mathbb{C}^N \longrightarrow E^{\otimes k}, \quad I e_n = b_n, \quad D_o = I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, \quad L = D_o^{-1} I^*. \quad (\text{SS.10})$$

Distinct tensor basis vectors are orthonormal for this coordinate calculation; hence D_o records exactly their orbit sizes. It is separate from the arithmetic Gram. One has $LI = 1$ and $IL = P_{\text{sym}} = k!^{-1} \sum P_{\pi}$, since either map replaces every coefficient on an orbit by its average. Here $\text{Sym}^k E$ denotes this invariant image. Its comparison with the symmetric tensor quotient is the quotient map restricted to that image: it sends b_n to $(k! / \prod_i n_i!) e_1^{n_1} \cdots e_d^{n_d}$. The inverse sends this quotient monomial to $(\prod_i n_i! / k!) b_n$. These formulas prove the isomorphism and retain the exact diagonal factors between the two presentations.

Permutation invariance of the arithmetic constraints and measure makes the unique constrained minimum satisfy $R_M P_{\pi} = P_{\pi} R_M$; applying the inverse permutation to a feasible minimizer gives a feasible vector of the same norm, and uniqueness identifies it. Thus G_M and G_M^{-1} commute with P_{sym} . Retain the matrices

$$G_M^s = I^* G_M I, \quad W_M^s = I^* W_M I, \quad A^s = L A^{(k)} I, \quad A^{(k)} I = I A^s, \quad (G_M^s)^{-1} = L G_M^{-1} L^*. \quad (\text{SS.11})$$

The last equality includes both D_o^{-1} factors in L . Indeed its product with $I^* G_M I$ is

$$L G_M^{-1} (L^* I^*) G_M I = L G_M^{-1} P_{\text{sym}} G_M I = LI = 1.$$

This proves the inverse formula without exchanging arbitrary compression and inversion.

In these unscaled coordinates, if $Ae_j = \sum_i A_{ij}e_i$, the full generator is

$$A^{(k)}b_n = \left(\sum_i n_i A_{ii} \right) b_n + \sum_{\substack{i \neq j \\ n_j > 0}} (n_i + 1) A_{ij} b_{n+e_i-e_j}. \quad (\text{SS.12})$$

For a specified target tensor of occupancy $n + e_i - e_j$, each of its $n_i + 1$ entries equal to e_i can have come from an e_j in a source tensor, and these are exactly the contributing preimages. The diagonal term counts its n_i unchanged positions. This proves every coefficient in (SS.12). Using the original quotient's complete CRT isomorphism, set $E_\rho = \mathbb{C}[x_\rho]/x_\rho^{m_\rho}$, $t = \rho + x_\rho$. The explicit CRT map is reduction modulo each $(t - \rho)^{m_\rho}$; its inverse is the unique degree $< d$ polynomial with the prescribed full jets. Applying its tensor power and restriction through I, L is the exact change from the original orbit coordinates to the following direct sum:

$$\text{Sym}^k E \simeq \bigoplus_{\substack{\ell_\rho \geq 0 \\ \sum_\rho \ell_\rho = k}} \bigotimes_\rho \text{Sym}^{\ell_\rho} E_\rho, \quad w_\ell = \prod_\rho \binom{\ell_\rho + m_\rho - 1}{m_\rho - 1}. \quad (\text{SS.13})$$

The isomorphism groups the unscaled tensor orbit sums by the number of factors in each CRT summand; these grouped basis vectors exhaust the full basis, which proves the direct sum and its dimensions.

On the summand ℓ , the generator is $\lambda_\ell + N_\ell$, where $\lambda_\ell = \sum_\rho \ell_\rho \rho$ and N_ℓ is the sum of the multiplications by the x_ρ in their original slots. Put $h_\ell = \sum_\rho \ell_\rho (m_\rho - 1)$. Every application raises the total local exponent by one, so $N_\ell^{h_\ell+1} = 0$. On the tensor of local constants,

$$N_\ell^{h_\ell} \left(\bigotimes_\rho 1^{\otimes \ell_\rho} \right) = \frac{h_\ell!}{\prod_\rho ((m_\rho - 1)!)^{\ell_\rho}} \bigotimes_\rho (x_\rho^{m_\rho-1})^{\otimes \ell_\rho} \neq 0. \quad (\text{SS.14})$$

The coefficient counts the ordered choices of increments in each slot, each of which must be incremented exactly $m_\rho - 1$ times. Therefore the nilpotent index is exactly $h_\ell + 1$. Even when different ℓ have the same λ_ℓ , their direct summands in (SS.13) are retained separately. Traces may kill individual nilpotent diagonal contributions, while the complete operators (SS.12)–(SS.14) retain them.

19.3 Complete symmetric spectral sums and their exact departure

List $\rho_1, \dots, \rho_d$ with every algebraic multiplicity and put

$$\delta_i = \text{Re } \rho_i - \frac{1}{2}, \quad \sigma_1 = \sum_i \delta_i, \quad \sigma_2 = \sum_i \delta_i^2, \quad a_k = \binom{k+d-1}{d}, \quad b_k = \binom{k+d-1}{d+1}. \quad (\text{SS.15})$$

Binomial coefficients with lower index exceeding the upper index are zero. Thus these formulas include $k = 1$ and $d = 1$. Triangularize A . The induced ordered orbit basis is also triangular under (SS.12), with diagonal $\lambda_n = \sum_i n_i \rho_i$, once for each $n \in \mathbb{N}^d$, $|n| = k$. For example, order the monomials compatibly with the triangular basis's index weight; each off-diagonal substitution in (SS.12) decreases that weight. Consequently

$$\det(z - A^s) = \prod_{|n|=k} (z - \lambda_n), \quad \sum_{k \geq 0} q^k \text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \prod_{i=1}^d (1 - qa^{\rho_i})^{-1}. \quad (\text{SS.16})$$

The second identity is an identity of formal power series: expand each geometric series and select its occupancy. On grouping repeated roots it is $\prod_\rho (1 - qa^\rho)^{-m_\rho}$, and the coefficient for grouped

occupancy ℓ is precisely w_ℓ from (SS.13). This gives the exact multiplicity dictionary between the cycle trace (SS.9) and the full local jets.

Define the symmetric absolute spectral cost and metric departure by

$$\begin{aligned}\mathcal{D}_k^s &= \sum_{|n|=k} \left| \sum_i n_i \delta_i \right| = \sum_{|\ell|=k} w_\ell \left| \sum_\rho \ell_\rho (\operatorname{Re} \rho - \tfrac{1}{2}) \right|, \\ \mathfrak{d}_M^s &= \operatorname{Tr}((G_M^s)^{-1} (A^s)^* G_M^s A^s) - \sum_{|n|=k} |\lambda_n|^2, \\ S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2}, \quad \epsilon_M^s = \|S_M^s\|_{\text{op}}.\end{aligned}\tag{SS.17}$$

The square root is used only as the displayed isometry for computing the original Hermitian pencil; it changes none of the orbit vectors, source functions, arithmetic constants, or entries of G_M^s .

Theorem 19.1 (Exact occupancy moments and symmetric departure). *The full unweighted occupancies satisfy*

$$\begin{aligned}\sum_{|n|=k} \sum_i n_i \delta_i &= a_k \sigma_1, \\ \sum_{|n|=k} \left(\sum_i n_i \delta_i \right)^2 &= (a_k + b_k) \sigma_2 + b_k \sigma_1^2.\end{aligned}\tag{SS.18}$$

In the original arithmetic orbit Gram,

$$\begin{aligned}\mathfrak{d}_M^s &\geq 0, \quad \operatorname{Tr} S_M^s = 2a_k \sigma_1, \\ \operatorname{Tr}((S_M^s)^2) &= 4\{(a_k + b_k) \sigma_2 + b_k \sigma_1^2\} + 2\mathfrak{d}_M^s.\end{aligned}\tag{SS.19}$$

In particular, if $\bar{\delta} = \sigma_1/d$, the exact mean-square spread of the occupancy offsets is

$$\sum_{|n|=k} \left(\sum_i n_i \delta_i - k\bar{\delta} \right)^2 = \frac{Nk(k+d)}{d(d+1)} \left(\sigma_2 - \frac{\sigma_1^2}{d} \right).\tag{SS.20}$$

Proof. The product $\prod_i (1 - q_i)^{-1}$ enumerates every occupancy. Apply $q_i \partial_{q_i}$ and then set all $q_i = q$. Its coefficient of q^k gives $\sum n_i = [q^k] q(1 - q)^{-d-1} = a_k$. For two distinct indices, applying the two derivatives gives $\sum n_i n_j = [q^k] q^2 (1 - q)^{-d-2} = b_k$. Applying $q_i^2 \partial_{q_i}^2$ gives $\sum n_i (n_i - 1) = 2b_k$. These coefficient identities follow by multiplying geometric series, or by induction from $(1 - q)^{-r} = \sum_{m \geq 0} \binom{m+r-1}{r-1} q^m$. Therefore $\sum n_i^2 = a_k + 2b_k$. Expanding the square, the cross terms have coefficient b_k ; collecting the diagonal terms gives (SS.18). The identities $a_k = Nk/d$ and $a_k + b_k = Nk(k+d)/(d(d+1))$ follow directly from the factorials, including the stated edge cases. Expanding the centered square proves (SS.20).

Let $T = (G_M^s)^{1/2} A^s (G_M^s)^{-1/2}$. A unitary upper triangularization exists by induction: choose a unit eigenvector as the first basis vector, complete it to an orthonormal basis, and repeat on the remaining diagonal block. If its strictly upper triangular part is N_T , its Hilbert–Schmidt squared norm is exactly $\mathfrak{d}_M^s$; the diagonal entries are the complete λ_n . This proves nonnegativity while retaining every nonzero off-diagonal entry. Since $S_M^s = T + T^* - kI$, its diagonal is $2 \sum_i n_i \delta_i$; each strictly upper triangular entry and its adjoint contribute twice their squared absolute value to the squared trace. Thus $\operatorname{Tr}((S_M^s)^2) = 4 \sum_n (n \cdot \delta)^2 + 2\mathfrak{d}_M^s$. Equation (SS.18) proves (SS.19). $\square$

For comparison with (CT.20)–(CT.21), the ordered tensor has an occupancy n exactly $k! / \prod_i n_i!$ times. The permutation complement therefore has that multiplicity minus one. Its characteristic

polynomial, trace, and absolute spectral cost are obtained with these nonnegative integer multiplicities. In particular

$$\mathcal{D}_k = \mathcal{D}_k^s + \sum_{|n|=k} \left(\frac{k!}{\prod_i n_i!} - 1 \right) |n \cdot \delta|. \quad (\text{SS.21})$$

This equality retains every contribution omitted by the symmetric inclusion. Moreover the permutation projection is orthogonal also for G_M , because it commutes with that Gram. Its image and kernel are invariant under $A^{(k)}$. Thus the full metric operator is the orthogonal direct sum of its symmetric and complementary restrictions; the squared Hilbert–Schmidt norms and the eigenvalue-square sums add separately. The full departure in (CT.20) is exactly $\mathfrak{d}_M^s$ plus the corresponding complementary departure, each nonnegative. This proves the full operator comparison as well as the scalar identity (SS.21).

19.4 The cost in the original symmetric relation layer

Restrict the original source maps without changing their Hilbert targets:

$$Y_M^s = Y_M I, \quad C_M^s = C_M I, \quad \Delta_M^s = G_M^s - G_{M+1}^s = (Y_M^s)^* Y_M^s, \quad W_M^s = -((Y_M^s)^* C_M^s + (C_M^s)^* Y_M^s). \quad (\text{SS.22})$$

Their images lie in the invariant original relation layer, whose dimension is $r_{\text{lay}} = p_{\leq k}(M+1)$. Indeed an unscaled symmetric polynomial basis in exact degree $M+1$ is indexed by its partitions into at most k parts. The invariant full-jet map is onto at every $M \geq k(d-1)$: average a complete tensor remainder to lift each invariant jet. The difference of the two kernel dimensions is therefore the number of the new degree $M+1$ basis vectors. The injective polynomial realization and orthogonal layer quotient transport this dimension to the actual source layer, proving the assertion with its original relation ideal. Put

$$r_M^s = \text{rank } Y_M^s, \quad \tau_M^s = \text{Tr}((G_M^s)^{-1} \Delta_M^s), \quad \chi_M^s = \text{Tr}((G_M^s)^{-1} (C_M^s)^* C_M^s). \quad (\text{SS.23})$$

Then the precise relative source costs are

$$\begin{aligned} n_{\pm}(W_M^s) &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}\}, \\ \mathcal{D}_k^s &\leq \frac{1}{2} \|S_M^s\|_1 \leq \sqrt{\tau_M^s \chi_M^s}, \quad \mathcal{D}_k^s \leq r_M^s \epsilon_M^s. \end{aligned} \quad (\text{SS.24})$$

To prove the inertia bound, the Hermitian form in (SS.22) vanishes on $\ker Y_M^s$. Any strictly positive or strictly negative subspace of dimension greater than $\text{rank } Y_M^s$ intersects that kernel nontrivially, a contradiction. Apply the same argument to C_M^s . In the triangular basis used in (SS.19), the absolute diagonal sum of S_M^s is $2\mathcal{D}_k^s$. Its spectral expansion gives $|S_{ii}| \leq \sum_j |\lambda_j| |U_{ij}|^2$; summing over i bounds this sum by the trace norm. Finally, for any two finite-domain Hilbert-space maps B, C , their singular-vector decomposition and entrywise Cauchy–Schwarz prove $\|B^* C\|_1 \leq \|B\|_{\text{HS}} \|C\|_{\text{HS}}$. Apply it to $B = Y_M^s (G_M^s)^{-1/2}$ and $C = C_M^s (G_M^s)^{-1/2}$ in (SS.22). The squared norms are τ_M^s, χ_M^s . The inertia bound gives at most $2r_M^s$ nonzero eigenvalues, proving the last assertion.

If the packet is stable, with full orders, under $\rho \mapsto 1 - \bar{\rho}$, its original antiunitary reflection commutes with permutations. Restricting (CT.16a) to the invariant space pairs every nonzero eigenvalue of the Hermitian pencil with its negative. Thus $\sigma_1 = 0$, and with $p_M^s = n_+(W_M^s) = n_-(W_M^s)$,

$$\begin{aligned} p_M^s &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}, \lfloor N/2 \rfloor\}, \\ \mathcal{D}_k^s &\leq p_M^s \epsilon_M^s, \\ p_M^s (\epsilon_M^s)^2 &\geq \frac{2Nk(k+d)}{d(d+1)} \sigma_2 + \mathfrak{d}_M^s. \end{aligned} \quad (\text{SS.25})$$

There are $2p_M^s$ nonzero eigenvalues, each of absolute value at most ϵ_M^s . Bounding their absolute and squared sums and using (SS.19) proves both inequalities. They involve no division by p_M^s ; when it vanishes, (SS.19) forces every displayed nonnegative term to vanish.

Let $\delta_* = \max_i |\delta_i|$. If it is positive, $|n \cdot \delta| \leq k\delta_*$ yields the additional explicit lower cost

$$\mathcal{D}_k^s \geq \frac{(a_k + b_k)\sigma_2 + b_k\sigma_1^2}{k\delta_*}. \quad (\text{SS.26})$$

Indeed square each occupancy offset, bound it by $k\delta_*|n \cdot \delta|$, and sum using (SS.18). For a reflected packet this becomes $\mathcal{D}_k^s \geq N(k + d)\sigma_2/(d(d + 1)\delta_*)$. Each repeated eigenvector $v^{\otimes k}$ is nonzero and invariant, and its exact original Rayleigh quotient is $2k\delta_i$. Hence $\epsilon_M^s \geq 2k\delta_*$. Equations (SS.25)–(SS.26) additionally measure the cumulative cost of every occupancy, with all multiplicities and original layer ranks. If any $m_\rho > 1$, (SS.14) gives a nonzero nilpotent on the all- ρ summand. An operator with that Jordan block cannot be unitarily diagonal, so the sum of squares defining $\mathfrak{d}_M^s$ is strictly positive for every positive G_M^s . This retains a further finite-metric contribution even when all $\delta_i = 0$.

Finally the density proved in (JD.8) passes to the invariant source by the bounded permutation projection. The canonical representatives are the restrictions $R_M I$; hence $G_M^s \downarrow 0$ and $\det G_M^s \rightarrow 0$, while the full jets remain the identity at every finite M . With $\pi_M^s = \det G_{M+1}^s / \det G_M^s$, the positive eigenvalues t_j of $(G_M^s)^{-1/2} \Delta_M^s (G_M^s)^{-1/2}$ lie in $(0, 1)$, their number is r_M^s , and $\tau_M^s = \sum t_j$, $\pi_M^s = \prod (1 - t_j)$. The arithmetic mean–geometric mean inequality therefore gives

$$(\mathcal{D}_k^s)^2 \leq \tau_M^s \chi_M^s \leq r_M^s (1 - (\pi_M^s)^{1/r_M^s}) \chi_M^s \quad (r_M^s > 0). \quad (\text{SS.27})$$

For $r_M^s = 0$, (SS.22) gives $\Delta_M^s = W_M^s = 0$, without using an exponent $1/r_M^s$. Since $-\log(1 - t) - t = \int_0^t u/(1 - u) du$, the same finite source sequence also obeys, for integers $k(d - 1) \leq a < b$ and $\mathcal{D}_k^s > 0$,

$$(\mathcal{D}_k^s)^2 \sum_{M=a}^{b-1} (\chi_M^s)^{-1} \leq \sum_{M=a}^{b-1} \tau_M^s \leq \log \frac{\det G_a^s}{\det G_b^s}. \quad (\text{SS.28})$$

Positivity of every denominator follows from (SS.24). Telescoping proves the last inequality. Moreover $\sum_{M=a}^\infty \tau_M^s = \infty$: otherwise the tail has $t_j \leq \tau_M^s \leq 1/2$, and $-\log(1 - t_j) \leq 2t_j$ would keep the telescoped logarithm bounded, contrary to $\det G_M^s \rightarrow 0$. These identities transfer the entire determinant and relative-energy calculation to the source's smaller invariant family through its exact inclusion; they supply no assumed arithmetic upper estimate.

20 Arithmetic quartet parity and the exact positive frontier operator

Let h be a nonempty packet of actual zeros with their full orders, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Their composition also preserves the packet. Retain $d = \deg h$, $g = 2\xi$, $v_h = g/h$, $v = j_h v_h$, and the actual measure $d\nu_h = |v_h(1/2 + it)|^2 dt/(2\pi)$. Throughout this section $k \geq 1$, $M \geq k(d - 1)$, and

$$E_k = (\mathbb{C}[s]/h)^{\otimes k}, \quad A_k = \sum_{j=1}^k A_{h,j}, \quad \mathcal{C}_M = G_M^{-1}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{AP.1})$$

The metric G_M is the actual total-degree minimum of (CT.12)–(CT.13). All coordinates below remain its original monomial tensor coordinates.

20.1 The actual two frontier matrices

Let q_j be the original monic orthogonal polynomials for ν_h , and $\kappa_j = \int |q_j|^2 d\nu_h$. Thus κ_0 retains the mass of the original measure. For a multi-index α , put

$$q_\alpha = \prod_{j=1}^k q_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad z_\alpha = U_{v^{\otimes k}}[q_\alpha] \in E_k. \quad (\text{AP.2})$$

The source's notation p_j, ω_j denotes these same q_j, κ_j : their monic leading terms and identical lower-degree orthogonality determine the polynomial uniquely, and then determine its norm. No polynomial or measure is divided by its norm. The triangular polynomial change of basis and product integration in (CT.12) give exactly

$$\mathcal{C}_M = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\kappa_\alpha}. \quad (\text{AP.3})$$

Indeed this basis has Gram diagonal κ_α , and its full interpolation matrix has columns z_α ; substitution in $JM^{-1}J^*$ proves (AP.3), retaining $v^{\otimes k}$ in every column.

Write $a_j = \kappa_j / \kappa_{j-1}$ for $j \geq 1$, $a_0 = 0$, and retain the ordered frontier $|\beta| = M + 1$. Its size is $r = \binom{M+k}{k-1}$. Define maps with the same declared source $\mathbb{C}^r$:

$$\begin{aligned} \Omega &= \text{diag}(\kappa_\beta)_{|\beta|=M+1}, & F &= [z_\beta]_{|\beta|=M+1}, \\ w_\beta &= \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, & E_+ &= [w_\beta]_{|\beta|=M+1}. \end{aligned} \quad (\text{AP.4})$$

The original recurrence $sq_j = q_{j+1} + b_j q_j - a_j q_{j-1}$, with $b_j + \bar{b}_j = 1$, applied to (AP.3) gives

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{AP.5})$$

To check the cancellation completely, an adjacent pair $\alpha, \alpha + e_j$ both of degree at most M has coefficients κ_α^{-1} and $-a_{\alpha_j+1} \kappa_{\alpha+e_j}^{-1}$, whose sum is zero. Its transposed conjugate cancels in the same way. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, which cancels the last term. The remaining frontier pair has $\beta = \alpha + e_j$ and coefficient $\kappa_\alpha^{-1} = a_{\beta_j} / \kappa_\beta$; summing these pairs is exactly (AP.4)–(AP.5). The same sum gives the full inverse-metric update

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (\text{AP.6})$$

20.2 The original linear reflection separates the two frontiers

Conjugation closure makes h real, and reflection closure then gives $h(1-s) = (-1)^d h(s)$. The original functional equation gives

$$v_h(1-s) = (-1)^d v_h(s), \quad q_j(1-s) = (-1)^j q_j(s). \quad (\text{AP.7})$$

For the second identity, the actual density is even in t . The polynomial $(-1)^j q_j(1-s)$ is monic, and substitution $t \mapsto -t$ verifies its orthogonality to every lower degree. The difference from q_j has degree less than j and is orthogonal to itself, so positivity proves it is zero. This proves (AP.7) with the original leading coefficient.

Let $\mathfrak{c} : E_k \rightarrow E_k$ be the complex-linear map substituting $s_j \mapsto 1 - s_j$ in every factor. It is well-defined because the ideal $(h(s_1), \dots, h(s_k))$ is preserved, and $\mathfrak{c}^2 = I$. Let C denote its matrix in the retained original basis. The original function map

$$(\mathfrak{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}) \quad (\text{AP.8})$$

is a linear unitary involution for $d^k x$, by the change of variables $y_j = x_j^{-1}$. Its Mellin transform replaces s_j by $1 - s_j$. Consequently, on an original numerator,

$$\mathfrak{t}_k \mathcal{T}_h^{(k)} P = (-1)^{kd} \mathcal{T}_h^{(k)} P(1 - s_1, \dots, 1 - s_k). \quad (\text{AP.9})$$

Equations (AP.7) and (AP.2) show that its full jet is $\mathfrak{c}\mathcal{J}P$, with exactly the displayed unit sign. Total degree and the original relation ideal are preserved. Applying $\mathfrak{t}_k$ to the unique minimum for a prescribed jet therefore gives the unique minimum for its $\mathfrak{c}$ -image. Uniqueness follows from the orthogonal decomposition (CT.5). Hence

$$\mathfrak{t}_k R_M = R_M C, \quad C^* G_M C = G_M, \quad C z_\alpha = (-1)^{kd+|\alpha|} z_\alpha. \quad (\text{AP.10})$$

In particular, with $\eta = (-1)^{kd+M+1}$,

$$CF = \eta F, \quad CE_+ = -\eta E_+. \quad (\text{AP.11})$$

Every term defining w_β has degree M , which proves the second identity without suppressing any coefficient.

Declare the following maps between the original positive spaces and their coordinate Hilbert spaces:

$$P = G_M^{1/2} C G_M^{-1/2}, \quad U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}. \quad (\text{AP.12})$$

Here $G_M^{1/2}$ and $\Omega^{1/2}$ are the explicit isometries from the declared original metric spaces to Euclidean coordinate spaces, with their displayed inverses. Every expression in (AP.12) retains both original matrices. From (AP.10), P is unitary and $P^2 = I$; hence $P^* = P$. Equations (AP.11) imply $PU = \eta U$, $PV = -\eta V$, and thus

$$U^* V = (PU)^*(PV) = -U^* V = 0. \quad (\text{AP.13})$$

20.3 The complete positive operator and the sharper arithmetic bound

Put $A = U^* U$, $B = V^* V$, retaining their source type $\mathbb{C}^r$. Equivalently,

$$A = \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, \quad B = \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}. \quad (\text{AP.14})$$

These positive-semidefinite matrices are built from the two original frontiers; A is distinct from the arithmetic generator A_k .

Theorem 20.1 (Exact positive frontier operator). *The actual relative weight satisfies*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = UV^* + VU^*, \\ \boxed{\epsilon_{k,M} &= \|UV^*\|, \quad \epsilon_{k,M}^2 = \lambda_{\max}(A^{1/2} B A^{1/2}),} \quad p_M = \text{rank}(A^{1/2} B^{1/2}), \\ \text{Tr}(S_M^2) &= 2 \text{Tr}(AB). \end{aligned} \quad (\text{AP.15})$$

Its positive and negative inertias both equal p_M . Define the original incidence bound and volume ratio by

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad \lambda_{k,M} = \|U\|^2, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M}. \quad (\text{AP.16})$$

Then

$$\boxed{\epsilon_{k,M}^2 \leq \Gamma_{k,M} \lambda_{k,M} \leq \Gamma_{k,M} (\pi_M^{-1} - 1).} \quad (\text{AP.17})$$

All formulas include the zero-rank cases and every full arithmetic jet. No growth estimate for $\Gamma_{k,M}$ or π_M is assumed.

Proof. Equation (AP.5) proves the first equality in (AP.15). Decompose the retained coordinate Hilbert space into the two orthogonal eigenspaces of the explicit Hermitian involution P . Relative to the order $(\eta, -\eta)$, the operator has blocks

$$S_M = \begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}, \quad T = UV^*|_{\ker(P+\eta I)}. \quad (\text{AP.18})$$

Indeed U^* vanishes on the opposite eigenspace and V^* vanishes on the first, by (AP.11). Squaring gives diagonal blocks TT^*, T^*T . For each nonzero singular value σ of T , choose unit singular vectors x, y with $Ty = \sigma x$, $T^*x = \sigma y$; then (x, y) and $(x, -y)$ have eigenvalues $\sigma, -\sigma$. The remaining eigenspaces are $\ker T^* \oplus \ker T$, with eigenvalue zero. These vectors form the full orthogonal decomposition by the spectral theorem for T^*T , including all its zero vectors. It follows that $\epsilon = \|T\| = \|UV^*\|$ and both inertias equal $\text{rank}(UV^*)$.

Now $(UV^*)(UV^*)^* = UBU^*$. Its nonzero eigenvalues and rank are those of $B^{1/2}AB^{1/2}$: the maps $UB^{1/2}$ and its adjoint transfer each nonzero eigenvector, with inverse divided by that eigenvalue. The latter matrix is $T_0^*T_0$, with $T_0 = A^{1/2}B^{1/2}$, while $T_0T_0^* = A^{1/2}BA^{1/2}$. Their full positive spectra agree by the same maps, and their ranks equal $\text{rank } T_0$. This proves the norm and rank formulas, including rank zero. Finally (AP.13) kills $(UV^*)^2$ and $(VU^*)^2$. Taking the trace of the remaining two products gives the last line of (AP.15).

For the incidence estimate, let $Z = [z_\alpha]_{|\alpha|=M}$ and $\Omega_M = \text{diag}(\kappa_\alpha)$. Define $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, and zero otherwise; summation applies if more than one specified incidence contributes. Then $E_+ = ZL$. The matrix $N = \Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is exactly

$$\sum_{j=1}^k \left(a_{\alpha_j+1} + \sum_{l \neq j} a_{\alpha_l} \right) = \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (\text{AP.19})$$

To verify each factor, its j -th contributing frontier is $\beta = \alpha + e_j$, and $\kappa_\alpha a_{\beta_j} / \kappa_\beta = 1$. The predecessors of this β supply $\sum_l a_{\beta_l}$, which is precisely the displayed summand. Every eigenvalue of N has modulus at most the largest row sum: in an eigenvector choose a coordinate of largest absolute value and apply that row equation. The matrix N is similar, by $\Omega_M^{1/2}$, to the Hermitian positive matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$. Its eigenvalues therefore lie in $[0, \Gamma_{k,M}]$. Undoing the congruence and using (AP.3) proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z \Omega_M^{-1} Z^* \preceq \Gamma_{k,M} \mathcal{C}_M, \quad VV^* \preceq \Gamma_{k,M} I. \quad (\text{AP.20})$$

Consequently $\epsilon^2 = \|UV^*\|^2 \leq \|U\|^2 \|V\|^2 \leq \Gamma_{k,M} \lambda_{k,M}$. Finally (AP.6) gives

$$\pi_M^{-1} = \det(I + UU^*) = \prod_j (1 + \lambda_j(UU^*)) \geq 1 + \lambda_{k,M}. \quad (\text{AP.21})$$

All eigenvalues here are nonnegative, proving the second bound in (AP.17). If $U = 0$ or $V = 0$, (AP.15) gives zero weight directly; all determinant denominators remain positive. $\square$

The arithmetic departure has an exact expression in these same two frontier Grams. List the original ρ_i with every algebraic multiplicity and put $\delta_i = \text{Re } \rho_i - 1/2$. Comparing (AP.15) with the fully proved tensor trace (CT.21) gives

$$\mathfrak{d}_M^{(k)} = \text{Tr}(AB) - 2 \sum_{i_1, \dots, i_k} (\delta_{i_1} + \dots + \delta_{i_k})^2. \quad (\text{AP.22})$$

This retains the complete nonsemisimple generator in the left-hand definition of (CT.20). For $k = 1$, a four-distinct-point tesserine orbit of common full order μ has the further exact chain, with $M = d + m$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_{1,M}^2 = \lambda_{\max}(A^{1/2}BA^{1/2}) \leq \Gamma_{1,M}(\pi_M^{-1} - 1). \quad (\text{AP.23})$$

The first inequality, original character, multiplicity and critical supported-zero case are proved in (R9); (AP.15)–(AP.17) prove the rest. Thus the original character observation is related to an explicit positive frontier operator through the original metric, full interpolation unit and declared isometries.

20.4 The same positive frontier on the actual symmetric summand

The cochain projector used here is exactly $S_k = (k!)^{-1} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, with the original Koszul permutations T_π . Its top-degree action is the ordinary permutation projection, because T_π there equals $\text{sgn}(\pi) P_\pi$. Equations (SS.1)–(SS.9) prove its continuous cochain maps, completed projective-tensor cohomology, and finite packet projector, including the complementary summand. We now use its finite invariant image with the original total-degree representatives R_M .

Write $e_1, \dots, e_d$ for the original power basis of E , with indices shifted by one, and retain each occupancy $n \in \mathbb{N}^d$, $|n| = k$. Let b_n be the literal sum of the distinct basis tensors of occupancy n . Define the maps and dimensions by

$$\begin{aligned} N_s &= \binom{k+d-1}{d-1}, & I : \mathbb{C}^{N_s} &\longrightarrow E_k, & Ie_n &= b_n, \\ D_o &= I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, & L &= D_o^{-1} I^*, & P_s &= IL = \frac{1}{k!} \sum_{\pi} P_\pi, & LI &= I_{N_s}. \end{aligned} \quad (\text{AP.24})$$

Indeed the distinct original tensor basis vectors are orthonormal for this coordinate calculation; hence their disjoint orbit sums have precisely the displayed diagonal Gram D_o . The map L averages coefficients on each orbit, proving all the compositions in (AP.24). In particular $P_s^* = P_s$ and $P_s^2 = P_s$. The factorials in D_o are retained separately from the arithmetic Gram.

Permutation invariance of the original product measure, relation ideal, total degree and full interpolation unit gives $R_M P_\pi = P_\pi R_M$, by applying the unitary inverse permutation to each constrained minimum. Thus G_M and G_M^{-1} commute with P_s . Put

$$\begin{aligned} R_M^s &= R_M I, & G_M^s &= I^* G_M I, & W_M^s &= I^* W_M I, \\ A_k^s &= L A_k I, & A_k I &= I A_k^s, & W_M^s &= (A_k^s)^* G_M^s + G_M^s A_k^s - k G_M^s, \\ I^* G_M &= G_M^s L, & G_M I &= L^* G_M^s, & \mathcal{K}_M^s &:= (G_M^s)^{-1} = L G_M^{-1} L^*. \end{aligned} \quad (\text{AP.25})$$

These are the same representatives and matrices as (SS.10)–(SS.11). For completeness, $I^* G_M = I^* P_s G_M = I^* G_M P_s = I^* G_M IL$, proving the first rectangular identity and then its adjoint. The generator commutes with every factor permutation, so it preserves the image of I , proving its intertwining identity and the displayed weight formula. Finally

$$(L G_M^{-1} L^*)(I^* G_M I) = L G_M^{-1} P_s G_M I = LI = I_{N_s},$$

which proves the inverse formula with both factors D_o^{-1} present. Positivity follows since I is injective and $G_M \succ 0$. The notation $\mathcal{K}_M^s$ refers only to this inverse kernel; it will not denote a reflection.

Keep the complete, possibly redundant frontier of (AP.4), with its unchanged domain $\mathbb{C}^r$ and norm matrix Ω . Its symmetric target maps are

$$\begin{aligned} F_s &= L F : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, & E_s &= L E_+ : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, \\ \mathcal{Z}_M^s &= F_s \Omega^{-1} E_s^* + E_s \Omega^{-1} F_s^*, & W_M^s &= G_M^s \mathcal{Z}_M^s G_M^s, \\ \mathcal{K}_{M+1}^s &= \mathcal{K}_M^s + F_s \Omega^{-1} F_s^*. \end{aligned} \quad (\text{AP.26})$$

The rectangular identities (AP.25), applied to $I^* G_M \mathcal{Z}_M G_M I$, prove the weight formula. Applying L and L^* to (AP.6) proves the inverse update. It uses the same orbit maps at both source degrees. Thus the update concerns the actual next symmetric minimum, not a newly selected positive form.

The original linear reflection C commutes with every factor permutation, since it substitutes the same $1 - s_j$ in each factor. Its invariant-coordinate matrix is

$$\begin{aligned} C_s &= L C I, & C I &= I C_s, & C_s L &= L C, & C_s^2 &= I_{N_s}, \\ C_s^* G_M^s C_s &= G_M^s, & C_s F_s &= \eta F_s, & C_s E_s &= -\eta E_s, & \eta &= (-1)^{kd+M+1}. \end{aligned} \quad (\text{AP.27})$$

For example $CI = P_s CI = ILCI$, and $C_s L = LCP_s = LP_s C = LC$; these prove the first intertwiners and then the involution. The metric equation follows by substituting $IC_s = CI$ into $I^* G_M I$ and using (AP.10). Applying L to (AP.11) proves both parities with the full arithmetic unit and its original sign.

Define the explicit isometries and the two retained frontier Grams by

$$\begin{aligned} P_M^s &= (G_M^s)^{1/2} C_s (G_M^s)^{-1/2}, \\ U_s &= (G_M^s)^{1/2} F_s \Omega^{-1/2}, \quad V_s = (G_M^s)^{1/2} E_s \Omega^{-1/2}, \\ A_s &= U_s^* U_s, \quad B_s = V_s^* V_s. \end{aligned} \tag{AP.28}$$

The superscript on P_M^s distinguishes this transported reflection from the fixed permutation projection P_s in (AP.24). Equation (AP.27) proves that P_M^s is a Hermitian involution and that U_s, V_s have opposite eigenvalues $\eta, -\eta$. Their ranges are consequently orthogonal, so $U_s^* V_s = 0$.

Theorem 20.2 (Exact symmetric arithmetic frontier). *For the actual signed symmetric cohomology summand, the original Hermitian pencil has*

$$\begin{aligned} S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2} = U_s V_s^* + V_s U_s^*, \\ \boxed{\epsilon_M^s &= \|U_s V_s^*\|, \quad (\epsilon_M^s)^2 = \lambda_{\max}(A_s^{1/2} B_s A_s^{1/2}),} \\ p_M^s &= \text{rank}(A_s^{1/2} B_s^{1/2}), \quad n_+(W_M^s) = n_-(W_M^s) = p_M^s, \\ \text{Tr}((S_M^s)^2) &= 2 \text{Tr}(A_s B_s). \end{aligned} \tag{AP.29}$$

With the original $\Gamma_{k,M}$ of (AP.16), put

$$\lambda_M^s = \|U_s\|^2, \quad \pi_M^s = \frac{\det G_{M+1}^s}{\det G_M^s}.$$

Then the same source sequence satisfies

$$\boxed{(\epsilon_M^s)^2 \leq \Gamma_{k,M} \lambda_M^s \leq \Gamma_{k,M} ((\pi_M^s)^{-1} - 1), \quad \lambda_M^s \leq \lambda_{k,M}.} \tag{AP.30}$$

All formulas retain every occupancy and nilpotent direction in (SS.12)–(SS.14), as well as the ungrouped frontier multiplicities.

Proof. Equation (AP.26) gives the first line of (AP.29). Decomposing $\mathbb{C}^{N_s}$ into the two orthogonal eigenspaces of P_M^s gives exactly the block operator (AP.18), now with $T_s = U_s V_s^*$ between those two spaces. The singular-vector maps and their inverses proved after (AP.18) apply to these declared matrices without any rank assumption. They give opposite eigenvalues for each positive singular value, $\text{rank rank}(A_s^{1/2} B_s^{1/2})$, and all remaining eigenvalues zero. The same two explicit composites $U_s B_s^{1/2}$ and its adjoint identify the positive eigenvalues with those of $A_s^{1/2} B_s A_s^{1/2}$. Finally $U_s^* V_s = 0$ kills the two square cross factors; the traces of the other two products both equal $\text{Tr}(A_s B_s)$. This proves every assertion in (AP.29).

Applying L and L^* to the original inequality (AP.20), and using (AP.25), proves

$$E_s \Omega^{-1} E_s^* \preceq \Gamma_{k,M} \mathcal{K}_M^s, \quad V_s V_s^* \preceq \Gamma_{k,M} I_{N_s}. \tag{AP.31}$$

Thus $\|U_s V_s^*\|^2 \leq \|U_s\|^2 \|V_s\|^2 \leq \Gamma_{k,M} \lambda_M^s$. Taking determinants of the exact inverse update in (AP.26) gives

$$(\pi_M^s)^{-1} = \det(I_{N_s} + U_s U_s^*) = \det(I_r + A_s) \geq 1 + \lambda_M^s, \quad 0 < \pi_M^s \leq 1. \tag{AP.32}$$

The determinant equality follows either by the two rectangular composites and their nonzero spectra, or by multiplying the two block factors for $\det(I + XY) = \det(I + YX)$. The inequality multiplies the nonnegative factors $1 + \lambda_j(A_s)$, exactly as in (AP.21). This proves the first two bounds in (AP.30).

For the final bound, the literal orbit maps give

$$\begin{aligned} L^* G_M^s L &= P_s G_M P_s \preceq G_M, \\ A_s &= \Omega^{-1/2} F^* P_s G_M P_s F \Omega^{-1/2} \preceq \Omega^{-1/2} F^* G_M F \Omega^{-1/2} = A. \end{aligned} \quad (\text{AP.33})$$

Indeed G_M commutes with the orthogonal projection P_s , so their omitted cross terms vanish and $G_M - P_s G_M P_s = (I - P_s) G_M (I - P_s) \succeq 0$. Taking the largest Rayleigh quotient of the two positive matrices in (AP.33) proves $\lambda_M^s \leq \lambda_{k,M}$. No comparison of arbitrary compressed inverse matrices is used. $\square$

There is also an exact comparison of the two relative operators. The map

$$Q_M = G_M^{1/2} I (G_M^s)^{-1/2} : \mathbb{C}^{N_s} \longrightarrow \mathbb{C}^{d^k}$$

has $Q_M^* Q_M = I_{N_s}$, and the inverse formula in (AP.25) gives $Q_M Q_M^* = P_s$. Direct multiplication therefore gives

$$S_M^s = Q_M^* S_M Q_M, \quad U_s = Q_M^* U, \quad V_s = Q_M^* V, \quad \epsilon_M^s \leq \epsilon_{k,M}. \quad (\text{AP.34})$$

Here $G_M^{1/2}$ commutes with P_s by its spectral decomposition. The full S_M also commutes with P_s , since both A_k and G_M do. Consequently (AP.34) is its invariant orthogonal restriction; the complementary operator remains on $\ker P_s$. The norm inequality follows directly by applying S_M to the unit vector $Q_M u$ and then the contraction Q_M^* .

If desired, the source frontier can be grouped with every factorial retained. For an orbit $\mathcal{O}$ of degree- $M + 1$ multi-indices, let $m_j(\beta) = |\{l : \beta_l = j\}|$ and

$$n_{\mathcal{O}} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!}, \quad \hat{f}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lz_{\beta}, \quad \hat{e}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lw_{\beta}, \quad \hat{\omega}_{\mathcal{O}} = n_{\mathcal{O}} \kappa_{\beta}. \quad (\text{AP.35})$$

The product norm κ_{β} is constant on an orbit. Equivariance of the full unit, polynomials and incidence sums implies that Lz_{β} and Lw_{β} are individually constant on the same orbit. Hence, for example,

$$\sum_{\beta \in \mathcal{O}} \frac{(Lz_{\beta})(Lw_{\beta})^*}{\kappa_{\beta}} = \frac{\hat{f}_{\mathcal{O}} \hat{e}_{\mathcal{O}}^*}{\hat{\omega}_{\mathcal{O}}}. \quad (\text{AP.36})$$

The same calculation with both columns of the first or second type proves the corresponding kernel and incidence identities. Equations (AP.35)–(AP.36) are the exact map to the source's orbit-factorial presentation. The ungrouped matrices in (AP.26)–(AP.34) continue to retain all their columns.

All zero-rank cases follow without inversion of A_s or B_s . If their square-root product vanishes, then $p_M^s = 0$, $W_M^s = 0$, and $\epsilon_M^s = 0$, even if the kernel update is nonzero. If $\pi_M^s = 1$, (AP.32) forces $U_s = 0$, with the same conclusion. The dimension N_s is always at least one; if it is one, a Hermitian involution has only one nonzero eigenspace, so one of U_s, V_s vanishes and the weight is zero. This argument also covers a zero-dimensional reflection eigenspace in a larger target. It neither removes the retained quotient nor asserts existence of a particular actual packet. For $k = 1$, every orbit cardinality is one and $I = L = I_d$; all the displayed formulas recover (AP.15)–(AP.21) exactly.

Finally the symmetric source is still $R_M I$, and its two actual boundary maps are $Y_M I$ and $C_M I$ in (SS.22). The original relation-graph map, followed by the bounded permutation projection on its Hilbert target, realizes the redundant frontier used here. Grouping it by (AP.35) gives the invariant relation layer of dimension $p_{\leq k}(M + 1)$, as proved in (SS.22)–(SS.24). Its primitive is the original fixed-order division primitive followed by S_k ; the cochain identity in (SS.3)–(SS.4) preserves its boundary and all signs. At invariant support fibres the same maps lift with that support; vanishing amplitudes remain the corresponding supported zero. For other masks the original orbit-joined transport and its kernel remain as stated after (SS.9). Thus (AP.30) is a proved bound for the same

finite arithmetic cohomology summand, with no assumed asymptotic estimate for either $\Gamma_{k,M}$ or π_M^s .

Section 21 carries these same two parities through the original graph frame into the actual derivative layer. The equality $-\widehat{Y}^*\widehat{C} = UV^*$ proved there supplies the exact original derivative cost for the positive operator here. Sections 22 and 24 calculate its one-factor and symmetric-square counterparts through explicit quadratic algebra maps, with all local nilpotents and both reflection components.

21 Quartet parity in the original derivative energy and volume loss

This continuation retains the original maps of (FE.3)–(FE.9) and the actual quartet reflection of (AP.7)–(AP.13). Their combination proves an exact block identity for the two maps into the original theta relation layer. It yields a stronger derivative energy estimate than (FE.24), including an operator-norm version and the resulting finite accumulated determinant bounds.

21.1 Original maps, their spaces, and the reflection

Fix a nonempty packet of actual zeros, with every selected zero's full multiplicity, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Keep

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad g = 2\xi, \quad v_h = g/h, \quad v = j_h v_h, \quad E_k = (\mathbb{C}[s]/h)^{\otimes k}.$$

Here $k \geq 1$, $M \geq k(d-1)$, and the original generator is $A_k = \sum_{j=1}^k A_{h,j}$. Let $G = G_M > 0$ be the actual degree- M minimum metric in the original monomial remainder basis, and let $R = R_M : E_k \rightarrow \mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$ be its original representative, so $R^*R = G$.

For the unchanged measure $d\nu_h(t) = |v_h(1/2 + it)|^2 dt / (2\pi)$, let q_j be the monic orthogonal polynomials and $\kappa_j = \|q_j\|_{\nu_h}^2$. Write $q_\beta = \prod_j q_{\beta_j}$, $\kappa_\beta = \prod_j \kappa_{\beta_j}$, and $z_\beta = U_{v \otimes k}[q_\beta]$. With the original ordering of $|\beta| = M+1$, retain

$$\begin{aligned} r &= \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\kappa_\beta) > 0, \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \\ E_+ &= [w_\beta] : \mathbb{C}^r \rightarrow E_k, \quad w_\beta = \sum_{j:\beta_j>0} \frac{\kappa_{\beta_j}}{\kappa_{\beta_j-1}} z_{\beta-e_j}, \\ T_+ &= [\mathcal{T}_h^{(k)} q_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k, \quad b = T_+ - RF : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M, \\ H &= b^*b = \Omega + F^*GF > 0. \end{aligned} \tag{QD.1}$$

The space $\mathcal{E}_M$ is the actual next theta relation layer with its inherited L^2 norm. The column Gram of T_+ is Ω ; its image is perpendicular to the old polynomial space containing $R(E_k)$. Thus expanding its Gram proves H in (QD.1). The leading degree map gives the inverse of b : each layer vector's highest polynomial coefficient determines it, and subtracting that graph image leaves a vector both in the old relation space and perpendicular to it. In particular b is a bijection onto the original layer, with no arbitrary Hilbert replacement.

The two original maps into that same layer and their weight are

$$\begin{aligned} Y &= -bH^{-1}F^*G : E_k \rightarrow \mathcal{E}_M, \quad C_D = b\Omega^{-1}E_+^*G : E_k \rightarrow \mathcal{E}_M, \\ \Delta &= Y^*Y = G - G_{M+1}, \quad W = A_k^*G + GA_k - kG = -(Y^*C_D + C_D^*Y). \end{aligned} \tag{QD.2}$$

The notation C_D is explicitly the map C_M of (FE.5), and does not denote the inverse interpolation matrix. Its formula records the top polynomial coefficient of the original Euler defect; the graph

projection and the original theta primitive in (FE.3)–(FE.5) supply its actual source realization. The weight identity also follows directly by inserting the two displayed maps and cancelling the middle $b^*b = H$.

Let $C : E_k \rightarrow E_k$ replace each residue variable s_j by $1 - s_j$, and let

$$(\mathbf{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}).$$

The original packet symmetries and functional equation give $h(1 - s) = (-1)^d h(s)$, $v_h(1 - s) = (-1)^d v_h(s)$, and $q_j(1 - s) = (-1)^j q_j(s)$. For the last identity, the reflected polynomial with its sign restored is monic and has the same lower-degree orthogonality for the even density; uniqueness of the monic orthogonal polynomial proves equality. The variable change $x_j \mapsto x_j^{-1}$ proves that $\mathbf{t}_k$ is a unitary involution for $d^k x$, with Mellin action $s_j \mapsto 1 - s_j$. Thus it preserves total degree, the relation ideal and the unique constrained minimum. Writing $\eta = (-1)^{kd+M+1}$, one obtains

$$\mathbf{t}_k R = RC, \quad C^2 = I, \quad C^*GC = G, \quad CF = \eta F, \quad CE_+ = -\eta E_+, \quad \mathbf{t}_k b = \eta b. \quad (\text{QD.3})$$

The last equality uses both $\mathbf{t}_k T_+ = \eta T_+$ and $CF = \eta F$. It states the exact target reflection on the whole original relation layer, since b is onto.

For explicit coordinate isometries put

$$\begin{aligned} P &= G^{1/2}CG^{-1/2}, \quad P^* = P, \quad P^2 = I, \quad U = G^{1/2}F\Omega^{-1/2}, \quad V = G^{1/2}E_+\Omega^{-1/2}, \\ \hat{Y} &= YG^{-1/2}, \quad \hat{C} = C_D G^{-1/2} : \mathbb{C}^{d^k} \rightarrow \mathcal{E}_M, \quad J_b = bH^{-1/2} : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M. \end{aligned} \quad (\text{QD.4})$$

The matrix $G^{1/2}$ is the isometry from (E_k, G) to its Euclidean coordinate space, with inverse $G^{-1/2}$. Similarly $\Omega^{1/2}$ records the original polynomial norms, and J_b is a unitary map with inverse $H^{-1/2}b^*$. No commutation of H with Ω is assumed. In these exact coordinates,

$$\hat{Y} = -J_b H^{-1/2} F^* G^{1/2}, \quad \hat{C} = J_b H^{1/2} \Omega^{-1} E_+^* G^{1/2}. \quad (\text{QD.5})$$

21.2 The single original boundary block

Theorem 21.1 (Exact quartet derivative block and its energy bound). *The two original domain parities and their entire block relation are*

$$YC = \eta Y, \quad C_D C = -\eta C_D, \quad \hat{Y}P = \eta \hat{Y}, \quad \hat{C}P = -\eta \hat{C}, \quad -\hat{Y}^* \hat{C} = UV^*. \quad (\text{QD.6})$$

Let

$$\begin{aligned} S &= G^{-1/2}WG^{-1/2}, \quad \epsilon = \|S\|_{\text{op}}, \quad \lambda = \|UU^*\|_{\text{op}}, \quad \pi = \frac{\det G_{M+1}}{\det G}, \\ \zeta &= \|\hat{C}\|_{\text{op}}^2, \quad \chi = \|\hat{C}\|_{\text{HS}}^2. \end{aligned} \quad (\text{QD.7})$$

Then all these actual finite quantities satisfy

$$\boxed{\epsilon = \|\hat{Y}^* \hat{C}\|_{\text{op}} = \|UV^*\|_{\text{op}}, \quad \epsilon^2 \leq \frac{\lambda}{1 + \lambda} \zeta \leq \frac{\lambda}{1 + \lambda} \chi.} \quad (\text{QD.8})$$

The operator-norm cost and the trace cost are exactly

$$\begin{aligned} \hat{C}^* \hat{C} &= V(I + U^*U)V^*, \quad \zeta = \lambda_{\max}(V(I + U^*U)V^*), \\ \chi &= \text{Tr}(V(I + U^*U)V^*) = \|V\|_{\text{HS}}^2 + \|VU^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{QD.9})$$

In particular the operator-norm replacement of χ in (QD.8) is valid without a rank assumption.

Proof. From $C^*GC = G$ and $C^2 = I$ one gets $GC = C^*G$. Thus $F^*GC = (CF)^*G = \eta F^*G$, and similarly $E_+^*GC = -\eta E_+^*G$. Substitution in (QD.2) proves the first two parities, and the coordinate change proves the next two. Equations (QD.2) and $b^*b = H$ give, in the order shown,

$$\hat{Y}^*\hat{C} = -G^{1/2}FH^{-1}H\Omega^{-1}E_+^*G^{1/2} = -UV^*.$$

Decompose $\mathbb{C}^{d^k} = \ker(P - \eta I) \oplus \ker(P + \eta I)$, orthogonally. The first map in (QD.6) annihilates the second summand and the second map annihilates the first. Consequently $X = \hat{Y}^*\hat{C}$ maps the second summand into the first and vanishes on the first. The actual weight is

$$S = -(X + X^*) = -\begin{pmatrix} 0 & X_{+-} \\ X_{+-}^* & 0 \end{pmatrix}.$$

Squaring gives diagonal blocks $X_{+-}X_{+-}^*$ and $X_{+-}^*X_{+-}$. Their largest eigenvalues both equal $\|X_{+-}\|^2$ when this map is nonzero, by its singular value decomposition; if it is zero both blocks vanish. Hence $\|S\| = \|X\|$, including empty parity summands and all zero ranks.

The retained Gram identity in (QD.1) is $H = \Omega^{1/2}(I + U^*U)\Omega^{1/2}$. Substitution gives

$$\hat{Y}^*\hat{Y} = U(I + U^*U)^{-1}U^* = UU^*(I + UU^*)^{-1}.$$

The second equality follows by multiplying $(I + UU^*)U = U(I + U^*U)$ by its positive invertible factors. Thus $\|\hat{Y}\|^2 = \lambda/(1 + \lambda)$, also when $\lambda = 0$. Submultiplicativity in the exact identity $\epsilon = \|\hat{Y}^*\hat{C}\|$ proves the first inequality of (QD.8). On the other hand $\hat{C} = b\Omega^{-1/2}V^*$, so its Gram is exactly $V\Omega^{-1/2}H\Omega^{-1/2}V^* = V(I + U^*U)V^*$. This proves (QD.9). A positive finite matrix has largest eigenvalue at most its trace, proving $\zeta \leq \chi$ and the second inequality. $\square$

This is an actual-source strengthening of (FE.24): its general factor 4 is replaced by 1 under the proved original quartet parity. The retained derivative map still lands in the original $\mathcal{E}_M$; only the displayed exact isometries were used to calculate its norm.

21.3 Complete determinant constraints, including zero cases

Set $\delta_* = \max_{\rho \in Z} |\operatorname{Re} \rho - 1/2|$ and $q_* = 4k^2\delta_*^2$. The name q_* denotes this nonnegative scalar, not an orthogonal polynomial. All original packet multiplicities remain in E_k, A_k, G_M .

Theorem 21.2 (Quartet derivative cost and accumulated volume). *At every admitted degree, including $\delta_* = 0$,*

$$q_* \leq \epsilon^2 \leq \frac{\lambda}{1 + \lambda} \zeta \leq (1 - \pi) \zeta \leq (1 - \pi) \chi. \quad (\text{QD.10})$$

If $\delta_ > 0$, the exact source quantities necessarily obey*

$$\begin{aligned} \zeta &> q_*, \quad \chi \geq \zeta > q_*, \quad \lambda \geq \frac{q_*}{\zeta - q_*} \geq \frac{q_*}{\chi - q_*}, \\ \log \frac{\det G_M}{\det G_{M+1}} &\geq -\log \left(1 - \frac{q_*}{\zeta_M} \right) \geq -\log \left(1 - \frac{q_*}{\chi_M} \right). \end{aligned} \quad (\text{QD.11})$$

For $\delta_ > 0$ and any integers $k(d - 1) \leq a < b$, this gives*

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} -\log \left(1 - \frac{4k^2\delta_*^2}{\zeta_M} \right) \geq \sum_{M=a}^{b-1} -\log \left(1 - \frac{4k^2\delta_*^2}{\chi_M} \right). \quad (\text{QD.12})$$

Every denominator in (QD.11)–(QD.12) is positive, and every logarithm has a strictly positive argument, by the preceding inequalities.

Proof. For every selected centre ρ , the original finite algebra has a nonzero eigenvector of A_h with eigenvalue ρ : in its local factor use $(s - \rho)^{m_\rho - 1}$ and put zero in the other full CRT factors. Its k -fold tensor is nonzero and has eigenvalue $k\rho$ for A_k . The quotient of $W = A_k^*G + GA_k - kG$ on that original vector is exactly $2k \operatorname{Re} \rho - k$. The operator norm of its Hermitian relative matrix is at least the absolute value of this Rayleigh quotient. Taking the largest displacement and squaring proves $q_* \leq \epsilon^2$, with no semisimplicity assumption.

The inverse update is $G_{M+1}^{-1} = G_M^{-1} + F\Omega^{-1}F^*$. Conjugating by the original metric isometry gives

$$\pi = \det(I + UU^*)^{-1} = \prod_i (1 + \lambda_i)^{-1},$$

where all positive eigenvalues of UU^* occur with their multiplicities; the empty product is one. Since the product is at least $1 + \lambda$, $\pi \leq (1 + \lambda)^{-1}$, and therefore $\lambda/(1 + \lambda) \leq 1 - \pi$. Insert (QD.8) and $\zeta \leq \chi$ to prove (QD.10).

When $q_* > 0$, (QD.10) implies $\lambda > 0$ and $\zeta > 0$. Because $\lambda/(1 + \lambda) < 1$, it then implies $\zeta > q_*$. Rearranging its first cost inequality gives $\lambda(\zeta - q_*) \geq q_*$, proving the lower bound for λ . The function $x \mapsto q_*/(x - q_*)$ decreases for $x > q_*$, proving its comparison with the bound using χ . Finally $\pi^{-1} \geq 1 + \lambda \geq \zeta/(\zeta - q_*)$. Apply the increasing logarithm, and use $\zeta \leq \chi$, to obtain (QD.11). Adding the finite inequalities telescopes their original determinants and proves (QD.12). $\square$

At $\delta_* = 0$, (QD.10) remains the complete undivided statement. No value is assigned to $0/0$ when a derivative cost is zero. If $F = 0$, then $U = Y = 0$, $\lambda = 0$, $\pi = 1$ and $W = 0$; the derivative map C_D and its cost can still be nonzero. Conversely $\zeta = 0$ means $C_D = 0$, and the injectivity of $b\Omega^{-1/2}$ gives $V = E_+ = 0$; then $W = 0$ while the metric can still contract through F . These are retained source cases, and neither is excluded by division.

The same proof also gives the stagewise estimate with q_* replaced by the actual ϵ_M^2 whenever $\epsilon_M > 0$, since then $\zeta_M > \epsilon_M^2$. Thus a positive boundary cost remains detected even when every selected centre has real part $1/2$; no nilpotent contribution has been removed by using a zero spectral displacement.

For comparison with (FE.22)–(FE.23), retain the actual positive recurrence quantity

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad a_j = \kappa_j/\kappa_{j-1}, \quad a_0 = 0.$$

The proved arithmetic incidence bound (AP.20) is $VV^* \preceq \Gamma_{k,M}I$, and therefore $\epsilon_M^2 = \|UV^*\|^2 \leq \Gamma_{k,M}\lambda_M$. It yields, for all $\delta_* \geq 0$, the finite consequence

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{4k^2\delta_*^2}{\Gamma_{k,M}} \right). \quad (\text{QD.13})$$

Here $\Gamma_{k,M} > 0$ because every $a_{\alpha_j+1} > 0$. This is the explicit accumulated determinant consequence of the already proved quartet operator estimate (AP.17), whereas (QD.8)–(QD.12) additionally use the original derivative energy and its stronger operator-norm cost. Neither chain asserts an upper asymptotic estimate for those arithmetic quantities.

21.4 Exact energy–volume equality for the original one-variable layer

Theorem 21.3 (One-variable arithmetic equality). *For $k = 1$, at every admitted original degree M ,*

$$\boxed{\zeta_M = \chi_M, \quad \epsilon_M^2 = \frac{\lambda_M}{1 + \lambda_M} \chi_M = (1 - \pi_M) \chi_M.} \quad (\text{QD.14})$$

These equalities include $F = 0$, $E_+ = 0$, and $\delta_* = 0$. Whenever $\chi_M > 0$, they give the exact original determinant identity

$$\log \frac{\det G_M}{\det G_{M+1}} = -\log \left(1 - \frac{\epsilon_M^2}{\chi_M} \right), \quad 0 \leq \epsilon_M^2 < \chi_M. \quad (\text{QD.15})$$

Proof. For $k = 1$, (QD.1) has $r = 1$ at every M . The actual graph isometry $J_b : \mathbb{C} \rightarrow \mathcal{E}_M$ therefore proves that this original relation layer is one-dimensional. Let $e = J_b 1$, so $\|e\| = 1$. There are unique row maps $y, c : \mathbb{C}^d \rightarrow \mathbb{C}$ with $\widehat{Y} = ey$ and $\widehat{C} = ec$. Their actual cross-pairing is $\widehat{Y}^* \widehat{C} = y^* c$. Its norm is $\|y\| \|c\|$: the upper bound is submultiplicativity, and if both rows are nonzero, evaluating on the unit vector $c^*/\|c\|$ attains it. If either row is zero the claimed equality is zero on both sides. The positive matrix $c^* c$ has its only possible positive eigenvalue $\|c\|^2$; thus its largest eigenvalue and trace agree, including $c = 0$. Equations (QD.8)–(QD.9) now prove $\epsilon_M^2 = \|\widehat{Y}\|^2 \chi_M = \lambda_M \chi_M / (1 + \lambda_M)$ and $\zeta_M = \chi_M$.

The matrix U has one column, so UU^* has rank at most one. Its determinant product consequently gives $\pi_M = (1 + \lambda_M)^{-1}$, including $U = 0$. This proves the last equality in (QD.14). If $\chi_M > 0$, division gives $1 - \epsilon_M^2/\chi_M = \pi_M > 0$, and applying the logarithm proves (QD.15). If $\chi_M = 0$, one retains (QD.14); it sets $\epsilon_M = 0$ and imposes no additional value on π_M . In particular the possible contraction from F remains present in the zero derivative-cost case. $\square$

The equality (QD.14) is proved for the actual one-variable arithmetic source and its original one-dimensional relation layer. Thus the coefficient one in (QD.8) is an exact source identity throughout this family, at all nonzero or zero frontier ranks; no auxiliary model is required to test it.

22 The exact two-Stieltjes-space representation of the theta packet

We retain the actual function $g(s) = 2\xi(s)$, a nonempty finite packet of its nontrivial zeros with their complete orders, and the original monic packet polynomial h . The packet is closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$, hence also under $\rho \mapsto 1 - \rho$. No zero in this packet is real: the argument with the alternating eta series in Section 2 applies to the whole real interval $0 < s < 1$. In particular $g(1/2) \neq 0$. Throughout this section all polynomial degrees, powers of $1/2$, and local nilpotent orders are retained.

22.1 The quadratic algebra and both reflection branches

Put

$$z = s - \frac{1}{2}, \quad x = -z^2, \quad d = \deg h = 2D. \quad (\text{SP.1})$$

The even degree follows by partitioning the packet into reflection pairs $\{\rho, 1 - \rho\}$; each pair has the same order at its two members and neither member is $1/2$. For each such pair choose one member, write $\alpha_\rho = \rho - 1/2$, and put

$$r_\rho = -\alpha_\rho^2, \quad H(x) = \prod_{\{\rho, 1-\rho\}} (x - r_\rho)^{m_\rho}. \quad (\text{SP.2})$$

The product is independent of the chosen member. Distinct reflection pairs have distinct r_ρ , because equal squares imply equal or opposite α_ρ . Thus H is monic of degree D , $H(0) \neq 0$, and every original order occurs unchanged in H . Multiplication of the two factors in each pair proves

$$h(s) = (-1)^D H\left(-\left(s - \frac{1}{2}\right)^2\right). \quad (\text{SP.3})$$

Conjugation permutes the factors of H , so H has real coefficients. Define $A_H = \mathbb{C}[x]/(H)$, let X denote multiplication by x in its basis $1, x, \dots, x^{D-1}$, and let c_H be the column of its unit.

Theorem 22.1 (A full algebra isomorphism with its original coordinates). *There is a unital complex algebra isomorphism*

$$\begin{aligned} \mathcal{T} : \mathbb{C}[s]/(h) &\longrightarrow \mathbb{C}[x, z]/(H(x), z^2 + x) = A_H \oplus zA_H, \\ [s] &\longmapsto \frac{1}{2} + z. \end{aligned} \quad (\text{SP.4})$$

The inverse sends x to $-(s - 1/2)^2$ and z to $s - 1/2$. The multiplication in the displayed direct sum is

$$(a, b)(c, e) = (ac - xbe, ae + bc). \quad (\text{SP.5})$$

Here the direct sum describes the underlying module, with the specified multiplication. In the ordered target basis $1, x, \dots, x^{D-1}, z, zx, \dots, zx^{D-1}$, the matrix T of $\mathcal{T}$ has column n , $0 \leq n < 2D$, equal to the coefficients of

$$\begin{aligned} a_n(x) &= \sum_{2j \leq n} \binom{n}{2j} 2^{-(n-2j)} (-x)^j, \\ b_n(x) &= \sum_{2j+1 \leq n} \binom{n}{2j+1} 2^{-(n-2j-1)} (-x)^j. \end{aligned} \quad (\text{SP.6})$$

Its determinant and the original multiplication matrix are exactly

$$\det T = (-1)^{D(D-1)/2}, \quad TAT^{-1} = \frac{1}{2}I_{2D} + \begin{pmatrix} 0 & -X \\ I_D & 0 \end{pmatrix}, \quad A = M_s. \quad (\text{SP.7})$$

Proof. The substitutions in (SP.4) and its inverse send the two defining relations into zero by (SP.3) and $z^2 = -x$. They compose to the identity on the generators and therefore on both algebras. Reduction first by the monic polynomial $z^2 + x$ and then by the monic polynomial $H(x)$ gives a unique expression $a + zb$ with $\deg a, \deg b < D$. Multiplication and the relation $z^2 = -x$ give (SP.5). Expanding $(1/2 + z)^n$ and separating even and odd powers proves (SP.6). In the interlaced basis $1, z, x, zx, \dots, x^{D-1}, zx^{D-1}$, this change of basis is triangular with diagonal entry $(-1)^{\lfloor n/2 \rfloor}$. Their product is $(-1)^{D(D-1)} = 1$. Moving all the odd basis members after the even members has $D(D-1)/2$ inversions, proving the stated determinant. The columns of the inverse are explicitly the original polynomials $-(s - 1/2)^2$ and $(s - 1/2)(-s - 1/2)^2$. Finally multiplication of $a + zb$ by $1/2 + z$ gives $a/2 - xb + z(a + b/2)$, proving (SP.7). $\square$

Here is the complete local meaning of the two branches, including all derivatives. In the factor belonging to $r = -\alpha^2$, of order m , write $\eta = x - r$ and

$$B_r = \mathbb{C}[\eta]/(\eta^m), \quad Z_r = \alpha \sum_{j=0}^{m-1} \binom{1/2}{j} \left(-\frac{\eta}{\alpha^2}\right)^j. \quad (\text{SP.8})$$

The formal binomial identity gives $Z_r^2 = \alpha^2 - \eta = -x$ in B_r . Its constant term is $\alpha \neq 0$, so Z_r is invertible. There are inverse algebra maps

$$\begin{aligned} B_r[z]/(z^2 + x) &\longrightarrow B_r \times B_r, & a + zb &\longmapsto (a + Z_rb, a - Z_rb), \\ (u, v) &\longmapsto \frac{u+v}{2} + z \frac{u-v}{2Z_r}. \end{aligned} \quad (\text{SP.9})$$

The two idempotents are $(1 \pm z/Z_r)/2$. Each component has length m , and corresponds respectively to the original zero $s = 1/2 + \alpha$ or $s = 1/2 - \alpha$. To check the identification on the full local algebras, in the first original factor put $w = s - (1/2 + \alpha)$. The map is $\eta = -2\alpha w - w^2$, with inverse $w = Z_r(\eta) - \alpha$. Its linear coefficient -2α is nonzero, and the two compositions are identities modulo the respective m -th powers. For the second factor replace α by $-\alpha$. This proves the full local isomorphisms without replacing a local algebra by evaluation at its centre. Taking their products gives (SP.4).

The entire functional equation $g(1-s) = g(s)$ also has an exact form in this variable. There is an entire function

$$\Phi(x) = \sum_{n=0}^{\infty} \frac{(-1)^n g^{(2n)}(1/2)}{(2n)!} x^n, \quad g(s) = \Phi\left(-\left(s - \frac{1}{2}\right)^2\right). \quad (\text{SP.10})$$

Indeed the Taylor series of the entire even function $g(1/2 + z)$ contains only even powers, and converges for every z ; its even-power series therefore has infinite radius in $x = -z^2$. By the local isomorphisms above, Φ vanishes to exactly order m_ρ at each r_ρ . Consequently

$$V(x) = (-1)^D \frac{\Phi(x)}{H(x)} \text{ is entire,} \quad v_h(s) := \frac{g(s)}{h(s)} = V(x), \quad v = j_h v_h, \quad \mathcal{T}(v) = (V \bmod H, 0). \quad (\text{SP.11})$$

The element $V \bmod H$ is a unit in A_H , because the removed orders were complete. We write U_V for its multiplication matrix, with all its Taylor coefficients modulo H retained.

22.2 The original measures and a Hilbert space isomorphism

The two original measures are

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad d\nu_h(t) = |v_h(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}. \quad (\text{SP.12})$$

Both densities are even. They have all polynomial moments, by the actual theta decay already established for the kernel construction. They are positive except at a discrete set, hence give a positive norm to every nonzero polynomial. Their pushforwards under $t \mapsto t^2$ are, on $x > 0$, exactly

$$\begin{aligned} d\lambda_0(x) &= \frac{|\Phi(x)|^2}{2\pi\sqrt{x}} dx, \\ d\lambda_h(x) &= \frac{|V(x)|^2}{2\pi\sqrt{x}} dx, \quad d\lambda_0(x) = |H(x)|^2 d\lambda_h(x). \end{aligned} \quad (\text{SP.13})$$

For example the two half-lines contribute $2 dt = dx/\sqrt{x}$; this proves the constants in (SP.13). The formulas use the entire quotients at removable zeros.

Theorem 22.2 (Both Hilbert components with an explicit inverse). *For either pair $(\nu, \lambda) = (\nu_h, \lambda_h)$ or (μ, λ_0) , the map*

$$\begin{aligned} \mathcal{U} : L^2(\lambda) \oplus L^2(x\lambda) &\longrightarrow L^2(\nu), \\ (a, b) &\longmapsto [t \mapsto a(t^2) + itb(t^2)] \end{aligned} \quad (\text{SP.14})$$

is unitary. Its inverse, defined for almost every $x > 0$, is

$$a(x) = \frac{f(\sqrt{x}) + f(-\sqrt{x})}{2}, \quad b(x) = \frac{f(\sqrt{x}) - f(-\sqrt{x})}{2i\sqrt{x}}. \quad (\text{SP.15})$$

Multiplication by the original coordinate $s = 1/2 + it$ is carried to $(a, b) \mapsto (a/2 - xb, a + b/2)$. Its exact domain is

$$a \in L^2(\lambda) \cap L^2(x\lambda), \quad b \in L^2(x\lambda) \cap L^2(x^2\lambda). \quad (\text{SP.16})$$

Reflection $f(t) \mapsto f(-t)$ becomes $(a, b) \mapsto (a, -b)$.

Proof. The two parity components are orthogonal. More explicitly their sum of squared absolute values at t and $-t$ is $2|a(t^2)|^2 + 2t^2|b(t^2)|^2$; integrating proves $\|\mathcal{U}(a, b)\|_\nu^2 = \|a\|_\lambda^2 + \|b\|_{x\lambda}^2$. The same identity applied to the even and odd parts of an arbitrary $f \in L^2(\nu)$ proves that (SP.15) belongs to the stated direct sum and is its inverse. There is no atom at $t = 0$, so no extra value is required there. Multiplication gives $(1/2 + it)(a + itb) = a/2 - xb + it(a + b/2)$. On the original space its domain is $f \in L^2(\nu)$ with $tf \in L^2(\nu)$, since $|1/2 + it|^2 = 1/4 + t^2$. Applying the same parity norm identity to tf gives exactly (SP.16), including both conditions on each component. The reflection formula follows by substitution. $\square$

Multiplication by the original h is also preserved: from $L^2(\mu)$ to $L^2(\nu_h)$ it is an isometry, and in (SP.14) it is $(a, b) \mapsto ((-1)^D H a, (-1)^D H b)$. It is onto as an L^2 map, since division by h off its finitely many real-line zeros has precisely the inverse norm identity; those points have measure zero. Its action on polynomial degree, however, remains the actual multiplication by the monic polynomial of degree $2D$. Thus no finite polynomial subspace is identified with one of a different degree without this explicit multiplication map.

22.3 The two monic families and their original norms

Let P_n be the monic orthogonal polynomial of degree n for λ_h , and let R_n be the monic polynomial for $x\lambda_h$. Set

$$p_n = \int |P_n|^2 d\lambda_h, \quad u_n = \int x |R_n|^2 d\lambda_h. \quad (\text{SP.17})$$

All these numbers are strictly positive; the zeroth ones retain the two masses. Orthogonal projection of x^n onto lower powers in the positive moment matrix proves existence and uniqueness. Since that matrix is real, all coefficients are real.

Theorem 22.3 (Monicity, the Christoffel transform, and exact norm ratios). *The original vertical-line monic polynomials are*

$$q_{2n}(s) = (-1)^n P_n(x), \quad q_{2n+1}(s) = (-1)^n z R_n(x), \quad \kappa_{2n} = p_n, \quad \kappa_{2n+1} = u_n. \quad (\text{SP.18})$$

For every $n \geq 0$, $P_n(0) \neq 0$, and

$$c_n = \frac{P_{n+1}(0)}{P_n(0)} < 0, \quad R_n(x) = \frac{P_{n+1}(x) - c_n P_n(x)}{x}, \quad u_n = -c_n p_n. \quad (\text{SP.19})$$

Consequently the original recurrence coefficients obey

$$a_{2n+1} = -c_n \quad (n \geq 0), \quad a_{2n} = \frac{p_n}{-c_{n-1} p_{n-1}} \quad (n \geq 1). \quad (\text{SP.20})$$

The same statements with $\lambda_0, P_n^0, R_n^0, p_n^0, u_n^0$ give the original theta norms $\mathfrak{h}_j$ exactly:

$$\mathfrak{h}_{2n} = p_n^0, \quad \mathfrak{h}_{2n+1} = u_n^0 = -c_n^0 p_n^0, \quad \frac{\mathfrak{h}_{2n+1}}{\mathfrak{h}_{2n}} = -c_n^0, \quad \frac{\mathfrak{h}_{2n+2}}{\mathfrak{h}_{2n+1}} = \frac{p_{n+1}^0}{-c_n^0 p_n^0}. \quad (\text{SP.21})$$

Proof. Since $x = -z^2$, the leading coefficient of each polynomial in (SP.18) in the original variable s is one. Lower-degree even polynomials correspond to lower-degree polynomials in x , and the odd ones correspond to z times lower-degree polynomials in x . Their mixed inner products vanish by parity. Equation (SP.14) therefore proves all the required orthogonality and norm identities, and uniqueness identifies the original q_j .

If $n > 0$ and $P_n(0) = 0$, write $P_n = xQ$, with Q real, monic and of degree $n - 1$. Orthogonality to Q would give $0 = \int x |Q|^2 d\lambda_h > 0$, a contradiction. For $n = 0$, $P_0(0) = 1$. The numerator in (SP.19) vanishes at zero; its quotient by x is monic of degree n . For each polynomial L of degree below n , its inner product with L for $x\lambda_h$ is $\int (P_{n+1} - c_n P_n) \bar{L} d\lambda_h = 0$. This proves the asserted polynomial formula. Since $R_n - P_n$ has lower degree, $\int R_n P_n d\lambda_h = p_n$, while $\int R_n P_{n+1} d\lambda_h = 0$. Multiplying its numerator by R_n and integrating proves $u_n = -c_n p_n$, and positivity proves $c_n < 0$. Taking successive quotients of (SP.18) gives (SP.20). The proof uses only positivity and the actual measure identity, so its application to λ_0 proves (SP.21). $\square$

22.4 Both full jet kernels and the exact weight operator

Fix an original source degree $N \geq 2D - 1$, and put $n_e = \lfloor N/2 \rfloor$, $n_o = \lfloor (N - 1)/2 \rfloor$. Use the actual residue map $p \mapsto v_j h p$ from degree at most N , with its norm from ν_h , onto the original $E_h = \mathbb{C}[s]/h$. Write G_N^{src} for its minimum metric in the original monomial coordinates. Thus at $N = d + m$ it is precisely the earlier G_m , including the fixed-section endpoint $m = -1$. Define the full columns

$$f_n = U_V P_n(X) c_H, \quad e_n = U_V R_n(X) c_H, \quad K_e = \sum_{n=0}^{n_e} \frac{f_n f_n^*}{p_n}, \quad K_o = \sum_{n=0}^{n_o} \frac{e_n e_n^*}{u_n}. \quad (\text{SP.22})$$

No column is replaced by scalar evaluation: $P_n(X)$, $R_n(X)$, and U_V act on the whole length- D algebra. Since $n_e, n_o \geq D - 1$, the first D monic polynomials in each family give a basis modulo H . The unit U_V is invertible, so $K_e, K_o > 0$. Set $G_e = K_e^{-1}$, $G_o = K_o^{-1}$.

Theorem 22.4 (The complete metric and a positive half-line comparison). *In the coordinates (SP.4), the actual minimum metric is*

$$T^{-*} G_N^{\text{src}} T^{-1} = \begin{pmatrix} G_e & 0 \\ 0 & G_o \end{pmatrix}, \quad \det G_N^{\text{src}} = \det G_e \det G_o. \quad (\text{SP.23})$$

For $W_N = A^* G_N^{\text{src}} + G_N^{\text{src}} A - G_N^{\text{src}}$, its full transported form and original relative allowance are

$$T^{-*} W_N T^{-1} = \begin{pmatrix} 0 & G_o - G_e X \\ G_o - X^* G_e & 0 \end{pmatrix}, \quad \epsilon_{N-d} = \left\| G_e^{-1/2} (G_o - G_e X) G_o^{-1/2} \right\|_{\text{op}}. \quad (\text{SP.24})$$

All zero eigenvalues and both signs of the relative form are retained: they are the positive and negative singular values of the displayed D -by- D matrix, with the remaining zero eigenvalues of total multiplicity twice its nullity.

Proof. For completeness, if a surjective linear map from a coefficient space with positive Gram matrix M to a target has matrix J , its minimum metric is $(JM^{-1}J^*)^{-1}$. Indeed the vector $M^{-1}J^*(JM^{-1}J^*)^{-1}y$ has image y ; it is orthogonal in M to $\ker J$, since its inner product with w in that kernel is $y^*(JM^{-1}J^*)^{-1}Jw = 0$. All other preimages differ by such a w , so Pythagoras proves the formula. The polynomial basis (SP.18) has diagonal source Gram matrix κ_j . Under T , its residue columns are $((-1)^n f_n, 0)$ or $(0, (-1)^n e_n)$. Thus the inverse minimum metric is exactly $\text{diag}(K_e, K_o)$, proving the first identity. The second follows from $|\det T| = 1$, proved in (SP.7).

Substitute (SP.7) and (SP.23) into the definition of W_N . The diagonal $1/2$ terms cancel its subtracted metric; multiplying the remaining two block matrices gives the first formula in (SP.24). Taking the congruence by $\text{diag}(G_e^{-1/2}, G_o^{-1/2})$ gives a Hermitian matrix $\begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$. If $Bv = \sigma u$ and $B^*u = \sigma v$, then (u, v) and $(u, -v)$ have eigenvalues σ and $-\sigma$. Singular value decomposition supplies all these vectors; the two kernels supply the remaining vectors. Its operator norm is consequently $\|B\|$. An invertible coordinate change preserves the quotient $|y^* W y| / (y^* G y)$; this proves the original relative allowance, including its zeros. $\square$

The positive comparison in (SP.24) can also be written entirely inside one copy of A_H . Define

$$R = G_e^{-1/2} G_o G_e^{-1/2} > 0, \quad L = G_e^{1/2} X G_e^{-1/2}. \quad (\text{SP.25})$$

Then

$$\epsilon_{N-d}^2 = \left\| (R - L) R^{-1} (R - L)^* \right\|_{\text{op}}. \quad (\text{SP.26})$$

To verify this without changing a metric implicitly, observe that the matrix B in (SP.24) is $(R - L) G_e^{1/2} G_o^{-1/2}$. The product of the last factor with its adjoint is R^{-1} , so $BB^* = (R - L) R^{-1} (R - L)^*$.

This proves (SP.26). In particular $\epsilon_{N-d} = 0$ holds precisely when $G_o = G_e X$. In that case X is similar to the positive Hermitian matrix R , hence is diagonalizable with positive real spectrum. The minimal polynomial of multiplication by x on A_H is H : a polynomial killing this operator kills the unit, and is therefore divisible by H . Diagonalizability then implies that H is squarefree. Thus a repeated original zero forces a strictly positive finite allowance for every N in this theorem. The complete operator in (SP.26) describes this effect of the retained nilpotents; no bound on its limiting size has been assumed.

Both finite updates are explicit. When N is even, passing to $N + 1$ leaves K_e fixed and adds $e_{n_o+1}e_{n_o+1}^*/u_{n_o+1}$ to K_o ; when N is odd it leaves K_o fixed and adds $f_{n_e+1}f_{n_e+1}^*/p_{n_e+1}$ to K_e . In either case, writing K for the changing block, v for the full added column and k for its positive norm, the determinant ratio is

$$\frac{\det G_{N+1}^{\text{src}}}{\det G_N^{\text{src}}} = \left(1 + \frac{v^* K^{-1} v}{k}\right)^{-1}. \quad (\text{SP.27})$$

Indeed $K + vv^*/k = K^{1/2}(I + ww^*)K^{1/2}$ with $w = K^{-1/2}v/\sqrt{k}$; the middle matrix is identity on $w^\perp$ and has eigenvalue $1 + \|w\|^2$ on its span. This argument also proves (SP.27) when $v = 0$, with ratio one.

22.5 The original tessierine coordinates and support lift

For an original zero $\rho = \beta + i\gamma$, let $\delta = \beta - 1/2$. The coordinate introduced here is exactly

$$r_\rho = \gamma^2 - \delta^2 - 2i\delta\gamma. \quad (\text{SP.28})$$

Retain the group-algebra element and its original characters

$$\begin{aligned} Q(\rho) &= \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa \quad \text{in } \mathbb{C}[K_4], \\ c_{++} &= 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1) = 4\delta, \quad c_{--} = 4i\gamma. \end{aligned} \quad (\text{SP.29})$$

Direct multiplication, with all constants unchanged, gives

$$r_\rho - \bar{r}_\rho = -4i\delta\gamma = -\frac{c_{-+}c_{--}}{4}. \quad (\text{SP.30})$$

Every actual γ here is nonzero. It follows that r_ρ is real if and only if $\beta = 1/2$; in that event $r_\rho = \gamma^2 > 0$. Thus the selected actual packet lies on the critical line exactly when the roots of its complete H are positive real numbers. Their multiplicities are still exactly the original ones. On an eigenvector of A at ρ , the relative Rayleigh quotient of W_N is $2\beta - 1$, because $u^* A^* G u + u^* G A u - u^* G u = (2\beta - 1)u^* G u$. Hence

$$|c_{-+}|/2 = 2|\delta| \leq \epsilon_{N-d}. \quad (\text{SP.31})$$

This is the one-factor spectral bound; the tensor and full-orbit multiplicity bounds proved earlier remain in their original scope.

Finally each algebra isomorphism (SP.4) and (SP.9) induces its exact split-zero isomorphism: send τ to τ and a supported amplitude a to the supported image of a . The semiring laws follow directly from the algebra homomorphism laws and absorption by τ ; the inverse is the same lift of the displayed inverse. In particular the target of the lift of (SP.9) is $G(B_r \times B_r)$, whose common external support is retained. The map to $G(B_r) \times G(B_r)$ sends τ to (τ, τ) and a supported pair (u, v) to its two supported coordinates. It is an injective semiring homomorphism, since both original amplitudes and the external point are recovered; its image consists exactly of the two coordinates both absent or both supported. The two mixed-support fibres remain explicit in the product target. Character evaluation $G(\mathbb{C}[K_4]) \rightarrow G(\mathbb{C})$ likewise sends the supported $Q(\rho)$ to supported c_{-+} . At a critical zero this image is the supported zero e , and the common external τ remains its separate absent fibre throughout these exact maps.

22.6 The complete rank-one half-line boundary and its derivative cost

The two positive kernels in (SP.22) have a further exact compatibility identity. Their finite discrepancy retains the full local algebra, even though its matrix rank is at most one. Use the kernels at the specified original degree N , and keep every column f_n, e_n defined in (SP.22). For the two possible degree parities one has

$$K_e - XK_o = \begin{cases} f_n e_n^* / p_n, & N = 2n, \\ -f_{n+1} e_n^* / u_n, & N = 2n + 1. \end{cases} \quad (\text{SP.32})$$

Here the matrices K_e, K_o in the first row have upper indices $n, n-1$, and those in the second row have upper indices n, n . In particular these are different finite kernels at the two parities.

Proof. The full residue recurrence and its conjugate give the one-factor inverse-metric boundary identity

$$AC_N + C_N A^* - C_N = \frac{z_{N+1}(a_{N+1} z_N)^* + (a_{N+1} z_N) z_{N+1}^*}{\kappa_{N+1}}, \quad C_N = (G_N^{\text{src}})^{-1}, \quad z_j = U_v[q_j]_h.$$

Indeed substitute $C_N = \sum_{j=0}^N z_j z_j^* / \kappa_j$ and $Az_j = z_{j+1} + b_j z_j - a_j z_{j-1}$. The diagonal cancels because $b_j + \bar{b}_j = 1$; each interior adjacent term cancels because $a_{j+1} / \kappa_{j+1} = 1 / \kappa_j$. Only the displayed last pair remains. Transform this identity by T on the left and T^* on the right. By (SP.7) and (SP.23), its upper right block is $K_e - XK_o$.

If $N = 2n$, the two transformed columns are $Tz_N = ((-1)^n f_n, 0)$ and $Tz_{N+1} = (0, (-1)^n e_n)$. Their signs cancel, and $a_{2n+1} / \kappa_{2n+1} = 1 / p_n$, giving the first row. If $N = 2n + 1$, they are $Tz_N = (0, (-1)^n e_n)$ and $Tz_{N+1} = ((-1)^{n+1} f_{n+1}, 0)$. Their product has the minus sign, and $a_{2n+2} / \kappa_{2n+2} = 1 / u_n$, giving the second row. No column has been assumed nonzero. This proves (SP.32) on the whole length- D algebra at both degree parities. $\square$

For the following scalar contractions use the same current metrics G_e, G_o . Equation (SP.32), multiplied on the left by G_e and on the right by G_o , gives the complete outer-product formula for $G_o - G_e X$. Its exact norm is

$$\epsilon_{N-d}^2 = \begin{cases} (f_n^* G_e f_n)(e_n^* G_o e_n) / p_n^2, & N = 2n, \\ (f_{n+1}^* G_e f_{n+1})(e_n^* G_o e_n) / u_n^2, & N = 2n + 1. \end{cases} \quad (\text{SP.33})$$

To prove this, the matrix in (SP.24) becomes the outer product of $G_e^{1/2} f$ and $G_o^{1/2} e$, divided by the displayed positive norm, with the displayed parity sign. For vectors a, b , the norm of ab^* is $\|a\| \|b\|$: Cauchy-Schwarz gives the upper bound, and the vector $b / \|b\|$ attains it when $b \neq 0$. If either vector is zero the equality remains true. This proves (SP.33). Thus the full relative weight has at most the two nonzero eigenvalues $\epsilon_{N-d}, -\epsilon_{N-d}$, and all its other eigenvalues are zero. The full packet and its nilpotents still enter each of the four quadratic contractions in (SP.33).

Let χ_N denote the trace cost of the original one-factor derivative map at source degree N , as in (QD.7), and put $\pi_N = \det G_{N+1}^{\text{src}} / \det G_N^{\text{src}}$. Define

$$\begin{array}{c|c|c} & \lambda_N & \chi_N \\ \hline N = 2n & \frac{e_n^* G_o e_n}{p_n} & \frac{u_n + e_n^* G_o e_n}{p_n^2} (f_n^* G_e f_n) \\ N = 2n + 1 & \frac{f_{n+1}^* G_e f_{n+1}}{p_{n+1}} & \frac{p_{n+1} + f_{n+1}^* G_e f_{n+1}}{u_n^2} (e_n^* G_o e_n). \end{array} \quad (\text{SP.34})$$

These formulas compute the actual cost, rather than choosing one. In the original one-column graph frame the next-layer Gram is $H_{\text{lay}} = \kappa_{N+1} + z_{N+1}^* G_N^{\text{src}} z_{N+1}$, and the derivative row has coefficient $a_{N+1} z_N^* G_N^{\text{src}} / \kappa_{N+1}$. Taking its squared Hilbert–Schmidt norm against the original metric gives $\chi_N = H_{\text{lay}}(z_N^* G_N^{\text{src}} z_N) / \kappa_N^2$. Inserting the two transformed columns just used proves (SP.34), including every original norm and sign. The determinant update (SP.27) now gives, without any division by a zero column,

$$\pi_N = (1 + \lambda_N)^{-1}, \quad \epsilon_{N-d}^2 = (1 - \pi_N) \chi_N. \quad (\text{SP.35})$$

Substitution from the two rows of (SP.34) verifies the second identity against (SP.33). If the new column vanishes then $\lambda_N = 0, \pi_N = 1$ while the derivative cost may persist; if the predecessor column vanishes then $\chi_N = 0$ while the new column may still contract the metric. These are the two original zero-frontier cases, carried through the full half-line kernel formulas.

23 The exact gamma reference and the arithmetic Gram determinants

The norm sequence in (KL.24) is attached to the actual measure

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi. \quad (\text{TG.1})$$

We construct an explicit gamma reference for this measure, prove its polynomial norms, and retain the full arithmetic multiplier in finite Gram determinants. This provides another exact formula for the original boundary allowance; no asymptotic estimate for an arithmetic determinant is assumed.

23.1 The original measure, phase, and packet cancellation

The defining identity for the completed function gives, on the original line $s = 1/2 + it$,

$$\begin{aligned} g(\tfrac{1}{2} + it) &= \Gamma(\tfrac{1}{4} + it/2) a(t), \\ a(t) &= -\pi^{-1/4} (t^2 + \tfrac{1}{4}) e^{-it \log \pi/2} \zeta(\tfrac{1}{2} + it). \end{aligned} \quad (\text{TG.2})$$

Indeed $s(s-1) = -(t^2 + 1/4)$ and $\pi^{-s/2} = \pi^{-1/4} e^{-it \log \pi/2}$. Both the minus sign and this phase remain in the multiplier.

For $\lambda > 0$, define

$$d\sigma_\lambda(t) = |\Gamma(\lambda + it/2)|^2 \frac{dt}{2\pi}. \quad (\text{TG.3})$$

In particular

$$d\mu = |a|^2 d\sigma_{1/4}, \quad d\nu_h = |a_h|^2 d\sigma_{1/4}, \quad a_h(t) = \frac{g(1/2 + it)/h(1/2 + it)}{\Gamma(1/4 + it/2)}. \quad (\text{TG.4})$$

The last expression defines the values at every zero of the original full packet h ; away from those points it equals $a(t)/h(1/2 + it)$. The function g/h is entire because the complete order of each selected zero is removed. Gamma has no zero and has no pole on the real t axis. Thus (TG.4) retains the actual cancellation at each critical packet centre, without introducing a pole into the measure.

Multiplication gives surjective Hilbert isometries

$$\begin{aligned} U_a : L^2(\mu) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto aF, \\ U_{a_h} : L^2(\nu_h) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto a_h F. \end{aligned} \quad (\text{TG.5})$$

The integral identity proves the isometry. The zero set of each multiplier on the real axis is discrete, since it comes from a nonzero analytic function. Given $H \in L^2(\sigma_{1/4})$, define $F = H/a$, respectively

H/a_h , off this null set and give F arbitrary finite values on it. Then $\int |F|^2 d\mu = \int |H|^2 d\sigma_{1/4}$, respectively the packet identity, proving surjectivity and the stated inverse. These are maps between the indicated Hilbert spaces. Their images of polynomial subspaces are exactly $a\mathbb{C}[t]$ and $a_h\mathbb{C}[t]$, with the displayed multipliers retained. Moreover the packet isometry from (KL.23) satisfies the exact diagram

$$U_{a_h}(h(1/2 + it)P(t)) = U_a P(t). \quad (\text{TG.6})$$

23.2 Gamma norms from a generating integral

We give the integral derivation to fix all constants independently of different conventions for Meixner–Pollaczek polynomials. The beta integral, with $r = e^x$, gives

$$\int_{\mathbb{R}} \frac{e^{\lambda x}}{(1 + e^x)^{2\lambda}} e^{iux} dx = \frac{\Gamma(\lambda + iu)\Gamma(\lambda - iu)}{\Gamma(2\lambda)}. \quad (\text{TG.7})$$

For completeness the beta integral follows by applying the change of variables $v = r_1 + r_2$, $y = r_1/(r_1 + r_2)$ to the product of the two absolutely convergent gamma integrals; the Jacobian is v . The conditions $\text{Re}(\lambda \pm iu) > 0$ hold for real u . The function on the left before Fourier transformation and its second derivative are integrable: at each end they are bounded by a constant times $e^{-\lambda|x|}$. Two integrations by parts therefore make its Fourier transform integrable. Fourier inversion applies. One direct justification is to multiply the transform by $e^{-\varepsilon u^2}$, apply Fubini, and use the Gaussian approximate identity; dominated convergence removes that factor because the transform is integrable. Using its evenness and then $t = 2u$, we obtain

$$\int_{\mathbb{R}} e^{iyt} d\sigma_{\lambda}(t) = \frac{2\Gamma(2\lambda)}{(2 \cosh y)^{2\lambda}} = 2^{1-2\lambda} \Gamma(2\lambda) (\cosh y)^{-2\lambda}. \quad (\text{TG.8})$$

The factor 2 in the numerator is the Jacobian of the original change $t = 2u$.

The following integral proves the required exponential integrability.

The function $(2 \cosh(x/2))^{-2\lambda}$ extends holomorphically to $|\text{Im } x| < \pi$, using the branch positive on the real axis. On each closed narrower strip it is bounded by $Ce^{-\lambda|\text{Re } x|}$. In (TG.7), for $u > 0$, shift the contour to $\mathbb{R} + ic$, where $0 < c < \pi$. The vertical sides of the truncated rectangle tend to zero by that bound. For $u < 0$, shift down instead. It follows that

$$|\Gamma(\lambda + iu)|^2 \leq C_{\lambda,c} e^{-c|u|}.$$

Consequently $\int e^{v|t|} d\sigma_{\lambda}(t) < \infty$ for every $0 < v < \pi/2$, by choosing $2v < c < \pi$. This proves holomorphic continuation of both sides of (TG.8) to $|\text{Im } y| < \pi/2$, where they agree by the identity theorem and use the branch inherited from the real axis.

For z near zero use the logarithms that vanish at $z = 0$ and set

$$F_{\lambda}(z, t) = (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} = \sum_{n \geq 0} P_n^{(\lambda)}(t/2; \pi/2) z^n. \quad (\text{TG.9})$$

For real z near zero this is $(1 + z^2)^{-\lambda} \exp(t \arctan z)$. Apply (TG.8) at $y = -i(\arctan z + \arctan w)$, where z, w are real and sufficiently small. The identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

then proves

$$\int_{\mathbb{R}} F_{\lambda}(z, t) F_{\lambda}(w, t) d\sigma_{\lambda}(t) = 2^{1-2\lambda} \Gamma(2\lambda) (1 - zw)^{-2\lambda}. \quad (\text{TG.10})$$

All derivatives in z, w at zero can be passed through the integral: on a smaller closed bidisc their integrands are bounded by a polynomial in $|t|$ times $e^{c|t|}$, with $c < \pi/2$. Thus comparison of the Taylor coefficients in (TG.10) is justified, not formal.

Define the monic polynomial and its unchanged squared norm by

$$b_n^{(\lambda)}(t) = n!P_n^{(\lambda)}(t/2; \pi/2), \quad \gamma_n^{(\lambda)} = 2^{1-2\lambda}n!\Gamma(n+2\lambda). \quad (\text{TG.11})$$

The coefficient of t^n in (TG.9) is $1/n!$, since $\arctan z = z + O(z^3)$. Consequently $b_n^{(\lambda)}$ is monic. Equation (TG.10) gives the full orthogonality identity

$$\int_{\mathbb{R}} b_n^{(\lambda)}(t)b_j^{(\lambda)}(t)d\sigma_{\lambda}(t) = \gamma_n^{(\lambda)}\delta_{nj}. \quad (\text{TG.12})$$

All these polynomials have real coefficients, as is also clear from the real form of the generating function. Differentiating (TG.9) gives $(1+z^2)\partial_z F_{\lambda} = (t-2\lambda z)F_{\lambda}$. Its coefficient of z^n proves

$$\begin{aligned} b_0^{(\lambda)} &= 1, & b_1^{(\lambda)} &= t, \\ b_{n+1}^{(\lambda)} &= tb_n^{(\lambda)} - n(n+2\lambda-1)b_{n-1}^{(\lambda)}, & n &\geq 1, \\ \gamma_n^{(\lambda)}/\gamma_{n-1}^{(\lambda)} &= n(n+2\lambda-1), & n &\geq 1. \end{aligned} \quad (\text{TG.13})$$

For the actual reference $\lambda = 1/4$, abbreviate $b_n = b_n^{(1/4)}$ and $\gamma_n = \gamma_n^{(1/4)}$. Thus

$$\boxed{\gamma_n = \sqrt{2}n!\Gamma(n+\frac{1}{2}), \quad \gamma_{n+1}/\gamma_n = (n+1)(n+\frac{1}{2}).} \quad (\text{TG.14})$$

These formulas agree, with the displayed Jacobian and monic factors, with NIST DLMF §18.19, equations (18.19.8)–(18.19.9), and §18.22, equation (18.22.8).

23.3 Finite arithmetic determinants in the retained metrics

For $N \geq 0$ let $N = \text{diag}(\gamma_0, \dots, \gamma_N)$, and keep both matrices

$$\begin{aligned} \mathbf{B}_N &= \left(\int_{\mathbb{R}} b_i(t)b_j(t)|a(t)|^2 d\sigma_{1/4}(t) \right)_{0 \leq i,j \leq N}, \\ \mathbf{J}_N &= {}_N^{-1/2} \mathbf{B}_N {}_N^{-1/2}, \quad D_N = \det \mathbf{J}_N, \quad D_{-1} = 1. \end{aligned} \quad (\text{TG.15})$$

The diagonal factors are explicit invertible coordinate maps; $\mathbf{B}_N = {}_N^{1/2} \mathbf{J}_N {}_N^{1/2}$ recovers every original inner product. The integrals converge with no unproved growth estimate. Indeed, integration of the step function $\lfloor x \rfloor$ gives, initially for $\text{Re } s > 1$, and then by analytic continuation for $\text{Re } s > 0, s \neq 1$,

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \{x\} x^{-s-1} dx.$$

The initial equality follows by integrating each constant step on $[n, n+1)$, or by summation by parts of the absolutely convergent Dirichlet series. The last integral converges absolutely and locally uniformly in the asserted half-plane since $0 \leq \{x\} < 1$. On $s = 1/2 + it$, it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4}$, because $|s/(s-1)| = 1$. Equation (TG.2) therefore bounds $|a(t)|$ by a polynomial of degree three. The exponential integrability just proved gives all its polynomial moments. The packet multiplier has the same property on the tail because h is monic, and is bounded on each compact interval by its cancelled analytic expression in (TG.4). Each matrix is positive definite: the integral of the squared modulus of a nonzero polynomial is positive, because a vanishes only at isolated real points.

Theorem 23.1 (The actual theta norms as arithmetic determinants). *For all $n \geq 0$, with the initial value included,*

$$\mathfrak{h}_n = \gamma_n \frac{D_n}{D_{n-1}}, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} = (n+1)(n+\frac{1}{2}) \frac{D_{n+1}D_{n-1}}{D_n^2}. \quad (\text{TG.16})$$

Proof. Let $Q_n(t)$ be the monic polynomial orthogonal to degrees below n in $L^2(\mu)$. Its existence and uniqueness follow from the positive definite Gram matrix. Expressing $Q_0, \dots, Q_N$ in $b_0, \dots, b_N$ gives an upper triangular matrix with diagonal entries one. The determinant of the Gram matrix is therefore unchanged, and orthogonality gives $\det \mathbf{B}_N = \prod_{j=0}^N \|Q_j\|_\mu^2$.

The algebra isomorphism $\Psi : \mathbb{C}[s] \rightarrow \mathbb{C}[t]$, $(\Psi p)(t) = p(1/2 + it)$, preserves the original integral inner product. On the affine space of degree- n monic polynomials the map $p \mapsto i^{-n} \Psi p$ is its monic version and preserves the norm because $|i^{-n}| = 1$. It takes all lower-degree polynomial subspaces onto themselves. Thus the original monic theta residual has squared norm $\mathfrak{h}_n = \|Q_n\|_\mu^2$. The degree-dependent phase is used on this affine space; it is not asserted to define a linear map on the entire polynomial algebra.

Finally $D_N = \det \mathbf{B}_N / \prod_{j=0}^N \gamma_j$. Divide this equality at $N = n$ by the equality at $N = n-1$, taking an empty product to be one. This proves the first formula. Dividing consecutive formulas and using (TG.14) proves the second. $\square$

Define $\mathbf{B}_N^{(h)}, \mathbf{J}_N^{(h)}, D_N^{(h)}$ by (TG.15) with the entire-cancelled multiplier a_h of (TG.4). The same proof, with no restriction on whether the packet centres are critical, gives

$$\kappa_n^{(h)} = \gamma_n \frac{D_n^{(h)}}{D_{n-1}^{(h)}}. \quad (\text{TG.17})$$

Combining (TG.17) and (TG.16) with (KL.24) yields the exact packet-to-reference determinant map, for every $m \geq 0$,

$$r_m^{(h)} = \frac{(d+m)! \Gamma(d+m+1/2)}{m! \Gamma(m+1/2)} \frac{D_{d+m}^{(h)}}{D_{d+m-1}^{(h)}} \frac{D_{m-1}}{D_m}. \quad (\text{TG.18})$$

Every determinant is strictly positive and the full degree $d = \sum_\rho m_\rho$ is retained.

For a packet stable under both ordinary conjugation and $\rho \mapsto 1 - \bar{\rho}$, and for every $m \geq 0$, (KL.32) becomes

$$\epsilon_m^2 = (m+1)(m+\frac{1}{2}) \frac{D_{m+1}D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{TG.19})$$

This is the exact same finite-stage allowance. Its factor $(m+1)(m+1/2)$ is explicit; the arithmetic determinant quotient remains in the formula.

23.4 Full confluent kernel and the original coordinates

Set $\hat{b}_j(s) = i^j b_j((s-1/2)/i)$. These are monic polynomials in the original variable s . With $\hat{\mathbf{r}}_N = \text{diag}(1, i, \dots, i^N)$, their actual packet Gram matrix is

$$\mathbf{B}_{N,s}^{(h)} = \hat{\mathbf{r}}_N^* \mathbf{B}_N^{(h)} \hat{\mathbf{r}}_N. \quad (\text{TG.20})$$

Indeed $\hat{b}_j(1/2 + it) = i^j b_j(t)$; taking the conjugate in the first argument gives exactly the displayed congruence, including every phase.

Let $\mathbf{C}_N$ have columns $[\hat{b}_j]_h \in E_h = \mathbb{C}[s]/(h)$, in the original monomial remainder coordinates. For $N \geq d-1$,

$$K_N = \mathbf{C}_N (\mathbf{B}_{N,s}^{(h)})^{-1} \mathbf{C}_N^*, \quad \mathbf{G}_{N-d} = U_{\varepsilon_h}^* K_N^{-1} U_{\varepsilon_h}. \quad (\text{TG.21})$$

Here $\mathbf{G}_{-1}$ is the fixed initial-section metric, and $\varepsilon_h = (j_h(g/h))^{-1}$ is the same full local unit as before. To prove the first identity, choose the actual monic orthogonal polynomial basis through degree N . If T is its invertible coefficient matrix in the $\hat{b}_j$ basis, then

$$T^* \mathbf{B}_{N,s}^{(h)} T = \text{diag}(\kappa_0, \dots, \kappa_N).$$

Inverting gives $(\mathbf{B}_{N,s}^{(h)})^{-1} = T \text{diag}(\kappa_j^{-1}) T^*$. The columns of $\mathbf{C}_N T$ are the full remainders $[p_j]_h$. Substitution proves exactly $K_N = \sum_{j=0}^N [p_j]_h [p_j]_h^* / \kappa_j$, the original interpolation kernel. The first d monic polynomials span all remainders, so $\mathbf{C}_N$ is onto and K_N is positive definite. The second identity is the fixed-unit minimum-lift formula already proved in (KL.27), with the exact index $N = d + m$. No single remainder column is required to be nonzero or invertible.

For a centre ρ , put $t_\rho = (\rho - 1/2)/i$ and write $f^{[\ell]} = f^{(\ell)}/\ell!$. The full Taylor coordinates of these columns are explicitly

$$\hat{b}_j^{[\ell]}(\rho) = i^{j-\ell} b_j^{[\ell]}(t_\rho), \quad 0 \leq \ell < m_\rho. \quad (\text{TG.22})$$

This follows by differentiating the affine argument ℓ times. CRT converts these full Taylor coordinates to the monomial remainder coordinates, and multiplication by U_{v_h} retains all derivatives of g/h . Thus (TG.21) computes the entire confluent kernel, including nilpotent orders and critical centres.

23.5 The finite map to Romik's gamma parameter

Romik's expansion uses $P_n^{(3/4)}(t/2; \pi/2)$, with the unchanged variable $t/2$; see his §3 and Appendix A.2. The exact finite change to the parameter in (TG.3) is

$$\begin{aligned} b_n^{(1/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{1/2}{k} b_{n-2k}^{(3/4)}(t), \\ b_n^{(3/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{-1/2}{k} b_{n-2k}^{(1/4)}(t). \end{aligned} \quad (\text{TG.23})$$

To prove these formulas, divide the two generating functions: $F_{1/4}(z, t) = (1 + z^2)^{1/2} F_{3/4}(z, t)$. The binomial series at zero and coefficient comparison prove the first formula. Multiplication by $(1 + z^2)^{-1/2}$ proves the inverse. Each finite coefficient matrix is triangular with diagonal one, preserves parity, and is invertible. Applied to the same measure, it transports Gram matrices by its full congruence and transports every Taylor column by the same coefficient matrix. Consequently the kernel expression (TG.21) is invariant under this change of polynomial basis.

The two reference measures themselves have the exact Hilbert-space map

$$L^2(\sigma_{1/4}) \longrightarrow L^2(\sigma_{3/4}), \quad F(t) \longmapsto \frac{\Gamma(1/4 + it/2)}{\Gamma(3/4 + it/2)} F(t). \quad (\text{TG.24})$$

The gamma quotient has neither zeros nor poles for real t . Cancellation of its squared modulus proves the isometry and its inverse; this isometry has a multiplier, whereas (TG.23) is the finite polynomial coordinate map. Their respective domains and metrics have both been given explicitly.

Romik's coefficient asymptotics concern the expansion of Ξ in a fixed gamma-orthogonal basis. Inoue's §2 similarly treats expansion coefficients and convergence. In the present construction the arithmetic norms are the determinants (TG.16). Equations (TG.20)–(TG.24) are the exact maps connecting those polynomial presentations to these determinants; no assertion that their coefficient asymptotics already estimate D_n is used.

23.6 Exact scope of the exponential-weight theorem

The original theorem of Lubinsky–Mhaskar–Saff does include tail order one. In their 1988 paper, Definition 2.1 requires $W = e^{-Q}$, with Q even and continuous, Q' existing on $(0, \infty)$, $xQ'(x)$ bounded at zero, and, eventually,

$$Q' > 0, \quad x^2|Q'''|/Q' \leq C, \quad 1 + xQ''/Q' \rightarrow \alpha > 0. \quad (\text{TG.25})$$

Theorem 2.3 admits a finite generalized Jacobi factor, a lower-degree polynomial exponential, and a bounded nonnegative factor $\Psi(t) \rightarrow 1$. Its Theorem 2.4 replaces the polynomial exponential by V , but requires $\log V(t)/Q(t) \rightarrow 0$, as well as bounded reciprocal approximation on the expanding interval by polynomials of degree $o(n)$. The precise statements are on printed pages 67–70; their proof is on pages 80–81. These hypotheses concern the full weight, not its gamma factor alone.

We prove a concrete obstruction to inserting the actual arithmetic multiplier into those displayed hypotheses. Hardy’s theorem supplies real numbers $T_j \rightarrow \infty$ with $g(1/2 + iT_j) = 0$. Removing a fixed full polynomial packet leaves an infinite subsequence of these zeros. At each remaining T_j , the multiplier a_h of (TG.4) has the local form

$$a_h(t) = (t - T_j)^{\ell_j} v_j(t), \quad \ell_j \geq 1, \quad v_j(T_j) \neq 0, \quad (\text{TG.26})$$

where v_j is analytic in a neighbourhood of T_j . This follows from the Taylor series at an isolated zero, with its entire order ℓ_j retained.

Let $B(t) > 0$ be any continuous comparison amplitude near all sufficiently large T_j , and suppose it has no zeros there. For every j and every $\eta > 0$, continuity and (TG.26) give a nonempty interval $I_{j,\eta}$ about T_j on which

$$|a_h(t)|/B(t) < \eta. \quad (\text{TG.27})$$

Thus a decomposition with this ratio as the bounded multiplier Ψ cannot satisfy $\Psi \rightarrow 1$. The obstruction holds on intervals of positive measure, so changing the value only at each zero does not repair it. A finite generalized Jacobi factor has only finitely many real zeros or singularities; choosing the subsequence beyond them leaves (TG.27) unchanged.

For the logarithmic hypothesis choose any eventually positive continuous Q . At a remaining T_j , shrink the neighbourhood so that $Q(t) \leq Q(T_j) + 1$, and choose $\eta = \exp(-j(Q(T_j) + 1))$ in (TG.27). On the resulting positive-measure interval,

$$\frac{\log(|a_h(t)|/B(t))}{Q(t)} < -j \quad (t \neq T_j). \quad (\text{TG.28})$$

Hence this ratio cannot be V with $\log V/Q \rightarrow 0$. An additional factor $\Psi \rightarrow 1$ is bounded away from zero and infinity on the tail and does not remove the contradiction. This disproves the indicated application of Theorems 2.3–2.4, including almost-everywhere modifications of the weight. It makes no claim that other asymptotic methods are unavailable. The exact positive Gram matrices (TG.15)–(TG.21) retain the arithmetic information at all those zeros and are available for such further analysis.

23.7 Primary source locators

The content-reading route and source hashes are recorded in `work/theta_norm_literature_route_20260912.json` and `work/theta_norm_literature_sources/download_receipt.json`. The principal public locators are:

- D. S. Lubinsky, H. N. Mhaskar and E. B. Saff, *A proof of Freud’s conjecture for exponential weights*, Constructive Approximation 4 (1988), 65–83, Definition 2.1, Theorems 2.3–2.5 and §4: <https://math.vanderbilt.edu/saffeb/texts/81.pdf>.

- Dan Romik, *Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases*, §3.1–3.3 and Appendix A.2: <https://www.math.ucdavis.edu/~romik/data/uploads/papers/riemannxi-final.pdf>.
- Hiroto Inoue, *Expansion of Dirichlet L-function on the critical line in Meixner–Pollaczek polynomials*, arXiv:1412.1220, §2.1–2.2: <https://arxiv.org/abs/1412.1220>.
- NIST DLMF, equations (18.19.8)–(18.19.9), (18.22.8): <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22>.
- G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, Comptes rendus 158 (1914), 1012–1014, §1–3. The exact page images and transcriptions are linked in the machine-readable source route.

24 The symmetric spectral sum and its unequal Stieltjes pair

The complete PR 17 note, *Spectral-sum descent with the relative arithmetic fibre retained*, supplies the invariant relation ideal and the original matrix weight used below. We prove their further reflection decomposition, its full paired-packet defect, and the minimum-metric and weight formulas. The coordinate x in this section belongs to the spectral sum. We write x_h for the one-factor coordinate of Section 22; their exact relation on the defect will be proved.

24.1 Original polynomial coordinates and the reflection ideals

Keep $g = 2\xi$, the same nonempty quartet-closed packet h with every full order, $d = 2D$, and

$$h(s) = (-1)^D H(-(s - \frac{1}{2})^2), \quad A_H = \mathbb{C}[x_h]/(H), \quad v_h = g/h, \quad v = j_h v_h. \quad (\text{SSP.1})$$

In particular $D \geq 1$ and $H(0) \neq 0$, as proved in (SP.1)–(SP.3). Write

$$\begin{aligned} S &= s_1 + s_2, & r &= s_1 - s_2, & \Delta &= r^2, & Z &= S - 1, & x &= -Z^2, \\ s_1 &= \frac{1}{2} + (Z + r)/2, & s_2 &= \frac{1}{2} + (Z - r)/2. \end{aligned} \quad (\text{SSP.2})$$

The swap fixes S and negates r . Thus substitution gives a filtered algebra isomorphism $\mathbb{C}[S, \Delta] \rightarrow \mathbb{C}[s_1, s_2]^{S_2}$, with $\deg S = 1$, $\deg \Delta = 2$: expansion in r keeps precisely its even powers, and the inverse coordinate map in (SSP.2) keeps the factors $1/2$.

Write $h(1/2 + y) = \sum_{j=0}^D b_j y^{2j}$, with $b_D = 1$. Define the following entire finite polynomials:

$$\begin{aligned} A_0(x, \Delta) &= \sum_{j=0}^D b_j 4^{-j} \sum_{\ell=0}^j \binom{2j}{2\ell} (-x)^{j-\ell} \Delta^\ell, \\ A_1(x, \Delta) &= \sum_{j=1}^D b_j 4^{-j} \sum_{\ell=0}^{j-1} \binom{2j}{2\ell+1} (-x)^{j-\ell-1} \Delta^\ell, \quad Q_\Delta = \Delta A_1. \end{aligned} \quad (\text{SSP.3})$$

Binomial expansion, with $x = -Z^2$, proves exactly

$$\begin{aligned} h(s_1) &= A_0 + rZ A_1, & h(s_2) &= A_0 - rZ A_1, \\ A_0(0, \Delta) &= (-1)^D H(-\Delta/4), & A_1(0, \Delta) &= \frac{(-1)^{D+1}}{2} H'(-\Delta/4). \end{aligned} \quad (\text{SSP.4})$$

For the last identity one can also differentiate the difference $h(1/2 + (Z + r)/2) - h(1/2 + (Z - r)/2)$ at $Z = 0$, and divide by $2r$; both sides are polynomials, so the identity includes $r = 0$. The polynomial $A_0(0, \Delta)$ has degree D and leading coefficient 4^{-D} . No localization at r , Δ , Z , or x has been made.

Theorem 24.1 (The original symmetric algebra with both parity modules). *Let $\mathcal{E} = (E_h \otimes E_h)^{S_2}$, $E_h = \mathbb{C}[s]/(h)$. There is an exact algebra isomorphism*

$$\mathcal{E} \cong \mathbb{C}[x, Z, \Delta]/(Z^2 + x, A_0, ZQ_\Delta). \quad (\text{SSP.5})$$

Put

$$B_e = \mathbb{C}[x, \Delta]/(A_0, xQ_\Delta), \quad B_o = \mathbb{C}[x, \Delta]/(A_0, Q_\Delta). \quad (\text{SSP.6})$$

The underlying module of (SSP.5) is $B_e \oplus ZB_o$. The two maps and the product are

$$\begin{aligned} \beta : B_e &\rightarrow B_o, & [a] &\mapsto [a], & \alpha : B_o &\rightarrow B_e, & [b] &\mapsto [xb], \\ (a, b)(c, e) &= (ac - \alpha(be), \beta(a)e + \beta(c)b), \\ \alpha\beta &= xI_{B_e}, & \beta\alpha &= xI_{B_o}. \end{aligned} \quad (\text{SSP.7})$$

The map β is a unital algebra homomorphism; α is a homomorphism of $\mathbb{C}[x, \Delta]$ -modules. Its full multiplicative relation is $\alpha(b)\alpha(c) = x\alpha(bc)$, and $\alpha(1) = x$. Reflection in both original variables acts as $(a, b) \mapsto (a, -b)$. The dimensions are

$$\dim B_e = D(D + 1), \quad \dim B_o = D^2. \quad (\text{SSP.8})$$

Proof. In the polynomial algebra $\mathbb{C}[S, r]$, the original ideal $(h(s_1), h(s_2))$ is (H_0, rH_1) , where $H_0 = A_0(-Z^2, \Delta)$ and $H_1 = ZA_1(-Z^2, \Delta)$. The factors $1/2$ in the sum and difference of the original generators give both ideal containments. An invariant polynomial in this ideal can be written $H_0a + rH_1b$. Average this expression under $r \mapsto -r$. The result is $H_0a_e + \Delta H_1b_o$, where a_e and b_o are polynomials in S, Δ ; the odd part of b is rb_o . Conversely both invariant generators lie in the original ideal. Every invariant class in the quotient has an invariant lift by this same averaging. Therefore

$$\mathcal{E} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1). \quad (\text{SSP.9})$$

Replace S by $1 + Z$ and adjoin $x = -Z^2$; this proves (SSP.5) with inverse given by the original substitutions.

Before imposing A_0, ZQ_Δ , division by $Z^2 + x$ gives the free module $\mathbb{C}[x, \Delta] \oplus Z\mathbb{C}[x, \Delta]$. Multiplying the two remaining generators by arbitrary elements gives

$$A_0(a + Zb) + ZQ_\Delta(c + Ze) = A_0a - xQ_\Delta e + Z(A_0b + Q_\Delta c). \quad (\text{SSP.10})$$

Its even and odd coefficient ideals are precisely those in (SSP.6). This proves the module decomposition, including both directions of its defining relations. Multiplication by Z is $(a, b) \mapsto (-\alpha b, \beta a)$, so direct multiplication proves (SSP.7) and the two compositions. All operations are well-defined by the displayed ideals.

Reflection sends Z, r to $-Z, -r$, and fixes x, Δ . The one-factor decomposition (SP.4) has reflection spaces of dimensions D, D . On the invariant tensor square the positive space is the sum of their two symmetric squares, with dimension $2D(D + 1)/2$. The negative space consists of their mixed symmetrized tensors, with dimension D^2 . They are exactly the even and odd coefficient modules in (SSP.10), proving (SSP.8), including $D = 1$. $\square$

24.2 The entire defect is the original paired packet

Theorem 24.2 (Two exact sequences with every original local order). *The map α is injective and β is surjective. Regard A_H as a $\mathbb{C}[x, \Delta]$ -module through $x \mapsto 0, \Delta \mapsto -4x_h$. The two exact sequences of modules are*

$$\begin{aligned} 0 &\longrightarrow B_o \xrightarrow{\alpha} B_e \xrightarrow{\rho_0} A_H \longrightarrow 0, & \rho_0[a] &= [a(0, -4x_h)], \\ 0 &\longrightarrow A_H \xrightarrow{\iota_0} B_e \xrightarrow{\beta} B_o \longrightarrow 0, & \iota_0[q] &= [Q_\Delta q(-\Delta/4)]. \end{aligned} \quad (\text{SSP.11})$$

In particular both defects retain the original polynomial H and every order in it. On the full algebra $\mathcal{E}$, multiplication by the original sum is

$$A_S = I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}. \quad (\text{SSP.12})$$

Its full kernel and cokernel at $S = 1$ are

$$\ker(A_S - I) = \{(\iota_0 q, 0) : q \in A_H\}, \quad \text{coker}(A_S - I) \cong A_H, \quad [(a, b)] \mapsto \rho_0 a. \quad (\text{SSP.13})$$

Proof. Let $\mathcal{R} = \mathbb{C}[x, \Delta]$. Since $A_0(0, \Delta) \neq 0$, the prime polynomial x does not divide A_0 . If $xb = A_0c + xQ_\Delta e$, then $x(b - Q_\Delta e) = A_0c$ forces $c = xc_1$, whence $b = A_0c_1 + Q_\Delta e$. This proves injectivity of α without dividing by x in either quotient. Surjectivity of β follows from its quotient definition. The image of α is xB_e , because every polynomial representative of an even class is also an odd coefficient representative. Therefore

$$\text{coker } \alpha \cong \mathcal{R}/(A_0, x) \cong \mathbb{C}[\Delta]/((-1)^D H(-\Delta/4)) \cong A_H. \quad (\text{SSP.14})$$

The last isomorphism sends Δ to $-4x_h$; its inverse sends x_h to $-\Delta/4$. This proves the first sequence, with all constants.

We justify the cancellation required for the second sequence. The quotient $\mathcal{R}/(A_0, Q_\Delta) = B_o$ has finite dimension D^2 , already proved in (SSP.8). If a nonconstant irreducible polynomial divided both A_0, Q_Δ , then it would divide every polynomial annihilator of x and every polynomial annihilator of Δ in this quotient. Such nonzero univariate annihilators exist by linear dependence of the powers in the finite vector space. Over $\mathbb{C}$, a common irreducible divisor of a nonzero polynomial in x alone and one in Δ alone is impossible: their irreducible factors are respectively $x - a$ and $\Delta - b$. Thus $\gcd(A_0, Q_\Delta) = 1$ in $\mathcal{R}$.

The kernel of β consists exactly of the classes $[Q_\Delta p]$ in B_e . Multiplication by Q_Δ gives a surjection

$$\mathcal{R}/(A_0, x) \longrightarrow \ker \beta, \quad [p] \longmapsto [Q_\Delta p]. \quad (\text{SSP.15})$$

It is well-defined since $A_0 Q_\Delta = x Q_\Delta = 0$ in B_e . If $Q_\Delta p = A_0 c + x Q_\Delta e$, the coprimality just proved forces $A_0 \mid p - xe$. Hence $[p] = 0$ in $\mathcal{R}/(A_0, x)$, proving injectivity. Compose (SSP.15) with the inverse last map of (SSP.14) to obtain exactly ι_0 in (SSP.11). Finally multiplication by Z is the off-diagonal part of (SSP.12). Its kernel has odd component zero by injectivity of α , and even component $\ker \beta$. Its cokernel has zero odd quotient by surjectivity of β , and even quotient $\text{coker } \alpha$. This proves (SSP.13). $\square$

The full multiplication on the second defect is also explicit. Put

$$\mathfrak{c}_H = 2(-1)^D x_h H'(x_h) \pmod{H}.$$

Then, for every $q, t \in A_H$,

$$\iota_0(q)\iota_0(t) = \iota_0(\mathfrak{c}_H qt). \quad (\text{SSP.15a})$$

Indeed $xQ_\Delta = 0$ gives $Q_\Delta^2 = Q_\Delta Q_\Delta(0, \Delta)$ in B_e . Equation (SSP.4), followed by $\Delta = -4x_h$, identifies $Q_\Delta(0, -4x_h)$ with $\mathfrak{c}_H$. Substitution in the two products proves (SSP.15a). Thus the module isomorphism ι_0 transports the inherited ideal multiplication to $q \star t = \mathfrak{c}_H qt$, retaining the derivative and its nilpotent values.

The generator in (SSP.11) is the full class $Q_\Delta = \Delta A_1$ modulo (A_0, xQ_Δ) . Reducing its coefficient at $x = 0$ before forming this class would change the map. In particular (SSP.4) can vanish on repeated local fibres through H' ; the proved injective map ι_0 still retains those fibres. Equation (SSP.15) specifies the complete mixed nilpotent class that carries them.

24.3 The original unit and all coordinate factorials

The original full unit on $\mathcal{E}$ is

$$v^{(2)} = [v_h(s_1)v_h(s_2)] \in \mathcal{E}^\times. \quad (\text{SSP.16})$$

This bracket denotes its complete double Taylor class modulo $(h(s_1), h(s_2))$, followed by the invariant identification; the values of all admitted derivatives are retained. Since $d = 2D$, $v_h(1-s) = v_h(s)$. Thus this unit has the form $(u_e, 0)$ under (SSP.5), and its two multiplication maps are

$$U_e = M_{u_e} : B_e \rightarrow B_e, \quad U_o = M_{\beta(u_e)} : B_o \rightarrow B_o. \quad (\text{SSP.17})$$

Both are invertible, by multiplication with the components of the original inverse unit. Their exact intertwiners and defect actions are

$$\begin{aligned} U_e \alpha &= \alpha U_o, & \beta U_e &= U_o \beta, & \rho_0 U_e &= M_{V(x_h)^2} \rho_0, \\ U_e \iota_0 &= \iota_0 M_{V(x_h)^2}, & V(x_h) &= (-1)^D \Phi(x_h)/H(x_h). \end{aligned} \quad (\text{SSP.18})$$

Indeed the first two equations are the module multiplication laws in (SSP.7). The algebra homomorphism $\mathcal{E} \rightarrow A_H$ given by $(a, b) \mapsto \rho_0 a$ is the original specialization $s_2 = 1 - s_1$, followed by its even quotient: $Z = 0$, $\Delta = 4(s_1 - 1/2)^2 = -4x_h$. It sends (SSP.16) to $v_h(s_1)v_h(1-s_1) = V(x_h)^2$, proving the third equation with all jets. For the last equation, $xQ_\Delta = 0$ implies

$$u_e Q_\Delta q(-\Delta/4) = Q_\Delta u_e(0, \Delta) q(-\Delta/4). \quad (\text{SSP.19})$$

The same specialization proves (SSP.18). In particular the factor $g = 2\xi$ occurs in each original unit factor and is present in the squared defect unit.

Here are explicit finite coordinate maps to the unscaled symmetric basis used earlier. Let $T : E_h \rightarrow A_H \oplus z_h A_H$ be the exact matrix (SP.6), with $z_h = s - 1/2$. Use the bases $a_i = x_h^i$ and $b_i = z_h x_h^i$, $0 \leq i < D$, in that target. The new symmetric basis is ordered in these three groups:

$$\begin{aligned} a_i \odot a_j \ (i \leq j), & \quad b_i \odot b_j \ (i \leq j), & \quad a_i \odot b_j \ (0 \leq i, j < D), \\ u \odot v = u \otimes v + v \otimes u \ (u \neq v), & \quad u \odot u = u \otimes u. \end{aligned} \quad (\text{SSP.20})$$

The first two groups are the basis of B_e , and the last, after removal of its displayed Z , is the basis of B_o . These identifications follow by the inverse substitutions in (SSP.2) and (SP.4): swap invariance leaves even powers of r , and reflection leaves even powers of Z in the first two groups and odd powers in the last. Replace $Z^{2j} r^{2a}$ by $(-x)^j \Delta^a$, and $Z^{2j+1} r^{2a}$ by $Z(-x)^j \Delta^a$. This finite coefficient prescription gives each basis polynomial explicitly, without choosing values at roots.

Let I_{old} and I_{new} insert the literal orbit sums in, respectively, the original monomial tensor basis and the basis (SSP.20) before passage to its invariant space. Put

$$\begin{aligned} D_{\text{old}} &= I_{\text{old}}^* I_{\text{old}}, & L_{\text{old}} &= D_{\text{old}}^{-1} I_{\text{old}}^*, \\ D_{\text{new}} &= I_{\text{new}}^* I_{\text{new}}, & L_{\text{new}} &= D_{\text{new}}^{-1} I_{\text{new}}^*, \\ T_s &= L_{\text{new}}(T \otimes T) I_{\text{old}}, & T_s^{-1} &= L_{\text{old}}(T^{-1} \otimes T^{-1}) I_{\text{new}}. \end{aligned} \quad (\text{SSP.21})$$

Each diagonal entry of D_{old} , D_{new} is one for a repeated basis vector and two for two distinct vectors. These are the original orbit cardinalities. The products IL are both the permutation average $(I + \text{swap})/2$; it commutes with $T \otimes T$. Hence the two displayed matrices compose to identity. This proves all factors in the coordinate maps. Moreover

$$|\det T_s| = |\det T|^{d+1} = 1. \quad (\text{SSP.22})$$

To verify the exponent, triangularize T over $\mathbb{C}$; on symmetric tensors its diagonal entries are the products of its diagonal entries indexed by $i \leq j$. Each original diagonal entry occurs $d + 1$ times. The similarity inducing that triangularization and the final permutation into (SSP.20) do not change the absolute determinant. Equation (SP.7) then proves (SSP.22). No tensor basis vector has been divided by its length.

24.4 The actual matrix weight and its positive half-line pair

Retain the original density

$$w_h(t) = |v_h(1/2 + it)|^2 / (2\pi). \quad (\text{SSP.23})$$

It is even and has the exponential and polynomial integrability proved for the theta source. For a fixed source degree M , put $L_M = \lfloor M/2 \rfloor$ and keep every relative direction $0 \leq a \leq L_M$. The PR 17 matrix weight is exactly

$$W_{ac}(u) = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv. \quad (\text{SSP.24})$$

For each u , this finite matrix is real Hermitian and strictly positive. Indeed its quadratic form on a nonzero vector c is one half of the integral of $|\sum_a c_a (-v^2)^a|^2$ against the two positive densities. Their zeros are discrete, while that polynomial is nonzero, proving strict positivity. The actual decay gives finite integrals. Evenness of w_h gives $W(-u) = W(u)$, with every matrix entry retained.

For a symmetric original numerator

$$P(S, \Delta) = \sum_{a=0}^{L_M} \Delta^a f_a(S), \quad \deg f_a \leq M - 2a,$$

the original coordinate change $u = t_1 + t_2$, $v = t_1 - t_2$ has $dt_1 dt_2 = du dv / 2$, and $S = 1 + iu$, $\Delta = -v^2$. Expansion and Fubini, justified by the actual moment bounds, give

$$\|\mathcal{T}_h^{(2)} P\|^2 = \int_{\mathbb{R}} f(1 + iu)^* W(u) f(1 + iu) du. \quad (\text{SSP.25})$$

Here $\mathcal{T}_h^{(2)} P = P(D_1 + D_2, (D_1 - D_2)^2) F_h^{\otimes 2}$; its Mellin multiplier is $P(S, \Delta) v_h(s_1) v_h(s_2)$. Thus (SSP.25) includes both factors $1/(2\pi)$, the real Jacobian $1/2$, and every relative-coordinate sign.

Define the matrix-valued measure on $x > 0$ by

$$d\Lambda(x) = W(\sqrt{x}) \frac{dx}{\sqrt{x}}. \quad (\text{SSP.26})$$

For a matrix measure $d\Lambda = W_\Lambda dx$, its squared L^2 norm is $\int f^* W_\Lambda f dx$; every component has the declared dimension $L_M + 1$. The full unitary map and inverse are

$$\begin{aligned} \mathcal{U}_\Sigma : L^2(\Lambda) \oplus L^2(x\Lambda) &\longrightarrow L^2(\mathbb{R}, W(u) du), \\ (A, B) &\longmapsto f(1 + iu) = A(u^2) + iuB(u^2), \\ A(x) &= \frac{f(1 + i\sqrt{x}) + f(1 - i\sqrt{x})}{2}, \\ B(x) &= \frac{f(1 + i\sqrt{x}) - f(1 - i\sqrt{x})}{2i\sqrt{x}}. \end{aligned} \quad (\text{SSP.27})$$

To prove the norm identity, add the two quadratic forms at u and $-u$. The cross terms cancel and give $2A^*WA + 2u^2B^*WB$. The substitution $x = u^2$ proves the direct-sum norm with exactly (SSP.26). The same calculation on the parity parts of an arbitrary f proves that the displayed inverse belongs to both asserted spaces. There is no atom at zero, so the almost-everywhere inverse requires no value there. Multiplication by $S = 1 + iu$ is

$$(A, B) \longmapsto (A - xB, A + B), \quad \begin{cases} A \in L^2(\Lambda) \cap L^2(x\Lambda), \\ B \in L^2(x\Lambda) \cap L^2(x^2\Lambda). \end{cases} \quad (\text{SSP.28})$$

Its exact domain follows by adding $\|uf\|^2$ to $\|f\|^2$, since $|1 + iu|^2 = 1 + u^2$; the same parity calculation gives the two additional weighted norms.

The finite source image in (SSP.27) is precisely

$$\deg A_a \leq \left\lfloor \frac{M-2a}{2} \right\rfloor, \quad \deg B_a \leq \left\lfloor \frac{M-2a-1}{2} \right\rfloor \quad (0 \leq a \leq L_M). \quad (\text{SSP.29})$$

A negative degree bound means the zero polynomial. In particular the last odd relative direction can have a zero allowed polynomial space while its ambient Hilbert component is still retained. The exact coefficient maps are obtained by expanding each original $S^j = (1+Z)^j$:

$$(1+Z)^j = \sum_{2\ell \leq j} \binom{j}{2\ell} (-x)^\ell + Z \sum_{2\ell+1 \leq j} \binom{j}{2\ell+1} (-x)^\ell. \quad (\text{SSP.30})$$

The inverse replaces x by $-(S-1)^2$, Z by $S-1$ in $\sum_a \Delta^a(A_a(x) + ZB_a(x))$. This proves both (SSP.29) and preservation of the original weighted total-degree filtration.

24.5 Full-jet kernels for the actual constrained minimum

Fix $M \geq 2(d-1) = 4D-2$. Let $\mathcal{I}_e = \{(a,j) \in \mathbb{N}^2 : 2a+2j \leq M\}$, and $\mathcal{I}_o = \{(a,j) \in \mathbb{N}^2 : 2a+2j+1 \leq M\}$. The positive source Gram matrices and full residue columns are

$$\begin{aligned} (M_e)_{(a,j),(c,\ell)} &= \int_0^\infty x^{j+\ell} d\Lambda_{ac}(x), & F_{a,j} &= u_e[\Delta^a x^j]_{B_e}, \\ (M_o)_{(a,j),(c,\ell)} &= \int_0^\infty x^{j+\ell+1} d\Lambda_{ac}(x), & E_{a,j} &= \beta(u_e)[\Delta^a x^j]_{B_o}. \end{aligned} \quad (\text{SSP.31})$$

Use the exact bases (SSP.20) for these columns and let J_e, J_o be the corresponding column matrices. Each Gram is strictly positive: a nonzero coefficient vector gives a nonzero polynomial vector in (SSP.29), whose squared norm is positive for the strictly positive matrix density. All its moments are finite by (SSP.24)–(SSP.26).

The two jet maps are onto at the stated degree. Indeed the original symmetric orbit basis $s_1^i s_2^j + s_1^j s_2^i$ for $i < j < d$, and the diagonal basis $s_1^i s_2^i$, has degree at most $2(d-1)$. Its full residue map is onto $\mathcal{E}$, and multiplication by the actual unit is invertible. Reflection preserves the degree filtration and this unit, so parity projection proves surjectivity onto each of B_e, B_o separately. Equivalently the concrete basis (SSP.20) has maximum degrees $4D-4, 4D-2$, and $4D-3$ in its three groups. This verifies the endpoint $M = 4D-2$, including $D=1, M=2$.

Define the actual kernels and their inverse metrics by

$$K_e = J_e M_e^{-1} J_e^*, \quad K_o = J_o M_o^{-1} J_o^*, \quad G_e = K_e^{-1} > 0, \quad G_o = K_o^{-1} > 0. \quad (\text{SSP.32})$$

All columns here are full polynomial remainders with the full unit. In particular their matrix sizes are $D(D+1)$ and D^2 , respectively.

Theorem 24.3 (The same original minimum and its unequal relative blocks). *Let R_M^s be the original symmetric canonical representative in the unscaled monomial orbit coordinates, $G_M^s = (R_M^s)^* R_M^s$, and A_M^s its original sum generator. Then*

$$\begin{aligned} T_s^{-*} G_M^s T_s^{-1} &= \text{diag}(G_e, G_o), & \det G_M^s &= \det G_e \det G_o, \\ T_s A_M^s T_s^{-1} &= I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}. \end{aligned} \quad (\text{SSP.33})$$

The original weight and its exact allowance satisfy

$$\begin{aligned} W_M^s &= (A_M^s)^* G_M^s + G_M^s A_M^s - 2G_M^s, \\ T_s^{-*} W_M^s T_s^{-1} &= \begin{pmatrix} 0 & \beta^* G_o - G_e \alpha \\ G_o \beta - \alpha^* G_e & 0 \end{pmatrix}, \end{aligned} \quad (\text{SSP.34})$$

$$\epsilon_M^s = \left\| G_e^{-1/2} (\beta^* G_o - G_e \alpha) G_o^{-1/2} \right\|_{\text{op}}.$$

If the displayed rectangular block has rank p , the relative weight has p positive and p negative eigenvalues, given by its positive singular values and their negatives. Its zero multiplicity is

$$D(D+1) + D^2 - 2p = D + 2(D^2 - p). \quad (\text{SSP.35})$$

Thus the dimension difference, all additional zero directions, and all nonzero multiplicities are retained.

Proof. The actual product source has the permutation-invariant Hilbert norm and full invariant jet map specified before (SSP.25). For a symmetric target, apply swap to its unique full minimum. The image and norm are unchanged, so uniqueness makes that minimum symmetric. Therefore minimizing over precisely the invariant degree- M polynomials gives the original R_M^s , not a new family of positive forms.

The exact coefficient transformation (SSP.30) has source Gram $\text{diag}(M_e, M_o)$; mixed terms vanish by the proved parity isometry. Its full jet map in coordinates (SSP.21) is $\text{diag}(J_e, J_o)$. For a positive source Gram M and surjective map J , the vector

$$M^{-1}J^*(JM^{-1}J^*)^{-1}y$$

has image y and is M -orthogonal to $\ker J$. Every other preimage differs by a kernel vector, and its squared norm is the squared norm of this vector plus the positive squared norm of that difference. Thus its minimum metric is exactly $(JM^{-1}J^*)^{-1}$. Apply this calculation to the two declared blocks to prove the metric congruence in (SSP.33). The coordinate representative is $R_M^s T_s^{-1}$, and its two source coefficient blocks are explicitly $M_e^{-1}J_e^*G_e$ and $M_o^{-1}J_o^*G_o$. Hence both its source function and full jets agree with the original minimum. Equation (SSP.22) proves the determinant equality. Equation (SSP.12) proves the generator similarity.

Multiplying the two generator blocks with the metric in (SSP.33) cancels the two identity terms against $2G_M^s$. This gives exactly the middle line of (SSP.34). The relative congruence by $\text{diag}(G_e^{-1/2}, G_o^{-1/2})$ has the form $\begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$ with the rectangular block displayed there. For each positive singular value σ and corresponding unit singular vectors u, v , its vectors (u, v) and $(u, -v)$ have eigenvalues $\sigma, -\sigma$. The remaining kernel is $\ker B^* \oplus \ker B$, with dimensions $D(D+1) - p$ and $D^2 - p$. This proves its norm, all signs, and (SSP.35). The coordinate map

$$\text{diag}(G_e^{1/2}, G_o^{1/2})T_s(G_M^s)^{-1/2}$$

is unitary by (SSP.33) and intertwines the two relative Hermitian matrices. Thus these are the complete relative eigenvalues of the original arithmetic weight. No inverse of α, β, B , or either frontier rank has been used. $\square$

In particular zero allowance is exactly the finite equation $\beta^*G_o = G_e\alpha$, with both rectangular maps retained. It does not follow from positivity of the two source Gram matrices. The map from those positive matrices to this equation is the fully proved difference block (SSP.34). Every original zero ρ , of any order m_ρ , has an eigenvector represented in its local factor by $(s - \rho)^{m_\rho - 1}$. Its square tensor is symmetric, nonzero, and has sum eigenvalue 2ρ . Its original Rayleigh quotient is therefore

$$\frac{u^*W_M^s u}{u^*G_M^s u} = 4 \operatorname{Re} \rho - 2, \quad 4 \left| \operatorname{Re} \rho - \frac{1}{2} \right| \leq \epsilon_M^s. \quad (\text{SSP.36})$$

This retains the exact tensor factor and every original local direction in the full metric; it makes no assumption about the limiting size of that metric difference.

24.6 Original primitives and supported maps

For any invariant relation written $H_0A + \Delta H_1B$, the original identity is

$$H_0A + \Delta H_1B = h(s_1) \frac{A + rB}{2} + h(s_2) \frac{A - rB}{2}. \quad (\text{SSP.37})$$

The two original cochain primitives apply the first coefficient to $\phi_* \otimes F_h$ and the negative of the second coefficient to $F_h \otimes \phi_*$. The second tensor differential contributes its own minus sign, so their total boundary is precisely (SSP.37). For an even relation $A_0a + xQ_\Delta b$, take $A = a$, $B = -Zb$; for an odd relation $Z(A_0a + Q_\Delta b)$, take $A = Za$, $B = b$. Substitution of $Z^2 = -x$ proves both assertions with their full signs. Thus the two coefficient ideals in (SSP.6) have the original theta primitives, including the factor xQ_Δ in the even ideal. At a finite admitted degree the polynomial identity is the same original relation. For completeness, divide a relation P of total degree at most M first by the monic $h(s_1)$, then divide its remainder by the monic $h(s_2)$. The final remainder is zero because P lies in the original ideal. This gives $P = h(s_1)C_1 + h(s_2)C_2$, with $\deg C_1, \deg C_2 \leq M - d$. Average under swap and put $B_1 = (C_1 + \text{swap}(C_2))/2$, $B_2 = \text{swap}(B_1)$. Then $A = B_1 + B_2$ is even in r , and $B = (B_1 - B_2)/r$ is a polynomial even in r . Their degrees satisfy $\deg A \leq M - d$ and $\deg B \leq M - d - 1$, and their coefficients in (SSP.37) are exactly B_1, B_2 . This proves the original finite-degree primitive bounds, including a zero coefficient when its degree bound is negative.

Every algebra isomorphism in (SSP.5), (SSP.9), and (SSP.14) has the exact G -lift: it fixes external τ and sends a supported amplitude to its supported algebra image. The algebra-homomorphism laws and the absorbing rule for τ prove the semiring laws; the inverse is the lift of the displayed inverse. The algebra maps β, ρ_0 have the same functorial lift. The module maps α, ι_0 and all Hilbert, jet, kernel and representative maps above have their fixed-label linear lifts $(\lambda, v) \mapsto (\lambda, Tv)$, fixing external absence. Their exact module actions are the ones proved in (SSP.7), (SSP.11), and (SSP.18). A class killed in either exact sequence goes to the zero in its receiving supported fibre; the represented kernel and cokernel A_H retain every local order and the unit V^2 . The direct sum in (SSP.6) is equipped with the multiplication (SSP.7); its parity components have not been assigned independent external supports by this map.

25 The sum connection on both half-line branches and its exact domain

This section integrates the complete sum-connection delivery, whose source note is retained in the appendices at its declared PR 20 revision 2144c358c2b41aa235b15cf0fa472289aec8673f. We prove its mass-dependent derivative identity in the original source and then carry the full relative connection into the quadratic coordinate. Both branches, all relative coefficients, and the condition where the two branches meet are retained. Only the stated dagger stability is needed until evenness is explicitly imposed.

25.1 The actual amplitude, its mass, and the original relative fibre

Keep $g = 2\xi$, $v_h = g/h$, the entire quotient at the full selected zero orders, and the original $F_h \in \mathcal{B}$ with Mellin transform v_h . Let the packet be stable under $\rho \mapsto 1 - \bar{\rho}$. The empty polynomial $h = 1$ is also allowed for this analytic calculation; no nonzero finite jet quotient is inferred in that case. Set

$$a_h^{\text{SC}}(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h^{\text{SC}}(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt = \|F_h\|_{L^2(dx)}^2. \quad (\text{FC.1})$$

The superscript distinguishes the delivered spectral amplitude from the gamma-reference multiplier in (TG.4). Their exact relation is $a_h^{\text{SC}}(t) = \Gamma(1/4 + it/2) a_h^{\text{TG}}(t) / \sqrt{2\pi}$. Taking its squared modulus gives precisely the two original measure formulas; neither mass nor phase is replaced.

For $k \geq 2$ use

$$u = \sum_i t_i, \quad y_i = t_i - u/k \ (i < k), \quad y_k = - \sum_{i < k} y_i, \quad t_i = u/k + y_i, \quad \Psi(u, \mathbf{y}) = \prod_i a_h^{\text{SC}}(t_i). \quad (\text{FC.2})$$

The Jacobian has absolute value one. One way to check its constant is to use coordinates $(u, t_1, \dots, t_{k-1})$, whose inverse sets $t_k = u - \sum_{i < k} t_i$ and has determinant of absolute value one. The replacement $t_i = y_i + u/k$, $i < k$, is triangular with diagonal one. For $k = 2$, $v = t_1 - t_2 = 2y_1$, so the earlier measure $du dv/2$ is unchanged.

The test-space estimates imply that $e^{r/2} F_h(e^r)$ and all its derivatives and polynomial multiples are Schwartz on $\mathbb{R}$. Indeed arbitrary powers of e^r bound both tails, and the logarithmic derivatives are the stated D -derivatives of F_h . Fourier differentiation and integration by parts therefore show that a_h^{SC} and all its derivatives are Schwartz. These facts justify the product integrals and differentiated integrals below.

Dagger stability gives $\overline{a_h^{\text{SC}}} = (-1)^d a_h^{\text{SC}}$. Thus this amplitude and its derivative have the same constant real or imaginary phase. Off the discrete zero set, $w'_h = 2\overline{a_h^{\text{SC}}}(a_h^{\text{SC}})'$, with the product real, and consequently

$$I_h := \int \frac{|w'_h|^2}{w_h} = 4 \int |(a_h^{\text{SC}})'|^2 = 4 \int_0^\infty (\log x)^2 |F_h(x)|^2 dx, \quad \int \overline{a_h^{\text{SC}}}(a_h^{\text{SC}})' = 0. \quad (\text{FC.3})$$

Values assigned to the quotient on its discrete zero set do not change these integrals. The second equality is the original Mellin Plancherel identity; the last integral is $\frac{1}{2} \int w'_h = 0$ by Schwartz decay. In particular every number in (FC.3) is finite.

Put $m(u) = \|\Psi(u, \cdot)\|^2 = w_h^{*k}(u) > 0$. Positivity holds at each u : the factors have only discrete zeros, whose inverse images in a relative fibre are a countable union of null hyperplanes. The integral of m is exactly μ_h^k . Let $\Pi_{0,u}$ be orthogonal projection onto the line spanned by $\Psi(u, \cdot)$, and let $n(u) = (1 - \Pi_{0,u})\partial_u \Psi$. Constant phase gives $\langle \Psi, \partial_u \Psi \rangle = m'/2$, so

$$n = \partial_u \Psi - \frac{m'}{2m} \Psi, \quad \boxed{\int \frac{|m'|^2}{m} + 4 \int \|n\|^2 = \frac{\mu_h^{k-1}}{k} I_h}. \quad (\text{FC.4})$$

To prove the equality, differentiate the product in (FC.2). Its derivative is

$$k^{-1} \sum_i (a_h^{\text{SC}})'(t_i) \prod_{j \neq i} a_h^{\text{SC}}(t_j).$$

Before the common factor k^{-2} , each of the k diagonal terms in its squared norm has integral $\mu_h^{k-1} \|(a_h^{\text{SC}})'\|^2$, and every distinct-index term contains $|\int \overline{a_h^{\text{SC}}}(a_h^{\text{SC}})'|^2 = 0$. Thus its integral is $\mu_h^{k-1} I_h/(4k)$. The orthogonal line and complementary components have squared norms $|m'|^2/(4m)$ and $\|n\|^2$; integration proves (FC.4).

25.2 The complete matrix connection and its actual graph

Choose a fixed finite independent homogeneous relative polynomial family θ_α , of degrees ℓ_α , in the $k-1$ independent variables $z_i = s_i - S/k$. At source degree M , retain exactly

$$P(\mathbf{s}) = \sum_\alpha \theta_\alpha(\mathbf{z}) f_\alpha(S), \quad \deg f_\alpha \leq M - \ell_\alpha, \quad S = k/2 + iu, \quad c_P(u) = (f_\alpha(k/2 + iu))_\alpha. \quad (\text{FC.5})$$

The family can be the whole admitted relative monomial family or the specified invariant family. Its cardinality r and all degree bounds remain part of the source; it is not held fixed while M changes. In $\mathcal{H} = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$, define

$$\begin{aligned} j_u c &= \Psi(u, \mathbf{y}) \sum_\alpha \theta_\alpha(i\mathbf{y}) c_\alpha, & W &= j_u^* j_u > 0, & B &= j_u^* j'_u, & T &= (j'_u)^* j'_u, \\ \Pi &= j_u W^{-1} j_u^*, & \Gamma &= W^{-1} B, & N &= (1 - \Pi) j'_u, & \mathcal{N} &= N^* N = T - B^* W^{-1} B. \end{aligned} \quad (\text{FC.6})$$

Positivity follows from independence of the relative polynomials and the almost-everywhere non-vanishing just used. Schwartz estimates on compact u -intervals prove smoothness of these finite matrices and of j_u in operator norm. Direct differentiation gives

$$\begin{aligned} \partial_u(j_u c) &= j_u(c' + \Gamma c) + Nc, & j_u^* N &= 0, \\ \Gamma^* W + W \Gamma &= W', & \mathcal{N} &\succeq 0, \\ \|\partial_u(j_u c)\|^2 &= (c' + \Gamma c)^* W (c' + \Gamma c) + c^* \mathcal{N} c. \end{aligned} \quad (\text{FC.7})$$

For example $j_u \Gamma = \Pi j'_u$; adding the complementary term proves the first identity. Differentiating $W = j_u^* j_u$ proves metric compatibility, and the projection identity proves the normal Gram and positivity. In this original fixed frame the common phase makes $B = B^*$, hence $B = W'/2$. After a variable change of frame $j^C = jC(u)$, one instead keeps the full formulas

$$\begin{aligned} W^C &= C^* W C, & B^C &= C^* B C + C^* W C', \\ \Gamma^C &= C^{-1} \Gamma C + C^{-1} C', & N^C &= N C, & \mathcal{N}^C &= C^* \mathcal{N} C. \end{aligned} \quad (\text{FC.8})$$

They follow by the product rule and the unchanged projection image. For instance $C = e^{iu} I$ gives $B^C = B + iW$, so the fixed-frame identity $B = W'/2$ is not silently imposed on a changed frame.

The actual derivative graph is the image of $P \mapsto (j_u(c'_P + \Gamma c_P), N c_P)$. Addition is an isometric injection on that graph by (FC.7); its image is precisely $\partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P$. Here $\mathcal{U}_k$ is the original Mellin-Fourier isometry followed by (FC.2). On the test space it obeys

$$\mathcal{U}_k D^{(k)} = (k/2 + iu) \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -I. \quad (\text{FC.9})$$

The first two formulas follow by integration by parts and differentiation of the Mellin integral. Applying $-x_i \partial_{x_i}$ to $\log x_i$ gives the last formula, including $1/k$. Therefore addition of both graph components, followed by $-i \mathcal{U}_k^{-1}$ and the original full jet, recovers $J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P$. The bare derivative lands in the image of graph addition; to land in the pair graph itself one uses the actual map $v \mapsto (\Pi_u \partial_u v, (1 - \Pi_u) \partial_u v)$.

25.3 A branch-preserving quadratic map for the entire matrix weight

Put $x = u^2 > 0$, $v = \sqrt{x}$, and retain the ordered branches $u = v$ and $u = -v$. These letters denote the spectral sum coordinate, not the original positive Mellin variables in (FC.9). For any coefficient function define

$$c(v) = a(x) + ivb(x), \quad c(-v) = a(x) - ivb(x), \quad \mathbf{M}(x) = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}. \quad (\text{FC.10})$$

Its inverse is $a = (c(v) + c(-v))/2$, $b = (c(v) - c(-v))/(2iv)$. No relation between $W(v)$ and $W(-v)$ is assumed. Write $W_{\pm} = W(\pm v)$. The full positive paired density is

$$\begin{aligned} \mathbf{H}(x) &= \mathbf{M}^* \frac{\text{diag}(W_+, W_-)}{2v} \mathbf{M} \\ &= \frac{1}{2\sqrt{x}} \begin{pmatrix} W_+ + W_- & i\sqrt{x}(W_+ - W_-) \\ -i\sqrt{x}(W_+ - W_-) & x(W_+ + W_-) \end{pmatrix} > 0. \end{aligned} \quad (\text{FC.11})$$

The inverse and the change of variable on each half-line prove the unitary equality

$$\int_{\mathbb{R}} c(u)^* W(u) c(u) du = \int_0^\infty \begin{pmatrix} a \\ b \end{pmatrix}^* \mathbf{H}(x) \begin{pmatrix} a \\ b \end{pmatrix} dx. \quad (\text{FC.12})$$

Here $\begin{pmatrix} a \\ b \end{pmatrix}^*$ means the adjoint of the two-block column. Positivity follows from the invertibility of M at every $x > 0$; surjectivity on these L^2 spaces follows from its pointwise inverse and the same norm identity. Both branch weights and their off-diagonal difference remain in (FC.11).

For the precise polynomial image in (FC.5), write $f_\alpha(S) = \sum_n f_{\alpha,n} S^n$. Then

$$\begin{aligned} a_\alpha(x) &= \sum_n f_{\alpha,n} \sum_{2j \leq n} \binom{n}{2j} (k/2)^{n-2j} (-x)^j, \\ b_\alpha(x) &= \sum_n f_{\alpha,n} \sum_{2j+1 \leq n} \binom{n}{2j+1} (k/2)^{n-2j-1} (-x)^j. \end{aligned} \quad (\text{FC.13})$$

Their degree bounds are respectively $\lfloor (M - \ell_\alpha)/2 \rfloor$ and $\lfloor (M - \ell_\alpha - 1)/2 \rfloor$, with a negative bound meaning a zero component. The inverse is exactly $f_\alpha(S) = a_\alpha(-(S - k/2)^2) + (S - k/2)b_\alpha(-(S - k/2)^2)$. Expanding the original powers proves both formulas and their inverse, including every $k/2$ coefficient. Thus (FC.12) retains the full filtered source.

Theorem 25.1 (The complete paired connection and normal energy). *Define*

$$\begin{aligned} \Gamma_b(x) &= \text{diag} \left(\frac{\Gamma(v)}{2v}, -\frac{\Gamma(-v)}{2v} \right), \\ A(x) &= M^{-1} \Gamma_b M + M^{-1} M' = M^{-1} \Gamma_b M + \text{diag}(0, (2x)^{-1} I), \\ Q(x) &= M^* \frac{\text{diag}(\mathcal{N}(v), \mathcal{N}(-v))}{2v} M \succeq 0, \quad \nabla_x = \partial_x + A. \end{aligned} \quad (\text{FC.14})$$

For $q = (a, b)^T$, the exact differentiated norm is

$$\int_{\mathbb{R}} \|\partial_u(j_u c(u))\|^2 du = 4 \int_0^\infty x (\nabla_x q)^* H \nabla_x q dx + \int_0^\infty q^* Q q dx. \quad (\text{FC.15})$$

The transformed coefficient derivative and its metric identities are

$$\begin{aligned} J(x) \nabla_x, \quad J &= \begin{pmatrix} 0 & 2ixI \\ -2iI & 0 \end{pmatrix}, \quad J^2 = 4xI, \quad J^* H J = 4xH, \\ A^* H + H A &= H' + \frac{1}{2x} H. \end{aligned} \quad (\text{FC.16})$$

Consequently $A - (4x)^{-1} I$ is the connection matrix compatible with the full density H . The explicit Jacobian term in (FC.16) remains in the transported original derivative.

Proof. Let $F = (c(v), c(-v))^T = Mq$ and $K = \text{diag}(2vI, -2vI)$. The two original covariant derivative values are exactly $K(F' + \Gamma_b F) = KM \nabla_x q$. The minus sign on the second branch is from $d(-v)/dx = -1/(2v)$. Multiplying the explicit inverse of M gives $M^{-1}KM = J$ and $M^{-1}M' = \text{diag}(0, 1/(2x)I)$. The block-scalar matrix K commutes with the two-block weight and its squared modulus is $4xI$. This proves the first energy term and the first line of (FC.16). Applying (FC.12) to the original normal Gram proves the second energy term; (FC.7) then proves (FC.15).

For metric compatibility set $D = \text{diag}(W_+, W_-)/(2v)$. The original equality $\Gamma^* W + W \Gamma = W'$, with the two opposite derivatives of v , gives $\Gamma_b^* D + D \Gamma_b = D' + D/(2x)$. The additional term is the derivative of the factor $1/(2v)$. Substitution of $H = M^* D M$ and the two terms defining A proves the last line of (FC.16). Subtracting $1/(4x)I$ from the matrix and its adjoint removes exactly that extra term, proving the last assertion. $\square$

Multiplication by the original spectral sum has the exact paired matrix

$$S = k/2 I + \begin{pmatrix} 0 & -xI \\ I & 0 \end{pmatrix}. \quad (\text{FC.17})$$

This follows by multiplying $a + iub$ by $k/2 + iu$. On smooth compactly supported sections in $x > 0$, the commutator is $[J\nabla_x, S] = iI$. To prove it with all variable-frame terms present, conjugate to the two branch coordinates used in the proof: multiplication is $\text{diag}(k/2 + iv, k/2 - iv)$, the connection part commutes with this block-scalar matrix, and multiplication of its derivative by K gives iI on both blocks. Conjugating back proves the claim. Through (FC.9) this is the same original pair $D^{(k)}, \mathcal{L}_k$, with its original sign.

25.4 The meeting point of the two branches is part of the domain

For the L^2 identity (FC.12) no value at zero is required. For the derivative domain it is essential. Define the original domain to consist of coefficient functions c that are locally absolutely continuous on $\mathbb{R}$, have finite norm in (FC.12), and have finite derivative energy in (FC.7). Equivalently these are the coefficient functions for which $j_u c(u)$ has an L^2 weak derivative: on each compact interval the smooth left inverse $W^{-1}j_u^*$ and strict positivity of W prove both directions by the product rule. Its exact image under (FC.10) consists of locally absolutely continuous q on $(0, \infty)$ with the two finite integrals in (FC.12), (FC.15), and the joining condition

$$\lim_{x \downarrow 0} (a(x) + i\sqrt{x} b(x)) = \lim_{x \downarrow 0} (a(x) - i\sqrt{x} b(x)). \quad (\text{FC.18})$$

These finite one-sided limits exist under the stated integrability conditions. Indeed W is bounded above and below by positive constants near zero and Γ is bounded there. The two finite integrals imply that each branch coefficient and its ordinary u -derivative are square-integrable near zero. The fundamental theorem for locally absolutely continuous functions and Cauchy–Schwarz then give their finite endpoint limits. Their union has a weak derivative in L^2 precisely when those limits agree; integration by parts against a compactly supported test function shows that a mismatch contributes their nonzero jump times a delta distribution. This proves (FC.18) and both directions of the domain correspondence. For the original polynomial image (FC.13), the condition holds automatically. In general, finite energy on the two open branches alone does not impose it: a function constant with different values on the two sides near zero has zero ordinary derivative there, while its joined distributional derivative has that nonzero jump. The matching condition preserves the original domain without discarding either branch.

25.5 The even specialization and the unchanged scalar contraction

If the original packet also has conjugation closure, w_h is even. For the symmetric square with relative frame $1, \Delta, \Delta^2, \dots$, the actual j_u is even in u , since its two amplitude factors are interchanged by $u \mapsto -u$ and amplitude evenness. Thus W and $\mathcal{N}$ are even and Γ is odd. Set $L(x) = W(\sqrt{x})/\sqrt{x}$, $L_N(x) = \mathcal{N}(\sqrt{x})/\sqrt{x}$, and $\Gamma_x = \Gamma(\sqrt{x})/(2\sqrt{x})$. Equations (FC.14)–(FC.16) reduce, by direct block substitution, to

$$\begin{aligned} H &= \text{diag}(L, xL), & Q &= \text{diag}(L_N, xL_N), & A &= \text{diag}(\Gamma_x, \Gamma_x + (2x)^{-1}I), \\ \int \|\partial_u(j_u c)\|^2 du &= \int_0^\infty [4x(a' + \Gamma_x a)^* L(a' + \Gamma_x a) + (b + 2x(b' + \Gamma_x b))^* L(b + 2x(b' + \Gamma_x b)) \\ &\quad + a^* L_N a + x b^* L_N b] dx. \end{aligned} \quad (\text{FC.19})$$

These are exactly the two measures and the complete relative family of the preceding symmetric-square section. Its unequal finite quotient components are obtained from this same filtered source, using its already proved full jet and actual minimum.

For any $k \geq 2$ under the same quartet assumption, the scalar sum density m is even. Define the actual half-line density $\rho_k(x) = m(\sqrt{x})/\sqrt{x}$. It retains total mass $\int_0^\infty \rho_k = \mu_h^k$. The source

contraction (FC.4) has the precise radial form

$$4 \int_0^\infty x \frac{|\rho'_k(x) + \rho_k(x)/(2x)|^2}{\rho_k(x)} dx + 4 \int_0^\infty \frac{\|n(\sqrt{x})\|^2}{\sqrt{x}} dx = \frac{\mu_h^{k-1}}{k} I_h. \quad (\text{FC.20})$$

To verify it, $m(u) = u\rho_k(u^2)$ for $u > 0$, so $m'(u) = \rho_k(x) + 2x\rho'_k(x)$. Changing variables in $2 \int_0^\infty m'^2/m du$ gives its first term. Reflection of all relative variables is a unitary map of each fibre and carries $n(u)$ to $-n(-u)$, so its norm is even; changing variables gives the second term. This proves (FC.20) with the stated drift and the entire relative contribution. It is the transported source estimate, with its mass and domain. The complete matrix energy remains (FC.15), on the actual graph and with its normal Gram, rather than being assigned the scalar constant-column bound.

All linear maps on the displayed filtered sources and graph images have the original fixed-support lift: they preserve the support label, send a supported zero to the corresponding supported zero, and send external absence to external absence. Their additive and scalar identities follow from the proved linear maps. In particular the arithmetic observation is taken after graph addition and the specified inverse Mellin isometry; no arithmetic jet is assigned to an arbitrary independent measurable normal section.

25.6 The actual arithmetic normal contribution is strictly positive

We now use the actual zero set of the amplitude in (FC.1). Hardy's theorem that infinitely many zeros of the Riemann zeta function lie on the critical line, with the original source and reading record retained in Section 23, implies that

$$\mathcal{Z}_h = \{T \in \mathbb{R} : a_h^{\text{SC}}(T) = 0\} \text{ is infinite and discrete.} \quad (\text{FC.21})$$

Indeed a finite polynomial h removes only its selected finite set of complete zero orders. Every remaining critical-line zero is still a zero of v_h , and the nonzero entire function v_h has a discrete zero set. This also applies to $h = 1$. At any $T \in \mathcal{Z}_h$, keep its entire order $\ell_T \geq 1$ and its original leading coefficient

$$a_h^{\text{SC}}(T + \delta) = c_T \delta^{\ell_T} + O(\delta^{\ell_T+1}), \quad (a_h^{\text{SC}})'(T + \delta) = \ell_T c_T \delta^{\ell_T-1} + O(\delta^{\ell_T}), \quad c_T = \frac{(a_h^{\text{SC}})^{(\ell_T)}(T)}{\ell_T!} \neq 0. \quad (\text{FC.22})$$

These are expansions of the original amplitude, including its fixed phase and its factor $1/\sqrt{2\pi}$.

Theorem 25.2 (Strict scalar residual and every finite normal direction). *For the actual amplitude and every integer $k \geq 2$,*

$$\int_{\mathbb{R}} \|n(u)\|^2 du > 0, \quad \int_{\mathbb{R}} \frac{|m'(u)|^2}{m(u)} du < \frac{\mu_h^{k-1}}{k} I_h. \quad (\text{FC.23})$$

For $k \geq 3$, $\|n(u)\| > 0$ for every real u . For any nonempty finite independent relative polynomial frame of (FC.5), its full normal Gram satisfies

$$\begin{aligned} \mathcal{N}(u) &> 0 \quad \text{for every } u \in \mathbb{R}, & k &\geq 3, \\ \mathcal{N}(u) &> 0 \quad \text{for every } u \notin \mathcal{Z}_h + \mathcal{Z}_h, & k &= 2. \end{aligned} \quad (\text{FC.24})$$

In particular $\mathcal{N}(u) > 0$ almost everywhere for every $k \geq 2$, since $\mathcal{Z}_h + \mathcal{Z}_h$ is countable. No lower bound uniform in u , the relative degree, or the packet is asserted by these strict inequalities.

Proof. First fix $T \in \mathcal{Z}_h$ and choose $b_2, \dots, b_k$ outside $\mathcal{Z}_h$. Put $u_0 = T + \sum_{i=2}^k b_i$ and $A_b = \prod_{i=2}^k a_h^{\text{SC}}(b_i) \neq 0$. Within that fixed original sum fibre consider the transverse path

$$t_1 = T + \delta, \quad t_i = b_i \quad (2 \leq i < k), \quad t_k = b_k - \delta.$$

It changes the relative variables while retaining $u = u_0$. The derivative ∂_u is still the original derivative at fixed relative variables in (FC.2), and its product formula gives

$$\begin{aligned}\Psi(u_0, \mathbf{y}(\delta)) &= c_T A_b \delta^{\ell_T} + O(\delta^{\ell_T+1}), \\ \partial_u \Psi(u_0, \mathbf{y}(\delta)) &= \frac{\ell_T}{k} c_T A_b \delta^{\ell_T-1} + O(\delta^{\ell_T}).\end{aligned}\tag{FC.25}$$

Only the derivative of the first amplitude has order $\ell_T - 1$; every other product term retains the full factor $a_h^{\text{SC}}(T + \delta)$ and has order at least ℓ_T . Since $m(u_0) > 0$ and m is smooth, subtracting $m'(u_0)\Psi/(2m(u_0))$ does not change that nonzero leading coefficient. Thus $n(u_0, \cdot)$ is nonzero at nearby points of this path and, by continuity, on a nonempty open subset of the entire relative fibre. Its squared fibre norm is strictly positive. Differentiated Schwartz estimates show that Ψ and $\partial_u \Psi$ are continuous as functions of u with values in $\mathcal{H}$. Smoothness and positivity of m therefore make n continuous in that Hilbert norm. Its squared norm stays positive on some interval around u_0 , proving positive integrated residual. Equation (FC.4) proves (FC.23).

For $k \geq 3$, the same construction works at every prescribed u_0 . Choose $b_3, \dots, b_{k-1}$ outside the zero set if that list is nonempty. Choose b_2 outside both $\mathcal{Z}_h$ and the translate-reflection of that set which would make $b_k = u_0 - T - \sum_{i=2}^{k-1} b_i$ a zero. These two discrete sets cannot exhaust the real line. This proves the pointwise scalar statement.

To prove the full matrix assertion, fix u and suppose $N_u c = 0$. The exact projection formula gives $j'_u c = j_u d$, where $d = \Gamma_u c$ is a finite coefficient vector. Write $\vartheta_c(\mathbf{y}) = \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}$, and similarly ϑ_d . Equality in $\mathcal{H}$ says

$$(\partial_u \Psi) \vartheta_c = \Psi \vartheta_d.$$

Both sides have smooth representatives in the relative variables, so their almost-everywhere equality is equality everywhere: a nonzero continuous difference at a point would remain nonzero on a set of positive measure.

Suppose $k \geq 3$. For each $T \in \mathcal{Z}_h$, restrict to the affine hyperplane $t_1 = u/k + y_1 = T$, of dimension $k - 2 \geq 1$. Each of the other original coordinates is a nonconstant affine function on this hyperplane. The points where one of their amplitude factors vanishes are consequently a countable union of proper affine hyperplanes of measure zero. Their complement is dense and is open near each of its points. At any such point p , apply (FC.25). In the asserted smooth identity the left side has leading coefficient $\ell_T c_T A_b \vartheta_c(p)/k$ in degree $\ell_T - 1$, whereas the right side has order at least ℓ_T . Hence $\vartheta_c(p) = 0$. Continuity proves its vanishing on the entire real hyperplane. Division in y_1 then shows that the polynomial ϑ_c has factor $y_1 - (T - u/k)$: the remainder polynomial in the other variables vanishes on all their real values, and therefore is identically zero. There are infinitely many distinct such T , while the degree in y_1 is finite. Thus $\vartheta_c = 0$, and independence of the original relative polynomials gives $c = 0$. This proves injectivity of N_u and therefore $\mathcal{N}(u) > 0$ on the whole finite coefficient space.

For $k = 2$, take $u \notin \mathcal{Z}_h + \mathcal{Z}_h$. At every $T \in \mathcal{Z}_h$, the other amplitude $a_h^{\text{SC}}(u - T)$ is nonzero. The same expansion forces the univariate polynomial ϑ_c to vanish at $y_1 = T - u/2$. These are infinitely many distinct points, so again $\vartheta_c = 0$ and $c = 0$. The discrete subset $\mathcal{Z}_h$ of $\mathbb{R}$ is countable, hence so is its set of pairwise sums. This proves every assertion in (FC.24). $\square$

The consequences hold for the actual full derivative graph and both quadratic branches. For every nonzero admitted source polynomial P ,

$$0 < \int_{\mathbb{R}} c_P(u)^* \mathcal{N}(u) c_P(u) du < \infty,\tag{FC.26}$$

$$\mathcal{Q}(x) > 0 \quad \text{for all } x > 0 \text{ when } k \geq 3, \quad \mathcal{Q}(x) > 0 \quad \text{for almost every } x > 0 \text{ when } k = 2.$$

Indeed a nonzero vector polynomial c_P is nonzero outside a finite set on the real line, and (FC.24) supplies strict positivity almost everywhere. The integral is finite because $\|N_u c_P\| \leq \|j'_u c_P\|$, and

the latter is a polynomial multiple of differentiated Schwartz columns. The assertion about Q follows from its invertible congruence (FC.14). For $k = 2$, the exceptional positive x for which either $\sqrt{x}$ or $-\sqrt{x}$ belongs to $\mathcal{Z}_h + \mathcal{Z}_h$ form a countable set. Under every declared smooth invertible frame, $N_u^C c_P^C(u) = N_u c_P(u)$ and $\mathcal{N}^C = C^* \mathcal{N} C$. Thus the full normal energy and strictness are preserved by the exact coefficient isomorphism $c_P^C = C^{-1} c_P$, including its original filtered image.

The excluded two-factor fibre can actually vanish. For the quartet packet, the original amplitude is even, so

$$\partial_u \Psi(0, y) = \frac{1}{2} ((a_h^{\text{SC}})'(y) a_h^{\text{SC}}(-y) + a_h^{\text{SC}}(y) (a_h^{\text{SC}})'(-y)) = 0. \quad (\text{FC.27})$$

Since the relative frame is independent of u , this gives $j'_0 = 0$, $N_0 = 0$, and $\mathcal{N}(0) = 0$. Also $m'(0) = 0$, hence $n(0) = 0$ exactly. The positive total residual and the almost-everywhere full matrix assertion therefore retain this original zero fibre. They prove that every nonzero actual finite polynomial direction has a positive normal contribution while leaving its complete size in (FC.15) and (FC.26).

25.7 Exact reconstruction from the actual finite normal image

Fix $k \geq 2$, the same actual packet, and one of the finite relative families in (FC.5). Retain its original degree bound M and the literal polynomial basis

$$\begin{aligned} \mathcal{I}_M &= \{(\alpha, n) : 0 \leq n \leq M - \ell_\alpha\}, \quad q_M = |\mathcal{I}_M|, \quad P_{\alpha, n}(\mathbf{s}) = \theta_\alpha(\mathbf{z}) S^n, \\ P_p &= \sum_{(\alpha, n) \in \mathcal{I}_M} p_{\alpha, n} P_{\alpha, n}, \quad c_{P_p}(u) = C_M(u) p, \quad (C_M(u))_{\beta, (\alpha, n)} = \delta_{\alpha\beta} (k/2 + iu)^n. \end{aligned} \quad (\text{FC.28})$$

Thus C_M is a specified rectangular polynomial matrix, not a change of relative frame. Each of its columns retains the original power of S . Its derivative is $(C'_M)_{\beta, (\alpha, n)} = \delta_{\alpha\beta} i n (k/2 + iu)^{n-1}$ for $n \geq 1$, and zero for $n = 0$. Independence of the relative family over $\mathbb{C}$, followed by comparison of the powers of S , proves uniqueness of these coefficients. The source is the exact space $\mathcal{P}_M^\theta = \text{span}\{P_{\alpha, n}\}$; it is the full original admitted source when that family is the complete admitted relative basis, and the declared summand otherwise. If $\mathcal{I}_M$ is empty, all the finite source and image spaces below are zero and their maps are the unique maps of zero spaces. We write the nonempty case explicitly.

Let $\mathcal{K} = L^2(\mathbb{R}_u; \mathcal{H})$, with the original measures $du d\mathbf{y}$. Define the normal and tangential maps by their actual amplitudes,

$$\begin{aligned} \mathbf{N}_M &: \mathbb{C}^{q_M} \longrightarrow \mathcal{K}, & (\mathbf{N}_M p)(u) &= N_u C_M(u) p, \\ \mathbf{T}_M &: \mathbb{C}^{q_M} \longrightarrow \mathcal{K}, & (\mathbf{T}_M p)(u) &= j_u (C'_M(u) + \Gamma_u C_M(u)) p, \\ H_M^N &= \mathbf{N}_M^* \mathbf{N}_M = \int_{\mathbb{R}} C_M(u)^* \mathcal{N}(u) C_M(u) du > 0, & \mathcal{R}_M^N &= \text{ran } \mathbf{N}_M. \end{aligned} \quad (\text{FC.29})$$

The coefficient adjoints in this subsection use the displayed literal basis and its ordinary Euclidean coordinate inner product. The computed metric pulled back from the normal image is H_M^N . The original source-amplitude metric and their exact comparison operator remain

$$\begin{aligned} H_M^0 &= \int_{\mathbb{R}} C_M(u)^* W(u) C_M(u) du > 0, & K_M^{N:0} &= (H_M^0)^{-1} H_M^N, \\ (K_M^{N:0})^* H_M^0 &= H_M^0 K_M^{N:0} = H_M^N, & p^* H_M^N q &= p^* H_M^0 K_M^{N:0} q. \end{aligned} \quad (\text{FC.29a})$$

Indeed $W > 0$ and a nonzero vector polynomial $C_M p$ give $H_M^0 > 0$, with finite entries by the original moment bounds. Direct multiplication proves the remaining identities, so $K_M^{N:0}$ is self-adjoint and strictly positive in the H_M^0 inner product. This is the exact operator between the two retained forms; no identity of their norms is assumed. All columns of $\mathbf{N}_M$ and $\mathbf{T}_M$ belong to $\mathcal{K}$. For the first map, projection contraction gives $\|N_u C_M p\| \leq \|j'_u C_M p\|$, whose square is integrable by the polynomially weighted Schwartz estimates. For the second, it is the projection by Π_u of $\partial_u(j_u C_M p)$, and the same

estimates apply. Equation (FC.26) proves strict positivity of H_M^N for every nonzero coefficient vector. In particular the normal map is injective on the whole original finite source, including nonzero polynomials whose original arithmetic class is a supported zero.

Theorem 25.3 (The actual normal image recovers the source and both graph parts). *The inverse on $\mathcal{R}_M^N$, and its precise ambient formula, are*

$$\begin{aligned} L_M^N &= (H_M^N)^{-1} \mathbf{N}_M^* : \mathcal{K} \rightarrow \mathbb{C}^{q_M}, & \mathbf{N}_M^* r &= \int_{\mathbb{R}} C_M(u)^* N_u^* r(u) du, \\ L_M^N \mathbf{N}_M &= I_{q_M}, & \Pi_M^N &= \mathbf{N}_M L_M^N \text{ is the orthogonal projection onto } \mathcal{R}_M^N. \end{aligned} \quad (\text{FC.30})$$

The full original derivative graph is $\mathcal{G}_M = \text{ran}(\mathbf{T}_M, \mathbf{N}_M)$. It is recovered from its normal image by the exact linear bijection

$$\begin{aligned} \text{Rec}_M : \mathcal{R}_M^N &\longrightarrow \mathcal{G}_M, & r &\longmapsto (\mathbf{T}_M L_M^N r, r), \\ \text{Rec}_M^{-1}(t, r) &= r, & p &= L_M^N r, & P &= P_{L_M^N r}. \end{aligned} \quad (\text{FC.31})$$

Its complete metric identity retains both components:

$$\begin{aligned} H_M^\partial &= \mathbf{T}_M^* \mathbf{T}_M + H_M^N, & \|(\mathbf{T}_M p, \mathbf{N}_M p)\|^2 &= p^* H_M^\partial p, \\ \|\text{Rec}_M r\|^2 &= \|r\|^2 + \|\mathbf{T}_M L_M^N r\|^2 = (L_M^N r)^* H_M^\partial (L_M^N r). \end{aligned} \quad (\text{FC.32})$$

In particular reconstruction is not asserted to preserve the normal norm alone; (FC.32) specifies its full metric.

Proof. Every entry of the integral in (FC.30) converges by Cauchy–Schwarz, since it pairs r with one of the finitely many L^2 columns of $\mathbf{N}_M$. Pairing against an arbitrary coefficient vector proves that this integral is its Hilbert adjoint. Therefore $L_M^N \mathbf{N}_M = (H_M^N)^{-1} H_M^N = I$. Multiplication shows that Π_M^N is self-adjoint and idempotent, that its image is $\mathcal{R}_M^N$, and that it fixes every vector of that image. This proves the projection and inverse claims. The image is finite dimensional and hence closed; all displayed maps are bounded. If $r = \mathbf{N}_M p$, (FC.30) makes its reconstructed pair exactly $(\mathbf{T}_M p, \mathbf{N}_M p)$. Conversely every pair in $\mathcal{G}_M$ has this form. Injectivity of $\mathbf{N}_M$ gives both uniqueness of p and the inverse in (FC.31). The two graph components are pointwise orthogonal by $j_u^* N_u = 0$; their squared norms give (FC.32). Their addition consequently is the original derivative amplitude and has the same full graph norm, as already proved in (FC.7). $\square$

The full arithmetic derivative observation can now be specified on this actual normal image. Put

$$\begin{aligned} I &= (h(s_1), \dots, h(s_k)), & E &= \mathbb{C}[\mathbf{s}]/I = E_h^{\otimes k}, & U(\mathbf{s}) &= \prod_i v_h(s_i), \\ v^{(k)} &= j_I U \in E^\times, & \partial_S^{\text{rel}} &= \frac{1}{k} \sum_i \partial_{s_i}, & \beta_h &= (v^{(k)})^{-1} j_I (\partial_S^{\text{rel}} U). \end{aligned} \quad (\text{FC.33})$$

Here j_I means the complete polynomial Hermite class in the original product remainder coordinates. It retains every order in each factor. The entire function U is a unit at the finite packet; its inverse and every derivative used in β_h are taken in those full local algebras. Thus this formula uses no global division across a zero of v_h . If $h = 1$, the target E is the zero algebra and all arithmetic observations below are its unique zero map; the preceding analytic reconstruction continues unchanged.

For a nonempty packet define the finite matrix $J_M^{\text{log}} : \mathbb{C}^{q_M} \rightarrow E$ by its original source columns. Mellin differentiation and (FC.9) give their exact formulas

$$\begin{aligned} (J_M^{\text{log}})_{\alpha, n} &= J^{(k)} \mathcal{L}_k T_h^{(k)} P_{\alpha, n} = j_I (\partial_S^{\text{rel}} (U P_{\alpha, n})) \\ &= n v^{(k)} [\theta_\alpha S^{n-1}]_I + \beta_h v^{(k)} [\theta_\alpha S^n]_I. \end{aligned} \quad (\text{FC.34})$$

For $n = 0$ the first term is defined to be zero, and no negative power is used. To verify the last equality, the original coordinate map gives $\partial_S^{\text{rel}} S = 1$ and $\partial_S^{\text{rel}} z_i = 0$. Hence the relative polynomial does not acquire a derivative term, while differentiating S^n gives precisely its original factor n . The other product-rule term is the full β_h just defined. Equivalently $\beta_h = k^{-1} \sum_i j_I(v'_h/v_h)(s_i)$, with the quotient taken only in its actual unit neighbourhoods. Both factors $g = 2\xi$ and every Taylor coefficient remain in these columns.

The exact observation and the complete commuting source path are

$$\begin{aligned} \text{Obs}_M^{\log} : \mathcal{R}_M^N &\rightarrow E, & \text{Obs}_M^{\log}(r) &= \mathbf{J}_M^{\log}(H_M^N)^{-1} \mathbf{N}_M^* r, \\ \text{Obs}_M^{\log}(\mathbf{N}_M p) &= J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P_p, \\ \mathcal{R}_M^N &\xrightarrow{\text{Rec}_M} \mathcal{G}_M \xrightarrow{\text{Add}} i \mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta, \\ i \mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta &\xrightarrow{-i \mathcal{U}_k^{-1}} \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta \xrightarrow{J^{(k)}} E. \end{aligned} \tag{FC.35}$$

Substitute $r = \mathbf{N}_M p$ in (FC.30)–(FC.31). Graph addition gives $\partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P_p = i \mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} P_p$, and the last two arrows give exactly the columns in (FC.34). This proves equality of the two descriptions of the observation. All functions in the penultimate space are actual finite tensor test functions, since logarithmic multiplication preserves $\mathcal{B}$. Consequently this is an arithmetic observation on the specified finite normal image with a proved source inverse. The ambient formula for L_M^N also defines an orthogonal projection, but no arithmetic-jet interpretation of an arbitrary unrelated measurable normal section is required by (FC.35).

The reconstruction is unchanged under a declared smooth invertible relative frame. In that frame its coefficient matrix is $C_M^C = C(u)^{-1} C_M$, and the exact identities in (FC.8) give

$$\begin{aligned} N^C C_M^C &= N C_M, & j^C((C_M^C)' + \Gamma^C C_M^C) &= j(C_M' + \Gamma C_M), \\ (H_M^N)^C &= H_M^N, & (\mathbf{N}_M)^C &= \mathbf{N}_M, & (\mathbf{T}_M)^C &= \mathbf{T}_M. \end{aligned} \tag{FC.36}$$

The derivative of C^{-1} cancels exactly against the $C^{-1} C'$ connection term. Thus the original normal-image inverse, the full reconstructed graph and the arithmetic observation are the same maps on the same source after its explicit coefficient transport. Their fixed-support linear lifts retain the original label and all supported zeros. In particular a nonzero polynomial relation can have a nonzero normal image whose reconstructed arithmetic derivative is the full conormal response (FC.34); the source relation is retained before either quotient observation.

26 The original conormal derivative and its finite section commutator

The sum-connection source constructs differentiation on the first thickening of the original packet ideal. We prove its algebra map to dual numbers, then compute the exact finite endomorphism obtained from the original monic remainder section and from its full arithmetic unit. The conormal row and the section correction have different domains; their explicit factorization, complete ranks, and residue comparison are retained throughout.

26.1 The first thickening and the dual-number map

Let $k \geq 1$, and retain a nonempty packet of actual zeros with all complete orders,

$$h(s) = \prod_{\rho \in \mathcal{Z}} (s - \rho)^{m_\rho}, \quad d = \deg h \geq 1, \quad r_0 = |\mathcal{Z}|, \quad g = 2\xi, \quad v_h = g/h. \tag{CW.1}$$

Write $h_i = h(s_i)$, and put

$$\begin{aligned} P &= \mathbb{C}[s_1, \dots, s_k], \quad I = (h_1, \dots, h_k), \quad E = P/I, \quad E^{[2]} = P/I^2, \quad N = I/I^2, \\ \pi : E^{[2]} &\rightarrow E, \quad \partial_\Sigma = \frac{1}{k} \sum_{i=1}^k \partial_{s_i}, \quad \delta : E^{[2]} \rightarrow E, \quad [p] \mapsto [\partial_\Sigma p]. \end{aligned} \quad (\text{CW.2})$$

The derivative maps I^2 into I by the product rule, so δ has exactly the stated type. For each i , the class of h_i maps to zero under π , while $\delta[h_i] = h'_i/k \neq 0$ in the retained remainder basis. Thus factoring δ through π would fail on this explicit class. The full algebra map (CW.6) below retains both its quotient value and derivative. Monic division in each original variable gives a unique expansion

$$p = \sum_{\mathbf{a} \in \mathbb{N}^k} h_1^{a_1} \cdots h_k^{a_k} p_{\mathbf{a}}, \quad \deg_{s_i} p_{\mathbf{a}} < d. \quad (\text{CW.3})$$

Only finitely many terms occur. Existence follows by repeated one-variable division, which commutes for distinct variables. For uniqueness, the leading monomials $\prod_i s_i^{da_i+b_i}$, $0 \leq b_i < d$, of the corresponding products are all distinct; comparison in a monomial order, followed by subtraction, proves independence. This also proves that the h_i are algebraically independent and that P is free over $\mathbb{C}[h_1, \dots, h_k]$ with the original remainder basis.

Consequently the original conormal coefficient map is the E -module isomorphism

$$\iota_N : E^k \xrightarrow{\sim} N, \quad (a_i) \mapsto \left[\sum_i h_i \tilde{a}_i \right]_{I^2}, \quad \dim E = d^k, \quad \dim E^{[2]} = (k+1)d^k. \quad (\text{CW.4})$$

Changing a lift $\tilde{a}_i$ adds an element of I^2 . Modulo I^2 , (CW.3) has exactly its constant coefficient and its k linear h_i -coefficients, proving every assertion in (CW.4). In particular $N^2 = 0$ in $E^{[2]}$. The derivative obeys

$$\delta(ab) = \pi(a)\delta(b) + \pi(b)\delta(a), \quad \delta\iota_N(a_i) = \frac{1}{k} \sum_i h'_i a_i. \quad (\text{CW.5})$$

These identities follow before taking the indicated quotients; the terms still containing h_i vanish only in E .

Theorem 26.1 (A full algebra bridge for the coefficient derivative). *Let $B = E[\epsilon]/(\epsilon^2)$, where ϵ is a formal infinitesimal, and let $\mathfrak{j} = (h'_1, \dots, h'_k)$ be the indicated ideal of E . There is a unital algebra map*

$$\Phi : E^{[2]} \rightarrow B, \quad a \mapsto \pi(a) + \epsilon\delta(a). \quad (\text{CW.6})$$

The derivative induces $\bar{\partial}_\Sigma : E \rightarrow E/\mathfrak{j}$, and its image and kernel are exactly

$$\begin{aligned} \text{im } \Phi &= \{q + \epsilon t : [t]_{\mathfrak{j}} = \bar{\partial}_\Sigma q\}, \\ \ker \Phi &= \iota_N \ker \left(E^k \xrightarrow{(a_i) \mapsto k^{-1} \sum_i h'_i a_i} E \right). \end{aligned} \quad (\text{CW.7})$$

There is the exact sequence of complex vector spaces

$$0 \longrightarrow \ker(\delta|_N) \longrightarrow E^{[2]} \xrightarrow{\Phi} B \xrightarrow{\Psi} E/\mathfrak{j} \longrightarrow 0, \quad \Psi(q + \epsilon t) = [t]_{\mathfrak{j}} - \bar{\partial}_\Sigma q. \quad (\text{CW.8})$$

The map Φ is an algebra homomorphism; Ψ is given its displayed complex-linear type. The complete ranks and dimensions are

$$\begin{aligned} \text{rank}(\delta|_N) &= d^k - (d - r_0)^k, \\ \dim \ker \Phi &= (k-1)d^k + (d - r_0)^k, \\ \dim \text{im } \Phi &= 2d^k - (d - r_0)^k. \end{aligned} \quad (\text{CW.9})$$

Proof. Equation (CW.5) gives

$$\Phi(ab) = \pi(a)\pi(b) + \epsilon\{\pi(a)\delta(b) + \pi(b)\delta(a)\} = \Phi(a)\Phi(b),$$

and $\delta(1) = 0$, proving the unital algebra claim. Changing a representative of $q \in E$ by $\sum_i h_i p_i$ changes its derivative modulo I by $k^{-1} \sum_i h'_i [p_i] \in \mathfrak{j}$. Thus $\bar{\partial}_\Sigma$ is well-defined, with its product rule along $E \rightarrow E/\mathfrak{j}$. Every $\Phi(a)$ consequently satisfies the condition in (CW.7).

Conversely let that condition hold. Take the unique original monic remainder $\sigma(q) \in P$, of degree below d in every variable. Then

$$t - \delta[\sigma(q)]_{I^2} = \frac{1}{k} \sum_i h'_i a_i$$

for some $a_i \in E$, by the definition of $\mathfrak{j}$. The element $[\sigma(q)]_{I^2} + \iota_N(a_i)$ has image $q + \epsilon t$. This proves the image equality. Its kernel requires $\pi(a) = 0$, hence $a \in N$, and then exactly the vanishing of the row in (CW.5). This proves the kernel statement. The map Ψ is onto, since $\Psi(\epsilon t) = [t]$, and its kernel is the already computed image of Φ . This proves (CW.8) with its exact map types.

For the ranks, retain an ordered local tensor block at centres ρ_i , of full lengths m_i . In its coordinates

$$L = \mathbb{C}[z_1, \dots, z_k] / (z_1^{m_1}, \dots, z_k^{m_k}), \quad z_i = s_i - \rho_i,$$

write $h(s_i) = z_i^{m_i} c_i(z_i)$, $c_i(0) \neq 0$. Modulo the original local ideal,

$$h'_i = m_i c_i(0) z_i^{m_i-1}, \quad h'_i a_i = m_i c_i(0) z_i^{m_i-1} a_i|_{z_i=0}. \quad (\text{CW.10})$$

Thus the original input constant coefficient in the i -th variable is sent into that variable's top nilpotent direction; all remaining variables and the factor $m_i c_i(0)$ are retained. The row image is the ideal spanned by monomials having at least one exponent $m_i - 1$. Its dimension is $\prod_i m_i - \prod_i (m_i - 1)$. The excluded monomials remain in the complementary vector space and in $E/\mathfrak{j}$. Sum over every ordered tuple of centres. The two sums are

$$\sum_{\rho_1, \dots, \rho_k} \prod_i m_{\rho_i} = d^k, \quad \sum_{\rho_1, \dots, \rho_k} \prod_i (m_{\rho_i} - 1) = (d - r_0)^k. \quad (\text{CW.11})$$

This proves the first line of (CW.9). Subtract it from $\dim N = kd^k$ for the kernel, then subtract that kernel from $\dim E^{[2]}$ for the image, proving the other two. These counts include every simple and repeated local factor. $\square$

When the packet has the original reflection $a^\dagger(\mathbf{s}) = \overline{a(1 - \mathbf{s})}$, its ideals are stable because $h^\dagger = (-1)^d h$. The derivative satisfies $\delta(a^\dagger) = -(\delta a)^\dagger$. Set $(q + \epsilon t)^\dagger = q^\dagger - \epsilon t^\dagger$. Then $\Phi(a^\dagger) = \Phi(a)^\dagger$, by (CW.6). This retains the sign of the infinitesimal and the original conjugate-linear reflection.

26.2 The original monic section and the exact finite correction

Let $\sigma : E \rightarrow P$ denote the multivariate monic remainder section of (CW.3), and set $j = [\sigma]_{I^2} : E \rightarrow E^{[2]}$. It is complex-linear, with $\pi j = I_E$. Let $S = \sum_i s_i$, and write Z and $\widehat{Z}$ for multiplication by $S - k/2$ on E and $E^{[2]}$, respectively. The original product rule gives the rectangular operator identity

$$Z\delta - \delta\widehat{Z} = -\pi, \quad D_0 = \delta j : E \rightarrow E. \quad (\text{CW.12})$$

The original finite arithmetic generator is $A_k = Z + (k/2)I_E$.

For each i , let $\ell_i : E \rightarrow E_{\widehat{i}} := \bigotimes_{j \neq i} E_h$ extract the coefficient of s_i^{d-1} in $\sigma(u)$. Let $c_i : E_{\widehat{i}} \rightarrow E$ insert the constant polynomial one in the i -th variable. If $k = 1$, the target of ℓ_1 is $\mathbb{C}$, and c_1 is its unit column. These maps give the exact section defect

$$\begin{aligned} \mathbf{t}_Z &:= \widehat{Z}j - jZ = \iota_N(c_1 \ell_1, \dots, c_k \ell_k) : E \rightarrow N, \\ [Z, D_0] &= -I_E + \mathbf{J}, \quad \mathbf{J} = \delta \mathbf{t}_Z = \frac{1}{k} \sum_i M_{h'_i} c_i \ell_i. \end{aligned} \quad (\text{CW.13})$$

Indeed multiplying the original remainder by s_i creates only its top s_i^d term outside the admitted coordinate range. Monic division subtracts exactly $h_i \sigma_i(\ell_i u)$. Adding these terms for $S - k/2$ proves the first formula. Compose (CW.12) with j and use $\widehat{Z}j = jZ + \mathbf{t}_Z$; this proves the commutator with its minus sign. Applying (CW.5) proves its last formula. In particular this is a specified endomorphism of the original finite vector space, obtained by a declared section of π .

Theorem 26.2 (All projectors, eigenvalues and ranks of the section correction). *The original coordinate endomorphisms*

$$P_i = \frac{1}{d} M_{h'_i} c_i \ell_i, \quad P_i^2 = P_i, \quad P_i P_j = P_j P_i, \quad \text{rank } P_i = d^{k-1} \quad (\text{CW.14})$$

are commuting idempotents. For $T \subset \{1, \dots, k\}$, put

$$P_T = \prod_{i \in T} P_i \prod_{i \notin T} (I - P_i). \quad (\text{CW.15})$$

Their pairwise products are zero when their labels differ, and $\sum_T P_T = I_E$. On their ranges the full correction has

$$J = \frac{d}{k} \sum_i P_i, \quad J|_{\text{ran } P_T} = \frac{d|T|}{k} I. \quad (\text{CW.16})$$

For $d > 1$, its eigenvalue dj/k has multiplicity $\binom{k}{j} (d-1)^{k-j}$, for every $0 \leq j \leq k$. For all $d \geq 1$,

$$\text{Tr } J = d^k, \quad \text{rank } J = d^k - (d-1)^k, \quad \ker J = \bigcap_i \ker \ell_i. \quad (\text{CW.17})$$

For $d = 1$, every $P_i = I$, $J = I$, and only the $j = k$ eigenspace is nonzero, of dimension one.

Proof. The polynomial h'_i has degree $d-1$ and leading coefficient d . Therefore

$$\ell_i M_{h'_i} c_i = d I_{E_i}. \quad (\text{CW.18})$$

Substitution proves $P_i^2 = P_i$. It also proves that $M_{h'_i} c_i$ is injective, and hence that the rank is d^{k-1} . Operators in distinct variables are tensor products of their respective one-variable maps, so they commute. The one-variable space is the direct sum of $\mathbb{C}h'$ and the polynomials of degree at most $d-2$: the former has nonzero top coefficient d , and every polynomial splits by subtracting that top coefficient times h'/d . These are the image of P_i and the kernel of ℓ_i , of dimensions one and $d-1$.

Taking tensor products of these direct decompositions proves all the identities for P_T and gives $\dim \text{ran } P_T = (d-1)^{k-|T|}$ when $d > 1$. On this range exactly the P_i for $i \in T$ act as identity, proving (CW.16). Summing over $|T| = j$ gives the asserted multiplicity. Its zero space is the all-complement range, of dimension $(d-1)^k$, proving the rank and kernel in (CW.17). Finally

$$\text{Tr } J = \frac{d}{k} \sum_i \text{rank } P_i = \frac{d}{k} k d^{k-1} = d^k. \quad (\text{CW.19})$$

For $d = 1$, the one-variable complement is zero and its image is the entire one-dimensional space. Only the all-image tensor survives. This proves the stated endpoint directly, without assigning a positive multiplicity to a zero-dimensional space. $\square$

These are algebraic idempotents in the retained coordinates. For any actual positive minimum metric G_M , their exact metric adjoints and coordinate transports are

$$P_i^{\dagger G_M} = G_M^{-1} P_i^* G_M, \quad \tilde{P}_i = G_M^{1/2} P_i G_M^{-1/2}, \quad \tilde{J} = G_M^{1/2} J G_M^{-1/2}. \quad (\text{CW.20})$$

The transports retain the products, ranks and algebraic spectrum. For each i , the individual $\tilde{P}_i$ is Hermitian precisely when $P_i^* G_M = G_M P_i$, whereas $\tilde{J}$ is Hermitian precisely when $J^* G_M = G_M J$. None of these metric equalities is assumed. The same isometry carries the original weight to

$$G_M^{-1/2} W_M G_M^{-1/2} = \tilde{Z}^* + \tilde{Z}, \quad \tilde{Z} = G_M^{1/2} Z G_M^{-1/2}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{CW.21})$$

It carries (CW.13) to the identical commutator with all three operators transported. Thus the algebraic correction and the original Hermitian weight have an explicit common coordinate map, with every adjoint still attached to its actual metric.

The full conormal row rank in (CW.9) and the section rank in (CW.17) are connected by exactly

$$E \xrightarrow{(c_i \ell_i)_i} E^k \xrightarrow{\iota_N} N \xrightarrow{\delta|_N} E. \quad (\text{CW.22})$$

Its composite is J . The first map retains the specific highest-coordinate coefficients of the monic section; the full conormal row accepts arbitrary k input coefficients. This proves the relationship between the two ranks without replacing one domain by the other.

26.3 The full arithmetic unit and the original finite representatives

Keep the complete Taylor classes

$$\hat{U} = \left[\prod_i v_h(s_i) \right]_{I^2} \in (E^{[2]})^\times, \quad U = \pi(\hat{U}) \in E^\times, \quad \beta_h = U^{-1} \delta \hat{U} \in E. \quad (\text{CW.23})$$

Finite multivariate Hermite interpolation supplies these classes from the entire original functions near every ordered packet point. Their values are nonzero at those points because the removed orders were complete. One can also prove invertibility in the thickening directly: lift U^{-1} to b ; then $\hat{U}b = 1 + n$, $n \in N$, and $b(1 - n)$ is the inverse since $N^2 = 0$. Thus this thickening retains the original unit and its first relation derivatives.

Theorem 26.3 (The unit-conjugated finite Weyl identity). *Define the actual unit section and its derivative by*

$$j_U = \hat{U} j U^{-1} : E \rightarrow E^{[2]}, \quad \pi j_U = I_E, \quad D_U = \delta j_U = M_{\beta_h} + M_U D_0 M_U^{-1}. \quad (\text{CW.24})$$

Then, with all multiplication operators in the original basis,

$$\begin{aligned} \mathbf{t}_U &:= \hat{Z} j_U - j_U Z = M_{\hat{U}} \mathbf{t}_Z M_U^{-1} : E \rightarrow N, \\ [Z, D_U] &= -I_E + J_U, \quad J_U = \delta \mathbf{t}_U = M_U J M_U^{-1}. \end{aligned} \quad (\text{CW.25})$$

The complete spectrum, multiplicities, ranks and trace in (CW.16)–(CW.19) therefore hold for J_U as well.

Proof. The section equation follows from $\pi j = I$ and $\pi(\hat{U}) = U$. Apply (CW.5) to $\hat{U} j(U^{-1}u)$. Its two terms are $(\delta \hat{U})U^{-1}u$ and $U D_0(U^{-1}u)$, which prove (CW.24). The multiplication maps $\hat{Z}$, $M_{\hat{U}}$ commute, as do Z , M_U^{-1} . This proves the formula for $\mathbf{t}_U$. Its image lies in N , so the product-rule term $(\delta \hat{U})\pi(\mathbf{t}_Z M_U^{-1}u)$ vanishes exactly. The other term is $U J M_U^{-1}u$, proving the last equality in (CW.25). Finally M_{β_h} commutes with multiplication by Z . Conjugating (CW.13) by M_U proves the commutator. Similarity gives the entire stated spectrum and its zero space, without making an orthogonality claim. $\square$

These sections have an original analytic realization. Let

$$\mathcal{T}_h^{(k)} p = p(D_1, \dots, D_k) F_h^{\otimes k}, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad R_{\text{ref}}^{(k)} u = \mathcal{T}_h^{(k)} \sigma(U^{-1}u). \quad (\text{CW.26})$$

The original Mellin transform gives $\mathcal{M}_k \mathcal{T}_h^{(k)} p = (\prod_i v_h(s_i))p$. Its thickened Taylor class is $\widehat{U}[p]_{I^2}$, so the thickened class of $R_{\text{ref}}^{(k)} u$ is exactly $j_U u$. Multiplication by $\mathcal{L}_k$ differentiates this Mellin transform by ∂_Σ . Hence

$$J^{(k)} \mathcal{L}_k R_{\text{ref}}^{(k)} = D_U. \quad (\text{CW.27})$$

This is the finite endomorphism of the specified original section, with its actual logarithmic source observable.

For the actual degree- M minimum, $M \geq k(d-1)$, retain its original coefficient Gram $\mathcal{M}_M > 0$ and full jet matrix $\mathcal{J}_M$, including multiplication by U . Its numerator and source function are exactly

$$p_{M,u} = \mathcal{M}_M^{-1} \mathcal{J}_M^* G_M u, \quad G_M = (\mathcal{J}_M \mathcal{M}_M^{-1} \mathcal{J}_M^*)^{-1}, \quad R_M u = \mathcal{T}_h^{(k)} p_{M,u}. \quad (\text{CW.28})$$

Indeed its jet is u , and its coefficient vector is $\mathcal{M}_M$ -orthogonal to $\ker \mathcal{J}_M$. Every other preimage differs by such a kernel vector; Pythagoras proves this same original constrained minimum. Define the concrete thickened section and admitted relation by

$$j_M u = \widehat{U}[p_{M,u}]_{I^2}, \quad a_M = j_M - j_U : E \rightarrow N, \quad D_M = \delta j_M = D_U + \delta a_M. \quad (\text{CW.29})$$

The image statement follows from $\pi j_M = \pi j_U = I_E$. Its numerator is the original allowed relation $p_{M,u} - \sigma(U^{-1}u) \in I$, not an independently chosen vector in the conormal layer. The corresponding exact identities are

$$\begin{aligned} J^{(k)} \mathcal{L}_k R_M &= D_M, & [Z, D_M] &= -I_E + J_M^{\text{sec}}, \\ J_M^{\text{sec}} &:= \delta(\widehat{Z} j_M - j_M Z) = J_U + [Z, \delta a_M], & \text{Tr } J_M^{\text{sec}} &= d^k. \end{aligned} \quad (\text{CW.30})$$

The first follows by the same original Mellin differentiation. For the second, compose (CW.12) with j_M . For the third, subtract (CW.25) and use $\delta \widehat{Z} a_M = Z a_M^\delta$, where $a_M^\delta = \delta a_M$, since $\pi a_M = 0$. This gives exactly the displayed commutator difference. Its trace is zero by the cyclic identity for finite matrices; (CW.19) gives the final trace. Only the specified fixed-section correction is assigned the ranks (CW.17). Every change to the actual minimum is retained through the computed map a_M and its commutator.

26.4 The one-variable residue functional and the original theta relation

Take $k = 1$, write $\ell : E_h \rightarrow \mathbb{C}$ for the coefficient of s^{d-1} , and let $U = j_h v_h$. The product derivative $g' = h' v_h + h v_h'$ gives

$$J_U = M_{j_h g'} c_1 \ell M_U^{-1}, \quad J_U u = (j_h g') \ell(U^{-1}u), \quad \text{rank } J_U = 1, \quad \text{Tr } J_U = d. \quad (\text{CW.31})$$

The rank follows either from (CW.25) or from the nonzero column $j_h g' = U h'$ and the nonzero functional. The trace of this rank-one map is $\ell(U^{-1} j_h g') = \ell(h') = d$.

For the original reflection-stable packet, put $f^\dagger(s) = \overline{f(1-\bar{s})}$. Its residue pairing is the well-defined form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{\widetilde{f}^\dagger(s) \widetilde{u}(s)}{g(s)} ds = \ell(U^{-1} f^\dagger u). \quad (\text{CW.32})$$

Changing either polynomial representative by h changes the integrand by a function holomorphic at all selected centres: $h^\dagger = (-1)^d h$, and $g = h v_h$, with v_h nonvanishing there. To prove the last equality, take $p = \text{rem}_h(U^{-1} f^\dagger u)$, of degree below d . The residues of the displayed quotient equal those of p/h . In the partial-fraction expansion of p/h , the coefficient of $1/s$ at infinity is the sum of its coefficients at the simple-pole terms; terms of pole order at least two have no $1/s$ coefficient. Since h is monic of degree d , that coefficient is exactly the original coefficient $\ell(p)$. This proves (CW.32),

keeping every higher local pole before the residue is taken. Consequently the finite correction is precisely

$$J_U u = (j_h g') \mathcal{R}_Z(1, u), \quad \text{Tr } J_U = \mathcal{R}_Z(1, j_h g') = \sum_{\rho \in Z} m_\rho = d. \quad (\text{CW.33})$$

The source map producing the same Jacobian is the original logarithmic theta observation:

$$J_h(\log x \Theta \phi) = j_h(g') j_h(H_\phi), \quad \mathcal{M} \Theta \phi = g H_\phi. \quad (\text{CW.34})$$

Differentiate the Mellin product before taking jets. Its $g H'_\phi$ term vanishes modulo the original h ; its other term is exactly (CW.34). For $\phi = P(D)\phi_*$, one has $H_\phi = P$, so the input ranges through all original finite coefficients. Thus this map's finite image is the full ideal $(j_h g')$, of dimension r_0 by (CW.10), rather than a selected one-dimensional input line. The exact connection to (CW.31) is its precomposition with

$$E_h \xrightarrow{\ell M_U^{-1}} \mathbb{C} \xrightarrow{c_1} E_h. \quad (\text{CW.35})$$

The rank-one correction and the rank- r_0 multiplication map therefore use the same Jacobian with their distinct input maps explicitly specified. Applying (CW.32) to (CW.34) gives, with the full conjugation,

$$\mathcal{R}_Z(f, j_h(g') j_h P) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho). \quad (\text{CW.36})$$

Indeed the residue of g'/g at a zero of order m_ρ is m_ρ , by differentiating its local product.

There is also an exact global source map

$$\nu : V \rightarrow Q = \mathcal{B}/\Theta V, \quad \nu(\phi) = q(\log x \Theta \phi). \quad (\text{CW.37})$$

Logarithmic multiplication preserves $\mathcal{B}$, by its defining two-sided polynomial bounds: $|\log x| \leq x + x^{-1}$ and $D^j((\log x)F) = (\log x)D^j F - jD^{j-1}F$ for $j \geq 1$ give every required bound from the original ones. Thus this map is defined. For the original $D = -x\partial_x$, $[D, \log x] = -I$ and $D\Theta = \Theta D$. Therefore

$$D_Q \nu - \nu D_V = q[D, \log x]\Theta = -q\Theta = 0. \quad (\text{CW.38})$$

For $a > 0$, let $\Lambda_a F(x) = F(ax)$. The identities $\Theta \Lambda_a = \Lambda_a \Theta$ and $(\log x)\Lambda_a - \Lambda_a(\log x) = -\log(a)\Lambda_a$ similarly give $\nu \Lambda_a = \Lambda_a \nu$. This proves scaling equivariance on the exact source V and target Q . Since $J_h \Theta = 0$, there is the induced full jet map $\bar{J}_h : Q \rightarrow E_h$ with $\bar{J}_h q = J_h$; its value on ν is exactly (CW.34).

26.5 Supported algebra and coefficient maps

Write $\text{pr}_0, \text{pr}_\epsilon : B \rightarrow E$ for the constant and infinitesimal coefficient maps. The first is a unital algebra homomorphism and the second is complex-linear, with the exact product rule

$$\text{pr}_\epsilon(bc) = \text{pr}_0(b) \text{pr}_\epsilon(c) + \text{pr}_0(c) \text{pr}_\epsilon(b). \quad (\text{CW.39})$$

The strongest typed lift of (CW.6) is the semiring map

$$G(E^{[2]}) \xrightarrow{G(\Phi)} G(B), \quad \pi^\tau = G(\text{pr}_0)G(\Phi), \quad \delta^\tau = (\text{pr}_\epsilon)^\tau G(\Phi). \quad (\text{CW.40})$$

Here $G(\Phi)$ and $G(\text{pr}_0)$ fix external τ and send supported amplitudes to their supported algebra images. The map $(\text{pr}_\epsilon)^\tau$ is the stated split-linear coefficient map; (CW.39) specifies its precise relationship to multiplication. Its value on the supported unit of B is the supported zero e_E .

Every vector in the algebraic kernel in (CW.7) maps under $G(\Phi)$ to supported zero, while external absence maps to external absence. A supported conormal element n maps to $(\epsilon \delta n)^\bullet$, retaining its

coefficient derivative; constant projection gives e_E . The sections, coefficient maps, exact sequences, and original representatives above lift at each fixed support by $(\lambda, u) \mapsto (\lambda, Tu)$. Thus the first thickening, its full kernel and quotient, and the source-selected endomorphism retain their distinct receiving fibres and the common external τ .

If $h = 1$, then $I = P$, $E = E^{[2]} = N = 0$, and all finite maps in (CW.2) have zero source and target. The source function is still the original theta seed $F_h = \Theta\phi_*$, whose map to this arithmetic quotient is zero. This states that case explicitly; the projector formulas dividing by d above have the declared domain $d \geq 1$.

26.6 The arithmetic unit and complete tensor partial residues

Return to the nonempty original packet of (CW.1), with $d \geq 1$ and $k \geq 1$. The unit itself has the following exact image under the algebra bridge already proved above.

Corollary 26.4 (The full unit and its first relation derivative). *For the original $\widehat{U}, U, \beta_h$ of (CW.23),*

$$\begin{aligned}\Phi(\widehat{U}) &= U + \epsilon\delta\widehat{U} = U(1 + \epsilon\beta_h), \\ \Phi(\widehat{U}^{-1}) &= \Phi(\widehat{U})^{-1} = U^{-1}(1 - \epsilon\beta_h).\end{aligned}\tag{CW.41}$$

Proof. The first identity is (CW.6) evaluated on the full thickened unit. The definition $\beta_h = U^{-1}\delta\widehat{U}$ gives its displayed factorization. Since $\epsilon^2 = 0$, $(1 + \epsilon\beta_h)(1 - \epsilon\beta_h) = 1$. The map Φ is unital and multiplicative, so applying it to $\widehat{U}\widehat{U}^{-1} = 1$ proves the second identity with its minus sign. In particular $\Phi(\widehat{U}a) = \Phi(\widehat{U})\Phi(a)$ transports the original arithmetic product through this same algebra map. $\square$

Retain all the original complete orders, $g = 2\xi$ and $v_h = g/h$, and write

$$\begin{aligned}E_h &= \mathbb{C}[s]/(h), & E &= E_h^{\otimes k}, & E_{\widehat{i}} &= \bigotimes_{j \neq i} E_h, \\ v_i &= j_h(v_h) \text{ in the } i\text{-th factor}, & \gamma_i &= j_h(g') \text{ in the } i\text{-th factor}, & U &= \prod_i v_i.\end{aligned}$$

Every v_i is the original complete invertible jet. Let $\ell_h : E_h \rightarrow \mathbb{C}$ extract the coefficient of s^{d-1} from the unique one-variable monic remainder. Let $c_i : E_{\widehat{i}} \rightarrow E$ insert the constant polynomial one in the i -th factor, and let $\ell_i : E \rightarrow E_{\widehat{i}}$ apply ℓ_h in that factor and identity in every other factor. Thus ℓ_i extracts the original coefficient of s_i^{d-1} , retaining every other-variable coordinate. For $k = 1$, $E_{\widehat{i}} = \mathbb{C}$, and c_i is the unit column. Let M_a denote multiplication by the displayed element in its declared algebra.

Define the coefficient-valued partial residue map

$$\mathcal{R}_i : E \longrightarrow E_{\widehat{i}}, \quad \mathcal{R}_i(u) = \sum_{\rho \in \mathbb{Z}} \text{Res}_{s_i=\rho} \frac{\widetilde{u}(s_1, \dots, s_k)}{g(s_i)} ds_i.\tag{CW.42}$$

The coefficients in the other original variables are taken modulo their original ideals after the indicated residue. Each residue uses the usual positively oriented complex s_i -coordinate; the residue notation carries no additional $2\pi i$ factor. Equivalently, each residue equals $1/(2\pi i)$ times the corresponding positively oriented contour integral.

This map is well-defined on E . A change by $h(s_i)p$ in its own variable changes the integrand by $p/v_h(s_i)$, which is holomorphic at every selected centre because the removed zero orders were complete. A change by $h(s_j)p$, $j \neq i$, produces that same factor $h(s_j)$ in the remaining coefficient algebra, hence zero in $E_{\widehat{i}}$. Linearity then covers the full original ideal. The finite sum of coefficient residues is complex-linear and $E_{\widehat{i}}$ -linear.

The exact coefficient formula is

$$\boxed{\mathcal{R}_i = \ell_i M_{v_i}^{-1}}.\tag{CW.43}$$

To prove it, let p_i be the degree-below- d remainder in s_i of $v_i^{-1}u$, retaining the full coefficient algebra $E_{\widehat{\gamma}}$. The residue in (CW.42) equals the corresponding residue of $p_i/h(s_i)$: the difference in the numerator vanishes to the complete selected orders, so division by h is locally holomorphic. Every coefficient of p_i/h has an ordinary partial-fraction expansion. The coefficient of s_i^{-1} at infinity is the sum of the finite simple-pole coefficients; higher-pole terms contribute no s_i^{-1} coefficient. Since h is monic and $\deg_{s_i} p_i < d$, that coefficient is precisely $\ell_i(p_i)$. This proof applies coefficientwise in the finite algebra $E_{\widehat{\gamma}}$ and gives (CW.43) with every other-variable nilpotent coefficient retained.

Define the original partial correction endomorphisms

$$C_i = M_{\gamma_i} c_i \mathcal{R}_i : E \longrightarrow E. \quad (\text{CW.44})$$

They satisfy

$$\begin{array}{l} C_i^2 = dC_i, \quad C_i C_j = C_j C_i, \\ \text{rank } C_i = d^{k-1}, \quad \text{Tr } C_i = d^k. \end{array} \quad (\text{CW.45})$$

Indeed $g' = h'v_h + hv'_h$ gives $\gamma_i = v_i h'_i$ in the original i -th quotient, so

$$\mathcal{R}_i M_{\gamma_i} c_i = \ell_i M_{v_i}^{-1} M_{\gamma_i} c_i = \ell_i M_{h'_i} c_i = dI_{E_{\widehat{\gamma}_i}}. \quad (\text{CW.46})$$

The last equality uses the literal degree $d - 1$ and leading coefficient d of the original derivative h' . This proves the first formula in (CW.45). It also proves that $M_{\gamma_i} c_i$ is injective and $\mathcal{R}_i$ is onto. The factorization (CW.44) therefore has rank $\dim E_{\widehat{\gamma}} = d^{k-1}$. Alternatively a specified right inverse for $\mathcal{R}_i$ is $a \mapsto v_i s_i^{d-1} c_i(a)$, using (CW.43). The two rectangular factors in (CW.44) have traces $\text{Tr}_E(AB) = \text{Tr}_{E_{\widehat{\gamma}}}(BA)$; apply (CW.46) to obtain $d d^{k-1} = d^k$. Operators belonging to distinct variables are tensor products of their respective one-variable maps with identity on the other factors. Their compositions therefore commute, proving the remaining identity.

These operators give precisely the correction of the original full-unit section:

$$J_U = \frac{1}{k} \sum_{i=1}^k C_i. \quad (\text{CW.47})$$

For proof, retain the complete thickened unit $\widehat{U}$, its residue $U = \prod_i v_i$, and its full first derivative $\beta_h = U^{-1} \delta \widehat{U}$. The original source-selected operator is still

$$\begin{aligned} D_U &= M_{\beta_h} + M_U D_0 M_U^{-1}, & [Z, D_U] &= -I_E + J_U, \\ J_U &= M_U \left(\frac{1}{k} \sum_i M_{h'_i} c_i \ell_i \right) M_U^{-1}. \end{aligned} \quad (\text{CW.48})$$

Write $U = v_i U_{\widehat{\gamma}_i}$, with $U_{\widehat{\gamma}_i} = \prod_{j \neq i} v_j$. The maps c_i, ℓ_i are $E_{\widehat{\gamma}_i}$ -linear, so the operator $M_{U_{\widehat{\gamma}_i}}$ commutes through their indicated source and target actions, and its inverse then cancels with exactly that same full other-variable unit. Thus each summand in (CW.48) is

$$M_U M_{h'_i} c_i \ell_i M_U^{-1} = M_{v_i h'_i} c_i \ell_i M_{v_i}^{-1} = M_{\gamma_i} c_i \mathcal{R}_i = C_i. \quad (\text{CW.49})$$

This proves (CW.47). Every v_i retains its original value, and M_{β_h} remains in D_U . It commutes with the original multiplication operator $Z = M_{\sum_i s_i - k/2}$, which is why it contributes zero to that specified commutator.

For completeness, retaining all k residue variables gives

$$\begin{aligned} \mathcal{R}_{\text{all}} : E &\longrightarrow \mathbb{C}, \\ \mathcal{R}_{\text{all}}(u) &= \sum_{\rho_1, \dots, \rho_k \in \mathcal{Z}} \text{Res}_{s_k = \rho_k} \cdots \text{Res}_{s_1 = \rho_1} \frac{\widetilde{u}(\mathbf{s})}{\prod_i g(s_i)} ds_1 \cdots ds_k. \end{aligned} \quad (\text{CW.50})$$

Here the formula means the ordered iterated coefficient residues, with each original variable positively oriented. Each variable operates on a different tensor factor, so swapping their evaluation order gives the identical scalar coefficient and introduces no sign. Formula (CW.43) in each factor proves

$$\mathcal{R}_{\text{all}} = \ell_h^{\otimes k} M_U^{-1}, \quad \boxed{C_1 \cdots C_k(u) = \left(\prod_i \gamma_i \right) \mathcal{R}_{\text{all}}(u).} \quad (\text{CW.51})$$

The map $\ell_h^{\otimes k} : E_h^{\otimes k} \rightarrow \mathbb{C}$ is the tensor product of the original one-variable highest-coefficient functionals. Tensoring the one-variable factorization (CW.44) proves the second equality directly, so it keeps all repeated-root and mixed nilpotent data before the final scalar functional is evaluated. The column and functional are both nonzero: their contraction is

$$\begin{aligned} \mathcal{R}_{\text{all}} \left(\prod_i \gamma_i \right) &= \prod_{i=1}^k \left(\sum_{\rho \in \mathcal{Z}} \text{Res}_{s_i=\rho} \frac{g'(s_i)}{g(s_i)} ds_i \right) \\ &= \prod_{i=1}^k \left(\sum_{\rho} m_{\rho} \right) = d^k. \end{aligned} \quad (\text{CW.52})$$

This is the rank-one full-product correction, with its exact nonzero eigenvalue and trace d^k . Its square is d^k times itself by (CW.51)–(CW.52). The residue of g'/g is m_{ρ} because $g(s) = (s - \rho)^{m_{\rho}} u_{\rho}(s)$ with $u_{\rho}(\rho) \neq 0$; hence $g'/g = m_{\rho}/(s - \rho) + u'_{\rho}/u_{\rho}$. This gives the complete multiplicity proof used in (CW.52).

At $d = 1$, (CW.46) gives $C_i = I_E$ on the one-dimensional E ; consequently $J_U = I_E$, and (CW.51) is the same identity. At $k = 1$, every formula reduces exactly to (CW.31)–(CW.33), including the original residue functional $\mathcal{R}_{\mathcal{Z}}(1, u)$, rank one and trace d . For $h = 1$, the original arithmetic quotient is zero, as already stated above; the positive-degree residue coefficient formulas have the declared domain $d \geq 1$. All repeated fibres, complete arithmetic units, and their original source and target maps have been retained.

27 Exact transitions between nested relative frames

This continuation starts with the original amplitude, relative coordinates, finite source powers, graph and arithmetic observation of FC.1–36. It proves the maps produced when the relative frame is enlarged. Every finite weight, original source coefficient and normal component is retained. The published source is used unchanged.

27.1 Original sources and coefficient inclusions

Fix the original packet polynomial $h, g = 2\xi$, the full entire quotient $v_h = g/h$, and the integer $k \geq 2$. Keep

$$a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad u = \sum_{i=1}^k t_i, \quad t_i = u/k + y_i, \quad y_k = - \sum_{i < k} y_i, \quad \Psi(u, \mathbf{y}) = \prod_{i=1}^k a(u/k + y_i). \quad (\text{NT.1})$$

The ambient fibre is $\mathcal{H} = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$, and the full ambient space is $\mathcal{K} = L^2(\mathbb{R}, du; \mathcal{H})$. Thus the original Jacobian $du d\mathbf{y}$ and the amplitude mass are unchanged. The original relative variables are $z_i = s_i - S/k$, $S = \sum_i s_i$, so $z_i = iy_i$ on this fibre.

Let the finite independent homogeneous relative families at levels $M \leq L$ have column counts r_M, r_L and degrees $\ell_{\alpha}^M, \ell_{\beta}^L$. Their inclusion is the literal constant matrix $E = E_{LM} : \mathbb{C}^{r_M} \rightarrow \mathbb{C}^{r_L}$

satisfying

$$\theta_\alpha^M(\mathbf{z}) = \sum_\beta \theta_\beta^L(\mathbf{z}) E_{\beta\alpha}, \quad j_M(u) = j_L(u)E, \quad j_a(u)c = \Psi(u, \mathbf{y}) \sum_\alpha \theta_\alpha^a(i\mathbf{y})c_\alpha. \quad (\text{NT.2})$$

Comparison of homogeneous components gives $E_{\beta\alpha} = 0$ unless $\ell_\beta^L = \ell_\alpha^M$. Independence of the smaller family makes E injective. For an actual nested monomial family, E is its original zero-extension matrix; no scaling of columns is introduced.

For $a = M, L$ retain the complete admitted source index set and its literal coefficient matrix,

$$\begin{aligned} \mathcal{I}_a &= \{(\alpha, n) : 0 \leq n \leq a - \ell_\alpha^a\}, & q_a &= |\mathcal{I}_a|, \\ P_p^a &= \sum_{(\alpha, n) \in \mathcal{I}_a} p_{\alpha, n} \theta_\alpha^a(\mathbf{z}) S^n, & (C_a(u))_{\beta, (\alpha, n)} &= \delta_{\alpha\beta} (k/2 + iu)^n. \end{aligned} \quad (\text{NT.3})$$

Define the source inclusion $F = F_{LM} : \mathbb{C}^{q_M} \rightarrow \mathbb{C}^{q_L}$ by $F_{(\beta, m), (\alpha, n)} = E_{\beta\alpha} \delta_{mn}$. Every nonzero entry is admitted since $L \geq M$ and the relative degrees agree. Direct expansion gives the exact identities

$$P_{Fp}^L = P_p^M, \quad C_L F = E C_M, \quad C'_L F = E C'_M. \quad (\text{NT.4})$$

Uniqueness of the relative-polynomial expansion and of each power of S proves that F is injective. Every factor $(k/2 + iu)^n$ and its derivative $in(k/2 + iu)^{n-1}$ is retained.

27.2 The original weights and their orthogonal transition

At every real u define, with the ambient Hilbert adjoint,

$$\begin{aligned} W_a &= j_a^* j_a > 0, & B_a &= j_a^* j'_a, & \Gamma_a &= W_a^{-1} B_a, & \Pi_a &= j_a W_a^{-1} j_a^*, \\ N_a &= (I - \Pi_a) j'_a, & \mathcal{N}_a &= N_a^* N_a, & a &= M, L. \end{aligned} \quad (\text{NT.5})$$

These are the original FC matrices. Their positivity and smoothness follow from independence of the relative family and the original Schwartz amplitude: a nonzero relative polynomial cannot vanish almost everywhere where Ψ is nonzero. The inverse matrices therefore exist at every real u , including $u = 0$.

The weighted coefficient retraction and coefficient projection are

$$R = W_M^{-1} E^* W_L : \mathbb{C}^{r_L} \rightarrow \mathbb{C}^{r_M}, \quad Q = ER : \mathbb{C}^{r_L} \rightarrow \mathbb{C}^{r_L}. \quad (\text{NT.6})$$

The symbol Q here is a coefficient projection; Π_a remains the ambient orthogonal projection. From (NT.2),

$$\begin{aligned} W_M &= E^* W_L E, & B_M &= E^* B_L E, & RE &= I, & Q^2 &= Q, & Q^* W_L &= W_L Q, \\ \Gamma_M &= R \Gamma_L E, & \Pi_L \Pi_M &= \Pi_M \Pi_L = \Pi_M. \end{aligned} \quad (\text{NT.7})$$

The first identities follow by multiplication of $j_M = j_L E$ and $j'_M = j'_L E$. For the projection identities, Π_a is self-adjoint, idempotent and has image $V_a = \text{ran } j_a$; $V_M \subseteq V_L$ implies that Π_L fixes $\text{ran } \Pi_M$, and taking adjoints proves the other order.

Theorem 27.1 (The complete connection transition). *Define the coefficient transition and its ambient image by*

$$K = K_{LM} = \Gamma_L E - E \Gamma_M = (I - Q) \Gamma_L E, \quad A = A_{LM} = j_L K. \quad (\text{NT.8})$$

Then $K : \mathbb{C}^{r_M} \rightarrow \mathbb{C}^{r_L}$ has image in the W_L -orthogonal complement of $\text{ran } E$, and

$$\begin{aligned} E^* W_L K &= 0, & A &= (\Pi_L - \Pi_M) j'_M, & N_M &= A + N_L E, \\ \text{ran } A &\subseteq V_L \cap V_M^\perp, & \text{ran } (N_L E) &\subseteq V_L^\perp, & A^* N_L E &= 0. \end{aligned} \quad (\text{NT.9})$$

Consequently the exact positive transition form is

$$\boxed{\mathcal{N}_M = E^* \mathcal{N}_L E + \Delta_{LM}}, \quad \begin{aligned} \Delta_{LM} &= A^* A = K^* W_L K \succeq 0 \\ &= E^* \Gamma_L^* W_L \Gamma_L E - \Gamma_M^* W_M \Gamma_M. \end{aligned} \quad (\text{NT.10})$$

For each $c \in \mathbb{C}^{r_M}$ the following equivalences and rank bound are exact:

$$c^* \Delta_{LM} c = 0 \iff Kc = 0 \iff Ac = 0, \quad \text{rank } \Delta_{LM} = \text{rank } K \leq \min(r_M, r_L - r_M). \quad (\text{NT.11})$$

Proof. Equation (NT.7) gives $E^* W_L \Gamma_L E = W_M \Gamma_M$; subtracting $E^* W_L E \Gamma_M$ proves $E^* W_L K = 0$. The original identity $j_a \Gamma_a = \Pi_a j'_a$ gives $j_L K = \Pi_L j'_L E - \Pi_M j'_M = (\Pi_L - \Pi_M) j'_M$. Splitting $I - \Pi_M = (\Pi_L - \Pi_M) + (I - \Pi_L)$ and using $j'_M = j'_L E$ proves $N_M = A + N_L E$. The commuting projection identities show that $\Pi_L - \Pi_M$ is the orthogonal projection onto $V_L \cap V_M^\perp$. This proves the two range statements and their orthogonality. Taking the Gram of this orthogonal sum proves the first line of (NT.10). Expanding $K = \Gamma_L E - E \Gamma_M$, using $E^* W_L \Gamma_L E = W_M \Gamma_M$ and its adjoint, gives the second line. Positivity of W_L implies that the quadratic form of $K^* W_L K$ vanishes exactly on $\ker K$. Injectivity of j_L gives $\ker A = \ker K$. The image of K lies in a space of dimension $r_L - r_M$; rank-nullity now proves (NT.11). $\square$

For a locally absolutely continuous original coefficient $c(u)$, write $\nabla_a = \partial_u + \Gamma_a$. The full derivative has the three mutually orthogonal components

$$\begin{aligned} \nabla_L(Ec) &= E \nabla_M c + Kc, \\ \partial_u(j_M c) &= j_M \nabla_M c + Ac + N_L Ec, \\ \|\partial_u(j_M c)\|^2 &= \|\nabla_M c\|_{W_M}^2 + c^* \Delta_{LM} c + c^* E^* \mathcal{N}_L Ec, \\ \|\nabla_L(Ec)\|_{W_L}^2 &= \|\nabla_M c\|_{W_M}^2 + c^* \Delta_{LM} c. \end{aligned} \quad (\text{NT.12})$$

The first equality is the product rule with constant E and (NT.8). The derivative formula follows from (NT.9) and the original connection decomposition. The image of its first term lies in V_M , the second in $V_L \cap V_M^\perp$, and the third in $V_L^\perp$. Their orthogonality proves every energy identity without discarding any coefficient derivative.

27.3 Full source matrices, graph maps and arithmetic observation

Define the actual maps on the original coefficient bases by

$$\begin{aligned} \mathbf{N}_a p &= N_a C_a p, & \mathbf{T}_a p &= j_a(C'_a + \Gamma_a C_a) p, & \mathbf{A}_{LM} p &= A_{LM} C_M p, \\ H_a^0 &= \int C_a^* W_a C_a du, & H_a^N &= \mathbf{N}_a^* \mathbf{N}_a, & H_a^\partial &= \mathbf{T}_a^* \mathbf{T}_a + H_a^N, \\ H_{LM}^{\text{tr}} &= \mathbf{A}_{LM}^* \mathbf{A}_{LM} = \int C_M^* \Delta_{LM} C_M du. \end{aligned} \quad (\text{NT.13})$$

Here the displayed equalities of functions take place in $\mathcal{K}$. All integrals are finite: each term is an orthogonal projection of an original polynomially weighted derivative amplitude, and the FC Schwartz estimates apply before any projection. Equations (NT.4), (NT.9), and (NT.12) give

$$\begin{aligned} \mathbf{N}_M &= \mathbf{A}_{LM} + \mathbf{N}_L F, & \mathbf{T}_L F &= \mathbf{T}_M + \mathbf{A}_{LM}, \\ \mathbf{T}_M^* \mathbf{A}_{LM} &= 0, & \mathbf{A}_{LM}^* \mathbf{N}_L F &= 0, & \mathbf{T}_M^* \mathbf{N}_L F &= 0, \\ H_M^0 &= F^* H_L^0 F, & H_M^N &= F^* H_L^N F + H_{LM}^{\text{tr}}, & F^* \mathbf{T}_L^* \mathbf{T}_L F &= \mathbf{T}_M^* \mathbf{T}_M + H_{LM}^{\text{tr}}, & H_M^\partial &= F^* H_L^\partial F. \end{aligned} \quad (\text{NT.14})$$

These identities follow first pointwise, then by integration. In particular, the positive amount transferred into the larger tangential image is precisely the amount removed from the normal form for the same original polynomial.

Let $\widehat{\Pi}_a$ denote the bounded orthogonal projection on $\mathcal{K}$ obtained by applying $\Pi_a(u)$ pointwise. Measurability follows from smoothness, and contraction follows by integrating the pointwise contraction. Define

$$\begin{aligned}\mathcal{U}_{LM}(t, n) &= (t + \widehat{\Pi}_L n, (I - \widehat{\Pi}_L)n), \quad t \in \text{ran } \widehat{\Pi}_M, \quad n \in \ker \widehat{\Pi}_M, \\ \mathcal{U}_{LM}^{-1}(t_L, n_L) &= (\widehat{\Pi}_M t_L, (I - \widehat{\Pi}_M)t_L + n_L).\end{aligned}\tag{NT.15}$$

The domain and codomain are respectively the orthogonal decompositions of $\mathcal{K}$ at levels M and L . The inverse formula is verified by the nested projection identities. Orthogonality of t , $\widehat{\Pi}_L n$ and $(I - \widehat{\Pi}_L)n$ shows that $\mathcal{U}_{LM}$ is unitary between these sum spaces. On the actual finite derivative graph it is exactly

$$\mathcal{U}_{LM}(\mathbf{T}_M p, \mathbf{N}_M p) = (\mathbf{T}_L F p, \mathbf{N}_L F p).\tag{NT.16}$$

Graph addition on either side is the same original function $\partial_u(j_M C_M p) = \partial_u(j_L C_L F p)$, so (NT.16) also supplies the exact map between the two presentations.

For completeness the finite normal maps are injective on nonempty actual polynomial sources. The FC.21–27 proof uses the infinite discrete set $\mathcal{Z}_h$ of unremoved critical-line zeros, retaining each original order ℓ_T and coefficient $c_T = a^{(\ell_T)}(T)/\ell_T!$. If $N_a(u)c = 0$, then $\Psi' \vartheta_c = \Psi \vartheta_d$ with finite relative polynomials and $d = \Gamma_a c$. On a fibre through $t_1 = T$ where all other amplitude factors are nonzero, the leading terms in the original transverse coordinate δ are

$$\Psi = c_T A_b \delta^{\ell_T} + O(\delta^{\ell_T+1}), \quad \Psi' = \frac{\ell_T}{k} c_T A_b \delta^{\ell_T-1} + O(\delta^{\ell_T}).\tag{NT.17}$$

Thus ϑ_c vanishes on that hyperplane. When $k \geq 3$ the other nonzero factors can be chosen in a dense subset of each hyperplane, so polynomial continuity gives vanishing on every such hyperplane; infinitely many distinct T force $\vartheta_c = 0$. For $k = 2$ and $u \notin \mathcal{Z}_h + \mathcal{Z}_h$, every remaining second factor $a(u - T)$ is nonzero; infinitely many distinct univariate roots give the same conclusion. Since $\mathcal{Z}_h + \mathcal{Z}_h$ is countable, $\mathcal{N}_a > 0$ almost everywhere for every finite nonempty frame. A nonzero vector polynomial $C_a p$ is nonzero outside a finite set: one nonzero component polynomial suffices to bound its common real zero set. Integrating proves $H_a^N > 0$. This argument also includes nonzero source relations whose arithmetic class is zero.

Set $\mathcal{R}_M^N = \text{ran } \mathbf{N}_M$ and $\mathcal{R}_{L|M}^N = \text{ran}(\mathbf{N}_L F)$. The concrete normal transition and its inverse on these specified images are

$$\begin{aligned}\mathcal{Q}_{LM} : \mathcal{R}_M^N &\longrightarrow \mathcal{R}_{L|M}^N, & \mathcal{Q}_{LM} r &= (I - \widehat{\Pi}_L) r, & \mathcal{Q}_{LM} \mathbf{N}_M p &= \mathbf{N}_L F p, \\ H_{L|M}^N &= F^* H_L^N F > 0, & \mathcal{Q}_{LM}^{-1} r_L &= \mathbf{N}_M (H_{L|M}^N)^{-1} F^* \mathbf{N}_L^* r_L, \\ \|r\|^2 - \|\mathcal{Q}_{LM} r\|^2 &= \|\mathbf{A}_{LM} p\|^2 \quad (r = \mathbf{N}_M p).\end{aligned}\tag{NT.18}$$

Positivity of $H_{L|M}^N$ follows from injectivity of F and of $\mathbf{N}_L$. Substitution of $r_L = \mathbf{N}_L F p$ proves the inverse formula in both orders. Thus $\mathcal{Q}_{LM}$ is a contraction and a linear isomorphism between these finite images, with its complete norm loss specified. No inverse bound independent of the frame level is inserted.

The bounded ambient extension obtained from the exact smaller normal image inverse is

$$\begin{aligned}V_{LM} &= \mathbf{N}_L F (H_M^N)^{-1} \mathbf{N}_M^* = (I - \widehat{\Pi}_L) \Pi_M^N : \mathcal{K} \rightarrow \mathcal{K}, \\ \Pi_M^N &= \mathbf{N}_M (H_M^N)^{-1} \mathbf{N}_M^*, & \|V_{LM}\| &\leq 1, & V_{LM}|_{\mathcal{R}_M^N} &= \mathcal{Q}_{LM}.\end{aligned}\tag{NT.18a}$$

The equality follows by multiplying the first line of (NT.18) by $(H_M^N)^{-1} \mathbf{N}_M^*$. The two factors on the right are orthogonal projections and hence contractions. The smaller normal image is fixed by Π_M^N , proving the restriction statement. For $L = M$ this ambient extension is Π_M^N and restricts to the identity precisely on its declared image.

Retain the original arithmetic target, full quotient unit and derivative,

$$\begin{aligned}\mathfrak{A}_h &= \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)), \\ U &= \prod_i v_h(s_i), \quad v = [U], \quad \delta = \frac{1}{k} \sum_i \partial_{s_i}, \quad \beta = v^{-1}[\delta U], \\ (j_a^{\log})_{\alpha, n} &= nv[\theta_\alpha^a S^{n-1}] + \beta v[\theta_\alpha^a S^n].\end{aligned}\tag{NT.19}$$

At $n = 0$ the first term is zero; no negative power is taken. The full Hermite class, multiplicities, $g = 2\xi$ and unit derivatives are retained. The identity $\delta S = 1, \delta z_i = 0$ proves these columns by the product rule applied before taking the quotient. Equation (NT.4) then proves $j_L^{\log} F = j_M^{\log}$ entry by entry. On the actual normal image put $\text{Obs}_a^{\log} r = j_a^{\log}(H_a^N)^{-1} \mathbf{N}_a^* r$. The exact arithmetic transition is

$$\boxed{\text{Obs}_L^{\log} \mathcal{Q}_{LM} = \text{Obs}_M^{\log} \quad \text{on } \mathcal{R}_M^N.}\tag{NT.20}$$

Indeed both sides on $r = \mathbf{N}_M p$ are $j_M^{\log} p$ by (NT.18)–(NT.19). Their original source path is graph addition, followed by $-i\mathcal{U}_k^{-1}$ and the complete original arithmetic jet; (NT.16) preserves that path exactly. For $h = 1$ the arithmetic target is the zero algebra and the observation is its unique zero map; every analytic identity above continues unchanged.

To retain any declared target metric as well, choose its fixed original quotient basis and let $G_M^{\text{tar}}, G_L^{\text{tar}} > 0$ be the two declared Hermitian matrices in that same basis. The coordinate identity $\tau_{LM} : \mathfrak{A}_h \rightarrow \mathfrak{A}_h$ has the exact comparison operator

$$\begin{aligned}D_{LM}^{\text{tar}} &= (G_M^{\text{tar}})^{-1} G_L^{\text{tar}}, \quad (D_{LM}^{\text{tar}})^* G_M^{\text{tar}} = G_M^{\text{tar}} D_{LM}^{\text{tar}} = G_L^{\text{tar}}, \\ \|\text{Obs}_L^{\log} \mathcal{Q}_{LM} \mathbf{N}_M p\|_{G_L^{\text{tar}}}^2 &= p^* (j_M^{\log})^* G_L^{\text{tar}} j_M^{\log} p, \\ \|\text{Obs}_L^{\log} \mathcal{Q}_{LM} \mathbf{N}_M p\|_{G_L^{\text{tar}}}^2 - \|\text{Obs}_M^{\log} \mathbf{N}_M p\|_{G_M^{\text{tar}}}^2 &= p^* (j_M^{\log})^* (G_L^{\text{tar}} - G_M^{\text{tar}}) j_M^{\log} p.\end{aligned}\tag{NT.20a}$$

Direct multiplication proves the comparison identities, and (NT.20) proves the two observed-form identities. This specifies both target forms through the actual coordinate map without assigning a sign to their difference. In the zero target all these are the unique zero forms and maps, with no positive-dimensional inverse needed.

27.4 Every variable-frame term

Let $G_M(u), G_L(u)$ be smooth invertible matrices of their declared sizes. The changed frame and coefficients are $j_a^G = j_a G_a, c_a^G = G_a^{-1} c_a$. The same inclusion now has the generally variable matrix

$$E^G = G_L^{-1} E G_M, \quad R^G = G_M^{-1} R G_L, \quad \Gamma_a^G = G_a^{-1} \Gamma_a G_a + G_a^{-1} G'_a.\tag{NT.21}$$

For any such variable inclusion the exact formulas are

$$\begin{aligned}\Gamma_M^G &= R^G (\Gamma_L^G E^G + (E^G)'), \quad K^G = (E^G)' + \Gamma_L^G E^G - E^G \Gamma_M^G, \\ K^G &= G_L^{-1} K G_M, \quad A^G = A G_M, \quad N_a^G = N_a G_a, \quad \Delta_{LM}^G = G_M^* \Delta_{LM} G_M.\end{aligned}\tag{NT.22}$$

To prove the first equality, differentiate $j_M^G = j_L^G E^G$ before taking its mixed Gram: $B_M^G = (E^G)^* B_L^G E^G + (E^G)^* W_L^G (E^G)'$. Multiplication by $(W_M^G)^{-1}$ gives it. In the second line, differentiate $G_L^{-1} E G_M$ and substitute both transformed connections. The terms containing G'_L and G'_M cancel separately, leaving $G_L^{-1} K G_M$. The original projection images are unchanged; applying $I - \Pi_a$ to $(j_a G_a)'$ therefore gives $N_a G_a$. This proves the remaining statements and every variable-inclusion version of (NT.9)–(NT.12). In particular the term $(E^G)'$ cannot be deleted from the transition.

For the same original polynomial source, $C_a^G = G_a^{-1} C_a$ and $C_L^G F = E^G C_M^G$. Its tangential and normal amplitudes obey

$$\mathbf{N}_a^G = \mathbf{N}_a, \quad \mathbf{T}_a^G = \mathbf{T}_a, \quad \mathbf{A}_{LM}^G = \mathbf{A}_{LM}.\tag{NT.23}$$

The first and third follow from (NT.22); in the second, the derivative of G_a^{-1} cancels the connection term $G_a^{-1}G'_a$. Thus the literal source matrices in (NT.14), the normal-image transition, its inverse and the arithmetic observation are all the same maps after their explicit coefficient transport. The transformed coefficient functions are exactly this image of the original filtered source; their degree bounds are not replaced by a new polynomial frame convention.

27.5 Nested chains and both quadratic branches

For a third level $T \geq L$ retain $E_{TM} = E_{TL}E_{LM}$ and $F_{TM} = F_{TL}F_{LM}$. Expansion of (NT.8) proves

$$\begin{aligned} K_{TM} &= K_{TL}E_{LM} + E_{TL}K_{LM}, & A_{TM} &= A_{TL}E_{LM} + A_{LM}, \\ \Delta_{TM} &= \Delta_{LM} + E_{LM}^* \Delta_{TL} E_{LM}, & H_{TM}^{\text{tr}} &= H_{LM}^{\text{tr}} + F_{LM}^* H_{TL}^{\text{tr}} F_{LM}. \end{aligned} \quad (\text{NT.24})$$

The two ambient summands of A_{TM} lie in $V_T \cap V_L^\perp$ and $V_L \cap V_M^\perp$; they are orthogonal. Taking their Gram proves the third identity, and integration with (NT.4) proves the fourth. Induction gives the same mutually orthogonal sum and positive telescoping formula along every finite nested chain. The actual graph maps and normal maps compose:

$$\mathcal{U}_{TL}\mathcal{U}_{LM} = \mathcal{U}_{TM}, \quad \mathcal{Q}_{TL}\mathcal{Q}_{LM} = \mathcal{Q}_{TM} \quad \text{on the corresponding original source image.} \quad (\text{NT.25})$$

For the normal maps this follows directly from $(I - \hat{\Pi}_T)(I - \hat{\Pi}_L) = I - \hat{\Pi}_T$. For the graph maps, both compositions are the two orthogonal projections of the same graph-added vector. These proofs retain the intermediate positive transition forms. The ambient normal extensions also compose on all of $\mathcal{K}$: $V_{TL}V_{LM} = V_{TM}$. Indeed their matrix product contains $(H_L^N)^{-1}\mathbf{N}_L^*\mathbf{N}_L = I_{q_L}$, leaving $\mathbf{N}_T F_{TL} F_{LM} (H_M^N)^{-1}\mathbf{N}_M^*$.

Now put $x = u^2 > 0$, $v = \sqrt{x}$ and, at each level, keep

$$M_a = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}, \quad H_a = M_a^* \frac{\text{diag}(W_a(v), W_a(-v))}{2v} M_a, \quad Q_a^n = M_a^* \frac{\text{diag}(\mathcal{N}_a(v), \mathcal{N}_a(-v))}{2v} M_a. \quad (\text{NT.26})$$

Let $\mathcal{A}_a = M_a^{-1} \text{diag}(\Gamma_a(v)/(2v), -\Gamma_a(-v)/(2v)) M_a + M_a^{-1} M'_a$ be the complete paired connection. Its derivative includes the original $M_a^{-1} M'_a = \text{diag}(0, I/(2x))$. Write $\tilde{E} = \text{diag}(E, E)$ and

$$\begin{aligned} \mathcal{K}_{LM} &= \mathcal{A}_L \tilde{E} - \tilde{E} \mathcal{A}_M = M_L^{-1} \text{diag}\left(\frac{K(v)}{2v}, -\frac{K(-v)}{2v}\right) M_M, \\ H_M &= \tilde{E}^* H_L \tilde{E}, \quad \tilde{E}^* H_L \mathcal{K}_{LM} = 0, \\ \boxed{Q_M^n &= \tilde{E}^* Q_L^n \tilde{E} + 4x \mathcal{K}_{LM}^* H_L \mathcal{K}_{LM}.} \end{aligned} \quad (\text{NT.27})$$

The identity $M_L \tilde{E} = \text{diag}(E, E) M_M$ proves the first two transport assertions by substitution. Applying (NT.10) on both branches gives the normal identity: multiplication by $4x = 4v^2$ cancels precisely the squares of $1/(2v)$ in $\mathcal{K}_{LM}$, leaving $M_M^* \text{diag}(\Delta_{LM}(v), \Delta_{LM}(-v)) M_M/(2v)$. Thus both original signs and both Jacobian factors are retained. For $\boldsymbol{\eta}_M = (a_M, b_M)^t$ the paired inclusion is $\boldsymbol{\eta}_L = \tilde{E} \boldsymbol{\eta}_M$ and

$$(\partial_x + \mathcal{A}_L) \tilde{E} \boldsymbol{\eta}_M = \tilde{E} (\partial_x + \mathcal{A}_M) \boldsymbol{\eta}_M + \mathcal{K}_{LM} \boldsymbol{\eta}_M. \quad (\text{NT.28})$$

Weighted orthogonality in (NT.27) shows that its tangential energy gain is exactly the normal energy loss in that equation. It follows pointwise, before integration, that the complete FC.15 energy is the same on the two included presentations.

In variable frames the paired inclusion is instead $\mathcal{E}(x) = M_L^{-1} \text{diag}(E^G(v), E^G(-v)) M_M$. The corresponding transition is exactly $\mathcal{E}' + \mathcal{A}_L^G \mathcal{E} - \mathcal{E} \mathcal{A}_M^G$; applying (NT.22) on both branches and the product rule proves this formula, including its derivative. The transported original density retains $(\mathcal{A}_a)^* H_a + H_a \mathcal{A}_a = H'_a + H_a/(2x)$ at both levels, since this identity is obtained by differentiating the same two original branch densities. No evenness assumption is used in (NT.26)–(NT.28).

27.6 Exact zero and endpoint scopes

The identities above hold at every real u . If a coefficient vector or admitted polynomial is zero, each displayed linear map sends it to its corresponding zero. If the smaller frame is empty, its coefficient and source spaces are zero and each inclusion, transition and inverse-on-zero-image is the unique such map. For equal frames with an invertible constant E , the weighted projection Q is the identity, so $K = A = \Delta_{LM} = 0$; the normal map is transported exactly by E . For unequal frames, (NT.11) is the precise equality criterion; strict positivity of the actual finite $\mathcal{N}_L$ does not force strict positivity of Δ_{LM} . The maps in (NT.9) and their Gram identity give the exact relation in every such case.

For a quartet packet and $k = 2$, the original amplitude is even. At $u = 0$ its product derivative is $\frac{1}{2}(a'(y)a(-y) + a(y)a'(-y)) = 0$. Every fixed relative frame therefore has $j'_a(0) = \Gamma_a(0) = N_a(0) = 0$ and $K_{LM}(0) = A_{LM}(0) = \Delta_{LM}(0) = 0$. Under a smooth variable frame, $\Gamma_a^G(0)$ can have the retained term $G_a(0)^{-1}G'_a(0)$; (NT.22) still gives $K^G(0) = N_a^G(0) = 0$. The integrated finite-source normal inverse in (NT.18) remains valid because its positivity is an almost-everywhere and integrated assertion, proved in (NT.17).

Let $\mathcal{D}_a$ denote the coefficient functions locally absolutely continuous on $\mathbb{R}$ for which $\int c^* W_a c$ and $\int \|\partial_u(j_a c)\|^2$ are finite. For the constant inclusion, the exact domain correspondence is

$$E\mathcal{D}_M = \mathcal{D}_L \cap \{c_L : c_L(u) \in \text{ran } E \text{ for almost every } u\}. \quad (\text{NT.29})$$

The forward inclusion follows from $j_L E c = j_M c$ and constant E . Conversely, use any constant left inverse of E , for example $(E^* E)^{-1} E^*$, to obtain a locally absolutely continuous c_M from c_L . The closed fixed subspace condition and continuity give $c_L = E c_M$ everywhere. The same amplitude and derivative identities then give both required finite integrals. This proves both inclusions without identifying distinct coefficient metrics.

On the quadratic branches the original domain has the exact trace condition

$$\lim_{x \downarrow 0} (a_M(x) + i\sqrt{x} b_M(x)) = \lim_{x \downarrow 0} (a_M(x) - i\sqrt{x} b_M(x)). \quad (\text{NT.30})$$

The analogous condition after $\tilde{E}$ is equivalent because E is injective. The finite norm and energy on each open branch give finite traces: near zero, W_M has positive upper and lower bounds and Γ_M is bounded, so the coefficient and its ordinary derivative are both locally square-integrable. The fundamental theorem and Cauchy-Schwarz give the one-sided limits. Integration by parts then shows that unequal limits give their jump times δ_0 , whereas equality joins them into the original weak-derivative domain. This proves the trace assertion. For smooth invertible gauges through zero the matching condition is transported by their common value at zero. For gauges specified only on the punctured branches one must retain the original condition with $G_M(v)c_M^G(v)$ and $G_M(-v)c_M^G(-v)$; it is exactly (NT.30) applied to the original coefficient. No new endpoint value is assigned by the half-line formulas alone.

Finally every displayed linear source, graph and observation map has the original fixed-support lift: on a fixed support label it acts by the proved linear map, preserves a supported zero as a supported zero, and sends external absence to external absence. Its additivity and scalar laws follow directly from linearity. In particular a nonzero polynomial relation is carried by $P_{Fp}^L = P_p^M$ before any full Hermite quotient, and its arithmetic derivative is the unchanged value (NT.19)–(NT.20).

27.7 One independent exact original-coordinate calibration

This finite calibration has ambient space $\mathbb{C}^5$ with its standard Hermitian form. It checks the proved transition using literal polynomial matrices, without identifying this finite model with the arithmetic

amplitude (NT.1). Keep

$$p(u) = 1 + u, \quad q(u) = 1 + u + u^2, \quad j_L(u) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & 2p & -p \\ 2q & -q & q \end{pmatrix}, \quad E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

$$j_M(u) = j_L(u)E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & p \\ 3q & 0 \end{pmatrix}. \quad (\text{NT.31})$$

The first three rows make j_L injective, and E is injective. For $a = (1, 2, -1)^t$ and $b = (2, -1, 1)^t$, direct multiplication gives

$$W_L = I_3 + p^2 aa^t + q^2 bb^t > 0, \quad W_M = \begin{pmatrix} 2 + 9q^2 & 1 \\ 1 & 2 + p^2 \end{pmatrix} > 0,$$

$$B_L = pp'aa^t + qq'bb^t = W'_L/2, \quad B_M = \text{diag}(9qq', pp') = W'_M/2,$$

$$W_M(1) - W_M(-1) = \begin{pmatrix} 72 & 0 \\ 0 & 4 \end{pmatrix}, \quad [W_M(0), W_M(1)] = \begin{pmatrix} 0 & -69 \\ 69 & 0 \end{pmatrix}, \quad (\text{NT.32})$$

$$W_L(1) - W_L(-1) = 4aa^t + 8bb^t, \quad [W_L(0), W_L(1)] = \begin{pmatrix} 0 & 25 & -15 \\ -25 & 0 & -5 \\ 15 & 5 & 0 \end{pmatrix}.$$

Positivity follows from the squared norm of the injective original maps. Every displayed difference is nonzero, so both original weights are non-even and have noncommuting values.

At $u = 1$, put $J = j_L(1)$ and $D = j'_M(1)$. Their matrices and the actual smaller connection are

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 4 & -2 \\ 6 & -3 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 9 & 0 \end{pmatrix}, \quad (\text{NT.33})$$

$$W_L(1) = \begin{pmatrix} 41 & -10 & 14 \\ -10 & 26 & -17 \\ 14 & -17 & 14 \end{pmatrix}, \quad W_M(1) = \begin{pmatrix} 83 & 1 \\ 1 & 6 \end{pmatrix}, \quad \Gamma_M(1) = \frac{1}{497} \begin{pmatrix} 486 & -2 \\ -81 & 166 \end{pmatrix}.$$

An independent construction of the transition projection starts with

$$v = (171, -277, -603, 440, 48)^t.$$

Multiplication verifies $v = J(171, -277, -603)^t$, $(JE)^*v = 0$, and $v^*v = 665483$. The larger and smaller column spaces have dimensions three and two. Their orthogonal difference is therefore exactly the line spanned by v , so

$$\Pi_L(1) - \Pi_M(1) = \frac{vv^*}{665483}, \quad A(1) = \frac{8}{665483}v \begin{pmatrix} 54 & 55 \end{pmatrix},$$

$$N_L(1)E = \frac{1}{1339} \begin{pmatrix} -1458 & -146 \\ 459 & -202 \\ -567 & 92 \\ 54 & 55 \\ 225 & 6 \end{pmatrix}. \quad (\text{NT.34})$$

For the last matrix define $T = 1339^{-1} \begin{pmatrix} 1458 & 146 \\ -459 & 202 \\ 567 & -92 \end{pmatrix}$. Direct multiplication gives $W_L(1)T = J^*D = \begin{pmatrix} 54 & 2 \\ -27 & 4 \\ 27 & -2 \end{pmatrix}$. Thus $D - JT$ is exactly the displayed matrix $N_L(1)E$. It is orthogonal to every column of J and hence to v . This verifies both normal components independently of the proposed connection difference. Their full Grams are

$$A(1)^*A(1) = \frac{1}{665483} \begin{pmatrix} 186624 & 190080 \\ 190080 & 193600 \end{pmatrix}, \quad E^*\mathcal{N}_L(1)E = \frac{1}{1339} \begin{pmatrix} 2025 & 54 \\ 54 & 55 \end{pmatrix}. \quad (\text{NT.35})$$

Take the original coefficient curve $c(u) = (1 + iu, u^2 - i)^t$. Its directly differentiated vector at one is

$$w = Dc(1) + JEc'(1) = (i, 2, 2 + i, 5 - i, 9 + 18i)^t, \quad w^*w = 441. \quad (\text{NT.36})$$

The three components in (NT.12), with their original values, are

$$t = \frac{1}{497} \begin{pmatrix} 484 + 985i \\ 1079 - 247i \\ 1563 + 738i \\ 2158 - 494i \\ 4356 + 8865i \end{pmatrix}, \quad N_L(1)Ec(1) = \frac{1}{1339} \begin{pmatrix} -1604 - 1312i \\ 257 + 661i \\ -475 - 659i \\ 109 - i \\ 231 + 219i \end{pmatrix}, \quad (\text{NT.37})$$

$$A(1)c(1) = \frac{8(109 - i)}{665483}v,$$

$$w = t + A(1)c(1) + N_L(1)Ec(1), \quad \boxed{441 = \frac{217065}{497} + \frac{58496}{51191} + \frac{320}{103}}.$$

Multiplication of the explicit vectors proves their sum and their squared norms; their orthogonality was proved from the original column spaces above. All three energies are strictly positive. The number 441 is this pointwise original energy. Multiplying c by any smooth compactly supported scalar function equal to one near $u = 1$ retains the same pointwise value and derivative and makes the corresponding integrated energy finite.

The companion exact script `nested_relative_frame_fixture_20260912.py` retains these original matrices and verifies 24 rational-function identities and nonzero witnesses, with identical successful records in ordinary and optimized Python. Its full matrix receipts and independent sample review are pinned in `nested_relative_frame_fixture_20260912_manifest.json`. The full general proof is (NT.1)–(NT.30); the single concrete family above supplies an independent nonzero calculation of the transition and the surviving normal component.

28 Density of the actual relative frames and the limiting derivative graph

This working continuation proves the nested-frame limit for the original arithmetic amplitude. The input is the sum-connection construction (FC.1)–(FC.36), retained at the source hash recorded in the accompanying receipt. All statements below use the original measure, full packet, relative coordinates, polynomial source, and derivative. The proof uses an exponential moment that is established directly from the arithmetic amplitude. No recurrence or Christoffel asymptotic is assumed.

28.1 An exponential bound with the original arithmetic factors

Let

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad h(s) = \prod_{\rho \in \mathcal{Z}} (s - \rho)^{m_\rho},$$

$$d = \deg h, \quad v_h(s) = g(s)/h(s), \quad a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w(t) = |a(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w(t) dt. \quad (\text{ND.1})$$

Here the set of selected zeros is finite and each selected multiplicity is the full multiplicity in g . The quotient is its entire extension at those zeros. The empty product $h = 1$ is allowed. The packet is stable under $\rho \mapsto 1 - \bar{\rho}$, as in FC.1. For a monic polynomial this says $h(1 - \bar{s}) = (-1)^d \bar{h}(s)$; the corresponding identity for g has factor one. Hence on the original line

$$\overline{a(t)} = (-1)^d a(t), \quad \overline{a^{(j)}(t)} = (-1)^d a^{(j)}(t). \quad (\text{ND.2})$$

There is no change of phase or division by the mass μ_h .

Theorem 28.1 (Actual amplitude and all derivative tails). *For every integer $j \geq 0$ and every real b with $0 < b < \pi/4$, there is a finite constant $C_{h,j,b}$ such that*

$$|a^{(j)}(t)| \leq C_{h,j,b} e^{-b|t|} \quad (t \in \mathbb{R}). \quad (\text{ND.3})$$

The amplitude is a nonzero entire function of t . In particular $0 < \mu_h < \infty$, all polynomially weighted derivative integrals used below exist, and its real zero set is discrete.

Proof. We give the tail argument, including the gamma factor. For $\lambda > 0$, $\tau \in \mathbb{R}$, and $0 < \phi < \pi/2$, rotate Euler's gamma integral in the sector with angle $\theta = \text{sgn}(\tau)\phi$, taking $\theta = 0$ when $\tau = 0$. The principal power is holomorphic on the open sector. On the circle of radius R the exponential has modulus at most $e^{-R \cos \phi}$; its product with the power and arc length tends to zero as $R \rightarrow \infty$. The small circular contribution is bounded by a constant times ε^λ and tends to zero. Cauchy's theorem therefore gives

$$\begin{aligned} \Gamma(\lambda + i\tau) &= e^{i\theta(\lambda+i\tau)} \int_0^\infty r^{\lambda+i\tau-1} e^{-e^{i\theta}r} dr, \\ |\Gamma(\lambda + i\tau)| &\leq e^{-\phi|\tau|} \Gamma(\lambda) (\cos \phi)^{-\lambda}. \end{aligned} \quad (\text{ND.4})$$

At $\tau = 0$ the second bound follows directly from Euler's integral. Its constant is bounded when λ ranges over a compact subinterval of $(0, \infty)$.

For completeness, summing the counting function $[x]$ on $\Re s > 1$ gives $\zeta(s) = s \int_1^\infty [x] x^{-s-1} dx$. Splitting $[x] = x - \{x\}$ yields

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (\text{ND.5})$$

The last integral is absolutely and locally uniformly convergent on $\Re s > 0$; its differentiated integrals converge on compact subsets because $\int_1^\infty x^{-1-\sigma} (\log x)^n dx$ is finite for every $\sigma > 0$. Thus (ND.5) continues the same zeta function to that half-plane away from $s = 1$. For $s = \sigma + it$, $1/4 \leq \sigma \leq 3/4$, we have $|s/(s-1)| \leq 5$, $|s|/\sigma \leq 1 + 4|t|$, and consequently $|\zeta(s)| \leq 6 + 4|t|$.

Write $z = t + iv$ with $|v| \leq 1/4$, so that $s = 1/2 + iz = (1/2 - v) + it$. In (ND.4) take $\lambda = (1/2 - v)/2 \in [1/8, 3/8]$ and $\tau = t/2$. The modulus of $\pi^{-s/2}$ is exactly $\pi^{-(1/2-v)/2}$. The two polynomial factors in g , followed by (ND.5), therefore give the bound

$$|g(1/2 + iz)| \leq C_\phi (1 + |t|)^3 e^{-\phi|t|/2} \quad (|\Im z| \leq 1/4). \quad (\text{ND.6})$$

For $|t| \geq \max(1, 2 \max_{\rho \in \mathcal{Z}} |\Im \rho|)$, each factor $|s - \rho|$ is at least $|t|/2$. Thus $|h(s)| \geq 2^{-d} |t|^d$; for the empty product the lower bound is one and the maximum over roots is unnecessary. Division by this original polynomial, without removing its remaining unit, bounds v_h on the unbounded part of the strip. The complementary closed rectangle is compact and the entire quotient is bounded there, including every cancelled root. Choose $2b < \phi < \pi/2$. The polynomial factor times $e^{-(\phi/2-b)|t|}$ is bounded. It follows that $|a(z)| \leq C_{h,b} e^{-b|\Re z|}$ on this strip, with the original factor $1/\sqrt{2\pi}$ included in $C_{h,b}$. The Cauchy estimate on the circle of radius $1/8$ about a real t now gives

$$|a^{(j)}(t)| \leq j! 8^j C_{h,b} e^{b/8} e^{-b|t|}.$$

This proves (ND.3). Entirety follows from cancellation at the specified finite roots. The function cannot vanish identically: the formula for g is nonzero on the real interval $s > 1$, and division by a nonzero polynomial preserves nonidentity. The identity theorem makes the real zeros isolated. Positivity and finiteness of the mass follow from nonidentity and (ND.3). $\square$

28.2 Every centred fibre and its exponential moment

Fix $k \geq 2$, put $r = k - 1$, and retain the real coordinates

$$u = \sum_{i=1}^k t_i, \quad y_i = t_i - u/k \quad (1 \leq i \leq r), \quad y_k = -\sum_{i=1}^r y_i, \quad t_i = u/k + y_i,$$

$$S = k/2 + iu, \quad z_i = s_i - S/k = iy_i, \quad \Psi_u(y) = \prod_{i=1}^k a(u/k + y_i), \quad d\mu_u(y) = |\Psi_u(y)|^2 dy. \quad (\text{ND.7})$$

The independent relative variables are $y_1, \dots, y_r$. The determinant of the map with ordered variables $(u, y_1, \dots, y_r) \mapsto (t_1, \dots, t_k)$ is $(-1)^r$. Indeed moving the first column to the last contributes $(-1)^r$, and the resulting block determinant is $\det \begin{pmatrix} I_r & \mathbf{1}/k \\ -\mathbf{1}^T & 1/k \end{pmatrix} = 1/k + r/k = 1$. In particular the original product measure is exactly $du dy$. For $k = 2$, $v = t_1 - t_2 = 2y_1$ gives $du dv/2$. Neither the centre $k/2$ nor this Jacobian is changed.

For every fixed u , each of the k affine functions $y \mapsto u/k + y_i$ is nonconstant. The inverse image of a real zero of a is a proper affine hyperplane. There are at most countably many such hyperplanes, each of Lebesgue measure zero. Consequently $\Psi_u \neq 0$ almost everywhere for every u , including any fibre on which a particular normal derivative vanishes. The orders of the zeros remain the orders of the original product; the measure assertion does not modify the product on a hyperplane.

For every $0 < \eta < \pi/2$, choose b so that $\eta < 2b < \pi/2$. Since $\sum_{i=1}^k |t_i| \geq \|y\|_1 - r|u|/k$, (ND.3) proves the explicit bound

$$\begin{aligned} \int_{\mathbb{R}^r} e^{\eta\|y\|_1} d\mu_u(y) &\leq C_{h,0,b}^{2k} e^{2br|u|/k} \int_{\mathbb{R}^r} e^{-(2b-\eta)\|y\|_1} dy \\ &= C_{h,0,b}^{2k} e^{2br|u|/k} \left(\frac{2}{2b-\eta} \right)^r < \infty. \end{aligned} \quad (\text{ND.8})$$

This bound is uniform on compact u -intervals. It proves both finite moments and $0 < m_k(u) := \mu_u(\mathbb{R}^r) < \infty$ for every u . The coordinate change and Tonelli's theorem retain the full mass identity

$$\int_{\mathbb{R}} m_k(u) du = \mu_h^k. \quad (\text{ND.9})$$

At fixed relative coordinate the exact derivative is

$$\Psi'_u(y) = \frac{1}{k} \sum_{i=1}^k a'(u/k + y_i) \prod_{j \neq i} a(u/k + y_j). \quad (\text{ND.10})$$

Every derivative of this product, multiplied by any polynomial in (u, y) , is square integrable in $du dy$. To verify this claim directly, every derivative is a finite sum of products of derivatives of the original $a(t_i)$, with the powers of $1/k$ from (ND.7) retained. Each product is bounded by a constant times $e^{-b \sum |t_i|}$ by (ND.3). A polynomial in (u, y) becomes a polynomial in $t_1, \dots, t_k$ under the displayed invertible linear map; each of its monomials is integrable against $e^{-2b \sum |t_i|}$. The same bound on a compact u -interval proves continuity, and all orders of differentiation, of the relevant functions with values in $L^2(dy)$, by dominated convergence.

28.3 A complete weighted polynomial density proof

Theorem 28.2 (Exponential moments imply the required density). *Let μ be a finite positive Borel measure on $\mathbb{R}^r$ such that $\int e^{\eta\|y\|_1} d\mu(y) < \infty$ for some $\eta > 0$. Complex polynomials in all r coordinates are dense in $L^2(\mu)$.*

Proof. All polynomials belong to this space by the exponential moment. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial, using inner products conjugate linear in the first entry, and define the finite complex measure $d\nu = \bar{f} d\mu$. For $0 \leq 2\tau < \eta$, Cauchy-Schwarz gives

$$\int e^{\tau\|y\|_1} d|\nu|(y) \leq \|f\|_{L^2(\mu)} \left(\int e^{2\tau\|y\|_1} d\mu(y) \right)^{1/2} < \infty. \quad (\text{ND.11})$$

Every monomial moment of ν is zero. Fix $\xi \in \mathbb{R}^r \setminus \{0\}$. The function

$$F_\xi(z) = \int e^{iz\xi \cdot y} d\nu(y)$$

is holomorphic on the connected strip $|\Im z| \|\xi\|_\infty < \eta/2$. To see the differentiability, restrict to a compact substrip and choose $\tau < \eta/2$ strictly larger than its bound on $|\Im z| \|\xi\|_\infty$. Every polynomial factor from differentiation is bounded by the remaining exponential slack in (ND.11), which justifies differentiation under the integral. In a disc about zero, the exponential series is absolutely integrable by the same bound. Its n -th coefficient is $i^n \int (\xi \cdot y)^n d\nu(y)/n! = 0$, since this is a polynomial moment. Thus F_ξ vanishes near zero and throughout the strip by the holomorphic identity theorem. In particular $\hat{\nu}(\xi) = F_\xi(1) = 0$. The zero moment also gives $\hat{\nu}(0) = 0$.

Here is the Fourier uniqueness argument needed for this finite complex measure. For $\varepsilon > 0$ put

$$G_\varepsilon(x) = (4\pi\varepsilon)^{-r/2} e^{-|x|^2/(4\varepsilon)} = (2\pi)^{-r} \int_{\mathbb{R}^r} e^{-\varepsilon|\xi|^2} e^{i\xi \cdot x} d\xi. \quad (\text{ND.12})$$

The equality follows in one variable by differentiating the Gaussian Fourier integral and integrating by parts: its derivative is $-x/(2\varepsilon)$ times itself and its value at zero is $\sqrt{\pi/\varepsilon}$. The latter Gaussian integral follows by squaring it and using polar coordinates. Taking products gives (ND.12). Since ν is finite and the frequency Gaussian is integrable, Fubini's theorem gives, with the same positive Fourier sign,

$$(G_\varepsilon * \nu)(x) = (2\pi)^{-r} \int_{\mathbb{R}^r} e^{-\varepsilon|\xi|^2} e^{i\xi \cdot x} \hat{\nu}(-\xi) d\xi = 0.$$

For every bounded uniformly continuous function b , another absolutely convergent Fubini integral yields $\int (b * G_\varepsilon) d\nu = 0$. Moreover $b * G_\varepsilon \rightarrow b$ uniformly: in its difference from b , the integration over $|x| < \delta$ is bounded by the uniform modulus of continuity at δ , while the complementary integral is at most $2\|b\|_\infty \int_{|x| \geq \delta} G_\varepsilon$, which tends to zero after the substitution $x = \sqrt{\varepsilon}z$. Thus $\int b d\nu = 0$ for all such b .

For each nonempty closed set $F \subset \mathbb{R}^r$, the bounded Lipschitz functions $b_n(x) = \max(0, 1 - n \text{dist}(x, F))$ converge pointwise to 1_F . Dominated convergence against $|\nu|$ proves $\nu(F) = 0$; this also holds for the empty closed set. In particular $\nu(\mathbb{R}^r) = 0$. The Borel sets with zero ν -measure form a Dynkin system: complements have measure zero by this total mass identity, and countable disjoint unions have measure zero by countable additivity. Closed sets form a pi-system generating the Borel sigma-algebra. The pi-lambda theorem therefore gives $\nu(A) = 0$ for every Borel set A . Applying this to the positive and negative level sets of $\Re f$ and $\Im f$ shows $f = 0$ almost everywhere. Finally, the orthogonal complement of the closed polynomial span is zero, so the Hilbert orthogonal decomposition proves that span is all of $L^2(\mu)$. $\square$

28.4 The multiplier, literal frames, and their strong limit

Let $\mathcal{H} = L^2(\mathbb{R}^r, dy)$. For each fixed u , the exact multiplication map is

$$\begin{aligned} \mathcal{M}_u : L^2(\mu_u) &\longrightarrow \mathcal{H}, & q &\longmapsto \Psi_u q, \\ \mathcal{M}_u^{-1} F(y) &= \begin{cases} F(y)/\Psi_u(y), & \Psi_u(y) \neq 0, \\ 0, & \Psi_u(y) = 0. \end{cases} \end{aligned} \quad (\text{ND.13})$$

It is a unitary map: its squared norm is exactly $\int |q|^2 |\Psi_u|^2 dy$, and the inverse has weighted squared norm $\int |F|^2 dy$ because the zero set is null. These are maps between the displayed weighted and unweighted spaces. No claim of bounded multiplication by $1/\Psi_u$ on unweighted $\mathcal{H}$ is made or used. Assigning zero on null hyperplanes only selects a representative of a Hilbert class; every analytic amplitude and all of its pointwise zero orders remain those of (ND.7).

For $m \geq 0$ use the full monomial family

$$\begin{aligned} \mathcal{A}_m &= \{\alpha \in \mathbb{N}^r : |\alpha| \leq m\}, \quad \theta_\alpha(z) = z^\alpha, \quad \theta_\alpha(iy) = i^{|\alpha|} y^\alpha, \\ j_m(u) : \mathbb{C}^{\mathcal{A}_m} &\rightarrow \mathcal{H}, \quad j_m(u)c = \Psi_u \sum_{\alpha \in \mathcal{A}_m} i^{|\alpha|} y^\alpha c_\alpha, \end{aligned} \quad (\text{ND.14})$$

$$V_m(u) = \text{ran } j_m(u), \quad W_m(u) = j_m(u)^* j_m(u), \quad \Pi_m(u) = j_m(u) W_m(u)^{-1} j_m(u)^*.$$

All moments defining these matrices are finite by (ND.8). If $c \neq 0$, its displayed polynomial is nonzero. By continuity it is nonzero in a nonempty open set, which has positive μ_u -measure because the density is positive almost everywhere. Thus $W_m(u) > 0$ for every u . Multiplication proves that Π_m is self-adjoint, idempotent, and has range V_m , so it is precisely the orthogonal projection onto that range. The inverse here is the positive source Gram W_m^{-1} ; no inverse of a possibly zero normal Gram is used.

For $n \geq m$, let E_{nm} be literal zero-padding of the coefficient array. The complete inclusion and Gram maps are

$$\begin{aligned} j_n E_{nm} &= j_m, \quad W_m = E_{nm}^* W_n E_{nm}, \quad E_{\ell n} E_{nm} = E_{\ell m}, \\ V_m &\subset V_n, \quad \Pi_n \Pi_m = \Pi_m \Pi_n = \Pi_m. \end{aligned} \quad (\text{ND.15})$$

The first line follows term by term in the original polynomial sum. The second line follows from the nested ranges and orthogonal complements. In particular $\Pi_n - \Pi_m$ is the orthogonal projection onto $V_n \cap V_m^\perp$.

Theorem 28.3 (Full relative frames on every actual fibre). *For every $u \in \mathbb{R}$ and every $F \in \mathcal{H}$,*

$$\lim_{m \rightarrow \infty} \|F - \Pi_m(u)F\|_{\mathcal{H}} = 0. \quad (\text{ND.16})$$

For every finite m , however, $\|I - \Pi_m(u)\|_{\mathcal{H} \rightarrow \mathcal{H}} = 1$.

Proof. The density theorem applies to μ_u by (ND.8), separately for each real u . The phases $i^{|\alpha|}$ are nonzero constants and hence the literal monomials in (ND.14) have precisely the full polynomial span. Applying the onto isometry (ND.13) proves $\bigcup_m V_m(u) = \mathcal{H}$. Given F and $\varepsilon > 0$, choose $v \in V_M(u)$ with $\|F - v\| < \varepsilon$. For $m \geq M$, orthogonal projection minimizes distance, so $\|F - \Pi_m F\| \leq \|F - v\| < \varepsilon$. This proves (ND.16). The Hilbert space $L^2(\mathbb{R}^r)$ is infinite dimensional: indicators of infinitely many disjoint finite positive-measure balls are linearly independent. Each V_m has finite dimension. Its orthogonal complement therefore contains a unit vector, on which $I - \Pi_m$ has norm one. Orthogonal projection contraction gives the opposite inequality, proving the asserted operator norm exactly. $\square$

28.5 A fixed original source and the complete derivative domain

Fix one original polynomial $P(s_1, \dots, s_k)$ of total degree D . In the exact linear coordinates $S = \sum s_i$, $z_i = s_i - S/k$, $\sum z_i = 0$, it has the unique expansion

$$P = \sum_{|\alpha| \leq D} z^\alpha f_\alpha(S), \quad \deg f_\alpha \leq D - |\alpha|. \quad (\text{ND.17})$$

Uniqueness is coefficient comparison in the independent variables $(S, z_1, \dots, z_r)$; substitution by an invertible linear map preserves total degree. For $m \geq D$, retain an admitted source degree $M_m \geq$

$\max(D, m)$ and the full relative family $\mathcal{A}_m$. The fixed source has coefficient column

$$c_{P,m}(u)_\alpha = \begin{cases} f_\alpha(k/2 + iu), & |\alpha| \leq D, \\ 0, & D < |\alpha| \leq m. \end{cases} \quad c_{P,n} = E_{nm} c_{P,m}. \quad (\text{ND.18})$$

Every original constraint $\deg f_\alpha \leq M_m - |\alpha|$ is respected; neither the fixed polynomial nor its coefficient degree is changed as the frame grows. One may in particular take $M_m = m$ for all $m \geq D$, which is the full degree- m source of FC.28.

Put

$$\begin{aligned} p_u(y) &= P(k/2 + iu, iy_1, \dots, iy_r), \quad A_P(u, y) = \Psi_u(y) p_u(y) = j_m(u) c_{P,m}(u), \\ \partial_S^{\text{rel}} &= \frac{1}{k} \sum_{i=1}^k \partial_{s_i}, \quad F_P(u, y) = \Psi'_u(y) p_u(y), \\ \partial_u A_P &= F_P + i \Psi_u(\partial_S^{\text{rel}} P)_u. \end{aligned} \quad (\text{ND.19})$$

The notation $P(S, z)$ in these formulas denotes the exact substitution $s_i = S/k + z_i$; $z_k = -\sum_{i < k} z_i$. At fixed relative coordinates $dS/du = i$, so the displayed derivative includes exactly the factor i/k on the sum of the original coordinate derivatives. By (ND.3) and (ND.10), $A_P, F_P, \partial_u A_P$ belong to $\mathcal{K} = L^2(\mathbb{R}_u; \mathcal{H})$. In fact all their polynomially weighted derivatives belong to $\mathcal{K}$. In particular A_P lies in the actual derivative domain

$$\mathcal{D}_u = \{A \in \mathcal{K} : \partial_u A \in \mathcal{K} \text{ in the distributional sense}\}. \quad (\text{ND.20})$$

The smooth derivative in (ND.19) equals the distributional derivative, as follows by integration by parts against compactly supported test functions and the preceding integrability bounds.

We also retain its original Mellin realization and sign. If $\mathcal{M}f(s) = \int_0^\infty f(x) x^{s-1} dx$, the original test vector F_h has $\mathcal{M}F_h = v_h$, and the polynomial source is $\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}$, where $D_i = -x_i \partial_{x_i}$. The normalized critical-line transform has factor $(2\pi)^{-k/2}$, followed by (ND.7). Integration by parts gives $\mathcal{M}(D_i f) = s_i \mathcal{M}f$, so its image is exactly A_P , with the original product of a 's. Writing $x_i = e^{q_i}$, the Mellin integral becomes the Fourier transform with kernel $e^{it_i q_i}$ of $e^{(q_1 + \dots + q_k)/2} f(e^{q_1}, \dots, e^{q_k})$. The change of variable has squared norm exactly $\int |f|^2 d\mathbf{x}$; Fourier Plancherel and the Jacobian one in (ND.7) give the isometry $\mathcal{U}_k$ in FC.9. On this original test domain,

$$\begin{aligned} \mathcal{U}_k \mathcal{T}_h^{(k)} P &= A_P, \quad \mathcal{U}_k D^{(k)} = S \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \\ D^{(k)} &= \sum_i D_i, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -I. \end{aligned} \quad (\text{ND.21})$$

The second derivative identity follows by differentiating the Fourier integral with $\partial t_i / \partial u = 1/k$; the commutator follows from $-x_i \partial_{x_i} \log x_i = -1$, summed with the displayed coefficient $1/k$. All these operations are on the original test vectors and their logarithmic derivatives. Their spectral images have the established L^2 derivatives, so no new domain or arbitrary assignment of an arithmetic jet is introduced.

In the degree- m frame define the original matrices and maps

$$\begin{aligned} B_m &= j_m^* j'_m, \quad \Gamma_m = W_m^{-1} B_m, \quad N_m = (I - \Pi_m) j'_m, \quad \mathcal{N}_m = N_m^* N_m, \\ t_m(P) &= j_m(c'_{P,m} + \Gamma_m c_{P,m}), \quad r_m(P) = N_m c_{P,m}. \end{aligned} \quad (\text{ND.22})$$

Every term is the FC.6 term in the literal enlarged frame. Since $j_m \Gamma_m = \Pi_m j'_m$, the product rule gives

$$\begin{aligned} t_m(P) &= \Pi_m \partial_u A_P, \quad r_m(P) = (I - \Pi_m) \partial_u A_P = (I - \Pi_m) F_P, \\ \partial_u A_P &= t_m(P) + r_m(P), \quad \langle t_m(P, u), r_m(P, u) \rangle_{\mathcal{H}} = 0, \\ \|\partial_u A_P\|_{\mathcal{H}}^2 &= \|t_m(P)\|_{\mathcal{H}}^2 + \|r_m(P)\|_{\mathcal{H}}^2. \end{aligned} \quad (\text{ND.23})$$

For the second equality, the coefficient derivative in (ND.19) belongs to $V_m(u)$, so its orthogonal complement is zero. This equality identifies a single fixed function F_P , independent of frame degree, which controls every residual.

Theorem 28.4 (Vanishing of the fixed-source actual normal residual). *For every fixed original polynomial P and every real u ,*

$$\begin{aligned} \|r_m(P, u)\|_{\mathcal{H}} &\longrightarrow 0, & \|t_m(P, u) - \partial_u A_P(u)\|_{\mathcal{H}} &\longrightarrow 0, \\ \|r_m(P)\|_{\mathcal{H}} &\longrightarrow 0, & \|t_m(P) - \partial_u A_P\|_{\mathcal{H}} &\longrightarrow 0. \end{aligned} \quad (\text{ND.24})$$

The fibre convergence of both norms is uniform on compact u -intervals. The derivative graph converges, in the original $\mathcal{H} \oplus \mathcal{H}$ norm, to $(\partial_u A_P, 0)$, while its sum is exactly $\partial_u A_P$ at every degree.

Proof. For each u , $F_P(u) \in \mathcal{H}$, and (ND.16) applied to this vector gives the first limit through (ND.23). Furthermore,

$$0 \leq \|r_m(P, u)\|^2 \leq \|F_P(u)\|^2, \quad \int_{\mathbb{R}} \|F_P(u)\|^2 du < \infty. \quad (\text{ND.25})$$

All functions here are measurable; in fact j_m, j'_m, W_m are smooth in u by the compact-interval bounds after (ND.10), and W_m^{-1} is smooth since its determinant is everywhere positive. The dominated convergence theorem proves the integrated residual limit. Since $t_m - \partial_u A_P = -r_m$, it proves the two tangential limits and the asserted graph convergence as well.

For $n \geq m$, (ND.15) gives the exact orthogonal decomposition

$$\begin{aligned} r_m(P, u) &= r_n(P, u) + (\Pi_n - \Pi_m)F_P(u), \\ \|r_m(P, u)\|^2 - \|r_n(P, u)\|^2 &= \|(\Pi_n - \Pi_m)F_P(u)\|^2 \geq 0. \end{aligned} \quad (\text{ND.26})$$

The two terms are orthogonal because the first lies in $V_n^\perp$ and the second in V_n . Each squared residual is continuous by the same smoothness argument. On a compact set K the open sets $\{u : \|r_m(P, u)\|^2 < \varepsilon\}$ are increasing and cover K . A finite subcover, followed by taking its largest index, makes the squared residual less than ε everywhere on K . This proves compact uniform convergence without a global uniform-in-source assertion. $\square$

For a fixed finite source core $\text{span}\{P_1, \dots, P_q\}$ let $R_m c = \sum_j c_j r_m(P_j)$. Cauchy-Schwarz gives

$$\|R_m\|^2 \leq \sum_{j=1}^q \|r_m(P_j)\|_{\mathcal{H}}^2 \rightarrow 0, \quad \|R_m^* R_m\| \rightarrow 0, \quad R_m^* R_m - R_n^* R_n \succeq 0 \ (n \geq m). \quad (\text{ND.27})$$

The last inequality is (ND.26) integrated for each linear combination of the fixed sources. These are exactly the normal Gram matrices restricted by literal inclusion of that source core. The core and its coefficient norm are fixed in this statement. The full degree-varying source dimension, its minimum arithmetic norm, and the size of its normal-image inverse are not replaced by this finite-core limit.

The exact translation to the original source-amplitude metric is also available on this same fixed core. For an independent core put $(G_0)_{ij} = \langle A_{P_i}, A_{P_j} \rangle_{\mathcal{H}}$. It is independent of m and strictly positive: a nonzero linear combination of the original polynomials is nonzero on an open subset of the real critical-line coordinates, and the original product amplitude is nonzero almost everywhere there. Its squared integral is therefore positive. With $H_m = R_m^* R_m$, define

$$K_m = G_0^{-1} H_m, \quad K_m^* G_0 = G_0 K_m = H_m, \quad \|R_m\|_{G_0 \rightarrow \mathcal{H}}^2 \leq \lambda_{\min}(G_0)^{-1} \sum_j \|r_m(P_j)\|_{\mathcal{H}}^2 \longrightarrow 0.$$

The first equalities follow by matrix multiplication, and the bound follows from $c^* G_0 c \geq \lambda_{\min}(G_0) \|c\|^2$ and the same column Cauchy-Schwarz inequality. Multiplication by the positive

square root $G_0^{1/2}$ is the isometry from this original coefficient metric to the Euclidean metric. It carries K_m to the positive matrix $G_0^{-1/2} H_m G_0^{-1/2}$; its largest eigenvalue is exactly $\|R_m\|_{G_0 \rightarrow \mathcal{H}}^2$. Thus K_m also tends to zero in its original metric operator norm. These explicit congruence and comparison maps preserve both the source-amplitude metric and the normal metric.

The arithmetic maps themselves are retained by the same inclusions. For a nonempty packet set $I = (h(s_1), \dots, h(s_k))$, let j_I denote the full finite jet map, and put $U(\mathbf{s}) = \prod_i v_h(s_i)$. At every selected root $v_h(\rho) \neq 0$, since its full zero order has been divided out; hence $v = j_I U$ is a unit in the full tensor jet algebra. The original source and logarithmic derivative observations are

$$J(\mathcal{T}_h^{(k)} P) = v[P], \quad J\mathcal{L}_k \mathcal{T}_h^{(k)} P = j_I(\partial_S^{\text{rel}}(UP)). \quad (\text{ND.28})$$

Indeed the Mellin transform of $\mathcal{T}_h^{(k)} P$ is UP ; insertion of $k^{-1} \sum \log x_i$ differentiates it by ∂_S^{rel} . At a selected tuple the algebra of germs modulo the full powers is finite; a germ with nonzero constant term is invertible by its finite geometric expansion in the nilpotent maximal ideal. This proves the asserted unit property including all mixed jet orders. Since (ND.18) is the identity on P , both quantities in (ND.28) are identical at every degree. No passage of this jet map through a Hilbert-space limit is required for, or asserted by, the density theorem. If $h = 1$, the quotient by $I = (1)$ is the zero algebra and both finite arithmetic targets are zero.

28.6 Invariant frames, sparse frames, and all original zero cases

Let G be any fixed subgroup of the factor permutation group S_k . Its action on $\sum y_i = 0$, expressed in the first r coordinates, is real linear with determinant of absolute value one. To check the determinant, the action on the full t variables is a permutation matrix with determinant ± 1 ; conjugating by (ND.7) leaves u fixed and acts linearly on y , so the remaining block has determinant ± 1 . The product Ψ_u is unchanged by this action. Therefore

$$(\mathbf{U}_g F)(y) = F(g^{-1}y), \quad \mathbf{R}_G = \frac{1}{|G|} \sum_{g \in G} \mathbf{U}_g \quad (\text{ND.29})$$

defines unitary group actions on both $\mathcal{H}$ and $L^2(\mu_u)$; they are intertwined by $\mathcal{M}_u$. The adjoint of $\mathbf{U}_g$ is $\mathbf{U}_{g^{-1}}$, so $\mathbf{R}_G^* = \mathbf{R}_G$. Counting the $|G|$ products $gh = \ell$ for each ℓ gives $\mathbf{R}_G^2 = \mathbf{R}_G$. Its image is the invariant subspace: its image is fixed under every group element, and its value on a fixed vector is that vector. Thus it is exactly the invariant orthogonal projection.

Let $V_m^G(u)$ be Ψ_u times all G -invariant relative polynomials of degree at most m , using any full independent basis of that space with its original coefficients. The degree is preserved by the linear action and by its finite average. Applying $\mathbf{R}_G$ to polynomial approximants in the density theorem proves density in the invariant weighted subspace. Intertwining by (ND.13) therefore gives

$$\overline{\bigcup_m V_m^G(u)} = \mathcal{H}^G, \quad \Pi_m^G(u) \xrightarrow{\text{strongly}} \mathbf{R}_G. \quad (\text{ND.30})$$

To verify the second formula for an arbitrary F , decompose it as $\mathbf{R}_G F + (I - \mathbf{R}_G)F$. The second part is orthogonal to every V_m^G , and the first is approximated by their dense union by the same best-approximation argument as (ND.16). If the original polynomial P is G -invariant, then $A_P, F_P, \partial_u A_P$ are invariant, since the group fixes u and Ψ_u . Its expansion in a full homogeneous invariant basis retains the bound $\deg f_\alpha \leq D - \ell_\alpha$: apply coefficient comparison in powers of S and the homogeneous relative grading to (ND.17). Consequently (ND.23)–(ND.27) hold for these actual invariant frames, with Π_m^G in place of Π_m . In particular their fixed-source residual also tends to zero pointwise, on compact u -intervals, and in $\mathcal{H}$. This includes full symmetric orbit frames without changing the unscaled orbit coefficients.

For an arbitrary nested family of finite polynomial spaces E_m , let $V_m^E(u) = \mathcal{M}_u E_m$ and $V_\infty^E(u) = \overline{\bigcup_m V_m^E(u)}$. Its exact strong projection limit is

$$\Pi_m^E(u) \xrightarrow{\text{strongly}} Q_u^E := \text{proj}_{V_\infty^E(u)}. \quad (\text{ND.31})$$

Here is a proof requiring no density assumption. For $n \geq m$, orthogonality of the nested projections gives $\|\Pi_n^E F - \Pi_m^E F\|^2 = \|\Pi_n^E F\|^2 - \|\Pi_m^E F\|^2$. The squared norms increase and are bounded by $\|F\|^2$. Thus the projections applied to F are Cauchy. Their limit belongs to V_∞^E , and its difference from F is orthogonal to each V_m^E , hence to their closure. This proves (ND.31) and identifies the exact projection map. A fixed source admitted from some level onward is a sum of fixed members of that relative polynomial space times polynomials in S . Its coefficient derivative replaces each such polynomial by i times its S -derivative and therefore belongs to the same relative space. The residual limit is consequently precisely $(I - Q_u^E)\partial_u A_P = (I - Q_u^E)F_P$. This convergence is also in $\mathcal{K}$, by pointwise strong convergence and the integrable bound $4\|F_P(u)\|^2$ for the squared difference from its limiting residual. The limit field is measurable as a pointwise norm limit of measurable projection fields.

The projection in (ND.31) need not be the identity. A concrete actual example uses $E_m = \mathbb{C}$ at every level. Set $b_u = m_k(u)^{-1} \int y_1 d\mu_u(y)$, which is finite by (ND.8). The nonzero vector $F = \Psi_u(y)(y_1 - b_u)$ has zero inner product with Ψ_u , so every $\Pi_m^E F = 0$. It is nonzero because μ_u has positive density almost everywhere and the affine hyperplane $y_1 = b_u$ is null. This gives the exact projection and an explicit failed approximation, without asserting that the sparse space is unrelated to the full one.

All zero and empty cases have concrete maps. The zero source has $A_P = F_P = t_m = r_m = 0$ at every degree. A nonzero source whose arithmetic class $[P]$ or $v[P]$ vanishes is still the same nonzero polynomial in (ND.17), with the same amplitude and convergence; (ND.28) retains its full arithmetic class. The packet $h = 1$ gives the actual $2\xi/\sqrt{2\pi}$ amplitude and satisfies every analytic estimate above. No actual fibre has zero mass for $k \geq 2$, by the hyperplane argument following (ND.7). One may append an initial empty frame $V_{-1} = \{0\}$ with $\Pi_{-1} = 0$; the full nonempty sequence starts with the constant frame at degree zero. Empty source arrays have the unique zero maps.

The original exceptional two-factor fibre is retained. If the packet makes a even, as for the quartet in FC.27, then a' is odd and (ND.10) gives

$$\Psi'_0(y) = \frac{1}{2}(a'(y)a(-y) + a(y)a'(-y)) = 0, \quad j'_m(0) = 0, \quad N_m(0) = 0, \quad r_m(P, 0) = 0. \quad (\text{ND.32})$$

The derivative of the coefficient polynomial may be nonzero; (ND.23) keeps that term in the tangential component. The density proof and the limiting derivative graph require no positive inverse for $\mathcal{N}_m(0)$. Their only fibre inverse is W_m^{-1} , which remains strictly positive there. No conclusion about one-factor relative frames is hidden in the assumption $k \geq 2$: for $k = 1$ the relative coordinate space has one point, the multiplication image is one-dimensional when $a(u) \neq 0$ and is zero when $a(u) = 0$. At such a zero it is not onto the scalar ambient Hilbert space, and the pointwise density-to-identity assertion (ND.16) was explicitly proved only for $k \geq 2$.

28.7 Frame covariance and the exact half-line transport

Let $C_m(u)$ be any smooth invertible change of relative frame. Use $\tilde{j}_m = j_m C_m$ and $\tilde{c}_{P,m} = C_m^{-1} c_{P,m}$. The product and projection remain the same, while the complete connection and normal formulas are

$$\begin{aligned} \tilde{W}_m &= C_m^* W_m C_m, & \tilde{B}_m &= C_m^* B_m C_m + C_m^* W_m C'_m, \\ \tilde{\Gamma}_m &= C_m^{-1} \Gamma_m C_m + C_m^{-1} C'_m, & \tilde{N}_m &= N_m C_m, & \tilde{N}_m \tilde{c}_{P,m} &= r_m(P), \\ \tilde{j}_m (\tilde{c}'_{P,m} + \tilde{\Gamma}_m \tilde{c}_{P,m}) &= t_m(P), & \tilde{E}_{nm} &= C_n^{-1} E_{nm} C_m. \end{aligned} \quad (\text{ND.33})$$

The first two lines follow by differentiating $j_m C_m$ and using the unchanged projection image; its complementary projection annihilates $j_m C'_m$. Differentiating $C_m^{-1} c_{P,m}$ cancels the $C_m^{-1} C'_m$ term and

proves the tangential identity. Finally $\tilde{j}_n \tilde{E}_{nm} = \tilde{j}_m$, so these are the exact typed inclusion maps for the same nested images. Arbitrary growth of C_m is not used to assign an independent coefficient domain: the admitted columns are precisely the pullbacks of (ND.18), and the actual amplitudes and graph norms remain unchanged.

For clarity the residual limit can also be transported into the original two-branch half-line representation without an evenness assumption. For $R \in \mathcal{K}$, let $x = u^2 > 0$, $v = \sqrt{x}$, and set

$$\begin{aligned} R_+(x) &= R(v), \quad R_-(x) = R(-v), \quad e_R(x) = \frac{1}{2}(R_+(x) + R_-(x)), \quad o_R(x) = \frac{R_+(x) - R_-(x)}{2i\sqrt{x}}, \\ R_+ &= e_R + i\sqrt{x} o_R, \quad R_- = e_R - i\sqrt{x} o_R, \\ \|R\|_{\mathcal{K}}^2 &= \int_0^\infty \frac{\|R_+\|^2 + \|R_-\|^2}{2\sqrt{x}} dx = \int_0^\infty \left(\frac{\|e_R\|^2}{\sqrt{x}} + \sqrt{x} \|o_R\|^2 \right) dx. \end{aligned} \tag{ND.34}$$

The first equality is the positive Jacobian on each branch; the second is the parallelogram identity after the displayed exact inverse. These formulas define an onto isometry between the full-line space and the displayed two weighted half-line spaces. Thus (ND.24) carries the actual residual to zero in the sum of both original weighted norms, and carries the full derivative graph to the same transported derivative. Pointwise fibre convergence holds at every $x > 0$ on both branches. At the meeting point $u = 0$, the original smooth amplitudes have their original trace; no new independent half-line boundary values are introduced. The proof does not assert convergence of $\partial_u r_m$, which would require control of a different derivative. It proves exactly the fixed-source convergence of the normal and tangential parts of the original first derivative, with their sum and arithmetic source fixed at every degree.

29 The actual normal-image limit, its graph, and its arithmetic observation

Keep the finite dagger-stable packet, its entire quotient $v_h = g/h$, $g = 2\xi$, and an integer $k \geq 2$. The empty packet $h = 1$ is included. We retain the original amplitude and coordinates

$$a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad t_i = u/k + y_i, \quad y_k = -\sum_{i < k} y_i, \quad \Psi(u, \mathbf{y}) = \prod_{i=1}^k a(t_i), \quad S = k/2 + iu. \tag{NN.1}$$

The original measures are $du d\mathbf{y}$, with the unit Jacobian proved in (FC.2). No mass, phase, residue unit or source coefficient is replaced. Write $\mathcal{H} = L^2(d\mathbf{y})$ and $\mathcal{K} = L^2(du; \mathcal{H})$.

For each integer $m \geq 0$, let $\mathcal{P}_m$ be the complete original polynomial source of total degree at most m . Its relative frame consists of every monomial $\theta_\alpha(\mathbf{z})$ of degree $\ell_\alpha \leq m$ in the $k-1$ independent variables $z_i = s_i - S/k$, $i < k$, with $z_k = -\sum_{i < k} z_i$ retained as the dependent coordinate. Its literal source basis consists of $\theta_\alpha S^n$, $0 \leq n \leq m - \ell_\alpha$, as in (FC.28). Keep its coefficient map $C_m(u)$, frame $j_m(u)$, orthogonal projection $\Pi_m(u)$, connection $\Gamma_m(u)$, and normal map $N_m(u)$ of (FC.6). Thus the actual amplitude, normal and tangential source maps are

$$\begin{aligned} A_m p &= j_m C_m p = \Psi P_p, \\ N_m p &= N_m C_m p = (I - \Pi_m) \partial_u A_m p, \\ T_m p &= j_m (C'_m + \Gamma_m C_m) p = \Pi_m \partial_u A_m p. \end{aligned} \tag{NN.2}$$

All three maps have values in the same original space $\mathcal{K}$. The last two equalities include the derivative of every original power of S ; its factor is inS^{n-1} , not nS^{n-1} .

Let $Q_{\ell m}$ be the literal coefficient inclusion for $\mathcal{P}_m \subseteq \mathcal{P}_\ell$, $\ell \geq m$, and $E_{\ell m}$ its relative-frame inclusion. The coefficient identities and amplitude identity are

$$C_\ell Q_{\ell m} = E_{\ell m} C_m, \quad j_\ell E_{\ell m} = j_m, \quad A_\ell Q_{\ell m} = A_m. \tag{NN.3}$$

They follow by comparing each literal column $\theta_\alpha S^n$. In particular these are inclusions of the original sources, and not an identification of coefficient spaces of different dimensions.

The weighted-density theorem for the amplitude (NN.1) proves $\Pi_m(u) \rightarrow I$ strongly for every u , and proves for every fixed original polynomial P that

$$\begin{aligned} r_m(P) &:= (I - \Pi_m)\partial_u(\Psi P) \longrightarrow 0 \quad \text{in } \mathcal{K}, \\ t_m(P) &:= \Pi_m\partial_u(\Psi P) \longrightarrow \partial_u(\Psi P) \quad \text{in } \mathcal{K}. \end{aligned} \quad (\text{NN.4})$$

Here and below $m \geq \deg P$. The full density proof, including its original exponential estimate, the division across the selected full zero orders, all fibre zero sets and the invariant-source variant, is given in the preceding relative-density section. Equivalently, the first statement of (NN.4) is obtained by applying those actual fibre projections to the one fixed function $\partial_u(\Psi P) \in \mathcal{K}$, then applying dominated convergence to the bound $\|(I - \Pi_m)\partial_u(\Psi P)\| \leq \|\partial_u(\Psi P)\|$. Thus no unproved estimate about a sequence of changing source polynomials is used here.

29.1 Exact contractions of the finite normal images

Put $\mathcal{R}_m^N = \text{ran } N_m$, and retain $H_m^N = N_m^* N_m > 0$ and $L_m^N = (H_m^N)^{-1} N_m^*$ of (FC.29)–(FC.30). Their positivity and the exact source inverse hold for the actual arithmetic amplitude, including $h = 1$. All these source spaces contain the constant monomial and are nonzero. Define the bounded orthogonal projection on $\mathcal{K}$

$$(\mathbb{P}_m f)(u) = \Pi_m(u) f(u). \quad (\text{NN.5})$$

The finite-frame entries are smooth, so this is a measurable contraction; self-adjointness and idempotence hold after integrating their pointwise identities. The relative inclusions give $\mathbb{P}_\ell \mathbb{P}_m = \mathbb{P}_m = \mathbb{P}_m \mathbb{P}_\ell$ for $\ell \geq m$.

The exact transition from one normal image to the next is

$$\begin{aligned} V_{\ell m} &: \mathcal{R}_m^N \longrightarrow \mathcal{R}_\ell^N, \\ V_{\ell m} r &= (I - \mathbb{P}_\ell) r = N_\ell Q_{\ell m} L_m^N r. \end{aligned} \quad (\text{NN.6})$$

To prove both its codomain and its second expression, write uniquely $r = N_m p = (I - \mathbb{P}_m)\partial_u A_m p$. The nested projection identity and (NN.3) give

$$(I - \mathbb{P}_\ell) r = (I - \mathbb{P}_\ell)\partial_u A_m p = N_\ell Q_{\ell m} p.$$

Injectivity follows from injectivity of N_ℓ and of the literal inclusion. Moreover

$$\|V_{\ell m}\| \leq 1, \quad V_{mm} = I, \quad V_{nm} = V_{n\ell} V_{\ell m} \quad (n \geq \ell \geq m). \quad (\text{NN.7})$$

Projection contraction proves the first assertion; the source formula and $L_\ell^N N_\ell = I$ prove the others. The source and transition square commutes exactly:

$$L_\ell^N V_{\ell m} = Q_{\ell m} L_m^N. \quad (\text{NN.8})$$

In particular these changing normal images have a specified algebraic relation even though their Hilbert subspaces are not asserted to be nested by the ambient identity map.

Their complete energy difference on the fixed source is

$$\begin{aligned} r_m(P) &= (\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P) + r_\ell(P), \\ \|r_m(P)\|^2 - \|r_\ell(P)\|^2 &= \|(\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P)\|^2, \\ \|t_\ell(P)\|^2 - \|t_m(P)\|^2 &= \|(\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P)\|^2. \end{aligned} \quad (\text{NN.9})$$

Indeed $\mathbb{P}_\ell - \mathbb{P}_m$, $\mathbb{P}_m$, and $I - \mathbb{P}_\ell$ are mutually orthogonal projections. Apply their orthogonal decomposition to the unchanged derivative amplitude. The full nested-frame connection identities in the preceding section identify this middle component in the original coefficients and also under each stated variable frame change.

29.2 The algebraic limit and its two exact metric presentations

Consider the algebraic direct limit of the injective transitions $V_{\ell m}$. A class represented by $r \in \mathcal{R}_m^N$ is sent to the original polynomial with coefficients $L_m^N r$. Equation (NN.8) proves that the map is well-defined. It is onto $\mathcal{P}_\infty = \mathbb{C}[s_1, \dots, s_k]$, since every polynomial lies in a finite source. If its polynomial is zero then $r = N_m L_m^N r = 0$, so it is injective. We have proved the exact complex-linear isomorphism

$$\operatorname{colim}_{m, V_{\ell m}} \mathcal{R}_m^N \xrightarrow{\sim} \mathcal{P}_\infty. \quad (\text{NN.10})$$

Because (NN.7) is contractive, the limit seminorm is well-defined by

$$\|[r]\|_{N, \infty} = \lim_{\ell \rightarrow \infty} \|V_{\ell m} r\| = \inf_{\ell \geq m} \|V_{\ell m} r\|. \quad (\text{NN.11})$$

Representing the same class at a later index leaves the limit unchanged; representing two classes at a common index proves homogeneity and the triangle inequality. For the fixed polynomial given by (NN.10), (NN.4) proves

$$\|[r]\|_{N, \infty} = 0 \quad \text{for every class.} \quad (\text{NN.12})$$

Thus its Hausdorff seminorm quotient, and its completion, are the zero vector space. This conclusion retains the nonzero algebraic space in (NN.10), together with the exact zero quotient map.

For comparison retain the complete derivative graph $\mathcal{G}_m = \operatorname{ran}(\mathbf{T}_m, \mathbf{N}_m)$ with its original sum norm. Its transitions are

$$W_{\ell m}(\mathbf{T}_m p, \mathbf{N}_m p) = (\mathbf{T}_\ell Q_{\ell m} p, \mathbf{N}_\ell Q_{\ell m} p). \quad (\text{NN.13})$$

Uniqueness of the source proves that these maps are well-defined, injective and compose. Orthogonality and (NN.3) show that both sides have squared norm $\|\partial_u A_m p\|^2$, so every transition is isometric. Addition of the two components commutes with transitions and gives the exact isometry

$$\operatorname{colim}_{m, W_{\ell m}} \mathcal{G}_m \xrightarrow{\text{Add}} \{\partial_u(\Psi P) : P \in \mathcal{P}_\infty\} \subset \mathcal{H}. \quad (\text{NN.14})$$

Its extension identifies the Hilbert completion with the closure of this specified derivative image in the original $\mathcal{H}$. The derivative image is injective in its polynomial source. Indeed $\Psi P \in L^2(du; \mathcal{H})$ and a zero weak derivative in u make every scalar pairing with an $\mathcal{H}$ vector a constant distribution in u . Such a constant belongs to $L^2(du)$ only when zero. Hence $\Psi P = 0$. The nonzero entire one-factor amplitude has a discrete real zero set, so Ψ is nonzero almost everywhere in the original coordinates; a polynomial vanishing almost everywhere there is the zero polynomial.

The algebraic reconstruction maps of (FC.31) give the commuting identity

$$W_{\ell m} \operatorname{Rec}_m = \operatorname{Rec}_\ell V_{\ell m}. \quad (\text{NN.15})$$

It follows by substituting $r = N_m p$. Thus the two algebraic limits are explicitly isomorphic through the original source, while (NN.12) and (NN.14) compute their two distinct retained metrics and their exact completion maps.

29.3 Original arithmetic jets and the size of reconstruction

For a nonempty packet retain the original finite algebra $E = \mathbb{C}[\mathbf{s}]/(h(s_1), \dots, h(s_k))$, its complete Hermite coordinates, and $U = \prod_i v_h(s_i)$. Its jet is a unit. The fixed polynomial observation is the actual derivative jet

$$\mathcal{J}^{\log} P = j_I \left(\frac{1}{k} \sum_i \partial_{s_i}(UP) \right) \in E. \quad (\text{NN.16})$$

It includes the entire product derivative of U at every retained nilpotent order. Formula (FC.34) gives its exact finite matrices $J_m^{\log}$, including the original Mellin factors and the $-i\mathcal{U}_k^{-1}$ arrow from the derivative amplitude. Their coefficient inclusions give $J_\ell^{\log} Q_{\ell m} = J_m^{\log}$, whence

$$\text{Obs}_\ell^{\log} V_{\ell m} = \text{Obs}_m^{\log}, \quad \text{Obs}_m^{\log} = J_m^{\log} L_m^N. \quad (\text{NN.17})$$

These equalities follow from (NN.8), so they define on (NN.10) exactly the original algebraic map $\mathcal{J}^{\log}$.

This map is nonzero, with an explicit original source witness. Choose any retained tuple $\rho = (\rho_1, \dots, \rho_k)$, set $\sigma = \sum_i \rho_i$, and take $P_\rho = S - \sigma$. The complete jet evaluation at the constant coordinate of that tuple is

$$(\mathcal{J}^{\log} P_\rho)(\rho) = U(\rho) \neq 0. \quad (\text{NN.18})$$

The product-rule term multiplying P_ρ vanishes at that tuple; the other term is $U \partial_S^{\text{rel}} S = U$, with its full $1/k$ retained. The nonvanishing follows from taking every selected zero to its full order in h .

There is also a permutation-invariant witness that is itself an original theta relation. Take $P_{\text{rel}} = \sum_i h(s_i) \in I$. Its original full jet is zero, while its retained derivative is

$$\mathcal{J}^{\log} P_{\text{rel}} = \frac{1}{k} U_k \sum_i [h'(s_i)]_I \neq 0. \quad (\text{NN.18a})$$

The term differentiating U still has the factor P_{rel} and is zero in E . If $d = 1$, the sum of derivatives is exactly k . If $d \geq 2$, its degree- $d - 1$ standard remainder terms are the distinct monomials ds_i^{d-1} ; their nonzero coefficients cannot cancel. Multiplication by the full unit U_k proves the nonvanishing in both cases, including all confluent orders. Each $h(s_i)$ term has the original theta primitive with the two matching Koszul signs, so this witness retains the actual relation before its supported-zero quotient. It is available in the full invariant source as well as in the full tensor source.

Fix a genuine original target metric as follows. Choose a degree $m_* \geq k(d - 1)$. The original jet map $J_* p = j_I(U P_p)$ on $\mathcal{P}_{m_*}$ is onto: full monic remainders have total degree at most $k(d - 1)$, and multiplication by $j_I U$ is invertible. With its original amplitude metric $H_{m_*}^0 = A_{m_*}^* A_{m_*}$, the retained minimum-source metric is

$$G_* = (J_*(H_{m_*}^0)^{-1} J_*^*)^{-1} > 0. \quad (\text{NN.19})$$

Indeed for a target vector q , the source $(H_{m_*}^0)^{-1} J_*^* G_* q$ maps to q ; it is orthogonal in the original source metric to $\ker J_*$. Its squared norm is $q^* G_* q$, and adding any kernel vector adds its nonnegative squared norm. This proves the stated actual minimum and its constants.

The exact sizes of the finite inverse, graph reconstruction and arithmetic observation are

$$\begin{aligned} \|L_m^N\|_{N \rightarrow 0}^2 &= \lambda_{\max}((H_m^N)^{-1/2} H_m^0 (H_m^N)^{-1/2}), \\ \|\text{Rec}_m\|^2 &= \lambda_{\max}((H_m^N)^{-1/2} H_m^\partial (H_m^N)^{-1/2}), \\ \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2 &= \lambda_{\max}((H_m^N)^{-1/2} (J_m^{\log})^* G_* J_m^{\log} (H_m^N)^{-1/2}). \end{aligned} \quad (\text{NN.20})$$

Here H_m^0 , H_m^N , and H_m^∂ are the three original forms in (FC.29)–(FC.32). For a nonzero coefficient vector p , the denominator of each norm ratio is $p^* H_m^N p$; the respective numerators are $p^* H_m^0 p$, $p^* H_m^\partial p$, and $p^* (J_m^{\log})^* G_* J_m^{\log} p$. The change of variable $b = (H_m^N)^{1/2} p$, with its exact inverse, proves the displayed eigenvalue formulas without altering any of these original forms.

All three norms tend to infinity as $m \rightarrow \infty$. For the first two use any fixed nonzero polynomial: its original amplitude and its derivative amplitude both have fixed positive norm, whereas (NN.4) makes its normal norm tend to zero and (FC.26) keeps that norm strictly positive at every finite degree. For the third use the single fixed polynomial P_ρ in (NN.18); its G_* target norm is a fixed positive number. Each operator norm is bounded below by this specified ratio for every sufficiently large m , proving divergence. Equivalently, (NN.17) is a nonzero algebraic observation on the limit, and cannot descend continuously through its zero Hausdorff quotient.

Every changing actual minimum metric is retained as well. Define G_m by the same minimum-source construction at degree $m \geq m_*$. Its original-source inclusion makes $G_m \preceq G_*$: the earlier minimum lift is an allowed later lift with exactly the same amplitude norm. Set $B_m = G_*^{-1/2} G_m G_*^{-1/2} > 0$. The precise comparison is

$$\begin{aligned} q^* G_m q &= (G_*^{1/2} q)^* B_m (G_*^{1/2} q), \quad 0 < B_m \preceq I, \\ \|\text{Obs}_m^{\log}\|_{N \rightarrow G_m}^2 &= \lambda_{\max}((H_m^N)^{-1/2} (J_m^{\log})^* G_m J_m^{\log} (H_m^N)^{-1/2}), \\ \lambda_{\min}(B_m) \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2 &\leq \|\text{Obs}_m^{\log}\|_{N \rightarrow G_m}^2 \leq \lambda_{\max}(B_m) \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2. \end{aligned} \quad (\text{NN.21})$$

The first equality is direct multiplication. Its minimum and maximum eigenvalue inequalities applied to $q = J_m^{\log} p$, followed by the same Rayleigh supremum, prove every remaining assertion. Thus the divergence proved for the fixed actual metric is transported to the changing actual metric by an explicit positive operator; its displayed eigenvalues have not been assigned an unproved asymptotic.

29.4 The split lift retains the algebraic source of the zero limit

For a complex vector space V , use the split semimodule $G(V) = \{\tau\} \sqcup V$ over $G(\mathbb{C})$. The external τ is the additive identity, addition of supported vectors is their original vector addition, supported scalar multiplication is the original one, and multiplication involving external τ is external τ . Its supported zero is $e_V = 0_V \neq \tau$. For a linear map $f : V \rightarrow W$, the map $G(f)$ sends τ to τ and a supported vector v to the supported vector $f(v)$. Checking separately external and supported inputs proves additivity and compatibility with every scalar. Composition is the original composition on supported inputs and fixes τ .

Consequently (NN.10), (NN.15), and (NN.17) give exact split-semimodule maps. In particular the seminorm quotient in (NN.12) has the split lift

$$G(\mathcal{P}_\infty) \longrightarrow G(0) = \{\tau, e_0\}, \quad \tau \longmapsto \tau, \quad P \longmapsto e_0 \quad (P \in \mathcal{P}_\infty). \quad (\text{NN.22})$$

Its two fibres are exactly $\{\tau\}$ and the entire supported polynomial source. The algebraic arithmetic map is simultaneously $G(\mathcal{J}^{\log}) : G(\mathcal{P}_\infty) \rightarrow G(E)$. For a nonempty packet it cannot factor through (NN.22): evaluating a proposed factorization on the supported zero polynomial forces the image of e_0 to be e_E , whereas (NN.18) supplies a polynomial whose supported image has nonzero amplitude. This is an exact factorization obstruction between the explicitly constructed maps, with the original source, supported zeros and external absence retained.

For $h = 1$, the algebraic target E is zero, so its arithmetic observation is the zero map and its split lift does factor through (NN.22). The amplitude, density theorem, injective finite normal maps, inverse/graph divergence and the complete graph limit remain unchanged. The arithmetic norm formulas involving a nonzero G_* target are then replaced by their unique zero-target maps. For a full invariant source under a fixed finite permutation group, all constructions restrict through the exact Reynolds maps of the density section; the derivative and source are invariant, $P_\rho = S - \sigma$ is invariant, and the arithmetic target is the corresponding invariant subspace with the restriction of its original target form. Thus the same proofs retain that invariant source and its original metric rather than adding noninvariant vectors. To verify the asserted minimum metric on this subspace, permutations act unitarily on the original tensor amplitude space because they preserve its product measure and its identical one-factor amplitudes. The full jet map is equivariant. Averaging an arbitrary lift of an invariant target with the literal factor $1/|\mathfrak{G}|$ gives an invariant lift of the same target, so the invariant jet map is onto. Uniqueness of the full minimum shows that every permutation fixes that minimum lift. It is consequently the minimum in the invariant source as well, with exactly the restriction of G_m or G_* .

30 The cyclic sum at every conormal depth

This section continues the complete cyclic-sum source, especially its Sections 1–3 and 7–8, and the separately supplied conormal-tower proof. It retains the original packet, arithmetic unit, tensor variables and reflection. The new constructions are the retraction and invariant dimensions at every depth, the exact scalar first-jet image, its full conormal comparison, and the finite cyclic Weyl correction with its complementary map.

30.1 Original coordinates and the exact collided annihilator

Let $k \geq 1$, and let the nonempty original monic packet be

$$h(s) = \prod_{\nu=1}^a (s - \rho_\nu)^{m_\nu}, \quad d = \sum_{\nu=1}^a m_\nu \geq 1. \quad (\text{CC.1})$$

The ρ_ν are distinct, and every m_ν is the full order retained by the packet. In the arithmetic application $g = 2\xi$, $v_h = g/h$, and the chosen packet consists of zeros of g with those full orders. Thus $v_h(\rho_\nu) \neq 0$. For the original arithmetic reflection and pairing, retain the delivered packet hypothesis: the map $\rho \mapsto 1 - \bar{\rho}$ permutes the packet and preserves each full multiplicity. This is the hypothesis used by the involution in (CC.42)–(CC.48). The polynomial-algebra statements below apply to the displayed original roots without any assumption that they lie on the critical line. Put

$$\begin{aligned} P &= \mathbb{C}[s_1, \dots, s_k], \quad h_i = h(s_i), \quad I = (h_1, \dots, h_k), \quad S = \sum_{i=1}^k s_i, \\ B_r &= P/I^r, \quad A_r = B_r^{\mathfrak{S}_k} \quad (r \geq 1), \\ U_r &= \left[\prod_{i=1}^k v_h(s_i) \right]_{I^r} \in A_r^\times. \end{aligned} \quad (\text{CC.2})$$

An entire function in (CC.2) means its full finite Taylor class on each local factor. These classes are compatible under every quotient map. Their invertibility follows from their nonzero constant terms: in a finite local algebra a class $u_0 + n$, with $u_0 \neq 0$ and $n^N = 0$, has inverse $u_0^{-1} \sum_{j=0}^{N-1} (-n/u_0)^j$. This fixes U_r without replacing any of its Taylor coefficients.

For an ordered tuple $\mathbf{t} = (\nu_1, \dots, \nu_k)$, write

$$\begin{aligned} \lambda_{\mathbf{t}} &= \sum_i \rho_{\nu_i}, & D_{\mathbf{t}} &= \sum_i (m_{\nu_i} - 1), & M_{\mathbf{t}} &= \max_i m_{\nu_i}, \\ L_{\mathbf{t},r} &= 1 + D_{\mathbf{t}} + (r-1)M_{\mathbf{t}}, & \ell_{\lambda,r} &= \max_{\lambda_{\mathbf{t}}=\lambda} L_{\mathbf{t},r}, \\ \Lambda_k &= \{\lambda_{\mathbf{t}}\}, & \chi_r(T) &= \prod_{\lambda \in \Lambda_k} (T - \lambda)^{\ell_{\lambda,r}}, & q_r &= \deg \chi_r. \end{aligned} \quad (\text{CC.3})$$

The letter T is the polynomial indeterminate in χ_r ; its evaluation in the original algebra is $T = S$. Equal sums are grouped before taking the maximum.

Theorem 30.1 (Full local depth and the exact scalar contraction). *There are algebra isomorphisms in the original shifted coordinates*

$$B_r \longrightarrow \prod_{\mathbf{t} \in \{1, \dots, a\}^k} \mathbb{C}[z_1, \dots, z_k] / (z_1^{m_{\nu_1}}, \dots, z_k^{m_{\nu_k}})^r, \quad s_i \longmapsto \rho_{\nu_i} + z_i. \quad (\text{CC.4})$$

In the $\mathbf{t}$ -factor an exact basis consists of the monomials

$$z^\alpha, \quad \sum_i \left\lfloor \frac{\alpha_i}{m_{\nu_i}} \right\rfloor \leq r-1. \quad (\text{CC.5})$$

The nilpotency order of $S - \lambda_{\mathbf{t}} = \sum_i z_i$ on that factor is exactly $L_{\mathbf{t},r}$. Consequently

$$I^r \cap \mathbb{C}[S] = (\chi_r(S)), \quad C_r := \mathbb{C}[T]/(\chi_r) \xrightarrow{\alpha_r} A_r, \quad [F] \mapsto [F(S)] \quad (\text{CC.6})$$

is an injective unital algebra map. The original arithmetic coefficient map on the cyclic source is $\eta_r = M_{U_r, \alpha_r} : C_r \rightarrow A_r$. It is injective and intertwines multiplication by T and S .

Proof. For $r = 1$, separate-variable Chinese remainder division gives $I = \bigcap_{\mathbf{t}} J_{\mathbf{t}}$, where $J_{\mathbf{t}} = ((s_i - \rho_{\nu_i})^{m_{\nu_i}})_i$. Two different tuples give comaximal ideals: in a coordinate where their roots differ the two one-variable powers are coprime, and their Bezout identity already gives 1 in the sum of the two ideals. For finitely many pairwise comaximal ideals, intersection equals product. Therefore $I^r = (\prod J_{\mathbf{t}})^r = \prod J_{\mathbf{t}}^r = \bigcap J_{\mathbf{t}}^r$. The same Bezout identity, or its power expansion, proves that the powers remain comaximal. This proves (CC.4).

The local ideal is monomial. A monomial belongs to its r -th power precisely when some non-negative integers b_i , with $\sum b_i = r$, satisfy $m_{\nu_i} b_i \leq \alpha_i$ for every i . Such b_i exist precisely when the sum of the displayed floors is at least r . This proves (CC.5), including independence. Write $\alpha_i = m_{\nu_i} b_i + c_i$, $0 \leq c_i < m_{\nu_i}$. The greatest surviving total degree is $D_{\mathbf{t}} + (r-1)M_{\mathbf{t}}$: put every allowed block b_i in one coordinate attaining $M_{\mathbf{t}}$, and put $c_i = m_{\nu_i} - 1$ in every coordinate. All monomials of larger degree vanish. In $(\sum z_i)^{D_{\mathbf{t}}+(r-1)M_{\mathbf{t}}}$ that chosen monomial has the nonzero coefficient $(D_{\mathbf{t}} + (r-1)M_{\mathbf{t}})! / \prod \alpha_i!$. Distinct monomials cannot cancel. The asserted exact order follows. The annihilator in $\mathbb{C}[T]$ of evaluation on this factor is therefore $((T - \lambda_{\mathbf{t}})^{L_{\mathbf{t},r}})$. Intersecting these ideals takes the maximum at each repeated λ , proving (CC.6). The class of $F(S)$ is invariant, and multiplication by the unit U_r preserves injectivity and the stated intertwining. $\square$

Proposition 30.2 (Degree, scalar powers and all endpoints). *For the nonempty packet,*

$$\begin{aligned} q_r &\geq k(d-1) + (r-1)d + 1, \\ \chi_r \mid \chi_1^r, \quad \chi_r = \chi_1^r &\iff (r=1) \text{ or } (k=1) \text{ or } (m_{\nu} = 1 \ \forall \nu). \end{aligned} \quad (\text{CC.7})$$

These are assertions about the complete collided polynomials.

Proof. Repeated monic division in the unchanged variables gives the unique finite expansion

$$F = \sum_{\mathbf{b} \in \mathbb{N}^k, 0 \leq c_i < d} a_{\mathbf{b}, \mathbf{c}} \prod_i h_i^{b_i} s_i^{c_i}. \quad (\text{CC.8})$$

The leading monomials have the pairwise distinct exponent vectors $d\mathbf{b} + \mathbf{c}$, so a finite dependence is impossible. Equivalently P is free over $\mathbb{C}[h_1, \dots, h_k]$ with the displayed remainder basis. Membership in I^r is exactly the vanishing of the coefficients with $\sum b_i < r$. It follows that the highest homogeneous part of every element of I^r belongs to $(s_1^d, \dots, s_k^d)^r$. Indeed each highest-degree basis term has that property, and its lower terms have smaller total degree.

Put $D = k(d-1) + (r-1)d$. The monomial with exponents $(rd-1, d-1, \dots, d-1)$ survives the corresponding monomial ideal and has degree D . For each $0 \leq j \leq D$, choose a divisor of this monomial of degree j . Its floor sum is still at most $r-1$, and its coefficient in S^j is a nonzero multinomial coefficient. Thus the leading homogeneous part S^j of a monic polynomial of degree j in S cannot belong to that monomial ideal. Such a polynomial cannot lie in I^r . Apply this to $\chi_r(S)$ to obtain the lower bound.

For every tuple, $M_{\mathbf{t}} \leq 1 + D_{\mathbf{t}} = L_{\mathbf{t},1}$, so $L_{\mathbf{t},r} \leq rL_{\mathbf{t},1}$. Taking maxima proves divisibility. Each of the three stated equality cases follows directly from (CC.3). Conversely suppose $r, k > 1$, and put $M = \max_{\nu} m_{\nu} > 1$. Choose a root ρ of order M and its pure k -tuple. At $\lambda = k\rho$, every tuple has $D_{\mathbf{t}} \leq k(M-1)$ and $M_{\mathbf{t}} \leq M$; the pure tuple attains both bounds. Therefore

$$r\ell_{\lambda,1} - \ell_{\lambda,r} = (r-1)(k-1)(M-1) > 0. \quad (\text{CC.9})$$

This proves strict divisibility even when other tuples collide there. $\square$

For one root of order m , (CC.3) reads $\chi_r(T) = (T - k\rho)^{1+k(m-1)+(r-1)m}$. For $k = 1$, $\chi_r = h^r$ and α_r is an isomorphism. For $d = 1$, $\chi_r = (T - k\rho)^r$, but the invariant algebra can have a complement to the cyclic image: for $k \geq 2$, $r = 3$, the two symmetric degree-two polynomials $\sum z_i^2$ and $\sum_{i < j} z_i z_j$ are independent. The empty packet $h = 1$ is treated separately: $I = P$, $B_r = A_r = C_r = 0$, $\chi_r = 1$, and every displayed finite linear map is the zero map. No formula involving a chosen root or q_r^{-1} is applied in that case. The arithmetic source itself need not be zero merely because its finite packet projection is zero.

The attaining tuple can change with depth. As a declared algebraic example, take roots $1/2 + it$, $t = 0, 1, 2, 3$, of orders $6, 4, 4, 1$, and $k = 2$. At $\lambda = 1 + 3i$ the only unordered pairs are $(0, 3)$ and $(1, 2)$; they give respectively $6r$ and $4r + 3$. Thus $\ell_{\lambda,r} = \max(6r, 4r + 3)$. This example asserts no location or multiplicity of an arithmetic zero.

30.2 Unscaled invariant bases and a full-unit retraction

For an occupation vector $\mathbf{n} = (n_1, \dots, n_a)$ with $\sum n_\nu = k$, put $\lambda_{\mathbf{n}} = \sum n_\nu \rho_\nu$. In the factor for a fixed ordered tuple of that occupation, its stabilizer is $\prod_\nu \mathfrak{S}_{n_\nu}$. For each orbit of the surviving monomials (CC.5), take the sum of the distinct monomials in that orbit, with coefficient one. These are the original unscaled invariant basis elements. Their exact number is

$$a_{\mathbf{n},r} = \sum_{p=0}^{r-1} [t^p x_1^{n_1} \cdots x_a^{n_a}] \prod_{\nu=1}^a \prod_{j=0}^{r-1} (1 - x_\nu t^j)^{-m_\nu}, \quad a_{\lambda,r} = \sum_{\lambda_{\mathbf{n}}=\lambda} a_{\mathbf{n},r}. \quad (\text{CC.10})$$

Coefficient extraction here is finite formal algebra. Each variable in a group of order m_ν has a pair of labels (j, c) , $j \geq 0$, $0 \leq c < m_\nu$, representing the original exponent $m_\nu j + c$. An orbit is a multiset of these labels, and the surviving condition is that their j 's sum to at most $r - 1$. The geometric series for each label proves (CC.10). An invariant in the full product (CC.4) is determined by its entry at one ordered tuple per occupation; the other entries are its variable-permuted translates. The stabilizer condition is precisely the one just imposed. This proves both the claimed basis and

$$\begin{aligned} \dim B_r &= d^k \binom{r+k-1}{k}, \\ \dim A_r &= \sum_{\lambda} a_{\lambda,r} = \sum_{p=0}^{r-1} [t^p x^k] \prod_{j=0}^{r-1} (1 - xt^j)^{-d}, \\ a_{\mathbf{n},1} &= \prod_{\nu} \binom{n_\nu + m_\nu - 1}{m_\nu - 1}. \end{aligned} \quad (\text{CC.11})$$

For the first equality use (CC.8): there are $\binom{r+k-1}{k}$ vectors $\mathbf{b}$ with $\sum b_i < r$, and d^k possible remainders. For the second alternative expression, permutation acts on that same basis by permuting its k pairs (b_i, c_i) ; its unscaled orbit sums give the stated series.

Theorem 30.3 (Constructive retraction at every depth). *There is an explicit $\mathbb{C}[T]$ -linear retraction $\Pi_r : A_r \rightarrow C_r$ of η_r , with the complete arithmetic unit retained. Put $K_r = \ker \Pi_r$ and $Q_r = 1 - \eta_r \Pi_r$. Then*

$$\begin{aligned} C_r \oplus K_r &\rightarrow A_r, & (c, v) &\mapsto \eta_r(c) + v, \\ A_r &\rightarrow C_r \oplus K_r, & w &\mapsto (\Pi_r w, Q_r w) \end{aligned} \quad (\text{CC.12})$$

are inverse $\mathbb{C}[T]$ -module isomorphisms. In particular $A_r/\eta_r C_r \rightarrow K_r$, $[w] \mapsto Q_r w$ is an isomorphism, inverse to inclusion followed by quotient.

Proof. For each λ , use the exact CRT idempotent $e_\lambda(T)$ in C_r , and let $A_{\lambda,r} = e_\lambda(S)A_r$. Write $N = S - \lambda$ on this summand. Choose an ordered tuple attaining $\ell = \ell_{\lambda,r}$. Choose a coordinate of maximal multiplicity in it, put all $r - 1$ allowed blocks in that coordinate, and put every remainder at its

maximum $m_{\nu_i} - 1$. Call the resulting exponent α^* . Thus $|\alpha^*| = \ell - 1$. Let $\theta : A_{\lambda,r} \rightarrow \mathbb{C}$ extract the coefficient of z^{α^*} in that specific local factor, in the basis (CC.5). Define, with $X = T - \lambda$,

$$\begin{aligned}\Pi_\lambda^0(v) &= \sum_{j=0}^{\ell-1} \theta(N^{\ell-1-j}v) X^j \pmod{X^\ell}, \\ w_\lambda &= e_\lambda(S)U_r, \quad p_\lambda(X) = \Pi_\lambda^0(w_\lambda), \\ \Pi_\lambda(v) &= p_\lambda(X)^{-1} \Pi_\lambda^0(v).\end{aligned}\tag{CC.13}$$

The equality $N^\ell = 0$ gives $\Pi_\lambda^0(Nv) = X\Pi_\lambda^0(v)$, coefficient by coefficient: the constant term on the left is zero and the other coefficients shift by one. Also

$$p_\lambda(0) = \left(\prod_i v_h(\rho_{\nu_i}) \right) \frac{(\ell-1)!}{\prod_i \alpha_i^*!} \neq 0.\tag{CC.14}$$

To verify (CC.14), the selected term in $N^{\ell-1}$ has that multinomial coefficient; multiplying by a positive-degree term of the unit would exceed the greatest surviving total degree. The idempotent $e_\lambda(S)$ is exactly one on this local factor. Set $t = p_\lambda/p_\lambda(0) - 1$. Its exact inverse is $p_\lambda(0)^{-1} \sum_{j=0}^{\ell-1} (-t)^j$. This retains every coefficient of p_λ . The intertwining now gives $\Pi_\lambda(e_\lambda(S)U_r F(S)) = F(\lambda + X) \pmod{X^\ell}$. Take the product of these maps and use scalar CRT to define Π_r . It is the asserted retraction. Its kernel is S -stable. The identities $Q_r \eta_r = 0$, $\Pi_r Q_r = 0$, and $Q_r|_{K_r} = 1$ prove every map and inverse in (CC.12). $\square$

For any polynomial or entire f , multiplication by $f(S)$ has on $A_{\lambda,r}$ the form $f(\lambda)1$ plus a polynomial without constant term in the nilpotent N . Powers of a nilpotent map have trace zero, as is seen in a basis adapted to its kernel filtration. The same statement holds on the two invariant summands in (CC.12). Thus

$$\begin{aligned}\mathrm{Tr}_{A_r} f(S) &= \sum_\lambda a_{\lambda,r} f(\lambda), \\ \mathrm{Tr}_{C_r} f(T) &= \sum_\lambda \ell_{\lambda,r} f(\lambda), \\ \mathrm{Tr}_{K_r} f(S) &= \sum_\lambda (a_{\lambda,r} - \ell_{\lambda,r}) f(\lambda).\end{aligned}\tag{CC.15}$$

This also proves $a_{\lambda,r} \geq \ell_{\lambda,r}$. All nilpotent coordinates remain in (CC.4), (CC.10) and (CC.13); the trace formula follows from those actual operators.

30.3 The derivative tower, scalar first jets, and the unit obstruction

Denote the quotient maps by $\pi_r : B_{r+1} \rightarrow B_r$ and $\rho_r : C_{r+1} \rightarrow C_r$. The original derivative is

$$\delta_r : B_{r+1} \longrightarrow B_r, \quad [F] \longmapsto \left[\frac{1}{k} \sum_i \partial_{s_i} F \right], \quad \partial_r : C_{r+1} \longrightarrow C_r, \quad [F] \longmapsto [F'].\tag{CC.16}$$

Each is a complex linear map. Differentiating a product of $r+1$ elements of I leaves at least r such factors, so $\partial_S I^{r+1} \subset I^r$. Formula (CC.3) gives $\ell_{\lambda,r+1} \geq \ell_{\lambda,r} + 1$. Thus the derivative of a multiple of χ_{r+1} is a multiple of χ_r , proving the second descent as well. On representatives,

$$\begin{aligned}\delta_r \alpha_{r+1} &= \alpha_r \partial_r, & \pi_r \alpha_{r+1} &= \alpha_r \rho_r, \\ \delta_r(xy) &= \pi_r(x) \delta_r(y) + \pi_r(y) \delta_r(x), & \pi_r \delta_{r+1} &= \delta_r \pi_{r+1}.\end{aligned}\tag{CC.17}$$

The last equality is a map $B_{r+2} \rightarrow B_r$. The symmetric subspaces are preserved because the derivative averages all k coordinates with the literal factor $1/k$. Every δ_r is onto: the invertible coordinates

$S, z_1, \dots, z_{k-1}$, where $z_i = s_i - S/k$, have $\partial_S S = 1$ and $\partial_S z_i = 0$. Integrating powers of S gives a polynomial primitive of any polynomial representative. Averaging such a primitive proves surjectivity on the invariant subspaces too.

The same monic expansion (CC.8) identifies

$$I^j/I^{j+1} \cong (P/I)[\zeta_1, \dots, \zeta_k]_j, \quad \zeta_i \mapsto [h_i], \quad \text{gr } \delta = \frac{1}{k} \sum_i [h'_i] \partial_{\zeta_i}. \quad (\text{CC.18})$$

Indeed differentiating $h^{\mathbf{b}} r_{\mathbf{b}}$ in degree $j = |\mathbf{b}|$ gives $b_i h^{\mathbf{b}-e_i} h'_i r_{\mathbf{b}}/k$ in degree $j-1$; differentiating $r_{\mathbf{b}}$ stays in degree j . Both the integer b_i and the original polynomial h'_i survive.

Introduce a formal square-zero parameter ϵ , with its ordinary coefficient algebra meaning. The two maps

$$\begin{aligned} \Phi_r : B_{r+1} &\longrightarrow B_r[\epsilon]/(\epsilon^2), & b &\longmapsto \pi_r b + \epsilon \delta_r b, \\ \phi_r : C_{r+1} &\longrightarrow C_r[\epsilon]/(\epsilon^2), & F &\longmapsto \rho_r F + \epsilon \partial_r F \end{aligned} \quad (\text{CC.19})$$

are unital algebra homomorphisms by (CC.17) and its one-variable product rule. Extending α_r coefficientwise to α_r^ϵ gives the exact square $\Phi_r \alpha_{r+1} = \alpha_r^\epsilon \phi_r$. The coefficient projections of these algebra maps are precisely the quotient and derivative maps in (CC.16).

Put $R(T) = \prod_{\lambda \in \Lambda_k} (T - \lambda)$ and $\psi_r = \chi_r R$. Then

$$\begin{aligned} \ker \phi_r &= (\psi_r)/(\chi_{r+1}), & \dim \text{im } \phi_r &= q_r + |\Lambda_k|, \\ \ker \rho_r &= (\chi_r)/(\chi_{r+1}) \cong \mathbb{C}[T]/(\chi_{r+1}/\chi_r), & \partial_r(\ker \rho_r) &= \chi'_r C_r. \end{aligned} \quad (\text{CC.20})$$

For the first assertion, F and F' vanish modulo $(T - \lambda)^{\ell_{\lambda,r}}$ exactly when F has order at least $\ell_{\lambda,r} + 1$ there. The order implication in (CC.16) gives $\psi_r \mid \chi_{r+1}$, so the asserted kernel is well defined and has dimension $q_{r+1} - q_r - |\Lambda_k|$. This proves the image dimension and the injective factorization $\mathbb{C}[T]/(\psi_r) \hookrightarrow C_r[\epsilon]/(\epsilon^2)$. Multiplication by χ_r gives the displayed isomorphism onto $\ker \rho_r$; its kernel is exactly (χ_{r+1}/χ_r) . Differentiating $\chi_r F$ modulo χ_r gives $\chi'_r F$. At each λ , multiplication by χ'_r maps the input constant to a nonzero multiple of $(T - \lambda)^{\ell_{\lambda,r}-1}$. Its image has one dimension per λ , proving all image and kernel claims even at repeated collided roots.

Write

$$\beta_r = U_r^{-1} \delta_r U_{r+1} \in A_r, \quad \Phi_r(U_{r+1}) = U_r(1 + \epsilon \beta_r), \quad \Phi_r(U_{r+1}^{-1}) = U_r^{-1}(1 - \epsilon \beta_r). \quad (\text{CC.21})$$

The first formula is the definition of the full logarithmic derivative class, the second follows directly from (CC.19), and the third follows by multiplication and $\epsilon^2 = 0$. In particular,

$$\delta_r \eta_{r+1} = \eta_r \partial_r + M_{\delta_r U_{r+1}} \alpha_r \rho_r. \quad (\text{CC.22})$$

This formula retains the full derivative of the arithmetic unit.

There is a complete decomposition of its last term using (CC.12):

$$\begin{aligned} b_r &= \Pi_r(\delta_r U_{r+1}) \in C_r, & n_r &= Q_r(\delta_r U_{r+1}) \in K_r, \\ \mathcal{N}_r : C_r &\longrightarrow K_r, & \mathcal{N}_r(F) &= \alpha_r(F) n_r, \\ \delta_r \eta_{r+1} &= \eta_r(\partial_r + M_{b_r} \rho_r) + \mathcal{N}_r \rho_r. \end{aligned} \quad (\text{CC.23})$$

Here K_r is a $\mathbb{C}[T]$ -submodule, so multiplication by $\alpha_r(F)$ preserves it. Both Π_r and Q_r intertwine this multiplication; applying them to the last term of (CC.22) proves (CC.23). To state the obstruction independently of the chosen retraction, let $\mathbf{q}_r : A_r \rightarrow A_r/\eta_r C_r$ be the module quotient and put

$$\Omega_r = \mathbf{q}_r M_{\delta_r U_{r+1}} \alpha_r : C_r \longrightarrow A_r/\eta_r C_r. \quad \mathbf{q}_r \delta_r \eta_{r+1} = \Omega_r \rho_r, \quad \Omega_r = 0 \iff \beta_r \in \alpha_r(C_r). \quad (\text{CC.24})$$

To prove the equivalence, evaluate Ω_r at 1. Its vanishing says $\delta_r U_{r+1} = U_r \alpha_r(\gamma_r)$ for a unique $\gamma_r \in C_r$, precisely the condition on β_r . Conversely that equality makes every value of Ω_r zero by closure of the subalgebra $\alpha_r(C_r)$. Under (CC.12), Ω_r corresponds exactly to $\mathcal{N}_r$. When it vanishes, $b_r = \gamma_r$, and the exact pulled-back derivative is $\partial_r + M_{\gamma_r} \rho_r$. The formulas construct both the obstruction and its strongest scalar derivative correspondence without deleting the complementary directions.

30.4 The finite cyclic derivative and its exact Weyl correction

The quotient derivative has two different depths in its domain and codomain. To compute its finite endomorphism on the prescribed scalar coordinates, let $s_r : C_r \rightarrow C_{r+1}$ send a class to the same polynomial of degree less than q_r given by monic division by χ_r . This is a complex-linear section of ρ_r . Put

$$\begin{aligned} L_r(F) &= [T^{q_r-1}]F_0(T), & d_r &= \partial_r s_r : C_r \rightarrow C_r, \\ D_r^{\text{red}} &= d_r + M_{b_r}, & A_r^C &= M_T : C_r \rightarrow C_r, \end{aligned} \quad (\text{CC.25})$$

where F_0 is that unchanged monic remainder. The scalar $L_r(F)$ retains its original coefficient; it is not a functional defined by dividing by a norm. The complete differentiated arithmetic map on this section is

$$\delta_r \eta_{r+1} s_r = \eta_r D_r^{\text{red}} + \mathcal{N}_r : C_r \longrightarrow A_r. \quad (\text{CC.26})$$

Thus the reduced derivative and its complementary component have explicit maps into the original arithmetic algebra.

Theorem 30.4 (Finite sum-coordinate Weyl relation). *For the original monic remainder coordinates,*

$$\begin{aligned} [A_r^C - k/2, D_r^{\text{red}}] &= -I_{C_r} + J_r, & J_r(F) &= [\chi_r'] L_r(F), \\ \text{rank } J_r &= 1, & J_r^2 &= q_r J_r, & \text{Tr } J_r &= q_r. \end{aligned} \quad (\text{CC.27})$$

The full arithmetic correction is the image of the original retained relation:

$$\begin{aligned} \rho_r([\chi_r]_{\chi_{r+1}} L_r(F)) &= 0, \\ \delta_r(U_{r+1} \alpha_{r+1}([\chi_r] L_r(F))) &= \eta_r([\chi_r'] L_r(F)). \end{aligned} \quad (\text{CC.28})$$

No complementary derivative is suppressed in (CC.26).

Proof. Since TF_0 has degree at most q_r , its exact monic remainder is $TF_0 - \chi_r L_r(F)$. Differentiating it and reducing gives

$$d_r A_r^C F = [F_0 + TF_0' - \chi_r' L_r(F)], \quad A_r^C d_r F = [TF_0'].$$

Their difference proves (CC.27), because multiplication by b_r commutes with multiplication by T . The vector $[\chi_r']$ has degree $q_r - 1$ and leading coefficient $q_r \neq 0$. Thus $L_r([\chi_r']) = q_r$; the operator is rank one, its square is q_r times itself, and its trace is that same scalar, either in the remainder basis or by the rank-one matrix formula. For (CC.28), the quotient kills χ_r , and the product rule leaves $U_r \alpha_r(\chi_r') L_r(F)$; the term with $\delta_r U_{r+1}$ still contains $\alpha_r(\chi_r) = 0$. The original relation itself remains in B_{r+1} before this application. Finally $\mathcal{N}_r$ is $\mathbb{C}[T]$ -linear by (CC.23). Consequently the full relation obtained by applying S to (CC.26) and subtracting (CC.26) applied to TF is exactly $-\eta_r + \eta_r J_r$; this proves the intertwining including the complementary map. $\square$

At $r = 1, k = 1, C_1 = \mathbb{C}[s]/(h)$, L_1 is the coefficient of s^{d-1} , and (CC.27) gives $[M_{s-1/2}, d_1 + M_{b_1}] = -I + [h'] L_1$. In the original weighted coordinates $u = v_h F$, its conjugated correction is $u \mapsto v_h h' L_1(v_h^{-1} u)$. Multiplication by v_h and by v_h^{-1} are the explicit inverse coordinate maps; the expression is the one-variable full-unit correction in the preceding finite Weyl calculation. Formula (CC.28) gives its all-depth extension using the actual relation layer rather than differentiating a quotient zero.

30.5 All first jets and the symmetric conormal kernel

Set $E = B_1, E^{[2]} = B_2, A_h = \mathbb{C}[s]/(h)$, and write $c_i = [h'(s_i)] \in E$. Let a continue to denote the number of distinct roots and put $b = d - a = \deg \gcd(h, h')$. The conormal coordinates are the actual

isomorphism

$$\begin{aligned} \iota : E^k &\longrightarrow I/I^2, & (v_i) &\longmapsto \left[\sum_i h_i \tilde{v}_i \right], \\ \partial_N : E^k &\longrightarrow E, & (v_i) &\longmapsto \frac{1}{k} \sum_i c_i v_i. \end{aligned} \quad (\text{CC.29})$$

Here $\tilde{v}_i$ is its original separate-variable monic remainder. Changing a lift changes the displayed sum by I^2 . Equation (CC.8) proves injectivity, surjectivity and the inverse coefficient extraction. Applying the product rule gives exactly $\delta_1 \iota = \partial_N$. The full conormal differential and the specified contraction are the explicit maps

$$E^k \longrightarrow \bigoplus_i E \, ds_i \longrightarrow E, \quad (v_i) \longmapsto \sum_i c_i v_i \, ds_i \longmapsto \frac{1}{k} \sum_i c_i v_i. \quad (\text{CC.30})$$

Thus the intermediate cotangent module and every coefficient of the sum direction are retained.

Theorem 30.5 (Exact ranks and complete first-jet observations). *Put $T_h = \mathbb{C}[s]/(h, h')$. Then*

$$\begin{aligned} \text{im } \partial_N &= (c_1, \dots, c_k), & \text{coker } \partial_N &\cong T_h^{\otimes k}, \\ \text{rank}_{\mathbb{C}} \partial_N &= d^k - b^k, & \dim \ker \partial_N &= (k-1)d^k + b^k. \end{aligned} \quad (\text{CC.31})$$

Equip $E \oplus E^k$ with multiplication $(x, v)(y, w) = (xy, xw + yv)$, unit $(1, 0)$. The maps

$$\begin{aligned} J_{\text{all}} : B_2 &\longrightarrow E \oplus E^k, & [F] &\longmapsto ([F], ([\partial_i F])_i), \\ J_S : B_2 &\longrightarrow E[\epsilon]/(\epsilon^2), & [F] &\longmapsto [F] + \epsilon \delta_1[F], \\ \mathcal{L} : E \oplus E^k &\longrightarrow E[\epsilon]/(\epsilon^2), & (x, v) &\longmapsto x + \frac{\epsilon}{k} \sum_i v_i \end{aligned} \quad (\text{CC.32})$$

are unital algebra homomorphisms and satisfy $J_S = \mathcal{L} J_{\text{all}}$. Their kernels and ranks are

$$\begin{aligned} \ker J_S &= \iota(\ker \partial_N), & \dim \text{im } J_S &= 2d^k - b^k, \\ \ker J_{\text{all}} &= \iota \left(\bigoplus_i \text{ann}_E(c_i) \right), & \dim \ker J_{\text{all}} &= kb d^{k-1}, \\ \dim \text{im } J_{\text{all}} &= (k+1)d^k - kb d^{k-1}. \end{aligned} \quad (\text{CC.33})$$

For squarefree h , J_{all} is an isomorphism onto the displayed square-zero algebra.

Proof. The row in (CC.29) has precisely the ideal of its entries as image. Quotienting by these entries in the separate tensor factors gives $T_h^{\otimes k}$, with the map induced by the original polynomial quotients. The Euclidean algorithm gives $(h, h') = (\gcd(h, h'))$, so $\dim T_h = b$. This proves the cokernel and rank in (CC.31), and rank-nullity on the kd^k -dimensional source proves its kernel dimension.

Each coordinate derivative maps I^2 into I ; hence the maps in (CC.32) are well defined. The product rule for each derivative gives exactly the specified square-zero multiplication. The factor $1/k$ gives $J_S = \mathcal{L} J_{\text{all}}$. An element in either jet kernel has zero ordinary quotient, so it is a conormal class. Applying (CC.29) gives its sum derivative, while applying each coordinate derivative gives $(c_i v_i)_i$. This proves the asserted kernels. Multiplication by h' on A_h has cokernel T_h , hence has rank $d - b$ and kernel dimension b . Tensoring the kernel and image with the other $k - 1$ vector-space factors shows that $\dim \text{ann}_E(c_i) = b d^{k-1}$. The dimension of B_2 is $(k+1)d^k$ by (CC.11), which proves both image dimensions. If $b = 0$, the full map has zero kernel and equal source and target dimensions; its inverse is multiplicative because it is the inverse of a bijective algebra homomorphism. $\square$

The images also have explicit coordinates. Let $j : E \rightarrow B_2$ choose the original remainder with $\deg_{s_i} < d$. Set $d_0(x) = \delta_1 j(x)$ and $D_0(x) = ([\partial_i j(x)])_i$. Directly from (CC.29),

$$\begin{aligned} \text{im } J_S &= \{x + \epsilon(d_0(x) + v) : x \in E, v \in (c_1, \dots, c_k)\}, \\ \text{im } J_{\text{all}} &= \{(x, D_0(x) + (c_i v_i)_i) : x \in E, (v_i) \in E^k\}. \end{aligned} \quad (\text{CC.34})$$

Every lift differs from $j(x)$ by exactly one conormal vector, which proves both inclusions of each image equality. In particular $\ker J_S = J_{\text{all}}^{-1}(\ker \mathcal{L})$, where $\ker \mathcal{L} = \{(0, v) : \sum_i v_i = 0\}$. This identifies precisely the relative directions removed by the sum contraction after all first derivatives have been retained.

The same calculation restricts to the permutation invariants. On E^k the simultaneous action is $(\sigma v)_i = \sigma(v_{\sigma^{-1}i})$; all the maps above intertwine this action. Evaluation at the first component gives the explicit vector-space isomorphism

$$(E^k)^{\mathfrak{S}_k} \longrightarrow A_h \otimes (A_h^{\otimes(k-1)})^{\mathfrak{S}_{k-1}}. \quad (\text{CC.35})$$

For its inverse, start with a tensor fixed by the stabilizer of index one. For each i , choose a permutation carrying one to i and use its translate as component i . Two choices differ by the stabilizer, so the component is independent of that choice. The resulting tuple is invariant, proving the inverse in full. When $k = 1$, the tensor with zero factors is $\mathbb{C}$ and both maps are the identity on A_h .

For a vector space of dimension n , the invariant tensors in its k -fold tensor power have a basis of unscaled sums over words with prescribed occupation numbers in a fixed basis. There are $\binom{n+k-1}{k}$ such occupations for $n \geq 1$, and zero for $n = 0, k \geq 1$. This proves the dimensions used here directly. Taking invariants in the exact cokernel sequence of (CC.31) remains exact: average a lift by $k!^{-1} \sum_{\sigma} \sigma$ to lift any invariant image. Consequently

$$\begin{aligned} \text{coker } \partial_N^{\text{sym}} &\cong (T_h^{\otimes k})^{\mathfrak{S}_k}, \\ \dim(E^k)^{\mathfrak{S}_k} &= d \binom{d+k-2}{k-1}, \\ \text{rank } \partial_N^{\text{sym}} &= \binom{d+k-1}{k} - \binom{b+k-1}{k}, \\ \dim \ker \partial_N^{\text{sym}} &= d \binom{d+k-2}{k-1} - \binom{d+k-1}{k} + \binom{b+k-1}{k}. \end{aligned} \quad (\text{CC.36})$$

The final binomial is defined to be zero when $b = 0$. For the declared polynomial fixture $h = s^2, k = 2$, the row is $(v_1, v_2) \mapsto s_1 v_1 + s_2 v_2$; its image is the span of $s_1, s_2, s_1 s_2$, of dimension three, and its kernel has dimension five. On invariants the domain has dimension four, target dimension three, image span $s_1 + s_2, s_1 s_2$, and kernel dimension two. These are exact quotient computations in the stated algebraic fixture, without an assertion about arithmetic roots.

In the original cochain convention a permutation of k factors of cochain degree one contributes $\text{sgn}(\sigma)$. Thus the top-degree action of $k!^{-1} \sum_{\sigma} \text{sgn}(\sigma) T_{\sigma}$ is $k!^{-1} \sum_{\sigma} \sigma$ on these tensor coefficients: the two signs multiply to one. This is the explicit identification with the prescribed symmetric cochain summand. Its top-degree supertrace multiplies each ordinary trace by the retained factor $(-1)^k$.

30.6 Cyclic relation response and the full product Jacobian

Fix the monic-division identity $\chi_1(S) = \sum_i h_i B_i(\mathbf{s})$ in the original polynomial ring; it exists by (CC.6) and (CC.8). Define

$$\begin{aligned} \mathcal{B} : C_1 &\longrightarrow E^k, & F &\longmapsto ([B_i]_I \alpha_1(F))_i, \\ \partial_N \mathcal{B}(F) &= \alpha_1([\chi'_1]F), & \delta_1[U \chi_1(S) F(S)]_{I^2} &= U_1 \alpha_1([\chi'_1]F). \end{aligned} \quad (\text{CC.37})$$

The first map is well defined and $\mathbb{C}[T]$ -linear because each coefficient multiplies the well-defined class $\alpha_1(F)$. Differentiate the displayed division identity with $k^{-1} \sum_i \partial_i$. Terms still containing h_i vanish modulo I ; the derivative of S is exactly one. This proves the first identity. Differentiating the full-unit relation proves the second, since terms with an undifferentiated χ_1 vanish only at the final reduction. These maps give the exact conormal-to-cyclic Jacobian comparison, including $\ker(\partial_N \mathcal{B}) = \ker M_{\chi_1'}$, since α_1 is injective. Its rank is $|\Lambda_k|$ by (CC.20), and its kernel has dimension $q_1 - |\Lambda_k|$.

The full product Jacobian has a different explicit factorization through the same original conormal row. Define

$$\begin{aligned} J_{\text{prod}} &= \prod_i c_i \in E, \\ \mathcal{P} : E &\longrightarrow E^k, & v &\longmapsto \left(v \prod_{j \neq i} c_j \right)_i, \\ \partial_N \mathcal{P} &= M_{J_{\text{prod}}} : E \rightarrow E. \end{aligned} \tag{CC.38}$$

Indeed each of the k summands in the row is $J_{\text{prod}} v / k$. This proves the factorization with the original factor $1/k$. It is permutation equivariant. Its exact kernel is $\mathcal{P}^{-1}(\ker \partial_N) = \text{ann}_E(J_{\text{prod}})$. Also $\ker \mathcal{P} = \bigcap_i \text{ann}_E(\prod_{j \neq i} c_j)$; these equalities follow component by component and retain the intermediate directions rather than declaring the two observations interchangeable.

In a local factor $\mathbb{C}[z]/(z^m)$, the literal factorization $h(\rho + z) = z^m u_\rho(z)$ gives $h'(\rho + z) = m u_\rho(0) z^{m-1}$ modulo z^m . Thus in an ordered product factor, multiplication by J_{prod} sends a constant to $\prod_i m_i u_{\rho_i}(0) \prod_i z_i^{m_i-1}$, with nonzero coefficient, and annihilates the maximal ideal. There is one image dimension for each of the a^k ordered root tuples. Therefore

$$\text{rank } M_{J_{\text{prod}}} = a^k, \quad \ker M_{J_{\text{prod}}} = \sqrt{0_E}, \quad \dim \sqrt{0_E} = d^k - a^k. \tag{CC.39}$$

To check the kernel assertion globally, a class is nilpotent precisely when its constant coefficient vanishes at each local factor: the local maximal ideals are nilpotent and the product is finite. This is exactly the computed annihilator. Under permutation invariants, one image dimension remains for each occupation of the distinct roots, giving rank $\binom{a+k-1}{k}$. The invariant kernel is $(\sqrt{0_E})^{\mathfrak{S}_k}$, either by restriction or by averaging a lift of each invariant image. The factorization (CC.38) continues to hold on these invariant spaces.

The arithmetic unit remains on every observed conormal vector:

$$\delta_1(U_2 \iota(v)) = U_1 \partial_N(v), \quad U_1 \partial_N \mathcal{P} = M_{U_1 J_{\text{prod}}}. \tag{CC.40}$$

The first formula follows from (CC.17), since $\pi_1 \iota(v) = 0$. Multiplication by U_1^{-1} is its explicit inverse change on the target, so the kernels and ranks remain those proved above while all actual Taylor coefficients stay in the map.

30.7 Trace pairings and all four complementary operator blocks

For a monic one-variable polynomial p , all finite residue loops below have positive orientation. The functional $\lambda_p(F) = \sum_{\zeta} \text{Res}_{T=\zeta} F(T) p(T)^{-1} dT$ is well defined on $\mathbb{C}[T]/(p)$: adding pQ adds an entire polynomial differential at every loop. At a root of order m , write $p = (T - \zeta)^m c(T)$, where $c(\zeta) \neq 0$. Then $p'/p = m/(T - \zeta) + c'/c$. Hence

$$\lambda_p(p'F) = \sum_{\zeta} m_{\zeta} F(\zeta) = \text{Tr}_{\mathbb{C}[T]/(p)} M_F. \tag{CC.41}$$

For the last equality, in each basis $1, T - \zeta, \dots, (T - \zeta)^{m-1}$, the multiplication matrix has diagonal $F(\zeta)$ repeated m times; positive nilpotent powers have zero diagonal. This proves the trace identity with all original multiplicities.

The packet reflection gives the conjugate-linear algebra involution

$$F^\dagger(\mathbf{s}) = \overline{F(1 - \bar{s}_1, \dots, 1 - \bar{s}_k)}$$

on each B_r , because it permutes the root factors of h with their full orders. On the scalar algebra its restriction is $F^{\dagger k}(T) = \overline{F(k - \bar{T})}$. The sum set and all the exponents in (CC.3) are permuted by $\lambda \mapsto k - \bar{\lambda}$; hence $\chi_r^{\dagger k} = (-1)^{q_r} \chi_r$. This proves well-definedness on each quotient and $\alpha_r(F)^\dagger = \alpha_r(F^{\dagger k})$. Consequently the cyclic pairing

$$R_{\chi_r}(F, G) = \lambda_{\chi_r}(F^{\dagger k} G) \quad \text{satisfies} \quad R_{\chi_r}(F, \chi'_r G) = \sum_{\lambda} \ell_{\lambda, r} \overline{F(k - \bar{\lambda})} G(\lambda). \quad (\text{CC.42})$$

This is (CC.41) applied to the same full quotient, with its specified involution.

For the original one-factor arithmetic pairing, put $v_h = [v_h]_h$, $\varepsilon_h = v_h^{-1}$, and $\mathcal{R}_Z(f, u) = \lambda_h(\varepsilon_h f^\dagger u)$. The identity $g' = h'v_h + hv'_h$ gives $j_h(g') = h'v_h$. Thus the complete unit calculation is

$$\begin{aligned} \mathcal{R}_Z(f, j_h(g')P) &= \lambda_h(\varepsilon_h f^\dagger v_h h'P) \\ &= \lambda_h(f^\dagger h'P) = \text{Tr}_{A_h} M_{f^\dagger P}. \end{aligned} \quad (\text{CC.43})$$

The product functional $\lambda_h^{\otimes k}$ consequently satisfies $\lambda_h^{\otimes k}(J_{\text{prod}} F) = \text{Tr}_E M_F$. To verify it without a tensor restriction on F , expand F in the original separate-variable monomial basis and apply (CC.41) factor by factor; both sides are linear in every coefficient. The product $U_1^{-1} U_1$ gives the same exact unit cancellation for its original weighted arithmetic version. This describes the full product trace and (CC.38) supplies its map through the sum-direction conormal observation.

Let $\chi_{A, r}(T) = \prod_{\lambda} (T - \lambda)^{a_{\lambda, r}}$. Equation (CC.15) gives the genuine quotient trace of every entire $f(S)$ on $T_r = A_r / \eta_r C_r$:

$$\text{Tr}_{T_r} f(S) = \sum_{\lambda} \text{Res}_{T=\lambda} f(T) \left(\frac{\chi'_{A, r}}{\chi_{A, r}} - \frac{\chi'_r}{\chi_r} \right) dT. \quad (\text{CC.44})$$

Its proof is the local logarithmic derivative calculation (CC.41) with exponents $a_{\lambda, r} - \ell_{\lambda, r}$. It retains the module T_r ; its formula is a trace of its induced operator, not a replacement of its nilpotent coordinates.

For an arbitrary linear endomorphism $M : A_r \rightarrow A_r$, use the fixed full-unit splitting (CC.12). Write $\iota_K : K_r \rightarrow A_r$ for inclusion and regard Q_r as a map with codomain K_r . Its full transformed matrix is

$$\begin{pmatrix} \Pi_r M \eta_r & \Pi_r M \iota_K \\ Q_r M \eta_r & Q_r M \iota_K \end{pmatrix} : C_r \oplus K_r \longrightarrow C_r \oplus K_r. \quad (\text{CC.45})$$

Applying the two inverse maps of (CC.12) on either side of M proves every block. In these original coordinates,

$$\text{Tr}_{A_r} M = \text{Tr}_{C_r}(\Pi_r M \eta_r) + \text{Tr}_{K_r}(Q_r M \iota_K). \quad (\text{CC.46})$$

The cross maps in (CC.45) remain part of the transformed operator. For completeness, invariance of trace under these inverse changes follows from $\text{Tr}(XY) = \text{Tr}(YX)$, proved by expanding $\sum_{i,j} X_{ij} Y_{ji}$. The quotient T_r has the induced map $[v] \mapsto [Mv]$ exactly when $M\eta_r C_r \subseteq \eta_r C_r$, equivalently $Q_r M \eta_r = 0$. Necessity follows by applying a well-defined quotient map to $[\eta_r c] = 0$; sufficiency follows by comparing any two representatives. In that case (CC.12) identifies its map with $Q_r M \iota_K$. This proves the exact quotient criterion and the always-defined complementary compression when it fails.

For a multiplier $M = M_w$ with $w \in A_r$, this criterion is

$$M_w \eta_r C_r \subseteq \eta_r C_r \iff w \in \alpha_r(C_r). \quad (\text{CC.47})$$

Indeed its value on $\eta_r(1) = U_r$ gives necessity after multiplication by U_r^{-1} . Conversely subalgebra closure proves sufficiency. If $w = f(S)$, both cross maps in (CC.45) are zero by $\mathbb{C}[T]$ -linearity, proving (CC.44) directly. The full weighted vector pairing is instead the explicitly typed trace

$$\mathrm{Tr}_{A_r} M_{(\eta_r F)^\dagger \eta_r G} = \mathrm{Tr}_{A_r} M_{U_r^\dagger U_r \alpha_r (F^\dagger_k G)}. \quad (\text{CC.48})$$

Multiplicativity of the involution proves this identity before any trace calculation. Insert that unchanged multiplier into all four blocks of (CC.45) to compute its full decomposition. In particular $U_r^\dagger U_r$ has an induced quotient multiplier precisely when it satisfies (CC.47); no membership is assumed from invariance under permutations. If an original positive Hermitian form is represented by H on A_r , the same coordinates give its full pullback

$$\begin{pmatrix} \eta_r^* H \eta_r & \eta_r^* H \iota_K \\ \iota_K^* H \eta_r & \iota_K^* H \iota_K \end{pmatrix}. \quad (\text{CC.49})$$

This follows by evaluating the form on $\eta_r c + \iota_K v$ and $\eta_r c' + \iota_K v'$; all four terms appear. The algebraic retraction entails no orthogonality assertion for that original metric.

30.8 Exact two-factor symmetric presentation at every depth

For $k = 2$, retain $S = s_1 + s_2$, $R = s_1 - s_2$ and $\Delta = R^2$, with inverse $s_1 = (S + R)/2$, $s_2 = (S - R)/2$. Write the original polynomial as $h(s) = \sum_{j=0}^d c_j s^j$, $c_d = 1$, and define

$$\begin{aligned} H_0(S, \Delta) &= \sum_{j=0}^d c_j 2^{-j} \sum_{2a \leq j} \binom{j}{2a} S^{j-2a} \Delta^a, \\ H_1(S, \Delta) &= \sum_{j=1}^d c_j 2^{-j} \sum_{2a+1 \leq j} \binom{j}{2a+1} S^{j-2a-1} \Delta^a. \end{aligned} \quad (\text{CC.50})$$

Binomial expansion gives exactly $h(s_1) = H_0 + R H_1$, $h(s_2) = H_0 - R H_1$, including $R = 0$. Therefore the linear change of generators with coefficients $1/2$ gives $I = (H_0, R H_1)$ in $\mathbb{C}[S, R]$. For every $r \geq 1$ the complete invariant relation ideal is

$$\begin{aligned} J_r &= \left(H_0^{r-j} \Delta^{\lceil j/2 \rceil} H_1^j : 0 \leq j \leq r \right) \subset \mathbb{C}[S, \Delta], \\ A_r &\cong \mathbb{C}[S, \Delta]/J_r, \quad C_r \hookrightarrow \mathbb{C}[S, \Delta]/J_r, \quad [F(T)] \mapsto [F(S)]. \end{aligned} \quad (\text{CC.51})$$

To prove the ideal identity, express any element of I^r as a sum of the generators $H_0^{r-j} R^j H_1^j$ times arbitrary polynomials in S, R . Average under $R \mapsto -R$. For even j , the coefficient's even part is a polynomial in S, Δ , giving $\Delta^{j/2}$. For odd j , its odd part is R times such a polynomial, giving $\Delta^{(j+1)/2}$. This proves containment in J_r . Each displayed generator is conversely an invariant member of I^r , proving equality. Every invariant quotient class has an invariant lift by the same average with coefficient $1/2$. Thus the induced algebra map is onto and has exactly this kernel. The injection in (CC.51) is precisely (CC.6) in those coordinates; its kernel before taking C_r is (χ_r) .

At depth one this recovers the literal presentation $\mathbb{C}[S, \Delta]/(H_0, \Delta H_1)$, with the relative factor Δ intact. Under the earlier quartet assumptions, its H_0, H_1 are exactly those of (SSP.9): both equal the even and odd coefficients of the same unchanged polynomial $h((S + R)/2)$. The further substitution $Z = S - 1$, $x = -Z^2$ is its original (SSP.2) map, so the cyclic injection sends $F(T)$ to $F(1 + Z)$ in the complete SSP algebra. Its weighted module injection sends it to $[v_h((S + R)/2) v_h((S - R)/2)] F(S)$; that product is invariant, and its finite Taylor class is exactly U_r . The inverse SSP coordinate formulas, and (CC.12) for the cyclic summand, retain the full relative and parity modules at every step.

The two parity modules themselves have exact formulas at every depth. Under those same quartet assumptions, put $\mathcal{R} = \mathbb{C}[x, \Delta]$ and use the unchanged polynomials A_0, A_1 of (SSP.3), so that $H_0 =$

$A_0(-Z^2, \Delta)$, $H_1 = ZA_1(-Z^2, \Delta)$. For all indices within $0 \leq j \leq r$, define

$$\begin{aligned} E_{2a} &= A_0^{r-2a} \Delta^a (-x)^a A_1^{2a}, \\ O_{2a+1} &= A_0^{r-2a-1} \Delta^{a+1} (-x)^a A_1^{2a+1}, \\ \mathfrak{a}_{e,r} &= (E_{2a}, xO_{2a+1}) \subset \mathcal{R}, & \mathfrak{a}_{o,r} &= (E_{2a}, O_{2a+1}) \subset \mathcal{R}, \\ B_{e,r} &= \mathcal{R}/\mathfrak{a}_{e,r}, & B_{o,r} &= \mathcal{R}/\mathfrak{a}_{o,r}. \end{aligned} \tag{CC.51a}$$

Then $A_r = B_{e,r} \oplus ZB_{o,r}$ as a module in these specified coordinates. Its exact maps and product are

$$\begin{aligned} \beta_r^{\text{par}} : B_{e,r} &\rightarrow B_{o,r}, & [a] &\mapsto [a], \\ \gamma_r^{\text{par}} : B_{o,r} &\rightarrow B_{e,r}, & [b] &\mapsto [xb], \\ (a, b)(c, e) &= (ac - \gamma_r^{\text{par}}(be), \beta_r^{\text{par}}(a)e + \beta_r^{\text{par}}(c)b), \\ \gamma_r^{\text{par}} \beta_r^{\text{par}} &= xI_{B_{e,r}}, & \beta_r^{\text{par}} \gamma_r^{\text{par}} &= xI_{B_{o,r}}. \end{aligned} \tag{CC.51b}$$

The superscript distinguishes these parity maps from the full logarithmic unit class β_r defined in (CC.21). For a direct proof, substitute $Z^2 = -x$ into each generator of (CC.51). Even $j = 2a$ gives E_{2a} , and odd $j = 2a + 1$ gives ZO_{2a+1} . Before imposing them the module is freely $\mathcal{R} \oplus Z\mathcal{R}$, by monic division in Z . Multiplication by $c + Zd$ shows that their even coefficient ideal is exactly (E_{2a}, xO_{2a+1}) , and their odd coefficient ideal is exactly (E_{2a}, O_{2a+1}) . This proves both directions of the claimed decomposition. The containments between those ideals and their multiples by x prove well-definedness of $\beta_r^{\text{par}}, \gamma_r^{\text{par}}$; multiplication in $\mathcal{R}[Z]/(Z^2 + x)$ proves every sign, product and composition in (CC.51b). At $r = 1$ these are exactly (SSP.6)–(SSP.7), because $E_0 = A_0$ and $O_1 = \Delta A_1 = Q_\Delta$. The original unit is still the full class of the product in (CC.2), carried through these inverse coefficient maps.

30.9 Split lifts, source provenance and quantitative scope

For a complex vector space V , retain $G(V) = \{\tau\} \sqcup \{v^\bullet : v \in V\}$, with $e_V = 0_V^\bullet$, τ the additive identity, and $v^\bullet + w^\bullet = (v + w)^\bullet$. The scalar action is $c^\bullet v^\bullet = (cv)^\bullet$, and an absent scalar or vector produces τ . For a complex-linear map $L : V \rightarrow W$, its exact lift sends $\tau \mapsto \tau$, $v^\bullet \mapsto (Lv)^\bullet$. It preserves addition and scalar action: on two supported inputs this is the linearity of L , and cases with τ follow from the given identity and absorbing rules. For an algebra homomorphism, the same formula also preserves supported products and units. These formulas apply to every map above with its proved type; a derivative uses the linear lift, while the maps (CC.19) and (CC.32) use the multiplicative lift.

In particular, for a conormal vector $n = \iota(v)$,

$$G(\pi_1)(n^\bullet) = e_E, \quad G(J_S)(n^\bullet) = (\epsilon \partial_N v)^\bullet, \quad G(\delta_1)(n^\bullet) = (\partial_N v)^\bullet. \tag{CC.52}$$

The common source is the retained thickening. Every arrow fixes external absence, and every supported kernel vector maps to the receiving supported zero. Equations (CC.37)–(CC.40) give the actual original-unit morphisms connecting cyclic, full conormal and product-Jacobian observations before this quotient.

The exact source map accompanying this section records hashes and line locators for the complete delivered cyclic note, the complete conormal pasted source, and the earlier proof fragments. The cyclic note's depth-one retraction and Sections 7–8 are proved here at every original depth, with complete invariant dimensions, the scalar-power equality classification, the higher-degree lower bound, and the full-unit complement in (CC.24). Equations (CC.45)–(CC.48) provide the precise operator interpretation for the general weighted trace. Equations (CC.29)–(CC.36) prove the first-jet and invariant rank calculations supplied in the pasted source. The uncompiled status of its embedded Lean text describes that historical delivered draft. The separately recorded PR 21 successor has its own checked CI provenance; it does not make the new written proofs of this chapter kernel-checked. No new Lean execution, zero-location calculation, or claimed arithmetic asymptotic is part

of these proofs. The formulas give the exact finite algebra, retained relation derivatives and trace maps used by the adjoining cyclic metric calculation. The quantitative interpolation metric remains the original one, with its degree cost from (CC.7) and all its cross terms from (CC.49).

31 The cyclic sum source inside the full connection

Keep the original nonempty dagger-stable packet h , its full orders, $g = 2\xi$, $v_h = g/h$, and $k \geq 2$. Use exactly (NN.1), $S = k/2 + iu$, and $m_k(u) = w_h^{*k}(u) > 0$, with $w_h = |a_h|^2$ and $a_h(t) = v_h(1/2 + it)/\sqrt{2\pi}$. The complete cyclic-sum source note is retained in the source appendices. Its annihilator is

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{\ell_{\lambda}}, \quad \ell_{\lambda} = \max_{\sum_i \rho_i = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) \right), \quad C_{\chi} = \mathbb{C}[S]/(\chi). \quad (\text{CM.1})$$

Keep the full algebra $E = E_h^{\otimes k}$, the original unit $U_k = v_h^{\otimes k}$, the injective algebra map $\alpha : C_{\chi} \rightarrow E^{S_k}$, and the separate module injection $\eta = M_{U_k} \alpha$. Their exact construction, retraction, complement, nilpotent orders and trace comparison are proved in the cyclic conormal section and the complete delivered source. In particular the original analytic source is the injective map

$$\begin{aligned} \mathcal{V}_k(P) &= P(D_1 + \cdots + D_k) F_h^{\otimes k}, \\ \mathcal{U}_k \mathcal{V}_k(P) &= \Psi_{c_P}, \quad c_P(u) = P(k/2 + iu), \\ J^{(k)} \mathcal{V}_k(P) &= \eta[P]_{\chi}. \end{aligned} \quad (\text{CM.2})$$

Here $\mathcal{U}_k$ is the original Mellin–Fourier isometry and the change to the original sum and relative coordinates. Its norm is $\int |c_P|^2 m_k du$, with its original mass μ_h^k .

31.1 Complete polynomial moments and the original finite metric

Put $\mu_j = \int t^j |a_h|^2 dt$, $\nu_j = \int t^j |a'_h|^2 dt$, and $\theta_j = \int t^j \overline{a_h} a'_h dt$. The fixed arithmetic phase makes $\overline{a_h} a'_h = w'_h/2$; integration by parts with the original decay gives

$$\theta_0 = 0, \quad \theta_j = -\frac{j}{2} \mu_{j-1} \quad (j \geq 1). \quad (\text{CM.3})$$

Define the formal series $\mathcal{M}(z) = \sum \mu_j z^j / j!$ and $\mathcal{N}(z) = \sum \nu_j z^j / j!$. The entire finite moment arrays are

$$\begin{aligned} M_{j,k} &= \int u^j m_k(u) du = j! [z^j] \mathcal{M}(z)^k, \\ B_{j,k} &= \int u^j \|\partial_u \Psi(u, \cdot)\|^2 du \\ &= j! [z^j] \left(\frac{1}{k} \mathcal{N}(z) \mathcal{M}(z)^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}(z)^k \right). \end{aligned} \quad (\text{CM.4})$$

For completeness, the multinomial expansion of $(t_1 + \cdots + t_k)^j$ and Fubini give the first identity. The derivative is $k^{-1} \sum_i a'_h(t_i) \prod_{r \neq i} a_h(t_r)$. There are k diagonal terms, each carrying k^{-2} , and $k(k-1)$ distinct-index terms with that same factor. Their formal series are respectively $\mathcal{N} \mathcal{M}^{k-1} / k$ and $(k-1) \Theta(z)^2 \mathcal{M}^{k-2} / k$, where $\Theta(z) = -z \mathcal{M}(z) / 2$ by (CM.3). This proves the second identity with every cross term and the original k^{-2} retained.

In the literal source coordinates $1, S, \dots, S^N$, write $c_j(u) = (k/2 + iu)^j$. Its original amplitude

and derivative Gram matrices are, entry by entry,

$$\begin{aligned} H_{ij}^0 &= \int \bar{c}_i c_j m_k du, \\ H_{ij}^\partial &= \int \bar{c}_i c_j \|\Psi'\|^2 du + \int \bar{c}_i c'_j m_k du \\ &\quad + \frac{1}{2} \int \bar{c}_i c'_j m'_k du + \frac{1}{2} \int \bar{c}'_i c_j m'_k du, \quad c'_j = ij(k/2 + iu)^{j-1} \quad (j \geq 1), \quad c'_0 = 0. \end{aligned} \quad (\text{CM.5})$$

Each integrand coefficient is obtained by multiplying the displayed original polynomials. Integrating its u^r coefficient uses $M_{r,k}$, $B_{r,k}$, or $\int u^r m'_k = -rM_{r-1,k}$, with zero for $r = 0$. This is an explicit finite matrix construction from the original one-factor arrays; it keeps the two separate cross terms. Both forms are positive definite. The amplitude map is injective by $\Psi \neq 0$ almost everywhere. A zero derivative would give an $L^2(du; \mathcal{H})$ function constant in u , hence zero, and the same argument then gives the zero polynomial.

Let p_j be the original monic orthogonal polynomials for m_k , with full norms $\omega_j = \int |p_j(k/2 + iu)|^2 m_k(u) du$, and $b_j = [p_j]_\chi$ in the fixed remainder coordinates. For $N \geq \deg \chi - 1$, put

$$K_N = \sum_{j=0}^N b_j b_j^* / \omega_j, \quad G_N = K_N^{-1}, \quad T_N u = \sum_{j=0}^N \text{coeff}(p_j) \frac{b_j^* G_N u}{\omega_j}. \quad (\text{CM.6})$$

Thus T_N is the actual coefficient matrix of the canonical cyclic representative, and its two complete forms are

$$\mathcal{R}_N = \mathcal{V}_k T_N, \quad \mathcal{R}_N^* \mathcal{R}_N = T_N^* H^0 T_N = G_N, \quad (\partial_u \mathcal{U}_k \mathcal{R}_N)^* (\partial_u \mathcal{U}_k \mathcal{R}_N) = T_N^* H^\partial T_N. \quad (\text{CM.7})$$

The first equality on forms follows by multiplying (CM.6) in the orthogonal polynomial coordinates; the second is the definition of the original derivative Gram (CM.5). The latter has not been replaced by the constant-column Fisher estimate.

31.2 A full real-exponential family of the same information identity

The original-amplitude density theorem proves $|a_h^{(j)}(t)| \leq C_{h,j,b} e^{-b|t|}$ for every $b < \pi/4$. It follows, by choosing $2b > |\theta|$, that for each real $|\theta| < \pi/2$ the following are finite actual integrals:

$$M(\theta) = \int e^{\theta t} |a_h(t)|^2 dt > 0, \quad N(\theta) = \int e^{\theta t} |a'_h(t)|^2 dt. \quad (\text{CM.8})$$

The same estimates with a slightly larger real exponential on compact subsets prove differentiation under the integral for every polynomial power. Hence M, N are holomorphic for $|\Re z| < \pi/2$, and their Taylor coefficients at zero are the original arrays in (CM.4). Using Fubini with this actual exponential, rather than only formal coefficients, gives

$$\begin{aligned} \int e^{\theta u} m_k(u) du &= M(\theta)^k, \\ B_k(\theta) &:= \int e^{\theta u} \|\Psi'(u)\|^2 du = \frac{1}{k} N(\theta) M(\theta)^{k-1} + \frac{k-1}{4k} \theta^2 M(\theta)^k. \end{aligned} \quad (\text{CM.9})$$

In this calculation $\int e^{\theta t} \bar{a}_h a'_h = -\theta M(\theta)/2$. The diagonal and distinct-index product terms are therefore exactly those used in (CM.4).

The exact Hilbert-space multiplier is

$$L^2(e^{\theta t} dt) \xrightarrow{f \mapsto e^{\theta t/2} f} L^2(dt), \quad f \mapsto e^{\theta t/2} f, \quad \text{inverse } q \mapsto e^{-\theta t/2} q. \quad (\text{CM.10})$$

It is unitary by direct integration. For the original amplitude set $a_\theta = e^{\theta t/2} a_h$. Its mass and derivative energy are

$$\|a_\theta\|^2 = M(\theta), \quad \|a'_\theta\|^2 = N(\theta) - \frac{\theta^2}{4} M(\theta) > 0. \quad (\text{CM.11})$$

Expansion of a'_θ , followed by the cross integral above, proves the equality. If its derivative norm were zero, the nonzero L^2 function a_θ would be constant, which is impossible; this proves strict positivity. The product amplitude is exactly $\Psi_\theta = e^{\theta u/2} \Psi$, its density is $m_{k,\theta} = e^{\theta u} m_k$, and its relative line projection is the same Π_0 , since the multiplier is a nonzero scalar on each fibre. For $n_k = (I - \Pi_0) \Psi'$, its normal component is consequently $n_{k,\theta} = e^{\theta u/2} n_k$. Applying the complete product-rule proof of (FC.4) to this specific amplitude, including its mass, proves

$$\int e^{\theta u} \frac{(m'_k + \theta m_k)^2}{m_k} du + 4 \int e^{\theta u} \|n_k(u)\|^2 du = \frac{4}{k} M(\theta)^{k-1} \left(N(\theta) - \frac{\theta^2}{4} M(\theta) \right). \quad (\text{CM.12})$$

All square quotients are taken away from the already stated fibre zero sets; $m_k > 0$ itself everywhere. The multiplier keeps the same zeros of the original one-factor amplitude. Its normal component is nonzero on the same fibres as that in (FC.21)–(FC.27), so the second integral is strictly positive. No mass has been assigned the value one. The arithmetic maps (CM.1)–(CM.2) still use the original U_k ; (CM.10) specifies the exact weighted Hilbert comparison used solely to evaluate the displayed original exponential integrals.

31.3 All cyclic normal energy is retained under frame enlargement

The cyclic source uses the single relative coefficient 1 at every scalar degree N . Its line projection is Π_0 . The complete relative frame of total degree m has projection Π_m . For any fixed cyclic polynomial P , $m \geq \deg P$, the actual derivative amplitude has the three mutually orthogonal components

$$\begin{aligned} d(P) &= \partial_u(\Psi c_P) = t_0(P) + b_m(P) + r_m(P), \\ t_0(P) &= \Pi_0 d(P) = \Psi \left(i c_{P'} + \frac{m'_k}{2m_k} c_P \right), \\ b_m(P) &= (\Pi_m - \Pi_0) d(P), \quad r_m(P) = (I - \Pi_m) d(P), \\ n_k c_P &= b_m(P) + r_m(P). \end{aligned} \quad (\text{CM.13})$$

Here $c_{P'}(u) = P'(k/2 + iu)$. The derivative of the cyclic coefficient is $i c_{P'}$; multiplication by c_P commutes with the fibre projections. Hence projecting its derivative gives precisely the first two formulas, and subtracting proves the last one. The nesting $\Pi_0 \leq \Pi_m$ proves orthogonality of all three components. With the original global norms this gives

$$\|n_k c_P\|^2 = \|b_m(P)\|^2 + \|r_m(P)\|^2, \quad \|d(P)\|^2 = \|t_0(P)\|^2 + \|b_m(P)\|^2 + \|r_m(P)\|^2. \quad (\text{CM.14})$$

The proved density limit (NN.4) now yields the exact convergence

$$r_m(P) \rightarrow 0, \quad b_m(P) \rightarrow n_k c_P \quad \text{in } \mathcal{K}. \quad (\text{CM.15})$$

Thus the complete fixed cyclic normal vector is the limit of the added relative tangential vectors, with its entire positive norm retained by (CM.14). This is the explicit map between the constant relative frame of the new cyclic construction and the enlarging full relative frames. It makes no substitution of one frame for another. For a fixed finite collection of original cyclic polynomials, the corresponding restricted Gram matrices converge as well: each entry is a pairing of finitely many convergent $\mathcal{K}$ vectors. No statement for a sequence of changing canonical polynomials is inferred from this fixed-source convergence.

31.4 The canonical metric difference has its actual conormal lift

For $N \geq \deg \chi - 1$, let R_N^{full} be the original full-source minimum on the whole arithmetic quotient E , and let $\mathcal{R}_N$ be (CM.7). This full minimum exists at this degree: $\deg \chi - 1 \geq k(d - 1)$. Here is a direct proof with the original monic polynomial retained. For every $0 \leq q \leq k(d - 1)$, choose $0 \leq \alpha_i < d$ with $\sum_i \alpha_i = q$. The monomial $\mathbf{s}^\alpha$ in S^q has coefficient $q! / \prod_i \alpha_i! \neq 0$. It is already a standard remainder monomial. Every term with some exponent at least d , when reduced by the original monic $h(s_i)$, strictly lowers total degree. Thus the remainder of S^q has total degree exactly q , with that same nonzero coefficient. Remainders of lower powers have strictly lower degree. A nonzero polynomial in S of degree at most $k(d - 1)$ therefore cannot annihilate S in the full algebra. Since χ is its exact annihilator, the asserted degree inequality follows. Every monic multivariate remainder now belongs to the original full source at degree N , so the full jet map is onto. Set

$$\Delta_N = \mathcal{R}_N - R_N^{\text{full}}\eta, \quad E_N = \Delta_N^* \Delta_N \succeq 0. \quad (\text{CM.16})$$

Both representatives have the same full original jets. Their difference is therefore an actual polynomial-source boundary. The full minimum is orthogonal to that boundary, proving the complete Gram and control identities

$$\begin{aligned} G_N &= \eta^* G_N^{\text{full}} \eta + E_N, \\ W_N - \eta^* W_N^{\text{full}} \eta &= A_\chi^* E_N + E_N A_\chi - k E_N, \quad A_\chi = M_S. \end{aligned} \quad (\text{CM.17})$$

The second equality follows by substituting the first and using $A_k \eta = \eta A_\chi$. For an actual eigenvector $A_\chi x = \lambda x$, its exact contribution is

$$x^*(W_N - \eta^* W_N^{\text{full}} \eta)x = (2\Re \lambda - k) \|\Delta_N x\|^2. \quad (\text{CM.18})$$

This is the full effect of the positive Gram difference on that original eigenline, including either sign of $2\Re \lambda - k$.

Write $\Delta_N x = \mathcal{T}_h^{(k)} Q_x$ in the original polynomial coordinates. Since its full jet is zero and U_k is a unit, $Q_x \in I = (h(s_1), \dots, h(s_k))$. Fixed-order monic division supplies $Q_x = \sum_i h(s_i) Q_{i,x}$, linearly in x . Denote the fixed conormal map of this ideal by $\partial_I : I/I^2 \rightarrow E$, $[Q] \mapsto [k^{-1} \sum_i \partial_{s_i} Q]_I$. It is independent of the source cutoff N . The retained conormal map and its derivative observation are

$$\begin{aligned} \kappa_N : C_\chi &\longrightarrow I/I^2, \quad x \longmapsto [Q_x]_{I^2}, \\ \partial_I \kappa_N(x) &= \frac{1}{k} \sum_i h'(s_i) [Q_{i,x}]_I, \\ J^{(k)} \mathcal{L}_k \Delta_N x &= U_k \partial_I \kappa_N(x), \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i. \end{aligned} \quad (\text{CM.19})$$

The first expression is independent of the displayed division because it is the actual class of the uniquely specified polynomial Q_x . The derivative of I^2 lies in I ; its product rule gives the second expression and proves independence of every conormal coordinate choice. For the last equality differentiate the full $U Q_x$: the term $(\partial_S^{\text{rel}} U) Q_x$ lies in I , whereas the other term is the stated U_k times the complete conormal row. Mellin differentiation supplies exactly $\mathcal{L}_k$, as in (FC.34). Thus the actual canonical-minimum difference has an explicit map into the original conormal layer and then to its full arithmetic derivative, with every selected order and unit retained.

The original quotient sends the supported class of Q_x to its supported zero. The conormal lift in (CM.19) is retained before that quotient, so its possibly nonzero differentiated observation is obtained from the stated original relation, not from differentiating the scalar supported zero. For $Q_x \in I^2$ the conormal class and this derivative vanish; for general $Q_x \in I$ their full kernels are related exactly by $\ker \kappa_N = \{x : Q_x \in I^2\}$ and $\ker(J^{(k)} \mathcal{L}_k \Delta_N) = \kappa_N^{-1}(\ker \partial_I)$, since U_k is invertible. The original conormal kernel is computed in the conormal section. The external τ maps to τ along every split lift.

32 Exterior trace equality, the spectral projector, and quantitative coupling

Date: 2026-09-12. This is an independent finite-dimensional proof for the exterior-amplification integration. All norms and adjoints below use the original Hermitian form. No arithmetic asymptotic estimate, Lean execution, or external-source verification is asserted.

32.1 1. Original objects and exact domains

Let $\mathcal{C}$ be a finite-dimensional complex vector space with the positive-definite Hermitian form

$$\langle x, y \rangle_G = x^* G y$$

(conjugate-linear in the first variable). Let $k \in \mathbb{R}$, let $A \in \text{End}_{\mathbb{C}}(\mathcal{C})$, and set

$$A^\sharp = G^{-1} A^* G, \quad H = A^\sharp + A - k I_{\mathcal{C}}.$$

Suppose the actual operator H has rank at most two and trace zero. Its eigenvalues, with zero eigenvalues retained, are $+\epsilon, -\epsilon, 0, \dots, 0$ for a uniquely determined $\epsilon \geq 0$.

Let $V = \mathcal{C}_{>}$ be the direct sum of the **full generalized eigenspaces** of A for the eigenvalues satisfying $\text{Re } \lambda > k/2$. Let $W = V^{\perp_G}$, and let $P : \mathcal{C} \rightarrow \mathcal{C}$ be the G -orthogonal projection with image V and kernel W . Let Q be the spectral projector with image V and kernel the direct sum of all the other full generalized eigenspaces of A .

For specificity, this last projector is an exact polynomial in the original operator. Factor the minimal polynomial as $m_A = m_+ m_-$, where m_+ contains exactly the selected eigenvalues with their original minimal-polynomial exponents and m_- contains all the remaining eigenvalues with their original exponents. The two factors are coprime. Choose polynomials a, b with

$$a(z)m_+(z) + b(z)m_-(z) = 1.$$

Then

$$Q = b(A)m_-(A).$$

Indeed, the polynomial $b m_-$ is congruent to one modulo m_+ and to zero modulo m_- ; applying it to each full generalized eigenspace gives the asserted identity and zero maps. This also proves $Q^2 = Q$, $AQ = QA$, and independence from the choice of the Bezout polynomials.

All block maps below use the original orthogonal direct sum $\mathcal{C} = V \oplus W$, without a change in either restricted Hermitian form. Since $A(V) \subseteq V$, define

$$B = A|_V : V \rightarrow V, \quad D = (1 - P)A|_W : W \rightarrow W, \quad T = PA|_W : W \rightarrow V.$$

Then the exact block identities are

$$A = \begin{pmatrix} B & T \\ 0 & D \end{pmatrix}, \quad A^\sharp = \begin{pmatrix} B^\sharp & 0 \\ T^\sharp & D^\sharp \end{pmatrix}, \quad H = \begin{pmatrix} B^\sharp + B - k I_V & T \\ T^\sharp & D^\sharp + D - k I_W \end{pmatrix}. \quad (\text{E1})$$

In particular, as maps $W \rightarrow V$,

$$PH|_W = PA|_W = T. \quad (\text{E2})$$

This equality includes the proof of the vanishing term: $PA^\sharp|_W = ((1 - P)AP)^\sharp|_W = 0$, because $(1 - P)AP = 0$.

Write

$$L = \text{Tr}_{\mathcal{C}}(PH) = \text{Tr}_V(B^\sharp + B - k I_V) = \sum_{\text{Re } \lambda > k/2} m_A^{\text{alg}}(\lambda)(2 \text{Re } \lambda - k), \quad (\text{E3})$$

where $m_A^{\text{alg}}(\lambda)$ is the original algebraic multiplicity of λ in A . The final equality follows by taking the trace of B , whose characteristic polynomial consists exactly of the selected full generalized blocks, and by $\text{Tr } B^\sharp = \overline{\text{Tr } B}$. Thus $L > 0$ exactly when $V \neq 0$.

If $\epsilon = 0$, then $H = 0$. For every eigenvector $x \neq 0$ of A ,

$$0 = \langle x, Hx \rangle_G = (2 \operatorname{Re} \lambda - k) \langle x, x \rangle_G,$$

so all eigenvalues have real part $k/2$, and $V = 0$. Hence the case $L > 0$ automatically has $\epsilon > 0$. Also $V \neq C$ in that case, since $P = I$ would give $L = \text{Tr } H = 0$.

32.2 2. Exact trace maximum and equality theorem

Theorem E4. With these definitions and $L > 0$,

$$0 < L \leq \epsilon,$$

and the following five statements are equivalent:

1. $L = \epsilon$.
2. $P = Q$.
3. V reduces A : both V and $V^\perp_G$ are A -invariant.
4. $PA = AP$.
5. $T = 0$.

Moreover, each statement is equivalent to the positive H -eigenline being contained in V and the negative H -eigenline being contained in W .

Proof. Choose vectors u_+, u_- on those two eigenlines with

$$\langle u_+, u_+ \rangle_G = \langle u_-, u_- \rangle_G = 1, \quad \langle u_+, u_- \rangle_G = 0.$$

This records the eigenline projectors using the existing form and does not replace that form. The full operator, including its zero eigenspace, is

$$Hx = \epsilon u_+ \langle u_+, x \rangle_G - \epsilon u_- \langle u_-, x \rangle_G. \quad (\text{E4a})$$

Put $a = \|Pu_+\|_G^2$, $b = \|Pu_-\|_G^2$. Then

$$L = \epsilon(a - b), \quad 0 \leq a, b \leq 1. \quad (\text{E4b})$$

This proves $L \leq \epsilon$. Equality holds precisely when $a = 1$ and $b = 0$. Orthogonality of P identifies these equalities with $u_+ \in V$ and $u_- \in W$. In that case (E4a) gives $PH(1 - P) = 0$, and (E2) gives $T = 0$.

By (E1), $T = 0$, reduction of A , and $PA = AP$ are equivalent. If $P = Q$, the identity $AQ = QA$ already proved implies $PA = AP$. Conversely, if $T = 0$, the decomposition in (E1) is a block diagonal decomposition of the original A . The spectrum of B , with multiplicities, is the selected spectrum, while the characteristic polynomial of D contains exactly the remaining factors: the full characteristic polynomial is the product of the two block characteristic polynomials. Applying the defining projector polynomial to these blocks gives $Q = I_V \oplus 0_W = P$.

Finally suppose $T = 0$. Equation (E1) shows that P commutes with H . The two nonzero eigenspaces of H are one-dimensional. The restriction of an idempotent commuting with H to either such eigenspace is therefore either identity or zero. It follows that $a, b \in \{0, 1\}$. Since $\epsilon(a - b) = L > 0$, necessarily $a = 1, b = 0$, and hence $L = \epsilon$. This proves every stated implication. $\square$

The hypothesis $L > 0$ matters for the converse from reduction to $L = \epsilon$. For example, an empty selected subspace has $P = Q = 0$ and reduces A , while $L = 0$ may hold with $\epsilon > 0$. The theorem concerns exactly the nonempty right-half-plane subspace that appears in (E3).

32.3 3. Sharp quantitative trace slack

For a map $F : W \rightarrow V$, set $\|F\|_{\text{HS},G}^2 = \text{Tr}_W(F^\sharp F)$. This is the Hilbert-Schmidt norm with the original restricted metrics on both its domain and codomain.

Keep u_+, u_- as above and set

$$c = \langle Pu_+, Pu_- \rangle_G, \quad M = \begin{pmatrix} a & c \\ \bar{c} & b \end{pmatrix}. \quad (\text{E5})$$

The matrix M is the compression of the original orthogonal projection P to $E = \text{span}(u_+, u_-)$, expressed in that specified eigenbasis. For $z = (z_+, z_-)$,

$$z^* M z = \|P(z_+ u_+ + z_- u_-)\|_G^2, \quad z^* (I_2 - M) z = \|(1 - P)(z_+ u_+ + z_- u_-)\|_G^2. \quad (\text{E6})$$

Both matrices are positive semidefinite. In particular,

$$ab - |c|^2 \geq 0, \quad (1 - a)(1 - b) - |c|^2 \geq 0. \quad (\text{E7})$$

Theorem E8. The exact identity is

$$\epsilon^2 - L^2 - \|T\|_{\text{HS},G}^2 = \epsilon^2 (\det M + \det(I_2 - M)). \quad (\text{E8})$$

Consequently,

$$\epsilon^2 - L^2 \geq \|T\|_{\text{HS},G}^2, \quad \epsilon - L \geq \frac{\|T\|_{\text{HS},G}^2}{\epsilon + L}. \quad (\text{E9})$$

Proof. By (E2) and selfadjointness of H ,

$$\begin{aligned} \|T\|_{\text{HS},G}^2 &= \text{Tr}_V(PH(1 - P)HP|_V) \\ &= \text{Tr}_\mathcal{C}(PH^2P) - \text{Tr}_\mathcal{C}((PHP)^2). \end{aligned} \quad (\text{E10})$$

Formula (E4a) gives $H^2x = \epsilon^2 u_+ \langle u_+, x \rangle_G + \epsilon^2 u_- \langle u_-, x \rangle_G$. Thus the first trace on the right is $\epsilon^2(a + b)$. The compressed operator is the difference of the two rank-one maps generated by Pu_+ and Pu_- , with coefficient ϵ . Squaring these maps and taking their traces gives

$$\text{Tr}_\mathcal{C}((PHP)^2) = \epsilon^2(a^2 + b^2 - 2|c|^2).$$

For clarity, the cross trace used here is $\text{Tr}((v \langle v, \cdot \rangle_G)(w \langle w, \cdot \rangle_G)) = |\langle v, w \rangle_G|^2$, directly by composition of the two rank-one maps. Therefore

$$\|T\|_{\text{HS},G}^2 = \epsilon^2(a + b - a^2 - b^2 + 2|c|^2). \quad (\text{E11})$$

Subtract this expression and $L^2 = \epsilon^2(a - b)^2$ from ϵ^2 . The result is

$$\epsilon^2(1 - a - b + 2ab - 2|c|^2) = \epsilon^2((ab - |c|^2) + ((1 - a)(1 - b) - |c|^2)),$$

which is exactly (E8). Inequalities (E7) prove its nonnegativity, and division by the positive number $\epsilon + L$ proves the second inequality in (E9). $\square$

Equality in the first inequality of (E9) holds precisely when M has eigenvalues zero and one. Indeed, its eigenvalues μ_1, μ_2 lie in $[0, 1]$, and the two nonnegative determinants both vanish precisely when $\mu_1 \mu_2 = 0$ and $(1 - \mu_1)(1 - \mu_2) = 0$. This is equivalent to the two eigenvalues being zero and one. It also means that E reduces P and that $P|_E$ has rank one: a compression eigenvector with eigenvalue zero satisfies $\|Px\|^2 = 0$, while a compression eigenvector with eigenvalue one satisfies $\|(1 - P)x\|^2 = 0$, by (E6). The reverse implication follows by restricting a rank-one orthogonal projection on E .

The same inequality can be expressed by exact commutators on $\mathcal{C}$:

$$[P, A] = \begin{pmatrix} 0 & T \\ 0 & 0 \end{pmatrix}, \quad [P, H] = \begin{pmatrix} 0 & T \\ -T^\sharp & 0 \end{pmatrix},$$

so

$$\epsilon^2 - L^2 \geq \|[P, A]\|_{\text{HS},G}^2 = \frac{1}{2} \|[P, H]\|_{\text{HS},G}^2. \quad (\text{E12})$$

32.4 4. Exact morphism from the projector difference to the coupling

Because Q has image V and restricts to identity on V , there is a unique linear map $X : W \rightarrow V$ such that

$$Q = \begin{pmatrix} I_V & X \\ 0 & 0 \end{pmatrix}, \quad Q - P = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}. \quad (\text{E13})$$

The equation $AQ = QA$, with (E1), yields the **exact Sylvester morphism**

$$\mathcal{S}_{B,D} : \text{Hom}_{\mathbb{C}}(W, V) \longrightarrow \text{Hom}_{\mathbb{C}}(W, V), \quad \mathcal{S}_{B,D}(X) = BX - XD = T. \quad (\text{E14})$$

It is an isomorphism. Here is an explicit proof and inverse retaining the original metrics. Set $\beta = \min_{\lambda \in \sigma(B)} \text{Re } \lambda$ and $\delta = \max_{\mu \in \sigma(D)} \text{Re } \mu$. By the full spectral selection, $\beta > k/2 \geq \delta$. The original operator exponentials have polynomial-times-exponential bounds because each nilpotent part in each generalized eigenspace has finite order. More explicitly, with the spectral projectors $\Pi_{\lambda}^B, \Pi_{\mu}^D$ and their nilpotency orders $\nu_{\lambda}^B, \nu_{\mu}^D$, the exact formulas are

$$\begin{aligned} e^{-tB} &= \sum_{\lambda \in \sigma(B)} e^{-t\lambda} \sum_{j=0}^{\nu_{\lambda}^B-1} \frac{(-t)^j}{j!} (B - \lambda I)^j \Pi_{\lambda}^B, \\ e^{tD} &= \sum_{\mu \in \sigma(D)} e^{t\mu} \sum_{\ell=0}^{\nu_{\mu}^D-1} \frac{t^{\ell}}{\ell!} (D - \mu I)^{\ell} \Pi_{\mu}^D. \end{aligned} \quad (\text{E15})$$

These formulas follow by expanding the exponential of scalar plus nilpotent on every generalized eigenspace; every sum in the nilpotent part terminates at the displayed order. Consequently the following integral converges in the original operator norm, for every $Y : W \rightarrow V$:

$$\mathcal{S}_{B,D}^{-1}(Y) = \int_0^{\infty} e^{-tB} Y e^{tD} dt. \quad (\text{E16})$$

Differentiating the integrand and integrating its derivative from zero to infinity proves

$$B \left(\int_0^{\infty} e^{-tB} Y e^{tD} dt \right) - \left(\int_0^{\infty} e^{-tB} Y e^{tD} dt \right) D = Y.$$

There is no upper-end boundary term, by $\beta - \delta > 0$ and the finite polynomial factors in (E15). If $BZ - ZD = 0$, differentiation shows that $e^{-tB} Z e^{tD}$ is constantly Z , while the same decay shows its limit is zero. Thus $Z = 0$, proving uniqueness and the asserted isomorphism.

Equip $\text{Hom}(W, V)$ with its original Hilbert-Schmidt norm, and define its actual positive smallest singular value

$$s_{B,D} = \min_{\|Y\|_{\text{HS},G}=1} \|BY - YD\|_{\text{HS},G} > 0. \quad (\text{E17})$$

The minimum exists because the unit sphere is compact, and is positive because (E14) is injective. The identity (E13) gives $\|X\|_{\text{HS},G} = \|Q - P\|_{\text{HS},G}$. Combining (E9), (E14), and (E17) proves

$$\epsilon^2 - L^2 \geq \|T\|_{\text{HS},G}^2 \geq s_{B,D}^2 \|Q - P\|_{\text{HS},G}^2. \quad (\text{E18})$$

Thus the difference between the algebraic spectral projector and the orthogonal projection is related to the trace defect by a proved, invertible, typed map.

An explicit bound on the inverse is also available with all original operator norms retained. Write

$$\kappa_{B,D} = \int_0^{\infty} \|e^{-tB}\|_{\text{op},G} \|e^{tD}\|_{\text{op},G} dt < \infty.$$

The ideal inequality $\|RYS\|_{\text{HS}} \leq \|R\|_{\text{op}}\|Y\|_{\text{HS}}\|S\|_{\text{op}}$ follows by diagonalizing the positive operators $R^\sharp R$ and $SS^\sharp$ in their respective existing Hermitian spaces and bounding their largest eigenvalues. Applying it under (E16) yields

$$\|Q - P\|_{\text{HS},G} \leq \kappa_{B,D}\|T\|_{\text{HS},G} \leq \kappa_{B,D}\sqrt{\epsilon^2 - L^2}. \quad (\text{E19})$$

If desired, (E15) gives the fully specified finite upper bound

$$\kappa_{B,D} \leq \sum_{\lambda,\mu} \sum_{j=0}^{\nu_\lambda^B-1} \sum_{\ell=0}^{\nu_\mu^D-1} \frac{(j+\ell)! \|(B-\lambda I)^j \Pi_\lambda^B\|_{\text{op},G} \|(D-\mu I)^\ell \Pi_\mu^D\|_{\text{op},G}}{j! \ell! (\text{Re } \lambda - \text{Re } \mu)^{j+\ell+1}}. \quad (\text{E20})$$

The scalar integral used is $\int_0^\infty t^n e^{-at} dt = n!/a^{n+1}$ for $a > 0$, proved by the substitution $v = at$ and n integrations by parts beginning with $\int_0^\infty e^{-v} dv = 1$. Every denominator in (E20) is positive. No bound on nonnormal operator norms has been replaced by a bound on eigenvalue real parts.

32.5 5. Sharp original two-dimensional family

Fix $k \in \mathbb{R}$, $\ell > 0$, and $t \in \mathbb{C}$. On $\mathbb{C}^2$ with its specified form $G = I_2$, define

$$A = \begin{pmatrix} k/2 + \ell/2 & t \\ 0 & k/2 - \ell/2 \end{pmatrix}, \quad H = A^* + A - kI_2 = \begin{pmatrix} \ell & t \\ \bar{t} & -\ell \end{pmatrix}. \quad (\text{E21})$$

Here $H^2 = (\ell^2 + |t|^2)I_2$, so $\epsilon = \sqrt{\ell^2 + |t|^2}$. The selected full generalized spectral subspace is exactly $V = \mathbb{C}(1, 0)$, since the two eigenvalues $k/2 \pm \ell/2$ are distinct. The original orthogonal projection is $P = \text{diag}(1, 0)$. Hence

$$L = \ell, \quad T = t : \mathbb{C} \rightarrow \mathbb{C}, \quad Q = \begin{pmatrix} 1 & t/\ell \\ 0 & 0 \end{pmatrix}, \quad s_{B,D} = \ell. \quad (\text{E22})$$

The formula for Q is also obtained directly from the polynomial $(A - (k/2 - \ell/2)I_2)/\ell$; thus its stated kernel is the full other eigenspace, with no orthogonality assumption. Direct calculation gives

$$\epsilon^2 - L^2 = |t|^2 = \|T\|_{\text{HS},G}^2 = s_{B,D}^2 \|Q - P\|_{\text{HS},G}^2. \quad (\text{E23})$$

Both quantitative constants in (E18) are attained by this family, including every $t \neq 0$. The equality theorem gives $L = \epsilon$ precisely when $t = 0$, exactly when $P = Q$ and the selected subspace reduces A .

32.6 6. Critical generalized blocks at the trace maximum

The equality theorem also constrains every full critical generalized block, including its original nilpotent structure.

Theorem E24. Suppose $L > 0$ and $L = \epsilon$. For every eigenvalue λ of the original operator A with $\text{Re } \lambda = k/2$, let

$$Z_\lambda = \ker(A - \lambda I)^{\nu_\lambda}$$

be its full generalized eigenspace, where ν_λ is its original exponent in the minimal polynomial. Then

$$HZ_\lambda = 0, \quad A|_{Z_\lambda} = \lambda I_{Z_\lambda}, \quad A^\sharp|_{Z_\lambda} = (k - \lambda)I_{Z_\lambda} = \bar{\lambda}I_{Z_\lambda}. \quad (\text{E24})$$

In particular, Z_λ reduces A , its spectral projector equals its G -orthogonal projection, and the original eigenvalue λ is semisimple. Consequently, the presence of even one critical-line Jordan chain of length greater than one implies the strict inequality

$$\epsilon > L \quad (\text{E25})$$

whenever $L > 0$.

Proof. At equality, Theorem E4 gives $P = Q$, $u_+ \in V$, and $u_- \in W$. Hence the restriction of the original Hermitian defect to the original orthogonal complement is

$$H|_W : w \mapsto -\epsilon u_- \langle u_-, w \rangle_G. \quad (\text{E26})$$

It is negative semidefinite and has rank one. Every critical generalized eigenspace is in $\ker Q = W$, by the defining polynomial for Q . This conclusion uses the proved equality $P = Q$; it makes no claim that the original unselected spectral subspace was orthogonal before that equality was proved.

Fix such a λ , and let R_λ be the original G -orthogonal projection onto Z_λ . The subspace Z_λ is A -invariant. With $A_\lambda = A|_{Z_\lambda}$, its compressed Hermitian form is therefore

$$R_\lambda H|_{Z_\lambda} = A_\lambda^\sharp + A_\lambda - kI_{Z_\lambda}. \quad (\text{E27})$$

The adjoint on the right uses the original form restricted to Z_λ ; (E27) follows from the defining adjoint identity for two vectors in that subspace. The trace is

$$\text{Tr}_{Z_\lambda}(R_\lambda H|_{Z_\lambda}) = 2 \text{Re Tr } A_\lambda - k \dim Z_\lambda = (2 \text{Re } \lambda - k) \dim Z_\lambda = 0. \quad (\text{E28})$$

Here $A_\lambda = \lambda I + N_\lambda$ with N_λ nilpotent, so $\text{Tr } A_\lambda = \lambda \dim Z_\lambda$. To justify the nilpotent trace directly, choose a basis compatible with the filtration $\ker N_\lambda^j$; the map N_λ lowers that filtration and its matrix has zero diagonal.

Choose any G -orthonormal basis $z_1, \dots, z_r$ of Z_λ , solely to evaluate the trace in its existing form. Equations (E26) and (E28) give

$$0 = \sum_{j=1}^r \langle z_j, H z_j \rangle_G = -\epsilon \sum_{j=1}^r |\langle u_-, z_j \rangle_G|^2. \quad (\text{E29})$$

Since $\epsilon > 0$, every displayed square is zero. Formula (E26) now gives $H z_j = 0$ for every basis vector, and hence $H Z_\lambda = 0$.

The defining equation $H = A^\sharp + A - kI$ implies, for every $z \in Z_\lambda$,

$$A^\sharp z = kz - Az \in Z_\lambda. \quad (\text{E30})$$

Thus Z_λ is invariant under both A and $A^\sharp$. Its orthogonal complement is A -invariant: if $w \perp_G Z_\lambda$ and $z \in Z_\lambda$, then $\langle z, Aw \rangle_G = \langle A^\sharp z, w \rangle_G = 0$. It is also $A^\sharp$ -invariant by the corresponding identity with A and $A^\sharp$ exchanged. This proves the asserted reduction and shows that the restricted adjoint in (E27) is the actual restriction of $A^\sharp$.

On Z_λ put $N = A_\lambda - \lambda I$. It is nilpotent by the definition of this full generalized eigenspace, and (E30) gives

$$N^\sharp = kI - A_\lambda - \bar{\lambda}I = \lambda I - A_\lambda = -N, \quad (\text{E31})$$

because $\lambda + \bar{\lambda} = k$. A nilpotent skew-adjoint map on a positive-definite Hermitian space is zero, as the following calculation proves. If $N \neq 0$, take its nilpotency order $r \geq 2$, choose v with $N^{r-1}v \neq 0$, and set $y = N^{r-1}v$, $z = N^{r-2}v$. Then $Nz = y \neq 0$ and $Ny = 0$, whereas

$$\|y\|_G^2 = \langle Nz, y \rangle_G = \langle z, N^\sharp y \rangle_G = -\langle z, Ny \rangle_G = 0, \quad (\text{E32})$$

contradicting positive definiteness. Consequently $N = 0$, proving (E24).

Because Z_λ reduces A , the original orthogonal direct sum $\mathcal{C} = Z_\lambda \oplus Z_\lambda^{\perp_G}$ gives a block diagonal matrix for A . The complement contains exactly the other generalized eigenspaces: its characteristic polynomial is the quotient of the full characteristic polynomial by that of the full λ -generalized block, so it has no eigenvalue λ . The polynomial spectral projector for λ is therefore identity on Z_λ and zero on its orthogonal complement. It equals R_λ , as claimed. Finally, (E4) already proves $L \leq \epsilon$; (E24)

rules out equality whenever a critical Jordan chain of length greater than one is present. This gives (E25). $\square$

Distinct critical eigenspaces are also mutually orthogonal in the original form. For $x \in Z_\lambda, y \in Z_\mu$, (E24) gives

$$\bar{\lambda}\langle x, y \rangle_G = \langle Ax, y \rangle_G = \langle x, A^\sharp y \rangle_G = \bar{\mu}\langle x, y \rangle_G.$$

If $\lambda \neq \mu$, the inner product vanishes. Therefore the full critical subspace $Z = \bigoplus_{\operatorname{Re} \lambda = k/2} Z_\lambda$ is an original orthogonal sum, reduces A , and has the exact diagonal action $\sum_\lambda \lambda R_\lambda$, with $H|_Z = 0$. The shifted operator $A|_Z - (k/2)I_Z$ is skew-adjoint. This describes the full critical block and its projectors, retaining all its original eigenvalues and multiplicities.

32.7 The original quartet, its critical blocks, and a canonical equality example

Apply the preceding maps to the actual cyclic module and canonical Gram G_N of the complete exterior source note. For the exact packet $h(s) = \prod_{\varepsilon, \eta = \pm 1} (s - (\frac{1}{2} + \varepsilon\delta + i\eta\gamma))^m$, $\delta, \gamma > 0$, its distinct sum roots and full lengths are

$$\lambda_{a,b} = k/2 + \delta(2a - k) + i\gamma(2b - k), \quad 0 \leq a, b \leq k, \quad \ell_k = 1 + k(m - 1), \quad q = \ell_k(k + 1)^2. \quad (\text{E33})$$

Indeed the four occupation counts $(t, a - t, b - t, k - a - b + t)$ exist for every integer $\max(0, a + b - k) \leq t \leq \min(a, b)$; conversely each tuple has these two margins. Equality of sums forces both margins to agree because δ, γ are nonzero real numbers. The top surviving local monomial has degree $k(m - 1)$ and coefficient $(k(m - 1))! / ((m - 1)!)^k$, proving the stated full nilpotent length. Summing positive real defects over these actual cyclic blocks gives

$$L = 2\delta\ell_k(k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}^{\text{cyc}}, \quad N \geq q - 1. \quad (\text{E34})$$

For $k = 2n$ the sum of $2a - k$ over $a > k/2$ is $n(n + 1)$; for $k = 2n + 1$ it is $(n + 1)^2$, proving the floor and its constant. If k is even and $m > 1$, each $a = k/2$ block is critical and has length $\ell_k > 1$. The full cyclic local algebra is a single Jordan block in powers of $S - \lambda_{a,b}$, so (E24)–(E25) prove the strict inequality $\epsilon_{h,k,N}^{\text{cyc}} > L$ at every admitted N . For a larger packet containing this quartet, the same sums survive as CRT factors and their maximal nilpotent lengths are at least ℓ_k . Thus (E34) remains a lower bound; its admission threshold is the larger packet's actual $N \geq \deg \chi_{h,k} - 1$.

Equality can occur for a specified positive canonical moment measure. Here is a complete calibration, with no claim about zeta zeros. Take $k = 1, \mu, \sigma > 0$ and

$$d\nu(u) = \frac{\mu}{\sqrt{2\pi\sigma}} e^{-u^2/(2\sigma^2)} du, \quad S = \frac{1}{2} + iu, \quad \chi = (S - \frac{1}{2})^2 - \sigma^2, \quad N = 1. \quad (\text{E35})$$

The Gaussian integral and one integration by parts give mass μ , mean zero and second moment $\mu\sigma^2$; odd moments vanish by the original reflection $u \mapsto -u$. Consequently the original monic orthogonal polynomials through degree two are $p_0 = 1, p_1 = S - \frac{1}{2}, p_2 = (S - \frac{1}{2})^2 + \sigma^2$, with $\omega_0 = \mu, \omega_1 = \mu\sigma^2$. In the fixed remainder coordinates $(1, S)$ the column map $B = [b_0, b_1] = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}$ is invertible, and the actual canonical Gram is $G = B^{-*} \operatorname{diag}(\mu, \mu\sigma^2) B^{-1}$. Reduction gives $b_2 = 2\sigma^2 b_0$, whence

$$a = \mu\sigma^2, \quad d = 4\mu\sigma^4, \quad c = 0, \quad \epsilon = 2\sigma = L. \quad (\text{E36})$$

The formula follows directly from $\epsilon^2 = (ad - |c|^2)/\omega_1^2$ and the two simple roots $1/2 \pm \sigma$. All original masses and coordinates are retained. For $0 < \sigma < 1/2$ both roots lie in the open critical strip, but they are declared calibration roots. Thus positive measure and canonical interpolation alone cannot supply universal strictness; (E14)–(E18) identify the exact projector and coupling maps governing equality for each original arithmetic measure.

The full exterior cochain idempotent also has its complementary cochain idempotent. For an idempotent e commuting with d , the maps $x \mapsto (ex, (1-e)x)$ and $(u, v) \mapsto u + v$ are inverse cochain maps, since $e(1-e) = 0$ and both summands are d -stable. They induce the same inverse maps on cycles modulo boundaries. Ordinary traces here are taken on the retained finite arithmetic submodules: the signed tensor permutation preserves the finite packet, so restriction of e and $1-e$ gives finite invariant summands and inclusion followed by projection is the stated idempotent. In a basis of these two summands, a commuting finite operator is block diagonal, proving additivity of its ordinary trace; its top cochain sign is the original $(-1)^{kp}$. The splitting of the entire possibly infinite-dimensional theta cohomology itself supplies these maps but no ordinary finite trace.

32.8 7. Integration scope

Equations (E3), (E4), (E8), (E18), and (E24) may be applied directly whenever the exterior-amplification construction supplies its actual finite Hermitian form, its original multiplication operator, and its proved rank-two trace-zero Hermitian defect. All constants $G, k, \epsilon, L, s_{B,D}$ remain attached to that specific construction and degree. The proof establishes an equality classification, an exact projector morphism, a sharp quantitative slack identity, and strictness in the presence of a retained critical Jordan chain. It supplies no uniform estimate for $s_{B,D}$, $\kappa_{B,D}$, or ϵ as arithmetic parameters change; such an estimate would require a further calculation on those original varying objects.

Independent algebra check: the determinant-slack calculation (E8)–(E11), its equality classification, and the sharp family (E21)–(E23) were separately recomputed by the bounded `determinant_slack_check` agent. That agent then independently checked all steps of (E24)–(E32), including the passage from zero compressed trace to ambient annihilation and the use of full primary spaces in identifying the projectors. Both derivations agreed with the formulas above. The projector conclusion applies to each full primary space, including every Jordan block at its eigenvalue; it asserts nothing about arbitrary coordinate projectors onto individually selected copies of a repeated eigenspace. These are second mathematical derivations, not Lean or numerical certificates.

33 The cyclic control as an exact change of arithmetic interpolation volume

Retain a nonempty reflection-stable packet with every original order, the tensor degree $k \geq 1$, the cyclic algebra $C = \mathbb{C}[S]/(\chi)$ of dimension $q \geq 1$, and its fixed monic remainder coordinates. Let $p_j(S)$ be the original monic orthogonal polynomials on $S = k/2 + iu$ for the measure $m_{h,k}(u) du$, and put

$$b_j = [p_j]_\chi, \quad \omega_j = \int_{\mathbb{R}} |p_j(k/2 + iu)|^2 m_{h,k}(u) du > 0, \quad K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad G_N = K_N^{-1}. \quad (\text{CV.1})$$

Here $N \geq q - 1$; the original $\omega_0 = \mu_h^k$ remains. All determinants below are taken in these same coordinates. Set $K_{-1} = 0$ and

$$D_j = \det K_j, \quad \tilde{K}_N = K_{N-1} + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad (\text{CV.2})$$

$$a = b_N^* G_N b_N, \quad d = b_{N+1}^* G_N b_{N+1}, \quad c = b_N^* G_N b_{N+1}.$$

The matrix $\tilde{K}_N$ records the actual polynomial subspace spanned by $p_0, \dots, p_{N-1}, p_{N+1}$, with its inherited norm: the degree- N orthogonal direction is omitted and the degree- $(N+1)$ direction is retained. It can be singular. No inverse for it is used.

Proposition 33.1 (Exact adjacent-volume identities). *The following equalities hold, including $N = q - 1$:*

$$\begin{aligned} 1 - \frac{D_{N-1}}{D_N} &= \frac{a}{\omega_N}, \\ \frac{D_{N+1}}{D_N} - 1 &= \frac{d}{\omega_{N+1}}, \\ \frac{\tilde{D}_N}{D_N} &= \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d}{\omega_{N+1}}\right) + \frac{|c|^2}{\omega_N \omega_{N+1}}. \end{aligned} \quad (\text{CV.3})$$

Consequently the original rank-two control allowance satisfies

$$\boxed{(\epsilon_{h,k,N}^{\text{cyc}})^2 = \frac{\omega_{N+1}}{\omega_N} \frac{D_{N+1} + D_{N-1} - D_N - \tilde{D}_N}{D_N}}. \quad (\text{CV.4})$$

Equivalently, retaining the entire complex cross term,

$$(\epsilon_{h,k,N}^{\text{cyc}})^2 = \frac{\omega_{N+1}}{\omega_N} \left[\left(1 - \frac{D_{N-1}}{D_N}\right) \left(\frac{D_{N+1}}{D_N} - 1\right) - \frac{|c|^2}{\omega_N \omega_{N+1}} \right]. \quad (\text{CV.5})$$

Proof. The first q monic polynomials give a triangular basis modulo χ , so K_N is positive definite in the stated range. For an invertible K and columns x, y , direct block elimination of $\begin{pmatrix} K & x \\ -y^* & 1 \end{pmatrix}$ in its two possible orders gives $\det(K + xy^*) = \det K(1 + y^* K^{-1}x)$. Applying this with $K = K_N$ and the displayed signed increments gives the first two equations, even if the matrix after subtraction is singular. For completeness the same block elimination with two columns proves $\det(K + UV^*) = \det K \det(I_2 + V^* K^{-1}U)$. Take

$$U = \left(-\frac{b_N}{\sqrt{\omega_N}}, \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}\right), \quad V = \left(\frac{b_N}{\sqrt{\omega_N}}, \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}\right).$$

The positive square roots only specify this auxiliary factorization; the original columns, coordinates and norms in (CV.1) are unchanged. The resulting two-by-two determinant is

$$\det \begin{pmatrix} 1 - a/\omega_N & c/\sqrt{\omega_N \omega_{N+1}} \\ -\bar{c}/\sqrt{\omega_N \omega_{N+1}} & 1 + d/\omega_{N+1} \end{pmatrix},$$

which proves the third equation with its plus sign. Subtract that equation from the sum of the first two determinant ratios as ordered in (CV.4). The resulting value is $(ad - |c|^2)/(\omega_N \omega_{N+1})$. The original control formula is $\epsilon^2 = (ad - |c|^2)/\omega_N^2$, proving (CV.4)–(CV.5). In particular its numerator is nonnegative by the Gram determinant of the two original vectors b_N, b_{N+1} in G_N . For $N = q - 1$, K_{N-1} has rank $q - 1$, so $D_{N-1} = 0$ and $a = \omega_N$; every equation still holds. For $q = 1$ the two-vector Gram determinant is zero, consistently with the trace-zero one-dimensional control. There is no positive-dimensional determinant or Gram inverse to apply when $h = 1$; the coefficient quotient is then zero and its unique maps have the split lift $\{\tau, e\}$. $\square$

33.1 The exact volume map and the reflected quartet bound

Let $\mathcal{P}_N$ be the span of $p_0, \dots, p_N$ with its original integral norm, and let $T_N : \mathcal{P}_N \rightarrow C$ be reduction modulo χ . In the displayed polynomial basis its matrix is $[b_0 \cdots b_N]$, and its source Gram is $\text{diag}(\omega_0, \dots, \omega_N)$. The adjoint is therefore the explicit map

$$T_N^* x = \sum_{j=0}^N p_j \frac{b_j^* x}{\omega_j}, \quad T_N T_N^* = K_N, \quad (\text{CV.6})$$

where the star uses the fixed Euclidean pairing on the written target coordinates, solely to specify this typed adjoint. The canonical right inverse is $T_N^* G_N$, whose original source Gram is G_N . Taking its q th exterior power maps the fixed unscaled coordinate wedge to a vector of squared norm $\det G_N = D_N^{-1}$ in $\bigwedge^q \mathcal{P}_N$. The exact exterior source map is

$$\bigwedge^q \mathcal{V}_{h,k} : \bigwedge^q \mathcal{P}_N \longrightarrow \bigwedge^q (\mathcal{B}^{\widehat{\otimes} k}).$$

It preserves the determinant norm D_N^{-1} , because the original source map preserves every pairwise integral inner product. To pass to the original tensor cochain source, use instead $\mathcal{V}_{h,k}^{\otimes q} \text{Alt}_q$, where $\text{Alt}_q(v_1 \wedge \cdots \wedge v_q) = \sum_{\pi \in S_q} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(q)}$. The exterior quotient satisfies $\text{wed Alt}_q = q! I$: every permuted summand contributes its permutation sign twice. Expanding the tensor inner product gives $q!$ copies of its determinant, so the literal tensor-source squared norm is $q! D_N^{-1}$. Its target is $\mathcal{B}^{\widehat{\otimes} kq}$ in top cochain degree kq . This proves the asserted interpolation-volume interpretation without changing its target metric or source mass. Replacing T_N by its restriction to the omitted-direction subspace gives exactly $\widetilde{K}_N$; the construction needs no assertion that this restricted map is onto.

Now take the packet polynomial to be exactly

$$h(s) = \prod_{\varepsilon, \eta \in \{+1, -1\}} \left(s - \left(\frac{1}{2} + \varepsilon \delta + i \eta \gamma \right) \right)^m$$

for a complete actual nonreal quartet $\frac{1}{2} \pm \delta \pm i\gamma$, $\delta, \gamma > 0$, of common order m , and set $\ell_k = 1 + k(m-1)$. Its cyclic annihilator has $q = \ell_k(k+1)^2$. The complete exterior trace calculation gives $L_k \leq \epsilon$, where

$$L_k = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{CV.7})$$

Substitution in (CV.4), with all quantities nonnegative, proves the exact necessary arithmetic-volume inequality

$$\boxed{D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N \geq \frac{\omega_N}{\omega_{N+1}} D_N L_k^2, \quad N \geq \ell_k(k+1)^2 - 1.} \quad (\text{CV.8})$$

The cross term has been transferred to the actual omitted-direction determinant by (CV.3); it has not been discarded. Thus (CV.4) supplies an exact expression for the still unestimated quantity

$$\frac{\epsilon^2}{k^6} = \frac{\omega_{N+1}}{k^6 \omega_N} \frac{D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N}{D_N}. \quad (\text{CV.9})$$

Every term comes from the specified arithmetic convolution measure and the complete quotient. No decay rate for (CV.9) is asserted here. The formula identifies which signed determinant combination, and which orthogonal-norm ratio at growing degree, the next analytic calculation has to estimate together. For a larger reflection-stable packet containing this quartet, the quartet sums remain present and their full nilpotent lengths are at least ℓ_k . Indeed the same ordered quartet factors embed as CRT summands of the larger tensor algebra, and the largest nilpotent length at a fixed sum is a maximum over all such summands. Their positive trace contribution is therefore at least (CV.7). Inequality (CV.8) remains valid, with every K_j, D_j, ω_j formed from that larger packet and the admission threshold replaced by its actual $N \geq \deg \chi_{h,k} - 1$. No equality for that larger degree is inferred from the quartet subpacket.

33.2 The positive exterior minor sum and its actual source vector

Write $\Delta_N = D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N$. For an increasing finite index set J , let B_J be the matrix whose columns are the original b_j , in that increasing order, and put $\Omega_J = \prod_{j \in J} \omega_j$, with the empty

product equal to one. For every $q \geq 2$ and $N \geq q - 1$,

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i}. \quad (\text{CV.10})$$

For $q = 2$ the index set I is empty and the sum has its single displayed two-column term. For $q = 1$ set this sum to zero; then $\Delta_N = 0$ exactly. Thus the signed combination in (CV.4) is an explicit sum of positive exterior contributions, retaining every original column and squared polynomial norm.

Proof. First take any finite set L of column indices and form $K_L = \sum_{j \in L} b_j b_j^* / \omega_j$. Multilinearity of its q columns expands its determinant as

$$\det K_L = \sum_{(j_1, \dots, j_q) \in L^q} \det[b_{j_1}, \dots, b_{j_q}] \prod_{r=1}^q \frac{\overline{(b_{j_r})_r}}{\omega_{j_r}}.$$

If two indices coincide, the determinant in that summand is zero. For a fixed increasing set $J = \{i_1 < \dots < i_q\}$, the remaining terms are indexed by its permutations. Their sum is

$$\frac{\det B_J}{\Omega_J} \sum_{\sigma \in \mathfrak{S}_q} \text{sgn}(\sigma) \prod_{r=1}^q \overline{(b_{i_{\sigma(r)}})_r} = \frac{|\det B_J|^2}{\Omega_J}.$$

This proves directly, including singular K_L , $\det K_L = \sum_{J \subseteq L, |J|=q} |\det B_J|^2 / \Omega_J$. Apply it to the four index sets

$$\{0, \dots, N+1\}, \quad \{0, \dots, N-1\}, \quad \{0, \dots, N\}, \quad \{0, \dots, N-1, N+1\}.$$

For each q -element subset of the first set, the coefficients in the signed combination defining Δ_N are respectively $1 + 1 - 1 - 1 = 0$ if it contains neither N nor $N+1$, $1 - 1 = 0$ if it contains exactly one of them, and 1 if it contains both. The surviving subsets have exactly the form $I \cup \{N, N+1\}$ in (CV.10). If $q = 1$, no subset contains both indices and the result is zero. No inverse for a smaller or omitted-direction Gram matrix has occurred, so this proof also applies at $N = q - 1$. $\square$

There is a precise source map for the positive sum. Let $e_C = [1] \wedge [S] \wedge \dots \wedge [S^{q-1}]$ be the fixed coordinate wedge in $\bigwedge^q C$, of squared norm one for the coordinate Euclidean pairing used in (CV.6). For $q \geq 2$ define

$$\mathcal{E}_N = \left(\bigwedge_{i=1}^{q-2} \mathcal{P}_{N-1} \right) \wedge p_N \wedge p_{N+1} \subseteq \bigwedge^q \mathcal{P}_{N+1}, \quad \Lambda_N = \left(\bigwedge^q T_{N+1} \right)|_{\mathcal{E}_N}.$$

The increasing vectors $e_I = \left(\bigwedge_{i \in I} p_i \right) \wedge p_N \wedge p_{N+1}$ form an orthogonal basis of $\mathcal{E}_N$ with their original squared norms $\Omega_{I \cup \{N, N+1\}}$. They satisfy $\Lambda_N e_I = \det B_{I \cup \{N, N+1\}} e_C$. Consequently the adjoint in these explicitly specified source and target pairings gives the original, unscaled vector

$$\zeta_N := \Lambda_N^* e_C = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\Omega_{I \cup \{N, N+1\}}} e_I, \quad (\text{CV.10a})$$

$$\|\zeta_N\|^2 = \Delta_N, \quad \Lambda_N \zeta_N = \Delta_N e_C, \\ \|\mathcal{V}_{h,k}^{\otimes q} \text{Alt}_q \zeta_N\|^2 = q! \Delta_N.$$

The first two equalities follow by inserting the displayed orthogonal basis and its full norms. The last equality follows from the pairwise isometry of $\mathcal{V}_{h,k}$ and the unscaled alternating-map calculation following (CV.6). Its target is the original tensor source $\mathcal{B}^{\widehat{\otimes} kq}$ in degree kq . For $q = 1$ take $\mathcal{E}_N = \{0\}$, $\Lambda_N = 0$, and $\zeta_N = 0$, so that all three zero identities remain exact. Thus (CV.10) comes with an explicitly typed original arithmetic source vector and the literal tensor factorial. Combining it with (CV.4) gives $\epsilon^2 = (\omega_{N+1}/\omega_N) \|\zeta_N\|^2 / D_N$; no estimate as k or N grows is inferred from this equality.

33.3 The exact first admitted degree

At $N = q - 1$ expand the original monic annihilator in the original orthogonal basis:

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\int \overline{p_j(k/2 + iu)} \chi(k/2 + iu) m_{h,k}(u) du}{\omega_j}. \quad (\text{CV.11})$$

This is a finite polynomial identity: both leading coefficients are one, and orthogonality gives every displayed coefficient, without a change of measure. The reduction map T_{q-1} is a linear isomorphism, since the q original monic remainders are a basis. Its inverse is therefore its canonical minimum section. Consequently $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $0 \leq i, j < q$. Reduction of (CV.11) gives $b_q = -\sum_{j<q} \gamma_j b_j$ and hence

$$a = \omega_{q-1}, \quad c = -\gamma_{q-1} \omega_{q-1}, \quad d = \sum_{j<q} |\gamma_j|^2 \omega_j, \quad (\text{CV.12})$$

$$\boxed{\epsilon_{h,k,q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\|\chi\|^2 - \omega_q - |\gamma_{q-1}|^2 \omega_{q-1}}{\omega_{q-1}}}.$$

The first expression follows by substituting the preceding three exact Gram entries into (CV.4); the second follows by Pythagoras applied to (CV.11). Reflection makes c purely imaginary, so the same is true of γ_{q-1} ; it has not been set to zero. Let $\mathcal{P}_{q-2}$ be the original integral orthogonal projection onto $\mathcal{P}_{q-2}$, with $\mathcal{P}_{-1} = 0$ for $q = 1$. Then the numerator in (CV.12) is exactly $\|\mathcal{V}_{h,k} \mathcal{P}_{q-2} \chi\|^2$ by the original source isometry. Meanwhile the full $\mathcal{V}_{h,k} \chi$ has zero cyclic and full jets, since $\chi(S) \in I$, and the original fixed-order polynomial division gives its theta primitive as in (CM.19). These are different specified source maps: projecting that boundary is an integral orthogonal operation and has the cyclic reduction $T_{q-2}(\mathcal{P}_{q-2} \chi) = \sum_{j<q-1} \gamma_j b_j$, where T_{q-2} is the restriction of T_{q-1} to $\mathcal{P}_{q-2}$; its actual full arithmetic jet is $J^{(k)} \mathcal{V}_{h,k} \mathcal{P}_{q-2} \chi = \eta(\sum_{j<q-1} \gamma_j b_j)$. For $q = 1$ the first map is the unique map from $\mathcal{P}_{-1} = 0$. This exact formula retains its possibly nonzero observation. For $q = 1$ the sum is empty and both sides are zero. The degree- q annihilator and its projection have thus given a complete original-source formula at the first admitted degree, with no large-degree estimate added.

34 The original source and relation Toda flows with their exact control maps

This is a standalone continuation of the identities CV.1–CV.12. It retains the original polynomial quotient, arithmetic measure, monic coordinates, arithmetic unit, source relations and complex phase. The delivered `Tau_Toda_Volume_Control/NOTE.tex` was read in full from its original archive; its SHA256 is

027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375.

The calculations below prove the exact maps underlying the determinant quotient, its first and second tilt derivatives, the omitted-direction determinant, the positive minor sum and the finite upper bound. No asymptotic arithmetic estimate is asserted.

34.1 Original arithmetic objects and the strip of real tilts

The original spaces and transforms are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx. \end{aligned}$$

The series and its Euler derivatives converge uniformly at large x with every power of decay. For small x , Poisson summation with Fourier kernel $e^{-2\pi ixy}$ gives $\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x)$; both zero terms vanish because $\phi(0) = \hat{\phi}(0) = 0$. The Fourier transform preserves the stated two zero constraints and evenness. This formula proves every small- x seminorm after Euler differentiation, hence $\Theta : V \rightarrow \mathcal{B}$. It also commutes with D on these spaces by termwise differentiation. Mellin transforms of elements of $\mathcal{B}$ are entire, and integration by parts gives $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$; both endpoint terms vanish with their original coefficients.

Let h be a monic, nonempty packet of actual nontrivial zeros of

$$g(s) = 2\zeta(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with their complete orders. Assume the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Set $d = \deg h$ and retain

$$v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = [v_h]_h \in E_h^\times, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}. \quad (\text{TVB.1})$$

Here the jet of an entire function f means its unique polynomial Hermite remainder of degree less than d : in the local factor $\mathbb{C}[t]/(t^{m_\rho})$, it is $\sum_{j < m_\rho} f^{(j)}(\rho) t^j / j!$, with $t = s - \rho$. The coprime factors of h give the isomorphism to their product by monic division and Bezout's identity, which supplies the unique global remainder. The constant $v_h(\rho)$ is nonzero at each selected zero because its full order was divided out. If the local jet is $\sum a_j t^j$, its inverse coefficients are $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$ for $n < m_\rho$. Multiplication verifies this inverse in every local factor, and the product isomorphism gives the inverse of v_h .

The original F_h has a construction within $\mathcal{B}$. For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy.$$

The equality is the vanishing Mellin value. For any desired power bound at infinity choose $A > \Re \rho$ with $|F(y)| \leq C_A y^{-A}$ for $y \geq 1$; the first integral is bounded by $C_A x^{-A}/(A - \Re \rho)$ there. At zero choose B with $B + \Re \rho > 0$ and $|F(y)| \leq C_B y^B$ for $0 < y \leq 1$; the second integral is bounded by $C_B x^B/(B + \Re \rho)$. The exact identity $(D - \rho)S_\rho F = F$ now gives all Euler derivative bounds inductively, so $S_\rho F \in \mathcal{B}$. Mellin integration by parts proves $\mathcal{M}(S_\rho F) = \mathcal{M}F/(s - \rho)$, with its removable value at $s = \rho$ supplied by the derivative of the numerator. For the original seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V$, direct substitution $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \Re s > -2.$$

For $\Re s > 1$, absolute convergence permits the two-sided theta sum to be integrated termwise, yielding $\mathcal{M}(\Theta\phi_0) = 2\zeta(s)\mathcal{M}\phi_0(s) = g(s)$; both sides continue to the same entire function. Apply S_ρ successively to $\Theta\phi_0$ over the full list of roots of h , including each repeated occurrence. At every stage the next vanishing Mellin value holds by its remaining order. This constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$

and $h(D)F_h = \Theta\phi_0$; the latter also follows by Mellin injectivity, which is Fourier injectivity after $x = e^y$. Fix $k \geq 1$, set $c = k/2$, $S = s_1 + \cdots + s_k$, and put $I = (h(s_1), \dots, h(s_k)) \subset \mathbb{C}[s_1, \dots, s_k]$. Let χ be the original monic generator of $\{P \in \mathbb{C}[S] : P(S) \in I\}$. Its degree is $q \geq 1$ and

$$C = \mathbb{C}[S]/(\chi), \quad A = M_S : C \longrightarrow C. \quad (\text{TVB.2})$$

The coordinate basis of C is always $1, S, \dots, S^{q-1}$. The polynomial exists because multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$ has an annihilating polynomial. Division by its least-degree monic annihilator proves that its entire annihilating ideal is (χ) . In particular the injection

$$\eta : C \longrightarrow (E_h^{\otimes k})^{\mathfrak{S}_k}, \quad [P] \longmapsto v_h^{\otimes k}[P(S)]_I \quad (\text{TVB.3})$$

has exactly zero kernel: multiplication by the displayed unit is invertible and the preceding annihilator ideal describes the other factor's kernel. The original tensor source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad D = -x\partial_x, \quad \mathcal{M}F_h = v_h.$$

On the original Mellin line its scalar observation has squared norm

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad \int m_k = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{TVB.4})$$

Indeed the tensor Mellin–Plancherel density is $\prod_i w_h(t_i) d^k t$. In coordinates $u = \sum_i t_i$, $y = (t_1, \dots, t_{k-1})$, the inverse is $(y, u) \mapsto (y_1, \dots, y_{k-1}, u - \sum y_i)$. With (y, u) in this order its determinant is 1; with (u, y) its determinant is $(-1)^{k-1}$. Integrating the original density over y gives (TVB.4), with no scalar divisor.

For every compact substrip $|\Re\theta| \leq r < \pi/2$, all integrals

$$M_h(\theta) = \int e^{\theta t} w_h(t) dt, \quad Z(\theta) = M_h(\theta)^k = \int e^{\theta u} m_k(u) du \quad (\text{TVB.5})$$

and all their polynomially weighted derivatives converge locally uniformly. Here is an elementary sufficient bound. Summation by parts for the alternating Dirichlet series, whose partial sums have modulus at most one, gives $|\eta(s)| \leq |s|/\Re s$ when $\Re s > 0$. On $\Re s = 1/2$, the denominator in $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ has modulus at least $\sqrt{2} - 1$. For $0 < \varphi < \pi/2$, rotate the defining gamma integral to the ray with argument $\text{sgn}(t)\varphi$. The large arc vanishes by $e^{-R\cos\varphi}$ and the small arc by the positive real part $1/4$ of its exponent. Its absolute value is bounded by

$$|\Gamma(1/4 + it/2)| \leq \Gamma(1/4)(\cos\varphi)^{-1/4} e^{-\varphi|t|/2}.$$

For sufficiently large $|t|$, the monic polynomial h satisfies $|h(1/2 + it)| \geq C_h(1 + |t|)^d$ with $C_h > 0$. The two factors $s, s - 1$, the preceding zeta bound and the gamma bound consequently give

$$w_h(t) \leq C_{h,\varphi}(1 + |t|)^{6-2d} e^{-\varphi|t|}$$

outside a compact interval. On that interval v_h is entire, since the selected orders are complete. Choose $r < \varphi < \pi/2$ and absorb every fixed polynomial in the remaining exponential decay. This proves the asserted domination and differentiation under the integral. Fubini gives the exact product M_h^k in (TVB.5). The nonzero entire function v_h has a discrete real-line zero set; therefore $w_h > 0$ almost everywhere. Its convolutions are positive almost everywhere (in fact at every u for $k \geq 2$), since their fibre integrands are positive off a finite union of measure-zero inverse images of this discrete set. Thus the norm of every nonzero polynomial is positive.

In what follows θ is real, $|\theta| < \pi/2$, and the source pairing, conjugate-linear in its first argument, is exactly

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (\text{TVB.6})$$

The map $f(u) \mapsto e^{\theta u/2} f(u)$ is an onto isometry from $L^2(e^{\theta u} m_k du)$ to $L^2(m_k du)$, with inverse $f \mapsto e^{-\theta u/2} f$. It commutes with multiplication by $c + iu$. This specifies the exact Hilbert comparison for the tilt. The polynomial algebra, its reduction and η are unchanged. When the target of $\mathcal{V}$ is observed at this tilt, its declared tensor norm is

$$\|F\|_{k,\theta}^2 = \int_{\mathbb{R}^k} e^{\theta \sum_i t_i} |\mathcal{M}_{\text{unit}} F(t_1, \dots, t_k)|^2 d^k t,$$

where $\mathcal{M}_{\text{unit}}$ is the original Mellin–Plancherel map with a factor $(2\pi)^{-1/2}$ in each coordinate. The map $f(t) \mapsto e^{\theta \sum_i t_i/2} f(t)$ is an onto isometry from this weighted Mellin Hilbert space to the original one, with the opposite exponential as inverse. The restriction to $\mathcal{V}P$ is exactly (TVB.6), by the same sum-fibre integral as (TVB.4). Thus every tilted exterior norm below has a specified target norm; at $\theta = 0$ it is the original tensor-source norm.

34.2 The source relation and its Laplace observation

Write $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$. For $N \geq q - 1$, put $n = N + 1$, $m = N - q + 1$. Literal monic division gives the exact sequence

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{T_N} C \longrightarrow 0, \quad M_\chi Q = \chi Q, \quad T_N P = [P]_\chi. \quad (\text{TVB.7})$$

Its kernel is precisely the indicated image, and its first q monomial columns prove surjectivity. Denote that kernel by $\mathcal{D}_N = \chi \mathcal{P}_{m-1}$.

This sequence has an explicit map to the original theta boundaries. To see every sign, choose the quotients Q_i from successive monic division, in the order $s_1, \dots, s_k$, so that

$$\chi(S) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{TVB.8})$$

The remainder is zero by the definition of χ and uniqueness of the tensor remainder basis. Put $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. It is even, vanishes at zero and has integral zero. Direct gamma integration of its two terms gives $\mathcal{M}(\Theta \phi_0) = g$, hence $h(D)F_h = \Theta \phi_0$ by Mellin injectivity on the original rapidly decreasing source. For every polynomial P , an actual primitive of $\mathcal{V}(\chi P)$ in the tensor complex $(V \xrightarrow{\Theta} \mathcal{B})^{\otimes k}$ is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{TVB.9})$$

The preceding $i - 1$ factors have cochain degree one. The tensor differential therefore contributes $(-1)^{i-1}$ at the i th factor, cancelling the written sign and yielding (TVB.8). Euler differentiation preserves the source conditions and commutes with Θ , so all terms have the indicated domains. Conversely the full jet of $\mathcal{V}P$ is $\eta T_N P$; injectivity in (TVB.3) shows that the admitted polynomial source has no further zero-jet elements. In particular every boundary vector in (TVB.7) retains its positive source norm before the quotient sends it to represented zero.

For any polynomial P , define its Laplace observation by

$$\mathcal{L}(P)(\theta) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Differentiation under (TVB.5) proves the typed intertwining $\mathcal{L}(SP) = (c + i\partial_\theta) \mathcal{L}(P)$. Coefficient-wise conjugation gives the exact Gram observation

$$Z_\chi = \bar{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0. \quad (\text{TVB.10})$$

Thus the two signs and both polynomial factors arise from $\langle M_\chi 1, M_\chi 1 \rangle_\theta$. This is a map on source functions, with no replacement of the higher contracted ideal by a power of χ .

34.3 Both monic determinants and their exact determinant-line map

Set

$$\mathfrak{D}_j = \det[Z^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{B}_j = \det[Z_\chi^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{D}_0 = \mathfrak{B}_0 = 1. \quad (\text{TVB.11})$$

All nonempty determinants are positive by polynomial positivity. The coefficient map from $1, S, \dots, S^{j-1}$ to $1, u, \dots, u^{j-1}$ is explicitly

$$S^a = (c + iu)^a = \sum_{b=0}^a \binom{a}{b} c^{a-b} i^b u^b. \quad (\text{TVB.12})$$

Its determinant is $i^{j(j-1)/2}$ and its Gram determinant multiplier is exactly one. Therefore $\mathfrak{D}_j$ is also the original S -monomial Gram determinant. The same calculation applies to the relation-weight pairing $\langle P, Q \rangle_{\chi, \theta} = \langle \chi P, \chi Q \rangle_\theta$ and $\mathfrak{B}_j$. This proves the coordinate correspondence without changing the original monic polynomial.

Let $p_j(S; \theta)$ be the monic source orthogonal polynomials, and $r_j(S; \theta)$ the monic polynomials for this relation-weight pairing. Retain their original squared norms

$$\omega_j = \|p_j\|_\theta^2 = \mathfrak{D}_{j+1}/\mathfrak{D}_j, \quad \nu_j = \|\chi r_j\|_\theta^2 = \mathfrak{B}_{j+1}/\mathfrak{B}_j. \quad (\text{TVB.13})$$

The equations follow by triangular monic Gram elimination, whose diagonal change is one. In particular $\omega_0 = M_h(\theta)^k$. The full mass has not been divided out.

Give C its written coordinate Euclidean pairing only when specifying adjoints. Let M_N be the original monomial source Gram, and J_N its reduction matrix. The minimum lift is

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = M_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N. \quad (\text{TVB.14})$$

Here the last expression specifies polynomial coefficients. It satisfies $T_N R_N = I$, $\text{im } R_N \perp \mathcal{D}_N$ and $R_N^* R_N = G_N$ in the original source pairing. Indeed the first equality is matrix multiplication, and pairing with any vector in $\ker J_N$ gives zero. Every other lift is $R_N x + B$ for $B \in \mathcal{D}_N$, so its squared norm is $x^* G_N x + \|B\|^2$. This proves the minimum assertion and uniqueness.

Use the source basis $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$. Its change from the n source monomials is triangular with diagonal one. Replacing its first q columns by $R_N[1], \dots, R_N[S^{q-1}]$ adds boundary columns and again has determinant one. The resulting Gram is the direct sum of G_N and the relation monomial Gram. Consequently

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_m}, \quad D_N^{\text{CV}} := \det K_N = V_N^{-1}. \quad (\text{TVB.15})$$

More precisely the map $\det C \otimes \det \mathcal{P}_{m-1} \rightarrow \det \mathcal{P}_N$ sends the written coordinate wedges to the wedge of these quotient lifts followed by M_χ of the relation monomials; its coordinate determinant is 1. Putting the kernel wedge first instead contributes the explicit sign $(-1)^{qm}$. Its norm identity is $\det G_N \mathfrak{B}_m = \mathfrak{D}_n$. Composing the wedge with $\mathcal{V}^{\otimes n} \text{Alt}_n$ multiplies its squared norm by exactly $n!$: expanding the unscaled alternating tensor sum gives $n!$ copies of the Gram determinant. The original tensor target is $\mathcal{B}^{\widehat{\otimes} kn}$ in cochain degree kn .

34.4 Toda derivatives and the full curvature operators

For either positive real u -weight in (TVB.11), write $Q_j(u)$ for its real monic orthogonal polynomials and h_j for their norms. Orthogonality and leading coefficients prove

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1} \ (j \geq 1), \quad a_0 = 0.$$

For $i \leq j-2$, differentiating $\langle Q_i, Q_j \rangle = 0$ shows that $\partial_\theta Q_j$ is orthogonal to Q_i , because uQ_i and $\partial_\theta Q_i$ have degree at most $j-1$. For $i = j-1$ the same identity gives its coefficient $-a_j$. Thus

$\partial_\theta Q_j = -a_j Q_{j-1}$ and $h'_j = b_j h_j$. To compute b'_j , differentiate $h_j b_j = \langle Q_j, u Q_j \rangle$. The two polynomial derivative terms contribute $-2a_j h_j$. The derivative of the measure contributes $\|u Q_j\|^2 = h_j(a_{j+1} + b_j^2 + a_j)$. After subtracting $h'_j b_j$ this gives $b'_j = a_{j+1} - a_j$. The monic Gram determinant is $\prod_{j < n} h_j$, hence

$$(\log \mathfrak{D}_n)'' = \frac{\omega_n}{\omega_{n-1}} \quad (n \geq 1), \quad (\log \mathfrak{B}_m)'' = \frac{\nu_m}{\nu_{m-1}} \quad (m \geq 1). \quad (\text{TVB.16})$$

Both 0th logarithms have second derivative zero. Multiplication by the square of the determinant proves each Hankel-Toda equation with coefficient one. Set

$$\alpha_{N+1} = \omega_{N+1}/\omega_N, \quad \beta_m = \nu_m/\nu_{m-1} \quad (m \geq 1), \quad \beta_0 = 0, \quad \ell_N = \log V_N.$$

Then $\ell''_N = \alpha_{N+1} - \beta_m$.

There is a finer exact source-map identity. Let $X = (S - c)/i$, which is multiplication by the real variable u in (TVB.6). Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ to $\mathcal{D}_N$ in that pairing. Differentiating $T_N R_N = I$ gives $R'_N x \in \mathcal{D}_N$. Differentiating orthogonality against each fixed boundary polynomial gives

$$R'_N = -P_{\mathcal{D}_N} X R_N =: -I_N, \quad O_N := (1 - P_N) X R_N. \quad (\text{TVB.17})$$

Every derivative is an actual boundary with the primitive (TVB.9). Differentiating its norm, including the derivative of the measure, gives

$$G'_N = R_N^* X R_N, \quad G''_N = R_N^* X^2 R_N - 2R_N^* X P_{\mathcal{D}_N} X R_N.$$

The projection to $\text{im } R_N$ is $R_N G_N^{-1} R_N^*$ and, together with $P_{\mathcal{D}_N}$, sums to P_N . Subtracting the square of the first derivative therefore proves

$$G''_N - G'_N G_N^{-1} G'_N = O_N^* O_N - I_N^* I_N. \quad (\text{TVB.18})$$

This is an equality of maps $C \rightarrow C^{\text{anti}*}$, where $C^{\text{anti}*}$ consists of conjugate-linear functionals on C . For example $O_N^* O_N$ sends x to the functional $y \mapsto \langle O_N y, O_N x \rangle_\theta$; this assignment is complex-linear in x . The fixed coordinate Euclidean pairing identifies that antidual with the written matrix coordinates.

Write $b_j = [p_j]_\chi$, and $a = b_N^* G_N b_N$, $d_+ = b_{N+1}^* G_N b_{N+1}$. The canonical next boundary has the particularly exact factorization

$$\mathfrak{b}_N := p_{N+1} - R_N b_{N+1} = \chi r_m, \quad \nu_m = \omega_{N+1} + d_+. \quad (\text{TVB.19})$$

To prove it, χr_m is monic of degree $N + 1$ and orthogonal to $\mathcal{D}_N$. Subtract p_{N+1} ; the difference lies in $\mathcal{P}_N \cap \mathcal{D}_N^\perp$ and has reduction $-b_{N+1}$. Uniqueness of R_N proves the first equality; the two terms in its definition are orthogonal, giving the second. The coefficient of p_N in $R_N x$ is $b_N^* G_N x / \omega_N$. The leading coefficient of X as an S -polynomial is $1/i = -i$. Consequently the full maps are

$$O_N x = -i p_{N+1} \frac{b_N^* G_N x}{\omega_N}, \quad I_N x = -i \chi r_{m-1} \frac{b_{N+1}^* G_N x}{\nu_{m-1}} \quad (m \geq 1), \quad I_N = 0 \quad (m = 0). \quad (\text{TVB.20})$$

For the second equation, $X \chi r_j$ is an old boundary if $j \leq m - 2$. For $j = m - 1$ its coefficient of χr_m is $-i$. Moving X across the pairing conjugates this coefficient to i , and $\langle \chi r_m, R_N x \rangle = -b_{N+1}^* G_N x$ by (TVB.19). Division by the original norm ν_{m-1} gives exactly the second sign in (TVB.20).

Both maps have rank at most one. For $N \geq q$, put $\delta_N = V_N/V_{N-1}$. The rank-one calculation in the next section gives $a/\omega_N = 1 - \delta_N$ and $\delta_N = \omega_N/\nu_{m-1}$. Taking traces in (TVB.20) gives the separate, positive source costs

$$\begin{aligned} \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1} (1 - \delta_N), \\ \text{Tr}(G_N^{-1} I_N^* I_N) &= \beta_m - \alpha_{N+1} \delta_N = \frac{\nu_m - \omega_{N+1}}{\nu_{m-1}}. \end{aligned} \quad (\text{TVB.21})$$

Their difference is $\alpha_{N+1} - \beta_m$ as required. Thus the individual positive map costs each retain the common quantity $\alpha_{N+1}\delta_N$; assigning them separately the two Toda coefficients would lose this quantity. At $N = q - 1$, the outer cost is α_q and the inner cost is zero. If a single notation for the cost identities is wanted at this endpoint, define only the scalar convention $\delta_{q-1}^{\text{end}} := \det K_{q-2} / \det K_{q-1} = 0$; the included column has leverage $a/\omega_{q-1} = 1$. Together with $\beta_0 = 0$ the cost identities are $\alpha_q(1 - \delta_{q-1}^{\text{end}}) = \alpha_q$ and $\beta_0 - \alpha_q\delta_{q-1}^{\text{end}} = 0$. This convention is not a quotient involving V_{q-2} and does not introduce an inverse for K_{q-2} or a norm ν_{-1} . All statements concern the original polynomial source and its original boundary subspace.

34.5 The rank-one update, trace phase and original rank-two radius

The source-orthogonal basis gives

$$K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad K_{N+1} = K_N + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}. \quad (\text{TVB.22})$$

The first q monic remainder columns are a triangular basis; K_N is positive definite precisely in the admitted range. Block elimination of $\begin{pmatrix} K^* & U \\ -V^* & I \end{pmatrix}$ in either order proves $\det(K + UV^*) = \det K \det(I + V^* K^{-1} U)$. Applying it to the positive update and the subtraction of b_N from K_N yields, for $N \geq q$,

$$\frac{a}{\omega_N} = 1 - \delta_N, \quad \frac{d_+}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1, \quad \delta_N = \frac{\omega_N}{\nu_{m-1}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_m}. \quad (\text{TVB.23})$$

The last two identities also follow directly from (TVB.15). Each δ is positive and at most one, since increasing the source space weakly decreases every minimum squared norm. The exact inverse update is

$$G_{N+1} = G_N - \frac{G_N b_{N+1} b_{N+1}^* G_N}{\omega_{N+1} + d_+}. \quad (\text{TVB.24})$$

Multiplying this expression by K_{N+1} proves it directly; the denominator is $\nu_m > 0$ by (TVB.19).

Reflection of the tensor algebra sends S to $k - \bar{S}$ and preserves its annihilator ideal. Monicity gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic. The exact algebra isomorphism

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P(S)] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q(u)] = [Q((S - c)/i)] \quad (\text{TVB.25})$$

transports the original ideal by its leading unit i^q . Its coefficient matrix E is (TVB.12) in dimension q . Thus A transports to $cI + iT$, where $T = M_u$ in the real quotient. Its Gram transports to $E^{-*} G_N E^{-1}$, its kernel to $E K_N E^*$, and $\det G_N$ is unchanged because $|\det E| = 1$. Write

$$\sigma = \text{Tr } T = \frac{\text{Tr } A - cq}{i} \in \mathbb{R}, \quad \psi_N = \sigma - \ell'_N.$$

Reality of $\hat{\chi}$, the weight and its monic orthogonal polynomials makes all the u -coordinate source and quotient Grams real. Let $Q_j(u) = i^{-j} p_j(c + iu)$ and $d_j = [Q_j]$. Then $E b_j = i^j d_j$ and both source norms are ω_j .

The leading coefficient of the u -polynomial minimum lift of x is $d_N^* G_N x / \omega_N$, where in this calculation G_N denotes the transported real-coordinate Gram. The difference $X R_N - R_N T$ has zero reduction, and subtracting $(Q_{N+1} - R_N d_{N+1}) d_N^* G_N / \omega_N$ cancels its highest coefficient. Its remainder is an old boundary. Pair with R_N ; old boundaries vanish and $R_N^* (Q_{N+1} - R_N d_{N+1}) = -G_N d_{N+1}$. Taking the trace after multiplication by G_N^{-1} proves

$$\ell'_N = \sigma - \frac{d_N^* G_N d_{N+1}}{\omega_N}, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i \psi_N. \quad (\text{TVB.26})$$

The second equation is back in the fixed original S coordinates: the phase is $\bar{i}^N i^{N+1} = i$. Both signs are retained.

For completeness, the u recurrence proved above transports to $Sp_j = p_{j+1} + (c + ib_j^{\text{rec}})p_j - \alpha_j p_{j-1}$ in the original source coordinates. Inserting this in (TVB.22) telescopes all interior terms and gives

$$AK_N + K_N A^* - kK_N = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}.$$

For $W_N = A^*G_N + G_N A - kG_N$, multiplication on both sides by G_N gives the corresponding identity for W_N . Its Hermitian relative operator is $H_N = G_N^{-1/2}W_N G_N^{-1/2}$. It has rank at most two and trace $2\Re \text{Tr } A - kq = 0$. The sum of the squares of its eigenvalues is $2(ad_+ - |z_N|^2)/\omega_N^2$, where $z_N = b_N^* G_N b_{N+1} = i\omega_N \psi_N$. This last assertion follows by expanding the square of $(xy^* + yx^*)/\omega_N$ for $x = G_N^{1/2}b_N, y = G_N^{1/2}b_{N+1}$; its trace is $[2ad_+ + 2\Re(z_N^2)]/\omega_N^2$. Its nonzero eigenvalues are therefore $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2. \quad (\text{TVB.27})$$

The first expression is valid already at $N = q - 1$.

34.6 Exact maps to the CV omitted determinant and positive minors

Retain $D_j^{\text{CV}} = \det K_j$, put $K_{-1} = 0$, and form

$$\tilde{K}_N = K_{N-1} + \frac{b_{N+1}b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad \Delta_N = D_{N+1}^{\text{CV}} + D_{N-1}^{\text{CV}} - D_N^{\text{CV}} - \tilde{D}_N.$$

This is the actual quotient kernel of the source with p_N omitted and p_{N+1} included. No inverse of it is used. The two-column determinant formula preceding (TVB.23), with one negative and one positive update, proves

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} = \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d_+}{\omega_{N+1}}\right) + \frac{|z_N|^2}{\omega_N \omega_{N+1}}, \quad \epsilon_N^2 = \alpha_{N+1} \frac{\Delta_N}{D_N^{\text{CV}}}. \quad (\text{TVB.28})$$

In particular, for $N \geq q$, the complete phase map is the exact nonnegative difference

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} - \frac{\delta_N}{\delta_{N+1}} = \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.29})$$

The sign of ψ_N is retained by the first derivative in (TVB.26); the omitted determinant observes its square.

For any finite column set L , determinant multilinearity gives

$$\det \left(\sum_{j \in L} \frac{b_j b_j^*}{\omega_j} \right) = \sum_{\substack{J \subseteq L \\ |J|=q}} \frac{|\det B_J|^2}{\prod_{j \in J} \omega_j}, \quad B_J = [b_j]_{j \in J \text{ increasing}}.$$

Terms with repeated indices vanish. The $q!$ remaining permutations of a fixed increasing index set yield its determinant times its conjugate, which proves the displayed identity including singular kernels. Apply it to the four column sets defining Δ_N . A set containing neither of the two last indices has coefficient $1 + 1 - 1 - 1 = 0$; one containing exactly one has coefficient $1 - 1 = 0$; one containing both has coefficient one. Hence

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \quad (\text{TVB.30})$$

For $q = 1$ it is zero. This proof applies at $N = q - 1$. For $q \geq 2$ define the actual source subspace $\mathcal{E}_N = (\bigwedge^{q-2} \mathcal{P}_{N-1}) \wedge p_N \wedge p_{N+1}$ and $\Lambda_N = (\bigwedge^q T_{N+1})|_{\mathcal{E}_N}$. For the fixed coordinate wedge $e_C = 1 \wedge S \wedge \cdots \wedge S^{q-1}$, whose coordinate Euclidean squared norm is one, the adjoint is

$$\zeta_N := \Lambda_N^* e_C = \sum_{|I|=q-2} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \left(\bigwedge_{i \in I} p_i \right) \wedge p_N \wedge p_{N+1}. \quad (\text{TVB.31})$$

The displayed source wedges are orthogonal with the written full product norms. Therefore $\|\zeta_N\|_\theta^2 = \Delta_N$ and $\Lambda_N \zeta_N = \Delta_N e_C$. At $\theta = 0$ its literal original tensor-source image has squared norm $q! \Delta_N$ under $\mathcal{V}^{\otimes q} \text{Alt}_q$. At other real tilts use the onto Hilbert comparison after (TVB.6) on each scalar source factor; the same determinant norm and factorial hold in that specified tilted observation. For $q = 1$ set $\mathcal{E}_N = 0$, $\Lambda_N = 0$, $\zeta_N = 0$.

34.7 Both exact losses, their equality cases and the telescope

For $N \geq q$ define

$$v_N = -\log \delta_N, \quad s_N = (v_N + v_{N+1})/2, \quad t_N = (v_{N+1} - v_N)/2, \quad \mathcal{R}_N = e^{2s_N} = \frac{V_{N-1}}{V_{N+1}} \geq 1.$$

Expanding $(1 - e^{-v_N})(e^{v_{N+1}} - 1)$ gives $2e^{t_N} \cosh s_N - e^{2t_N} - 1$. Completing this scalar square, with both terms retained, yields

$$\boxed{\epsilon_N^2 = \alpha_{N+1} \sinh^2 s_N - \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 - \psi_N^2.} \quad (\text{TVB.32})$$

Thus the exact upper bound and its equality conditions are

$$\epsilon_N \leq \sqrt{\alpha_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}, \quad \text{equality} \iff \psi_N = 0 \text{ and } \delta_N(2 - \delta_{N+1}) = 1. \quad (\text{TVB.33})$$

Indeed $\alpha_{N+1} > 0$ and both lost squares in (TVB.32) are nonnegative. Multiplication of $e^{t_N} = \cosh s_N$ by $2\sqrt{\delta_N \delta_{N+1}}$ gives precisely the displayed equation. This includes the case $\mathcal{R}_N = 1$, when both contractions equal one and $\epsilon_N = \psi_N = 0$ by (TVB.27). The corresponding equality with the full positive minor source is

$$\frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N} - \frac{\|\zeta_N\|_\theta^2}{D_N^{\text{CV}}} = (e^{t_N} - \cosh s_N)^2 + \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.34})$$

It follows from (TVB.28), (TVB.31) and (TVB.32), so its minor equality has the same two necessary and sufficient conditions. In particular equality is a fully specified source-volume equation, not a consequence of positivity alone.

There is a direct formulation in the actual volumes, without logarithmic variables. Put $U_N = \sqrt{\alpha_{N+1}}(\mathcal{R}_N - 1)/(2\sqrt{\mathcal{R}_N})$. Since

$$e^{t_N} = \frac{V_N}{\sqrt{V_{N-1} V_{N+1}}}, \quad \cosh s_N = \frac{V_{N-1} + V_{N+1}}{2\sqrt{V_{N-1} V_{N+1}}},$$

equation (TVB.32) gives the exact two losses

$$U_N^2 - \epsilon_N^2 = \alpha_{N+1} \frac{(2V_N - V_{N-1} - V_{N+1})^2}{4V_{N-1} V_{N+1}} + \frac{|z_N|^2}{\omega_N^2}. \quad (\text{TVB.34a})$$

Thus the volume equality is the arithmetic midpoint $2V_N = V_{N-1} + V_{N+1}$ together with $z_N = 0$. Equal consecutive logarithmic contractions alone do not give this midpoint: their extra equation $V_N^2 = V_{N-1} V_{N+1}$ implies it only if all three positive volumes are equal, by the square $(\sqrt{V_{N-1}} -$

$\sqrt{V_{N+1}})^2$. In the original inverse-volume coordinates $D_- = D_{N-1}^{\text{cv}}$, $D_0 = D_N^{\text{cv}}$, $D_+ = D_{N+1}^{\text{cv}}$ and $E_N = \tilde{D}_N$, the same identities become

$$\begin{aligned} E_N D_0 - D_- D_+ &= \frac{D_0^2 |z_N|^2}{\omega_N \omega_{N+1}}, \\ \frac{D_0(D_+ - D_-)^2}{4D_- D_+} - \Delta_N &= \frac{(2D_- D_+ - D_0(D_- + D_+))^2}{4D_0 D_- D_+} + E_N - \frac{D_- D_+}{D_0}. \end{aligned} \quad (\text{TVB.34b})$$

Substitute $D_j = V_j^{-1}$ in (TVB.34a), or multiply out its two squares, to verify every denominator. Both terms on the right are nonnegative by the first equation. Equality is equivalently $D_0 = 2D_- D_+ / (D_- + D_+)$ and $E_N = (D_- + D_+)/2$; it then gives $\Delta_N = E_N - D_0$. In particular phase zero at these degrees records a positive omitted-direction determinant $E_N = D_- D_+ / D_0$, with its exact map specified in (TVB.28), rather than a zero determinant.

The algebraic exterior lower bound applies at each real tilt as well as at zero. Here is a direct proof. In the Hermitian matrix H_N above let P be the orthogonal projection to the image, under $G_N^{1/2}$, of the sum of all full generalized eigenspaces of A with $\Re \lambda > c$. Invariance of that subspace makes the transformed A upper triangular in its orthogonal direct sum with the complement, without asserting that the spectral projection itself is orthogonal. Taking the trace of its first diagonal block gives

$$\text{Tr}(PH_N) = L := \sum_{\Re \lambda > c} \ell_\lambda (2\Re \lambda - k).$$

Since H_N has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros, $\text{Tr}(PH_N) \leq \epsilon_N$: its positive eigenline contributes at most ϵ_N and its negative eigenline contributes a nonpositive number. All algebraic multiplicities ℓ_λ are retained. Thus $L \leq \epsilon_N$ for the same tilted minimum metric, without choosing another quotient metric.

Combining this with (TVB.33), using monotonicity of $\sinh$ on $[0, \infty)$ and its inverse, proves

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.35})$$

For $N = N_0 + 2j$, $0 \leq j < r$, with $N_0 \geq q$, every denominator in one logarithm is the numerator in the next. Summing the exact logs and then their lower bounds proves

$$\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N_0+2j+1}}} \right), \quad r \geq 1. \quad (\text{TVB.36})$$

Every term is at the same fixed real tilt, packet and tensor degree. No relation between different k or a limiting value of either determinant is used. Under $\mathcal{V}$ the minimum representatives at all degrees retain jets η and their original quotient classes; each difference of two such lifts is the explicit boundary in (TVB.7)–(TVB.9).

34.8 The first admitted degree and the empty quotient

At $N = q - 1$, the first q monic remainder columns are a basis, so $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Expand the original polynomial, with its coefficients retained,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle_\theta}{\omega_j}. \quad (\text{TVB.37})$$

Reduction gives $b_q = -\sum_{j<q} \gamma_j b_j$. Hence

$$a = \omega_{q-1}, \quad z_{q-1} = -\gamma_{q-1} \omega_{q-1}, \quad d_+ = \sum_{j<q} |\gamma_j|^2 \omega_j = \nu_0 - \omega_q, \quad \gamma_{q-1} = -i\psi_{q-1}.$$

The last equality follows from (TVB.26), which does not require K_{q-2}^{-1} . Substitution gives the exact CV.12-Toda endpoint

$$\epsilon_{q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - \psi_{q-1}^2. \quad (\text{TVB.38})$$

There is no V_{q-2} here. The omitted determinant instead records $\tilde{D}_{q-1}/D_{q-1}^{\text{CV}} = \psi_{q-1}^2/\alpha_q$. The polynomial projection $\mathbf{P}_{q-2}\chi = \sum_{j<q-1} \gamma_j p_j$ has the actual jet $\eta \sum_{j<q-1} \gamma_j b_j$, whereas χ itself has zero jet and primitive (TVB.9). Its squared source norm in (TVB.6) is the numerator in (TVB.38). This is the exact map from the whole relation to its lower orthogonal projection, including the possibly nonzero resulting observation. For $q = 1$, the sum is empty, the lower source is $\mathcal{P}_{-1} = 0$, and $\epsilon_N = 0$ at every admitted degree because its one-dimensional relative control is trace zero.

The endpoint projection and the minor vector have a stronger exact typed correspondence. Assume $q \geq 2$, let $D = D_{q-1}^{\text{CV}} = \prod_{j<q} \omega_j^{-1}$, and set $I_j = \{0, \dots, q-2\} \setminus \{j\}$ for $0 \leq j \leq q-2$. The determinant of the first q monic remainder columns is 1. Substituting $b_q = -\sum_{l<q} \gamma_l b_l$ in the minor with columns $I_j \cup \{q-1, q\}$ leaves only $l = j$. Moving b_j from its last position to position j takes $q-1-j$ transpositions; with the preceding minus sign this proves

$$\det B_{I_j \cup \{q-1, q\}} = (-1)^{q-j} \gamma_j. \quad (\text{TVB.38a})$$

Write $e_j = (\bigwedge_{i \in I_j} p_i) \wedge p_{q-1} \wedge p_q$. The explicit conjugate-linear isomorphism

$$\mathcal{H} : \mathcal{P}_{q-2} \longrightarrow \mathcal{E}_{q-1}, \quad \mathcal{H} \left(\sum_{j<q-1} a_j p_j \right) = \sum_{j<q-1} (-1)^{q-j} \overline{a_j} \frac{\omega_j D}{\omega_q} e_j \quad (\text{TVB.38b})$$

has inverse $\sum t_j e_j \mapsto \sum (-1)^{q-j} \omega_q \overline{t_j} p_j / (\omega_j D)$. The source and target basis vectors are orthogonal and $\|e_j\|_\theta^2 = \omega_q / (D \omega_j)$. Consequently

$$\|\mathcal{H}P\|_\theta^2 = \frac{D}{\omega_q} \|P\|_\theta^2, \quad \zeta_{q-1} = \mathcal{H}(\mathbf{P}_{q-2}\chi), \quad \Delta_{q-1} = \frac{D}{\omega_q} \|\mathbf{P}_{q-2}\chi\|_\theta^2. \quad (\text{TVB.38c})$$

This follows term by term from (TVB.31) and (TVB.38a). The actual alternating tensor image has squared norm $q! D \|\mathbf{P}_{q-2}\chi\|_\theta^2 / \omega_q$ in the specified tilted tensor observation, and in the original source norm at $\theta = 0$. The top coefficient is retained separately: $\tilde{D}_{q-1} = D |\gamma_{q-1}|^2 \omega_{q-1} / \omega_q$. For $q = 1$ the map $\mathcal{H}$ is the unique map between zero spaces, with the same zero identities; the top-coefficient formula still records the entire omitted determinant.

For completeness the finite upper bound is strict when $q = 1$ and $N \geq 1$. Consecutive source polynomials have no common root: their three-term recurrence with its nonzero coefficient $\alpha_j > 0$ would propagate such a root backwards to $p_0 = 1$. For any nonzero quotient $q \geq 1$, both b_N, b_{N+1} therefore cannot vanish, since a root of χ would then be common to both polynomials. When $N \geq q$, K_{N-1} is positive definite and $K_{N+1} - K_{N-1}$ is a nonzero positive semidefinite matrix. Conjugating by $K_{N-1}^{-1/2}$ and multiplying its eigenvalues proves $D_{N+1}^{\text{CV}} > D_{N-1}^{\text{CV}}$, hence $\mathcal{R}_N > 1$ and $U_N > 0$. At $q = 1$ this is strictly larger than the already proved $\epsilon_N = 0$.

Finally let $h = 1$, so $\chi = 1$, $q = 0$, and $C = 0$. The analytic function g , w_1 and all source norms still exist. For every $N \geq 0$ the source sequence is multiplication by one from $\mathcal{P}_N$ onto itself followed by the zero quotient. Then $m = N+1$, $Z_\chi = Z$, $\mathfrak{B}_m = \mathfrak{D}_m$, and the determinant of the zero-dimensional Gram is the empty determinant $V_N = 1$. Its logarithm and derivatives are zero. The minimum lift, its curvature maps, and the relative control are the unique zero maps, with $\epsilon_N = L = 0$. The two Toda coefficients in their curvature difference are equal, so that difference is zero. No positive-dimensional inverse is introduced. The support lift retains $G(0) = \{\tau, e\}$: every represented polynomial, including each nonzero source relation, maps to e , while τ maps to τ .

34.9 The complete join with the original spectral coupling

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Its two inputs are the delivered Toda equation (28), proved here as (TVB.32), and the complete original-metric exterior identity E8. The latter source is retained in the cyclic companion at commit 161089942cce70a09fbc441d08ba4d36c1fd810c. We reproduce the finite identity and its typed coupling map here to make the join complete; no novelty or limiting estimate is claimed.

All quantities in this section are at the original observation $\theta = 0$, with $N \geq q$. Write $A^\sharp = G_N^{-1} A^* G_N$ and $\mathcal{H}_N = A^\sharp + A - kI$, acting on the original (C, G_N) . Let C_+ be the full generalized spectral subspace with $\Re \lambda > c$, let $C_- = C_+^{\perp_{G_N}}$, and let $P : C \rightarrow C$ be their orthogonal projection onto C_+ . Suppose first $C_+ \neq 0$. Its trace L from (TVB.35) is positive, so $\epsilon_N > 0$. Choose G_N -unit eigenvectors u_+, u_- of $\mathcal{H}_N$ at $\epsilon_N, -\epsilon_N$; selfadjointness makes them orthogonal. They specify only the coefficient injection

$$E : \mathbb{C}^2 \longrightarrow C, \quad E(z_+, z_-) = z_+ u_+ + z_- u_-, \quad E^\sharp E = I_2, \quad M = E^\sharp P E. \quad (\text{TVB.39})$$

Both M and $I_2 - M$ are positive semidefinite, since their quadratic forms are respectively the squared original norms of PEz and $(1 - P)Ez$. In the original orthogonal decomposition define

$$B = A|_{C_+}, \quad D^{\text{op}} = (1 - P)A|_{C_-}, \quad C = PA|_{C_-} : C_- \longrightarrow C_+.$$

Invariance of C_+ gives $A = \begin{pmatrix} B & C \\ 0 & D^{\text{op}} \end{pmatrix}$. Its adjoint has the conjugate-transpose blocks in the original restricted metrics, so $P\mathcal{H}_N|_{C_-} = C$ exactly.

Put $a_+ = \|Pu_+\|^2$, $a_- = \|Pu_-\|^2$, and $z = \langle Pu_+, Pu_- \rangle_{G_N}$. The compression M has these two diagonal entries and off-diagonal entries $z, \bar{z}$. The original rank-two spectral expansion gives $L = \epsilon_N(a_+ - a_-)$ and

$$\begin{aligned} \|C\|_{\text{HS}, G_N}^2 &= \text{Tr}(P\mathcal{H}_N^2 P) - \text{Tr}((P\mathcal{H}_N P)^2) \\ &= \epsilon_N^2(a_+ + a_- - a_+^2 - a_-^2 + 2|z|^2). \end{aligned}$$

The cross trace in the second equation is $|\langle v, w \rangle|^2$ for the two rank-one maps $v\langle v, \cdot \rangle$ and $w\langle w, \cdot \rangle$, as follows by composing them. Subtract the displayed norm and $\epsilon_N^2(a_+ - a_-)^2$ from ϵ_N^2 ; direct expansion proves

$$\epsilon_N^2 - L^2 = \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}. \quad (\text{TVB.40})$$

Both determinants are nonnegative by the quadratic forms after (TVB.39). Adding (TVB.40) to (TVB.32) cancels precisely the intermediate ϵ_N^2 and gives all four nonnegative losses

$$\boxed{\alpha_{N+1} \sinh^2 s_N - L^2 = \alpha_{N+1}(e^{t_N} - \cosh s_N)^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}.} \quad (\text{TVB.41})$$

Every adjoint, projection, norm and coefficient is for the same original minimum metric as the Toda volumes.

The spectral projector has an exact map to the coupling. Factor the minimal polynomial of A as $m_+ m_-$ over the selected and remaining full generalized blocks, with their original exponents. Bezout polynomials a, b with $am_+ + bm_- = 1$ give $Q = b(A)m_-(A)$: it is identity on each selected block and zero on each other block. Thus $Q^2 = Q$, $AQ = QA$, and in the preceding orthogonal decomposition $Q = \begin{pmatrix} I & Y \\ 0 & 0 \end{pmatrix}$ for a unique map $Y : C_- \rightarrow C_+$. Block multiplication proves

$$\mathcal{S} : \text{Hom}(C_-, C_+) \longrightarrow \text{Hom}(C_-, C_+), \quad \mathcal{S}(Y) = BY - YD^{\text{op}} = C, \quad Q - P = \begin{pmatrix} 0 & Y \\ 0 & 0 \end{pmatrix}. \quad (\text{TVB.42})$$

This morphism is invertible with its full original nilpotent parts:

$$\mathcal{S}^{-1}(Z) = \int_0^\infty e^{-tB} Z e^{tD^{\text{op}}} dt. \quad (\text{TVB.43})$$

Indeed every selected eigenvalue has real part greater than c and every remaining one has real part at most c . On a full block at λ the first exponential is exactly $e^{-t\lambda} \sum_{j < \ell_\lambda} (-t)^j (B - \lambda I)^j / j!$, with its corresponding projector, and the second has the same formula with positive t on its remaining blocks. Summing these finite expressions gives a polynomial times a strictly decaying exponential for their product in (TVB.43). This proves convergence in the original operator norm. Differentiating the integrand gives $-B e^{-tB} Z e^{tD^{\text{op}}} + e^{-tB} Z e^{tD^{\text{op}}} D^{\text{op}}$. Its integral has lower endpoint Z and zero upper endpoint, proving the inverse identity. If $BY - Y D^{\text{op}} = 0$, the same product with $Z = Y$ is constant and decays to zero, hence $Y = 0$. This proves uniqueness as well. The space C_- is nonzero: otherwise $C_+ = C$ and $L = \text{Tr } \mathcal{H}_N = 0$, contradicting $L > 0$. Together with $C_+ \neq 0$ this makes the unit sphere in $\text{Hom}(C_-, C_+)$ nonempty. The actual smallest singular value $s_{B,D} > 0$ of $\mathcal{S}$ in the original Hilbert-Schmidt pairing exists by compactness of its unit sphere and injectivity. Consequently $\|C\|_{\text{HS}, G_N} \geq s_{B,D} \|Q - P\|_{\text{HS}, G_N}$. No bound on $s_{B,D}$ independent of the arithmetic inputs is used.

Retaining the phase and coupling in (TVB.41) gives the finite strengthening

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{\sqrt{L^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2}}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.44})$$

It follows by subtracting only the nonnegative imbalance and compression-determinant terms and applying the same inverse sinh as in (TVB.35). Summation over the disjoint indices $N = N_0 + 2j$ has the same exact endpoints as (TVB.36), with this degree-dependent square root in each summand. If $C_+ = 0$, use (TVB.32) directly; no positive-eigenline compression is asserted when $\epsilon_N = 0$. At $N = q - 1$, the scalar term instead uses the separate endpoint expression (TVB.38). These identities supply no asymptotic estimate and change no arithmetic class.

34.10 Crosswalk and exact continuing arithmetic quantity

CV.1–CV.3 are (TVB.22)–(TVB.24), with the additional proved identification $D_N^{\text{CV}} = \mathfrak{B}_m / \mathfrak{D}_{N+1}$. CV.4–CV.5 become (TVB.27)–(TVB.29), where the whole complex cross term is $i\omega_N(\sigma - \ell'_N)$. CV.6 is the minimum lift (TVB.14) and its determinant-line map after (TVB.15). CV.10–CV.10a are (TVB.30)–(TVB.31), now with the exact two-loss identity (TVB.34). CV.11–CV.12 are precisely (TVB.37)–(TVB.38). The new operator refinement is (TVB.20)–(TVB.21): it computes both individual rank-one curvature costs, including their shared retained quantity $\alpha_{N+1} \delta_N$.

For an actual quartet polynomial the exterior multiplicities and their lower trace L remain those of that exact polynomial; for a larger packet they remain a lower contribution to its actual full trace. All admission thresholds here use its actual $q = \deg \chi$. The finite formulas prove an upper estimate by the exact source and relation determinants and prove the telescope (TVB.36). They do not estimate the growth of α_{N+1} or the joint determinant ratio as k and N grow. That continuing arithmetic question is the original expression (TVB.27), equivalently the positive-minor source norm in (TVB.28)–(TVB.31), with both losses (TVB.32) available without changing any of its inputs.

35 The original theta input and certified infinite-series tails

This continuation supplies the omitted-tail and ball-arithmetic certificate for the analytic input of Section 9 of *Source-boundary volume dynamics for the arithmetic control*. Every theta coefficient, Fourier factor, conjugation, and complex branch used below is specified. The original seed has total

mass $M_1(0)$; the measure is never divided by that mass. The result computes analytic input of the source determinant flow. Its exact connection with a finite zero packet is given below by Mellin division and the multiplier $h(1/2 + it)$.

35.1 The original theta function and its Mellin transform

Put $D = -z\partial_z$, let $\mathcal{S}_{\pi/4} = \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \pi/4\}$, and use the restriction of the principal logarithm to this sector. Define

$$\vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{TI.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n=1}^{\infty} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{TI.2})$$

On a compact subset of the sector, each differentiated summand is bounded by a polynomial in n times e^{-an^2} with $a > 0$. Consequently both series define holomorphic functions there, and all the displayed derivatives are termwise valid.

Lemma 35.1 (The exact theta inversion). *On $\mathcal{S}_{\pi/4}$,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{TI.3})$$

For each $0 \leq \alpha < \pi/4$, every integer $j \geq 0$, and every real $A > 0$, $D^j f_0(z) = O(|z|^{-A})$ as $|z| \rightarrow \infty$ and $D^j f_0(z) = O(|z|^A)$ as $|z| \rightarrow 0$, uniformly for $|\arg z| \leq \alpha$.

Proof. For $x > 0$, the Fourier transform with kernel $e^{-2\pi i \xi t}$ of $e^{-\pi x^2 t^2}$ is $x^{-1}e^{-\pi \xi^2/x^2}$. For completeness, the value at $\xi = 0$ follows from $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$, whose square is evaluated in polar coordinates. Differentiation in ξ and integration by parts give the differential equation $I'(\xi) = -2\pi \xi I(\xi)/x^2$; the stated transform is its solution with the stated initial value. Periodize the Gaussian over integer translates. The resulting smooth periodic function has the absolutely convergent Fourier series with coefficients $x^{-1}e^{-\pi n^2/x^2}$. Evaluation at zero gives $\vartheta(x) = x^{-1}\vartheta(1/x)$. Both sides are holomorphic in the connected sector, so the identity theorem gives the first equation in (TI.3).

For the linear involution $(Rf)(z) = z^{-1}f(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this identity to $\vartheta = R\vartheta$ proves the second equation. The real coefficients in (TI.2) give the third. On a closed subsector, $\Re(z^2) \geq |z|^2 \cos(2\alpha) > 0$; the series and each derivative therefore have Gaussian decay at infinity, with a fixed polynomial factor. The second equation in (TI.3), differentiated using $DR = R(1 - D)$, transports that bound to zero and proves all the endpoint assertions. $\square$

For functions with these endpoint bounds set

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{TI.4})$$

The Mellin integral is entire. When $\Re s > 1$, the sum of the absolute integrals of the terms in (TI.2) is finite, because it is bounded by a constant depending on s times $\sum n^{-\Re s}$. Thus

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2}\zeta(s) (4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)) \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) =: g(s). \end{aligned} \quad (\text{TI.5})$$

The first line contains the factor 2 of (TI.2) and the factor $1/2$ from the substitution $u = \pi n^2 x^2$. The second follows from $\Gamma(z+1) = z\Gamma(z)$, which follows by integration by parts. Equation (TI.5) gives the entire continuation of its right side. In particular $g = 2\zeta$ with the original convention in the source note. Substituting $x \mapsto 1/x$ in (TI.4), using (TI.3), proves

$$g(1-s) = g(s), \quad \overline{g(\bar{s})} = g(s). \quad (\text{TI.6})$$

35.2 Mellin division, the dagger sign, and the tilted observation

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a monic finite product of actual zeros of g , with the selected orders at most their actual orders and with the selected multiset invariant under $\rho \mapsto 1 - \bar{\rho}$. The complete-order packets used in the source note satisfy these conditions. Write $d = \deg h$ and

$$v_h = g/h, \quad h^{\dagger}(s) = \overline{h(1 - \bar{s})}, \quad v_h^{\dagger}(s) = \overline{v_h(1 - \bar{s})}. \quad (\text{TI.7})$$

Removable singularities are filled in. The leading coefficient of $h^{\dagger}$ is $(-1)^d$, and its zero multiset is that of h ; therefore

$$h^{\dagger} = (-1)^d h, \quad v_h^{\dagger} = (-1)^d v_h. \quad (\text{TI.8})$$

Proposition 35.2 (Construction without an endpoint assumption). *For every $\sigma \in \mathbb{R}$ the integral*

$$F_h(z) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} v_h(s) z^{-s} ds \quad (\text{TI.9})$$

converges on the sector, is independent of σ , defines a holomorphic function there, and has the rapid endpoint bounds in the first lemma on each closed subsector. It satisfies

$$\mathcal{M}F_h = v_h, \quad h(D)F_h = f_0, \quad F_h(1/\bar{z}) = (-1)^d \bar{z} \overline{F_h(z)}. \quad (\text{TI.10})$$

Proof. Rotate the Mellin integral of f_0 to the ray $e^{i\beta}\mathbb{R}_{>0}$, where $|\beta| < \pi/4$. The endpoint estimates make both closing arcs vanish, and give

$$g(s) = e^{i\beta s} \int_0^{\infty} f_0(e^{i\beta} x) x^{s-1} dx.$$

For σ in any compact interval, the absolute integral on the right is uniformly bounded. Choosing the sign of β to be that of t proves $|g(\sigma + it)| \leq C e^{-|\beta||t|}$. For sufficiently large $|t|$, the explicit finite product h is bounded below by $c(1 + |t|)^d$ on the same strip. On the remaining compact rectangle, v_h is bounded because its singularities are removable. These facts give the same exponential estimate for v_h , with a different constant. It is valid for every $|\beta| < \pi/4$ and every compact interval of σ .

In (TI.9), choose $|\beta| > |\arg z|$. Absolute convergence and local uniform convergence of every differentiated integral follow. Horizontal contour segments tend to zero by the same estimate, proving independence of σ . Multiplication by s^j computes $D^j F_h$; moving σ arbitrarily far to the right or left gives respectively the asserted infinity or zero bounds.

The change $x = e^y$ turns (TI.9) on the line $\Re s = \sigma$ into Fourier inversion for $e^{\sigma y} F_h(e^y)$. The exponentially decreasing function $v_h(\sigma + it)$ and its derivatives have the required integrability, and inversion gives $\mathcal{M}F_h = v_h$. The Fourier inversion and Plancherel conventions here can be verified by inserting $e^{-\varepsilon t^2}$ in the inverse integral: Fubini gives convolution with the Gaussian approximate identity of integral one; letting $\varepsilon \downarrow 0$ gives inversion, and applying the same calculation to the inner product gives the factor $1/(2\pi)$ in Plancherel. Thus no change of Fourier normalization is implicit. Multiplication by $h(s)$ in (TI.9) is $h(D)$ on the source. Fourier inversion of (TI.5) proves $h(D)F_h = f_0$.

For the conjugate-linear involution $(\mathcal{R}F)(x) = x^{-1} \overline{F(1/\bar{x})}$, substitution gives $\mathcal{M}(\mathcal{R}F)(s) = \overline{\mathcal{M}F(1 - \bar{s})}$. Equations (TI.8) and Mellin uniqueness imply $\mathcal{R}F_h = (-1)^d F_h$ on the positive axis. Both sides extend holomorphically to the sector as $z^{-1} \overline{F_h(1/\bar{z})}$ and $(-1)^d F_h(z)$. Conjugating the resulting identity gives the last equation in (TI.10). $\square$

Here are the exact Hilbert-space maps. The change of variables

$$U : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}, dy), \quad (UF)(y) = e^{y/2} F(e^y) \quad (\text{TI.11})$$

is unitary, with inverse $u \mapsto [x \mapsto x^{-1/2} u(\log x)]$. On the rapid sector functions constructed above define, for real $|\theta| < \pi/2$,

$$(E_{\theta}F)(y) = e^{(y+i\theta/2)/2} F(e^{y+i\theta/2}). \quad (\text{TI.12})$$

This map has codomain $L^2(\mathbb{R}, dy)$; it is the analytic translation of UF on this specified domain. Its norm is not asserted to equal the untilted norm. Shifting the y contour by $i\theta/2$ in (TI.4), with the closing vertical edges vanishing by the proved endpoint bounds, gives

$$\mathcal{F}_+(E_\theta F_h)(t) = e^{\theta t/2} v_h(1/2 + it). \quad (\text{TI.13})$$

Indeed $y = z - i\theta/2$ contributes $e^{ity} = e^{\theta t/2} e^{itz}$. In the source expression the factor $e^{i\theta/4}$ remains present.

Define the original, unscaled density and its mass by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad \mu_h = M_h(0). \quad (\text{TI.14})$$

The exponential estimate above proves that the integral defining M_h and each of its derivatives converge locally uniformly on $|\Re\theta| < \pi/2$: for a compact substrip choose $2|\beta| > \sup |\Re\theta|$. By (TI.11)–(TI.13) and Plancherel,

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2} x)|^2 dx = 2 \int_1^\infty |F_h(e^{i\theta/2} x)|^2 dx \quad (\theta \text{ real}). \quad (\text{TI.15})$$

For the second equality the last identity in (TI.10) gives

$$F_h(e^{i\theta/2}/x) = (-1)^d e^{-i\theta/2} x \overline{F_h(e^{i\theta/2} x)}.$$

After taking absolute squares, the factor x^2 cancels the Jacobian of $x \mapsto 1/x$. The dagger sign is retained before that operation. Dagger stability alone gives this endpoint identity. If the selected multiset is additionally stable under ordinary conjugation, then $v_h(\bar{s}) = \overline{v_h(s)}$, so $w_h(-t) = w_h(t)$ and $M_h(-\theta) = M_h(\theta)$. This proves the precise extra symmetry needed for evenness.

The typed relation between seed and packet input is also explicit:

$$F_h \xrightarrow{h(D)} f_0 \quad v_h(1/2 + it) \xrightarrow{h(1/2 + it)} g(1/2 + it), \quad w_1(t) = |h(1/2 + it)|^2 w_h(t). \quad (\text{TI.16})$$

The arrow in the first display means the linear differential map on the rapid sector-function space; (TI.10) and (TI.13) prove its commutation with the transform. The scalar equality of densities holds even at selected line zeros by continuity; it does not require division of a measured zero value by zero.

35.3 The complete double series and an explicit omitted tail

For $h = 1$ we have $F_1 = f_0$. Set $c_1 = -6$, $c_2 = 4$ and, for real θ first,

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}), \quad \alpha_{rs} = r + s + \frac{1}{2}. \quad (\text{TI.17})$$

For $\Re C > 0$ define

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}. \quad (\text{TI.18})$$

The power uses the holomorphic logarithm of the right half-plane, real on its positive axis. To check (TI.18), first substitute $u = Cx^2$ for positive real C . Both sides extend holomorphically to the right half-plane: the integral by dominated convergence and the right side by the same continued incomplete gamma integral with its ray deformable inside that half-plane. The identity theorem then gives (TI.18) there with the stated branch.

Theorem 35.3 (The retained factor and the locally uniform tail). *For every θ with $|\Re\theta| < \pi/2$,*

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(\alpha_{rs}, C_{mn}(\theta))}{C_{mn}(\theta)^{\alpha_{rs}}}. \quad (\text{TI.19})$$

Let $M_{1,J}$ be exactly this sum with $1 \leq m, n \leq J$, for an integer $J \geq 0$. Fix $0 \leq \theta_0 < \pi/2$ and $B \geq 0$, and put

$$d_0 = \pi e^{-B} \cos \theta_0, \quad L = J + 1, \quad \rho_a = e^{-d_0(2a+1)}, \quad (\text{TI.20})$$

$$H_p(a, d_0) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{matrix} j \\ b \end{matrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{TI.21})$$

$$A_a(d_0, B) = 6\pi e^B H_2(a, d_0) + 4\pi^2 e^{2B} H_4(a, d_0). \quad (\text{TI.22})$$

Here $\left\{ \begin{matrix} j \\ b \end{matrix} \right\}$ is determined by $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$, $\left\{ \begin{matrix} j \\ 0 \end{matrix} \right\} = 0$ for $j > 0$, $\left\{ \begin{matrix} 0 \\ b \end{matrix} \right\} = 0$ for $b > 0$, and $\left\{ \begin{matrix} j+1 \\ b \end{matrix} \right\} = b \left\{ \begin{matrix} j \\ b \end{matrix} \right\} + \left\{ \begin{matrix} j \\ b-1 \end{matrix} \right\}$, with out-of-range indices zero. For every $|\Re\theta| \leq \theta_0$ and $|\Im\theta| \leq B$,

$$|M_1(\theta) - M_{1,J}(\theta)| \leq \mathcal{E}_J(\theta_0, B) := 16A_1(d_0, B)A_L(d_0, B)I_8(d_0(1+L^2)). \quad (\text{TI.23})$$

Every quantity on the right is explicit and positive. It tends to zero as $J \rightarrow \infty$. In particular (TI.23) supplies the requested locally uniform real-tilt bound by taking $B = 0$.

Proof. For real θ , substitute (TI.2) into the last expression in (TI.15). A product contributes $2 \cdot 2 \cdot 2 = 8$ before integration: the endpoint factor 2 and the two original theta coefficients 2. Equation (TI.18) contributes $1/2$. The remaining factor is precisely 4, and the powers of the two dilations give $e^{i\theta r} e^{-i\theta s}$. This establishes the coefficients and phases in (TI.19), subject to the absolute convergence now proved.

For integers $a \geq 1$, $p \geq 0$, and $x \geq 1$,

$$\begin{aligned} \sum_{n \geq a} n^p e^{-d_0 n^2 x^2} &\leq e^{-d_0 a^2 x^2} \sum_{j \geq 0} (a+j)^p e^{-d_0(2a+1)j} \\ &= e^{-d_0 a^2 x^2} H_p(a, d_0). \end{aligned}$$

The inequality uses $(a+j)^2 - a^2 = 2aj + j^2 \geq (2a+1)j$ and $x^2 \geq 1$. For the equality expand $(a+j)^p$ by the binomial theorem and use $j^r = \sum_b \left\{ \begin{matrix} r \\ b \end{matrix} \right\} (j)_b$. That last identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$. Thus (TI.21) is a finite positive rational function, not an unevaluated infinite sum.

For complex θ in the rectangle, let $f^\pm(x) = f_0(e^{\pm i\theta/2} x)$ and let $f_J^\pm$ contain only $1 \leq n \leq J$ in the original series. Since

$$\Re(\pi n^2 e^{\pm i\theta}) = \pi n^2 e^{\mp \Im\theta} \cos(\Re\theta) \geq d_0 n^2,$$

and $|e^{\pm i\theta}| \leq e^{rB}$, the positive bound just proved implies

$$\begin{aligned} |f^\pm(x)|, |f_J^\pm(x)| &\leq 2A_1(d_0, B) x^4 e^{-d_0 x^2}, \\ |f^\pm(x) - f_J^\pm(x)| &\leq 2A_L(d_0, B) x^4 e^{-d_0 L^2 x^2}. \end{aligned}$$

Here $x^2 \leq x^4$ retains a bound for the original -6 term; no coefficient of the exact series is changed. These inequalities prove absolute integrability of the full expanded double series, locally uniformly on the rectangle. They also prove

$$2 \int_1^\infty |f^+(x)f^-(x) - f_J^+(x)f_J^-(x)| dx \leq 16A_1(d_0, B)A_L(d_0, B) \int_1^\infty x^8 e^{-d_0(1+L^2)x^2} dx,$$

because the product difference is $(f^+ - f_J^+)f^- + f_J^+(f^- - f_J^-)$. The right side is (TI.23). For fixed $d_0 > 0$, formula (TI.21) bounds $H_p(L, d_0)$ by a fixed constant times $(1 + L)^p$: its denominators are bounded away from zero uniformly for $L \geq 1$. Moreover $I_8(b) \leq e^{-b} \int_1^\infty x^8 e^{-d_0(x^2-1)} dx$ for $b \geq d_0$. Hence (TI.23) tends to zero exponentially in L^2 .

For complex θ , $C_{mn}(\theta)$ stays in the right half-plane by the displayed inequality. The integral $2 \int_1^\infty f^+(x)f^-(x) dx$ is holomorphic on the whole strip. It equals M_1 on the real interval by (TI.15) and conjugation of f_0 . Holomorphic uniqueness proves equality on the strip. The same locally uniform bounds justify (TI.19) there and finish the proof. $\square$

The tail controls differentiated moments as well. If the closed disc $\{z : |z - \theta| \leq r\}$ lies in the rectangle of the theorem, Cauchy's integral formula gives, for every integer $j \geq 0$,

$$|M_1^{(j)}(\theta) - M_{1,J}^{(j)}(\theta)| \leq \frac{j!}{r^j} \mathcal{E}_J(\theta_0, B). \quad (\text{TI.24})$$

The bound includes arbitrary finite derivative orders used by a Hankel determinant. It does not assert a useful estimate when the derivative order or determinant dimension itself grows without bound.

35.4 The second moment and the first tensor coefficient

Let $H = (D - 1/2)f_0$. With $y = \pi n^2 x^2$ and $P(y) = 8y^2 - 12y$, literal differentiation gives

$$\begin{aligned} (D - 1/2)(P(y)e^{-y}) &= (-2yP'(y) + 2yP(y) - P(y)/2) e^{-y} \\ &= (16y^3 - 60y^2 + 30y)e^{-y}. \end{aligned} \quad (\text{TI.25})$$

Thus

$$H(x) = \sum_{n \geq 1} \sum_{r=1}^3 d_r (\pi n^2)^r x^{2r} e^{-\pi n^2 x^2}, \quad (d_1, d_2, d_3) = (30, -60, 16). \quad (\text{TI.26})$$

The unitary map (TI.11) satisfies $U(D - 1/2)F = -(UF)'$ by differentiation. Integration by parts in the Fourier transform therefore gives $\mathcal{F}_+(UH)(t) = it g(1/2 + it)$. It follows, with the original $1/(2\pi)$ factor, that

$$M_1''(0) = \frac{1}{2\pi} \int_{\mathbb{R}} t^2 |g(1/2 + it)|^2 dt = \int_0^\infty |H(x)|^2 dx = 2 \int_1^\infty H(x)^2 dx. \quad (\text{TI.27})$$

For the last equality, $DR = R(1 - D)$ and $Rf_0 = f_0$ give $RH = -H$, that is $H(1/x) = -xH(x)$. Its square gives precisely the same Jacobian cancellation as before, with the new minus sign retained before squaring.

Let H_J restrict (TI.26) to $n \leq J$ and put $T_J = 2 \int_1^\infty H_J(x)^2 dx$. Then

$$T_J = \sum_{m,n=1}^J \sum_{r,s=1}^3 d_r d_s (\pi m^2)^r (\pi n^2)^s \frac{\Gamma(r+s+1/2, \pi(m^2+n^2))}{(\pi(m^2+n^2))^{r+s+1/2}}. \quad (\text{TI.28})$$

There is no coefficient 4 in this last expression: (TI.26) already contains its fully differentiated coefficients, and the endpoint factor 2 cancels the $1/2$ in (TI.18). Set

$$B_a = 30\pi H_2(a, \pi) + 60\pi^2 H_4(a, \pi) + 16\pi^3 H_6(a, \pi). \quad (\text{TI.29})$$

The proof of (TI.23), now with degree six in x , gives

$$|M_1''(0) - T_J| \leq \mathcal{T}_J := 4B_1 B_{J+1} I_{12}(\pi(1 + (J+1)^2)). \quad (\text{TI.30})$$

In detail, $|H|, |H_J| \leq B_1 x^6 e^{-\pi x^2}$ and $|H - H_J| \leq B_{J+1} x^6 e^{-\pi(J+1)^2 x^2}$; apply $|H^2 - H_J^2| \leq |H - H_J|(|H| + |H_J|)$ and the factor 2 of (TI.27).

For tensor degree $k \geq 1$, let $m_k = w_1^{*k}$ with its total mass μ_1^k , where $\mu_1 = M_1(0)$. Absolute integrability and Fubini give $Z_k(\theta) = \int e^{\theta u} m_k(u) du = M_1(\theta)^k$. The function w_1 is even by (TI.6), so $Z'_k(0) = M'_1(0) = 0$. The degree-zero monic source polynomial is 1, of squared norm $h_0 = \mu_1^k$, and its monic degree-one orthogonal polynomial in the original sum variable u is u , of squared norm $h_1 = Z''_k(0) = k\mu_1^{k-1}M''_1(0)$. Therefore the first recurrence coefficient is exactly

$$a_1^{(k)} = \frac{h_1}{h_0} = k \frac{M''_1(0)}{M_1(0)}. \quad (\text{TI.31})$$

The quotient of norms in (TI.31) is the coefficient in the monic three-term recurrence. It leaves both norms and the measure mass specified, and performs no replacement of m_k by a probability law.

35.5 Certified numerical enclosures and their executable scope

The accompanying script `toda_theta_input_tail_check_20260912.py` evaluates (TI.19), (TI.28), (TI.23), and (TI.30) at $J = 8$ with `python-flint 0.9.0`, using Arb real balls and Acb rectangular complex balls at 256-bit precision and one arithmetic thread. All integer and rational input, including $\theta = 1/10$, is supplied exactly; π , exponentials, powers, and incomplete gamma functions are enclosed by those ball operations. The incomplete gamma call has argument C and parameter α , that is `C.gamma_upper(alpha)`. Its branch is the one in (TI.18). The documented ball and endpoint semantics used here are in [the Arb interface](#) and [the Acb interface](#).

The script adds a symmetric error ball whose radius is the upward rounded upper endpoint of the proved tail bound. Thus the result encloses the infinite sum, rather than only its finite arithmetic. The finite real-tilt sum is real by the proved conjugation identity; the script additionally checks that its imaginary ball contains zero. The positive real mass enclosure is then used in interval division for (TI.31). The following are rational decimal intervals:

$$\begin{aligned} M_1(0) &\in [1.2790072478464851404795335922671932744916, \\ &\quad 1.2790072478464851404795335922671932744919], \\ M_1(1/10) &\in [1.3459071784582497969682205923576298051604, \\ &\quad 1.3459071784582497969682205923576298051607], \\ M''_1(0) &\in [13.0555493025705584353926846581231936824343, \\ &\quad 13.0555493025705584353926846581231936824346], \\ a_1^{(k)}/k &\in [10.2075647534857216888657612522375551749964, \\ &\quad 10.2075647534857216888657612522375551749967]. \end{aligned}$$

Every width is $3 \cdot 10^{-40}$. The computed positive tail bounds satisfy, respectively,

$$\mathcal{E}_8(0,0) < 7 \cdot 10^{-107}, \quad \mathcal{E}_8(1/10,0) < 3 \cdot 10^{-106}, \quad \mathcal{T}_8 < 4 \cdot 10^{-103}. \quad (\text{TI.32})$$

The receipt preserves both the displayed decimal intervals and exact rational dyadic lower and upper endpoints of each resulting ball. Its decimal endpoints are generated by outward rounding at 40 decimal places, with one additional unit on each side, and are subsequently checked by rigorous ball comparisons. Normal and optimized execution produce identical mathematical records. The requested negative-control mode inserts the false inequality $a_1^{(k)}/k < 1$ and must raise an error; it uses an explicit checked exception rather than a Python assertion, so optimization cannot remove that check.

The numerical certificate uses the standard correctness contract of Arb/Acb ball arithmetic together with the written infinite-tail proof. It is not inferred from agreement of two numerical precisions. Its output concerns the three specified infinite analytic-seed quantities and their recurrence

ratio. For $h = 1$ the cyclic arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the actual quotient map sends each polynomial to zero, while the positive seed source norm and (TI.31) remain. For a selected packet h , the precise passage from the present input to w_h is (TI.16), obtained from the globally defined Mellin division g/h . No packet determinant or bound as k tends to infinity is certified by these four scalar intervals.

36 An exact comparison of equality, volume contraction, and its discarded square

This calculation extends the positive Gaussian calibration of E35–E36 through two further source degrees. Every measure, phase, coefficient and quotient basis is specified. It is a calibration of the same constructions; the roots below are declared coefficients, with no assertion that they are zeros of the arithmetic function.

Fix $\mu > 0, \sigma > 0, k = 1$ and $c = 1/2$. Put

$$d\nu(u) = \frac{\mu}{\sqrt{2\pi\sigma}} e^{-u^2/(2\sigma^2)} du, \quad x = S - c, \quad \chi(S) = x^2 - \sigma^2, \quad C = \mathbb{C}[S]/(\chi). \quad (\text{TC1})$$

The original source pairing is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) d\nu(u)$. We use both the original quotient basis $(1, S)$ and the displayed basis $(1, x)$. The exact coordinate matrix, sending columns in the latter basis to columns in the former, is

$$L_{\text{coord}} = \begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix}, \quad A_S = \begin{pmatrix} 0 & \sigma^2 - c^2 \\ 1 & 2c \end{pmatrix}, \quad A_x = \begin{pmatrix} c & \sigma^2 \\ 1 & c \end{pmatrix}, \quad A_S L_{\text{coord}} = L_{\text{coord}} A_x. \quad (\text{TC2})$$

Here A_x represents multiplication by S in the basis $(1, x)$; multiplication by x is $A_x - cI$. Thus $G_{N,x} = L_{\text{coord}}^* G_{N,S} L_{\text{coord}}$ and $K_{N,S} = L_{\text{coord}} K_{N,x} L_{\text{coord}}^*$. In particular $\det L_{\text{coord}} = 1$ preserves the specified determinant, while the matrices and source polynomials retain their full coordinate transformations.

Integration by parts, with the Gaussian boundary term zero, gives $\int u^{2j} d\nu = \mu(2j-1)!!\sigma^{2j}$ and all odd moments zero: indeed the zeroth integral is μ , and differentiating $u^{2j-1}e^{-u^2/(2\sigma^2)}$ gives the recurrence $m_{2j} = (2j-1)\sigma^2 m_{2j-2}$. The first four monic source polynomials and their full squared norms are consequently

j	$p_j(S)$	ω_j
0	1	μ
1	x	$\mu\sigma^2$
2	$x^2 + \sigma^2$	$2\mu\sigma^4$
3	$x^3 + 3\sigma^2 x$	$6\mu\sigma^6$

(TC3)

For example $p_2(c + iu) = -(u^2 - \sigma^2)$ and $p_3(c + iu) = -i(u^3 - 3\sigma^2 u)$. Substitution of the moments proves every pairwise orthogonality and every displayed norm; the phases -1 and $-i$ are retained before taking the pairings.

In the basis $(1, x)$ the quotient columns $b_j = [p_j]_\chi$ are

$$b_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 2\sigma^2 \\ 0 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 0 \\ 4\sigma^2 \end{pmatrix}. \quad (\text{TC4})$$

For completeness, the minimum-source metric is obtained by writing a source polynomial as $\sum_{j=0}^N z_j p_j$. Its norm is $\sum \omega_j |z_j|^2$ and its quotient is $\sum b_j z_j$. The constrained minimizer is $z_j = \omega_j^{-1} b_j^* G_{N,x} v$, where $K_{N,x} = \sum b_j b_j^* / \omega_j$ and $G_{N,x} = K_{N,x}^{-1}$. Indeed these coefficients have quotient v , and their inner product with every quotient-kernel coefficient vector is zero. Pythagoras then

proves uniqueness and minimum norm $v^*G_{N,x}v$. Set $V_N = \det G_{N,x} = \det G_{N,S}$. Directly from (TC3)–(TC4),

$$\begin{aligned} G_{1,x} &= \text{diag}(\mu, \mu\sigma^2), \\ G_{2,x} &= \text{diag}(\mu/3, \mu\sigma^2), & (V_1, V_2, V_3) &= \mu^2\sigma^2(1, 1/3, 1/11). \\ G_{3,x} &= \text{diag}(\mu/3, 3\mu\sigma^2/11), \end{aligned} \quad (\text{TC5})$$

This is also exactly the source/relation determinant formula. On the original vertical line $\chi(c + iu) = -(u^2 + \sigma^2)$, so its zeroth relation norm and its next monic relation norm are

$$\nu_0 = \int (u^2 + \sigma^2)^2 d\nu = 6\mu\sigma^4, \quad \nu_1 = \int u^2(u^2 + \sigma^2)^2 d\nu = 22\mu\sigma^6. \quad (\text{TC6})$$

The two relation columns $\chi, \chi x$ are orthogonal by parity. For $n = 2, 3, 4$, the source Gram determinant is $\mathfrak{D}_n = \prod_{j < n} \omega_j$; the corresponding relation determinants are $\mathfrak{B}_0 = 1, \mathfrak{B}_1 = \nu_0, \mathfrak{B}_2 = \nu_0\nu_1$. These identities follow by the triangular monic changes of basis. They give $\mathfrak{D}_{N+1}/\mathfrak{B}_{N-1} = V_N$ for each $N = 1, 2, 3$, including the literal empty determinant at $N = 1$.

Let $H_N = A_x + G_{N,x}^{-1}A_x^*G_{N,x} - 2cI$. Multiplication of the displayed matrices yields

$$H_1 = \begin{pmatrix} 0 & 2\sigma^2 \\ 2 & 0 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0 & 4\sigma^2 \\ 4/3 & 0 \end{pmatrix}, \quad \epsilon_1^2 = 4\sigma^2, \quad \epsilon_2^2 = 16\sigma^2/3. \quad (\text{TC7})$$

Each H_N is self-adjoint in its stated positive metric; its squared matrix is the corresponding scalar multiple of I . Thus these are the squares of its operator norm, or equivalently $\text{Tr}(H_N^2)/2$. The spectral aggregate for A_x is $L_{\text{spec}} = 2\sigma$, since its eigenvalues are $c + \sigma, c - \sigma$. Hence equality $\epsilon_1 = L_{\text{spec}}$ holds at the first admitted degree, whereas $\epsilon_2 > L_{\text{spec}}$ holds at the next. No freedom to choose another metric was used in either calculation.

To compute the auxiliary derivative at the original observation, replace $d\nu$ by $e^{\theta u}d\nu$ for real θ and set $\ell_N(\theta) = \log \det G_N(\theta)$ in either specified quotient basis. The precise real-coordinate moment Grams are $H_{rs}(\theta) = \int u^{r+s}e^{\theta u}d\nu$ and $H_{rs}^\chi(\theta) = \int u^{r+s}(u^2 + \sigma^2)^2e^{\theta u}d\nu$. Their matrices at $-\theta$ are conjugated by $\text{diag}(1, -1, 1, \dots)$ from those at θ . The exact map to original monomials is $S^j = \sum_{r=0}^j \binom{j}{r} c^{j-r} i^r u^r$. On dimension n its triangular determinant is $i^{n(n-1)/2}$, of modulus one. It therefore identifies these determinants with the original source and relation determinants, including the relation multiplier χ . Both determinants are even functions of θ . Their positive determinant ratio is therefore even, giving $\ell'_N(0) = 0$. Also $\sigma_T := \text{Tr}((A_x - cI)/i) = 0$. This proves the vanishing of the phase subtraction at zero tilt; it does not impose evenness at a nonzero tilt.

Finally set $\delta_N = V_N/V_{N-1}$ at the two degrees where both metrics exist, and set $\mathcal{R}_2 = V_1/V_3 = (\delta_2\delta_3)^{-1}$. Equations (TC3) and (TC5) give

$$\delta_2 = 1/3, \quad \delta_3 = 3/11, \quad a_3 = \omega_3/\omega_2 = 3\sigma^2, \quad \mathcal{R}_2 = 11. \quad (\text{TC8})$$

The complete Toda expression and its finite upper bound are respectively

$$a_3(1 - \delta_2)(\delta_3^{-1} - 1) = 16\sigma^2/3, \quad a_3 \frac{(\mathcal{R}_2 - 1)^2}{4\mathcal{R}_2} = 75\sigma^2/11. \quad (\text{TC9})$$

To identify the exact lost term, retain $s = (\log 3 + \log(11/3))/2 = \log(11)/2$ and $t = (\log(11/3) - \log 3)/2$. Then $e^t = \sqrt{11}/3$ and $\cosh s = 6/\sqrt{11}$, whence

$$a_3(e^t - \cosh s)^2 = 49\sigma^2/33 = 75\sigma^2/11 - 16\sigma^2/3. \quad (\text{TC10})$$

Thus the entire gap between the exact arithmetic-construction radius in this specified calibration and its Toda upper expression is precisely the retained imbalance square. The phase square is zero

for the proved parity reason. Both metric degrees, the quotient matrix, the source and relation volumes, and the original mass μ occur explicitly in this comparison.

There is also an explicit comparison with the spectral and orthogonal projectors in the exterior equality proof. In the $(1, x)$ coordinates let $v_+ = (\sigma, 1)^\top$ and $v_- = (-\sigma, 1)^\top$. They are eigenvectors of A_x for $c + \sigma$ and $c - \sigma$. Directly,

$$v_+^* G_{1,x} v_- = 0, \quad v_+^* G_{2,x} v_- = 2\mu\sigma^2/3. \quad (\text{TC11})$$

At $N = 2$ the orthogonal projector to $\mathbb{C}v_+$ and the algebraic spectral projector along $\mathbb{C}v_-$ are respectively

$$P = \frac{v_+ v_+^* G_{2,x}}{v_+^* G_{2,x} v_+} = \begin{pmatrix} 1/4 & 3\sigma/4 \\ 1/(4\sigma) & 3/4 \end{pmatrix}, \quad Q = \frac{A_x - (c - \sigma)I}{2\sigma} = \begin{pmatrix} 1/2 & \sigma/2 \\ 1/(2\sigma) & 1/2 \end{pmatrix}. \quad (\text{TC12})$$

These formulas prove $P^2 = P$, $P^{\dagger G_{2,x}} = P$, $Q^2 = Q$, and $\text{im } P = \text{im } Q = \mathbb{C}v_+$. Their exact kernels are $\ker P = \mathbb{C}(-3\sigma, 1)^\top = v_+^{\perp G_{2,x}}$ and $\ker Q = \mathbb{C}v_-$: direct multiplication proves the two zero values, while $Pv_+ = Qv_+ = v_+$ proves rank one. In particular $Pv_- = v_+/2$. They also retain the change to original coordinates: both matrices are sent to $L_{\text{coord}}(\cdot)L_{\text{coord}}^{-1}$ and paired with $G_{2,S} = L_{\text{coord}}^{-*} G_{2,x} L_{\text{coord}}^{-1}$. For $X = Q - P$, the exact Hilbert–Schmidt pairing gives

$$\|X\|_{\text{HS}, G_{2,x}}^2 = \text{Tr}(G_{2,x}^{-1} X^* G_{2,x} X) = 1/3, \quad \epsilon_2^2 - L_{\text{spec}}^2 = L_{\text{spec}}^2 \|X\|_{\text{HS}, G_{2,x}}^2 = 4\sigma^2/3. \quad (\text{TC13})$$

For example the two diagonal entries of $G_{2,x}^{-1} X^* G_{2,x} X$ are $1/4$ and $1/12$, which proves the trace value without changing its pairing. The last equality follows also from (TC7). Thus the two explicit gaps have different exact source maps: (TC13) measures the spectral/orthogonal projector discrepancy, while (TC10) measures the square discarded by the Toda upper bound. Both maps and both positive quantities are calculated in the original specified metric.

37 The original gamma convolution and the exact arithmetic coefficient maps

37.1 Source, attribution, and original coordinates

This chapter proves the coefficient and cochain maps needed to integrate *Tau Gamma Convolution Descent*, whose original NOTE.tex has SHA256

7cf984d87de405b815250b44454c4ee27c8ab46574da245b377ebc6e49327fa2.

The analytic generating identities (GC.24)–(GC.26) were supplied by the *Zeta Function Literature & Local Corpus Foundation* task on 13 September 2026. The proofs here verify and integrate that supplied join; they do not claim its discovery. The original gamma generating integral and monic family are the existing TG construction, with the classical Meixner–Pollaczek parameters retained.

All scalar source coordinates remain

$$s = \frac{1}{2} + it, \quad S = s_1 + \cdots + s_k = \frac{k}{2} + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GC.1})$$

Here $k \geq 1$ is an integer. We use inner products conjugate-linear in the first argument. For a polynomial, an overline over its name means coefficientwise conjugation. In particular $\overline{P(c + iu)} = \overline{P(c - iu)}$ when c, u are real.

37.1.1 The analytic source and its actual theta boundary

Let $D = -x\partial_x$. Let V be the complex vector space of even Schwartz functions ϕ on $\mathbb{R}$ with $\phi(0) = 0$ and $\int_{\mathbb{R}} \phi(x) dx = 0$. Let $\mathcal{B}$ be the smooth functions on $\mathbb{R}_{>0}$ such that for every $a > 0$ and every integer $j \geq 0$,

$$D^j F(x) = O(x^a) \quad (x \downarrow 0), \quad D^j F(x) = O(x^{-a}) \quad (x \rightarrow \infty).$$

The original theta map and cochain grading are

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1. \quad (\text{GC.2})$$

The sum and its differentiated sums converge at infinity by the Schwartz bounds. Poisson summation gives its corresponding bounds at zero: the zero Fourier coefficient vanishes by $\int \phi = 0$, and the original zero term vanishes by $\phi(0) = 0$. Differentiation preserves both conditions, because $D\phi(0) = 0$ and integration by parts gives $\int D\phi = \int \phi = 0$. Thus $D\Theta = \Theta D$ on these spaces. The rapid bounds make the Mellin integral

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx$$

entire for $F \in \mathcal{B}$ and give $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$, including the vanished endpoint terms.

For clarity the seed and all its constants can be constructed directly. Put

$$\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad f_0 = D(D-1)\vartheta = \Theta\phi_0. \quad (\text{GC.3})$$

Gaussian integration gives $\int 4\pi^2 x^4 e^{-\pi x^2} dx = 3$ and $\int 6\pi x^2 e^{-\pi x^2} dx = 3$ on $\mathbb{R}$, and $\phi_0(0) = 0$, so $\phi_0 \in V$. The Gaussian Fourier transform in the convention $\widehat{f}(y) = \int f(x)e^{-2\pi ixy} dx$ is $e^{-\pi y^2}$: differentiation under the integral and integration by parts give $\widehat{f}' = -2\pi y\widehat{f}$, and the integral at zero is one. Periodizing this Gaussian gives an absolutely convergent Fourier series with those computed coefficients. Evaluating it at zero proves $\vartheta(1/x) = x\vartheta(x)$ for $x > 0$. Holomorphic continuation gives the same identity for $|\arg z| < \pi/4$, where both theta series converge normally on compact subsets.

For $RF(x) = x^{-1}F(1/x)$, direct differentiation gives $DR = R(1-D)$. Hence $D(D-1)$ commutes with R and

$$f_0(1/z) = zf_0(z). \quad (\text{GC.4})$$

On every closed narrower sector, each derivative of f_0 at infinity is bounded by a polynomial times $e^{-c|z|^2}$ for some $c > 0$. Equation (GC.4), differentiated finitely many times, proves rapid decay at zero too. In particular $f_0 \in \mathcal{B}$.

For $\Re s > 1$, absolute convergence permits termwise Mellin integration of $2 \sum_{n \geq 1} \phi_0(nx)$. The change $y = \pi n^2 x^2$ gives

$$\begin{aligned} \mathcal{M}f_0(s) &= \pi^{-s/2} \zeta(s) \{4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)\} \\ &= s(s-1)\pi^{-s/2} \Gamma(s/2) \zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GC.5})$$

The first line retains the theta factor two and the integration Jacobian $1/2$. The gamma recurrence proves the second line. The Mellin integral gives its entire continuation.

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a finite monic packet of actual nontrivial zeros of g , each selected with its complete order. The empty packet is $h = 1$. Write $d = \deg h$ and $v_h = g/h$. At each selected centre, cancelling its exact local factor constructs the analytic value of v_h ; hence v_h is entire. It is nonzero at each selected centre and is not identically zero.

We record the bounds that justify the integrals below. For any closed real interval of σ , any $0 < \beta < \pi/4$, and any $j \geq 0$,

$$|v_h(\sigma + it)| \leq C_{j,\beta,h}(1 + |t|)^{-j} e^{-\beta|t|}. \quad (\text{GC.6})$$

To prove this first for g , rotate its Mellin contour to $z = e^{i\beta}x$ for $t \geq 0$ and to $z = e^{-i\beta}x$ for $t < 0$. The sector estimates after (GC.4) make the two closing arcs vanish, uniformly for the stated σ interval. The rotation contributes $e^{\mp\beta t}$ and a unit-modulus factor depending on σ . Repeated integration by parts along that ray gives every displayed power of $(1 + |t|)^{-1}$; all ray endpoint terms vanish by the same estimates. For $|t|$ larger than twice the moduli of all packet centres relative to that σ interval, monicity gives a lower bound for $|h(\sigma + it)|$ by a positive constant times $|t|^d$. On the remaining compact rectangle the cancelled entire quotient is bounded. This proves (GC.6) for v_h .

For any real σ define

$$F_h(x) = \frac{1}{2\pi} \int_{\mathbb{R}} v_h(\sigma + it) x^{-\sigma - it} dt. \quad (\text{GC.7})$$

Horizontal rectangle estimates from (GC.6) prove independence of σ . Differentiating in x multiplies the integrand by a power of $\sigma + it$ and remains absolutely convergent. Choosing σ arbitrarily large positive at infinity and arbitrarily large negative at zero proves $F_h \in \mathcal{B}$. Fourier inversion in $\log x$, with the displayed $1/(2\pi)$ constant, gives $\mathcal{M}F_h = v_h$. Consequently Mellin injectivity gives

$$h(D)F_h = f_0 = \Theta\phi_0. \quad (\text{GC.8})$$

Define the actual density, mass, and Laplace integral by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt. \quad (\text{GC.9})$$

Equation (GC.6) proves holomorphy of M_h on $|\Re\theta| < \pi/2$, with $M_h^{(j)}(\theta) = \int t^j e^{\theta t} w_h(t) dt$ there. Indeed any compact substrip is dominated using a β for which 2β is larger than its real width. The mass is positive because v_h is a nonzero analytic function; its real-line zeros are discrete. Every original spectral moment and its 2π factor are retained.

37.2 The gamma reference with its actual mass

Fix $\lambda > 0$ and write $\alpha = 2\lambda$. Define

$$d\sigma_\lambda(t) = r_\lambda(t) dt, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda). \quad (\text{GC.10})$$

The product of two gamma integrals, changed by $(a, b) = (ry, (1 - r)y)$ with Jacobian y , proves the beta integral. Its substitution $r/(1 - r) = e^x$ gives

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{\Gamma(\lambda + iv)\Gamma(\lambda - iv)}{\Gamma(2\lambda)}.$$

The integrand before Fourier transformation and its second derivative are integrable. Two integrations by parts make its transform integrable, so Fourier inversion is valid. Alternatively, multiplying that transform by $e^{-\varepsilon v^2}$ gives an absolutely integrable Gaussian approximation and proves the same inversion by dominated convergence. The substitution $t = 2v$, with Jacobian two, therefore proves

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GC.11})$$

The beta integrand extends holomorphically to $|\Im x| < \pi$ using the branch positive on $\mathbb{R}$. On every closed narrower strip it is bounded by a constant times $e^{-\lambda|\Re x|}$. Shifting the contour up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_{\lambda,b} e^{-b|v|}$ for every $b < \pi$. Thus σ_λ has exponential moments of every order less than $\pi/2$ in the original t variable.

Fourier uniqueness for finite measures and multiplication of (GC.11) give the literal sum density

$$r_{\lambda,k} := r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (\text{GC.12})$$

Its mass is c_λ^k . Analytic continuation justified by the proved exponential moments gives

$$Z_{\lambda,k}^\Gamma(\theta) = \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-k\alpha}, \quad |\Re \theta| < \pi/2. \quad (\text{GC.13})$$

The power uses the analytic logarithm of $\cos \theta$ equal to zero at zero; this cosine has no zeros on the stated strip.

For $|z| < 1$, let

$$\begin{aligned} \mathcal{G}_\lambda(z, t) &= (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} \\ &= \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n. \end{aligned} \quad (\text{GC.14})$$

Both logarithms are the principal logarithms and vanish at zero, since $1 \pm iz$ have positive real part. For real z, w near zero, the Laplace transform and the elementary cosine addition identity give

$$\int \mathcal{G}_\lambda(z, t) \mathcal{G}_\lambda(w, t) d\sigma_\lambda(t) = c_\lambda (1 - zw)^{-\alpha}.$$

On a sufficiently small closed bidisc all derivatives of the integrands are bounded by a polynomial times $e^{a|t|}$ with $a < \pi/2$. The exponential moment proof permits their integration. Coefficient comparison then proves, with the monic leading coefficient obtained from $\arctan z = z + O(z^3)$,

$$\begin{aligned} \int b_n^{(\lambda)} b_j^{(\lambda)} d\sigma_\lambda &= c_\lambda n! (\alpha)_n \delta_{nj}, \\ b_0^{(\lambda)} &= 1, \quad b_1^{(\lambda)} = t, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \end{aligned} \quad (\text{GC.15})$$

The recurrence follows directly by differentiating (GC.14): $(1 + z^2) \partial_z \mathcal{G}_\lambda = (t - \alpha z) \mathcal{G}_\lambda$. The polynomials have real coefficients.

They are complete in $L^2(r_{\lambda,k} du)$ at parameter $k\lambda$. In fact, an orthogonal vector f defines the finite complex measure $f(u) r_{\lambda,k}(u) du$. Cauchy–Schwarz and exponential integrability give its Fourier transform on a nonempty complex strip. All derivatives at zero vanish by polynomial orthogonality, so analytic uniqueness and Fourier uniqueness imply $f = 0$ almost everywhere. This argument uses the full measure mass.

37.3 The sum projection and exact coefficient powers

On the actual spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du),$$

let $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$. Fubini and (GC.12) prove that it is an isometry. The affine fibre chart has Jacobian one, so its adjoint for $k \geq 2$ is

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GC.16})$$

The denominator is positive everywhere. Fibre Cauchy–Schwarz and Fubini give $\|C_\Sigma F\| \leq \|F\|$, so the formula defines the indicated adjoint on all of $\mathcal{H}_k$. When $k = 1$, both maps are the identity. In every case $C_\Sigma U_\Sigma = 1$ and $P_\Sigma = U_\Sigma C_\Sigma$ is the orthogonal projection. The maps

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n, \quad n \in \ker C_\Sigma \quad (\text{GC.17})$$

are inverse Hilbert isometries. This retains the entire relative fibre.

Multiplying the k generating functions gives a finite identity at each coefficient:

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1+\dots+n_k=n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \quad (\text{GC.18})$$

Pairing it with a product polynomial and using (GC.15) leaves only its total degree $n = \sum_i n_i$. The pairing at that degree is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$ and the sum norm is $c_\lambda^k n! (k\alpha)_n$. Completeness therefore proves

$$C_\Sigma \prod_i b_{n_i}^{(\lambda)} = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}. \quad (\text{GC.19})$$

The squared norm of the retained complementary polynomial is exactly

$$c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right], \quad (\text{GC.20})$$

by the orthogonal decomposition (GC.17).

For the original arithmetic density define, for every $\lambda > 0$,

$$\begin{aligned} A_{h,\lambda}(t) &= \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda} = |A_{h,\lambda}|^2, \quad w_h = B_{h,\lambda} r_\lambda, \\ c_{h,n} &= \frac{\int b_n^{(\lambda)}(t) w_h(t) dt}{c_\lambda n! (\alpha)_n}, \quad \mathcal{A}_h(z) = \sum_{n \geq 0} (\alpha)_n c_{h,n} z^n, \\ d_{h,k,n} &= [z^n] \mathcal{A}_h(z)^k. \end{aligned} \quad (\text{GC.21})$$

The notation $\mathcal{A}_h$ retains the fixed parameter λ . The complex amplitude $A_{h,\lambda}$ and the generating function $\mathcal{A}_h$ are different specified maps. The integrals in (GC.21) converge by (GC.6), before any convergence claim about the power series. Each $d_{h,k,n}$ is the finite expression

$$d_{h,k,n} = \sum_{n_1+\dots+n_k=n} \prod_i (\alpha)_{n_i} c_{h,n_i}. \quad (\text{GC.22})$$

Let $m_{h,k} = w_h^{*k}$. Absolute moments and Fubini give, by (GC.18),

$$\int_{\mathbb{R}} b_n^{(k\lambda)}(u) m_{h,k}(u) du = c_\lambda^k n! d_{h,k,n}. \quad (\text{GC.23})$$

This finite identity holds for every $\lambda > 0$ and every n ; it does not require an infinite product to be interchanged with an integral. In particular $c_{h,0} = \mu_h/c_\lambda$ and the total mass is μ_h^k .

37.4 The supplied analytic generating join, with its branches

Theorem 37.1 (The exact analytic generating function). *For the original arithmetic measure and every $\lambda > 0$, the series in (GC.21) converges holomorphically on $|z| < 1$ and is exactly*

$$\mathcal{A}_h(z) = \frac{1}{c_\lambda} (1 + z^2)^{-\lambda} M_h(\arctan z). \quad (\text{GC.24})$$

Here $\arctan z = (\log(1 + iz) - \log(1 - iz))/(2i)$, and every logarithm is the branch inherited from zero. For real $|\theta| < \pi/4$,

$$M_h(\theta) = c_\lambda \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta). \quad (\text{GC.25})$$

With the original tensor mass and reference mass,

$$\frac{Z_{h,k}(\theta)}{Z_{\lambda,k}^\Gamma(\theta)} = \mathcal{A}_h(\tan \theta)^k, \quad Z_{h,k}(\theta) = \int e^{\theta u} m_{h,k}(u) du = M_h(\theta)^k. \quad (\text{GC.26})$$

Proof. For $|z| < 1$, both $1 \pm iz$ have positive real part. Their principal logs are analytic, and their sum is the analytic logarithm of $1 + z^2$ vanishing at zero. Direct substitution in (GC.14) gives

$$\mathcal{G}_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GC.27})$$

There is no omitted phase in this identity: the $it/2$ coefficient multiplies $\log(1 - iz) - \log(1 + iz) = -2i \arctan z$.

The ratio $w = (1 + iz)/(1 - iz)$ has

$$\Re w = \frac{1 - |z|^2}{|1 - iz|^2} > 0, \quad \tan(2\Re \arctan z) = \frac{2\Re z}{1 - |z|^2}.$$

The argument of w lies in $(-\pi/2, \pi/2)$ and equals $2\Re \arctan z$. Thus for every $0 \leq r < 1$,

$$|z| \leq r \implies |\Re \arctan z| \leq \arctan r < \pi/4, \quad |\mathcal{G}_\lambda(z, t)| \leq (1 - r^2)^{-\lambda} e^{\arctan(r)|t|}. \quad (\text{GC.28})$$

For the angle inequality use monotonicity of arctangent and $2|\Re z|/(1 - |z|^2) \leq 2r/(1 - r^2)$, together with $2\arctan r \in [0, \pi/2)$. For the modulus inequality use $|1 + z^2| \geq 1 - r^2$ and real $\lambda > 0$.

Equation (GC.6) makes the right side of (GC.28) integrable against $w_h(t) dt$, uniformly on every compact subdisc. Dominated integration therefore defines a holomorphic function $c_\lambda^{-1} \int \mathcal{G}_\lambda(z, t) w_h(t) dt$ on the whole unit disc. One explicit justification is Morera's theorem: domination permits exchanging the integral with each triangle contour, and the inner holomorphic contour integral is zero. For its derivatives on $|z| \leq r$, choose $r < R < 1$ and apply Cauchy's derivative formula on circles of radius $R - r$; the kernel derivative $|\partial_z^n \mathcal{G}_\lambda(z, t)|$ is bounded by $n!(R - r)^{-n}(1 - R^2)^{-\lambda} e^{\arctan(R)|t|}$. This integrable bound justifies every derivative interchange.

At zero, (GC.14) now gives

$$\mathcal{A}_h^{(n)}(0) = \frac{1}{c_\lambda} \int b_n^{(\lambda)}(t) w_h(t) dt = n!(\alpha)_n c_{h,n}. \quad (\text{GC.29})$$

Thus the Taylor series of this holomorphic integral is exactly the original coefficient series. Substitution of (GC.27) in the integral proves (GC.24). All its constants occur before the integration.

For real $|\theta| < \pi/4$, $|\tan \theta| < 1$, $\arctan(\tan \theta) = \theta$ on the specified branch, and $1 + \tan^2 \theta = \sec^2 \theta > 0$. Solving (GC.24) for $M_h(\theta)$ gives (GC.25). Finally the absolute integral of $\prod_i e^{\theta t_i} w_h(t_i)$ is finite throughout $|\Re \theta| < \pi/2$. Fubini gives $Z_{h,k} = M_h^k$ with no mass division. Divide the k th power of (GC.25) by (GC.13) to obtain (GC.26). At zero this ratio is explicitly $(\mu_h/c_\lambda)^k$. $\square$

The inverse also has a precise complex domain. If $|\Re \theta| < \pi/4$, then

$$|\tan \theta|^2 = \frac{\cosh(2\Im \theta) - \cos(2\Re \theta)}{\cosh(2\Im \theta) + \cos(2\Re \theta)} < 1.$$

The denominator is positive and the branch satisfies $\arctan(\tan \theta) = \theta$, first on a neighbourhood of zero and then on this connected strip by analytic uniqueness. Hence (GC.25)–(GC.26) extend to that strip with the analytic branch of the cosine power in (GC.13). No logarithm of $\mathcal{A}_h$ is asserted: it can have complex zeros, whereas all real-tilt integrals $M_h(\theta)$ are strictly positive.

For each finite derivative there is also an exact moment formula. Write $b_n^{(\lambda)}(t) = \sum_{j=0}^n b_{n,j} t^j$. Then

$$n!(\alpha)_n c_{h,n} = \frac{1}{c_\lambda} \sum_{j=0}^n b_{n,j} M_h^{(j)}(0), \quad d_{h,k,n} = \frac{1}{n!} \left. \frac{d^n}{dz^n} \mathcal{A}_h(z)^k \right|_{z=0}. \quad (\text{GC.30})$$

The first identity follows by the finite polynomial expansion and (GC.9); the second is Taylor's formula, whose Leibniz sum is exactly (GC.22). The triangular coefficients $b_{n,n} = 1$ give an invertible finite map between moments through degree n and coefficients through degree n , with the full diagonal factors $c_\lambda n!(\alpha)_n$ retained.

37.5 The complete density expansion for the original bounded references

The finite identities and analytic join above hold for every $\lambda > 0$. For the original full-quartet reference choose $\lambda = 1/4$. More generally choose the actual shifted reference

$$\lambda = \frac{1}{4} + L, \quad L \text{ an integer}, \quad L \geq \max(0, 3 - d). \quad (\text{GC.31})$$

These choices admit the following proved bounded multiplier and hence the complete L^2 density expansion.

Indeed at $s = 1/2 + it$ the defining g identity gives

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (\text{GC.32})$$

This expression is continued through each selected zero by the already defined entire quotient. Integrating the step function $[x]$, initially for $\Re s > 1$, gives $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$. The last integral is holomorphic for $\Re s > 0$, so the identity continues there away from 1. On the original line it gives $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For a nonempty packet put $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$; for $h = 1$ put $T_h = 1$. For $|t| \geq T_h$, every original packet factor has modulus at least $|t|/2$. Gamma recurrence then gives

$$\begin{aligned} A_{h,1/4+L}(t) &= \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}, \\ B_{h,1/4+L}(t) &\leq \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} |t|^{6-2d-2L}. \end{aligned} \quad (\text{GC.33})$$

For the constant 25, the two squared numerator estimates at $|t| \geq 1$ are $(t^2 + 1/4)^2 \leq 25t^4/16$ and $|\zeta(s)|^2 \leq 16t^2$. Each shift factor has modulus at least $|t|/2$. The exponent in (GC.33) is nonpositive for (GC.31). Thus a completely specified finite bound is

$$C_{h,L} = \max \left\{ \sup_{|t| \leq T_h} B_{h,1/4+L}(t), \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} T_h^{6-2d-2L} \right\} > 0. \quad (\text{GC.34})$$

The compact term is finite by continuity of the cancelled multiplier. This is a mathematical bound with a specified supremum, not a claim that a numerical packet supremum has already been enclosed.

For these references $B \in L^2(d\sigma_\lambda)$, because the reference has finite mass and $0 \leq B \leq C_{h,L}$. Fibre integration of $w_h = Br_\lambda$ proves

$$B_{h,\lambda;k} = C_\Sigma(B^{\otimes k}), \quad m_{h,k} = r_{\lambda,k} B_{h,\lambda;k}, \quad 0 \leq B_{h,\lambda;k} \leq C_{h,L}^k. \quad (\text{GC.35})$$

For $k \geq 2$ it is positive at every sum coordinate: the zeros of B form a countable discrete set, and their forbidden fibre coordinates are a countable union of affine hyperplanes of measure zero. Every remaining reference factor is positive. For $k = 1$ the original isolated zeros remain.

Combining (GC.23), completeness, and the sum norm in (GC.15) gives the full, convergent $L^2(r_{\lambda,k} du)$ identity

$$B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (\text{GC.36})$$

In particular the norm of the retained relative density vector is

$$\|B^{\otimes k} - U_\Sigma B_{h,\lambda;k}\|^2 = \left(\int B^2 d\sigma_\lambda \right)^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (\text{GC.37})$$

This is Pythagoras for (GC.17), so every norm and mass factor is retained. A truncation of (GC.36) is not asserted pointwise positive.

37.6 Original-coordinate finite Grams and every tilt derivative

Let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ for $N \geq 0$, with original ordered basis $1, S, \dots, S^N$, and set $c = k/2$. The arithmetic source Gram and its reference are

$$\begin{aligned} (\mathbf{M}_{h,k,N}(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} m_{h,k}(u) du, \\ (\mathbf{M}_{\lambda,k,N}^\Gamma(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} r_{\lambda,k}(u) du, \quad 0 \leq i, j \leq N. \end{aligned} \quad (\text{GC.38})$$

They are holomorphic entrywise for $|\Re \theta| < \pi/2$ and positive definite at every real tilt in that strip: a nonzero polynomial has only finitely many real-line roots and each measure is positive almost everywhere. Let $\mathbf{H}_{h,k,N}$ denote the corresponding real- u monomial Hankel matrix. The exact matrix of the unchanged coordinate substitution is

$$(\mathbf{L}_{N+1})_{aj} = \begin{cases} \binom{j}{a} c^{j-a} i^a, & 0 \leq a \leq j \leq N, \\ 0, & a > j, \end{cases} \quad \det \mathbf{L}_{N+1} = i^{N(N+1)/2}. \quad (\text{GC.39})$$

Expanding each original S monomial proves

$$\mathbf{M}_{h,k,N} = \mathbf{L}_{N+1}^* \mathbf{H}_{h,k,N} \mathbf{L}_{N+1} \quad (\text{GC.40})$$

and the identical congruence for the reference. Since the displayed determinant has modulus one, it also proves exact equality of the two coordinate determinants. The congruence, rather than a relabeling of a matrix, preserves every complex phase in its entries.

For an integer $r \geq 0$, expand the specified polynomial uniquely as

$$u^r (c - iu)^i (c + iu)^j = \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} b_n^{(k,\lambda)}(u). \quad (\text{GC.41})$$

The monic triangular basis makes these coefficients finite, unique, and computable by polynomial subtraction in the written order. Differentiation of (GC.38) and (GC.23) now give the exact formula

$$\partial_\theta^r (\mathbf{M}_{h,k,N})_{ij} \big|_{\theta=0} = c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.42})$$

The derivative interchange is justified by the exponential bounds after (GC.9), or directly by Fubini on the k -fold product. Since $d_{h,k,n}$ uses exactly the finite coefficient power (GC.22), the entire r th derivative matrix is determined by $c_{h,0}, \dots, c_{h,2N+r}$. At zero tilt $r = 0$, the list ends at $2N$. There is no truncation error in this finite statement.

Equation (GC.42) supplies each derivative at zero from its stated finite coefficient list. Evaluation at a nonzero tilt is related to those derivatives by the following exact analytic Taylor identity, valid for $|\theta| < \pi/2$:

$$(\mathbf{M}_{h,k,N}(\theta))_{ij} = \sum_{r=0} \frac{\theta^r}{r!} c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.43})$$

Absolute convergence follows by bounding the absolute exponential series by $e^{|\theta||u|}$ times the original polynomial modulus. For larger imaginary components within the full holomorphic strip, analytic continuation of these entries is the defined original Laplace integral (GC.38). This specifies the relation between all finite coefficient jets and the full tilted metric.

37.7 The original cyclic quotient and the exact cochain correction

For a nonempty packet let $E_h = \mathbb{C}[s]/(h)$ and $\mathcal{E}_h = E_h^{\otimes k}$. Let a be the element $s_1 + \cdots + s_k$ of this finite algebra. The kernel of $\mathbb{C}[S] \rightarrow \mathcal{E}_h, P \mapsto P(a)$, is generated by its unique monic polynomial χ . It is nonzero since the powers of a lie in a finite-dimensional vector space. Put $q = \deg \chi \geq 1$, $C = \mathbb{C}[S]/(\chi)$, and $A = M_S : C \rightarrow C$. The induced map $C \rightarrow \mathcal{E}_h$ is injective by the definition of the kernel. For each selected zero, the complete Taylor value of g/h is nonzero at its centre; its residue $v_h = j_h(g/h) \in E_h$ is consequently a unit in every local factor and in E_h . Thus the original arithmetic injection

$$\eta_{h,k} : C \hookrightarrow \mathcal{E}_h, \quad [P] \mapsto v_h^{\otimes k} P(a) \quad (\text{GC.44})$$

retains the entire local unit and every selected multiplicity.

The original analytic polynomial map is

$$\mathcal{V}_{h,k} P = P(D_1 + \cdots + D_k) F_h^{\otimes k}. \quad (\text{GC.45})$$

Its tensor Mellin transform is $P(s_1 + \cdots + s_k) \prod_i v_h(s_i)$. Taking complete selected jets gives precisely (GC.44) after the quotient π_χ . No modulus operation has been applied to that arithmetic jet map.

Successive monic divisions, in the order $s_1, \dots, s_k$, provide specific polynomials $Q_i(s_1, \dots, s_k)$ with

$$\chi(s_1 + \cdots + s_k) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{GC.46})$$

To prove this, divide in s_1 , retain its remainder, divide that remainder in s_2 , and continue. The final remainder has degree less than d in every coordinate. Its residue in $\mathcal{E}_h$ is zero by the defining property of χ , and the tensor remainder monomials form a basis, so every coefficient is zero. This proves the displayed identity and fixes the division order.

The operators D_i commute on the tensor complex. For a polynomial P , define the actual cochain of degree $k - 1$ by

$$\begin{aligned} \mathcal{P}_{\chi,h,k}(P) = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \cdots + D_k) \\ \cdot (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \end{aligned} \quad (\text{GC.47})$$

Every summand belongs to the indicated tensor source, since D preserves V and $\mathcal{B}$. The first $i - 1$ factors have cochain degree one. The tensor differential on the i th factor contributes $(-1)^{i-1}$, cancelling the written sign. The other factors have zero differential. Equations (GC.8) and (GC.46) therefore give

$$d_{\text{tot}} \mathcal{P}_{\chi,h,k}(P) = \mathcal{V}_{h,k}(\chi P). \quad (\text{GC.48})$$

This identifies the cochain primitive, its sign, and its boundary in their different cochain degrees.

For the full quotient at finite source degree take $N \geq q - 1$ and $m = N - q + 1$. With $\mathcal{P}_{-1} = 0$, monic division gives the exact sequence in original coordinates

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad J_N P = [P]_\chi. \quad (\text{GC.49})$$

The kernel assertion is divisibility, and the first q monomial columns prove surjectivity. Let B_χ be the $(N+1) \times m$ matrix of M_χ in the original ordered bases. Its entries are $[S^i](\chi(S)S^j)$; it is independent of the metric and tilt. The relation Gram and every zero-tilt derivative are therefore

$$\mathbf{B}_{h,k,N}(\theta) = B_\chi^* \mathbf{M}_{h,k,N}(\theta) B_\chi, \quad \mathbf{B}_{h,k,N}^{(r)}(0) = B_\chi^* \mathbf{M}_{h,k,N}^{(r)}(0) B_\chi. \quad (\text{GC.50})$$

Equivalently its polynomial integrand is $u^r |\chi(c+iu)|^2 (c-iu)^i (c+iu)^j$, where $0 \leq i, j < m$. Its degree is at most $r + 2q + 2(m-1) = r + 2N$, so the relation derivative uses the same list through $2N + r$

as the source derivative. Both complex polynomial factors, their conjugates, and all χ coefficients remain in this formula.

For a fixed real admitted tilt write $M_h = \mathbf{M}_{h,k,N}(\theta)$ and $M_\Gamma = \mathbf{M}_{\lambda,k,N}^\Gamma(\theta)$. Their canonical quotient Grams and minimum lifts are

$$G_h = (J_N M_h^{-1} J_N^*)^{-1}, \quad R_h = M_h^{-1} J_N^* G_h, \quad G_\Gamma = (J_N M_\Gamma^{-1} J_N^*)^{-1}, \quad R_\Gamma = M_\Gamma^{-1} J_N^* G_\Gamma. \quad (\text{GC.51})$$

Surjectivity of J and positivity of the source Grams prove every inverse exists. These formulas give $JR_h = JR_\Gamma = I_C$ and make each lift orthogonal to $\ker J$ in its own metric, proving its minimum property by the orthogonal Pythagorean identity. For $m > 0$ put

$$K = (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_\Gamma : C \longrightarrow \mathbb{C}^m. \quad (\text{GC.52})$$

The relation Gram is positive definite because B_χ is injective. The section $R_\Gamma - B_\chi K$ still has quotient identity and is orthogonal to the relation space in M_h . The preceding uniqueness gives the exact section map

$$\boxed{R_h = R_\Gamma - B_\chi K.} \quad (\text{GC.53})$$

At $N = q - 1$, $m = 0$, take K to be the zero map to the zero vector space and omit the inverse in (GC.52); (GC.53) remains the identity of the unique degree-less-than- q representatives.

For $u_0 \in C$, let $\kappa_{u_0}(S)$ be the polynomial whose coefficient vector is Ku_0 . Combining (GC.48) and (GC.53) proves the exact cochain correction

$$\boxed{\mathcal{V}_{h,k}(R_h u_0 - R_\Gamma u_0) = -\mathcal{V}_{h,k}(\chi \kappa_{u_0}) = d_{\text{tot}}(-\mathcal{P}_{\chi,h,k}(\kappa_{u_0})).} \quad (\text{GC.54})$$

Thus applying $\mathcal{V}$ gives the boundary in degree k ; the explicit negative cochain on the right is its primitive in degree $k - 1$. Passing to the original arithmetic quotient sends this boundary to zero. Explicitly, for the original set of labels $\mathcal{L}$ and vector spaces E_ℓ , write the supplied carrier as $\{\tau\} \sqcup \bigsqcup_{\ell \in \mathcal{L}} (\{\ell\} \times E_\ell)$. A family of linear maps $L_\ell : E_\ell \rightarrow F_\ell$ has the exact support lift $\tau \mapsto \tau$ and $(\ell, v) \mapsto (\ell, L_\ell v)$, with $e_\ell = (\ell, 0)$ in each target fibre. Substitution proves that these lifts preserve identities and composition. Applied to the original quotient, the relation boundary in (GC.54) therefore has image e_ℓ , while the external τ has image τ . The source vector and its primitive remain available before that quotient.

37.8 Determinant indices and the original Toda substitution

For real admitted θ , write $\mathfrak{D}_n$ for the original degree-less-than- n source Gram determinant and $\mathfrak{B}_m$ for the relation Gram determinant on m multiplication columns. Empty determinants are one. The analogous reference determinants carry the superscript Γ . For $N \geq q - 1$, set

$$V_N = \det G_h, \quad V_N^\Gamma = \det G_\Gamma, \quad X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_m = \mathfrak{B}_m / \mathfrak{B}_m^\Gamma.$$

The quotient-first coefficient matrix with columns $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$ has determinant one: its columns have successively distinct degrees and all their leading coefficients are one. Replacing its first q columns by the corresponding minimum sections only subtracts relation columns, so it still has determinant one. Orthogonality then gives the exact block determinant formula

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}} = V_N^\Gamma T_N, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}, \quad N \geq q - 1. \quad (\text{GC.55})$$

In particular the first full quotient has denominator $\mathfrak{B}_0 = 1$ and requires no relation inverse.

For completeness the reference source determinant is

$$\mathfrak{D}_n^\Gamma(\theta) = c_\chi^{kn} \prod_{j=0}^{n-1} j!(k\alpha)_j (\cos \theta)^{-n(k\alpha+n-1)}, \quad a_n^\Gamma(\theta) = n(n + k\alpha - 1) \sec^2 \theta. \quad (\text{GC.56})$$

Here a_n is the ratio of consecutive monic squared norms in the real- u coordinate; the norm-preserving phase of (GC.39) gives the same ratios in the original S coordinate. A direct proof of the Hankel identity used to obtain (GC.56) is as follows. Let p_j be the monic orthogonal polynomials for an $e^{\theta u}$ -tilted positive measure, let h_j be their squared norms, and write $up_j = p_{j+1} + b_j p_j + a_j p_{j-1}$, with $p_{-1} = 0$ and $a_0 = 0$. For $j \geq 1$, pairing with p_{j-1} gives $a_j = h_j/h_{j-1}$, and pairing with p_j gives $h'_j = b_j h_j$ for every $j \geq 0$. Differentiating orthogonality for $j \geq 1$ shows $\partial_\theta p_j = -a_j p_{j-1}$: its pairings with degrees at most $j-2$ vanish, and the remaining pairing is $-h_j$. For $j \geq 1$, write $p_j = u^j + c_j u^{j-1} + \dots$, and set $c_0 = 0$ for $p_0 = 1$. The derivative statement for p_0 is $\partial_\theta p_0 = 0$. Thus $c'_j = -a_j$ and $b_j = c_j - c_{j+1}$ hold for every $j \geq 0$, and consequently $b'_j = a_{j+1} - a_j$. Since $\mathfrak{D}_n = \prod_{j < n} h_j$,

$$(\log \mathfrak{D}_n)'' = \sum_{j < n} b'_j = a_n = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1} / \mathfrak{D}_n^2, \quad n \geq 1.$$

For the reference, the determinant values at $n=0$ and $n=1$ are 1 and (GC.13). The right side of the proposed formula (GC.56) has second logarithmic derivative $n(k\alpha + n - 1) \sec^2 \theta$. Its consecutive ratio has exactly the same value, with constants $n!(k\alpha)_n / ((n-1)!(k\alpha)_{n-1})$ and cosine exponent -2 . Positive induction in the displayed determinant identity proves (GC.56) for every n .

The reference relation determinant is the determinant of the moment matrix with entries

$$\partial_\theta^{i+j} \{ \bar{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \}, \quad 0 \leq i, j < m. \quad (\text{GC.57})$$

Differentiation under the integral gives the unchanged weight $|\chi(c + iu)|^2 r_{\lambda,k}(u)$ in this u -coordinate matrix. The relation version of (GC.39)–(GC.40) has determinant of modulus one, proving that (GC.57) is the original S -coordinate relation determinant as stated. It contains the actual χ with all coefficients.

At $\theta = 0$ put

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

The consecutive full-quotient formulas have domain $N \geq q$ and give, by literal substitution in (GC.55)–(GC.56),

$$a_{N+1} = (N+1)(N+k\alpha)Q_N, \quad \frac{V_{N-1}}{V_{N+1}} = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}. \quad (\text{GC.58})$$

The domain $N \geq q$ is necessary because V_{N-1} occurs. The separate original endpoint at $N = q - 1$ is not replaced by a q -dimensional metric at degree $q - 2$. At every fixed admissible real tilt the phase derivative remains the exact equality

$$\partial_\theta \log V_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N, \quad N \geq q - 1. \quad (\text{GC.59})$$

All these derivatives are of positive real-tilt determinants. Formula (GC.42) gives their underlying source and relation entry derivatives with the precise finite coefficient indices; their quotient and determinant operations use those original matrices.

For $h = 1$ the finite arithmetic quotient is zero and $\chi = 1$. Its relation and source spaces coincide at every degree N , so $\mathfrak{B}_{N+1} = \mathfrak{D}_{N+1}$, $V_N = V_N^\Gamma = T_N = 1$. The quotient lift has zero-dimensional domain and the cochain correction (GC.54) is the zero map on that domain. The nonzero analytic density, its mass μ_1^k , its coefficient series and (GC.24)–(GC.43) remain as constructed. The support lift is $G(0) = \{\tau, e\}$; it does not produce a nonzero finite spectral quotient from the analytic seed.

38 Certified theta moments and the original gamma arithmetic coefficients

This continuation proves and evaluates the fourth and sixth moments of the original theta seed, then transports the enclosures through the exact coefficient map for the reference parameter $\lambda = 13/4$

in *An exact gamma pushforward for the arithmetic Toda control*. The seed retains $g = 2\xi$, the density $|g(1/2 + it)|^2/(2\pi)$, the Fourier coordinate t , and its actual total mass. The reference retains the argument $\lambda + it/2$ and its different actual mass. We give the full analytic formulas, an explicit infinite-square tail for every finite derivative order, and the exact rational enclosures used by the executable certificate.

38.1 The theta input and its transform, with the original factors

On $\mathcal{S} = \{z \neq 0 : |\arg z| < \pi/4\}$ use the principal logarithm, and put

$$D = -z\partial_z, \quad L = D - \frac{1}{2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{GS.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{GS.2})$$

The series and each differentiated series converge locally uniformly: on a compact subset of $\mathcal{S}$, every summand is bounded by a polynomial in n times e^{-an^2} for some $a > 0$.

Lemma 38.1 (Inversion and both endpoints). *On the specified sector,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{GS.3})$$

Every D -derivative of f_0 decays faster than every power both at zero and at infinity on a closed proper subsector.

Proof. For $x > 0$, Fourier transformation with kernel $e^{-2\pi i \eta u}$ maps $e^{-\pi x^2 u^2}$ to $x^{-1}e^{-\pi \eta^2/x^2}$. The value at $\eta = 0$ follows from the Gaussian integral, whose square in polar coordinates is one. Differentiation in η and integration by parts give $I'(\eta) = -2\pi \eta I(\eta)/x^2$, which proves the stated formula with that initial value. Periodize this Gaussian over integer translates. Its Fourier coefficients are the displayed rapidly decreasing transform values; its absolutely convergent Fourier series evaluated at zero proves $\vartheta(x) = x^{-1}\vartheta(1/x)$. Holomorphic uniqueness extends the identity to $\mathcal{S}$.

For $(RF)(z) = z^{-1}F(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this to $R\vartheta = \vartheta$ proves the second equality in (GS.3). The real series coefficients prove the third. On $|\arg z| \leq \beta < \pi/4$ one has $\Re z^2 \geq |z|^2 \cos(2\beta) > 0$; the series and each derivative therefore have Gaussian decay at infinity with a fixed polynomial factor. Inversion and $DR = R(1 - D)$ transport all these bounds to zero, with every derivative retained. $\square$

Define, on these rapid sector functions,

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (UF)(y) = e^{y/2}F(e^y), \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{GS.4})$$

Here $U : L^2(\mathbb{R}_{>0}, dx) \rightarrow L^2(\mathbb{R}, dy)$ is a surjective linear isometry: $x = e^y$ proves the norm identity, and its inverse is $u \mapsto [x \mapsto x^{-1/2}u(\log x)]$. The endpoint bounds prove that the Mellin integral is entire. For $\Re s > 1$, absolute integration of (GS.2) is justified by $\sum_{n \geq 1} n^{-\Re s} < \infty$. Substitution $v = \pi n^2 x^2$ then gives

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2}\zeta(s)\{4\Gamma(s/2+2) - 6\Gamma(s/2+1)\} \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GS.5})$$

The factor 2 in (GS.2) and the substitution factor $1/2$ both occur before the first equality. Integration by parts proves the gamma recurrence used in the second equality. The entire Mellin integral continues the right side. Substitution $x \mapsto 1/x$ and real conjugation in (GS.4) prove

$$g(1-s) = g(s), \quad g(\bar{s}) = \overline{g(s)}. \quad (\text{GS.6})$$

In particular the following original density is even:

$$w(t) = \frac{|g(1/2 + it)|^2}{2\pi}, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt. \quad (\text{GS.7})$$

All moments exist. To see the exponential bound directly, rotate the Mellin contour to the ray $e^{i\beta}\mathbb{R}_{>0}$ with $|\beta| < \pi/4$. The endpoint bounds make the closing arcs vanish, and the rotated formula has the factor $e^{i\beta s}$. On the line $\Re s = 1/2$, choosing the sign of β to match $\Im s$ yields $|g(1/2 + it)| \leq C_\beta e^{-|\beta||t|}$. Thus $\int e^{a|t|} w(t) dt < \infty$ for every $0 < a < \pi/2$. Evenness gives $\mu_{2r+1} = 0$.

38.2 Every derivative polynomial and the exact higher-moment map

Let $H_r = L^r f_0$ for integers $r \geq 0$. Direct differentiation of (GS.4) gives $ULF = -(UF)'$ on the rapid sector functions. Integration by parts in $\mathcal{F}_+$, with the endpoint bounds already proved, therefore gives the commuting transform identity

$$\mathcal{F}_+(UH_r)(t) = (it)^r g(1/2 + it). \quad (\text{GS.8})$$

Every sign and power i^r belongs to this map.

Theorem 38.2 (Exact higher-moment formula). *Define polynomials $P_r \in \mathbb{Q}[y]$ by*

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y), \quad (\text{GS.9})$$

$$P_r(y) = \sum_{a=1}^{r+2} p_{r,a} y^a, \quad p_{r+1,a} = 2p_{r,a-1} - (2a + \tfrac{1}{2})p_{r,a}, \quad (\text{GS.10})$$

where coefficients outside their indicated range are zero. Then

$$H_r(x) = \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \mu_{2r} = \int_0^\infty |H_r(x)|^2 dx = 2 \int_1^\infty H_r(x)^2 dx. \quad (\text{GS.11})$$

The first four complete polynomials are

$$\begin{aligned} P_0 &= 8y^2 - 12y, \\ P_1 &= 16y^3 - 60y^2 + 30y, \\ P_2 &= 32y^4 - 224y^3 + 330y^2 - 75y, \\ P_3 &= 64y^5 - 720y^4 + 2116y^3 - 1635y^2 + \tfrac{375}{2}y. \end{aligned} \quad (\text{GS.12})$$

Proof. With $y = \pi n^2 x^2$, the operator D is $-2y\partial_y$. Consequently $L(P(y)e^{-y}) = \{-2yP'(y) + (2y - 1/2)P(y)\}e^{-y}$. This proves (GS.9), the series identity in (GS.11), and (GS.10) by collecting each specified coefficient. It also proves inductively that the degree is $r+2$, the leading coefficient is 2^{r+3} , and the constant coefficient is zero. Applying the recurrence three times gives the complete rational lists (GS.12).

Fourier Plancherel in the convention (GS.4) has factor $1/(2\pi)$: $\int |u|^2 dy = (2\pi)^{-1} \int |\mathcal{F}_+ u|^2 dt$. For the Schwartz functions here this follows by inserting $e^{-\varepsilon t^2}$ into Fourier inversion, applying Fubini, and evaluating the Gaussian kernel of integral one; letting $\varepsilon \downarrow 0$ gives inversion and the inner-product identity. Applying it to (GS.8) and the unitary U proves the first integral equality in (GS.11).

The relation $DR = R(1 - D)$ gives $RL = -LR$, while $Rf_0 = f_0$. Thus $RH_r = (-1)^r H_r$, or $H_r(1/x) = (-1)^r x H_r(x)$ for $x > 0$. In the integral over $(0, 1)$ substitute $x = 1/v$. The factor v^2 from the squared identity cancels the Jacobian v^{-2} and yields exactly the integral over $(1, \infty)$. The sign $(-1)^r$ is retained before squaring. All values $H_r(x)$ are real because their series has real coefficients, which finishes (GS.11). $\square$

The numerical driver in the delivered Gamma note writes $q_0 = 4y^2 - 6y$ and uses the same linear recurrence for q_r . Its exact coefficient map to this presentation is $q_r \mapsto P_r = 2q_r$, with inverse $P_r \mapsto P_r/2$. Thus $H_r = 2 \sum_n q_r(\pi n^2 x^2) e^{-\pi n^2 x^2}$. At the integral level (GS.11) consequently gives a factor 8 in front of the sum using $q_r q_r I_{2a+2b}$. At the gamma-function level below it gives a factor 4 in front of $q_r q_r \Gamma/C^\alpha$. The coefficient map accounts for both presentations without changing the theta input or its mass.

38.3 The incomplete-gamma square and a proved omitted tail

For real $C > 0$ and integers $p \geq 0$ put

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}, \quad \Gamma(a, C) = \int_C^\infty v^{a-1} e^{-v} dv. \quad (\text{GS.13})$$

Substitution $v = Cx^2$ proves the equality. Only positive real arguments and their positive real powers occur here. The companion recurrence, including its endpoint term, is

$$I_{p+2}(C) = \frac{p+1}{2C} I_p(C) + \frac{e^{-C}}{2C}. \quad (\text{GS.14})$$

It follows by integration by parts in (GS.13), whose upper endpoint vanishes exponentially.

Let $H_{r,J}$ restrict the original series (GS.11) to $1 \leq n \leq J$, where $J \geq 0$ is an integer. Put $T_{r,J} = 2 \int_1^\infty H_{r,J}^2 dx$. The entire finite expression is

$$T_{r,J} = \sum_{m,n=1}^J \sum_{a,b=1}^{r+2} p_{r,a} p_{r,b} (\pi m^2)^a (\pi n^2)^b \frac{\Gamma(a+b+1/2, \pi(m^2+n^2))}{[\pi(m^2+n^2)]^{a+b+1/2}}. \quad (\text{GS.15})$$

Here the endpoint factor 2 in (GS.11) cancels the factor 1/2 in (GS.13); the coefficients $p_{r,a}$ already contain the complete original theta factor. Empty sums at $J = 0$ have value zero.

To specify the tail by finite positive expressions, define Stirling numbers of the second kind by $\left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} = 1$, zero when one index is zero and the other positive, zero outside $0 \leq b \leq j$, and

$$\left\{ \begin{smallmatrix} j+1 \\ b \end{smallmatrix} \right\} = b \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} j \\ b-1 \end{smallmatrix} \right\}.$$

For $a \geq 1, d > 0$, integers $p \geq 0$, set

$$\rho_a = e^{-d(2a+1)}, \quad \mathbf{H}_p(a, d) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{GS.16})$$

$$\mathbf{A}_r(a) = \sum_{b=1}^{r+2} |p_{r,b}| \pi^b \mathbf{H}_{2b}(a, \pi). \quad (\text{GS.17})$$

Theorem 38.3 (Every finite derivative order has an explicit square tail). *For every integer $r \geq 0$ and $J \geq 0$, with $K = J + 1$,*

$$|\mu_{2r} - T_{r,J}| \leq E_{r,J} := 4\mathbf{A}_r(1)\mathbf{A}_r(K)I_{4r+8}(\pi(1+K^2)). \quad (\text{GS.18})$$

The right side is a finite positive expression and tends to zero as $J \rightarrow \infty$. In particular the infinite version of (GS.15) is absolutely convergent and equals μ_{2r} .

Proof. For $x \geq 1$, integers $a \geq 1$ and $p \geq 0$, write $n = a + j$. The inequality $(a + j)^2 - a^2 = 2aj + j^2 \geq (2a + 1)j$ proves

$$\sum_{n \geq a} n^p e^{-\pi n^2 x^2} \leq e^{-\pi a^2 x^2} \sum_{j \geq 0} (a + j)^p \rho_a^j = e^{-\pi a^2 x^2} H_p(a, \pi). \quad (\text{GS.19})$$

For the equality, expand $(a + j)^p$ and then use $j^v = \sum_b \left\{ \begin{smallmatrix} v \\ b \end{smallmatrix} \right\} (j)_b$, where $(j)_b = j(j-1) \cdots (j-b+1)$. This identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed Stirling recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$, which proves the last equality in (GS.19).

Since $x^{2b} \leq x^{2r+4}$ for every original exponent $b \leq r+2$, (GS.19) gives the three explicit inequalities

$$\begin{aligned} |H_r(x)|, |H_{r,J}(x)| &\leq A_r(1) x^{2r+4} e^{-\pi x^2}, \\ |H_r(x) - H_{r,J}(x)| &\leq A_r(K) x^{2r+4} e^{-\pi K^2 x^2}. \end{aligned}$$

The same estimate for the sum of absolute values of all summands proves absolute integrability of the expanded double square. In particular termwise integration in the infinite formula is legitimate. Now use $|H_r^2 - H_{r,J}^2| \leq |H_r - H_{r,J}|(|H_r| + |H_{r,J}|)$ and the endpoint factor 2 in (GS.11). The resulting constant is exactly 4, the power is $4r + 8$, and integration gives (GS.18).

For fixed r , the denominators in (GS.16) are bounded away from zero uniformly for integers $a \geq 1$; thus $H_p(a, \pi)$ is at most a constant times $(1 + a)^p$, and $A_r(a)$ has at most polynomial growth. For $C \geq \pi$,

$$I_p(C) \leq e^{-C} \int_1^\infty x^p e^{-\pi(x^2-1)} dx.$$

The last integral is finite. Hence the factor with $C = \pi(1 + K^2)$ dominates that polynomial growth and forces $E_{r,J} \rightarrow 0$, proving the remaining assertions. $\square$

In the two new cases the coefficients in (GS.17) are completely specified by

$$\begin{aligned} A_2(a) &= 75\pi H_2 + 330\pi^2 H_4 + 224\pi^3 H_6 + 32\pi^4 H_8, \\ A_3(a) &= \frac{375}{2}\pi H_2 + 1635\pi^2 H_4 + 2116\pi^3 H_6 + 720\pi^4 H_8 + 64\pi^5 H_{10}, \end{aligned} \quad (\text{GS.20})$$

where each H_p on the right means $H_p(a, \pi)$. These absolute coefficients enter the error estimate only. Every original negative coefficient stays in the exact square (GS.15).

38.4 The coefficient morphism at the actual gamma reference

For $\lambda > 0$ put

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad \alpha = 2\lambda. \quad (\text{GS.21})$$

We prove the constants needed for the coefficient map. The beta integral follows from the product of two gamma integrals by the substitution $v = x_1 + x_2$, $u = x_1/(x_1 + x_2)$ with Jacobian v . After $u/(1-u) = e^x$ it reads

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{|\Gamma(\lambda + iv)|^2}{\Gamma(2\lambda)}. \quad (\text{GS.22})$$

The integrand before transformation and its first two derivatives decay exponentially at both ends. Two integrations by parts therefore give an integrable Fourier transform, so Fourier inversion applies with the same Gaussian justification as above. Using its evenness and substituting $t = 2v$ gives

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GS.23})$$

The factor 2 from $t = 2v$ is present in c_λ . The function $(2 \cosh(x/2))^{-2\lambda}$ is holomorphic on $|\Im x| < \pi$ with the branch positive on the real line and decays exponentially on each closed narrower strip. Shifting the contour in (GS.22) up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_c e^{-c|v|}$ for each $0 < c < \pi$. Thus the density in (GS.21) has exponential moments for every exponent of absolute value less than $\pi/2$. Analytic continuation of (GS.23) yields

$$\int e^{\theta t} r_\lambda(t) dt = c_\lambda (\cos \theta)^{-\alpha} \quad (|\theta| < \pi/2). \quad (\text{GS.24})$$

Define the monic polynomials by the local generating function

$$F_\lambda(z, t) = (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} = \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n, \quad (\text{GS.25})$$

$$b_0 = 1, \quad b_1 = t, \quad b_{n+1} = t b_n - n(n + \alpha - 1) b_{n-1}. \quad (\text{GS.26})$$

The logarithms in (GS.25) vanish at zero. For real z close to zero, $F_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}$. Equation (GS.24) and $\cos(\arctan z + \arctan v) = (1 - zv)/\sqrt{(1 + z^2)(1 + v^2)}$ give

$$\int F_\lambda(z, t) F_\lambda(v, t) r_\lambda(t) dt = c_\lambda (1 - zv)^{-\alpha}. \quad (\text{GS.27})$$

Exponential integrability permits every derivative at $(0, 0)$ under the integral. Coefficient comparison proves

$$\int b_n^{(\lambda)}(t) b_j^{(\lambda)}(t) r_\lambda(t) dt = c_\lambda n! (\alpha)_n \delta_{nj}. \quad (\text{GS.28})$$

The highest coefficient of (GS.25) is $t^n/n!$, so these polynomials are monic. Finally differentiating (GS.25) gives $(1 + z^2) \partial_z F_\lambda = (t - \alpha z) F_\lambda$; coefficient comparison proves (GS.26), including all its constants.

The actual amplitude linking the two measures is

$$A_{1,\lambda}(t) = \frac{g(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{1,\lambda}(t) = |A_{1,\lambda}(t)|^2, \quad w(t) = B_{1,\lambda}(t) r_\lambda(t). \quad (\text{GS.29})$$

Gamma has no zeros and no poles at the displayed arguments. Here is a direct proof on their half-plane. For $\Re z > 0$, repeated integration by parts gives $B(z, n + 1) = n!/[z(z + 1) \cdots (z + n)]$. Substitution $u = v/n$ in the beta integral and $(1 - v/n)^n \leq e^{-v}$ prove $n^z B(z, n + 1) \rightarrow \Gamma(z)$ by dominated convergence. The reciprocal finite products converge to

$$ze^{\gamma z} \prod_{j \geq 1} (1 + z/j) e^{-z/j}, \quad \gamma = \lim_{n \rightarrow \infty} \left(\sum_{j=1}^n j^{-1} - \log n \right).$$

Integral comparison proves existence of the real limit. The logarithm of the tail product converges absolutely because $\log(1 + z/j) - z/j = O(j^{-2})$, and every factor is nonzero on $\Re z > 0$. The product is therefore finite and nonzero; by the previous limit it is $1/\Gamma(z)$. The original gamma integral is holomorphic on this half-plane, which also excludes poles there. The phase-preserving source map is multiplication by $A_{1,\lambda}$ from $L^2(w dt)$ to $L^2(r_\lambda dt)$; the exact norm identity follows by (GS.29). It is onto: the zero set of g on the line is discrete because g is entire and $g(2) > 0$ by (GS.5). Division by the amplitude off that null set gives the inverse and the same norm equality.

For completeness the full source phase at the chosen parameter is

$$A_{1,13/4}(t) = \frac{-\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it)}{(1/4 + it/2)(5/4 + it/2)(9/4 + it/2)}. \quad (\text{GS.30})$$

It follows from (GS.5), $s(s-1) = -(t^2 + 1/4)$, and three gamma recurrences. The elementary identity $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ first follows by summation by parts for $\Re s > 1$ and extends to $\Re s > 0$, $s \neq 1$, because its integral converges absolutely and locally uniformly there. On the line in (GS.30) it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 4\sqrt{t^2 + 1/4}$. The squared modulus of the denominator in (GS.30) is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, so

$$0 \leq B_{1,13/4}(t) \leq \frac{1024}{\sqrt{\pi}} \frac{(t^2 + 1/4)^2}{(t^2 + 25/4)(t^2 + 81/4)} \leq \frac{1024}{\sqrt{\pi}}. \quad (\text{GS.31})$$

The last denominator exceeds the numerator by $26t^2 + 253/2 > 0$ after expansion. Thus $B_{1,13/4}$ belongs to $L^2(r_{13/4} dt)$, and its gamma coefficients have their literal Hilbert-space type.

Write $c = c_{13/4}$ and $c_j = c_{1,j}$. Equations (GS.28)–(GS.29) give the linear coefficient map on the moment functional:

$$c_j = \frac{\int b_j^{(13/4)}(t) w(t) dt}{c j! (13/2)_j}, \quad c = 2^{-11/2} \Gamma(13/2) = \frac{10395\sqrt{\pi}}{2048\sqrt{2}}. \quad (\text{GS.32})$$

The last identity follows by six gamma recurrences from $\Gamma(1/2) = \sqrt{\pi}$. At this exact parameter the complete polynomials needed are

$$\begin{aligned} b_0 &= 1, & b_1 &= t, & b_2 &= t^2 - \frac{13}{2}, \\ b_3 &= t^3 - \frac{43}{2}t, & b_4 &= t^4 - 47t^2 + \frac{663}{4}, \\ b_5 &= t^5 - 85t^3 + \frac{3931}{4}t, \\ b_6 &= t^6 - \frac{275}{2}t^4 + \frac{13801}{4}t^2 - \frac{69615}{8}. \end{aligned} \quad (\text{GS.33})$$

Substituting successively into (GS.26) proves every line. Evenness gives $c_1 = c_3 = c_5 = 0$, and the exact even coefficient map becomes

$$\begin{aligned} c_0 &= \mu_0/c, \\ c_2 &= (2\mu_2 - 13\mu_0)/(195c), \\ c_4 &= (4\mu_4 - 188\mu_2 + 663\mu_0)/(377910c), \\ c_6 &= (8\mu_6 - 1100\mu_4 + 27602\mu_2 - 69615\mu_0)/(2737957950c). \end{aligned} \quad (\text{GS.34})$$

Here $2!(13/2)_2 = 195/2$, $4!(13/2)_4 = 188955/2$, and $6!(13/2)_6 = 1368978975/4$. These factors show directly how every denominator in (GS.34) arises. The invertible triangular moment map has the complete inverse

$$\begin{aligned} \mu_0 &= cc_0, \\ \mu_2 &= (195cc_2 + 13\mu_0)/2, \\ \mu_4 &= (377910cc_4 + 188\mu_2 - 663\mu_0)/4, \\ \mu_6 &= (2737957950cc_6 + 1100\mu_4 - 27602\mu_2 + 69615\mu_0)/8. \end{aligned} \quad (\text{GS.35})$$

Substitution in (GS.34) proves that the displayed maps are mutual inverses. In particular the mass in (GS.7) remains μ_0 , while the reference mass in (GS.21) remains c .

38.5 Certified infinite quantities and exact rational error transport

The companion checker evaluates (GS.15) for $r = 0, 1, 2, 3$ and $J = 8$ in Arb real balls at 256-bit precision, with one arithmetic thread and python-flint 0.9.0. All polynomial input is exact rational arithmetic; π , exponentials, powers and gamma functions are enclosed by Arb. The upper-incomplete-gamma call is `C.gamma_upper(alpha)`, with argument C and order α , precisely as in (GS.13). Each finite

ball is enlarged by a symmetric ball with radius the upward-rounded upper endpoint of (GS.18). It therefore encloses the infinite quantity. The computed tails satisfy the rational inequalities

$$E_{0,8} < 7 \cdot 10^{-107}, \quad E_{1,8} < 4 \cdot 10^{-103}, \quad E_{2,8} < 2 \cdot 10^{-99}, \quad E_{3,8} < 14 \cdot 10^{-96}. \quad (\text{GS.36})$$

Each inequality is tested as a strict rigorous ball comparison.

The resulting exact rational decimal intervals are:

$$\begin{aligned} \mu_0 &\in [1.2790072478464851404795335922671932744916, \\ &\quad 1.2790072478464851404795335922671932744919], \\ \mu_2 &\in [13.0555493025705584353926846581231936824343, \\ &\quad 13.0555493025705584353926846581231936824346], \\ \mu_4 &\in [383.2743417173642425544072374791308324300288, \\ &\quad 383.2743417173642425544072374791308324300291], \\ \mu_6 &\in [17946.5273016052076235667646919834204772848931, \\ &\quad 17946.5273016052076235667646919834204772848934], \\ c &\in [6.3614260045872192926298535487331092297380, \\ &\quad 6.3614260045872192926298535487331092297383], \\ c_0 &\in [0.2010566886927858658099161632202187798614, \\ &\quad 0.2010566886927858658099161632202187798617], \\ c_2 &\in [0.0076454429994864011881338514807939967436, \\ &\quad 0.0076454429994864011881338514807939967439], \\ c_4 &\in [-0.0000305169208037620437467248395269085390, \\ &\quad -0.0000305169208037620437467248395269085387], \\ c_6 &\in [-0.0000003851554876709816618220743079695913, \\ &\quad -0.0000003851554876709816618220743079695910]. \end{aligned}$$

Each interval has width $3 \cdot 10^{-40}$. The coefficient balls are computed directly from the infinite-moment balls in (GS.34), before their conversion to the displayed decimal intervals. Interval dependency can enlarge enclosures; it cannot invalidate inclusion. The signs $c_4 < 0$ and $c_6 < 0$ follow from their negative upper endpoints and are preserved.

Here is the exact outward conversion used by the checker. If the resulting Arb ball is Q , let ℓ and u be its outward dyadic endpoints and set $d = 40$,

$$L_Q = 10^{-d}(\lfloor 10^d \ell \rfloor - 1), \quad U_Q = 10^{-d}(\lceil 10^d u \rceil + 1). \quad (\text{GS.37})$$

The checker then verifies $Q > L_Q$ and $Q < U_Q$ by rigorous ball comparison. The receipt stores L_Q, U_Q both as decimal strings and exact rational strings, and stores the dyadic endpoints as exact rationals. Thus every display can be checked with integer arithmetic after ball generation. The same receipt records the tail's exact upward dyadic rational.

The executable finite checks additionally verify all exact rational polynomial lists in (GS.12) and (GS.33) and the four exact rational denominator factors following (GS.34). They compare the ball evaluations of the polynomial and integer numerator forms in (GS.34) for overlap, and similarly compare the endpoint-containing recurrence (GS.14) at each argument pair used. These checks detect convention mistakes; the analytic certificate is the ball enclosure enlarged by the proved infinite tail, independently of any truncation agreement.

Four fault modes inject specific false mathematical assertions: discarding the inversion factor two preserves μ_4 ; assigning reference mass one preserves c_0 ; c_4 is nonnegative; and the cutoff-zero sum with zero error encloses the full positive seed mass. The already certified disjoint intervals reject each assertion. These checks use explicit conditionals and a nonzero exit status; Python optimization

cannot remove them. Normal and optimized runs produce the same mathematical results and each fault produces one recorded rejection.

The source note's two floating truncations remain its original numerical evidence. The intervals above independently certify their displayed values through the original theta series and (GS.18). The correctness contract for the elementary ball operations and outward endpoints is documented in the [python-flint Arb interface](#). No proof of that arithmetic library's implementation is asserted.

38.6 Exact tensor coefficient interface through degree six

The map from the certified one-factor data to every finite tensor degree can also be stated without changing the mass or sum variable. Let $k \geq 1$, $u = t_1 + \cdots + t_k$, and $m_k = w^{*k}$. Fubini gives $\int m_k = \mu_0^k$, and exponential integrability proves all the moments below are finite. Multiplying the generating functions (GS.25) gives the finite polynomial identity

$$b_n^{(k\lambda)}(u) = \sum_{n_1 + \cdots + n_k = n} \frac{n!}{n_1! \cdots n_k!} \prod_{j=1}^k b_{n_j}^{(\lambda)}(t_j). \quad (\text{GS.38})$$

Its proof is literal multiplication: the exponents of $1 \mp iz$ add to $-k\lambda \pm iu/2$, and the coefficient of degree n has only finitely many terms.

Use $\lambda = 13/4$, $\alpha = 13/2$, $a_j = (\alpha)_j c_j$, and the formal coefficient series $\mathcal{A}(z) = \sum_{j \geq 0} a_j z^j$. For any coefficient appearing below only finitely many c_j are needed. Integration of (GS.38), using (GS.32) separately in each factor, proves

$$\int b_n^{(k\lambda)}(u) m_k(u) du = c^k n! d_{k,n}, \quad d_{k,n} = [z^n] \mathcal{A}(z)^k. \quad (\text{GS.39})$$

No interchange of an infinite formal series with an integral is required. Evenness gives the complete first four nonzero expressions

$$\begin{aligned} d_{k,0} &= c_0^k, \\ d_{k,2} &= k c_0^{k-1} a_2, \\ d_{k,4} &= k c_0^{k-1} a_4 + \binom{k}{2} c_0^{k-2} a_2^2, \\ d_{k,6} &= k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3. \end{aligned} \quad (\text{GS.40})$$

For degree six the partitions into even positive degrees are 6, 4 + 2 and 2 + 2 + 2; their distinct-factor choices have respectively k , $k(k-1)$ and $\binom{k}{3}$ possibilities. This proves all three coefficients, including the mixed term and its sign. The analogous enumeration proves degree two and four. Since $c_0 > 0$, even the displayed negative powers at $k = 1, 2$ are defined; their accompanying zero combinatorial factors give the correct absent terms.

Equations (GS.34)–(GS.35), (GS.39)–(GS.40) are the exact map from the certified original moments to these all- k low-degree arithmetic gamma moments, retaining c^k and the coordinate u . For example their degree-zero identity is $c^k d_{k,0} = \mu_0^k$. They give interval evaluations at every specified finite integer k by rational and ball arithmetic with the already enclosed inputs. They assert no uniform estimate when polynomial degree grows. For this actual seed $h = 1$, the finite arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the quotient map sends every polynomial to zero because every polynomial lies in the ideal (1). The analytic source measure and all the coefficients above remain nonzero before that quotient. No nonzero packet quotient is inferred from these seed calculations.

39 The first three original source costs at every tensor degree

Retain the exact theta seed, its Fourier coordinate and its measure

$$w(t) dt = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt.$$

Equations (GS.1)–(GS.18) prove that w is even, has positive mass μ_0 , and has finite moments of every order. The certificate of (GS.36)–(GS.37) encloses the original $\mu_0, \mu_2, \mu_4, \mu_6$. This section transports those existing certified integrals by exact rational arithmetic. It performs no further quadrature.

For each integer $k \geq 1$ put $u = t_1 + \cdots + t_k$, $m_k = w^{*k}$ and $M_j(k) = \int u^j m_k(u) du$. Fubini and evenness give the full first four even sum moments

$$\begin{aligned} M_0(k) &= \mu_0^k, \\ M_2(k) &= k\mu_2\mu_0^{k-1}, \\ M_4(k) &= k\mu_4\mu_0^{k-1} + 3k(k-1)\mu_2^2\mu_0^{k-2}, \\ M_6(k) &= k\mu_6\mu_0^{k-1} + 15k(k-1)\mu_4\mu_2\mu_0^{k-2} + 15k(k-1)(k-2)\mu_2^3\mu_0^{k-3}. \end{aligned} \quad (\text{GS.41})$$

Here every missing term integrates to zero because at least one factor has an odd moment. For M_4 , the surviving exponent partitions are 4 and 2 + 2. The second has $\binom{k}{2} 4!/(2!2!) = 3k(k-1)$ terms. For M_6 the partitions are 6, 4 + 2 and 2 + 2 + 2; their numbers of terms are respectively k , $k(k-1)6!/(4!2!) = 15k(k-1)$ and $\binom{k}{3} 6!/(2!2!2!) = 15k(k-1)(k-2)$. The second-moment expansion has precisely k surviving terms. These finite counts prove (GS.41) with its full mass factors. Because $\mu_0 > 0$, the displayed powers are defined even when a zero combinatorial coefficient accompanies a negative exponent.

All nonzero polynomial norms in $L^2(m_k du)$ are strictly positive. Indeed the nonnegative density m_k has positive mass μ_0^k , so its positivity set has positive Lebesgue measure. A nonzero polynomial has finitely many real roots, as repeated division by its distinct linear factors cannot exceed its degree. Its squared modulus times m_k is therefore positive on a set of positive measure, and its integral is positive. This proves every positivity and division below for the actual source.

Put $\nu = \mu_2/\mu_0$. With the norms taken in the unchanged measure $m_k du$, the first four monic orthogonal polynomials and all their squared norms are

$$\begin{aligned} p_0(u) &= 1, & h_0(k) &= M_0(k), \\ p_1(u) &= u, & h_1(k) &= M_2(k), \\ p_2(u) &= u^2 - k\nu, & h_2(k) &= M_4(k) - \frac{M_2(k)^2}{M_0(k)}, \\ p_3(u) &= u^3 - \frac{M_4(k)}{M_2(k)}u, & h_3(k) &= M_6(k) - \frac{M_4(k)^2}{M_2(k)}. \end{aligned} \quad (\text{GS.42})$$

To verify orthogonality of p_2 , its pairing with 1 is $M_2 - k\nu M_0 = 0$, and its pairing with u is zero by evenness. For p_3 , parity gives zero against 1 and p_2 ; its pairing with u is $M_4 - (M_4/M_2)M_2 = 0$. Expanding each squared polynomial gives the norms in (GS.42). Uniqueness follows because the difference of two monic orthogonal polynomials of the same degree lies in the lower-degree space, is orthogonal to itself, and hence is zero by the preceding positivity argument.

Define the positive original monic source costs $a_j^{(k)} = h_j(k)/h_{j-1}(k)$. These are exactly the lower coefficients in the real-coordinate three-term recurrence: in up_j , its pairing with p_{j-1} equals the pairing of p_j with up_{j-1} , which is h_j because $up_{j-1} - p_j$ has lower degree. Pairings with lower degrees vanish by the same argument. The degree- j coefficient vanishes by evenness. Therefore $up_j = p_{j+1} + a_j^{(k)} p_{j-1}$. For $j = 3$, the next monic polynomial p_4 exists by finite Gram orthogonalization, whose Gram is positive definite by the proved polynomial norm property; its full formula is not

needed to determine the already specified coefficient h_3/h_2 . Substituting (GS.41) into (GS.42) gives

$$\begin{aligned} a_1^{(k)} &= k\nu, \\ a_2^{(k)} &= \frac{\mu_4}{\mu_2} + (2k-3)\nu, \\ a_3^{(k)} &= \frac{M_6(k) - M_4(k)^2/M_2(k)}{M_4(k) - M_2(k)^2/M_0(k)}. \end{aligned} \quad (\text{GS.43})$$

For the second equality one obtains $M_4/M_2 = \mu_4/\mu_2 + 3(k-1)\nu$ and $M_2/M_0 = k\nu$, whose exact difference is the displayed cost. For an explicit expanded version of the third equality, write $r_4 = \mu_4/\mu_0$ and $r_6 = \mu_6/\mu_0$, solely as the indicated scalar quotients of the original moments. Then

$$a_3^{(k)} = \frac{r_6 - r_4^2/\nu + 9(k-1)r_4\nu + 3(k-1)(2k-7)\nu^3}{r_4 + (2k-3)\nu^2}. \quad (\text{GS.44})$$

To check every term, divide the numerator of (GS.43) by $k\mu_0^k$ using (GS.41). The square term contributes $r_4^2/\nu + 6(k-1)r_4\nu + 9(k-1)^2\nu^3$; subtracting it from $r_6 + 15(k-1)r_4\nu + 15(k-1)(k-2)\nu^3$ gives the numerator in (GS.44). The denominator of (GS.43), divided by the same $k\mu_0^k$, is $r_4 + (2k-3)\nu^2$. Thus both original norms and all mass factors are recovered from these scalar abbreviations.

The original spectral coordinate is $S = k/2 + iu$. Let $c_k = k/2$ and define the algebra isomorphism

$$\Psi_k : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad (\Psi_k P)(u) = P(c_k + iu), \quad \Psi_k^{-1}p(S) = p((S - c_k)/i). \quad (\text{GS.45})$$

Direct substitution proves the two inverse identities. Give $\mathbb{C}[S]$ the source norm $\int |P(c_k + iu)|^2 m_k(u) du$; then Ψ_k is a linear isometry onto the polynomial source in the real coordinate. On the degree- j monic affine space, its phase-preserving monic inverse is $p \mapsto P_j = i^j \Psi_k^{-1}p$. Since $|i^j| = 1$, this map preserves the original norm and has inverse $P \mapsto i^{-j} \Psi_k P$ on that affine space. In particular the complete first four spectral monic polynomials are

$$\begin{aligned} P_0 &= 1, \quad P_1 = S - c_k, \quad P_2 = (S - c_k)^2 + k\nu, \\ P_3 &= (S - c_k)^3 + \left\{ \frac{\mu_4}{\mu_2} + 3(k-1)\nu \right\} (S - c_k). \end{aligned} \quad (\text{GS.46})$$

Their squared norms are exactly h_0, h_1, h_2, h_3 from (GS.42). Multiplying the real recurrence by i^{j+1} proves the spectral recurrence with its actual lower sign,

$$SP_j = P_{j+1} + c_k P_j - a_j^{(k)} P_{j-1}. \quad (\text{GS.47})$$

Thus the positive norm ratio $a_j^{(k)}$ and the negative lower coefficient in the S presentation have their exact phase map.

To connect these costs to the existing Gamma determinant corrections, let D_n be the determinant of the source Gram on $1, u, \dots, u^{n-1}$, with $D_0 = 1$, and let D_n^Γ use the Gamma sum measure $r_{13/4}^{*k} du$. Its mass is $c_{13/4}^k$, and its monic norms are $c_{13/4}^k j!(13k/2)_j$, by (GS.23), Fourier uniqueness and (GS.28). Specifically Fourier transformation of the convolution gives $c_{13/4}^k \cosh^{-13k/2} y$; division by the mass constant in (GS.23) identifies its density as a positive scalar times $r_{13k/4}$, proving the asserted norms without any change to u . Monic orthogonalization is triangular with diagonal one, so $D_n = \prod_{j < n} h_j$ and the analogous Gamma product holds. Define $X_n = D_n/D_n^\Gamma$ and

$$Q_N = \frac{X_{N+2}X_N}{X_{N+1}^2}, \quad a_{N+1}^{(k)} = (N+1)(N+13k/2)Q_N. \quad (\text{GS.48})$$

Dividing the two consecutive determinant products proves the second identity. In particular, with $\eta = \mu_4/\mu_2$,

$$\begin{aligned} Q_0 &= \frac{2\nu}{13}, \\ Q_1(k) &= \frac{\eta + (2k-3)\nu}{13k+2} = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \eta - \frac{43}{13}\nu, \\ Q_2(k) &= \frac{2a_3^{(k)}}{39k+12}. \end{aligned} \tag{GS.49}$$

Every equality is obtained by substituting (GS.43) into (GS.48). No volume of a nonzero packet quotient enters these source costs.

The pure rational checker reads the exact dyadic moment endpoints from the pinned GS.36 certificate and propagates them through (GS.43)–(GS.49) using `fractions.Fraction` exclusively. For intervals $[a, b]$ and $[c, d]$, it uses endpoint addition, $[a - d, b - c]$ for subtraction, the minimum and maximum of the four endpoint products for multiplication, and multiplication by $[1/d, 1/c]$ for division after checking $c > 0$. These rules follow from linearity in each variable and the decreasing map $x \mapsto 1/x$ on the positive axis. Thus they preserve inclusion without any floating-point operation. Applying (GS.37) with exact rational floor and ceiling yields, among the saved intervals,

$$\begin{aligned} \nu &\in [10.2075647534857216888657612522375551749964, \\ &\quad 10.2075647534857216888657612522375551749967], \\ Q_0 &\in [1.5703945774593417982870401926519315653839, \\ &\quad 1.5703945774593417982870401926519315653842], \\ a_2^{(1)} &\in [19.1496329281085602472067784658376980738791, \\ &\quad 19.1496329281085602472067784658376980738794], \\ \rho &\in [-4.4062857337815667270988244239412754068822, \\ &\quad -4.4062857337815667270988244239412754068819]. \end{aligned}$$

All four widths are $3 \cdot 10^{-40}$. The receipt also encloses μ_4/μ_2 , $\mu_4\mu_0 - \mu_2^2$, $a_2^{(2)}$, $Q_1(1)$, $a_3^{(1)}$ and $Q_2(1)$. In particular $\rho < 0$ and (GS.49) give the exact fixed-degree all- k comparison

$$0 < Q_1(1) \leq Q_1(k) < Q_0, \quad Q_1(k+1) - Q_1(k) = \frac{-13\rho}{(13k+2)(13k+15)} > 0. \tag{GS.50}$$

The positive lower bound follows either from $h_2/h_1 > 0$ or its certified $k = 1$ enclosure; the difference follows by literal subtraction in (GS.49). The computation concerns original source polynomial degrees at most three, at every specified tensor degree. The analytic seed still has quotient $\mathbb{C}[S]/(1) = 0$, its quotient determinant is the empty determinant one, and every nonzero source norm above remains before that zero quotient.

40 Finite arithmetic Gram transfer with the original minimum quotient metric

This continuation proves a finite coefficient-to-metric map, its exact boundary-resolvent remainder, and a two-degree determinant formula for the consecutive volume ratio in the arithmetic Toda control. The original Gamma mass, tensor sum, arithmetic amplitude, quotient coordinates and relation polynomial remain in every formula. A bounded positive-weight calibration at the end proves that the relative Gamma correction can increase with degree even while the actual minimum volume decreases. That calibration is explicitly an auxiliary measure, with no assertion that its polynomial is an annihilator of actual zeta zeros.

The sources read for this calculation are the complete `RESEARCH_NOTE.md` and `NOTE.tex` in the delivered Tau Gamma Convolution Descent, and the original arithmetic definitions and proofs TVB.1–TVB.38 in `toda_cv_exact_bridge_20260912.tex`. The finite transfer below is at the original observation $\theta = 0$. Its proof does not replace a nonzero exponential tilt by a polynomial test. All interval claims here are finite rational inequalities; no uniform estimate as packet, tensor degree or polynomial degree grow is asserted.

40.1 The original source, relation and coordinate maps

Let $h(s)$ be a monic nonempty packet of actual zeros of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s),$$

including the complete order at each selected zero. Put $v_h = g/h$, $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$, and $\mu_h = \int_{\mathbb{R}} w_h(t) dt$. The quotient is entire: local Taylor division removes exactly the stated vanishing order. Its value at each selected zero is nonzero. The full Taylor remainder $[v_h]_h$ is therefore a unit of $E_h = \mathbb{C}[s]/(h)$. Indeed in $\mathbb{C}[t]/(t^m)$ the inverse of $\sum_{j < m} a_j t^j$, $a_0 \neq 0$, has coefficients $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$; the Chinese remainder map from coprime factors is the exact algebra isomorphism obtained by Bezout's identity. This retains every jet and multiplicity.

The original tensor coordinate is

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GMT.1})$$

The source observation is the conjugate-linear-first pairing

$$\langle P, Q \rangle_h = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k. \quad (\text{GMT.2})$$

The change $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Thus Fubini gives (GMT.2) from the literal tensor Mellin observation without a divisor or an average coordinate. The complex Mellin amplitude remains

$$\prod_{i=1}^k \frac{v_h(1/2 + it_i)}{\sqrt{2\pi}} P(c + i \sum_i t_i).$$

Let $I = (h(s_1), \dots, h(s_k))$ and let χ be the monic generator of $I \cap \mathbb{C}[S]$, of degree q . Existence follows from an annihilating polynomial of multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$. Division by its least-degree monic member proves that every member of the intersection is a multiple of χ . For $N \geq q - 1$ define $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and the exact maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0, \quad M_\chi P = \chi P, \quad \mathcal{C} = \mathbb{C}[S]/(\chi), \quad J_N P = [P]_\chi. \quad (\text{GMT.3})$$

Here $\mathcal{P}_{-1} = 0$. All coordinate vectors of $\mathcal{C}$ are in the written basis $1, S, \dots, S^{q-1}$. Its map to the original tensor jet is

$$\eta[P] = [v_h]_h^{\otimes k} [P(S)]_I.$$

Its kernel is zero by the preceding intersection ideal and invertible unit. No Gamma coefficient calculation changes this map.

Write B_N for the coefficient matrix of the columns $\chi, \chi S, \dots, \chi S^{N-q}$ in $\mathcal{P}_N$. The matrix $[E_N, B_N]$, where E_N includes the first q monomials, is triangular with diagonal one. Thus its determinant is exactly one in this order; putting the relation block first contributes $(-1)^{q(N-q+1)}$.

The original theta map is $\Theta\phi(x) = \sum_{n \neq 0} \phi(nx)$, with $D = -x\partial_x$, and F_h satisfies $\mathcal{M}F_h = v_h$, $h(D)F_h = \Theta\phi_0$, where $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. The exact primitive attached to a column χP is

retained as follows. Successive monic division in $s_1, \dots, s_k$ gives $\chi(S) = \sum_i h(s_i) Q_i(\mathbf{s})$. Its tensor primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) P(\sum_j D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GMT.4})$$

The differential at the i th position has sign $(-1)^{i-1}$, so differentiation gives precisely the original source vector $\mathcal{V}_{h,k}(\chi P)$. This proves the relation-to-boundary map, including its sign. The construction of F_h uses the actual zero-division operator $S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ at a vanishing Mellin value. The two integrals give respectively every large- and small- x power bound. Differentiation gives $(D - \rho) S_\rho F = F$, hence all Euler derivative bounds and the Mellin quotient by $s - \rho$. Starting from $\Theta \phi_0$ and successively dividing the complete root list constructs the displayed F_h . The seed identity follows by substituting $y = \pi x^2$ in its Mellin integral, which gives $\mathcal{M}\phi_0(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2)$ for $\Re s > -2$, followed by the absolutely convergent theta sum for $\Re s > 1$ and analytic continuation.

On each original support label ℓ , a boundary still maps to $(\ell, 0) = e_\ell$; external absence τ still maps to τ . The linear polynomial maps (GMT.3)–(GMT.4) precede the quadratic map $R \mapsto R^* H R$ from source representatives to Hermitian forms. For the empty packet $h = 1$, the analytic source remains, but $\chi = 1, q = 0, \mathcal{C} = 0, G_N$ is the zero-dimensional form and its determinant is one. Every quotient formula below then has its empty-block interpretation.

40.2 Full Gamma mass and the finite coefficient evaluation

For $\lambda > 0$ set

$$\alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{GMT.5})$$

The beta integral with its coordinate $x = 2y$ gives

$$|\Gamma(\lambda + it/2)|^2 = 2\Gamma(2\lambda) \int_{\mathbb{R}} e^{ity} (2 \cosh y)^{-2\lambda} dy.$$

Fourier inversion therefore gives $\int e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}$. Both functions are integrable: the right integrand and all its derivatives decay exponentially, so integration by parts bounds its Fourier transform by every inverse power. Exponential integrability of r_λ on $|a| < \pi/2$ follows by rotating the gamma integral by any φ with $|a| < \varphi < \pi/2$; the squared modulus has a bound $\Gamma(\lambda)^2 (\cos \varphi)^{-2\lambda} e^{-\varphi|t|}$. Analytic continuation now proves

$$\int e^{au} r_{\lambda,k}(u) du = C_k (\cos a)^{-\beta}, \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{C_k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GMT.6})$$

The last identity is Fourier uniqueness with the exact product mass.

Let $b_j^{(\lambda)}$ denote the monic polynomials with

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GMT.7})$$

These agree with $j! P_j^{(\lambda)}(t/2; \pi/2)$ in the classical generating function at [DLMF 18.23.7](#). The next proof fixes their scale without relying on that identification. Differentiating (GMT.7) gives

$$b_{j+1}^{(\lambda)} = t b_j^{(\lambda)} - j(j + \alpha - 1) b_{j-1}^{(\lambda)}.$$

For small real z, w , integration of the product of generating functions by (GMT.6) gives $c_\lambda(1 - zw)^{-\alpha}$, since $\cos(\arctan z + \arctan w) = (1 - zw)/\sqrt{(1 + z^2)(1 + w^2)}$. Local exponential domination justifies every coefficient derivative. Thus

$$\int b_i^{(\lambda)} b_j^{(\lambda)} r_\lambda = \delta_{ij} c_\lambda j! (\alpha)_j, \quad \int b_i^{(k\lambda)} b_j^{(k\lambda)} r_{\lambda,k} = \delta_{ij} C_k j! (\beta)_j. \quad (\text{GMT.8})$$

Put $B(t) = |v_h(1/2 + it)/\Gamma(\lambda + it/2)|^2$, so that $w_h = Br_\lambda$ with its full complex amplitude still present in the Mellin source. One may use the bounded reference constructed in the Gamma descent: $\lambda = 1/4$ for degree at least three, or $\lambda = 1/4 + L$, $L \geq \max(0, 3 - \deg h)$ otherwise. The relevant boundedness follows directly from the compact entire quotient, $|h(1/2 + it)| \geq (|t|/2)^{\deg h}$ beyond twice the root radius, $|\zeta(1/2 + it)| \leq 2(1 + |t|)$, and the nonvanishing factors $\prod_{j < L} (j + 1/4 + it/2)$ in the gamma shift. In particular $B \in L^2(r_\lambda dt)$. All statements of this section also hold for any specified nonnegative nonzero $B \in L^2(r_\lambda dt)$. For that domain write $w_B = Br_\lambda$ and $m_{B,k} = w_B^{*k}$; at the arithmetic input these equal the original $w_h, m_{h,k}$. Cauchy-Schwarz with (GMT.6) gives all polynomial moments. The calibration below uses this latter domain before it is explicitly capped to a bounded function.

The sum adjoint is the actual conditional integral

$$C_\Sigma F(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GMT.9})$$

Its adjoint inclusion $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$ is isometric by Fubini; hence $C_\Sigma U_\Sigma = 1$ and $F \mapsto (C_\Sigma F, (1 - U_\Sigma C_\Sigma)F)$ has inverse $(f, n) \mapsto U_\Sigma f + n$ on $L^2(r_{\lambda,k}) \oplus \ker C_\Sigma$. At $k = 1$ both maps are the identity. The normal component and every support label remain in this paired map. The specified sum density is $m_{B,k} = r_{\lambda,k} B_k$, $B_k = C_\Sigma B^{\otimes k}$; in particular it is $m_{h,k}$ at the arithmetic input.

Define the real coefficients $c_{B,j}$ (the original $c_{h,j}$ for the arithmetic multiplier $B = B_{h,\lambda}$) and formal coefficient power

$$c_{B,j} = \frac{\int b_j^{(\lambda)} w_B}{c_\lambda j! (\alpha)_j}, \quad A(z) = \sum_{j \geq 0} (\alpha)_j c_{B,j} z^j, \quad d_n = [z^n] A(z)^k. \quad (\text{GMT.10})$$

Multiplying the finite generating coefficients in (GMT.7) gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing with $B^{\otimes k}$ gives the exact finite identity

$$\int b_n^{(k\lambda)}(u) m_{B,k}(u) du = C_k n! d_n. \quad (\text{GMT.11})$$

Every term is integrable by Cauchy-Schwarz; no infinite formal-series product was interchanged with an integral.

For a specified finite list of polynomials $p_0, \dots, p_r$ in $\mathcal{P}_N$, expand the actual tests

$$\overline{p_a(c + iu)} p_b(c + iu) = \sum_{n=0}^{2N} L_{ab,n} b_n^{(k\lambda)}(u).$$

This expansion is a finite algorithm: subtract the top coefficient times the monic polynomial of the same degree, then continue in descending degree through zero. The resulting remainder is exactly zero, by the decreasing degree. Equations (GMT.10)–(GMT.11) prove

$$\boxed{H(p)_{ab} = C_k \sum_{n=0}^{2N} n! L_{ab,n} d_n.} \quad (\text{GMT.12})$$

The coefficient-to-form map is homogeneous of degree k in $c_{B,0}, \dots, c_{B,2N}$. Its full coordinate type is $\mathbb{R}^{2N+1} \rightarrow \text{Herm}(\mathcal{P}_N)$ on the actual moment vectors, followed by congruence $H \mapsto P^*HP$ for a polynomial coefficient matrix P . The nonnegative density has positive mass. Its positivity set has positive Lebesgue measure, and a nonzero polynomial has only finitely many real roots. Thus the integral of its squared modulus is strictly positive, proving positivity of every finite source Gram and of every injective relation Gram. Choosing $p_a = S^a$ gives H_N ; choosing the columns of B_N gives exactly $B_N^* H_N B_N$. Choosing actual representatives gives their actual Gram. Every relation column has degree at most N , including its complete χ factor.

One may instead evaluate in the monic Gamma columns $p_j^\Gamma(S) = i^j b_j^{(k\lambda)}((S - c)/i)$. Their coefficient matrix U_N is triangular with diagonal one, and their reference Gram is

$$U_N^* H_N^\Gamma U_N = \text{diag}\{C_k j! (\beta)_j : 0 \leq j \leq N\}. \quad (\text{GMT.13})$$

This is a stated invertible coordinate map, with all squared norms retained. In particular it divides neither the measure nor the polynomials by their mass or norms. The original reduction matrix becomes $J_N U_N$; the original source form is recovered by $U_N^{-*}(\cdot) U_N^{-1}$.

40.3 Finite coefficient errors and polynomial-test bounds

Finite errors propagate before any inverse is taken. For real central coefficients $\widehat{c}_{B,j}$ and radii $e_j \geq 0$ enclosing the actual $c_{B,j}$ for $0 \leq j \leq 2N$, define the finite polynomials

$$\widehat{A} = \sum_{j=0}^{2N} (\alpha)_j \widehat{c}_{B,j} z^j, \quad A_+ = \sum_{j=0}^{2N} (\alpha)_j |\widehat{c}_{B,j}| z^j, \quad E_+ = \sum_{j=0}^{2N} (\alpha)_j e_j z^j,$$

and set

$$\widehat{d}_n = [z^n] \widehat{A}^k, \quad e_n^{(k)} = [z^n] \{(A_+ + E_+)^k - A_+^k\}. \quad (\text{GMT.14})$$

The multinomial expansion of each difference of products groups terms by the nonempty set of factors containing a coefficient error. Taking absolute values term by term proves $|d_n - \widehat{d}_n| \leq e_n^{(k)}$ for $n \leq 2N$. Thus the Hermitian central matrix $\widehat{H}$ from (GMT.12) satisfies

$$|(H - \widehat{H})_{ab}| \leq \varepsilon_{ab} := C_k \sum_{n=0}^{2N} n! |L_{ab,n}| e_n^{(k)}. \quad (\text{GMT.15})$$

The masses in this formula are exact constants. An interval evaluation of those constants is propagated by the same product rule; it cannot be replaced by setting $C_k = 1$. Exact polynomial coefficients of χ remain fixed in $L_{ab,n}$; any numerical enclosure of such coefficients must likewise be propagated before reporting a matrix certificate.

There is also a strictly polynomial-test refinement of the L^2 tail bound in the supplied descent. Write $B_L = \sum_{j \leq L} c_{B,j} b_j^{(\lambda)}$ and $E_k = B_k - C_\Sigma(B_L^{\otimes k})$. Orthogonality proves $\|B_L\| \leq \|B\|$. Expanding the tensor difference into k terms and using contraction of C_Σ proves $\|E_k\| \leq k \|B\|^{k-1} \|B - B_L\|$ with the original measures. Equation (GMT.11) applied degree by degree shows that E_k is orthogonal to all polynomials of degree at most L . Therefore

$$\left| \int \overline{p_a(c + iu)} p_b(c + iu) E_k(u) r_{\lambda,k}(u) du \right| \leq \|E_k\| \left(C_k \sum_{n > L} n! (\beta)_n |L_{ab,n}|^2 \right)^{1/2}. \quad (\text{GMT.16})$$

This follows by discarding only the orthogonal test components whose pairings are proved zero, and then applying Cauchy–Schwarz and (GMT.8). It is exactly zero at $L \geq 2N$. In particular a signed truncated density has the same finite positive Gram at that degree; pointwise positivity of the truncation is not required by this map.

An intrinsic finite polynomial-test bound is also available. The reference evaluation kernel is

$$\mathcal{K}_N^\Gamma(u) = \sum_{j=0}^N \frac{b_j^{(k\lambda)}(u)^2}{C_k j! (\beta)_j}.$$

Expansion in the original monic basis and Cauchy-Schwarz give $|P(c + iu)|^2 \leq \|P\|_\Gamma^2 \mathcal{K}_N^\Gamma(u)$. It follows that the form induced by a real error E_k obeys

$$|\langle P, P \rangle_{E_k}| \leq \|P\|_\Gamma^2 \|E_k\|_{L^2(r_{\lambda,k})} \|\mathcal{K}_N^\Gamma\|_{L^2(r_{\lambda,k})}. \quad (\text{GMT.17})$$

The last squared norm is an exact Gamma integral of a degree- $4N$ polynomial, evaluable by (GMT.8) or (GMT.12) with the reference weight. This is a finite evaluation-functional estimate, retaining every Christoffel-kernel denominator and the full mass.

40.4 Exact least lifts and the boundary-resolvent remainder

For a positive definite source matrix H in the fixed source coordinates, define

$$K(H) = J_N H^{-1} J_N^*, \quad G(H) = K(H)^{-1}, \quad R(H) = H^{-1} J_N^* G(H). \quad (\text{GMT.18})$$

Surjectivity of J_N makes $K(H)$ positive definite. Multiplication gives $J_N R(H) = 1$ and $B_N^* H R(H) = 0$. Every other lift is $R(H)x + B_N y$, whose squared norm is $x^* G(H)x + y^* B_N^* H B_N y$. This proves minimum, uniqueness and $R(H)^* H R(H) = G(H)$.

Let H_0 and H_1 be two actual positive finite source forms on the same polynomial space. In particular they may be the Gamma and arithmetic forms already constructed. Put

$$H_t = H_0 + tE, \quad E = H_1 - H_0, \quad 0 \leq t \leq 1, \quad A_0 = B_N^* H_0 B_N, \quad F = B_N^* E B_N, \quad C = B_N^* E R_0,$$

where $R_0 = R(H_0)$ and $G_0 = G(H_0)$. The whole interval of forms is positive because it is the convex combination of the two actual positive forms. Thus every nonempty $A_t = A_0 + tF$ is positive. The complete, signed resolvent formulas are

$$\boxed{R_t = R_0 - tB_N A_t^{-1} C, \quad G_t = G_0 + tR_0^* E R_0 - t^2 C^* A_t^{-1} C.} \quad (\text{GMT.19})$$

To prove the first equation, its reduction is the identity and $B_N^* H_t R_0 = tC$, while $B_N^* H_t B_N = A_t$. Consequently its boundary pairing is zero, so (GMT.18) identifies it. Expanding its full H_t norm gives $R_0^* H_t R_0 - 2t^2 C^* A_t^{-1} C + t^2 C^* A_t^{-1} A_t A_t^{-1} C$. The two negative terms and one positive term yield exactly (GMT.19). In particular the unscaled matrix $C^* A_t^{-1} C$ is positive semidefinite and vanishes precisely when $C = 0$. Pairing $A_t^{-1} > 0$ with every column of C proves this criterion. The full quadratic loss $t^2 C^* A_t^{-1} C$ vanishes precisely when $t = 0$ or $C = 0$; for an empty boundary block it is the zero form throughout.

The representative difference is a literal boundary with coefficient vector $-tA_t^{-1} Cx$; inserting that polynomial into (GMT.4) gives its full original theta primitive. The same jets ηx persist. No statement of metric nonidentity leaves its comparison map unspecified.

For completeness the exact derivative identities are

$$R'_t = -B_N A_t^{-1} B_N^* E R_t, \quad G'_t = R_t^* E R_t, \quad G''_t = -2(B_N^* E R_t)^* A_t^{-1} (B_N^* E R_t). \quad (\text{GMT.20})$$

Differentiate $J_N R_t = 1$ and $B_N^* H_t R_t = 0$ to obtain the first formula. The derivative R'_t is a boundary, hence its two pairings with R_t in the derivative of $R_t^* H_t R_t$ vanish. This proves the second formula. Differentiating it and substituting the first proves the third. This is a proved concavity statement along the actual finite metric segment, with the entire boundary coupling retained.

The determinant-line calculation has no hidden source-size exponent. The change $[E_N, B_N] \mapsto [R(H), B_N]$ adds boundary columns to the first q columns and has determinant one. Its Gram is block diagonal. Therefore

$$V_N(H) = \det G(H) = \frac{\det H}{\det(B_N^* H B_N)}. \quad (\text{GMT.21})$$

For the original arithmetic and Gamma forms this is exactly $V_N = V_N^\Gamma T_N$, with the same $T_N = X_{N+1}/Y_{N-q+1}$ as the supplied descent. Formula (GMT.19) gives its complete matrix-level map and its nonlinear remainder.

40.5 A two-degree map for the actual consecutive ratio

Fix $N \geq q$. In original source monomials split degree $N + 1$ into degrees $0, \dots, N - 1$ followed by S^N, S^{N+1} , and write

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_-^{-1} C_D, \quad J_+ = [J_-, J_D], \quad F_D = J_D - J_- H_-^{-1} C_D. \quad (\text{GMT.22})$$

Here $H_- = H_{N-1}$, $H_+ = H_{N+1}$, W is 2×2 , and $F_D : \mathbb{C}^2 \rightarrow \mathcal{C}$. Its columns are the quotient classes of the two specified polynomials after subtracting their actual orthogonal projections onto $\mathcal{P}_{N-1}$. The block map

$$(x, y) \mapsto (x - H_-^{-1} C_D y, y)$$

has determinant one and carries the source Gram to $\text{diag}(H_-, W)$. Positivity proves $W > 0$ and, by applying the reduction map to this exact orthogonal direct sum,

$$K_+ = K_- + F_D W^{-1} F_D^*, \quad Z_D = F_D^* G_- F_D \succeq 0, \quad G_- = K_-^{-1}. \quad (\text{GMT.23})$$

Block elimination of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in its two orders proves $\det(I + AB) = \det(I + BA)$. Applying this identity to (GMT.23) gives the exact scalar maps

$$\boxed{\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W} = 1 + \text{Tr}(W^{-1} Z_D) + \frac{\det Z_D}{\det W}.} \quad (\text{GMT.24})$$

The last equality is the complete characteristic polynomial of a 2×2 matrix. All denominators are positive. This gives the actual monotonicity because both eigenvalues of $W^{-1/2} Z_D W^{-1/2}$ are non-negative.

The same block also gives the adjacent source recurrence coefficient. In the order in (GMT.22), the monic residual of S^N has squared norm W_{11} , and eliminating it from the residual of S^{N+1} gives squared norm $\det W / W_{11}$. Therefore

$$\boxed{a_{N+1} = \frac{\omega_{N+1}}{\omega_N} = \frac{\det W}{W_{11}^2}.} \quad (\text{GMT.25})$$

The first residual is monic because the subtracted projection has smaller degree; so is the second after its additional elimination. The norms have not been rescaled. The coefficient map $S^j = \sum_{l \leq j} \binom{j}{l} c^{j-l} i^l u^l$ has determinant $i^{j(j-1)/2}$ in dimension j and hence Gram determinant multiplier one, which proves that these are also the source Hankel-Toda norms.

Apply (GMT.22)–(GMT.25) separately to the Gamma reference and the arithmetic source, retaining their same J_+ and χ . Then

$$\boxed{\frac{T_{N-1}}{T_{N+1}} = \frac{\det(W + Z_D) / \det W}{\det(W^\Gamma + Z_D^\Gamma) / \det W^\Gamma}.} \quad (\text{GMT.26})$$

Every entry is a finite polynomial in the one-factor coefficients through degree $2N + 2$, followed by explicitly stated inverses of positive matrices. In the supplied Toda upper estimate, substitute the two positive scalars $a_{N+1} = \det W / W_{11}^2$ and $\mathcal{R}_N = \det(W + Z_D) / \det W$. Its full right side becomes

$$\sqrt{\frac{\det W}{W_{11}^2} \frac{\det(W + Z_D) / \det W - 1}{2\sqrt{\det(W + Z_D) / \det W}}}. \quad (\text{GMT.27})$$

This is an exact substitution into that existing inequality; its discarded phase and volume-imbalance squares remain the ones proved in TVB.32–TVB.34a. Formula (GMT.27) alone does not remove them or assert equality. The two-degree determinant formula itself also works at $q = 0$, where F_D and Z_D are zero and the ratio is one.

40.6 Certified finite envelopes and quotient-dimensional bounds

Suppose a finite coefficient calculation supplies the entry bounds (GMT.15), or the proved polynomial-test bounds (GMT.16). Let $v_i > 0$ be any specified positive rational weights and set

$$\Delta = \text{diag}(\delta_i), \quad \delta_i = \sum_j \varepsilon_{ij} \frac{v_j}{v_i}, \quad H^{\text{lo}} = \hat{H} - \Delta, \quad H^{\text{hi}} = \hat{H} + \Delta. \quad (\text{GMT.28})$$

The radii may be replaced by symmetric rational upper radii. For each pair i, j , the inequality $2|x_i||x_j| \leq (v_j/v_i)|x_i|^2 + (v_i/v_j)|x_j|^2$ gives $|x^*(H - \hat{H})x| \leq x^*\Delta x$ after summation. This proves $H^{\text{lo}} \preceq H \preceq H^{\text{hi}}$ without using a pointwise lower bound on B .

The positive-pivot certificate for H^{lo} is entirely finite. Eliminate successively its first diagonal pivot, recording

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \mapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \mapsto (x - a^{-1}b^*y, y).$$

The displayed congruence is invertible. Positive recorded rational pivots prove positivity by induction; nonpositive pivots reject that particular enclosure. Actual matrix positivity is already proved by the source integral. For any sequence of valid entry intervals tending to their actual entries, the lower envelopes tend to that positive matrix. Its least unit-sphere quadratic value is positive by compactness, and convergence in the finite sum of entry errors eventually makes every lower envelope positive. Hence the certificate process succeeds at every fixed finite degree when its actual input intervals are refined; no success or interval refinement is asserted for absent arithmetic data.

For a positive certified lower envelope, calculate the actual finite quotient forms $G^{\text{lo}}, G^{\text{hi}}$ by (GMT.18), or by the equivalent Schur formula

$$G(H) = E_N^* H E_N - E_N^* H B_N (B_N^* H B_N)^{-1} B_N^* H E_N. \quad (\text{GMT.29})$$

The formula follows by substituting $E_N - B_N (B_N^* H B_N)^{-1} B_N^* H E_N$ for the unique least lift. The same radii certify the representative coefficients themselves. Write $\hat{R} = R(\hat{H})$, $A^{\text{lo}} = B_N^* H^{\text{lo}} B_N$ and define, using any rational upper bounds for the displayed complex moduli,

$$\begin{aligned} s_{ai} &= \sum_{r,l} |(B_N)_{ra}| \varepsilon_{rl} |\hat{R}_{li}|, \\ k_{ab} &= \frac{1}{2} ((A^{\text{lo}})_{aa}^{-1} + (A^{\text{lo}})_{bb}^{-1}), \\ e_{ri}^R &= \sum_{a,b} |(B_N)_{ra}| k_{ab} s_{bi}. \end{aligned}$$

Then the full original-coordinate error and its quadratic loss obey

$$\boxed{|(R(H) - \hat{R})_{ri}| \leq e_{ri}^R, \quad \hat{R}^* H \hat{R} - G(H) = (R(H) - \hat{R})^* H (R(H) - \hat{R})}. \quad (\text{GMT.29a})$$

To prove the entry bound set $A = B_N^* H B_N$ and $S_E = B_N^* (H - \widehat{H}) \widehat{R}$. Formula (GMT.19) gives $R(H) - \widehat{R} = -B_N A^{-1} S_E$. The triangle inequality gives $|(S_E)_{ai}| \leq s_{ai}$. Inversion reverses positive order: conjugating $A \succeq A^{\text{lo}}$ by $(A^{\text{lo}})^{-1/2}$, inverting its eigenvalues, and conjugating back gives $A^{-1} \preceq (A^{\text{lo}})^{-1}$. Positivity of each two-coordinate Gram minor of A^{-1} gives $|A_{ab}^{-1}| \leq \sqrt{A_{aa}^{-1} A_{bb}^{-1}} \leq k_{ab}$, where the last step is the arithmetic-geometric mean inequality and the proved diagonal bounds. Matrix multiplication now proves the coefficientwise bound in (GMT.29a). For its second identity, $\widehat{R} - R(H)$ is an actual boundary, orthogonal to $R(H)$ in H . Expanding $\widehat{R}^* H \widehat{R}$ proves the equality.

This loss has an explicit q -dimensional rational upper form. Choose positive rational weights v_i in the original quotient coordinates and set

$$\tau_{ij} = \sum_{a,b} s_{ai} |(A^{\text{lo}})^{-1}|_{ab} s_{bj}, \quad \Delta_Q = \text{diag} \left\{ \sum_j \tau_{ij} \frac{v_j}{v_i} \right\}.$$

The same entrywise quadratic inequality as (GMT.28) proves

$$0 \preceq \widehat{R}^* H \widehat{R} - G(H) \preceq \Delta_Q. \quad (\text{GMT.29b})$$

Indeed the middle matrix is $S_E^* A^{-1} S_E \preceq S_E^* (A^{\text{lo}})^{-1} S_E$. Its value on x is at most $\sum_{i,j} \tau_{ij} |x_i| |x_j| \leq x^* \Delta_Q x$. Every source row, relation coefficient and quotient coordinate is retained in these sums. At an empty boundary block both error forms are zero. Applying $\mathcal{V}_{h,k}$ to each representative difference gives the theta primitive (GMT.4), so the certified coefficient error also has its original source and quotient interpretation.

Minimizing the two proven source inequalities over the same affine space of representatives gives

$$G^{\text{lo}} \preceq G(H) \preceq G^{\text{hi}}, \quad \det G^{\text{lo}} \leq V_N(H) \leq \det G^{\text{hi}}. \quad (\text{GMT.30})$$

For the determinant assertion, conjugate by the positive square root of the lower form and multiply the eigenvalues, all of which are at least one; apply the same argument for the upper form.

Construct the degree $N - 1$ envelope by taking the corresponding principal submatrix of the same degree $N + 1$ envelope. Then

$$\frac{\det G_-^{\text{lo}}}{\det G_+^{\text{hi}}} \leq \frac{V_{N-1}}{V_{N+1}} \leq \frac{\det G_-^{\text{hi}}}{\det G_+^{\text{lo}}}. \quad (\text{GMT.31})$$

Division by the exact positive Gamma ratio from (GMT.26) gives the certified interval for T_{N-1}/T_{N+1} . These are determinants of size q , with no independent numerator/denominator error estimate on the larger source and relation determinants.

For example a relative certificate $(1 - \eta)\widehat{H} \preceq H \preceq (1 + \eta)\widehat{H}$, with a recorded rational $0 \leq \eta < 1$ and $\widehat{H} > 0$, gives exactly

$$(1 - \eta)^q \det G(\widehat{H}) \leq V_N(H) \leq (1 + \eta)^q \det G(\widehat{H}),$$

and a common such certificate for the two nested degrees gives ratio multipliers $((1 - \eta)/(1 + \eta))^q$ and its reciprocal. This follows because $R(aH) = R(H)$ and $G(aH) = aG(H)$ for $a > 0$, as direct substitution in (GMT.18) proves. The mass is scaled only in this explicitly declared comparison, never assigned a new value in the original arithmetic source.

40.7 Exact calibration and a bounded relative-volume counterexample

Take the explicitly auxiliary one-factor weight

$$\lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = \frac{1}{2}, \quad C_2 = \frac{1}{4}, \quad B(t) = 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = (S - 1)^2 - 2. \quad (\text{GMT.32})$$

The positive polynomial B lies in $L^2(r_1)$ by (GMT.6). It is not the function $|g/h\Gamma|^2$ of a claimed zeta packet. Since $b_2^{(1)} = t^2 - 2$, its coefficients are $c_{B,0} = 3/2$, $c_{B,2} = 1/4$ and all others zero. Therefore

$$A(z) = \frac{3}{2} + \frac{3}{2}z^2, \quad (d_0, d_2, d_4) = \left(\frac{9}{4}, \frac{9}{2}, \frac{9}{4}\right), \quad B_2(u) = \frac{54}{35} + \frac{39}{280}u^2 + \frac{3}{1120}u^4. \quad (\text{GMT.33})$$

To verify the last equality, use $b_2^{(2)} = u^2 - 4$, $b_4^{(2)} = u^4 - 32u^2 + 72$ in $\sum_n d_n b_n^{(2)}/(4)_n$ and expand all three terms. The one-factor mass is $3/4$ and the sum mass is $9/16$. Repeated multiplication by t in the recurrence (GMT.8) gives the one-factor reference even moments through degree eight as $(1/2, 1, 8, 136, 3968)$. Inserting $B = 1 + t^2/4$ and expanding $(t_1 + t_2)^n$ with the full binomial coefficients gives sum moments

$$(\mu_0, \mu_1, \dots, \mu_8) = (9/16, 0, 9/2, 0, 117, 0, 5472, 0, 385272). \quad (\text{GMT.34})$$

This is also (GMT.11)–(GMT.12); the two calculations independently fix its mass and tensor cross term.

The original degree-three source Gram and reference Gram are

$$H_3 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix}, \quad H_3^\Gamma = \frac{1}{4} \begin{pmatrix} 1 & 1 & -3 & -11 \\ 1 & 5 & 5 & -55 \\ -3 & 5 & 65 & 65 \\ -11 & -55 & 65 & 1685 \end{pmatrix}. \quad (\text{GMT.35})$$

Every entry follows by expanding $(1 - iu)^a(1 + iu)^b$ against (GMT.34), or against the reference moments. The complete original relation and reduction matrices are

$$B_3 = \begin{pmatrix} -1 & 0 \\ -2 & -1 \\ 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{pmatrix}. \quad (\text{GMT.36})$$

These follow from $S^2 = 2S + 1$ and $S^3 = 5S + 2$ modulo χ ; multiplication verifies $J_3 B_3 = 0$ and $\det[E_3, B_3] = 1$. Direct substitution into (GMT.18) gives

$$G_1 = \frac{1}{16} \begin{pmatrix} 9 & 9 \\ 9 & 81 \end{pmatrix}, \quad G_3 = \begin{pmatrix} 81/244 & 81/244 \\ 81/244 & 175041/80764 \end{pmatrix},$$

$$R_3 = \begin{pmatrix} 117/122 & -801/40382 \\ -5/61 & 18963/20191 \\ 5/122 & -907/40382 \\ 0 & 7/331 \end{pmatrix}, \quad G_3^\Gamma = \begin{pmatrix} 5/38 & 5/38 \\ 5/38 & 2255/4142 \end{pmatrix}. \quad (\text{GMT.37})$$

These values can be checked without trusting an inverse algorithm: multiply to obtain $J_3 R_3 = I$, $B_3^* H_3 R_3 = 0$ and $R_3^* H_3 R_3 = G_3$, then use the uniqueness proved in (GMT.18).

For the degree-one to degree-three split, the full two-degree data are

$$W = \begin{pmatrix} 81 & 243 \\ 243 & 3159 \end{pmatrix}, \quad F_D = \begin{pmatrix} 10 & 2 \\ 0 & 28 \end{pmatrix}, \quad Z_D = \frac{1}{4} \begin{pmatrix} 225 & 675 \\ 675 & 16137 \end{pmatrix}. \quad (\text{GMT.38})$$

Equation (GMT.22), evaluated with (GMT.35)–(GMT.36), proves these entries by ordinary rational multiplication. Consequently

$$\frac{V_1}{V_3} = \frac{20191}{4860}, \quad a_3 = 30, \quad \frac{V_1^\Gamma}{V_3^\Gamma} = \frac{2071}{450}, \quad \boxed{\frac{T_1}{T_3} = \frac{100955}{111834} < 1}. \quad (\text{GMT.39})$$

Here $V_1^\Gamma = 1/4$, $V_3^\Gamma = 225/4142$, obtained by taking the two displayed 2×2 determinants. Thus no unrecorded cancellation of reference masses is used.

The phenomenon has a completely specified bounded positive version. Define $B^{[200]}(t) = \min(1 + t^2/4, 10001)$ and $w^{[200]} = B^{[200]}r_1$. This changes the auxiliary weight only; it leaves $S = 1 + iu$, χ , the reference Gamma mass and the unscaled convolution operation fixed. For $0 \leq j \leq 6$,

$$\int |t|^j (w - w^{[200]})(t) dt \leq \frac{1}{4} \int_{|t| > 200} |t|^{j+2} r_1(t) dt < \frac{8!}{2^{92}} =: \epsilon_*. \quad (\text{GMT.40})$$

Here is a proof with no floating evaluation. For every integer $r \geq 0$, $x^r \leq r! 2^r e^{x/2}$; on $x > 200$ multiply by $e^{(x-200)/2} \geq 1$. Equation (GMT.6) and evenness give

$$\int_{|t| > 200} |t|^r r_1(t) dt \leq 2c_1 r! 2^r e^{-100} (\cos 1)^{-2} \leq 4r! 2^r e^{-100}.$$

The estimate $\cos 1 \geq 1/2$ follows from $1 - \cos 1 = \int_0^1 \sin x dx \leq \int_0^1 x dx = 1/2$, using $|\sin x| \leq x$ on this interval. Since $e > 2$, $e^{-100} < 2^{-100}$. Use $r = j + 2 \leq 8$ and the factor $1/4$ to obtain (GMT.40).

For $0 \leq j \leq 6$, the absolute j th one-factor moment of w is at most $A_* = \mu_0(w) + \mu_6(w) = 3/4 + 1128 = 4515/4$, because $|t|^j \leq 1 + |t|^6$. The exact tensor difference is $(w - w^{[200]}) \otimes w + w^{[200]} \otimes (w - w^{[200]})$. Expanding $|t_1 + t_2|^n \leq \sum_j \binom{n}{j} |t_1|^j |t_2|^{n-j}$ proves a sum-moment difference at degree $n \leq 6$ of at most $2^{n+1} A_* \epsilon_* \leq 128 A_* \epsilon_*$. The coefficient absolute sum in $(1 - iu)^a (1 + iu)^b$ is at most $2^{a+b} \leq 64$ for $a, b \leq 3$. Hence every source entry changes by at most $8192 A_* \epsilon_* < 1/40000$. The last inequality is an integer comparison after substitution of A_*, ϵ_* . Equation (GMT.28) with four weights equal to one now proves

$$H_3 - 10^{-4}I \prec H_3^{[200]} \prec H_3 + 10^{-4}I. \quad (\text{GMT.41})$$

Strictness follows because the error in each row sum is strictly less than 10^{-4} . The four leading principal minors of the lower matrix are

$$\frac{703}{1250}, \quad \frac{253068751}{100000000}, \quad \frac{102481960035937}{500000000000}, \quad \frac{4980299350501627807501}{10000000000000000}.$$

They are positive; their successive ratios are exactly the positive pivots in the elimination proving (GMT.28). Strict source inequalities give strict quotient inequalities: evaluate the lower form on the actual minimum lift, and the actual form on the upper minimum lift. Each lift of a nonzero quotient vector is nonzero, so the positive source difference has a strictly positive quadratic value. Computing (GMT.29) for these two rational endpoint matrices, and the degree-one principal submatrices, gives the strict ratio enclosure

$$\frac{20694424798404596925036}{4984219548001747192501} < \frac{V_1^{[200]}}{V_3^{[200]}} < \frac{20703435010591353075036}{4980299350501627807501} < \frac{2071}{450}. \quad (\text{GMT.42})$$

The first two strict comparisons follow from that strict minimization and (GMT.31); the final comparison follows by positive-denominator cross multiplication. The rational envelope computation is reproduced by the exact checker accompanying this proof. Division by the fixed positive Gamma ratio therefore proves $T_1^{[200]}/T_3^{[200]} < 1$ for a function bounded between 1 and 10001. This is an explicit obstruction to deriving relative volume monotonicity from positivity and boundedness of a multiplier alone. Its exact relation to the arithmetic problem is the same coefficient, relation, least-lift and determinant maps (GMT.12), (GMT.18), (GMT.24) and (GMT.26). It does not assert that the actual arithmetic multiplier has this behavior.

40.8 Verification scope

The accompanying exact checker constructs moments in two ways: Jacobi recurrence and the literal two-factor binomial expansion, then compares them with the Gamma coefficient formula. It verifies the original S -coordinate Gram, every displayed lift and relation identity, the full boundary-resolvent loss, five points on its actual positive matrix segment, both two-degree formulas and the rational envelope calculation. Its deliberate formula changes remove the mass, omit the tensor cross term, reverse the boundary-loss sign, remove the conjugation, or remove one power in the Schur denominator. Each change is rejected by the corresponding exact check in ordinary and optimized Python. The written proofs establish the all-degree identities; these finite checks provide reproducible regression evidence. There is no Lean execution or analytic interval claim for a nonempty zeta packet in this verification.

41 The complete complex amplitude in gamma sum descent

41.1 Original objects and the question answered

The delivered Gamma Convolution Descent note constructs the conditional expectation of the squared arithmetic amplitude. We construct the exact comparison for its full complex amplitude. The two resulting source Grams add to the original arithmetic Gram. We prove a strict positive contribution of the retained relative component for every full quartet, and give its exact map into the original minimum quotient metric. All measures below retain their masses. The variable is the original sum $u = t_1 + \cdots + t_k$, and the arithmetic variable remains $S = k/2 + iu$.

Fix $\lambda > 0$ and an integer $k \geq 1$. Retain

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GP.1})$$

Thus $\int r_{\lambda,k} = c_\lambda^k$. The beta-integral proof of this identity, with its original Jacobian $t = 2v$, is retained in the cumulative reader, TG.7–TG.8, and the delivered note, (5)–(9). In particular

$$\int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}, \quad |\theta| < \pi/2. \quad (\text{GP.2})$$

Let h be the original finite polynomial packet of complete orders of actual zeros of $g = 2\xi$, and include $h = 1$ when considering the analytic seed. Define the entire quotient $v_h = g/h$ and keep

$$A(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B(t) = |A(t)|^2, \quad w(t) = B(t)r_\lambda(t), \quad \mu_h = \int w(t) dt. \quad (\text{GP.3})$$

Use one of the references proved in the delivered note for which $B \leq C_h < \infty$. In particular a full quartet of degree $d \geq 4$ uses its original $\lambda = 1/4$. The seed uses $\lambda = 13/4$ and $C_1 = 1024/\sqrt{\pi}$. The statements involving decay explicitly use the quartet reference. The function A is nonzero and analytic on the real line: v_h is a nonzero entire function and reciprocal gamma is entire and nonzero on this line. Its real zeros form a discrete set. No value at a removed packet zero is assigned using an uncanceled fraction.

Put $d\nu(\mathbf{t}) = \prod_i r_\lambda(t_i) d\mathbf{t}$ and $d\nu_\Sigma(u) = r_{\lambda,k}(u) du$. In $\mathcal{H} = L^2(\nu)$ and $\mathcal{H}_\Sigma = L^2(\nu_\Sigma)$ write

$$Uf(\mathbf{t}) = f(\sum_i t_i), \quad C = U^*, \quad P = UC, \quad a(\mathbf{t}) = \prod_i A(t_i), \quad (\text{GP.4})$$

$$b_k = C(|a|^2), \quad a_k = C(a), \quad d_k = b_k - |a_k|^2.$$

For $k = 1$, $C = U = I$, $a_1 = A$, $b_1 = B$, $d_1 = 0$. For $k \geq 2$ the conditional integral CF is precisely

$$(CF)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GP.5})$$

These are probability measures on individual fibres, derived by dividing by their stated fibre mass; neither original measure is replaced by a probability measure. Fubini proves $CU = I$ and the isometry of U . Jensen's inequality in (GP.5) proves $d_k \geq 0$ and $0 \leq b_k \leq C_h^k$. For $k \geq 2$, $b_k(u) > 0$ at every real u : the discrete zeros of each factor exclude only countably many hyperplanes in this fibre, which have zero Lebesgue measure.

41.2 Unitary transport of the actual conditional projection

Theorem 41.1 (Complete amplitude comparison). *Use the actual arithmetic spaces*

$$\mathcal{H}^h = L^2(\mathbb{R}^k, |a|^2 d\nu), \quad \mathcal{H}_\Sigma^h = L^2(\mathbb{R}, b_k d\nu_\Sigma).$$

The maps

$$T : \mathcal{H}^h \longrightarrow \mathcal{H}, \quad F \longmapsto aF, \quad T_\Sigma : \mathcal{H}_\Sigma^h \longrightarrow \mathcal{H}_\Sigma, \quad f \longmapsto \sqrt{b_k}f \quad (\text{GP.6})$$

are surjective isometries. The positive square root in T_Σ is a specified comparison map; the original amplitude in T remains complex. The actual arithmetic sum inclusion $U_h f = f(\sum t_i)$ and its adjoint C_h^{sum} satisfy

$$C_h^{\text{sum}} F = \frac{C(|a|^2 F)}{b_k}, \quad J = TU_h T_\Sigma^{-1} = M_a U M_{b_k^{-1/2}}, \quad J^* = M_{b_k^{-1/2}} C M_{\bar{a}}, \quad J^* J = I. \quad (\text{GP.7})$$

All quotients here are defined almost everywhere. In particular their operator extensions are bounded even when the individual inverse multiplier is unbounded. The transported arithmetic projection is

$$Q = TP_h T^{-1} = JJ^*, \quad QF = aU \left(\frac{C(\bar{a}F)}{b_k} \right), \quad P_h = U_h C_h^{\text{sum}}. \quad (\text{GP.8})$$

Thus P and Q are related by the displayed exact transport from the original arithmetic spaces.

Proof. The squared norms in (GP.6) agree by direct substitution. The zero set of a is a union of null hyperplanes; b_k is positive almost everywhere, including $k = 1$. Division off those null sets constructs the two inverses with their exact squared norms. Applying (GP.5) to the arithmetic fibre density gives C_h^{sum} . Pairing $M_a U M_{b_k^{-1/2}} f$ with F proves the stated adjoint. The identity $C(|a|^2 U f) = b_k f$ then gives $J^* J = I$. Conjugating P_h gives (GP.8). These norm identities first hold for bounded functions supported on finite-measure sets on which the denominators are bounded below; truncation and the isometries extend them to the indicated complete Hilbert spaces. This also proves that no unbounded multiplier composition is being asserted on a larger domain. $\square$

Theorem 41.2 (Exact fibre angle and retained defect). *Define $\gamma_k = a_k / \sqrt{b_k}$ almost everywhere. Then*

$$U^* J = M_{\gamma_k}, \quad J = U M_{\gamma_k} + N, \quad U^* N = 0, \quad N^* N = M_{1-|\gamma_k|^2} = M_{d_k/b_k}. \quad (\text{GP.9})$$

For every real u at which $b_k > 0$, the two fibre projections onto 1 and $a/\sqrt{b_k}$ obey

$$\text{Tr}_u(P_u - Q_u)^2 = 2(1 - |\gamma_k(u)|^2), \quad \|[P_u, Q_u]\|_{\text{HS},u}^2 = 2|\gamma_k(u)|^2(1 - |\gamma_k(u)|^2). \quad (\text{GP.10})$$

These are finite-rank traces in the specified fibre Hilbert space, not a claim that the global decomposable operators are trace class. The complete unscaled variance also has the formula

$$d_k(u) = \frac{1}{2} \iint |a(\mathbf{t}) - a(\mathbf{s})|^2 d\nu_u(\mathbf{t}) d\nu_u(\mathbf{s}), \quad (\text{GP.11})$$

where both ν_u are the exact conditional measures in (GP.5).

Proof. Applying C to Jf gives $\gamma_k f$. Orthogonal decomposition with P gives (GP.9), and taking squared norms uses $J^*J = I$. On a fibre let $e_0 = 1$ and $v = a/\sqrt{b_k}$, so both have norm one and $\langle e_0, v \rangle = \gamma_k$, using the inner product antilinear in its first argument. If $|\gamma_k| < 1$, put $e_1 = (v - \gamma_k e_0)/\sqrt{1 - |\gamma_k|^2}$. In the displayed orthonormal comparison frame, without altering either original vector, the two projections are

$$P_u = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q_u = \begin{pmatrix} |\gamma_k|^2 & \gamma_k \sqrt{1 - |\gamma_k|^2} \\ \overline{\gamma_k} \sqrt{1 - |\gamma_k|^2} & 1 - |\gamma_k|^2 \end{pmatrix}.$$

Matrix multiplication proves (GP.10); both vanish when $|\gamma_k| = 1$. Expanding the square in (GP.11), and using the unit total mass of each conditional measure, gives $C(|a|^2) - |Ca|^2$. No phase is discarded in this calculation. $\square$

41.3 Strictness for an actual quartet, with its exact constant

For a full quartet retain $\lambda = 1/4$, degree $d \geq 4$ and $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. The delivered proof gives

$$|A(t)|^2 \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d} \quad (|t| \geq T_h). \quad (\text{GP.12})$$

Here is its direct calculation. In the unchanged amplitude,

$$A(t) = -\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it) / h(1/2 + it).$$

The continuation formula $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ gives $|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For $|t| \geq T_h$, every retained root factor has modulus at least $|t|/2$, so $|h| \geq (|t|/2)^d$. Also $(t^2 + 1/4)^2 \leq 25t^4/16$ and $4(1 + |t|)^2 \leq 16t^2$ for $|t| \geq 1$. Multiplication proves (GP.12). Consequently $A(t) \rightarrow 0$ at both real ends.

Theorem 41.3 (Strict relative contribution on every sum fibre). *For every such actual quartet, every $k \geq 2$, and every real u ,*

$$d_k(u) > 0, \quad |\gamma_k(u)| < 1. \quad (\text{GP.13})$$

Thus the relative component is nonzero at every fibre. It is given exactly by (GP.9) and (GP.11); strictness does not assign a uniform positive lower bound to it.

Proof. Fix u . If $d_k(u) = 0$, (GP.11), or equality in the variance identity, says that the continuous function a is constant almost everywhere on the fibre. Its conditional density is strictly positive everywhere, so continuity makes it constant everywhere. For $k = 2$, this would make $A(t)A(u - t)$ constant in t . Its limit is zero by (GP.12) and boundedness of A , whereas some value is nonzero: avoid the two discrete sets of zeros of $A(t)$ and $A(u - t)$. For $k > 2$, choose fixed $t_2, \dots, t_{k-1}$ at nonzeros of A and vary $t_1 = t$, $t_k = u - t - \sum_{2 \leq i < k} t_i$. The identical argument makes the product both zero and nonzero, a contradiction. The assertion for γ_k follows by dividing by the strictly positive b_k already proved. This proof uses only the explicit quartet tail, not a conjecture about positions of unselected zeros. $\square$

41.4 Full complex coefficient transform and exact variance norm

Put $\alpha = 2\lambda$ and keep the original monic polynomials

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}, \quad \|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n. \quad (\text{GP.14})$$

The specified gamma measure has exponential moments. Its polynomials are complete in L^2 : for a vector orthogonal to them, multiplying it by the measure gives a finite complex measure by Cauchy-Schwarz; the same inequality with exponential moments makes its Fourier transform holomorphic

on a strip about the real axis. All derivatives at zero vanish. Analytic uniqueness and Fourier uniqueness make the measure, hence the vector, zero. The same proof applies to the sum measure with mass c_λ^k .

Define complex coefficients, retaining conjugation conventions,

$$a_j = \frac{\int b_j^{(\lambda)}(t)A(t)r_\lambda(t)dt}{c_\lambda j!(\alpha)_j}, \quad \mathfrak{A}(z) = \sum_{j \geq 0} (\alpha)_j a_j z^j, \quad e_{k,n} = [z^n] \mathfrak{A}(z)^k. \quad (\text{GP.15})$$

The square-integrable vector A has these orthogonal coefficients. The series in (GP.15) may be used coefficientwise; no interchange of an infinite product with an integral is required below.

Theorem 41.4 (Complex convolution and its exact lost norm). *With all original masses,*

$$a_k(u) = \sum_{n \geq 0} \frac{e_{k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u) \quad \text{in } L^2(r_{\lambda,k} du),$$

$$\int d_k r_{\lambda,k} = \mu_h^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |e_{k,n}|^2. \quad (\text{GP.16})$$

For every polynomial F in the original S coordinate and every real $|\theta| < \pi/2$, the exact weighted version is

$$\|a U(e^{\theta u/2} F(k/2 + iu))\|^2 = \int e^{\theta u} |F(k/2 + iu)|^2 |a_k(u)|^2 r_{\lambda,k}(u) du$$

$$+ \int e^{\theta u} |F(k/2 + iu)|^2 d_k(u) r_{\lambda,k}(u) du. \quad (\text{GP.17})$$

Proof. Multiplication of the finite-degree coefficients of the generating functions gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing this finite identity with a gives $c_\lambda^k n! e_{k,n}$. Division by the original sum norm $c_\lambda^k n! (k\alpha)_n$, followed by completeness, gives the first identity. Pythagoras for $P = UC$ gives the second since $\|a\|^2 = \mu_h^k$. Finally conditional expectation commutes with multiplication by a function of the sum. Apply the same orthogonal decomposition to $aU(e^{\theta u/2} F)$ to obtain (GP.17). Boundedness by C_h^k , (GP.2), and polynomial growth prove all integrals finite, including any fixed number of theta derivatives. $\square$

41.5 The exact induced change of the original finite quotient metric

Fix the actual annihilator $\chi = \chi_{h,k}$, degree $q > 0$, and $N \geq q - 1$. Use original polynomial spaces $E = \mathcal{P}_N(S)$, $E_\partial = \mathcal{P}_{N-q}(S)$, where a negative degree means the zero space, and $C_\chi = \mathbb{C}[S]/(\chi)$. Let $B_\chi : E_\partial \rightarrow E$ be multiplication by χ and $\pi : E \rightarrow C_\chi$ the quotient. Use the original monomial bases, or any explicitly transported fixed bases, in every matrix below. For a coefficient column x denote its polynomial by F_x . Define three source matrices by

$$x^* M_h x = \int e^{\theta u} |F_x(k/2 + iu)|^2 b_k r_{\lambda,k} du,$$

$$x^* M_{\text{coh}} x = \int e^{\theta u} |F_x(k/2 + iu)|^2 |a_k|^2 r_{\lambda,k} du,$$

$$x^* M_{\text{rel}} x = \int e^{\theta u} |F_x(k/2 + iu)|^2 d_k r_{\lambda,k} du. \quad (\text{GP.18})$$

These matrices are elements of $\text{Herm}(E)$, equivalently linear maps $E \rightarrow E^\vee$ with $E^\vee$ the conjugate dual under $x \mapsto [y \mapsto y^* M x]$. The associated linear observation maps are $F \mapsto aU(e^{\theta u/2} F)$, $F \mapsto$

$U(e^{\theta u/2} a_k F)$, and $F \mapsto (I - P)aU(e^{\theta u/2} F)$. The Gram operation itself has type $\text{Hom}_{\mathbb{C}}(E, \mathcal{H}) \rightarrow \text{Herm}(E)$, $R \mapsto R^* R$. Thus

$$M_h = M_{\text{coh}} + M_{\text{rel}}. \quad (\text{GP.19})$$

The original arithmetic matrix M_h is unchanged.

Lemma 41.5. M_{coh} is positive definite for every packet/reference admitted above, every $k \geq 1$, every finite N , and real $|\theta| < \pi/2$. For the quartets of (GP.12) and $k \geq 2$, M_{rel} is also positive definite.

Proof. Put $f = Ar_{\lambda}$. It is a nonzero L^1 function, since A is bounded and nonzero almost everywhere. The numerator of a_k is f^{*k} . If this convolution were zero almost everywhere, its Fourier transform $(\hat{f})^k$ would vanish everywhere, whence $\hat{f} = 0$ and $f = 0$ by Fourier uniqueness, a contradiction. Consequently $|a_k|^2 r_{\lambda,k}$ is positive on a set of positive Lebesgue measure. A nonzero polynomial on the indicated affine line has only finitely many real zeros, so its squared norm is strictly positive. This proves the first assertion. The second follows from $d_k(u) > 0$ for every u , proved in (GP.13). $\square$

Let $R_h, R_{\text{coh}} : C_{\chi} \rightarrow E$ be the least-norm sections for M_h, M_{coh} , and write $G_h = R_h^* M_h R_h$, $G_{\text{coh}} = R_{\text{coh}}^* M_{\text{coh}} R_{\text{coh}}$. They exist uniquely: for any section R_0 , subtract $B_{\chi}(B_{\chi}^* M B_{\chi})^{-1} B_{\chi}^* M R_0$; the boundary Gram is invertible because B_{χ} is injective and M positive definite. An empty boundary contributes the zero map.

Theorem 41.6 (Exact two positive terms in the quotient change). *In these original coordinates put*

$$K = (B_{\chi}^* M_h B_{\chi})^{-1} B_{\chi}^* M_{\text{rel}} R_{\text{coh}}. \quad (\text{GP.20})$$

It is a map $C_{\chi} \rightarrow E_{\partial}$, or the unique zero map when the boundary is empty. Then

$$R_h = R_{\text{coh}} - B_{\chi} K, \quad \boxed{G_h = G_{\text{coh}} + K^* B_{\chi}^* M_{\text{coh}} B_{\chi} K + R_h^* M_{\text{rel}} R_h.} \quad (\text{GP.21})$$

For an actual full quartet, $k \geq 2$, and $q > 0$, this gives $G_h - G_{\text{coh}} > 0$ for every admitted N and theta. More precisely, with the actual positive matrix

$$E_q = G_{\text{coh}}^{-1/2} (K^* B_{\chi}^* M_{\text{coh}} B_{\chi} K + R_h^* M_{\text{rel}} R_h) G_{\text{coh}}^{-1/2}, \quad (\text{GP.22})$$

the original determinants satisfy

$$\det G_h = \det G_{\text{coh}} \det(I + E_q). \quad (\text{GP.23})$$

These comparison conjugations do not replace either quotient metric.

Proof. $B_{\chi}^* M_{\text{coh}} R_{\text{coh}} = 0$, so the formula for the M_h section using R_{coh} as initial section gives (GP.20) and the first identity. Expand $R_h^* M_{\text{coh}} R_h$ about R_{coh} . The two cross terms vanish by the stated orthogonality. Add $R_h^* M_{\text{rel}} R_h$ using (GP.19); this proves (GP.21) with both terms retained. R_h is injective because $\pi R_h = I$. Thus strict positivity of M_{rel} makes its displayed pullback positive definite on the nonzero quotient. Conjugation by the actual positive square root of G_{coh} proves (GP.23). For $N = q - 1$ the first positive term is zero and the second remains as stated. For $q = 0$ the original quotient is zero: both section maps are zero, and the empty determinant is one. The nonzero analytic source does not supply an extra quotient dimension. $\square$

41.6 Support, the original unit, and the domain of arithmetic observation

The original cyclic source is $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$. In its Mellin realization its full image is

$$\prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} a(\mathbf{t}) P(k/2 + i \sum_i t_i). \quad (\text{GP.24})$$

Multiplication by the displayed gamma factors is an isometry $\mathcal{H} \rightarrow L^2(d\mathbf{t})$. Equations (GP.9) and (GP.17) split the vector before this multiplication and reconstruct it by addition. On the image graph

$$\mathcal{G}_N = \{(U(a_k F), (I - P)aUF) : F \in \mathcal{P}_N(S)\}$$

addition $\text{add}_N : \mathcal{G}_N \rightarrow aU(\mathcal{P}_N)$ is injective, since the squared norm of the reconstructed vector is the strictly positive polynomial norm M_h . It is onto the displayed image by construction. Denote the multiplication in (GP.24) by $M_\Gamma : \mathcal{H} \rightarrow L^2(d\mathbf{t})$ and the original unitary Mellin map by $\mathcal{M}_{\text{unit}}$. The graph observation is exactly $J^{(k)}\mathcal{M}_{\text{unit}}^{-1}M_\Gamma \text{add}_N$, and likewise with $q^{(k)}$. These are well-defined linear maps because the reconstructed image is precisely the original admitted Mellin source. Their values on the graph of P are exactly

$$J^{(k)}\mathcal{V}P = \eta_{h,k}\pi P, \quad q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}\pi P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{GP.25})$$

This specifies the domain of the observation; no arithmetic jet is assigned to an arbitrary independent measurable relative vector. The graph in this display uses theta zero. At a nonzero admitted theta, multiply both graph components by the same $e^{\theta u/2}$; its inverse multiplies by $e^{-\theta u/2}$ before the original inverse Mellin map. Both maps are isometries between the declared tilted and untilted spaces.

Here is the full cochain map for the relation. In the original tensor algebra

$$\mathbb{C}[s_1, \dots, s_k]/(h(s_1), \dots, h(s_k)),$$

the cyclic annihilator satisfies $\chi(s_1 + \dots + s_k) = 0$. Successive monic division in the fixed order $s_1, \dots, s_k$ gives unique quotients at each division step and polynomials Q_i with

$$\chi(s_1 + \dots + s_k) = \sum_{i=1}^k h(s_i)Q_i(s_1, \dots, s_k).$$

The final remainder is zero because the monomials of degree less than $\deg h$ in each variable are a basis of that tensor algebra. In $[V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ put $D = -x\partial_x$ and $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. It is even, $\phi_0(0) = 0$ and $\int_{\mathbb{R}} \phi_0 = 0$: the two even Gaussian integrals are $4\pi^2 \cdot 3/(4\pi^2)$ and $6\pi \cdot 1/(2\pi)$, both equal to three. For $\Re s > 1$, termwise Mellin integration of $\Theta\phi_0(x) = \sum_{n \neq 0} \phi_0(nx)$ yields $2\zeta(s)\mathcal{M}\phi_0(s)$. Direct integration with $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \pi^{-s/2}(2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

Thus $\mathcal{M}(\Theta\phi_0) = g$ there and by continuation on the original Mellin domain. The original inverse Mellin definition of F_h then gives $h(D)F_h = \Theta\phi_0$. The precise primitive is

$$\mathcal{P}_\chi F = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) F(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GP.26})$$

Euler differentiation preserves the even Schwartz source and its two conditions: its value at zero vanishes and its integral is the original integral by integration by parts. It also preserves rapid decay at both multiplicative ends and commutes with Θ . Every term therefore lies in the asserted cochain domain. The preceding $i - 1$ degree-one factors give the tensor differential sign $(-1)^{i-1}$, cancelling the written sign. Hence, with no sign omitted,

$$d\mathcal{P}_\chi F = \mathcal{V}(\chi F), \quad d[-\mathcal{P}_\chi K] = \mathcal{V}(R_h - R_{\text{coh}}). \quad (\text{GP.27})$$

This proves the cochain correction map on C_χ in (GP.21), including its zero value when the boundary is empty.

Every linear map on this graph and every original source or quotient has its original fixed-support lift: $(\ell, x) \mapsto (\ell, Lx)$, and external absence maps to external absence. The maps preserve addition and scalar action because the underlying maps are linear. For an actual relation the quotient value is $(\ell, 0)$, its supported zero, with the source and the two graph components still reconstructible. The norm has type $\mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ and is positively homogeneous; its square is quadratic. Trace has type $\text{End}_{\mathbb{C}}(E) \rightarrow \mathbb{C}$ and is linear; its restriction to Hermitian endomorphisms is real-linear. The Gram map and squared modulus are quadratic on the underlying real vector spaces, and trace composed with a Gram map is quadratic. The exact linear observation maps producing the source Grams and the quotient values are (GP.18), (GP.21), and (GP.24)–(GP.25).

41.7 Exact polynomial calibration with the original gamma mass

The following auxiliary calibration tests the complete complex-phase identities by rational arithmetic. It is a declared polynomial amplitude, with all fixed moments finite in the original gamma measure. It does not invoke the bounded packet estimate (GP.12). Take $\lambda = 1$, $k = 2$, $c_1 = 1/2$, $c_1^2 = 1/4$, and $A(t) = 1 + it/2$. The original sum coordinate is $S = 1 + iu$. Direct conditional polynomial calculation gives

$$\begin{aligned} a_2(u) &= \frac{6}{5} + \frac{iu}{2} - \frac{u^2}{20}, \\ b_2(u) &= \frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120}, \\ d_2(u) &= \frac{18}{175} + \frac{13u^2}{1400} + \frac{u^4}{5600} > 0. \end{aligned} \tag{GP.28}$$

Here are enough exact polynomial identities to prove the calculation. For the one-factor parameter $\alpha = 2$, $b_2^{(1)}(t) = t^2 - 2$. For the sum parameter 4, the second and fourth monic polynomials are $u^2 - 4$ and $u^4 - 32u^2 + 72$. The conditional addition formula yields

$$\begin{aligned} C(t_1 t_2) &= (u^2 - 4)/5, \\ C(t_1^2) &= C(t_2^2) = 3(u^2 - 4)/10 + 2, \\ C(t_1^2 t_2^2) &= \frac{3}{70}(u^4 - 32u^2 + 72) + \frac{6}{5}(u^2 - 4) + 4. \end{aligned}$$

Indeed the respective coefficients are $2^2/(4)_2$, $(2)_2/(4)_2$, and $(2)_2^2/(4)_4$. Expanding $(1 + it_1/2)(1 + it_2/2)$ and $(1 + t_1^2/4)(1 + t_2^2/4)$ using these identities gives the first two lines of (GP.28). Subtracting the squared modulus gives its last line. This is the complete complex subtraction, including its imaginary linear coefficient.

For the auxiliary annihilator $\chi(S) = (S - 1)^2 - 2$, use the original S monomials at $N = 1$, where the boundary is empty. The sum gamma moments of orders zero, two, four, six are respectively $1/4, 1, 14, 376$, obtained by differentiating $\frac{1}{4} \cos^{-4} \theta$. Substitution in (GP.18) gives

$$\begin{aligned} G_h &= \begin{pmatrix} 9/16 & 9/16 \\ 9/16 & 81/16 \end{pmatrix}, \quad G_{\text{coh}} = \begin{pmatrix} 21/40 & 21/40 \\ 21/40 & 189/40 \end{pmatrix}, \\ G_h - G_{\text{coh}} &= \begin{pmatrix} 3/80 & 3/80 \\ 3/80 & 27/80 \end{pmatrix}, \\ \det G_h &= 81/32, \quad \det G_{\text{coh}} = 441/200, \quad \det(G_h - G_{\text{coh}}) = 9/800. \end{aligned} \tag{GP.29}$$

For example $\int b_2 r_{1,2} = 9/16$ and $\int u^2 b_2 r_{1,2} = 9/2$, while the corresponding coherent moments are $21/40$ and $21/5$. Odd moments vanish, so the original off-diagonal S entries equal the masses. These calculations prove every entry above. They exhibit the strictly positive contribution in the unchanged quotient coordinates.

The accompanying checker additionally derives moments from the Laplace function, solves the independent product-moment equations, and checks all moments through degree eight and the section/quotient identities through $N = 4$. It uses a separate two-atom conditional fibre of original masses two and three to test the complex adjoint and both projection traces. Its mathematical controls change a conjugate transpose, omit an original mass factor, or omit the relative quotient term. Those finite checks supplement the proofs above and do not estimate an actual growing-degree zeta quotient.

41.8 Provenance and consequence

The source constants, packet, conditional density, and coefficient addition identity come from the complete delivered Gamma Convolution Descent note, equations (2), (5)–(19), (24)–(27), and (32). The original unit and theta relation primitive come from the retained cyclic source construction and TVB in the cumulative reader. The classical generating function is [NIST DLMF 18.23.7](#), with $b_n^{(\lambda)}(t) = n!P_n^{(\lambda)}(t/2; \pi/2)$. The exact specialization and norms are proved in TG.7–TG.14. The bounded amplitude is used here to prove the full phase transport, fibre projection formulas, strict quartet variance, complex coefficient transform, and two-term positive quotient identity. These equations provide additional exact data for the same Toda quotient. They do not give a growing-degree estimate for its consecutive volume ratio. That ratio remains the actual determinant expression of the retained metric, now with the explicit contribution (GP.21)–(GP.23).

42 Exact endpoint windows: Gamma coefficients, paired Schur volumes, and intrinsic sensitivity

This continuation evaluates the original endpoint quantity of the complete written criterion at source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f of PR 23. The source criterion and the supplied *Endpoint degree selection in the actual gamma-corrected coordinates* were read completely, together with GC.1–59 and GMT.1–42, including GMT.29a–b. The present work proves the arbitrary-window Schur map, its exact determinant cancellations, a finite rational interval certificate, and a new sensitivity estimate through the actual angles between canonical least-lift spaces. The finite input list grows through degree $2(n+r)$. Every statement below is finite. No estimate for that growing list or its arithmetic conditioning is presumed.

42.1 The original source and the complete finite input map

Let $h(s)$ be the given monic packet of actual zeros, with their complete orders, of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s).$$

Retain its entire quotient $v_h = g/h$ and original positive density

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (\text{EW.1})$$

The complete analytic construction in GC.2–9 proves that w_h has every polynomial moment and exponential moments of every order less than $\pi/2$. Its mechanism is the theta seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, the equality $\mathcal{M}\Theta\phi_0 = g$, contour rotation through every angle $|\beta| < \pi/4$, and exact local division by h . In particular $|v_h(\sigma + it)|$ is bounded, uniformly for σ in each fixed compact real interval, by $C_{j,\beta,h}(1 + |t|)^{-j}e^{-\beta|t|}$ for every $j \geq 0$ and $0 < \beta < \pi/4$. These proofs and their original constants are retained in the accompanying complete GC source. The zero set of $v_h(1/2 + it)$ is discrete, so $w_h > 0$ almost everywhere. Its convolution has positive mass μ_h^k .

Throughout this paper

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k, \\ \langle P, Q \rangle_H = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du. \quad (\text{EW.2})$$

The pairing is conjugate-linear in its first argument. The sum chart $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Fubini therefore proves both the convolution in (EW.1) and the full mass μ_h^k , with no average coordinate or mass division. Every nonzero polynomial has finitely many real-line zeros, so its squared norm in (EW.2) is strictly positive.

Fix the original monic relation $\chi \in \mathbb{C}[S]$, of degree $q \geq 1$. For the packet source it generates the exact intersection $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ under $S = \sum s_i$. Define $\mathcal{C} = \mathbb{C}[S]/(\chi)$ in the ordered coordinates $1, S, \dots, S^{q-1}$. The quotient map is the actual remainder map. Its injection into the tensor arithmetic jets is

$$\eta : \mathcal{C} \longrightarrow (\mathbb{C}[s]/(h))^{\otimes k}, \quad [P] \longmapsto [v_h]_h^{\otimes k} P(s_1 + \cdots + s_k). \quad (\text{EW.3})$$

Complete local cancellation makes the constant term of v_h nonzero at every selected centre. Each local truncated power series is invertible by recursively solving its product with an inverse to equal one. The Chinese remainder isomorphism then proves that $[v_h]_h$ is a unit. The unfactored polynomial map $\mathbb{C}[S] \rightarrow (\mathbb{C}[s]/(h))^{\otimes k}$ in (EW.3) consequently has kernel equal to the stated intersection ideal (χ) . Factoring by that ideal gives the displayed η with kernel zero, proving injectivity without discarding a jet.

For $N \geq q - 1$, put $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{P}_{-1} = 0$. Let H_N be its source matrix in the original monomials. Let $J_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}^q$ be monic remainder and $B_N : \mathbb{C}^{N-q+1} \rightarrow \mathbb{C}^{N+1}$ be multiplication by χ . Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0. \quad (\text{EW.4})$$

In particular B_N is injective, J_N is surjective, and $\ker J_N = \text{im } B_N$.

For later coefficient evaluation keep all Gamma constants:

$$\lambda > 0, \quad \alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{EW.5})$$

Let the real monic polynomials $b_j^{(\lambda)}$ be specified by

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 + z^2)^{-\lambda} e^{t \arctan z}, \quad |z| < 1, \quad b_{j+1} = t b_j - j(j + \alpha - 1) b_{j-1}, \quad (\text{EW.6})$$

where the logarithms defining the generating function are the branches from zero, $b_0 = 1$, and $b_1 = t$. Differentiation of the displayed generating function proves the recurrence and monicity. The beta integral and Fourier inversion give the literal convolution mass and Laplace transform

$$r_{\lambda,k} = r_{\lambda}^{*k} = \frac{c_{\lambda}^k}{c_{k\lambda}} r_{k\lambda}, \quad \int e^{\theta u} r_{\lambda,k}(u) du = C_k (\cos \theta)^{-\beta}, \quad |\Re \theta| < \pi/2. \quad (\text{EW.7})$$

Their complete beta-integral proof with the changes $t = 2v$ and $u = \sum t_i$ is GC.10–15. In particular the reference monic norms at $\theta = 0$ are $C_k j! (\beta)_j$. They also follow directly by integrating the product of two functions (EW.6), which gives $C_k (1 - zw)^{-\beta}$, and comparing its finite derivatives.

Define the given one-factor coefficients and their finite powers by

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_{\lambda} j! (\alpha)_j}, \quad d_l = \sum_{j_1 + \cdots + j_k = l} \prod_{a=1}^k (\alpha)_{j_a} c_{h,j_a}. \quad (\text{EW.8})$$

Multiplying the k generating functions (EW.6) and comparing the coefficient of z^l gives the finite polynomial identity

$$b_l^{(k\lambda)}(\sum t_i) = \sum_{j_1+\dots+j_k=l} \frac{l!}{\prod j_i!} \prod_i b_{j_i}^{(\lambda)}(t_i).$$

Integrating every term against the full product measure proves

$$\int b_l^{(k\lambda)}(u) m_{h,k}(u) du = C_k l! d_l. \quad (\text{EW.9})$$

All sums here are finite and all their terms are integrable. For any specified polynomials $p_a \in \mathcal{P}_M$, perform monic polynomial subtraction in the actual u variable to find the unique coefficients $L_{ab,l}$ in

$$\overline{p_a(c+iu)} p_b(c+iu) = \sum_{l=0}^{2M} L_{ab,l} b_l^{(k\lambda)}(u).$$

The complete coefficient-to-Gram map is therefore

$$H(p)_{ab} = C_k \sum_{l=0}^{2M} l! L_{ab,l} d_l. \quad (\text{EW.10})$$

It is a degree- k real polynomial map from the actual coefficient vector to Hermitian forms. The precise tests contain $c = k/2$, the two phases $\pm i$, and, when a relation column is used, both $\chi(c+iu)$ and $\overline{\chi}(c-iu)$. In particular (EW.10) constructs H_M from $c_{h,0}, \dots, c_{h,2M}$ and the stated k, λ . At $l = 0$, $c_{h,0} = \mu_h/c_\lambda$, so the same map gives $H_{00} = \mu_h^k$.

42.2 Canonical lifts and one arbitrary-window block

For each admitted degree define the exact matrices and maps

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : \mathbb{C}^q \longrightarrow \mathbb{C}^{N+1}, \quad V_N = \det G_N. \quad (\text{EW.11})$$

Positivity of H_N and surjectivity of J_N prove $K_N > 0$. Multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. For every $x \in \mathbb{C}^q$, any lift is $R_N x + B_N y$, and its norm is exactly $x^* G_N x + y^* B_N^* H_N B_N y$. This proves the unique minimum property and $R_N^* H_N R_N = G_N$.

Let E_N include the first q monomials. The square coefficient matrix $[E_N, B_N]$ has determinant one: its columns are ordered by successive degree and each leading coefficient is one. Subtracting boundary columns to replace E_N by R_N retains determinant one. Its Gram is block diagonal, so

$$V_N = \frac{\det H_N}{\det(B_N^* H_N B_N)}. \quad (\text{EW.12})$$

Empty determinants are one. This fixes the relation determinant and its index $N - q + 1$ in the original quotient coordinates.

Take integers

$$n \geq q, \quad r \geq 1, \quad m = n + r, \quad d = r + 1. \quad (\text{EW.13})$$

In $\mathcal{P}_m$ split the ordered monomials into degrees $0, \dots, n-1$ and the d appended columns $S^n, \dots, S^m$. Write

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad J_m = [J_{n-1}, J_D], \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EW.14})$$

The map

$$\Psi : \mathbb{C}^n \oplus \mathbb{C}^d \longrightarrow \mathbb{C}^{m+1}, \quad (x, y) \longmapsto (x - H_{n-1}^{-1} C_D y, y) \quad (\text{EW.15})$$

is invertible with determinant one. Direct multiplication gives $\Psi^* H_m \Psi = \text{diag}(H_{n-1}, W)$ and $J_m \Psi = [J_{n-1}, F]$. Thus $W > 0$; F maps the appended orthogonal residual polynomials to their original quotient classes. No quotient is independently chosen.

Let W_j be the first j rows and columns of W , F_j its first j quotient columns, and $Z_j = F_j^* G_{n-1} F_j$ for $1 \leq j \leq d$. The same map (EW.15), restricted to these columns, gives the exact positive-kernel update

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*. \quad (\text{EW.16})$$

Eliminating the two diagonal blocks of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in the two orders proves $\det(I + AB) = \det(I + BA)$. Use this identity in (EW.16) to obtain

$$\mathcal{R}_j := \frac{V_{n-1}}{V_{n+j-1}} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad \mathcal{R}_0 = 1. \quad (\text{EW.17})$$

All the determinants on the right are of size j , whereas its equivalent kernel calculation has size q . The matrices Z_j are positive semidefinite. Diagonalizing $W_j^{-1/2} Z_j W_j^{-1/2}$ proves $\mathcal{R}_j \geq 1$. Nested minimization also proves $G_{N+1} \preceq G_N$, so $\mathcal{R}_j$ is nondecreasing in j .

Let $p_N(S)$ be the unique monic polynomial of degree N orthogonal to $\mathcal{P}_{N-1}$, and let ω_N be its original squared norm. Existence and uniqueness follow by solving the positive lower-degree projection problem; its leading coefficient remains one. The first diagonal entry $w = W_1$ is ω_n . Eliminating the first r residual columns from the final one gives $\omega_m = \det W_d / \det W_r$. Consequently

$$U := \frac{\omega_m}{\omega_n} = \frac{\det W_d}{\det W_r w}, \quad \mathcal{R} := \frac{V_{n-1} V_n}{V_{m-1} V_m} = \frac{\mathcal{R}_r \mathcal{R}_d}{\mathcal{R}_1} \geq 1. \quad (\text{EW.18})$$

The last inequality also follows by multiplying $V_{n-1}/V_{m-1} \geq 1$ and $V_n/V_m \geq 1$. Both numerator and denominator norms carry their literal masses.

For the Gamma source (EW.7), the same J_m and χ give matrices W^Γ , F^Γ , Z_j^Γ by the identical maps. For their original source determinants put $\mathfrak{D}_j = \det H_{j-1}$ for $j \geq 1$ and $\mathfrak{D}_0 = 1$. For the relation determinant on exactly j columns put $\mathfrak{B}_j = \det(B_{q+j-1}^* H_{q+j-1} B_{q+j-1})$ for $j \geq 1$ and $\mathfrak{B}_0 = 1$. The reference determinants use the same columns and carry the superscript Γ . Writing the supplied corrections as $X_j = \mathfrak{D}_j / \mathfrak{D}_j^\Gamma$, $Y_j = \mathfrak{B}_j / \mathfrak{B}_j^\Gamma$ and $T_N = X_{N+1} / Y_{N-q+1}$, the relation (EW.12) proves

$$U = (n+1)_r (n+\beta)_r \frac{X_{m+1} X_n}{X_m X_{n+1}}, \quad (\text{EW.19})$$

$$\mathcal{R} = \frac{V_{n-1}^\Gamma V_n^\Gamma}{V_{m-1}^\Gamma V_m^\Gamma} \frac{T_{n-1} T_n}{T_{m-1} T_m}.$$

Indeed $\omega_j = C_k j! (\beta)_j X_{j+1} / X_j$ follows by taking successive source determinants in the original monic basis. Taking the ratio gives the first line with its exact rising factorials. The new symbol $\mathcal{R}_j$ in (EW.17) has no effect on the original positive arithmetic correction T_N .

42.3 Endpoint roots and exact paired-volume cancellations

Put $a = 1/(2r)$. The original endpoint expression is

$$\mathcal{E}_{n,r} = U^a \sinh(a \log \mathcal{R}) = \frac{(U\mathcal{R})^a - (U/\mathcal{R})^a}{2}. \quad (\text{EW.20})$$

All roots are the unique positive real roots. The equality follows from the definition of the real hyperbolic sine and positivity. It evaluates the existing quantity, preserving its two original norm endpoints and all four original volume endpoints.

There is a further exact cancellation inside its two root arguments. Set

$$p = \det W_r, \quad s = \det W_d, \quad e = w + Z_1, \quad D = \det(W_r + Z_r) \det(W_d + Z_d).$$

The scalar Z_1 is real nonnegative, and all four quantities p, s, e, D are strictly positive. Substituting (EW.17)–(EW.18) without changing a coordinate gives

$$U = \frac{s}{pw}, \quad \mathcal{R} = \frac{Dw}{pse}, \quad \boxed{U\mathcal{R} = \frac{D}{p^2e}, \quad \frac{U}{\mathcal{R}} = \frac{s^2e}{w^2D}}. \quad (\text{EW.21})$$

Thus the common highest source determinant cancels from the first argument; the common prefix source determinant cancels from the second. The first and last original monic norms have not been set equal or removed from the underlying source. Formula (EW.21) is the exact rational map explaining these cancellations.

It is sometimes useful to retain the small difference in (EW.20) as one positive numerator. With $P = (U\mathcal{R})^a$ and $Q = (U/\mathcal{R})^a$, the finite factorization $P^{2r} - Q^{2r} = (P - Q) \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j$ proves

$$\boxed{\mathcal{E}_{n,r} = \frac{U(\mathcal{R}^2 - 1)}{2^{2r-1} \sum_{j=0}^{2r-1} P^{2r-1-j} Q^j}}. \quad (\text{EW.22})$$

The denominator remains positive at $\mathcal{R} = 1$, and the numerator is then exactly zero. This representation preserves the sign of the endpoint quantity and avoids subtracting two enclosures before their common positive factor has been identified.

Every entry in (EW.14) is determined by (EW.10) through degree $2m = 2(n + r)$. Each J_N and B_N is computed by monic division or multiplication by the same χ . Equations (EW.11)–(EW.22) therefore prove the complete finite map

$$\boxed{(c_{h,0}, \dots, c_{h,2(n+r)}; k, \lambda, \chi) \mapsto (H_m, J_m, W, F, G_{n-1}) \mapsto \mathcal{E}_{n,r}}. \quad (\text{EW.23})$$

It consists of stated polynomial operations, inverses of matrices whose positivity has been proved, determinants, and positive roots.

For reference the endpoint selection follows from the original degree-wise estimate by a finite argument. Write $x_j = \frac{1}{2} \log(V_{n+j-1}/V_{n+j+1}) \geq 0$ and $A_j = \sqrt{\omega_{n+j+1}/\omega_{n+j}}$. The local estimate is $0 \leq \epsilon_{n+j} \leq A_j \sinh x_j$. If an x_j is zero, the minimum allowance is zero. Otherwise $(\log \sinh x)'' = -1/\sinh^2 x < 0$ proves finite Jensen, hence $\prod_j \sinh x_j \leq \sinh(\sum_j x_j/r)^r$. The actual products give $\prod_j A_j = \sqrt{U}$ and $2 \sum_j x_j = \log \mathcal{R}$. Multiplying the local estimates and taking the positive r th root therefore proves

$$\min_{0 \leq j < r} \epsilon_{n+j} \leq \mathcal{E}_{n,r}. \quad (\text{EW.24})$$

The local arithmetic trace inequality and its retained loss terms are proved in the complete source TVB.32–34a; (EW.24) is their exact finite selection, not an added local estimate or an asymptotic claim.

42.4 Exact projection between the canonical representative spaces

The following result supplies more information than the separate determinant values. Use the common ambient space $(\mathcal{P}_m, \langle \cdot, \cdot \rangle_H)$, with its original source inner product. Let $\iota_j : \mathbb{C}^{j+1} \rightarrow \mathbb{C}^{m+1}$ be monomial inclusion. The least-lift space and its exact isometry are

$$L_j = \iota_j R_j \mathbb{C}^q \subset \mathcal{P}_m, \quad \mathcal{J}_j : (\mathbb{C}^q, G_j) \longrightarrow L_j, \quad x \longmapsto \iota_j R_j x. \quad (\text{EW.25})$$

Here $\mathcal{J}_j^* H_m \mathcal{J}_j = G_j$ proves the word isometry; its inverse on L_j is the original remainder map. The original arithmetic correction T_N retains its meaning in (EW.19).

Theorem 42.1 (Canonical projection and all its moving directions). *For $q - 1 \leq a < b \leq m$, let $\Pi_b : \mathcal{P}_m \rightarrow \mathcal{P}_m$ be the ambient orthogonal projection with range L_b , including that range by its fixed polynomial inclusion. It obeys*

$$\boxed{\Pi_b \iota_a R_a x = \iota_b R_b x \quad (x \in \mathbb{C}^q)}. \quad (\text{EW.26})$$

Its restriction $L_a \rightarrow L_b$ is invertible. Its squared singular values, with precisely the source inner products in (EW.25), are the eigenvalues σ_i^2 of

$$G_a^{-1/2} G_b G_a^{-1/2}, \quad 0 < \sigma_i \leq 1. \quad (\text{EW.27})$$

At most $s_{a,b} = \min(q, b - a)$ of these values are less than one, and

$$\begin{aligned} \prod_i \sigma_i^2 &= V_b/V_a, \\ \|\Pi_a - \Pi_b\|_1 &= 2 \sum_i \sqrt{1 - \sigma_i^2}, \\ \|\Pi_a - \Pi_b\|_1 &\leq \min\{2\sqrt{s_{a,b} \log(V_a/V_b)}, 2s_{a,b}\}. \end{aligned} \quad (\text{EW.28})$$

The trace norm is for operators in the stated ambient inner product.

Proof. Both representatives in (EW.26) have remainder x . Their difference belongs to $\chi\mathcal{P}_{b-q}$ and is therefore orthogonal to L_b by (EW.11). The second representative belongs to L_b ; this proves the orthogonal projection identity. The same identity and injectivity of each lift show that the restriction is invertible. In the quotient coordinates it is the identity map from metric G_a to metric G_b . The explicit maps from Euclidean coordinates are $u \mapsto \iota_a R_a G_a^{-1/2} u$ and $v \mapsto \iota_b R_b G_b^{-1/2} v$. Their matrices give the squared singular values (EW.27), and no original form or mass has been changed. A projection is a contraction, so these values lie in $(0, 1]$. Taking its determinant proves the first line of (EW.28).

The boundary space at degree b extends the one at degree a by exactly $b - a$ independent monic columns. Vectors in L_a are already orthogonal to the older boundary. Imposing orthogonality to these extra $b - a$ columns has rank at most $b - a$. Its kernel is $L_a \cap L_b$: any vector in that kernel is in $\mathcal{P}_b$ and orthogonal to its full boundary, hence is a minimum representative. The intersection therefore has dimension at least $q - (b - a)$ when this number is positive. A unit vector in L_a is unchanged in norm by Π_b precisely when it belongs to L_b , by orthogonal Pythagoras. Thus at most $s_{a,b}$ singular values are less than one.

For a complete trace-norm calculation choose orthonormal eigenvectors $x_i \in L_a$ for the squared restriction and put $y_i = \Pi_b x_i / \sigma_i \in L_b$. Orthogonality gives $\langle x_i, y_j \rangle = \sigma_j \delta_{ij}$ and the y_i form an orthonormal basis of L_b . If $\sigma_i = 1$, then $y_i = x_i$. If $\sigma_i < 1$, set $z_i = (y_i - \sigma_i x_i) / \sqrt{1 - \sigma_i^2}$. Directly taking all these inner products proves that the planes $\text{span}(x_i, z_i)$ are mutually orthogonal and orthogonal to the common directions. In such a plane the matrix of $\Pi_a - \Pi_b$ is exactly

$$\begin{pmatrix} 1 - \sigma_i^2 & -\sigma_i \sqrt{1 - \sigma_i^2} \\ -\sigma_i \sqrt{1 - \sigma_i^2} & -(1 - \sigma_i^2) \end{pmatrix}.$$

Its eigenvalues are $\pm \sqrt{1 - \sigma_i^2}$. On the common directions and the orthogonal complement of $L_a + L_b$ the difference is zero. This proves the exact trace-norm sum. Finally $1 - x \leq -\log x$ for $0 < x \leq 1$, because the derivative of $x - 1 - \log x$ is $1 - 1/x$ and its value at one is zero. Cauchy-Schwarz on the at most $s_{a,b}$ nonzero terms gives the first bound in (EW.28); bounding each square root by one gives the second bound independently. $\square$

42.5 Window sensitivity and exact cancellation of mass changes

The next calculation holds for every continuously differentiable family of positive source matrices $H_m(t)$, using all their principal submatrices and the fixed χ, J_j . Thus it applies both to the actual coefficient forms and to a segment between their certified finite enclosures. Write $E = \dot{H}_m$ and define the real spectral spread

$$\mathfrak{o}_H(E) = \lambda_{\max}(H_m^{-1/2} E H_m^{-1/2}) - \lambda_{\min}(H_m^{-1/2} E H_m^{-1/2}) \geq 0. \quad (\text{EW.29})$$

The two congruences in this expression specify its exact coordinate map; the source H_m remains in every evaluation.

Embed the coefficient vector of p_j in $\mathcal{P}_m$ and set

$$P_j = \frac{\iota_j p_j p_j^* \iota_j^*}{\omega_j}, \quad Q_j = \iota_j R_j G_j^{-1} R_j^* \iota_j^*. \quad (\text{EW.30})$$

These are contravariant matrices; their products $P_j H_m$ and $Q_j H_m$ are respectively the ambient orthogonal projections onto the monic orthogonal line and L_j . To verify this, multiply each by itself and use $p_j^* H_j p_j = \omega_j$ or $R_j^* H_j R_j = G_j$. The resulting operators are idempotent and self-adjoint in H_m and have the stated ranges. The matrices $H_m^{1/2} P_j H_m^{1/2}$ and $H_m^{1/2} Q_j H_m^{1/2}$ are their exact Euclidean isometric conjugates.

Differentiating the norm of the monic polynomial gives $\dot{\omega}_j = p_j^* \dot{H}_j p_j$: its derivative has degree less than j , and both cross terms vanish by its original orthogonality. Differentiating $J_j R_j = I$ makes $\dot{R}_j$ a boundary, so the same expansion gives $\dot{G}_j = R_j^* \dot{H}_j R_j$. The determinant derivative follows by expanding the determinant multilinearly in its columns, or by differentiating $\det(G_j + t \dot{G}_j) = \det G_j \det(I + t G_j^{-1} \dot{G}_j)$. Thus

$$\frac{d}{dt} \log \omega_j = \text{Tr}(P_j E), \quad \frac{d}{dt} \log V_j = \text{Tr}(Q_j E). \quad (\text{EW.31})$$

All these formulas use the original coordinates before the final cyclic trace.

Put $\mathcal{L} = \log \mathcal{R} \geq 0$ and $s = \min(q, r)$. The endpoint derivatives are exactly

$$\begin{aligned} (\log U)' &= \text{Tr}((P_m - P_n)E), \\ \mathcal{L}' &= \text{Tr}((Q_{n-1} + Q_n - Q_{m-1} - Q_m)E), \\ \mathcal{E}'_{n,r} &= \frac{1}{2r} \left\{ \mathcal{E}_{n,r} (\log U)' + U^{1/(2r)} \cosh(\mathcal{L}/(2r)) \mathcal{L}' \right\}. \end{aligned} \quad (\text{EW.32})$$

These equalities also hold at $\mathcal{R} = 1$. They identify the exact paired sensitivity rather than separate errors of large source and relation determinants.

Theorem 42.2 (Intrinsic finite sensitivity). *Throughout every such positive continuously differentiable family,*

$$|(\log U)'| \leq \mathfrak{o}_H(E), \quad |\mathcal{L}'| \leq \mathfrak{o}_H(E) \min\{2s, \sqrt{2s\mathcal{L}}\}. \quad (\text{EW.33})$$

For any two positive source matrices $H^{(0)}, H^{(1)}$ in the same original coordinates, let

$$\kappa = \frac{\lambda_{\max}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})}{\lambda_{\min}((H^{(0)})^{-1/2} H^{(1)} (H^{(0)})^{-1/2})} \geq 1. \quad (\text{EW.34})$$

Their original window values satisfy

$$|\log(U^{(1)}/U^{(0)})| \leq \log \kappa, \quad |\sqrt{\mathcal{L}^{(1)}} - \sqrt{\mathcal{L}^{(0)}}| \leq \sqrt{s/2} \log \kappa. \quad (\text{EW.35})$$

In particular these sensitivity bounds contain no independent source-size determinant exponent. They vanish for any scalar change $H^{(1)} = t H^{(0)}$, $t > 0$, exactly as the original endpoint does.

Proof. Let $\Lambda = H_m^{-1/2} E H_m^{-1/2}$. Subtracting $\frac{1}{2}(\lambda_{\max}\Lambda + \lambda_{\min}\Lambda)I$ leaves operator norm $\mathfrak{o}_H(E)/2$. This scalar subtraction does not change either trace in (EW.32): the two monic-line projections each have trace one and the four least-lift projections each have trace q , with the displayed signs summing to zero.

The two monic orthogonal lines at degrees $n < m$ are orthogonal in the common ambient source. Their projection difference has eigenvalues 1, -1 and zeros, hence trace norm two. For any self-adjoint A and B , diagonalizing A gives $|\mathrm{Tr}(AB)| \leq \sum_i |\lambda_i(A)| \|B\|$. This proves the first bound in (EW.33).

For the second trace split its four projections as $(\Pi_{n-1} - \Pi_{m-1}) + (\Pi_n - \Pi_m)$. Both index gaps are exactly r . Put $\ell_1 = \log(V_{n-1}/V_{m-1})$ and $\ell_2 = \log(V_n/V_m)$, so $\ell_1 + \ell_2 = \mathcal{L}$. Equation (EW.28), the triangle inequality for the trace norm, and $\sqrt{\ell_1} + \sqrt{\ell_2} \leq \sqrt{2\mathcal{L}}$ give

$$\|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 2\sqrt{2s\mathcal{L}}, \quad \|\Pi_{n-1} + \Pi_n - \Pi_{m-1} - \Pi_m\|_1 \leq 4s.$$

Multiplying either bound by $\mathfrak{o}_H(E)/2$ proves the second bound of (EW.33).

For a continuously differentiable path and $\varepsilon > 0$, the exact bound $|\mathcal{L}'| \leq \mathfrak{o}_H(E)\sqrt{2s\mathcal{L}}$ implies

$$\left| \frac{d}{dt} \sqrt{\mathcal{L} + \varepsilon} \right| \leq \sqrt{s/2} \mathfrak{o}_H(E).$$

Integrate on the compact path and let ε decrease to zero. This proves the square-root Lipschitz assertion along a path, including points at which $\mathcal{L} = 0$; no division by zero is used.

Use the straight positive segment $H(t) = (1-t)H^{(0)} + tH^{(1)}$, $0 \leq t \leq 1$. Let $\ell_i > 0$ be the eigenvalues of $(H^{(0)})^{-1/2}H^{(1)}(H^{(0)})^{-1/2}$. Congruence by $(H^{(0)})^{1/2}$ and then its displayed orthonormal eigenbasis gives the generalized eigenvalues of $(E, H(t))$ exactly as

$$\frac{\ell_i - 1}{1 + t(\ell_i - 1)}.$$

These are the eigenvalues of $H(t)^{-1/2}EH(t)^{-1/2}$: both follow from the same equation $Ex = \lambda H(t)x$. The displayed function is increasing in ℓ_i , with derivative $[1 + t(\ell_i - 1)]^{-2}$. Hence

$$\int_0^1 \mathfrak{o}_{H(t)}(E) dt = \log \ell_{\max} - \log \ell_{\min} = \log \kappa.$$

Integrating the already proved differential inequalities proves (EW.35). Finally direct substitution in (EW.11) gives $R_N(tH) = R_N(H)$, $G_N(tH) = tG_N(H)$, and $\omega_j(tH) = t\omega_j(H)$. Thus U and $\mathcal{R}$ are exactly unchanged by that explicitly stated common scaling. The source mass remains $t\mu_h^k$ in the scaled comparison; it was never reassigned the value one. $\square$

The independent projection review supplied a stronger exact version from the same angles. For $s \geq 1$, it is

$$|\mathcal{L}'| \leq 2s \mathfrak{o}_H(E) \sqrt{1 - e^{-\mathcal{L}/(2s)}}. \quad (\text{EW.35a})$$

To prove it, list $x_i = -\log \sigma_i^2$ for each of the two projection pairs used above, omitting their zero terms and then appending zeros to make exactly $2s$ entries. Their sum is $\mathcal{L}$. The exact trace-norm formula and the scalar cancellation prove $|\mathcal{L}'| \leq \mathfrak{o}_H(E) \sum_i f(x_i)$, where $f(x) = \sqrt{1 - e^{-x}}$. For $x > 0$,

$$f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} < 0.$$

Continuity at zero extends concavity there: apply the concavity inequality to positive arguments increased by ε and then let $\varepsilon \downarrow 0$. Finite Jensen on the full $2s$ -term list proves (EW.35a). Since $1 - e^{-x} \leq x$ and $1 - e^{-x} \leq 1$, it implies both bounds on $|\mathcal{L}'|$ in (EW.33).

Define the explicit increasing map on the actual loss variable by

$$\Phi_s : [0, \infty) \longrightarrow [0, \infty), \quad \Phi_s(L) = 2 \operatorname{arcosh}(e^{L/(4s)}), \quad \Phi_s^{-1}(v) = 4s \log \cosh(v/2).$$

Here $\operatorname{arcosh} x = \log(x + \sqrt{x^2 - 1})$ for $x \geq 1$, so both maps and their inverses have specified real branches. For $L > 0$, $\Phi'_s(L) = [2s\sqrt{1 - e^{-L/(2s)}}]^{-1}$. Apply (EW.35a) to $\Phi_s(\mathcal{L} + \varepsilon)$. Its derivative in absolute value is at most $\mathfrak{o}_H(E)\sqrt{(1 - e^{-\mathcal{L}/(2s)})/(1 - e^{-(\mathcal{L}+\varepsilon)/(2s)})} \leq \mathfrak{o}_H(E)$. Integrating and letting $\varepsilon \downarrow 0$, then using the exact straight-segment length already proved, gives

$$|\Phi_s(\mathcal{L}^{(1)}) - \Phi_s(\mathcal{L}^{(0)})| \leq \log \kappa. \quad (\text{EW.35b})$$

This conclusion includes zero losses by continuity. For $s = 0$, the only loss is zero and define the one-point map $\Phi_0(0) = 0$; the projection and endpoint comparisons are their zero identities.

For $s \geq 1$, the stronger enclosure itself has an exact algebraic form. Given any $\rho \geq \kappa$ and the original $\mathcal{R}^{(0)} \geq 1$, put

$$z_0 = (\mathcal{R}^{(0)})^{1/(4s)} + \sqrt{(\mathcal{R}^{(0)})^{1/(2s)} - 1}, \quad z_- = \max\{1, z_0/\sqrt{\rho}\}, \quad z_+ = \sqrt{\rho} z_0, \quad (\text{EW.35c})$$

$$\left(\frac{z_- + z_-^{-1}}{2} \right)^{4s} \leq \mathcal{R}^{(1)} \leq \left(\frac{z_+ + z_+^{-1}}{2} \right)^{4s}.$$

Indeed $z_0 = \exp(\Phi_s(\mathcal{L}^{(0)})/2)$, (EW.35b) encloses its counterpart between z_- and z_+ , and $z \mapsto (z + z^{-1})/2$ is increasing for $z \geq 1$. The identity $\mathcal{R} = \cosh(\Phi_s(\log \mathcal{R})/2)^{4s}$ proves (EW.35c). At the same time $U^{(0)}/\rho \leq U^{(1)} \leq \rho U^{(0)}$ by (EW.35). Thus monotonicity in (EW.20) supplies an endpoint enclosure using only nonnegative real roots, rational operations and integer powers when the central $U, \mathcal{R}$ and a bound ρ are rational. For example any proved rational Loewner comparison $aH^{(0)} \preceq H^{(1)} \preceq bH^{(0)}$, $0 < a \leq b$, gives $\rho = b/a$: the extreme Rayleigh quotients lie between a and b . This is the exact map from a positive-matrix certificate to the sharper window certificate, with the original mass and relation unchanged.

The square-root assertion is a Lipschitz assertion, and does not assert ordinary differentiability at every zero of $\mathcal{L}$. For an exact auxiliary illustration take $q = 1$, $\chi = S$, $n = 1$, $r = 1$, and the positive original-coordinate matrix

$$H_2(t) = \begin{pmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ t & 0 & 1 \end{pmatrix}, \quad |t| < 1.$$

Its leading pivots are $1, 1, 1 - t^2$, its remainder is evaluation at $S = 0$, and (EW.11) gives $V_0 = V_1 = 1$, $V_2 = 1 - t^2$. Thus $\mathcal{L} = -\log(1 - t^2)$ and $\sqrt{\mathcal{L}}/|t| \rightarrow 1$ as $t \rightarrow 0$. Its square root has a corner. This positive-form example belongs exactly to the finite family domain of (EW.33)–(EW.35); it is not claimed to be an arithmetic moment source. The regularized proof above includes it.

For completeness, (EW.35) has a directly usable endpoint enclosure. Writing $d_H = \log \kappa$ and $b = \sqrt{s/2} d_H$, it proves

$$e^{-d_H} U^{(0)} \leq U^{(1)} \leq e^{d_H} U^{(0)}, \quad (\text{EW.36})$$

$$(\max\{0, \sqrt{\mathcal{L}^{(0)}} - b\})^2 \leq \mathcal{L}^{(1)} \leq (\sqrt{\mathcal{L}^{(0)}} + b)^2.$$

The function $U^{1/(2r)} \sinh(\mathcal{L}/(2r))$ is increasing in each positive U and nonnegative $\mathcal{L}$, as direct differentiation shows. Substitution of the two endpoints in (EW.36) therefore bounds the actual endpoint expression. This is a finite stability theorem, including its exact $s = \min(q, r)$ dependence.

42.6 Rational interval certificates for the original input

Here is an explicit finite algorithm which uses enclosures of the original coefficients; it does not replace them by an arbitrary moment matrix. For clarity first suppose the constants in (EW.5) and the polynomial tests in (EW.10) are exact. Given real intervals $|c_{h,j} - \hat{c}_j| \leq e_j$ for $0 \leq j \leq 2m$, set

$$\begin{aligned}\hat{A}(z) &= \sum_{j=0}^{2m} (\alpha)_j \hat{c}_j z^j, \\ A_+(z) &= \sum_{j=0}^{2m} (\alpha)_j |\hat{c}_j| z^j, \quad E_+(z) = \sum_{j=0}^{2m} (\alpha)_j e_j z^j, \\ \hat{d}_l &= [z^l] \hat{A}(z)^k, \quad e_l^{(k)} = [z^l] \{(A_+(z) + E_+(z))^k - A_+(z)^k\}.\end{aligned}\tag{EW.37}$$

The multinomial expansion groups each difference of products by the nonempty set of factors carrying an error. Taking absolute values term by term proves $|d_l - \hat{d}_l| \leq e_l^{(k)}$. Thus the Hermitian central matrix and entry radii

$$\hat{H}_{ab} = C_k \sum_l l! L_{ab,l} \hat{d}_l, \quad \varepsilon_{ab} = C_k \sum_l l! |L_{ab,l}| e_l^{(k)}\tag{EW.38}$$

enclose the actual source. Rational upper bounds for the moduli give rational radii. An uncertain real constant, including C_k , is propagated by interval multiplication before the final matrix is reported. For $|x - \hat{x}| \leq e_x$ and $|y - \hat{y}| \leq e_y$, the exact elementary rule is $|xy - \hat{x}\hat{y}| \leq |\hat{x}|e_y + |\hat{y}|e_x + e_x e_y$. Repeated use handles every finite product in (EW.8)–(EW.10).

The original relation need not have rational coefficients. To retain it exactly while producing rational certificates, use its quotient-first coordinate map at each endpoint degree N :

$$A_N = [E_N, B_N] : \mathbb{C}^q \oplus \mathbb{C}^{N-q+1} \longrightarrow \mathbb{C}^{N+1}, \quad \det A_N = 1, \quad \tilde{H}_N = A_N^* H_N A_N.\tag{EW.39}$$

The first component of $A_N^{-1}P$ is exactly $J_N P$ by monic division. Supply rational enclosures of the real and imaginary parts of the actual χ coefficients when they are not exact rational numbers. Ordinary complex interval addition, multiplication and conjugation in (EW.39) give rational entry enclosures of the actual $\tilde{H}_N$. For a complex interval a modulus radius is bounded rationally, for example by the sum of its two coordinate radii. Conjugation changes the sign of the imaginary coordinate; it is applied to the actual first factor. This procedure never changes χ , its degree, its monicity or the original quotient coordinates. Intervals for λ and its positive constants, if required, are propagated in the same finite operations and in the specified recurrence (EW.6).

Here is a rational matrix certificate valid for either H_m or any of these $\tilde{H}_N$. Choose a Hermitian rational centre $\hat{M}$ and symmetric rational radii ε_{ab} enclosing its actual error. Choose any positive rational v_a and form

$$\Delta = \text{diag} \left(\sum_b \varepsilon_{ab} \frac{v_b}{v_a} \right), \quad M^- = \hat{M} - \Delta, \quad M^+ = \hat{M} + \Delta.\tag{EW.40}$$

The pairwise inequality $2|x_a||x_b| \leq (v_b/v_a)|x_a|^2 + (v_a/v_b)|x_b|^2$ proves $M^- \preceq M \preceq M^+$. Rational complex arithmetic now eliminates each leading pivot by the congruence

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \longmapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \longmapsto (x - a^{-1}b^*y, y).\tag{EW.41}$$

Recording strictly positive rational pivots proves $M^- > 0$ by induction and hence also $M^+ > 0$. A nonpositive pivot rejects that particular proposed certificate. Every actual matrix is already positive by its source integral and, for (EW.39), its invertible congruence. At a fixed finite degree, convergent valid entry intervals eventually give positive lower matrices: the least unit-sphere value of the

actual quadratic form is positive by compactness, and the finite sum of entry errors tends to zero. This proves the finite certificate's convergence without assuming or claiming that a growing-degree arithmetic input has already been enclosed.

For the two norm endpoints take the corresponding principal submatrices of the certified H_m^-, H_m^+ . Define ω_j^-, ω_j^+ by their exact monic Schur complements. The squared norm of any fixed monic polynomial is between its two enclosing forms. Minimizing over the same affine monic set proves

$$0 < \omega_j^- \leq \omega_j \leq \omega_j^+, \quad j = n, m. \quad (\text{EW.42})$$

For each of $N = n - 1, n, m - 1, m$, use (EW.39)–(EW.41) and split its first q and last $N - q + 1$ coordinates. The Schur complements of the latter block in $\tilde{H}_N^-, \tilde{H}_N^+$ give rational positive forms G_N^-, G_N^+ . Their affine fibre has fixed first coordinate x , by the exact map (EW.39). Minimization therefore proves

$$G_N^- \preceq G_N \preceq G_N^+, \quad 0 < V_N^- := \det G_N^- \leq V_N \leq \det G_N^+ =: V_N^+. \quad (\text{EW.43})$$

For the determinant inequality, conjugate by the square root of the smaller positive form and multiply the resulting eigenvalues, each at least one. At $N = q - 1$ the boundary block is empty and the Schur complement is the whole q -dimensional form.

All numbers in the following enclosure are rational and positive:

$$\begin{aligned} U_- &= \omega_m^- / \omega_n^+, & U_+ &= \omega_m^+ / \omega_n^-, \\ \mathcal{R}_- &= \max \left\{ 1, \frac{V_{n-1}^- V_n^-}{V_{m-1}^+ V_m^+} \right\}, & \mathcal{R}_+ &= \frac{V_{n-1}^+ V_n^+}{V_{m-1}^- V_m^-}. \end{aligned} \quad (\text{EW.44})$$

They satisfy $U_- \leq U \leq U_+$ and $1 \leq \mathcal{R}_- \leq \mathcal{R} \leq \mathcal{R}_+$. The use of the lower bound one follows from the proved nested minimization, rather than an interval rounding choice.

Positive roots require no floating logarithm certificate. For any positive rational x and integer $2r$, find rational numbers $l, u > 0$ such that $l^{2r} \leq x \leq u^{2r}$. Doubling upper or lower powers of two supplies an initial interval, and ordinary rational bisection makes it arbitrarily narrow. Each decision is a positive-denominator integer comparison. Strict monotonicity of $t \mapsto t^{2r}$ for $t > 0$ proves the enclosure and convergence. Apply this algorithm to

$$U_- \mathcal{R}_-, \quad U_- / \mathcal{R}_-, \quad U_+ \mathcal{R}_+, \quad U_+ / \mathcal{R}_+.$$

Write their certified root intervals as $[l_1, u_1], \dots, [l_4, u_4]$. The function in (EW.20) is increasing in $U > 0$ and $\mathcal{R} \geq 1$: its U derivative is its nonnegative value times $1/(2rU)$, and its $\mathcal{R}$ derivative is $U^{1/(2r)} \cosh(\log \mathcal{R} / (2r)) / (2r\mathcal{R}) > 0$. Therefore the complete rational endpoint certificate is

$$\boxed{\max\{0, (l_1 - u_2)/2\} \leq \mathcal{E}_{n,r} \leq (u_3 - l_4)/2.} \quad (\text{EW.45})$$

Alternatively (EW.22) gives a positive numerator and denominator certificate from the same root brackets. Every matrix pivot, integer power comparison, relation coefficient and source mass used in this algorithm is inspectable. It makes no claim for coefficient intervals which have not been provided.

42.7 Boundary cases and the retained cochain map

At $r = 1$, the block has two columns, $W_r = w$, and $\mathcal{R} = \mathcal{R}_2$ because the two copies of $\mathcal{R}_1$ cancel. Equations (EW.18)–(EW.22) reduce exactly to the two-degree formula GMT.24–27. At $n = q$, the base source degree is $q - 1$ and its boundary dimension is zero; its quotient metric is simply the source Gram. All formulas above still use positive matrices of their specified sizes. No degree $q - 2$ full-quotient form is inserted.

The actual positive polynomial source has an additional strictness property for $q \geq 1$: every two-degree quotient-volume ratio is strictly greater than one. Here is the complete algebraic reason. Let $\pi_j(u)$ be the real monic orthogonal family of the original positive real- u measure. Multiplication by u is self-adjoint. Pairing $u\pi_j$ with lower orthogonal polynomials and using their degrees gives

$$u\pi_j = \pi_{j+1} + b_j\pi_j + a_j\pi_{j-1}, \quad b_j \in \mathbb{R}, \quad a_j = \omega_j/\omega_{j-1} > 0 \quad (j \geq 1).$$

For $j = 0$ the last term is absent. The exact monic phase map $p_j(S) = i^j \pi_j((S - c)/i)$ has the same squared norm, because $|i^j| = 1$, and gives

$$Sp_j = p_{j+1} + (c + ib_j)p_j - a_j p_{j-1}.$$

If a nonconstant χ divided p_j and p_{j+1} , this recurrence and $a_j \neq 0$ would make it divide p_{j-1} . Induction would make it divide $p_0 = 1$, a contradiction. For the two appended columns at degrees $N, N+1$ over $\mathcal{P}_{N-1}$, the residuals are p_N and $p_{N+1} + zp_N$, where the original coefficient is exactly $z = \langle p_N, S^{N+1} \rangle_H / \omega_N$. Indeed their difference from those polynomials has smaller degree, and both residuals are orthogonal to $\mathcal{P}_{N-1}$. If their full quotient residual map F_2 were zero, both p_N and p_{N+1} would have zero remainder, contradicting the divisibility argument. Thus $Z_2 = F_2^* G_{N-1} F_2$ is a nonzero positive semidefinite matrix, and (EW.17) gives $V_{N-1}/V_{N+1} > 1$ for every $N \geq q$. The finite product

$$\mathcal{R} = \prod_{j=0}^{r-1} \frac{V_{n+j-1}}{V_{n+j+1}} > 1 \quad (q \geq 1, n \geq q, r \geq 1) \quad (\text{EW.46a})$$

is its exact overlapping telescope. This proves strict positivity of the actual $\mathcal{E}_{n,r}$ and of $\mathcal{L}$ at each admitted nonempty arithmetic window; it supplies no uniform lower bound as degrees or packets grow. The more general positive-form sensitivity theorem includes the zero-loss boundary separately.

The empty packet is a separate declared case: $h = 1$, $\chi = 1$, $q = 0$, and $\mathcal{C} = 0$. The analytic density and mass remain those of (EW.1). Put $V_N = 1$ even at the empty degree $N = -1$; all quotient matrices, projections and F_j have their empty dimensions. For $n \geq 0$, $r \geq 1$ the source norms exist, $\mathcal{R} = 1$, and $\mathcal{E}_{n,r} = 0$. If $n = 0$, use the empty base source H_{-1} , with determinant one and no inverse operation; the block $W = H_r$ then has its original monomial coordinates. Its first entry is the full mass. The projection and volume sensitivity statements become their zero-map identities, while the source-norm sensitivity remains the proved rank-two calculation.

Finally the projection comparison (EW.26) has an explicit original theta representative. For $a < b$ and $x \in \mathcal{C}$, monic division gives a unique polynomial $\kappa_x \in \mathcal{P}_{b-q}$ with

$$\iota_a R_a x - \iota_b R_b x = \chi \kappa_x. \quad (\text{EW.46})$$

Its coefficients are obtained by the same exact division, so the map $x \mapsto \kappa_x$ is linear. Let $\mathcal{V}_{h,k} P = P(D_1 + \cdots + D_k) F_h^{\otimes k}$, where $D = -x\partial_x$, $MF_h = v_h$, and $h(D)F_h = \Theta\phi_0$. Successive division in $s_1, \dots, s_k$ supplies specific Q_i with $\chi(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$. The final tensor remainder vanishes because its monomials form a basis of $(\mathbb{C}[s]/(h))^{\otimes k}$ and χ annihilates the original sum. Thus the exact cochain of degree $k-1$

$$\mathcal{P}_\chi(\kappa_x) = \sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) \kappa_x(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}) \quad (\text{EW.47})$$

satisfies

$$d_{\text{tot}} \mathcal{P}_\chi(\kappa_x) = \mathcal{V}_{h,k} (\iota_a R_a x - \iota_b R_b x). \quad (\text{EW.48})$$

Indeed the tensor differential at position i contributes $(-1)^{i-1}$ because its preceding factors have cochain degree one; this cancels the written sign, and $h(D_i)F_h = \Theta\phi_0$ then proves the equality term

by term. The arithmetic jet of both representatives is the same ηx of (EW.3). In the original support carrier $\{\tau\} \sqcup \bigsqcup_{\ell}(\{\ell\} \times E_{\ell})$, a fibrewise linear map sends (ℓ, v) to $(\ell, L_{\ell}v)$ and τ to τ . Thus the displayed boundary maps to $e_{\ell} = (\ell, 0)$ while external absence remains τ . The endpoint's representative changes, projection angles, and volume changes have therefore all been related to the original typed maps and their full theta primitives.

42.8 Verification scope and exact auxiliary calibration

The companion checker uses the explicitly auxiliary weight

$$\begin{aligned} \lambda &= 1, \quad k = 2, \quad c = 1, \quad c_1 = 1/2, \quad C_2 = 1/4, \\ B(t) &= 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = S^2 - 2S - 1. \end{aligned} \tag{EW.49}$$

This is a specified positive $L^2(r_1)$ multiplier and the same finite coefficient-to-Gram domain as (EW.8)–(EW.10). It is not asserted to be an actual zeta packet. Its exact connection to the arithmetic problem is the complete sequence of maps (EW.10), (EW.11), (EW.14)–(EW.23), and (EW.39)–(EW.45), with its auxiliary inputs written explicitly.

For this fixture $b_2^{(1)} = t^2 - 2$, so $c_{B,0} = 3/2$, $c_{B,2} = 1/4$, and every other coefficient is zero. Consequently $\sum_j (2)_j c_{B,j} z^j = 3/2 + 3z^2/2$ and its square has coefficients $d_0 = 9/4$, $d_2 = 9/2$, $d_4 = 9/4$. Its one-factor mass is $3/4$, the literal sum mass is $9/16$, and the reference sum mass is $1/4$. The checker independently constructs Gamma moments from the monic recurrence, inserts $1 + t^2/4$ in each factor, and expands every power of $t_1 + t_2$ with its full binomial coefficient. It also constructs the coefficient-side moments from (EW.9); agreement compares two exact finite calculations, including the tensor cross terms.

In this auxiliary example gamma recurrence gives a further exact all-degree check. Since $|\Gamma(2 + it/2)|^2 = (1 + t^2/4)|\Gamma(1 + it/2)|^2$, one has $Br_1 = r_2$. The displayed masses give $c_2 = 3/4$, $c_4 = 315/8$, and hence

$$m = r_2 * r_2 = r_4/70, \quad r_1 * r_1 = r_2/3, \quad \omega_j = \frac{9}{16} j!(8)_j, \quad U = (n+1)_r(n+8)_r. \tag{EW.50}$$

The first two equalities follow from the literal convolution law (EW.7), with coefficients $(3/4)^2/(315/8) = 1/70$ and $(1/2)^2/(3/4) = 1/3$. The monic original-coordinate map is $p_j(S) = i^j b_j^{(4)}((S-1)/i)$, whose squared norm is exactly the third quantity by (EW.7). Taking the stated endpoint ratio proves the final formula.

The exact selected windows computed in the accompanying rational receipt are

(n, r)	U	$\mathcal{R}$
(2, 1)	30	20191/4860
(2, 2)	1320	3073295611/216513000
(3, 2)	2640	295913127219469/36104825790000

(EW.51)

Here is a full independent finite derivation of the entries in that table, using the original quotient kernel rather than a Schur block. Let $x = S - 1$ solely in the stated quotient-coordinate map

$$A : \mathbb{C}_{(1,x)}^2 \longrightarrow \mathbb{C}_{(1,S)}^2, \quad (a, b) \longmapsto (a - b, b), \quad A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \det A = 1.$$

It takes the identical relation to $x^2 - 2$, so its residue equation is exactly $x^2 = 2$. It preserves the determinant of the quotient form by the congruence $G_N^x = A^* G_N A$. The monic source family in (EW.50) obeys $p_{j+1} = xp_j + j(j+7)p_{j-1}$, by applying the exact phase map to (EW.6). Orthogonal expansion of the original source inverse in (EW.11) proves the finite quotient kernel formula

$$(G_N^x)^{-1} = \sum_{j=0}^N \frac{[p_j]_{(1,x)} [p_j]_{(1,x)}^*}{\omega_j}. \tag{EW.52}$$

Indeed the matrix of the monic orthogonal columns in the original source is invertible triangular with diagonal one, takes H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, and sends the reduction columns to their written remainders. Substituting that inverse congruence into $J_N H_N^{-1} J_N^*$ gives (EW.52) term by term. The recurrence and $x^2 = 2$ give the complete required list

j	$[p_j]_{(1,x)}$	ω_j
0	(1, 0)	9/16
1	(0, 1)	9/2
2	(10, 0)	81
3	(0, 28)	2430
4	(356, 0)	106920
5	(0, 1588)	6415200

(EW.53)

For example the successive remainders are obtained from $x^2 + 8 = 10, 10x + 18x = 28x, 28x^2 + 300 = 356$, and $356x + 44 \cdot 28x = 1588x$. The positive kernels from (EW.52) are diagonal in this explicitly transported quotient basis, with

N	$((G_N^x)^{-1})_{00}$	$((G_N^x)^{-1})_{11}$	V_N
1	16/9	2/9	81/32
2	244/81	2/9	729/488
3	244/81	662/1215	98415/161528
4	56102/13365	662/1215	16238475/37139524
5	56102/13365	376069/400950	2679348375/10549111519

(EW.54)

Each diagonal entry is the indicated finite sum of rational terms in (EW.52), and V_N is the reciprocal of their product because $\det A = 1$. Substituting (EW.53)–(EW.54) in (EW.18) gives all three rows of (EW.51) by positive-denominator multiplication.

There is also an executed coefficient-box certificate, before any matrix error is chosen. Put $\varepsilon = 10^{-12}$ and retain $n = 2, r = 1, m = 3$, the same reference mass and the same χ . Allow the complete specified one-factor input box

$$|c_{B,0} - 3/2| \leq \varepsilon, \quad |c_{B,2} - 1/4| \leq \varepsilon, \quad c_{B,j} = 0 \quad (j = 1, 3, 4, 5, 6). \quad (\text{EW.55})$$

Every point of this box is realized by the specified positive multiplier $B_{a,b}(t) = a + bt^2$, where $a = c_{B,0} - 2c_{B,2}$ and $b = c_{B,2}$. They obey $a \geq 1 - 3\varepsilon > 0$ and $b \geq 1/4 - \varepsilon > 0$. Its coefficient formula follows by $a + bt^2 = (a + 2b)b_0^{(1)} + bb_2^{(1)}$. The interior input $a = 1 + \varepsilon, b = 1/4 - \varepsilon/4$ has one-factor mass $3/4 + \varepsilon/4$ and sum mass $(3/4 + \varepsilon/4)^2$. Thus neither the box nor its test input replaces the zeroth coefficient by a unit mass.

For (EW.55), the complete coefficient-power error formula (EW.37) is exactly

$$\begin{aligned} e_0^{(2)} &= 3\varepsilon + \varepsilon^2, & e_2^{(2)} &= 21\varepsilon + 12\varepsilon^2, & e_4^{(2)} &= 18\varepsilon + 36\varepsilon^2, \\ e_1^{(2)} &= e_3^{(2)} = e_5^{(2)} = e_6^{(2)} = 0. \end{aligned} \quad (\text{EW.56})$$

Indeed the central polynomial is $3/2 + (3/2)z^2$ and its error polynomial is $\varepsilon + 6\varepsilon z^2$; expanding the full square difference gives the three displayed coefficients. The upper corner of the coefficient box attains all three positive differences, so its quadratic error terms cannot be omitted.

Expanding all sixteen original tests $(1 - iu)^a(1 + iu)^b, 0 \leq a, b \leq 3$, in the monic Gamma basis of (EW.6), then using (EW.38)–(EW.40) with four weights equal to one, gives

$$\hat{H}_3 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix},$$

$$\begin{aligned}\Delta_{00} &= 54\varepsilon + 28\varepsilon^2, & \Delta_{11} &= (1029\varepsilon + 873\varepsilon^2)/2, \\ \Delta_{22} &= (2109\varepsilon + 1773\varepsilon^2)/2, & \Delta_{33} &= 29754\varepsilon + 32878\varepsilon^2.\end{aligned}\tag{EW.57}$$

These diagonal sums retain the absolute value of every test coefficient. The real coefficients occur at even degrees and the imaginary coefficients at odd degrees; the latter pair exactly to zero against every member of the specified even coefficient box. The complete complex expansions, including those zero pairings, are recorded in the certificate.

Here is an elementary exact positive-definiteness certificate for the whole box. Rational multiplication verifies

$$\hat{H}_3^{-1} = \frac{1}{2430} \begin{pmatrix} 8019 & -1863 & 351 & -27 \\ -1863 & 1501 & -147 & 29 \\ 351 & -147 & 39 & -3 \\ -27 & 29 & -3 & 1 \end{pmatrix}, \quad \text{Tr } \hat{H}_3^{-1} = 956/243 < 4.$$

The central form is positive by its actual measure, or by the four original monic norms in (EW.53). Its inverse is positive, so each of its eigenvalues is at most its trace. Consequently $\hat{H}_3 \succ I_4/4$. Substitution of $\varepsilon = 10^{-12}$ into the four full polynomials (EW.57) gives $0 < \Delta_{jj} < 10^{-6}$ by integer comparison. Hence

$$\hat{H}_3 - \Delta \succ (1/4 - 10^{-6})I_4 \succ 0, \quad \hat{H}_3 - \Delta \preceq H_3 \preceq \hat{H}_3 + \Delta \tag{EW.58}$$

for every actual multiplier in (EW.55). This proves the positive matrix enclosure before its inverses are evaluated. The accompanying receipt also records each exact rational elimination pivot separately.

Use the original J_N and χ at $N = 1, 2, 3$ in (EW.42)–(EW.45). The resulting rational interval for every endpoint value in this whole coefficient box is

$$\frac{10247872526420120792621067}{2^{81}} \leq \mathcal{E}_{2,1} \leq \frac{10247872551311149445602506}{2^{81}}. \tag{EW.59}$$

The four endpoint arguments in (EW.45) are computed from the written rational matrices $\hat{H}_3 \pm \Delta$ by their monic and quotient Schur formulas; the receipt gives every one of these exact rational values and its two squared-root comparisons. Thus (EW.59) is certified by integer powers and the minimum principle, with no rounding assertion about a transcendental logarithm. The interior positive multiplier and all four box corners are independently reconstructed by the literal one-factor moment recurrence and tensor binomial expansion, and pass the same original-coordinate entry bounds.

All original- S Gram entries are computed from $(1 - iu)^a(1 + iu)^b$. The checker's window certificates include the exact reduction matrix, source norms, quotient metrics, the $r + 1$ Schur block and all its prefixes, both cancellations in (EW.21), positive rational matrix envelopes, and the integer power comparisons for (EW.45). The written proof establishes the all-degree statements; each executed finite check is reported separately in the accompanying receipt. The base fixture executes 416 distinct mathematical checks across fifteen windows; its normal and optimized runs pass and six changed-formula runs in each mode are rejected after executing the complete suite. The separate supplement executes 329 mathematical checks and two dependency-hash checks; its four changed-formula runs in each mode are also rejected after the complete suite, including the omitted quadratic coefficient error at an attained box corner. The independent remainder-kernel calibration executes the six scalar comparisons in (EW.51). These counts describe their separate scopes; repeated modes and changed-formula runs do not add new mathematical checks. Numerical examples do not stand in for the arithmetic input or a growing-window bound.

43 Complete arithmetic endpoint derivation and exact source audit

Source role. Full nine-section derivation, constants, endpoint domains and metric maps. Current provenance correction: fresh normal and optimized JSON outputs have identical bytes, but the delivered arithmetic

file has 989 LF bytes and the fresh file has 1,016 bytes with 27 CRLF endings. Replacing each CRLF by LF gives the delivered bytes exactly; parsed JSON values agree. Thus the later reproduced claim of byte identity with the delivered file is superseded. The full byte-to-text map and separately retained hashes are in `endpoint_restriction_json_equality_correction_20260913.md` and its JSON companion.

Exact revision. `d0afdde7ebb41d1415ffca8ba61a6099a30bbaf4593576388ddb85895b6241b9`

Complete source. `sources/local_endpoint_continuations/arithmetic_endpoint_intake_20260913.md`

Source SHA-256. `d0afdde7ebb41d1415ffca8ba61a6099a30bbaf4593576388ddb85895b6241b9`

43.1 Arithmetic endpoint delivery: complete source intake and mathematical audit

This audit covers the complete 13 September 2026 arithmetic endpoint delivery. It proves the displayed dyadic norm estimate for the original arithmetic measure, supplies the intermediate constants and typed maps, and records exactly what the finite replay establishes. The source's equation numbers are retained as references. This file does not modify the source archive, the 478-page Gamma paper, a release, or a publication branch.

43.1.1 1. Exact source and execution record

The archive `Tau_Arithmetic_Endpoint_Bounds_2026-09-13.zip` has 781925 bytes and SHA-256 `c32ca4bebc86c6b7c7ffb58533a49b75e2140b00b31cdb6602cea6d8537ad9f6`. All 32 archive members were extracted with path traversal, absolute path, and conflicting-existing-byte checks into `sources/web_arithmetic_endpoint_delivery/Tau_Arithmetic_Endpoint_Bounds`. Every extracted byte agrees with its archive member. The original archive's 31-entry manifest checks every member other than that manifest itself.

The complete mathematical Markdown (28116 bytes, SHA `8fa97624a55c2f1baad54e465e4637c584329f24f6c198bb5d168a443811a280`) and complete 840-line TeX (31714 bytes, SHA `f776e7934f026342f48732a74f14d57470b8aae1cef76ebdee5f831bd49dedf8`) were read. The complete second user paste, README, handoff, programme state, source review, checks description, Python checker, reader builder, manifest, all JSON receipts and all text logs were read. The add-only patch was completely parsed; all 15 added files reconstruct their corresponding delivered files byte for byte, including the explicitly recorded missing final newline of `checks/source-reading.json`. No patch was applied to any checkout.

The supplied HTML contains all 225 mathematical nodes from the Markdown in the same order and with exactly the same TeX annotation strings. Its 226th MathML element is the duplicate g/h in the table of contents. All 48 numbered displays are present. The supplied desktop, mobile and theorem PNGs were each actually viewed; the title and theorem regions shown are readable. Those three PNGs and browser-layout measurements remain historical evidence; this intake does not claim a fresh browser rendering or inspection of portions absent from those screenshots.

The checker `check_endpoint_bounds.py` has SHA `dcf57f151fd34865907bfa80f8fc063e5d46dbf69201eb ee7e0a9b19b26fd2b7`. Complete inspection found only finite SymPy/unittest calculations and stdout output. Four sequential fresh processes were run with one thread in the BLAS/OpenMP environment, with bytecode writing disabled and an isolated working directory:

- Normal Python: all 18 named test methods passed.
- `python -O`: the same 18 methods passed; the full JSON stdout is byte-identical to the normal run and to the delivered successful payload.
- Normal `--fail-control`: exit 1 with the declared explicit exception.
- Optimized `--fail-control`: the same declared exit 1.

The failure option executes before the test suite is loaded. These are two guard-failure controls with **zero test methods executed and zero mathematical formula mutants**. They do not test mutation sensitivity of the analytic argument. All 37 entries in the inherited Gamma manifest were independently checked against the retained original Gamma files. The inherited 17-method executions are historical in this intake; their earlier root replay remains a separate record.

The fresh receipts are `work/arithmetic_endpoint_intake_replay_20260913/REPLAY.json` and `HTML_MATH_AUDIT.json`. The replay helper is `work/replay_arithmetic_endpoint_intake_20260913.py`. Its first byte-parser attempt stopped before all executions on the standard no-final-newline marker; the parser was repaired to retain that marker's exact semantics, after which all byte comparisons and all four executions completed. No delivered code or proof was repaired to obtain passing tests.

43.1.2 2. Original measure and the two exact quotient maps

Let

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s), \quad h(s) = \prod_{\rho \in Z} (s-\rho)^{m_\rho}, \quad d = \deg h.$$

The packet is fixed and finite, and each selected multiplicity is the actual full order of the zero of the entire nonzero function g . Hence the unique holomorphic continuation v_h defined by $g = hv_h$ is entire and nonzero. Define the original density and mass

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k.$$

No symmetry of the selected packet is required for the analytic norm theorem. Put $c = k/2$. The source norm of $P \in \mathbb{C}[S]$ is

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du.$$

It is positive for every nonzero polynomial: the density is positive almost everywhere, and a nonzero polynomial has only finitely many real-line zeros. All moments exist by the exponential estimate proved below. Finite-dimensional projection therefore gives a unique monic minimizer p_n , with positive squared norm $\omega_{h,k,n}$, for each $n \geq 0$.

Writing $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$ and $\mathcal{P}_j = 0$ for $j < 0$, the leading coefficient map gives

$$0 \longrightarrow \mathcal{P}_{n-1} \longrightarrow \mathcal{P}_n \xrightarrow{\text{lc}_n} \mathbb{C} \longrightarrow 0, \quad n \geq 0.$$

For monic χ of degree $q \geq 1$, Euclidean division proves, for every $n \geq q-1$,

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times \chi} \mathcal{P}_n \xrightarrow{\pi_\chi} \mathbb{C}[S]/(\chi) \longrightarrow 0.$$

The first arrow is injective because $\mathbb{C}[S]$ is a domain; its image consists exactly of the degree- $\leq n$ multiples of χ . The remainder has degree $< q \leq n+1$, proving surjectivity of the last arrow at the stated endpoint. For $n \geq q$, multiplication by χ induces the exact isomorphism

$$\overline{\times \chi} : \mathcal{P}_{n-q}/\mathcal{P}_{n-q-1} \xrightarrow{\sim} \mathcal{P}_n/\mathcal{P}_{n-1}, \quad \text{lc}_n(\chi Q) = \text{lc}_{n-q} Q.$$

Required typing correction to the sentence after source (13): this graded isomorphism requires $n \geq q$. At $n = q-1$ its proposed source is zero while the target leading line has dimension one. The exact sequence at $n = q-1$ remains valid, with zero relation source and $\mathcal{P}_{q-1} \cong \mathbb{C}[S]/(\chi)$. Thus there is an exact endpoint description, not an identification of unequal spaces.

In each original admitted support fibre λ , the lifted map is $(\lambda, P) \mapsto (\lambda, [P]_\chi)$, and it sends τ to τ . Its kernel inside that fibre is precisely the polynomial relation space above, and a relation maps to $(\lambda, 0)$. The leading coefficient square commutes because multiplication by the monic polynomial preserves the literal coefficient. The original arithmetic map remains

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h).$$

These are the delivered original theta-source identities. The present norm proof uses their explicitly specified norm integral and does not replace the Taylor unit or the target of either map.

43.1.3 3. Full local mass argument and all compact cases

For $|T| \geq 10$ let $f_T(z) = \zeta(1/2 + iT + iz)$. The Joukowski image of $|w| = 8$ is the ellipse with semiaxes $a = 65/16$, $b_0 = 63/16$. On and inside it, the image coordinate s satisfies

$$-55/16 \leq \Re s \leq 71/16, \quad |\Im s - T| \leq 65/16, \quad |\Im s| \geq 95/16 > 0.$$

The pole at $s = 1$ is consequently outside an open neighbourhood of the filled ellipse. For $\sigma = \Re s \geq 1/4$, summing $\lfloor x \rfloor$ on $\sigma > 1$, then using locally uniform absolute convergence on $\sigma > 0$, gives

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx, \quad \left| \int_1^\infty \{x\} x^{-s-1} dx \right| \leq \frac{1}{\sigma} \leq 4.$$

The denominator $|s-1|$ stays bounded away from zero on this set and $|s| = O(1+|T|)$, so this part has a uniform $O(1+|T|)$ bound. For $\sigma \leq 1/4$, the functional equation and vertical-strip Gamma bound give

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s),$$

$$|\sin(\pi s/2) \Gamma(1-s)| \leq C(1+|\Im s|)^{1/2-\sigma}.$$

Here $\Re(1-s) \in [3/4, 71/16]$, so the preceding integral estimate bounds $\zeta(1-s)$ by $C'(1+|T|)$. The power is at most $3/2 + 55/16 = 79/16 < 5$. All factors $2^\sigma \pi^{\sigma-1}$ have fixed compact real range. Consequently one finite $C_0 \geq 1$ gives

$$|f_T(z)| \leq M_T = C_0(1+|T|)^6$$

on the full ellipse. The open neighbourhood noted above validates Cauchy's formula on each closed unit disc centred in $[-1, 1]$: the discs lie in $|\Re z| \leq 2$, $|\Im z| \leq 1$, and $4/a^2 + 1/b_0^2 < 1$ (both semiaxes exceed three). Hence $\sup_{[-1,1]} |f'_T| \leq M_T$.

Let $r_* = (3 + \sqrt{13})/2$, so $r_* - r_*^{-1} = 3$, $r_* < 4$, and $z_* = (-ir_* + (-ir_*)^{-1})/2 = -3i/2$. The absolutely convergent Möbius series at real part two gives

$$|\zeta(2+iT)| \geq c_* := \zeta(2)^{-1} > 0.$$

The holomorphic function $F_T(w) = f_T((w+w^{-1})/2)$ on $1 < |w| < 8$ has inner-boundary bound $m_T = \max_{[-1,1]} |f_T| > 0$ and outer-boundary bound M_T . Subtract from $\log |F_T|$ the harmonic function linear in $\log |w|$ with those boundary values. The subharmonic maximum principle, including any zeros of F_T , gives

$$c_* \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}.$$

Therefore $m_T \geq c_*^{1/\vartheta} M_T^{-(1/\vartheta-1)} \geq c_*^3 M_T^{-2}$, since $0 < c_* < 1$, $M_T \geq 1$, $1/\vartheta < 3$. A maximizing point of the interval has a direction into the interval with at least one unit of available length. Since $m_T \leq M_T$, the length $\ell = m_T/(2M_T) \leq 1/2$ fits in that direction. The derivative bound implies $|f_T| \geq m_T/2$ on it. Thus

$$I(T) := \int_{T-1}^{T+1} |\zeta(1/2+it)|^2 dt \geq \ell(m_T/2)^2 = \frac{m_T^3}{8M_T} \geq \frac{c_*^9}{8C_0^7} (1+|T|)^{-42}.$$

On $[-10, 10]$, write $I(T) = \int_{-1}^1 |\zeta(1/2+i(T+x))|^2 dx$. Continuity and uniform boundedness on a compact set give continuity in T . If any such integral were zero, its continuous nonnegative integrand would vanish throughout a whole interval, and the identity theorem on the pole-free neighbourhood of the critical line would force ζ identically zero, contradicting $\zeta(2) > 0$. Hence $I_{\min} = \min_{[-10,10]} I(T) > 0$. The literal choice

$$c_\zeta = \min\{I_{\min}, c_*^9/(8C_0^7)\} > 0$$

proves source (23) for every real T , including the transition endpoints.

43.1.4 4. Carry every original factor to the convolution lower bound

Stirling's formula on the vertical line and the fact that Gamma has neither zeros nor poles at $1/4 + it/2$ show that

$$c_\Gamma := \inf_{t \in \mathbb{R}} |\Gamma(1/4 + it/2)|^2 (1+|t|)^{1/2} e^{\pi|t|/2} > 0.$$

Indeed the expression has a positive finite limit as $|t| \rightarrow \infty$; on the intervening compact set it is continuous and strictly positive. This supplies the stated global Gamma lower bound, including $t = 0$.

Let $H_h = \prod_\rho (1 + |\rho - 1/2|)^{m_\rho}$. The triangle inequality gives $|h(1/2 + it)| \leq H_h(1 + |t|)^d$. Applying this to the entire identity $g = hv_h$ gives $|v_h|^2 \geq |g|^2/[H_h^2(1 + |t|)^{2d}]$ even at a selected centre: the denominator used is strictly positive and is never the zero value of h . On the line, $|s(s-1)|^2 = (t^2 + 1/4)^2$ and $|\pi^{-s/2}|^2 = \pi^{-1/2}$. Moreover

$$\frac{(t^2 + 1/4)^2}{(1 + |t|)^{1/2}} \geq \frac{1}{16\sqrt{2}}.$$

For $|t| \leq 1$ this follows by numerator $\geq 1/16$ and denominator $\leq \sqrt{2}$; for $|t| \geq 1$ it follows from $|t|^4/\sqrt{2|t|} \geq 1/\sqrt{2}$. Thus a literal permissible source constant is

$$c_h^{(0)} = \frac{c_\Gamma}{32\sqrt{2}\pi^{3/2}H_h^2}, \quad \alpha_0 = \pi/2,$$

and

$$w_h(t) \geq c_h^{(0)} e^{-\alpha_0|t|} (1 + |t|)^{-2d} |\zeta(1/2 + it)|^2.$$

When $|t - T| \leq 1$, $|t| \leq |T| + 1$ and $1 + |t| \leq 2(1 + |T|)$. Set

$$B_h = 42 + 2d, \quad c_h^{(1)} = c_h^{(0)} e^{-\alpha_0} 2^{-2d} c_\zeta > 0.$$

Integrating proves the exact lower local mass

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\alpha_0|T|} (1 + |T|)^{-B_h}.$$

For completeness, the analytic upper properties used next also follow from the actual density. For sufficiently large $|t|$, each factor $|1/2 + it - \rho| \geq |t|/2$. The upper Gamma estimate, the real-part-1/2 integral estimate for zeta above, and this lower bound for h give an exponential $e^{-\alpha_0|t|}$ times a fixed polynomial upper bound for w_h . On compact sets its entire definition is continuous and bounded. It is therefore bounded and integrable and has finite Laplace values for $|b| < \alpha_0$. Its zero set on the real line is discrete, because v_h is nonzero entire, so $w_h > 0$ almost everywhere.

For each real u , the product $w_h(t)w_h(u - t)$ is positive almost everywhere, integrable, and has a positive integral. In addition

$$\|(w_h * w_h)(\cdot + a) - (w_h * w_h)\|_\infty \leq \|w_h\|_\infty \|w_h(\cdot + a) - w_h\|_1 \rightarrow 0.$$

Hence $m_{h,2}$ is continuous and strictly positive. The quantities

$$b_h = \min_{|v| \leq 1} m_{h,2}(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0, \quad c_h = b_h c_h^{(1)} > 0$$

are actual compact observations of the original measure. Restricting the positive convolution integral gives

$$m_{h,3}(u) = \int w_h(t) m_{h,2}(u - t) dt \geq b_h \int_{u-1}^{u+1} w_h(t) dt \geq c_h e^{-\alpha_0|u|} (1 + |u|)^{-B_h}.$$

For $k > 3$, the exact convolution integral is over $v_4, \dots, v_k \in \mathbb{R}$, with first factor $m_{h,3}(u - \sum v_a)$. Tonelli applies. Restrict it to $[-1, 1]^{k-3}$, where $|\sum v_a| \leq k - 3$, and integrate each remaining factor. This proves, including $k = 3$ with an empty product,

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(|u|+k-3)} (1 + |u| + k - 3)^{-B_h}.$$

Every factor in the omitted complement remains in the original nonnegative integral. There is no conclusion at $k = 1, 2$ by substituting a negative number of factors. The density still has mass μ_h^k .

43.1.5 5. Exact monic norm and literal uniform constant

Fix $0 < b < \alpha_0$, and define $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$. Tonelli and the definition of convolution give

$$\int e^{\pm bu} m_{h,k}(u) du = M_h(\pm b)^k, \quad \int e^{b|u|} m_{h,k}(u) du \leq M_h(b)^k + M_h(-b)^k.$$

The nonnegative exponential series implies $|u|^{2j} \leq (2j)! b^{-2j} e^{b|u|}$. The literal monic trial polynomial $(S - c)^j$ becomes $i^j u^j$ on the original integration line. Consequently

$$\omega_{h,k,j} \leq (2j)! b^{-2j} [M_h(b)^k + M_h(-b)^k], \quad j \geq 0.$$

The Legendre constant can be derived without dropping its leading coefficient. Let

$$P_j(x) = \frac{1}{2^j j!} \frac{d^j}{dx^j} (x^2 - 1)^j, \quad \ell_j = \frac{(2j)!}{2^j (j!)^2}.$$

Integration by parts j times proves its orthogonality against every polynomial of degree below j , because the boundary terms vanish to the required orders. The same integration gives

$$\int_{-1}^1 P_j(x)^2 dx = \frac{\ell_j}{2^j} \int_{-1}^1 (1 - x^2)^j dx = \frac{\ell_j}{2^j} \frac{2^{2j+1} (j!)^2}{(2j+1)!} = \frac{2}{2j+1}.$$

The beta integral in the middle equality follows by $x = 2y - 1$ and integration by parts for $\int_0^1 y^j (1 - y)^j dy = (j!)^2 / (2j+1)!$. Thus the monic polynomial $L^j \ell_j^{-1} P_j(u/L)$ has squared norm

$$\mathfrak{l}_j(L) = \frac{2L^{2j+1}}{2j+1} \left(\frac{2^j (j!)^2}{(2j)!} \right)^2.$$

Any other polynomial with the same leading coefficient differs by a lower-degree polynomial. Orthogonality gives an exact sum of squared norms, proving minimality and uniqueness. A monic S -polynomial maps to a u -polynomial with leading coefficient i^j . The explicit inverse image of the interval minimizer is

$$\frac{i^j L^j}{\ell_j} P_j \left(\frac{S - c}{iL} \right),$$

whose S^j coefficient is $i^j i^{-j} = 1$. This proves the claimed lower bound at the original coordinate type. Multiplication by the inside and outside characteristic functions is the exact isometry

$$L^2(m_{h,k} du) \longrightarrow L^2([-L, L], m_{h,k} du) \oplus L^2(\mathbb{R} \setminus [-L, L], m_{h,k} du),$$

with inverse addition after extension by zero. Applying the lower envelope to its first component gives

$$\omega_{h,k,j} \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(L+k-3)} (1 + L + k - 3)^{-B_h} \mathfrak{l}_j(L), \quad k \geq 3, j \geq 0, L > 0.$$

Taking $j = n$, $L = n$ here and $j = 2n$ above gives exactly source (35). To verify the constant in source (36), put

$$M_* = \max\{1, M_h(b), M_h(-b)\}, \quad C_c = \max(1, c_h^{-1}), \quad C_\vartheta = \max(1, \vartheta_h^{-1}).$$

The central binomial estimate $\binom{2n}{n} \leq 4^n$ gives $\mathfrak{l}_n(n) \geq 2n^{2n+1} / ((2n+1)4^n)$. Combining the factor 2 from the two Laplace values with this prefactor, before taking roots, yields

$$\left(\frac{2(4n)!}{\mathfrak{l}_n(n)} \right)^{1/(2n)} \leq 32n(2 + 1/n)^{1/(2n)} \leq 64n.$$

The first inequality uses $(4n)! \leq (4n)^{4n}$. For the second, $2 + 1/n \leq 4^n$ for every integer $n \geq 1$. For $n \geq k \geq 3$, the remaining factors satisfy

$$M_*^{k/(2n)} \leq \sqrt{M_*}, \quad c_h^{-1/(2n)} \leq \sqrt{C_c}, \quad \vartheta_h^{-(k-3)/(2n)} \leq \sqrt{C_\vartheta},$$

$$e^{\alpha_0(n+k-3)/(2n)} \leq e^{\alpha_0}, \quad (1+n+k-3)^{B_h/(2n)} \leq (2n)^{B_h/(2n)} \leq 2^{B_h/2} e^{B_h/(2e)}.$$

The final inequality follows by differentiating $\log x/x$, whose maximum on $x > 0$ is $1/e$. The factor b^{-4n} becomes exactly b^{-2} . Multiplying proves the literal sufficient constant

$$C_h = 64b^{-2} e^{\alpha_0} 2^{B_h/2} e^{B_h/(2e)} \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}$$

and the theorem

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n, \quad n \geq k \geq 3.$$

The packet and the chosen b determine a finite constant; neither k nor n enters it. The compact minima are proved positive, and they have not been numerically enclosed. The same argument gives source (37) for integers $k \geq 3, n \geq 0, r \geq 1$ and $L > 0$, with numerator degree $n+r$, exponent $1/(2r)$, and all factors exactly as printed. An endpoint selection additionally requires $n \geq q$.

43.1.6 6. Exact composition with the delivered endpoint theorem

The complete immutable PR23 note was read at commit c720f40530eed2f5969dbabe94dfd3fddc0f507f, from work/endpoint_criterion_pr23_c720f405_20260913.md. Only its endpoint theorem is used here. Its original local source estimate is

$$0 \leq \epsilon_N \leq A_N \sinh x_N, \quad A_N = \sqrt{\omega_{N+1}/\omega_N} > 0, \quad x_N = \frac{1}{2} \log(V_{N-1}/V_{N+1}) \geq 0.$$

For a window $N = n, \dots, n+r-1, n \geq q, r \geq 1$, if one $x_N = 0$, the corresponding nonnegative allowance is zero. Otherwise $(\log \sinh x)'' = -1/\sinh^2 x < 0$, so finite Jensen gives

$$\prod \sinh x_N \leq \sinh \left(\frac{1}{r} \sum x_N \right)^r.$$

The minimum allowance raised to r is bounded by the product of the allowances. Both exact products telescope:

$$\prod A_N = \sqrt{\omega_{n+r}/\omega_n}, \quad 2 \sum x_N = \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}.$$

Taking nonnegative roots proves source (38). The finite minimum is attained inside the existing family. Choosing $r = n \geq \max(q, k), k \geq 3$, and using the just-proved norm theorem yields source (39) with its original C_h .

For the packet consisting exactly of a nonreal off-line quartet with common actual multiplicity m , put $\ell = 1 + k(m-1)$. The inherited aggregate exterior result supplies

$$q_k = \ell(k+1)^2, \quad L_{h,k} = 2\delta\ell(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}$$

at every admitted degree. The elementary finite sum behind the count is $\sum_{a > k/2} (2a-k) = \lfloor (k+1)^2/4 \rfloor$. If k is even, $4 \lfloor (k+1)^2/4 \rfloor = k(k+2) \geq k(k+1)$; if k is odd, it equals $(k+1)^2 \geq k(k+1)$. Hence $q_k \geq k$ and $L_{h,k}/q_k \geq \delta k/2$. Applying (39) at $n = q_k$ gives

$$\mathcal{B}_{h,k} := \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}} \geq 2q_k \operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right).$$

For $a > 0$, $\operatorname{arsinh}(ak) = \log k + \log(a + \sqrt{a^2 + k^{-2}})$; its second summand tends to $\log(2a)$. Dividing proves the source's threshold $\liminf \mathcal{B}_{h,k}/(q_k \log k) \geq 2$. This implication uses the existing exterior lower theorem; it does not assert the existence of such a quartet. Later exact-radius improvements belong to the parallel product lane and are not asserted to have been proved by the present delivered source.

43.1.7 7. Complete block boundary isomorphism and determinant transport

Here $q \geq 1$, $i < j$, and $i \geq q - 1$. In the polynomial source let $\mathcal{D}_a = \ker(\pi_\chi|_{\mathcal{P}_a})$. In the actual Hilbert source these are their images under the injective source map. Let $R_i : E \rightarrow \mathcal{P}_i$ be the unique minimum norm section of π_χ , and $G_i = R_i^* R_i$, with all adjoints taken in the stated original source metric and the fixed remainder coordinates. Let

$$Tz = \sum_{a=i+1}^j z_a p_a, \quad F = \pi_\chi T = [b_{i+1}, \dots, b_j], \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j).$$

The original orthogonality gives $T^*T = \Omega$, $T^*R_i = 0$, and $T^*\mathcal{D}_i = 0$. Since $R_i \perp \mathcal{D}_i$, the map $B = T - R_i F$ takes values in $\mathcal{D}_j \cap \mathcal{D}_i^\perp$. It has zero remainder because $\pi_\chi R_i = I$. Its coefficients along the basis $p_{i+1}, \dots, p_j$ are exactly z , so it is injective. For any d in that intersection, choose z to be these upper coefficients of d . Then $d - Bz \in \mathcal{P}_i$, has zero remainder, and is perpendicular to $\mathcal{D}_i$. Thus $d - Bz = 0$. This proves surjectivity and its explicit inverse.

The exact Gram and pairing identities are

$$B^*B = H := \Omega + F^*G_iF > 0, \quad B^*R_i = -F^*G_i.$$

The section

$$\tilde{R} = R_i + BH^{-1}F^*G_i$$

has remainder identity, is perpendicular to $\mathcal{D}_i$, and satisfies $B^*\tilde{R} = -F^*G_i + HH^{-1}F^*G_i = 0$. Since $\mathcal{D}_j = \mathcal{D}_i \oplus \text{im } B$ is the proved orthogonal decomposition, $\tilde{R} = R_j$. Expanding its Gram gives

$$G_j = G_i - G_iFH^{-1}F^*G_i.$$

An alternative direct kernel derivation retains the same maps: in the orthogonal monic basis, $K_i = G_i^{-1} = \sum_{a=0}^i b_a b_a^* / \omega_a$, hence

$$K_j = K_i + F\Omega^{-1}F^*, \quad \frac{V_i}{V_j} = \det(I + \Omega^{-1}F^*G_iF).$$

The determinant identity follows from $V_i/V_j = \det K_j / \det K_i$ and $\det(I + AB) = \det(I + BA)$. To specify its positive spectral type, $\Omega^{-1}F^*G_iF$ is conjugate by $\Omega^{1/2}$ to the Hermitian positive semidefinite matrix $\Omega^{-1/2}F^*G_iF\Omega^{-1/2}$. Its eigenvalues are precisely the generalized eigenvalues of the pair (F^*G_iF, Ω) . Therefore the determinant equals $\prod_a (1 + \lambda_a) > 0$. Applying it to $(i, j) = (q - 1, 2q - 1)$ and $(q, 2q)$, then taking real logarithms, proves source (47). Integrating $(1 + x)^{-1} \leq 1$ from 0 to λ_a gives $\log(1 + \lambda_a) \leq \lambda_a$, proving (48) with its two separately typed matrices.

In original theta source coordinates, each column of B is the source image of a uniquely divisible polynomial χQ . Thus its original cochain primitive is the inherited fixed-order division primitive for that Q , and linear combination gives a primitive for every block column. The exact transport is

$$\mathbb{C}^{j-i} \xrightarrow{B} \mathcal{D}_j / \mathcal{D}_i \hookrightarrow \mathcal{B}^{\hat{\otimes} k} / \mathcal{D}_i \longrightarrow \mathcal{B}^{\hat{\otimes} k} / \mathcal{D}_j.$$

The final composite vanishes because the representative lies in $\mathcal{D}_j$. Its marked lift remains the receiving fibre's supported zero. The direct-sum proof and the displayed Gram keep all source norms and pairings before this projection.

43.1.8 8. The confluent formula's exact domain and full-order bridge

The complete immutable PR24 note was fetched from commit dfcbb55cbf7fec8c9301fe242c13e741d762013e into work/pr24_confluent_transfer_dfcbb55_20260913.md (14007 bytes, SHA d1046b75762e4248ded8f0b901c1ac12dc01065e4e684ca2d166b6a535425763) and read in full. Its source determinant identity, source (41), gives source (42) simply by substitution of $N = q - 1, q, 2q - 1, 2q$, for which $a = N - q + 1$ is exactly $0, 1, q, q + 1$. Each numerator and denominator retains its product of q original monic norms. The common raw Vandermonde cancels in this quotient because it occurs twice on each side and is nonzero.

There is a domain precision needed when applying that raw-jet presentation to arbitrary finite packets. Define the exact ring map $\Psi P = P(c + iu)$, and put $\psi = i^{-q}\Psi\chi$, which is monic. The norm multiplier is always

$$\Pi(u) = \overline{\chi}(c - iu)\chi(c + iu) = \overline{\psi}(u)\psi(u),$$

of degree $2q$, real and nonnegative on the real line. The formula $\Pi = \psi^2$ with every node length doubled requires ψ real, equivalently invariance of the original root multiset under $S \mapsto k - \overline{S}$. The exact quartet satisfies this. For an arbitrary packet, take the union of the roots of ψ and their conjugates, adding their orders whenever nodes coincide. These are the actual full orders of Π . The raw jet map then has kernel exactly (Π) by polynomial divisibility and has dimension $2q$. Its Vandermonde is

$$\det V = \left(\prod_a \prod_{d=0}^{r_a-1} d! \right) \prod_{a < b} (z_b - z_a)^{r_a r_b}$$

in the fixed displayed node and derivative order. The coefficient d in multiplication-by- u on raw jets remains $(Xv)_{a,d} = z_a v_{a,d} + dv_{a,d-1}$.

The transfer proof itself applies to this general Π : if a linear combination of $Q_a, \dots, Q_{a+2q-1}$ has all these raw jets zero, it equals ΠH with $\deg H < a$. Orthogonality gives $0 = \int \Pi |H|^2 m_{h,k}$, which forces $H = 0$. Thus the actual jet matrix F_a is invertible. Dividing the unique raw-jet cancellation of Q_{a+2q} by Π gives the monic relation-weight polynomial. Its norm is $-d_{a,0}\omega_a > 0$, and shifting the $2q$ source columns gives $\det F_{a+1} / \det F_a = -d_{a,0}$, with sign $(-1)^{2q-1} = -1$. Multiplication yields the relation/source determinant ratio; the original Schur determinant then gives (41). Therefore the analytic theorem remains valid for all fixed finite packets, and the literal doubled-node description is applied only on its proven symmetry domain. The parallel endpoint-product proof is expanding this transfer in its complete original coordinate dictionary.

43.1.9 9. Source-review claims and research integration status

The delivered source review records selected readings from a six-part S20 archive, its 216580466-byte concatenation and hash, and the discrepancy between an old TeX <=2 and a current page record <2 in Deligne 3.2.10, together with an explicit squared-eigenvalue correction. That external archive and its selected passages are **not included** among these 32 members. This intake has verified the complete delivered report and its manifest bytes; it does not independently reassert reassembly or inspection of files that it has not opened. The arithmetic estimate proved here uses the stated classical zeta/Gamma/Legendre facts and no finite-field purity theorem.

The resulting deliverable establishes a full written norm theorem at the original measure and a finite exact block-volume comparison. Its only newly identified literal source correction is the endpoint domain of the graded multiplication isomorphism; the raw-confluent symmetry domain is made explicit when applying the cited upstream presentation beyond quartets. Neither correction changes the dyadic norm proof or its constant. The theorem composes with the existing finite endpoint criterion exactly as shown. The growing-degree upper estimate for the quotient volumes is not a theorem of this delivery. The independently developing exact-radius and balanced-window improvements are separate additions; they should receive their own proof bodies and provenance before the next cumulative publication.

Classical primary references for the exact functions used above are [DLMF 25.4, Reflection Formulas](#), [DLMF 5.11, Asymptotic Expansions](#), and [DLMF 18.5, Explicit Representations](#). The independent analytic review supplies its actual source-access record and a second full derivation.

44 The original leading-coefficient line and its arithmetic relation image

This proof supplies the exact endpoint domain and metric of the multiplication map in the paragraph following equation (13) of the delivered *Arithmetic Endpoint Bounds*. The arithmetic monic-norm theorem retains all of its original constants and its domain $n \geq k \geq 3$.

Fix a finite packet h , a positive integer k , and the original source map

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad D = -x\partial_x, \quad \mathcal{M}F_h = (2\xi)/h.$$

Its original Hermitian form, conjugate-linear in the first argument, is

$$\langle P, Q \rangle_{h,k} = \int_{\mathbb{R}} \overline{P(k/2 + iu)} Q(k/2 + iu) m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}, \quad w_h(t) = \frac{|((2\xi)/h)(1/2 + it)|^2}{2\pi}. \quad (\text{AL.1})$$

The actual mass is $\int m_{h,k} = \mu_h^k > 0$. The exponential upper bound proved in the delivered arithmetic note makes every polynomial norm finite. The density is positive almost everywhere; a nonzero polynomial has finitely many zeros on the integration line. Thus (AL.1) is positive definite on every finite polynomial space.

Let $\mathcal{P}_a = \mathbb{C}[S]_{\leq a}$ for $a \geq 0$ and $\mathcal{P}_a = 0$ for $a < 0$. For $a \geq 0$ put $\mathcal{L}_a = \mathcal{P}_a/\mathcal{P}_{a-1}$ with its original quotient norm. The leading-coefficient map is denoted lc_a . Let p_a be the unique monic minimizer in (AL.1), and put $\omega_a = \|p_a\|_{h,k}^2 > 0$. Its existence follows by projecting S^a onto the finite-dimensional subspace $\mathcal{P}_{a-1}$, and its uniqueness follows from positivity. The defining orthogonality is

$$p_a \perp \mathcal{P}_{a-1}, \quad \|z[p_a]\|_{\mathcal{L}_a}^2 = |z|^2 \omega_a, \quad z \in \mathbb{C}. \quad (\text{AL.2})$$

Here $[p_a] = [S^a]$; this equality is in the degree quotient and does not identify their polynomial representatives in the original source.

Fix the original monic arithmetic relation polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$ and the original remainder algebra $E = \mathbb{C}[S]/(\chi)$. Write π_χ for polynomial reduction.

Proposition 44.1 (Exact multiplication and leading-coefficient maps). *For $n \geq q - 1$ the following sequence is exact:*

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{AL.3})$$

For $n \geq q$ there is a commuting diagram of exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{P}_{n-q-1} & \longrightarrow & \mathcal{P}_{n-q} & \xrightarrow{\text{lc}_{n-q}} & \mathbb{C} \longrightarrow 0 \\ & & \downarrow \times\chi & & \downarrow \times\chi & & \downarrow \text{id} \\ 0 & \longrightarrow & \mathcal{P}_{n-1} & \longrightarrow & \mathcal{P}_n & \xrightarrow{\text{lc}_n} & \mathbb{C} \longrightarrow 0. \end{array} \quad (\text{AL.4})$$

It induces the coefficient-preserving isomorphism

$$\overline{m}_\chi : \mathcal{L}_{n-q} \xrightarrow{\sim} \mathcal{L}_n, \quad [Q] \longmapsto [\chi Q]. \quad (\text{AL.5})$$

At $n = q - 1$ the relation source in (AL.3) is zero and $\pi_\chi : \mathcal{P}_{q-1} \rightarrow E$ is an isomorphism. The degree map from that relation source to $\mathcal{L}_{q-1}$ is the zero map from 0 to a one-dimensional space.

Proof. Multiplication by a nonzero polynomial in the integral domain $\mathbb{C}[S]$ is injective. Its degree formula is $\deg(\chi Q) = q + \deg Q$ for $Q \neq 0$, so its image in $\mathcal{P}_n$ consists exactly of the multiples of χ in that space. Euclidean division by the monic polynomial χ gives a unique remainder of degree below q . That remainder belongs to $\mathcal{P}_n$ precisely throughout the admitted range $n \geq q - 1$. This proves (AL.3), including its endpoint.

For $n \geq q$, both displayed leading-coefficient rows are exact by their definitions, and the degree formula shows that the two multiplication maps have the indicated targets. Since χ is monic,

$$\text{lc}_n(\chi Q) = \text{lc}_{n-q}(Q), \quad Q \in \mathcal{P}_{n-q}. \quad (\text{AL.6})$$

This proves commutativity of the right square; the inclusion square commutes because both composites are literal multiplication by χ . Lower-degree inputs map to $\mathcal{P}_{n-1}$, so (AL.5) is well-defined.

The two leading-coefficient identifications with $\mathbb{C}$ take it to the identity in (AL.4), proving that it is an isomorphism and specifying its inverse: $z[S^n] \mapsto z[S^{n-q}]$.

At $n = q - 1$, $\mathcal{P}_{-1} = \mathcal{P}_{-2} = 0$. The exact endpoint diagram instead is

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow 0 & & \\ 0 & \longrightarrow & \mathcal{P}_{q-2} & \longrightarrow & \mathcal{P}_{q-1} & \xrightarrow{\text{lc}_{q-1}} & \mathbb{C} & \longrightarrow & 0. \end{array} \quad (\text{AL.7})$$

Every square commutes, and the degree target has dimension one even when $q = 1$, for which it is the constant line $\mathcal{P}_0/0$. This proves the claimed endpoint map and the exact domain restriction $n \geq q$ for (AL.5). $\square$

Proposition 44.2 (Exact source metric and arithmetic image). *For $n \geq q$, the norm, adjoint, and original polynomial representative of (AL.5) are given by*

$$\|\overline{m}_\chi\|^2 = \frac{\omega_n}{\omega_{n-q}}, \quad \overline{m}_\chi^*[p_n] = \frac{\omega_n}{\omega_{n-q}}[p_{n-q}], \quad \overline{m}_\chi[p_{n-q}] = [p_n]. \quad (\text{AL.8})$$

With the actual source correction

$$r_n = \chi p_{n-q} - p_n \in \mathcal{P}_{n-1}, \quad (\text{AL.9})$$

one has the exact identities

$$\|\mathcal{V}_{h,k}(\chi p_{n-q})\|^2 = \omega_n + \|\mathcal{V}_{h,k}r_n\|^2, \quad \pi_\chi(r_n) = -\pi_\chi(p_n), \quad \pi_\chi(\chi p_{n-q}) = 0. \quad (\text{AL.10})$$

The unquotiented leading coefficient of every relation is transported by (AL.6), and its arithmetic image is the original supported zero.

Proof. Both χp_{n-q} and p_n are monic of degree n . Their difference therefore has degree at most $n - 1$, proving (AL.9), and their classes in $\mathcal{L}_n$ agree. Formula (AL.2) gives

$$\frac{\|\overline{m}_\chi(z[p_{n-q}])\|^2}{\|z[p_{n-q}]\|^2} = \frac{|z|^2 \omega_n}{|z|^2 \omega_{n-q}} = \frac{\omega_n}{\omega_{n-q}} \quad (z \neq 0).$$

This proves the operator norm in (AL.8). The adjoint identity follows by testing the defining equation against an arbitrary scalar multiple of $[p_{n-q}]$ and using the positive real quotient weights in (AL.2). Orthogonality $p_n \perp r_n$ gives the first identity of (AL.10) with the literal original source norms. Reducing (AL.9) modulo χ gives its other two identities.

To state the supported image with its complete type, for each original admitted label λ and the fixed admitted degree n , define the map on $\{\tau\} \sqcup \{(\lambda, P) : P \in \mathcal{P}_n\}$ with target $\{\tau\} \sqcup \{(\lambda, x) : x \in E\}$ by

$$\tilde{\pi}_\chi(\lambda, P) = (\lambda, \pi_\chi P), \quad \tilde{\pi}_\chi(\tau) = \tau.$$

It sends a relation $(\lambda, \chi Q)$ to $(\lambda, 0_E)$, while external absence is fixed. The kernel inside this finite source fibre is exactly $\chi \mathcal{P}_{n-q}$, the image of multiplication in (AL.3). The same formula on the full fibre $\mathbb{C}[S]$ has kernel $\chi \mathbb{C}[S]$; its restriction along $\mathcal{P}_n \hookrightarrow \mathbb{C}[S]$ is the finite map just defined, because $(\chi \mathbb{C}[S]) \cap \mathcal{P}_n = \chi \mathcal{P}_{n-q}$ by the degree formula. The original arithmetic map satisfies

$$J^{(k)} \mathcal{V}_{h,k} = \eta_{h,k} \pi_\chi, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_k) 1, \quad v_h = j_h((2\xi)/h). \quad (\text{AL.11})$$

Composing (AL.3) with this displayed original map sends each relation to zero at that receiving fibre; every Taylor-unit coefficient stays in $\eta_{h,k}$. Before that projection, (AL.10) records its full source norm and the exact correction to the minimum-norm leading representative. $\square$

For the empty arithmetic packet $\chi = 1, q = 0$, the algebra E is zero. Multiplication is the identity $\mathcal{P}_n \rightarrow \mathcal{P}_n$ and the induced line map is the identity for every $n \geq 0$, with norm one; (AL.8)–(AL.10) reduce to $\omega_n/\omega_n = 1$ and $r_n = 0$. Its analytic source and actual mass μ_h^k remain in (AL.1). This case uses no positive-rank quotient inverse.

The coefficient-preserving isomorphism, its original metric in (AL.8), the representative correction in (AL.10), and the endpoint map (AL.7) give the complete relation between the degree-line norm and its arithmetic relation image. All identities hold at the original coordinate $S = k/2 + iu$; no change to the norm estimate or the source generator is required.

45 Exact window products, sharp endpoint optimization, and original relation costs

This continuation proves the product refinement in the delivered four-endpoint calculation, then solves the associated scalar problem with only the combined logarithmic budget retained. It gives explicit maps from these quantities to the original source, relation and Schur determinants. The new arithmetic consequence uses the proved norm estimate of the accompanying *Arithmetic endpoint bounds* source. Its analytic proof remains an explicit dependency; the finite polynomial calculations below do not certify that analytic theorem or an upper estimate for the growing arithmetic relation volumes.

45.1 Original source and a proof of its exact radius identity

Fix the original monic packet h of actual selected zeros, with all their orders, of $g = 2\xi$, and put

$$v_h = g/h, \quad w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad m_{h,k} = w_h^{*k}, \quad c = k/2, \quad \int m_{h,k} = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{EP.1})$$

The complete source construction proves positivity almost everywhere and all polynomial moments, also after multiplication by $e^{\theta u}$ for real $|\theta| < \pi/2$. Retain that literal density and the pairing

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_{h,k}(u) du. \quad (\text{EP.2})$$

It is conjugate-linear in its first entry. Every nonzero polynomial has positive squared norm since its zero set on the line is finite. Let χ be the original degree- q monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$, where $S = \sum s_i$. In this section $q \geq 1$ and the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Reflection preserves this ideal and gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$ by monicity. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic.

Write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and $C = \mathbb{C}[S]/(\chi)$ in the coordinates $1, S, \dots, S^{q-1}$. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0 \quad (N \geq q-1). \quad (\text{EP.3})$$

Let H_N be the Gram in the stated monomial coordinates. The exact least-lift matrices, with their full types, are

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N, \quad V_N = \det G_N. \quad (\text{EP.4})$$

Surjectivity of J_N proves $K_N > 0$. If B_N is the multiplication matrix for χ , direct multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Every lift of x is $R_N x + B_N y$ and has squared norm $x^* G_N x + y^* B_N^* H_N B_N y$. This proves both the minimum property and $R_N^* H_N R_N = G_N$. Inclusions of the polynomial spaces therefore give $G_{N+1} \preceq G_N$ and $V_{N+1} \leq V_N$.

Let p_j be the monic orthogonal family in S , with squared norms $\omega_j > 0$, and let $b_j = [p_j]_X$. The monic triangular coefficient matrix conjugates H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, so

$$K_N = \sum_{j=0}^N b_j b_j^* / \omega_j. \quad (\text{EP.5})$$

For $N \geq q$ define $\delta_N = V_N / V_{N-1} \in (0, 1]$. The determinant identity $\det(I + XY) = \det(I + YX)$ follows by eliminating the two diagonal blocks of $\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ in the two orders. Applied to (EP.5), it gives

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - \delta_N, \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1. \quad (\text{EP.6})$$

Both follow respectively by subtracting the last term of K_N and adding the next term; all the earlier kernels inverted here are positive.

The exact algebra map

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q] = [Q((S - c)/i)] \quad (\text{EP.7})$$

has coefficient matrix E triangular with diagonal $1, i, \dots, i^{q-1}$. Thus $|\det E| = 1$. The quotient Gram becomes $E^{-*} G_N E^{-1}$ and its determinant stays V_N . Put $A = M_S : C \rightarrow C$ and $T = M_u$ in the real quotient, so $\Phi A \Phi^{-1} = cI + iT$ and $\sigma = \text{Tr } T = (\text{Tr } A - cq)/i \in \mathbb{R}$. The real monic polynomials $Q_j = i^{-j} p_j(c + iu)$ have norms ω_j . Pairing uQ_j with Q_l proves

$$uQ_j = Q_{j+1} + a_j^{(0)} Q_j + (\omega_j / \omega_{j-1}) Q_{j-1}, \quad a_j^{(0)} \in \mathbb{R}; \quad Sp_j = p_{j+1} + (c + ia_j^{(0)}) p_j - (\omega_j / \omega_{j-1}) p_{j-1}. \quad (\text{EP.8})$$

The last term is absent for $j = 0$. Indeed pairing with degrees below $j - 1$ vanishes by self-adjoint multiplication by u , the leading coefficient of Q_{j+1} is one, and pairing with Q_{j-1} gives the displayed norm ratio.

For completeness the phase has its original derivative interpretation. In the real coordinates let $d_j = [Q_j]$, and keep the transported matrices denoted temporarily by R, G . Differentiating the minimum norm shows $G' = R^* u R$: the two terms containing R' vanish because $J_N R' = 0$ and R is orthogonal to $\ker J_N$. The leading coefficient of Rx is $d_N^* Gx / \omega_N$. The polynomial

$$uRx - RTx - (Q_{N+1} - Rd_{N+1}) d_N^* Gx / \omega_N$$

is in $\ker J_N$: its leading degree cancels and its reduction is zero. Pair it with R and use $R^* Q_{N+1} = 0$. Taking the trace gives

$$\partial_\theta \log V_N = \sigma - d_N^* G d_{N+1} / \omega_N, \quad \phi_N := \sigma - \partial_\theta \log V_N, \quad b_N^* G_N b_{N+1} = i \omega_N \phi_N. \quad (\text{EP.9})$$

The factor i is $i^N i^{N+1}$; no phase has been removed. Differentiation is justified by the exponential moment domination of the original source.

Substitute (EP.8) into (EP.5). Terms with adjacent indices cancel because ω_j / ω_{j-1} is the exact recurrence coefficient; diagonal terms cancel against $2cK_N$. Thus

$$AK_N + K_N A^* - 2cK_N = (b_{N+1} b_N^* + b_N b_{N+1}^*) / \omega_N. \quad (\text{EP.10})$$

The relative Hermitian defect $\mathcal{H}_N = G_N^{-1/2} (A^* G_N + G_N A - 2cG_N) G_N^{-1/2}$ has rank at most two and trace zero. For $x = G_N^{1/2} b_N$, $y = G_N^{1/2} b_{N+1}$ its matrix is $(xy^* + yx^*) / \omega_N$. Expanding its square gives trace $2(\|x\|^2 \|y\|^2 - |x^* y|^2) / \omega_N^2$ because $x^* y = i \omega_N \phi_N$. Its nonzero eigenvalues, if present, are $\epsilon_N, -\epsilon_N$ with $\epsilon_N > 0$; at rank zero define $\epsilon_N = 0$. Consequently (EP.6) proves the full identity

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1). \quad (\text{EP.11})$$

For $q = 1$ the trace-zero defect is the zero map; the identity retains its possibly nonzero phase. We never form V_{q-2} .

45.2 Window product, endpoint extremum, and phases

Fix $n \geq q$, $r \geq 1$, $m = n + r$, and put

$$d_j = \delta_{n+j} \ (0 \leq j \leq r), \quad U = \omega_m / \omega_n, \quad \mathcal{A} = U^{1/(2r)}, \quad E_* = \min_{0 \leq j < r} \epsilon_{n+j}. \quad (\text{EP.12})$$

Multiplying every term of (EP.11), including both endpoint terms, gives the exact overlapping product

$$\mathcal{K} := \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = U \frac{V_n}{V_m} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (\text{EP.13})$$

Indeed the norm ratios telescope to U and $\prod_{j=1}^r d_j = V_m / V_n$. Each interior numerator occurs twice. As $E_*^{2r} \leq \prod \epsilon_{n+j}^2 \leq \mathcal{K}$ and $0 \leq 1 - d_j < 1$, this proves

$$E_* \leq (UV_n / V_m)^{1/(2r)}. \quad (\text{EP.14})$$

For $r \geq 2$ write $s = r - 1$ and retain all four endpoint ratios

$$\alpha = d_0 = V_n / V_{n-1}, \quad \beta = d_r = V_m / V_{m-1}, \quad t = \prod_{j=1}^s d_j = V_{m-1} / V_n. \quad (\text{EP.15})$$

For $0 < t < 1$, set $v_j = -\log d_j$. The function $f(v) = \log(1 - e^{-v})$ has $f''(v) = -e^{-v} / (1 - e^{-v})^2 < 0$ on $v > 0$. The tangent-line inequality $f(v) \leq f(\bar{v}) + f'(\bar{v})(v - \bar{v})$ follows by integrating the decreasing derivative on either side of $\bar{v}$. Summing at $\bar{v} = (-\log t) / s$ and exponentiating proves

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (\text{EP.16})$$

If one $d_j = 1$, the left side is zero; if $t = 1$, all are one and both sides are zero. For $t < 1$ with all $d_j < 1$, strict concavity shows equality exactly when all $d_j = t^{1/s}$. Insert this result in (EP.13) to obtain

$$E_* \leq \mathcal{U} := \mathcal{A} \left(\frac{(1 - \alpha)(1 - \beta)}{\beta t} \right)^{1/(2r)} (1 - t^{1/(r-1)})^{(r-1)/r}. \quad (\text{EP.17})$$

At $r = 1$ the separate, zero-safe formula is $E_* \leq \mathcal{A} \sqrt{(1 - \alpha)(\beta^{-1} - 1)}$. The four endpoint slots then overlap and no $t^{1/0}$ is defined.

This bound is attained in the relaxed scalar data class: choose the equal interior contractions and, when $\alpha, \beta, t < 1$, put $f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0$ and

$$\omega_{n+j+1} / \omega_{n+j} = \mathcal{U}^2 / f_j, \quad \epsilon_{n+j} = \mathcal{U}, \quad \phi_{n+j} = 0. \quad (\text{EP.18})$$

Their product is U by (EP.17), so both norm endpoints are respected and every radius identity holds. If an endpoint contraction is one, or $t = 1$, every compatible sequence has a zero radius at some step, hence $E_* = 0$ and the zero bound is attained. At $r = 1$ choose the sole norm ratio U . These scalar data are related to the arithmetic objects by the evaluation map (EP.4)–(EP.12); the proof of scalar attainment does not assert that this evaluation map is surjective.

For strictly interior contractions define the exact loss

$$\mathcal{J} = s \log(1 - t^{1/s}) - \sum_{j=1}^s \log(1 - d_j) \geq 0. \quad \mathcal{K} = \mathcal{U}^{2r} e^{-2\mathcal{J}}. \quad (\text{EP.19})$$

It also has the proved quantitative lower bound

$$\mathcal{J} \geq \frac{t}{2(1-t)^2} \sum_{j=1}^s \left(v_j + \frac{\log t}{s} \right)^2. \quad (\text{EP.20})$$

To prove this, all v_j and their mean lie in $(0, -\log t]$, where $-f''(v) \geq t/(1-t)^2$. Integrating the derivative twice gives the tangent inequality with the displayed quadratic remainder. The linear terms sum to zero. This proves (EP.20), including $s = 1$, where both sides vanish. If an interior $d_j = 1$ the exact product (EP.13) applies with value zero; no logarithm of zero is evaluated.

Define the actual phase polynomial

$$P(y) = \prod_{j=0}^{r-1} (y + \phi_{n+j}^2), \quad y \geq 0. \quad P(E_*^2) \leq \mathcal{K}. \quad (\text{EP.21})$$

It is continuous, tends to infinity, and is strictly increasing on $[0, \infty)$: for $y_2 > y_1 \geq 0$ every factor increases, and if a factor vanishes at y_1 , the product at y_2 is strictly positive. The actual product guarantees $\mathcal{K} \geq P(0)$. There is therefore exactly one $y_* \geq 0$ with $P(y_*) = \mathcal{K}$, and $E_* \leq \sqrt{y_*}$. If $\mathcal{K} = P(0)$, then $y_* = 0$. If an externally supplied number were smaller than $P(0)$, it would be incompatible with the displayed radius data; it would not define a nonnegative threshold. In particular at $\mathcal{K} = 0$ at least one phase vanishes and $y_* = 0$. Positive rational enclosures for the phases' squares and $\mathcal{K}$ lead to polynomial evaluation and rational bisection; positive powers and products suffice for each comparison. Using $\mathcal{U}^{2r}$ in place of $\mathcal{K}$ gives the valid endpoint-only version, also with its phase retained.

45.3 The sharp scalar bound with only the combined budget

Put

$$\mathcal{B} = \log \frac{V_{n-1}V_n}{V_{m-1}V_m} = -\log \left(d_0 d_r \prod_{j=1}^{r-1} d_j^2 \right) \geq 0. \quad (\text{EP.22})$$

The product in (EP.13), divided by U , is the function

$$F(d) = \frac{(1-d_0)(1-d_r) \prod_{j=1}^{r-1} (1-d_j)^2}{d_r \prod_{j=1}^{r-1} d_j}. \quad (\text{EP.23})$$

For $\mathcal{B} > 0$ its unique maximum over these scalar contractions with this budget is attained at

$$d_0 = \frac{1}{1+2x}, \quad d_r = 1-2x, \quad d_j = \frac{1-x}{1+x} \quad (1 \leq j < r), \quad (\text{EP.24})$$

where $x \in (0, 1/2)$ is the unique solution of

$$\mathcal{B} = 2 \operatorname{artanh}(2x) + 4(r-1) \operatorname{artanh} x. \quad (\text{EP.25})$$

Here $\operatorname{artanh} x = \frac{1}{2} \log((1+x)/(1-x))$. The exact maximum and consequent bound are

$$F_{\max} = \frac{4^r x^{2r}}{(1-4x^2)(1-x^2)^{r-1}}, \quad \boxed{E_* \leq \mathcal{U} \leq \mathcal{A} \frac{2x}{[(1-4x^2)(1-x^2)^{r-1}]^{1/(2r)}}}. \quad (\text{EP.26})$$

The middle expression for $r = 1$ uses the separate expression after (EP.17).

Here is a full extremization proof. Write $v_j = -\log d_j$. The feasible simplex is $v_0 + v_r + 2 \sum_{j=1}^{r-1} v_j = \mathcal{B}$, $v_j \geq 0$, a compact set. The product vanishes at every boundary point and is positive at interior points, so a maximum is interior. Its logarithm is

$$f(v_0) + f(v_r) + 2 \sum_{j=1}^{r-1} f(v_j) + v_r + \sum_{j=1}^{r-1} v_j. \quad (\text{EP.27})$$

Its Hessian is strictly negative diagonal, so it is strictly concave. The Lagrange multiplier equations are

$$\frac{1}{e^{v_0} - 1} = \lambda, \quad 1 + \frac{1}{e^{v_r} - 1} = \lambda, \quad 1 + \frac{2}{e^{v_j} - 1} = 2\lambda \quad (1 \leq j < r). \quad (\text{EP.28})$$

They imply $\lambda > 1$ and yield (EP.24) with $x = 1/(2\lambda)$. The right side of (EP.25) is strictly increasing from zero to infinity because its derivative is $4/(1 - 4x^2) + 4(r - 1)/(1 - x^2) > 0$. Thus it has exactly one solution and determines the unique maximum. Substitution into (EP.23) gives (EP.26). The value $\mathcal{B} = 0$ forces all $d_j = 1$, and all bounds are zero; it is represented by the continuous endpoint $x = 0$. The norm-ratio construction (EP.18), now using these contractions and $\mathcal{A}F_{\max}^{1/(2r)}$, proves sharpness in the same scalar class for this reduced data.

We prove the comparison with the earlier hyperbolic-sine bound directly. For $a, b \in (0, 1]$ and $z = -\frac{1}{2} \log(ab)$,

$$\sinh^2 z - (1 - a)(b^{-1} - 1) = \frac{(1 - 2a + ab)^2}{4ab} \geq 0. \quad (\text{EP.29})$$

Expansion after multiplying by $4ab$ proves the equality term by term. Also $(\log \sinh z)'' = -1/\sinh^2 z < 0$ for $z > 0$, so the same tangent-line proof used for (EP.16) shows $\prod \sinh z_j \leq \sinh((\sum z_j)/r)^r$. If any $z_j = 0$, the product is zero and the inequality is immediate. Taking $a = d_j, b = d_{j+1}$ gives $\sum z_j = \mathcal{B}/2$, and hence

$$\mathcal{A}F_{\max}^{1/(2r)} \leq \mathcal{A} \sinh(\mathcal{B}/(2r)). \quad (\text{EP.30})$$

For $r > 1, \mathcal{B} > 0$ the inequality is strict. At a maximizer every $z_j > 0$. Equality in the concavity step requires all $z_j = z > 0$, so $d_j d_{j+1} = e^{-2z} =: p < 1$. Equality in (EP.29) requires $d_j = (1 + p)/2$ for every $0 \leq j < r$. For $r > 1$ this implies $d_0 = d_1$, whence $p = ((1 + p)/2)^2$ and $(1 - p)^2 = 0$, a contradiction. For $r = 1$, (EP.25)–(EP.26) give exactly $\sinh(\mathcal{B}/2)$. For geometric contractions $d_j = e^{-v}$ the fixed-endpoint bound is $2\mathcal{A} \sinh(v/2)$ and the earlier bound is $\mathcal{A} \sinh v$; their ratio is exactly $\cosh(v/2)$.

The optimizer can be certified without a transcendental equation. The real endpoint datum $R = e^{\mathcal{B}} \geq 1$ gives

$$(1 + 2x)(1 + x)^{2(r-1)} = R(1 - 2x)(1 - x)^{2(r-1)}. \quad (\text{EP.31})$$

The ratio of the two positive sides before subtraction is strictly increasing on $[0, 1/2)$ by (EP.25), so rational bisection isolates its unique root using integer products when R is rational. For a general actual endpoint, certified rational bounds $1 \leq R_- \leq R \leq R_+$ give enclosing roots $x(R_-) \leq x(R) \leq x(R_+)$ by the same monotonicity. Positive-power bisection then encloses (EP.26). This specifies the computational map exactly.

45.4 Schur windows and the reduced source-relation determinant

Split the source monomials of $\mathcal{P}_m$ at degree $n - 1$ and set

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EP.32})$$

The triangular map $(x, y) \mapsto (x - H_{n-1}^{-1} C_D y, y)$, from $\mathbb{C}^n \oplus \mathbb{C}^{r+1}$ onto $\mathbb{C}^{m+1}$, has determinant one and takes H_m to $\text{diag}(H_{n-1}, W)$ and the remainder map to $[J_{n-1}, F]$. Consequently $W > 0$. For its first j columns put $W_j = W[1 : j, 1 : j]$, $F_j = F[:, 1 : j]$, $Z_j = F_j^* G_{n-1} F_j$. Block inversion and the determinant identity from (EP.6) prove

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*, \quad R_j := V_{n-1}/V_{n+j-1} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad R_0 = 1. \quad (\text{EP.33})$$

This R_j is a scalar and is distinct from the typed least-lift R_N by its explicit definition; the scalar is obtained from that least-lift Gram by determinant. Write

$$\begin{aligned} w &= (W_1)_{11}, \quad p = \det W_r, \quad s_+ = \det W_{r+1}, \quad e = w + (Z_1)_{11}, \\ H_+ &= \det(W_{r+1} + Z_{r+1}), \quad H_- = \det(W_r + Z_r). \end{aligned} \quad (\text{EP.34})$$

All six quantities are positive. Successive monic Schur complements give $w = \omega_n$ and $\omega_m = s_+/p$. Thus the exact bridge from the original EW matrices to the new endpoint quantities is

$$\begin{aligned} U &= s_+/(pw), \quad \alpha = w/e, \quad \beta = H_-s_+/(pH_+), \quad t = ep/(wH_-), \\ U \frac{V_n}{V_m} &= \frac{H_+}{pe}, \quad \mathcal{B} = \log \frac{H_-H_+w}{ps_+e}. \end{aligned} \quad (\text{EP.35})$$

For example $\alpha = 1/R_1$, $\beta = R_r/R_{r+1}$ and $t = R_1/R_r$, and substitution gives each displayed equality. Every individual contraction can also be recovered as $d_j = R_j/R_{j+1}$ for $0 \leq j \leq r$, so the exact product, the loss $\mathcal{J}$ and every comparison use the same block. For $r = 1$, $p = w$, $H_- = e$, $t = 1$, and the separate single-step formula applies. At $n = q$ the base relation space is empty and $G_{q-1} = H_{q-1}$; no inverse of an empty deficient kernel occurs.

To prove the determinant cancellation independently, define

$$\mathfrak{D}_j = \det H_{j-1}, \quad \mathfrak{D}_0 = 1, \quad \mathfrak{B}_l = \det(B_{q+l-1}^* H_{q+l-1} B_{q+l-1}), \quad \mathfrak{B}_0 = 1. \quad (\text{EP.36})$$

The coefficient matrix with first columns the q quotient monomials and last columns $\chi, S\chi, \dots, S^{N-q}\chi$ is triangular with diagonal one. Subtracting boundary columns to replace the first ones by R_N keeps its determinant one and makes the Gram block diagonal. Hence

$$\omega_N = \mathfrak{D}_{N+1}/\mathfrak{D}_N, \quad V_N = \mathfrak{D}_{N+1}/\mathfrak{B}_{N-q+1}, \quad \Lambda_N := \omega_N/V_N = \mathfrak{B}_{N-q+1}/\mathfrak{D}_N. \quad (\text{EP.37})$$

The norm identity uses the analogous monic orthogonal triangular basis. Equations (EP.14), (EP.35), and (EP.37) now prove

$$E_* \leq \left(\frac{\Lambda_m}{\Lambda_n} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{m-q+1} \mathfrak{D}_n}{\mathfrak{D}_m \mathfrak{B}_{n-q+1}} \right)^{1/(2r)} = \left(\frac{H_+}{pe} \right)^{1/(2r)}. \quad (\text{EP.38})$$

For $N \geq q$, multiplication by χ in (EP.3) also sends the original leading-coefficient line at degree $N - q$ isomorphically to the leading-coefficient line of its relation at degree N . Its leading coefficient is one; the source norm of that relation is exactly the form defining $\mathfrak{B}$. The quotient J_N sends the same relation to zero. These give the maps relating the determinant and quotient observations.

For comparison only, multiplying the original source measure by a positive scalar a sends $H_N \mapsto aH_N$, $G_N \mapsto aG_N$, $\omega_N \mapsto a\omega_N$, $V_N \mapsto a^q V_N$, and $\Lambda_N \mapsto a^{1-q} \Lambda_N$. This follows directly from (EP.4) and (EP.37). All window ratios displayed above are unchanged; the actual measure throughout remains (EP.1).

45.5 Full confluent transfer and its two-determinant window

We now construct the transfer determinants in (EP.38) from their full raw derivative maps. Retain $\psi = \hat{\chi} = i^{-q}\chi(c + iu)$ and $\Pi = \psi^2$ under the reflection hypothesis of (EP.7). If χ has distinct roots λ_a of complete lengths ℓ_a , let $\zeta_a = (\lambda_a - c)/i$ and $r_a = 2\ell_a$. The width is $d = \sum_a r_a = 2q$. The map

$$\mathcal{J}_2 f = (f^{(e)}(\zeta_a))_{a, 0 \leq e < r_a} : \mathbb{C}[u] \longrightarrow \bigoplus_a \mathbb{C}^{r_a} \quad (\text{EP.38a})$$

uses the fixed order of nodes and then increasing raw derivative orders. Its kernel is (II) by Taylor divisibility and the coprimality of distinct linear factors. In these coordinates a product uses

$\sum_{b=0}^e \binom{e}{b} f^{(b)} g^{(e-b)}$, and multiplication by u has the block formula $(X_{\Pi}v)_{a,e} = \zeta_a v_{a,e} + e v_{a,e-1}$. The original coordinate map has the exact raw-jet expression

$$(\partial_u^e \Phi P)(\zeta_a) = i^e P^{(e)}(\lambda_a), \quad \mathbb{C}[S]/(\chi^2) \xrightarrow{\sim} \mathbb{C}[u]/(\Pi), \quad [P] \mapsto [P(c + iu)], \quad (\text{EP.38b})$$

since $\Phi(\chi^2) = i^{2q}\Pi$. Every $e!$ and i^e is retained.

Let $v_j = \mathcal{J}_2 Q_j$ for the same source polynomials used in (EP.8), and form

$$\mathbf{F}_a = [v_a, \dots, v_{a+d-1}], \quad \mathbf{V} = [\mathcal{J}_2 1, \dots, \mathcal{J}_2 u^{d-1}]. \quad (\text{EP.38c})$$

Their full raw Vandermonde determinant is

$$\det \mathbf{V} = \left(\prod_a \prod_{e=0}^{r_a-1} e! \right) \prod_{a < b} (\zeta_b - \zeta_a)^{r_a r_b} \neq 0. \quad (\text{EP.38d})$$

To prove the formula without changing node orders, replace the monomial columns successively by the monic polynomials $\prod_{b < a} (u - \zeta_b)^{r_b} (u - \zeta_a)^e$. This is a triangular column change of determinant one, ordered by total degree. Its jet matrix is block lower triangular. At its (a, e) diagonal entry, the derivative is $e! \prod_{b < a} (\zeta_a - \zeta_b)^{r_b}$. Multiplying these entries proves (EP.38d), including repeated orders. The matrix $\mathbf{F}_0$ equals $\mathbf{V}$ times the actual monic coefficient matrix; the determinants, rather than all entries, agree.

Every $\mathbf{F}_a$ is invertible. A vector in its kernel gives a polynomial $Q \in \text{span}(Q_a, \dots, Q_{a+d-1})$ with $Q = \Pi H$ and $\deg H < a$. Source orthogonality gives $0 = \langle H, Q \rangle = \int \Pi |H|^2 d\mu_\theta$. The polynomial Π is positive off its finitely many real roots, and the original source density is positive almost everywhere, so $H = 0$. For $a = 0$ divisibility and degree already force $Q = 0$. Solving $\mathbf{F}_a z_a = v_{a+d}$ yields a monic polynomial

$$\hat{Q}_a = \frac{Q_{a+d} - \sum_{j=0}^{d-1} (z_a)_j Q_{a+j}}{\Pi}. \quad (\text{EP.38e})$$

It has degree a and is orthogonal for the weight $\Pi d\mu_\theta$: pair its numerator with any polynomial of smaller degree. The unique monic orthogonal polynomial for this real positive weight is real, so this quotient is real. Pairing its numerator with itself after division, or with $\hat{Q}_a$, leaves only the Q_a term and gives

$$\nu_a := \int \Pi |\hat{Q}_a|^2 d\mu_\theta = -(z_a)_0 \omega_a > 0. \quad (\text{EP.38f})$$

All source monic polynomials have been left at their literal norms. The original monic relation polynomial is $\chi(S) i^a \hat{Q}_a((S - c)/i)$, whose value on the source line is $i^{a+q} \psi(u) \hat{Q}_a(u)$ and whose norm is exactly ν_a . This is the explicit source map to the relation norm in (EP.36).

The column shift gives $\mathbf{F}_{a+1} = \mathbf{F}_a[e_1, \dots, e_{d-1}, z_a]$. Its determinant multiplier is $(-1)^{d-1} (z_a)_0 = -(z_a)_0$ because $d = 2q$. Multiplying (EP.38f) from index zero proves

$$\boxed{\frac{\det \mathbf{F}_a}{\det \mathbf{V}} = \prod_{j=0}^{a-1} \frac{\nu_j}{\omega_j} = \frac{\mathfrak{B}_a}{\mathfrak{D}_a}}. \quad (\text{EP.38g})$$

The source and relation determinant identities follow from their monic orthogonal triangular matrices; the S - u coefficient matrices have determinant of modulus one, including the common i^q on each relation column. Equation (EP.38g) therefore holds for the original determinants, with their complex node factors intact.

For $a = N - q + 1$, substitution into (EP.37) gives

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det \mathbf{V}}{\det \mathbf{F}_a}, \quad \Lambda_N = \frac{\det \mathbf{F}_a}{\det \mathbf{V} \prod_{j=a}^{N-1} \omega_j}. \quad (\text{EP.38h})$$

In particular with $a = n - q + 1$ the two-volume norm criterion is

$$E_* \leq \left[\frac{\det F_{a+r}}{\det F_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}. \quad (\text{EP.38i})$$

For $q = 1$ the product has zero terms and value one. The expression inside the root is positive by its exact equality to Λ_m/Λ_n ; no absolute value conceals a complex phase.

The complete transfer of the jet multiplication map is also explicit. Put $K_a = [e_1, \dots, e_{d-1}, z_a]$ and $T_a = F_a^{-1} X_{\Pi} F_a$. Then $T_{a+1} = K_a^{-1} T_a K_a$, and the source recurrence gives $z_a = (T_a - a_{a+d-1}^{(0)} I) e_{d-1} - (\omega_{a+d-1}/\omega_{a+d-2}) e_{d-2}$. Here $d \geq 2$, so both indices exist. Differentiating monic orthogonality gives $\partial_{\theta} Q_j = -(\omega_j/\omega_{j-1}) Q_{j-1}$ for $j \geq 1$ and $\partial_{\theta} Q_0 = 0$: its pairings below $j - 1$ vanish and its Q_{j-1} pairing is $-\omega_j$. The norm derivative is $\partial_{\theta} \log \omega_j = a_j^{(0)}$ by the same minimum argument as (EP.9). Internal shifted columns have zero trace, so

$$\partial_{\theta} \log V_N = \sum_{j=a}^N a_j^{(0)} + \frac{\omega_a}{\omega_{a-1}} e_0^T F_a^{-1} v_{a-1}. \quad (\text{EP.38j})$$

At $a = 0$ omit the last term entirely. This follows by writing $F'_a = F_a L_a - (\omega_a/\omega_{a-1}) v_{a-1} e_0^T$, where $(L_a)_{j-1,j} = -\omega_{a+j}/\omega_{a+j-1}$ and all other entries vanish. Taking the trace of $F_a^{-1} F'_a$ in (EP.38h) proves (EP.38j). It supplies the phase in (EP.9) through the same complete jet map.

For the Gamma reference use the same Π , nodes, row order, and raw derivatives, and denote the reference by a superscript Γ . With $X_a = \mathfrak{D}_a/\mathfrak{D}_a^{\Gamma}$ and $Y_a = \mathfrak{B}_a/\mathfrak{B}_a^{\Gamma}$, division of (EP.38g) gives

$$\frac{\det F_a}{\det F_a^{\Gamma}} = \frac{Y_a}{X_a}, \quad \frac{V_N}{V_N^{\Gamma}} = \frac{X_{N+1}}{Y_{N-q+1}}. \quad (\text{EP.38k})$$

The invariant endpoint values still depend only on original moments through degree $2m$ by (EP.32)–(EP.38). In contrast the individual entries of F_a , with $a = N - q + 1$, involve Q_{N+q} , which is obtained by a projection using moments through $2N + 2q - 1$. The determinant identity proves their exact cancellation in the invariant answer. It does not identify every raw entry with a function of the shorter input list.

For a non-reflection-stable polynomial, define $\chi^{\dagger}(S) = (-1)^q \bar{\chi}(k - S)$, where bars conjugate coefficients. Then $\Phi\chi = i^q \psi$ and $\Phi\chi^{\dagger} = i^q \bar{\psi}$, so the exact extension of the norm-jet map has domain $\mathbb{C}[S]/(\chi\chi^{\dagger}) \xrightarrow{\sim} \mathbb{C}[u]/(\bar{\psi}\psi)$ under the same Φ . It uses $\Pi = \bar{\psi}\psi$ instead of ψ^2 . Its nodes are the union of the roots of ψ and their conjugates, and the orders add at every overlap; their sum is again $2q$. The positivity, division, raw-jet, and determinant proofs above apply to this actual monic Π , with the just-proved changed quotient domain. The original relation map still multiplies by χ , whose squared norm weight is $|\psi|^2$. This supplies the exact general norm map. The original radius and phase argument (EP.7)–(EP.11) is stated under its reflection hypothesis and is not extended by this norm observation alone.

45.6 An arithmetic lower cost on one relation block

The complete analytic theorem in the delivered arithmetic endpoint source proves, for the actual density (EP.1) at $\theta = 0$,

$$(\omega_{h,k,2n}/\omega_{h,k,n})^{1/(2n)} \leq C_h n \quad (n \geq k \geq 3). \quad (\text{EP.39})$$

Its retained literal sufficient constant, from equation (36) there, is

$$C_h = 64b^{-2} e^{\pi/2} 2^{B_h/2} e^{B_h/(2e)} \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}, \quad 0 < b < \pi/2, \quad (\text{EP.40})$$

where $B_h = 42 + 2 \deg h$, $M_* = \max(1, M_h(b), M_h(-b))$, $M_h(b) = \int e^{bt} w_h(t) dt$, $\vartheta_h = \int_{-1}^1 w_h(t) dt$, and $c_h = b_h c_h^{(1)} > 0$ with $b_h = \min_{|u| \leq 1} (w_h * w_h)(u)$ and the original local-mass constant $c_h^{(1)}$ of that source. No norm estimate is assumed in this paper in place of that proof. Its complete source is retained in the dependency record, with the independent analytic audit.

For a packet consisting exactly of an off-line quartet $1/2 \pm \delta \pm i\gamma$, $\delta, \gamma > 0$, with common full multiplicity m_0 , the original exterior calculation gives

$$\begin{aligned} q_k &= [1 + k(m_0 - 1)](k + 1)^2, \\ L_{h,k} &= 2\delta[1 + k(m_0 - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor, \\ L_{h,k}/q_k &\geq \delta k/2, \quad L_{h,k} \leq \epsilon_{h,k,N}. \end{aligned} \tag{EP.41}$$

Here is the complete finite spectral calculation behind (EP.41). In each tensor local algebra write $s_j = \rho_j + z_j$ with $z_j^{m_0} = 0$. The sum nilpotent $\sum z_j$ has exact order $1 + k(m_0 - 1)$: its top nonzero power has coefficient $[k(m_0 - 1)]! / [(m_0 - 1)!]^k$ at $\prod z_j^{m_0-1}$, and every higher monomial has an exponent at least m_0 . The distinct sum centres are $c + \delta(2a - k) + i\gamma(2b - k)$, $0 \leq a, b \leq k$. Every such pair occurs: the number of simultaneous plus signs can be any integer between $\max(0, a + b - k)$ and $\min(a, b)$, a nonempty interval. Distinct pairs give distinct centres since $\delta, \gamma > 0$. The cyclic annihilator is the least common multiple of these local annihilators, giving the displayed q_k with the full common order.

For any admitted G_N , conjugate $A - cI$ to the Euclidean metric by $G_N^{1/2}$ and put it in unitary upper triangular form, ordering the eigenvalues with positive real part first. Such a form follows inductively by choosing an eigenvector and applying induction to the quotient orthogonal block. Its Hermitian sum has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros by (EP.10)–(EP.11). The trace of its compression onto the positive-real diagonal slots is twice the sum of those positive real parts. This trace is at most ϵ_N : in an eigenbasis of the Hermitian sum it equals $\epsilon_N \langle P e_+, e_+ \rangle - \epsilon_N \langle P e_-, e_- \rangle \leq \epsilon_N$ for the orthogonal projection P onto those slots. The sum of the positive integers $2a - k$, $0 \leq a \leq k$, is $\lfloor (k + 1)^2/4 \rfloor$, by separately summing the even and odd arithmetic progressions. Multiplying by 2δ , the $k + 1$ imaginary centres, and the full local length proves exactly $L_{h,k}$ and its bound by ϵ_N . Finally $\lfloor (k + 1)^2/4 \rfloor \geq k(k + 1)/4$, checked in the same two parities, proves $L_{h,k}/q_k \geq \delta k/2$. Thus the last inequality in (EP.41) holds at every admitted degree. Retaining the exact $L_{h,k}$ makes the following deduction explicit. Set $n = r = q_k$ in (EP.13), and abbreviate $q = q_k$, $L = L_{h,k}$. Every $d_j < 1$ in this window because $L > 0$ and (EP.11). Taking logarithms in (EP.13) and using (EP.39) proves

$$\begin{aligned} \log \frac{V_q}{V_{2q}} &\geq 2q \log \frac{L}{C_h q} - \log(1 - d_0) - \log(1 - d_q) - 2 \sum_{j=1}^{q-1} \log(1 - d_j) \\ &\quad + \sum_{j=0}^{q-1} \log(1 + \phi_{q+j}^2/L^2). \end{aligned} \tag{EP.42}$$

Indeed $\prod(\epsilon^2 + \phi^2) \geq L^{2q} \prod(1 + \phi^2/L^2)$, whereas $\log U \leq 2q \log(C_h q)$. All added terms on the right of (EP.42) after its first one are nonnegative, with the contraction penalty strictly positive. For $q > 1$, (EP.16) gives the additional endpoint-only lower penalty $-\log(1 - \alpha) - \log(1 - \beta) - 2(q - 1) \log(1 - t^{1/(q-1)})$. In particular the actual two-volume threshold is

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k}/V_{2q_k})}{q_k \log k} \geq 2. \tag{EP.43}$$

This follows by replacing L/q_k by $\delta k/2$ in (EP.42) and dividing by $q_k \log k$. The constants are independent of k . The first norm window also gives the exact determinant lower bound $\Lambda_{2q}/\Lambda_q \geq L^{2q}$ by (EP.38).

Here is the exact receiving relation layer. For $i \geq q - 1, j > i$, let $\mathsf{T} = [p_{i+1}, \dots, p_j]$ be source columns, $\mathsf{F} = [b_{i+1}, \dots, b_j]$, and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. Then

$$\mathsf{b} = \mathsf{T} - R_i \mathsf{F} : \mathbb{C}^{j-i} \rightarrow \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathcal{D}_N = \chi \mathcal{P}_{N-q}, \quad \mathsf{b}^* \mathsf{b} = \Omega + \mathsf{F}^* G_i \mathsf{F}. \quad (\text{EP.44})$$

Reduction vanishes termwise. Each column is orthogonal to $\mathcal{D}_i$ because $p_{i+a} \perp \mathcal{P}_i$ and $R_i \perp \mathcal{D}_i$. Their highest degrees are distinct and monic. Successive subtraction of these highest coefficients spans the dimension- $j - i$ new relation layer and proves that b is an isomorphism onto that layer. Orthogonality of the two summands gives its exact Gram. Kernel addition as in (EP.5) yields

$$\frac{V_i}{V_j} = \det(I + \Omega^{-1/2} \mathsf{F}^* G_i \mathsf{F} \Omega^{-1/2}) = \prod_{a=1}^{s_0} (1 + \lambda_a), \quad s_0 = \text{rank } \mathsf{F} \leq \min(q, j - i). \quad (\text{EP.45})$$

The positive eigenvalues λ_a are those of the written positive semidefinite matrix; zero eigenvalues contribute one. At $(i, j) = (q, 2q)$, (EP.42) thus gives

$$\sum_a \log(1 + \lambda_a) \geq 2q \log \frac{L}{C_h q}, \quad \lambda_{\max} \geq \left(\frac{L}{C_h q} \right)^2 - 1 \geq \frac{\delta^2 k^2}{4C_h^2} - 1. \quad (\text{EP.46})$$

For the second assertion, pad the eigenvalue list to length q with zeros, define $\lambda_{\max}$ as the maximum of this full list, and bound the determinant by $(1 + \lambda_{\max})^q$. If its right side is negative, the assertion still follows from $\lambda_{\max} \geq 0$. For sufficiently large k the determinant lower bound exceeds one, so the layer has positive rank.

The representative change is determined, not discarded: projection onto the new orthogonal relation layer in (EP.44) gives

$$R_j = R_i + \mathsf{b}(\Omega + \mathsf{F}^* G_i \mathsf{F})^{-1} \mathsf{F}^* G_i. \quad (\text{EP.47})$$

Indeed $\mathsf{b}^* R_i = -\mathsf{F}^* G_i$, so the right side is a lift orthogonal to both old and new relations, hence the unique minimum lift. Its arithmetic jet remains $\eta x = [v_h]_h^{\otimes k} x(S)$, whose injectivity follows because the full local unit has a recursive power-series inverse and (χ) is exactly the sum-polynomial kernel. For an actual relation $\chi \kappa$, repeated monic division gives polynomials Q_a with $\chi(\sum s_a) = \sum_a h(s_a) Q_a(\mathbf{s})$. The original cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa \left(\sum D_b (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \right) \quad (\text{EP.48})$$

has differential $\mathcal{V}_{h,k}(\chi \kappa)$, since $h(D)F_h = \Theta \phi_0$ and the tensor differential contributes the same sign $(-1)^{a-1}$, with all other differentials zero. Thus its quotient image is the supported zero $(\ell, 0)$ at the same label ℓ . The fibre map sends (ℓ, x) to $(\ell, R_N x)$ and the external absence τ to τ . This identifies the retained source cost in (EP.44)–(EP.46) with the relation sent to that supported zero through the actual maps.

The four-volume budget is linked exactly to the two-volume cost by

$$\mathcal{B} = 2 \log(V_n/V_m) + \log(\beta/\alpha). \quad (\text{EP.49})$$

The boundary term remains signed. The two-volume threshold alone does not bound it from below. A different, balanced window connects the original four endpoints without that signed comparison: taking $n = q, r = q - 1$ yields the central cost $\log(V_q/V_{2q-1})$, and monotonicity proves

$$\log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} \geq 2 \log \frac{V_q}{V_{2q-1}}. \quad (\text{EP.50})$$

The complete proof in the accompanying *Four-volume threshold* continuation, read in full, establishes

$$(\omega_{h,k,n+r}/\omega_{h,k,n})^{1/(2r)} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n, \quad (\text{EP.50a})$$

with the literal sufficient constant

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) M_* \max(1, \vartheta_h^{-1}) \max(1, b^{-3}) 6^{B_h/3}. \quad (\text{EP.50b})$$

It derives this from the original lower convolution envelope and the exact monic Legendre norm, retaining every factor before its $2r$ -th root. Its full source proof is included as a dependency. Applying (EP.13) now at $n = q, r = q - 1$ gives the following complete version with its retained penalties:

$$\begin{aligned} C_{\text{mid}} := \log \frac{V_q}{V_{2q-1}} &\geq 2(q-1) \log \frac{L}{C_h^{\text{bal}} q} - \log(1 - d_0) - \log(1 - d_{q-1}) \\ &\quad - 2 \sum_{j=1}^{q-2} \log(1 - d_j) + \sum_{j=0}^{q-2} \log(1 + \phi_{q+j}^2/L^2), \quad d_j = \delta_{q+j}. \end{aligned} \quad (\text{EP.50c})$$

All $d_j < 1$ follow again from $L > 0$ in (EP.11). The original four-volume budget has the exact positive decomposition

$$\log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = 2C_{\text{mid}} + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}}. \quad (\text{EP.50d})$$

Both extra terms are nonnegative. Dropping only nonnegative terms from (EP.50c)–(EP.50d) and using $L/q \geq \delta k/2$ proves

$$\boxed{\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 4.} \quad (\text{EP.50e})$$

Indeed the numerator is at least $4(q_k - 1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q_k - 1)/q_k \rightarrow 1$. For the actual central relation block $(i, j) = (q, 2q - 1)$, its full matrix in (EP.45) has size $q - 1$, so the same determinant argument gives

$$\lambda_{\text{max}}^{\text{mid}} \geq \left(L / (C_h^{\text{bal}} q) \right)^2 - 1 \geq \delta^2 k^2 / (4(C_h^{\text{bal}})^2) - 1. \quad (\text{EP.50f})$$

The norm bound, the new product penalties, the central relation eigenvalues, and the original four-volume budget are therefore connected by these explicit maps. No upper estimate for those growing arithmetic relation matrices is proved by this lower-cost calculation.

45.7 Mass-seven polynomial calibration and zero cases

The independent finite fixture is the measure $7e^{-u^2/2} du / \sqrt{2\pi}$ and quotient $\mathbb{C}[u]/(u^2 + 1)$. Integration by parts gives monic Hermite polynomials $Q_0 = 1, Q_1 = u, Q_{j+1} = uQ_j - jQ_{j-1}$ and norms $\omega_j = 7j!$: the derivative identity $Q'_j = jQ_{j-1}$ follows by induction in the recurrence, and integration by parts proves orthogonality and $\|Q_j\|^2 = j\|Q_{j-1}\|^2$ starting from mass seven. In the quotient basis $(1, u)$ use

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A = \frac{1}{2}I + iT. \quad (\text{EP.51})$$

Its original- S relation is $\chi(S) = (S - 1/2)^2 - 1$. The exact map from the $(1, S)$ coefficients to $(1, u)$ is $E = \begin{pmatrix} 1 & 1/2 \\ 0 & i \end{pmatrix}$; $G_N^S = E^* G_N^u E$ and $|\det E| = 1$. Thus all fixture volumes are the original- S volumes as well. Kernel summation (EP.5), in the transported coordinates, gives

$$\begin{aligned} G_2^u &= \text{diag}(7/3, 7), \quad G_3^u = \text{diag}(7/3, 21/11), \quad G_4^u = \text{diag}(42/43, 21/11), \\ (V_1, V_2, V_3, V_4, V_5) &= (49, 49/3, 49/11, 882/473, 980/1333). \end{aligned} \quad (\text{EP.52})$$

For a diagonal $G = \text{diag}(g_0, g_1)$, $A^*G + GA - G = i(GT - T^T G)$ has relative squared radius $(g_0 + g_1)^2 / (g_0 g_1)$. The resulting exact values are

$$(\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = (16/3, 400/99, 4225/946), \quad U_{2,3} = 60, \quad (\alpha, \beta, t) = (1/3, 110/279, 54/473). \quad (\text{EP.53})$$

Reflection $u \mapsto -u$ conjugates the source and quotient matrices at tilt θ to those at $-\theta$ by diagonal sign matrices. Their determinants are even, and $\text{Tr } T = 0$, so $\phi_N = 0$ at zero tilt for this fixture. This proves that the direct defect and the radius identities test the same allowance.

All finite values in (EP.52)–(EP.53) are derived in the checker from literal Gaussian moments $7(2j - 1)!!$, monomial Gram matrices, monic polynomial division, and (EP.4), independently of assigned quotient matrices. Further windows check (EP.13), (EP.35), and (EP.38), while exact scalar extremizers check the new formula and phase boundary cases. These are finite auxiliary calculations. They do not assert that the Gaussian fixture is the arithmetic convolution measure; (EP.1)–(EP.7), (EP.32)–(EP.38) give the explicit common maps and the distinct specified source inputs.

For the mass-seven window, the exact inputs to the new reduced-budget root calculation are $U = 60$ and $e^B = 630509/1080$. The rational procedure (EP.31), followed by the positive sixth-root comparisons in (EP.26), gives the explicit certificate

$$\frac{719736229856370310835738215}{2^{88}} \leq \mathcal{A}F_{\max}^{1/6} \leq \frac{719736229856370310835738217}{2^{88}}. \quad (\text{EP.54})$$

Its value is approximately 2.325593185504684 , with that decimal used only for orientation. The check record contains the two rational brackets for x , their exact comparisons in (EP.31), and the two sixth-power comparisons determining (EP.54).

The primary checker executes 415 exact auxiliary mathematical checks normally and under python -0. Its six actual formula changes alter the literal mass, the repeated interior contraction, the terminal ratio orientation, the Schur cancellation, the retained phase, and a raw derivative factorial; each changed run executes the full suite and is rejected in both modes. An independent scalar checker executes 2,453 exact checks in its unmodified runs, and its five changed-formula runs are rejected in both modes. Repeated modes and changed-formula runs do not add mathematical checks to those scopes. Their full source, complete records, independent mathematical reviews, and the eight exact proof dependencies accompany this article.

For the empty quotient, declare $q = 0$, $\chi = 1$, $C = 0$ and $V_N = 1$ including $N = -1$. Then $\epsilon_N = \phi_N = 0$, all quotient maps and relation-kernel update columns have their empty dimensions, and the exact product is zero. The formulas requiring $q \geq 1$ are used only in their stated range; the cochain boundary map and original source mass still exist. For $q = 1$ the transfer norm bands, when used, are empty products equal to one. At an endpoint contraction equal to one all zero-radius and phase cases are handled by (EP.11), (EP.13), and (EP.21) before logarithms.

46 The complete balanced four-volume threshold calculation

Source role. Complete owner calculation with the original arithmetic norm and volume.

Exact revision. 50cf16bc0ccd955d2dccac62b7fb8d8f08623cfe6436c4fff7374ac5ea3f3ec2

Complete source. sources/owner_endpoint_terminal/published/workbenches/tau-arithmetic-endpoint-bounds/FOUR_VOLUME_THRESHOLD.md

Source SHA-256. 50cf16bc0ccd955d2dccac62b7fb8d8f08623cfe6436c4fff7374ac5ea3f3ec2

46.1 The two endpoint calculations give a four-volume threshold of four

46.1.1 Result and motivation

The arithmetic endpoint note bounds the monic source norms. The subsequent window-product calculation retains the repeated occurrence of every interior volume contraction. Composing them on the window one degree shorter than the original window improves the original four-volume lower threshold from 2 to 4:

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 4.$$

This statement concerns the original arithmetic family belonging to a fixed packet consisting exactly of a full off-line quartet, with every multiplicity retained. It is a necessary volume-growth consequence for such a packet, not an assertion that an off-line zero exists. No upper bound contradicting this growth is established here.

This note is a written continuation of the two supplied calculations. It is not included in the frozen Toda-Gamma edition DOI [10.5281/zenodo.22730756](https://doi.org/10.5281/zenodo.22730756).

46.1.2 1. Original source, constants, and coordinates

Keep $g = 2\xi$, the fixed monic packet polynomial h , and

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad M_h(b) = \int e^{bt} w_h(t) dt.$$

The mass remains μ_h^k . The monic degree- j source norm is $\omega_{h,k,j}$ for the observation $S = k/2 + iu$. In this coordinate map the leading coefficient acquires the phase i^j , of squared modulus one. Neither the source measure nor the generator is replaced.

The arithmetic endpoint proof gives constants $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, and $B_h = 42 + 2 \deg h$, such that

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\pi(|u|+k-3)/2} (1 + |u| + k - 3)^{-B_h}, \quad k \geq 3. \quad (1)$$

Here (1) is the proved lower envelope of the actual threefold-and-higher convolution, not a newly postulated comparison measure. Its proof is in the [supplied complete arithmetic note](#).

Choose one $0 < b < \pi/2$ and define bounding constants

$$X_h = \max\{1, M_h(b), M_h(-b)\}, \quad K_h = \max\{1, \vartheta_h^{-1}\}.$$

Taking maxima here only bounds constants; it performs no rescaling of the source, its quotient, or either Gram determinant.

46.1.3 2. The balanced-window norm estimate, with an explicit constant

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq n$, one has

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq C_h^{\text{bal}} n, \quad (2)$$

where the following literal finite constant suffices:

$$C_h^{\text{bal}} = 256e^\pi \max\{1, 3/c_h\} X_h K_h \max\{1, b^{-3}\} 6^{B_h/3}. \quad (3)$$

To prove this, the trial polynomial $(S - k/2)^j$ and exponential moments give the unchanged upper estimate

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k).$$

The exact monic Legendre norm on $[-n, n]$ is

$$\mathfrak{l}_n(n) = \frac{2n^{2n+1}}{2n+1} \left(\frac{2^n}{\binom{2n}{n}} \right)^2 \geq \frac{2n^{2n+1}}{(2n+1)4^n}.$$

This follows from the Legendre leading coefficient and squared norm in [DLMF Table 18.3.1](#). Combining (1) with this least-norm calculation gives the exact finite bound

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b)^k + M_h(-b)^k) e^{\pi(n+k-3)/2} (1+n+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2r)}. \quad (4)$$

For completeness, use $(2(n+r))! \leq (4n)^{2(n+r)}$, $M_h(b)^k + M_h(-b)^k \leq 2X_h^k$, $\vartheta_h^{-(k-3)} \leq K_h^k$, and $1+n+k-3 \leq 2n$. After taking the root, the factorial and Legendre factors are bounded by

$$4n 8^{n/r} \left(\frac{2n+1}{c_h n} \right)^{1/(2r)}.$$

The remaining factors are

$$b^{-(n+r)/r} (X_h K_h)^{k/(2r)} e^{\pi(n+k-3)/(4r)} (2n)^{B_h/(2r)}.$$

Now $n/r \leq 2$, $(n+r)/r \leq 3$, $k/(2r) \leq 1$, and $(n+k-3)/(4r) \leq 1$. Also $2n+1 \leq 3n$, and $\log(2n)/n$ decreases for $n \geq 3$, so $(2n)^{B_h/(2r)} \leq 6^{B_h/3}$. These inequalities prove (2)–(3). The diagonal $r = n$ recovers the scale of the supplied theorem. The needed new case is $r = n-1$, which lies in the same balanced range.

46.1.4 3. The exact product on the same canonical metrics

Let $V_N = \det G_N$, $\delta_N = V_N/V_{N-1} \in (0, 1]$, and retain the calculated real phase $\phi_N = \sigma - \partial_\theta \log V_N$. The original radius identity is

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N)(\delta_{N+1}^{-1} - 1), \quad N \geq q.$$

Multiply at $N = n, \dots, n+r-1$, without discarding intermediate factors:

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - \delta_n)(1 - \delta_{n+r}) \prod_{j=1}^{r-1} (1 - \delta_{n+j})^2. \quad (5)$$

The original phase remains on the left. Each factor on the right after the ratio belongs to $[0, 1]$. The exterior theorem bounds the aggregate spectral defect $L_{h,k}$ by every ϵ_N . Thus (5) proves

$$L_{h,k}^{2r} \leq \frac{\omega_{h,k,n+r} V_n}{\omega_{h,k,n} V_{n+r}}. \quad (6)$$

This deduction needs neither a new metric nor an assumption that the phase vanishes. A unit contraction makes a factor in (5) zero and directly forces $L_{h,k} = 0$; the inequality (6) still holds. The first admitted degree $q-1$ is not used in any radius identity here.

46.1.5 4. Central window and the factor four

For the exact quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$ and common order m , keep

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

In particular $L_{h,k}/q_k \geq \delta k/2$. Put $q = q_k$ and choose $n = q, r = q-1$. All hypotheses of the proved balanced-window estimate are satisfied: $q \geq k \geq 3$, $q-1 \geq q/2$, and every radius index is at least q . Equations (2) and (6) yield

$$\log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q} \geq 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (7)$$

Write $C_k = \log(V_q/V_{2q-1})$. The original four-volume budget is exactly

$$\begin{aligned} \mathcal{B}_{h,k} &= \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} \\ &= 2C_k + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}} \geq 2C_k. \end{aligned} \quad (8)$$

Both last terms are nonnegative by the original rank-one kernel update. Substitution of (7) proves

$$\mathcal{B}_{h,k} \geq 4(q_k - 1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (9)$$

Since h is fixed and $q_k \rightarrow \infty$, division by $q_k \log k$ gives the claimed liminf of at least 4. An upper limit strictly below 4 would already conflict with (9); no such upper estimate has been obtained here. The two-volume conclusion at $n = r = q$ is also retained: $\liminf \log(V_q/V_{2q})/(q \log k) \geq 2$.

46.1.6 5. Source maps and verification scope

All representatives remain those of the supplied arithmetic construction: $J^{(k)}R_N = \eta_{h,k}$ and $q^{(k)}R_N = \sigma_h^{\otimes k} \eta_{h,k}$. Restricting the degree window does not alter their unit, jets, fibre labels, or cohomology classes. The supported-zero relation and the external absence τ retain their original separate maps. In particular, no repeated exterior iteration or erased multiplicity cost occurs in (5)–(9).

The analytic envelope and original norm theorem are from the supplied Arithmetic Endpoint Bounds note. Equation (5) is the subsequent supplied window-product calculation. Equations (2)–(4) and their composition (7)–(9) are the present continuation. The two independent review lanes record finite tests separately from written analytic proof; no new Lean certificate or interval enclosure of C_h^{bal} is asserted.

47 The complete balanced-window arithmetic proof

Source role. Complete owner balanced-window proof and its exact parameter domain.

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Complete source. sources/owner_endpoint_terminal/published/workbenches/tau-arithmetic-endpoint-bounds/BALANCED_WINDOW_PROOF.md

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47.1 Balanced arithmetic norm windows and the strengthened endpoint threshold

This is a written analytic extension of the supplied arithmetic endpoint note, not a Lean certificate or a numerical enclosure. The fixed packet, original source, Taylor unit, measure and total mass are unchanged. It is intended for the next integration cut, not the already frozen E publication.

47.1.1 Original measure and constants

Fix a monic finite divisor h of $g=2x$, removing its selected zeros with their full orders, and put $d=\deg h$. On the original line $S=k/2+iu$, use

$$w_h(t) = |(g/h)(1/2+it)|^{1/2/(2\pi i)}, \quad m_h(k) = w_h^{\otimes k}, \\ \omega_h(k, j) = \inf_{P \text{ monic, degree } j} \int |P(k/2+iu)|^{1/2} m_h(k)(u) du.$$

The mass is μ_h^k , not one. The original note proves, for $a=\pi/2$ and $B=42+2d$,

$$m_h(k)(u) \geq c_h \vartheta_h^{k-3} \exp[-a(|u|+k-3)] (1+|u|+k-3)^{-B}, \quad k \geq 3,$$

where $c_h = b_h c_h^{1/2} > 0$, $b_h = \min_{|v| \leq 1} (w_h^* w_h)(v) > 0$, and $\vartheta_h = \int_{-1,1} w_h > 0$. Fix $0 < b < a$ and retain both actual finite exponential moments $M_+ = \int \exp(bt) w_h(t) dt$ and $M_- = \int \exp(-bt) w_h(t) dt$. Define $X = \max(1, M_+, M_-)$ and $K = \max(1, \vartheta_h^{-1})$. None of these definitions normalizes the source.

47.1.2 Exact preceding estimate

The monic Lebesgue minimum, allowing any leading coefficient of modulus one, is

$$\text{ell}_n(L) = [2 L^{(2n+1)/(2n+1)}] [2^n (n!)^2/(2n)!]^2.$$

This includes the original leading phase i^n under $S=k/2+iu$. For integers $n \geq 0$, $r \geq 1$, $k \geq 3$ and any $L > 0$, the trial-monomial exponential upper bound and interval Legendre lower bound give

$$\begin{aligned} A_{(h,k,n,r)} &:= (\omega_{(h,k,n+r)}/\omega_{(h,k,n)})^{1/(2r)} \\ &\leq [(2(n+r))! b^{-(2(n+r))} (M_+^{k+M_-^k}) \\ &\quad \exp[a(L+k-3)] (1+L+k-3)^B \\ &\quad / (c_h \text{vartheta}_h^{k-3} \text{ell}_n(L))]^{1/(2r)}. \end{aligned} \quad (\text{BW1})$$

For the balanced-window theorem take integers $n \geq k \geq 3$ and $n/2 \leq r \leq n$, and set $L=n$. Then

$$A_{(h,k,n,r)} \leq \text{Cbal}_h n, \quad (\text{BW2})$$

$$\begin{aligned} \text{Cbal}_h &= 256 \exp(\pi) \max(1, 3/c_h) X K \\ &\quad \max(1, b^{(-3)}) 6^{(B/3)}. \end{aligned} \quad (\text{BW3})$$

Proof of the uniform constant The central-binomial bound gives $\text{ell}_n(n) \geq 2 n^{(2n+1)/[(2n+1)4^n]}$. Also $(2(n+r))! \leq (4n)^{(2(n+r))}$ and $M_+^{k+M_-^k} \leq 2X^k$. Combining these before taking roots cancels the factor two. The factorial, Legendre and c_h factors after the $2r$ -th root are at most

$$\begin{aligned} n * 4 * 8^{(n/r)} * [(2n+1)/(c_h n)]^{1/(2r)} \\ \leq 256 n \max(1, 3/c_h). \end{aligned}$$

Here $n/r \leq 2$, $(2n+1)/n \leq 3$, and $0 < 1/(2r) \leq 1$. The remaining factors obey

$$\begin{aligned} X^{k/(2r)} &\leq X, \\ \text{vartheta}_h^{-(k-3)/(2r)} &\leq K, \\ b^{-(n+r)/r} &\leq \max(1, b^{(-3)}), \\ \exp[a(n+k-3)/(2r)] &\leq \exp(2a) = \exp(\pi). \end{aligned}$$

Finally $1+n+k-3 \leq 2n$ and $2r \geq n$. The function $\log(2x)/x$ decreases for $x \geq 3$, since its derivative is $[1 - \log(2x)]/x^2 < 0$. Hence

$$(1+n+k-3)^{(B/(2r))} \leq (2n)^{(B/n)} \leq 6^{(B/3)}.$$

Multiplication proves BW2-BW3. This proof works for the original arithmetic density without an evenness hypothesis. The constants depend on fixed h and b only, never n, k, r .

47.1.3 Composition with the exact consecutive-window product

This paragraph assumes the separately reviewed exact-window theorem for the original canonical quotient family, including its nonnegative contractions and zero-contraction cases. Write $V_N = \det G_N > 0$ and $\Lambda_N = \omega_N/V_N$. Its simple product corollary is

$$\begin{aligned} \min_{(n \leq N \leq n+r)} \epsilon_N \\ \leq (\Lambda_N^{(n+r)}/\Lambda_N)^{1/(2r)} \\ = A_{(h,k,n,r)} (V_N/V_{(n+r)})^{1/(2r)}. \end{aligned} \quad (\text{BW4})$$

Do not infer BW4 from BW1 alone. It uses the original rank-two Toda identities and the retained source/quotient maps. If every allowance on the window has the exterior lower bound $L_{(h,k)}$, BW2-BW4 imply

$$\log(V_N/V_{(n+r)}) \geq 2r \log[L_{(h,k)}/(\text{Cbal}_h n)]. \quad (\text{BW5})$$

This holds even if the right side is negative; canonical volume monotonicity independently gives a nonnegative left side.

For the packet consisting exactly of the hypothetical off-line quartet $1/2 \pm \delta \pm i\gamma$, with fixed $\delta, \gamma > 0$ and common multiplicity $m \geq 1$, retain

$$\begin{aligned} q_{k,m} &= [1+k(m-1)](k+1)^2, \\ L_{(h,k)} &= 2\delta [1+k(m-1)](k+1)^{\lfloor (k+1)^2/4 \rfloor}, \\ q &\geq k \geq 3, \quad L_{(h,k)}/q \geq \delta k/2. \end{aligned}$$

Take $n=q$ and $r=q-1$. Since $q \geq 3$, $q/2 \leq q-1 \leq q$; the degrees $q, \dots, 2q-2$ are admitted and the norm endpoint is $2q-1$, not $2q$. Define the central two-volume cost

$$C_k = \log(V_q/V_{(2q-1)}).$$

Then

$$C_k \geq 2(q-1) \log[\delta k / (2 C_{bal_h})]. \quad (BW6)$$

Canonical monotonicity gives $V_{(q-1)} \geq V_q$ and $V_{(2q-1)} \geq V_{(2q)}$. Consequently each outer cost dominates this same central cost:

$$\begin{aligned} \log(V_{(q-1)}/V_{(2q-1)}) &\geq C_k, \\ \log(V_q/V_{(2q)}) &\geq C_k. \end{aligned}$$

For the original four-volume budget $B_k = \log[V_{(q-1)}V_q/(V_{(2q-1)}V_{(2q)})]$,

$$\begin{aligned} B_k &\geq 2 C_k \geq 4(q-1) \log[\delta k / (2 C_{bal_h})], \\ \liminf_{k \rightarrow \infty} B_k / (q_k \log k) &\geq 4. \end{aligned} \quad (BW7)$$

Indeed $q_k \geq (k+1)^2$ tends to infinity, so $(q_k-1)/q_k$ tends to one; fixed positive $\delta/(2C_{bal_h})$ contributes only $O_h(q_k)$. The diagonal choice $r=q$ also yields $\log(V_q/V_{(2q)}) \geq 2q \log[\delta k / (2C_{bal_h})]$, but the overlapping shorter window proves the stronger four-volume threshold.

The supplied note's older threshold 2 remains correct under its weaker sinh endpoint estimate. BW7 is a strengthened synthesis using BW4, not a correction to that valid implication. No upper estimate on B_k or C_k follows. A separately proved $\limsup B_k / (q_k \log k) < 4$ would contradict this quartet; no such upper estimate and no RH proof is asserted here.

47.1.4 Reference and evidence boundary

The analytic ingredients are the zeta reflection formula (DLMF 25.4.2), the gamma vertical-strip estimate, uniform for bounded real part (DLMF 5.11.9), and the Legendre orthogonality/leading coefficient (DLMF 18.3). The annulus propagation, convolution restriction, BW1-BW3 and BW5-BW7 are written derivations. Finite regression checks support algebraic fixtures only, not the analytic quantifiers or arithmetic zero hypotheses.

48 PR 24: complete implemented corrected proof

Source role. Complete corrected proof. The sealed proposal bytes equal the terminal adopted corrected note; both provenance identities are retained with their exact adoption receipts.

Exact revision. 47eb717484d92c96a062a0990a4567a0929e33e1038742e538b610fcc157b9d6

Complete source. [sources/owner_endpoint_review/review_pr24/proposed/RESEARCH_NOTE.md](#)

Source SHA-256. 47eb717484d92c96a062a0990a4567a0929e33e1038742e538b610fcc157b9d6

48.1 Confluent transfer for the original source-relation volume

48.1.1 Scope and current-main integration

This add-only analytic continuation starts from main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f, after the merge of PR #21 and its workflow-trigger correction. The PR #22 mathematical note was read at 811210d24b80813a08972ca23f919db015383137; its strict verification was in progress at that reading. This contribution neither modifies that branch nor treats pending checks as a certificate.

The source is the delivered Toda-volume continuation, following the cyclic and exterior-trace notes. Its full LaTeX was read directly. The new result is a fixed-width, full-jet transfer for the SAME canonical source/relation quotient metric. It applies the classical Christoffel transformation; it is not a claim to a new general orthogonal-polynomial theorem or to RH.

Keep $G(Z) \rightarrow Z$, with its infinite target, and its coefficient square to $G(C) \rightarrow C$. The structural base remains b_τ . Every supported coefficient map below retains $e=0^\bullet$ and external τ separately. The original theta complex is $C_+ = [V \rightarrow \Theta \rightarrow B]$, with $D = -x \, d/dx$, $Q = B/\Theta$ and $g=2$. No scalar, source function or measure is assigned a new normalization.

For a nonempty finite packet h of actual zeros with full orders, retain F_h with Mellin transform g/h . For the doubled-original-root transfer below, require the same dagger stability as the original source: its cyclic annihilator satisfies the polynomial identity $\text{conj}(\chi)(k-S) = (-1)^q \chi(S)$. The complete reflection-stable quartet packets used in the final target satisfy this hypothesis. It does not follow from being an arbitrary finite subpacket of zeros. At tensor degree $k \geq 1$, keep the actual cyclic sum module

```
C=C[S]/(chi),  q=deg chi>=1,  c=k/2,  A=M_S,
eta[P]=epsilon_h^(tensor k) P(sum_i M(s_i))1,
epsilon_h=j_h(g/h),
Vcal P=P(D_1+...+D_k)(F_h^(tensor k)).
```

The original equations are

```
J^(k) Vcal=eta pi_chi,
q^(k) Vcal=sigma_h^(tensor k) eta pi_chi.
```

Its norm is the actual integral against $m_k = w_h^{\{k\}}$, where

```
w_h(t)=|(g/h)(1/2+i t)|^2/(2 pi),  integral m_k=mu_h^k.
```

The auxiliary measure is $dm_\theta(u) = \exp(\theta u) m_k(u) du$, $|\theta| < \pi/2$. Its mass is $M_h(\theta)^k$. θ in this expression is the real tilt parameter, not the theta-summation map. The exponential-moment bound from the source makes all finite derivatives below valid. At tilt zero the metric is the original arithmetic metric.

For $h=1$ the finite arithmetic quotient is zero. Its determinant is the empty determinant 1; the analytic seed and its mass remain, and $G(0)$ still has e and τ . The positive-dimensional inverses below require $q \geq 1$.

48.1.2 1. Exact coordinate and raw-jet maps

Use the ring isomorphism

```
Psi:C[S]->C[u],  P |-> P(c+i u),  u |-> (S-c)/i.
```

Thus $d_u \Psi(P) = i \Psi(P')$, and multiplication by S remains $c+i M_u$. Define the original norm polynomial

```
Pi(u)=conj(chi)(c-i u) chi(c+i u).
```

Its leading coefficient is literally $(-i)^q i^q = 1$. It is real and nonnegative on $\mathbb{R}$. Reflection gives $\chi^\dagger = (-1)^q \chi$, hence

```
psi(u)=i^(-q) chi(c+i u) is real and monic,
Pi=psi^2,  Psi(chi)=i^q psi.
```

The factor i^q remains in the source multiplication map. For distinct roots λ_a of χ , of complete lengths ℓ_a , put

```
zeta_a=(lambda_a-c)/i,  r_a=2 ell_a,  r=sum r_a=2q.
```

Explicitly, the coordinate isomorphism Ψ induces $C[S]/\chi^2 \rightarrow C[u]/\Pi$ because $\Psi(\chi^2) = i^{2q} \Pi$, and the raw jet conversion is $(d_u^d \Psi(P))(zeta_a) = i^d P^{(d)}(\lambda_a)$. This is the actual map used below, including every derivative factor. Without the displayed dagger identity, Π need not equal ψ^2 and its roots need not be the original ζ_a with doubled orders; that unsupported general-packet version is not asserted here.

Define raw, NOT factorial-divided, derivative coordinates

```
J_2 f=(f^(d)(zeta_a))_(a, 0<=d<=r_a).
```

Their kernel is (Π) . Their product uses the full binomial rule. Multiplication by u has matrix

$$(X_{\Pi} v)_{(a,d)} = \zeta_a v_{(a,d)} + d v_{(a,d-1)}, \quad v_{(a,-1)} = 0.$$

All nilpotent orders and the coefficient d are retained.

Let Q_j be the real monic orthogonal source polynomials for $d\mu_{\theta}$, obtained by subtracting lower-degree projections, and write

$$\begin{aligned} \omega_j &= \int Q_j^2 d\mu_{\theta} > 0, \\ a_j &= \omega_j / \omega_{j-1}, \quad a_0 = 0, \\ b_j &= \int u Q_j^2 d\mu_{\theta} / \omega_j. \end{aligned}$$

The source recurrence is

$$u Q_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad Q_{-1} = 0.$$

No Q_j is divided by its norm. Set

$$\begin{aligned} v_j &= J_2 Q_j, \\ F_n &= [v_n, \dots, v_{n+r-1}], \\ V &= [J_2 1, \dots, J_2 u^{r-1}]. \end{aligned}$$

Both are r by r . The raw confluent determinant is

$$\det V = (\text{product}_a \text{product}_{(d=0)}^{(r_a-1)} d!) \text{product}_{(a < b)} (\zeta_b - \zeta_a)^{(r_a - r_b)}.$$

The fixed row order is part of this formula. $F_0 = V$ times the actual unit-triangular polynomial coefficient matrix; only their determinants are equal.

48.1.3 2. Invertibility and the original relation polynomial

Invertibility proof. Suppose F lies in $\text{span}(Q_n, \dots, Q_{n+r-1})$ and $J_2 F = 0$. Then $F = \Pi H$ with $\deg H < n$. Orthogonality gives

$$0 = \langle H, F \rangle = \int \Pi |H|^2 d\mu_{\theta}.$$

The integrand is nonnegative and its weight is positive almost everywhere except finitely many points. Thus $H=0$, then $F=0$. For $n=0$ the degree argument suffices. Hence every F_n is invertible. This does NOT estimate its inverse uniformly.

Solve

$$d_n = F_n^{(-1)} v_{n+r}, \quad \text{vartheta}_n = -(d_n)_0.$$

The polynomial

$$\text{qhat}_n = (Q_{n+r} - \sum_{j=0}^{r-1} (d_n)_j Q_{n+j}) / \Pi$$

exists by exact raw-jet divisibility and is monic of degree n . Pairing its numerator with every polynomial of degree below n proves that qhat_n is precisely the monic orthogonal polynomial for the ORIGINAL relation weight $\Pi d\mu_{\theta}$. Its norm is

$$\begin{aligned} \nu_n &= \int \text{qhat}_n^2 \Pi d\mu_{\theta} \\ &= -(d_n)_0 \omega_n = \omega_n \text{vartheta}_n > 0. \end{aligned} \quad (1)$$

Indeed only the Q_n column pairs with the degree- n quotient; its coefficient is $-(d_n)_0$. This proves $\text{vartheta}_n > 0$ with arbitrary nonreal and repeated nodes.

Let $K_n = [e_1, \dots, e_{r-1}, d_n]$. Then

$$\begin{aligned} F_{n+1} &= F_n K_n, \\ \det K_n &= (-1)^{(r-1)} (d_n)_0 = \text{vartheta}_n. \end{aligned} \quad (2)$$

The sign uses the displayed even $r=2q$. On the same jet algebra,

$$\begin{aligned} T_n &= F_n^{(-1)} X_{\Pi} F_n, \\ T_{n+1} &= K_n^{(-1)} T_n K_n, \\ d_n &= (T_n - b_{n+r-1} I) e_{(r-1)} - a_{n+r-1} e_{(r-2)}. \end{aligned} \quad (3)$$

Thus the full operator and its jets have an explicit discrete transport, rather than only a determinant observation.

48.1.4 3. Fixed-width determinant and quotient-volume formulas

Let D_n be the n by n source Hankel determinant, and B_n the corresponding determinant for Π dmu_{θ} . $D_0=B_0=1$. Orthogonality gives

$$D_n = \text{product}_{(j < n)} \omega_j, \quad B_n = \text{product}_{(j < n)} \nu_j.$$

Multiplying (1) and (2) proves the complete confluent Christoffel identity

$$B_n/D_n = \text{product}_{(j < n)} \vartheta_j = \det F_n / \det V. \quad (4)$$

This is a proof from the source division and norm, not a distinct-root approximation. Raw derivative factorials remain in $\det V$.

For $N \geq q-1$ set $n=N-q+1$. The source sequence is still

$$\emptyset \rightarrow P_{(N-q)} \xrightarrow{\chi} P_N \rightarrow C[S]/\chi \rightarrow \emptyset.$$

Its canonical least-norm quotient Gram G_N satisfies the existing Schur identity $\det G_N = D_{(N+1)}/B_n$. Consequently

$$V_N = \det G_N = (\text{product}_{(j=n)}^{(n+q-1)} \omega_j) \det V / \det F_n. \quad (5)$$

The constant coefficient map $S=c+i u$ has determinant of modulus one on each remainder basis, so these are the original quotient determinants, with that coordinate map retained.

For $n \geq 1$,

$$\begin{aligned} \delta_N &= V_N/V_{(N-1)} = \omega_N/\nu_{(n-1)} \\ &= \omega_N/(\omega_{(n-1)} \vartheta_{(n-1)}), \end{aligned} \quad (6)$$

$$\begin{aligned} R_N &= V_{(N-1)}/V_{(N+1)} \\ &= \omega_{(n-1)} \omega_n \vartheta_{(n-1)} \vartheta_n \\ &\quad / (\omega_N \omega_{(N+1)}). \end{aligned} \quad (7)$$

The transfer width $2q$ does not grow with N . The source recurrence and high-degree polynomial evaluation are still required; no computational-complexity or conditioning claim is inferred from the width alone.

48.1.5 4. One additional solve retains the full phase

Differentiating actual source orthogonality under $\exp(\theta u)$ gives

$$d_{\theta} Q_j = -a_j Q_{(j-1)}, \quad d_{\theta} \log \omega_j = b_j.$$

For the first identity, every coefficient below $j-1$ vanishes by differentiated orthogonality; pairing with $Q_{(j-1)}$ gives $-\omega_j$. Let L_n have entries $(L_n)_{(j-1,j)} = -a_{(n+j)}$ for $1 \leq j \leq r$, all other entries zero. Then

$$F_n' = F_n L_n - a_n v_{(n-1)} e_{\theta}^T. \quad (8)$$

At $n=0$ the last term is zero. Trace of L_n is zero. Hence

$$\begin{aligned} (\log \det F_n)' &= -a_n e_{\theta}^T F_n^{(-1)} v_{(n-1)}, \\ (\log V_N)' &= \sum_{(j=n)}^{(n+q-1)} b_j \\ &\quad + a_n e_{\theta}^T F_n^{(-1)} v_{(n-1)}. \end{aligned} \quad (9)$$

Retain $T=(A-cl)/i$ and $\sigma=\text{Tr } T$, which is real. The previous source cross term is exactly

$$(b_N^{*G_N} b_{(N+1)})/\omega_N=i(\sigma-(\log V_N)').$$

Here b_N in that formula denotes the remainder vector of the S-monic source polynomial, not the scalar recurrence b_j . The factor i and the distinction of types remain. The phase is $\phi_N=\sigma-(\log V_N)'$. At zero tilt for a full quartet, evenness gives $\phi_N=0$; it is not set to zero at a general tilt.

48.1.6 5. The positive gaps are norms of actual source vectors

Put

$$Z_j=\psi \hat{Q}_j-Q_{(j+q)} \text{ in } P_{(j+q-1)}.$$

The leading terms cancel. Source orthogonality therefore proves

$$\Delta_j:=||Z_j||^2=\nu_j-\omega_{(j+q)}>0. \quad (10)$$

The first term $\psi \hat{Q}_j$ is, through the explicit unit $i^{(-q)}$, an original chi relation. Its quotient is its supported zero. The difference Z_j instead has class $-[Q_{(j+q)}]$ in the original quotient. Its norm is not assigned to an absent element.

The SAME canonical rank-two control is $W_N=A^*G_N+G_N A-kG_N$. Combining its previously proved radius identity with (6) gives, for $n \geq 1$,

$$\epsilon_N^2=\Delta_{(n-1)}\Delta_n/(\nu_{(n-1)}\omega_N)-\phi_N^2. \quad (11)$$

Thus every factor in the upper bound

$$\epsilon_N \leq \sqrt{\Delta_{(n-1)}\Delta_n/(\nu_{(n-1)}\omega_N)}$$

comes from a specified positive source norm. Both gaps are computed by the transfer:

$$\begin{aligned} \Delta_{(n-1)} &= \omega_{(n-1)} \vartheta_{(n-1)} - \omega_N, \\ \Delta_n &= \omega_n \vartheta_n - \omega_{(N+1)}. \end{aligned}$$

The previous two-step bound is now

$$\epsilon_N \leq \sqrt{\omega_{(N+1)}/\omega_N} \cdot (R_N-1)/(2 \sqrt{R_N}), \quad (12)$$

with R_N from (7). At the first degree $N=q-1$, use the separate expression

$$\epsilon_{(q-1)}^2=(\nu_0-\omega_q)/\omega_{(q-1)}-\phi_{(q-1)}^2.$$

No inverse at the previous rank-deficient degree is introduced.

48.1.7 6. Keep the actual conormal quotient, not a repeated first ideal

The helper algebra for the norm computation is $C[u]/\Pi$, isomorphic through $S=c+iu$ to $C[S]/\chi_1^2$. The actual depth-two object is $C[S]/\chi_2$, defined using the literal pullback of the original I^2 . The merged conormal work gives $\chi_2 \mid \chi_1^2$. The actual maps are

$$C[S]/\chi_1^2 \twoheadrightarrow C[S]/\chi_2 \twoheadrightarrow C[S]/\chi_1. \quad (13)$$

Their kernels are respectively $(\chi_2)/(\chi_1^2)$ and $(\chi_1)/(\chi_2)$. In raw jets they truncate at the indicated full orders and intertwine multiplication by S . Differentiation into depth one also commutes, since $\chi_1 \mid \chi_2'$. The coordinate relation $d_u \Psi = i \Psi d_S$ is retained.

For $h=(s-\rho)^2$ and tensor degree two, $X=S-2\rho$ gives

$$C[X]/X^6 \twoheadrightarrow C[X]/X^5 \twoheadrightarrow C[X]/X^3.$$

The first kernel is the class of X^5 . The retained depth-two class X^3 has derivative $3X^2$ at depth one. These maps use one original scalar support. Their split lifts send quotient-killed classes to e at the receiving label and fix external τ .

The full multiplier $U = \text{product}_i(g/h)(s_i)$ retains the checked equation $\delta(U)P = U \delta(P) + (\delta U)P$. The second term is not discarded or presumed cyclic. The $2q$ -jet norm calculation is not counted as construction of new actual spectral multiplicities.

48.1.8 7. Residual-controlled numerical interface

For an actual F and a specified candidate inverse Y , suppose a justified bound gives $\|I - YF\|_\infty \leq \eta < 1$. For candidate $\hat{d}$ and residual $r = v - F \hat{d}$, the exact Neumann argument proves

$$\|\hat{d} - d\|_\infty \leq \|Y\|_\infty \|r\|_\infty / (1 - \eta). \quad (14)$$

This bounds the first coordinates used in (1) and (9). Arithmetic application requires certified input enclosures for the moments, source polynomials and zeros. Floating-point output alone is not a certificate.

Positive enclosures of ω and ϑ bound (7). Since $(R-1)^2/(4R)$ is increasing for $R \geq 1$, (12) yields a valid upper enclosure of ϵ^2 . The full phase can improve it through (11). Finite positivity/invertibility is not substituted for a uniform estimate of these residuals or inverse norms.

48.1.9 8. Exact calibration and analytic target

For a declared Gaussian measure of literal mass seven and variance one, $\psi = u^2 + 1$, $q = 2, n = 2, N = 3$, the calculation gives

```
(omega_0, ..., omega_4) = (7, 7, 14, 42, 168),
vartheta_1 = 22, vartheta_2 = 86/3,
Qhat_1 = u, Qhat_2 = u^2 - 11/3,
Delta_1 = 112, Delta_2 = 700/3,
G_3 = diag(7/3, 21/11), epsilon_3^2 = 400/99.
```

An independent quotient-Gram calculation with $A = (1/2)I + i M_u$ gives the same result. The nonreal u -roots are algebraic fixtures, not zeros of ξ ; the nonzero allowance explicitly tests that the method does not manufacture purity. Repeated nodes, complex quartets, asymmetric atomic tilt derivatives and mass changes are covered by the expanded checks.

For the actual quartet retain

```
q_k = [1 + k(m-1)](k+1)^2,
L_(h,k) = 2 delta [1 + k(m-1)](k+1) floor((k+1)^2/4).
```

The source exterior/filtration result gives $L_{(h,k)} \leq \epsilon_{(h,k,N)}$, and $L_{(h,k)}/(k q_k) \geq \delta/2$. Thus $\epsilon_{(h,k)} = o(k q_k)$, with fixed h and admitted $N \geq q_k$, is sufficient for the intended contradiction. The new representation attaches that exact target to TWO consecutive source-to-relation transfer coefficients and ONE phase solve. It has NOT proved that asymptotic bound.

No new Lean execution or arithmetic interval certificate is claimed. All new mathematical claims have the written proofs above at the retained source inputs. The original arithmetic function, unit, action, θ relations, residue pairing and trace remain unchanged.

48.1.10 References and checks

Classical background: C. Krattenthaler (2021), 'A determinant identity for moments of orthogonal polynomials that implies Uvarov's formula for the orthogonal polynomials of rationally related densities', arXiv:2103.03969; NIST DLMF Section 18.2. The proofs above supply the particular confluent and arithmetic interfaces rather than attributing them wholesale to those references.

The expanded conversation delivery includes the complete LaTeX/HTML reader and twenty exact finite regression methods; both normal and optimized runs passed with matching records, and both deliberate-failure controls failed. The preceding Toda archive's 35 manifest entries verified and its 24-method checker

passed fresh normal and optimized runs. These are stated verification scopes, not new Lean certificates. The public core checker in this directory independently tests the principal raw-jet, transfer, norm, determinant, volume and phase interfaces.

49 An arithmetic lower-envelope comparison for central quotient volumes: full jets, finite determinant bounds, and an exact quadratic benchmark

This calculation constructs upper bounds for the original central volume $\log(V_q/V_{2q-1})$. Its input is the proved lower envelope and Laplace estimate of the complete *Arithmetic Endpoint Bounds* source, equations (29)–(32). The comparison acts on the original polynomial and quotient maps, including all multiplicities and phases. The strongest displayed bound retains an explicit positive matrix: its arithmetic moments stop at degree $2q$, while its comparison moments continue to $4q - 2$. A second bound uses only the supplied envelope constants and two Laplace values. The explicit scalar estimate has quadratic, rather than logarithmic, growth per quotient direction. An exact auxiliary calculation determines that scale and supplies a monic family relating its full jets to the original quartet jets. No bound contradicting the quartet lower estimate is asserted.

49.1 Original source and every jet coordinate

Fix a complete quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$, and common complete zero order $m \geq 1$ of $g = 2\xi$. Keep its monic polynomial h , its entire quotient $v_h = g/h$, and

$$w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad k \geq 3. \quad (\text{AU.1})$$

Write $c = k/2$ and keep the original observation $S = c + iu$. The source form is $\int \overline{P(c + iu)} Q(c + iu) m_k(u) du$, conjugate-linear in its first argument. Its constant norm is μ_h^k . The coefficient map and its inverse are

$$\Psi : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad P \longmapsto P(c + iu), \quad f \longmapsto f((S - c)/i). \quad (\text{AU.2})$$

These are inverse ring homomorphisms by substitution. Multiplication by S is carried to $c + iM_u$, and $\partial_u^d \Psi(P) = i^d \Psi(P^{(d)})$. On the ordered degree- N coefficient spaces the matrix is triangular with diagonal $1, i, \dots, i^N$, so its determinant has modulus one. The original quotient volumes therefore have exactly the same values in the transported remainder basis; no mass or norm is divided out.

Let $\ell = 1 + k(m - 1)$ and use the ordered distinct grid

$$\lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \zeta_{ab} = (\lambda_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{AU.3})$$

The original cyclic relation and its transported monic polynomial are

$$q = \ell(k + 1)^2, \quad \chi(S) = \prod_{a,b} (S - \lambda_{ab})^\ell, \quad \psi(u) = i^{-q} \chi(c + iu) = \prod_{a,b} (u - \zeta_{ab})^\ell. \quad (\text{AU.4})$$

The grid is invariant under complex conjugation in the u chart, so ψ is real monic. Equivalently the original coefficient identity is $\overline{\chi}(k - S) = (-1)^q \chi(S)$. In particular the original norm multiplier is $\overline{\chi}(c - iu) \chi(c + iu) = \psi(u)^2$, with the source multiplication still carrying the unit i^q .

For clarity the preceding cyclic degree also follows from the original tensor algebra. In a tensor of k length- m local factors, the sum of the k nilpotent generators has power $k(m - 1)$ nonzero: its coefficient on the product of their top powers is $(k(m - 1))!/((m - 1)!)^k$, nonzero in $\mathbb{C}$. Its next power is zero by total degree. Every grid sum in (AU.3) occurs, since the two signs in each quartet

factor can be chosen independently. Direct products of local factors give the same maximal length ℓ at every grid sum. Distinct pairs have distinct real and imaginary parts. The minimal polynomial of the original sum is therefore exactly (AU.4), with no lost multiplicity.

Put $\mathcal{E} = \mathbb{C}[u]/(\psi)$, with ordered remainder coefficients $1, u, \dots, u^{q-1}$. For $N \geq q - 1$ let J_N be remainder and B_N multiplication by ψ in the ordered monomials. Monic division gives the exact sequence

$$0 \longrightarrow \mathbb{C}[u]_{\leq N-q} \xrightarrow{B_N} \mathbb{C}[u]_{\leq N} \xrightarrow{J_N} \mathcal{E} \longrightarrow 0. \quad (\text{AU.5})$$

The source on the left is zero if $N = q - 1$. Precisely, the coefficient isomorphisms are $\iota_N : \mathbb{C}^{N+1} \rightarrow \mathbb{C}[u]_{\leq N}$, $a \mapsto \sum_{j=0}^N a_j u^j$, and $\kappa : \mathbb{C}^q \rightarrow \mathcal{E}$, $x \mapsto [\sum_{j < q} x_j u^j]$. The polynomial quotient map is $\kappa J_N \iota_N^{-1}$ and the polynomial multiplication map is $\iota_N B_N \iota_N^{-1}$. When a matrix formula is evaluated on a polynomial below, these displayed coefficient isomorphisms are the specified identifications. The original S chart uses the same coefficient isomorphisms with u^j replaced by S^j . The complete raw derivative map is

$$\mathbf{E}_N f = (f^{(d)}(\zeta_{ab}))_{a,b, 0 \leq d < \ell}, \quad (\mathbf{E}_N)_{(a,b,d),j} = \begin{cases} j! \zeta_{ab}^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d. \end{cases} \quad (\text{AU.6})$$

Taylor expansion at each distinct centre shows that its kernel is $(\psi) \cap \mathbb{C}[u]_{\leq N}$. Let $\mathbf{Z} = \mathbf{E}_{q-1}$. Then $\mathbf{E}_N = \mathbf{Z} J_N$ and $\mathbf{Z}$ is invertible. Indeed a degree below q polynomial with all those zero raw jets is divisible by degree- q ψ and is zero. For the fixed row order,

$$\det \mathbf{Z} = \left(\prod_{a,b} \prod_{d=0}^{\ell-1} d! \right) \prod_{(a,b) < (a',b')} (\zeta_{a'b'} - \zeta_{ab})^{\ell^2}. \quad (\text{AU.7})$$

One proof first regards all roots as distinct variables in the ordinary Vandermonde determinant, groups each cluster, and replaces its rows by successive divided differences. Each row operation divides by exactly the corresponding within-cluster difference. Sending the nodes in a cluster to its centre replaces its divided-difference row of order d by $f^{(d)}/d!$, by the finite Taylor expansion of each monomial. Multiplying those rows by $d!$ proves (AU.7). This is an identity of polynomials after removal of the indicated factors; no source norm or zero multiplicity is approximated. In the original S coordinates the same raw jet of order d has its literal factor i^d from (AU.2).

For any positive source matrix H_N , the least quotient form is

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \quad R_N = H_N^{-1} J_N^* G_N. \quad (\text{AU.8})$$

Surjectivity proves $K_N > 0$. Multiplication proves $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Thus every lift $R_N x + B_N y$ has norm $x^* G_N x + y^* B_N^* H_N B_N y$, proving the minimum property. In raw jets the kernel is $\mathbf{K}_N = \mathbf{E}_N H_N^{-1} \mathbf{E}_N^*$, and $\det \mathbf{K}_N = |\det \mathbf{Z}|^2 / V_N$. Hence

$$C_k := \log(V_q / V_{2q-1}) = \log \det \mathbf{K}_{2q-1} - \log \det \mathbf{K}_q. \quad (\text{AU.9})$$

All factorials in (AU.7) cancel only in this specified ratio.

49.2 The lower envelope as an explicit positive comparison form

The supplied arithmetic proof establishes $c_h > 0$, $\vartheta_h = \int_{-1}^1 w_h > 0$, $B = 42 + 2 \deg h = 42 + 8m$, and

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad \alpha = \pi/2. \quad (\text{AU.10})$$

Set $a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}$ and $d_k = k - 2 > 0$. Define the comparison measure, retaining its literal mass, by

$$d\nu_k(u) = a_k e^{-\alpha|u|} (d_k + |u|)^{-B} du. \quad (\text{AU.11})$$

The identity map $L^2(m_k du) \rightarrow L^2(\nu_k)$ is well defined and a contraction, since (AU.10) gives the squared-norm inequality for every measurable function. It restricts to the identity on each polynomial space and commutes with multiplication by $c + iu$, remainder (AU.5), and every raw jet (AU.6). The comparison is therefore a map on the same quotient, not a declaration that the measures coincide.

Every moment of ν_k is finite, and the following integral evaluates it with positive integrands:

$$\nu_{k,j} := \int_{\mathbb{R}} u^j d\nu_k(u) = \begin{cases} \frac{2a_k j!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-d_k t}}{(\alpha+t)^{j+1}} dt, & j \text{ even}, \\ 0, & j \text{ odd}. \end{cases} \quad (\text{AU.12})$$

To prove it, repeated integration by parts gives $\int_0^\infty t^{B-1} e^{-yt} dt = (B-1)! y^{-B}$ for $y > 0$. Insert $y = d_k + u$ on the positive half-line. Tonelli applies because each integrand is nonnegative there. Integrating $u^j e^{-(\alpha+t)u}$ gives $j!(\alpha+t)^{-j-1}$ by the same integration by parts. Evenness supplies the factor two or the vanishing odd moment. The bound $\nu_{k,j} \leq 2a_k d_k^{-B} j! \alpha^{-j-1}$ for even j also proves finiteness directly. In particular $\nu_{k,0} > 0$ is retained, and generally differs from μ_h^k .

Let $H_N^\nu = (\nu_{k,i+j})_{0 \leq i,j \leq N}$. Its quadratic form is the ν_k integral of the polynomial's squared modulus. The weight is strictly positive everywhere, so a nonzero polynomial has positive norm. Thus $0 < H_N^\nu \preceq H_N$. Minimizing on the identical affine fibres in (AU.5) proves

$$0 < G_N^\nu \preceq G_N, \quad \mathbf{K}_N \preceq \mathbf{K}_N^\nu, \quad \mathbf{K}_N^\nu = \mathbf{E}_N (H_N^\nu)^{-1} \mathbf{E}_N^*. \quad (\text{AU.13})$$

The inverse inequality follows also by conjugating H_N by $(H_N^\nu)^{-1/2}$ and inverting its positive eigenvalues.

Theorem 49.1 (Actual low-degree data control the higher quotient volume). *For the original quartet source and $k \geq 3$, the finite upper bound is*

$$0 \leq C_k \leq \log \det \mathbf{K}_{2q-1}^\nu - \log \det \mathbf{K}_q. \quad (\text{AU.14})$$

Its actual arithmetic inputs are the moments through degree $2q$ and the stated envelope constants. The comparator moments through $4q - 2$ are the explicit positive integrals (AU.12).

Proof. The first inequality follows by minimizing the same quotient norm in successively larger polynomial spaces. For the second use (AU.13) at $N = 2q - 1$ and (AU.9). Determinant is increasing on positive Hermitian forms: conjugate by the square root of the smaller one and multiply eigenvalues at least one. H_q uses precisely the original moments $0, \dots, 2q$, and all the matrices in the second term of (AU.14) have already specified finite algebraic maps from them. No higher actual moment enters. Equivalently, if $\mathfrak{D}_j^\nu = \det H_{j-1}^\nu$ and $\mathfrak{B}_j^\nu$ is the Gram determinant of the original ψ times $1, u, \dots, u^{j-1}$, the quotient-first coefficient matrix $[1, u, \dots, u^{q-1}, \psi, u\psi, \dots]$ is triangular with diagonal one. Orthogonal elimination of its relation block gives

$$V_{2q-1}^\nu = \mathfrak{D}_{2q}^\nu / \mathfrak{B}_q^\nu, \quad C_k \leq \log V_q + \log \mathfrak{B}_q^\nu - \log \mathfrak{D}_{2q}^\nu. \quad (\text{AU.15})$$

This proves the same upper bound in the exact source/relation determinants. Both the source mass and relation multiplication factors remain in their respective determinants. $\square$

The existing Gamma coefficient map also makes the cutoff in (AU.14) explicit. With $\lambda > 0$, $\alpha_\lambda = 2\lambda$, $c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda)$ and $C_\lambda = c_\lambda^k$, define the original monic Gamma polynomials by

$$\sum_{j \geq 0} b_j^{(\lambda)}(u) z^j / j! = (1 + z^2)^{-\lambda} e^{u \arctan z}, \quad c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha_\lambda)_j}.$$

Multiplying the k generating functions and taking finite derivatives at zero gives

$$d_l = \sum_{j_1+\dots+j_k=l} \prod_a (\alpha_\lambda)_{j_a} c_{h,j_a}, \quad \int b_l^{(k\lambda)} m_k = C_\lambda l! d_l. \quad (\text{AU.16})$$

All finite terms are integrable by the supplied exponential moments. Expand $u^{i+j} = \sum_{l \leq i+j} L_{ij,l} b_l^{(k\lambda)}(u)$ by monic subtraction. Then $(H_q)_{ij} = C_\lambda \sum_l l! L_{ij,l} d_l$. Thus the actual coefficient list $c_{h,0}, \dots, c_{h,2q}$ suffices for (AU.14). At zero degree it gives $C_\lambda c_{h,0}^k = \mu_h^k$, retaining the source mass.

49.3 An upper determinant depending only on the two Laplace values

Fix $0 < b < \pi/2$ and set $M_k = M_h(b)^k + M_h(-b)^k$, with the original $M_h(\pm b) = \int e^{\pm bu} w_h(u) du$. Positivity and convolution give $\int e^{b|u|} m_k(u) du \leq M_k$. The exponential series therefore gives $\int |u|^{2j} m_k(u) du \leq M_k (2j)! b^{-2j}$. For arbitrary $\tau_0, \dots, \tau_q > 0$ define

$$D_q(\tau) = M_k \left(\sum_{j=0}^q \tau_j \right) \text{diag} \left(\frac{(2j)!}{b^{2j} \tau_j} \right)_{j=0}^q. \quad (\text{AU.17})$$

For $f = \sum a_j u^j$, Cauchy–Schwarz applied to $\sqrt{\tau_j}$ and $a_j u^j / \sqrt{\tau_j}$ proves

$$|f(u)|^2 \leq \left(\sum \tau_j \right) \sum |a_j|^2 |u|^{2j} / \tau_j.$$

Integration proves $H_q \preceq D_q(\tau)$. Inverting and applying the same full jet map proves

$$\mathbf{K}_q \succeq \frac{1}{M_k \sum \tau_j} \mathbf{A}_q(\tau), \quad \mathbf{A}_q(\tau) = \mathbf{E}_q \text{diag} \left(\frac{\tau_j b^{2j}}{(2j)!} \right) \mathbf{E}_q^* > 0. \quad (\text{AU.18})$$

Consequently an upper bound without any unevaluated actual polynomial moment is

$$C_k \leq q \log \left(M_k \sum_{j=0}^q \tau_j \right) + \log \det \mathbf{K}_{2q-1}^\nu - \log \det \mathbf{A}_q(\tau). \quad (\text{AU.19})$$

Every positive choice of τ gives a proved bound, and the infimum over such choices remains a bound. Multiplying all τ_j by the same positive number cancels between the two written determinant factors; it changes neither measure. No optimizer or numerical enclosure is assumed in (AU.19).

The lower determinant has the exact positive expansion

$$\det \mathbf{A}_q(\tau) = \sum_{\substack{I \subset \{0, \dots, q\} \\ |I|=q}} |\det(\mathbf{E}_q)_{[:,I]}|^2 \prod_{j \in I} \frac{\tau_j b^{2j}}{(2j)!}. \quad (\text{AU.20})$$

Expand each determinant multilinearly in columns in the product $\mathbf{E}_q D \mathbf{E}_q^*$: terms with repeated column indices vanish by alternation, and grouping permutations of the same subset gives (AU.20). Its subset $I = \{0, \dots, q-1\}$ is strictly positive by (AU.7). This expression retains full raw-jet minors and factorials without estimating an inverse by its individual entries.

49.4 The interval kernel and the actual comparison map

For $T > 0$, set

$$a_k(T) = a_k e^{-\alpha T} (d_k + T)^{-B}. \quad (\text{AU.21})$$

Restriction to $[-T, T]$ followed by the identity on polynomials maps the actual source to $L^2([-T, T], a_k(T)du)$ contractively. The same is true from $L^2(\nu_k)$, by the pointwise lower bound on that interval. Let $H_N^I(T)$ be the Lebesgue interval Gram. Then $a_k(T)H_N^I(T) \preceq H_N^\nu \preceq H_N$. Let P_j denote the Legendre polynomial specified by Rodrigues' formula, $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Integration by parts j times proves its orthogonality to lower degrees; direct evaluation of the resulting beta integral gives its squared norm $2/(2j+1)$. Its highest coefficient is $(2j)!/(2^j (j!)^2)$. The endpoint boundary terms vanish because $(x^2 - 1)^j$ has zeros of order j at both endpoints. Applying these constants to $P_j(u/T)$ gives the full interval jet kernel

$$(\mathbf{K}_N^I(T))_{(a,b,d),(a',b',e)} = \sum_{j=0}^N \frac{2j+1}{2T^{1+d+e}} P_j^{(d)}(\zeta_{ab}/T) \overline{P_j^{(e)}(\zeta_{a'b'}/T)}. \quad (\text{AU.22})$$

This follows by writing the inverse Gram in the orthogonal Legendre basis and applying (AU.6); each derivative contributes its stated T^{-d} . In particular

$$\mathbf{K}_{2q-1} \preceq \mathbf{K}_{2q-1}^\nu \preceq a_k(T)^{-1} \mathbf{K}_{2q-1}^I(T). \quad (\text{AU.23})$$

Equations (AU.14) and (AU.19) remain valid if their $\log \det \mathbf{K}_{2q-1}^\nu$ is replaced by $\log \det \mathbf{K}_{2q-1}^I(T) - q \log a_k(T)$. The global envelope version is at least as strong as each such interval version by the proved positive-matrix inequality (AU.23).

The restriction map has the exact orthogonal decomposition $f \mapsto (f1_{[-T,T]}, f1_{\mathbb{R} \setminus [-T,T]})$ in the original weighted L^2 source; its two squared norms add to the original norm. The interval bound drops a nonnegative term in that equality. It does not assign the outside component to external absence.

49.5 The explicit scalar bound and its finite gap

The following explicit scalar estimate is the independent full calculation VC.1–43, composed here with the preceding original quotient maps. Put $R_0 = \sqrt{\delta^2 + \gamma^2}$ and define

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = kR_0/T \leq 1/2, \quad N = 2q - 1, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \quad A_N = \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \\ B_q(r) &= \left(\frac{1+r}{1-r} \right)^q, \quad \mathcal{C}_{k,q} = \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{AU.24})$$

All values are positive. In particular $q \geq k$ and $D \geq 2R_0$ give the stated bound on r . These choices bound constants; they do not change the original density.

Theorem 49.2 (A full finite Hermitian upper bound). *In the transported ordered quotient coordinates, and by the explicit congruence below also in the original S coordinates,*

$$\boxed{K_q \preceq K_{2q-1} \preceq \mathcal{C}_{k,q} K_q, \quad 0 \leq C_k \leq q \log \mathcal{C}_{k,q}.} \quad (\text{AU.25})$$

Proof. Let $\mathbf{T}_j$ be the coefficient matrix of the original map $\Psi(P) = P(c + iu)$ in degree j . Write $H_j^S, J_j^S, K_j^S, G_j^S$ for matrices in the original S monomials. Then

$$H_j^S = \mathbf{T}_j^* H_j \mathbf{T}_j, \quad J_j \mathbf{T}_j = \mathbf{T}_{q-1} J_j^S, \quad K_j = \mathbf{T}_{q-1} K_j^S \mathbf{T}_{q-1}^*.$$

The first identity is the source substitution, the second is monic division, and substitution in (AU.8) gives the third. Each $\det \mathbf{T}_j$ has modulus one, as proved in (AU.2). All source norms on S polynomials in this proof use H_j^S . The exact auxiliary coordinate map used in this proof is $\Psi_T P(x) = P(c + iTx)$,

with inverse $f((S - c)/(iT))$. Its target measure is $Tm_k(Tx)dx$ and retains mass μ_h^k by substitution. The target monic relation is $\chi_T(x) = (iT)^{-q}\chi(c + iTx)$, so $\Psi_T(\chi Q) = (iT)^q\chi_T\Psi_T Q$. Monic division therefore proves $\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_{\chi} P$. At every original centre λ , the full order- d raw jet is $(\Psi_T P)^{(d)}((\lambda - c)/(iT)) = (iT)^d P^{(d)}(\lambda)$. The coefficient matrix has diagonal $(iT)^j$ at degree j ; its quotient determinant is $(iT)^{q(q-1)/2}$. Consequently the target quotient form is the original form transported by inverse congruence, and its determinant is $T^{-q(q-1)}V_j$ at every source degree j . This factor cancels in the central ratio, with all phases and masses still present in the map.

For a polynomial $f(x) = \sum f_j x^j$ put $\|f\|_{\text{coeff},1} = \sum |f_j|$. The Rodrigues formula above, expanded by the product rule, gives

$$P_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a.$$

The coefficient norm of a product is at most the product of the two coefficient norms by convolution and the triangle inequality. Since $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$, this proves $\|P_j\|_{\text{coeff},1} \leq \sum_a \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j$. The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$. In the orthogonal expansion $f = \sum_{j=0}^N f_j^L P_j$, Cauchy-Schwarz and the squared norm $2/(2j+1)$ give $|f_j^L| \leq \sqrt{(2j+1)/2} \|f\|_{L^2([-1,1])}$. Summing 4^j therefore proves

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}, \quad \deg f \leq N. \quad (\text{AU.26})$$

List all roots $z_1, \dots, z_q$ of χ_T , with repetitions; each has modulus at most r . Define e_a, h_j by

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j.$$

The second identity is formal multiplication of geometric series. It gives $|e_a| \leq \binom{q}{a} r^a$ and $|h_j| \leq \binom{q+j-1}{j} r^j$: the latter counts exponent lists of total degree j by their $q-1$ separating bars. For $0 \leq s \leq q-1$, multiply those inverse series through order s . It proves that the quotient of x^{q+s} by χ_T is exactly $Q_s = \sum_{j=0}^s h_j x^{s-j}$ and its remainder is $x^{q+s} - \chi_T Q_s$, of degree below q . Removing the leading term of $\chi_T Q_s$ removes precisely one from its coefficient norm. All its remaining coefficients are the negative remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \\ &\leq ((1+r)/(1-r))^q = B_q(r). \end{aligned}$$

At degrees below q the remainder is the original monomial and has coefficient norm one, also bounded by $B_q(r)$. Linearity proves

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{AU.27})$$

Every root was retained in this proof. Its smaller literal finite constant $\max\{1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1\}$ can also replace $B_q(r)$.

For v of degree at most $q-1$, each monomial is bounded by $\max\{1, |x|^{q-1}\}$. Using $\max(1, y) \leq 1+y$ for $y \geq 0$ and the proved Laplace moment bound gives

$$\int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2. \quad (\text{AU.28})$$

On the other hand, restriction to $[-T, T]$ and (AU.21) prove $\|P\|_{H_N^S}^2 \geq T a_k(T) \|\Psi_T P\|_{L^2([-1,1])}^2$. Take $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (AU.26)–(AU.28) and the exact division map now give

$$\|\text{rem}_{\chi} P\|_{H_{q-1}^S}^2 \leq C_{k,q} \|P\|_{H_N^S}^2. \quad (\text{AU.29})$$

The map estimated here is $s\pi_{\chi,N} : \mathcal{P}_N^S \rightarrow \mathcal{P}_{q-1}^S$, where $\pi_{\chi,N}$ is the polynomial quotient map and $s : \mathbb{C}[S]/(\chi) \rightarrow \mathcal{P}_{q-1}^S$ selects the unique remainder representative. Their exact identities are $\pi_{\chi,q-1}s = I$ and $\pi_{\chi,q-1}s\pi_{\chi,N} = \pi_{\chi,N}$. The composite $s\pi_{\chi,N}$ equals the identity on polynomials of degree below q .

Apply (AU.29) to the least lift $R_N^S x$ from the original matrices. Its remainder lifts the same x already at degree $q-1$, hence $x^* G_q^S x \leq x^* G_{q-1}^S x \leq \mathcal{C}_{k,q} x^* G_N^S x$. Conversely degree inclusion gives $G_N^S \preceq G_q^S$. Conjugating by $(G_q^S)^{-1/2}$, bounding its positive eigenvalues and then inverting proves (AU.25), including the determinant exponent q , first for K_j^S and then by the displayed congruence for K_j . It also proves $\mathcal{C}_{k,q} \geq 1$. $\square$

Retain the fixed constants

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_{\text{env}} = \max\{1, 2/(3c_h D)\}.$$

Because $bT \geq 2q$, the factorial ratio in (AU.24) is at most one. Also $M_k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that $U_k(T) \leq 3X^k$. The displayed A_N satisfies $A_{2q-1}^2 \leq (2q/9)256^q$. On $[0, 1/2]$ the derivative of $\log((1+r)/(1-r))$ is $2/(1-r^2) \leq 8/3$, so

$$\log B_q(r)^2 \leq \frac{16}{3}qr = \frac{16R_0}{3D}k. \quad (\text{AU.30})$$

Use also $\vartheta_h^{-(k-3)} \leq K^k$, $1 + Dq + k - 3 \leq 1 + (D+1)q$, and $e^{-3\alpha} \leq 1$. Substitution in (AU.24)–(AU.25) proves the finite scalar bound

$$\begin{aligned} 0 \leq C_k \leq \mathcal{U}_k &:= (\alpha D + \log 256)q^2 \\ &\quad + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &\quad + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}. \end{aligned} \quad (\text{AU.31})$$

All of its constants are independent of k and q for fixed h, b . The exact finite bounds (AU.14), (AU.19), and $q \log \mathcal{C}_{k,q}$ can be retained together by taking their minimum; each independently bounds the same C_k with the proved maps.

The complete central lower estimate in the balanced-window/factor-four source is $C_k \geq \mathcal{L}_k = 2(q-1)(\log k + \log(\delta/(2C_h^{\text{bal}})))$. The current explicitly computed gap is the literal function

$$\begin{aligned} \mathcal{U}_k - \mathcal{L}_k &= (\alpha D + \log 256)q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &\quad + Bq \log(1 + (D+1)q) + q \log C_{\text{env}} \\ &\quad - 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \end{aligned} \quad (\text{AU.32})$$

This quantity concerns the two displayed upper and lower estimates. It is not an asserted exact value of the unknown C_k . Since $q = \ell(k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. Dividing every term of (AU.31)–(AU.32) by q^2 proves

$$\lim_{k \rightarrow \infty} \frac{\mathcal{U}_k - \mathcal{L}_k}{q_k^2} = \alpha D + \log 256 > 0, \quad \limsup_{k \rightarrow \infty} C_k/q_k^2 \leq \alpha D + \log 256. \quad (\text{AU.33})$$

Dividing the upper estimate by $q \log k$ gives the exact leading factor $(\alpha D + \log 256)q/\log k$. It grows as $(\alpha D + \log 256)k^2/\log k$ for $m = 1$ and as $(\alpha D + \log 256)(m-1)k^3/\log k$ for $m > 1$. The division step itself has only the linear-in- k logarithmic cost (AU.30). The two quadratic contributions in this scalar proof come from the full coefficient estimate (AU.26) and the least density over the interval in (AU.21). The positive matrices in (AU.14), (AU.19), and (AU.22) retain the directional information discarded in those scalar estimates, with their exact source and quotient maps already proved. No upper asymptotic for those sharper determinants is inserted.

49.6 A full-jet quadratic benchmark and its exact relation family

Here the auxiliary source is explicitly Lebesgue measure on $[-1, 1]$ and the auxiliary relation is u^q . Its mass is two. These data are inserted into the same sequence (AU.5), kernel (AU.8), and central ratio (AU.9). No assertion that they are an arithmetic density is made. Write D_d for its d by d monomial determinant, $D_0 = 1$, and $B_j(q)$ for the j by j determinant of the relation measure $u^{2q}du$ on that interval. The quotient-first triangular determinant calculation proving (AU.15) gives exactly

$$\frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} = \frac{D_{q+1}B_q(q)}{B_1(q)D_{2q}}, \quad B_1(q) = \frac{2}{2q+1}. \quad (\text{AU.40})$$

The orthogonal Legendre calculation gives

$$D_d = \prod_{j=0}^{d-1} \frac{2}{2j+1} \left(\frac{2^j}{\binom{2j}{j}} \right)^2. \quad (\text{AU.41})$$

For $q = 2s$, parity splits the relation Gram into two s by s blocks, and

$$B_{2s}(2s) = \mathcal{C}_s(2s+1/2)\mathcal{C}_s(2s+3/2), \quad \mathcal{C}_s(x) = \frac{\prod_{j=0}^{s-1} (j!)^2}{\prod_{a,b=0}^{s-1} (x+a+b)}. \quad (\text{AU.42})$$

Indeed the even block has entries $1/(2s+a+b+1/2)$ and the odd block entries $1/(2s+a+b+3/2)$. The Cauchy determinant formula used here has numerator $\prod_{a<b} (b-a)^2 = \prod_j (j!)^2$. It follows by multiplying the determinant by all its denominators: the numerator is alternating separately in its two lists of variables, so contains their two Vandermonde factors. Its degree is their total degree, and comparing one residue inductively gives constant one. Substitution gives (AU.42) with no absolute-value or sign ambiguity.

Theorem 49.3 (An explicit quadratic scale in the full-origin jet fibre). *For every even integer $q \geq 2$,*

$$\left| \log \frac{V_q^{\text{aux}}}{V_{2q-1}^{\text{aux}}} - \kappa_* q^2 \right| \leq 12q \log(8q), \quad \kappa_* := \frac{9}{2} \log 3 - 6 \log 2 = \frac{3}{2} \log(27/16) > 0. \quad (\text{AU.43})$$

Proof. The elementary binomial bounds $4^j/(2j+1) \leq \binom{2j}{j} \leq 4^j$ hold because the central coefficient is the largest of the $2j+1$ coefficients summing to 4^j . Therefore (AU.41) gives

$$\log D_d = -d(d-1) \log 2 + E_d, \quad |E_d| \leq d \log(4d) \quad (d \geq 1). \quad (\text{AU.44})$$

At $s \geq 1$, expand the numerator of $\mathcal{C}_s(2s)$ as $2 \sum_{i=1}^{s-1} (s-i) \log i$. The function $f(x) = (1-x) \log x$ is increasing on $(0, 1]$, has integral $-3/4$, and is nonpositive. Its right Riemann sum differs from its integral by a number between zero and $(\log s + 1)/s$: the difference is bounded above by $-\int_0^{1/s} f(x) dx \leq (\log s + 1)/s$, using $-f(x) \leq -\log x$. It follows that this numerator is $s^2 \log s - 3s^2/2$ plus an error of absolute value at most $s \log s + 2s$.

The denominator's logarithm is $s^2 \log s + \sum_{a,b=0}^{s-1} \log(2+a/s+b/s)$. Both partial derivatives of the latter function lie in $[0, 1/2]$. Its left-square Riemann sum differs from its integral by at most $1/s$, hence the sum by at most s . Two integrations give

$$\int_0^1 \int_0^1 \log(2+x+y) dx dy = 18 \log 2 - 9 \log 3 - 3/2.$$

For example use the second antiderivative $F(t) = t^2 \log t/2 - 3t^2/4$ and calculate $F(4) - 2F(3) + F(2)$. Thus

$$|\log \mathcal{C}_s(2s) - 9s^2 \log(3/4)| \leq s \log s + 3s.$$

For $c > 0$, the change to $\mathcal{C}_s(2s + c)$ has logarithm $-\sum_{a,b} \log(1 + c/(2s + a + b))$, between $-cs/2$ and zero. Using $c = 1/2, 3/2$ in (AU.42) proves

$$\left| \log B_q(q) - \frac{9}{2}q^2 \log(3/4) \right| \leq q \log(q/2) + \frac{7}{2}q. \quad (\text{AU.45})$$

Insert (AU.44)–(AU.45) into (AU.40). The quadratic coefficient is $3 \log 2 + (9/2) \log(3/4) = \kappa_*$. The remaining absolute error is at most

$$3q \log 2 + (q + 1) \log(4(q + 1)) + 2q \log(8q) \\ + q \log(q/2) + \frac{7}{2}q + \log((2q + 1)/2).$$

For $q \geq 2$, use $q + 1 \leq 3q/2$, $4(q + 1) \leq 8q$, $\log(q/2) \leq \log(8q)$ and $\log((2q + 1)/2) \leq \log(8q)$. The two constant multiples of q are bounded together by $3q \log(8q)$, since $3 \log(16) > 3 \log 2 + 7/2$. Their sum is then at most $8q \log(8q)$, hence certainly the displayed $12q \log(8q)$. This proves (AU.43) for every indicated finite q , not only along a limit. $\square$

There is an exact algebraic relation of this benchmark's quotient to every original grid quotient. Define

$$\psi_t(u) = \prod_{a,b} (u - t\zeta_{ab})^\ell, \quad \mathcal{A} = \mathbb{C}[t, u]/(\psi_t(u)). \quad (\text{AU.46})$$

Monic division over $\mathbb{C}[t]$ proves that $1, u, \dots, u^{q-1}$ form a free $\mathbb{C}[t]$ basis: division gives spanning and a nonzero multiple of a monic degree- q polynomial cannot have smaller degree. Every fibre has dimension q . The fibre at $t = 1$ is the original transported quotient; at $t = 0$ it is $\mathbb{C}[u]/u^q$. For $t \neq 0$,

$$\mathcal{A}_t \xrightarrow{\sim} \mathcal{A}_1, \quad [f(u)] \mapsto [f(tu)], \quad [g(u)] \mapsto [g(u/t)] \quad (\text{AU.47})$$

are inverse maps, since $\psi_t(tu) = t^q \psi_1(u)$. Their full raw jet equation is $(d/du)^d f(tu)|_{u=\zeta_{ab}} = t^d f^{(d)}(t\zeta_{ab})$, so the coefficient d in multiplication's raw-jet Jordan blocks and each order ℓ remain for $t \neq 0$. At $t = 0$ monic division gives the complete merged order q , rather than an assertion that separate evaluation rows stay invertible at coincident points.

For any fixed positive polynomial source H_N , remainder matrices $J_N(t)$ are polynomial in t by the monic subtraction algorithm. They contain I_q in their first q columns. Thus $K_N(t) = J_N(t)H_N^{-1}J_N(t)^*$ is positive for every real t , including zero, and the exact central function obeys

$$C(t) = \log \det K_{2q-1}(t) - \log \det K_q(t), \quad C'(t) = \text{Tr}(K_{2q-1}^{-1}K'_{2q-1} - K_q^{-1}K'_q), \quad (\text{AU.48})$$

where $K'_N = J'_N H_N^{-1} J_N^* + J_N H_N^{-1} (J'_N)^*$. The determinant derivative follows by its multilinear expansion. This gives both the free family and the exact finite source-dependent map along it. The estimate (AU.43) belongs to its stated Lebesgue special fibre. Equations (AU.46)–(AU.48) do not transfer that estimate to $t = 1$ without the displayed change in the full kernel.

49.7 Source representatives and support

The source polynomial maps throughout are the original $\mathcal{V}_{h,k}P = P(D_1 + \dots + D_k)F_h^{\otimes k}$, $D = -x\partial_x$, $\mathcal{M}F_h = g/h$. The arithmetic inclusion is the original $\eta[P] = [g/h]_h^{\otimes k} P(\sum s_i)$, after the explicit chart (AU.2). Each local Taylor constant of g/h at a selected complete zero is nonzero. Recursively solving multiplication of truncated Taylor series therefore proves this factor is a unit. It does not alter the original cyclic kernel (AU.4). The comparison to ν_k changes the norm used in an estimate, while its identity polynomial map leaves η , every raw jet, and the original relations unchanged.

For two polynomial representatives with the same quotient, monic division gives their difference χQ . In the tensor variables the original annihilator identity supplies $\chi(\sum s_i) = \sum_i h(s_i)Q_i(\mathbf{s})$, with Q_i obtained by successive fixed-order monic divisions. The final remainder is zero because

the monomial tensor basis is independent and the left side annihilates the sum. If $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$ and $h(D)F_h = \Theta\phi_0$, the actual primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) Q(\sum D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{AU.49})$$

The tensor differential at position i contributes the second $(-1)^{i-1}$, cancelling the displayed one and giving exactly $\mathcal{V}_{h,k}(\chi Q)$. Thus every quotient comparison retains the same cohomology class with an explicit original primitive. On the supported carrier a fibre map sends (λ, v) to (λ, Lv) and sends external τ to τ . The killed relation has image $(\lambda, 0) = e_\lambda$. No comparison identifies that supported zero with external absence.

49.8 Verification and provenance

The complete arithmetic source (including the proof of its envelope), the exact endpoint-window source, the full PR24 note and scope correction/review, and the owner's complete balanced-window/factor-four proof were read for this calculation. Its new work is (AU.11)–(AU.23), the scalar estimate (AU.24)–(AU.33), and the full calculation (AU.40)–(AU.48). The constants $c_h, \vartheta_h, M_h(\pm b)$ retain their supplied definitions; no numerical enclosure of actual quartet data is claimed. The accompanying finite rational checker verifies the displayed maps and determinant formulas on declared auxiliary positive sources. It does not certify the analytic source by sampled values and makes no Lean claim.

50 Reflected relation norms and the exact common refinement of conormal algebras

Fix an integer $k \geq 1$, retain the original coordinate S and the real number $c = k/2$, and let $\chi \in \mathbb{C}[S]$ be monic of degree $q \geq 1$. The intended application uses the actual cyclic annihilator with every root order retained. All the algebra below holds for every such χ . It completes the comparison raised by the full-jet transfer construction: the squared norm of an original relation, its first conormal algebra, and their common refinement have explicit maps even for a packet which is not closed under reflection. The reflected factors and their phases are displayed throughout.

50.1 The original reflection and the norm polynomial

Let a bar over a polynomial mean conjugation of its coefficients. Define the conjugate-linear involution and its reflected monic polynomial by

$$f^*(S) = \overline{f}(k - S), \quad \chi^\dagger(S) = (-1)^q \chi^*(S), \quad N_\chi(S) = \chi(S) \chi^\dagger(S). \quad (\text{RF.1})$$

The factor $(-1)^q$ remains part of this definition. In particular $\chi^* = (-1)^q \chi^\dagger$ is the original involution image. The change of polynomial coordinates and the full leading phase are

$$\Psi : \mathbb{C}[S] \xrightarrow{\sim} \mathbb{C}[u], \quad \Psi(f)(u) = f(c + iu), \quad \Psi^{-1}(F)(S) = F((S - c)/i), \quad \psi = i^{-q} \Psi(\chi). \quad (\text{RF.2})$$

Proposition 50.1. *The polynomial $\chi^\dagger$ is monic of degree q , $(\chi^\dagger)^\dagger = \chi$, and $N_\chi^* = N_\chi$. The original-coordinate identities are*

$$\Psi(f^*) = \overline{\Psi(f)}, \quad \Psi(\chi) = i^q \psi, \quad \Psi(\chi^\dagger) = i^q \overline{\psi}, \quad \Psi(N_\chi) = i^{2q} \psi \overline{\psi}. \quad (\text{RF.3})$$

Consequently the monic polynomial $\Pi = \psi\bar{\psi}$ has real coefficients and satisfies $\Pi(u) = |\psi(u)|^2 \geq 0$ for real u . The following three conditions are equivalent:

$$\chi^\dagger = \chi, \quad \bar{\chi}(k - S) = (-1)^q \chi(S), \quad \bar{\psi} = \psi. \quad (\text{RF.4})$$

Under these equivalent conditions, $N_\chi = \chi^2$ and $\Pi = \psi^2$.

Proof. The leading coefficient of $\bar{\chi}(k - S)$ is $(-1)^q$, which proves monicity with its stated phase. Since k is real, $(f^*)^* = f$ and $(fg)^* = f^*g^*$. Applying these equations twice, including both real factors $(-1)^q$, gives $(\chi^\dagger)^\dagger = \chi$ and $N_\chi^* = (-1)^{2q} \chi^\dagger \chi = N_\chi$. Direct substitution gives $\Psi(f^*)(u) = \bar{f}(c - iu) = \overline{\Psi(f)}(u)$. Conjugating $\Psi(\chi) = i^q \psi$ therefore gives $\Psi(\chi^*) = (-i)^q \bar{\psi}$. Multiplication by $(-1)^q$ yields the third equation of (RF.3); multiplication by $\Psi(\chi)$ yields the fourth. The two leading coefficients in Π equal one, and conjugating its product reverses its factors, proving that its coefficients are real. Evaluation on the real axis proves its value there. Finally (RF.3) and bijectivity of Ψ prove the equivalence (RF.4), including the factor i^q . $\square$

The exact norm map can also be written without suppressing its phase. For the original positive density $m_{h,k}(u)$ and any polynomial Q , put $\tilde{Q} = \Psi(Q)$. Polynomial integrability is supplied by the original arithmetic source construction. Pointwise (RF.3) gives

$$\Psi(\chi Q) = i^q \psi \tilde{Q}, \quad \int_{\mathbb{R}} |\chi(c + iu)Q(c + iu)|^2 m_{h,k}(u) du = \int_{\mathbb{R}} \Pi(u) |\tilde{Q}(u)|^2 m_{h,k}(u) du. \quad (\text{RF.5})$$

Thus multiplication by χ on the original polynomial source is carried by Ψ to i^q times multiplication by ψ . Both integrals use the original density and its literal mass. The equality of integrands is $|i^q|^2 |\psi \tilde{Q}|^2 = \Pi |\tilde{Q}|^2$, and proves the complete norm comparison.

50.2 Full jets at the union of the original and reflected roots

Let a_λ be the order of λ as a root of χ , with value zero at other points. The order of λ in $\chi^\dagger$ is

$$b_\lambda = a_{k-\bar{\lambda}}, \quad r_\lambda = a_\lambda + b_\lambda. \quad (\text{RF.6})$$

To prove this, write $\chi(S) = \prod_\alpha (S - \alpha)^{a_\alpha}$ and substitute it in (RF.1). Each reflected factor is $k - S - \bar{\alpha} = -(S - (k - \bar{\alpha}))$; their q minus signs cancel the displayed $(-1)^q$. Orders at coincident roots add when the two polynomials are multiplied. In particular $\sum_\lambda r_\lambda = 2q$.

Theorem 50.2. *There are explicit isomorphisms*

$$\mathcal{A}_N := \mathbb{C}[S]/(N_\chi) \xrightarrow{\Psi_N} \mathbb{C}[u]/(\Pi), \quad [P] \mapsto [P(c + iu)], \quad (\text{RF.7})$$

and, for any fixed ordering of the distinct roots in (RF.6),

$$\mathcal{J}_N : \mathcal{A}_N \xrightarrow{\sim} \bigoplus_{\lambda: r_\lambda > 0} \mathbb{C}^{r_\lambda}, \quad [P] \mapsto (P^{(d)}(\lambda))_{\lambda, 0 \leq d < r_\lambda}. \quad (\text{RF.8})$$

The product in each raw-derivative block is $(v \cdot w)_d = \sum_{j=0}^d \binom{d}{j} v_j w_{d-j}$. If $\zeta_\lambda = (\lambda - c)/i$, the same raw jets in the u coordinate are exactly

$$(\Psi P)^{(d)}(\zeta_\lambda) = i^d P^{(d)}(\lambda). \quad (\text{RF.9})$$

Multiplication by S in (RF.8) is the block map $(M_S v)_d = \lambda v_d + d v_{d-1}$, with $v_{-1} = 0$.

Proof. Equation (RF.3) maps the ideal (N_χ) onto (Π) because i^{2q} is an invertible constant. Applying Ψ^{-1} proves both well-definedness and inverse for (RF.7). A polynomial has its first r raw derivatives zero at λ exactly when $(S - \lambda)^r$ divides it: its Taylor expansion is $\sum_d P^{(d)}(\lambda)(S - \lambda)^d/d!$. For distinct roots the corresponding factors are relatively prime. Repeated polynomial division, or Bezout's identity applied successively to those factors, shows that simultaneous divisibility is divisibility by their product N_χ . This proves the exact kernel in (RF.8). Representatives of degree less than $2q$ are unique by monic division, so the source and target of the induced injection both have dimension $2q$; it is therefore onto. Leibniz's rule gives the stated block product, including every binomial coefficient. The chain rule proves (RF.9), and differentiating SP gives $(SP)^{(d)}(\lambda) = \lambda P^{(d)}(\lambda) + dP^{(d-1)}(\lambda)$. These calculations also show that all maps retain the entire nilpotent root order. $\square$

There is a canonical original arithmetic quotient map $\pi_N : \mathcal{A}_N \rightarrow \mathcal{A}_\chi := \mathbb{C}[S]/(\chi)$. Its exact kernel and its coefficient-module parametrization are

$$0 \longrightarrow \mathbb{C}[S]/(\chi^\dagger) \xrightarrow{\iota_N} \mathcal{A}_N \xrightarrow{\pi_N} \mathcal{A}_\chi \longrightarrow 0, \quad \iota_N[f] = [\chi f]. \quad (\text{RF.10})$$

Indeed $N_\chi \mid \chi f$ is equivalent to $\chi^\dagger \mid f$ by cancellation in $\mathbb{C}[S]$. The image consists precisely of the multiples of χ modulo N_χ , and reduction modulo χ is surjective. This proves exactness, injectivity and all dimensions. The map ι_N is a $\mathbb{C}[S]$ -module map; the quotient maps are algebra homomorphisms.

50.3 A common refinement with the original first conormal algebra

Retain the original first conormal algebra $\mathcal{A}_2 = \mathbb{C}[S]/(\chi^2)$ and its exact sequence

$$0 \longrightarrow \mathbb{C}[S]/(\chi) \xrightarrow{[f] \mapsto [\chi f]} \mathcal{A}_2 \longrightarrow \mathcal{A}_\chi \longrightarrow 0. \quad (\text{RF.11})$$

Let g be the monic greatest common divisor of $\chi, \chi^\dagger$, and put

$$d = \chi g, \quad \ell = \chi^2 \chi^\dagger / g. \quad (\text{RF.12})$$

These polynomials retain all original roots: at a point λ their orders are respectively $a_\lambda + \min(a_\lambda, b_\lambda)$ and $a_\lambda + \max(a_\lambda, b_\lambda)$.

Theorem 50.3. *The following sequence is exact as a sequence of $\mathbb{C}[S]$ -modules:*

$$0 \longrightarrow \mathbb{C}[S]/(\ell) \xrightarrow{j} \mathcal{A}_2 \oplus \mathcal{A}_N \xrightarrow{t} \mathbb{C}[S]/(d) \longrightarrow 0, \quad j[P] = ([P], [P]), \quad t([P], [Q]) = [P - Q]. \quad (\text{RF.13})$$

The first map is an algebra isomorphism onto the fibre product

$$\boxed{\mathbb{C}[S]/(\ell) \simeq \mathcal{A}_2 \times_{\mathbb{C}[S]/(\chi g)} \mathcal{A}_N.} \quad (\text{RF.14})$$

The projections of this fibre product followed by reduction modulo χ agree. All the maps in (RF.10)–(RF.14) commute with the original multiplication operator M_S .

Proof. In the polynomial ring, the intersection of two principal ideals has at every root the maximum of their orders; their sum has the minimum. The intersection assertion follows immediately from divisibility at every root. For the sum, divide the two polynomials by their monic greatest common divisor. The remaining factors are coprime, and the Euclidean algorithm supplies a polynomial linear combination equal to one. Multiplying back proves that their ideal sum is the gcd ideal. Applying this to χ^2 and $\chi\chi^\dagger$ gives the polynomials ℓ and d in (RF.12). Thus j is well-defined and its kernel is zero. The difference map t is well-defined since d divides both defining polynomials; it is onto because $t([P], 0) = [P]$. If $P - Q$ belongs to their ideal sum, write $P - Q = \chi^2 A + N_\chi B$. Then $T = P - \chi^2 A = Q + N_\chi B$ maps to $([P], [Q])$ under j . This proves $\ker t = \text{im } j$ without choosing different roots or

changing any order. The image consists of exactly those pairs with equal reduction modulo d , and its multiplication is componentwise. Therefore the injection is the algebra isomorphism (RF.14). Since χ divides d , the final reductions agree. Multiplication by S commutes with taking every polynomial residue, with $P \mapsto \chi P$, and with the difference of representatives; this proves the final assertion directly. $\square$

On the two kernel $\mathbb{C}[S]$ -modules of the quotient maps to $\mathcal{A}_\chi$, the comparison map has an especially explicit description. Reduction to $\mathbb{C}[S]/(\chi g)$ sends $[\chi f]$ from either algebra to the same residue class. Its image is $(\chi)/(\chi g)$, canonically isomorphic to $\mathbb{C}[S]/(g)$ by $[f] \mapsto [\chi f]$. Thus the exact diagram of kernel maps is the pair of quotient maps

$$\mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[S]/(g) \longleftarrow \mathbb{C}[S]/(\chi^\dagger), \quad (\text{RF.15})$$

with their literal multiplication-by- χ parametrizations in (RF.11) and (RF.10). This is the relation between the first conormal kernel and the kernel recorded by the relation norm.

50.4 Square-zero structure, reflected sectors, and supported zero

The kernel in (RF.11) has square zero, because products of its representatives contain χ^2 . The kernel in (RF.10) has square zero exactly when $\chi^\dagger = \chi$. To prove necessity, its element $[\chi]$ would have square zero, requiring $\chi\chi^\dagger \mid \chi^2$, and hence $\chi^\dagger \mid \chi$. Equal degree and monicity force equality. Sufficiency follows from the same divisibility. The exact common refinement for both cases remains (RF.13)–(RF.15).

For an explicit complementary case suppose $g = 1$. Choose the polynomials u, v supplied by the Euclidean algorithm with $u\chi + v\chi^\dagger = 1$. Then the algebra map

$$\mathcal{A}_N \xrightarrow{\sim} \mathcal{A}_\chi \times \mathcal{A}_{\chi^\dagger}, \quad [P] \longmapsto ([P]_\chi, [P]_{\chi^\dagger}), \quad e_{\text{ref}} = [u\chi]_{N_\chi} \quad (\text{RF.16})$$

is an isomorphism. Its kernel before factoring is the intersection $(\chi) \cap (\chi^\dagger) = (N_\chi)$, and for arbitrary residues represented by a, b , the polynomial $av\chi^\dagger + bu\chi$ has those two residues. This proves surjectivity and the inverse formula. The element e_{ref} has coordinates $(0, 1)$, proving $e_{\text{ref}}^2 = e_{\text{ref}}$. Its ideal is the kernel of π_N and is isomorphic as a unital algebra with its local unit e_{ref} to $\mathcal{A}_{\chi^\dagger}$. In these coordinates M_S acts by the original residue of S in each factor. Equations (RF.14)–(RF.16) therefore calculate the reflected component and its exact map to the original arithmetic quotient.

To state the Split-Zero observation explicitly, attach any original support label λ_0 to each of these vector spaces. A linear map $L : E \rightarrow F$ has the support-preserving lift

$$\tilde{L}(\tau) = \tau, \quad \tilde{L}(\lambda_0, x) = (\lambda_0, Lx). \quad (\text{RF.17})$$

Composition satisfies $\widetilde{\tilde{L}_2 \tilde{L}_1} = \tilde{L}_2 \tilde{L}_1$ by evaluation on τ and on each labelled vector. Addition and scalar action within a fibre are preserved because L is linear. Every vector of the kernels in (RF.10), (RF.11), or (RF.13) maps to the receiving supported zero $(\lambda_0, 0)$; the input and all its pre-quotient coordinates remain defined. In particular, the nonzero algebra idempotent e_{ref} in (RF.16) has zero amplitude in $\mathcal{A}_\chi$, and its labelled image is $(\lambda_0, 0)$. These two values and the external point τ have been related by the explicit quotient, rather than identified in the source.

The first conormal algebra used here is exactly $\mathbb{C}[S]/(\chi^2)$. If a tensor construction supplies a different annihilator, its divisibility maps require that actual polynomial. No replacement of a tensor annihilator is made by (RF.12): (RF.13) is the complete comparison for the two polynomials explicitly stated here.

50.5 Reflected kernel transport and every power of the norm kernel

Write $I_2 = \ker(\mathcal{A}_2 \rightarrow \mathcal{A}_\chi)$ and $I_N = \ker \pi_N$. The coefficient parametrizations in (RF.10) and (RF.11) prove

$$\text{Ann}_{\mathbb{C}[S]} I_2 = (\chi), \quad \text{Ann}_{\mathbb{C}[S]} I_N = (\chi^\dagger). \quad (\text{RF.18})$$

Indeed a polynomial p annihilates the cyclic generator $[\chi]$ of the second kernel precisely when $\chi\chi^\dagger \mid p\chi$, equivalently $\chi^\dagger \mid p$; the first assertion follows with χ^2 in place of $\chi\chi^\dagger$. Thus the original $\mathbb{C}[S]$ -action on I_N descends to $\mathcal{A}_\chi$ exactly when $\chi^\dagger \mid \chi$, or equivalently $\chi^\dagger = \chi$. The original action always descends to $\mathcal{A}_{\chi^\dagger}$.

There is nevertheless a direct reflected identification of both kernel spaces, retaining the exact scalar action:

$$F : I_2 \xrightarrow{\sim} I_N, \quad F([\chi f]_{\chi^2}) = [\chi f^*]_{N_\chi}, \quad G([\chi h]_{N_\chi}) = [\chi h^*]_{\chi^2}. \quad (\text{RF.19})$$

If $\chi \mid f_1 - f_2$, then $\chi^\dagger \mid (f_1 - f_2)^*$ by (RF.1), proving that F is well-defined. If $\chi^\dagger \mid h_1 - h_2$, then $\chi \mid (h_1 - h_2)^*$, proving the assertion for G . The identity $\star^2 = 1$ proves $GF = 1$ and $FG = 1$. Both maps preserve addition and conjugate complex scalars. For every $p \in \mathbb{C}[S]$ their complete intertwining law is

$$F(p(S)x) = p^*(S)F(x), \quad FM_S = (kI - M_S)F. \quad (\text{RF.20})$$

This follows by applying $\star$ to the product pf . The labelled lift sends τ to τ and (λ_0, x) to (λ_0, Fx) ; its scalar law is $F(\alpha x) = \bar{\alpha}F(x)$. It is therefore a conjugate-linear fibre map with the original support retained.

The exact product on the coefficient module of I_N is

$$[f] \diamond [h] = [\chi fh]_{\chi^\dagger}, \quad \iota_N([f] \diamond [h]) = \iota_N[f] \iota_N[h]. \quad (\text{RF.21})$$

Changing f or h by a multiple of $\chi^\dagger$ changes the right-hand representative by a multiple of that polynomial. The second equality is the identity $\chi(\chi fh) = \chi^2 fh$ modulo N_χ . This proves the full product transport; no algebra-isomorphism assertion is imposed on the semilinear map F .

Theorem 50.4. *For every integer $t \geq 1$, retaining all root orders from (RF.6),*

$$\dim_{\mathbb{C}} I_N^t = \sum_{\lambda: r_\lambda > 0} \max\{b_\lambda - (t-1)a_\lambda, 0\}. \quad (\text{RF.22})$$

The kernel is nilpotent exactly when every reflected root occurs among the original roots. In this case its exact nilpotence index is

$$\max_{\lambda: b_\lambda > 0} \left\lceil \frac{a_\lambda + b_\lambda}{a_\lambda} \right\rceil. \quad (\text{RF.23})$$

In general its persistent part is the algebra

$$I_N^\infty := \bigcap_{t \geq 1} I_N^t \simeq \prod_{\lambda: a_\lambda = 0, b_\lambda > 0} \mathbb{C}[S]/((S - \lambda)^{b_\lambda}). \quad (\text{RF.24})$$

The isomorphism is the restriction of the original CRT residue map. An empty product here denotes the zero algebra. Its unit when nonzero is the idempotent with value one in those reflected-only blocks and zero in all other blocks. Projection to $\mathcal{A}_\chi$ sends it to zero, and its labelled image is supported zero.

Proof. In the block at λ put $z = S - \lambda$ and write $\chi = z^{a_\lambda} U_\lambda$ with $U_\lambda(\lambda) \neq 0$. The original polynomial U_λ remains part of the parametrization. It is invertible in $\mathbb{C}[S]/(z^{a_\lambda + b_\lambda})$: its constant coefficient is nonzero, and the finite geometric expansion in its nilpotent nonconstant part gives an inverse. Therefore the ideal generated by χ^t in this block is exactly (z^{ta_λ}) . Its basis is $z^{ta_\lambda}, \dots, z^{a_\lambda + b_\lambda - 1}$ when the lower index is smaller than the upper length, and is empty otherwise. This proves (RF.22), including $a_\lambda = 0$.

If $a_\lambda = 0 < b_\lambda$, that ideal is the whole nonzero block for every t , so nilpotence is impossible. Otherwise each block with $b_\lambda > 0$ vanishes precisely when $ta_\lambda \geq a_\lambda + b_\lambda$. Taking the largest of these exact integer thresholds proves (RF.23). The maximum is over a nonempty set because $\deg \chi^\dagger = q \geq 1$. There are finitely many blocks. The blocks with $a_\lambda > 0$ all vanish after a common finite threshold, whereas the others persist unchanged. The CRT isomorphism and its inverse from Bezout consequently give (RF.24) and the stated idempotent. Reduction modulo χ omits precisely the blocks where $a_\lambda = 0$, proving its amplitude image and the assertion about its support-preserving lift. $\square$

For a complete overlapping example keep $k = 2$, $c = 1$ and put $A = S - (2 + i)$, $B = S - i$, so $A = B - 2$. For $\chi = A^2B$ the retained polynomials are

$$\chi^\dagger = AB^2, \quad g = AB, \quad N_\chi = A^3B^3, \quad d = A^3B^2, \quad \ell = A^4B^3. \quad (\text{RF.25})$$

In particular $\chi^2 - N_\chi = A^3B^2(A - B) = -2A^3B^2$ has degree five, so the square of $[\chi]$ in the degree-six norm algebra is nonzero. Its cube is zero because $N_\chi \mid A^6B^3$. Formula (RF.22) gives dimensions 3, 1, 0 for the first three powers, with every original root order retained. If representatives P, Q satisfy $P - Q = dR$, their common refinement is explicitly

$$T = P + \frac{1}{2}R\chi^2 = Q + \frac{1}{2}RN_\chi, \quad (\text{RF.26})$$

since $N_\chi - \chi^2 = 2d$. Its residues are P modulo χ^2 and Q modulo N_χ . Thus the full mixed fibre-product map is available even though the norm kernel and the square-zero conormal kernel have different products.

50.6 Provenance and verification scope

This derivation follows the explicit dagger-stability correction to the full-jet transfer note at revision dfcbba5cbf7fec8c9301fe242c13e741d762013e. The endpoint product continuation retains the full transfer calculation for $\Pi = \psi\bar{\psi}$. The present note proves the original-coordinate helper algebra, both kernel modules, and their exact gcd/lcm refinement, including their raw jets and original action. These are polynomial algebra proofs with all domains and maps displayed. The source norm identity (RF.5) uses the unchanged arithmetic density; it does not estimate the growth of any quotient volume or establish a zero-location claim.

51 Exact optimization of the arithmetic full-jet determinant weights

This is a separate continuation of the sealed arithmetic volume upper estimate AU.1–33 and AU.40–49. It solves the weight optimization in AU.19 completely, including boundary optima, zero polynomial coefficients, the sole nonunique case, and an evaluable one-dimensional characterization. The signed raw-jet minors, source coordinate phases, factorials and arithmetic masses remain explicit. The improvement over equal weights is proved to be less than one natural-logarithm unit for every degree within that specified family. The continuation (OW.30)–(OW.40) proves a stronger source estimate by keeping the even moments together. It then solves the entire resulting diagonal family using the actual moments through degree $2q$. Its exact comparison with AU.19 removes a term of order $q \log q$; the earlier less-than-one theorem retains its precise, narrower domain.

51.1 The original source and the signed full-jet minors

Keep $g = 2\xi$, the original complete monic zero packet h , and

$$w_h(t) = |(g/h)(1/2 + it)|^2 / (2\pi), \quad m_k = w_h^{*k}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad S = c + iu, \quad c = k/2. \quad (\text{OW.1})$$

The source polynomial norm is $\int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du$ and its mass is μ_h^k . For the original degree- q monic relation χ put

$$\psi(u) = i^{-q} \chi(c + iu) = \sum_{j=0}^q \psi_j u^j, \quad \psi_q = 1, \quad \psi_j = \frac{i^{j-q}}{j!} \chi^{(j)}(c). \quad (\text{OW.2})$$

The last identity is the polynomial Taylor expansion at the original centre c . The exact ring map $P(S) \mapsto P(c + iu)$ has inverse $f(u) \mapsto f((S - c)/i)$ and intertwines multiplication by S with $c + iM_u$. Its raw derivative of order d at $\zeta = (\lambda - c)/i$ is $i^d P^{(d)}(\lambda)$. No coefficient phase or derivative factorial has been reassigned.

List the distinct roots ζ_a of ψ , in the original order, with their complete lengths ℓ_a , so $\sum_a \ell_a = q \geq 1$. For $N \geq q - 1$ the full raw-jet matrix is

$$(E_N)_{(a,d),j} = \begin{cases} j! \zeta_a^{j-d} / (j-d)!, & j \geq d, \\ 0, & j < d, \end{cases} \quad 0 \leq d < \ell_a, \quad 0 \leq j \leq N. \quad (\text{OW.3})$$

Taylor expansion proves that its kernel consists of polynomials divisible by ψ . Thus $Z := E_{q-1}$ is invertible by degree, and $E_q(\psi_0, \dots, \psi_q)^T = 0$. Write $\Delta_j = \det E_q[:, \{0, \dots, q\} \setminus \{j\}]$, with the retained increasing column order. The precise identity is

$$\Delta_j = (-1)^{q-j} \psi_j \det Z \quad (0 \leq j \leq q). \quad (\text{OW.4})$$

To prove the sign, append any row of E_q as a final row and expand the resulting zero determinant along that row. It shows that $((-1)^j \Delta_j)_{j=0}^q$ lies in the kernel of E_q . That kernel has dimension one because its first q columns form Z . It is therefore a scalar multiple of the displayed monic coefficient vector. Its last coordinate is $(-1)^q \Delta_q = (-1)^q \det Z$, which proves (OW.4). In particular a zero coefficient gives exactly a zero minor, rather than a deleted column or shortened jet space.

The raw confluent determinant, in the same row order, is

$$\det Z = \left(\prod_a \prod_{d=0}^{\ell_a-1} d! \right) \prod_{a < a'} (\zeta_{a'} - \zeta_a)^{\ell_a \ell_{a'}}. \quad (\text{OW.5})$$

For completeness take the ordinary Vandermonde determinant with temporarily distinct nodes grouped into the prescribed clusters. Successive divided-difference row operations remove its within-cluster Vandermonde factors. The finite Taylor expansion of each monomial shows that its order- d divided-difference row tends to the raw derivative divided by $d!$. Multiplication of these rows by $d!$ gives (OW.5), with the cross-cluster factors still in their original order. Equivalently this proves the identity after cancellation of polynomial factors, so no analytic source approximation is involved.

Fix $0 < b < \pi/2$ and retain the two literal Laplace values

$$M_k = M_h(b)^k + M_h(-b)^k, \quad M_h(b) = \int_{\mathbb{R}} e^{bt} w_h(t) dt, \quad \beta_j = \frac{b^{2j}}{(2j)!} > 0. \quad (\text{OW.6})$$

For positive auxiliary weights τ_j the matrix used in AU.19 is $A_q(\tau) = E_q \text{diag}(\tau_j \beta_j) E_q^*$. Cauchy–Binet and (OW.4) prove

$$\det A_q(\tau) = |\det Z|^2 \sum_{j=0}^q a_j \prod_{l \neq j} \tau_l, \quad a_j = |\psi_j|^2 \prod_{l \neq j} \beta_l = \left(\prod_{l=0}^q \beta_l \right) \eta_j, \quad \eta_j = \frac{|\psi_j|^2}{\beta_j} = \frac{(2j)!}{(j!)^2 b^{2j}} |\chi^{(j)}(c)|^2. \quad (\text{OW.7})$$

In particular $a_q > 0$ and $\eta_q = (2q)! b^{-2q} > 0$. To verify Cauchy–Binet directly, expand the determinant multilinearly in the columns of E_q and then its adjoint. Repeated column choices vanish by

alternation; collecting each q -element set gives its minor times its conjugate and the product of its diagonal weights. This yields exactly the first line of (OW.7). The signed identity is proved before taking the squared modulus.

For arbitrary positive τ , let $s = \sum_j \tau_j$ and $t_j = \tau_j/s$. The map $(\tau_j) \mapsto (s, (t_j))$ is a bijection from the positive cone to $(0, \infty)$ times the open simplex, with inverse $\tau_j = st_j$. The determinant in (OW.7) acquires exactly the factor s^q . Thus the $q \log s$ in AU.19 cancels exactly. This changes auxiliary optimization coordinates and does not change m_k , its mass, or either source or quotient norm. Multiplication of all a_j by the common positive product in (OW.7) multiplies the objective by that product, with the maximizing weights unchanged. It suffices to solve the following finite optimization with all its original coefficients retained.

51.2 Global maximizers, including every zero coefficient

For $a_0, \dots, a_q \geq 0$ with $a_q > 0$, define

$$S_q = \{\tau \in \mathbb{R}^{q+1} : \tau_j \geq 0, \sum_j \tau_j = 1\}, \quad D_a(\tau) = \sum_{j=0}^q a_j \prod_{l \neq j} \tau_l, \quad d(a) = \max_{\tau \in S_q} D_a(\tau). \quad (\text{OW.8})$$

The objective is a polynomial and the simplex is compact. Its value at equal weights is positive, so a positive maximum exists. Put $I = \{j : a_j > 0\}$, $p = |I|$, $A = \max_j a_j$, and $S_a = \sum_j a_j$.

Theorem 51.1 (Complete boundary and uniqueness classification). *For $q \geq 2$ the maximizing weights are precisely the following.*

1. *If $A \geq S_a - A$, choose its index j . Unless exactly two coefficients are positive and equal, the maximizer is unique: $\tau_j = 0$, $\tau_i = 1/q$ for every $i \neq j$, and $d(a) = A/q^q$.*
2. *If exactly two coefficients $a_j = a_l = A$ are positive, every maximizer, and only a maximizer, satisfies $\tau_j + \tau_l = 1/q$ and $\tau_i = 1/q$ at the other $q - 1$ indices. Its value is again A/q^q .*
3. *If $A < S_a - A$, then $p \geq 3$ and the maximizer is unique and has every coordinate strictly positive. At a zero coefficient it satisfies $\tau_j = 1/q$.*

For $q = 1$, $D_a = a_0\tau_1 + a_1\tau_0$ is linear: it is maximized at the endpoint selecting its larger coefficient, or everywhere if the two coefficients are equal. Its value is $\max(a_0, a_1)$.

Proof. An elementary determinant representation proves the needed concavity. Set $v_i = \sqrt{a_i/a_q}$ for $0 \leq i < q$ and

$$M(\tau) = \text{diag}(\tau_0, \dots, \tau_{q-1}) + \tau_q vv^*. \quad D_a(\tau) = a_q \det M(\tau). \quad (\text{OW.9})$$

For positive first q weights, the rank-one determinant expansion gives $\det M = \prod_{i < q} \tau_i (1 + \tau_q \sum_{i < q} |v_i|^2 / \tau_i)$. This is (OW.9); since both sides are polynomials it holds at boundary weights too. Every $M(\tau)$ is positive semidefinite on the simplex, and $D_a > 0$ precisely when it is positive definite. For a real affine line $M(t) = M_0 + tE$ through positive definite matrices, differentiating the inverse identity gives $(M^{-1})' = -M^{-1}EM^{-1}$. The determinant's multilinear derivative therefore proves

$$(\log \det M)' = \text{Tr}(M^{-1}E), \quad (\log \det M)'' = -\text{Tr}((M^{-1/2}EM^{-1/2})^2) \leq 0. \quad (\text{OW.10})$$

The final matrix is Hermitian, so its squared eigenvalues are nonnegative, and equality holds exactly when $E = 0$.

If $p \geq 3$, at least two coordinates of v are nonzero. The linear map from τ to $M(\tau)$ is injective: an off-diagonal entry between two such coordinates forces a zero increment of τ_q whenever the matrix increment is zero; then its diagonal forces every other increment to be zero. Thus $\log D_a$ is strictly concave on its positive domain, which is convex as the inverse image of the positive definite cone. There can be at most one maximizer with positive objective.

If a simplex point has at least two zero coordinates, every monomial in D_a contains a zero, so its objective is zero. A positive boundary maximum must therefore have exactly one zero coordinate, say $\tau_j = 0$, and then $D_a = a_j \prod_{i \neq j} \tau_i$. It requires $a_j > 0$. For q nonnegative numbers of sum one, concavity of $\log x$ (whose second derivative is $-x^{-2}$) gives product at most q^{-q} , with equality exactly when they all equal $1/q$; a zero factor has product zero. Hence every positive boundary maximum is the candidate just described. At this point

$$\partial_i D_a = a_j / q^{q-1} \quad (i \neq j), \quad \partial_j D_a = (S_a - a_j) / q^{q-1}. \quad (\text{OW.11})$$

If $S_a - a_j > a_j$, the feasible path $\tau_j = \varepsilon$, $\tau_i = (1 - \varepsilon)/q$ for $i \neq j$ has positive derivative $(S_a - 2a_j)/q^{q-1}$ at zero. That boundary point cannot maximize. Conversely, if $S_a - a_j \leq a_j$, the gradient of $\log D_a$ at the candidate has all active entries equal and the inactive entry no larger. Its scalar product with any feasible displacement is nonpositive. Concavity along the segment to any point with positive objective gives $\log D_a(\tau) \leq \log D_a(\tau^*) + \nabla \log D_a(\tau^*) \cdot (\tau - \tau^*) \leq \log D_a(\tau^*)$. This derivative is valid at the boundary because $M(\tau^*) > 0$ and remains positive in a neighborhood. Points of zero objective cannot exceed it. This proves the boundary criterion and, for $p \geq 3$, uniqueness from strict concavity.

If $p = 1$, $D_a = A \prod_{i \neq j} \tau_i$ and the same product inequality gives the stated unique maximizer. If $p = 2$, with positive coefficients at j, l , factor the polynomial exactly as

$$D_a(\tau) = \left(\prod_{i \notin \{j, l\}} \tau_i \right) (a_j \tau_l + a_l \tau_j). \quad (\text{OW.12})$$

At fixed $\tau_j + \tau_l$, unequal coefficients allocate that sum entirely to the larger coefficient's complementary variable. The remaining q factors of total sum one have the unique equal product maximum. If the two coefficients coincide, their sum is one factor instead, and equality in that same product inequality gives exactly the listed segment of maximizers.

When $A < S_a - A$, no boundary candidate can maximize, so compactness gives an interior maximum and strict concavity makes it unique. It remains to verify the stated zero coordinate value; that follows from the complete interior stationarity calculation below. The case $q = 1$ is immediate from its displayed linear polynomial. $\square$

No artificial coefficient phase is required to apply (OW.9) to the original raw-jet matrix. In fact $E_q = Z[I_q, -\psi_{<q}]$ by monic remainder, so

$$A_q(\tau) = ZB_0^{1/2} (\text{diag}(\tau_0, \dots, \tau_{q-1}) + \tau_q z z^*) B_0^{1/2} Z^*, \quad B_0 = \text{diag}(\beta_0, \dots, \beta_{q-1}), \quad z_i = -\psi_i \sqrt{\beta_q / \beta_i}. \quad (\text{OW.13})$$

Its $|z_i|^2$ equals η_i / η_q . Define the diagonal unitary $U_{ii} = -\psi_i / |\psi_i|$ when $\psi_i \neq 0$ and $U_{ii} = 1$ otherwise. Then $z = Uv$, with $v_i = \sqrt{\eta_i / \eta_q}$, and its central matrix is $UM(\tau)U^*$. This is the exact phase-preserving congruence connecting the auxiliary determinant proof to the original columns; all the phase entries remain in U .

51.3 A complete monotone scalar description of the interior optimum

Assume $q \geq 2$ and $A < S_a - A$, so the preceding theorem gives its unique interior maximizer. For positive weights put $P = \prod_j \tau_j$ and $S = \sum_j a_j / \tau_j$; then $D_a = PS$. Differentiating every factor gives $\partial_i D_a = P(S / \tau_i - a_i / \tau_i^2)$. The Lagrange multiplier for $\sum \tau_i = 1$ is qD_a because $\sum_i \tau_i \partial_i D_a = qD_a$, each monomial having degree q . Consequently its complete equations are

$$a_i = S \tau_i (1 - q \tau_i). \quad (\text{OW.14})$$

If $a_i = 0$, positivity gives $\tau_i = 1/q$. At a positive coefficient let $y_i = 1 - q\tau_i$. The equation shows $0 < y_i < 1$, and counting the zero coordinates gives

$$\sum_{i \in I} y_i = 1, \quad a_i = c_* y_i (1 - y_i), \quad c_* = S/q > 0, \quad \tau_i = (1 - y_i)/q. \quad (\text{OW.15})$$

Here c_* is the optimizer's scalar, distinct from the source centre $c = k/2$ in (OW.1). Conversely any positive solution of (OW.15), together with $\tau_i = 1/q$ at zero coefficients, has sum one and satisfies (OW.14): indeed $\sum_i a_i/\tau_i = qc_* \sum_{i \in I} y_i = qc_*$. Concavity then proves that this stationary point is the global maximizer, and uniqueness has already been established.

For $0 \leq t \leq 1/A$ and $i \in I$ define the continuous functions

$$f_i(t) = \frac{1 - \sqrt{1 - a_i t}}{2} = \frac{a_i t}{2(1 + \sqrt{1 - a_i t})}, \quad F(t) = \sum_{i \in I} f_i(t), \quad t = 4/c_*. \quad (\text{OW.16})$$

All square roots are the nonnegative real roots. The second formula is its exact rationalization, valid also at $t = 0$. Each f_i is strictly increasing on this interval, has value zero at zero, and takes values at most $1/2$.

Theorem 51.2 (The unique scalar root and the complete maximizing value). *Under $A < S_a - A$ exactly one of the following descriptions applies.*

1. If $F(1/A) \geq 1$, there is a unique $t_* > 0$ in $(0, 1/A]$ with $F(t_*) = 1$. Set $y_i = f_i(t_*)$ at every positive coefficient.
2. If $F(1/A) < 1$, the largest coefficient has a unique index j . There is a unique $t_* \in (0, 1/A)$ solving

$$H(t) := \sum_{i \in I \setminus \{j\}} \frac{a_i}{A} \frac{1 + \sqrt{1 - At}}{1 + \sqrt{1 - a_i t}} = 1. \quad (\text{OW.17})$$

The function H is strictly decreasing, with $H(0) = (S_a - A)/A > 1$ and $H(1/A) = 2F(1/A) - 1 < 1$. Set $y_j = 1 - f_j(t_*)$ and $y_i = f_i(t_*)$ for $i \neq j$.

In either case the weights are those in (OW.15), with zeros handled as stated, and the exact value is

$$d(a) = \frac{4}{q^q t_*} \prod_{i \in I} (1 - y_i). \quad (\text{OW.18})$$

Proof. In (OW.15) at most one y_i can exceed $1/2$ because their sum is one. If none does, its quadratic equations have precisely the smaller roots $f_i(t)$, and their sum is $F(t) = 1$. Continuity, strict increase and $F(0) = 0$ prove existence and uniqueness exactly in the first listed case.

If $y_j > 1/2$, each other y_i is strictly smaller than $1 - y_j < 1/2$, because $p \geq 3$ gives at least two other positive terms. The function $x(1 - x)$ is strictly increasing on $(0, 1/2)$. It follows that $a_i < c_* y_j (1 - y_j) = a_j$, so the large root must occur at the unique largest coefficient. Its sum equation is $\sum_{i \neq j} f_i(t) = f_j(t)$. At $t > 0$ dividing by the positive f_j and using the second formula in (OW.16) gives exactly (OW.17).

If $F(1/A) < 1$, the maximum is unique already from the endpoint values: two largest coefficients would contribute $1/2 + 1/2$ and the other positive terms would make $F(1/A) > 1$. For $a_i < A$ and $0 < t < 1/A$, direct logarithmic differentiation gives

$$\frac{d}{dt} \log \frac{1 + \sqrt{1 - At}}{1 + \sqrt{1 - a_i t}} = \frac{(1 - a_i t)^{-1/2} - (1 - At)^{-1/2}}{2t} < 0. \quad (\text{OW.19})$$

Thus every term of H decreases strictly. Its value at zero is given by continuity, and $f_j(1/A) = 1/2$ gives the second endpoint value. The intermediate value theorem proves its unique root. The

constructed y has positive entries summing to one and solves the quadratic equations, so (OW.15) proves it is the optimum. Conversely if $F(1/A) \geq 1$, an index with a large root would have to be the unique largest one; its decreasing H would have $H(1/A) = 2F(1/A) - 1 \geq 1$ and hence no root inside $(0, 1/A)$. At equality, the largest root is exactly $1/2$ and coincides with the first construction. This proves completeness of the cases.

Finally $\prod_j \tau_j = q^{-(q+1)} \prod_{i \in I} (1 - y_i)$ and $S = qc_* = 4q/t_*$. Multiplication gives (OW.18), with the full original dimension exponent q retained. $\square$

The zero-coefficient assertion can also be proved globally without interior differentiation. If $p \geq 2$ and $z = q + 1 - p > 0$, write $s = \sum_{i \in I} \tau_i$ and $\tau_i = s\sigma_i$ on I . The polynomial factors as $(\prod_{i \notin I} \tau_i) s^{p-1} D_{a|I}(\sigma)$. For fixed s , the first product is at most $((1-s)/z)^z$. The derivative of $\log((1-s)^z s^{p-1})$ vanishes only at $s = (p-1)/q$, and its second derivative is strictly negative. The endpoints have zero product. Hence every maximum has $\tau_i = 1/q$ at zero coefficients, and

$$d(a) = \frac{(p-1)^{p-1}}{q^q} d_{p-1}(a|I), \quad (\text{OW.20})$$

where the last maximum uses the explicitly displayed positive-index simplex of dimension $p-1$. If $z = 0$, the same formula has factor one. For $p = 1$, the first theorem gives the value directly, with the positive coordinate zero and the other q weights $1/q$. Thus the reduction has an explicit map and determinant power, without changing the dimension of the original arithmetic quotient.

For three positive coefficients a, b, d satisfying the strict triangle inequalities, the solution even has a rational expression. Put $\Delta = 2ab + 2ad + 2bd - a^2 - b^2 - d^2 > 0$. Then

$$\tau_a = \frac{2a(b+d-a)}{q\Delta}, \quad \tau_b = \frac{2b(a+d-b)}{q\Delta}, \quad \tau_d = \frac{2d(a+b-d)}{q\Delta}, \quad (\text{OW.21})$$

$$\tau_i = 1/q \quad (a_i = 0), \quad \boxed{d(a) = \frac{4abd}{q^q \Delta}}.$$

Indeed $\Delta = 4ab - (a+b-d)^2 > 0$ because $0 < a+b-d < 2\min(a, b) \leq 2\sqrt{ab}$. On the positive-index simplex the proposed coordinates $\sigma_a = a(b+d-a)/\Delta$ and its cyclic counterparts are positive and sum to one. For its quadratic polynomial $a\sigma_b\sigma_d + b\sigma_a\sigma_d + d\sigma_a\sigma_b$, direct substitution makes every gradient coordinate $2abd/\Delta$. Euler's degree-two identity then gives its value abd/Δ ; concavity proves maximality. Equation (OW.20) supplies exactly the powers and the weights in (OW.21).

51.4 Evaluable enclosures and the exact maximum gain

For exact rational nonnegative input coefficients, the branch comparison at $1/A$ uses only finitely many nonnegative square roots. Equality is permitted as the first branch's endpoint. In the strict cases, (OW.16) or (OW.17) has a continuous strictly monotone scalar function on a known compact interval with opposite endpoint signs. Rational bisection therefore gives its unique root. Square roots themselves admit rational brackets by squaring positive rationals and bisecting. If an exact trial point is a root, it already solves the problem. Otherwise convergent radical brackets eventually resolve the nonzero sign at that trial point. A positive lower bracket for t_* is obtained after finitely many bisections because $t_* > 0$. Substitution in (OW.18) then yields convergent value brackets; every denominator is bounded away from zero on those root brackets. Boundary maximizers require only the exact coefficient comparison $A \geq S_a - A$ and their explicit product formula.

There is also a uniform certificate that avoids deciding a branch from uncertain coefficients. For two nonnegative coefficient lists, the product inequality used in (OW.11) gives, at every simplex point,

$$|D_a(\tau) - D_{\hat{a}}(\tau)| \leq q^{-q} \sum_j |a_j - \hat{a}_j|.$$

Taking a maximizing point for each list in turn proves

$$\boxed{|d(a) - d(\hat{a})| \leq q^{-q} \sum_j |a_j - \hat{a}_j|.} \quad (\text{OW.22})$$

The maximum is also increasing in each coefficient because each monomial is nonnegative. Thus nonnegative rational boxes $a^- \leq a \leq a^+$ give $d(a^-) \leq d(a) \leq d(a^+)$, and their value width is at most $q^{-q} \sum (a_j^+ - a_j^-)$. These statements remain valid if a corner list is all zero, where its maximum is exactly zero, or its reference last coefficient vanishes: a positive reference coefficient can be selected by the explicit simultaneous permutation of coefficient and simplex indices, whose inverse permutation restores every original index. No unresolved arithmetic equality must be assumed to construct the enclosure. For the original list $a_q > 0$ its lower value eventually stays positive, so logarithms can also be enclosed by their ordinary monotone integral $\log x = \int_1^x dt/t$.

A rational candidate has a direct exact value certificate, so no radical equality needs to be decided merely to enclose the answer. At any simplex point τ with $D := D_a(\tau) > 0$, calculate the full polynomial gradient $g_i = \partial_i D_a(\tau)$ and set

$$h = \max_i g_i - qD \geq 0.$$

The sign follows from $\sum_i \tau_i g_i = qD$. Concavity and the simplex constraint give, for every point of positive objective,

$$\log D_a(\sigma) \leq \log D + \frac{\sum_i g_i(\sigma_i - \tau_i)}{D} \leq \log D + h/D.$$

Therefore

$$\boxed{D \leq d(a) \leq De^{h/D}, \quad h < D \implies d(a) \leq \frac{D^2}{D - h}.} \quad (\text{OW.22a})$$

The last inequality follows from $e^x = \sum_{n \geq 0} x^n/n! \leq \sum_{n \geq 0} x^n = (1 - x)^{-1}$ for $0 \leq x < 1$. The polynomial gradient is defined also at a boundary point; the argument is valid there because its matrix (OW.9) is positive definite. Thus an exactly stationary rational candidate, including a boundary maximum with the correct inactive gradient, has $h = 0$ and certifies its exact value.

Here is a terminating way to combine this certificate with scalar evaluation, including an unresolved equality $F(1/A) = 1$. Use rational brackets for every square root. If the endpoint interval separates from one, select the proved branch and bisect its scalar root. If an endpoint interval still contains one, also form the candidate from its smaller-root values at $1/A$. At any scalar trial whose comparison interval contains zero, form its candidate as well, instead of making an unsupported sign decision. Rational midpoints of the root brackets give nonnegative y approximations; replace the corresponding nonnegative auxiliary weights by their ratios to their positive sum, making an exact rational simplex point. One may keep zero-coefficient weights exactly $1/q$ and apply the positive-index rescaling (OW.20) only to the remaining coordinates. These are the explicit positive-cone coordinate maps already proved, and no source measure changes. Test (OW.22a) at each such candidate.

As square-root intervals narrow, a strictly nonzero branch or bisection sign is eventually determined. If a tested equality is exact, its candidate tends instead to the already proved optimum. In a fixed strict branch, successive root brackets tend to its unique root. In all cases the resulting candidates tend to the maximizer or to a member of the stated maximizing segment. There $D > 0$ and $h = 0$ by (OW.11) or (OW.14), and D, h are continuous functions of the candidate. Thus eventually $h < D$ and the rational certificate width $D^2/(D - h) - D$ is smaller than any prescribed positive rational tolerance. Every returned interval is certified solely by the rational polynomial gradient and (OW.22a), independently of the numerical branch-selection history.

Theorem 51.3 (The finite gain is uniformly less than one). *Let $\tau_j^{\text{eq}} = 1/(q+1)$ and $S_a = \sum a_j > 0$. For every $q \geq 1$,*

$$D_a(\tau^{\text{eq}}) = \frac{S_a}{(q+1)^q}, \quad \frac{S_a}{(q+1)^q} \leq d(a) \leq \frac{S_a}{q^q}, \quad (\text{OW.23})$$

$$0 \leq \mathfrak{g}(a) := \log \frac{d(a)}{D_a(\tau^{\text{eq}})} \leq q \log(1 + 1/q) < 1.$$

The lower equality holds precisely when all coefficients are equal. The upper non-strict equality holds precisely when exactly one coefficient is positive.

Proof. The equal-weight value is obtained by substituting each weight. Every product of q selected nonnegative weights has sum at most one and is therefore at most q^{-q} . Summing with coefficients a_j gives the upper bound. If just one coefficient is positive, its boundary maximizer attains that bound. If two or more are positive, equality would require every corresponding product to attain q^{-q} . For the term omitting j , that requires $\tau_j = 0$ and $\tau_i = 1/q$ at every other index. Two distinct positive terms cannot meet those two incompatible conditions. Thus the upper equality has exactly the stated case.

If equal weights maximize, they are an interior point. Differentiating the polynomial there gives $\partial_i D_a = (q+1)^{-(q-1)}(S_a - a_i)$, which must be constant in i under the sum constraint. Thus every a_i is equal. Conversely equal coefficients give equal gradient coordinates at equal weights, and the determinant concavity proves maximality for $q \geq 2$; for $q = 1$ it is the constant linear polynomial. Finally $\log(1+x) < x$ for $x > 0$, since $\log(1+x) = \int_0^x (1+t)^{-1} dt < \int_0^x dt = x$. Apply this with $x = 1/q$ to prove the strict upper bound one. $\square$

One additional fully explicit lower bracket for the gain is

$$\mathfrak{g}(a) \geq \max\{0, q \log(1 + 1/q) + \log(A/S_a)\}. \quad (\text{OW.24})$$

It follows by testing the best boundary point. In the boundary regime its second expression is the exact gain. When $p \geq 2$, testing equal weights on the positive-index simplex in (OW.20) also gives $\mathfrak{g}(a) \geq q \log(1 + 1/q) + (p-1) \log((p-1)/p)$, with every zero retained through the factor in (OW.20).

51.5 The optimized original arithmetic bound, including boundary weights

To specify the analytic comparator used in the original bound, retain the proved constants $c_h > 0$, $\vartheta_h > 0$, $B = 42 + 2 \deg h$, $\alpha = \pi/2$, and $k \geq 3$. Its measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du \leq m_k(u) du. \quad (\text{OW.25})$$

Let H_N^ν and H_N be the two monomial source Grams and $K_N^{\nu, \text{raw}} = E_N(H_N^\nu)^{-1} E_N^*$, with the original E_N and ψ . Their positive definiteness follows from their positive almost-everywhere densities and the finite zero set of every nonzero polynomial. Contraction of the identity polynomial map gives $H_N^\nu \preceq H_N$, so $K_N^{\text{raw}} \preceq K_N^{\nu, \text{raw}}$. If J_N is monic remainder, the original quotient metric is $G_N = (J_N H_N^{-1} J_N^*)^{-1}$, $V_N = \det G_N$. Every least lift is $H_N^{-1} J_N^* G_N x$; its difference from any other lift is orthogonal to it, as multiplication by J_N verifies. Thus these are the original quotient norms. Since $E_N = Z J_N$, $\det K_N^{\text{raw}} = |\det Z|^2 / V_N$, exactly as in the signed jet map above. The comparison measure defines V_N^ν by the same formula, with its own literal mass.

The exponential-series inequality and convolution give $\int |u|^{2j} m_k(u) du \leq M_k / \beta_j$. Weighted Cauchy-Schwarz, with positive weights of sum one, then gives $H_q \preceq M_k \text{diag}(1/(\tau_j \beta_j))$. Inversion and the original raw-jet map prove $K_q^{\text{raw}} \succeq M_k^{-1} A_q(\tau)$. Together with the upper comparison at degree $2q-1$ this proves the original AU.19 bound on the open simplex:

$$C_k := \log(V_q/V_{2q-1}) \leq q \log M_k + \log \det K_{2q-1}^{\nu, \text{raw}} - \log \det A_q(\tau). \quad (\text{OW.26})$$

At a boundary maximizer in the first theorem its omitted coefficient is positive, so (OW.7) shows $\det A_q(\tau^*) > 0$. Use the explicit positive regularization $\tau^\varepsilon = (1 - \varepsilon)\tau^* + \varepsilon\tau^{\text{eq}}$ for $0 < \varepsilon < 1$. The raw matrices converge entry by entry to the positive definite $A_q(\tau^*)$. Taking the limit of their quadratic-form inequality preserves $K_q^{\text{raw}} \succeq M_k^{-1} A_q(\tau^*)$. Continuity of the positive determinant therefore extends (OW.26) to the boundary optimum. No expression $1/\tau_j$ is evaluated at zero; the boundary statement is proved on the dual raw kernel.

Let $B_\beta = \prod_{j=0}^q \beta_j$. Substitution of the exact maximum (OW.7), followed by the already proved raw-jet determinant conversion, gives the sharpened original bound

$$C_k \leq \mathcal{U}_{\text{opt}} := q \log M_k - \log B_\beta - \log d(\eta) - \log V_{2q-1}^\nu. \quad (\text{OW.27})$$

Here $d(\eta)$ is completely evaluated by the first theorem and (OW.16)–(OW.18), including every coefficient zero and the actual factorials in (OW.7). The factor $|\det Z|^2$ cancels between the two displayed raw kernels only after its signed minors and phase map have been proved. The original source mass stays in M_k , and the comparison mass stays in V_{2q-1}^ν . If desired the latter determinant is computed in its exact source/relation form $V_{2q-1}^\nu = \det H_{2q-1}^\nu / \det(B_{2q-1}^* H_{2q-1}^\nu B_{2q-1})$, where B_{2q-1} is multiplication by the same monic ψ . The quotient-first monomial-plus-relation coefficient matrix has determinant one; eliminating its relation block proves this formula.

The equal-weight upper expression from the identical estimate is

$$\mathcal{U}_{\text{eq}} = q \log M_k - \log B_\beta - \log \left(\frac{\sum_j \eta_j}{(q+1)^q} \right) - \log V_{2q-1}^\nu.$$

Thus its exact improvement is

$$\mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} = \mathfrak{g}(\eta), \quad 0 \leq \mathfrak{g}(\eta) \leq q \log(1 + 1/q) < 1. \quad (\text{OW.28})$$

This is an absolute bound, not a bound per coordinate or per degree. For the original complete quartet, $q = [1 + k(m-1)](k+1)^2$ tends to infinity. Dividing (OW.28) by $q \log k$ or by q^2 proves that the difference tends to zero on either scale. This conclusion requires no bound at all on the unknown comparison determinant's own asymptotic size.

For that quartet the gain is strictly positive and strictly smaller than the stated bound. Indeed its sum roots in the u chart are $(2b-k)\gamma - i(2a-k)\delta$, $0 \leq a, b \leq k$, each with the full length $1 + k(m-1)$. They are stable under negation, so comparing leading coefficients gives $\psi(-u) = (-1)^q \psi(u)$. Hence coefficients of the opposite parity vanish, while $\psi_q = 1$ is nonzero. The coefficients η_j cannot all be equal. Also at least one root is nonzero because $\delta, \gamma > 0$, so ψ is not the monomial u^q and at least two coefficients are nonzero. Applying both equality classifications in (OW.23) proves

$$0 < \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} < q \log(1 + 1/q) < 1 \quad \text{for every admitted complete quartet.} \quad (\text{OW.29})$$

The nonmonomial assertion follows directly from the factorizations: the roots of u^q are all zero, whereas the displayed root list contains a nonzero root. Its exact replacement here is the signed co-factor map (OW.4), whose nonzero extra coefficient yields the additional positive determinant term. No relation or jet is removed.

51.6 Joint even moments and exact optimization of their diagonal family

Keep the source, chart, polynomial and every raw-jet row from (OW.1)–(OW.7). For $0 \leq j \leq q$ write

$$\mu_{2j} := \int_{\mathbb{R}} u^{2j} m_k(u) du > 0, \quad \mu_0 = \mu_h^k, \quad S_q(b) := \sum_{j=0}^q \beta_j \mu_{2j}. \quad (\text{OW.30})$$

The inequalities are strict because the source density is positive almost everywhere and $u^{2j} > 0$ off zero. Finiteness follows from (OW.6) and its exponential-series majorant. These are moments in the original u chart; the zeroth moment remains the full k -fold source mass.

For a coefficient vector $p = (p_0, \dots, p_q)^T$ the ordinary complex Cauchy–Schwarz inequality gives the pointwise statement

$$\left| \sum_{j=0}^q p_j u^j \right|^2 \leq \left(\sum_{j=0}^q \frac{|p_j|^2}{\beta_j} \right) \left(\sum_{j=0}^q \beta_j u^{2j} \right) \leq \left(\sum_{j=0}^q \frac{|p_j|^2}{\beta_j} \right) \cosh(bu).$$

The first inequality uses the two vectors $p_j / \sqrt{\beta_j}$ and $\sqrt{\beta_j} u^j$, with complex conjugation in the inner product; the second uses the full nonnegative even exponential series. Its finite tail is strictly positive for every $u \neq 0$. Integration, including both original Laplace values, therefore proves

$$\begin{aligned} H_q &\preceq S_q(b) \operatorname{diag}(\beta_j^{-1}) \prec \frac{M_k}{2} \operatorname{diag}(\beta_j^{-1}), \\ 0 < S_q(b) &< \int_{\mathbb{R}} \cosh(bu) m_k(u) du = \frac{M_h(b)^k + M_h(-b)^k}{2} = \frac{M_k}{2}. \end{aligned} \quad (\text{OW.31})$$

The second matrix inequality is strict positive definiteness because it is a strictly positive scalar difference times a positive diagonal matrix. The convolution identity here follows by integrating $e^{b(t_1 + \dots + t_k)}$ against $\prod_i w_h(t_i) dt_i$; its nonnegative integrand permits iterated integration. The identical calculation applies at $-b$.

Inverting (OW.31) and applying the unchanged full raw-jet matrix gives

$$K_q^{\text{raw}} \succeq S_q(b)^{-1} E_q \operatorname{diag}(\beta_j) E_q^*.$$

The determinant identity (OW.7) at the literal weights all equal to one is $|\det Z|^2 B_\beta \sum_j \eta_j$. The original numerator comparison $K_{2q-1}^{\text{raw}} \preceq K_{2q-1}^{\nu, \text{raw}}$ is unchanged. Hence

$$\begin{aligned} C_k &\leq \mathcal{U}_{\text{fin}} := q \log S_q(b) - \log B_\beta - \log \left(\sum_j \eta_j \right) - \log V_{2q-1}^\nu, \\ \mathcal{U}_{\text{fin}} &< \mathcal{U}_{\text{cosh}} := q \log(M_k/2) - \log B_\beta - \log \left(\sum_j \eta_j \right) - \log V_{2q-1}^\nu. \end{aligned} \quad (\text{OW.32})$$

Thus the pointwise finite sum supplies its own proved source bound; it does not use the older scalar estimate $M_k \sum \tau_j$ from AU.19. Direct subtraction, with no moment or factorial discarded, gives

$$\begin{aligned} \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{cosh}} &= q \log(2(q+1)), \\ \mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{fin}} &= q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} > 0. \end{aligned} \quad (\text{OW.33})$$

The positivity follows from $\mathfrak{g}(\eta) < 1$ and $q \log(2(q+1)) \geq \log 4 > 1$ for $q \geq 1$. For the last elementary inequality, $\log 4 = 2 \int_1^2 dt/t > 2(2-1)/2 = 1$.

The same argument determines a larger family completely. To avoid identifying these new diagonal weights with the cofactor coefficients a_j in (OW.7), denote them by $r_j > 0$ and put

$$s(r) := \sum_{j=0}^q r_j \mu_{2j}, \quad H_q \preceq s(r) \operatorname{diag}(r_j^{-1}), \quad K_q^{\text{raw}} \succeq s(r)^{-1} E_q \operatorname{diag}(r_j) E_q^*. \quad (\text{OW.34})$$

Indeed pointwise Cauchy–Schwarz with $p_j / \sqrt{r_j}$ and $\sqrt{r_j} u^j$, followed by integration, gives the middle inequality for every coefficient vector p . Inversion reverses the positive definite matrix order: if $X \preceq Y$, then $Y^{-1/2} X Y^{-1/2} \preceq I$ and its positive eigenvalues are at most one, so $X^{-1} \succeq Y^{-1}$. Congruence by E_q proves the final inequality. This explicitly specifies the source-to-raw-jet map.

The positive-cone map and its inverse are

$$(r_j) \mapsto \left(s(r), t_j = \frac{r_j \mu_{2j}}{s(r)} \right), \quad (s, t) \mapsto r_j = \frac{s t_j}{\mu_{2j}}. \quad (\text{OW.35})$$

They are mutually inverse bijections between $\mathbb{R}_{>0}^{q+1}$ and $(0, \infty) \times \mathcal{S}_q^\circ$. Positivity of every moment proves each division is defined; summing the displayed t_j proves the simplex constraint, and substituting either composite restores every original index. This map acts only on auxiliary diagonal weights. It does not divide the source measure by its mass or alter a polynomial, jet, source coordinate, or quotient coordinate.

Put $\kappa_j = |\psi_j|^2 \mu_{2j}$, so $\kappa_q = \mu_{2q} > 0$. Applying (OW.4) and Cauchy–Binet before any phase cancellation gives

$$\begin{aligned} \frac{\det(E_q \operatorname{diag}(r_j) E_q^*)}{s(r)^q} &= |\det Z|^2 \sum_{j=0}^q |\psi_j|^2 \prod_{l \neq j} \frac{r_l}{s(r)} \\ &= |\det Z|^2 \left(\prod_{j=0}^q \mu_{2j}^{-1} \right) D_\kappa(t). \end{aligned} \quad (\text{OW.36})$$

The product has exactly $q+1$ moments. In each degree- q monomial its missing moment is supplied by the factor μ_{2j} in κ_j ; thus it includes, without a changed exponent, $\mu_0 = \mu_h^k$. Every original coefficient phase still occurs in (OW.4) and (OW.13), and each zero ψ_j gives its precise zero coefficient κ_j .

Combining (OW.34)–(OW.36) with the numerator comparison and the exact raw-jet determinant conversion gives the complete optimized bound

$$C_k \leq \mathcal{U}_{\text{mom}} := \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log V_{2q-1}^\nu. \quad (\text{OW.37})$$

The earlier global optimizer theorems apply to the literal list $\kappa_0, \dots, \kappa_q$, so (OW.8)–(OW.22a) calculate $d(\kappa)$ with all its boundary cases, scalar root, moment dependence and raw polynomial multiplicities retained. They also prove that (OW.37) is the infimum over the entire positive diagonal family (OW.34). For completeness at a boundary optimum t^* the limiting raw matrix is $E_q \operatorname{diag}(t_j^*/\mu_{2j}) E_q^*$, whose determinant is positive by (OW.36). Regularize t^* by $(1 - \varepsilon)t^* + \varepsilon t^{\text{eq}}$ and use $r_j = t_j/\mu_{2j}$, so $s(r) = 1$. For every $\varepsilon > 0$ (OW.34) holds, and the raw matrices converge entrywise to that positive definite limit. Their quadratic-form inequalities and determinants pass to the limit exactly as in (OW.26). No inverse of a zero boundary weight is evaluated.

The finite even-series weights $r_j = \beta_j$ correspond under (OW.35) to the explicit interior point

$$t_j^\beta = \frac{\beta_j \mu_{2j}}{S_q(b)}, \quad D_\kappa(t^\beta) = \left(\prod_{j=0}^q \mu_{2j} \right) \frac{B_\beta \sum_j \eta_j}{S_q(b)^q}. \quad (\text{OW.38})$$

This equality follows term by term from the defining polynomial; it also follows by setting $r = \beta$ in (OW.36). Since its value is positive, all the following logarithms are defined. Their exact comparison is

$$\begin{aligned} \mathcal{U}_{\text{fin}} - \mathcal{U}_{\text{mom}} &= \log \frac{d(\kappa)}{D_\kappa(t^\beta)} \geq 0, \\ \mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{mom}} &= q \log(2(q+1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} + \log \frac{d(\kappa)}{D_\kappa(t^\beta)}. \end{aligned} \quad (\text{OW.39})$$

The last logarithm is exactly computable from the proved optimizer; no less-than-one claim is made for it relative to the generally nonuniform point t^β . If the moment simplex instead uses its own equal

point t^{eq} , its gain is precisely $\mathfrak{g}(\kappa) < 1$ by (OW.23). These different reference points are related by the explicit map (OW.35), rather than silently identified.

The exact removed term $q \log(2(q+1))$ has order $q \log q$. For the original $q = [1+k(m-1)](k+1)^2$ and fixed multiplicity m , its ratio to $q \log k$ tends to 2 if $m = 1$ and to 3 if $m > 1$. Indeed $q = (k+1)^2$ in the first case; in the second $q/k^3 = (m-1+1/k)(1+1/k)^2 \rightarrow m-1 > 0$. Taking logarithms proves both limits, while $\log(2(q+1))/q \rightarrow 0$ proves its ratio to q^2 tends to zero. The literal comparison (OW.39) retains all additional finite moment gains. It does not determine their growth or the comparison determinant's growth. In particular it does not subtract a term from AU's separately derived interval scalar bound without a proved comparison to that bound. It gives a stronger, fully specified source inequality and its original quotient-volume consequence using the actual finite moments.

There is also an exact finite comparison to the full original low-degree kernel, retaining all mixed moments. Let $\psi = (\psi_0, \dots, \psi_q)^T$ and let the coefficient matrix T have columns $1, u, \dots, u^{q-1}, \psi$ in the monomial basis of degree at most q . Its upper block is the identity and its last diagonal entry is $\psi_q = 1$, so $\det T = 1$. The last vector spans the relation kernel of J_q . In the Gram $T^* H_q T$, eliminate that last block by its positive scalar $\psi^* H_q \psi$. The resulting Schur complement is the Gram of the least lifts of the first q quotient coordinates: subtracting the orthogonal projection onto the relation is exactly the minimization defining G_q . Taking determinants therefore proves

$$V_q = \frac{\det H_q}{\psi^* H_q \psi}, \quad \mathcal{U}_{\text{low}} := \log V_q - \log V_{2q-1}^{\nu}, \quad (\text{OW.40})$$

$$0 \leq \mathcal{U}_{\text{mom}} - \mathcal{U}_{\text{low}} = \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log \det H_q + \log(\psi^* H_q \psi).$$

The sign follows directly from the limiting raw-kernel inequality used in (OW.37): its determinant gives $|\det Z|^2/V_q \geq |\det Z|^2(\prod_j \mu_{2j}^{-1})d(\kappa)$. Its cancellation is legitimate because Z is invertible by (OW.3). The numerator comparison also gives $C_k \leq \mathcal{U}_{\text{low}}$. Thus (OW.40) records the exact nonnegative finite loss of the fully optimized diagonal source estimate relative to the full original H_q kernel. All entries of H_q are actual moments through degree $2q$ and every determinant is on its declared original space; the formula imposes no estimate on the loss's asymptotic growth.

51.7 Verification boundary and continuity

The sealed AU article is an input and remains unmodified. All new mathematics is written in (OW.1)–(OW.40): a signed full-jet identity, global maximum classification, monotone scalar evaluation, complete uncertain-input enclosures, the exact gain within AU.19, and the stronger joint-moment source family with its complete optimum. The auxiliary weight simplex and all its coordinate maps are specified separately from the original arithmetic source; its supported relations, Taylor unit, and quotient action are unchanged because every matrix uses the same E_N, J_N, ψ and (OW.2). A quotient-killed class remains the supported zero at its original label, and the induced fibre map fixes the external absence τ ; the optimization variables τ_j in this paper are real auxiliary weights with their own declared domain (OW.8).

Finite fixtures accompany the written proof and test its exact maps and optimum formulas on declared coefficient lists. They do not infer an arithmetic asymptotic from numerical optimization, assert new zero data, or replace the complete global proofs above.

52 Original relation moments and their positive-form comparison

This supplement gives the complete finite proof behind the incoming relation-moment certificate, records its two domain corrections, and constructs the exact source factorization for an upper comparison kernel. Every quotient coordinate, original source norm, zero eigenvalue, and source label

is retained. The comparison kernel is additional data on the existing coefficient space; it does not replace the arithmetic norm. The arithmetic construction of that kernel is supplied by the cumulative paper's arithmetic upper-comparison proof.

52.1 The original Hilbert objects and their maps

Fix a monic polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$ and retain $E = \mathbb{C}[S]/(\chi)$ with ordered remainder basis $1, S, \dots, S^{q-1}$. The original multiplication endomorphism is $A = M_S$. Fix the original monic orthogonal polynomial columns p_n , their literal positive squared norms ω_n , and $b_n = [p_n]_\chi$. The degree- N source $\mathcal{P}_N = (\mathbb{C}^{N+1}, O_N)$ has $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and quotient map $B_N = [b_0, \dots, b_N]$. For $N \geq q - 1$, its first q columns form an invertible triangular matrix, so B_N is surjective and

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad V_N = \det G_N > 0. \quad (\text{RM.1})$$

Here K_N is a positive form on the conjugate dual coefficient space and G_N is the original quotient inner product, conjugate-linear in its first argument. Direct multiplication proves $B_N R_N = 1$ and $R_N^* O_N R_N = G_N$. If $u \in \ker B_N$, then $u^* O_N R_N x = u^* B_N^* G_N x = 0$. Thus R_N is the actual least-norm lift.

Fix $q - 1 \leq i \leq j$, set $r = j - i$, and write

$$F = [b_{i+1}, \dots, b_j], \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j), \quad D = K_j - K_i = F \Omega^{-1} F^*. \quad (\text{RM.2})$$

For $r = 0$ these maps have their literal zero-dimensional domains. The identity $I : E_i = (E, G_i) \rightarrow E_j = (E, G_j)$ has adjoint $T = K_i G_j : E_j \rightarrow E_i$, since $G_i T = G_j$. Consequently

$$Q = T I \in \text{End}(E_i), \quad Z = Q^{-1} - 1 = D G_i = F F^\dagger, \quad F^\dagger = \Omega^{-1} F^* G_i. \quad (\text{RM.3})$$

In particular $G_i Z = G_i D G_i \succeq 0$, so Z is positive self-adjoint in the original G_i . With its full q eigenvalues $\lambda_a \geq 0$,

$$g_a = (1 + \lambda_a)^{-1}, \quad \frac{V_i}{V_j} = \det(1 + Z) = \prod_{a=1}^q (1 + \lambda_a). \quad (\text{RM.4})$$

All zero λ_a contribute the factor one and remain in the dimension. If $Z = 0$, then $V_i = V_j > 0$, the ratio equals one, and its logarithm equals zero. This corrects the incoming note's phrase "the volume is zero."

For the real scalar k in the arithmetic action identity, put $H_N = K_N A^* G_N + A - k1$. Regard H_j on E_i via the specified coordinate identity and denote that endomorphism by $\hat{H}_j$. Calculation in the original coordinates gives

$$\begin{aligned} H_i Q - Q \hat{H}_j &= (K_i A^* G_i + A - k1) K_i G_j - K_i G_j (K_j A^* G_j + A - k1) \\ &= A K_i G_j - K_i G_j A = [A, Q]. \end{aligned}$$

Multiplying $[A, Q^{-1}] = -Q^{-1}[A, Q]Q^{-1}$ out proves

$$[A, Z] = \hat{H}_j(1 + Z) - (1 + Z)H_i. \quad (\text{RM.5})$$

Thus the cost and action are related even when their eigenspaces differ.

52.2 The source correction, all cross-pairings, and exterior area

The map $\Phi = R_i F : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{P}_i$ has image Gram $\Phi^* O_i \Phi = F^* G_i F$. Trace cyclicity and Hermitian symmetry give

$$t_1 = \text{Tr } Z = \sum_{n=i+1}^j \frac{b_n^* G_i b_n}{\omega_n}, \quad (\text{RM.6})$$

$$t_2 = \text{Tr } Z^2 = \sum_{n,m=i+1}^j \frac{|b_n^* G_i b_m|^2}{\omega_n \omega_m}, \quad (\text{RM.7})$$

$$\mathcal{A}_2 = \frac{t_1^2 - t_2}{2} = \sum_{i < n < m \leq j} \frac{(b_n^* G_i b_n)(b_m^* G_i b_m) - |b_n^* G_i b_m|^2}{\omega_n \omega_m}. \quad (\text{RM.8})$$

The diagonal terms cancel in $t_1^2 - t_2$; each unordered pair occurs twice, proving the last identity with its literal factor $1/2$. In unscaled exterior bases endowed with determinant Gram matrices,

$$\|\Phi\|_{\text{HS}}^2 = t_1, \quad \left\| \bigwedge^2 \Phi \right\|_{\text{HS}}^2 = \mathcal{A}_2. \quad (\text{RM.9})$$

Indeed the source vector $e_n \wedge e_m$ has squared norm $\omega_n \omega_m$ and its image has exactly the numerator in (RM.8). The alternating tensor map $u \wedge v \mapsto u \otimes v - v \otimes u$ has image Gram twice the exterior Gram. Applying this map in both source and target yields the commutative exterior/tensor square and cancels that same factor in the Hilbert–Schmidt ratio; neither Gram is identified with the other without the factor.

Let $L : \mathcal{P}_i \rightarrow \mathcal{P}_j$ pad by zero and let $N : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{P}_j$ be the actual new-column inclusion. Their images are orthogonal and $B_j N = F$, $B_j L = B_i$. Hence the retained relation map

$$\mathfrak{b} = N - L R_i F, \quad B_j \mathfrak{b} = 0, \quad \mathfrak{b}^* O_j \mathfrak{b} = \Omega + F^* G_i F. \quad (\text{RM.10})$$

This proves the relation identity and its full norm. Under the original source realization $\mathcal{V}$, the equality $J^{(k)} \mathcal{V} = \eta \pi_{\chi}$ makes both components have the same arithmetic jets. The original internal quotient sends $(\ell_j, \mathfrak{b}v)$ to $(\ell_j, 0)$, preserving the receiving label. Its lift sends external absence τ to τ . The coefficient amplitude map $p_R : G(R) \rightarrow R$ sends both $0_R^\bullet$ and τ to 0_R , and the exact fibre $p_R^{-1}(0_R) = \{0_R^\bullet, \tau\}$ records precisely the information lost by this subsequent observation. Likewise a dependent admitted wedge remains an admitted source vector whose observed squared area is $0^\bullet$.

52.3 Sharp determinant bound, exact remainder, and all boundary cases

For $q \geq 2$, positivity gives $t_1^2/q \leq t_2 \leq t_1^2$ by Cauchy–Schwarz and the nonnegative pair products. Define

$$d = \sqrt{\frac{q t_2 - t_1^2}{q-1}}, \quad a = \frac{t_1 + (q-1)d}{q}, \quad b = \frac{t_1 - d}{q}. \quad (\text{RM.11})$$

Thus $0 \leq d \leq t_1$, $0 \leq b \leq a$, and direct expansion proves $a + (q-1)b = t_1$ and $a^2 + (q-1)b^2 = t_2$. For any eigenvalue x , the other $q-1$ eigenvalues give $t_2 \geq x^2 + (t_1 - x)^2/(q-1)$. Solving that quadratic yields $x \leq a$. Consequently $Q \succeq (1+a)^{-1}1$ in G_i .

Here is a direct algebraic proof of the entire interpolation remainder. For $s \geq 1$, let

$$H_s(x) = \frac{1}{s+b} - \frac{x-b}{(s+b)^2} + \frac{(x-b)^2}{(s+a)(s+b)^2}. \quad (\text{RM.12})$$

Its value and derivative agree with $1/(s+x)$ at b , and its value agrees at a . Multiplication by the common denominator proves, identically,

$$\frac{1}{s+x} - H_s(x) = \frac{(a-x)(x-b)^2}{(s+x)(s+b)^2(s+a)}. \quad (\text{RM.13})$$

Integrate $1/s - 1/(s+x)$ over $[1, \infty)$ to obtain $\log(1+x)$. Integration of $1/s - H_s(x)$ gives its quadratic interpolant

$$P(x) = \log(1+b) + \frac{x-b}{1+b} + c(x-b)^2, \\ c = \frac{\log(1+a) - \log(1+b) - (a-b)/(1+b)}{(a-b)^2} < 0 \quad (a > b). \quad (\text{RM.14})$$

The coefficient integrals converge: the quadratic coefficient is $-\int_1^\infty ((s+a)(s+b)^2)^{-1} ds$, and the other two are the displayed logarithm and reciprocal. Therefore, for $0 \leq x \leq a$,

$$P(x) - \log(1+x) = (a-x)(x-b)^2 \int_1^\infty \frac{ds}{(s+x)(s+b)^2(s+a)} \geq 0. \quad (\text{RM.15})$$

The trace of any quadratic depends only on q, t_1, t_2 . Those values equal the moments of $a, b, \dots, b$, so summing (RM.15) proves

$$\log(V_i/V_j) \leq U_2 := \log(1+a) + (q-1)\log(1+b). \quad (\text{RM.16})$$

The remainder in (RM.15) is strictly positive away from $x = a, b$. If m eigenvalues equal a , the first trace identity gives $ma + (q-m)b = a + (q-1)b$, hence $(m-1)(a-b) = 0$. Equality for $a > b$ therefore occurs exactly once at a and $q-1$ times at b . That spectrum realizes the prescribed moments, proving sharpness from these data. For $a = b$, zero variance forces $Z = b1$ and equality already holds. For $q = 1$, the ratio is exactly $1 + t_1$. For $q = 0$, its empty determinant is one. The case $i = j$ gives $Z = 0$ at every admitted dimension.

With $t_3 = \text{Tr } Z^3$, set

$$C_3 = -t_3 + (a+2b)t_2 - (2ab+b^2)t_1 + qab^2 = \text{Tr}((a1-Z)(Z-b1)^2) \geq 0. \quad (\text{RM.17})$$

Finite spectral summation of (RM.15) proves the exact loss

$$U_2 - \log \det(1+Z) = \int_1^\infty \frac{\text{Tr}((a1-Z)(Z-b1)^2(s1+Z)^{-1})}{(s+b)^2(s+a)} ds. \quad (\text{RM.18})$$

Every factor is a function of the same G_i -self-adjoint endomorphism. Since $(s+x)(s+b)^2(s+a) \leq (s+a)^4$, the loss is at least $C_3 \int_1^\infty (s+a)^{-4} ds = C_3/(3(1+a)^3)$. Also Jensen's scalar inequality, whose strictness follows from $\frac{d^2}{dx^2} \log(1+x) = -(1+x)^{-2} < 0$, gives $U_2 \leq q \log(1+t_1/q)$, strictly unless $a = b$.

For $q \geq 2$ and $t_1 > 0$, $\mathcal{A}_2 = (q-1)b(t_1 - qb/2)$ and $0 \leq b \leq t_1/q$ imply

$$\frac{\mathcal{A}_2}{(q-1)t_1} \leq b \leq \frac{2\mathcal{A}_2}{(q-1)t_1}, \quad \frac{V_i}{V_j} \leq (1+t_1) \left(1 + \frac{2\mathcal{A}_2}{(q-1)t_1}\right)^{q-1}. \quad (\text{RM.19})$$

When $\mathcal{A}_2 = 0 < t_1$, all eigenvalue pair products vanish, so exactly one eigenvalue is positive and the ratio is $1 + t_1$.

52.4 The comparator join in the original metric

Let U_j be the actual explicit upper kernel supplied by the arithmetic comparison construction, expressed in the same quotient coordinates. That construction proves $U_j \succeq K_j \succeq K_i$ as dual forms. Define from the available lower-degree arithmetic data and this comparator

$$D_+ = U_j - K_i, \quad Z_+ = D_+ G_i, \quad t_1^+ = \text{Tr } Z_+, \quad t_2^+ = \text{Tr } Z_+^2. \quad (\text{RM.20})$$

Then $G_i(Z_+ - Z) = G_i(U_j - K_j)G_i \succeq 0$. Thus both operators are positive self-adjoint in G_i and $Z_+ \succeq Z$ there. For completeness, ordered eigenvalues increase under this order: for an orthonormal

eigenbasis of either operator, the top- m span gives the lower inequality below; every m -dimensional subspace intersects the span of eigenvectors numbered $m, \dots, q$, which gives the upper inequality. This proves the max-min characterization

$$\lambda_m(X) = \max_{\dim M=m} \min_{v \in M \setminus 0} \frac{\langle v, Xv \rangle_{G_i}}{\langle v, v \rangle_{G_i}}.$$

Applying the pointwise form inequality inside this characterization yields $\lambda_m(Z_+) \geq \lambda_m(Z)$ for each of the original q indices. In particular, for $q \geq 2$, with a_+, b_+ obtained by (RM.11) from t_1^+, t_2^+ ,

$$\frac{V_i}{V_j} \leq \det(1 + Z_+) = \det(U_j G_i) \leq (1 + a_+)(1 + b_+)^{q-1}. \quad (\text{RM.21})$$

No monotonicity of the sharp bound as a function of two separately varied trace numbers was used; the first inequality follows from the actual form order, and the second follows from (RM.16) for Z_+ . For $q \geq 2$, $Q \succeq (1 + a_+)^{-1}1$ is a derived finite gap. For $q = 1$, the comparator determinant is $1 + t_1^+$ and $Q \succeq (1 + t_1^+)^{-1}1$; for $q = 0$, the empty determinant is one. When $i \leq q$, the actual input to K_i, G_i uses moments only through $2i \leq 2q$; U_j uses the explicit comparison measure's moments through $2j \leq 4q$ in the two endpoint blocks. Formula (RM.21) supplies a certificate without introducing the unknown actual high-degree moments into its inputs.

There is an exact source map behind this comparison. Interpret D_+ as a positive Hermitian form on the conjugate dual coefficient space $E^\vee$. Let

$$\mathcal{C}_+ = E^\vee / \ker D_+, \quad \langle [\alpha], [\beta] \rangle_+ = \alpha^* D_+ \beta, \quad J_+[\alpha] = D_+ \alpha \in E_i. \quad (\text{RM.22})$$

These expressions are independent of representatives. Positivity implies the only zero-norm classes are the removed radical: diagonalize the positive Hermitian form in the stated coordinate space to prove $\alpha^* D_+ \alpha = 0 \iff D_+ \alpha = 0$. This diagonalization proves a fact about that form and does not alter G_i . The adjoint is $J_+^\dagger v = [G_i v]$, hence $J_+ J_+^\dagger = Z_+$ and

$$\Phi_+ = R_i J_+ : \mathcal{C}_+ \rightarrow \mathcal{P}_i, \quad \|\Phi_+\|_{\text{HS}}^2 = t_1^+, \quad \|\bigwedge^2 \Phi_+\|_{\text{HS}}^2 = ((t_1^+)^2 - t_2^+)/2. \quad (\text{RM.23})$$

To prove the second identity in any Gram basis, diagonalize the positive endomorphism $\Phi_+^\dagger \Phi_+$ on its existing source Hilbert space. Its exterior eigenvalues are all unordered products. Its nonzero eigenvalues agree, with multiplicity, with those of $\Phi_+ \Phi_+^\dagger$ by the mutually inverse maps Φ_+ and $\lambda^{-1} \Phi_+^\dagger$ on each nonzero eigenspace. Finally R_i is an isometry from E_i into $\mathcal{P}_i$, which proves (RM.23), retaining the zero eigenspaces.

The original correction factors through this source by an exact contraction. For $\alpha \in \ker D_+$, (RM.2) and $D \preceq D_+$ imply $\alpha^* F \Omega^{-1} F^* \alpha = 0$, so $F^* \alpha = 0$. Thus $Fv \in \text{im } D_+$ for every v . There is a unique class $Cv = [u]$ solving $D_+ u = Fv$. This defines a linear map

$$C : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{C}_+, \quad J_+ C = F, \quad \Phi_+ C = \Phi. \quad (\text{RM.24})$$

Let $s = u^* D_+ u \geq 0$. The equality $D_+ u = Fv$ and Cauchy-Schwarz in Ω give $s = |u^* Fv| \leq (u^* F \Omega^{-1} F^* u)^{1/2} (v^* \Omega v)^{1/2} \leq s^{1/2} (v^* \Omega v)^{1/2}$. Dividing when $s > 0$, with the case $s = 0$ immediate, proves $\|Cv\|_+^2 \leq v^* \Omega v$. The full kernel is $\ker C = \ker F$ by (RM.24) and uniqueness of the class. Exterior functoriality gives $\bigwedge^2 \Phi = (\bigwedge^2 \Phi_+) (\bigwedge^2 C)$, preserving every admitted pair. This is the exact morphism from the old source correction to the new positive-form source. The original relation $N - L\Phi$ from (RM.10) is retained, with its original supported-zero image.

The cost-action defect of the comparator also has an explicit form. Put $\Delta = U_j - K_j$. Then $Z_+ = Z + \Delta G_i$, and direct expansion using $\hat{H}_j = A + K_j A^* G_j - k1$ gives

$$\begin{aligned} [A, Z_+] &= \hat{H}_j (1 + Z_+) - (1 + Z_+) \hat{H}_j + \mathcal{E}, \\ \mathcal{E} &= [A, \Delta G_i] - \hat{H}_j \Delta G_i + \Delta G_i \hat{H}_j \\ &= \Delta A^* G_i - K_j A^* G_j \Delta G_i. \end{aligned} \quad (\text{RM.25})$$

The second line defines the exact defect; substituting the displayed control matrices and cancelling equal terms proves the last expression. It is not discarded or asserted small.

52.5 Literal Gaussian calibration and source corrections

For the declared polynomial calibration, retain $S = 1 + ix$, literal Gaussian mass seven, variance one, $p_{n+1} = (S - 1)p_n + np_{n-1}$ and $\omega_n = 7n!$. For $q = 3$, write $y = S - 1$ only as the explicit polynomial $S - 1$, without changing the remainder basis. The relevant columns, expressed in the fixed basis $1, S, S^2$, are

$$\begin{aligned} b_0 &= (1, 0, 0)^t, \quad b_1 = (-1, 1, 0)^t, \quad b_2 = (2, -2, 1)^t, \quad b_3 = (-3, 3, 0)^t, \\ b_4 &= (9, -12, 6)^t, \quad b_5 = (-15, 15, 0)^t, \quad b_6 = (60, -90, 45)^t. \end{aligned}$$

These follow by the literal recurrence and division by $(S - 1)^3$. Substitution into (RM.1)–(RM.3) gives

$$Z_{3,6} = \frac{1}{8} \begin{pmatrix} -29 & -35 & 84 \\ 42 & 48 & -126 \\ -21 & -21 & 69 \end{pmatrix}. \quad (\text{RM.26})$$

The direct recurrence gives $G_3 = \begin{pmatrix} 7 & 7 & 0 \\ 7 & 49/5 & 28/5 \\ 0 & 28/5 & 126/5 \end{pmatrix}$, so (RM.26) is its literal coefficient matrix in the retained metric. Its invariant conclusions are

$$t_1 = 11, \quad t_2 = 3365/32, \quad t_3 = 273947/256, \quad \mathcal{A}_2 = 507/64, \quad \det(1 + Z) = 5145/256. \quad (\text{RM.27})$$

The characteristic polynomial factors with roots $3/4, (41 - 7\sqrt{34})/8, (41 + 7\sqrt{34})/8$; all are positive, distinct, and retain $q = 3$. The two-trace extremal roots are $a = 11/3 + 7\sqrt{127}/12, b = 11/3 - 7\sqrt{127}/24$. Since the actual three roots are distinct and the equality spectrum in (RM.16) has two equal roots, its inequality is strict. Since $a > b$, its improvement over $3 \log(1 + 11/3)$ is also strict. Rational logarithm enclosures in the companion replay give the numerical values in the incoming note without relying on decimals for either strictness argument.

In dimension two, the same literal recurrence gives $(t_1, t_2) = (2, 5/2)$ for $(i, j) = (1, 3)$ and $(7/4, 37/16)$ for $(2, 4)$. The respective (a, b) are $(3/2, 1/2)$ and $(3/2, 1/4)$, so (RM.16) is exact and yields ratios $15/4$ and $25/8$.

The incoming window expressions contain $1/(2r)$ and a minimum over $i \leq N < j$. Their domain is $i < j$. The original and comparator maps retain $i = j$ as well; scalar formulas keep their explicitly stated dimension and positive-trace domains, with zero cost handled separately. No minimum over an empty window is defined, and no inverse below the first admissible degree $q - 1$ is introduced. The complete first-degree and consecutive-window proofs remain in the cumulative reader.

52.6 Rank information with every zero direction retained

Let $W = \text{im } F \subset E_i$ and $\rho = \dim W$. Since $\langle v, Zv \rangle_{G_i} = \|F^\dagger v\|_\Omega^2$, one has

$$\ker Z = \ker F^\dagger = W^{\perp_{G_i}}, \quad E_i = W \oplus^{\perp_{G_i}} W^{\perp_{G_i}}. \quad (\text{RM.28})$$

To check invariance and positivity on W , $Z = FF^\dagger$ has image in W , and the only vector of W in its kernel is zero by (RM.28). The restriction Z_W is therefore positive definite on its inherited metric, and the literal inclusion/projection maps give

$$Z = Z_W \oplus 0_{q-\rho}, \quad Q = (1_W + Z_W)^{-1} \oplus 1_{q-\rho}, \quad \det(1_E + Z) = \det(1_W + Z_W) 1^{q-\rho}. \quad (\text{RM.29})$$

The source kernel is also retained. Put $\mathcal{U} = (\ker F)^{\perp_\Omega}$ in the original source. Its orthogonal projection $P_\mathcal{U}$ satisfies $F = F|_\mathcal{U} P_\mathcal{U}$. The map $F|_\mathcal{U} : \mathcal{U} \rightarrow W$ is a bijection: injectivity is the orthogonal intersection with $\ker F$, and surjectivity follows by decomposing any source preimage. Thus it identifies

the source Hilbert quotient $(\mathbb{C}^r, \Omega)/\ker F$ with W by a specified map; the norms on these objects are related by that map's retained positive Gram, rather than being asserted equal. Its composition with R_i is precisely the old least-lift correction on the quotient source. For $v \in \ker F$, the original relation (RM.10) is Nv , so the kernel directions remain actual newly admitted theta relations, with their norm $v^*\Omega v$ and their receiving supported-zero labels.

Any proved integer upper bound $\rho \leq R \leq q$ gives a sharper finite determinant certificate. The ρ positive eigenvalues and $R - \rho$ expressly retained zeros have the same t_1, t_2 . For $R \geq 2$ set

$$d_R = \sqrt{\frac{Rt_2 - t_1^2}{R-1}}, \quad a_R = \frac{t_1 + (R-1)d_R}{R}, \quad b_R = \frac{t_1 - d_R}{R}. \quad (\text{RM.30})$$

Applying the already proved scalar argument to that R -entry list gives

$$\det(1_E + Z) \leq (1 + a_R)(1 + b_R)^{R-1} 1^{q-R}. \quad (\text{RM.31})$$

This operation does not identify or remove the $q - R$ original directions. The full decomposition (RM.29) records their action by Q and Z . For $R = 1$, the exact ratio is $1 + t_1$; for $R = 0$, it is one. One always has $\rho \leq \min(q, j - i)$ from the literal dimensions of F . Additional parity or relation identities may give a better proved bound; their full proofs must supply that information. The present argument makes no such unproved rank assertion for an arithmetic packet.

For fixed feasible t_1, t_2 , this bound is at least as sharp as the q -dimensional bound. Indeed the R -entry extremizer $a_R, b_R, \dots, b_R$ with $q - R$ additional zeros is feasible for the q -dimensional moment problem. The optimality already proved for that problem yields

$$\log(1 + a_R) + (R - 1) \log(1 + b_R) \leq \log(1 + a) + (q - 1) \log(1 + b).$$

The known rank bound belongs to the actual cost Z . To apply (RM.30)–(RM.31) to Z_+ , its own rank must be proved; the inequality $Z_+ \succeq Z$ does not imply $\text{rank } Z_+ \leq \text{rank } Z$. Its exact source in (RM.22) gives $\text{rank } Z_+ = \dim \mathcal{C}_+ = \text{rank } D_+$, which is the correct replacement.

52.7 The entire comparison excess is the source contraction defect

The contraction in (RM.24) has a concrete adjoint:

$$C^\dagger[\alpha] = \Omega^{-1} F^* \alpha. \quad (\text{RM.32})$$

It is well-defined because $\ker D_+ \subset \ker F^*$ was proved above. For $Cv = [u]$, its defining equation gives $\langle Cv, [\alpha] \rangle_+ = u^* D_+ \alpha = v^* F^* \alpha = \langle v, \Omega^{-1} F^* \alpha \rangle_\Omega$, proving the adjoint identity. Define $\mathcal{L} = 1_{\mathcal{C}_+} - CC^\dagger$. Then

$$\langle [\alpha], \mathcal{L}[\beta] \rangle_+ = \alpha^* (D_+ - F\Omega^{-1}F^*)\beta = \alpha^* (U_j - K_j)\beta, \quad (\text{RM.33})$$

$$\mathcal{L} \succeq 0, \quad J_+ \mathcal{L} J_+^\dagger = (U_j - K_j)G_i = Z_+ - Z. \quad (\text{RM.34})$$

The first equality follows by (RM.32) and the definition of the adjoint; the second follows by (RM.2) and (RM.20). The right-hand form is nonnegative by the actual upper-kernel comparison. Multiplying out the left side of (RM.34) gives $J_+ J_+^\dagger - J_+ C C^\dagger J_+^\dagger = Z_+ - F F^\dagger$, proving its last equality. Thus the complete excess of the comparator over the arithmetic cost is retained as a specific source contraction defect.

For clarity, the arithmetic upper kernel itself is formed on the original polynomial coordinates. Write $X_j = [[S^0]_\chi, \dots, [S^j]_\chi]$ and let H_j^m be the full monomial Gram for the actual source measure $m_{h,k}(u)du$, with $S = k/2 + iu$. The arithmetic upper-comparison proof constructs the explicit positive measure

$$d\nu_{h,k}(u) = a_k e^{-\alpha|u|} (k - 2 + |u|)^{-B} du, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h = 42 + 8m,$$

$$a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}, \quad k \geq 3,$$

where $m \geq 1$ is the full common multiplicity of the four zeros in the quartet and $\deg h = 4m$. Its literal constants are $c_h > 0$ and $\vartheta_h = \int_{-1}^1 w_h(t) dt > 0$. That complete proof establishes $d\nu_{h,k} \leq m_{h,k}(u) du$. For any polynomial coefficient vector v , $v^* H_j^\nu v = \int |\sum v_n(k/2 + iu)^n|^2 d\nu_{h,k} \leq v^* H_j^m v$. To prove the inverse inequality without a metric change, the exact completion of the square gives, for either positive form H ,

$$\ell^* H^{-1} \ell = \max_v \{2 \operatorname{Re}(\ell^* v) - v^* H v\},$$

the maximum attained at $v = H^{-1} \ell$. The form inequality therefore implies $(H_j^m)^{-1} \preceq (H_j^\nu)^{-1}$. Consequently

$$U_j = X_j (H_j^\nu)^{-1} X_j^*, \quad K_j = X_j (H_j^m)^{-1} X_j^* \preceq U_j. \quad (\text{RM.35})$$

The expression for K_j agrees exactly with (RM.1): the invertible triangular monic-polynomial coefficient matrix transports H_j^m to O_j and X_j to B_j , so the middle matrices cancel in this composite. All factors $(k/2 + iu)^n$ remain in these Grams. The measure inequality and its constants are supplied by the complete arithmetic upper-comparison proof retained with this supplement; no bound on the actual omitted high-degree moments is assumed.

52.8 Attribution and exact source scope

The general trace-constrained determinant bound is classical. The author-hosted copy of Grone, Johnson, Marques de Sá and Wolkowicz, *Improving Hadamard's inequality*, Linear and Multilinear Algebra 16 (1984), 305–322, identifies the dimension/trace/trace-square optimization; see <https://www.math.uwaterloo.ca/~hwolkowi/henry/reports/PAPER17.pdf> and <https://doi.org/10.1080/03081088408817634>. The complete argument (RM.11)–(RM.19) above is independently supplied here. The added comparison consists of the exact maps (RM.20)–(RM.25).

The unchanged incoming archive has SHA256

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Its full note, markdown, handoff, checker, patch, source metadata and all 37 original file members remain preserved. The note's actual high-degree moment requirements remain stated; its unproved asymptotic example is recorded as motivation in the source, without supplying any missing arithmetic estimate. No Lean, actual zeta-packet quadrature, uniform trace-growth bound, or RH conclusion is asserted by this finite supplement.

53 The first canonical window, its full phase product, and both arithmetic block budgets

This chapter joins the original first-degree source calculation (CV.11)–(CV.12) and (TVB.37)–(TVB.38) to the consecutive-window proof of PR 25 at the exact head 66fac5e7885a40cd4873e74b754907320f05b067. The complete original and proposed corrected sources accompany this chapter. Their correction retains all zero generalized eigenvalues. The source radius, original norm envelope and finite propagation are proved below with their original domains. The full first-window phase product has a single terminal contraction and two copies of each interior contraction; its first step contains no preceding quotient inverse.

53.1 Original source and the actual first-degree step

Retain $g = 2\xi$, the complete fixed packet h , integer tensor degree $k \geq 1$, and the original density

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad S = c + iu, \quad c = k/2. \quad (\text{CJ.1})$$

Its source inner product is the integral against $m_{h,k}(u)du$; the original monic orthogonal polynomials p_n have squared norms $\omega_n > 0$, including $\omega_0 = \mu_h^k$. Use the original dagger-stable monic cyclic polynomial χ of degree $q \geq 1$, the full quotient $E = \mathbb{C}[S]/(\chi)$ and the original remainder basis. Its columns, kernels and metrics are

$$\begin{aligned} b_n &= [p_n]_\chi, & B_N &= [b_0, \dots, b_N], & O_N &= \text{diag}(\omega_0, \dots, \omega_N), \\ K_N &= B_N O_N^{-1} B_N^*, & G_N &= K_N^{-1}, & V_N &= \det G_N \quad (N \geq q-1). \end{aligned} \quad (\text{CJ.2})$$

Surjectivity of B_N and positivity of O_N prove $K_N > 0$. The section $R_N = O_N^{-1} B_N^* G_N$ satisfies $B_N R_N = I$ and $R_N^* O_N R_N = G_N$; its orthogonality to $\ker B_N$ proves that it is the unique minimum lift. Source inclusion proves $G_{N+1} \preceq G_N$ and $0 < V_{N+1} \leq V_N$.

Write the original expansion, retaining every coefficient,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle}{\omega_j}, \quad \nu_0 = \|\chi\|^2 = \omega_q + \sum_{j<q} |\gamma_j|^2 \omega_j. \quad (\text{CJ.3})$$

The first q remainder columns form an invertible triangular matrix, so $b_a^* G_{q-1} b_b = \delta_{ab} \omega_b$ for $a, b < q$. Reduction of (CJ.3) gives $b_q = -\sum_{j<q} \gamma_j b_j$ and hence $b_q^* G_{q-1} b_q = \nu_0 - \omega_q$. The exact rank-one determinant formula therefore gives

$$\delta_q := \frac{V_q}{V_{q-1}} = \left(1 + \frac{b_q^* G_{q-1} b_q}{\omega_q} \right)^{-1} = \frac{\omega_q}{\nu_0}. \quad (\text{CJ.4})$$

The rank-one determinant identity follows by taking a basis whose first vector is the update vector u when it is nonzero. In that basis uv^* has entries only in its first row; $I + uv^*$ is therefore upper triangular with diagonal entries $1 + v^*u, 1, \dots, 1$. Its determinant is $1 + v^*u$. The zero update gives the same identity directly.

Let ϕ_N denote the retained real phase of the original theta deformation, with the sign convention of (TVB.26): $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. Its reality follows in the literal chart $S = c + iu$ as follows. Write $Q_n(u) = i^{-n} p_n(c + iu)$ and $\hat{\chi}(u) = i^{-q} \chi(c + iu)$. Dagger stability gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$, so $\hat{\chi}$ is real monic. The source measure is real and positive; the monic orthogonality equations therefore give real Q_n . If E is the exact coefficient map $P(S) \mapsto P(c + iu)$, then $E b_n = i^n [Q_n]_{\hat{\chi}}$, while the transported kernel and metric are $E K_N E^*$ and $E^{-*} G_N E^{-1}$. The latter has real entries because the phases in each kernel summand occur together with their conjugates. Thus the original cross pairing is i times a real number, with its original factor ω_N retained. This proves the stated reality and specifies the exact chart map rather than changing the metric. At the first degree that identity reads $\gamma_{q-1} = -i\phi_{q-1}$. The original two-column Hermitian control formula (CV.4), before specialization, is

$$\epsilon_N^2 = \frac{a_N d_N - |z_N|^2}{\omega_N^2}, \quad a_N = b_N^* G_N b_N, \quad z_N = b_N^* G_N b_{N+1}, \quad d_N = b_{N+1}^* G_N b_{N+1}. \quad (\text{CJ.5})$$

Here z_N is purely imaginary in the stated dagger-stable source. The original boundary-control identity on (E, G_N) is

$$H_N = G_N^{-1} (A^* G_N + G_N A - k G_N) = \frac{b_{N+1} b_N^* G_N + b_N b_{N+1}^* G_N}{\omega_N}, \quad A = M_S.$$

For completeness this identity follows directly from the original monic recurrence. Self-adjointness of multiplication by u in the integral pairing and orthogonality give $u Q_n = Q_{n+1} + \beta_n Q_n + \alpha_n Q_{n-1}$,

where $\beta_n \in \mathbb{R}$ and $\alpha_n = \omega_n/\omega_{n-1}$ for $n \geq 1$; the lower term is zero at $n = 0$. Indeed all coefficients below degree $n - 1$ vanish by pairing with uQ_j of degree at most $n - 1$, and pairing with Q_{n-1} gives the displayed norm ratio. In the original S coordinates this is $Sp_n = p_{n+1} + (c + i\beta_n)p_n - \alpha_n p_{n-1}$. Reduce it modulo χ and insert it into $AK_N + K_N A^* - kK_N$. The diagonal coefficients are $c + i\beta_n + c - i\beta_n - k = 0$. For each internal adjacent pair the coefficient is $\omega_n^{-1} - \alpha_{n+1}\omega_{n+1}^{-1} = 0$. The remaining two terms are $(b_{N+1}b_N^* + b_N b_{N+1}^*)/\omega_N$. Multiplication on the right by the original G_N proves the written boundary-control formula with its signs. It has rank at most two, is self-adjoint in G_N , and has trace $(z_N + \bar{z}_N)/\omega_N = 0$. Squaring its two summands and taking the trace gives $[z_N^2 + \bar{z}_N^2 + 2a_N d_N]/\omega_N^2 = 2(a_N d_N - |z_N|^2)/\omega_N^2$. For $q \geq 2$ its full spectrum is $\epsilon_N, -\epsilon_N$ and $q - 2$ zeros, with $\epsilon_N \geq 0$; these first two entries may themselves be zero. For $q = 1$ trace zero makes the entire control zero. This proves (CJ.5), including the cross-pairing. At $N = q - 1$ the three entries are respectively $\omega_{q-1}, -\gamma_{q-1}\omega_{q-1}, \nu_0 - \omega_q$. Substitution proves

$$\epsilon_{q-1}^2 + \phi_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}. \quad (\text{CJ.6})$$

For $q = 1$, the lower projection in (CV.12) is the zero source and $\epsilon_{q-1} = 0$; the retained phase identity and (CJ.6) still hold. No G_{q-2} occurs in this calculation.

Define $\Lambda_N = \omega_N/V_N$. Combining (CJ.4)–(CJ.6) gives the exact first step

$$(\epsilon_{q-1}^2 + \phi_{q-1}^2)\Lambda_{q-1} = \Lambda_q(1 - \delta_q). \quad (\text{CJ.7})$$

Indeed its left side is $(\nu_0 - \omega_q)/V_{q-1}$, and $\nu_0/V_{q-1} = \omega_q/V_q$ by (CJ.4). This proves the identity with its single contraction and full original phase.

53.2 The complete first-window product and both finite budgets

At each regular degree $N \geq q$, the proved original radius identity (EP.11), equivalently (TVB.27), is

$$(\epsilon_N^2 + \phi_N^2)\Lambda_N = \Lambda_{N+1}(1 - \delta_N)(1 - \delta_{N+1}), \quad \delta_N = V_N/V_{N-1}. \quad (\text{CJ.8})$$

Every δ_N lies in $(0, 1]$. Multiply (CJ.7) by the consecutive identities (CJ.8) at $N = q, \dots, j - 1$. All Λ_N are positive, so canceling their identical intermediate factors is valid. It yields, for every $j \geq q$,

$$\prod_{N=q-1}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_{q-1}}{\omega_{q-1} V_j} (1 - \delta_j) \prod_{N=q}^{j-1} (1 - \delta_N)^2. \quad (\text{CJ.9})$$

For $j = q$ the interior product is empty and equal to one, so this is precisely the first step. For $j > q$, each $\delta_N, q \leq N < j$, occurs once from the step ending at N and once from the step beginning there. The terminal δ_j occurs once. This proves every exponent in (CJ.9). For a regular initial degree $i \geq q$, the identical multiplication instead gives

$$\prod_{N=i}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_i}{\omega_i V_j} (1 - \delta_i)(1 - \delta_j) \prod_{N=i+1}^{j-1} (1 - \delta_N)^2, \quad j > i. \quad (\text{CJ.10})$$

These two expressions explicitly relate the first and regular windows to the same source, phase and quotient volumes.

For the original off-line quartet let its common multiplicity be $m \geq 1$, its real displacement be $\delta > 0$ and its positive imaginary coordinate be $\gamma > 0$. The complete tensor calculation gives, with no omitted repeated root,

$$q = [1 + k(m - 1)](k + 1)^2, \quad L = 2\delta[1 + k(m - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2. \quad (\text{CJ.11})$$

The original Hermitian compression proof gives $0 < L \leq \epsilon_N$ at every admitted degree: on the full sum of generalized eigenspaces with $\Re \lambda > k/2$, its trace is L . An orthogonal projection onto that subspace in the original metric has trace pairing at most ϵ_N with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N$ and zeros. This is the same source metric as (CJ.2). Thus each contraction occurring in (CJ.9)–(CJ.10) is strictly between zero and one. A value one for any δ_N would make a positive radius product zero.

For either original length- q block $(i, j) = (q-1, 2q-1)$ or $(q, 2q)$, taking logarithms of its exact product now gives

$$\log(V_i/V_j) = \sum_{N=i}^{j-1} \log(\epsilon_N^2 + \phi_N^2) - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j}, \quad (\text{CJ.12})$$

where the entire contraction penalty is

$$\begin{aligned} \mathcal{P}_{q-1, 2q-1} &= -\log(1 - \delta_{2q-1}) - 2 \sum_{N=q}^{2q-2} \log(1 - \delta_N), \\ \mathcal{P}_{q, 2q} &= -\log(1 - \delta_q) - \log(1 - \delta_{2q}) - 2 \sum_{N=q+1}^{2q-1} \log(1 - \delta_N). \end{aligned} \quad (\text{CJ.13})$$

Both sums retain their literal first and last degrees and are nonnegative. Since $\epsilon_N \geq L$, the exact finite bound is

$$\log(V_i/V_j) \geq 2q \log L - \log(\omega_j/\omega_i) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (\text{CJ.14})$$

Every phase term remains; none is assigned the value zero.

53.3 Actual comparable norm windows with complete constants

The original arithmetic lower-envelope proof gives, for $k \geq 3$,

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h. \quad (\text{CJ.15})$$

Its complete analytic proof is retained in the arithmetic source appendix and AU.1–AU.12. Fix $0 < b < \pi/2$, and keep $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$. Set the explicit positive constants

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ M &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{M}/b. \end{aligned} \quad (\text{CJ.16})$$

For every integer $n \geq k \geq 3$ the literal monic norms obey

$$(\ell_h n)^{2n} \leq \omega_n \leq (U_h n)^{2n}. \quad (\text{CJ.17})$$

Here is the full envelope calculation. The exact minimum squared norm of a monic polynomial on $[-n, n]$ is

$$\frac{2n^{2n+1}}{2n+1} \left[\frac{2^n (n!)^2}{(2n)!} \right]^2 \geq \frac{2}{3} n^{2n} 4^{-n}.$$

The last inequality uses $(2n)!/(n!)^2 \leq 4^n$ and $2n/(2n+1) \geq 2/3$. The original chart's factor i^n has modulus one; its monic polynomial space is exactly this extremal space through the inverse substitution. On $[-n, n]$, (CJ.15) is at least $c_h \vartheta_h^{k-3} e^{-\alpha(k-3+n)} (k-2+n)^{-B}$. Use, in their stated directions,

$$\vartheta_h^{k-3} \geq t_h^n, \quad e^{-\alpha(k-3+n)} \geq e^{-2\alpha n}, \quad (k-2+n)^{-B} \geq (2n)^{-B} \geq 2^{-nB}, \quad 2c_h/3 \geq c_0^n.$$

The elementary $2n \leq 2^n$ holds at $n = 1$ and follows by induction from $2(n+1) \leq 4n$ for $n \geq 1$. Multiplication proves the lower bound (CJ.17). For the upper bound use the monic trial polynomial $(S - c)^n$, the nonnegative exponential series, and convolution to obtain $\omega_n \leq (2n)!b^{-2n}(M_h(b)^k + M_h(-b)^k)$. This is at most $(2n)^{2n}b^{-2n}2M^n$, which is at most $(4\sqrt{M}n/b)^{2n}$ since $2 \leq 4^n$. This proves both inequalities with the original mass retained.

For integers $n \geq k \geq 3$ and $n/2 \leq r \leq 2n$, divide the two instances of (CJ.17) and take the positive $2r$ -th root:

$$\left(\frac{\omega_{n+r}}{\omega_n}\right)^{1/(2r)} \leq nU_h^{1+n/r}\ell_h^{-n/r}(1+r/n)^{1+n/r} \leq C_h^{\text{win}}n, \quad (CJ.18)$$

$$C_h^{\text{win}} = 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2.$$

Indeed $1 + n/r \leq 3$, $n/r \leq 2$, and $1 + r/n \leq 3$. For each block above, $r = q$ and $n = q$ or $q - 1$. Equation (CJ.11) gives $q - 1 \geq k$ for $k \geq 3$, and $q/2 \leq q \leq 2(q - 1)$ for these q . Both norm factors are therefore at most $C_h^{\text{win}}q$. Combining (CJ.11), (CJ.14) and (CJ.18), with $D_h = \delta/(2C_h^{\text{win}}) > 0$, proves for each block

$$\log(V_i/V_j) \geq 2q \log(D_h k) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2/L^2). \quad (CJ.19)$$

Adding the two bounds and dividing by $q \log k$ proves the four-volume lower limit at least four. This includes the complete finite penalties before taking that consequence.

53.4 The full relation cost and its retained zero directions

For any admitted $i < j$, set $F = [b_{i+1}, \dots, b_j]$ and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. The exact update is $K_j - K_i = F\Omega^{-1}F^*$, so

$$V_i/V_j = \det(I + F\Omega^{-1}F^*G_i) = \det(I + \Omega^{-1}F^*G_iF). \quad (CJ.20)$$

The rectangular determinant equality follows by eliminating either off-diagonal block of the original block matrix

$$\begin{pmatrix} I & F \\ -\Omega^{-1}F^*G_i & I \end{pmatrix}.$$

On the new-column source $\mathbb{C}^{j-i}$ with metric Ω , the second cost is positive self-adjoint. Its eigenvalues are nonnegative, including their zero multiplicities. Write $L_{i,j} : \mathbb{C}^{i+1} \rightarrow \mathbb{C}^{j+1}$ for padding in the same monic orthogonal basis, and $T_{\text{new}} : \mathbb{C}^{j-i} \rightarrow \mathbb{C}^{j+1}$ for the coordinate inclusion into positions $i+1, \dots, j$. The original relation map from $(\mathbb{C}^{j-i}, \Omega)$ to $(\mathcal{P}_j, O_j)$ is $\mathfrak{b} = T_{\text{new}} - L_{i,j}R_iF$. The new columns and the lower source are orthogonal, and $R_i^*O_iR_i = G_i$, so

$$B_j\mathfrak{b} = 0, \quad \mathfrak{b}^*O_j\mathfrak{b} = \Omega + F^*G_iF. \quad (CJ.21)$$

Set $\mathcal{D}_N = \ker B_N = \chi\mathcal{P}_{N-q}$, with the zero space when $N = q - 1$. The new-coordinate component of $\mathfrak{b}$ is the identity; it is therefore injective. Its image lies in $\mathcal{D}_j \cap (L\mathcal{D}_i)^\perp$, whose dimension is $j - i$, by the dimensions of the two nested relation spaces. Thus it is an isomorphism onto that entire relation layer. Reduction sends its labelled vectors to their receiving supported zero; the lifted reduction fixes external absence τ .

At zero tilt the full quartet gives an even original density. Its u -polynomial $\psi = i^{-q}\chi(c + iu)$ satisfies $\psi(-u) = (-1)^q\psi(u)$ because its complete root multiset is stable under negation. Uniqueness of the monic orthogonal polynomial gives $Q_n(-u) = (-1)^nQ_n(u)$: substituting $-u$ preserves all original integral pairings and the adjusted leading coefficient. Monic division modulo ψ preserves parity. When k is even, q in (CJ.11) is odd. In the first block the q new polynomials $Q_q, \dots, Q_{2q-1}$ contain $(q+1)/2$ odd polynomials, while the odd quotient subspace has dimension $(q-1)/2$. Their reduction columns are dependent. The original $S = c + iu$ basis map and the nonzero column

phases i^n preserve rank. Thus at least one of the q cost eigenvalues in that block is zero. It remains in the full determinant as a factor one; (CJ.19)–(CJ.21) remain valid with every original dimension and nilpotent direction retained.

For either length- q block let $\lambda_a \geq 0$ be all its cost eigenvalues. Equations (CJ.19)–(CJ.20) imply

$$\prod_{a=1}^q (1 + \lambda_a) \geq (D_h k)^{2q}, \quad \sum_{a=1}^q (1 + \lambda_a)^s \geq q(D_h k)^{2s} \quad (s > 0). \quad (\text{CJ.22})$$

The second assertion is the arithmetic-geometric mean inequality for the q positive numbers $(1 + \lambda_a)^s$; it also obeys the trivial lower bound q . Here is its exact restriction map. Give the same coefficient space E its two original metrics and let $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$ be the identity coefficient map. Its Hilbert adjoint is $T_{i,j} : (E, G_j) \rightarrow (E, G_i)$ with matrix $G_i^{-1} G_j$, because $x^* G_j y = x^* G_i (G_i^{-1} G_j) y$ for every x, y . Thus $Q = T_{i,j} I_{i,j}$ acts on (E, G_i) and satisfies

$$Q^{-1} = G_j^{-1} G_i = K_j G_i = I + F \Omega^{-1} F^* G_i.$$

It is positive self-adjoint in that original metric; its quadratic form is $x^* G_i K_j G_i x > 0$ for $x \neq 0$. Consequently its eigenvalues are exactly the $1 + \lambda_a$ in (CJ.22), with all zero cost directions giving eigenvalue one. Hence (CJ.22) is also an inverse-restriction trace bound in its specified original metric. No individual positive cost lower bound or strict-positive rank claim is inferred.

53.5 Provenance and verification scope

The complete cumulative sources CV and TVB prove the radius and phase map used here, and the complete arithmetic source and AU prove the actual density envelope (CJ.15). PR 25 supplies the comparable-window constants and first-degree propagation. The full first-window phase product (CJ.9), its finite penalties (CJ.13)–(CJ.19), and their connection to the retained relation costs are written out here. The owner review at the cited head requires a nonnegative-eigenvalue correction and workflow-trigger corrections; this chapter preserves that status and the complete proposed parity proof. It performs no local Lean execution and makes no new formal-verification claim. Its finite algebra and analytic envelope calculations belong to this written proof.

54 Exact arithmetic restriction certificates and projection products

This article joins the original arithmetic lower-envelope comparison to the canonical restriction maps. It proves a positive arithmetic return gap at every finite degree, carries entry errors through the actual quotient, and identifies all loss traces with observations of two original source projections. The balanced product gives a complementary lower estimate on every inverse spectral power in both endpoint windows. The complete arithmetic envelope and balanced monic-norm proofs are retained as explicit source dependencies; their constants appear below. Every finite matrix and restriction consequence used here is proved.

54.1 The original source, quotient, and canonical lifts

Retain $g = 2\xi$, the monic packet h of selected zeros with their complete orders, $v_h = g/h$, and

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad c = k/2. \quad (\text{ERJ.1})$$

The original pairing, conjugate-linear in its first entry, is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) m_k(u) du$. Its constant norm is μ_h^k . The arithmetic construction supplies positivity almost everywhere and all polynomial moments. A nonzero polynomial has finitely many zeros on this line, so its norm is positive.

Let $S = s_1 + \cdots + s_k$, let the monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ be χ , and put $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, $A = M_S$. The ordered quotient coordinates are $1, S, \dots, S^{q-1}$. For $N \geq q - 1$ write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$, with negative degree spaces zero. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{ERJ.2})$$

In monomial coordinates let H_N be the source Gram. In the original monic orthogonal basis $p_0, \dots, p_N$, put $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and $B_N = [b_0, \dots, b_N]$, $b_n = [p_n]_\chi$. All p_n stay monic; $\omega_0 = \mu_h^k$. The monic triangular coefficient change has determinant one and gives the equal constructions

$$\begin{aligned} K_N &= J_N H_N^{-1} J_N^* = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \\ R_N &= O_N^{-1} B_N^* G_N : (E, G_N) \longrightarrow (\mathcal{P}_N, O_N), \quad K_N = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n}. \end{aligned} \quad (\text{ERJ.3})$$

Here and below a formula with O_N uses monic source coordinates; its monomial version conjugates by that explicit coefficient change. Surjectivity of B_N proves $K_N > 0$. Multiplication gives $B_N R_N = I$ and $R_N^* O_N R_N = G_N$. For $z \in \ker B_N$, $z^* O_N R_N = z^* B_N^* G_N = 0$. Thus every lift $R_N x + z$ has squared norm $x^* G_N x + z^* O_N z$. This proves the unique minimum property, $\text{im } R_N = \mathcal{D}_N^\perp$, and, for $i \leq j$, $G_j \preceq G_i$ and $K_i \preceq K_j$.

The actual analytic representative is $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$, where $D = -x\partial_x$, $\mathcal{M}F_h = v_h$, and $h(D)F_h = \Theta\phi_0$. Its norm is the pairing in (ERJ.1). Its full-jet map is $J^{(k)}\mathcal{V} = \eta J_N$, where $\eta[P] = [v_h]_h^{\otimes k} P(A_k)1$. Each local factor of $[v_h]_h$ has nonzero constant coefficient, so its inverse is the finite geometric expansion in its nilpotent part. The kernel of $P \mapsto P(A_k)1$ is exactly (χ) by its definition. These two facts prove that η is injective, retaining every local derivative and arithmetic unit.

54.2 Return, action, and the entire original relation correction

Let $L_{i,j} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$ pad monic coefficients, and let $P_{i,j}$ be the orthogonal projection retaining their first $i + 1$ entries. Then $B_j L_{i,j} = B_i$ and $L_{i,j}^* O_j L_{i,j} = O_i$. The original identity transport $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$, $x \mapsto x$, is a contraction and intertwines A . Its metric adjoint is

$$T_{i,j} = K_i G_j : (E, G_j) \longrightarrow (E, G_i), \quad L_{i,j} R_i T_{i,j} = P_{i,j} R_j, \quad T_{i,j} T_{j,l} = T_{i,l}. \quad (\text{ERJ.4})$$

Indeed $x^* G_j y = x^* G_i T_{i,j} y$. The first $i + 1$ coordinates of R_j are $O_i^{-1} B_i^* G_j$, exactly those of $R_i T_{i,j}$. The composition identity is $K_i G_j K_j G_l = K_i G_l$. For endomorphism types put $U_{i,j} = T_{i,j} I_{i,j}$ on (E, G_i) and $\widehat{U}_{i,j} = I_{i,j} T_{i,j}$ on (E, G_j) . Their coordinate matrices are $K_i G_j$, and the original invertible identity transport satisfies $I_{i,j} U_{i,j} = \widehat{U}_{i,j} I_{i,j}$. An endomorphism power or eigenvalue of T below means these specified similar endomorphisms, with all metrics retained. As an underlying endomorphism of E , $T = T_{i,j}$ is self-adjoint and positive in both displayed metrics: $G_i T = G_j$ and $G_j T = G_j K_i G_j$. It obeys $0 < T \preceq I$ in each metric. Its eigenvalues g_a lie in $(0, 1]$. The squared singular values of $I_{i,j}$, of its adjoint in the reverse Hilbert types, and of the restricted canonical lift in (ERJ.4) are these same g_a . Consequently

$$H_{i,j}^{\text{loss}} = I - T_{i,j}, \quad \mathcal{L}_{i,j} := \log(V_i/V_j) = -\log \det T_{i,j} = -\sum_{a=1}^q \log g_a. \quad (\text{ERJ.5})$$

Write $W_N = A^* G_N + G_N A - k G_N$, $H_N = K_N W_N$, and $V_N = A K_N + K_N A^* - k K_N$. Direct expansion proves

$$AT - TA = H_i T - T H_j = V_i G_j - K_i W_j. \quad (\text{ERJ.6})$$

The two $K_i A^* G_j$ terms and the two $k K_i G_j$ terms cancel. If $\epsilon_N = \|H_N\|_{G_N}$, contraction of the adjoint gives $\|AT - TA\|_{(E, G_j) \rightarrow (E, G_i)} \leq \epsilon_i + \epsilon_j$. Thus the original action accompanies the return through

its exact commutator, including on any chosen return eigenspace. The representative change is the entire map

$$\begin{aligned} \Delta_{i,j} &= L_{i,j}R_i - R_j = L_{i,j}R_i H_{i,j}^{\text{loss}} - (I - P_{i,j})R_j, \\ B_j \Delta_{i,j} &= 0, \quad \text{im } \Delta_{i,j} \subseteq \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \Delta_{i,j}^* O_j \Delta_{i,j} = G_i - G_j. \end{aligned} \quad (\text{ERJ.7})$$

Both terms in the difference have remainder $H_{i,j}^{\text{loss}}$. Orthogonality to $\mathcal{D}_i$ follows for both complete lifts. Since R_j is orthogonal to $\mathcal{D}_j$, the equality $L_{i,j}R_i = R_j + \Delta_{i,j}$ is an orthogonal sum and proves the Gram.

For the explicit theta primitive keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, with F_h of degree one and ϕ_0 of degree zero. The original compatible chain action D commutes with this differential. Successive monic division gives $\chi(\sum s_a) = \sum_a h(s_a)Q_a(\mathbf{s})$ in the tensor ideal. For any relation $\chi\kappa$ the cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa(\sum_b D_b) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \quad (\text{ERJ.8})$$

has differential $\mathcal{V}(\chi\kappa)$: applying the tensor differential gives a second factor $(-1)^{a-1}$ and $h(D_a)F_h = \Theta\phi_0$; all other terms have zero differential. Apply this fixed linear division primitive to every column of (ERJ.7). On an active split fibre the maps are $x^\bullet \mapsto x^\bullet$ and $x^\bullet \mapsto (Tx)^\bullet$, with $\tau \mapsto \tau$. They are $G(\mathbb{C})$ -linear by the original fibre operations and linearity of the displayed maps. The relation image in the internal quotient is the supported zero $(\ell_j, 0)$; the external τ is retained. The reverse maps compose by (ERJ.4), while their action comparison is (ERJ.6). No relation, support label, or primitive is deleted.

54.3 The exact source projection and principal-angle morphism

Use the common original Hilbert space $\mathcal{P}_j$ and its isometries $\ell_i = L_{i,j}R_i : (E, G_i) \rightarrow \mathcal{P}_j$ and $\ell_j = R_j : (E, G_j) \rightarrow \mathcal{P}_j$. Let Π_i, Π_j be the orthogonal endomorphism projections onto these two images. Because $R_j \perp \mathcal{D}_j \supseteq \mathcal{D}_i$, its lower-degree projection lies in $\ell_i E$. Also $\ell_i x - \ell_j x \in \mathcal{D}_j$. Hence

$$\Pi_i \ell_j = P_{i,j} \ell_j = \ell_i T, \quad \Pi_j \ell_i = \ell_j, \quad (\Pi_i \Pi_j)|_{\ell_i E} = \ell_i T \ell_i^{-1}, \quad (\Pi_j \Pi_i)|_{\ell_j E} = \ell_j T \ell_j^{-1}. \quad (\text{ERJ.9})$$

These are similarities on the indicated full images. The range of $\Pi_i \Pi_j$ is $\ell_i E$ because its restriction there is invertible. Its kernel is $\ker \Pi_j$: the reverse inclusion follows from invertibility of $\Pi_i : \ell_j E \rightarrow \ell_i E$. This gives its direct sum into that image and kernel, with exactly q nonzero eigenvalues g_a and $j+1-q$ zero eigenvalues. Reversing the pair proves the corresponding assertion for $\Pi_j \Pi_i$.

Choose a G_i -orthonormal eigenbasis x_a of T . Put $u_a = \ell_i x_a$ and $v_a = \ell_j x_a / \sqrt{g_a}$. Since $G_j = G_i T$, both families are orthonormal and $\langle u_a, v_b \rangle = \delta_{ab} \sqrt{g_a}$ by (ERJ.9). If $g_a = 1$ then $u_a = v_a$. Otherwise $w_a = (v_a - \sqrt{g_a} u_a) / \sqrt{1 - g_a}$ makes mutually orthonormal pairs (u_a, w_a) . On each such plane,

$$\Pi_i = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Pi_j = \begin{pmatrix} g_a & \sqrt{g_a(1-g_a)} \\ \sqrt{g_a(1-g_a)} & 1-g_a \end{pmatrix}. \quad (\text{ERJ.10})$$

Their difference has trace zero and square $(1 - g_a)I_2$. Both vanish on the common orthogonal complement, and the shared directions with $g_a = 1$ contribute zero. Therefore, for every real $s > 0$ and in particular each integer $m \geq 1$,

$$\text{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2s} = 2 \sum_a (1 - g_a)^s, \quad \text{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2m} = 2 \text{Tr}_E ((H_{i,j}^{\text{loss}})^m). \quad (\text{ERJ.11})$$

Functional powers here use the proved positive spectral decomposition. The restricted map $\mathcal{R} = \Pi_i|_{\ell_j E}$ is the bijection $\ell_i T_{i,j} \ell_j^{-1} : \ell_j E \rightarrow \ell_i E$. For every $s > 0$ the exact inverse singular-value-power observation is

$$\sum_a \sigma_a(\mathcal{R}^{-1})^{2s} = \sum_a \sigma_a(T_{i,j}^{-1})^{2s} = \sum_a \sigma_a(I_{i,j}^{-1})^{2s} = \text{Tr } U_{i,j}^{-s} = \sum_a g_a^{-s}. \quad (\text{ERJ.11a})$$

This follows from $I_{i,j}^\dagger I_{i,j} = U_{i,j}$ and the equal singular values of a map and its adjoint; inversion reciprocates the singular values. These are Schatten observations of the original typed maps, including when $0 < 2s < 1$, where no norm triangle inequality is asserted. Only the bijection between the canonical images is inverted; the full projection products retain their ambient zeros. For a return eigenvector x of eigenvalue g the original lower portion satisfies

$$\frac{\|P_{i,j}R_jx\|_{O_j}^2}{\|R_jx\|_{O_j}^2} = g, \quad \frac{\|(I - P_{i,j})R_jx\|_{O_j}^2}{\|R_jx\|_{O_j}^2} = 1 - g. \quad (\text{ERJ.12})$$

Indeed its first squared norm is $x^*G_jK_iG_jx = gx^*G_jx$; orthogonal decomposition proves the second. Its old quotient coordinate is gx . Applying the invertible T^{-1} recovers x ; the complete representative difference is exactly (ERJ.7).

54.4 The arithmetic remainder bound at every finite degree

For the actual quartet $1/2 \pm \delta \pm i\gamma$ of common order $m \geq 1$, $\delta, \gamma > 0$, retain

$$\ell = 1 + k(m-1), \quad q = \ell(k+1)^2, \quad \chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^\ell. \quad (\text{ERJ.13})$$

The top nonzero power of the sum of k local length- m nilpotents is $k(m-1)$, with coefficient $(k(m-1))!/((m-1)!)^k$ on the product of their top powers; its next power vanishes by total degree. Each displayed grid sum occurs and the distinct pairs have distinct real and imaginary parts. This proves the complete order and degree in (ERJ.13) by taking the least common multiple of local annihilators.

The complete arithmetic-envelope source proves, for $k \geq 3$,

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h, \quad \vartheta_h = \int_{-1}^1 w_h > 0. \quad (\text{ERJ.14})$$

Here $c_h > 0$ is its original local-mass convolution constant. Fix $0 < b < \pi/2$, $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$ and $M_k = M_h(b)^k + M_h(-b)^k$. Positivity and convolution give $\int e^{b|u|} m_k \leq M_k$, and hence $\int |u|^{2d} m_k \leq M_k (2d)! b^{-2d}$. Set $R_0 = \sqrt{\delta^2 + \gamma^2}$, $D = \max(1, 2/b, 2R_0)$, $T = Dq$, $r_0 = kR_0/T \leq 1/2$, and retain

$$\begin{aligned} a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(k-3+T)} (k-2+T)^{-B}, \\ U_k(T) &= \mu_h^k + M_k \frac{(2q-2)!}{(bT)^{2q-2}}, \quad A_N = \sqrt{(2N+1)/2} \frac{4^{N+1}-1}{3}, \\ B_q(r_0) &= \left(\frac{1+r_0}{1-r_0} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r_0)^2}{T a_k(T)}. \end{aligned} \quad (\text{ERJ.15})$$

The coordinate map $\Psi_T P(x) = P(c + iTx)$ has inverse $f \mapsto f((S-c)/(iT))$ and target measure $Tm_k(Tx)dx$, whose mass remains μ_h^k . Put $\chi_T = (iT)^{-q} \chi(c + iTx)$. Then $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$, so remainder commutes with this map. Every original raw derivative of order d is multiplied by $(iT)^d$. The quotient coefficient determinant is $(iT)^{q(q-1)/2}$, and its transported form determinant is $T^{-q(q-1)} V_N$; this identical factor cancels in a volume ratio.

List all q roots z_a of χ_T with repetition. Their moduli are at most r_0 . Write $E(t) = \prod_a (1 - z_a t)$ and $E(t)^{-1} = \sum_{l \geq 0} h_l t^l$. For every integer $s \geq 0$, multiplication of these formal inverse series through order s proves that $Q_s(x) = \sum_{l=0}^s h_l x^{s-l}$ is the quotient of x^{q+s} by χ_T , and that $x^{q+s} - \chi_T Q_s$ has degree below q . Convolution gives $\|\chi_T\|_1 \leq (1+r_0)^q$ and $|h_l| \leq \binom{q+l-1}{l} r_0^l$, where $\|\cdot\|_1$ is the coefficient absolute sum. The sole coefficient of $\chi_T Q_s$ at degree at least q is its leading coefficient one. Its remaining coefficients are the negative remainder. Therefore

$$\|\text{rem}_{\chi_T} x^{q+s}\|_1 \leq (1+r_0)^q \sum_{l=0}^s \binom{q+l-1}{l} r_0^l - 1 \leq B_q(r_0). \quad (\text{ERJ.16})$$

The last bound sums the q geometric series. At degrees below q the remainder has norm one. Linearity proves $\|\text{rem}_{\chi_T} f\|_1 \leq B_q(r_0)\|f\|_1$ for every polynomial f , without a degree restriction on s .

Rodrigues' formula for Legendre polynomials gives orthogonality, squared norm $2/(2n+1)$, and coefficient norm at most $\binom{2n}{n} \leq 4^n$. For the norm, integrate by parts n times; the endpoint terms vanish because $(x^2 - 1)^n$ has order n there, and the remaining beta integral gives $2/(2n+1)$. For the coefficient bound use $P_n(x) = 2^{-n} \sum_{a=0}^n \binom{n}{a}^2 (x-1)^{n-a} (x+1)^a$, then sum the coefficient absolute values. Cauchy-Schwarz on each orthogonal coefficient and summation of 4^n yield $\|f\|_1 \leq A_N \|f\|_{L^2[-1,1]}$ for $\deg f \leq N$. For $\deg v \leq q-1$, each monomial at u/T has modulus at most $\max(1, |u/T|^{q-1})$, so $\int |v(u/T)|^2 m_k \leq U_k(T) \|v\|_1^2$. Finally (ERJ.14) restricted to $[-T, T]$ gives $\|P\|^2 \geq T a_k(T) \|\Psi_T P\|_{L^2[-1,1]}^2$. Composing these actual maps proves, for every $N \geq q-1$,

$$\|\text{rem}_\chi P\|^2 \leq C_N \|P\|^2, \quad G_{q-1} \preceq C_N G_N, \quad C_j^{-1} K_j \preceq K_i \preceq K_j \quad (q-1 \leq i \leq j). \quad (\text{ERJ.17})$$

Apply the first inequality to the least lift $R_N x$ to obtain the second. Then use $G_j \preceq G_i \preceq G_{q-1}$ and inversion of positive forms. Taking $i = j$ proves $C_j \geq 1$. Thus the two original endpoint pairs $(q-1, 2q-1)$ and $(q, 2q)$ have the explicit positive gaps C_{2q-1}^{-1} and C_{2q}^{-1} .

54.5 Finite comparison matrices and the complete trace error

All following comparisons use the original quotient coordinates. For positive source brackets $H_N^- \preceq H_N \preceq H_N^+$, inversion and congruence give

$$L_N = J_N (H_N^+)^{-1} J_N^* \preceq K_N \preceq U_N = J_N (H_N^-)^{-1} J_N^*. \quad (\text{ERJ.18})$$

Their positivity follows from surjectivity of J_N . One actual upper comparator uses the density $\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}$ from (ERJ.14). Its even moment of degree d is

$$\int u^d \nu_k(u) du = \frac{2c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} d!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-(k-2)t}}{(\alpha+t)^{d+1}} dt; \quad \int u^d \nu_k(u) du = 0 \quad (d \text{ odd}). \quad (\text{ERJ.19})$$

Insert $(k-2+u)^{-B} = (B-1)!^{-1} \int_0^\infty t^{B-1} e^{-(k-2+u)t} dt$ on the positive half-line; Tonelli and $\int_0^\infty u^d e^{-yu} du = d! y^{-d-1}$ prove the formula. Thus $H_N^\nu \preceq H_N$ is an explicit strictly positive source. It supplies U_j using comparison moments through $2j$, at most $4q$ at the second endpoint; actual low-degree source data for i stop at $2i \leq 2q$. These cutoffs keep the actual and comparison integrals attached to their stated maps.

Given $0 < L_i \preceq K_i \preceq K_j \preceq U_j$, set $\tau = \text{Tr}(L_i^{-1} U_j)$. The Hermitian matrix $L_i^{-1/2} U_j L_i^{-1/2}$ has all q eigenvalues at least one. Each is at most $\tau - q + 1$ because the remaining $q-1$ contribute at least $q-1$. It follows that

$$U_j \preceq (\tau - q + 1) L_i, \quad g := \frac{1}{\tau - q + 1} > 0, \quad K_i \succeq g K_j. \quad (\text{ERJ.20})$$

This coefficient is optimal from the trace and the lower bound one: $\text{diag}(\tau - q + 1, 1, \dots, 1)$ attains it. An upper rational $\tau_+ \geq \tau$ gives the certified rational gap $(\tau_+ - q + 1)^{-1}$.

Now take rational Hermitian brackets for both K_i, K_j as in (ERJ.18), and define the exact rational quantities

$$H_* = I - L_i U_j^{-1}, \quad t_m^* = \text{Tr } H_*^m, \quad d = \text{Tr}(U_i L_j^{-1}) - \text{Tr}(L_i U_j^{-1}), \quad r = 1 - g. \quad (\text{ERJ.21})$$

Here g is certified for the reference pair (L_i, U_j) itself. For increasing generalized eigenvalues the exact minimum-maximum formula is

$$\lambda_a(F, G) = \min_{\dim W=a} \max_{0 \neq x \in W} \frac{x^* F x}{x^* G x}. \quad (\text{ERJ.22})$$

For $G = I$, the span of the first a eigenvectors attains λ_a ; every a -dimensional W intersects the span of eigenvectors $a, \dots, q$, whose dimension is $q - a + 1$, and hence its maximum is at least λ_a . The invertible substitution $y = G^{1/2}x$ proves the general case with subspace dimensions preserved. The pointwise Rayleigh quotients for (L_i, U_j) , (K_i, K_j) , and (U_i, L_j) are ordered. Their extrema are therefore ordered. If $\alpha_a = \lambda_a(L_i, U_j)$ and $\beta_a = \lambda_a(K_i, K_j)$, this proves

$$g \leq \alpha_a \leq \beta_a \leq 1, \quad 0 \leq \sum_a (\beta_a - \alpha_a) \leq d. \quad (\text{ERJ.23})$$

The sum identity uses similarity of FG^{-1} to its positive Hermitian representative. The kernels F, G are forms on the dual quotient coefficient space: $G^{-1}F$ there and FG^{-1} on the quotient are similar through the invertible map G . These β_a are the actual g_a of the return. For $r > 0$, integration of the derivative of x^m on $[0, r]$ gives the complete moment enclosure

$$a_m := \max(0, t_m^* - mr^{m-1}d) \leq s_m := \text{Tr}((H_{i,j}^{\text{loss}})^m) \leq t_m^* \leq qr^m. \quad (\text{ERJ.24})$$

Indeed apply $0 \leq u^m - v^m \leq mr^{m-1}(u - v)$ to $u = 1 - \alpha_a$, $v = 1 - \beta_a$ and sum (ERJ.23). If $r = 0$, all $\alpha_a = \beta_a = 1$ and both loss operators vanish; the exact certificate is zero.

Integrating the geometric series on $[0, x]$ proves $-\log(1 - x) = \sum_{m \geq 1} x^m/m$ for $0 \leq x < 1$. All terms are nonnegative. Its tail is bounded by $x^{p+1}/((p+1)(1-x))$. Thus for every integer $p \geq 1$,

$$\sum_{m=1}^p \frac{a_m}{m} \leq \mathcal{L}_{i,j} \leq \sum_{m=1}^p \frac{t_m^*}{m} + \frac{t_{p+1}^*}{(p+1)g}. \quad (\text{ERJ.25})$$

The width of the two stated endpoints is at most

$$\frac{d(1 - r^p)}{g} + \frac{qr^{p+1}}{(p+1)g}. \quad (\text{ERJ.26})$$

The first term sums $r^{m-1}d$ using (ERJ.24); the second uses $t_{p+1}^* \leq qr^{p+1}$. This accounts for the finite matrix error and series tail separately with their original dimension.

The sharper coefficient

$$c_p(r) = \sum_{l \geq 0} \frac{r^l}{p+1+l} = \frac{-\log g - \sum_{m=1}^p r^m/m}{r^{p+1}} \leq \frac{1}{(p+1)g} \quad (\text{ERJ.27})$$

can replace the last coefficient in (ERJ.25): the tail at $x \leq r$ is at most $c_p(r)x^{p+1}$. To certify its logarithm, write exactly $x = 2^e y$ for rational $x > 0$, integer e , and $1 \leq y < 2$. For $t = (y-1)/(y+1)$, integration of $2/(1-t^2)$ gives

$$\log y = 2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} + E_M, \quad 0 \leq E_M \leq \frac{2t^{2M+1}}{(2M+1)(1-t^2)}. \quad (\text{ERJ.28})$$

At $t = 1/3$ the same formula encloses $\log 2$. Multiply that interval by e , reversing endpoints when $e < 0$, and add the $\log y$ interval. If $\log g \in [\ell_-, \ell_+]$ has width ε , replacing $-\log g$ by $-\ell_-$ in (ERJ.27) adds at most $\varepsilon t_{p+1}^*/r^{p+1} \leq q\varepsilon$ to the upper endpoint. The $r = 0$ case was handled before this division.

54.6 Constructing rational brackets for the actual relation

Let an actual Hermitian source matrix M have a rational Hermitian center M_0 and symmetric rational radii $\varepsilon_{ab} \geq 0$ with $|(M - M_0)_{ab}| \leq \varepsilon_{ab}$. For fixed positive rational w_a set $D_{aa} = \sum_b \varepsilon_{ab} w_b / w_a$.

Squaring $\sqrt{t}x - y/\sqrt{t}$ proves $2xy \leq tx^2 + t^{-1}y^2$. Therefore

$$\begin{aligned} |z^*(M - M_0)z| &\leq \sum_{a,b} \varepsilon_{ab} |z_a| |z_b| \\ &\leq \frac{1}{2} \sum_{a,b} \varepsilon_{ab} \left(\frac{w_b}{w_a} |z_a|^2 + \frac{w_a}{w_b} |z_b|^2 \right) = z^* D z. \end{aligned} \quad (\text{ERJ.29})$$

Hence $M_0 - D \preceq M \preceq M_0 + D$. Exact Hermitian elimination tests positivity: a first pivot $d_0 > 0$ gives $d_0|x + d_0^{-1}b^*y|^2 + y^*(C - bb^*/d_0)y$. Induction proves positivity precisely when all successive pivots and their Schur complements are positive.

For the actual, possibly nonrational coefficients of χ , use the exact ordered coefficient matrix

$$\mathcal{A}_N = [1, S, \dots, S^{q-1}, \chi, S\chi, \dots, S^{N-q}\chi]. \quad (\text{ERJ.30})$$

It is triangular with diagonal one. At $N = q - 1$ its relation block is empty. Monic division gives $J_N \mathcal{A}_N = [I_q, 0] =: J_0$. Consequently, in the original remainder coordinates,

$$M_N = \mathcal{A}_N^* H_N \mathcal{A}_N, \quad K_N = J_0 M_N^{-1} J_0^*. \quad (\text{ERJ.31})$$

Intervals for the original source entries and original coefficients of χ enter the literal finite products for M_N ; interval addition, multiplication, and conjugation contain those products. The sum of the real and imaginary radii bounds a complex error. Use conjugate centers and symmetrically enlarged radii, then (ERJ.29) and exact positive pivots. The two inverse compressions of the resulting source brackets yield rational L_N, U_N by (ERJ.18). No center polynomial replaces the actual quotient map in (ERJ.31).

For a fixed actual pair and fixed w_a , refining source and coefficient intervals makes the radii and D tend to zero. Positivity of the actual M_N ensures eventual positive lower pivots. The cofactor formula proves continuity of inversion where the determinant is nonzero; thus $L_N, U_N \rightarrow K_N, d \rightarrow 0$, and the reference gap in (ERJ.20) tends to a positive number. For any requested positive width, first make d/g sufficiently small and then choose p so the geometric tail in (ERJ.26) is smaller than the remaining width. The rational logarithm error can be decreased independently. This proves convergence of the finite algebraic certificate for each fixed actual pair. The quantities controlling that convergence remain explicit when the degree is allowed to grow.

54.7 The four-volume comparison with its literal finite constants

The arithmetic pair losses add to $\mathcal{B}_{h,k} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$. Equations (ERJ.15)–(ERJ.17) prove

$$\mathcal{B}_{h,k} \leq q \log C_{2q-1} + q \log C_{2q}, \quad \frac{C_{2q}}{C_{2q-1}} = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2 \leq \frac{147}{5} < 32. \quad (\text{ERJ.32})$$

Indeed the first ratio is at most $5/3$, and the ratio inside the square is $4 + 3/(4^{2q}-1) \leq 21/5$. Set $X_0 = \max(1, M_h(b), M_h(-b))$, $K_0 = \max(1, \vartheta_h^{-1})$ and $C_{\text{env}} = \max(1, 2/(3c_h D))$. Since $bT \geq 2q$, the factorial ratio in U_k is at most one; also $M_k = 2 \int \cosh(bu) m_k \geq 2\mu_h^k$. Thus $U_k(T) \leq 3X_0^k$. The stated coefficient constant obeys $A_{2q-1}^2 \leq (2q/9)256^q$. On $0 \leq r_0 \leq 1/2$, the derivative of $\log((1+r_0)/(1-r_0))$ is at most $8/3$, so $\log B_q(r_0)^2 \leq 16R_0 k/(3D)$. Substitution into C_{2q-1} , using $\vartheta_h^{-(k-3)} \leq K_0^k$ and $k-2+Dq \leq 1+(D+1)q$, proves

$$\begin{aligned} q \log C_{2q-1} &\leq \mathcal{U}_k, \quad \mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32, \\ \mathcal{U}_k &= (\alpha D + \log 256)q^2 + \left(\log(X_0 K_0) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}. \end{aligned} \quad (\text{ERJ.33})$$

All constants are independent of k for the fixed packet and b . As $q = [1 + k(m-1)](k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. The displayed scalar majorant divided by q^2 tends to $2(\alpha D + \log 256)$; divided by $q \log k$ it grows. The finite matrix and trace bounds (ERJ.18)–(ERJ.28) observe the same original four volumes and retain the directional data before this scalar estimate.

54.8 Central rank, both endpoint returns, and all inverse powers

Keep the original phase ϕ_N and relative Hermitian radius ϵ_N . For a reflection-stable packet the exact monic recurrence is $Sp_n = p_{n+1} + (c + ia_n)p_n - (\omega_n/\omega_{n-1})p_{n-1}$, $a_n \in \mathbb{R}$. It follows by transporting to u with $p_n(c + iu) = i^n Q_n(u)$ and pairing self-adjoint multiplication by u with every lower monic orthogonal polynomial. Substitution into (ERJ.3) cancels the internal adjacent terms and gives

$$V_N = (b_{N+1}b_N^* + b_Nb_{N+1}^*)/\omega_N, \quad b_N^*G_Nb_{N+1} = i\omega_N\phi_N. \quad (\text{ERJ.34})$$

The pairing in the second formula is pure imaginary: in the real u coordinates both the quotient Gram and the remainder columns are real, and the source factors contribute $i^N i^{N+1} = i$. This defines the retained real phase; it equals the original phase in the monic norm derivative formula. The relative Hermitian matrix has rank at most two and trace zero. Its square has trace $2[(b_N^*G_Nb_N)(b_{N+1}^*G_Nb_{N+1}) - |b_N^*G_Nb_{N+1}|^2]/\omega_N^2$, giving eigenvalues $\epsilon_N, -\epsilon_N$ and the remaining zeros. The rank-one determinant update in (ERJ.3) gives, for $N \geq q$, $b_N^*G_Nb_N/\omega_N = 1 - \delta_N$ and $b_{N+1}^*G_Nb_{N+1}/\omega_{N+1} = \delta_{N+1}^{-1} - 1$, where $\delta_N = V_N/V_{N-1}$. Therefore

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N}(1 - \delta_N)(\delta_{N+1}^{-1} - 1). \quad (\text{ERJ.35})$$

The determinant update follows by eliminating $\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ in its two block orders to prove $\det(I + XY) = \det(I + YX)$. Multiplying (ERJ.35) for $N = q, \dots, 2q-2$ proves

$$\prod_{N=q}^{2q-2} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_{2q-1}V_q}{\omega_qV_{2q-1}}(1 - d_0)(1 - d_{q-1}) \prod_{a=1}^{q-2} (1 - d_a)^2, \quad d_a = \delta_{q+a}. \quad (\text{ERJ.36})$$

Each internal contraction occurs twice and both endpoint contractions occur once.

For (ERJ.13) the complete positive-real spectral compression gives

$$L = 2\delta\ell(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2, \quad 0 < L \leq \epsilon_N. \quad (\text{ERJ.37})$$

Here twice the sum of positive real parts of the eigenvalues of $A - cI$ equals L : sum the positive integers $2a - k$, giving $\lfloor (k+1)^2/4 \rfloor$, then include all $k+1$ imaginary centres and their full length ℓ . A unitary upper triangular form in the actual G_N metric, with these eigenvalues first, shows that this is the trace of the Hermitian defect compressed to their invariant subspace. A projection paired with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N, 0, \dots, 0$ has trace at most ϵ_N , proving the stated bound. The integer inequality $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$ follows in each parity.

The complete balanced monic-norm source proves

$$(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{bal}} n \quad (n \geq k \geq 3, n/2 \leq r \leq n), \quad (\text{ERJ.38})$$

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) X_0 K_0 \max(1, b^{-3}) 6^{B/3}.$$

This is its literal sufficient constant from the original envelope and monic Legendre minimum, with its full analytic proof included in the dependency record. Apply it at $n = q, r = q-1$ and put $X = L/(C_h^{\text{bal}}q)$. Every d_a in (ERJ.36) is strictly below one, since otherwise its right side is zero while

every left factor is at least $L^2 > 0$. Define the entire nonnegative penalty

$$P_{\text{mid}} = -\log(1 - d_0) - \log(1 - d_{q-1}) - 2 \sum_{a=1}^{q-2} \log(1 - d_a) + \sum_{a=0}^{q-2} \log(1 + \phi_{q+a}^2/L^2). \quad (\text{ERJ.39})$$

Taking logarithms in (ERJ.36) proves

$$C_{\text{mid}} := \log(V_q/V_{2q-1}) \geq 2(q-1) \log X + P_{\text{mid}}. \quad (\text{ERJ.40})$$

Indeed every factor $\epsilon_N^2 + \phi_N^2$ is at least $L^2(1 + \phi_N^2/L^2)$, and (ERJ.38) bounds the norm ratio.

The central update is the full matrix

$$K_{2q-1} - K_q = \sum_{a=q+1}^{2q-1} b_a b_a^* / \omega_a, \quad \text{rank}(K_{2q-1} - K_q) \leq q-1. \quad (\text{ERJ.41})$$

Its positive Hermitian representative shows that the central return has at least one exact unit eigenvalue. List one such value separately and the other $q-1$ as $\gamma_1, \dots, \gamma_{q-1}$, keeping any additional unit values. Their product is $e^{-C_{\text{mid}}}$. For every real $s > 0$, convexity of e^{sx} gives

$$\begin{aligned} \text{Tr } T_{q,2q-1}^{-s} &= 1 + \sum_{a=1}^{q-1} \gamma_a^{-s} \\ &\geq 1 + (q-1)e^{sC_{\text{mid}}/(q-1)} \\ &\geq \max \left\{ q, 1 + (q-1)X^{2s}e^{sP_{\text{mid}}/(q-1)} \right\}. \end{aligned} \quad (\text{ERJ.42})$$

For completeness $e^y \geq 1 + y$ follows by differentiating $e^y - 1 - y$, whose minimum is zero at $y = 0$. Apply it to $y_a = -s \log \gamma_a - \bar{y}$, where $\bar{y} = -(s/(q-1)) \sum_a \log \gamma_a$: $\sum_a e^{-s \log \gamma_a} = e^{\bar{y}} \sum_a e^{y_a} \geq (q-1)e^{\bar{y}}$. The q bound also follows termwise from $\gamma_a \leq 1$. The actual rank $\rho = \text{rank}(K_{2q-1} - K_q)$ preserves exactly $q - \rho$ unit eigenvalues, because $I - T = (K_{2q-1} - K_q)G_{2q-1}$ has rank ρ . Write $A_+ = \max\{0, 2(q-1) \log X + P_{\text{mid}}\}$. If $\rho > 0$, applying the same exponential argument to just the ρ eigenvalues below one proves the stronger exact-rank consequence

$$\text{Tr } T_{q,2q-1}^{-s} \geq q - \rho + \rho e^{sC_{\text{mid}}/\rho} \geq q - \rho + \rho e^{sA_+/\rho}. \quad (\text{ERJ.42a})$$

For fixed $a \geq 0$, $r(e^{a/r} - 1)$ decreases as $r > 0$ increases: its derivative is $e^{a/r}(1 - a/r) - 1 \leq 0$, since the derivative of $e^t(1 - t)$ is $-te^t \leq 0$ for $t \geq 0$. Thus $\rho \leq q-1$ proves that (ERJ.42a) improves (ERJ.42). At $\rho = 0$ the central return is the identity, its cost is zero, and the trace is exactly q . There is no division by ρ in that case. By (ERJ.22), the ordered eigenvalues of each pair (K_{q-1}, K_{2q-1}) and (K_q, K_{2q}) are at most those of (K_q, K_{2q-1}) : their Rayleigh quotients decrease respectively in the numerator and by increase of the denominator. Since x^{-s} is decreasing, *each* original endpoint return obeys the final bound in (ERJ.42), and the stronger rank-refined bound (ERJ.42a), by the same ordered comparison. Its minimum eigenvalue also obeys

$$g_{\min}^{(a)} \leq \min \left\{ 1, X^{-2}e^{-P_{\text{mid}}/(q-1)} \right\} \leq \min \left\{ 1, \frac{4(C_h^{\text{bal}})^2}{\delta^2 k^2} \right\}, \quad a = 0, 1. \quad (\text{ERJ.43})$$

The first inequality follows from $\gamma_{\min}^{q-1} \leq \prod \gamma_a$ and (ERJ.40), followed by the same ordered comparison. Equations (ERJ.9)–(ERJ.12) and (ERJ.7)–(ERJ.8) give its exact source, action, and theta interpretation. The identity

$$\mathcal{B}_{h,k} = 2C_{\text{mid}} + \log(V_{q-1}/V_q) + \log(V_{2q-1}/V_{2q}) \quad (\text{ERJ.44})$$

has nonnegative endpoint terms. Combining (ERJ.37)–(ERJ.40) proves

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4,$$

because its lower bound is $4(q-1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q-1)/q \rightarrow 1$. The consecutive-first-window continuation separately proves the length- q products at the two actual endpoints, including the single terminal factor at initial degree $q-1$, and gives its stronger per-window inverse-power estimate. Its original constant and phase formula stay in that complete companion proof; (ERJ.42) retains the exact central unit eigenvalue and full central penalty.

54.9 Exact auxiliary calculation and retained proof dependencies

The accompanying complete fixture proof uses $S = 1 + iu$, $\chi = S^2 - 2S - 1$ and $r_\lambda(u) = |\Gamma(\lambda + iu/2)|^2/(2\pi)$. Its source is $m = r_4/70$ of mass $9/16$; its smaller source is $\nu = (54/35)(r_2/3)$ of mass $27/70$. The full identity

$$m(u) = \left(\frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120} \right) \frac{r_2(u)}{3} \quad (\text{ERJ.45})$$

follows from two applications of $\Gamma(z+1) = z\Gamma(z)$, and proves their source contraction. The complete fixture appendix derives all moments, original S Grams, remainder matrices, extrema, and rational logarithm endpoints. Its remainder norm squared is $763/108$, attained by the original polynomial $-245/36 + (791/108)S - (7/9)S^2 + (7/27)S^3$ in both degree three and degree four. The resulting source gap is $108/763$, while the two actual minimum return eigenvalues equal $135/331$. The exact four-volume ratio is $3073295611/216513000$. These quantities are proved in the full fixture text retained with this article, rather than inferred from a decimal computation.

The existing fixture executed 303 distinct mathematical checks in each unchanged normal and optimized run. Seven literal formula changes in each mode were rejected after the complete calculation, giving 16 actual jobs in total. The preserved source and full execution JSON have hashes recorded in this article's manifest. This completion did not rerun that unchanged suite. Its exact source/Gamma fixture is connected to the arithmetic construction by (ERJ.2)–(ERJ.4), (ERJ.17)–(ERJ.20); the density (ERJ.45) and its masses remain its declared auxiliary inputs.

For the empty packet $h = 1$, $E = 0$, the quotient determinants are empty determinants one and every quotient trace and four-volume loss is zero. The original analytic source and mass still exist. The positive inverses and $q-1$ spectral averages above are used only in their stated domains; the actual quartet has $q \geq (k+1)^2$ and $k \geq 3$. At $i = j$, return is the identity, its relation correction and projection difference vanish, and the exact loss is zero.

The complete source dependencies are the original *Canonical restriction and the arithmetic endpoint volume*; *An arithmetic lower-envelope comparison for central quotient volumes* (AU.1–49); *Exact window products, sharp endpoint optimization, and original relation costs* (EP.1–54), with its complete balanced-window proof dependency; the endpoint-window proof (EW.1–59); the complete 303-check fixture proof; and the *Consecutive first-window join* (CJ.1–22). The accompanying manifest pins their actual files and the new independent mathematical review. The global cumulative reader carries these full proofs and the new article in their stated order.

55 Deligne machinery inside Split-Zero cohomology

55.1 Deligne's image comparison inside the Split-Zero source complexes

This sidebar integrates the delivered specialization-curvature, exponential-comparison, and logarithmic-constituent-control notes. Their logarithmic cohomology and constituent-coupling calculations are reproduced with complete proofs and explicitly joined to the existing Split-Zero complexes; they are attributed to those web contributions. The sidebar supplies the maps connecting the

compact-support comparison, its dual, the Split-Zero internal quotient, and the compactified curve of arithmetic representatives. It also calculates the logarithmic de Rham complex of that curve. In particular, it identifies which cohomology groups the exceptional directions occupy and the divisor twist in their support-compatible dual.

The source reference is P. Deligne, *La conjecture de Weil II*, Publications Mathématiques de l'IHÉS 52 (1980), 137–252, https://www.numdam.org/item/PMIHES_1980_52_137_0/. The specific weight comparison is in (3.3.4)–(3.3.6), printed page 206; the preceding geometric argument is in (3.2.8)–(3.2.13). The source-language S20 records for printed pages 201–207 were read in full, including the strict inequality in (3.2.10) and the squared eigenvalue in (3.2.13). The complete S20 typed transcription is retained separately from this integration. The symbol Q_0 in (DS7)–(DS8) denotes a finite-field cardinality; later Q_0 denotes the explicitly defined quotient Gram, as in the supplied logarithmic note; q denotes the dimension of the actual arithmetic packet.

55.1.1 The comparison image, its entire cone, and its dual

Let $c : C^\bullet \rightarrow D^\bullet$ be a cochain map of complexes of vector spaces over a field K . All differentials raise degree by one. Write $Z_C^m = \ker d_C^m$, $B_C^m = \operatorname{im} d_C^{m-1}$, and similarly for D . Define

$$K_c^m = \{x \in Z_C^m : c(x) \in B_D^m\}, \quad I_c^m = \operatorname{im}(H^m(c) : H^m(C) \rightarrow H^m(D)). \quad (\text{DS1})$$

There are exact sequences with the displayed maps:

$$\begin{aligned} 0 \longrightarrow K_c^m / B_C^m &\xrightarrow{[x] \mapsto [x]} H^m(C) \xrightarrow{[x] \mapsto [cx]} I_c^m \longrightarrow 0, \\ 0 \longrightarrow I_c^m &\longrightarrow H^m(D) \longrightarrow H^m(D) / I_c^m \longrightarrow 0. \end{aligned} \quad (\text{DS2})$$

Indeed, $B_C^m \subset K_c^m$ because c commutes with the differentials. The kernel of the second arrow in the first sequence is exactly the classes represented by K_c^m , and every element of I_c^m has the indicated preimage. The second sequence is the defining inclusion and quotient. Thus both the lost source classes and the extra target classes are present.

The single complex retaining both sides is

$$\operatorname{Cone}(c)^m = D^m \oplus C^{m+1}, \quad d(y, x) = (d_D y + cx, -d_C x). \quad (\text{DS3})$$

Expansion gives $d^2(y, x) = (d_D^2 y + d_D cx - cd_C x, d_C^2 x) = 0$. The maps $y \mapsto (y, 0)$ and $(y, x) \mapsto x$ give the exact sequence of complexes $0 \rightarrow D \rightarrow \operatorname{Cone}(c) \rightarrow C[1] \rightarrow 0$, where $C[1]^m = C^{m+1}$ has differential $-d_C$. To make the relevant cohomology segment explicit, a cone cocycle obeys $d_C x = 0$ and $cx = -d_D y$, so its projection represents a class in K_c^{m+1} / B_C^{m+1} . Conversely, if $cx = d_D v$, then $(-v, x)$ is a cone cocycle. A projected class is zero precisely when subtracting a cone boundary leaves a representative $(y, 0)$ with $d_D y = 0$. Finally $(y, 0)$ is a cone boundary precisely when $y = d_D u + cv$ with $d_C v = 0$. This proves the exact segment

$$H^m(C) \xrightarrow{H^m(c)} H^m(D) \longrightarrow H^m(\operatorname{Cone}(c)) \longrightarrow H^{m+1}(C) \xrightarrow{H^{m+1}(c)} H^{m+1}(D). \quad (\text{DS4})$$

Every chain action commuting with c acts on (DS1)–(DS4), since it preserves both cycles and boundaries; on the cone its action is the direct sum of the two specified actions.

For a finite-dimensional linear map $f : V \rightarrow W$, its ordinary linear transpose $f^\vee : W^\vee \rightarrow V^\vee$ gives a perfect pairing

$$\operatorname{im} f \times \operatorname{im} f^\vee \longrightarrow K, \quad (f(v), f^\vee \lambda) \longmapsto \lambda(f(v)). \quad (\text{DS5})$$

If v is changed by $\ker f$, the value is unchanged. If λ is changed by $\ker f^\vee$, its restriction to $\operatorname{im} f$ is unchanged. A nonzero $f(v)$ is detected by some functional on W , and a nonzero $f^\vee \lambda$ is detected by some v , proving nondegeneracy on both sides. In particular

$$(\operatorname{im} f)^\vee \simeq \operatorname{im} f^\vee, \quad \ker f^\vee = (\operatorname{im} f)^\perp, \quad \operatorname{im} f^\vee = (\ker f)^\perp. \quad (\text{DS6})$$

For invertible actions T_V, T_W with $fT_V = T_W f$, the dual actions are $(T_V^{-1})^\vee, (T_W^{-1})^\vee$; their induced actions on the two images make the evaluation in (DS5) invariant. For an additive action A and a specified scalar b , the action $bI - A^\vee$ on the dual instead satisfies $\langle Av, \lambda \rangle + \langle v, (bI - A^\vee)\lambda \rangle = b\langle v, \lambda \rangle$. These are two explicitly defined action conventions, with different types of weight parameter.

Deligne's comparison takes

$$C = R\Gamma_c(X, \mathcal{F}), \quad D = R\Gamma(X, \mathcal{F}), \quad c = c_{X, \mathcal{F}}, \quad I_c^m = \text{im } c_m. \quad (\text{DS7})$$

Here $X = X_0 \otimes_{\mathbb{F}_{Q_0}} \overline{\mathbb{F}}_{Q_0}$, X_0 is smooth of pure dimension N and finite type, and $\mathcal{F}_0$ is a smooth $\overline{\mathbb{Q}}_\ell$ -sheaf, $\ell \neq \text{char } \mathbb{F}_{Q_0}$, pure of weight n . In the source theorem the upper weight bound on H_c^m is $n + m$. The dual of $\mathcal{F}$ has weight $-n$, and Poincaré duality gives the exact degree and twist

$$H^m(X, \mathcal{F})^\vee \simeq H_c^{2N-m}(X, \mathcal{F}^\vee)(N). \quad (\text{DS8})$$

The right side has upper weight $-n + (2N - m) - 2N = -n - m$ by the already established compact-support theorem. Dualizing gives the lower weight bound $n + m$ on H^m . Since c_m is Frobenius equivariant, choose a basis first adapted to its kernel in the source and then to its image in the target. The corresponding action matrices are block triangular. Every eigenvalue on I_c^m therefore occurs on both H_c^m and H^m , with its algebraic conjugates. The two bounds force its weight to equal $n + m$. This proves the image step (3.3.6) from the two stated source bounds and proves exactly why (DS7) retains the cone (DS3).

The tensor comparison has an equally explicit algebraic map. For finite-dimensional $f_a : V_a \rightarrow W_a$ ($1 \leq a \leq r$),

$$\text{im} \left(\bigotimes_{a=1}^r f_a \right) = \bigotimes_{a=1}^r \text{im } f_a \subseteq \bigotimes_{a=1}^r W_a. \quad (\text{DS9})$$

The tensor map sends a pure tensor to the pure tensor of its images, so its image is contained in the right side. Conversely these image pure tensors span the right side and each has a preimage; inclusivity into the target follows by extending a basis of each image to a basis of W_a . Thus tensor powers preserve the same comparison image. For complexes, the tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{\deg x} x \otimes dy$; swapping the factors has the sign $(-1)^{\deg x \deg y}$. In characteristic zero a specified symmetry projection is $\frac{1}{r!} \sum_{\pi \in \mathfrak{S}_r} \pi$ with those graded signs. Its idempotence follows because each group element occurs exactly $r!$ times in the double sum. No unspecific tensor eigenvalue is substituted for the eigenvalue on this actual image.

55.1.2 The internal Split-Zero lift of the same maps

For a complex vector space V , set $S(V) = \{\tau\} \sqcup V^\bullet$. Supported addition and scalar multiplication are the original operations: $v^\bullet + w^\bullet = (v + w)^\bullet$ and $a^\bullet v^\bullet = (av)^\bullet$. External τ is the additive identity and is absorbing for scalar multiplication. The supported zero is $e_V = 0_V^\bullet$. Every linear map $f : V \rightarrow W$ has the canonical lift

$$S(f)(\tau) = \tau, \quad S(f)(v^\bullet) = f(v)^\bullet. \quad (\text{DS10})$$

Checking the two supported inputs and inputs containing τ proves additivity, the scalar law and composition. For $B \subseteq V$, $S(V) \rightarrow S(V/B)$ sends $b^\bullet$ to $e_{V/B}$ for every $b \in B$, and sends τ to τ . Two supported vectors have the same image exactly when their difference lies in B . This is the coequalizer of inclusion $S(B) \rightarrow S(V)$ and the map $b^\bullet \mapsto e_V, \tau \mapsto \tau$: any map equalizing these has equal values on $v^\bullet$ and $(v + b)^\bullet$, by additivity, and hence has a unique factorization through the displayed quotient.

Applying (DS10) to all maps in (DS1)–(DS4) lifts their source kernels, images, quotients and actions. A differential squared is the supported-zero map on represented vectors and still sends τ to τ . The ordinary kernel $\ker H^m(c)$ is carried by the represented fibre of $S(H^m(c))^{-1}(e)$; the preimage of external absence is only $\{\tau\}$. These two fibres are disjoint. Their exact common carrier is $S(\ker H^m(c)) = \{\tau\} \sqcup (\ker H^m(c))^\bullet$, with the inclusion of each of the two fibres. If $p_V : S(V) \rightarrow V$

sends $v^\bullet$ to v and τ to 0, this carrier is precisely $p_V^{-1}(\ker H^m(c))$. Thus the amplitude projection proves the two inclusions while retaining the two distinct receiving zeros. With a join semilattice of supports, carry out the construction in each fibre and use the original transition maps; (DS1)–(DS4) commute with those transitions because their definitions use the commuting cochain maps. No support label is discarded.

The coefficient comparison remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad p_R(a^\bullet) = a, \quad p_R(\tau) = 0. \quad (\text{DS11})$$

It commutes on every supported integer and on τ . In particular the linear/coherent calculation below is the amplitude image of a specified Split-Zero diagram; its supported exceptional kernel remains available before applying p .

55.1.3 The original source relation produces the curve

Fix the actual cyclic object $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, and the original action $A = M_S$. The monic source spaces $\mathcal{P}_N$, their positive Gram matrices O_N , and their full remainder maps $B_N : \mathcal{P}_N \rightarrow E$ give

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N. \quad (\text{DS12})$$

Take $i \leq j$ with both remainder maps onto, and retain the source inclusion L_{ij} with $B_j L_{ij} = B_i$ and $L_{ij}^* O_j L_{ij} = O_i$. Put

$$\Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j, \quad R(z) = R_j + z \Delta. \quad (\text{DS13})$$

The equalities follow directly: $B_N R_N = I$, $R_j^* O_j \Delta = G_j B_j \Delta = 0$, and expansion of $(L_{ij} R_i)^* O_j (L_{ij} R_i)$ gives $G_j + \Delta^* O_j \Delta$. For the kernel order, the actual source projection $P = L_{ij} O_i^{-1} L_{ij}^* O_j$ obeys $P^2 = P$ and $P^* O_j = O_j P$ by the retained source isometry. Therefore $K_j - K_i = B_j (I - P) O_j^{-1} (I - P)^* B_j^* \succeq 0$; expanding the right side gives $B_j O_j^{-1} B_j^* - B_i O_i^{-1} B_i^*$. Therefore

$$B_j R(z) = I, \quad R(z)^* O_j R(z) = G_j + |z|^2 C. \quad (\text{DS14})$$

The source is the original theta source with its arithmetic coefficient unit and every full jet; the new variable z parametrizes only the comparison and does not replace S .

Let $E_0 = \ker \Delta$, $E_1 = E_0^{\perp G_j}$, $r = \dim E_1$, and let P_0, P_1 be the specified projections. On $X = \mathbb{P}_{\mathbb{C}}^1$ with coordinates $[u : v]$, finite coordinate $z = v/u$ and infinity coordinate $w = u/v$, set $D = \{\infty\}$ and

$$\mathcal{E} = (E_0 \otimes \mathcal{O}_X) \oplus (E_1 \otimes \mathcal{O}_X(-D)). \quad (\text{DS15})$$

The source morphism is

$$\Psi(x_0 + x_1 \otimes (u, v)) = R_j x_0 + u R_j x_1 + v \Delta x_1. \quad (\text{DS16})$$

It is fibrewise injective. At $u \neq 0$, apply B_j and use $E_0 \cap E_1 = 0$; at $u = 0$, apply B_j to force $x_0 = 0$ and then use injectivity of $\Delta|_{E_1}$. Thus the original source Gram defines a smooth positive metric on this bundle, including infinity. The observation $\alpha = B_j \Psi$ is identity in the finite frame and $P_0 + w P_1$ in the infinity frame. Consequently

$$0 \rightarrow \mathcal{E} \xrightarrow{\alpha} E \otimes \mathcal{O}_X \rightarrow i_* E_1 \rightarrow 0, \quad (\text{DS17})$$

where $i : D \hookrightarrow X$. In fact its image lattice is $\mathcal{M} = E_0 \mathcal{O}_X + E \mathcal{O}_X(-D)$; at infinity it is $E_0 \mathcal{O} + w E \mathcal{O}$.

Tensoring the two locally free terms of (DS17) with the residue field at infinity gives $[E \xrightarrow{P_0} E]$ in degrees $-1, 0$. Its cohomology is E_1 in both degrees. Intrinsically the negative term is $E_1 \otimes \mathcal{O}(-D)|_D$. This calculation is the derived fibre of the skyscraper cokernel, and explicitly retains the Tor term that ordinary right-exact specialization would omit. Dualizing (DS17) gives

$$0 \rightarrow E^* \mathcal{O}_X \rightarrow \mathcal{E}^\vee \rightarrow i_*(E_1^* \otimes \mathcal{O}(D)|_D) \rightarrow 0. \quad (\text{DS18})$$

To check its terminal term, resolve $\mathcal{O}_D$ by $[\mathcal{O}(-D) \rightarrow \mathcal{O}]$ and apply $\mathcal{H}om(-, \mathcal{O})$. The resulting cokernel is $\mathcal{O}(D)|_D$. The two boundary factors pair by evaluation and $\mathcal{O}(-D)|_D \otimes \mathcal{O}(D)|_D \rightarrow \mathbb{C}_D$.

55.1.4 A calculated compact-support/ordinary de Rham comparison

Transport the trivial connection through α on $X \setminus D$. All connection complexes here are complexes of sheaves of $\mathbb{C}$ -vector spaces: the differentials satisfy the Leibniz rule and are not asserted to be $\mathcal{O}_X$ -linear. In the infinity frame it is

$$\nabla = d + P_1 \frac{dw}{w}, \quad \mathcal{K} = [\mathcal{E} \xrightarrow{\nabla} \mathcal{E} \otimes \Omega_X^1(\log D)] \quad \text{in degrees } 0, 1. \quad (\text{DS19})$$

Indeed $\alpha^{-1}d\alpha = P_1 dw/w$. Its residue is P_1 , its horizontal monodromy is $\exp(-2\pi i P_1) = I$, and the map α is a morphism to $\mathcal{K}_{\text{ord}} = [E\mathcal{O}_X \xrightarrow{d} E\Omega_X^1(\log D)]$.

Here is the complete local calculation of the extension at infinity. On analytic power series the E_0 differential, with dw/w as the one-form frame, is $\sum a_n w^n \mapsto \sum n a_n w^n$. Its kernel and cokernel are the constant terms, both E_0 . The E_1 differential is

$$\sum_{n \geq 0} b_n w^n \mapsto \sum_{n \geq 0} (n+1) b_n w^n. \quad (\text{DS20})$$

It is invertible: its inverse divides the n th coefficient by $n+1$, which preserves every disk of convergence. Thus the E_1 part has zero derived stalk at infinity. On a punctured disk, integrating a Laurent series term by term divides its w^n coefficient by n for $n \neq 0$; the only one-form class remaining is dw/w with integral $2\pi i$ around a positively oriented w -circle. These statements, together with the usual local primitive on a disk, identify the E_0 part with $Rj_* E_0$ and the E_1 part with $j_! E_1$, where $j : X \setminus D \hookrightarrow X$. The identification for $j_!$ is the extension-by-zero map of flat sections; its stalk cohomology is an isomorphism on the open disk and is zero on both sides at infinity by (DS20). The identification for Rj_* is the restriction map to the punctured disk and the two Laurent calculations. This constructs the comparison on the present curve rather than assigning compact support by name.

Its entire global cohomology is

$$\mathbb{H}^0(X, \mathcal{K}) = E_0, \quad \mathbb{H}^1(X, \mathcal{K}) = 0, \quad \mathbb{H}^2(X, \mathcal{K}) = E_1 \otimes \mathbb{T}, \quad \mathbb{T} = H^1(X, \Omega_X^1). \quad (\text{DS21})$$

For a direct verification, $\Omega_X^1(\log D) = \mathcal{O}_X(-1)$. The two terms of $\mathcal{K}$ are therefore $E_0 \mathcal{O} \oplus E_1 \mathcal{O}(-1)$ and $E_0 \mathcal{O}(-1) \oplus E_1 \mathcal{O}(-2)$. Using the two affine charts, polynomial division of Laurent polynomials proves

$$\begin{aligned} H^0(\mathcal{O}) &= \mathbb{C}, & H^1(\mathcal{O}) &= 0, & H^0(\mathcal{O}(-1)) &= H^1(\mathcal{O}(-1)) = 0, \\ H^0(\mathcal{O}(-2)) &= 0, & H^1(\mathcal{O}(-2)) &= \mathbb{C}. \end{aligned} \quad (\text{DS22})$$

More explicitly, after identifying $\mathcal{O}(-2)$ with Ω_X^1 , a Laurent one-form on the overlap has every term except dw/w removable by a one-form on one of the charts. The coefficient of dw/w is not removable and is its stated one-dimensional quotient. Retain $\mathbb{T}$ as this quotient line, with its original generator $[dz/z] = -[dw/w]$. The double complex for these two terms has only the entries E_0 in total degree zero and $E_1 \otimes \mathbb{T}$ in total degree two. There is no differential to or from them, proving (DS21) without an uncomputed higher differential.

Taking the quotient of the two complexes in (DS19) gives

$$\mathcal{Q}_D = [i_* E_1 \xrightarrow{0} i_* E_1 dw/w] \quad \text{in degrees } 0, 1. \quad (\text{DS23})$$

The zero differential follows by reducing $w\partial_w$ modulo w . The termwise short exact sequence $0 \rightarrow \mathcal{K} \rightarrow \mathcal{K}_{\text{ord}} \rightarrow \mathcal{Q}_D \rightarrow 0$ and the fact $\mathbb{H}^0(\mathcal{K}_{\text{ord}}) = E$, $\mathbb{H}^{>0}(\mathcal{K}_{\text{ord}}) = 0$ give

$$0 \rightarrow E_0 \rightarrow E \rightarrow E_1 \rightarrow 0, \quad E_1 dw/w \xrightarrow{\delta} E_1 \otimes \mathbb{T} \quad \text{an isomorphism.} \quad (\text{DS24})$$

Fix the Čech convention $s_V - s_U$ for the overlap, with U the finite chart and V the infinity chart, and the total-complex differential $\delta + (-1)^p \nabla$ on Čech degree p . Lifting a boundary one-form as $(0, v dw/w)$ makes its connecting class $v[dw/w]$; thus $\delta(v dw/w) = v \otimes [dw/w] = -v \otimes [dz/z]$. In the original finite coordinate $dw/w = -dz/z$. Both that sign and the chosen positive orientation of the w -circle are retained. Under (DS21), the ordinary comparison image is exactly E_0 in degree zero and zero in degrees one and two. The full E_1 occurs in its boundary comparison and in degree two.

55.1.5 The divisor twist in the support-compatible dual

The bundle dual in (DS18) and the dual de Rham comparison are joined by the exact additional map of line bundles $\mathcal{O}(-D) \hookrightarrow \mathcal{O}$. Define

$$\mathcal{E}^{\text{cd}} = \mathcal{E}^\vee \otimes \mathcal{O}(-D), \quad \mathcal{K}^{\text{cd}} = [\mathcal{E}^{\text{cd}} \xrightarrow{\nabla^{\text{cd}}} \mathcal{E}^{\text{cd}} \otimes \Omega_X^1(\log D)]. \quad (\text{DS25})$$

The dual connection has residue $-P_1^\vee$. Multiplication of its local frame by w , the local generator of $\mathcal{O}(-D)$, adds I to the residue. Hence the residue of (DS25) is $P_0^\vee$, and

$$\mathcal{E}^{\text{cd}} = E_0^* \mathcal{O}(-D) \oplus E_1^* \mathcal{O}, \quad \frac{m}{\mathbb{H}^m(\mathcal{K}^{\text{cd}})} \left| \begin{array}{ccc} 0 & 1 & 2 \\ E_1^* & 0 & E_0^* \otimes \mathbb{T} \end{array} \right. \quad (\text{DS26})$$

The proof is exactly the power-series calculation (DS20) and the four line-bundle groups (DS22), now with the two summands exchanged.

Evaluation is a morphism $\mathcal{E} \otimes \mathcal{E}^{\text{cd}} \rightarrow \mathcal{O}(-D)$. The connections obey the Leibniz identity for evaluation, since the dual connection cancels the original connection term. Wedge product therefore gives a cochain map

$$\mathcal{K} \otimes_{\mathbb{C}} \mathcal{K}^{\text{cd}} \longrightarrow [\mathcal{O}(-D) \xrightarrow{d} \Omega_X^1] =: \mathcal{K}_c(\mathbb{C}). \quad (\text{DS27})$$

On a degree-zero tensor it is evaluation; on degree-one tensors it is evaluation times the retained one-form; on two one-forms it is zero because $\Omega_X^2 = 0$. These prescriptions and the tensor sign in (DS9) verify the cochain identity in every degree. The degreewise evaluation and wedge maps factor through their $\mathcal{O}_X$ -balanced tensor products; the displayed complex tensor product is over $\mathbb{C}$, so its differential is defined. Define the trace on $\mathbb{H}^2(\mathcal{K}_c(\mathbb{C}))$ by

$$\text{Tr}_D([dz/z]) = 1, \quad \text{Tr}_D([dw/w]) = -1 = -\frac{1}{2\pi i} \int_{|w|=\epsilon}^{\text{counterclockwise}} \frac{dw}{w}. \quad (\text{DS28})$$

Holomorphic one-forms on either chart have zero overlap residue, so this is well-defined on (DS22). The induced pairings are

$$E_0 \times E_0^* \rightarrow \mathbb{C}, \quad E_1 \times E_1^* \rightarrow \mathbb{C}, \quad (\text{DS29})$$

ordinary evaluation after applying the retained trace on $\mathbb{T}$, in complementary degrees 0, 2 and 2, 0. They are perfect, since a nonzero vector is detected by a linear functional. Thus (DS25)–(DS29) prove the support-compatible duality, including the dimension-one degree change and its actual boundary line factor. The finite-field analogue of the trace target is $\overline{\mathbb{Q}}_\ell(-1)$; the characteristic-zero calculation here has the explicit residue normalization (DS28).

55.1.6 Retaining the arithmetic action through the exceptional fibre

Transporting the original A through α gives

$$\hat{A}_\infty = (P_0 + w^{-1}P_1)A(P_0 + wP_1) = \begin{pmatrix} A_{00} & wA_{01} \\ w^{-1}A_{10} & A_{11} \end{pmatrix}. \quad (\text{DS30})$$

It extends regularly on $\mathcal{E}$ exactly when $AE_0 \subseteq E_0$, since its sole possible negative power is $w^{-1}P_1AP_0$. The explicit repair on the same ambient arithmetic target is

$$F_A = \sum_{a=0}^{q-1} A^a E_0, \quad \mathcal{M}_A = F_A \mathcal{O}_X + E \mathcal{O}_X(-D). \quad (\text{DS31})$$

Cayley–Hamilton puts $A^q E_0$ in the displayed sum, proving invariance; every invariant subspace containing E_0 contains this sum. Similarly every coherent A -stable submodule of $E \mathcal{O}$ containing $\mathcal{M}$ contains $\sum_{a=0}^{q-1} A^a \mathcal{M}$, which is exactly $\mathcal{M}_A$ in (DS31). There are exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{M} \rightarrow \mathcal{M}_A \rightarrow i_*(F_A/E_0) \rightarrow 0, \\ 0 \rightarrow \mathcal{M}_A \rightarrow E \mathcal{O}_X \rightarrow i_*(E/F_A) \rightarrow 0. \end{aligned} \quad (\text{DS32})$$

The second is A equivariant on every term. The connection d on $E \mathcal{O}$ preserves $\mathcal{M}_A$ logarithmically, because $d(wv) = w dv + v dw$ lies in $\mathcal{M}_A \otimes \Omega^1(\log D)$. It commutes with the constant original A . Define its two-term complex $\mathcal{K}_A$ with the same degree convention as (DS19). Calculation with any vector-space complement of F_A proves, intrinsically,

$$\mathbb{H}^0(\mathcal{K}_A) = F_A, \quad \mathbb{H}^1(\mathcal{K}_A) = 0, \quad \mathbb{H}^2(\mathcal{K}_A) = (E/F_A) \otimes \mathbb{T}, \quad (\text{DS33})$$

with actions $A|_{F_A}$ and $\bar{A}$, respectively. The identifications are independent of that auxiliary complement: the degree-zero map is inclusion into E and the degree-two map is the connecting quotient map in (DS32), as in (DS24).

Its dual lattice has a coordinate-free formula. Put $F_A^\perp = \{\lambda \in E^* : \lambda(F_A) = 0\}$. Then

$$\mathcal{M}_A^\vee(-D) = F_A^\perp \mathcal{O}_X + E^* \mathcal{O}_X(-D). \quad (\text{DS34})$$

Indeed a meromorphic functional belongs to the left side precisely when its value on every element of $F_A \mathcal{O} + wE \mathcal{O}$ lies in $w \mathcal{O}$. This first forces it to be holomorphic, by testing wE , and then forces its constant term to annihilate F_A . This is exactly the right side. Fix the additive dual action as $-A^\vee$, so evaluation has the zero action on its one-dimensional target. The displayed lattice is invariant under this action because F_A is invariant. Equations (DS27)–(DS29) therefore become the perfect equivariant pairings $F_A \times F_A^*$ and $(E/F_A) \times F_A^\perp$. For the original real integer k , replacing the dual action by $kI - A^\vee$ puts the scalar k action on the evaluation target, as checked in (DS6). The passage to the metric antidual is the specified conjugate-linear map $\mathfrak{c} : E^\vee \rightarrow E^\vee$, $\mathfrak{c}(\lambda)(v) = \bar{\lambda}(v)$. In the fixed polynomial basis it conjugates the functional's coefficient column, and hence $\mathfrak{c}(kI - A^T) = (kI - A^*)\mathfrak{c}$. This proves the exact coefficient map between the linear dual and the original metric antidual convention. For every polynomial p , an adapted basis to F_A proves

$$\text{Tr } p(A) = \text{Tr } p(A|_{F_A}) + \text{Tr } p(\bar{A}). \quad (\text{DS35})$$

This retains the complete arithmetic trace across the calculated cohomological degrees. There is no need to presume that the metric kernel E_0 was an arithmetic constituent.

The source cost of (DS31) remains visible. If $s_A = \dim F_A - \dim E_0$, choose a basis first for E_0 , then for F_A/E_0 , and then for E/F_A . The local inclusion $\mathcal{M} \rightarrow \mathcal{M}_A$ has diagonal blocks I, wI_{s_A}, I and determinant w^{s_A} . If the metric on $\mathcal{M}_A$ is transported on the punctured chart from the original source metric on $\mathcal{M}$, the determinant identity is

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w). \quad (\text{DS36})$$

It follows by taking determinants of $G_{\mathcal{M}} = J^* G_A J$. Thus the action-stable complex (DS33) retains the exceptional contribution through an explicitly measured metric pole.

55.1.7 The same curvature, its source map, and the weight interface

For $\mathcal{L} = (\det \mathcal{E})^\vee$ with its original source metric, put

$$Z = G_j^{-1}(G_i - G_j) = (K_j - K_i)G_i. \quad (\text{DS37})$$

The second equality is $K_j G_i - I = (K_j - K_i)G_i$. The first expression proves self-adjointness and positivity in G_j ; the second proves the same in G_i , because $G_i Z = G_i(K_j - K_i)G_i \succeq 0$ and $K_j - K_i \succeq 0$. No zero eigenvalue is removed. Differentiating $\log \det(G_j + |z|^2 C)$ yields

$$c_1(\mathcal{L}) = \frac{i}{2\pi} \kappa(|z|^2) dz \wedge d\bar{z}, \quad \kappa(t) = \text{Tr}(Z(I + tZ)^{-2}). \quad (\text{DS38})$$

To verify every coefficient, set $f(t) = \log \det(I + tZ)$. Then $f'(t) = \text{Tr}(Z(I + tZ)^{-1})$ and $f'(t) + t f''(t) = \kappa(t)$. Since $\partial \bar{\partial} f(|z|^2) = (f'(|z|^2) + |z|^2 f''(|z|^2)) dz \wedge d\bar{z}$, this proves (DS38) with the stated sign. For a positive eigenvalue λ , $\int_0^\infty \lambda(1 + t\lambda)^{-2} dt = 1$; for a zero eigenvalue the integrand is zero. Since $i dz \wedge d\bar{z} = 2dx \wedge dy$, polar integration gives the degree and the logarithmic moment

$$\int_X c_1(\mathcal{L}) = r, \quad \log \frac{\det G_i}{\det G_j} = \int_0^1 \log(1/t) \kappa(t) dt. \quad (\text{DS39})$$

For the second equality, write $M(t) = t f'(t)$, so $M' = \kappa$ and $M(0) = 0$. Nonnegative Fubini gives $\int_0^1 \log(1/t) \kappa(t) dt = \int_0^1 t^{-1} M(t) dt = f(1) - f(0)$.

The full source observation is also retained. With

$$P_z = R(z)G(z)^{-1}R(z)^*O_j, \quad \mathcal{N}_z = (I - P_z)\Delta, \quad G(z) = G_j + |z|^2 C, \quad (\text{DS40})$$

orthogonal projection gives

$$\mathcal{N}_z^* O_j \mathcal{N}_z = C - |z|^2 C G(z)^{-1} C, \quad \text{Tr}(G(z)^{-1} \mathcal{N}_z^* O_j \mathcal{N}_z) = \kappa(|z|^2). \quad (\text{DS41})$$

Indeed $R(z)^* O_j \Delta = \bar{z} C$; substitute into $\Delta^* O_j (I - P_z) \Delta$. Factoring $G(z) = G_j(I + |z|^2 Z)$ in the trace proves the second equality. Applying the actual remainder map gives

$$B_j P_z \Delta = \bar{z} G(z)^{-1} C, \quad B_j \mathcal{N}_z = -\bar{z} G(z)^{-1} C. \quad (\text{DS42})$$

Their sum is the original theta relation. In the split lift its quotient value is the supported zero e , with both prequotient pieces still given by (DS42).

The curvature jets are the retained relation moments:

$$\kappa^{(n)}(0) = (-1)^n (n+1)! \text{Tr}(Z^{n+1}), \quad t_1 = \kappa(0), \quad t_2 = -\frac{1}{2} \kappa'(0), \quad \mathcal{A}_2 = \frac{1}{2} (t_1^2 - t_2). \quad (\text{DS43})$$

These follow by differentiating each rational eigenvalue term in (DS38); finiteness of the matrix makes the differentiation exact. The dimension-one divisor degree in (DS39), the arithmetic trace in (DS35), and the source metric pole in (DS36) are now related by specified maps on the same compactified curve.

For completeness, exponentiation supplies the precise multiplicative observer of the original additive action, for every real t :

$$F_t = \exp(tA) : E \rightarrow E, \quad F_t|_{F_A} = \exp(tA|_{F_A}), \quad \bar{F}_t = \exp(t\bar{A}). \quad (\text{DS44})$$

On the full generalized eigenspace of A at ρ , write $A = \rho I + N$, $N^m = 0$ with its actual order. Then $F_t = e^{t\rho} \sum_{a=0}^{m-1} t^a N^a / a!$; in particular it retains the nilpotent terms and has eigenvalue modulus $e^{t \text{Re } \rho}$. Choosing $t = \log Q_0$ gives the exact number $Q_0^{\text{Re } \rho}$. This is the proved operator map connecting the additive spectral parameter and a multiplicative weight observer. It does not provide a finite-field descent or an étale Frobenius structure. Those structures in (DS7) have the specific types $X_0/\mathbb{F}_{Q_0}$, a

smooth $\overline{\mathbb{Q}}_\ell$ -sheaf and its geometric Frobenius action; the constructed characteristic-zero object has instead the fully calculated complexes (DS19), (DS25), (DS33) and the commuting action (DS44). Its complete comparison to those source operations is (DS1)–(DS9), (DS24), (DS27), and (DS34), with their actual kernels.

Finally the original positive metric supplies its precise additive weight measurement on every invariant inclusion $I_F : F \hookrightarrow E$:

$$G_F = I_F^* G(z) I_F, \quad W_F = A_F^* G_F + G_F A_F - k G_F, \quad \text{Tr}(G_F^{-1} W_F) = 2 \text{Re Tr } A_F - k \dim F. \quad (\text{DS45})$$

The identity follows by cyclicity of the finite trace, with $A I_F = I_F A_F$. Thus k remains the original tensor parameter, and all constituents of E are retained through (DS32)–(DS35). Equations (DS38)–(DS43) make the current endpoint estimate an estimate on a specified source-induced curvature measure. Its total degree alone does not bound its logarithmic moment: for the explicitly declared matrix family $G_j = I_r$, $C = L I_r$, $L > 0$, these quantities are respectively r and $r \log(1 + L)$, and the mass on $0 \leq t \leq 1/L$ is exactly $r/2$ by direct integration. The original arithmetic source, its relation map (DS13), its full dual comparison (DS25)–(DS34), and the actual moment observers (DS43) therefore remain the concrete data for the continuing curvature calculation.

55.1.8 Every divisor depth and the unchanged conormal directions

For a specified A -invariant subspace $F \subseteq E$ and an integer $m \geq 1$, retain

$$\mathcal{M}_{F,m} = F \mathcal{O}_X + E \mathcal{O}_X(-mD), \quad \mathcal{K}_{F,m} = [\mathcal{M}_{F,m} \xrightarrow{d} \mathcal{M}_{F,m} \otimes \Omega_X^1(\log D)]. \quad (\text{DS46})$$

The local coefficient differential on $w^m(E/F)$ is $w \partial_w + m$. The inclusion $\mathcal{K}_{F,m} \hookrightarrow \mathcal{K}_{F,1}$ has quotient supported at D , with coefficient basis $w, w^2, \dots, w^{m-1}$ in each of degrees zero and one and differential

$$w^a v \mapsto a w^a v dw/w \quad (1 \leq a < m). \quad (\text{DS47})$$

Define $h(w^a v dw/w) = a^{-1} w^a v$ on degree one and $h = 0$ on degree zero. Then $dh + hd = I$ on each displayed basis element, so this entire quotient is contractible. Thus the inclusion is a quasi-isomorphism, equivariant for the original A , and

$$\mathbb{H}^0(\mathcal{K}_{F,m}) = F, \quad \mathbb{H}^1(\mathcal{K}_{F,m}) = 0, \quad \mathbb{H}^2(\mathcal{K}_{F,m}) = (E/F) \otimes \mathbb{T}. \quad (\text{DS48})$$

This preserves every higher coefficient until its explicit differential (DS47) contracts it. In original arithmetic degree k the shifted complex is $\mathcal{K}_{F,m}[-k]$, so the two surviving degrees are $k, k+2$ and

$$\text{Str}(p(A) \mid R\Gamma \mathcal{K}_{F,m}[-k]) = (-1)^k \text{Tr}_E p(A). \quad (\text{DS49})$$

The equality follows from (DS35); multiplication by $p(A)$ acts trivially on the factor $\mathbb{T}$.

The exact higher-depth dual is

$$\mathcal{M}_{F,m}^\vee(-D) = F^{\text{ann}} \mathcal{O}_X((m-1)D) + E^* \mathcal{O}_X(-D) \subset E^* \otimes \mathcal{O}_X(*D). \quad (\text{DS50})$$

At infinity, testing a meromorphic functional on $w^m E$ permits poles of order at most $m-1$, and testing on F forces every coefficient of powers $w^{1-m}, \dots, w^0$ to annihilate F . Its remaining coefficients are $w E^* \mathcal{O}$. This proves (DS50), including its ambient meromorphic sheaf. In a coefficient complement its residues are 1 on F^* and $1-m$ on $(E/F)^*$. The natural inclusion of the depth-one dual into (DS50) has quotient basis $w^{1-m}, \dots, w^{-1}$ in both degrees, with differential $w^a v \mapsto a w^a v dw/w$ and inverse a^{-1} , so it is contractible. Therefore its dual cohomology is F^{ann} in degree zero and $F^* \otimes \mathbb{T}$ in degree two. Evaluation into $\mathcal{O}(-D)$ and (DS27) give the same perfect complementary-degree pairings. All negative powers, residues and the divisor twist have been retained.

The independent conormal depth has the explicit map

$$\frac{\mathbb{C}[w] \otimes \mathcal{P}}{(w^m) + I^a(\mathbb{C}[w] \otimes \mathcal{P})} \xrightarrow{\sim} (\mathbb{C}[w]/(w^m)) \otimes (\mathcal{P}/I^a), \quad [f(w)P] \mapsto [f] \otimes [P], \quad (\text{DS51})$$

where $\mathcal{P} = \mathbb{C}[s_1, \dots, s_k]$ and $I = (h(s_1), \dots, h(s_k))$. Reduction of the w -degree writes the left side uniquely as $\sum_{b=0}^{m-1} w^b [P_b]_{I^a}$. This supplies the inverse and proves injectivity as well as surjectivity. Restriction in either depth commutes with the other. Differentiation in w has the well-defined type from depth $m+1$ to depth m ; differentiation in s_j has type from I^{a+1} quotients to I^a quotients, because $\partial_{s_j} I^{a+1} \subset I^a$. On pure tensors the two derivatives commute, hence on their full spans. Lifting these maps by (DS10) retains the same support label and the receiving supported zero in both depth directions.

Pullback of (DS38) under the actual map $z \mapsto z^m$ has radial density

$$\kappa_m(t) = m^2 t^{m-1} \kappa(t^m), \quad t = |z|^2. \quad (\text{DS52})$$

Indeed differentiate $f(t^m)$ twice in the combination $f'_m + t f''_m$ used in (DS38). Substitution $s = t^m$ proves $\int_0^\infty \kappa_m(t) dt = mr$ and $\int_0^1 \log(1/t) \kappa_m(t) dt = \int_0^1 \log(1/s) \kappa(s) ds$. Thus increasing divisor depth multiplies degree by m while leaving the original endpoint logarithmic moment fixed.

55.1.9 The invariant quotient gives an exact coupling subtraction

The following quantitative calculation is the delivered logarithmic constituent-control result, now proved in the same source notation. Take $I : F \hookrightarrow E$ and $\pi : E \twoheadrightarrow E/F$ with $AI = IA_F$ and $\pi A = A_Q \pi$, and put $G(t) = G_0 + tC$, $G_0 = G_j$, $0 \leq t \leq 1$. The cases $F = 0, E$ use the empty determinant 1, empty trace 0, and the unique maps on the zero-dimensional space; all formulas below still hold. For any initial coefficient lift $\overline{H}$, define

$$H_0 = \overline{H} - I(I^* G_0 I)^{-1} I^* G_0 \overline{H}, \quad \pi H_0 = I_{E/F}, \quad I^* G_0 H_0 = 0. \quad (\text{DS53})$$

These last two identities follow from $\pi I = 0$ and matrix multiplication. Changing $\overline{H}$ by IU cancels the change in (DS53); thus the orthogonal lift itself is independent of the initial lift. In the original bases define

$$\begin{aligned} G_{F,0} &= I^* G_0 I, \quad C_F = I^* C I, \quad D_F = I^* C H_0, \\ Q_0 &= H_0^* G_0 H_0, \quad C_Q = H_0^* C H_0, \quad G_F(t) = G_{F,0} + t C_F, \\ H_t &= H_0 - t I G_F(t)^{-1} D_F. \end{aligned} \quad (\text{DS54})$$

Here Q_0 is the original quotient Gram, not the finite-field cardinality used in (DS7). Every inverse is defined: I and H_0 are injective on their respective coefficient spaces, and $G(t) > 0$. In coordinates $[I, H_0]$ the original metric is exactly

$$\begin{pmatrix} G_F(t) & t D_F \\ t D_F^* & Q_0 + t C_Q \end{pmatrix}. \quad (\text{DS55})$$

The replacement $H_0 \mapsto H_t$ is the upper triangular coefficient change $\begin{pmatrix} I & -t G_F(t)^{-1} D_F \\ 0 & I \end{pmatrix}$ of determinant 1. It gives

$$I^* G(t) H_t = 0, \quad \pi H_t = I_{E/F}, \quad Q(t) := H_t^* G(t) H_t = Q_0 + t C_Q - t^2 D_F^* G_F(t)^{-1} D_F > 0. \quad (\text{DS56})$$

Expansion proves the formula, and injectivity of H_t proves positivity. It is also the quotient norm: any other lift of v is $H_t v + Ix$, whose squared norm is $v^* Q(t) v + x^* G_F(t) x$ by orthogonality. The unique minimum is therefore attained at $H_t v$.

Define the actual coupling map

$$\beta_t = G_F(t)^{-1} I^* C H_t : E/F \rightarrow F. \quad (\text{DS57})$$

Differentiate $\pi H_t = I$ to write $H'_t = IU_t$; differentiating $I^*G(t)H_t = 0$ then gives $I^*CH_t + G_F(t)U_t = 0$. Therefore $H'_t = -I\beta_t$. In the derivative of $Q(t) = H_t^*G(t)H_t$ the two terms containing H'_t vanish by orthogonality. Differentiating the remaining term gives

$$Q'(t) = H_t^*CH_t, \quad Q''(t) = -2\beta_t^*G_F(t)\beta_t. \quad (\text{DS58})$$

The factor two follows because $I^*CH_t = G_F(t)\beta_t$ and its adjoint supply the two equal terms. The exact integral remainder for $Q(1)$ is $Q(1) = Q(0) + Q'(0) + \int_0^1 (1-t)Q''(t)dt$; insert (DS56) at the two endpoints to get

$$\Pi_F := D_F^*G_F(1)^{-1}D_F = 2 \int_0^1 (1-t)\beta_t^*G_F(t)\beta_t dt. \quad (\text{DS59})$$

This identifies the finite subtraction with an integrated squared map on the original quotient source.

The action defect is the unique map $a_t : E/F \rightarrow F$ with

$$AH_t - H_tA_Q = Ia_t, \quad a'_t = \beta_t A_Q - A_F \beta_t. \quad (\text{DS60})$$

The first left side has zero image under π , so it lies in $\text{im } I$, which proves existence and uniqueness. Differentiating and inserting $H'_t = -I\beta_t$ proves the second formula, using injectivity of I . Thus the nilpotent extension data are retained in a_t even when the two constituent actions have equal eigenvalues.

For clarity the entire Sylvester operator has a finite exact description. On a pair of primary spaces let $A_F = \lambda I + N_F$, $A_Q = \mu I + N_Q$, with $N_F^a = 0$ and $N_Q^b = 0$. Put $\mathcal{N}(X) = XN_Q - N_F X$. The two multiplications commute, so the binomial expansion makes $\mathcal{N}^{a+b-1} = 0$. If $\mu \neq \lambda$, the inverse of $X \mapsto XA_Q - A_F X$ is

$$Y \mapsto \sum_{r=0}^{a+b-2} \frac{(-1)^r}{(\mu - \lambda)^{r+1}} \mathcal{N}^r(Y). \quad (\text{DS61})$$

Multiplying the finite geometric series by $(\mu - \lambda)I + \mathcal{N}$ leaves identity and a zero terminal power. At $\mu = \lambda$, the operator is exactly $\mathcal{N}$; its kernel is the space $\{X : XN_Q = N_F X\}$ and its image is the span of the explicit matrices $E_{uv}N_Q - N_F E_{uv}$. Matrix-unit expansion proves that this span is its whole image. This keeps the kernel and every coefficient of the image without a nonexistent inverse.

The same original control $W(t) = A^*G(t) + G(t)A - kG(t)$ obeys

$$\begin{aligned} I^*W(t)I &= A_F^*G_F(t) + G_F(t)A_F - kG_F(t), \\ H_t^*W(t)H_t &= A_Q^*Q(t) + Q(t)A_Q - kQ(t). \end{aligned} \quad (\text{DS62})$$

The first identity uses invariance of F . For the second insert (DS60); both terms containing Ia_t vanish by (DS56). On the original theta source write $r(z) = \mathcal{V}R(z)$ and retain its actual primitive $D^{(k)}r(z) - r(z)A = dK(z)$, with $K(z) = (1-z)K_j + zK_i$ in the inherited primitive notation. Then exact multiplication gives

$$D^{(k)}r(z)H_t - r(z)H_tA_Q = dK(z)H_t + r(z)Ia_t, \quad t = |z|^2. \quad (\text{DS63})$$

Here $D^{(k)}$ differentiates the original source variables, not the comparison parameter z or t . The first term is killed by the original theta quotient. The second has its retained image in the arithmetic constituent F and is killed by the further quotient $E \rightarrow E/F$. Formula (DS63) proves the precise map between these two successive supported-zero operations.

Put $Q_+ = Q_0 + C_Q$, $\Xi_F = Q_+^{-1}\Pi_F$. Then Ξ_F is self-adjoint and nonnegative for Q_+ , and $Q(1) = Q_+ - \Pi_F > 0$ proves that all its eigenvalues belong to $[0, 1)$. Taking determinants in (DS55)–(DS56),

with the fixed coefficient determinant $|\det[I, H_0]|^2$ cancelled only between the same two original endpoints, gives

$$\begin{aligned} \log \frac{V_i}{V_j} &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} + \log \det(I - \Xi_F) \\ &= \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}. \end{aligned} \quad (\text{DS64})$$

For each eigenvalue $0 \leq x < 1$, integrating the geometric series $\sum_{a \geq 0} x^a = (1 - x)^{-1}$ from zero to x proves the logarithm series. There are finitely many eigenvalues, so their sum yields the second equality and every term is nonnegative. Its first p terms therefore give the stated upper estimate, and these estimates decrease to the exact original endpoint volume.

There is also an explicit remainder from the same matrices. If $\rho = \lambda_{\max}(\Xi_F) < 1$, then for every $p \geq 0$,

$$0 \leq \sum_{r=p+1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r} \leq \frac{\text{Tr}(\Xi_F^{p+1})}{(p+1)(1-\rho)}. \quad (\text{DS65})$$

Indeed $\text{Tr}(\Xi_F^{p+1+a}) \leq \rho^a \text{Tr}(\Xi_F^{p+1})$ on each nonnegative eigenvalue, and $p+1+a \geq p+1$; summing the resulting geometric series proves the inequality. The value ρ is the largest generalized eigenvalue of (Π_F, Q_+) of this same source pair, with no independently assumed uniform gap. On the zero-dimensional quotient $E/F = 0$, define $\rho = 0$; every trace and both sides of (DS65) are then zero, so the same formula includes that endpoint.

The two positive terms in (DS64) are the logarithmic determinants of the compressed positive operators $Z_F = G_{F,0}^{-1} C_F$ and $Z_Q = Q_0^{-1} C_Q$. Consequently the already proved two-trace majorant, applied to each of these specified matrices in its own full dimension, can be combined with every finite subtraction in (DS64). This integration retains the extension map (DS60), the quotient source (DS63), and the original total curvature (DS39) together; the cohomological constituent decomposition does not remove its measurable coupling.

55.1.10 The degree-one exponential family joins the same coefficient object

The delivered exponential-comparison note constructs the additional degree-one family from the unchanged polynomial $\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a$. Its immediate join to the present cohomology calculation is as follows. Use new independent parameters u, t and retain every coefficient of

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1}, \quad \Phi_t(S) = \Phi(S) - tS, \quad R = \mathbb{C}[u, t]. \quad (\text{DS66})$$

The two-term coefficient complex is

$$\mathcal{C} = [R[S] \xrightarrow{\mathcal{D}} R[S]dS], \quad \mathcal{D}P = (uP' + (\chi - t)P)dS. \quad (\text{DS67})$$

For a nonzero polynomial P of degree a and leading coefficient $b \in R$, $\mathcal{D}P/dS$ has degree $a + q$ and leading coefficient b . Successively subtracting $\mathcal{D}(bS^a)/dS$ cancels each highest term in a polynomial of degree at least q and terminates. Uniqueness follows from the same degree calculation. Hence $R[S] = \text{im}(\mathcal{D}/dS) \oplus R[S]_{<q}$ as coefficient modules, $H^0(\mathcal{C}) = 0$, and $\mathcal{H} = H^1(\mathcal{C})$ is free with the original basis $[1], \dots, [S^{q-1}]$. At $u = t = 0$ the differential is multiplication by the literal χ , so $\mathcal{H}/(u, t)\mathcal{H} = E$ with every nilpotent jet.

The commutators are $[\mathcal{D}/dS, \partial_t] = I$ and $[\mathcal{D}/dS, -S/u] = -I$ after inverting u . They prove that $\nabla_t = \partial_t - S/u$ descends through the differential. Define $\mathcal{R} : E \rightarrow E$ by sending a remainder to its S^{q-1} coefficient times $[1]$. Reducing S^q with the relation $[\chi - t] = 0$ shows in the displayed basis that

$$\nabla_t = \partial_t - \frac{A + t\mathcal{R}}{u}, \quad (-u\nabla_t) \bmod (u, t) = A. \quad (\text{DS68})$$

The regular operator $-u\partial_t + A + t\mathcal{R}$ preserves $(u, t)\mathcal{H}$: on uv its derivative contribution is $-u^2\partial_t v$, and on tv it is $-uv - ut\partial_t v$. This verifies the asserted specialization as an actual induced operator, rather than specializing a singular connection symbol.

The logarithmic comparison retains the original exact coefficient sequence $0 \rightarrow F \rightarrow E \rightarrow E/F \rightarrow 0$, with the outer terms occurring in (DS48). The common full E is obtained through its actual outer comparison complexes: $\mathbb{H}^0(\mathcal{K}_{E,1}) = E$ and $\mathbb{H}^2(\mathcal{K}_{0,1}) = E \otimes \mathbb{T}$. The maps to and from the mixed complex are $0 \rightarrow F \rightarrow E$ and $E \otimes \mathbb{T} \rightarrow (E/F) \otimes \mathbb{T} \rightarrow 0$. At the exponential special fibre the cochain map $[\mathbb{C}[S] \xrightarrow{\chi} \mathbb{C}[S]dS] \rightarrow E[-1]$ is zero in degree zero and $P dS \mapsto [P]_\chi$ in degree one. Its kernel complex is the isomorphism $\mathbb{C}[S] \xrightarrow{\chi} \chi\mathbb{C}[S]dS$ and is contractible by division by χ on its image. This proves the special-fibre quasi-isomorphism. The resulting map to the full coefficient object of the logarithmic outer complexes is the identity on $[1], \dots, [S^{q-1}]$, intertwining A . Composing with $E \rightarrow E\mathcal{O}_X$, $v \mapsto v \otimes 1$, gives the map into the original constant target of (DS32). It retains the coefficient extension and its map a_t in (DS60). The original theta comparison $j_E r_N = I$, $j_E D^{(k)} = A j_E$ makes this identity a source comparison as well:

$$D_t^{(k)} = D^{(k)} + tr_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t\mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t\mathcal{R}) = dK_N. \quad (\text{DS69})$$

Both equalities follow by multiplying and using $j_E r_N = I$; the added terms cancel exactly in the second equality. Thus the source primitive is unchanged and the full kernel $\ker j_E$ remains. The exponential note's contour periods and finite-field specializations develop this new family further; equations (DS66)–(DS69) are its complete coefficient, cohomological and source-level join to the logarithmic machinery in this sidebar.

56 Exact determinant of the polynomial-exponential period comparison

56.1 Exact determinant of the polynomial-exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix $u \in \mathbb{C}^*$ and one argument of u . All the following comparisons retain this u , the original coordinate S , and the ordered remainder basis $1, S, \dots, S^{q-1}$. Set

$$D_{u,t}P = uP' + (\chi - t)P, \quad H_{u,t} = \mathbb{C}[S]/D_{u,t}\mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let $\text{red}_{u,t}$ be its unique representative of degree less than q . Let $A(t)$ be the multiplication matrix of S on $\mathbb{C}[S]/(\chi - t)$ in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case $q = 1$.

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{jb}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

56.1.1 Uniform estimates on the fixed unbounded contours

For a compact set $K \subset \mathbb{C}^{q+1}$ of the parameters $(c_0, \dots, c_{q-1}, t)$, put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies $S^d/(du) = -r^d/(d|u|)$. For $r \geq 1$ every other exponent has degree at most $d-1$, so

$$\operatorname{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in S : a derivative in c_a contributes $S^{a+1}/((a+1)u)$ and a derivative in t contributes $-S/u$. For any fixed derivative order its absolute value on the tail is bounded by $C r^L e^{-\alpha r^d}$, with C, L uniform on K . The required majorant is integrable, since substitution $x = \alpha r^d$ gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and t ; every parameter derivative may be taken under the integral. The same bound gives $e^{\Phi_t(S)/u} P(S) \rightarrow 0$ at the infinite endpoint of each ray, for every polynomial P . Consequently

$$\int_{\Gamma_j} e^{\Phi_t/u} D_{u,t} P dS = u \int_{\Gamma_j} d(e^{\Phi_t/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is $-e^{\Phi_t(0)/u} P(0) = -P(0)$; the coefficients $-1, +1$ of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

56.1.2 An exact trace comparison for twisted reduction

For $1 \leq m \leq q$ and $0 \leq b < q$, ordinary monic division gives unique polynomials $Q_{m,b}, R_{m,b}$ satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If $m+b < q$, take $Q_{m,b} = 0$. Otherwise $\deg Q_{m,b} = m+b-q \leq q-1$, and the differential itself gives

$$\operatorname{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below q . In every nonzero case,

$$\deg Q'_{m,b} \leq m+b-q-1 \leq b-1. \quad (\text{XD10})$$

Thus the coefficient of S^b in the derivative correction vanishes. Let $T_m(u, c, t)$ have column b equal to the coordinates of the left side of (XD9). The columns $R_{m,b}$ are exactly those of $A(t)^m$. Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\operatorname{Tr} T_m(u, c, t) = \operatorname{Tr} A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set $B_a = T_{a+1}/((a+1)u)$ for $0 \leq a < q$. Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \operatorname{Tr} B_a = \frac{\operatorname{Tr} A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in c, t for this fixed nonzero u .

56.1.3 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix A and integer $n \geq 1$, the product rule and cyclic invariance of trace give

$$d \text{Tr } A^n = \sum_{r=0}^{n-1} \text{Tr}(A^r (dA) A^{n-1-r}) = n \text{Tr}(A^{n-1} dA). \quad (\text{XD14})$$

Because $A(t)$ acts on the actual quotient by $\chi - t$, it satisfies $\chi(A(t)) - tI_q = 0$. Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt + \text{Tr}((\chi(A(t)) - tI_q) dA(t)) = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt. \quad (\text{XD15})$$

At $c = t = 0$ the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u) e^{il\theta_0} (\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d} \Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting $x = r^d/(d|u|)$ on each oriented ray in (XD3). In particular all $a_l(u)$ are positive real numbers. For $V_{jl} = \zeta^{jl} - 1$ one may include $j = 0$ in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here $1 \leq l, m < d$, so the two individual nonconstant sums vanish, and the first sum is $d\delta_{lm}$. The all-ones rank-one matrix has eigenvalue q on the all-ones vector and zero on its $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[\frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase: $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{il\theta_0}$.

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join $(0, 0)$ to (c, t) by (sc, st) , $0 \leq s \leq 1$. Equations (XD12) give $\Pi' = \Pi B(s)$, with continuous $B(s)$, and (XD15) gives $\text{Tr } B(s) = \mathcal{F}'(sc, st)/u$. For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = \det \Pi \text{Tr } B. \quad (\text{XD20})$$

Consequently $\det \Pi \exp(-\mathcal{F}/u)$ has derivative zero. At the starting point A is nilpotent and $\mathcal{F}(0, 0) = 0$, so its value is $\det \Pi^{\text{mon}}(u)$. This proves (XD19). Equation (XD18) proves its nonvanishing for every c, t , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \text{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by Π^{-1} followed by the specified degree-below- q section.

56.1.4 Exact determinant transport to the arithmetic representative

Retain the original positive Gram G_N on the cyclic coefficient space E , its determinant $V_N = \det G_N$, and the original maps $r_N : E \rightarrow \mathcal{B}^{\otimes k}$ and j_E on the admitted test-function domain with $j_E r_N = I_E$. Identify E with $H_{u,t}$ by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \rightarrow \mathbb{C}^q, \quad x \mapsto \Pi j_E x, \quad \mathbb{C}^q \rightarrow r_N(E), \quad y \mapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively $r_N j_E$ on $r_N(E)$ and $\Pi(j_E r_N) \Pi^{-1}$ on $\mathbb{C}^q$. On the full admitted domain the first observation has precisely $\ker j_E$ as its kernel; all original theta boundaries killed by j_E remain in that kernel.

With the explicitly fixed coordinate metric I_q on the target, put

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (\text{XD23})$$

These operators satisfy $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_\Pi > 0$. Taking determinants in (XD19) gives the exact identities

$$\det H_\Pi = C_q(u) \exp\left(2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\boxed{\log \det B_N^{\text{per}} = \log C_q(u) + 2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N.} \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same u, c, t for two admitted degrees $i \leq j$, subtraction proves

$$\boxed{\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j).} \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on V_i/V_j is inferred from this determinant identity.

57 Translation of the exponential comparison: full coefficient, period, metric, and finite-field maps

57.1 The original fibre and the auxiliary translation

The September 13 exponential comparison starts with the actual arithmetic polynomial and sum operator

$$\chi(S) = S^q + \sum_{b=0}^{q-1} c_b S^b, \quad q > 0, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S,$$

and the specified primitive

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{b=0}^{q-1} \frac{c_b S^{b+1}}{b+1}.$$

Its cohomology is the cokernel of $D_{u,t} = u \partial_S + \chi(S) - t$, with the omitted differential symbol understood as the coefficient identification with one-forms. All formulas below retain this original object at $a = 0$. The parameter $a \in \mathbb{C}$ defines an auxiliary comparison family; a translated polynomial is not asserted to be a new zeta zero packet.

Set

$$\chi_a(S) = \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \quad T_a P(X) = P(X + a). \quad (\text{DT.1})$$

Thus $\Phi_a(0) = 0$, and no additive constant in the potential is lost. Use the same substitution on coefficients of one-forms, with $dS = dX$.

Proposition 57.1 (The coefficient and connection maps). *The map T_a is an isomorphism of the two-term polynomial complexes for χ_a and χ , over $\mathbb{C}[u, t]$. It induces an isomorphism of their free rank- q cohomology modules. On the special fibre $(u, t) = (0, 0)$,*

$$T_a : E_a = \mathbb{C}[S]/(\chi_a) \xrightarrow{\sim} E, \quad T_a A_a T_a^{-1} = A + aI. \quad (\text{DT.2})$$

Every primary multiplicity and nilpotent Jordan block is retained. For $u \neq 0$, the descended parameter connections satisfy

$$T_a \nabla_t^a T_a^{-1} = \nabla_t - \frac{a}{u} I, \quad \nabla_t = \partial_t - \frac{A(t)}{u}, \quad (\text{DT.3})$$

where $A(t)$ is the original companion matrix for $\chi - t$.

Proof. Substitution gives, for every polynomial P ,

$$T_a(uP' + (\chi_a - t)P) = u\partial_X(T_a P) + (\chi(X) - t)T_a P.$$

The inverse is substitution by $-a$. A polynomial of degree n with leading coefficient $b \neq 0$ is sent by either differential to a polynomial of degree $n + q$ with the same leading coefficient. Successive subtraction of such images therefore gives a unique representative of degree less than q , over the coefficient ring. This proves the stated freeness and the base-change identification. On the special fibre, $T_a(SP) = (X + a)T_a P$, which proves DT.2. In particular, $(A_a - (\lambda + a)I)^m$ is conjugate to $(A - \lambda I)^m$ for every m , proving the multiplicity statement without discarding the nilpotent part. On arbitrary polynomial representatives, the connection is $\partial_t - S/u$. Since T_a is independent of t , its conjugate is $\partial_t - (X + a)/u$. The commutator with the differential vanishes, so this equality descends and yields DT.3. $\square$

In the fixed bases $1, S, \dots, S^{q-1}$ and $1, X, \dots, X^{q-1}$, the matrix of T_a is

$$(C_a)_{rb} = \begin{cases} \binom{b}{r} a^{b-r}, & 0 \leq r \leq b < q, \\ 0, & r > b, \end{cases} \quad C_a^{-1} = C_{-a}, \quad \det C_a = 1. \quad (\text{DT.4})$$

The inverse and composition rules follow by composing the literal substitutions. In particular, $C_a A_a(t) C_a^{-1} = A(t) + aI$.

57.2 Periods and the retained exponential factor

Fix $u \neq 0$, put $d = q + 1$, and use the original rays

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \Gamma_j = -\ell_0 + \ell_j, \quad 1 \leq j \leq q,$$

oriented as in the exponential comparison. Let $\Pi(u, t)$ denote its period matrix, whose (j, b) entry is $\int_{\Gamma_j} e^{(\Phi(X) - tX)/u} X^b dX$. Let Π_a use the identical labelled ray prescription for Φ_a in the S -plane.

Proposition 57.2 (Exact period translation). *Define*

$$h_a(t) = -\Phi(-a) - ta, \quad f_a(u, t) = \exp(h_a(t)/u). \quad (\text{DT.5})$$

Then

$$\Pi_a(u, t) = f_a(u, t) \Pi(u, t) C_a. \quad (\text{DT.6})$$

The map $f_a T_a$ intertwines the two parameter connections in DT.3.

Proof. The substitution $S = X + a$ gives the exact identity

$$\Phi_a(S) - tS = \Phi(X) - tX + h_a(t).$$

It changes the contour to $\Gamma_j - a$. This translated contour may be moved to Γ_j without changing the integral. Indeed, truncate the two rays at radius R , join their translated endpoints by segments of length $|a|$, and apply Cauchy's theorem to the resulting closed path. The entire integrand has no finite singularity. On each joining segment its leading exponent is $-R^d/(d|u|) + O(R^{d-1})$, uniformly along that segment; a polynomial factor grows at most polynomially. Its integral tends to zero. The two portions at the finite starting point have opposite orientation and cancel. Absolute convergence on the original and translated rays follows from the same estimate. Finally, $(X + a)^b$ is the column expansion DT.4. This proves DT.6 including the scalar and all phases.

Since $\partial_t f_a = -(a/u)f_a$, for every section v one has

$$\nabla_t(f_a T_a v) = f_a \left(\nabla_t T_a v - \frac{a}{u} T_a v \right) = f_a T_a \nabla_t^a v.$$

The scalar f_a is nonzero on $u \neq 0$; no extension of this exponential gauge to $u = 0$ is used. DT.2 is the separately proved coefficient specialization at that fibre. $\square$

57.3 The original positive forms and their control operators

The period matrices used below are invertible on the stated domain $u \neq 0$. Here is a complete verification needed for the determinant ratios. At the monomial potential, the substitution $x = r^d/(d|u|)$ in the absolutely convergent ray integral gives

$$\Pi_{j,b}^{\text{mon}} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} (\zeta^{j(b+1)} - 1), \quad \zeta = e^{2\pi i/d},$$

where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx > 0$ for $s > 0$. The column factors are all nonzero. A dependence among the remaining columns would give a polynomial $\sum_{l=1}^q b_l (Z^l - 1)$ of degree at most q , vanishing at all $q+1$ roots of unity, including 1. This polynomial is zero, so every $b_l = 0$.

Join the monomial coefficients to the given (c, t) by a straight path with real parameter $s \in [0, 1]$, keeping u fixed. On this compact coefficient path the leading negative ray term dominates all lower terms uniformly: for sufficiently large r the absolute value of any fixed coefficient derivative of the integrand is at most $C r^L \exp[-r^d/(2d|u|)]$. This is integrable by the same gamma substitution; hence differentiation under every contour integral is valid. A derivative in c_b multiplies the integrand by $S^{b+1}/((b+1)u)$, and one in t multiplies it by $-S/u$. The unique reduction proved above, together with the vanishing of $\int_{\Gamma_j} e^{(\Phi-tS)/u} D_{u,t} P dS$, thus gives $\Pi'(s) = \Pi(s)B(s)$ with a continuous matrix $B(s)$. That integral vanishes because it is u times an exact derivative; both infinite endpoint values are zero and the two finite endpoint values cancel with the specified contour signs. The determinant identity

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = (\det \Pi) \text{Tr} B$$

holds also at singular matrices. Therefore $\det \Pi(s) = \det \Pi(0) \exp(\int_0^s \text{Tr} B(v) dv) \neq 0$. This proves invertibility without removing repeated-root parameters. Equation DT.6 proves it for the translated matrix as well.

Let G_N be any one of the original canonical arithmetic Gram matrices, and retain its index N , cochain degree k , and control form $W_N = A^* G_N + G_N A - k G_N$. The exact pullback along DT.2 is

$$G_N^{[a]} = C_a^* G_N C_a. \quad (\text{DT.7})$$

This is a transported specified form. No claim identifies it with a newly computed canonical theta form for a translated polynomial.

Proposition 57.3 (Control and determinant transport). *For the transported form and the original value of k ,*

$$W_N^{[a]} = A_a^* G_N^{[a]} + G_N^{[a]} A_a - k G_N^{[a]} = C_a^* (W_N + (a + \bar{a}) G_N) C_a, \quad (\text{DT.8})$$

$$\text{Tr}((G_N^{[a]})^{-1} W_N^{[a]}) = \text{Tr}(G_N^{-1} W_N) + 2q \text{Re } a. \quad (\text{DT.9})$$

Write $H_\Pi = \Pi^* \Pi$ and $B_N^{\text{per}} = G_N^{-1} H_\Pi$. Then

$$B_N^{\text{per},[a]} := (G_N^{[a]})^{-1} \Pi_a^* \Pi_a = |f_a|^2 C_a^{-1} B_N^{\text{per}} C_a. \quad (\text{DT.10})$$

Consequently, for the same (u, t) and the same original indices,

$$\frac{\det B_j^{\text{per},[a]}}{\det B_i^{\text{per},[a]}} = \frac{\det B_j^{\text{per}}}{\det B_i^{\text{per}}} = \frac{\det G_i}{\det G_j}. \quad (\text{DT.11})$$

Proof. Insert $C_a A_a C_a^{-1} = A + aI$ into the first definition of DT.8. Taking the inverse Gram and using invariance of trace under similarity gives DT.9. Equation DT.6 gives $\Pi_a^* \Pi_a = |f_a|^2 C_a^* H_\Pi C_a$, proving DT.10. Its determinant contains the factor $|f_a|^{2q}$; this same factor occurs at both indices and cancels in their ratio. The remaining period Gram also cancels, giving DT.11. Thus the original endpoint determinant cost is retained by this exact comparison. $\square$

57.4 The finite-field comparison with its scalar character

Let $R_0 \subset \mathbb{C}$ be the finitely generated coefficient ring of the original exponential construction, and adjoin a . For a specified map $R_0[a] \rightarrow \mathbb{F}_Q$ of characteristic $p > d$, and specified $u_0 \in \mathbb{F}_Q^\times$, $t_0 \in \mathbb{F}_Q$, reduce every displayed coefficient through that map. Put

$$f(X) = \frac{\Phi(X) - t_0 X}{u_0}, \quad \gamma_a = \frac{-\Phi(-a) - t_0 a}{u_0} \in \mathbb{F}_Q.$$

Use the Artin–Schreier rank-one sheaf convention in which the geometric-Frobenius trace at $x \in \mathbb{F}_{Q^n}$ is $\psi(\text{Tr}_{\mathbb{F}_{Q^n}/\mathbb{F}_p} f(x))$, for a fixed nontrivial additive character ψ valued in $\overline{\mathbb{Q}_\ell}$, $\ell \neq p$.

Proposition 57.4 (Translation and the constant character line). *For the isomorphism $\sigma_a : X \mapsto S = X + a$,*

$$\sigma_a^* \mathcal{L}_\psi((\Phi_a - t_0 S)/u_0) \simeq \mathcal{L}_\psi(f) \otimes \mathcal{L}_\psi(\gamma_a). \quad (\text{DT.12})$$

Hence compactly supported cohomology in every degree is identified with that of $\mathcal{L}_\psi(f)$, tensored with the displayed constant character line. Each geometric-Frobenius eigenvalue is multiplied by

$$\zeta_a = \psi(\text{Tr}_{\mathbb{F}_Q/\mathbb{F}_p} \gamma_a). \quad (\text{DT.13})$$

This is a root of unity, so it preserves each complex absolute value and hence the purity weights of this finite-field realization.

Proof. The potential identity in the period proof holds after the specified coefficient reduction. The tensor identity of Artin–Schreier character sheaves can be obtained directly from their covers: if $y_1^p - y_1 = f$ and $y_2^p - y_2 = \gamma_a$, then $(y_1 + y_2)^p - (y_1 + y_2) = f + \gamma_a$. The addition map identifies the contracted product of the two $\mathbb{F}_p$ -torsors with the third torsor. On the character lines, the addition action becomes the product of their characters, which gives DT.12. Pullback by the base-defined isomorphism σ_a preserves compact supports and commutes with Frobenius. Tensoring with a line pulled back from the base tensors the compact-support complex with that line: this can be checked on a cochain resolution before taking cohomology. Our stated trace convention makes its Frobenius scalar DT.13. The additive group of $\mathbb{F}_p$ has exponent p , so $\zeta_a^p = 1$, proving the absolute-value assertion. $\square$

These formulas compose with the exponential-family application of Deligne, *La conjecture de Weil II*, §§3.7.2–3.7.4, to the degree- d polynomial with nonzero leading coefficient. The source for that application, including the boundary chart and its hypotheses, is retained in the exponential comparison note; the present proof establishes the additional translation maps themselves.

57.5 Specialization and split support

The polynomial chain isomorphism T_a is defined before either complex or finite-field realization. Its coefficient map, the specialization $u = t = 0$, and the inclusion of the original fibre $a = 0$ induce maps under the same split-zero functor G . For a ring map $\varphi : B \rightarrow B'$, this functor acts by

$$G(\varphi)(\tau) = \tau, \quad G(\varphi)(b^\bullet) = \varphi(b)^\bullet.$$

Therefore a supported coefficient specialized to zero maps to $e = 0^\bullet$, and an absent coefficient maps to τ . The square with the amplitude maps $p_B, p_{B'}$ commutes, since both composites take τ to zero and $b^\bullet$ to $\varphi(b)$. On each already specified receiving module fibre, the coefficient chain relation specializes to the relation $\chi P = 0$ without changing its label. The functorial map thus retains both the supported kernel and external absence.

The translation formulas specify exactly how the two realizations respond to the same coefficient operation: DT.2 shifts the arithmetic sum operator, DT.9 changes its metric trace by $2q \operatorname{Re} a$, DT.10 retains the analytic exponential scalar, and DT.13 supplies the finite-field character scalar. These are complete comparison maps. They do not identify a finite-field Frobenius with A , and they leave the original theta metric estimate as the same determinant quantity in DT.11. The original arithmetic construction remains the specified fibre $a = 0$ throughout.

58 Primary-source correction to the Deligne comparison

58.1 Primary-source correction to the Deligne comparison

This dated correction supersedes the earlier reported internal numbering mismatch in the final citation of Weil II, Section 3.7.4. The original delivered note, S20 page records, historical typed witnesses, and sealed intake remain preserved. The correction is based on the Zeta literature task's complete retained primary-source review of 13 September 2026.

The final citation on printed page 216 is **(I.8.11), with Roman I**. It refers to **Weil I, Section 8.11, printed page 306**, rather than to Weil II, Corollary 1.8.11. Weil I, Section 8.10, reduces to one variable and the monomial $Q = x^d$; Section 8.11 handles that case, and Sections 8.12–8.13 continue the Euler-characteristic calculation. The citation is resolved. This does not assert that Section 8.11 alone states the entire general purity theorem used in Weil II.

The printed page 216 itself uses S'_0 in the connectedness sentence. The preceding parameter sheaves are defined on the good open S_0 . That open is connected: it is a nonempty open subset of an irreducible affine parameter space, hence is irreducible. The printed prime remains part of the literal source typography while the application retains the explicitly defined good open.

The high-resolution printed display has the three terms

$$\begin{aligned} R^i f_! \mathcal{F}_0(\Psi Q_S) &= R^i \bar{f}_* j_! \mathcal{F}_0(\Psi Q_S) \\ &= R^i \bar{f}_* j_! \mathcal{F}_0(\Psi Q_S). \end{aligned}$$

The two final terms both have $\bar{f}_*$. The identity $\bar{f}_! = \bar{f}_*$ for the proper compactification explains a possible intermediate step, but replacing one of the printed stars by an exclamation mark would reconstruct that step. It would not transcribe a visible distinction. The repeated display and the prime are already present in the printed source.

The same primary review confirms the strict weight inequality < 2 in Lemma 3.2.10 on printed page 203. Section 3.2.13 explicitly uses $w_q(\alpha^2)$ and then divides the resulting bound by two. Both the strict inequality and the tensor square of the same eigenvalue remain essential to those deductions; losses in older typed witnesses must not enter the current proof.

On printed page 178, Corollary 1.8.12 requires a lisse ι -mixed sheaf on a connected scheme of finite type over the finite field, pure at one point, and concludes pointwise purity. Normality is not a hypothesis of that corollary; mixedness is. Constant rank alone does not supply its hypotheses.

The verified exponential theorem (3.7.2.3), printed pages 215–216, applies to the actual specialized potential $(\Phi - tS)/u$. Its degree is $d = q + 1$, its leading coefficient is the unit $1/(du)$ on $u \neq 0$, and the specified characteristic $p > d$ makes d prime to p . The leading projective zero locus in $\mathbb{P}^0$ is empty and smooth. Thus compact-support cohomology is concentrated in degree one, has dimension q , and is pure of weight one. Repeated critical roots do not obstruct these hypotheses. Lemma 3.7.3 supplies the lisse family through the actual compactification and its local-acyclicity calculation, including infinity. These are applications of the stated historical theorem. The exact source and period comparisons in the adjoining proofs retain their own operators and original theta metrics.

The retained primary review records visual reading of complete Weil II printed pages 178, 202, 203, 206, 215 and 216, including a higher-resolution check of page 216. Its bounded collaborator checked complete Weil I printed pages 305–306 and the cited case. This correction records their actual inspection scope; it does not claim a new audit of every foundational theorem in either paper. The evidence includes the complete written review, source URLs and exact source hashes, with no reproduced scan or rendered page.

Primary sources: [Deligne, *La conjecture de Weil II*, printed pages 178, 203, 215–216](#); [Deligne, *La conjecture de Weil I*, printed pages 305–306](#).

59 Which completion retains the arithmetic jets?

This section extends the Hilbert/jet calculation of the September 12 orthogonal-boundary note. All changes of topology are specified maps on the same arithmetic functions. The theta relation space, the operator $D = -x\partial_x$, and $g = 2\xi$ are retained.

59.1 Logarithmic coordinates and the original measure

Let $\mathcal{B}, V, \Theta$ have the definitions in Section 6. Put $\mathcal{H} = L^2(\mathbb{R}_+, dx)$. The map

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{L}^{-1}f(x) = x^{-1/2}f(\log x)$$

is unitary from $\mathcal{H}$ to $L^2(\mathbb{R}, dy)$, since $dx = e^y dy$. Direct differentiation gives

$$\mathcal{L}D\mathcal{L}^{-1} = \frac{1}{2} - \partial_y, \quad \mathcal{M}F(\rho) = \int_{\mathbb{R}} (\mathcal{L}F)(y)e^{(\rho-1/2)y} dy. \quad (\text{JT1})$$

On $\mathcal{H}$, D denotes the closed maximal distributional operator which $\mathcal{L}$ identifies with $1/2 - \partial_y$ on the Sobolev domain $H^1(\mathbb{R})$. For an integer $r \geq 0$, let $\mathcal{H}^{(r)}$ be the domain of the first r powers of D in $\mathcal{H}$, with norm $\|F\|_r^2 = \sum_{k=0}^r \|D^k F\|_{\mathcal{H}}^2$. The Mellin-line transform makes this the norm with density $w_r(t) = \sum_{k=0}^r |1/2 + it|^{2k}$ relative to $dt/(2\pi)$.

Theorem 59.1 (Finite differential graph norms and the complete jet graph). *Let Z be a nonempty finite packet of actual nontrivial zeros with full multiplicities, $E_Z = \mathbb{C}[t]/h_Z$, and $J_Z : \mathcal{B} \rightarrow E_Z$ the complete Mellin-jet map. For every finite r ,*

$$\overline{\text{span}\{D^j f_0 : j \geq 0\}}^{\mathcal{H}^{(r)}} = \mathcal{H}^{(r)}, \quad f_0 = \Theta\phi_*, \quad (\text{JT2})$$

$$\overline{\{(F, J_Z F) : F \in \mathcal{B}\}}^{\mathcal{H}^{(r)} \oplus E_Z} = \mathcal{H}^{(r)} \oplus E_Z. \quad (\text{JT3})$$

Thus J_Z is not closable in this norm. The completed theta-boundary graph is $\mathcal{H}^{(r)} \oplus 0$; its quotient in (JT3) is E_Z through $[(F, u)] \mapsto u$.

Proof. For each $0 < c < \pi/4$, the actual source gives $|g(1/2 + it)| \leq C_c e^{-c|t|}$. Indeed $e^{z/2} f_0(e^z)$ is holomorphic in $|\operatorname{Im} z| < \pi/4$: its Gaussian series and derivatives converge uniformly on compact subsets since $\operatorname{Re}(e^{2z}) > 0$. On smaller horizontal strips it decays faster than every exponential at the positive end. Poisson gives $J f_0 = f_0$, so analytic continuation gives $e^{z/2} f_0(e^z) = e^{-z/2} f_0(e^{-z})$; this controls the negative end. Shift the Fourier contour to height c or $-c$. The vertical integrals vanish by that decay, proving the bound.

The measure $d\mu_r = w_r(t)|g(1/2 + it)|^2 dt / (2\pi)$ has a positive exponential moment. If $u \in L^2(\mu_r)$ is orthogonal to every polynomial, $d\nu = \bar{u} d\mu_r$ is a finite complex measure with a smaller positive exponential moment, by Cauchy–Schwarz. Its Fourier transform is analytic in a strip, and every derivative at zero vanishes. Analytic continuation makes it zero there. Fourier uniqueness gives $\nu = 0$, hence $u = 0$. For this uniqueness step, convolve with Gaussian approximate identities: each convolution is in L^2 , since $\|\nu * \gamma_\epsilon\|_2 \leq \|\nu\|_{\text{TV}} \|\gamma_\epsilon\|_2$; its transform is zero, so Plancherel makes it zero, and weak convergence then gives $\nu = 0$. Polynomials are therefore dense in $L^2(\mu_r)$. Multiplication by $g(1/2 + it)$ is an isometry onto $L^2(w_r dt / (2\pi))$: divide almost everywhere, since this nonzero analytic function vanishes on a discrete, measure-zero set. The Mellin transform of $D^j f_0$ is $(1/2 + it)^j g(1/2 + it)$, proving (JT2).

Every theta polynomial has zero J_Z -jet. Thus the closure in (JT3) contains $(h, 0)$ for every h . The actual packet section $R_{\text{ref}} : E_Z \rightarrow \mathcal{B}$ satisfies $J_Z R_{\text{ref}} = 1$. Approximate $R_{\text{ref}} u$ in $\mathcal{H}^{(r)}$ by theta polynomials $p_n(D) f_0$. Then $(R_{\text{ref}} u - p_n(D) f_0, u) \rightarrow (0, u)$. This proves (JT3). For $u \neq 0$ it prevents closability. The boundary and quotient statements follow from the coordinate projections. $\square$

Corollary 59.2 (Differential on the completed graph). *The continuous map*

$$(D, A) : \mathcal{H}^{(r+1)} \oplus E_Z \longrightarrow \mathcal{H}^{(r)} \oplus E_Z, \quad (F, u) \mapsto (DF, Au) \quad (\text{JT4})$$

extends the original $(F, J_Z F) \mapsto (DF, J_Z DF)$. On the quotient by the boundary closure it is exactly $A = M_t$.

Proof. The norm definitions bound $D : \mathcal{H}^{(r+1)} \rightarrow \mathcal{H}^{(r)}$; the jet identity $J_Z D = A J_Z$ follows by retaining the complete Taylor polynomial of $s\mathcal{M}F(s)$. Apply (JT3). $\square$

Its split observation records $(e_{\mathcal{H}}, u^\bullet)$ when the Hilbert component tends to its supported zero and the jet survives. The projections compare this point to $e_{\mathcal{H}}$ and to $u^\bullet$; they do not identify it with (τ, τ) .

59.2 Exponential weights and exact jet norms

For $a \geq 0$, retain the original functions and introduce

$$\mathcal{H}_a = L^2(\mathbb{R}_+, e^{2a|\log x|} dx), \quad \|F\|_a^2 = \int_{\mathbb{R}} |\mathcal{L}F(y)|^2 e^{2a|y|} dy. \quad (\text{JT5})$$

The identity inclusions $\mathcal{B} \hookrightarrow \mathcal{H}_a \hookrightarrow \mathcal{H}_0$ are continuous. Source seminorms with powers exceeding $a + 1/2$ control both ends. Smooth functions compactly supported in log coordinates prove density.

Theorem 59.3 (Exact threshold, including the boundary). *Put $\ell_{\rho,k}(F) = (\mathcal{M}F)^{(k)}(\rho)/k!$ and $b = \operatorname{Re} \rho - 1/2$, with $k \geq 0$. The functional on $\mathcal{B}$ extends continuously to $\mathcal{H}_a$ if and only if $|b| < a$. Its exact squared norm is*

$$\|\ell_{\rho,k}\|^2 = \frac{(2k)!}{(k!)^2} \left(\frac{1}{[2(a-b)]^{2k+1}} + \frac{1}{[2(a+b)]^{2k+1}} \right). \quad (\text{JT6})$$

Proof. Inner products are conjugate-linear in their first argument. The Riesz vector in log coordinates must be

$$r_{\rho,k}(y) = \frac{y^k}{k!} e^{(\bar{\rho}-1/2)y} e^{-2a|y|}.$$

Its norm squared is $(k!)^{-2} \int y^{2k} e^{2by-2a|y|} dy$. The two half-lines converge exactly when $|b| < a$; their gamma integrals give (JT6). If $|b| \geq a$, testing any putative Riesz representative against compactly supported smooth functions forces it to be this same non-square-integrable density, a contradiction. The equality case is included: one end has no exponential decay. $\square$

Theorem 59.4 (Complete confluent Cauchy Gram). *For distinct centres ρ_i with $|\operatorname{Re} \rho_i - 1/2| < a$ and orders $0 \leq k < m_i$, let $J_a : \mathcal{H}_a \rightarrow \mathbb{C}^d$ be their Taylor-jet map. In the original Taylor coordinates its Gram $C_a = J_a J_a^*$ is*

$$(C_a)_{(i,k),(j,l)} = \frac{(k+l)!}{k!l!} \left(\frac{1}{(2a - z_{ij})^{k+l+1}} + \frac{(-1)^{k+l}}{(2a + z_{ij})^{k+l+1}} \right), \quad z_{ij} = \rho_i + \bar{\rho}_j - 1. \quad (\text{JT7})$$

It is positive definite. The exact right inverse $S_a = J_a^ C_a^{-1}$ satisfies*

$$J_a S_a = 1, \quad \|S_a u\|_a^2 = u^* C_a^{-1} u, \quad \|F\|_a^2 = \|(1 - S_a J_a)F\|_a^2 + (J_a F)^* C_a^{-1} (J_a F). \quad (\text{JT8})$$

Proof. The entry is $\langle r_{\rho_i,k}, r_{\rho_j,l} \rangle_a$; integrating $y^{k+l} e^{z_{ij}y-2a|y|} / (k!l!)$ on both half-lines proves (JT7) with its stated sign. The vectors are independent. After removing their common nonzero factor, a dependence is an exponential polynomial $\sum_i P_i(y) e^{\bar{\rho}_i y} = 0$. For each i , apply $\prod_{j \neq i} (\partial_y - \bar{\rho}_j)^{m_j}$. On the i -th polynomial factor every nonzero constant plus derivative is invertible on the finite-degree polynomial space; hence $P_i = 0$. Thus $C_a > 0$. Now $S_a J_a$ is the orthogonal projection onto $\operatorname{range} J_a^* = (\ker J_a)^\perp$. Multiplying the specified matrices proves all of (JT8). $\square$

For an actual packet the unchanged theta subspace lies in $\ker J_a$. Consequently there is an onto continuous quotient map

$$\mathcal{H}_a / \overline{\Theta V}^{\mathcal{H}_a} \longrightarrow \mathbb{C}^d, \quad [F] \mapsto J_a F, \quad \ker = (\ker J_a) / \overline{\Theta V}^{\mathcal{H}_a}. \quad (\text{JT9})$$

Its surjectivity follows from (JT8). Define the retained coordinate map

$$T : E_Z \longrightarrow \mathbb{C}^d, \quad T[p] = (p^{(k)}(\rho_i) / k!)_{i,k}.$$

Then $J_a|_{\mathcal{B}} = T J_Z$, and the exact representative difference is $R_{\text{ref}} T^{-1} - S_a : \mathbb{C}^d \rightarrow \ker J_a$. Here R_{ref} is the original source map; S_a is a Hilbert map, with no claim it lies in $\mathcal{B}$. In E_Z coordinates the observation Gram is $C_E = T^{-1} C_a (T^{-1})^*$ and the right inverse is $S_a T$.

59.3 The energy term carried by the weighted topology

Theorem 59.5 (Weighted boundary identity on the actual representatives). *On $\mathcal{B}$, with $(\Sigma H)(x) = \operatorname{sgn}(\log x) H(x)$,*

$$\langle DF, H \rangle_a + \langle F, DH \rangle_a - \langle F, H \rangle_a = 2a \langle F, \Sigma H \rangle_a. \quad (\text{JT10})$$

For an actual packet representative $R : E_Z \rightarrow \mathcal{B}$ with $DR - RA = B = \Theta k$, put $G_a = R^ R$, $T_a = R^* \Sigma R$. Adjoints use exactly $\mathcal{H}_a$. Then*

$$A^* G_a + G_a A - G_a = 2a T_a - (B^* R + R^* B), \quad -G_a \leq T_a \leq G_a. \quad (\text{JT11})$$

For $Au = \rho u \neq 0$,

$$2 \operatorname{Re} \rho - 1 = 2a \frac{u^* T_a u}{u^* G_a u} - 2 \operatorname{Re} \frac{\langle Ru, Bu \rangle_a}{\|Ru\|_a^2}. \quad (\text{JT12})$$

Proof. In log coordinates integrate $-(\bar{f}h)'e^{2a|y|}$ on the two half-lines. Source decay kills the terms at infinity; those at zero cancel because the weight is continuous there. The weak derivative of the weight is $2a \operatorname{sgn}(y)e^{2a|y|}$. This proves (JT10) with its positive sign. Substitute $DR = RA + B$ and move both boundary terms to obtain (JT11). Multiplication by sign is a self-adjoint contraction, proving the form bounds. Evaluation on u proves (JT12). $\square$

At $a = 0$, (JT11) is the source boundary-control formula. At $a > 0$, the topology making a jet continuous has its explicitly retained energy term $2aT_a$. Off-line jets cannot remain continuous as $a \downarrow 0$. For line-centred jets, all $a > 0$ work but (JT6) diverges as $a \downarrow 0$. These are exact comparisons on the same Mellin map and differential, without a weight theorem assumed.

59.4 The tesserine coordinate in the actual weight form

Retain $c_{-+} = 4(\beta - 1/2)$, $c_{--} = 4i\gamma$ for $\rho = \beta + i\gamma$. For $p > 1$ and $|\psi| = 1$,

$$z = \psi \exp\left(\frac{\log p}{4}(c_{-+} + c_{--})\right), \quad \frac{u^*(A^*G + GA - G)u}{u^*Gu} = \frac{c_{-+}}{2}, \quad |b| < a \iff |c_{-+}| < 4a. \quad (\text{JT13})$$

The first equality is exactly $z = \psi p^{\rho-1/2}$; the second is evaluation on $Au = \rho u$. Thus the original sector, local holonomy radius, arithmetic weight form and jet-continuity strip have specified comparison maps. The scalar evaluations do not replace the original matrix or its support-labelled source.

60 The current arithmetic route and its next calculation

The source chain now supplies an actual $\mathbb{C}[t]$ -module cohomology Q , a finite invariant packet inside that cohomology, a specified nonzero extension class, an original theta boundary for its representative defect, and an exact arithmetic formula for the relative weight form. The local continuation makes that last formula an explicit confluent interpolation problem. These constructions are logically prior to any purity conclusion. The subsequent continuation now supplies certified entries of the original arithmetic moment matrix, the exact finite-compression and infinite-tail resolvent comparison, and a convergence rate for the initial resolvent including its complete off-line jets.

60.1 A single chain of exact comparison maps

For each actual finite packet, retaining every full order, the maps proved in this reader are

$$E = \mathbb{C}[t]/h \xrightarrow{R_{\text{ref}}} \mathcal{B} \xrightarrow{q} Q, \quad J_Z R_{\text{ref}} = 1, \quad DR_{\text{ref}} - R_{\text{ref}}A = \Theta\phi_*\ell.$$

Every admitted polynomial correction keeps these same quotient and jet maps. At its distinguished orthogonal level let $B_{N,u}(t) = (p_{N,u}(t) - R_Z(u)(t))/h(t)$, a polynomial of degree at most m . The complete calculation is

$$\begin{aligned} d\nu_Z(t) &= \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad N = d + m, \quad C_N P = j_h P, \quad K_N = C_N (H_N^\nu)^{-1} C_N^*, \\ p_{N,u} &= (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad R_m u = R_{\text{ref}} u + \Theta(B_{N,u}(D)\phi_*), \\ G_m &= U_\varepsilon^* K_N^{-1} U_\varepsilon, \quad W_m = U_\varepsilon^* K_N^{-1} (A K_N + K_N A^* - K_N) K_N^{-1} U_\varepsilon. \end{aligned} \quad (\text{R1})$$

The middle line evaluates the specified polynomial at the original D ; every source vector lies in V .

Write $v_n = j_h q_n$ and $\kappa_n = \|q_n\|_{\nu_Z}^2$. The original arithmetic measure now determines the exact quantity

$$\epsilon_m = \frac{|\operatorname{Re}(v_N^* K_N^{-1} v_{N+1})| + \sqrt{(v_{N+1}^* K_N^{-1} v_{N+1})(v_N^* K_N^{-1} v_N) - (\operatorname{Im}(v_N^* K_N^{-1} v_{N+1}))^2}}{\kappa_N}. \quad (\text{R2})$$

The proofs in Section 9 establish (R1)–(R2) before passing to cohomology. For a reflection-stable packet the first absolute-value term is zero by the exact trace identity. The same sequence obeys

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}|. \quad (\text{R3})$$

Thus the rank-two boundary detects the combined displacement of the entire packet, not only its largest individual displacement. The estimate concerns the distinguished R_m sequence. The larger family of all polynomial source corrections has the separately proved metric image and sharp infimum in Section 8; those results do not identify its arbitrary member with R_m .

The full two-row source observation in Section 9 is injective after the original A -iterates are retained. It therefore offers a finite arithmetic calculation preserving every nilpotent jet: compute the source moments, the adjacent monic polynomials and their complete residue vectors, rather than discard a component whose scalar trace vanishes.

60.2 Topology, duality and the two sides of a weight estimate

The actual source moments now have the absolutely convergent incomplete-gamma expression and explicit double-tail bound of Section 7. Its certified six-by-six Gram matrix concerns the original $|2\xi|^2 dt/(2\pi)$ measure. The exact identity (AM.19) supplies the original monic recurrence norms through degree five, with interval enclosures.

For the $|g/h|^2$ kernel, Section 10 proves a universal polynomial quadrature formula, including centres that coincide with a quadrature node. On the precise nonintersection domain its alternate expression is

$$K_{n-1} = \kappa_{n-1}^{-1} q_n(A) S_n q_n(A)^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*, \quad r_n = F_n(A)c.$$

The exact Schur comparison (KR.15)–(KR.20) then joins the initial and terminal observations to the actual infinite-dimensional tail. Section 11 proves the explicit initial-resolvent rate (GC.8) and its complete finite-algebra bound (GC.23). The projection to that off-line domain retains the critical summand as its stated kernel. All factors in the Schur identity remain in converting the initial estimate to the terminal form that determines ϵ_m .

In particular, Section 13 identifies exactly the two contributions in a reflection-stable packet:

$$\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).$$

Its full-jet inequalities also quantify the required metric ratios along each retained nilpotent chain. These identities do not infer a relative estimate from absolute matrix shrinkage.

The theta-polynomial span is dense in every finite differential graph norm. The completed jet graph, however, is $\mathcal{H}^{(r)} \oplus E$, with quotient E , not a quotient of $\mathcal{H}^{(r)}$ in which the arithmetic component has vanished. This supplies the exact comparison between the shrinking Hilbert representative and its unchanged cohomological class.

The exponential observation norm supplies a second exact comparison. A Mellin jet at ρ is bounded precisely in the strip

$$|\operatorname{Re} \rho - \tfrac{1}{2}| < a,$$

with the factorial norm in Theorem 59.3. The original differential then has the energy term $2a \operatorname{sgn}(\log x)$ in Theorem 59.5. Consequently stronger topology and weight control are related by that explicit term; choosing a positive observation norm does not remove it.

For any packet Z , reflection gives the conjugate-linear ring isomorphism

$$\dagger_Z : E_Z \longrightarrow E_{Z^\dagger}, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad Z^\dagger = \{1 - \bar{\rho} : \rho \in Z\},$$

with all multiplicities retained. In fact $h_Z^\dagger = (-1)^d h_{Z^\dagger}$; this proves well-definedness, and applying the same operation twice is the identity. Restrict the following internal residue pairing to reflection-stable Z , so that this map is an involution on E_Z . The arithmetic residue form is the source form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger u}{g} ds.$$

Its Jacobian contraction is

$$\mathcal{R}_Z(f, g'u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho). \quad (\text{R4})$$

Indeed locally $g = z^{m_\rho} w(z)$ gives $g'/g = m_\rho/z + w'/w$, proving (R4) with multiplicities. The trace form pairs reflected centres and can retain nontrivial null directions from nilpotents. In the retained basis $1, t, \dots, t^{d-1}$, define

$$S_{ij} = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{(s^i)^\dagger s^j}{g(s)} ds, \quad 0 \leq i, j < d, \quad J_g = M_{j_h g'}.$$

Changing representatives by h makes the local quotient by g holomorphic, proving that this matrix represents the intrinsic form. By (R4) the trace pairing has matrix SJ_g . Its exact comparison to the positive source metric is therefore $G_m^{-1} SJ_g$, since $f^* G_m (G_m^{-1} SJ_g) u = f^* SJ_g u$. Its positivity is an arithmetic question about this operator, not a consequence of the positive Gram construction.

The finite-field two-sided weight theorem remains a source for methods, with its compact-support comparison, duality and tensor hypotheses. This reader has not constructed a realization meeting those geometric hypotheses for the entire arithmetic quotient. It has instead calculated the current characteristic-zero objects and their original maps, including the local-cohomology spectral extraction and meromorphic complement documented in the complete finite-comparison appendix.

60.3 Provenance and mathematical status

The coherent interpolation now has the full coefficient identity $J_{h,N} = U_{v_h} C_N$, with $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$. Equations (CT.3)–(CT.7) prove equality with (R1), including the original representative, and compatibility with the complete CRT injections between finite packets. The actual reflection preserves the unique minimum and proves the same-metric signed pairing of (CT.10) and (CT.16a).

On the original tensor complex, the complete calculation is

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, & A^{(k)} &= \sum_{j=1}^k A_{h,j}, & \mathcal{I}_{h,k,M} &= \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)), \\ R_M &= (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, & G_M &= (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}, \\ W_M &= -(Y_M^* C_M + C_M^* Y_M), & G_M - G_{M+1} &= Y_M^* Y_M, & Y_M, C_M &: E^{(k)} \longrightarrow \mathcal{E}_M, \\ \dim \mathcal{E}_M &= \binom{M+k}{k-1}. \end{aligned} \quad (\text{R5})$$

Section 14 proves the original ordered division, both cochain signs, every layer map, and the extra orthogonal projection required when tensor packets are enlarged. Its formula (CT.18) calculates the resulting change in both G and W .

For a reflection-stable packet, let p_M be either signed inertia of W_M , and let $\mathfrak{d}_M^{(k)}$ be its complete tensor departure from (CT.20). The same source layers give

$$\begin{aligned} \mathcal{D}_k &\leq p_M \epsilon_M, & p_M \epsilon_M^2 &\geq 2kd^{k-1} \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_M^{(k)}, \\ p_M &\leq \min\{\operatorname{rank} Y_M, \operatorname{rank} C_M, \lfloor d^k/2 \rfloor\}, & \mathcal{D}_k^2 &\leq \operatorname{Tr}(G_M^{-1}(G_M - G_{M+1})) \operatorname{Tr}(G_M^{-1} C_M^* C_M). \end{aligned} \quad (\text{R6})$$

The zero-inertia case is proved directly in (CT.25). The full ordered sum $\mathcal{D}_k$ and all multiplicities are retained in (CT.20). The joint density and exact graph closure in Section 15 show $G_M \downarrow 0$ for this actual total-degree sequence and identify every retained jet in the vertical fibre of (JD.11). The same section proves the coercive inverse-interpolation limit with its arithmetic unit, the supported-zero limit morphism, and the accumulated determinant cost (JD.14)–(JD.15). These formulas connect the relative layer energy to the exact rate of determinant loss through the same original matrices.

The September 11 support, Rees, comparison and audit packages are retained in the companion source directory. The current analytic baseline is the September 12 sequence: finite operator comparison (PR 8), chain descent (PR 9), global retraction (PR 10), homotopy control (PR 11), formal tau recovery (PR 12) orthogonal boundary control (PR 13), and coherent interpolation with joint tensor boundaries (PR 14). Part II includes the complete selected source proof notes with their exact revisions. Historical statements that a map remains unconstructed belong to their stated source dates; a later explicit construction in this reader or the source chain supersedes that particular statement.

The new local proofs are the full support-index equivalence, the corrected carrier and sector comparisons, the arithmetic confluent-kernel minimum and complete two-row source observation, the classification of source-metric images and aggregate control, and the graph/weighted-jet comparison. The next tranche adds the certified original theta-moment series and its full truncation bound, the confluent quadrature and finite/terminal/tail resolvent identities, the explicit initial convergence rate, and the metric-departure and nilpotent-chain identities. The recovered June 24 historical packet calculation is reproduced with a full-jet comparison in Section 2. Its exact A -invariant saturation is every off-axis local factor; the quotient, involutions, source-metric transport and split-zero endomorphism descent are proved there. The original transcript excerpt and its response locators are retained in the source package. This is an explicitly recorded historical recovery, with its new arithmetic saturation proof distinguished from the original permutation calculation. The subsequent historical reconstruction in Section 3 proves the inverse map from a complete local Hurwitz zero-cluster germ to every moment of its original compact probability measure, followed by the explicit Bernstein recovery of that measure. The monic cluster polynomial retains arbitrary zero multiplicity. The same section proves the complete Laurent period maps and their collision kernels. For two distinct primes, the joint Laurent map is surjective onto the full finite spectral algebra; Theorem 3.1 gives every primary kernel and the inverse original spectral coordinate, including its truncated logarithm on each nilpotent factor. The factorwise Split-Zero lifts and their exact comparison with the synchronized global lift are proved in (S4)–(S5). Their elementary algebraic and analytic steps are written here. No claim of global mathematical priority follows from finding or proving an identity in this research programme.

The finite validation records distinguish exact symbolic identities, rational calibration models, numerical quadratures and intentionally rejected negative controls. They do not count test methods as theorems or imply analytic certification from an L^2 numerical experiment. The supplied formalization records are also kept separate: successful exact-head CI metadata does not turn the unformalized analytic theta arguments into Lean proofs. No Lean run was performed for this continuation.

The current research calculation includes (R2) on the actual $|g/h|^2$ -moments, with its determinant update and two-row recurrence, together with the topology constants of Theorem 59.3. No uniform arithmetic estimate forcing (R2) to vanish has been established in this reader. RH, GRH and a global purity theorem are not conclusions of it. The exact maps and source constraints above specify the live route on which tandem work can now continue. The next estimate must propagate (GC.8) through (KR.16), retaining q_n^2 , κ_n , a_n , the terminal ratio F_n and the actual tail denominator.

The arithmetic moment formulas give an executable source for those finite data. The exact departure identity and the full-jet metric bounds specify what a resulting relative estimate would control in the original packet; no component is removed by changing observation.

The next step of that propagation is now calculated in Section 12. For $u_n = q_n(A)c$, $w_n = q_{n-1}(A)c$, and $M_n = \kappa_{n-1}^{-1}K_{n-1}^{-1}$, let $\beta_n = u_n^* M_n u_n$, and let P_n be the exact M_n -orthogonal projection with kernel $\mathbb{C}u_n$ when $\beta_n > 0$. Put $p_n = P_n w_n$, $z_n = P_n A u_n$. The universal step is

$$\begin{aligned} M_{n+1} &= a_n^{-1} \left(M_n - \frac{M_n u_n u_n^* M_n}{a_n + \beta_n} \right), & \frac{\det G_{m+1}}{\det G_m} &= \frac{a_n}{a_n + \beta_n}, \\ \Delta_{n+1} &= \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}, & \epsilon^{(n)} &= |g_Z| + \sqrt{g_Z^2 + \Delta_n}, \quad g_Z = \operatorname{Re} \operatorname{tr} A - d/2. \end{aligned} \quad (\text{R7})$$

Here $m = n - d - 1$. The zero-vector exceptional step is given completely in (KT.14), so critical centres and every polynomial coincidence remain in (R7)'s source calculation. The alternate terminal Gram follows by the exact isometry (KT.15) and recurrence (KT.16). The full scalar-to-matrix extraction uses the doubled off-line algebra and the explicit sequence (KT.23); at a reflected pair its matrix coefficient is exactly $(-1)^l F_n^{[k+l+1]}(\rho)$. Finally (KT.24)–(KT.25) recover these coefficients from the original initial error and tail, with both reciprocal series and all polynomial/norm factors displayed. The remaining energy estimate is therefore attached to the three explicit terms of (KT.11), including their actual cross pairing, on the same arithmetic moment sequence.

For a packet closed under both conjugation and reflection, the original real-coefficient calculation in (KT.28) closes this step in terms of consecutive determinant ratios. With the fixed section metric G_{-1} retained, put $r_m = \det G_m / \det G_{m-1}$ for $m \geq 0$. Then the actual arithmetic allowance obeys exactly

$$\epsilon_m^2 = a_{d+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{R8})$$

The coefficient reality and full reflection trace prove $\gamma_n = 0$ in the original coordinates; (KT.7)–(KT.8) then prove (R8), including zero-step cases. This places the relative quartet control directly on the same monic norm ratios and consecutive Gram determinants as the certified moment problem.

This also gives a direct arithmetic bound for the original tesserine violation channel. Retain the original quartet element

$$Q(\rho) = \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa, \quad \rho = \beta + i\gamma,$$

and its character $c_{-+}(\rho) = 2(2\beta - 1)$, as calculated in the carrier section. If $\gamma \neq 0$, $\beta \neq 1/2$, and Z is this entire four-point zero orbit with common full order μ , then $d = 4\mu$ and for every $m \geq 0$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = a_{4\mu+m+1} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{R9})$$

To prove every coefficient, conjugation and the functional equation preserve the full order, and the four real displacements are $\beta - 1/2, \beta - 1/2, -(\beta - 1/2), -(\beta - 1/2)$. The proved aggregate bound (R3) is therefore $\epsilon_m \geq 4\mu|\beta - 1/2| = \mu|2(2\beta - 1)|$. Squaring and using (R8) gives (R9). The sector morphism on the original supported quartet is precisely

$$G(\chi_{-+}) : G(\mathbb{C}[K_4]) \longrightarrow G(\mathbb{C}), \quad Q(\rho)^\bullet \longmapsto c_{-+}(\rho)^\bullet, \quad \tau \longmapsto \tau.$$

Here $\chi_{-+} = \operatorname{ev}_{-1,+1}$ is the original group character extended linearly; it preserves multiplication because the character does, and hence its displayed G -lift is a semiring morphism. At $\beta = 1/2$ this morphism takes the supported value e ; the same original quartet has phase channel $c_{--} = 4i\gamma$. The orbit then has two distinct points when $\gamma \neq 0$, so its degree is 2μ and (R8) applies with that exact degree; the left-hand side of the sector bound is zero. Thus the arithmetic volume comparison and the supported critical value use the same typed character map, with the phase coordinate retained.

60.4 The original theta norm and the positive arithmetic frontier

The two newly delivered source notes, reproduced completely in the appendices, admit the following exact common formula with (R8). Keep $u_j = (I - P_{L_{j-1}})D^j f_0$, $\mathfrak{h}_j = \|u_j\|^2$, and the original measure $|g|^2 dt/(2\pi)$. The polynomial division and original minimum in (KL.23)–(KL.27) prove, at every full packet,

$$\kappa_{d+j} = \mathfrak{h}_j r_j \quad (j \geq 0), \quad \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1} |c_m|^2 \quad (\text{R10})$$

for reflection-stable packets and $m \geq 0$, where $c_m = d_m G_m^{-1} d_{m+1}^*$ uses the original theta projection rows. The first identity includes $j = 0$ and the fixed section G_{-1} : the actual new-layer vector at that endpoint is exactly f_0 . The row dictionary (KL.30) retains its separate domain $m \geq 0$; its cross term is $\mathfrak{h}_{m+1} c_m = \overline{\gamma_n}$, with $n = d + m + 1$. Both signs and the conjugation are obtained by multiplying those rows against the original inverse metric, including both local-unit matrices. Equation (KL.25) substitutes the first identity in (R10) into the complete terminal energy, proving its second identity. Under conjugation closure the original coefficients are real, so γ_n is real; the reflection trace makes its real part zero. Thus $c_m = 0$ in this case, by precisely the same calculation, with no discarded cross term.

For the full quartet in (R9), this gives the proved chain

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{R11})$$

Indeed the order-preserving symmetries establish the hypotheses of (R10) and its vanishing cross term; the lower bound was proved with every multiplicity in (R9). The norms $\mathfrak{h}_j$ are independent of the packet. For $h \mid H$, the original polynomial map $P \mapsto (H/h)P$ is an isometry because $(g/H)(H/h)P = (g/h)P$. Its jets are the old complete jets and zero jets at each added centre. Its fixed representative and all its corrected representatives agree under this insertion, as proved in (KL.28). Thus the common theta sequence in (R11) connects the actual nested-packet maps; the packet dependence remains in the two displayed determinant ratios.

For the tensor frontier, retain the original full column maps $F, E_+ : \mathbb{C}^r \rightarrow E_h^{\otimes k}$ and its original norm diagonal Ω in (AP.4). With G_M the actual total-degree minimum, define

$$\begin{aligned} \mathbf{A} &= \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, & \mathbf{B} &= \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}, \\ \Gamma_{k,M} &= \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), & \pi_M &= \frac{\det G_{M+1}}{\det G_M}. \end{aligned} \quad (\text{R12})$$

All complete conjugation-and-reflection packets satisfy

$$\epsilon_{k,M}^2 = \lambda_{\max}(\mathbf{A}^{1/2} \mathbf{B} \mathbf{A}^{1/2}) \leq \Gamma_{k,M} \lambda_{\max}(\mathbf{A}) \leq \Gamma_{k,M} (\pi_M^{-1} - 1). \quad (\text{R13})$$

Here the square roots are the explicit metric isometries, with their inverses in (AP.12); the original matrices in (R12) remain part of the formula. The full proof in (AP.7)–(AP.21) starts from the actual linear reflection of the original Mellin source. It places the two frontiers in opposite orthogonal eigenspaces and gives $U^*V = 0$. The relative weight is consequently the two off-diagonal blocks UV^* and its adjoint. Its complete singular-vector decomposition identifies its norm squared with the positive operator in (R13), including all zero vectors. The explicit incidence row sum gives $VV^* \preceq \Gamma I$; the inverse-metric update gives $\pi_M^{-1} = \det(I + UU^*)$. Multiplication of its nonnegative eigenvalue factors proves the two inequalities. For $k = 1$, $M = d + m$, (R13), (R11), and (AP.23) calculate the same original tessarine bound through its norm, volume, and positive-operator presentations.

60.5 The symmetric summand, the actual relation layer, and support

Every factorial in the symmetric comparison is retained. The literal orbit-sum injection I , its diagonal $D_o = I^*I$, and $L = D_o^{-1}I^*$ in (AP.24) give $G_M^s = I^*G_M I$ and the exact identity $(G_M^s)^{-1} = LG_M^{-1}L^*$. The proof uses permutation invariance of the actual minimum, so that its metric commutes with the projection IL . Hence the original redundant frontier has symmetric maps $F_s = LF$, $E_s = LE_+$ with the unchanged Ω , and

$$\begin{aligned} A_s &= \Omega^{-1/2} F_s^* G_M^s F_s \Omega^{-1/2}, & B_s &= \Omega^{-1/2} E_s^* G_M^s E_s \Omega^{-1/2}, \\ (\epsilon_M^s)^2 &= \lambda_{\max}(A_s^{1/2} B_s A_s^{1/2}) \leq \Gamma_{k,M}((\pi_M^s)^{-1} - 1), & \pi_M^s &= \frac{\det G_{M+1}^s}{\det G_M^s}. \end{aligned} \quad (\text{R14})$$

Equations (AP.25)–(AP.36) prove all these maps, the full positive and zero spectra, the opposite reflection parities, and the orbit-factorial grouping. In particular the isometry $Q_M = G_M^{1/2} I (G_M^s)^{-1/2}$ satisfies $Q_M^* Q_M = I$ and $Q_M Q_M^* = IL$; it carries the symmetric relative operator to the invariant restriction of the original one, retaining the complementary operator on $\ker L$. The continuous cochain projector in (SS.1)–(SS.9) has exactly this cohomology image. Its finite packet version uses the original section and full jet, and all Koszul signs in that construction remain in its primitive.

The actual next relation layer underlying these matrices is proved in (KL.7)–(KL.20). Its complete graph frame, Gram, fixed-order division primitive, derivative image, and derivative cokernel are given there. The comparison with a homogeneous shell is the exact polynomial identity (KPC.7), with every lower coefficient in (KPC.6). Its full graph is the fibre product (KPC.12), and its two projection kernels are specified in (KPC.11). For two tensor coordinates the original ratio algebra embeds by (KPC.16), with exponent $\max_{\rho/\sigma=r}(m_\rho + m_\sigma - 1)$ at each coincident ratio. Its adjacent-shell generator action is (KPC.22a); the missing second-coordinate multiplier remains in that formula. The finite example (KPC.23)–(KPC.28a) calculates both original shell ranks, all nine node values, and the actual minimum, establishing the exact scope of that comparison.

For every linear map in these graph, kernel, quotient and cochain comparisons, the fixed-support lift at the end of the projective-comparison section maps (λ, v) to (λ, Tv) and maps external absence to absence. Its scalar and additive identities are proved there, including supported zero. For measured support, the historical branch chapter proves the separate full connection from ordinary branches to frame points through $\text{Idl}(L) \rightarrow L$, with its exact kernel congruence and descent criterion. Its L^∞ correspondence identifies every Boolean branch with a multiplicative state and its normal branches with actual measure atoms. The factor $\mu(a)^{-1} \int_a f d\mu$ is retained. These explicit maps specify the support observations available to the analytic construction and preserve the nonatomic support component alongside its atomic quotient.

60.6 The original derivative cost and both positive-half-line kernels

The same original derivative layer has the stronger arithmetic identity proved in (QD.6)–(QD.15). In its exact metric coordinates, $-\hat{Y}^* \hat{C} = UV^*$ occupies one off-diagonal reflection block. Thus its norm is the full relative allowance, and the original trace cost χ_M , operator cost ζ_M and largest update eigenvalue λ_M satisfy

$$\epsilon_{k,M}^2 \leq \frac{\lambda_M}{1 + \lambda_M} \zeta_M \leq \frac{\lambda_M}{1 + \lambda_M} \chi_M, \quad \epsilon_{1,M}^2 = (1 - \pi_M) \chi_M. \quad (\text{R15})$$

The first two inequalities follow from the norm of the product of the two actual layer maps and the eigenvalues of $UU^*(I + UU^*)^{-1}$. For one factor, the layer is one-dimensional, so that product is the outer product of its two original rows and its norm is exactly their norm product. Its rank-one determinant update is $\pi_M = (1 + \lambda_M)^{-1}$, proving the last equality. This argument includes either row zero. The full lower bounds $4k^2\delta_*^2$ and all finite accumulated volume constraints follow in (QD.10)–(QD.13), with their positive denominators established before division.

For one factor, put $z = s - 1/2$, $x_h = -z^2$, and retain the monic polynomial H defined by $h(s) = (-1)^D H(x_h)$, $d = 2D$. The exact algebra is $A_H \oplus zA_H = \mathbb{C}[x_h, z]/(H, z^2 + x_h)$, with its two local length- m_ρ branches explicitly inverted in (SP.8)–(SP.9). The original source is the Hilbert direct sum for $d\lambda_h = |V(x_h)|^2 dx_h / (2\pi\sqrt{x_h})$ and $x_h d\lambda_h$. Here $v_h(s) = V(x_h)$ is the entire original quotient and its full unit modulo H enters both kernels. With those kernels and their inverse metrics, (SP.23)–(SP.35) give

$$\begin{aligned} \epsilon_{N-d} &= \left\| G_e^{-1/2} (G_o - G_e X) G_o^{-1/2} \right\|, \\ K_e - X K_o &= \begin{cases} f_n e_n^* / p_n, & N = 2n, \\ -f_{n+1} e_n^* / u_n, & N = 2n + 1. \end{cases} \end{aligned} \quad (\text{R16})$$

The first line follows by substituting the exact multiplication matrix $\frac{1}{2}I + \begin{pmatrix} 0 & -X \\ I & 0 \end{pmatrix}$ in the original weight form. The second is its full inverse-metric recurrence: all interior adjacent terms cancel with their original norm ratios, and the last pair has the two displayed parity signs. The finite kernels have indices $(n, n-1)$ and (n, n) , respectively. Hence the norm is given by the full quadratic contractions in (SP.33), with at most one nonzero singular value. Every nilpotent jet remains inside f_n, e_n, X and both metrics. The exact derivative cost (SP.34) identifies (R16) with (R15), including the two zero-frontier cases.

60.7 The gamma determinant and the retained spectral-sum fibre

The original completed function has the exact factorization $g(1/2 + it) = \Gamma(1/4 + it/2)a(t)$, with $a(t) = -\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)$. The beta integral and generating-function calculation (TG.7)–(TG.14) prove the reference monic norms $\gamma_n = \sqrt{2} n! \Gamma(n + 1/2)$. Keeping the actual multiplier in the full positive arithmetic Gram matrix gives its determinant D_n , with $D_{-1} = 1$. Monic triangular Gram congruence proves $\mathfrak{h}_n = \gamma_n D_n / D_{n-1}$, so the actual quartet formula is exactly

$$\epsilon_m^2 = (m+1)(m + \tfrac{1}{2}) \frac{D_{m+1} D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}), \quad m \geq 0. \quad (\text{R17})$$

This follows by substituting the consecutive norm ratio into (R10), whose cross term vanishes by the proved quartet symmetry. The full packet determinants are independently expressed in (TG.17)–(TG.18), and the original confluent kernel is the complete Gram congruence (TG.20)–(TG.22). The explicit triangular maps to the parameter 3/4 family and their inverses retain every phase and degree. The theorem-hypothesis analysis (TG.25)–(TG.28) proves the obstruction from the infinitely many retained critical zeros on positive-measure intervals; it also supplies these exact Gram maps for continuing beyond that obstruction.

The web spectral-sum construction retains an additional relative coordinate $\Delta = (s_1 - s_2)^2$. Write $S = s_1 + s_2$, $Z = S - 1$ and $x_\Sigma = -Z^2$. Its complete reflection components, proved in Section 24, are

$$B_e = \frac{\mathbb{C}[x_\Sigma, \Delta]}{(A_0, x_\Sigma \Delta A_1)}, \quad B_o = \frac{\mathbb{C}[x_\Sigma, \Delta]}{(A_0, \Delta A_1)}. \quad (\text{R18})$$

Here the original polynomial identities are $H_0(S, \Delta) = A_0(x_\Sigma, \Delta)$ and $H_1(S, \Delta) = Z A_1(x_\Sigma, \Delta)$; all coefficients come from $h((S+r)/2) = H_0 + r H_1$, $r^2 = \Delta$. The map $\alpha : B_o \rightarrow B_e$ multiplies by x_Σ , and $\beta : B_e \rightarrow B_o$ is the quotient. They satisfy $\alpha\beta = x_\Sigma$ and $\beta\alpha = x_\Sigma$. The original generator is $S = I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}$. Their dimensions are $D(D+1)$ and D^2 ; both values follow by taking the symmetric square of the original two D -dimensional reflection eigenspaces. The exact maps, rather than an identification of these unequal spaces, enter the weight.

The full kernel and cokernel of $S - 1$ both recover A_H . Indeed $A_0(0, \Delta) = (-1)^D H(-\Delta/4) \neq 0$ makes α injective, while β is onto. Its cokernel map evaluates $x_\Sigma = 0$ and sends Δ to $-4x_h$. The kernel injection sends $q(x_h)$ to $\Delta A_1 q(-\Delta/4)$ in B_e , with the full polynomial ΔA_1 retained before taking the quotient. The ideal calculation and the dimension count prove both exact sequences in

the cited section, including all orders. On both defect modules the actual tensor unit acts as $V(x_h)^2$, because at $S = 1$ the two original arguments are s and $1 - s$. Thus the complete paired packet, its nilpotents and its arithmetic unit remain in the relative fibre.

The exact matrix-weight pushforward retains all powers of Δ , the Jacobian $dx_\Sigma/\sqrt{x_\Sigma}$, and both vector Hilbert measures. Its actual least-norm quotient gives the two metrics in (R18); the upper block of the original weight is $\beta^*G_o - G_e\alpha$, with its full rectangular singular spectrum. All maps preserve external absence and carry a vanishing amplitude to its supported zero through the displayed fixed-support lifts. These finite identities and complete source maps are established without an asymptotic estimate for (R17).

60.8 The sum derivative, its strict normal energy and the full conormal algebra

For the actual sum amplitude retain $a_h^{\text{SC}}(t) = v_h(1/2 + it)/\sqrt{2\pi}$, $w = |a_h^{\text{SC}}|^2$, $\mu_h = \int w$, and $I_h = 4 \int |(a_h^{\text{SC}})'|^2$. In the original coordinates $t_i = u/k + y_i$, $y_k = -\sum_{i < k} y_i$, the full product amplitude is $\Psi = \prod_i a_h^{\text{SC}}(t_i)$ and its squared fibre norm is $m = w^{*k}$. The exact Jacobian has absolute value one. Differentiating before integration gives $\partial_u \Psi = k^{-1} \sum_i a_i' \prod_{j \neq i} a_j$. The mixed integrated terms vanish because $\int \bar{a} a' = 0$. Orthogonal projection onto the actual product-amplitude line gives

$$\int_{\mathbb{R}} \frac{|m'|^2}{m} du + 4 \int_{\mathbb{R}} \|n(u)\|^2 du = \frac{\mu_h^{k-1}}{k} I_h, \quad n = \partial_u \Psi - \frac{m'}{2m} \Psi. \quad (\text{R19})$$

The proof in (FC.1)–(FC.4) retains the original mass and phase, and establishes every required integrability statement. For every finite independent relative polynomial frame, $j_u c = \Psi \sum_\alpha \theta_\alpha(i\mathbf{y}) c_\alpha$, the corresponding exact derivative is

$$\partial_u(j_u c) = j_u(c' + \Gamma_u c) + N_u c, \quad \Gamma = W^{-1} j^* j', \quad N = (I - j W^{-1} j^*) j', \quad j^* N = 0. \quad (\text{R20})$$

This identity follows by inserting the orthogonal projection onto $\text{im } j_u$, with $W = j^* j > 0$. Its full normal Gram is $\mathcal{N} = N^* N$.

Every finite packet leaves infinitely many critical-line zeros of the original amplitude. At such a zero T of complete order ℓ_T , the transverse expansion in (FC.25) has coefficients $c_T A_b \delta^{\ell_T}$ for Ψ and $(\ell_T/k) c_T A_b \delta^{\ell_T-1}$ for $\partial_u \Psi$, with $c_T A_b \neq 0$. Thus an equality $N_u c = 0$ forces the relative polynomial to vanish on the entire hyperplane $y_1 = T - u/k$ whenever the other factors can be chosen nonzero. For $k \geq 3$ this holds at every u ; for $k = 2$ it holds outside the countable sumset $\mathcal{Z}_h + \mathcal{Z}_h$. Infinitely many such hyperplanes force the finite polynomial to vanish. Consequently (FC.21)–(FC.27) prove

$$\begin{aligned} \mathcal{N}(u) &> 0 \text{ everywhere for } k \geq 3, \\ \mathcal{N}(u) &> 0 \text{ almost everywhere for } k = 2, \quad \int \frac{|m'|^2}{m} < \frac{\mu_h^{k-1}}{k} I_h. \end{aligned} \quad (\text{R21})$$

The full proof retains the orders, coefficient domains and the quartet two-factor zero fibre $N_0 = 0$.

The two-branch transformation is equally exact for a matrix weight which is not even. At $x > 0$, put $v = \sqrt{x}$ and

$$M = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}, \quad H = \frac{M^* \text{diag}(W(v), W(-v)) M}{2v}.$$

The original coefficient is $c(\pm v) = a(x) \pm ivb(x)$. Equations (FC.10)–(FC.18) prove the full unitary correspondence, the covariant derivative $\nabla_x = \partial_x + \mathcal{A}$, and its energy

$$\begin{aligned} \mathcal{A} &= M^{-1} \text{diag}\left(\frac{\Gamma(v)}{2v}, -\frac{\Gamma(-v)}{2v}\right) M + \text{diag}(0, I/(2x)), \\ \int \|\partial_u(jc)\|^2 du &= 4 \int_0^\infty x(\nabla_x q)^* H \nabla_x q dx + \int_0^\infty q^* Q q dx, \quad q = (a, b). \end{aligned} \quad (\text{R22})$$

Here $Q = M^* \text{diag}(\mathcal{N}(v), \mathcal{N}(-v))M/(2v)$. The retained Jacobian drift is $\mathcal{A}^*H + H\mathcal{A} = H' + H/(2x)$. The two endpoint traces $a + i\sqrt{x}b$ and $a - i\sqrt{x}b$ must agree at $x = 0$; this is the proved joining condition for the original derivative domain.

The arithmetic derivative has the unital algebra bridge

$$\Phi : P/I^2 \longrightarrow (P/I)[\epsilon]/(\epsilon^2), \quad [p] \longmapsto [p] + \epsilon[\partial_\Sigma p], \quad I = (h(s_1), \dots, h(s_k)), \quad \partial_\Sigma = k^{-1} \sum_i \partial_{s_i}. \quad (\text{R23})$$

The product rule modulo I proves multiplicativity. Its conormal restriction is the row $(a_i) \mapsto k^{-1} \sum_i h'_i a_i$ on $I/I^2 \cong (P/I)^k$. On an ordered local block of lengths m_i , multiplication by h'_i sends the input constant in variable i to $m_i c_i(0) z_i^{m_i-1}$ times that constant. Counting the monomials having at least one top exponent proves its full rank $d^k - (d-r_0)^k$, r_0 the number of distinct packet roots. The kernel and image of the algebra map, including its linear cokernel, are proved in (CW.6)–(CW.11).

For the original monic remainder section $j : P/I \rightarrow P/I^2$, $Z = S - k/2$ and $D_0 = \delta j$, the exact section defect and correction are

$$[Z, D_0] = -I + J, \quad J = \frac{1}{k} \sum_i M_{h'_i c_i} \ell_i = \frac{d}{k} \sum_i P_i, \quad P_i = \frac{1}{d} M_{h'_i c_i} \ell_i. \quad (\text{R24})$$

Here ℓ_i extracts the original s_i^{d-1} coefficient and c_i inserts the constant polynomial in that variable. The identity $\ell_i M_{h'_i c_i} = dI$ proves that the P_i are commuting idempotents. Their complete tensor decomposition gives eigenvalues dj/k with multiplicities $\binom{k}{j} (d-1)^{k-j}$ for $d > 1$, and $J = I$ for $d = 1$. Its trace is d^k and its rank $d^k - (d-1)^k$. The explicit factorization (CW.22) through the full conormal row proves the relationship between these two rank formulas.

The complete arithmetic unit $\hat{U} = [\prod_i v_h(s_i)]_{I^2}$ gives $j_U = \hat{U} j U^{-1}$, $D_U = M_{U^{-1} \delta \hat{U}} + M_U D_0 M_U^{-1}$, and $[Z, D_U] = -I + U J U^{-1}$. For the actual minimum section j_M , the difference $a_M = j_M - j_U$ is its admitted original relation in I/I^2 ; the correction is exactly $U J U^{-1} + [Z, \delta a_M]$, with trace d^k . These are the full source maps proved in (CW.23)–(CW.30). In one variable they also give

$$J_U u = (j_h g') \mathcal{R}_Z(1, u), \quad \mathcal{R}_Z(1, u) = \ell(U^{-1} u), \quad \mathcal{R}_Z(1, j_h g') = \sum_\rho m_\rho = d. \quad (\text{R25})$$

The partial-fraction proof in (CW.32)–(CW.36) retains the full poles before taking residues. Finally $G(\Phi)$ is the proved semiring lift; the derivative is its infinitesimal coefficient map. Thus every relation killed by the quotient retains its supported zero and its explicit derivative observation.

60.9 The full relative limit and the original arithmetic observation

Retain $k \geq 2$ and the nonempty original packet $d = \deg h \geq 1$ throughout this subsection. For two nested relative frames, keep the literal coefficient inclusion E_{LM} and the original matrices of (NT.1)–(NT.37). Their exact connection difference $K_{LM} = \Gamma_L E_{LM} - E_{LM} \Gamma_M$ obeys

$$N_M = j_L K_{LM} + N_L E_{LM}, \quad N_M^* N_M = K_{LM}^* W_L K_{LM} + E_{LM}^* N_L^* N_L E_{LM}. \quad (\text{R26})$$

The cross terms vanish because $j_L^* N_L = 0$. Thus enlarging a frame transfers the first displayed component into its tangent part, with its complete positive Gram retained. The arithmetic exponential bounds and weighted polynomial density proved in (ND.1)–(ND.34) give $\Pi_M \rightarrow I$ strongly on every original relative fibre. For each fixed original source polynomial P , (NN.1)–(NN.22) prove $r_M(P) = (I - \Pi_M) \partial_u(\Psi P) \rightarrow 0$ in the original global L^2 norm. Their normal-image transitions are the explicit contractions $r_M(P) \mapsto (I - \Pi_L) r_M(P) = r_L(P)$; their full derivative-graph transitions keep $\partial_u(\Psi P)$ unchanged and are isometric.

The same proofs retain the actual logarithmic arithmetic observation. The invariant relation $P_{\text{rel}} = \sum_i h(s_i)$ has zero ordinary jet but derivative jet $(U_k/k) \sum_i [h'(s_i)] \neq 0$. For $d = 1$ the sum is k ;

for $d > 1$ its distinct highest standard monomials have coefficients d , so it cannot vanish. Its normal seminorm tends to zero, while its algebraic observation remains this specified nonzero vector. The exact split quotient of that zero limiting seminorm sends every supported source to e and external absence to τ ; the nonzero observation therefore cannot factor through that quotient. Its original finite maps and full derivative graph remain. For the changing canonical target metrics the complete comparison is

$$G_M \preceq G_*, \quad B_M = G_*^{-1/2} G_M G_*^{-1/2}, \quad x^* G_M x = (G_*^{1/2} x)^* B_M (G_*^{1/2} x). \quad (\text{R27})$$

The fixed-source limit has not estimated this changing B_M . For $h = 1$ the finite coefficient quotient is zero, so the arithmetic observation above is its zero map; the original analytic source and the fixed-source density and derivative-graph maps remain as in (NN.22). The cyclic/full-frame derivative split and canonical-section difference are proved with all four metric terms in (CM.1)–(CM.19).

60.10 Every relation depth and its full cyclic complement

For the same monic h and $I = (h(s_1), \dots, h(s_k))$, the complete depth calculation (CC.1)–(CC.52) proves

$$\begin{aligned} I^r \cap \mathbb{C}[S] &= (\chi_r), \quad \chi_r(T) = \prod_{\lambda} (T - \lambda)^{\ell_{\lambda,r}}, \\ \ell_{\lambda,r} &= \max_{\sum_i \rho_i = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) + (r - 1) \max_i m_{\rho_i} \right), \\ C_r &= \mathbb{C}[T]/\chi_r \xrightarrow{\eta_r = U_r \alpha_r} A_r = (P/I^r)^{\mathfrak{S}_k}, \quad \Pi_r \eta_r = I, \quad A_r \cong C_r \oplus \ker \Pi_r. \end{aligned} \quad (\text{R28})$$

The coefficient-extraction formula (CC.13) constructs this retraction, and (CC.10) counts every invariant orbit before taking its trace. The quotient derivative $\delta_r : P/I^{r+1} \rightarrow P/I^r$ and its scalar counterpart satisfy the commuting squares (CC.17). Their full unit term has the proved decomposition $\delta_r \eta_{r+1} = \eta_r (\partial_r + M_{b_r} \rho_r) + \mathcal{N}_r \rho_r$ of (CC.23); its complementary quotient map vanishes exactly when $U_r^{-1} \delta_r U_{r+1} \in \alpha_r C_r$. The first-jet maps, their full kernels, the product-Jacobian factorization, and the exact two-factor parity ideals at every depth are proved in (CC.29)–(CC.51b). An arbitrary weighted multiplier has all four blocks (CC.45); its lower diagonal block is a genuine quotient operator precisely when the lower left block is zero. Thus the earlier weighted-trace language has the exact operator interpretation (CC.45)–(CC.49), with no complementary direction lost.

60.11 Exterior amplification and the arithmetic volume to estimate

For the actual cyclic quotient, keep $A = M_S$, its canonical G_N and $H = G_N^{-1}(A^* G_N + G_N A - k G_N)$. The complete exterior source note proves that H has spectrum $\epsilon, -\epsilon, 0, \dots, 0$. The additive exterior action $A^{[p]}$ and the original unscaled representative $R_N^{\otimes p} \text{Alt}_p$ have source Gram $p!(\det(G_N)_{I,J})_{I,J}$; their control has the same literal factorial and allowance ϵ . On the determinant of the full positive generalized spectral subspace this gives

$$L_{h,k} = \sum_{\Re \lambda > k/2} \ell_{\lambda}(2\Re \lambda - k) \leq \epsilon_{h,k,N}^{\text{cyc}}. \quad (\text{R29})$$

The full cochain primitive carries the sign $(-1)^{k(j-1)}$ in block j , which the tensor differential repeats; their product is one. Hence this exterior comparison uses the original theta boundary. The cochain idempotent splits full cohomology, while the ordinary trace in (R29) is on the specified finite arithmetic packet.

For the exact complete nonreal quartet polynomial of (E33), with common order m , let $\ell_k = 1 + k(m-1)$. The counted sum grid yields

$$q = \ell_k(k+1)^2, \quad L_{h,k} = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad N \geq q-1. \quad (\text{R30})$$

The explicit four occupation counts in (E33)–(E34) prove this formula without a noncollision assumption on ordered tuples. In a larger packet it remains a lower bound and the threshold is its actual $\deg \chi_{h,k} - 1$. The equality analysis (E1)–(E36) further proves, in the original metric, with positive spectral projector Q , orthogonal range projection P , and $T = PA(1 - P)$,

$$\begin{aligned} \epsilon^2 - L^2 &= \|T\|_{\text{HS}, G_N}^2 + \epsilon^2 \{\det M + \det(I_2 - M)\}, \\ BX - XD &= T, \quad X = (Q - P)|_{\ker P}, \quad X = \int_0^\infty e^{-tB} T e^{tD} dt. \end{aligned} \quad (\text{R31})$$

Here M is the explicit positive compression in (E5)–(E7), and B, D are the original diagonal blocks of A in the orthogonal sum. The finite nilpotent exponential formulas (E15) prove convergence of the integral and that this Sylvester map is invertible. For $L > 0$, equality $L = \epsilon$ is equivalent to $P = Q$, or to $T = 0$. A retained critical Jordan block forces strict inequality. The canonical Gaussian equality calibration (E35)–(E36) proves why positivity of a moment measure alone supplies no further strictness.

Finally the arithmetic matrices themselves give the exact calculation (CV.1)–(CV.12). Set $D_N = \det K_N$ and $\tilde{K}_N = K_{N-1} + b_{N+1}b_{N+1}^*/\omega_{N+1}$. Then

$$\begin{aligned} \epsilon^2 &= \frac{\omega_{N+1}}{\omega_N D_N} \Delta_N, \quad \Delta_N = D_{N+1} + D_{N-1} - D_N - \det \tilde{K}_N, \\ \Delta_N &= \sum_{\substack{I \subset \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det[b_I, b_N, b_{N+1}]|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \end{aligned} \quad (\text{R32})$$

Determinant multilinearity proves the positive sum, including the singular first-degree boundary, and (CV.10a) constructs its original source vector with norm squared Δ_N and tensor norm squared $q! \Delta_N$. At $N = q - 1$, (CV.11)–(CV.12) give the same control as $\sum_{j < q-1} |\gamma_j|^2 \omega_j / \omega_{q-1}$ in the unchanged orthogonal expansion of χ . Thus the current quantitative expression is $\omega_{N+1} \Delta_N / (k^6 \omega_N D_N)$ along admitted growing degrees. Its limit has not been estimated. The complete lower bound (R30), the positive exterior data (R32), and the changing metric (R27) are retained together for that calculation.

60.12 The original Toda volumes and the complete four-term loss

The Toda continuation now calculates this same quantity from the original source and relation volumes. In the notation proved in (TVB.1)–(TVB.38c), use $k \geq 1$, a nonempty dagger-stable packet with its full zero orders, $q \geq 1$, real $|\theta| < \pi/2$, and $N \geq q$, except at the separately stated $N = q - 1$ endpoint, throughout (R33)–(R34). We retain $c = k/2$, $Z(\theta) = M_h(\theta)^k$ and $Z_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta)Z$. The Hankel determinants $\mathfrak{D}_n = \det[Z^{(a+b)}]_{a,b < n}$ and $\mathfrak{B}_m = \det[Z_\chi^{(a+b)}]_{a,b < m}$ are the actual source and relation Gram determinants. With $m = N - q + 1$ their exact determinant-line map and the rank-one update give

$$\begin{aligned} V_N &= \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}, \quad \delta_N = \frac{V_N}{V_{N-1}}, \quad \alpha_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \\ \psi_N &= \frac{\text{Tr } A - cq}{i} - \partial_\theta \log V_N, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i\psi_N, \\ \epsilon_N^2 &= \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2 = \alpha_{N+1} \frac{\Delta_N}{\det K_N}, \quad N \geq q. \end{aligned} \quad (\text{R33})$$

Every equality has its complete source, phase and determinant proof in the cited bridge. At $N = q - 1$, the exact replacement remains $\epsilon_{q-1}^2 = (\nu_0 - \omega_q)/\omega_{q-1} - \psi_{q-1}^2$, with $\nu_0 = \|\chi\|^2$; no preceding inverse occurs. The explicit conjugate-linear map (TVB.38b) sends the lower orthogonal projection of χ to the actual positive-minor exterior source vector, with the full squared-norm factor $\det K_{q-1}/\omega_q$.

The two scalar Toda curvatures have full source maps. For $X = (S - c)/i$, the actual least-norm lift satisfies $R'_N = -I_N$, where $I_N = P_{D_N} X R_N$ and $O_N = (1 - P_N) X R_N$ in the current tilted pairing, where P_N is the tilted source projection onto $\mathcal{P}_N$. With $\beta_m = \nu_m/\nu_{m-1}$ for $m \geq 1$, the proofs of (TVB.17)–(TVB.21) give

$$\begin{aligned} G''_N - G'_N G_N^{-1} G'_N &= O_N^* O_N - I_N^* I_N, \\ \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1}(1 - \delta_N), \quad \text{Tr}(G_N^{-1} I_N^* I_N) = \beta_m - \alpha_{N+1} \delta_N. \end{aligned} \quad (\text{R34})$$

The common subtracted term is present in both positive costs. At the first admitted degree the two costs are α_q and zero, with no negative-index relation norm. Every derivative boundary has the literal cochain primitive (TVB.9), including its tensor signs.

The coordinating Zeta contribution joins the two volume losses with the original spectral coupling. At $\theta = 0$, $N \geq q$ and $L > 0$, let $\mathbf{P}$ be the G_N -orthogonal projection onto the full positive spectral subspace, $\mathbf{C} = \mathbf{P}A(1 - \mathbf{P})$ its restricted coupling, and $M = E^\# \mathbf{P}E$, where E embeds the two G_N -unit eigenvectors of $A^\# + A - kI$ at $\pm \epsilon_N$. Set $v_N = -\log \delta_N$, $s_N = (v_N + v_{N+1})/2$ and $t_N = (v_{N+1} - v_N)/2$. The complete derivation in (TVB.39)–(TVB.43), using the same metric on both sides, proves

$$\begin{aligned} \alpha_{N+1} \sinh^2 s_N - L^2 &= \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 + \psi_N^2 \\ &\quad + \|\mathbf{C}\|_{\text{HS}, G_N}^2 + \epsilon_N^2 \{\det M + \det(I_2 - M)\}. \end{aligned} \quad (\text{R35})$$

All four terms are nonnegative. In particular the two-step volume lower bound retains $L^2 + \psi_N^2 + \|\mathbf{C}\|_{\text{HS}, G_N}^2$ inside its inverse-hyperbolic-sine expression. The exact Sylvester inverse maps $\mathbf{C}$ to the difference between spectral and orthogonal projectors; its full original nilpotent factors and singular value remain specified. The complete supplied join is also reproduced as a source appendix.

60.13 Certified theta input and a calculated comparison of the two gaps

The theta continuation proves the sector construction and the exact maps

$$h(D)F_h = f_0, \quad w_1(t) = |h(1/2 + it)|^2 w_h(t), \quad M_h(\theta) = 2 \int_1^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (\text{R36})$$

The absolute-square integral is for real $|\theta| < \pi/2$; its holomorphic strip continuation is proved in the theta input chapter. The dagger sign, complex Fourier phase, endpoint Jacobian and source factor $g = 2\xi$ are all retained in (TI.1)–(TI.16). The exact incomplete-gamma double series and its explicit infinite-square-tail bounds (TI.19)–(TI.30) supply rigorous interval enclosures for $M_1(0)$, $M_1(1/10)$, $M_1''(0)$ and $\alpha_1^{(k)}/k = M_1''(0)/M_1(0)$. Each displayed rational interval in that proof has width $3 \cdot 10^{-40}$. The certificate adds the proved infinite tail to outward-rounded ball arithmetic; it does not infer an enclosure from precision agreement. The source norm is positive for $h = 1$ while its arithmetic cyclic quotient is zero. The density identity in (R36), and the globally constructed Mellin division, specify the exact passage to packet input without division of a measured zero value by zero. No packet determinant estimate follows from these four seed intervals alone. For the empty packet $h = 1$, the arithmetic quotient is zero and $V_N = 1$ by the empty-determinant convention. For $q = 1$, $\epsilon_N = 0$ at every admitted degree, whereas the upper Toda expression is strictly positive for $N \geq 1$, as proved in the bridge.

Finally (TC1)–(TC13) calculate both gaps in the original Gaussian calibration with mass μ , variance σ^2 , $c = 1/2$ and $\chi = (S - c)^2 - \sigma^2$. The source and quotient coordinate maps are fully displayed there. They give

$$\begin{aligned} (V_1, V_2, V_3) &= \mu^2 \sigma^2 (1, 1/3, 1/11), \quad \epsilon_1^2 = L_{\text{spec}}^2 = 4\sigma^2, \quad \epsilon_2^2 = 16\sigma^2/3, \\ \alpha_3 \frac{(\mathcal{R}_2 - 1)^2}{4\mathcal{R}_2} - \epsilon_2^2 &= 49\sigma^2/33, \quad \epsilon_2^2 - L_{\text{spec}}^2 = L_{\text{spec}}^2 \|Q - P\|_{\text{HS}, G_2}^2 = 4\sigma^2/3. \end{aligned} \quad (\text{R37})$$

The first gap is precisely the Toda imbalance square; the second has the explicitly calculated spectral/orthogonal projector map. This calibration uses declared polynomial coefficients. The actual growing packet calculation continues with (R33)–(R36), retaining its arithmetic measure and all four terms of (R35).

60.14 Gamma convolution, certified higher input, and the complete quotient comparison

Result R38: an exact analytic and finite coefficient interface. Keep the original $w_h(t) = |g(1/2 + it)/h(1/2 + it)|^2/(2\pi)$ and $r_\lambda(t) = |\Gamma(\lambda + it/2)|^2/(2\pi)$, with $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda)$. The full Gamma source appendix and the coefficient-join proof construct

$$r_{\lambda,k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad m_{h,k} = w_h^{*k} = r_{\lambda,k} C_\Sigma(B_{h,\lambda}^{\otimes k}), \quad B_{h,\lambda} = |v_h/\Gamma(\lambda + it/2)|^2.$$

The Hilbert maps $U_\Sigma f = f(\sum t_i)$ and its conditional adjoint C_Σ have the inverse paired decomposition $F \leftrightarrow (C_\Sigma F, (I - U_\Sigma C_\Sigma)F)$. All masses and the relative vector remain. With $\alpha = 2\lambda$,

$$c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_\lambda j! (\alpha)_j}, \quad \mathcal{A}_h(z) = \sum_{j \geq 0} (\alpha)_j c_{h,j} z^j, \quad d_{h,k,n} = [z^n] \mathcal{A}_h(z)^k,$$

the exact finite moment is $\int b_n^{(k\lambda)} m_{h,k} = c_\lambda^k n! d_{h,k,n}$. The original degree- N source and relation Grams at theta zero use $c_{h,0}, \dots, c_{h,2N}$; their derivative of order r uses the list through $2N+r$, through the fully written original- S polynomial test. The analytic join supplied by the Zeta task and proved in the coefficient chapter is

$$\mathcal{A}_h(z) = c_\lambda^{-1} (1 + z^2)^{-\lambda} M_h(\arctan z), \quad |z| < 1, \quad M_h(\theta) = \int e^{\theta t} w_h(t) dt.$$

The specified branches at zero and exponential domination justify all coefficient derivatives. Its inverse uses $M_h(\theta) = c_\lambda \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta)$ for $|\operatorname{Re} \theta| < \pi/4$, with the analytic branches continued from zero. The complete source appendix is preserved alongside the explicit cochain correction; applying $\mathcal{V}$ to a section difference gives the boundary, and (GP.26)–(GP.27) construct its primitive.

Result R39: certified original moments and signed coefficients. Equations (GS.1)–(GS.40) prove the original theta derivative recurrence

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y),$$

$$\mu_{2r} = 2 \int_1^\infty \left| \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2} \right|^2 dx.$$

Their fully specified positive square tail (GS.18) bounds the omitted infinite integer sum at every fixed order. At order two and three, the rigorous rational enclosures are

$$\begin{aligned} \mu_4 &\in [383.2743417173642425544072374791308324300288, \\ &\quad 383.2743417173642425544072374791308324300291], \\ \mu_6 &\in [17946.5273016052076235667646919834204772848931, \\ &\quad 17946.5273016052076235667646919834204772848934]. \end{aligned}$$

The original reference $c_{13/4} = 10395\sqrt{\pi}/(2048\sqrt{2})$ is enclosed separately from the actual mass μ_0 . The complete invertible moment/coefficient map (GS.34)–(GS.35) certifies c_0, c_2, c_4, c_6 and proves $c_4 < 0, c_6 < 0$. Its tensor coefficient of degree six retains all three terms

$$d_{k,6} = k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3, \quad a_j = (13/2)_j c_j.$$

The accompanying source-cost continuation (GS.41)–(GS.50) proves the first four monic norms h_0, h_1, h_2, h_3 and all three positive ratios $a_j^{(k)} = h_j/h_{j-1}$ from these same moments. In particular, with $\nu = \mu_2/\mu_0$ and the original sum moments $M_j(k)$,

$$a_1^{(k)} = k\nu, \quad a_2^{(k)} = \frac{\mu_4}{\mu_2} + (2k-3)\nu, \quad a_3^{(k)} = \frac{M_6 - M_4^2/M_2}{M_4 - M_2^2/M_0}.$$

The exact polynomial isometry $P(S) \mapsto P(k/2 + iu)$ and its monic phase i^j preserve these norms and prove the original spectral recurrence $SP_j = P_{j+1} + (k/2)P_j - a_j^{(k)}P_{j-1}$. For the source determinant corrections, put

$$Q_N = \frac{X_{N+2}X_N}{X_{N+1}^2}, \quad Q_0 = \frac{2\nu}{13}, \quad Q_1(k) = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \frac{\mu_4}{\mu_2} - \frac{43\nu}{13}.$$

The certified interval for ρ is strictly negative; hence $0 < Q_1(1) \leq Q_1(k) < Q_0$ and $Q_1(k+1) - Q_1(k) = -13\rho/((13k+2)(13k+15)) > 0$. These are exact statements at every tensor degree for the specified source polynomial degrees. The analytic seed has the original zero arithmetic quotient $\mathbb{C}[S]/(1)$ throughout.

Result R40: finite error propagation to the actual consecutive ratio. The exact polynomial coefficient power and test expansion in (GMT.12)–(GMT.17) propagate finite coefficient enclosures into Hermitian source envelopes. The full boundary-resolvent formula (GMT.19)–(GMT.20) carries those actual forms to their least lifts, including the signed quadratic loss and its original theta primitive. Equations (GMT.29a)–(GMT.29b) also certify every coefficient of the least lift and the full representative loss $\widehat{R}^*H\widehat{R} - G(H)$ by an explicit positive q -dimensional rational upper form. The positive finite envelopes induce quotient-dimensional determinant enclosures by minimizing on the same affine space of representatives. For $N \geq q$, retain the exact two-degree source block

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^*H_-^{-1}C_D, \quad F_D = J_D - J_-H_-^{-1}C_D, \quad Z_D = F_D^*G_-F_D.$$

Equations (GMT.22)–(GMT.27) prove, in the original coordinates,

$$\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W}, \quad a_{N+1} = \frac{\det W}{W_{11}^2}.$$

Applying the same maps to the Gamma reference gives the exact relative correction T_{N-1}/T_{N+1} as the quotient of these two positive ratios. The fully proved auxiliary bounded multiplier $\min(1 + t^2/4, 10001)$ with $\lambda = 1$, $k = 2$, and the declared $\chi(S) = (S-1)^2 - 2$ has $T_1/T_3 < 1$. Its exact finite tail, rational envelopes, relation matrices, and source/quotient maps are (GMT.32)–(GMT.42). Thus positivity and boundedness alone do not imply relative correction monotonicity. This explicit counterexample is connected to the arithmetic problem by those same coefficient, least-lift, and determinant maps; its auxiliary weight is not assigned the value of an actual zeta packet weight.

Result R41: the complete phase and a strict relative contribution. For the actual complex product $a = \prod_i A_{h,\lambda}(t_i)$, put $a_k = C_{\Sigma}a$, $b_k = C_{\Sigma}|a|^2$, and $d_k = b_k - |a_k|^2$. Equations (GP.6)–(GP.11) prove the exact unitary transport of the arithmetic sum projection and its fibre defect. Equations (GP.12)–(GP.13) use the original quartet tail, with its full constant $25 \cdot 2^{2d}/\sqrt{\pi}$, to prove $d_k(u) > 0$ at every real u for $\lambda = 1/4$, $d \geq 4$, and $k \geq 2$. The full complex coefficient transform and its Parseval identity preserve the factors c_{λ}^k and $n!$. The source matrices and the original quotient satisfy

$$\begin{aligned} M_h &= M_{\text{coh}} + M_{\text{rel}}, & R_h &= R_{\text{coh}} - B_{\chi}K, \\ K &= (B_{\chi}^*M_hB_{\chi})^{-1}B_{\chi}^*M_{\text{rel}}R_{\text{coh}}, \\ G_h &= G_{\text{coh}} + K^*B_{\chi}^*M_{\text{coh}}B_{\chi}K + R_h^*M_{\text{rel}}R_h. \end{aligned}$$

For every admitted nonzero quartet quotient the last difference is positive definite. The empty boundary at $N = q - 1$ and zero quotient at $q = 0$ retain their explicit separate values. The graph-to-source addition, gamma multiplication, inverse Mellin map, and full jet unit in (GP.24)–(GP.27) reconstruct the original arithmetic observation and its exact supported zero.

The new finite coefficients, strict fibre contribution, and interval certificates therefore enter the same original Toda and exterior quantities in R33–R37. In particular the lower spectral cost and the upper expression are compared in the same G_N , with its volume-imbalance square, phase square, spectral coupling, and projection terms retained. The following arithmetic endpoint continuation proves a norm bound for the original measure and a finite upper comparison at the full amplified degree. Results R42–R46 below give their complete maps and the explicitly calculated gap between the resulting upper and lower volume estimates.

60.15 Arithmetic endpoint estimates and the full reflected relation map

Result R42: the actual monic norm and its exact leading line. The complete arithmetic endpoint source and its analytic audit prove, for the original density and every integer $n \geq k \geq 3$,

$$(\omega_{h,k,2n}/\omega_{h,k,n})^{1/(2n)} \leq C_h n.$$

Here the literal constant C_h is (EP.40), and the original lower convolution envelope is (AU.10), with $B = 42 + 2 \deg h$ and the compact mass v_h^{k-3} unchanged. The complete balanced-window proof strengthens the admitted norm comparison to

$$(\omega_{h,k,n+r}/\omega_{h,k,n})^{1/(2r)} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n,$$

with every factor of C_h^{bal} stated in (EP.50b). The analytic proof derives the local zeta mass from its exact ellipse, the entire identity $g = hv_h$ at the selected centres, the three-factor positive convolution, and the monic Legendre norm. It includes the two Laplace values and their factor two before the final root.

Equations (AL.3)–(AL.10) prove the endpoint and metric of the underlying degree map. Multiplication by the original monic χ gives $\mathcal{L}_{n-q} \simeq \mathcal{L}_n$ for $n \geq q$, with squared operator norm ω_n/ω_{n-q} and adjoint coefficient equal to that same positive ratio. At $n = q - 1$ its source is zero and its target is the original one-dimensional leading line. For $n \geq q$, its representative has the complete Pythagorean decomposition

$$\|\mathcal{V}_{h,k}(\chi p_{n-q})\|^2 = \omega_n + \|\mathcal{V}_{h,k}(\chi p_{n-q} - p_n)\|^2.$$

The correction reduces to $-[p_n]_\chi$ and the relation reduces to zero. Thus the degree-line norm, the original source cost and the supported arithmetic zero are connected by the explicit commutative diagrams of the leading-line chapter.

Result R43: exact products, their optimizer, and full window transport. Keep $n \geq q$, $r \geq 1$, $m = n + r$ and $d_j = V_{n+j}/V_{n+j-1}$. The full radius calculation in (EP.1)–(EP.11), including the original complex phase $b_N^* G_N b_{N+1} = i\omega_N \phi_N$, gives

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_m}{\omega_n} \frac{V_n}{V_m} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2.$$

For the combined budget $\mathcal{B} = -\log(d_0 d_r \prod_{j=1}^{r-1} d_j^2) > 0$, the unique maximizing contraction sequence for this scalar product is

$$d_0 = (1 + 2x)^{-1}, \quad d_r = 1 - 2x, \quad d_j = (1 - x)/(1 + x) \quad (1 \leq j < r),$$

where $0 < x < 1/2$ solves $\mathcal{B} = 2 \operatorname{artanh}(2x) + 4(r-1) \operatorname{artanh} x$. The proof in (EP.22)–(EP.31) establishes existence, uniqueness, the exact maximum, its strict improvement over the earlier hyperbolic-sine bound when $r > 1$, and rational root certificates. The fixed-four-endpoint optimum, the phase polynomial and the quantitative concavity loss are separately proved in (EP.15)–(EP.21). No assertion that the scalar optimizer is realized by an arithmetic source is needed for the proved bound on the original evaluation map.

The full Hermitian window chapter constructs its Schur matrices W_j, F_j, Z_j and their determinant ratios from the same source coefficients. Its finite metric transport retains the eigenvalue spread and the exact hyperbolic coordinate for the four-volume logarithm. Equations (EP.32)–(EP.38) give the literal evaluation map from those matrices to every d_j , the budget and the product. With H_+, p, e defined in (EP.34), the complete cancellation is

$$\frac{\omega_m}{\omega_n} \frac{V_n}{V_m} = \frac{H_+}{pe} = \frac{\mathfrak{B}_{m-q+1} \mathfrak{D}_n}{\mathfrak{D}_m \mathfrak{B}_{n-q+1}}.$$

The confluent transfer proof (EP.38a)–(EP.38k) constructs the raw-derivative matrices for every node and order, proves their invertibility, and proves $\det F_a / \det V = \mathfrak{B}_a / \mathfrak{D}_a$. The sign of the column shift, all derivative factorials and every power of i remain in that construction. It also proves the theta derivative giving the phase and the exact Gamma-reference ratio $\det F_a / \det F_a^\Gamma = Y_a / X_a$.

Result R44: the necessary central and four-volume costs. For an actual off-line quartet with full common multiplicity m_0 , the retained degree and allowance are

$$q_k = [1 + k(m_0 - 1)](k + 1)^2, \\ L_{h,k} = 2\delta[1 + k(m_0 - 1)](k + 1) \lfloor (k + 1)^2 / 4 \rfloor.$$

The complete local tensor calculation and Hermitian compression proof in (EP.41) give $\epsilon_{h,k,N} \geq L_{h,k}$ and $L_{h,k}/q_k \geq \delta k/2$. At $n = q_k, r = q_k - 1$, write $C_k = \log(V_{q_k}/V_{2q_k-1})$. Combining the just-proved balanced norm estimate with the exact product yields

$$C_k \geq 2(q_k - 1) \log \frac{L_{h,k}}{C_h^{\text{bal}} q_k}$$

and the stronger full phase and contraction penalties of (EP.50c). The exact four-volume identity is

$$\log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}} = 2C_k + \log \frac{V_{q_k-1}}{V_{q_k}} + \log \frac{V_{2q_k-1}}{V_{2q_k}}.$$

Both additional logarithms are nonnegative by minimization over nested lift spaces. Substitution of $L_{h,k}/q_k \geq \delta k/2$, division by $q_k \log k$, and $(q_k - 1)/q_k \rightarrow 1$ prove the necessary four-volume lower limit 4 in (EP.50e). Equations (EP.44)–(EP.48) construct the original relation layer and the exact change of least lift. Its central Gram has largest relative eigenvalue at least $\delta^2 k^2 / (4(C_h^{\text{bal}})^2) - 1$ by (EP.50f). The cochain primitive and supported quotient image are written there for the same representative.

Result R45: an actual arithmetic upper comparison and its calculated gap. The new arithmetic upper-volume chapter constructs the positive measure

$$d\nu_k(u) = a_k e^{-(\pi/2)|u|} (k - 2 + |u|)^{-B} du, \quad a_k = c_h \vartheta_h^{k-3} e^{-(\pi/2)(k-3)},$$

which is dominated by the original $m_{h,k}(u) du$ for $k \geq 3$. The identity on polynomials is contractive and commutes with every original raw jet, remainder and spectral multiplication map. Its moments are the complete positive Gamma-mixture integrals (AU.12). Writing K_N for the original raw-jet kernel, minimization and positive inversion prove

$$0 \leq C_k \leq \log \det K_{2q_k-1}^\nu - \log \det K_{q_k}.$$

The actual arithmetic moments needed by this bound stop at degree $2q_k$; the comparator moments through $4q_k - 2$ have the explicit positive integral in (AU.12). The exact finite Gamma coefficient map (AU.16) gives the same cutoff on $c_{h,j}$. The two-Laplace-value majorant, positive Cauchy–Binet minor sum, and interval Legendre jet kernel in (AU.17)–(AU.23) supply further proved upper expressions for this same C_k .

The full scalar proof (AU.24)–(AU.33) gives

$$\begin{aligned} C_k \leq \mathcal{U}_k &= (\alpha D + \log 256) q_k^2 \\ &\quad + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k q_k \\ &\quad + B q_k \log(1 + (D + 1) q_k) + q_k \log C_{\text{env}}, \end{aligned}$$

where $\alpha = \pi/2$, $R_0 = \sqrt{\delta^2 + \gamma^2}$, $D = \max(1, 2/b, 2R_0)$, $0 < b < \pi/2$, $X = \max(1, M_h(b), M_h(-b))$, $K = \max(1, \vartheta_h^{-1})$, and $C_{\text{env}} = \max(1, 2/(3c_h D))$. All constants and the literal source mass are retained. For $\mathcal{L}_k = 2(q_k - 1) \log(\delta k / (2C_h^{\text{bal}}))$, (AU.32) computes every term of $\mathcal{U}_k - \mathcal{L}_k$, and (AU.33) proves that its quotient by q_k^2 tends to $\alpha D + \log 256 > 0$. This upper estimate leaves the stated gap to the quartet lower estimate.

The complete auxiliary Lebesgue calculation proves

$$|\log(V_q^{\text{aux}}/V_{2q-1}^{\text{aux}}) - (\tfrac{9}{2} \log 3 - 6 \log 2) q^2| \leq 12q \log(8q)$$

for every even $q \geq 2$. Its full-origin-jet quotient is connected to the original grid by the free monic family $\mathbb{C}[t, u] / \prod_{a,b} (u - t\zeta_{ab})^\ell$ in (AU.46)–(AU.48). The original full jets at $t \neq 0$, the merged jet at zero and the source-dependent kernel derivative are all explicit. The displayed auxiliary estimate belongs to its Lebesgue source; the exact derivative gives the change along the family, without assigning the same asymptotic to the arithmetic source.

Result R46: reflected norm, conormal algebra and their complete kernels. For any original monic χ of degree $q \geq 1$, set $\chi^\dagger = (-1)^q \overline{\chi}(k - S)$, $N_\chi = \chi\chi^\dagger$ and $g = \gcd(\chi, \chi^\dagger)$. The reflected relation chapter proves the algebra isomorphism

$$\mathbb{C}[S]/(\chi^2\chi^\dagger/g) \simeq \mathbb{C}[S]/(\chi^2) \times_{\mathbb{C}[S]/(\chi g)} \mathbb{C}[S]/(\chi\chi^\dagger),$$

with its exact sequence, inverse residue construction and original M_S action. The norm map retains $\chi(k/2 + iu) = i^q \psi(u)$ and $N_\chi(k/2 + iu) = i^{2q} \psi(u) \overline{\psi}(u)$. Thus the full union of reflected and original roots, with added orders at overlaps, is the precise general norm-transfer domain.

The kernel coefficient modules have annihilators (χ) and $(\chi^\dagger)$. Equations (RF.19)–(RF.21) prove their conjugate-linear bijection $F([\chi f]_{\chi^2}) = [\chi f^*]_{N_\chi}$ with inverse, $FM_S = (kI - M_S)F$, and the exact transported norm-kernel product. If a_λ, b_λ are the respective root orders,

$$\dim I_N^t = \sum_{\lambda} \max(b_\lambda - (t - 1)a_\lambda, 0), \quad t \geq 1.$$

Equations (RF.22)–(RF.26) prove the exact nilpotence index, the persistent reflected-only CRT algebra, and the fully worked mixed example of kernel dimensions 3, 1, 0. Projection sends each specified kernel vector, including the persistent sector's idempotent when present, to supported zero in its original fibre. The same lifted map fixes the external point τ .

Result R47: restriction spectra, actual arithmetic gaps and finite certificates. For every $q - 1 \leq i \leq j$, retain the original quotient kernels K_N , metrics $G_N = K_N^{-1}$ and minimum source lifts R_N . In the actual monic orthogonal coordinates, let $L_{i,j}$ pad source coefficients and let $P_{i,j}$ be the

orthogonal endomorphism projection onto its range in $\mathcal{P}_j$. The complete restriction source and the arithmetic join prove

$$T_{i,j} = K_i G_j : (E, G_j) \longrightarrow (E, G_i), \quad L_{i,j} R_i T_{i,j} = P_{i,j} R_j.$$

Let $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$ denote the original identity transport, and let A_i, A_j denote the same original action A on these two specified metric spaces. The return is its metric adjoint. The actual endomorphism $U_{i,j} = T_{i,j} I_{i,j}$ of (E, G_i) has coordinate matrix $K_i G_j$. Its eigenvalues $0 < g_a \leq 1$ are the squared singular values of the source restriction between the two canonical minimum-lift ranges. For the orthogonal projections Π_i, Π_j onto those ranges in the same original source, the join constructs both restricted-product similarities and proves

$$\mathrm{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2m} = 2 \mathrm{Tr}_E ((I - U_{i,j})^m), \quad m \geq 1.$$

Its explicit two-dimensional principal-angle blocks retain all common-range and ambient-zero components.

The original arithmetic action A has the exact reverse-map defect

$$A_i T_{i,j} - T_{i,j} A_j = H_i T_{i,j} - T_{i,j} H_j, \quad H_N = K_N (A^* G_N + G_N A - k G_N).$$

The change of representatives is the complete relation

$$\mathfrak{d}_{i,j} = L_{i,j} R_i - R_j I_{i,j} = L_{i,j} R_i (I - U_{i,j}) - (I - P_{i,j}) R_j I_{i,j}, \quad \mathfrak{d}_{i,j}^* O_j \mathfrak{d}_{i,j} = G_i - G_j.$$

Here $\mathfrak{d}_{i,j} : (E, G_i) \rightarrow \mathcal{P}_j$ has the original theta image. Both terms on the right have the same remainder $I_{i,j} (I - U_{i,j}) : (E, G_i) \rightarrow (E, G_j)$: indeed $B_j L_{i,j} R_i = I_{i,j}$ and $B_j P_{i,j} R_j I_{i,j} = I_{i,j} U_{i,j}$. The full division primitive and theta image of their difference are supplied in the join. Reduction sends its labelled relation vector to the supported zero at that receiving label; the lifted map sends external τ to τ .

The arithmetic remainder estimate is extended to every admitted degree with its entire monic quotient formula. In the original notation of the upper-comparison chapter, it gives

$$G_{q-1} \preceq C_N G_N, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}, \quad N \geq q - 1,$$

with $T = Dq, r = kR_0/T < 1$, the unchanged original density mass, and all constants defined in the complete proof. Thus the actual endpoint gaps C_{2q-1}^{-1} and C_{2q}^{-1} are proved arithmetic inequalities. The exact ratio satisfies $C_{2q}/C_{2q-1} \leq 147/5 < 32$, so the same construction gives $\mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32$ with $\mathcal{U}_k$ from R45.

The finite matrix construction improves these directional bounds. For rational Hermitian enclosures of the actual kernels,

$$0 < L_i \preceq K_i \preceq U_i, \quad 0 < L_j \preceq K_j \preceq U_j,$$

define

$$g = (\mathrm{Tr}(L_i^{-1} U_j) - q + 1)^{-1}, \quad r_* = 1 - g, \quad H_* = I - L_i U_j^{-1}, \quad t_m^* = \mathrm{Tr}(H_*^m),$$

$$d = \mathrm{Tr}(U_i L_j^{-1}) - \mathrm{Tr}(L_i U_j^{-1}).$$

The complete ordered generalized-eigenvalue proof gives $0 < g \leq 1, d \geq 0$ and the rational intervals

$$\max\{0, t_m^* - m r_*^{m-1} d\} \leq \mathrm{Tr}((I - U_{i,j})^m) \leq t_m^*.$$

The exact quotient-first triangular basis constructs these brackets from certified source and relation-coefficient intervals, retaining the actual polynomial rather than replacing it by an interval centre.

For $r_* > 0$, summing the moment intervals and bounding the positive logarithmic tail gives a certificate for $\log(V_i/V_j)$ with width at most

$$\frac{d(1 - r_*^p)}{g} + \frac{qr_*^{p+1}}{(p+1)g}.$$

The source's sharper coefficient and a rational logarithm enclosure give the further proved refinement. At $g = 1$ the loss is exactly zero. Entry refinement followed by increasing p proves convergence at each fixed pair with every error term displayed.

The balanced central product of R44 now has a consequence for each of the two original endpoint pairs. Put $X = L_{h,k}/(C_h^{\text{bal}}q)$ and retain the full nonnegative phase/contraction penalty P_{mid} from the central product. Since $\text{rank}(K_{2q-1} - K_q) \leq q-1$, at least one central return eigenvalue is exactly one. The remaining $q-1$ eigenvalues have product at most $X^{-2(q-1)}e^{-P_{\text{mid}}}$. The ordered Rayleigh-quotient comparison with each endpoint therefore proves

$$\min \text{Spec } U_{q-1,2q-1}, \quad \min \text{Spec } U_{q,2q} \leq \min \left\{ 1, X^{-2}e^{-P_{\text{mid}}/(q-1)} \right\} \leq \min \left\{ 1, \frac{4(C_h^{\text{bal}})^2}{\delta^2 k^2} \right\}.$$

Every such eigenvalue is the squared norm fraction of the actual higher-degree minimum representative retained by the lower-degree source projection. The inverse return recovers its original coordinate, with precisely the relation correction proved above.

The whole inverse spectrum has the corresponding finite lower bound. To retain endomorphism types, write $U_{i,j} = T_{i,j}I_{i,j}$ on (E, G_i) , where $I_{i,j}$ is the original identity transport. Its coordinate matrix is $K_i G_j$. For every real $s > 0$, each endpoint pair separately obeys

$$\text{Tr } U_{i,j}^{-s} \geq \max \left\{ q, 1 + (q-1)X^{2s}e^{sP_{\text{mid}}/(q-1)} \right\}.$$

Indeed the central trace is one plus the inverse-power sum of its remaining $q-1$ eigenvalues. The tangent inequality $e^y \geq 1 + y$ at their logarithmic mean bounds this sum from below by their geometric mean, and the ordered comparison transfers every eigenvalue. If the exact central update rank is $\rho = \text{rank}(K_{2q-1} - K_q) > 0$, put $A_+ = \max\{0, 2(q-1) \log X + P_{\text{mid}}\}$. Retaining all $q - \rho$ unit returns gives the stronger bound

$$\text{Tr } U_{i,j}^{-s} \geq q - \rho + \rho e^{sA_+/\rho}.$$

At $\rho = 0$ the central trace is exactly q and both endpoint traces are at least q . The complete proof and rank comparison are (ERJ.42)–(ERJ.42a). Under the canonical isometries the source restriction is the bijection $\mathcal{R} = \Pi_i|_{\ell_j E}$ between the two canonical images. Its inverse singular-value-power sum is exactly $\sum_a \sigma_a(\mathcal{R}^{-1})^{2s} = \text{Tr } U_{i,j}^{-s}$ by (ERJ.11a). The ambient zero eigenspaces of the full projection products remain outside this restricted inverse and are retained by (ERJ.9).

Result R48: the complete arithmetic weight optimizer and a stronger moment family. For the original relation χ , coordinate $S = k/2 + iu$ and full raw-jet matrix E_q , put $\psi(u) = i^{-q}\chi(k/2 + iu)$, $Z = E_{q-1}$ and $\beta_j = b^{2j}/(2j)!$ for $0 < b < \pi/2$. The weight-optimization chapter proves the signed cofactor map

$$\det E_q[:, \{0, \dots, q\} \setminus \{j\}] = (-1)^{q-j} \psi_j \det Z, \quad \psi_j = \frac{i^{j-q}}{j!} \chi^{(j)}(k/2).$$

In particular every vanishing coefficient has its corresponding zero minor, with the full jet space retained. For auxiliary weights $\tau_j \geq 0$ of sum one, set

$$\eta_j = \frac{(2j)!}{(j!)^2 b^{2j}} |\chi^{(j)}(k/2)|^2, \quad D_\eta(\tau) = \sum_{j=0}^q \eta_j \prod_{l \neq j} \tau_l, \quad d(\eta) = \max_{\tau \geq 0, \sum \tau_j = 1} D_\eta(\tau).$$

Equations (OW.4)–(OW.13) prove the precise determinant identity

$$\det(E_q \operatorname{diag}(\tau_j \beta_j) E_q^*) = |\det Z|^2 \left(\prod_{j=0}^q \beta_j \right) D_\eta(\tau)$$

and its original phase-preserving congruence. The chapter proves every maximizer: a largest coefficient at least the sum of the others gives the corresponding zero weight and the remaining weights $1/q$; exactly two equal positive coefficients give the explicit segment of maximizers. Otherwise the maximizer is unique and interior. Its zero-coefficient weights are $1/q$, and its positive-coefficient weights are $(1 - y_j)/q$, where

$$\sum_{\eta_j > 0} y_j = 1, \quad \eta_j = c_* y_j (1 - y_j), \quad 0 < y_j < 1.$$

The two monotone scalar constructions (OW.16)–(OW.19) determine the unique solution, including the branch with one large root. Equations (OW.21)–(OW.22a) prove the rational three-coefficient formula, coefficient-error bounds and terminating rational value certificates. The linear case $q = 1$ and all boundary cases are included in the same complete proof.

With the literal original $M_k = M_h(b)^k + M_h(-b)^k$ and comparator volume V_{2q-1}^ν from R45, the resulting arithmetic bound is

$$C_k \leq \mathcal{U}_{\text{opt}} = q \log M_k - \log \left(\prod_{j=0}^q \beta_j \right) - \log d(\eta) - \log V_{2q-1}^\nu.$$

The limit of the original positive raw-kernel inequality proves this bound at boundary optima; no reciprocal zero weight is used. For $\mathcal{U}_{\text{eq}}$ obtained from the same inequality with equal weights, (OW.23)–(OW.29) prove

$$0 \leq \mathcal{U}_{\text{eq}} - \mathcal{U}_{\text{opt}} = \log \frac{d(\eta)}{\sum_j \eta_j / (q+1)^q} \leq q \log(1 + 1/q) < 1.$$

Both non-strict inequalities are strict for the admitted complete quartet: its parity gives a zero coefficient, while its nonzero roots and monicity give at least two nonzero coefficients. Thus the finite optimization within this specified family is solved, and its improvement divided by $q_k \log k$ or q_k^2 tends to zero.

The continuation (OW.30)–(OW.40) proves a stronger source inequality. Write $\mu_{2j} = \int u^{2j} m_{h,k}(u) du$, retaining $\mu_0 = \mu_h^k$, and set $S_q(b) = \sum_{j=0}^q \beta_j \mu_{2j}$. Pointwise Cauchy–Schwarz and the full positive even exponential series, integrated against the original source, give

$$H_q \preceq S_q(b) \operatorname{diag}(\beta_j^{-1}) \prec \frac{M_k}{2} \operatorname{diag}(\beta_j^{-1}), \quad S_q(b) < M_k/2.$$

The corresponding upper expression $\mathcal{U}_{\text{cosh}}$ improves the earlier $\mathcal{U}_{\text{eq}}$ by exactly $q \log(2(q+1))$. Using $S_q(b)$ gives the further strict improvement $q \log(M_k/(2S_q(b)))$. Every constant follows from the integrated finite sum and both original Laplace values.

For every positive auxiliary list r_j , the same calculation proves

$$H_q \preceq \left(\sum_{j=0}^q r_j \mu_{2j} \right) \operatorname{diag}(r_j^{-1}).$$

The exact positive-cone map $t_j = r_j \mu_{2j} / \sum_l r_l \mu_{2l}$, with inverse $r_j = s t_j / \mu_{2j}$, transforms its determinant optimization to the already solved polynomial D_κ , where $\kappa_j = |\psi_j|^2 \mu_{2j}$. Thus the entire stronger diagonal family has the proved optimum

$$C_k \leq \mathcal{U}_{\text{mom}} = \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log V_{2q-1}^\nu.$$

All $q + 1$ moment factors, every zero coefficient and the boundary raw-kernel limit are retained. The ray $r_j = \beta_j$ maps to $t_j^\beta = \beta_j \mu_{2j} / S_q(b)$, and the exact total gain is

$$\mathcal{U}_{\text{opt}} - \mathcal{U}_{\text{mom}} = q \log(2(q + 1)) - \mathfrak{g}(\eta) + q \log \frac{M_k}{2S_q(b)} + \log \frac{d(\kappa)}{D_\kappa(t^\beta)}.$$

The last term is nonnegative and is evaluated by the same complete optimizer. The removed term $q_k \log(2(q_k + 1))$, divided by $q_k \log k$, tends to 2 for a simple quartet and to 3 for fixed common multiplicity greater than one.

Finally the original quotient-first source Gram proves $V_q = \det H_q / (\psi^* H_q \psi)$. Equation (OW.40) therefore calculates the exact remaining comparison loss against the full low-degree kernel:

$$\mathcal{U}_{\text{mom}} - (\log V_q - \log V_{2q-1}^\nu) = \sum_{j=0}^q \log \mu_{2j} - \log d(\kappa) - \log \det H_q + \log(\psi^* H_q \psi) \geq 0.$$

These identities retain all actual moments through degree $2q$. They determine the finite improvement and its precise comparison map without assigning an unproved growth rate to the last loss or the comparator determinant. The original source, support label and quotient action remain connected through the same cofactor and kernel maps.

Result R49: the first-degree phase product and each full block budget. The complete first-window chapter joins the original first-degree radius to the comparable-window theorem of PR 25, retaining the nonnegative generalized-eigenvalue correction. At $N = q - 1$, expand $\chi = p_q + \sum_{j < q} \gamma_j p_j$ and keep $\nu_0 = \|\chi\|^2 = \omega_q + \sum_{j < q} |\gamma_j|^2 \omega_j$. The rank-one update and the original phase identity give

$$\frac{V_q}{V_{q-1}} = \frac{\omega_q}{\nu_0}, \quad \gamma_{q-1} = -i\phi_{q-1}, \quad \epsilon_{q-1}^2 + \phi_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}.$$

Equations (CJ.3)–(CJ.7) prove each equality in the actual source. For $\Lambda_N = \omega_N / V_N$, the first step consequently has exactly one contraction:

$$(\epsilon_{q-1}^2 + \phi_{q-1}^2) \Lambda_{q-1} = \Lambda_q (1 - \delta_q).$$

Multiplication by the regular consecutive identities gives the full first-window formula (CJ.9), for every $j \geq q$:

$$\prod_{N=q-1}^{j-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_j V_{q-1}}{\omega_{q-1} V_j} (1 - \delta_j) \prod_{N=q}^{j-1} (1 - \delta_N)^2.$$

The original source contains no quotient inverse below degree $q - 1$. The complete two-sided monic norm envelope (CJ.15)–(CJ.18) supplies the explicit C_h^{win} for $n/2 \leq r \leq 2n$. For $D_h = \delta / (2C_h^{\text{win}})$ and each original block $(i, j) = (q - 1, 2q - 1)$ or $(q, 2q)$, it proves the finite bound

$$\log(V_i/V_j) \geq 2q \log(D_h k) + \mathcal{P}_{i,j} + \sum_{N=i}^{j-1} \log(1 + \phi_N^2 / L_{h,k}^2).$$

The penalties $\mathcal{P}_{i,j}$ are written completely in (CJ.13), including the single terminal factor for the first block and both boundary factors for the regular block. All terms added to the per-block $2q$ budget are nonnegative.

The original relation map in (CJ.20)–(CJ.21) has Gram $\Omega + F^* G_i F$ and is an isomorphism onto the whole newly admitted relation layer. At zero tilt and even tensor degree, the exact quartet's first increment matrix has a kernel by the full parity calculation. Every zero generalized eigenvalue stays in the q -dimensional cost spectrum, contributing the factor one to the determinant. For all $s > 0$, (CJ.22) also proves

$$\text{Tr } Q_{i,j}^{-s} \geq \max\{q, q(D_h k)^{2s}\}, \quad Q_{i,j} = T_{i,j} I_{i,j} \text{ on } (E, G_i).$$

These are traces of the same original metric endomorphism. The full cost, source relation, and its supported-zero quotient image remain connected by the maps in that chapter.

Result R50: the two relation traces and their actual comparison source. The complete relation-moment chapter proves the sharp finite determinant majorant from the two traces of the original positive cost operator $Z = (K_j - K_i)G_i$, keeping its full dimension and every zero eigenvalue. Writing $t_1 = \text{Tr } Z$, $t_2 = \text{Tr } Z^2$, for $q \geq 2$ define

$$d = \sqrt{\frac{qt_2 - t_1^2}{q-1}}, \quad a = \frac{t_1 + (q-1)d}{q}, \quad b = \frac{t_1 - d}{q}.$$

The complete rational Hermite remainder in that chapter gives

$$\log(V_i/V_j) = \log \det(I + Z) \leq \log(1 + a) + (q-1) \log(1 + b).$$

For dimension one the exact expression is $\log(1 + t_1)$, and for dimension zero it is zero. The proof identifies both traces with the original relation map and its exterior square, including every cross pairing. For the arithmetic upper kernel U_j , the positive space $\mathcal{C}_+ = (E^\vee / \ker(U_j - K_i), U_j - K_i)$ and its map $J_+[\alpha] = (U_j - K_i)\alpha$ realize the comparison cost $Z_+ = (U_j - K_i)G_i$. The constructed contraction from the original relation source has the complete positive defect

$$J_+(I - CC^\dagger)J_+^\dagger = (U_j - K_j)G_i.$$

This retains the precise map and the excess in the bound, together with the original arithmetic action defect and all unit return factors. The relation-moment chapter contains the full definitions and proofs.

Result R51: the full Deligne comparison and the constituent coupling. The detailed Deligne sidebar proves the image, kernel and cone maps for the same compact-to-ordinary comparison, and the complementary degree duality with its actual boundary line. For every specified invariant $F \subseteq E$ and divisor depth $m \geq 1$, its original curve has the logarithmic complex

$$\mathcal{H}_{F,m} = [F\mathcal{O} + E\mathcal{O}(-m\infty) \xrightarrow{d} (F\mathcal{O} + E\mathcal{O}(-m\infty)) \otimes \Omega^1(\log \infty)].$$

Its complete cohomology is F in degree zero, $(E/F) \otimes \mathbb{T}$ in degree two and zero in degree one, with $\mathbb{T} = H^1(\Omega^1)$ and the retained trace $\text{tr}[dz/z] = 1$. The higher-depth quotient contracts by division by the displayed nonzero integer coefficients; its dual retains all meromorphic powers and the twist by $\mathcal{O}(-\infty)$. Each split lift distinguishes the represented kernel from external absence through their disjoint inclusions into the full kernel carrier.

The same original metric, $G(t) = G_j + t(G_i - G_j)$, gives the invariant quotient metric and coupling map proved in (DS53)–(DS65). With those specified maps, its endpoint identity is

$$\log(V_i/V_j) = \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}.$$

The series has nonnegative terms before subtraction, and the sidebar proves its finite remainder. It also proves the exact original theta source defect $dK(z)H_t + r(z)Ia_t$, so the theta boundary and the retained constituent pass through their specified successive quotients. The two positive determinants can use R50 on their respective compressed matrices; the coupling subtraction remains in the same upper estimate.

Result R52: a degree-one exponential family with exact periods. For the original cyclic polynomial χ and its specified primitive Φ , the delivered exponential note proves

$$\mathcal{H} = \text{coker}(u\partial_S + \chi(S) - t), \quad \mathcal{H} \cong \mathbb{C}[u, t]^q, \quad \mathcal{H}/(u, t)\mathcal{H} \cong E, \quad (-u\nabla_t) \bmod (u, t) = A.$$

Its coefficient-module division, original theta observation, finite-rank source deformation, infinity chart and dual residue pairing are all retained with complete proofs in the exponential note and sidebar. The new determinant proof (XD1)–(XD26) computes the entire period determinant for the stated oriented rays:

$$\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp\left(\frac{\text{Tr } \Phi_t(A(t))}{u}\right).$$

All Gamma factors and monomial phases are given in (XD16)–(XD18). The proof uses the actual twisted-reduction matrices, keeps their off-diagonal corrections, and proves that their trace corrections vanish. It establishes nonvanishing across every repeated-root parameter. On the original canonical metric it gives

$$\log \det B_N^{\text{per}} = \log C_q(u) + 2 \operatorname{Re} \frac{\text{Tr } \Phi_t(A(t))}{u} - \log V_N.$$

Subtracting at the same parameters retains the exact original endpoint cost. The translation companion (DT1)–(DT13) further proves the complete maps under $\chi_a(S) = \chi(S - a)$: the coefficient action shifts by aI , the period gains the scalar $\exp[(-\Phi(-a) - ta)/u]$ and the full Pascal matrix, and the finite-field realization tensors the corresponding constant Artin–Schreier character line. The original arithmetic fibre remains $a = 0$. These explicit maps preserve the canonical metric comparison and the full inherited endpoint inequalities.

The arithmetic norm theorem, sharp endpoint products and upper comparison therefore enter the existing Toda, Gamma and exterior maps with complete proof texts. The finite upper determinants in R45 are the concrete next quantities to estimate at the amplified degree; the explicit scalar estimate and the lower limit 4 currently coexist with the gap proved above.

Part II**Complete source proof notes and chronology**

S1 PR 8: Finite operator comparison

Source attribution. Web contribution, pull request #8, [Zeta research repository](#).

Exact revision. 25689c5376166df37801229bc275ff5a896007d9

Source file. workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 57502c7fe6a7e1de239c94c3479abbb6a70b4718b551ad1bb1e26e2d3e4f4c22

S1.1 Exact finite operator quotients of the tau-base theta cohomology

Owner-directed continuation, 12 September 2026. Mathematical derivation and research draft; no new analytic theorem below is claimed to have been checked by Lean.

S1.1.1 0. Inputs, provenance, and the new result

This note continues TAU_BASE_MODEL.md, rather than moving the programme back to a finite-field variety. Its arithmetic coefficient object is the original split semiring

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

The geometric base remains the one-point absolute pointed base described there. The quotient

$$p_{\mathbb{Z}} : G(\mathbb{Z}) \longrightarrow \mathbb{Z}$$

and its square with $G(\mathbb{C}) \rightarrow \mathbb{C}$ remain part of the marked object. No ideal norm, Euler factor, scaling parameter, Gaussian, or completion factor is changed here.

The supplied derived formalization (PR #6, 5f6ee6575550d8a20c6fc3990359cb4f67855f80) provides the actual reconstruction, internal quotient, homology-kernel equivalence, and compatible kernel actions. Its finite-jet module supplies the balanced tensor-quotient comparison with the split-reflection square. Those results are reused; this note does not duplicate their Lean implementation.

The new analytic step computes the previously only surjective finite Mellin-jet observation as an exact operator quotient. For every polynomial h whose roots lie in the open critical strip, it constructs

$$Q/h(D)Q \xrightarrow{\sim} E_h/g_h E_h, \quad Q[h(D)] \xrightarrow{\sim} \ker(g_h : E_h \rightarrow E_h),$$

with $E_h = \mathbb{C}[t]/(h)$ and $g = 2\xi$. It then constructs finite spectral projectors on the original Q , not only quotients of Q , and identifies the torsion subsheaf of the balanced analytic realization with $\mathcal{O}/(2\xi)$. Their kernels, original supported zeros, nilpotent actions, and trace-producing residue pairing are all retained.

The general existence of spectral realizations of zeta zeros is prior literature (not a novelty claim here): see R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. The contribution of this note is the explicit comparison in the specified workbench, together with its original SplitZero quotients and its complete finite kernel calculation.

S1.1.2 1. The exact arithmetic complex and its operator

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}.$$

The source maps, with their original factors, are

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The Fourier transform is $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx$. Poisson and the two moment conditions put ΘV in $\mathcal{B}$. Also $D\Theta = \Theta D$, and D preserves V .

Set

$$A = \mathbb{C}[t], \quad t \cdot F = DF, \quad t \cdot \phi = D\phi,$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1, \quad Q = \mathcal{B}/\Theta V.$$

The existing injectivity proof of Θ gives $H^0(C_+) = 0$ and $H^1(C_+) = Q$.

This is the one-leg subcomplex of the actual tau-base chart complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta\phi - \Theta\widehat{\psi}} \mathcal{B}].$$

Its inclusion is $\phi \mapsto (\phi, 0)$ in degree zero and the identity in degree one. The quotient complex is the second copy of V in degree zero. The explicit section is $\psi \mapsto (\psi, \psi)$. The operator on that second copy is $1 - D$, because $D\widehat{\psi} = (1 - D)\psi$. Thus the additional joint-support degree-zero cohomology is retained in this split exact sequence; it is not included in Q and then silently discarded.

The actual source multiplier is

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$H_\phi(s) = c(s)M_+\phi(s), \quad c(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

On the open strip $\mathcal{S} = \{0 < \Re s < 1\}$, $c(s)$ is holomorphic and nowhere zero. The source identity is

$$\mathcal{M}\Theta\phi = gH_\phi.$$

The specific function

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$$

has $H_{\phi_*} = 1$. Consequently

$$H_{P(D)\phi_*}(s) = P(s), \quad \mathcal{M}\Theta(P(D)\phi_*) = g(s)P(s).$$

These are identities with the stated Gaussian and $g = 2\xi$; no rescaling is performed.

S1.1.3 2. A continuous inverse to the actual Euler differential

For any $\rho \in \mathbb{C}$, define on the subspace $\{H \in \mathcal{B} : \mathcal{M}H(\rho) = 0\}$

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1} dt.$$

All powers use the real logarithm on $x, t > 0$. Direct differentiation gives

$$(D - \rho)S_\rho H = H.$$

Endpoint control. Write $x = e^y$, $h(y) = H(e^y)$, and $\sigma = \Re \rho$. At $y \geq 0$, the displayed tail and a bound $|h(t)| \leq C_N e^{-Nt}$ give

$$|S_\rho H(e^y)| \leq \frac{C_N}{N - \sigma} e^{-Ny} \quad (N > \sigma).$$

At $y \leq 0$, the zero moment changes the same expression into the negative lower tail. A bound $|h(t)| \leq C'_N e^{Nt}$ gives

$$|S_\rho H(e^y)| \leq \frac{C'_N}{N + \sigma} e^{Ny} \quad (N + \sigma > 0).$$

Differentiating uses $(S_\rho H)'_y = -\rho S_\rho H - H(e^y)$, so every derivative has the same family of estimates. This proves membership in $\mathcal{B}$ and continuity in its stated seminorms. The homogeneous solution $x^{-\rho}$ fails the required rapid decrease unless its coefficient is zero. Thus $D - \rho$ is injective, and

$$(D - \rho)\mathcal{B} = \ker(F \mapsto \mathcal{M}F(\rho)).$$

Mellin intertwining gives

$$\mathcal{M}(S_\rho H)(s) = \frac{\mathcal{M}H(s)}{s - \rho},$$

with the removable value filled at $s = \rho$.

There is also a source inverse. For $\rho \in \mathcal{S}$ and $\phi \in V$ with $H_\phi(\rho) = 0$, use the same tail for $x > 0$ and extend evenly. The nonvanishing of $c(\rho)$ gives $M_+\phi(\rho) = 0$. Near zero the formula is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Differentiation under this integral proves smoothness at zero in every order. The function is even and its value at zero is zero. The upper-tail formula proves Schwartz decay at infinity. Finally

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi(x) dx = \int_{\mathbb{R}} \phi(x) dx = 0,$$

so the second moment condition is preserved. This proves the source range identity

$$(D - \rho)V = \ker(\phi \mapsto H_\phi(\rho)) \quad (\rho \in \mathcal{S}).$$

At the level of Taylor coefficients $\phi(x) = \sum_{j \geq 1} a_{2j} x^{2j}$, the local expression is $-\sum_{j \geq 1} a_{2j} x^{2j} / (\rho + 2j)$, interpreted to each finite Taylor order with its smooth remainder. All denominators and the minus sign are retained. Endpoints 0, 1 are not included in this source range theorem.

S1.1.4 3. Every finite operator test, without choosing zeros first

Choose any finite set of distinct centres $\rho_1, \dots, \rho_r \in \mathcal{S}$ and positive integers m_i . Define the particular polynomial

$$h(t) = \prod_{i=1}^r (t - \rho_i)^{m_i}, \quad d = \deg h, \quad E_h = A/(h).$$

This defines the test; it does not presume that any centre is a zeta zero. The Chinese-remainder map is

$$\theta_h : E_h \xrightarrow{\sim} \prod_i \mathbb{C}[z_i]/(z_i^{m_i}), \quad [P(t)] \mapsto [P(\rho_i + z_i)].$$

For a holomorphic germ family f at the centres, $j_h f$ denotes its Taylor jets transported back by θ_h^{-1} . In particular $g_h = j_h g$ retains the original local units of g .

Iterate Section 2 through the factors of h . Uniqueness makes the inverse independent of their chosen order. This gives the continuous inverse

$$S_h : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B}, \quad h(D)S_h = \text{id}.$$

The exact range identities are

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H).$$

Surjectivity on the source follows from $H_{P(D)\phi_*} = P$. Surjectivity of the target jet map follows directly from finite Mellin interpolation: on an interval in $y = \log x$, the functions $e^{\rho_i y} y^j / j!$ are linearly independent. For a nonnegative smooth bump η positive on a subinterval, the Gram matrix

$$G_{(i,j),(k,l)} = \int \eta(y) e^{(\rho_i + \bar{\rho}_k)y} \frac{y^{j+l}}{j!l!} dy$$

is positive definite. Multiplying its inverse into the prescribed jet vector produces a compactly supported smooth preimage. This is only an interpolation device; it does not define or alter the arithmetic Weil pairing.

Consequently the vertical arrows in

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\bar{\theta}} & \mathcal{B}/h(D)\mathcal{B} \\ \alpha_h \downarrow \wr & & \wr \downarrow \beta_h \\ E_h & \xrightarrow{\times g_h} & E_h \end{array}$$

are actual A -module isomorphisms:

$$\alpha_h[\phi] = j_h H_\phi, \quad \beta_h[F] = j_h \mathcal{M}F.$$

An inverse to α_h sends the unique polynomial representative P of degree less than d to $[P(D)\phi_*]$. An inverse to β_h is any of the above explicit interpolation lifts, taken modulo $h(D)\mathcal{B}$; the class is independent of that choice by the range theorem.

S1.1.5 4. The kernel and quotient of the global arithmetic operator

Both V and $\mathcal{B}$ are torsion-free A -modules: every factor $D - \rho$ is injective, since its homogeneous solution cannot be Schwartz at infinity. Over the PID A , they are flat. The bounded complex C_+ is therefore a valid flat model for derived tensoring. The displayed isomorphism computes

$$C_+ \otimes_A^{\mathbf{L}} E_h \simeq [E_h \xrightarrow{\times g_h} E_h]$$

in degrees zero and one. It is the base-change of the actual theta complex, not a newly declared spectral complex with an assigned differential.

The original exact sequence $0 \rightarrow V \xrightarrow{\Theta} \mathcal{B} \rightarrow Q \rightarrow 0$ also gives

$$0 \rightarrow Q[h(D)] \xrightarrow{\kappa_h} E_h \xrightarrow{\times g_h} E_h \rightarrow Q/h(D)Q \rightarrow 0.$$

Here $Q[h(D)] = \ker(h(D) : Q \rightarrow Q)$. The maps are explicit:

- Given $[F] \in Q[h(D)]$, uniquely write $h(D)F = \Theta\phi$. Then $\kappa_h[F] = j_h H_\phi$.
- The quotient observation sends $[F]$ to $j_h \mathcal{M}F$ modulo $g_h E_h$.
- For $P \in \ker(g_h : E_h \rightarrow E_h)$, choose its polynomial representative of degree less than d and set

$$\partial_h(P) = [S_h \Theta(P(D)\phi_*)] \in Q[h(D)].$$

The last expression is defined because its theta input has zero jets modulo h . Replacing P by $P + hR$ changes the result by $\Theta(R(D)\phi_*)$. Thus it is independent of representatives and is the inverse of κ_h onto its kernel. The two injectivity and surjectivity assertions also follow directly by lifting representatives in the preceding exact sequence; no dimension-only argument is needed.

All maps commute with D , with $U_a F(x) = F(x/a)$, and with compactly supported multiplicative convolution. On E_h , the scaling action is multiplication by $j_h(a^s)$, including its entire finite nilpotent Taylor polynomial.

At one centre, retain $g(\rho + z) = u(z)z^r$, where $u(0) \neq 0$ (and $r = 0$ is allowed). For test order m , the two cohomology modules have dimension $\min(m, r)$. When $r = 0$, multiplication by g_h is invertible. The derived amplitude complex is then acyclic, while its reconstructed supported homology still has each labelled internal zero.

The degree-zero term is $Q[h(D)]$; the degree-one term is $Q/h(D)Q$. Their finite alternating traces cancel. The arithmetic packet trace below is the degree-one action, or equivalently its explicitly constructed direct summand in the original Q . The canonical truncation triangle retains the degree-zero term rather than relabelling it as zero.

S1.1.6 5. Exact finite spectral projectors on the original cohomology

Now let Z be any finite set of actual nontrivial zeros and retain their full multiplicities r_ρ . Put

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{r_\rho}, \quad E_Z = A/(h_Z), \quad v_Z(s) = g(s)/h_Z(s).$$

The quotient v_Z is entire and its germs are nonzero at every point of Z . Hence $v_{Z,h} = j_{h_Z} v_Z$ is an invertible element of E_Z . This inverse retains the actual Taylor coefficients of g/h_Z ; it is not set to one.

Here $g_{h_Z} = 0$, and the exact sequence becomes

$$Q[h_Z(D)] \xrightarrow[\kappa_Z]{\sim} E_Z, \quad Q/h_Z(D)Q \xrightarrow[\beta_Z]{\sim} E_Z.$$

There is a further map that must be calculated, not assumed: the inclusion of the kernel into Q , followed by the quotient. For $P \in E_Z$,

$$\mathcal{M}(S_{h_Z}\Theta(P(D)\phi_*)) = \frac{gP}{h_Z} = v_Z P.$$

Therefore this kernel-to-quotient comparison is precisely

$$\boxed{\beta_Z q_Z \partial_Z = M_{v_{Z,h}} : E_Z \xrightarrow{\sim} E_Z.}$$

Define

$$\sigma_Z = \partial_Z \circ M_{v_{Z,h}}^{-1} : E_Z \longrightarrow Q,$$

$$\boxed{\Pi_Z = \sigma_Z J_Z : Q \longrightarrow Q, \quad J_Z[F] = j_{h_Z} \mathcal{M}F.}$$

Then

$$\boxed{J_Z \sigma_Z = \text{id}, \quad \Pi_Z^2 = \Pi_Z, \quad \text{im } \Pi_Z = Q[h_Z(D)], \quad \ker \Pi_Z = h_Z(D)Q.}$$

This gives the actual equivariant direct sum

$$\boxed{Q = Q[h_Z(D)] \oplus h_Z(D)Q.}$$

A representative formula is completely determined. For $F \in \mathcal{B}$, let $P_{Z,F}$ be the unique polynomial of degree less than $\deg h_Z$ representing $v_{Z,h}^{-1} j_{h_Z} \mathcal{M}F$. Then

$$\boxed{\Pi_Z[F] = [S_{h_Z}\Theta(P_{Z,F}(D)\phi_*)].}$$

For a simple actual zero ρ , this specializes to

$$\Pi_{\{\rho\}}[F] = \frac{\mathcal{M}F(\rho)}{g'(\rho)} \left[x^{-\rho} \int_x^\infty \Theta\phi_*(t) t^{\rho-1} dt \right].$$

The displayed denominator is the actual $g'(\rho) = 2\xi'(\rho)$; it is not omitted. The bracketed class is an actual nonzero eigenclass in Q , proved by the exact sequence, even though its representative is only an eigenvector modulo the theta relation.

For finite packets the projectors satisfy

$$\Pi_Z \Pi_W = \Pi_{Z \cap W}, \quad \Pi_{Z \cup W} = \Pi_Z + \Pi_W \quad (Z \cap W = \emptyset).$$

These follow from the direct sums, coprimality of disjoint packet polynomials, and commutation with D . Inclusions of finite packets therefore give compatible invariant subspaces and quotient observations. No convergence of an infinite sum of projectors is asserted.

S1.1.7 6. Carry every projector through the original split quotient

For any vector space M the single-active-fibre lift is $M^\tau = \{\tau\} \sqcup \{m^\bullet : m \in M\}$. For a submodule N use exactly

$$q_N^{\text{sp}} : M^\tau \rightarrow (M/N)^\tau, \quad m^\bullet \mapsto [m]^\bullet, \quad \tau \mapsto \tau.$$

This is the original coequalizer of the relation inclusion and its internal-zero companion. Its universal property is the one implemented in PR #6, including arbitrary noncancellative target semi-modules.

Consequently the finite observation has the split-linear equivalence

$$(Q/h(D)Q)^\tau \xrightarrow{\sim} G(E_h/g_h E_h), \quad [F]^\bullet \mapsto (j_h \mathcal{M} F \bmod g_h E_h)^\bullet.$$

The domain is being used as a split semimodule, not as an unconstructed ring. A class in $h(D)Q$ maps to the supported zero, and external absence alone maps to external absence.

Lift the projector as

$$\tilde{\Pi}_Z(\lambda, [F]) = (\lambda, \Pi_Z[F]).$$

This formula applies to the actual nonempty support faces of the tau-base theta model; their degree-one transports are the stated identities of Q . At bottom it is the original zero map of the bottom module. The full reconstruction interface permits a nonzero bottom fibre when a subsequent diagram requires one; it is not silently specialized away on an opposite-index dual diagram.

For disjoint packets,

$$\tilde{\Pi}_Z \tilde{\Pi}_W = \varepsilon, \quad \varepsilon(\lambda, x) = (\lambda, 0).$$

This is the supported lift of the zero linear operator, not the constant external-zero map. More uniformly, take $\Pi_\emptyset = 0_Q$ and lift it to ε . The scalar representation

$$G(\text{End}_{\mathbb{C}} Q) \longrightarrow \text{End}_{G(\mathbb{C})}(Q^\tau)$$

sends $f^\bullet$ to its split lift and sends τ to the constant- τ map. It is a unital semiring morphism and sends the supported scalar zero to ε .

Retain smaller admitted source labels For a subspace $W \subset V$, first use its specified enlargement $W_D = \mathbb{C}[D]W$ and keep the map

$$\mathcal{B}/\Theta W \longrightarrow Q_{W_D} := \mathcal{B}/\Theta W_D, \quad \ker = \Theta W_D/\Theta W.$$

For a D -stable W , define the ideal

$$J_{W,h} = \{j_h H_\phi : \phi \in W\} \subseteq E_h.$$

It is an ideal because $H_{P(D)\phi} = PH_\phi$ and every element of E_h has a polynomial representative. The same target range calculation gives

$$Q_W/h(D)Q_W \cong E_h/(g_h J_{W,h}).$$

Enlarging to $W + \mathbb{C}[D]\phi_*$ makes $J_{W,h} = E_h$. The exact finite comparison kernel is

$$\boxed{g_h E_h / (g_h J_{W,h}).}$$

Thus the finite equality for the full arithmetic relation space has a precise smaller-label comparison. It is not asserted for every original W without this kernel.

S1.1.8 7. The Hausdorff observation and the remaining global kernel

The projectors just constructed have continuous finite-rank representatives on $\mathcal{B}$: J_Z is continuous, E_Z is finite dimensional, and S_{h_Z} is continuous on its zero-jet domain. They annihilate ΘV , hence also its closure. They therefore descend to

$$Q^H = \mathcal{B} / \overline{\Theta V}.$$

The finite observation still has kernel exactly $h_Z(D)Q^H$. Indeed

$$\ker J_Z = h_Z(D)\mathcal{B} + \Theta V,$$

and continuity gives $\overline{\Theta V} \subset \ker J_Z$. Hence adding the closure to the right side leaves it unchanged. The same finite section yields a continuous direct sum

$$Q^H = \sigma_Z(E_Z) \oplus h_Z(D)Q^H.$$

The algebraic packet $Q[h_Z(D)]$ injects into this Hausdorff quotient: its composite with J_Z is already an isomorphism onto E_Z . No conclusion about vanishing of the entire algebraic-to-Hausdorff kernel follows from this finite statement.

Let

$$\mathcal{R}_\infty = \bigcap_{Z \text{ finite}} \ker \Pi_Z = \bigcap_{Z \text{ finite}} h_Z(D)Q.$$

The map $Q \rightarrow \prod_\rho \mathcal{O}_\rho / (g)$ has exactly this kernel. Every nonzero polynomial whose roots are in $\mathcal{S}$ acts bijectively on $\mathcal{R}_\infty$. To verify this, its kernel on Q lies in the finite packet at its zero centres; its cokernel is supported there as calculated in Section 4. On the complementary summand it is bijective, and its inverse preserves all other commuting finite projectors. This proves injectivity and surjectivity on the intersection.

The continuation retains $\mathcal{R}_\infty$ and its scaling action. Finite packets do not prove it zero. A topological spectral-synthesis theorem or a global cohomological trace argument must address that specific invariant module rather than infer its disappearance from finite-jet isomorphisms.

7.1 The global balanced kernel has a computed local-cohomology complement Work now on the analytic strip $\mathcal{S}$, with its sheaf $\mathcal{O}$ and sheaf $\mathcal{K}$ of meromorphic functions. Form the sheafified balanced realization

$$\mathcal{M}_Q = \mathcal{O} \otimes_A Q, \quad \beta : \mathcal{M}_Q \longrightarrow \mathcal{O}/(g), \quad b \otimes [F] \longmapsto [bMF].$$

Let $\text{tors}_{\mathcal{O}}(\mathcal{M}_Q)$ denote its torsion subsheaf: a germ is in it when some nonzero holomorphic germ annihilates it. This is torsion over the specified analytic coefficient ring; its connection to the original split support is the reconstruction of the inclusion and quotient maps below.

Theorem. The actual Mellin map restricts to an isomorphism

$$\boxed{\beta|_{\text{tors}} : \text{tors}_{\mathcal{O}}(\mathcal{M}_Q) \xrightarrow{\sim} \mathcal{O}/(g).}$$

Moreover, $\mathcal{N} = \ker \beta$ has a canonical $\mathcal{K}$ -module structure, and the displayed map canonically splits the sequence:

$$\boxed{\mathcal{M}_Q \cong \mathcal{O}/(g) \oplus \mathcal{N}, \quad \mathcal{K} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{N}.}$$

Proof at every stalk. Fix $\rho \in \mathcal{S}$ and set $x = s - \rho$. If $g(\rho) \neq 0$, Section 4 with $h = t - \rho$ proves that $D - \rho$ is bijective on Q . Consequently x is invertible on $\mathcal{M}_{Q,\rho}$, and this module is a module over the meromorphic field $\mathcal{O}_\rho[1/x]$. Its torsion is zero, as is $\mathcal{O}_\rho/(g)$.

If $r = \text{ord}_\rho g > 0$, the finite projector for this one full packet gives

$$Q = Q[(D - \rho)^r] \oplus (D - \rho)^r Q.$$

On the second summand, $(D - \rho)^r$ is bijective: it is injective by the direct-sum intersection, and

$$(D - \rho)^r Q = (D - \rho)^{2r} Q$$

follows by applying $(D - \rho)^r$ to the direct sum. Hence $D - \rho$ itself is bijective there. Tensoring the direct sum with $\mathcal{O}_\rho$ gives a first summand $\mathcal{O}_\rho/(x^r)$, on which β is the isomorphism calculated in Section 5, and a second summand on which x is invertible. The map β kills the second summand because it is in $x^r \mathcal{M}_{Q,\rho}$, while x^r annihilates its target. Every nonzero germ of $\mathcal{O}_\rho$ is a unit times a power of x , so that second summand is a module over $\mathcal{K}_\rho$ and is torsion-free.

This proves the stated torsion isomorphism at every stalk. It glues without choosing neighbourhood projectors: both the torsion subsheaf and the restricted map β are intrinsic. The inverse of that isomorphism, followed by the torsion inclusion, is the global section of β . Local inverses to nonzero holomorphic scalars on $\mathcal{N}$ are unique, so they glue to its $\mathcal{K}$ -module structure. This proves the theorem.

7.2 The extraction is an actual local-cohomology map At the same stalk the principal-ideal local-cohomology complex is the explicit localization complex

$$R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) = [\mathcal{M}_{Q,\rho} \longrightarrow \mathcal{M}_{Q,\rho}[1/x]], \quad \text{degrees } 0, 1.$$

The decomposition just proved computes it:

$$\boxed{R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) \xrightarrow{\sim} (\mathcal{O}_\rho/(g))[0].}$$

The finite torsion summand maps to zero upon localization; the meromorphic summand maps isomorphically to itself. The map to the displayed target is β in degree zero and zero in degree one. Thus the supported arithmetic divisor is the actual cohomological image of the localization kernel, not a prescribed deletion of a list of spectral points.

Before analytic extension there is the corresponding operator-localization map

$$Q \longrightarrow A[h_Z^{-1}] \otimes_A Q,$$

with kernel $Q[h_Z(D)]$ and cokernel zero, because $h_Z(D)$ is invertible on its complementary summand. Its supported lift sends that kernel to the internal supported zero, not to external absence.

At every original label λ , reconstruct the localization complex using

$$d(\lambda, m) = (\lambda, m/1), \quad \varepsilon(\lambda, m) = (\lambda, 0).$$

Its internal degree-zero cycles are exactly the reconstructed torsion module. Its internal first cohomology is the support skeleton, since the ordinary first cohomology vanishes. The map $p_{\mathbb{Z}}$ remains

in the structural coefficient square throughout; x is the analytic image of the operator $t - \rho$, not a replacement for the scalar e or τ .

The theorem does not assert $\mathcal{N} = 0$. It computes what local cohomology does to it: localization is an isomorphism on that module, so it contributes no cohomology to the stated local-support complex. Its possible contribution to a different global trace functor has not been declared absent.

S1.1.9 8. The residue duality is now inside an actual derived finite comparison

Assume the centres and their orders in h are stable under $\iota(s) = 1 - \bar{s}$. For germs define

$$f^\dagger(s) = \overline{f(1 - \bar{s})}.$$

The chosen monic polynomial satisfies $h^\dagger = (-1)^d h$. Retain that factor. Define the perfect finite algebra pairing

$$R_h(f, u) = \sum_i \operatorname{Res}_{s=\rho_i} \frac{f^\dagger(s)u(s)}{h(s)} ds.$$

It satisfies $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$. No phase is inserted to change this sign. Since $g^\dagger = g$, multiplication by g_h is self-adjoint for this pairing. Therefore it induces a perfect pairing

$$\overline{\operatorname{coker} g_h} \otimes \ker g_h \longrightarrow \mathbb{C}.$$

Well-definedness is the identity $R_h(ga, u) = R_h(a, gu)$; nondegeneracy follows by taking the annihilator of the image in the finite perfect algebra pairing. Through Section 4 this pairs the two actual cohomology groups of $C_+ \otimes_A^{\mathbf{L}} E_h$.

For an exact packet h_Z , set $v_{Z,h} = j_h(g/h_Z)$ as above and

$$j_Z : A_Z \rightarrow \ker g_h = E_Z, \quad [f] \mapsto [j_h(h_Z/g)f].$$

Here $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g) \cong E_Z$ by the specified local quotient maps; h_Z/g is holomorphic at every centre. The local units are not replaced by one. The exact contraction is

$$R_h(f, j_Z k) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger k}{g} ds =: R_Z(f, k).$$

Thus the previously constructed arithmetic residue form is the duality pairing of the actual finite operator comparison after the displayed unit morphism.

The Jacobian map is still $J_g[k] = [g'k]$. Its contraction is

$$R_h(f, j_Z J_g k) = R_Z(f, J_g k) = \sum_{\rho \in Z} r_\rho \overline{f(1 - \bar{\rho})} k(\rho) = \operatorname{Tr}(M_{f^\dagger k} | A_Z).$$

In particular the factor 2 in $g = 2\xi$ is retained in both the denominator and the Jacobian. The trace contraction is invariant under their combined explicitly specified transformation, not by deleting that factor in only one place.

Every map lifts to the original support fibres. Dual support arrows are precomposition maps on the opposite index category. A vanishing pairing value is the supported scalar $e_{\mathbb{C}}$; an absent argument gives τ .

S1.1.10 9. Cohomological placement and arithmetic traces

On the tau-base chart use the one-leg sheaf $\mathcal{T}_+$ and its A -action. The bounded projective resolution from TAU_BASE_MODEL.md gives the canonical map

$$\mathbf{A}_\tau(\mathcal{T}_+ \otimes_A^{\mathbf{L}} E_h) \xrightarrow{\sim} \mathbf{A}_\tau(\mathcal{T}_+) \otimes_A^{\mathbf{L}} E_h \xrightarrow{\sim} [E_h \xrightarrow{g_h} E_h].$$

These isomorphisms use finite evaluation sums and the displayed free/flat resolutions. They are not compact-support base-change assertions for a different arithmetic scheme.

The packet projector is now an actual finite-rank endomorphism of the original $H_\tau^1 = Q$ (and of each of its retained support fibres). For an admissible smooth compactly supported multiplicative test k ,

$$H_k F(x) = \int_0^\infty k(a) F(x/a) \frac{da}{a}$$

commutes with the projector. Its ordinary finite-rank trace is

$$\boxed{\mathrm{Tr}(H_k \Pi_Z | Q) = \sum_{\rho \in Z} r_\rho \mathcal{M}k(\rho).}$$

This is a finite-rank trace on the original cohomology, using its explicit finite invariant image. It is stronger than declaring that a surjective finite observation has a trace. Infinite trace summation still requires the specified topology and admissible test class.

On r -fold products over the tau-point, the already calculated external tensor map gives $H^r = Q^{\otimes r}$. The operator $\Pi_Z^{\otimes r}$ is a finite projector onto $Q[h_Z]^{\otimes r}$. The full nilpotent scaling action there is

$$a^{\rho_1 + \cdots + \rho_r} \exp \left((\log a) \sum_{j=1}^r N_{\rho_j, j} \right).$$

For a repeated exponent the exact modulus comparison remains

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re\rho - 1) \log a.$$

This supplies actual finite invariant objects on which a Deligne-style tensor estimate would have to act. It does not prove that estimate. The source's opposite weight bounds, its global dualizing theory, and the admissibility of its geometric hypotheses have not been inferred from these finite isomorphisms.

S1.1.11 10. Status and publication boundary

Established in this note, with written proofs: the differential range identities on the stated spaces; the exact finite operator quotient and torsion kernel; the derived finite comparison; the analytic torsion-subsheaf and local-cohomology identification with $\mathcal{O}/(2\xi)$; finite spectral direct summands with explicit projectors; the original split lifts and smaller-label kernels; the residue/Jacobian contraction on that comparison; and compatible finite-rank arithmetic traces and tensor actions.

Not claimed: a global injectivity theorem for the full balanced analytic realization; vanishing of $\mathcal{R}_\infty$; identification of the entire four-point chart category with an arithmetic curve; a full compact-support/ordinary-cohomology realization satisfying the hypotheses of Weil II; positivity of the unshifted Weil form; RH or GRH. The finite tests are not a Lean or analytic proof certificate.

The prior base note and this continuation are published as additive research material beside PR #6. The original scalar core, formal modules, frozen reader and all existing branches remain unchanged. New derivations are AI-assisted work under the owner's programme; the structural source and its Lean implementation retain their own attribution. No copyrighted source corpus or private conversation is included.

References and source interfaces

- Owner's SplitZero programme: the original scalar, support fibre and quotient constructions; PR #6 at 5f6ee6575550d8a20c6fc3990359cb4f67855f80, especially SplitZeroHomology.lean, SplitZeroComplex.lean, SplitZeroInternalQuotient.lean, and SplitZeroFiniteJets.lean.
- TAU_BASE_MODEL.md: the actual characteristic-zero chart, cohomology, right-adjoint and tensor constructions used here. It remains an authored research model, not an accepted geometric RH realization.
- Deligne, P., *La conjecture de Weil. II* (1980), sections 3.3.4-3.3.6 and 6.2.1-6.2.7. Selected supplied TeX passages were reread for the intended comparison/duality architecture; no source proof is republished or claimed fully audited.
- Meyer, R., *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. Prior spectral-realization context; no theorem from it is silently used to identify the test spaces here.
- The Stacks Project, Derived tensor product, Tag 06XY. The general bounded-flat-complex construction used in Section 4.

S2 PR 9: Chain descent

Source attribution. Web contribution, pull request #9, [Zeta research repository](#).

Exact revision. 8f99b94d306c3b1aaa817d52c57fa6fd298d511c

Source file. workbenches/tau-chain-descent/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 1ab370cf2db9fce70020bebbab76d3dbd3bd37d39e28c83114eeb0eccfa2f0b3

S2.1 Chain-level arithmetic descent over the tau base

Owner-directed research continuation, 12 September 2026. Draft proofs for review; no new Lean certification or RH claim.

S2.1.1 0. Fixed inputs and publication scope

This is an additive continuation of Zeta PR #8 at 25689c5376166df37801229bc275ff5a896007d9, particularly workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md (Git blob 68f59d098d60708f8e899bb960abbd1135ebb3b4). Its sections 1–5 supply the stated analytic range theorem and the finite packet maps. The new result here is their chain-level descent, its actual extension class, and the connection of that class to residue duality and the arithmetic trace.

The source programme retains

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The absolute pointed base and the coefficient square remain

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the first vertical map remains the infinite ring of integers. Every new arrow below is a coefficient or cochain construction over this marked base. No base field, ideal norm, Euler factor, or completion constant is changed.

The supplied derived note and PR #6 provide `Recovery.recoverEquiv`, `Relations.coequalizer_universal`, the internal-homology coequalizer, `ChainMap.homologyKernelEquiv`, and `homology_int_ertwines`. Their algebraic conclusions are used at their stated types. No existing formal module is modified. Analytic inputs from PR #8 remain written research arguments; they are not promoted to Lean theorems.

S2.1.2 1. The actual arithmetic source and packet

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Retain

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad A = \mathbb{C}[t], \quad t \cdot F = DF.$$

The one-leg tau-chart complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad |V| = 0, \quad |\mathcal{B}| = 1, \quad Q = H^1(C_+).$$

The injectivity of theta gives $H^0(C_+) = 0$. The two-leg complex is

$$C_{+-} = [V \oplus V \longrightarrow \mathcal{B}], \quad (\phi, \psi) \longmapsto \Theta\phi - \Theta\widehat{\psi}.$$

Its source operator is $(D, 1 - D)$; the inclusion $C_+ \rightarrow C_{+-}$ is $\phi \mapsto (\phi, 0)$ and the identity in degree one. It leaves the extra degree-zero copy of V in C_{+-} present.

Use the unchanged arithmetic multiplier

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The input function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

and the established identities are

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s).$$

Fix a nonempty finite packet Z of actual nontrivial zeros, with full orders m_ρ . Set

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = A/(h), \quad v(s) = g(s)/h(s).$$

The specific monic polynomial defines this finite test; it does not rescale g . Write j_h for the complete Taylor-jet/Chinese-remainder map at these centres, and retain the invertible element

$$v_h = j_h v \in E^\times, \quad \varepsilon_Z = v_h^{-1}.$$

For $u \in E$, let $R_Z(u) \in \mathbb{C}[t]_{<d}$ be the unique polynomial representing $\varepsilon_Z u$.

The input range theorem supplies the inverse $S_h^{\mathcal{B}}$ of $h(D)$ on the zero-jet range, and the analogous source inverse S_h^V . For one factor,

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

The stated source inverse uses $0 < \Re \rho < 1$ and the zero moment. Iteration supplies S_h . Define the actual maps

$$J : \mathcal{B} \rightarrow E, \quad JF = j_h \mathcal{M}F,$$

$$s : E \rightarrow \mathcal{B}, \quad s(u) = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*).$$

The inverse is defined because $gR_Z(u)$ has all required zero jets. Directly,

$$\mathcal{M}s(u) = vR_Z(u), \quad Js = 1_E, \quad J\Theta = 0, \quad h(D)s(u) = \Theta(R_Z(u)(D)\phi_*). \quad (1)$$

These equations fix the representative map, including all derivatives of g/h . On Q , the existing projector is $\Pi_Z = \bar{s}\bar{J}$, with image $Q[h(D)]$ and kernel $h(D)Q$.

S2.1.3 2. The exact equivariant extension and its class

Define an actual subspace of the original test-function space:

$$\mathcal{B}_Z = \{F \in \mathcal{B} : [F] \in Q[h(D)]\}.$$

Then

$$\mathcal{B}_Z = \Theta V \oplus s(E) \quad \text{as complex vector spaces,}$$

with maps

$$(\phi, u) \mapsto \Theta\phi + s(u), \quad F \mapsto (\Theta^{-1}(F - sJF), JF). \quad (2)$$

For the inverse, $F - sJF$ has zero class in Q by the packet projector theorem; injectivity of theta makes its source unique. The intersection is zero by $Js = 1$ and $J\Theta = 0$.

The original operator D preserves $\mathcal{B}_Z$, and there is a short exact sequence of A -modules

$$\boxed{0 \longrightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J} E \longrightarrow 0.} \quad (3)$$

The arrows in (3), not just the three modules, specify its extension class. Compute it using the resolution

$$0 \longrightarrow A \xrightarrow{\times h} A \longrightarrow E \longrightarrow 0.$$

The resulting isomorphism $\text{Ext}_A^1(E, V) \cong V/h(D)V$ sends an extension to $h(D)b$ in the source, where b is any lift of $1 \in E$. Fixing this explicit rule fixes its sign. Under the existing map $[\phi] \mapsto j_h H_\phi$, the class of (3) is

$$\boxed{[\mathcal{B}_Z] = \varepsilon_Z = j_h(h/g) \in E^\times.} \quad (4)$$

Proof: use the lift $s(1)$ and (1). Its source boundary is $R_Z(1)(D)\phi_*$, whose H -jet is ε_Z .

In particular, (3) has no A -linear section. This also follows immediately from the injectivity of $h(D)$ on $\mathcal{B}$: an A -linear map $E \rightarrow \mathcal{B}$ has image killed by $h(D)$ and is therefore zero. The map s in (2) is the explicit complex-linear section, and the next section computes exactly how it fails to intertwine the arithmetic action. This non-splitting occurs for every nonempty packet, including critical-line packets; it is not an RH-failure criterion.

The general use of extension classes is standard homological algebra (Stacks, Tags 010I and 06XP). Formula (4) calculates the specific extension arising from this theta complex.

S2.1.4 3. The scaling defect is an original theta boundary

For $P \in A$ and $u \in E$, define

$$q_P(u) = \frac{P(t)R_Z(u) - R_Z(Pu)}{h(t)} \in A, \quad k_P(u) = q_P(u)(D)\phi_* \in V. \quad (5)$$

The numerator is divisible by h , since both of its residues are $P\varepsilon_Z u$. Using (1) and the injectivity of $h(D)$ on $\mathcal{B}$ gives

$$\boxed{P(D)s - sM_P = \Theta k_P.} \quad (6)$$

No quotient has yet been applied in (6). The right side is an actual theta function. The composition law is

$$\boxed{k_{PQ} = P(D)k_Q + k_P M_Q.} \quad (7)$$

It follows either by expanding the numerator of (5), or by expanding the commutator in (6) and using injectivity of theta.

For $P = t$, the numerator of (5) has degree at most d . Hence

$$\ell_Z(u) = [t^{d-1}]R_Z(u), \quad q_t(u) = \ell_Z(u), \quad \boxed{Ds(u) - s(tu) = \ell_Z(u)\Theta\phi_*} \quad (8)$$

Thus, in the exact coordinates (2), the original operator is

$$\boxed{D_{\mathcal{B}_Z} = \begin{pmatrix} D_V & \phi_*\ell_Z \\ 0 & M_t \end{pmatrix}.} \quad (9)$$

Its off-diagonal map is rank one and its coefficient is determined by the retained local arithmetic unit. It is not chosen to improve a spectral estimate.

For one simple zero,

$$h = t - \rho, \quad \varepsilon_Z = 1/g'(\rho),$$

$$s(1) = \frac{1}{g'(\rho)} x^{-\rho} \int_x^\infty \Theta\phi_*(y) y^{\rho-1} dy,$$

$$\boxed{(D - \rho)s(1) = \frac{1}{g'(\rho)} \Theta\phi_*} \quad (10)$$

The coefficient is $1/(2\xi'(\rho))$, with the original factor two retained.

The full dilation group For $a > 0$, retain $U_a F(x) = F(x/a)$ and $A_a = M_{j_h(a^s)}$ on E . Define

$$k_a(u) = S_h^V (U_a(R_Z(u)(D)\phi_*) - R_Z(A_a u)(D)\phi_*). \quad (11)$$

The source of S_h^V has zero H -jets because $a^s \varepsilon_Z u$ and $\varepsilon_Z A_a u$ have equal jets. All powers use the real logarithm of a . The exact identities are

$$U_a s - s A_a = \Theta k_a, \quad k_{ab} = U_a k_b + k_a A_b, \quad k_1 = 0. \quad (12)$$

The same construction applies to a smooth compactly supported multiplicative test f : replace U_a by $H_f = \int f(a) U_a da/a$ and A_a by $M_{j_h \mathcal{M} f}$. The source integral belongs to V ; its H -jets give the same cancellation. These are actual linear maps, not a choice of a formal matrix on E alone.

S2.1.5 4. A derived equivariant inclusion with its full homotopy

Put $P_Z = sJ : \mathcal{B} \rightarrow \mathcal{B}$. It is a finite-rank idempotent, and $P_Z\Theta = 0$. Consequently

$$\mathfrak{p}_Z = (0, P_Z) : C_+ \rightarrow C_+ \quad (13)$$

is a cochain idempotent. Its image is the two-term complex $[0 \rightarrow s(E)]$, complex-linearly isomorphic to $E[-1]$.

For the operator $P(D)$ on C_+ , define a degree-minus-one map with sole component

$$\mathbf{H}_P^1 = k_P J : \mathcal{B} \rightarrow V.$$

Equation (6) gives

$$\boxed{P(D)\mathfrak{p}_Z - \mathfrak{p}_Z P(D) = d\mathbf{H}_P + \mathbf{H}_P d.} \quad (14)$$

In degree one this is $\Theta k_P J$; in degree zero it is zero because $J\Theta = 0$. For dilation use $\mathbf{H}_a^1 = k_a J$. Equation (12) is the coherence law of those homotopies.

There is also a strictly A -linear derived presentation. Set

$$C_Z = [V \xrightarrow{\Theta} \mathcal{B}_Z].$$

In the coordinates (2) its differential is $\phi \mapsto (\phi, 0)$ and its action is (9). The maps

$$\iota : C_Z \hookrightarrow C_+, \quad \pi : C_Z \rightarrow E[-1], \quad \pi^0 = 0, \pi^1 = J \quad (15)$$

are strictly A -linear cochain maps, and π is a quasi-isomorphism. Thus the actual equivariant inclusion is the roof

$$\boxed{E[-1] \xleftarrow[\simeq]{\pi} C_Z \xrightarrow{\iota} C_+.} \quad (16)$$

The complex-linear deformation retraction also has an exact contracting homotopy: on $\mathcal{B}_Z$, send $\Theta\phi + s(u)$ to $\phi \in V$. Its equation is $1 - s\pi = dH + Hd$ (with s interpreted as the degree-one cochain section). This homotopy is not claimed A -linear; the A -linear replacement is the roof (16).

The strictly A -linear map $j : C_+ \rightarrow E[-1]$ is J in degree one and zero in degree zero. It satisfies $j\iota = \pi$. This gives a split retract in $D(A)$ without falsely asserting an A -linear section into $\mathcal{B}$.

On the two-leg complex use $\mathfrak{p}_Z^0 = 0$ on $V \oplus V$, $\mathfrak{p}_Z^1 = P_Z$, and $\mathbf{H}_a^1(F) = (k_a JF, 0)$. The source dilation is $(U_a, aU_{1/a})$, so (14) holds with its original two-leg differential. The inclusion $C_+ \rightarrow C_{+-}$ commutes with these cochain projectors. The extra degree-zero cohomology of C_{+-} is killed by this projector; it remains in the complementary complex.

S2.1.6 5. The mixed-support homotopies require the actual synchronization map

Let $L = \mathcal{P}(\{+, -\})$ be the original chart-support lattice. Its three active complexes are

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad C_- = [V \xrightarrow{-\Theta\mathcal{F}} \mathcal{B}], \quad C_{+-} = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}].$$

Here $\mathcal{F}$ is the specified Fourier transform, with $\mathcal{F}^2 = 1$ on even functions. The source inclusions into the joint fibre are $v \mapsto (v, 0)$ and $v \mapsto (0, v)$; degree-one transports are identities. The bottom complex is zero in this particular model. The new formalization allows a nonzero bottom fibre in the general reconstruction; no assertion about all diagrams is obtained by specializing this model.

The reconstructed operations are

$$(\lambda, x) + (\mu, y) = (\lambda \vee \mu, \rho_{\lambda, \lambda \vee \mu} x + \rho_{\mu, \lambda \vee \mu} y),$$

$$e(\lambda, x) = (\lambda, 0), \quad \tau(\lambda, x) = (\perp, 0). \quad (17)$$

The cochain projector $\tilde{p}_Z$ is zero-in-its-fibre in degree zero and P_Z in degree one. It commutes with the support transports. So does the actual action: U_a on the plus source, $aU_{1/a}$ on the minus source, their direct sum on the joint source, and U_a at every active overlap.

The objectwise homotopies are, however, different maps. With $b = k_a JF$, they are

$$H_{a,+}^1(F) = b, \quad H_{a,-}^1(F) = -\hat{b}.$$

Their images in the joint source are $(b, 0)$ and $(0, -\hat{b})$. The difference is the explicit cycle

$$(b, \hat{b}) \in \ker d_{+-} = H^0(C_{+-}). \quad (18)$$

For nonzero k_a , there is no label-preserving family of homotopies compatible with both source inclusions: the joint component would have to equal both $(b, 0)$ and $(0, -\hat{b})$. This says exactly which naturality square fails. It does not discard the cohomology or the arithmetic action.

The replacement is the programme's actual synchronization. Define the support map

$$r_L(\perp) = \perp, \quad r_L(\lambda) = \{+, -\} \quad (\lambda \neq \perp).$$

In degree zero use literal multiplication by $E = (1, e)$ on the original two-coordinate support module:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea).$$

In degree one define $\mathfrak{r}^1(\lambda, F) = (r_L \lambda, F)$ for active labels and preserve global zero. An absent source coordinate becomes its supported vector zero, so the source differential has unchanged amplitude. Hence

$$d\mathfrak{r}^0 = \mathfrak{r}^1 d, \quad \mathfrak{r}^2 = \mathfrak{r}. \quad (19)$$

This uses the operation verified in the synchronization attachment, now on the actual complex. The fixed active fibre is exactly C_{+-} . On degree-one homology, the map is

$$(\lambda, [F]) \mapsto (\{+, -\}, [F]).$$

It has no nonzero amplitude kernel, because all degree-one transports are the identity of Q . The original mask is retained through

$$x \mapsto (\text{supp}(x), \mathfrak{r}x).$$

This map is injective in each degree of this chart complex: in degree zero the coordinate inclusions are injective, and in degree one the transport is the identity. This injectivity is a proved property of these maps, not an assumption for general diagrams with noninjective transports.

There are now two actual split-linear homotopies with the same support map r_L . Define

$$\tilde{H}_a^{\text{left},1}(\lambda, F) = (\{+, -\}, (k_a JF, 0)),$$

$$\tilde{H}_a^{\text{right},1}(\lambda, F) = (\{+, -\}, (0, -\mathcal{F}k_a JF)),$$

and send global zero to global zero. Both satisfy

$$\tau(\tilde{U}_a \tilde{p}_Z + (-1)^{\bullet} \tilde{p}_Z \tilde{U}_a) = \tilde{d} \tilde{H}_a + \tilde{H}_a \tilde{d}. \quad (20)$$

In degree one this is (6); in degree zero it follows from $Jd = 0$. The two choices differ by the retained cycle (18). No averaging or removal of that cycle is performed. This is the precise role of synchronization here: it permits a common supported comparison fibre for the homotopy, while the original mask and the difference of homotopies remain recorded.

In the simple-zero case (10), the defect in any source support is the nonzero chain amplitude $\Theta(\phi_*/g'(\rho))$. Its internal homology class is the zero of that support, not external absence.

For every chain comparison $f : C \rightarrow C'$, the supplied formalized kernel equivalence remains

$$\{z \in Z(C) : f(z) \in B(C')\} / B(C) \xrightarrow{\sim} \ker H(f).$$

Applied to π in (15), it computes the zero kernel in degree one from its actual exact sequence. Applied to support synchronization, it yields the degree-one assertion just proved and the separately retained degree-zero data. It is not applied to a different global comparison without carrying that comparison's maps.

For any smaller D -stable admitted source $W \subset V$, the map $Q_W \rightarrow Q$ has kernel $\Theta V / \Theta W$. A homotopy value may not already lie in W . The explicit replacement is enlargement by the actual source values, with this quotient map and its kernel retained. The present chart uses full V , where every displayed homotopy is defined.

S2.1.7 6. The same extension class produces the arithmetic duality factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$. Then

$$f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad h^\dagger = (-1)^d h.$$

The perfect finite pairing is

$$R_h(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) u(s)}{h(s)} ds,$$

with exact symmetry $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$.

Identify E with the actual divisor algebra $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g)$ by polynomial evaluation and the complete Chinese-remainder jets. The nonzero local units in g/h make this an isomorphism. The operator represented by the extension class is

$$M_{\varepsilon_Z} : E \rightarrow E.$$

Substituting (4) into the residue formula gives

$$R_h(f, M_{\varepsilon_Z} u) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger u}{g} ds = R_Z(f, u). \quad (21)$$

This is independent of representatives: changing ε_Z by h times a germ adds a holomorphic differential. Thus the actual extension class is precisely the unit morphism converting the polynomial-resolution residue form into the arithmetic residue form.

Let $J_g = M_{j_h g'}$. Then

$$\boxed{R_h(f, M_{\varepsilon_Z} J_g u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho) = \text{Tr}(M_{f^\dagger u} \mid E).} \quad (22)$$

Locally $g = z^m w(z)$ gives $g'/g = m/z + w'/w$. This proves the formula and retains every multiplicity. The product $\varepsilon_Z J_g$ carries both factors associated to $g = 2\xi$.

The pairing-valued support map sends a zero value to $e_{\mathbb{C}}$ and an absent argument to τ . Its original input support and relation fibre remain available upstream. Formula (22) is a calculated trace, not a positivity assertion.

Reflection is strict for this specified section Define conjugate-linear involutions

$$j_B F(x) = x^{-1} \overline{F(1/x)}, \quad j_V \phi = \bar{\phi}, \quad j_E u = u^\dagger.$$

They satisfy $\Theta j_V = j_B \Theta$ and $J j_B = j_E J$. The actual source function is Fourier-fixed: writing $\phi_* = D(D-1)e^{-\pi x^2}$, Fourier sends the two factors to $(1-D)(-D) = D(D-1)$ and fixes the Gaussian. It is real, so $j_V \phi_* = \phi_*$.

Since $\varepsilon_Z^\dagger = (-1)^d \varepsilon_Z$ and reflection preserves polynomial degree,

$$R_Z(j_E u) = (-1)^d (R_Z(u))^\dagger.$$

Uniqueness of the inverse of $h(D)$ on its range gives

$$j_B S_h^{\mathcal{B}} = (-1)^d S_h^{\mathcal{B}} j_B.$$

Both factors $(-1)^d$ are retained in the substitution, and their product is one. Therefore

$$\boxed{j_B s = s j_E, \quad j_B P_Z = P_Z j_B.} \quad (23)$$

This is an actual equality for this section; it would not follow for an arbitrary choice of representatives. Its compatibility with the nonzero dilation homotopy is

$$\boxed{j_V k_a = a k_{1/a} j_E.} \quad (24)$$

Indeed, apply j_B to (12), use $j_B U_a = a U_{1/a} j_B$ and (23), then use injectivity of theta. The split lift is semilinear through $r^\bullet \mapsto \bar{r}^\bullet$, which fixes both e and τ .

S2.1.8 7. A finite-rank Lefschetz trace on the actual chain complex

For $f \in C_c^\infty(\mathbb{R}_{>0})$, let U_f be its convolution action on C_+ . The cochain endomorphism

$$p_Z U_f = (0, s J H_f)$$

has finite-rank components. Its cochain supertrace is therefore defined without an infinite-dimensional regularization:

$$\boxed{\text{Str}(p_Z U_f \mid C_+) = -\text{Tr}(s J H_f \mid \mathcal{B}) = -\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho).} \quad (25)$$

The finite-rank identity $\text{Tr}(AB) = \text{Tr}(BA)$ reduces the middle expression to $\text{Tr}(J H_f s \mid E)$; this is multiplication by $j_h \mathcal{M}f$. Equation (25) equals the supertrace of the induced operator on homology.

On C_{+-} its degree-zero contribution is still zero, since the same cochain projector has zero degree-zero component.

The boundary representatives in (6), (14), and (20) are retained even though their contribution to homology is zero. The finite-rank trace does not delete their complex or assign its zero to external absence.

For the n -fold tensor complex over the tau-base product, use the original coefficient tensor over $\mathbb{C}$ and its differential

$$d(x_1 \otimes \cdots \otimes x_n) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_n.$$

The projector $\mathfrak{p}_Z^{\otimes n}$ has image $E^{\otimes n}[-n]$. For independently specified tests $f_1, \dots, f_n$,

$$\text{Str} \left(\bigotimes_j (\mathfrak{p}_Z \cup_{f_j}) \right) = (-1)^n \prod_j \sum_{\rho \in Z} m_\rho \mathcal{M} f_j(\rho). \quad (26)$$

Let $A = U_a \mathfrak{p}_Z$ and $B = \mathfrak{p}_Z U_a$ as cochain maps. A homotopy for $A^{\otimes n} - B^{\otimes n}$ is

$$\sum_{j=1}^n B^{\otimes(j-1)} \otimes H_a \otimes A^{\otimes(n-j)},$$

with the displayed Koszul sign on homogeneous inputs when the degree-minus-one map crosses the preceding factors. The telescoping identity and (14) prove it. The full support mask maps to the product of its input masks; vanishing products of represented amplitudes stay supported.

S2.1.9 8. What now enters the Weil II programme

The structural base, the infinite quotient, and the actual theta complex are unchanged. The new objects are the extension (3), its unit class (4), its synchronized homotopy (20), its strict equivariant derived roof (16), and the chain trace (25). The arithmetic residue unit in (21) is now supplied by that very extension rather than appended without its source.

Deligne's inspected comparison-image argument (Weil II 3.3.4–3.3.6) requires two weight bounds on the same cohomological image. The inspected derived formulation (6.2.1–6.2.7) retains the dualizing operation, its degree and Tate shifts, and its exchange of direct-image functors. This continuation supplies coherent arithmetic maps and finite trace operations for the tau model; it does not transfer the finite-field hypotheses or assert those two bounds.

The retained error is explicit: it is the original boundary Θ_{k_a} on representatives, with the nonzero Ext class (4). It is removed at homology by the original supported quotient, not by making its input absent. A geometric comparison can now be tested for compatibility with this class and the residue map (21). A complex-linear section that ignores (6) fails that test.

No infinite sum of packet projectors is asserted to converge. The existing meromorphic/kernel summand from PR #8 remains in its own exact sequence, and a different global trace functor must retain or analyze its contribution. The source's finite-jet isomorphisms and the new finite-rank chain trace do not by themselves set that summand to zero.

S2.1.10 9. Verification and references

The new finite checker tests polynomial remainder identities, the extension class, rank-one scaling defects, cochain homotopies, dilation cocycles, residue contraction, tensor signs, trace identities, and

original support operations. These finite algebraic models are not numerical tests of zeta zeros and not certificates for the analytic range theorem. No Lean execution is claimed for this continuation.

Read source interfaces:

- PR #8 at the pinned revision above, `FINITE_OPERATOR_COMPARISON.md`, sections 1–10; `TAU_BASE_MODEL.md` is its already-published model.
- Owner’s `SplitZero_Derived_Mathematics_2026-09-12(1).md`, sections 1–7. In particular its allowance of a nonzero bottom fibre and the exact homology-kernel/action theorems are retained.
- `formal/splitzero/SplitZeroHomology.lean`, lines 175–end, read at the same revision; blob `7b9e5c49a494dd0c751133e0234b3e01048f7b67`.
- Supplied Deligne TeX excerpts: French segment 2, original lines 882–950; segment 3, original lines 675–756. The used equations are the image bounds and duality operations, not every assertion in the historical transcription. No source corpus is republished with this contribution.
- The Stacks Project, Tags 010I and 06XP, for extension classes and their projective-resolution interpretation.
- R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, `arXiv:math/0412277`, for prior spectral-realization context. No global priority claim is made here.

New calculations are AI-assisted work under the owner’s programme. Existing mathematical files, formalizations, licensing, and source attribution remain untouched.

S3 PR 10: Global retraction

Source attribution. Web contribution, pull request #10, [Zeta research repository](#).

Exact revision. d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7

Source file. workbenches/tau-global-retraction/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859

S3.1 A global retraction of the original theta complex over the tau base

Owner-directed continuation of Zeta PR #9, 12 September 2026. New derivations are submitted as research proofs for review, not as Lean-certified analytic theorems.

S3.1.1 0. Fixed source, result, and provenance

The base is PR #9 at 8f99b94d306c3b1aaa817d52c57fa6fd298d511c; its chain-descent note has Git blob 86db838e79d10a8fe7d4376303706c4ea64570a1. Its source and the owner's derived SplitZero attachment remain the inputs. This continuation constructs an inverse on the entire original theta image, not only on a chosen finite spectral packet.

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The first vertical target is still infinite. The geometric base, the original additive laws, and the arithmetic quotient are unchanged. The auxiliary cutoffs below are parameters of an explicit continuous representative map; they do not change an Euler factor, the theta relation space, or the completed zeta function.

Result. For every pair of the cutoffs specified in section 3 there is an explicitly constructed continuous complex-linear map

$$\Lambda_{\alpha,\beta} : \mathcal{B} \longrightarrow V, \quad \boxed{\Lambda_{\alpha,\beta} \Theta = \operatorname{id}_V}. \quad (2)$$

Consequently the actual image ΘV is closed and complemented in the stated strong-Schwartz topology. The original quotient $Q = \mathcal{B}/\Theta V$ is Hausdorff. The note constructs its global section, the full supported cochain homotopy, the exact change of the finite sections in #9, its continuous dual sequence, and the completed tensor powers. The proof does not assume RH, simplicity of any zero, or vanishing of the separate balanced-Mellin kernel.

Spectral realizations on nuclear Frechet spaces are prior work, notably R. Meyer, arXiv:math/0412277. That paper's abstract was consulted for scope, but no theorem from an unread body of that paper is imported. The proof below is given for the workbench's precise spaces. No global novelty claim is made.

S3.1.2 1. Spaces and all transforms

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) := \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \quad (3)$$

These seminorms give $\mathcal{B}$ its Frechet topology; under $x = e^y$ they are the weighted uniform derivative seminorms on smooth functions of y . Completeness follows by uniform convergence on compact sets of every derivative, followed by the weighted bounds. V has its closed-subspace Schwartz topology.

The original maps are

$$\begin{aligned} \Theta\phi(x) &= \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{F}\phi(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \mathcal{F}^{-1}u(x) = \int_{\mathbb{R}} u(\xi) e^{2\pi i x \xi} d\xi, \\ JF(x) &= x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (4)$$

Poisson summation, with both source moments zero, gives

$$\Theta\mathcal{F} = J\Theta, \quad J^2 = 1, \quad DJ = J(1 - D). \quad (5)$$

The theta sum and each derivative converge absolutely on closed subsets of $x > 0$. At infinity, Schwartz estimates give all $p_{b,k}$ bounds. At zero use (5) and the same estimates on $\hat{\phi}$. This proves continuity $\Theta : V \rightarrow \mathcal{B}$. Also J is continuous on $\mathcal{B}$: $p_{b,k}(JF)$ is bounded by the binomial sum of $p_{1-b,j}(F)$, $0 \leq j \leq k$.

The original complex is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, over $A = \mathbb{C}[t]$ with t acting as D . The two-leg tau chart retains $C_{+-} = [V^2 \rightarrow \mathcal{B}]$ with differential $\Theta(\phi - \hat{\psi})$ and source action $(D, 1 - D)$.

S3.1.3 2. Arithmetic inversion on the two exterior regions

Let μ be the ordinary Mobius function, characterized by

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For $F \in \mathcal{B}$ define, for $x \neq 0$,

$$\boxed{\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|)}. \quad (6)$$

The factor $1/2$ is the inverse of the full nonzero-integer theta sum in (4). The map is complex-linear into smooth even functions on $\mathbb{R} \setminus \{0\}$. It is not asserted to be smooth at zero for arbitrary F .

Here are the required estimates. For $j \geq 0$, let $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, with $P_0 = 1$. Then $x^j \partial_x^j = P_j(D)$. If $C_{N,j}(F)$ is the sum of the $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (7)$$

Indeed each summand is bounded by $C_{N,j}(F) n^{-N} x^{-N-j}/2$. This proves convergence of every derivative uniformly for $|x| \geq \delta > 0$, and gives rapid decay there. It also proves continuity of the cut-off exterior maps used below.

For every $\phi \in V$,

$$\boxed{\mathcal{U}\Theta\phi(x) = \phi(x), \quad \mathcal{U}J\Theta\phi(x) = \widehat{\phi}(x) \quad (x \neq 0).} \quad (8)$$

For the first equality the double series is absolutely convergent for each $x > 0$: its absolute value is bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which is finite by Schwartz decay. Rearrange by $k = mn$ and use the displayed divisor identity. Evenness handles negative x . The second equality follows from (5) and the first equality applied to $\widehat{\phi} \in V$.

In the split lift, every $\mu(n)$ is the supported integer $\mu(n)^\bullet$, including $\mu(n) = 0$. A convergent sum within a specified active fibre begins at that fibre's zero and is taken in that fibre's vector-space topology. Formula (8) thus lifts to a support-preserving equality. A computed cancellation is not external absence.

S3.1.4 3. The reconstruction operator, with the strict contraction proved

Choose and retain real, even $\alpha, \beta \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq \alpha, \beta \leq 1$ and equal to 1 on neighborhoods of zero. Their supports and all values are part of the auxiliary data. Work on $\mathcal{H} = L^2(\mathbb{R}, dx)$ with the Fourier convention in (4).

Define bounded operators

$$A_\alpha = M_\alpha, \quad B_\beta = \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad T_{\alpha,\beta} = A_\alpha B_\beta. \quad (9)$$

Lemma. $T_{\alpha,\beta}$ is compact and $c_{\alpha,\beta} := \|T_{\alpha,\beta}\| < 1$.

Proof. Its kernel is $\alpha(x)\check{\beta}(x-y)$, whose squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$. Hence it is Hilbert-Schmidt. Both factors are contractions. If the norm were 1, compactness would supply a unit vector f attaining it. Equality in

$$\|A_\alpha B_\beta f\| \leq \|B_\beta f\| \leq \|f\|$$

forces $\widehat{f}$ to be supported where $\beta = 1$; there $B_\beta f = f$. The first equality forces f to be supported where $\alpha = 1$. Both supports are compact. The inverse Fourier transform of compactly supported $\widehat{f} \in L^2$ is entire, by differentiation under its finite-interval integral. Its continuous restriction agrees with the compactly supported f , so it vanishes on an open real interval and hence identically. This contradicts $\|f\| = 1$. Thus $c_{\alpha,\beta} < 1$.

Consequently the exact Neumann inverse exists:

$$\boxed{(1 - T_{\alpha,\beta})^{-1} = \sum_{k \geq 0} T_{\alpha,\beta}^k, \quad \|(1 - T_{\alpha,\beta})^{-1}\| \leq \frac{1}{1 - c_{\alpha,\beta}}.} \quad (10)$$

No uniform bound as the cutoffs vary is claimed. In particular the gap $1 - c_{\alpha,\beta}$ is retained, not replaced by a positive universal constant.

The operator $T_{\alpha,\beta}$ maps L^2 continuously to the Schwartz space. For each derivative, compact Fourier support and Cauchy-Schwarz bound $\partial^j B_\beta f$ uniformly by a constant times $\|f\|_2$; multiplication by the smooth compactly supported α gives every Schwartz seminorm.

Define the two exterior data and their combination:

$$a_F = (1 - \alpha)\mathcal{U}F, \quad b_F = (1 - \beta)\mathcal{U}JF,$$

$$Y_{\alpha,\beta}F = a_F + \alpha\mathcal{F}^{-1}b_F. \quad (11)$$

The exterior terms are extended by zero near the origin, where their cutoff is identically zero. Estimates (7) prove that both are even Schwartz functions and depend continuously on F . Therefore $Y : \mathcal{B} \rightarrow \mathcal{S}(\mathbb{R})^{\text{even}}$ is continuous.

Put

$$\Phi_{\alpha,\beta}F = (1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta}F. \quad (12)$$

This initially defines an L^2 vector, but the identity $\Phi F = YF + T\Phi F$ proves it is Schwartz. If q is any Schwartz seminorm and $q(Tu) \leq C_q\|u\|_2$, then

$$q(\Phi F) \leq q(YF) + \frac{C_q}{1 - c_{\alpha,\beta}}\|YF\|_2. \quad (13)$$

This is an explicit continuity estimate. Both sides are even because the cutoff operators preserve parity.

For $F = \Theta\phi$, equations (8) give

$$Y\Theta\phi = (1 - A_\alpha)\phi + A_\alpha(1 - B_\beta)\phi = (1 - T_{\alpha,\beta})\phi,$$

hence

$$\boxed{\Phi_{\alpha,\beta}\Theta\phi = \phi.} \quad (14)$$

S3.1.5 4. Keep both original moment conditions

For arbitrary F , (12) need not satisfy the two source moments. The correction is the following specified projection, not an alteration of V .

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int \phi)$ are $(1, 1)$ and $(0, 1)$. Define

$$\mathbf{m}(\phi) = (\phi(0), \int_{\mathbb{R}} \phi), \quad j_0(a, b) = a\gamma_0 + (b - a)\gamma_2,$$

$$P_V\phi = \phi - j_0\mathbf{m}(\phi) = \phi - \phi(0)\gamma_0 - \left(\int \phi - \phi(0)\right)\gamma_2. \quad (15)$$

Then $\mathbf{m}j_0 = 1$, $P_V^2 = P_V$, $\text{im } P_V = V$, and $P_V|_V = 1$. All maps are continuous.

The promised map is

$$\boxed{\Lambda_{\alpha,\beta} = P_V(1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta} : \mathcal{B} \rightarrow V.} \quad (16)$$

Equation (14) and $P_V|_V = 1$ prove (2). The discarded moment vector has the recorded map $m\Phi : \mathcal{B} \rightarrow \mathbb{C}^2$ and the explicit correction $j_0 m\Phi$. It vanishes on the actual theta image. No endpoint constant was absorbed into the zeta function.

S3.1.6 5. The full quotient is closed and has a continuous section

Write $\Lambda = \Lambda_{\alpha,\beta}$, and define

$$P_\Theta = \Theta\Lambda, \quad K = 1_{\mathcal{B}} - \Theta\Lambda. \quad (17)$$

Direct substitution gives

$$P_\Theta^2 = P_\Theta, \quad K^2 = K, \quad \ker K = \Theta V, \quad \text{im } K = \ker \Lambda. \quad (18)$$

Since K is continuous, its kernel is closed. Thus $Q = \mathcal{B}/\Theta V$ with its actual quotient topology is Frechet and Hausdorff. If $q : \mathcal{B} \rightarrow Q$ is the original quotient map, the continuous section is

$$\boxed{s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0.} \quad (19)$$

The explicit topological direct-sum isomorphism is

$$\boxed{V \oplus Q \xrightarrow{\sim} \mathcal{B}, \quad (\phi, u) \mapsto \Theta\phi + su, \quad F \mapsto (\Lambda F, qF).} \quad (20)$$

For a convergent sequence $\Theta\phi_n \rightarrow F$ in $\mathcal{B}$, one can check the closedness without any abstract theorem: $\phi_n = \Lambda\Theta\phi_n \rightarrow \Lambda F$ in V , so $F = \Theta\Lambda F$ by continuity of theta.

This computes the earlier Hausdorff arrow:

$$Q^{\text{alg}} = \mathcal{B}/\Theta V \longrightarrow Q^{\text{H}} = \mathcal{B}/\overline{\Theta V}, \quad [F] \mapsto [F], \quad (21)$$

as an isomorphism with zero amplitude kernel. It does not say the map from Q into its product of spectral jets is injective. The latter map and the balanced analytic map still have their own named kernels:

$$Q \rightarrow \prod_{\rho} \mathcal{O}_{\rho}/(g), \quad \mathcal{O} \otimes_{\mathbb{C}[t]} Q \xrightarrow{\beta} \mathcal{O}/(g).$$

No claim that $\mathcal{R}_{\infty}$ or $\ker \beta$ vanishes is inferred from (21).

Smaller admitted relations For any specified subspace $W \subset V$, retain

$$Q_W = \mathcal{B}/\Theta W \longrightarrow Q, \quad \ker = \Theta V/\Theta W \cong V/W. \quad (22)$$

The isomorphism sends $[\phi] \mapsto [\Theta\phi]$. It is algebraic for arbitrary W ; when W is closed, (20) makes it a topological isomorphism onto the kernel. The pullback of the global section has the representative relation kernel in (22); the full-source result is not silently asserted with W in place of V .

S3.1.7 6. Perform the original internal quotient and synchronization

On the single-leg map, or a fixed-index diagram of that same map, the split lifts are

$$\tilde{\Theta}(\lambda, \phi) = (\lambda, \Theta\phi), \quad \tilde{\Lambda}(\lambda, F) = (\lambda, \Lambda F), \quad \tilde{K}(\lambda, F) = (\lambda, KF). \quad (23)$$

They preserve external bottom and commute with scalar reflection. In particular

$$\tilde{K}\tilde{\Theta}(\lambda, \phi) = (\lambda, 0_{\mathcal{B}}) = z_{V, \mathcal{B}}(\lambda, \phi).$$

Here $z_{V, \mathcal{B}}$ is the label-preserving zero map from the source fibre to the target fibre. The mixed chart, whose minus restriction is $-\Theta\mathcal{F}$, is treated separately by the explicit maps (25)-(28), without changing that restriction. The actual chart has zero bottom fibre; a general nonzero bottom fibre is acted on by its stated linear map, rather than identified with the single global zero.

The supported image projection is idempotent, and the quotient map is exactly the coequalizer of the theta relation inclusion and its supported-zero companion. The implementation's universal property applies against all split-linear targets. A theta boundary is sent to $e_\lambda = (\lambda, 0)$; it is not sent to external absence.

For the single-leg complex there is an actual continuous contraction

$$C_+ \xrightleftharpoons[s]{q} Q[-1], \quad H^1 = \Lambda,$$

$$\boxed{qs = 1, \quad 1_{C_+} - sq = dH + Hd.} \quad (24)$$

In degree zero this is $\Lambda\Theta = 1_V$; in degree one it is $\Theta\Lambda = 1 - K$. In split notation the subtraction means addition of the supported scalar $(-1)^\bullet$ times the second map. It yields the correct labelled zero wherever the amplitude vanishes.

For the two-leg chart set $H_0 = V[0] \oplus Q[-1]$ with zero differential, and define

$$p_L^0(\phi, \psi) = \psi, \quad p_L^1 = q, \quad i^0(\psi) = (\hat{\psi}, \psi), \quad i^1 = s,$$

$$H_L^1(F) = (\Lambda F, 0). \quad (25)$$

Then $p_L i = 1$, and $1 - ip_L = dH_L + H_L d$. The degree-zero identity is $(\phi, \psi) - (\hat{\psi}, \psi) = (\phi - \hat{\psi}, 0)$. The other leg gives

$$p_R^0(\phi, \psi) = \hat{\phi}, \quad p_R^1 = q, \quad H_R^1(F) = (0, -\widehat{\Lambda F}). \quad (26)$$

It obeys $1 - ip_R = dH_R + H_R d$. Their homotopies differ by

$$\boxed{H_L^1 F - H_R^1 F = (\Lambda F, \widehat{\Lambda F}),} \quad (27)$$

the actual joint degree-zero cycle, which is retained.

On the complete original support diagram, use literal multiplication by $E_{\text{sync}} = (1, e)$ in degree zero and the source's active-label-to-joint map in degree one:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea), \quad \mathfrak{r}^1(\lambda, F) = (\{+, -\}, F) \quad (\lambda \neq \perp).$$

The absent input remains absent. The original differential obeys $d\mathfrak{r} = \mathfrak{r}d$ and $\mathfrak{r}^2 = \mathfrak{r}$. Consequently (25) lifts to the fully typed equation

$$\mathfrak{r} - \tilde{i}\tilde{p}_L\mathfrak{r} = \tilde{d}\tilde{H}_L + \tilde{H}_L\tilde{d}, \quad (28)$$

where $\tilde{H}_L(\lambda, F) = (\{+, -\}, (\Lambda F, 0))$ for each active label. The right-leg equation is analogous with (26). The original mask is recorded with the injection $x \mapsto (\text{supp } x, \mathfrak{r}x)$ already proved for this chart. Neither synchronization nor the contraction identifies the three supported origins with τ .

S3.1.8 7. Global scaling defects, and their agreement with the finite extensions

Let $U_a F(x) = F(x/a)$, $a > 0$, and denote its induced continuous action on Q by $\overline{U}_a$. The section (19) gives a global continuous boundary map

$$k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\overline{U}_a = \Theta k_a. \quad (29)$$

The proof applies $K + \Theta\Lambda = 1$ to $U_a s$, then uses $qU_a s = \overline{U}_a$. Therefore

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (30)$$

The same calculation works for D and compactly supported multiplicative convolution, with $k_D = \Lambda D s$. In the explicit coordinates (20),

$$U_a|_{\mathcal{B}} = \begin{pmatrix} U_a|_V & k_a \\ 0 & \overline{U}_a \end{pmatrix}, \quad D|_{\mathcal{B}} = \begin{pmatrix} D|_V & k_D \\ 0 & D_Q \end{pmatrix}. \quad (31)$$

Thus the topological splitting is not substituted for an equivariant splitting. Equation (29) is the precise comparison between them.

For two cutoff choices, let s_0, s_1 be their sections and set $b_{10} = \Lambda_0 s_1 : Q \rightarrow V$. Then

$$s_1 - s_0 = \Theta b_{10}, \quad k_a^{(1)} = k_a^{(0)} + U_a b_{10} - b_{10} \overline{U}_a. \quad (32)$$

All dependence on the chosen representative map is carried by this displayed coboundary. No canonical cutoff or average over choices is declared.

Restrict back to PR #9 without changing its extension class Use its existing maps $s_Z : E_Z \rightarrow \mathcal{B}$, $\sigma_Z = q s_Z : E_Z \rightarrow Q$, and $J_Z : \mathcal{B} \rightarrow E_Z$. Put $b_Z = \Lambda s_Z$. The new continuous representative of the same finite arithmetic class is

$$s_Z^{\text{new}} = s \sigma_Z = K s_Z = s_Z - \Theta b_Z. \quad (33)$$

If $P(D)s_Z - s_Z M_P = \Theta k_{P,Z}$, direct substitution gives

$$k_{P,Z}^{\text{new}} = k_{P,Z} - P(D)b_Z + b_Z M_P. \quad (34)$$

For $P = h_Z$, this changes the source boundary by $-h_Z(D)b_Z$. It is zero in $V/h_Z(D)V$, so the extension class remains exactly

$$\varepsilon_Z = j_{h_Z}(h_Z/g), \quad g = 2\xi. \quad (35)$$

The polynomial residue form still becomes the arithmetic one by M_{ε_Z} ; its Jacobian contraction remains $M_{\varepsilon_Z} M_{j_h g'}$. No unit, derivative, sign, or factor of 2 is changed.

The two finite cochain projectors differ by the explicit homotopy $-b_Z J_Z$ in degree minus one. Their finite-rank trace difference is zero because $J_Z \Theta = 0$. Hence their cochain Lefschetz trace is the unchanged

$$-\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (36)$$

S3.1.9 8. Continuous duality and completed tensor powers of the full object

Use continuous duals, denoted E'_b with the strong topology. Transpose the actual topological splitting (20). It gives

$$\boxed{0 \rightarrow Q'_b \xrightarrow{q'} \mathcal{B}'_b \xrightarrow{\Theta'} V'_b \rightarrow 0, \quad \Theta' \Lambda' = 1.} \quad (37)$$

The continuous cochain-dual complex is $[\mathcal{B}'_b \xrightarrow{\Theta'} V'_b]$ in degrees $-1, 0$. Its cohomology is Q'_b in degree -1 and zero in degree 0 . The transpose contraction proves this directly; no general exactness theorem for arbitrary locally convex quotients is assumed.

The actual finite residue map into that continuous dual is

$$\overline{E_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto (u \mapsto R_Z(f, j_Z u)), \quad (38)$$

where $j_Z q = J_Z$. Continuity follows from the finite Mellin jets and finite dimensionality of E_Z . It is injective by their surjectivity and perfect residue duality. Multiplication by g' then gives the arithmetic finite trace pairing, with its previously calculated radical retained.

Let $\widehat{\otimes}_\pi$ denote completed projective tensor product. The comparison from the previous algebraic tensor is its canonical completion map. All chain operators in (24) are continuous. Thus their finite tensor products and homotopies extend to the completed products. With $P = sq$ on C_+ , the homotopy

$$H_n = \sum_{j=1}^n P^{\widehat{\otimes}(j-1)} \widehat{\otimes} H \widehat{\otimes} 1^{\widehat{\otimes}(n-j)} \quad (39)$$

includes $(-1)^{\sum_{i < j} |x_i|}$ on homogeneous inputs. It satisfies

$$dH_n + H_n d = 1 - P^{\widehat{\otimes} n}.$$

Consequently, for the completed external coefficient complex on the same tau-base chart product,

$$\boxed{H^k(C_+^{\widehat{\otimes} n}) = \begin{cases} Q^{\widehat{\otimes} n}, & k = n, \\ 0, & k \neq n. \end{cases}} \quad (40)$$

This follows from a split continuous contraction, not from a blanket claim that completed tensor product preserves every exact sequence.

For the joint chart use (25). Its full cohomology in degree k is the direct sum, indexed by subsets $J \subseteq \{1, \dots, n\}$ with $|J| = k$, of the ordered completed tensors having Q in the slots of J and V in the other slots. In those V slots the scaling action is $aU_{1/a}$, exactly as in the original minus-coordinate representative. The degree-zero contribution is not discarded. Split reconstruction retains each product mask and each internal zero.

S3.1.10 9. The pairing calculation is now a specified global boundary problem

The section $s : Q \rightarrow \mathcal{B} \subset L^2(dx)$ supplies a positive definite auxiliary form

$$\mathcal{H}_s(u, v) = \langle su, sv \rangle_{L^2(dx)}.$$

This is not assigned to the arithmetic Weil form. The explicit weight-covariance defect is calculated using (29) and $\langle U_a f, U_a g \rangle = a \langle f, g \rangle$:

$$\boxed{\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a \mathcal{H}_s(u, v) = -\langle U_a s u, \Theta k_a v \rangle - \langle \Theta k_a u, U_a s v \rangle + \langle \Theta k_a u, \Theta k_a v \rangle.} \quad (41)$$

The original source boundary and both of its cross-pairings remain. Being a boundary in Q does not assert these pairings are zero in the representative Hilbert space. In a split lift each computed zero is its supported value e .

For a reduced eigenline of the actual dilation representation with exponent ρ , the left side at $u = v$ is $(a^{2\rho} - a) \mathcal{H}_s(u, u)$. Formula (41) locates that defect in the full global boundary cocycle, with the original operator retained.

A concrete comparison explaining which topology must be kept In fact ΘV is dense in $L^2(\mathbb{R}_{>0}, dx)$, although it is closed in $\mathcal{B}$. Here is a proof retaining the map between them. Set $b_* = \Theta \phi_*$ and

$$W : L^2(dx) \xrightarrow{\sim} L^2(dy), \quad W F(y) = e^{y/2} F(e^y).$$

The Fourier transform in y , with kernel $e^{-2\pi i \nu y}$, gives

$$\mathcal{F}_y W b_*(\nu) = g(1/2 - 2\pi i \nu).$$

This is nonzero almost everywhere, because g is a nonzero entire function. The span of all translates of $W b_*$ is dense in $L^2(dy)$: a vector orthogonal to all translates has Fourier transform whose product with $\overline{\mathcal{F}_y W b_*}$ is an L^1 function with zero Fourier transform, hence vanishes almost everywhere. The nonvanishing just stated then makes that vector zero. Since $W U_a b_* = a^{1/2} (W b_*)(y - \log a)$ and $U_a b_* \in \Theta V$, density follows.

The exact quotient comparison is therefore

$$\boxed{Q = \mathcal{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2} = 0.} \quad (42)$$

Its kernel is all of Q . Its split lift sends every supported input to the supported zero in $G(0) = \mathbb{B}$, while keeping external absence separate. The constructive replacement is the Frechet quotient and the explicit global section (19), not a claim that the ordinary Hilbert quotient secretly retains those classes.

S3.1.11 10. Scope of the progress toward the Deligne comparison

The selected supplied Weil II passages 3.3.4-3.3.6 put two bounds on the same compact-to-ordinary image; 6.2 retains the dualizing operation and its shifts. This continuation does not apply those bounds over a new base without their geometric proof. It makes the existing characteristic-zero arithmetic complex globally split in its precise topological category, gives its continuous dual and completed products, and computes the operator defects on those same objects.

The formerly unresolved algebraic-to-Hausdorff kernel is now zero by (16)-(21). The finite extension class and its residue/Jacobian trace are carried by (33)-(36). The full scaling boundary is (29), its dependence on the auxiliary cutoffs is (32), and its metric defect is (41). These are concrete maps available for a global comparison.

Still retained: the common kernel of all finite spectral observations, the meromorphic summand of the balanced analytic realization, and the requirement to identify the correct arithmetic polarization/weight control on the global geometric image. The density map (42) explains why discarding the stronger topology would remove the entire arithmetic cohomology, not why the supported programme should be abandoned.

The reconstruction estimate contains $1 - c_{\alpha,\beta}$; no asymptotic uniform estimate for it is proved. No global positivity, RH, GRH, or infinite trace-class theorem is claimed. No theorem from the earlier finite-range work is upgraded to a Lean certificate. The new global retraction proof itself uses only the original theta/Poisson identities, absolute Mobius inversion, the displayed Fourier compactness argument, and the two retained moment maps.

S3.1.12 11. Verification and references

The written analytic proofs are the evidence for the new analytic statements. The accompanying unittest script tests exact finite identities: divisor cancellation with supported zeros, cutoff-resolvent identities in a finite unitary model, moment corrections, contractions, synchronization masks, operator cocycles, changes of sections, preservation of extension classes, finite trace changes, and completed-tensor algebra before completion. It does not numerically certify a Fourier operator norm, an analytic convergence statement, or a Riemann zero.

Source interfaces read:

- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1-7; the source permits nonzero bottom fibres in its general interface.
- PR #9, workbenches/tau-chain-descent/RESEARCH_NOTE.md, at the commit above; local bytes match its Git blob.
- PR #8's finite comparison as retained and invoked in #9; the sections using s_Z and J_Z are explicitly dependencies of (33)-(38), not inputs to the proof of (2).
- Supplied Weil II TeX: the actual 3.3.1-3.3.6 excerpt, including its finite-type base and duality assumptions. It was read as TeX, not obtained by OCR. No full source-proof audit is claimed.
- Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277: prior-context abstract only in this continuation. A source-archive retrieval failed; no unread detailed theorem is cited as a premise.

The contribution is AI-assisted under the owner's mathematical direction. Original mathematical files, formal sources, attribution and licenses are unchanged. Only authored exposition, test code and validation records are proposed for public addition; no private transcript or source-literature corpus is published.

S4 PR 11: Homotopy control

Source attribution. Web contribution, pull request #11, [Zeta research repository](#).

Exact revision. d37eade4f9e6a82c1a88aeff710a3b851a34eb49

Source file. workbenches/tau-homotopy-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9924b358c0365a8cbdcddb3a7496f610c65d64617cd1860679b42e371d2fe88d

S4.1 Homotopy choices and source-preserving arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #10, d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7. This integration note records the new derivations; the conversation delivery also contains a longer equation-by-equation exposition. No original scalar definitions or formal modules are changed.

S4.1.1 1. Reviewed theorem and analytic specialization

The supplied other-session proof classifies all homotopies for

$$C = [V \oplus V \xrightarrow{d} B], \quad d(v, w) = \Theta(v - \mathcal{F}w), \quad Q = B/\Theta V,$$

where Θ is injective and $\mathcal{F}$ is an R -linear automorphism. For a fixed $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, every solution of BOTH equations

$$dH = \Theta \kappa, \quad Hd = 0$$

has the unique form

$$H_\alpha(b) = (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b), \quad \alpha : Q \rightarrow_R V.$$

Proof: writing $H=(u,v)$, injectivity gives $u-\mathcal{F}v=\kappa$. Since $d(x,0)=\Theta x$, $Hd=0$ gives $H \Theta = 0$. Hence $\mathcal{F}v$ factors uniquely through q , and supplies α . Conversely substitution proves both equations. The reference solution is $H_0=(\kappa,0)$; the family is an affine space under $\text{Hom}_R(Q,V)$, relative to that reference.

Let $\gamma(v)=(v, \mathcal{F}^{-1} v)$. Then

$$H_\alpha - H_\beta = \gamma(\alpha - \beta)q.$$

The difference is a map $H_1(C) \rightarrow H_0(C)$. There are no degree-minus-two endomorphisms of this two-term coefficient complex. Thus it is a genuine degree-minus-one Hom-complex class, not an erased difference.

The supplied theorem is mathematically correct. Its Lean draft explicitly remains uncompiled, and its earlier failed chart build is not relabelled successful. No Lean or Lake executable is installed in the review environment.

For the actual analytic spaces instantiate $R=C$. With the quotient topology on Q , the continuous version follows: $\alpha q = \mathcal{F} \text{pr}_2 H$ is continuous, so the quotient property makes α continuous. Conversely the displayed formula sends continuous α to continuous H .

For the polynomial action $A_0 = \mathbb{C}[t]$, Fourier has the exact type

$$\mathcal{F} : V^j \xrightarrow{\sim} V, \quad t \cdot_{V^j} v = (1 - D)v, \quad j(t) = 1 - t,$$

because $F D = (1 - D)F$. Thus d is A_0 -linear from $V + V^\wedge j$ to B . A complex-linear homotopy is not automatically A_0 -linear. The reflected module above is the actual comparison, not a change to the Fourier definition.

S4.1.2 2. Original split support on every homotopy

Retain the base b_τ and the square $G(Z) \rightarrow G(C)$ over $Z > C$. The quotient $G(Z) \rightarrow Z$ remains infinite. Both lifted coefficient maps preserve τ and the supported scalar e separately.

Use the original support lattice $L = \{\text{bottom}, +, -, J\}$, $J = \{+, -\}$. Degree-zero fibres are $0, V, V, V + V$ with the coordinate inclusions. Degree-one active fibres are all B , with identity transports. Their differentials are Θ , $-\Theta$, F and d . Retain zero modules in the other degrees; their reconstructed zero map is $z(\lambda, x) = (\lambda, 0)$.

Literal multiplication by $E_{\text{sync}} = (1, e)$ induces

$$r^0(a, b) = (a + eb, b + ea), \quad r_L(\perp) = \perp, \quad r_L(\lambda) = J(\lambda \neq \perp).$$

Every homotopy lifts by

$$\tilde{H}_\alpha(\lambda, b) = (J, (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b)), \quad \lambda \neq \perp; \quad \tilde{H}_\alpha(\tau) = \tau.$$

This is split-linear: two active inputs add at their joined label, both outputs add at J , and their amplitudes obey the two displayed linear formulas. A supported scalar zero gives $(J, (0, 0))$; external τ gives the global zero. Therefore

$$\tilde{d} \tilde{H}_\alpha + \tilde{H}_\alpha \tilde{d} = r \tilde{\Delta}.$$

The difference is $(J, \gamma(\alpha - \beta)qb)$, retaining its degree-zero class. For this chart, the pair $(\text{support}(x), r(x))$ is injective. No such injectivity is assumed for arbitrary diagrams with noninjective transports. For a smaller relation space W , keep $Q_W \rightarrow Q$ with kernel $\Theta V / \Theta W$; q in the formulas means the displayed quotient by the full admitted V .

S4.1.3 3. Coherence for all dilations

Let $S_a = U_a$ on V , $T_a = U_a$ on B , $U_a F(x) = F(x/a)$, and T_{bar}_a the quotient action. The actual cochain action is

$$\mathsf{T}_a^0 = (S_a, \mathcal{F}^{-1} S_a \mathcal{F}) = (U_a, a U_{1/a}), \quad \mathsf{T}_a^1 = T_a.$$

It satisfies $T_a^0 \gamma = \gamma S_a$ and $q T_a = T_{\text{bar}}_a q$. With the PR #10 section s , take $P = (0, sq)$, and write

$$[\mathsf{T}_a, P]^1 = \Theta \kappa_a, \quad \kappa_a = k_a q, \quad T_a s - s \overline{T}_a = \Theta k_a.$$

The commutator product identity gives $\kappa_{ab} = S_a \kappa_b + \kappa_a T_b$. The family of homotopies is composition-coherent,

$$H_{ab} = \mathsf{T}_a H_b + H_a \mathsf{T}_b,$$

exactly when

$$\alpha_{ab} = S_a \alpha_b + \alpha_a \bar{T}_b.$$

Proof: the reference family $(\kappa_a, 0)$ is coherent. Subtract it and use injectivity of γ and surjectivity of q . No action or boundary is omitted.

For a smoothly parameterized continuous-linear family with $L = (d/dt)|_0 \alpha(e^t)$, the exact formula is

$$\alpha_{e^t} = \int_0^t S_{e^{t-r}} L \bar{T}_{e^r} dr.$$

This is a vectorwise integral in the complete Frechet source, with the original oriented integral for negative t . Differentiating the composition equation gives the formula; splitting the integral proves its converse. In particular $\alpha_a = S_a b - b \bar{T}_a$ has infinitesimal parameter $D_V b - b D_Q$.

S4.1.4 4. A homotopy parameter leaves the control form fixed

Fix an actual finite arithmetic packet with full multiplicities,

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

Use the existing $R: E_Z \rightarrow B$ and $J_Z: B \rightarrow E_Z$ with

$$qR = \sigma_Z, \quad J_Z R = 1, \quad J_Z \Theta = 0, \quad J_Z D = A J_Z, \quad DR - RA = \Theta k.$$

The cochain projector $P_R = (0, R|_Z)$ has commutator $(0, \Theta k|_Z)$, so the reviewed theorem applies with $\kappa = k|_Z$. The quotient $\text{jet } J_{\text{bar}}_Z$ satisfies $J_{\text{bar}}_Z \sigma_Z = 1$, hence any $\alpha_E: E_Z \rightarrow V$ extends to $\alpha_E J_{\text{bar}}_Z$ on Q . Restriction to R gives

$$H_{\alpha_E} R = (k + \alpha_E, \mathcal{F}^{-1} \alpha_E), \quad dH_{\alpha_E} R = \Theta k.$$

Take $H = L^2(R_+, dx)$ with the original inner product, conjugate-linear in the first variable. Define

$$G_R = R^* R, \quad B_R = \Theta k, \quad W_R = -(R^* B_R + B_R^* R).$$

The composite of actual maps

$$\text{Hom}_{\mathbb{C}}(E_Z, V) \longrightarrow \text{Hom}(E_Z, V^2) \xrightarrow{d \circ (-)} \text{Hom}(E_Z, H) \xrightarrow{B \mapsto -(R^* B + B^* R)} \text{Herm}(E_Z)$$

is constant at W_R . Its last map is real-linear. The retained H^0 ambiguity does not change this boundary contraction.

Integration by parts on the actual source functions gives

$$A^* G_R + G_R A - G_R = W_R.$$

The boundary term $[x \text{ conjugate}(F)H]_0^\infty$ vanishes by the stated decay. No claim that a θ boundary is orthogonal to a representative is used.

S4.1.5 5. The actual representative-change family

Every representative of the same inclusion σ_Z is uniquely

$$R_b = R + \Theta b, \quad b : E_Z \rightarrow_{\mathbb{C}} V.$$

Indeed $q(R'-R)=0$ gives values in ΘV , and the PR #10 retraction gives the unique $b=\Lambda(R'-R)$. Finite source dimension gives continuity. Global continuous sections have the identical formula with $b:Q \rightarrow V$ continuous.

The complete changes are

$$k_b = k + D_V b - bA,$$

$$\delta G_b = R^* \Theta b + (\Theta b)^* R + (\Theta b)^* \Theta b,$$

$$\boxed{W_{R_b} - W_R = A^* \delta G_b + \delta G_b A - \delta G_b.}$$

Their extension-coordinate map is $U_b(v,u)=(v+bu,u)$, and

$$U_b^{-1} \begin{pmatrix} D_V & k \\ 0 & A \end{pmatrix} U_b = \begin{pmatrix} D_V & k + D_V b - bA \\ 0 & A \end{pmatrix}.$$

Lift that automorphism on the common active fibre $(V+E_Z)^{\tau}$. For the original masks, first keep the pair $(\text{support}(x), r(x))$; apply the shear on the joint fibre, and recover the original amplitudes by U_b^{-1} and the original mask by its recorded component. No inverse to the unrecorded support collapse is asserted.

The projectors obey $P_{(R_b)} - P_R = \text{partial } L_b$ with $L_b^{-1}(F)=(b|_Z F, 0)$. Their finite cochain traces agree because

$$\text{Tr}(\Theta b J_Z H_f) = \text{Tr}(J_Z H_f \Theta b) = 0.$$

The cochain supertrace sign remains -1 in degree one. The arithmetic extension class changes by $h_Z(D)b(1)$, so its class in $V/h_Z(D)V$ remains $\epsilon_Z = j_h(h_Z/(2\pi))$. The residue/Jacobian trace form is consequently unchanged.

S4.1.6 6. Finite actual source ansatz and one master Gram matrix

Retain $\phi = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$, $\phi_j = D^j \phi$, and

$$\Phi_m : \mathbb{C}^{m+1} \rightarrow V, \quad (c_j) \mapsto \sum_{j=0}^m c_j \phi_j.$$

Their actual Mellin identities are $M \Theta \phi_j(s) = 2\pi i(s)^j$. They remain in the original two-moment source and are linearly independent. The source section $R_0 = s_Z$ has the exact rank-one defect $DR_0 - R_0 A = \Theta \phi^* \ell_Z$, with the unchanged arithmetic remainder functional ℓ_Z .

Define maps, including the target enlargement,

$$I_m e_j = e_j, \quad S_m e_j = e_{j+1}, \quad I_m, S_m : \mathbb{C}^{m+1} \rightarrow \mathbb{C}^{m+2}, \quad D\Phi_m = \Phi_{m+1} S_m.$$

For B in $\text{Mat}_{((m+1) \times d)}(\mathbb{C})$, $d = \dim E_Z$, put

$$R_B = R_0 + \Theta \Phi_m B, \quad K_B = e_0 \ell_Z + S_m B - I_m B A.$$

Then

$$DR_B - R_B A = \Theta \Phi_{m+1} K_B, \quad q R_B = \sigma_Z, \quad J_Z R_B = 1.$$

Take the actual column map and its Gram matrix

$$\mathcal{T}_m : E_Z \oplus \mathbb{C}^{m+2} \rightarrow L^2(\mathbb{R}_+, dx), \quad (u, c) \mapsto R_0 u + \Theta \Phi_{m+1} c, \quad M_m = \mathcal{T}_m^* \mathcal{T}_m.$$

M_m is positive definite. Applying J_Z to a vanishing combination gives $u=0$; Mellin then gives $c=0$. Its entries are the convergent integrals of the displayed arithmetic functions, with the original measure.

Define

$$X_B = \begin{pmatrix} I_d \\ I_m B \end{pmatrix}, \quad Y_B = \begin{pmatrix} 0 \\ K_B \end{pmatrix}.$$

The complete control calculation is

$$\boxed{G_B = X_B^* M_m X_B, \quad W_B = -(X_B^* M_m Y_B + Y_B^* M_m X_B) = A^* G_B + G_B A - G_B.}$$

This restricts representative choice to an actual finite family of theta boundaries. It is not an arbitrary positive matrix ansatz, and it is not asserted to exhaust all continuous source corrections. All packet coordinates, multiplicities, local units and finite source-degree changes are retained.

S4.1.7 7. Error transport, duality, and the remaining estimate

With actual enclosures $||M - \hat{M}|| \leq \eta_M$, $||X - \hat{X}|| \leq \eta_X$, $||Y - \hat{Y}|| \leq \eta_Y$, put $x = ||\hat{X}||$, $y = ||\hat{Y}||$, $m_0 = ||\hat{M}||$. Product estimates give

$$\eta_G = \eta_M (x + \eta_X)^2 + m_0 (2x\eta_X + \eta_X^2),$$

$$\eta_W = 2[\eta_M (x + \eta_X)(y + \eta_Y) + m_0 (x\eta_Y + y\eta_X + \eta_X \eta_Y)].$$

These include coefficient errors, not just quadrature errors. Exact certificates for $\hat{G} - \eta_G I \succ 0$ and

$$\epsilon \hat{G} \pm \hat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0$$

imply the actual two-sided control. This is an error-transfer theorem; arithmetic enclosures have not been computed here.

The unchanged residue matrix S satisfies $S^* = -S$ and $A^* S + S A = S$. Its exact dual transport is

$$G_B^D = S^* G_B^{-1} S, \quad W_B^D = -S^* G_B^{-1} W_B G_B^{-1} S.$$

Thus an upper certificate gives a lower certificate on the same arithmetic packet. The Weil form stays $S J_g$, with metric comparison $G_B^{-1} S J_g$; no positivity is assigned by choosing G_B .

For tensor degree n , retain $A_n = \sum_j 1$ tensor A tensor 1 and $G_{(B,n)} = G_B^{\wedge}(\text{tensor } n)$. The product control is $\sum_j G_B$ tensor W_B tensor G_B . Changing a homotopy by $\gamma \alpha q$ changes

the tensor homotopy by its Koszul-signed insertions; its differential is zero, hence it does not change that control form.

The existing orbit construction on this actual tensor image gives

$$G_{B,n,T} = \int_0^T e^{-nt} e^{tA_n^*} G_{B,n} e^{tA_n} dt,$$

$$A_n^* G_{B,n,T} + G_{B,n,T} A_n - n G_{B,n,T} = e^{-nT} e^{TA_n^*} G_{B,n} e^{TA_n} - G_{B,n}.$$

Every occurrence of $G_{B,n}$ now has the explicitly constrained source in section 6.

A nonzero boundary need not have nonzero control. The exact calibration $D_{\text{amb}} = \text{diag}(1/2+i, 1/2+2i)$, $\Theta(v) = (v, 0)$, $A = 1/2+2i$, $R(u) = (u, u)$ has boundary $\Theta(-i)$, Gram 2, and control zero. It is an algebraic calibration, not a zeta packet. The real-linear map $B \mapsto (R^*B + B^*R)$ explains which nonzero boundaries can have zero control.

Conversely every eigenvector $Av = \rho v$ obeys

$$\frac{v^* W_B v}{v^* G_B v} = 2\Re \rho - 1.$$

This ensures that changing representatives has not changed a spectral coordinate. The analytic target is now to bound the actual matrices from section 6, or their tensor-orbit endpoints, with all packet and tensor dependence retained. No uniform purity estimate, RH, GRH, or global spectral-kernel vanishing is proved here.

S4.1.8 8. Validation and attribution

The attachment supplies the homotopy-classification theorem and the uncompiled draft. This note verifies its proof and adds the continuous/twisted interface, supported reconstruction, coherent-family condition, and the connection to the actual control matrices. The original owner-directed SplitZero construction remains the source.

The independent checker has sixteen exact unittest methods, including finite parameter families. They passed normally and under python -O with identical successful JSON records. Intentional negative controls failed in both modes. The previous control checker also passed its sixteen methods. These finite checks cover algebraic models and polynomial-times-exponential integral calibrations, not evaluated zeta packets or analytic/Lean certification.

General references: The Stacks Project, Tags 0A8H and 0115 for the Hom-complex differential and homotopy, and 010I and 06XP for extension classes. The formulas above are proved on the specified programme objects; those references are not imported as a number-field weight theorem. Only authored notes, test code and validation records are published; no private transcript, source-literature corpus, or font files are included.

S5 PR 13: Orthogonal boundary control

Source attribution. Web contribution, pull request #13, [Zeta research repository](#).

Exact revision. e7ba4c29d4503bf20d06cb70c93c3347f199f1b4

Source file. workbenches/tau-orthogonal-boundary-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9d364d9cc2f6840af8eb39555e4b89622acd7714fbcab7ca135e8ec0ce25ab0b

S5.1 Orthogonal theta boundaries and rank-two arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #11, d37eade4f9e6a82c1a88aeff710a3b851a34eb49. This is an additive research contribution with written proofs for review, not a new Lean certificate for the analytic results. The conversation delivery contains a longer numbered LaTeX/HTML version, the full reading record, and exploratory arithmetic moments.

S5.1.1 1. Reviewed input and exact types

The supplied other-session theorem is correct: for injective $\Theta: V \rightarrow B$, an R -linear automorphism F , $Q = B/\Theta V$, and $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, all homotopies satisfying BOTH $dH = \Theta \kappa$ and $Hd = 0$, for $d(v, w) = \Theta(v - Fw)$, are

$$H_\alpha(b) = (\kappa b + \alpha \kappa b, F^{-1} \alpha \kappa b), \quad \alpha: Q \rightarrow R V.$$

Writing $H = (u, v)$ gives $u - Fv = \kappa$. Since $d(x, 0) = \Theta x$, $Hd = 0$ gives $H \Theta = 0$; hence Fv descends uniquely to α . Substitution proves the converse. The difference is $\gamma(\alpha - \beta)q$ with $\gamma(v) = (v, F^{-1}v)$. The solution set is affine relative to H_0 , not generally a vector space with the zero homotopy as origin.

The attachment's uncompiled-draft status has now been superseded by the other session's successful PR #12 at 5d2772ea0a16d177ac3e01ff70394b90ba95a232. The actual SplitZeroTauHomotopy.lean, completed job 103535438942, and its full log were read. Both cochain equations and the inverse equivalence passed the specified kernel checks. Retraction and quotient-topology interfaces also passed with their explicit hypotheses. This is the other session's formalization; neither the actual analytic theta construction nor the new Gram/control results below are certified by it. No local Lean/Lake execution is claimed. See COORDINATION_UPDATE.md for exact scope. PR #11 already supplies the continuous specialization, the Fourier twist over $C[t]$, support-changing lifts, coherent dilations and representative-change formulas. Those are retained inputs, not new results here.

Keep the original tau base and the square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$, with $p: G(Z) \rightarrow Z$ still infinite, $e = \text{ofR}(0)$, and tau the external scalar zero. No Boolean localization replaces the base quotient. All quotient relations below use the original internal coequalizer and preserve the support label.

Retain V, B and the actual operators of PRs #8-#11:

```
V={even Schwartz phi on R: phi(0)=0 and integral(phi)=0};
B={smooth F on R_+: sup x^b |D^k F(x)| < infinity for all integer b and k>=0};
D=-x d/dx; Theta phi(x)=2 sum_(n>=1) phi(nx);
M F(s)=integral_0^infinity F(x)x^s dx/x;
phi_*= (4 pi^2 x^4 - 6 pi x^2) exp(-pi x^2);
phi_j=D^j phi_*; f_j=Theta phi_j=D^j f_0;
M f_j(s)=s^j g(s), g=2 xi.
```

The Hilbert observation is $H=L^2(\mathbb{R}_+, dx)$, conjugate-linear in the first argument. Polynomial packet coordinates are retained, not changed to orthonormal coordinates.

For an actual finite packet with all multiplicities, let $E=C[t]/h_Z$, $A=M_t$. The previous analytic constructions supply

$R_\theta: E \rightarrow B$, $J_Z: B \rightarrow E$, $qR_\theta = \sigma_Z$, $J_{ZR_\theta} = 1$, $J_Z \theta = 0$,
 $DR_\theta - R_\theta A = f_\theta \text{ ell}_Z$.

The exact ell_Z comes from $\epsilonpsilon_Z = j_h(h/(2\xi))$; at a simple zero its value on 1 is $1/(2\xi'(\rho))$. These analytic constructions remain the cited research-proof dependencies, not theorems certified by the finite tests here.

S5.1.2 2. An actual minimum over the admitted theta relations

For $m \geq 0$ define

$W_m = \text{span}(\phi_0, \dots, \phi_m)$;
 $\phi_m: C^{(m+1)} \rightarrow V$, $c \mapsto \sum c_j \phi_j$;
 $F_m = \theta \phi_m: C^{(m+1)} \rightarrow H$;
 $H_m = F_m^* F_m$; $C_m = F_m^* R_\theta$; $G_\theta = R_\theta^* R_\theta$.

The ϕ_j are independent: Mellin sends a relation to $g(s)P(s)=0$. Thus H_m is positive definite. Here C_m is a cross-Gram matrix $E \rightarrow C^{(m+1)}$, not the complex of that name in the endpoint workbench.

Set

$B_m = -H_m^{-1} C_m$;
 $\pi_m = F_m H_m^{-1} F_m^*$;
 $R_m = R_\theta + \theta \phi_m B_m = (1 - \pi_m) R_\theta$.

The correction is a finite original theta boundary, so R_m lands in B , $qR_m = \sigma_Z$, and $J_{ZR_m} = 1$ at every m . Direct multiplication proves

$F_m^* R_m = 0$;
 $G_m = R_m^* R_m = G_\theta - C_m^* H_m^{-1} C_m > 0$.

Positive definiteness follows from $J_{ZR_m} = 1$. For any $B: E \rightarrow C^{(m+1)}$,

$(R_\theta + F_m B)^* (R_\theta + F_m B)$
 $= G_\theta + (B - B_m)^* H_m (B - B_m)$.

This is the minimum of the Hilbert Gram within the stated source family, not yet a minimum of its relative weight error. Also $(1 - \pi_m)(R_\theta + F_m B') = R_m$: changing the initial representative by a relation already in W_m leaves the result unchanged.

S5.1.3 3. The complete control form has rank at most two

Let $I_m, S_m: C^{(m+1)} \rightarrow C^{(m+2)}$ be $I_m e_j = e_j$ and $S_m e_j = e_{(j+1)}$. Define

$K_m = e_\theta \text{ ell}_Z + S_m B_m - I_m B_m A$.

Then

$DR_m - R_m A = F_{(m+1)} K_m = \theta (\phi_{(m+1)} K_m)$.

No boundary component is discarded in this identity. Put $b_m = e_m^* B_m: E \rightarrow C$, and $z_m = R_m^* f_{(m+1)}: C \rightarrow E$. Integration by parts on the stated source functions gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

with vanishing boundary term $[x \text{ conjugate}(F)H]_0^\infty$. Hence

$$\begin{aligned} W_m &= A^* G_m + G_m A - G_m \\ &= -(R_m^* F_{(m+1)} K_m + K_m^* F_{(m+1)}^* R_m) \\ &= -z_m b_m - b_m^* z_m^*. \end{aligned}$$

The last equality uses $F_m^* R_m = 0$ and the exact last row of K_m , which is b_m . Thus $\text{rank}(W_m) \leq 2$ for every packet dimension and every finite m . The pairings with the first $m+1$ theta directions vanish; those supported vectors themselves are retained.

Define the actual scalar quantities

$$\begin{aligned} u_m &= z_m^* G_m^{-1} z_m \geq 0; \\ v_m &= b_m^* G_m^{-1} b_m \geq 0; \\ c_m &= b_m^* G_m^{-1} z_m. \end{aligned}$$

The nonzero spectrum of $G_m^{-1} W_m$ is the nonzero spectrum of

$$-\begin{bmatrix} c_m & v_m \\ u_m & \text{conjugate}(c_m) \end{bmatrix}.$$

This follows by factoring through the two-column map $[z_m, b_m^*]$ and using the equality of nonzero spectra of the two compositions. Its trace is $-2 \text{Re}(c_m)$, determinant $|c_m|^2 - u_m v_m$. Therefore the least symmetric control allowance is

$$\begin{aligned} \epsilonpsilon_m &= |\text{Re}(c_m)| + \sqrt{u_m v_m - (\text{Im}(c_m))^2}, \\ -\epsilonpsilon_m G_m &\leq W_m \leq \epsilonpsilon_m G_m. \end{aligned}$$

Cauchy-Schwarz makes the radicand nonnegative. The formula also covers rank one. A reflection-stable full packet has

$$\text{trace}(G_m^{-1} W_m) = 2 \text{Re}(\text{trace } A) - \dim E = 0,$$

so $\text{Re}(c_m) = 0$. This is an exact trace identity, not a choice of coordinates. For every eigenvector $Av = \rho v$,

$$(v^* W_m v) / (v^* G_m v) = 2 \text{Re}(\rho) - 1.$$

Thus the relative control still detects precisely the original spectral weight.

S5.1.4 4. The next relation is a rank-one update with its own retained coordinate

Set

$$\begin{aligned} a_{(m+1)} &= (1 - \Pi_m) f_{(m+1)} = \Theta \psi_{(m+1)}, \\ \psi_{(m+1)} &= \phi_{(m+1)} - \Pi_m H_m^{-1} F_m^* f_{(m+1)}, \\ \eta_{(m+1)} &= \|a_{(m+1)}\|^2 > 0. \end{aligned}$$

Then $R_m^* a_{(m+1)} = z_m$, and orthogonal decomposition gives

$$\begin{aligned} \Pi_{(m+1)} &= \Pi_m + a_{(m+1)} \eta_{(m+1)}^{-1} a_{(m+1)}^*; \\ R_{(m+1)} &= R_m - a_{(m+1)} \eta_{(m+1)}^{-1} z_m^*; \\ G_{(m+1)} &= G_m - \eta_{(m+1)}^{-1} z_m z_m^*. \end{aligned}$$

The control update is obtained by applying the same real-linear operator $X \mapsto A^*X + XA - X$ to the last identity.

The original quotient transport is

$$Q_W = B / \Theta W_m \rightarrow Q_{W(m+1)} = B / \Theta W_{(m+1)}, \\ \text{kernel} = \Theta W_{(m+1)} / \Theta W_m \simeq W_{(m+1)} / W_m.$$

That kernel is one-dimensional. The square $E \rightarrow Q_W$ by R_m and $E \rightarrow Q_{W(m+1)}$ by $R_{(m+1)}$ commutes. The new relation becomes the supported zero only through the actual next-level quotient; it is not another scalar adjunction at the bottom.

There is a form on the entire quotient:

$$H_{\text{cal}_m}([F], [G]) = \langle (1 - \Pi_m)F, (1 - \Pi_m)G \rangle.$$

It is positive definite since the kernel of $1 - \Pi_m$ in B is exactly ΘW_m . Let $\lambda_{(m+1)}[F] = \langle a_{(m+1)}, F \rangle$. This is well defined on Q_W and

$$H_{\text{cal}_m}(x, y) = H_{\text{cal}_{(m+1)}}(qx, qy) \\ + \eta_{(m+1)}^{-1} \text{conjugate}(\lambda_{(m+1)}(x)) \lambda_{(m+1)}(y).$$

The map $(q, \lambda_{(m+1)}): Q_W \rightarrow Q_{W(m+1)}$ plus C is a linear isomorphism. Its inverse is

$$([F], c) \mapsto [F - a \eta_{(m+1)}^{-1} \langle a, F \rangle + a \eta_{(m+1)}^{-1} c]_W.$$

Changing F by a relation at level $m+1$ changes this expression only by a relation at level m . This supplies the exact metric comparison between levels instead of treating the quotient transports as isometries.

Every displayed linear map has its original fixed-support lift. On the full source chain, reconstructed addition uses the join $\max(m, n)$ and the above transports. The scalar e sends $(m, [F])$ to $(m, [0])$; τ sends it to external absence. The two-leg primitives are $(k_m u, 0)$ and $(0, \text{Fourier}^{-1} k_m u)$, where $k_m = \Phi_{(m+1)} K_m$. Synchronization by the original $E_{\text{sync}} = (1, e)$ supplies their common joint support. Their difference remains the H_0 cycle $(k_m u, \text{Fourier}^{-1} k_m u)$. The other session's $\text{Hom}(Q, H_0)$ freedom leaves their differential and hence W_m unchanged.

S5.1.5 5. Actual arithmetic moments and their truncation allowance

The matrices H_m come from the original arithmetic functions. Since $JF(x) = x^{-1} F(1/x)$ is unitary on $L_2(dx)$, $Jf_0 = f_0$ and $JD = (1 - D)J$,

$$H_{ij} = \int_{-\infty}^{\infty} [\text{conjugate}(f_i) f_j \\ + \text{conjugate}(f_i^{\text{ref}}) f_j^{\text{ref}}] dx, \\ f_j^{\text{ref}} = (1 - D)^j f_0.$$

This retains both multiplicative ends and permits absolutely convergent Gaussian sums on $x \geq 1$. Write

$$\phi_j(x) = P_j(x) \exp(-\pi x^2), \\ P_0 = 4 \pi^2 x^4 - 6 \pi x^2, \\ P_{(j+1)} = 2 \pi x^2 P_j - x P_j', \\ P_j = \sum_r c_{(j, r)} x^{(2r)}, \\ c_{(j+1, r)} = 2 \pi c_{(j, r-1)} - 2r c_{(j, r)}.$$

For $f_j^{\text{N}} = 2 \sum_{(n=1)}^N P_j(nx) \exp(-\pi n^2 x^2)$, put

$$C_j = \sum_r |c_j(r)| \sqrt{\Gamma(2r+1/2) / (2 \pi^{(2r+1/2)})}.$$

Then

$$\begin{aligned} & \|f_j - f_j^N\|_{L^2([1, \infty))} \\ & \leq 2 C_j (N+1)^{-1/2} \exp(-\pi(N+1)^2/2) \\ & \quad / [1 - \exp(-\pi(2N+3)/2)]. \end{aligned}$$

Proof: on $x \geq 1$, $2 \pi n^2 x^2 \geq \pi n^2 + \pi n^2 x^2$. Integrating each monomial's squared bound gives $n^{-1/2} \exp(-\pi n^2/2)$ times the displayed gamma constant. Minkowski and a geometric bound on consecutive n terms give the formula. The reflected derivative retains the absolute binomial sum of these allowances.

The finite integral entries use exactly

$$\begin{aligned} & \text{integral}_1^{\infty} x^{(2k)} \exp(-c x^2) dx \\ & = \Gamma(k+1/2, c) / (2 c^{(k+1/2)}), \quad c = \pi(n^2 + n'^2). \end{aligned}$$

For truncation tail norms δ_i, δ_j , each inner-product error is bounded by $\delta_i \|f_j^N\| + \|f_i^N\| \delta_j + \delta_i \delta_j$, plus its reflected contribution. Numerical evaluation errors for the finite expression still require their own enclosure. The optional conversation computation is explicitly exploratory, not interval-certified.

The exact consistency check is $H_{(i+1,j)} + H_{(i,j+1)} = H_{ij}$. Cross-Gram columns $F_m \cdot R_0$ remain their actual source integrals and are not replaced by arbitrary matrices.

S5.1.6 6. The same source proves a strong topology comparison

For this actual f_0 ,

$$\text{closure span}\{D^j f_0 : j \geq 0\} \text{ in } L^2(\mathbb{R}_+, dx) = H.$$

Proof. The explicit unitary Mellin-line map sends F to $M F(1/2+it)$ in $L^2(\mathbb{R}, dt/(2\pi))$. It sends D to $1/2+it$ and f_j to $(1/2+it)^j g(1/2+it)$. The log-space function $h(z) = \exp(z/2) f_0(\exp z)$ is holomorphic in $|\text{Im } z| < \pi/4$ by its Gaussian series. For $|\text{Im } z| \leq a < \pi/4$ it decays superexponentially as $\text{Re } z$ tends to positive infinity. Poisson gives $h(z) = h(-z)$, so the same holds at the other end. Fourier contour displacement therefore gives $|g(1/2+it)| \leq C_a \exp(-a|t|)$.

The measure $d\mu = |g(1/2+it)|^2 dt/(2\pi)$ has a positive exponential moment. If an $L^2(\mu)$ function is orthogonal to all polynomials, multiplying μ by its conjugate gives a finite complex measure with a smaller positive exponential moment. Its Fourier transform is analytic in a strip and all derivatives at zero vanish. Analyticity and uniqueness of the Fourier transform imply that measure is zero. Thus polynomials are dense in $L^2(\mu)$. Multiplication by g is unitary from this weighted space onto $L^2(dt/(2\pi))$, because the nonzero analytic function g on the line is nonzero almost everywhere. This proves the assertion without RH.

As a result, for a fixed finite packet,

$$\begin{aligned} & R_m \rightarrow 0 \text{ in } \text{Hom}(E, H), \quad G_m \rightarrow 0 \text{ and } W_m \rightarrow 0, \\ & \text{while } q_{R_m} = \sigma_Z \text{ and } J_{ZR_m} = 1 \text{ for every } m. \end{aligned}$$

This specifies why unscaled smallness of W_m cannot be used alone as weight control: its denominator G_m is also shrinking. The exact relative formula in section 3 must be retained.

The corrective map is the Hilbert/jet graph:

$$\begin{aligned} & \text{closure}\{(F, J_{ZF}) : F \text{ in } B\} = H \text{ plus } E; \\ & \text{closure}\{(\Theta \phi, 0) : \phi \text{ in } V\} = H \text{ plus } 0. \end{aligned}$$

Indeed finite theta polynomials give every $(h,0)$ in the closure, and the sequence $(R_m u, u)$ gives $(0, u)$. For a specified positive metric on E , completion in $\|F\|_2^2 + \|J_Z F\|_E^2$ is this graph closure, and the quotient by the boundary closure is E by $[(h, u)] \rightarrow u$. The added metric is retained supplied data, not an implied positivity theorem for the Weil pairing.

Its split observation is

$$((R_m u)^{\bullet}, u^{\bullet}) \rightarrow (e_H, u^{\bullet})$$

in the product of the stated support-fibre topologies. The arithmetic component survives above a supported Hilbert zero. External absence is (τ, τ) .

S5.1.7 7. Duality, trace and scope

The original residue map has matrix S with $S^* = -S$ and $A^*S + SA = S$. Its pullback of the dual metric is

$$\begin{aligned} G_m^* D &= S^* G_m^{-1} S, \\ W_m^* D &= -S^* G_m^{-1} W_m G_m^{-1} S. \end{aligned}$$

Thus the upper-to-lower comparison is retained on the same arithmetic packet. The Weil pairing stays $S J_g$, with explicit metric comparison $G_m^{-1} S J_g$. Its positivity is not imposed by choosing a positive Gram matrix.

All representative changes above are original theta boundaries. Their extension class $j_h(h/(2\xi))$ is unchanged; the induced finite cochain projectors differ by the homotopy $(b)_Z, 0$. The trace difference is zero because $J_Z \Theta = 0$. Degree-one and tensor Koszul signs remain those of the source complex. On a raw tensor metric, the inserted W_m terms give the copied allowance $n \epsilon_m$; tensoring alone does not improve it.

The new results are the actual orthogonal source choice, rank-two boundary formula, exact relative certificate, rank-one source-kernel recurrence, arithmetic moment tail bound, and the full Hilbert/jet topology comparison. No uniform arithmetic purity estimate, global spectral-kernel vanishing, RH or GRH is asserted.

The thirteen finite exact tests check the encoded algebra and declared calibration models, not infinite analytic arguments or actual zeta zeros. The complete cochain-homotopy theorem remains the other session's result, now supported by the independently inspected successful PR #12 run. Its certificate does not extend to the new analytic control results. Existing Lean modules and earlier validation receipts are untouched.

References: live PR #11 at the pinned head; successful PR #12 and source/job references in COORDINATION_UPDATE.md; the supplied `Tau_Deligne_Control` and `Tau_Global_Comparison` source notes; the supplied other-session homotopy proof; the previously retained Weil II 3.3.4-3.3.6 and 6.2 duality excerpts. Standard orthogonal-polynomial context is NIST DLMF 18.2; the density argument needed here is proved above for the actual arithmetic measure.

S6 PR 17: Spectral-sum descent and the complete relative arithmetic fibre

Source attribution. Web contribution, pull request #17, [Zeta research repository](#).

Exact revision. 33b29f706008124886614ba4bd55bffc489df9e2

Source file. workbenches/tau-spectral-sum/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 551e3fe89b4409d5ed1eeacfaac4cb7d9c1291978127de1b4f989b272bd24a47

S6.1 Spectral-sum descent with the relative arithmetic fibre retained

Owner-directed continuation, 12 September 2026. Draft for review; no purity or RH theorem is claimed. Base: PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc.

S6.1.1 1. Original source and scope

Keep $G(R) = \{\tau\} \text{ disjoint-union } \{r^\bullet\}$, $e_R = 0_R^\bullet$, and the original square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$. The first arithmetic quotient remains infinite. Fixed-label linear maps lift by $(\lambda, v) \rightarrow (\lambda, Tv)$ and fix external absence; a vanishing value remains the original labelled supported zero.

Use the existing $C_+ = [V \rightarrow \Theta_B]$, $Q = B/\Theta \backslash V$, $D = -x/d/dx$, $g = 2/x$, and the actual source function $\phi_* = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$. For a finite packet h of actual zeros, including full orders, the prior analytic construction supplies F_h with Mellin transform g/h . Retain

$$T_h(P) = P(D)F_h, \quad T_h(hP) = \Theta(P(D)\phi_*), \quad J_h T_h(P) = v_h[P]_h,$$

where $\text{upsilon}_h = j_h(g/h)$ is the full invertible jet, not the scalar e . These analytic range statements are inputs from the previous work, not newly Lean-certified here.

For k factors, the actual numerator is $P_M = C[s_1, \dots, s_k]_{(\text{degree} \leq M)}$, the relation space is its intersection with $(h(s_1), \dots, h(s_k))$, and its theta realization is $B_-(k, M)$. The canonical representative $R_-(k, M)$ is the unique least-norm lift with exactly these relations allowed. Its full jets, arithmetic class, Gram G and control W remain those of #14.

The supplied Kernel Layer Integration supplement agrees with the Symmetric Frontier note under $K = C_{\text{inverse}}$, $U = F$, $D = \Omega$, $E_{\text{layer}} = b_M$, and $F_{\text{derivative}} = \Omega^{(-1)} E_{\text{plus}}$. Its sharper rank and original-layer identities are retained as source results, not counted again as new work.

S6.1.2 2. An actual one-parameter pushforward and its resolution

Define

$$\iota_k : \mathbb{C}[S] \rightarrow E_h^{\otimes k}, \quad S \mapsto [s_1 + \dots + s_k].$$

Restriction of modules through this map is the exact quasi-coherent pushforward along the finite scheme morphism. Its adjunction maps are $f \rightarrow (1 \rightarrow f(1 \otimes 1))$ and $a \otimes 1 \rightarrow a f(1)$. Applying G gives $p \backslash G(\iota) = \iota \backslash p$, with e sent to e and τ to τ ; the morphism is over the same absolute pointed base. No finite field is substituted.

For the actual finite image E , with sum operator A , there is the free resolution

$$0 \rightarrow \mathbb{C}[S] \otimes E \xrightarrow{SI-A} \mathbb{C}[S] \otimes E \xrightarrow{\text{ev}_A} E_A \rightarrow 0.$$

Its augmentation sends $P \backslash \text{tensor} \backslash v$ to $P(A)v$. Injectivity follows from the highest polynomial coefficient. Exactness follows from

$$S^j v - A^j v = (SI - A) \sum_{i=0}^{j-1} S^{j-1-i} A^i v.$$

Dualizing with $\mathbb{C}[S] dS$ gives $SI - A^{\vee}$ in degrees 0,1, where $A^{\vee}(\lambda) = \lambda$ compose A is the algebraic transpose, not a Hermitian adjoint. Thus

$$\text{Ext}_{\mathbb{C}[S]}^1(E_A, \mathbb{C}[S]dS) \cong E^{\vee},$$

with $[\sum S^j \lambda_j dS] \rightarrow \sum (A^{\vee})^j \lambda_j$ and inverse the constant class. With $\mathbb{C}[b]^i = \mathbb{C}^{i+b}$, the exact shifted statement is

$$R\text{Hom}(E_A[-k], \mathbb{C}[S]dS[1]) \simeq E^{\vee}[k].$$

The right adjoint is the specified module functor $R\text{Hom}_{\mathbb{C}[S]}(E_h^{\text{tensor } k}, -)$. This is not an assertion about pushforward on all sheaves of modules. See Stacks Tags 0AVV and 0AWZ.

The same duality has the concrete pairing

$$\langle v, [P(S)dS] \rangle = \sum_{\lambda \in \text{Spec } A} \text{Res}_{S=\lambda} P(S)((SI - A)^{-1}v)dS.$$

Expansion at infinity gives $\sum \lambda_j (A^j v)$, proving the formula and perfectness. On a length- l block,

$$(SI - A)^{-1} = \sum_{j=0}^{l-1} \frac{N^j}{(S - \lambda)^{j+1}}.$$

All poles are retained before taking the trace. The further observation is

$$\text{Tr}(\psi(A)) = \sum_{\lambda} \text{Res}_{S=\lambda} \psi(S) \frac{\chi'_A(S)}{\chi_A(S)} dS.$$

Here $\chi_A = \det(SI - A)$ is the finite characteristic polynomial, not a replacement for $g=2x$. With $\psi(S) = a^S$ this is the finite trace of the same diagonal scaling correspondence; its cochain sign is $(-1)^k$.

S6.1.3 3. Exact symmetric-square relation algebra

Retain the coordinate morphisms

$$S = s_1 + s_2, \quad r = s_1 - s_2, \quad \Delta = r^2, \quad s_1 = (S + r)/2, \quad s_2 = (S - r)/2.$$

Swap fixes S and negates r . Even/odd polynomial expansion gives the filtered algebra isomorphism $\Phi: \mathbb{C}[S, \Delta] \rightarrow \mathbb{C}[s_1, s_2]^{\text{swap}}$, with $\deg(S)=1$, $\deg(\Delta)=2$.

Define H_0, H_1 without localizing away from $r=0$:

$$h((S+r)/2) = H_0(S, r^2) + rH_1(S, r^2), \quad h((S-r)/2) = H_0(S, r^2) - rH_1(S, r^2).$$

Then

$$(E_h \otimes E_h)^{S_2} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1).$$

Proof: the original ideal in $\mathbb{C}[S, r]$ is (H_0, rH_1) . Averaging an expression $H_0 \setminus a + r \setminus H_1 \setminus b$ under $r \rightarrow -r$ gives exactly $H_0 \setminus a_{\text{even}} + \Delta \setminus H_1 \setminus b_{\text{odd}}$. Conversely both generators lie in the original ideal. Reynolds averaging is exact over $\mathbb{C}$, so the invariant quotient has precisely this ideal, including all nilpotents.

The unscaled orbit sums of $s_1^{i_1} \dots s_d^{i_d}$, $0 \leq i_1 \leq \dots \leq i_d < d$, give dimension $d(d+1)/2$. The associated graded Hilbert series for the original degree filtration is

$$\sum_{0 \leq i_1 \leq \dots \leq i_d < d} z^{i_1 + \dots + i_d} = \frac{(1-z^d)(1-z^{d+1})}{(1-z)(1-z^2)}.$$

No grading of a nonhomogeneous h is asserted.

The original primitive map is explicit:

$$H_0 A_0 + \Delta H_1 A_1 = h(s_1) \frac{A_0 + r A_1}{2} + h(s_2) \frac{A_0 - r A_1}{2}.$$

Its two cochain primitives use $\phi_* \setminus \text{tensor} \setminus F_h$ and $F_h \setminus \text{tensor} \setminus \phi_*$, respectively. The second primitive has a minus sign, matched by the second tensor-differential sign. For a relation of degree M , ordered division followed by swap averaging gives $\deg A_0 \leq M-d$, $\deg A_1 \leq M-d-1$.

The unit $\epsilon_h \setminus \text{tensor} \setminus \epsilon_h$ descends to the invariant algebra and remains in its full jet map. All finite relation quotients and the transitions $M \rightarrow M+1$ commute with Φ . A class killed by the next relation maps to the receiving e -zero, never to external absence.

S6.1.4 4. Push the actual norm to a matrix weight, not a scalar observation

Set

$$w_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{1}{2\pi}.$$

The exact real coordinate map is

$$u = t_1 + t_2, \quad v = t_1 - t_2, \quad dt_1 dt_2 = \frac{1}{2} du dv.$$

It gives $S=1+iu$, $\Delta=-v^2$. For

$$P(S, \Delta) = \sum_{a=0}^{\lfloor M/2 \rfloor} \Delta^a f_a(S), \quad \deg f_a \leq M - 2a,$$

define

$$W_{ac}(u) = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv.$$

Then the original arithmetic norm is exactly

$$\|T_h^{(2)} \Phi P\|^2 = \int_{\mathbb{R}} f(1+iu)^* \mathbf{W}(u) f(1+iu) du.$$

Proof: substitute the coordinate map and expand the finite polynomial. Exponential moments justify Fubini. Each finite matrix $W(u)$ is strictly positive: the integrand is the squared modulus of a nonzero polynomial in $-v^2$ against a density positive almost everywhere. The entire nonzero g/h has only discrete zeros on these real lines. The source exponential bound also gives convergence for each fixed u .

Its scalar entry is $(w_h^* w_h)(u)$. Its other entries retain the relative-coordinate moments; they are not replaced by that scalar entry.

For general k retain $S = \sum_{i=1}^k s_i$, $x_i = s_i$ for $i < k$ and $s_k = S - \sum_{i=1}^{k-1} s_i$. This is a polynomial isomorphism and the real Jacobian is one. Writing $P = \sum x^\alpha f_\alpha(S)$ gives

$$W_{\alpha\beta}(u) = \int_{\mathbb{R}^{k-1}} \prod_{i<k} (1/2 + it_i)^{\alpha_i} \prod_{i<k} (1/2 + it_i)^{\beta_i} \prod_{i<k} w_h(t_i) w_h(u - \sum_{i<k} t_i) d^{k-1}t.$$

The outer coordinate remains $S = k/2 + iu$, and $\deg f_\alpha \leq M - |\alpha|$. This supplies an all- k isometry retaining the entire filtered source. Constants in exponential-moment bounds depend on h, k and the degrees; no uniform tensor bound is inferred.

S6.1.5 5. Canonical control commutes with the descent

For the explicit source-coordinate matrix C and quotient-coordinate matrix U , keep

$$M' = C^* M C, \quad J' = U^{-1} J C, \quad K' = U^{-1} K (U^{-1})^*, \quad G' = U^* G U,$$

$$A' = U^{-1} A U, \quad W' = U^* W U, \quad R' = R U.$$

These follow by substituting into $K = J M^* (-1) J^*$ and the exact constrained-minimum formula. Thus no positive metric has been freely selected. The same isometry carries the actual next relation layer and both boundary maps Y, C_{rel} ; their cross-pairing is unchanged.

There is a further exact base/relative map. Write $w(u) = [m_0, b], [b^*, c_{\text{rel}}]$, with $m_0 > 0$. Then

$$f \mapsto (f_0 + m_0^{-1} b f_{\text{rel}}, f_{\text{rel}})$$

is an isometry into the direct sum weighted by m_0 and $c_{\text{rel}} - b^* m_0^{-1} b$. Its inverse is $(a, z) \mapsto (a - m_0^{-1} b z, z)$. The precise image of the polynomial source is retained; the ratio need not be polynomial. Multiplication by S commutes with this map. Neither the scalar mass nor positivity of this representative norm is substituted for the arithmetic Weil pairing.

S6.1.6 6. Full nilpotent fibres for every tensor degree

For an ordered tuple of roots with orders m_i , the local algebra is

$$L = \mathbb{C}[z_1, \dots, z_k] / (z_i^{m_i}), \quad A = \rho_\Sigma I + N, \quad N = M_{\sum z_i}, \quad D_0 = \sum_i (m_i - 1).$$

Let $c_r = [t^r] \prod_{i=1}^k (1 + t + \dots + t^{m_i-1})$, $c_{-1} = 0$. Then the complete module pushforward is

$$\pi_* L \cong \bigoplus_{r=0}^{\lfloor D_0/2 \rfloor} (\mathbb{C}[S]/(S - \rho_\Sigma)^{D_0-2r+1})^{c_r - c_{r-1}}.$$

Proof and maps: on $z_i \wedge^j$ define $F_i = j(m_i - j)z_i \wedge^{(j-1)}$ and $H_i = (2j - (m_i - 1))z_i \wedge^j$. Direct calculation gives $[N, F] = H$, $[H, N] = 2N$, $[H, F] = -2F$. The explicit auxiliary diagonal Gram

$$\|z^\alpha\|_0^2 = \prod_i \frac{\alpha_i! (m_i - 1)!}{(m_i - 1 - \alpha_i)!}$$

makes $F = N^*$. A degree- r primitive p with $Fp = 0$ has ladder length $w+1$, $w = D_0 - 2r$, since

$$FN^j p = j(w - j + 1)N^{j-1}p, \quad \|N^j p\|_0^2 = \frac{j!w!}{(w - j)!} \|p\|_0^2.$$

Orthogonal invariant complements construct all ladders; degree counts give $c_r - c_{(r-1)}$. The actual module map is $[P(S)] \rightarrow P(\rho_\Sigma + N)_p$ on each chosen primitive. This auxiliary Gram is related to the arithmetic one by the explicit operator $G_{\text{arithmetic}}^{(-1)} G_0$; it is used for the algebraic decomposition only. All cross-ladder entries $U^* G_{\text{arithmetic}} U$ remain in control estimates.

For an occupation list n_ρ with sum k , the symmetric component has graded dimensions given by

$$\prod_{\rho} \prod_{j=1}^{n_{\rho}} \frac{1 - t^{m_{\rho} + j - 1}}{1 - t^j}.$$

These are polynomial orbit-count identities, not evaluations of a rational function at roots of unity. Apply the same primitive construction to invariants, with $D_0 = \sum n_\rho (m_\rho - 1)$ and $\rho_\Sigma a = \sum n_\rho \rho$. Central occupation idempotents are retained even when different occupations have the same spectral sum.

For an order-three local input taken twice, the symmetric fibre has ladders

$$1, X, X^2, X^3, X^4, \quad p = z_1^2 - z_1 z_2 + z_2^2, \quad X = z_1 + z_2, \quad Xp = 0.$$

It is $\mathbb{C}[S]/(S - 2\rho)^5$ direct-sum $\mathbb{C}[S]/(S - 2\rho)$, dimension six. This is a local algebra calculation, not an assertion that ζ has a triple zero. The full arithmetic unit commutes with A and carries the ladder coordinates to the original jets.

Each occupation trace is its full dimension times $a^{(\sum n_\rho \rho)}$, with dimension product $\text{binomial}(n_\rho + m_\rho - 1, n_\rho)$. The cochain trace retains $(-1)^k$. Quotienting a top ladder vector, when explicitly requested as a module quotient, makes it the supported zero; the full pushforward itself keeps it.

S6.1.7 7. Exact calibration and scope

For the declared measure $\exp(-t^2/2)dt$, not the arithmetic measure, the three relative directions $1, \Delta, \Delta^2$ give

$$W(u) = \sqrt{\pi} e^{-u^2/4} \begin{pmatrix} 1 & -2 & 12 \\ -2 & 12 & -120 \\ 12 & -120 & 1680 \end{pmatrix}.$$

The unscaled coordinate columns $1, \Delta + 2, \Delta^2 + 12\Delta + 12$ turn this into $\sqrt{\pi} \exp(-u^2/4) \text{diag}(1, 8, 384)$. The full two-variable mass is still 2π . The published core

checker verifies these signs, constants, algebraic primitives, sum resolution and nilpotent blocks. The longer conversation package has a separate 24-method suite, plus independent reruns of the source 18- and 22-method suites; those are not all claimed to be tests in this smaller repository checker.

The new reduction isolates the convolution mass, the relative covariance, and full-jet interpolation on one explicit filtered image. It does not prove their required uniform sublinear tensor estimate. The finite pushforward and matrix-weight model are not asserted to be lisse sheaves on Deligne's Lefschetz pencil. They supply a precisely defined product-to-one-parameter operation to study next, with its adjoint, cohomology, arithmetic trace and original supported relations retained.

References: Stacks Tags 0AVV and 0AWZ; NIST DLMF 18.2; Deligne, La conjecture de Weil II, IHES 52 (1980), 137-252. A bounded supplied French TeX window around 3.2.11–3.2.15 was reread. Its historical line 872 contains an inconsistent numerical implication and is not used as a theorem. No complete new audit of Weil II, analytic integral certificate, Lean execution, or global novelty claim is made.

S7 Derived Split-Zero mathematics

Source attribution. Web contribution retained on the pre-integration main revision.

Exact revision. 14c69d604044b848d64318751a7bee19edfd9aba

Source file. formal/splitzero/DERIVED_MATHEMATICS.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 6a290e193161eb09f49b6faba578193fdaf28b5ae3a97340d1c41a40aa99e695

S7.1 SplitZero: reconstruction, internal homology, and balanced finite jets

Owner-directed formalization continuation, 12 September 2026.

S7.1.1 Scope and provenance

This contribution extends formal/splitzero on the owner's Zeta workbench. Its base is the synchronization PR #5, commit 236f0a8370477991e2d0cf334ad1ebfe4f3f8b17, itself based on the structural PR #4. The original scalar core and all five preceding extension sources remain unchanged. There is no Sneed-side submission or integration in this continuation.

The mathematical source is the owner's SplitZero programme, including the supplied continuation's internal homology and balanced-Mellin sections. The implementation below distinguishes source-derived structure, ordinary reusable algebra, and the new explicit comparison example. It makes no global priority claim and does not treat the number of Lean declarations as a number of mathematical discoveries. No private conversation or source-literature corpus is republished.

The exact build certificate for a revision is the completed SplitZero\derived\constructions workflow on that commit, including direct source checks and its generated AuditDerived.lean. The PR records the accepted execution revision and run. A failed earlier development run is not a certificate for later source.

S7.1.2 1. Reverse reconstruction, with actual scalar action

Let R be a commutative ring, L a join-semilattice with bottom, and let D consist of R -modules V_i with linear transition maps

$$\begin{aligned} \text{rho_ij} &: V_i \rightarrow V_j \quad (i \leq j), \\ \text{rho_ii} &= \text{id}, \quad \text{rho_jk} \text{ rho_ij} = \text{rho_ik}. \end{aligned}$$

No finiteness, field, injectivity, or nontriviality assumption is imposed. Construct

$$\text{Total}(D) = \text{disjoint union of } V_i \text{ over } i \text{ in } L.$$

The global zero is $(\text{bottom}, 0)$, and

$$\begin{aligned} (i, x) + (j, y) &= (i \text{ join } j, \text{rho_i, i-join-j}(x) + \text{rho_j, i-join-j}(y)). \\ \text{tau} \cdot (i, x) &= (\text{bottom}, 0). \\ \text{ofR}(r) \cdot (i, x) &= (i, r \cdot x). \end{aligned}$$

`LinearDiagram.totalAddCommMonoid` and `LinearDiagram.totalModule` construct the actual instances and prove every law. For associativity, transport both sides to $i \text{ join } j \text{ join } k$. Linearity and transition composition reduce both to the same three-term sum. The scalar laws follow in the supported case from the R -module laws and in the absent cases from the chosen global zero. The supported-zero scalar sends (i,x) to $(i,0)$.

The bottom fibre is allowed to be nonzero. Specializing it to the zero module recovers the source's carrier consisting of one absent point and the non-bottom fibres. This is an explicit generalization, not an assumption that the bottom fibre in every later dual diagram vanishes. Opposite-index duality itself is not constructed here.

Fixed-index natural families of linear maps produce actual $G(R)$ -linear maps between reconstructed totals. Identity and composition are proved. More substantially, `Recovery.recoverEquiv` reconstructs the intrinsic support fibres of **any** existing $G(R)$ -semimodule M and gives a full $G(R)$ -linear equivalence back to M :

$$(1,x) \mapsto x, \quad m \mapsto (e\ m, m).$$

Its addition proof uses the original absorption identity $m + e\ m = m$. It is not merely a bijection of carriers. A bundled equivalence of the two full categories, including arbitrary changes of support index and both natural inverse comparisons, is a separate library interface not claimed by this implementation.

S7.1.3 2. The internal quotient is characterized by a universal property

Let B_i be transition-stable submodules of V_i . The quotient diagram has fibres V_i/B_i and the genuinely induced quotient transition maps. The projection

$$q : \text{Total}(D) \rightarrow \text{Total}(D/B), \quad q(i,x) = (i,[x])$$

is a surjective $G(R)$ -linear map. The code proves the exact equality criterion

$$q(i,x)=q(i,y) \text{ iff } x-y \text{ is in } B_i.$$

Different labels cannot become equal under q . In particular a relation maps to its own fibre zero.

The main point is stronger than that observation. Let N be **any** $G(R)$ -semimodule, not assumed additively cancellative, and let $f:\text{Total}(D)\rightarrow N$ be split-linear. If

$$f(i,b)=f(i,0) \text{ for every } b \text{ in } B_i,$$

then there exists a unique split-linear $g:\text{Total}(D/B)\rightarrow N$ with $g\ q=f$. The proof does not subtract elements in N . If $x-y$ is in B_i , use the identity $y+(x-y)=x$ inside V_i and then apply additivity of f . This proves constancy on q -fibres. Surjectivity constructs the descended function; linearity and uniqueness follow by lifting representatives.

`Relations.coequalizer_universal` states this against the actual inclusion of the relation diagram and its supported-zero companion. Thus the universal property is proved against arbitrary semimodule targets, not only against a selected class of fixed-support maps. The implementation gives the complete unbundled universal property; it does not merely define the desired quotient and label it a coequalizer.

S7.1.4 3. Actual homology and the exact transported-class kernel

At a degree i of a cochain complex, use the window

$C^{(i-1)} \xrightarrow{d_{\text{Prev}}} C^i \xrightarrow{d_{\text{Next}}} C^{(i+1)}, \quad d_{\text{Next}} d_{\text{Prev}} = 0.$

Window.boundaryToCycles constructs the incoming map into $\ker(d_{\text{Next}})$. Boundaries are its actual linear range, and H is the module quotient of cycles by that range. Window.ofCochain imports these data at every integer degree of a Mathlib CochainComplex; ChainMap.ofCochain imports an actual cochain map without replacing its differentials.

For a chain map $f:C \rightarrow D$, the code constructs

$f_Z : Z(C) \rightarrow Z(D),$
 $H(f) : Z(C)/B(C) \rightarrow Z(D)/B(D).$

It proves identity, composition, and

$H(f)([z]) = 0 \iff \text{there exists } y \text{ with } d_{\text{Prev}_D}(y) = f_i(z).$

Define the module of killed representatives and its original relation submodule:

$\text{Krep} = \{z \text{ in } Z(C) : f_Z(z) \text{ is in } B(D)\},$
 $\text{Bold} = B(C), \text{ viewed as a submodule of Krep.}$

There is a constructed linear equivalence

$\text{Krep}/\text{Bold} \simeq \ker H(f).$

Forward map: the class of z goes to the kernel element $[z]$. It is well-defined because Bold consists exactly of original source boundaries. It is injective because equality of source classes is exactly a difference in Bold. It is surjective because every kernel class has a cycle representative whose image is a boundary. This is ChainMap.homologyKernelEquiv, obtained from a general quotient-kernel equivalence proved in the same file.

The calculation retains actions as well. Given chain endomorphisms a of C and b of D satisfying $f \circ a = b \circ f$, homology_intertwines proves $H(f) \circ H(a) = H(b) \circ H(f)$. kernelAction constructs the restriction of $H(a)$ to $\ker H(f)$. Thus passing to the comparison kernel does not discard a compatible operator or require asserting that the kernel is zero.

S7.1.5 4. Internal homology on the complete support diagram

For coherent support-indexed chain windows, the code constructs the diagrams of preceding terms, middle terms, following terms, cycles, boundaries, and homology. Let R_L denote the reverse reconstruction.

The differential identity is typed as

$R_L(d_{\text{Next}}) \circ R_L(d_{\text{Prev}}) = z_L : R_L(C^{(i-1)}) \rightarrow R_L(C^{(i+1)}),$
 $z_L(1, x) = (1, 0).$

The source and target degrees are explicit. A theorem shows that z_L differs from the constant global-zero map whenever L has a non-bottom element. Cycles satisfy the actual equalizer equation $d(x) = e.d(x)$.

The internal homology projection

$q : R_L(Z(C)) \rightarrow R_L(H(C))$

is the coequalizer of the **original incoming differential into cycles** and its supported-zero companion. `homology_coequalizes` and `homology_coequalizer` establish both parts of that universal property against every split-linear target. This is not an assumed homology compatibility of reconstruction.

For every transition $i \leq j$,

$\text{rho}^{\wedge} H_{ij}([z]) = 0$ iff $\text{rho}_{ij}(z)$ is a boundary at j .

If z is not a boundary at i , its retained class $(i, [z])$ is not $(i, 0)$, regardless of whether that transition kills it. The quotient-kernel equivalence of section 3 computes all such classes as a module, not just a yes/no predicate.

S7.1.6 5. A concrete comparison that detects more than the support labels

Fix any nontrivial commutative ring k . In the displayed degree let

$C: k \xrightarrow{-\theta} k \xrightarrow{-\theta} k,$
 $D: k \xrightarrow{-a} (a, \theta) \xrightarrow{-} k^2 \xrightarrow{-\theta} k.$

Use zero maps in the outside degrees. The two middle chain maps are

$f(a) = (a, \theta),$ $g(a) = (\theta, a).$

The code proves that the source class $[1]$ is nonzero. Its f -image is the **nonzero vector** $(1, 0)$, but that vector is a boundary in D , so $H(f)([1]) = 0$. In fact $H(f) = 0$ on every source class. The g -image $(0, 1)$ cannot be a boundary, because every boundary has second coordinate zero. Hence $H(g)([1])$ is nonzero and $H(f)$ is not $H(g)$.

The ordinary calculation gives $H(C) = k$ and $H(D) = k$ (the latter through second-coordinate projection). These identifications explain the example; the Lean acceptance tests directly prove the nonzero source class, the nonzero chain image, the zero induced map, the surviving class, and the distinction of induced maps rather than relying on a dimension computation.

Consequently the source and target complexes, individual fibre homology, and terminal homology can be identical while the transition kernel differs. This example is parametric over all nontrivial commutative rings, not a finite numerical experiment. Together with the general assembly theorem it supplies an explicit test of the information retained by support-indexed homology. It is an algebraic test case, not a claimed model of the global theta comparison.

S7.1.7 6. Balanced finite jets, including the original split quotient square

Let k and B be commutative rings and fix a $k[t]$ -algebra structure on B . Let $u: k[t] \rightarrow B$ denote that map and

$I_{-}(\text{rho}, m) = ((t - \text{rho})^{\wedge} m), \quad m \geq 0.$

There is an explicit B -algebra equivalence

$B \otimes_{k[t]} (k[t] / I_{-}(\text{rho}, m)) \simeq B / ((u(t) - u(\text{rho}))^{\wedge} m).$

It maps $b \otimes [p]$ to $[u(p)b]$, with inverse $[b]$ to $b \otimes 1$. No flatness assumption is used for this quotient base-change theorem. This is a specialization of the existing Mathlib theorem `Algebra.tensorProduct.quotIdealMapEquivTensorQuot`, not a claim to have independently developed general tensor-product quotient theory.

splitBaseChange lifts this equivalence to the original G-construction on both sides and proves the commutative square with reflect. It preserves τ and e separately and is a genuine semiring equivalence, not just an amplitude comparison. Thus every element of the finite-jet quotient has an inverse image before forgetting split support.

For any $k[t]$ -module V and v in V , the exact balancing calculation is

$$(1-u(t)) \text{ tensor } (t.v) - u(t) \text{ tensor } (v-t.v) = 0.$$

Expanding gives $1 \text{ tensor } (t.v) - u(t) \text{ tensor } v$, which vanishes by the tensor balancing relation. This is the algebraic equation of the source's Mellin repair. The choice $u(t)=s$ and $t.v=Dv$ is an application once those analytic module structures and intertwining maps have been constructed. It does not assert that the entire global balanced kernel vanishes, or that analytic scalar extension is flat without proof.

S7.1.8 7. Where Deligne enters the intended RH route

None of sections 1-6 imports a Deligne purity theorem. They provide the algebraic maps with which a proposed realization can now be specified and its kernel retained.

For this programme, the subsequent dependency diagram should distinguish the following mathematical obligations:

1. Construct the actual operator-balanced theta complex at the relevant support labels, and its action-compatible analytic comparison β to the spectral sheaf $\mathcal{O}/(2\xi)$. Identify its kernel or cone using the same maps; a finite-jet isomorphism is not a global injectivity argument.
2. Construct the geometric compact-support and ordinary complexes, their actual comparison c , their correspondence action, and a compatible realization identifying the required spectral sheaf with the image of $H^1(c)$. Any additional kernel/cone contributions to the trace must be kept or proved to vanish.
3. Establish two-sided weight control on that **same image**. In Deligne's geometric setting the compact-support upper estimate and the duality-derived ordinary-cohomology lower estimate meet on the image. The dualizing operation, degree and Tate twists, tensor-power maps, and hypotheses of the estimates are part of this step, not consequences stipulated in the definition of support.
4. Transport the spectral normalization without changing the arithmetic function. For a dilation parameter $a>1$, an eigenvalue a^ρ has modulus $a^{(\text{Re } \rho)}$. A proved weight-one modulus $a^{(1/2)}$, for the representation identified with every relevant spectral zero, then forces $\text{Re } \rho=1/2$. This last calculation does not supply the missing weight estimate; it states the precise final implication of this route.

These are dependencies of the specified route, not a claim that every possible proof of RH must follow it. The newly implemented kernel action is useful in steps 1-2 precisely because it keeps the part killed by a comparison available to the operator and trace calculation.

The supplied Connes-Consani article has an actual quotient-sheaf realization with a specified closure and scaling action. Its Theorem 1.2 realizes **critical zeros**. That theorem alone cannot be substituted for an identification representing all nontrivial zeros in the desired purity argument. This is a concrete source/target distinction for a future comparison map, not a dismissal of the construction.

S7.1.9 8. Reading record and references

The six supplied S20 chunks were checked against their individual declared hashes and concatenated in order. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256

e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The current top-level readers/english_standalone.html and readers/source_language.html were used for the selected readings, not the nested zero-accepted salvage corpus. The inspected windows include the tensor-square improvement in 3.2.13, the upper-bound statement in 3.3.1, and the duality/image argument in 3.3.4-3.3.6. This is not an independent audit of every page or of the entire historical proof. The Connes-Consani introduction and Theorem 1.2 were read in the supplied volume, printed pages 83-85 (physical PDF pages 94-96). No copyrighted source corpus is included in this contribution.

Deligne, P. (1980). La conjecture de Weil. II. *Publications Mathématiques de l'IHES*, 52, 137-252. <https://doi.org/10.1007/BF02684780>

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and zeta-cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83-102). American Mathematical Society.

Mathlib contributors. (2026). *Mathlib4*, revision fabf563a7c95a166b8d7b6efca11c8b4dc9d911f. In particular, LinearAlgebra/Quotient/{Defs,Basic}.lean, RingTheory/TensorProduct/Quotient.lean, Algebra/Homology/HomologicalComplex.lean, and Algebra/Category/ModuleCat/Basic.lean.

S7.1.10 9. Reproduction

From formal/splitzero on this branch, with elan and Python installed:

```
python3 prepare.py
python3 check_derived.py --prepare
python3 -m unittest -v test_derived_checker.py
lake update
# Verify the resolved Mathlib revision before fetching its cache.
test "$(git -C .lake/packages/mathlib rev-parse HEAD)" = fabf563a7c95a166b8d7b6efca11c8b4dc9d911f
lake exe cache get
lake -KmaxJobs=1 build
for name in SplitZeroReconstruction SplitZeroRecovery SplitZeroHomology SplitZeroInternalQuotient
↪ SplitZeroComplex SplitZeroHomologyExample SplitZeroFiniteJets; do
  lake env lean --trust=0 -DwarningAsError=true "$name.lean"
done
lake env lean --trust=0 -DwarningAsError=true AuditDerived.lean > AuditDerived.log
python3 check_derived.py AuditDerived.log
```

The manifest enumerates the selected declarations, including the principal structures, universal properties, equivalences, and example proofs. Missing, duplicate, unexpected, or nonstandard axiom reports fail. Source checks reject unfinished proofs and native-reduction escapes. The Python tests validate this checking machinery; the mathematical certificate is the successful Lean execution. Build-job totals include dependencies and are not theorem counts.

S8 PR 14: Coherent arithmetic interpolation and joint tensor boundary control

Source attribution. Complete TeX witness delivered with [web pull request #14](#); the associated Git revision is recorded in its delivery receipt.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Coherent_Interpolation_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 01a4fa76072ca4f3d8266c9fdbf61c816e53d09c673f8b8776b981b736c09e26

Original archive. Tau_Coherent_Interpolation_Control_2026-09-12.zip

Archive SHA-256. a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26

S8.1 Coherent arithmetic interpolation and joint tensor boundary control

Owner-directed continuation, 12 September 2026. Written mathematical arguments for review; the new analytic and matrix results in this note have not been Lean-checked.

S8.1.1 0. Accepted inputs and the change in the continuation

The new formalization receipt reports—and the repository metadata and workflow step summaries confirm—a successful check at 5d2772ea0a16d177ac3e01ff70394b90ba95a232 in PR #12. That certificate covers the specified retraction’s algebraic/topological consequences, the complete coefficient homotopy family, and the stated finite-chart dual maps. It does not certify the analytic construction of the theta retraction, support-changing synchronization homotopies, or the matrix estimates developed here. Its branch is left untouched.

The second attachment proves, for an unrestricted finite correction matrix B , the exact formula

$$G_B = G_{\perp,m} + (B + H_m^{-1}C_m)^* H_m (B + H_m^{-1}C_m). \quad (0.1)$$

Once $m+1 \geq \dim E_Z$, every positive metric ray is realized by an actual theta-boundary correction, with the arithmetic jets unchanged. Its further conclusion

$$\inf_B \epsilon_A(G_B) = \max_{\rho \in Z} |2\Re \rho - 1| \quad (0.2)$$

therefore has to be retained. This note does not count unrestricted metric optimization as a new arithmetic estimate. It instead uses the unique least-norm representative for a fixed, increasing space of the original arithmetic boundaries. It proves compatibility of these representatives for all finite packets, an exact full-jet moment-matrix formula, and a joint total-degree tensor construction.

All conclusions are derived on the existing characteristic-zero theta complex over the marked tau-base. None is asserted to be an application of finite-field purity to a space merely renamed tau.

S8.1.2 1. Fixed scalar, cochain, and arithmetic maps

Retain the original scalar construction

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the entire square

$$\boxed{\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}} \quad (1.1)$$

The target of $p_{\mathbb{Z}}$ is infinite. This quotient remains part of the marked geometry over $\mathfrak{b}_\tau$. Coefficient maps below fix external absence and carry a represented scalar zero to the represented scalar zero of the target.

Use the unchanged spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \forall b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{m \neq 0} \phi(mx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Write $q : \mathcal{B} \rightarrow Q$. Its action is by the original D , not a shifted operator. Let

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}f_0 = g. \quad (1.2)$$

The established range identities are used with these exact constants. They give $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ entire. In particular every finite jet map used below annihilates ΘV .

For $n \geq 0$, retain

$$V_n = \text{span}\{\phi_*, D\phi_*, \dots, D^n\phi_*\}, \quad L_n = \Theta V_n,$$

$$Q_n = \mathcal{B}/L_n \longrightarrow Q_{n+1}, \quad \ker = L_{n+1}/L_n. \quad (1.3)$$

Their split reconstructions have labels V_n . The transport of a class in L_{n+1}/L_n is the supported zero at V_{n+1} ; it is not external absence. The derivative has type $Q_n \rightarrow Q_{n+1}$ because $DL_n \subseteq L_{n+1}$.

S8.1.3 2. A polynomial presentation of the actual finite representatives

Take a nonempty finite set Z of actual zeros of g , with their complete orders m_ρ , and put

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[t]/(h), \quad v_h(s) = g(s)/h(s). \quad (2.1)$$

The polynomial variable t acts on the source by D and is evaluated at s by Mellin transformation. Full orders make v_h nonzero at all selected centres. Let j_h be the complete Hermite-jet/Chinese-remainder map into E_h and retain

$$v_h = j_h v_h \in E_h^\times, \quad \varepsilon_h = v_h^{-1}. \quad (2.2)$$

The unit ε_h is not the scalar e . Its split scalar is $\varepsilon_h^\bullet$.

There is an actual $F_h \in \mathcal{B}$ with $h(D)F_h = f_0$ and $\mathcal{M}F_h = v_h$. It can be obtained successively from

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad (2.3)$$

whenever the required Mellin value vanishes. At zero, replace the integral by its negative from 0 to x , using that vanishing; at infinity use the displayed integral. The defining estimates of $\mathcal{B}$ prove rapid decay of every scaling derivative at both ends. Repeated zero jets permit iteration. Uniqueness follows from injectivity of Mellin and $h\mathcal{M}F = 0 \Rightarrow F = 0$.

Define the linear map

$$\boxed{\mathcal{T}_h : \mathbb{C}[t] \longrightarrow \mathcal{B}, \quad P \longmapsto P(D)F_h.} \quad (2.4)$$

It is injective and satisfies

$$\mathcal{M}\mathcal{T}_h(P) = \frac{g}{h}P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (2.5)$$

Here $J_h F = j_h \mathcal{M}F$. The first equation retains the whole completed function, not its values only on the critical line.

For $N = n + d$, let $\mathcal{P}_N = \mathbb{C}[t]_{\leq N}$, and define

$$\mathcal{J}_{h,N} : \mathcal{P}_N \twoheadrightarrow E_h, \quad P \longmapsto v_h[P]_h.$$

Its exact sequence is

$$\boxed{0 \longrightarrow \mathcal{P}_n \xrightarrow{\times h} \mathcal{P}_{n+d} \xrightarrow{\mathcal{J}_{h,n+d}} E_h \longrightarrow 0.} \quad (2.6)$$

Applying $\mathcal{T}_h$ sends the left map to the original theta relations L_n . Thus its split internal quotient has exactly the relation $P - P' \in h\mathcal{P}_n$, with the support label retained.

The original source section is

$$s_h(u) = \mathcal{T}_h(\text{rem}_h(\varepsilon_h u)), \quad J_h s_h = 1, \quad q s_h = \sigma_h. \quad (2.7)$$

No arbitrary metric or section has entered (2.6).

S8.1.4 3. Exact arithmetic moment matrices and the full-jet interpolation formula

The Hilbert observation is $\mathcal{H} = L^2((0, \infty), dx)$. The precise transform map is

$$\mathcal{H} \xrightarrow{\mathcal{L}} L^2(\mathbb{R}, dy) \xrightarrow{\mathcal{F}_+} L^2(\mathbb{R}, dt/(2\pi)),$$

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{F}_+\psi(t) = \int_{\mathbb{R}} \psi(y)e^{ity}dy, \quad (\mathcal{F}_+\mathcal{L}F)(t) = \mathcal{M}F(\tfrac{1}{2} + it). \quad (3.1)$$

Both arrows are unitary with these measures. The inverse Fourier factor is $1/(2\pi)$; no mass of a measure below is set equal to one. The same map carries D to multiplication by $s_t = 1/2 + it$.

Define the finite positive measure

$$d\nu_h(t) = \left| \frac{g(\frac{1}{2} + it)}{h(\frac{1}{2} + it)} \right|^2 \frac{dt}{2\pi}. \quad (3.2)$$

Any apparent singularity at a selected critical-line zero is removed by the entire function v_h . All moments are finite. In the fixed basis $1, t, \dots, t^N$ of $\mathcal{P}_N$, put

$$(M_{h,N})_{ij} = \int_{\mathbb{R}} \overline{s_t^i} s_t^j d\nu_h(t), \quad 0 \leq i, j \leq N. \quad (3.3)$$

In the integrand t is the real integration variable; the polynomial basis is evaluated at s_t . This prevents the polynomial variable from being confused with the spectral-height coordinate.

The matrix $M_{h,N}$ is positive definite. A polynomial with zero norm vanishes almost everywhere on a line on which v_h is nonzero almost everywhere, hence is zero. Write $J_{h,N}$ for the matrix of $\mathcal{J}_{h,N}$ in the fixed quotient basis $1, t, \dots, t^{d-1}$. It includes multiplication by the actual unit v_h .

Set

$$\mathcal{C}_{h,N} = J_{h,N} M_{h,N}^{-1} J_{h,N}^*, \quad C_{h,N} = M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1}. \quad (3.4)$$

These are finite matrices on declared coordinate spaces. $J_{h,N} C_{h,N} = 1$. The actual test-function map is

$$R_{h,n} = \mathcal{T}_h C_{h,n+d} : E_h \longrightarrow \mathcal{B}. \quad (3.5)$$

Theorem. This is precisely the preceding orthogonal representative $(1 - P_n)s_h$, where P_n is the $L^2(dx)$ orthogonal projection onto L_n . In particular,

$$qR_{h,n} = \sigma_h, \quad J_h R_{h,n} = 1, \quad G_{h,n} := R_{h,n}^* R_{h,n} = \mathcal{C}_{h,n+d}^{-1}. \quad (3.6)$$

Proof. Every polynomial with jet u is $C_{h,N}u + z$ with $J_{h,N}z = 0$. The cross term vanishes because

$$z^* M_{h,N} C_{h,N} u = z^* J_{h,N}^* \mathcal{C}_{h,N}^{-1} u = 0.$$

Its norm is therefore $u^* \mathcal{C}_{h,N}^{-1} u + z^* M_{h,N} z$. This proves the unique minimizer. By (2.6), its difference from the degree-below- d lift (2.7) is exactly a member of L_n . It must be $(1 - P_n)s_h$. This also proves (3.6).

Thus the subtraction-of-nearby-Grams formula of the previous note has the exact alternative (3.6): invert the full-jet interpolation matrix. This is a matrix-valued Christoffel construction with the arithmetic local unit retained. The finite least-norm identity itself is standard linear algebra; its application here identifies the actual original representative, source relation, and completed multiplier.

For one simple zero ρ , the same formula is

$$G_{\rho,n} = \frac{1}{|g'(\rho)|^2 \mathcal{K}_{\nu_{\rho,n+1}}(\rho, \rho)}, \quad \mathcal{K}_{\nu,N}(z, w) = (1, z, \dots, z^N) M_N^{-1} (1, w, \dots, w^N)^*. \quad (3.7)$$

The complex conjugate in the second evaluation is part of the displayed kernel. Higher orders use (3.4), not just point evaluations. The factor $g' = 2\xi'$ is retained.

S8.1.5 4. All finite packets fit the same source filtration

Let $h \mid H$ be polynomials for nested packets with full orders and set $m = H/h$. Since the newly added centres are disjoint, h and m are coprime. There is a $\mathbb{C}[t]$ -linear CRT injection

$$\iota_{hH} : E_h \hookrightarrow E_H,$$

which inserts zero jets at the new centres. It is multiplication by the appropriate CRT idempotent after lifting a residue; it sends the unit of E_h to that idempotent, so it is not being asserted to be a unital ring map.

The polynomial map $P \mapsto mP$ has the exact properties

$$\mathcal{T}_H(mP) = \mathcal{T}_h(P), \quad J_H \mathcal{T}_h(P) = \iota_{hH} \mathcal{J}_h(P), \quad (4.1)$$

and is isometric from ν_h to ν_H because $|v_H mP| = |v_h P|$. The CRT equations and degree bounds prove

$$s_H \iota_{hH} = s_h.$$

The original L_n is independent of the packet, so

$$\boxed{R_{H,n} \iota_{hH} = R_{h,n}, \quad \iota_{hH}^* G_{H,n} \iota_{hH} = G_{h,n}, \quad \iota_{hH}^* W_{H,n} \iota_{hH} = W_{h,n}.} \quad (4.2)$$

The action also intertwines: $A_H \iota_{hH} = \iota_{hH} A_h$, where $A_h = M_t$. Thus a certificate on the larger packet restricts by this actual congruence to the smaller packet. The inverse interpolation matrices satisfy the corresponding inverse-compression formula; inverse and compression are not commuted.

The finite sections consequently glue on the algebraic direct limit of the actual packet injections. This statement does not assert convergence of an infinite sum of packet projectors or identify the whole Q with that direct limit.

At every support label, lift these maps as $(\lambda, u) \mapsto (\lambda, \iota_{hH} u)$ and $(\lambda, u) \mapsto (\lambda, R_{h,n} u)$. Added zero jets remain represented zeros. The label map in (1.3) is still the map that admits one more source relation.

S8.1.6 5. A global density result, with its precise cohomological consequence

The single derivative family $\{D^n f_0 : n \geq 0\}$ is already dense in $\mathcal{H}$.

Proof. Put $d\nu(t) = |g(1/2 + it)|^2 dt / (2\pi)$. It has an exponential moment of every order $c < \pi/2$. To verify this, the alternating-series formula and partial summation give

$$|\zeta(\tfrac{1}{2} + it)| \leq \frac{2|\tfrac{1}{2} + it|}{\sqrt{2} - 1}.$$

The gamma vertical-line asymptotic gives, for sufficiently large $|t|$,

$$|g(\tfrac{1}{2} + it)|^2 \leq C(1 + |t|)^{11/2} e^{-\pi|t|/2}.$$

The constant C is a fixed bound, not a numerical enclosure claimed here. The gamma fact used is NIST DLMF 5.11.9. The same exponential-moment property holds for ν_h , since v_h is entire at the finitely many possible denominator zeros and has polynomially modified tails.

If $f \in L^2(\nu)$ is orthogonal to every polynomial, the finite measure $\bar{f} d\nu$ has a holomorphic Fourier transform in a nonempty horizontal strip, by Cauchy–Schwarz and the exponential moment. All

derivatives at the origin vanish. Analytic uniqueness makes its transform zero; Fourier uniqueness for finite measures makes the measure zero. Thus polynomials are dense in $L^2(\nu)$.

Multiplication by the actual function $g(1/2 + it)$ is a unitary map $L^2(\nu) \rightarrow L^2(dt/(2\pi))$: its inverse divides by g away from its measure-zero set of real-line zeros. The inverse is a map of weighted L^2 spaces, not holomorphic division through an arithmetic zero. Combining it with (3.1) sends polynomials to $P(D)f_0$. This proves the assertion.

It follows for each fixed packet that

$$\boxed{G_{h,n} \longrightarrow 0 \quad \text{in every matrix entry and in operator norm.}} \quad (5.1)$$

Indeed $P_n \rightarrow 1$ strongly on $\mathcal{H}$, and the finite number of columns of s_h is fixed. Since each $G_{h,n}$ is positive definite, the least eigenvalue of its inverse tends to infinity. Every nonzero arithmetic row functional $\ell : E_h \rightarrow \mathbb{C}$ therefore has

$$\boxed{\|\ell\|_{G_{h,n}^\vee}^2 = \ell \mathcal{C}_{h,n+d} \ell^* \longrightarrow \infty.} \quad (5.2)$$

These facts do not kill a cohomology class. The actual maps are

$$\mathcal{B}/L_n \longrightarrow \mathcal{H}/L_n, \quad Q = \mathcal{B}/\Theta V \longrightarrow \mathcal{H}/\overline{\Theta V}^{\mathcal{H}} = 0. \quad (5.3)$$

The first map is injective because L_n is finite-dimensional and consists of the same smooth functions in both spaces. The second map forgets the stronger topology and has the whole Q as kernel. Its split lift lands at the represented zero in $G(0)$, not at τ . $G(0)$ retains both elements. An active class may therefore tend to zero under the L^2 observation while remaining fixed by $qR_{h,n} = \sigma_h$ and $J_h R_{h,n} = 1$.

The constructive consequence is to estimate relative boundary flux against the complete interpolation cost (3.4), not to use the absolute smallness of $G_{h,n}$ as a weight bound.

S8.1.7 6. Tail representation and same-metric reflection control

Retain the unscaled orthogonal vectors from the previous calculation:

$$u_0 = f_0, \quad u_n = (1 - P_{n-1})D^n f_0, \quad \mathfrak{h}_n = \|u_n\|^2, \quad (6.1)$$

$$Du_n = u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1).$$

The diagonal $1/2$ and the negative last coefficient follow from $D^* = 1 - D$ on these functions. Neither the u_n nor their norms are rescaled. Define $r_n = \mathfrak{h}_n^{-1}u_n^* s_h$. The new density result makes the entire tail formula exact:

$$\boxed{G_{h,n} = \sum_{j>n} \mathfrak{h}_j r_j^* r_j.} \quad (6.2)$$

This series converges in finite matrix norm. Its source map is $u \mapsto (r_j u)_{j>n}$ into the weighted sequence space with norm $\sum_{j>n} \mathfrak{h}_j |c_j|^2$. The map retains every coefficient; it is an isometry for $G_{h,n}$.

The previous boundary calculation, or telescoping (6.1), gives

$$\boxed{W_{h,n} = A_h^* G_{h,n} + G_{h,n} A_h - G_{h,n} = \mathfrak{h}_{n+1}(r_{n+1}^* r_n + r_n^* r_{n+1}).} \quad (6.3)$$

It pairs two explicit maps into L_{n+1}/L_n . At the original support V_n the derivative defect is $-[u_{n+1}]r_n$; at V_{n+1} it becomes the supported zero. This is also the input to the now-checked coefficient homotopy family, with $\kappa_n = K_n J_h$ and $\kappa_n \Theta = 0$.

For a reflection-stable packet define the actual antilinear involution

$$j_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}.$$

On functions use $j_B F(x) = x^{-1} \overline{F(1/x)}$. It is antiunitary on $\mathcal{H}$ and satisfies $Dj_B = j_B(1 - D)$. The specific section (2.7) satisfies $j_B s_h = s_h j_h$: since $h^\dagger = (-1)^d h$, both the unit and the numerator have the corresponding $(-1)^d$ factor. Also $j_B L_n = L_n$, so uniqueness of orthogonal projection gives $j_B R_{h,n} = R_{h,n} j_h$.

Write $j_h u = C_h \bar{u}$ in the retained coordinates. Then

$$\boxed{C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}.} \quad (6.4)$$

Consequently the generalized spectrum of $(W_{h,n}, G_{h,n})$ is symmetric under $\lambda \mapsto -\lambda$ **for this same canonical metric**, with full multiplicities retained. An upper inequality $W_{h,n} \preceq \epsilon G_{h,n}$ therefore implies the lower inequality $W_{h,n} \succeq -\epsilon G_{h,n}$ by applying j_h to the input vector.

The residue duality is still the separate explicit morphism with matrix S_h , and its pulled-back metric is $S_h^* G_{h,n}^{-1} S_h$. The relation to the current metric is the linear map $G_{h,n}^{-1} S_h$ into its antilinear dual coordinates. The arithmetic Weil form remains $S_h J_g$; no positivity is supplied merely by (6.4).

For completeness, retain the full rank-two formula

$$a_n = r_n G_{h,n}^{-1} r_n^*, \quad b_n = r_{n+1} G_{h,n}^{-1} r_{n+1}^*, \quad c_n = r_n G_{h,n}^{-1} r_{n+1}^*.$$

The least control excess is

$$\epsilon_{h,n} = \mathfrak{h}_{n+1} \left(|\Re c_n| + \sqrt{a_n b_n - (\Im c_n)^2} \right). \quad (6.5)$$

For reflection-stable Z , the additional proved identity is $\Re c_n = 0$: take the trace of $G_{h,n}^{-1} W_{h,n}$, which is $2\Re \operatorname{Tr} A_h - d = 0$. Thus (6.5) also equals $\mathfrak{h}_{n+1} \sqrt{a_n b_n - (\Im c_n)^2}$. Both equations are retained. Equation (3.6) supplies every occurrence of $G_{h,n}^{-1}$ without a difference of nearly equal Gram matrices.

S8.1.8 7. Joint total-degree control on the actual product complex

The tensor degree will be denoted k , independently of the relation degree n and the packet multiplicities. Put

$$E_h^{(k)} = E_h^{\otimes k}, \quad A_h^{(k)} = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A_h \otimes I^{\otimes(k-j)}.$$

The existing product complex over $\mathfrak{b}_\tau$ has differential

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k.$$

Define $\mathcal{P}_M^{(k)} = \mathbb{C}[t_1, \dots, t_k]_{\text{total degree} \leq M}$ for $M \geq k(d-1)$ and

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) (F_h^{\otimes k}). \quad (7.1)$$

Its Mellin transform is $P(s_1, \dots, s_k) \prod_j g(s_j)/h(s_j)$. Its full jet map $J_{k,M}$ is reduction modulo each $h(t_j)$ followed by multiplication by $v_h^{\otimes k}$. The target is $E_h^{(k)}$ and this map is surjective. The tensor remainder basis has total degree at most $k(d-1)$.

The exact relation space is

$$\mathcal{I}_{k,M} = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)). \quad (7.2)$$

Polynomial division in the fixed order $t_1, \dots, t_k$ represents each member as $\sum_j h(t_j)P_j$, with $\deg P_j \leq M-d$. It terminates because every h is monic and replacing a leading power lowers total degree. Under (7.1), each term is an actual top cochain boundary: in the j -th source factor use ϕ_* in place of F_h and give its primitive the factor $(-1)^{j-1}$. The differential contributes the second $(-1)^{j-1}$. The resulting boundary is $+h(t_j)P_j$ with no removed sign.

Thus these joint corrections are maps in the original product complex, not freely chosen positive tensor metrics. Set $\mathcal{B}_{k,M} = \mathcal{T}_h^{(k)} \mathcal{I}_{k,M}$.

The moment matrix on the monomials of total degree at most M is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s_t^{\alpha_j}} s_t^{\beta_j} d\nu_h(t). \quad (7.3)$$

Its positive definiteness follows because a polynomial cannot vanish almost everywhere on the full real product unless it is zero. It is a principal submatrix of the stated product moment matrix, not a declaration that the total-degree space is a tensor product of smaller univariate spaces.

The actual joint least-norm representative and its metric are

$$R_{k,M} = \mathcal{T}_h^{(k)} M_{k,M}^{-1} J_{k,M}^* (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1},$$

$$G_{k,M} = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (7.4)$$

The proof is the same explicit constrained-minimum calculation as (3.6), now with the relation ideal (7.2). In particular its arithmetic class is exactly $\sigma_h^{\otimes k}$ and every jet is fixed.

For a rectangular degree budget N one has the actual inclusion

$$\mathcal{P}_N^{\otimes k} \hookrightarrow \mathcal{P}_{kN}^{(k)}.$$

It carries the old product representative into the feasible set of the new problem. Therefore

$$G_{k,kN} \preceq G_{h,N-d}^{\otimes k} \quad (N \geq d). \quad (7.5)$$

This is an inequality from an inclusion of admissible original boundaries. It is not asserted to imply the same ordering of control forms, whose dependence on A is retained separately.

S8.1.9 8. The product error is one new total-degree layer

Let $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, and let $P_{\mathcal{B}_{k,M}}$ be its orthogonal projection onto the finite-dimensional actual boundary space. With the fixed low-degree section $s_h^{\otimes k}$,

$$R_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) s_h^{\otimes k}.$$

Put

$$\mathcal{E}_{k,M} = (1 - P_{\mathcal{B}_{k,M}})\mathcal{B}_{k,M+1}.$$

Because the jet map is already onto at both degrees,

$$\dim \mathcal{E}_{k,M} = \binom{M+k}{k-1}. \quad (8.1)$$

This is $\dim \mathcal{P}_{M+1}^{(k)} - \dim \mathcal{P}_M^{(k)}$; the d^k jet constraints cancel from the dimension difference.

Let $D^{(k)} = \sum_j D_j$. Its induced action on jets is $A_h^{(k)}$. The actual derivative defect lies in $\mathcal{B}_{k,M+1}$:

$$B_{k,M} = D^{(k)} R_{k,M} - R_{k,M} A_h^{(k)}.$$

Define the two explicitly typed maps into the same new relation layer

$$Y_{k,M} = P_{\mathcal{E}_{k,M}} R_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M},$$

$$C_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) B_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M}.$$

Integration by parts gives $(D^{(k)})^* + D^{(k)} = kI$ on these functions. Orthogonality then proves

$$W_{k,M} := (A_h^{(k)})^* G_{k,M} + G_{k,M} A_h^{(k)} - k G_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad (8.2)$$

$$G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}, \quad \text{rank } W_{k,M} \leq 2 \binom{M+k}{k-1}. \quad (8.3)$$

For (8.3), use the orthogonal decomposition $\mathcal{B}_{k,M+1} = \mathcal{B}_{k,M} \oplus \mathcal{E}_{k,M}$. Its projection sends $R_{k,M}$ to $Y_{k,M}$ and the new representative is $R_{k,M} - Y_{k,M}$.

The source quotient map is the same degree transport as before:

$$\mathcal{B}_{k,M+1}/\mathcal{B}_{k,M} \xrightarrow{\sim} \mathcal{E}_{k,M}, \quad [z] \mapsto (1 - P_{\mathcal{B}_{k,M}})z.$$

A derivative class in this kernel becomes the supported zero only at the next total-degree label. The primitive retains which tensor factor supplied the theta boundary and its cochain sign. Synchronization by $(1, e)$ in each required two-leg factor supplies the common joint support, while the original mask remains recorded.

Thus the product estimate need not simply copy k identical one-factor errors: (8.2) gives a new jointly optimized boundary-layer expression with completely specified moment data and maps. Its rank bound alone does not bound its generalized eigenvalues.

S8.1.10 9. Same-metric reflection and residue duality on the product

For a reflection-stable packet, $j_h^{\otimes k}$ is the antilinear involution on $E_h^{(k)}$, defined on pure tensors by applying j_h to every factor. The corresponding function reflection is $F(x_1, \dots, x_k) \mapsto \prod_j x_j^{-1} \overline{F(x_1^{-1}, \dots, x_k^{-1})}$.

It preserves both the total-degree numerator space and its relation ideal. Since $(g/h)^\dagger = (-1)^d (g/h)$, its action on a numerator is $(-1)^{kd} P^\dagger$; this factor is retained. Uniqueness of the minimum in (7.4) proves reflection compatibility of $R_{k,M}$. Therefore the analogue of (6.4) gives symmetry of the generalized spectrum of $(W_{k,M}, G_{k,M})$ on the same metric.

Residue duality remains the explicitly specified map with matrix $S_h^{\otimes k}$ and target line $\mathcal{L}_k$, on which dilation acts by a^k . It satisfies

$$(A_h^{(k)})^* S_h^{\otimes k} + S_h^{\otimes k} A_h^{(k)} = k S_h^{\otimes k}.$$

Any fixed degree-orientation scalar in the cochain evaluation is retained separately; the ordered product of the scalar residue pairings here has matrix exactly $S_h^{\otimes k}$.

Its dual metric and control are

$$G_{k,M}^D = (S_h^{\otimes k})^* G_{k,M}^{-1} S_h^{\otimes k},$$

$$W_{k,M}^D = -(S_h^{\otimes k})^* G_{k,M}^{-1} W_{k,M} G_{k,M}^{-1} S_h^{\otimes k}. \quad (9.1)$$

The Jacobian contraction is still the original multiplication trace, with every multiplicity, and is unchanged by the source boundaries used to construct (7.4). Neither (9.1) nor reflection alone establishes an arithmetic upper bound.

S8.1.11 10. Quantitative certificate and the Deligne comparison

For one or several variables write M for the positive moment matrix, J for its fixed full-jet map, and $\mathcal{C} = JM^{-1}J^*$. These actual matrices now replace an unrestricted metric search.

A useful deterministic enclosure is as follows. Suppose a Hermitian approximation $\widehat{M}$ has a verified bound $\|M - \widehat{M}\| \leq \eta$ and $\widehat{M} \succeq \mu I$ for a retained $\mu > \eta$. Then

$$\|M^{-1} - \widehat{M}^{-1}\| \leq \frac{\eta}{\mu(\mu - \eta)}, \quad \|\mathcal{C} - J\widehat{M}^{-1}J^*\| \leq \|J\|^2 \frac{\eta}{\mu(\mu - \eta)}. \quad (10.1)$$

This is the resolvent identity for inverses. The positive lower bound on $\mathcal{C}$ must also be retained before a second inversion. With $\widehat{G}$ and $\widehat{W}$ and verified errors η_G, η_W , the sufficient matrix certificate is

$$\widehat{G} - \eta_G I \succ 0, \quad \epsilon \widehat{G} \pm \widehat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0. \quad (10.2)$$

No numerical interval enclosure for the actual infinite arithmetic moments is claimed merely from a floating-point calculation. The previous exterior-theta sums give another exact representation of their one-variable inputs; all tails and roundoff must be supplied to (10.1).

Deligne's retained Corollaries 3.3.4–3.3.6 place upper and dual lower bounds on the same compact-to-ordinary image; the tensor step in 3.2.13 improves an estimate only after additional geometric control. This note supplies a canonical coherent family, its exact source relation ideal on products, and the same-metric reflection/dual operations with which such an upper estimate can now be tested. It does not import finite-field hypotheses into the tau chart.

For an actual eigenvector $A_h v = \rho v$, every constructed tensor metric still has the exact diagnostic

$$\frac{(v^{\otimes k})^* W_{k,M} v^{\otimes k}}{(v^{\otimes k})^* G_{k,M} v^{\otimes k}} = k(2\Re \rho - 1). \quad (10.3)$$

The quantity to bound is therefore the generalized boundary form (8.2) relative to (7.4). A dimension/rank estimate, a decrease of a Gram matrix, or unrestricted metric realizability is not substituted for that relative estimate. The new constructive route is the exact finite full-jet moment problem together with its next-layer flux, not another homotopy choice.

S8.1.12 11. Reading, provenance, and validation

Newly read: both supplied formalization/control notes in full; the preceding local `Tau_Homotopy_Boundary_Control/NOTE.tex`, especially sections 4–10; the retained Deligne English TeX excerpt 3.3.4–3.3.6 and French 6.2.4–6.2.7. This is not a new audit of every page of Weil II. No OCR or PDF extraction was used. Source hashes are in the package reading record.

External context checked: NIST DLMF 5.11.9 for the gamma bound; Connes–Consani–Moscovici, arXiv:2403.01247, abstract, for prior moment/Jacobi constructions from squared absolute local factors; Romik, arXiv:1902.06330, abstract, for prior orthogonal-polynomial expansions of Ξ ; Lasserre–Pauwels, arXiv:1606.03858, abstract, for inverse-moment/Christoffel context. Only the gamma asymptotic is used as an external estimate; the other papers are priority/context references, not sources of an unread theorem. The arguments above prove the specific maps used here.

The accompanying exact checker uses declared finite discrete calibration measures and rational complex polynomial data. It checks the general finite matrix identities, joint relation ranks, signs, full repeated-root reductions, reflection, naturality, and comparison metrics. Its model polynomial is never labelled an arithmetic zeta counterexample. Analytic density, asymptotic gamma bounds, and the original theta range identities are written arguments, not certified by those finite tests. No new Lean execution is claimed. No RH or GRH conclusion is claimed.

S9 Full-jet interpolation kernels and the actual next relation layer

Source attribution. Complete TeX witness from the dated kernel-layer supplement, delivered 12 September 2026. Its handoff identifies PR 14 as the arithmetic source and the existing PR 15 revision as its formal baseline. The supplement’s add-only patch was verified locally; its receipt records no remote branch modification. These NOTE bytes are identified by the source and archive hashes below; neither baseline revision is represented as their Git head.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Kernel_Layer_Integration/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Arithmetic source baseline PR 14

Cited existing PR 15 formal baseline. 21970bbf4760d0bbca512a4a7996a2e38f968947

Source SHA-256. be8485471fc7dac41d97e4a9fd52413da14e83fbc42c6c55c01a5f0908caa06c

Original archive. Tau_Kernel_Layer_Integration_2026-09-12(1).zip

Archive SHA-256. ebd927c8a9afa90600f5320bd4189ecb48085889835205057b1bfafdb68d23d2

S9.1 Full-jet interpolation kernels and the actual next relation layer

S9.1.1 Integration supplement — 12 September 2026

This supplement reviews the supplied **Coherent arithmetic interpolation and joint tensor boundary control** package and derives an additional finite-stage bridge. The source’s canonical representative, original theta boundaries, full multiplicities, and support transport are retained. The new formulas below have written proofs and exact finite regression tests; they are **not new Lean certificates**.

The supplied ZIP has SHA-256 a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26. All 29 entries of its manifest verified. Its note, handoff, and checker match the published Git blobs at PR #14, commit 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc. The source checker was independently rerun: nineteen tests passed normally and under optimized Python, with identical result records; the intentionally false twentieth test failed in both modes. See SOURCE_REVIEW_RECEIPT.json.

The comparison with PR #15 is limited precisely: its existing five-module receipt certifies its stated support-changing, relation-layer, finite-Krylov and rank-two interfaces. This supplement does not extend that receipt to the source’s theta analytics, all-degree orthogonal-polynomial construction, or the new tensor-kernel calculations.

S9.1.2 1. Source results retained, rather than reconstructed differently

Write the source polynomial degree as N , to distinguish it from a moment matrix. Fix tensor degree $k \geq 1$, a nonempty packet polynomial h of degree d , and $N \geq k(d - 1)$. The packet includes the complete orders of the selected actual zeros. Retain

$$g = 2\xi, \quad E = E_h^{\otimes k}, \quad A = A_h^{(k)}, \quad \mathcal{J}_N : \mathcal{P}_N^{(k)} \twoheadrightarrow E. \quad (1)$$

The last map is reduction modulo the original ideal $(h(t_1), \dots, h(t_k))$, followed by multiplication by the original local unit $v_h^{\otimes k}$. Its kernel is $\mathcal{I}_{k,N}$. The source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k) F_h^{\otimes k}, \quad \mathcal{MT}(P) = P(s_1, \dots, s_k) \prod_i \frac{g(s_i)}{h(s_i)}. \quad (2)$$

Let $\mathcal{L}_N = \mathcal{TI}_{k,N}$, $P_N = P_{\mathcal{L}_N}$, and let $\sigma : E \hookrightarrow Q^{\widehat{\otimes} k}$ be the source arithmetic inclusion. Its canonical representative and forms are

$$R_N = (I - P_N) s_h^{\otimes k}, \quad G_N = R_N^* R_N \succ 0, \quad q^{(k)} R_N = \sigma, \quad J^{(k)} R_N = I, \quad (3)$$

$$B_N = D^{(k)} R_N - R_N A, \quad W_N = A^* G_N + G_N A - k G_N. \quad (4)$$

The supplied note proves $B_N(E) \subseteq \mathcal{L}_{N+1}$ and, with $\mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp$,

$$Y_N = P_{\mathcal{E}_N} R_N, \quad C_N = (I - P_N) B_N, \quad W_N = -(Y_N^* C_N + C_N^* Y_N). \quad (5)$$

These are source results (source note, §§2–4 and 7–9), not claimed as discoveries of this supplement. Their finite-dimensional derivations check: the constrained minimum is over precisely the original ideal, the derivative increases total degree by one, and orthogonal projection removes only the admitted relation space. The source's exact sequence and packet naturality require full orders. Its total-degree statements require the threshold in (1). Its analytic function-space identities remain the stated analytic inputs, not consequences of a finite fixture.

The original coefficient square $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ over $\mathbb{Z} \rightarrow \mathbb{C}$, with projections $p_{\mathbb{Z}}, p_{\mathbb{C}}$, is unchanged. A vector in an admitted relation maps to the receiving fibre's supported zero, never to external absence τ .

S9.1.3 2. Calculate the inverse interpolation metric by full-jet columns

Use exactly the source measure

$$d\nu_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}. \quad (6)$$

Let $p_j^{(h)}(s)$ be its monic orthogonal polynomials, with the polynomial variable evaluated at $s = 1/2 + it$, and retain their positive squared norms

$$\kappa_j^{(h)} = \int |p_j^{(h)}(1/2 + it)|^2 d\nu_h(t) > 0. \quad (7)$$

These are **not** the earlier theta-Krylov vectors u_j , nor their norms $\mathfrak{h}_j$. In particular they depend on h ; they are not assigned norm one. Existence through any finite degree follows from the source measure's positive-definite moment matrices. No limiting assertion is needed here.

Suppress h in notation. For a multi-index α , define

$$p_\alpha = \prod_i p_{\alpha_i}, \quad \kappa_\alpha = \prod_i \kappa_{\alpha_i}, \quad a_j = v_h[p_j]_h \in E_h, \quad a_\alpha = \bigotimes_i a_{\alpha_i} \in E. \quad (8)$$

Every a_j is the full jet, including the local unit. Product orthogonality gives $\langle \mathcal{T} p_\alpha, \mathcal{T} p_\gamma \rangle = \kappa_\alpha \delta_{\alpha\gamma}$. These polynomials form a basis of $\mathcal{P}_N^{(k)}$, because the monic triangular basis change preserves total degree.

Set

$$K_N := G_N^{-1} = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha}. \quad (9)$$

Proof. In this basis the source moment matrix is $\text{diag}(\kappa_\alpha)$ and the full-jet matrix has columns a_α . Substitute into the source identity $G_N^{-1} = J_N M_N^{-1} J_N^*$. A change of polynomial basis does not change that product. This proves (9) for the same canonical metric, not a newly selected metric.

Its representative has the explicit coefficient formula

$$R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (10)$$

In particular $R_N K_N$ is an actual map from finite antidual coordinates into the original test-function space. It satisfies $J^{(k)} R_N K_N = K_N$ and $q^{(k)} R_N K_N = \sigma K_N$. The invertible coordinate change K_N is retained, not identified with the identity.

S9.1.4 3. An explicit basis of the original next relation layer

Let $\Sigma_{N+1} = \{\beta \in \mathbb{N}^k : |\beta| = N + 1\}$. Define actual test functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \beta \in \Sigma_{N+1}. \quad (11)$$

They are in $\mathcal{E}_N$. First, their full jets vanish by (3), so their numerator polynomials lie in the original ideal at degree at most $N + 1$; thus $e_\beta \in \mathcal{L}_{N+1}$. Second, $\mathcal{T}p_\beta$ is orthogonal to all degree-at-most- N polynomials and $R_N a_\beta$ is orthogonal to $\mathcal{L}_N$. Hence $e_\beta \perp \mathcal{L}_N$.

Define

$$\mathcal{E}_N : \mathbb{C}^{\Sigma_{N+1}} \longrightarrow \mathcal{E}_N, \quad c \longmapsto \sum_{\beta} c_\beta e_\beta, \quad (12)$$

$$U_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad D_N = \text{diag}(\kappa_\beta)_{\beta \in \Sigma_{N+1}}. \quad (13)$$

The map $\mathcal{E}_N$ is an isomorphism. Independence follows by taking the degree- $N + 1$ polynomial part of (11). For spanning, subtract from a relation its degree- $N + 1$ combination of the e_β . The remainder is a relation of degree at most N , hence belongs to $\mathcal{L}_N$. An element of $\mathcal{E}_N$ has no such remaining component.

Its Gram matrix and pairing with the representative are exactly

$$H_N := \mathcal{E}_N^* \mathcal{E}_N = D_N + U_N^* G_N U_N, \quad \mathcal{E}_N^* R_N = -U_N^* G_N. \quad (14)$$

The cross terms vanish because the new-degree polynomials are orthogonal to the old polynomial space. In particular $H_N \succ 0$, and no original norm has been suppressed.

This identifies the existing quotient equivalence in coordinates:

$$\mathbb{C}^{\Sigma_{N+1}} \xrightarrow{\mathcal{E}_N} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1} / \mathcal{L}_N. \quad (15)$$

The inverse quotient-to-layer map is the source's $[z] \mapsto (I - P_N)z$. For each nonzero coordinate vector, its class in this relative quotient is nonzero, and its image in the next admitted quotient is zero. At the split level this remains the next supported zero.

For an explicit theta primitive, form $p_\beta - \sum_{|\alpha| \leq N} p_\alpha a_\alpha^* G_N a_\beta / \kappa_\alpha$. Its full jet is zero. Fixed-order monic division expresses it as $\sum_i h(t_i) Q_{i,\beta}$, with $\deg Q_{i,\beta} \leq N + 1 - d$. In the i -th cochain factor use the original ϕ_* in place of F_h , and multiply its primitive by $(-1)^{i-1}$. The differential supplies the same sign, producing the displayed relation with positive sign. This supplies the actual primitive, not just a dimension count.

S9.1.5 4. Both boundary maps now have explicit coordinates

Define the matrix $F_N : E \rightarrow \mathbb{C}^{\Sigma_{N+1}}$ by its rows

$$(F_N)_\beta = \sum_{i: \beta_i > 0} \frac{a_{\beta - \mathbf{e}_i}^*}{\kappa_{\beta - \mathbf{e}_i}}. \quad (16)$$

Then

$$\boxed{Y_N = -\mathcal{E}_N H_N^{-1} U_N^* G_N, \quad C_N = \mathcal{E}_N F_N G_N.} \quad (17)$$

The first equation is the actual orthogonal projection formula using (14). For the second, multiply the finite expansion (10) by $s_1 + \cdots + s_k$. Monicity implies that its degree- $N + 1$ coefficient at p_β is exactly $(F_N)_\beta G_N u$. The term $R_N A u$ has degree at most N . Subtract $\mathcal{E}_N F_N G_N u$ from $B_N u$: the result has degree at most N and zero full jets, hence is in $\mathcal{L}_N$. Projecting by $I - P_N$ proves the equation.

Consequently the old quotient records

$$\boxed{[B_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N} = [\mathcal{E}_N F_N G_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N}.} \quad (18)$$

It becomes the supported zero under the next relation transition. The symbol $\mathcal{B}^{(k)}$ denotes the original product test-function space used by the source; no derivative on all of its L^2 completion is defined here.

S9.1.6 5. The control form is a top-degree full-jet endpoint

Define

$$\boxed{V_N := A K_N + K_N A^* - k K_N.} \quad (19)$$

Substituting (17) into the source boundary identity (5), and using $\mathcal{E}_N^* \mathcal{E}_N = H_N$, gives

$$\boxed{W_N = G_N (U_N F_N + F_N^* U_N^*) G_N.} \quad (20)$$

Meanwhile direct multiplication of (4) by K_N on both sides gives $K_N W_N K_N = V_N$. Thus

$$\boxed{V_N = U_N F_N + F_N^* U_N^* = \sum_{|\alpha|=N} \sum_{i=1}^k \frac{a_{\alpha + \mathbf{e}_i} a_\alpha^* + a_\alpha a_{\alpha + \mathbf{e}_i}^*}{\kappa_\alpha}.} \quad (21)$$

This is a finite endpoint identity derived through the actual theta relation layer. It does not assume a small boundary, discard mixed terms, or replace the packet operator. It can also be obtained by finite Christoffel–Darboux telescoping. The standard scalar orthogonal-polynomial kernel and telescoping identities are recorded by NIST (n.d., §18.2(v)); the full-jet, original-layer identification (11)–(21) is proved here rather than imported from that scalar formula.

For this alternative telescoping proof, multiplication satisfies $sp_j = p_{j+1} + \beta_j p_j - (\kappa_j / \kappa_{j-1}) p_{j-1}$, with $\operatorname{Re} \beta_j = 1/2$. The imaginary part need not vanish for the packet-dependent measure. Only $\beta_j + \overline{\beta_j} = 1$ is needed. Interior adjacent terms cancel; (21) is the uncanceled total-degree boundary. The new tests include an asymmetric measure with a nonzero imaginary diagonal coefficient.

Regroup by α and put $b_\alpha = \sum_i a_{\alpha + \mathbf{e}_i}$. Equation (21) factors through the two sets of columns $(a_\alpha)_{|\alpha|=N}$ and $(b_\alpha)_{|\alpha|=N}$. Therefore

$$\boxed{\text{rank } W_N = \text{rank } V_N \leq \min \left\{ d^k, 2 \binom{N+k-1}{k-1} \right\}.} \quad (22)$$

This sharpens the source's general next-layer bound $2 \binom{N+k}{k-1}$ for this product-polynomial construction. It does **not** change the actual dimension of $\mathcal{E}_N$, which remains $\binom{N+k}{k-1}$. The sharpening comes from the scalar summed derivative and regrouping its top-degree columns. For $k = 2, d = 4, N = 6$, the two structural bounds are 14 and 16 on a 16-dimensional packet; the exact regression tests cover this nonvacuous case. Neither rank bound is a small-eigenvalue estimate.

S9.1.7 6. Estimate the same form without a second inversion

Since K_N is Hermitian positive definite,

$$\boxed{-\epsilon G_N \preceq W_N \preceq \epsilon G_N \iff -\epsilon K_N \preceq V_N \preceq \epsilon K_N.} \quad (23)$$

This is congruence by the specified invertible map K_N . More explicitly, for $u = K_N \lambda$, $u^* G_N u = \lambda^* K_N \lambda$ and $u^* W_N u = \lambda^* V_N \lambda$. This identifies the generalized Rayleigh quotients, including multiplicities of generalized eigenvalues. The original generator A is not replaced.

The source's same-metric reflection therefore still transfers an upper estimate to a lower estimate for precisely the original G_N . Under the natural weight- k antidual action $A^D = kI - A^*$, the defect of the antidual metric K_N is $-V_N$, not V_N . The dual sign and weight are both retained.

The reflection map itself can be retained explicitly. If the source involution is $j(u) = C\bar{u}$, its antidual involution is $j^\#(\lambda) = C^T \bar{\lambda}$. From $C\bar{C} = I$ and $C^* G_N C = \overline{G_N}$, inversion gives $(C^T)^* K_N C^T = \overline{K_N}$. The control identity transports to $(C^T)^* V_N C^T = -\overline{V_N}$. Thus the two-sided equivalence retains the actual involution, not only spectral symmetry.

Equation (23) avoids the *second* inversion $G_N = K_N^{-1}$ in the final inequality test. It does not eliminate the need to enclose moments, full-jet coefficients, or the conditioning of the first construction of K_N .

A deterministic finite certificate can therefore work directly with $\hat{K}$, $\hat{A}$ and error bounds $\|K_N - \hat{K}\| \leq \eta_K$, $\|A - \hat{A}\| \leq \eta_A$, where $\hat{K}$ is Hermitian. Put

$$\hat{V} = \hat{A}\hat{K} + \hat{K}\hat{A}^* - k\hat{K}, \quad \eta_V = (2\|\hat{A}\| + k)\eta_K + 2\eta_A(\|\hat{K}\| + \eta_K). \quad (24)$$

Expanding both perturbations proves $\|V_N - \hat{V}\| \leq \eta_V$. Sufficient conditions, for $\epsilon \geq 0$, are

$$\hat{K} - \eta_K I \succ 0, \quad \epsilon \hat{K} \pm \hat{V} - (\epsilon \eta_K + \eta_V) I \succeq 0. \quad (25)$$

No enclosure of the actual infinite arithmetic moments is asserted in this supplement.

S9.1.8 7. The metric update and the univariate connection

The positive update from (9) and its inverse are

$$\boxed{K_{N+1} = K_N + U_N D_N^{-1} U_N^*,} \quad (26)$$

$$\boxed{G_{N+1} = G_N - G_N U_N (D_N + U_N^* G_N U_N)^{-1} U_N^* G_N.} \quad (27)$$

Equation (27) follows either by direct multiplication with (26), or from $R_{N+1} = R_N - Y_N$ and the orthogonal decomposition. The middle matrix is exactly the actual layer Gram matrix H_N in (14), not an unrelated auxiliary matrix. The lost Gram form is $Y_N^* Y_N$.

For $k = 1$ and $N = n + d$, (11) recovers the earlier monic theta vector:

$$\boxed{e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1}.} \quad (28)$$

Indeed the numerator of (11) is monic of degree $n+d+1$, divisible by monic h , and its quotient has degree $n+1$ and leading coefficient one. Applying $\mathcal{T}_h$ gives f_{n+1} plus an element of L_n , orthogonal to L_n . This uniquely identifies u_{n+1} . Thus the packet-dependent polynomial construction is connected to the packet-independent original theta norm by an exact identity; those two norm families were not conflated.

S9.1.9 8. Degree recursion and the remaining estimate

For $A_j = a_j a_j^* / \kappa_j$, define finite matrix sums

$$Q_{k,m} = \sum_{|\alpha|=m} \bigotimes_i A_{\alpha_i}, \quad Q_{k,m} = \sum_{j=0}^m A_j \otimes Q_{k-1,m-j}, \quad K_{k,N} = \sum_{m=0}^N Q_{k,m}. \quad (29)$$

These are finite degree convolutions; they do not assert convergence of an infinite generating series. They compute the product full-jet interpolation matrix using one-variable polynomial norms and jets, without inverting a total-degree moment matrix of dimension $\binom{N+k}{k}$. The target dimension remains d^k ; no polynomial-time complexity or unconditional numerical stability is claimed.

In univariate packet inclusions, the source's exact equality $R_{H,n} \iota_{hH} = R_{h,n}$ remains in force. The packet-dependent polynomial bases used here need not themselves commute under inclusion. No extension of this univariate equality to different joint total-degree budgets is presumed.

Finally, on $Av = k\rho v$, with $v = v_0^{\otimes k}$, and $\lambda = G_N v$,

$$\frac{\lambda^* V_N \lambda}{\lambda^* K_N \lambda} = k(2 \operatorname{Re} \rho - 1). \quad (30)$$

The additional work is therefore the explicit layer basis, derivative-coordinate map, endpoint identity, and direct relative certificate for the same source-controlled object. It is not a proof of a uniform sublinear excess. No arithmetic block or nilpotent jet has been removed.

S9.1.10 Verification and literature scope

The independent new checker contains eighteen exact test methods. They cover the kernel basis change, repeated jets, two-variable layers, actual derivative-class coordinates, Gram updates, the original univariate theta vector, endpoint factorization, a nonvacuous improved rank bound, antidual signs, generalized characteristic polynomials, nonzero off-line *polynomial model* diagnostics, nonreal recurrence diagonals, fixed-order theta primitives, and deterministic matrix perturbation certificates. Normal and optimized result records agree. Both intentional-failure modes return failure.

These are finite regression tests, not proofs of the general analytic inputs and not Lean execution. The general finite identities have the written proofs above. The source's nineteen tests and this supplement's eighteen tests are separate suites, not counts of novel theorems.

The matrix-kernel and Christoffel–Darboux methods have established antecedents. NIST's general orthogonal-polynomial formulas supply the standard scalar comparison. Lasserre and Pauwels (2016) connect inverse moment matrices with orthogonal polynomials and the Christoffel function; only their abstract was examined here, so no theorem from that paper is used as a premise. No global priority claim is made for standard kernel or inverse-update identities.

S9.1.11 References

Coherent arithmetic interpolation and joint tensor boundary control. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #14, commit 3bff1b2bd87dfa2592f65c5cc231f6ee9541dfc. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/14>

Formalize support-changing relation layers and the Krylov boundary-control calculation. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #15, verified source commit 21970bbf4760d0bbca512a4a7996a2e38f968947. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/15>

Lasserre, J.-B., & Pauwels, E. (2016). *Sorting out typicality with the inverse moment matrix SOS polynomial* [Preprint]. arXiv. <https://arxiv.org/abs/1606.03858>

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S10 Arithmetic frontier estimates and symmetric tensor control

Source attribution. Complete TeX witness from the dated symmetric-frontier continuation, delivered 12 September 2026. Its delivery receipt records an add-only local patch based on PR 14, with no remote write. The exact revision below is that patch base; it is not a new remote head for this NOTE. The source and archive hashes identify the complete delivered witness.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Symmetric_Frontier_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Recorded add-only patch base, PR 14

Source SHA-256. 734b6c935892ebd9d18b506f6b592f1ca3489afcb95d5c669a489f4ca04656cf

Original archive. Tau_Symmetric_Frontier_Control_2026-09-12(1).zip

Archive SHA-256. a024ee4bbb4df9555bc6f94c2d74a2e90c077533ad2ce312863342b758539796

S10.1 Arithmetic frontier estimates and symmetric tensor control

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. New written deductions from the precise arithmetic source of PR #14, submitted for mathematical review. The supplied Lean receipt certifies its stated coefficient, quotient, homotopy, and topology interfaces; it does not certify the analytic inputs or the results in this note. No purity estimate is assumed. No claim of global novelty is made for the underlying orthogonal-polynomial, finite-group, or matrix identities.

S10.1.1 0. Source and purpose

The source is PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, research-note Git blob 4ec40c15397128552bc276ec903fe59c89657d4e. Its actual total-degree representative, full-jet map, and theta relation ideal are used without changing them.

The source obtained two maps into the next relation layer, with

$$W_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}. \quad (0.1)$$

This note constructs coordinates on that *actual relation layer*, computes the relative control in them, proves a finite arithmetic upper bound, and constructs a smaller symmetric tensor cohomology summand that still contains every repeated eigenline used in amplification. Its exact complement and its trace are retained.

The decisive new coordinate map is

$$b_M(q) = \mathcal{T}_h^{(k)} q - R_{k,M} \mathcal{J} q. \quad (0.2)$$

It converts the highest orthogonal polynomial degree into an original theta boundary and has a calculated inverse and Gram matrix. A highest-degree polynomial alone need not be a boundary; its full-jet value is subtracted by the specific source representative in (0.2).

S10.1.2 1. Unchanged split base, arithmetic source, and quotients

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.1)$$

The base is the marked absolute point $\mathfrak{b}_\tau$. The first arithmetic target in (1.1) is still infinite. Every fixed-label linear map below is reconstructed by $(\lambda, x) \mapsto (\lambda, Tx)$; its value on external absence is external absence. A vanishing amplitude at an active label remains that label's supported zero.

Use the original $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, where

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= \sum_{n \neq 0} \phi(nx), \quad q : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V. \end{aligned}$$

The input is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_*(s) = g(s) = 2\xi(s).$$

Fix a nonempty finite packet of actual nontrivial zeros of g , with their full orders, and define

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[t]/(h), \quad v_h = g/h, \quad v = j_h v_h \in E^\times. \quad (1.2)$$

No squarefree assumption is made. The source constructs $F_h \in \mathcal{B}$ with $h(D)F_h = \Theta\phi_*$ and $\mathcal{M}F_h = v_h$. It supplies

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{M}\mathcal{T}_h(P) = v_h P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v[P]_h. \quad (1.3)$$

For tensor degree k , put $E_k = E^{\otimes k}$ and $A_k = \sum_{j=1}^k A_j$, where $A = M_t$. Let

$$\begin{aligned} \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, \quad M \geq k(d-1), \\ \mathcal{T}^{(k)}P &= P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{J}P = v^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \quad (1.4)$$

The relation space and its realization are

$$\mathcal{I}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)}\mathcal{I}_M. \quad (1.5)$$

The source's least-norm representative $R_M : E_k \rightarrow \mathcal{B}^{\otimes k} \subset \mathcal{H}_k$, with $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, satisfies

$$J^{(k)}R_M = 1, \quad q^{\otimes k}R_M = \sigma_h^{\otimes k}, \quad R_M(E_k) \perp \mathcal{B}_M. \quad (1.6)$$

Its finite Gram is $G_M = R_M^* R_M \succ 0$. Every adjoint here is of a finite-domain map into the declared Hilbert observation; no bounded Hilbert-space extension of the full theta map is assumed.

S10.1.3 2. Unscaled polynomial coordinates with every moment retained

The arithmetic measure is exactly

$$d\nu_h(y) = |v_h(\tfrac{1}{2} + iy)|^2 \frac{dy}{2\pi}. \quad (2.1)$$

The source's log-coordinate and Fourier maps give $\|\mathcal{T}_h P\|_{L^2(dx)}^2 = \int |P(\tfrac{1}{2} + iy)|^2 d\nu_h(y)$. The mass of this measure is not set to one.

Starting from $1, t, t^2, \dots$, subtract its orthogonal projections onto the preceding span:

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathcal{P}_{n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \quad (2.2)$$

No polynomial is divided by its norm. Let T_N be the triangular matrix with columns $p_0, \dots, p_N$ in the original power basis. The exact coordinate relation is

$$T_N^* M_{h,N} T_N = \text{diag}(\omega_0, \dots, \omega_N). \quad (2.3)$$

The earlier seed-boundary vectors $u_n = P_n(D)\Theta\phi_*$ are related to these coordinates by $\mathcal{T}_h(hP_n) = u_n$. Their measure is $|g|^2 dy/(2\pi)$; equation (1.3) is the map to the present $|g/h|^2$ measure. The two orthogonalizations have not been interchanged.

Recurrence. Define $a_0 = 0$ and $a_n = \omega_n/\omega_{n-1}$ for $n \geq 1$. Then

$$tp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad b_n = \frac{\int (\tfrac{1}{2} + iy) |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, \quad \Re b_n = \tfrac{1}{2}. \quad (2.4)$$

Here $p_{-1} = 0$. The possibly nonzero $\Im b_n$ remains. The packet measure need not be even in the height coordinate y .

Proof. Multiplication by t has adjoint multiplication by $1 - t$ for this polynomial inner product. Pairing tp_n with $p_j, j < n-1$, gives zero. Its coefficient of p_{n+1} is one by its leading term; pairing with p_{n-1} gives $-\omega_n/\omega_{n-1}$. The diagonal is the displayed integral. This proves every coefficient. These are the usual orthogonal-polynomial operations, applied through the specified coordinate map rather than changing the arithmetic generator.

For $\alpha \in \mathbb{N}^k$, define

$$p_\alpha = \prod_j p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}p_\alpha \in E_k. \quad (2.5)$$

The columns z_α contain the complete unit $v^{\otimes k}$ and all nilpotent jets.

Inverse-metric formula. In the fixed original basis of E_k ,

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}. \quad (2.6)$$

Indeed (2.3) applied to the tensor polynomial basis transforms $JM^{-1}J^*$ exactly into this sum. The representative coefficient on p_α is

$$c_\alpha(u) = \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (2.7)$$

These formulas avoid inversion of a large raw monomial moment matrix, without imposing a new metric.

S10.1.4 3. The highest polynomial degree is isomorphic to the actual new relation layer

Let $\mathcal{U}_M = \text{span}\{p_\beta : |\beta| = M + 1\}$ and retain its ordered basis. Put

$$\Omega = \text{diag}(\omega_\beta)_{|\beta|=M+1}, \quad F = [z_\beta]_{|\beta|=M+1} : \mathbb{C}^r \rightarrow E_k, \quad r = \binom{M+k}{k-1}. \quad (3.1)$$

Write $T_+ : \mathbb{C}^r \rightarrow \mathcal{H}_k$ for the map with columns $\mathcal{T}^{(k)} p_\beta$. Its image is orthogonal to the entire degree- M polynomial image, including $R_M(E_k)$.

The prior actual new relation layer is

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M})\mathcal{B}_{M+1}.$$

Theorem: relation-graph isomorphism. The map

$$b_M = T_+ - R_M F : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M \quad (3.2)$$

is an isomorphism and has

$$H_M = b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (3.3)$$

Proof. Its full jet is $F - F = 0$, so (1.5) makes it an actual boundary of degree at most $M + 1$. Both terms are orthogonal to $\mathcal{B}_M$. If $b_M u = 0$, its component in the highest orthogonal polynomial degree is $T_+ u = 0$, so $u = 0$. The source calculation gives $\dim \mathcal{E}_M = r$, proving surjectivity. Alternatively every degree- $M + 1$ boundary has a unique highest orthogonal component; subtract its b_M image and the result lies in $\mathcal{B}_M$. This also constructs the inverse. Orthogonality $T_+^* R_M = 0$ proves (3.3).

A primitive is fully specified: form the numerator $p_\beta - c_M(z_\beta)$, where c_M is (2.7), divide it in the fixed variable order by $h(t_1), \dots, h(t_k)$, and use the original theta primitive in the factor where $h(t_j)$ appears. Give that primitive coefficient $(-1)^{j-1}$; the tensor differential supplies the same sign. No relation is inferred solely from a dimension count.

The comparison with the original internal quotient is

$$\mathbb{C}^r \xrightarrow{b_M} \mathcal{E}_M \xrightarrow{v \mapsto [v]} \mathcal{B}_{M+1}/\mathcal{B}_M \hookrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_M \longrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_{M+1}. \quad (3.4)$$

The last composite is zero. Its split lift sends (λ_M, u) to $(\lambda_{M+1}, 0)$, not to τ . At λ_M , $u \neq 0$ represents a nonzero kernel class. The map b_M describes the entire class that is about to become a supported zero.

Representative update. Orthogonal projection onto $b_M(\mathbb{C}^r)$ gives

$$Y_M = -b_M H_M^{-1} F^* G_M, \quad R_{M+1} = R_M + b_M H_M^{-1} F^* G_M. \quad (3.5)$$

Consequently

$$G_{M+1} = G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \quad (3.6)$$

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (3.7)$$

The complete volume identity is

$$\frac{\det \mathcal{C}_{M+1}}{\det \mathcal{C}_M} = \frac{\det(\Omega + F^* G_M F)}{\det \Omega}. \quad (3.8)$$

The full matrices in (3.6)–(3.7) remain available; their determinant in (3.8) is an additional observation, not their replacement.

S10.1.5 4. Relative control is a highest-degree Christoffel–Darboux identity

For each $|\beta| = M + 1$, put

$$w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (4.1)$$

The name E_+ denotes this finite matrix, not the scalar e or the arithmetic quotient E_h ; its type is $\mathbb{C}^r \rightarrow E_k$.

Theorem: exact displacement.

$$\boxed{\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*}. \quad (4.2)$$

Proof. Apply (2.4) in each factor of (2.6). A pair of adjacent indices inside $|\alpha| \leq M$ cancels because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, precisely the term subtracted in (4.2). The remaining ordered pairs have $|\beta| = M + 1$ and predecessor $\beta - e_j$; collecting them gives (4.1)–(4.2). The imaginary diagonal coefficients cancel with their conjugates here; they were retained in the recurrence and are not assigned zero.

This identity is the tensor total-degree version of the Christoffel–Darboux cancellation. The standard one-variable formula is background; the maps to the full arithmetic quotient and its source boundaries are specified above.

The derivative defect also has an exact expression in the same relation-graph coordinates:

$$B_M = D^{(k)} R_M - R_M A_k, \quad \boxed{C_M^{\text{rel}} := (1 - P_{\mathcal{B}_M}) B_M = b_M \Omega^{-1} E_+^* G_M}. \quad (4.3)$$

Proof. The coefficient of p_β in $D^{(k)} R_M u$ is $\sum_j \omega_{\beta-e_j}^{-1} z_{\beta-e_j}^* G_M u = \omega_\beta^{-1} w_\beta^* G_M u$. The defect is an original boundary. The inverse of (3.2), followed by projection away from $\mathcal{B}_M$, therefore gives (4.3).

Substituting (3.5) and (4.3) into the previous cross-pairing formula proves

$$\boxed{W_M = A_k^* G_M + G_M A_k - k G_M = G_M \mathcal{Z}_M G_M = -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M)}. \quad (4.4)$$

Thus the new highest-degree computation is connected to both maps in the preceding note, including their minus sign and their support-valued boundary primitive.

S10.1.6 4.1 Exact finite certificate and a proved upper estimate

For every real ϵ , congruence by the actual invertible G_M gives

$$\boxed{-\epsilon G_M \preceq W_M \preceq \epsilon G_M \iff -\epsilon \mathcal{C}_M \preceq \mathcal{Z}_M \preceq \epsilon \mathcal{C}_M}. \quad (4.5)$$

This transports the inequality; no eigenvalue or generator has been rescaled.

Put $\mathcal{B} = [F, E_+]$ and

$$J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

The nonzero generalized eigenvalues of $(\mathcal{Z}_M, \mathcal{C}_M)$ are the nonzero eigenvalues of

$$\boxed{J \mathcal{B}^* G_M \mathcal{B}}. \quad (4.6)$$

The proof is the pair of maps $\mathcal{B} : \mathbb{C}^{2r} \rightarrow E_k$ and $J\mathcal{B}^* G_M : E_k \rightarrow \mathbb{C}^{2r}$ and the equality of nonzero spectra of their two composites. These eigenvalues are real because the first composite is the operator of the Hermitian pencil in (4.5). The $2r$ -matrix itself need not be Hermitian in its displayed coordinates.

There is also a direct upper bound. Let

$$\lambda_{k,M} = \max_{v \neq 0} \frac{v^* F \Omega^{-1} F^* v}{v^* \mathcal{C}_M v},$$

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (4.7)$$

Then the actual arithmetic control excess satisfies

$$\boxed{\epsilon_{k,M} \leq 2\sqrt{\Gamma_{k,M} \lambda_{k,M}}}. \quad (4.8)$$

Proof. Let Z_M be the columns z_α , $|\alpha| = M$, and $\Omega_M = \text{diag}(\omega_\alpha)$. The lowering-incidence matrix L has $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, summed over matching j . Then $E_+ = Z_M L$. The matrix $\Omega_M L \Omega^{-1} L^*$ is similar to a positive semidefinite matrix and has nonnegative entries. Its row sum at α is exactly the sum in (4.7), so its spectral radius is at most $\Gamma_{k,M}$. This proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z_M \Omega_M^{-1} Z_M^* \preceq \Gamma_{k,M} \mathcal{C}_M.$$

Cauchy–Schwarz in the same Ω^{-1} pairing now gives

$$|v^* \mathcal{Z}_M v| \leq 2\sqrt{v^* F \Omega^{-1} F^* v} \sqrt{v^* E_+ \Omega^{-1} E_+^* v} \leq 2\sqrt{\lambda_{k,M} \Gamma_{k,M}} v^* \mathcal{C}_M v.$$

This proves (4.8). The phase-sensitive formula (4.6) is retained for a sharper estimate. Equation (4.8) is not claimed sublinear in k ; its two arithmetic factors are now fully specified.

S10.1.7 5. A symmetric cohomology summand that retains the amplification witnesses

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$ with its original tensor differential. For $\pi \in S_k$, the chain permutation is

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i| |x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (5.1)$$

The exponent uses the original inputs crossed by π . Adjacent transpositions verify the cochain equation and the permutation relations. In top degree every input has degree one, so $T_\pi = \text{sgn}(\pi) P_\pi$, where P_π is ordinary factor permutation.

Define the actual idempotent

$$\boxed{S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi}. \quad (5.2)$$

The unscaled numerator squares to $k!$ times itself. The displayed $1/k!$ is necessary to make the specified morphism idempotent and is not absorbed into a pairing or an arithmetic function. In degree k the two signs multiply to $+1$, hence

$$\boxed{H^k(S_k \mathcal{C}_k) \cong (Q^{\widehat{\otimes} k})^{S_k}}. \quad (5.3)$$

On a finite packet this contains $(E^{\otimes k})^{S_k}$ with its entire nilpotent action. If $Av = \rho v$, then $v^{\otimes k}$ is nonzero, lies in this summand, and has generator eigenvalue $k\rho$. Using the unsigned chain average instead would give the alternating top-degree summand and would kill a repeated vector for $k > 1$; equation (5.2) is the precise replacement.

The complement is retained through the actual cochain decomposition

$$\mathcal{C}_k \xrightarrow{x \mapsto (S_k x, (1-S_k)x)} \text{im } S_k \oplus \text{im}(1 - S_k), \quad (u, v) \mapsto u + v. \quad (5.4)$$

At an invariant joint support this is reconstructed fibrewise. On all support masks the lift of (5.2) has the specified support map $\lambda \mapsto \bigvee_{\pi} \pi \lambda$. Its formula is the sum of the actual supported scalars $(\text{sgn } \pi / k!) \bullet T_{\pi}$. A vanishing projected amplitude lands at the orbit-joined supported zero, not at τ . Original transports into that joint fibre remain recorded. No injectivity of arbitrary support transports is assumed.

S10.1.8 5.1 The canonical representatives commute with this operation

Permutation of variables preserves the total-degree numerator, its relation ideal, the product arithmetic measure, and the full unit $v^{\otimes k}$. Uniqueness of the constrained minimum gives

$$R_M P_{\pi} = P_{\pi} R_M. \quad (5.5)$$

Consequently R_M restricts to the invariant arithmetic module

$$E_k^{\text{sym}} = (E^{\otimes k})^{S_k}, \quad \boxed{\dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}}. \quad (5.6)$$

This is not a change of the spectral packet. The inclusion $I_k : E_k^{\text{sym}} \hookrightarrow E_k$ gives

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad W_M^{\text{sym}} = I_k^* W_M I_k, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (5.7)$$

All repeated eigenlines remain, so this smaller family still contains every witness needed by the tensor-amplification upper estimate. For example, $d = 3, k = 8$ gives dimension 45 rather than $3^8 = 6561$. This is a dimension reduction, not a bound on W_M^{sym} .

S10.1.9 6. Orbit sums give the symmetric matrices with all factorials retained

For a nondecreasing tuple $\lambda \in \mathbb{N}^k$, let $\text{Orb } \lambda$ denote its distinct permutations and

$$n_{\lambda} = \frac{k!}{\prod_{j \geq 0} m_j(\lambda)!}, \quad p_{[\lambda]} = \sum_{\alpha \in \text{Orb } \lambda} p_{\alpha}, \quad \omega_{[\lambda]} = n_{\lambda} \prod_j \omega_{\lambda_j}. \quad (6.1)$$

Distinct product basis vectors are orthogonal, proving the last formula directly. The orbit sum is not divided by its norm or its orbit size.

The target basis is likewise the unscaled orbit sums of $e_{a_1} \otimes \cdots \otimes e_{a_k}$, $0 \leq a_j < d$, where $e_i = [t^i]$. Let I_k be their column inclusion and

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_i n_i!} \right), \quad L_k = D_k^{-1} I_k^*. \quad (6.2)$$

The explicit L_k extracts the common coefficient on an orbit. For each orbit polynomial, define

$$z_{[\lambda]} = L_k \mathcal{J} p_{[\lambda]}.$$

Then

$$\mathcal{C}_M^{\text{sym}} = \sum_{|\lambda| \leq M} \frac{z^{[\lambda]} z_{[\lambda]}^*}{\omega_{[\lambda]}}, \quad G_M^{\text{sym}} = (\mathcal{C}_M^{\text{sym}})^{-1}. \quad (6.3)$$

The comparison with full coordinates is

$$\mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad G_M^{\text{sym}} = I_k^* G_M I_k. \quad (6.4)$$

Inversion and compression have not been commuted without the displayed D_k factors. Equation (6.4) follows because the full form commutes with the orthogonal permutation projection and by substituting $L_k = D_k^{-1} I_k^*$.

The symmetric polynomial numerator has one basis vector per partition of its degree into at most k parts. For $M \geq k(d-1)$ the jet map is onto, because a tensor remainder has total degree at most $k(d-1)$ and averaging makes a symmetric lift of every invariant jet. Thus

$$\dim(\mathcal{E}_M^{S_k}) = p_{\leq k}(M+1), \quad (6.5)$$

where $p_{\leq k}(n)$ is the number of partitions of n into at most k parts. The maps $b_M, Y_M, C_M^{\text{rel}}$ are permutation-equivariant and therefore restrict to that very subspace. Fixed-order polynomial division yields a primitive; applying S_k to it gives a symmetric cochain primitive because $dS_k = S_k d$.

The highest-layer formulas in sections 3–4 hold on the invariant coordinates by grouping F, E_+ into orbit sums, applying L_k , and using the diagonal entries $\omega_{[\lambda]}$. This computes the symmetric bound directly, instead of building a d^k -dimensional matrix first. The upper-bound row-sum estimate can still use the fully retained $\Gamma_{k,M}$; the exact grouped incidence matrix can give a smaller constant.

S10.1.10 6.1 Reflection and the original trace

For a reflection-stable packet the source involution j_h induces $j_h^{\otimes k}$ on both summands in (5.4), commutes with permutations, and preserves the canonical representatives. Hence on the symmetric metric an upper control implies the opposite lower control by the same involution. The residue pairing and its Jacobian map are still transported through their specified inclusions; no auxiliary positive Gram is declared to be the Weil form.

For a finite packet action $U(a) = \exp((\log a)A)$, the exact trace on the invariant summand is

$$\text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|} | E). \quad (6.6)$$

The proof expands the permutation operator in the tensor basis; each cycle forces the corresponding contracted indices and gives the indicated trace. It holds for nonsemisimple U . The cohomological supertrace is $(-1)^k$ times (6.6). The complement trace remains

$$(-1)^k \left(\text{Tr}(U(a) | E)^k - \text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) \right). \quad (6.7)$$

These are traces of finite-rank cochain projectors on the original complex. An infinite packet sum and its trace topology are not supplied by these formulas.

S10.1.11 7. An exact joint-control calculation with the nilpotent direction retained

This section is a declared algebraic calibration, not a packet of zeta zeros. Take the polynomial quotient

$$E = \mathbb{C}[s]/((s - \frac{1}{2})^2), \quad d\nu(y) = e^{-y^2/2} dy, \quad c = \sqrt{2\pi}.$$

The measure has its literal mass c , and the unit in the calibration jet map is 1. The polynomial basis comparison $x = s - 1/2$ is given by

$$B : \text{span}(1, x) \xrightarrow{\sim} \text{span}(1, s), \quad B = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}. \quad (7.1)$$

The operator remains multiplication by $s = 1/2 + x$. In the symmetric basis

$$1 \otimes 1, \quad 1 \otimes x + x \otimes 1, \quad x \otimes x,$$

its two-fold generator is

$$A_2^{\text{sym}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}. \quad (7.2)$$

The two-fold total-degree-4 calculation gives

$$\boxed{G_{2,4}^{\text{sym}} = c^2 \text{diag}(1/3, 2/3, 1/4),} \quad (7.3)$$

$$\boxed{W_{2,4}^{\text{sym}} = c^2 \begin{pmatrix} 0 & 2/3 & 0 \\ 2/3 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{pmatrix}.} \quad (7.4)$$

Its generalized characteristic polynomial is

$$\frac{\det(zG_{2,4}^{\text{sym}} - W_{2,4}^{\text{sym}})}{\det G_{2,4}^{\text{sym}}} = z(z^2 - 7/2),$$

so the exact two-sided excess is $\sqrt{7/2}$. The independent rectangular degree-2-per-factor construction gives $\sqrt{6}$ at the same containing total degree. This proves an actual improvement for the specified joint construction in this calibration; the nilpotent operator in (7.2) is unchanged.

For comparison, the smaller joint total degree $M = 2$ has excess $\sqrt{6}$, whereas the rectangular degree-1 construction contained in it has excess 2. Consequently a larger correction space alone does not monotonically improve the control constant. The typed comparison is inclusion of the two polynomial numerators and the resulting Gram compression; the direct formula (4.2) supplies the actual control value. Both calibrations are retained so that the first improvement is not promoted into a false monotonicity theorem.

S10.1.12 8. Exact target, counterfactuals, and handoff

The next estimate is now an arithmetic highest-layer estimate, either by the phase-sensitive pencil (4.5)–(4.6), or by bounding the two factors in (4.8). On the symmetric cohomology family it concerns the same G_M^{sym} and W_M^{sym} , with all relation primitives retained. The source of every matrix entry is the actual measure $|2\xi/h|^2 dy/(2\pi)$, not a fitted metric.

A sublinear excess in tensor degree has not been proved. For every actual eigenvector $Av = \rho v$, the retained witness in the symmetric summand satisfies

$$\frac{(v^{\otimes k})^* W_M(v^{\otimes k})}{(v^{\otimes k})^* G_M(v^{\otimes k})} = k(2\Re\rho - 1). \quad (8.1)$$

The symmetric reduction therefore retains the witness needed by Deligne-style amplification; it does not assume its disappearance. The input Deligne passages use tensoring *and* a separate geometric estimate. Our new work supplies a computable smaller arithmetic target and a proved finite bound, not that geometric estimate by renaming a category.

Constructive continuations are recorded with their exact maps:

- Raw moment inversion: use (2.3)–(2.7), retaining every ω_j and the full jet unit.
- Highest polynomials that are not relations: use $b_M = T_+ - R_M F$, not an assertion that high degree is zero.
- Absolute Gram decay: use the congruence (4.5), retaining $\mathcal{C}_M = G_M^{-1}$ and the whole highest-layer update.
- Excessive tensor dimension: use the chain projector (5.2) and the symmetric numerator/quotient (6.1)–(6.5), while retaining the complementary cohomology and trace.
- A Cauchy–Schwarz bound too large: use the exact signed matrix (4.6); its phases and cross terms have not been replaced by their absolute values in the stored data.
- Ordinary zero observations: reconstruct the maps and kernels in (3.4). A new relation is *e*-zero at the next label; the original class and the full kernel remain upstream.

The supplied quotient/homotopy Lean results can formalize the finite algebra of (3.2)–(4.6) and the chain-permutation calculation. The original analytic range theorem, arithmetic moment bounds, and any uniform tensor estimate remain separate written-analysis inputs.

S10.1.13 References and evidence scope

The current input was read in the mounted NOTE.tex and checked against the pinned public mark-down (especially sections 7–9). The prior formalization receipt was read at its recorded scope: its section-change and internal-quotient statements are used, without treating its analytic assumptions as proved Lean theorems.

The classical orthogonal-polynomial recurrence and Christoffel–Darboux formula are documented in NIST DLMF sections 18.2(iv)–(v), <https://dlmf.nist.gov/18.2>. The tensor complex and its signed commutativity are documented in the Stacks Project, Tag 0GWN, <https://stacks.math.columbia.edu/tag/0GWN>. These are background mechanisms; sections 3–6 above prove their specific maps on this arithmetic source.

The retained Deligne reading for the intended amplification is the supplied account of Weil II 3.2.13 and 3.3.4–3.3.6. No new full rereading of Weil II is claimed in this turn. No external source corpus is redistributed in this package.

The checker uses exact finite Gaussian moment fixtures, including repeated-root quotients, non-constant units, a nonzero imaginary recurrence diagonal, and both signed chain degrees. They are not arithmetic measurements. All derived infinite-dimensional statements rest on the written arguments and the explicitly named prior analytic inputs, not on the finite tests.

S11 Complete spectral-sum descent: relative arithmetic fibres, dual resolution and all nilpotents

Source attribution. Complete spectral-sum TeX witness retained inside the sum-connection delivery archive. The earlier public PR 17 Markdown note is separately preserved in this reader. This longer witness includes the original sum-action resolution, linear dual and antidual comparison, Ext and resolvent calculations, full matrix weight and constrained minimum. The associated public revision below is not a Git identity assigned to these longer TeX bytes; the source and archive hashes identify them.

Exact revision. 33b29f706008124886614ba4bd55bffc489df9e2

Source file. Tau_Sum_Connection_Control/sources/Tau_Spectral_Sum_Descent/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Associated public PR 17 note revision; full delivered TeX identified by its separate hash

Source SHA-256. a3becf8455ddf96e8e382762acbc0e9d0548ea7342c01d70aecdad744ae79552

Original archive. Tau_Sum_Connection_Control_2026-09-12.zip

Archive SHA-256. f2de513041b7336b495a338e90325e84c5ce53cfa98799e1b89c53af308356de

S11.1 Spectral-sum descent with the relative arithmetic fibre retained

Owner-directed continuation of the split-zero programme. 12 September 2026.

Status. Written deductions for review. These results use the actual theta range and representative constructions stated in the preceding packages. Their analytic inputs are not newly Lean-certified. The finite polynomial, quotient, and matrix statements below have independent proofs and executable finite tests. No RH, GRH, or sublinear arithmetic weight bound is asserted.

S11.1.1 1. Integration and purpose

The new Kernel-Layer Integration supplement and the preceding Symmetric Frontier note overlap in their explicit relation basis and inverse-metric control. We retain the dictionary

$$K_N = \mathcal{C}_M, \quad U_N = F, \quad D_N = \Omega, \quad \mathcal{E}_N = b_M, \quad F_N = \Omega^{-1}E_+^*, \quad V_N = \mathcal{Z}_M, \quad N = M. \quad (1.1)$$

The supplement's sharper rank bound follows by grouping the summed derivative at degree M rather than counting all degree- $M+1$ coordinates. Its identities are not counted again as new work. In particular its one-variable map

$$e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1} \quad (1.2)$$

is retained: it identifies the packet-dependent interpolation layer with the original packet-independent theta relation by an actual polynomial-divisibility map.

The new step is to push the arithmetic tensor system along its **sum coordinate**, retaining the relative coordinates in a matrix-valued arithmetic measure. We calculate the entire symmetric-square relation algebra, its original theta primitives, the exact norm under the pushforward, and the full

nilpotent fibres of the sum action. This is a product-to-one-parameter construction on the existing characteristic-zero object over the tau base. It is not a use of finite-field purity with relabelled coefficients.

S11.1.2 2. Unchanged source, split quotient, and tensor cohomology

Keep

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (2.1)$$

The first quotient remains infinite. All spaces and morphisms are marked over the existing absolute pointed base $\mathfrak{b}_\tau$. At an active coefficient label λ , the lift of a linear map T is $(\lambda, x) \mapsto (\lambda, Tx)$; it fixes external absence. Its amplitude-zero value is $(\lambda, 0)$, the image of the original supported-zero action, not τ .

Use precisely

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad \mathcal{B} = \{F : \sup_{x>0} x^b |D^j F(x)| < \infty (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \\ \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g, \quad g(s) = 2\xi(s). \quad (2.2)$$

For a finite nonempty packet of actual zeros with their full orders, put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times.$$

The prior analytic construction supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. Retain its maps

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h. \quad (2.3)$$

For tensor degree k , use $\mathcal{P}_M = \mathbb{C}[s_1, \dots, s_k]_{\deg \leq M}$ and

$$E_k = E_h^{\otimes k}, \quad A_k = M_{s_1 + \dots + s_k}, \quad \mathcal{T}_k(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \\ I_{k,M} = \mathcal{P}_M \cap (h(s_1), \dots, h(s_k)), \quad B_{k,M} = \mathcal{T}_k I_{k,M}. \quad (2.4)$$

The jet map $\mathcal{J}_k P = v_h^{\otimes k}[P]$ has this exact kernel and is onto when $M \geq k(d-1)$. This map is linear, with an invertible unit multiplication after the algebra quotient; it is not called a unital algebra homomorphism. The source representative $R_{k,M}$ is its unique least-norm lift with the specified theta relations allowed. Thus

$$J^{(k)} R_{k,M} = 1, \quad q^{(k)} R_{k,M} = \sigma_h^{\otimes k}, \quad G_{k,M} = R_{k,M}^* R_{k,M}, \quad W_{k,M} = A_k^* G_{k,M} + G_{k,M} A_k - k G_{k,M}. \quad (2.5)$$

The tensor differential retains $(-1)^{j-1}$ when it applies the differential in the j th factor to a degree- $(k-1)$ primitive with all other factors of degree one. A primitive for the positive relation $h(s_j)P_j$ has the matching sign $(-1)^{j-1}$.

In top degree the signed chain projector

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi \quad (2.6)$$

selects ordinary symmetric tensors: the Koszul sign in T_π supplies the second permutation sign. The factor $1/k!$ is retained. We work on this image while retaining the complementary chain projector $1 - S_k$ and its trace. A source mask is transported to the union of its permutation orbit before averaging; its cancelled coordinates remain supported at that joined label.

S11.1.3 3. The spectral-sum direct image is an exact functor with its action fixed

Define the actual ring map

$$\iota_k : \mathbb{C}[S] \longrightarrow E_k, \quad S \longmapsto [s_1 + \cdots + s_k]. \quad (3.1)$$

For an E_k -module M , let $\pi_{k*}M$ be that same module with the $\mathbb{C}[S]$ action restricted through ι_k . The functor is exact because its maps, kernels, and underlying abelian groups are unchanged. It has the explicit adjunction

$$\text{Hom}_{E_k}(E_k \otimes_{\mathbb{C}[S]} L, M) \cong \text{Hom}_{\mathbb{C}[S]}(L, \pi_{k*}M), \quad (3.2)$$

with forward map $f \mapsto (l \mapsto f(1 \otimes l))$ and inverse $a \otimes l \mapsto af(l)$. On quasi-coherent sheaves this is pushforward along the finite morphism $\text{Spec } E_k \rightarrow \text{Spec } \mathbb{C}[S]$. The affine pushforward statement concerns quasi-coherent complexes, not all sheaves of modules (Stacks, Tag 0AVV).

The split map is $G(\iota_k)$, with

$$p_{E_k} G(\iota_k) = \iota_k p_{\mathbb{C}[S]}, \quad G(\iota_k)(e_{\mathbb{C}[S]}) = e_{E_k}, \quad G(\iota_k)(\tau) = \tau. \quad (3.3)$$

Consequently the geometric arrow is over the same $\mathfrak{b}_\tau$. It does not replace that base by a finite field. The additional line $\text{Spec } \mathbb{C}[S]$ is the recipient of a specified arithmetic correspondence.

At the original cochain level, S acts by $D_1 + \cdots + D_k$. Its finite representative equation remains

$$D^{(k)} R_{k,M} - R_{k,M} A_k = dK_{k,M} \in B_{k,M+1}. \quad (3.4)$$

The jet and cohomology maps intertwine that action exactly. Restriction of scalars preserves their kernels and this boundary equation. It does not declare the chosen representative $R_{k,M}$ equivariant.

On the symmetric summand the ring is $B_k = (E_k)^{S_k}$, its inclusion is the original invariant inclusion, and S acts by the same sum. Exactness of taking invariants here follows from the specified Reynolds projector $k!^{-1} \sum P_\pi$ over $\mathbb{C}$.

S11.1.4 3.1 The pushforward has a computed free resolution and dualizing comparison

Let E denote the actual finite ordered or symmetric image, with its exact sum operator A . Put $R = \mathbb{C}[S]$ and $n = \dim_{\mathbb{C}} E$. There is the actual free resolution

$$0 \longrightarrow R \otimes_{\mathbb{C}} E \xrightarrow{SI-A} R \otimes_{\mathbb{C}} E \xrightarrow{\text{ev}_A} \pi_* E \longrightarrow 0, \quad \text{ev}_A(P \otimes v) = P(A)v. \quad (3.5)$$

Injectivity of $SI - A$ follows either from its monic nonzero determinant or by inspecting the highest coefficient of a polynomial vector. To identify the kernel of ev_A , use

$$S^j v - A^j v = (SI - A) \sum_{i=0}^{j-1} S^{j-1-i} A^i v.$$

Subtracting these displayed relations leaves precisely the constant vector $\text{ev}_A(f)$. This proves exactness with both original arrows retained.

Let $E^\vee = \text{Hom}_{\mathbb{C}}(E, \mathbb{C})$ and $A^\vee \lambda = \lambda \circ A$. This is the algebraic dual; its matrix is a transpose. The coefficient conjugation $\overline{E^\vee} \rightarrow E^\#$, $\bar{\lambda} \mapsto (v \mapsto \overline{\lambda(v)})$, connects it explicitly to the antidual used in the control matrices.

Applying $\text{Hom}_R(-, R dS)$ to (3.5) gives

$$R \text{Hom}_R(\pi_* E, R dS) \simeq [R \otimes E^\vee dS \xrightarrow{SI - A^\vee} R \otimes E^\vee dS], \quad (3.6)$$

in degrees 0, 1. Its degree-zero group is zero, and

$$\boxed{\text{Ext}_R^1(\pi_* E, R dS) \cong E^\vee, \quad [P(S) dS] \mapsto P(A^\vee).} \quad (3.7)$$

Here $P(A^\vee)$ means $\sum_j (A^\vee)^j \lambda_j$ for a polynomial dual vector $P(S) = \sum_j S^j \lambda_j$. The inverse is the constant class $\lambda \mapsto [\lambda dS]$.

For the explicit cohomological shift, $C[b]^i = C^{i+b}$. Thus

$$R \text{Hom}_R(\pi_* E[-k], R dS[1]) \simeq E^\vee[k]. \quad (3.8)$$

The degree- k arithmetic image becomes degree $-k$ under the stated dualizing object of the sum line. These shifts are obtained from (3.6), not inferred by calling the finite packet a curve.

The adjoint behind this calculation is the actual module-category map

$$\text{Hom}_R(\pi_* M, L) \cong \text{Hom}_{B_k}(M, \text{Hom}_R(B_k, L)), \quad f \mapsto (m \mapsto (b \mapsto f(bm))). \quad (3.9)$$

The inverse evaluates the inner map at 1. Passing to derived Hom gives the corresponding derived right adjoint. This concerns the exact restriction functor on module categories; no exactness of pushforward on all sheaves is asserted (Stacks, Tag 0AWZ).

S11.1.5 3.2 Resolve the finite arithmetic trace through that same duality

The pairing in (3.7) is equivalently the concrete resolvent-residue pairing

$$\boxed{\langle v, [P(S) dS] \rangle = \sum_{\lambda \in \text{Spec } A} \text{Res}_{S=\lambda} P(S) ((SI - A)^{-1} v) dS.} \quad (3.10)$$

Relations from $SI - A^\vee$ produce polynomial expressions with no finite residue. Expanding the resolvent at infinity proves that the sum is $P(A^\vee)(v)$: the coefficient of S^{-1} is $\sum_j \lambda_j (A^j v)$. This proves perfectness without deleting nilpotents.

On a block $A = \lambda I + N$ the retained resolvent is

$$(SI - A)^{-1} = \sum_{j=0}^{\ell-1} \frac{N^j}{(S - \lambda)^{j+1}}, \quad N^\ell = 0. \quad (3.11)$$

The trace observation of that resolvent is a further specified map. For a function ψ holomorphic at the finite spectrum,

$$\boxed{\mathrm{Tr}(\psi(A)) = \sum_{\lambda} \mathrm{Res}_{S=\lambda} \psi(S) \mathrm{Tr}((SI - A)^{-1}) dS = \sum_{\lambda} \mathrm{Res}_{S=\lambda} \psi(S) \frac{\chi'_A(S)}{\chi_A(S)} dS,} \quad (3.12)$$

where $\chi_A(S) = \det(SI - A)$ is the exact finite characteristic polynomial. It is not substituted for the original $g = 2\xi$. With $\psi(S) = a^S$, (3.12) is the finite Lefschetz contribution of the same diagonal scaling correspondence. The cochain image is in degree k , so its supertrace includes $(-1)^k$.

The constant-class maps in (3.5)–(3.7), and the linear functional in (3.10), lift fibrewise to the original split objects. Every zero functional or zero residue remains supported. At fixed label the equations forming the resolution have composite the supported zero map. The residue pairing is an ordinary bilinear pairing in that coefficient fibre, with its split lift separately additive, not a linear map on its Cartesian product.

S11.1.6 4. The symmetric tensor-square has an explicit two-generator relation algebra

Use three distinguished polynomials

$$S = s_1 + s_2, \quad r = s_1 - s_2, \quad \Delta = r^2. \quad (4.1)$$

The inverse map before taking invariants is

$$s_1 = (S + r)/2, \quad s_2 = (S - r)/2.$$

Swap fixes S and sends r to $-r$. Therefore the algebra map

$$\Phi : \mathbb{C}[S, \Delta] \xrightarrow{\sim} \mathbb{C}[s_1, s_2]^{S_2}, \quad S \mapsto s_1 + s_2, \quad \Delta \mapsto (s_1 - s_2)^2 \quad (4.2)$$

is an isomorphism. Its proof is even/odd expansion in r , using the displayed inverse with both factors $1/2$. Give S degree one and Δ degree two; then Φ preserves the entire original total-degree filtration.

Define actual polynomials H_0, H_1 by

$$\begin{aligned} h((S + r)/2) &= H_0(S, r^2) + rH_1(S, r^2), \\ h((S - r)/2) &= H_0(S, r^2) - rH_1(S, r^2). \end{aligned} \quad (4.3)$$

In particular H_0 is the half-sum and H_1 is the difference divided by $2r$. That difference is an odd polynomial, so this is polynomial division, not localization away from $r = 0$.

Theorem (complete symmetric relation ideal).

$$\boxed{B_2 = (E_h \otimes E_h)^{S_2} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1).} \quad (4.4)$$

Proof. In $\mathbb{C}[S, r]$, the original ideal $(h(s_1), h(s_2))$ is (H_0, rH_1) . Its even part is exactly (H_0, r^2H_1) : average an expression $H_0a + rH_1b$ under $r \mapsto -r$. Writing $a = a_0 + ra_1$, $b = b_0 + rb_1$ gives the even expression $H_0a_0 + \Delta H_1b_1$. Conversely both generators are in the original ideal. Invariants of the quotient equal the quotient of invariants because Reynolds averaging is an exact projector. This proves (4.4), including coincident roots and all nilpotents.

The quotient has dimension $d(d+1)/2$. Indeed the unscaled orbit sums of the basis $s_1^i s_2^j$, $0 \leq i, j < d$, give that many independent invariant vectors. The associated graded Hilbert series is

$$\sum_{0 \leq i \leq j < d} z^{i+j} = \frac{(1-z^d)(1-z^{d+1})}{(1-z)(1-z^2)}. \quad (4.5)$$

Here associated graded refers to the original degree filtration; (4.5) is not a claim that a nonhomogeneous h defines a graded algebra. Polynomial remainder lowers degree, and Reynolds averaging respects the filtration, proving the formula for every monic h of degree d .

S11.1.7 4.1 All original primitives and their degrees survive

Let a symmetric relation of degree at most M have, after original ordered division, the expression $h(s_1)Q_1 + h(s_2)Q_2$ with $\deg Q_i \leq M-d$. Averaging yields paired coefficients

$$C_1 = (Q_1 + Q_2 \circ \text{swap})/2, \quad C_2 = C_1 \circ \text{swap}.$$

Put $A_0 = C_1 + C_2$ and $A_1 = (C_1 - C_2)/r$. Then A_0, A_1 are polynomials in S, Δ and

$$H_0 A_0 + \Delta H_1 A_1 = h(s_1) \frac{A_0 + r A_1}{2} + h(s_2) \frac{A_0 - r A_1}{2}. \quad (4.6)$$

Their weighted degree bounds are $\deg A_0 \leq M-d$ and $\deg A_1 \leq M-d-1$. The two source primitives are respectively

$$\frac{A_0 + r A_1}{2} (D_1, D_2) (\phi_* \otimes F_h), \quad -\frac{A_0 - r A_1}{2} (D_1, D_2) (F_h \otimes \phi_*). \quad (4.7)$$

The negative sign in the second primitive meets the negative cochain sign of the second differential. Applying d gives exactly (4.6) through $\mathcal{T}_2$.

The unit $v_h \otimes v_h$ is symmetric. Under (4.4) denote its full class by $v_{2,h}$. The jet map in these coordinates is exactly

$$\mathcal{J}'_{2,M} : \mathbb{C}[S, \Delta]_{\deg S + 2 \deg \Delta \leq M} \longrightarrow B_2, \quad P \longmapsto v_{2,h}[P]. \quad (4.8)$$

Its kernel is the full weighted-degree part of $(H_0, \Delta H_1)$. It is onto for $M \geq 2(d-1)$. This is the original full-jet map conjugated by Φ , not a scalar evaluation at the sum coordinate.

On the split totals, $\tilde{\Phi}(\lambda_M, P) = (\lambda_M, \Phi P)$ and $\tilde{\Phi}(\tau) = \tau$. The induced isomorphism of relation quotients commutes with every $M \rightarrow M+1$. A nonzero new relation therefore becomes $(\lambda_{M+1}, 0)$ under exactly the same degree transition as before.

S11.1.8 5. The arithmetic norm becomes a one-variable matrix weight

Let

$$w_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{1}{2\pi}, \quad d\nu_h(t) = w_h(t) dt. \quad (5.1)$$

No mass is assigned the value one. The source norm of a two-variable polynomial is

$$\|\mathcal{T}_2 P\|^2 = \int_{\mathbb{R}^2} |P(1/2 + it_1, 1/2 + it_2)|^2 w_h(t_1) w_h(t_2) dt_1 dt_2.$$

Make the exact real coordinate change

$$u = t_1 + t_2, \quad v = t_1 - t_2, \quad (t_1, t_2) = ((u+v)/2, (u-v)/2), \quad dt_1 dt_2 = \frac{1}{2} du dv. \quad (5.2)$$

Then $S = 1 + iu$, $r = iv$, and $\Delta = -v^2$. If

$$P(S, \Delta) = \sum_{a=0}^b \Delta^a f_a(S), \quad b = \lfloor M/2 \rfloor, \quad \deg f_a \leq M - 2a,$$

define the full matrix

$$W_{h,b}(u)_{ac} = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv. \quad (5.3)$$

Theorem (isometric spectral-sum realization).

$$\|\mathcal{T}_2 \Phi P\|^2 = \int_{\mathbb{R}} f(1 + iu)^* W_{h,b}(u) f(1 + iu) du. \quad (5.4)$$

This follows by (5.2), finite polynomial expansion, and Fubini. All integrals converge by the source's exponential moments. For every real u , the matrix $W_{h,b}(u)$ is positive definite: its quadratic form is the integral of the squared modulus of a nonzero polynomial in $-v^2$ against a density positive almost everywhere. Zeros of the nonzero entire g/h are discrete and cannot remove a positive-measure set of the v -line. The exponential bounds prove convergence also for each fixed u .

For $b = 0$ the scalar coefficient is the convolution

$$W_{h,0}(u) = (w_h * w_h)(u). \quad (5.5)$$

For $b > 0$, its additional rows are the actual relative-coordinate moments. They have not been replaced by (5.5).

The map in (5.4) sends a polynomial to its vector of coefficient functions; its inverse is $f \mapsto \sum \Delta^a f_a(S)$ on the specified filtered image. Multiplication by $s_1 + s_2$ is exactly multiplication by S on every component. Hence its generator is still A_2 , its degree increase is still one, and its boundary is still (3.4).

S11.1.9 5.1 General tensor degree

For any k , set $S = \sum s_i$, retain $x_i = s_i$ for $i < k$, and put $s_k = S - \sum_{i < k} x_i$. This is a polynomial algebra isomorphism with determinant one on the real variables $(t_1, \dots, t_k) \leftrightarrow (t_1, \dots, t_{k-1}, u = \sum t_i)$.

Write $P = \sum_{\alpha} x^{\alpha} f_{\alpha}(S)$, with $\deg f_{\alpha} \leq M - |\alpha|$. Its matrix weight is

$$W_{k,M}(u)_{\alpha\beta} = \int_{\mathbb{R}^{k-1}} \prod_{i < k} \overline{(1/2 + it_i)^{\alpha_i}} \prod_{i < k} (1/2 + it_i)^{\beta_i} \prod_{i < k} w_h(t_i) w_h(u - \sum_{i < k} t_i) d^{k-1}t. \quad (5.6)$$

The outer coordinate is $S = k/2 + iu$. The same proof gives an isometry on the actual degree- M source, with every original relation substituted through this algebra isomorphism. Its entries have exponential moments in u for every exponent below the source threshold $\pi/2$: use $|\sum t_i| \leq \sum |t_i|$, the polynomial factors, and the one-variable exponential moment. Constants depend on the retained packet, k , and degrees; no uniform-in- k estimate is inferred.

S11.1.10 6. The canonical representative, kernel control and relation layers commute with the descent

Let C_M be the explicit coefficient matrix of Φ from $S^a \Delta^b$ to the original symmetric polynomial basis. It is invertible between those two filtered spaces. Let U be the exact coordinate map of (4.4) to the original unscaled invariant quotient basis. Retain its orbit coefficients.

For the source moment and full-jet matrices,

$$M'_M = C_M^* M_M C_M, \quad J'_M = U^{-1} J_M C_M. \quad (6.1)$$

The matrix in (5.3) gives exactly M'_M by one-dimensional integration of its entries against the retained powers of S .

The inverse interpolation kernel and its metric satisfy

$$K'_M = J'_M (M'_M)^{-1} (J'_M)^* = U^{-1} K_M (U^{-1})^*, \quad G'_M = U^* G_M U. \quad (6.2)$$

These identities follow by multiplying the stated inverses, not by selecting a new metric. The least-norm representatives obey $R'_M = R_M U$ after their polynomial-source identification. Their action and control are

$$A'_2 = U^{-1} A_2 U, \quad W'_M = U^* W_M U, \quad \mathcal{Z}'_M = U^{-1} \mathcal{Z}_M (U^{-1})^*. \quad (6.3)$$

Thus every control inequality is exactly transported. The same isometry sends the actual next boundary layer, its Gram matrix, and its two maps Y_M, C_M^{rel} to their images in (5.4). It preserves the finer rank bound supplied by the new attachment.

The arithmetic representative defect remains a relation with an explicit primitive (4.7). Its cross-pairing is evaluated by (5.3); it is not erased when the relation becomes a supported zero in the next internal quotient.

S11.1.11 6.1 Retain the base and relative parts of the norm separately through a triangular map

Write the finite matrix weight as

$$W(u) = \begin{pmatrix} m_0(u) & b(u) \\ b(u)^* & C(u) \end{pmatrix}, \quad m_0(u) > 0.$$

There is the exact isometric map into the stated direct-sum weighted space

$$\mathcal{A} : f \mapsto (f_0 + m_0^{-1} b f_{\text{rel}}, f_{\text{rel}}), \quad (6.4)$$

with inverse $(a, z) \mapsto (a - m_0^{-1} b z, z)$ and weights

$$m_0(u), \quad C(u) - b(u)^* m_0(u)^{-1} b(u). \quad (6.5)$$

Completing the square proves the isometry and positivity of the second weight. No measure is renormalized: m_0 and its reciprocal are both in the formulas. The image $\mathcal{A}(\mathcal{P}_M^{\text{sym}})$ is retained, not enlarged to arbitrary functions in the target. The ratio $m_0^{-1} b$ need not be polynomial. Multiplication by $S = 1 + iu$ commutes with this pointwise map, so the derivative action is still specified without a new connection term.

This supplies two concrete analytic quantities for the next estimates: the convolution mass and the full relative covariance matrix. Neither positivity of this representative norm nor the scalar mass alone is being substituted for the arithmetic Weil pairing.

S11.1.12 7. Calculate every nilpotent fibre of the sum map

The pushforward must not replace a multiple jet by its residue. We now give its complete Jordan lengths.

For an ordered root tuple $(\rho_1, \dots, \rho_k)$ with retained orders m_i , its local algebra is

$$L_\rho = \mathbb{C}[z_1, \dots, z_k]/(z_1^{m_1}, \dots, z_k^{m_k}), \quad A_k = \rho_\Sigma I + N, \quad \rho_\Sigma = \sum \rho_i, \quad N = M_{\sum z_i}. \quad (7.1)$$

Let $D_0 = \sum (m_i - 1)$ and $c_r = [t^r] \prod_i (1 + t + \dots + t^{m_i-1})$, with $c_{-1} = 0$.

Theorem (full ordered sum-fibre decomposition). As a $\mathbb{C}[S]$ -module,

$$\pi_{k*} L_\rho \cong \bigoplus_{r=0}^{\lfloor D_0/2 \rfloor} (\mathbb{C}[S]/(S - \rho_\Sigma)^{D_0-2r+1})^{\oplus (c_r - c_{r-1})}. \quad (7.2)$$

Here the isomorphism is constructed by primitive vectors, as follows. On z_i^j set

$$F_i z_i^j = j(m_i - j) z_i^{j-1}, \quad H_i z_i^j = (2j - (m_i - 1)) z_i^j.$$

For $F = \sum F_i$ and $H = \sum H_i$, direct substitution gives

$$[N, F] = H, \quad [H, N] = 2N, \quad [H, F] = -2F. \quad (7.3)$$

An auxiliary positive inner product making $F = N^*$ is explicitly diagonal:

$$\langle z^\alpha, z^\beta \rangle_0 = \delta_{\alpha\beta} \prod_i \frac{\alpha_i! (m_i - 1)!}{(m_i - 1 - \alpha_i)!}. \quad (7.4)$$

These are specified positive numbers. For the actual arithmetic Gram $G_{k,M}$ and the displayed diagonal Gram G_0 , their exact comparison is the invertible operator $C_0 = G_{k,M}^{-1} G_0$, with inverse $G_0^{-1} G_{k,M}$:

$$\langle v, w \rangle_0 = \langle v, C_0 w \rangle_{G_{k,M}}. \quad (7.4a)$$

Equation (7.4) is used only to prove the algebraic decomposition. All arithmetic control remains with $G_{k,M}$. In the primitive-ladder basis its entries are $(N^i p)^* G_{k,M} (N^j q)$, including all cross-ladder terms. They are not discarded because the auxiliary Gram admits orthogonal ladders.

A primitive vector p has degree r and satisfies $Fp = 0$. Put $w = D_0 - 2r$. The commutators prove

$$F N^j p = j(w - j + 1) N^{j-1} p, \quad \|N^j p\|_0^2 = \frac{j! w!}{(w - j)!} \|p\|_0^2 \quad (0 \leq j \leq w). \quad (7.5)$$

Thus its ladder has length $w + 1$ and $N^{w+1} p = 0$. Conversely a nonzero lowest vector has $r \leq D_0/2$, since $\|Np\|_0^2 = (D_0 - 2r) \|p\|_0^2$. Orthogonal complements of the constructed ladders are invariant under N, F, H , so induction decomposes the full finite space. Counting degree- r vectors before the middle gives $c_r - c_{r-1}$ new primitive ladders. The isomorphism in (7.2) maps $[q(S)]$ in a chosen primitive copy to $q(\rho_\Sigma + N)p$.

S11.1.13 7.1 All symmetric packets, including collisions of sums

For an occupation list $\mathbf{n} = (n_\rho)_\rho$, $\sum n_\rho = k$, the corresponding invariant local component is

$$B_{\mathbf{n}} = \bigotimes_{\rho} \left(\mathbb{C}[z_1, \dots, z_{n_\rho}] / (z_j^{m_\rho}) \right)^{S_{n_\rho}}.$$

It retains its own canonical central idempotent even when two different occupation lists have the same sum $\sum n_\rho \rho$. Its graded dimensions c_r are counted by unscaled orbit monomials; equivalently their generating polynomial is

$$\prod_{\rho} \prod_{j=1}^{n_\rho} \frac{1 - t^{m_\rho + j - 1}}{1 - t^j}. \quad (7.6)$$

For $n_\rho = 0$ the factor is one. This is a polynomial identity for the orbit counts; no division at a root of unity is needed. The same operators preserve invariants and the same primitive proof gives (7.2) with

$$D_0 = \sum n_\rho (m_\rho - 1), \quad \rho_\Sigma = \sum n_\rho \rho$$

and these invariant c_r . It includes all k and all finite multiplicities.

For one order-three zero taken twice, write $z_i = s_i - \rho$, $X = z_1 + z_2$. The symmetric component has the exact ladders

$$1, X, X^2, X^3, X^4, \quad p = z_1^2 - z_1 z_2 + z_2^2, \quad Xp = 0, \quad (7.7)$$

with $X^3 = 3(z_1^2 z_2 + z_1 z_2^2)$ and $X^4 = 6z_1^2 z_2^2$. Its pushforward is

$$\mathbb{C}[S]/(S - 2\rho)^5 \oplus \mathbb{C}[S]/(S - 2\rho). \quad (7.8)$$

The actual total dimension is six. Merely retaining one evaluation at 2ρ would lose five dimensions; the displayed injective primitive maps and direct sum retain all six.

The original $v_h^{\otimes k}$ acts by an invertible multiplication map on every component and commutes with A_k . It carries these coordinates to the original full-jet coordinates. Its coefficients are not assigned one.

S11.1.14 8. Pairings, traces, and the support-layer meaning of the ladders

The algebraic direct image in (3.1) keeps the full underlying vector space. Therefore its trace of a^{A_k} is unchanged. On the occupation component,

$$\mathrm{Tr}(a^{A_k} \mid B_{\mathbf{n}}) = \left[\prod_{\rho} \binom{n_\rho + m_\rho - 1}{n_\rho} \right] a^{\sum n_\rho \rho}. \quad (8.1)$$

This is valid with all nonzero nilpotents, since their exponential is triangular with diagonal one. The cochain summand has degree k , so its supertrace is $(-1)^k$ times (8.1). The complementary symmetry summands retain their previously calculated traces. No packet is deleted by pushing to the sum line.

Each ladder map in (7.2) has a split lift fixing τ . In the module $\mathbb{C}[S]/(S - \rho_\Sigma)^\ell$, the transition to the quotient of length $\ell - 1$ takes the nonzero top jet to the supported zero. On its original side the quotient map is the corresponding quotient by $\mathbb{C}N^{\ell-1}p$; it is a $\mathbb{C}[S]$ -module quotient, not a declaration

that this line is an algebra ideal. The full direct-image theorem retains the line before that optional observation.

The residue pairing and arithmetic Jacobian trace form are transported by the same full coordinate matrices. For an invertible coordinate map U , retain

$$S' = U^* S U, \quad J'_g = U^{-1} J_g U, \quad G' = U^* G U, \quad S' J'_g = U^* S J_g U. \quad (8.2)$$

On product cohomology one additionally retains the original tensor-duality signs. No auxiliary inner product from (7.4) is substituted in (8.2). A supported zero of the trace and a full vanishing jet are connected only through the stated Jacobian/trace map.

S11.1.15 9. A complete unscaled arithmetic-moment calibration

For a declared Gaussian calibration, not an asserted zeta packet, take the one-variable measure $e^{-t^2/2} dt$. The change (5.2) gives

$$\mathbf{W}_2(u) = \sqrt{\pi} e^{-u^2/4} \begin{pmatrix} 1 & -2 & 12 \\ -2 & 12 & -120 \\ 12 & -120 & 1680 \end{pmatrix} \quad (9.1)$$

on coefficient functions of $1, \Delta, \Delta^2$. This uses

$$\frac{1}{2} \int v^{2j} e^{-v^2/4} dv = 2^j (2j-1)!! \sqrt{\pi},$$

with $(-1)!! = 1$. The actual triangular coordinate matrix with columns $1, \Delta + 2, \Delta^2 + 12\Delta + 12$ is

$$T = \begin{pmatrix} 1 & 2 & 12 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{pmatrix}, \quad T^* \mathbf{W}_2(u) T = \sqrt{\pi} e^{-u^2/4} \text{diag}(1, 8, 384). \quad (9.2)$$

All three fibre directions and their norms survive. For the first two directions, the exact base/relative decomposition is

$$\|f_0 + \Delta f_1\|^2 = \sqrt{\pi} \int e^{-u^2/4} (|f_0 - 2f_1|^2 + 8|f_1|^2) du. \quad (9.3)$$

The source metric has total two-variable mass 2π , retained by these formulas. Exact symbolic tests compare the moments of $S^a \Delta^b$ computed directly in (t_1, t_2) and through (9.1), and compare the resulting constrained representatives, jet maps, Grams and controls.

For the actual arithmetic measure, (5.3) supplies the corresponding nonconstant matrix. Its relative covariance need not have the constant Gaussian form. Neither the calibration's triangular coefficients nor its control are imported into the arithmetic case.

S11.1.16 10. Position in the Deligne-style programme

The source's tensor-square step is followed by geometric pushforward and control on that pushforward, not by tensoring alone. The construction here now supplies a **specified sum-coordinate pushforward** of the actual finite arithmetic image and its original filtered representatives. Its metric realization is the exact one-dimensional matrix weight (5.3) or (5.6), with the relative directions retained. The symmetric relation presentation (4.4) and original primitives (4.7) say exactly which zero-amplitude relations are present in that fibre.

The result does not identify this finite spectral scheme or matrix-weight model with a lisse sheaf on a Lefschetz pencil. The map actually constructed is the quasi-coherent finite pushforward (3.1), connected to the original analytic representatives by (2.3), (4.8), and (5.4). Deligne’s geometric hypotheses and estimates have not thereby been imported.

The immediate control task now has an explicit base/relative split: estimate the actual convolution mass, the full relative covariance, and the full-jet interpolation on the **image** of the filtered polynomial family in (6.4). These determine the same $K_M, \mathcal{Z}_M$ as the preceding work. The sum map retains A_k and every repeated eigenline, so a bound sublinear in k would still have the intended force. No such bound is proved in this note.

The concrete advances are the exact filtered symmetric relation algebra, an all- k spectral-sum norm realization, and a full nilpotent direct-image decomposition that retains multiplicities and colliding occupations. They are new applications to this specified programme; the reusable algebra and integration tools are standard and are not asserted to be globally novel.

S11.1.17 References and reading scope

1. Supplied **Tau Kernel Layer Integration**, RESEARCH_NOTE.md and NOTE.tex; read in full in this continuation. Its finite-layer identifications and sharper rank are retained as source results.
2. Supplied **Tau Symmetric Frontier Control**, RESEARCH_NOTE.md; read in full. Its signed projector, orbit factors, next-layer map, and finite bound are retained.
3. Supplied **Tau Coherent Interpolation Control**, especially polynomial/jet definitions and joint total-degree sections; used as stated analytic input.
4. The Stacks Project, Tag **0AVV**, affine pushforward of quasi-coherent complexes; Tag **0AWZ**, the precise categories for the finite-pushforward adjoint. <https://stacks.math.columbia.edu/tag/0AVV> and <https://stacks.math.columbia.edu/tag/0AWZ>.
5. NIST DLMF, §18.2, polynomial moment and kernel background. <https://dlmf.nist.gov/18.2>.
6. Deligne, *La conjecture de Weil II*, IHES 52 (1980), 137–252. The retained French TeX window around §§3.2.11–3.2.15 was reread. Its line 872 contains an inconsistent numerical implication in the historical transcription; that literal line is not used as a proof step here. The source-based tensor/pushforward architecture is used, not a certification of the entire 116-page proof. Original bibliographic record: <https://archive.numdam.org/articles/10.1007/BF02684780/>.

S12 PR 20: Arithmetic sum density, the full relative connection and conormal derivatives

Source attribution. Complete TeX proof witness delivered in the sum-connection archive on 12 September 2026. The verified PR 20 head contains its shorter public note and checker. This full NOTE is identified by its own source and archive hashes, and is not assigned the public note's Git blob. The local conormal continuation proves the input-constant to socle map with its full multiplicities. The local connection continuation supplies the exact projected derivative pair for the graph map. The reproduced source witness is preserved with its original statements.

Exact revision. 2144c358c2b41aa235b15cf0fa472289aec8673f

Source file. Tau_Sum_Connection_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Verified public PR 20 head; the complete delivered witness has the separate hash below

Source SHA-256. 11870208e3a08df186e42629ae1a6c956faaf28f79dfce0dc2f2d44dc64b4687

Original archive. Tau_Sum_Connection_Control_2026-09-12.zip

Archive SHA-256. f2de513041b7336b495a338e90325e84c5ce53cfa98799e1b89c53af308356de

S12.1 Quantitative sum-fibre connection and the conormal arithmetic trace

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. Written analytic and algebraic deductions for review. The original theta source, its asserted range theorem, and the canonical finite representatives are inputs from the preceding work. No new Lean execution or arithmetic purity theorem is claimed. The new analytic estimate is a mass-retaining contraction for the actual spectral-sum density. The full matrix connection, its normal component, and the conormal derivative of the original theta relations are constructed as well; the contraction is not assigned to the full Weil control form without a comparison.

S12.1.1 1. Source intake and the unchanged marked object

The newly uploaded Kernel Layer Integration archive is byte-identical to its earlier uploaded copy. Its thirty-one manifest entries and the fifty-three entries in the Spectral Sum Descent archive were independently verified. The kernel-layer note's identities are used as source results, not counted as new deductions:

$$K_M = G_M^{-1}, \quad V_M = A_k K_M + K_M A_k^* - k K_M, \quad W_M = G_M V_M G_M. \quad (1.1)$$

In particular the same control problem is transported by the actual map $v = K_M \lambda$. The source's next theta-relation layer, its full-jet columns, and its map into B_{M+1}/B_M remain attached. Its formalization interface is the existing internal quotient, not a replacement quotient.

Retain the marked absolute base $\mathfrak{b}_\tau$, the scalar semiring, and the original infinite quotient:

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.2)$$

The target of $p_{\mathbb{Z}}$ is infinite. Every ring map below is lifted through G and this square. A complex-linear map $T : V \rightarrow W$ is lifted instead by the explicitly defined split-linear map $T^\tau(v^\bullet) = (Tv)^\bullet$, $T^\tau(\tau) = \tau$. These are different types of construction, connected by their respective arithmetic projection squares. In the full support diagram the same formula is $(\lambda, v) \mapsto (\varphi(\lambda), T_\lambda v)$, with the displayed support transport φ . In particular amplitude zero is the receiving fibre's zero, not external absence.

The original test spaces and operators are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ C_+ &= [V \xrightarrow{\Theta} \mathcal{B}], \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \\ \phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad g(s) = 2\xi(s), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1. \end{aligned} \quad (1.3)$$

Fix a finite packet of actual zeros with full multiplicities, stable under $\rho \mapsto 1 - \bar{\rho}$. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h v_h \in E_h^\times. \quad (1.4)$$

The source supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. It also supplies

$$\mathcal{T}_h P = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (1.5)$$

The statements below also apply to $h = 1$ as the actual theta-seed calculation, but that case has zero-dimensional arithmetic quotient and is not substituted for a nonempty packet.

S12.1.2 2. The sum coordinate and its conjugate logarithmic operator

Let $k \geq 2$, $S = \sum_{i=1}^k s_i$, and use the explicit centered relative coordinates

$$z_i = s_i - S/k \quad (i < k), \quad z_k = -\sum_{i < k} z_i, \quad s_i = S/k + z_i. \quad (2.1)$$

On the original integration lines set

$$s_i = \frac{1}{2} + it_i, \quad S = \frac{k}{2} + iu, \quad u = \sum_i t_i, \quad y_i = t_i - u/k \ (i < k), \quad y_k = -\sum_{i < k} y_i. \quad (2.2)$$

The inverse is $t_i = u/k + y_i$. The real Jacobian has absolute value one. For $k = 2$, the earlier relative coordinate was $v = t_1 - t_2 = 2y_1$, so $dt_1 dt_2 = du dy_1 = \frac{1}{2} du dv$. Thus the earlier factor $1/2$ remains exactly in that coordinate comparison. Its Δ is $4z_1^2 = -4y_1^2$.

Write

$$a_h(t) = \frac{v_h(\frac{1}{2} + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt = \|F_h\|_{L^2(dx)}^2. \quad (2.3)$$

The factor $\sqrt{2\pi}$ is the stated Mellin–Fourier Plancherel factor. No mass has been set to one. Define the actual fibre amplitude

$$\Psi_k(u, \mathbf{y}) = \prod_{i=1}^k a_h(u/k + y_i). \quad (2.4)$$

Mellin–Fourier followed by (2.2) is the unitary map

$$\mathcal{U}_k : L^2(\mathbb{R}_{>0}^k, d^k x) \longrightarrow L^2(\mathbb{R}_u \times \mathbb{R}_{\mathbf{y}}^{k-1}, du d\mathbf{y}).$$

On the given test functions it satisfies

$$\begin{aligned} \mathcal{U}_k(D^{(k)}F) &= (\tfrac{k}{2} + iu)\mathcal{U}_kF, \quad D^{(k)} = \sum_i D_i, \\ \partial_u \mathcal{U}_kF &= i\mathcal{U}_k(\mathcal{L}_kF), \quad \mathcal{L}_k = \frac{1}{k} \sum_{i=1}^k \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -1. \end{aligned} \quad (2.5)$$

Here $\mathcal{L}_k$ is multiplication by the displayed function; $1/k$ comes from the inverse coordinate map keeping the relative z_i fixed. It does not change the scaling generator $D^{(k)}$. The identities follow by differentiating the Mellin integral. Logarithmic multiplication preserves $\mathcal{B}$ and the corresponding tensor test space.

For the polynomial source,

$$\mathcal{U}_k \mathcal{T}_h^{(k)} P = \Psi_k(u, \mathbf{y}) P(S/k + iy_1, \dots, S/k + iy_k). \quad (2.6)$$

Thus the operator whose new estimate is computed is related to the original scaling operator by the precise commutator (2.5), not equated with the generator or its weight defect.

S12.1.3 3. A quantitative contraction on the actual arithmetic density

Define the convolution mass and its logarithmic-derivative energy by

$$\begin{aligned} m_k(u) &= \int_{\mathbb{R}^{k-1}} |\Psi_k(u, \mathbf{y})|^2 d\mathbf{y} = w_h^{*k}(u), \quad \int m_k = \mu_h^k, \\ \mathcal{I}_h &= \int_{\mathbb{R}} \frac{|w'_h(t)|^2}{w_h(t)} dt, \quad \mathcal{I}_{h,k} = \int_{\mathbb{R}} \frac{|m'_k(u)|^2}{m_k(u)} du. \end{aligned} \quad (3.1)$$

At isolated zeros of w_h , the integrand is assigned its limiting value where applicable, or any value on that null set. The construction and proofs use the smooth amplitude a_h , so no division at a zero is required. For $k \geq 2$, $m_k(u) > 0$ for every real u : each factor w_h is positive almost everywhere, and its convolution integral has a positive integrand almost everywhere on its fibre.

The original functional equation gives

$$g^\dagger = g, \quad h^\dagger = (-1)^d h, \quad v_h^\dagger = (-1)^d v_h, \quad \overline{a_h(t)} = (-1)^d a_h(t). \quad (3.2)$$

Accordingly a_h and a'_h have the same fixed real/imaginary phase and

$$\boxed{\mathcal{I}_h = 4 \int_{\mathbb{R}} |a'_h(t)|^2 dt = 4 \int_0^\infty (\log x)^2 |F_h(x)|^2 dx < \infty, \quad \int \overline{a_h} a'_h = 0.} \quad (3.3)$$

Proof: outside the discrete zero set, $w'_h = 2\overline{a_h}a'_h$ is real, giving $w_h'^2/w_h = 4|a'_h|^2$. Mellin Plancherel and $\partial_t v_h(\frac{1}{2} + it) = i v'_h(\frac{1}{2} + it)$ give the second equality. The zero integral is the integral of $w'_h/2$. The decay needed at both endpoints follows from $F_h \in \mathcal{B}$: $e^{y/2} F_h(e^y)$ is Schwartz, as are its polynomially weighted versions. Reflection (3.2) is essential for the equality rather than just an upper bound.

Let $\mathcal{H}_u = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$. The line inclusion and projection are

$$j_{0,u} : \mathbb{C} \rightarrow \mathcal{H}_u, \quad c \mapsto c\Psi_k(u, \cdot), \quad \Pi_{0,u}Z = \Psi_k \frac{\langle \Psi_k, Z \rangle}{m_k(u)}.$$

Put

$$n_k(u) = (1 - \Pi_{0,u})\partial_u \Psi_k = \partial_u \Psi_k - \frac{m'_k}{2m_k} \Psi_k. \quad (3.4)$$

The inner product is antilinear in its first argument. Equation (3.2) makes $\langle \Psi_k, \partial_u \Psi_k \rangle = m'_k/2$ real. Therefore (3.4) is precisely an orthogonal decomposition, including at zeros of individual amplitude factors.

Theorem 3.1 (mass-retaining sum contraction and exact relative term).

$$\mathcal{I}_{h,k} + 4 \int_{\mathbb{R}} \|n_k(u)\|_{\mathcal{H}_u}^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h. \quad (3.5)$$

In particular

$$\mathcal{I}_{h,k} \leq \frac{\mu_h^{k-1}}{k} \mathcal{I}_h. \quad (3.6)$$

Proof. Differentiate the actual product amplitude in the centered coordinates:

$$\partial_u \Psi_k = \frac{1}{k} \sum_{i=1}^k a'_h(t_i) \prod_{j \neq i} a_h(t_j).$$

Integrating its squared modulus over all original t_i , the k diagonal terms each give $\mu_h^{k-1} \|a'_h\|^2$. Every distinct-index cross term is $\mu_h^{k-2} |\int \overline{a_h} a'_h|^2 = 0$. Thus

$$\int \|\partial_u \Psi_k\|^2 du = \frac{\mu_h^{k-1}}{4k} \mathcal{I}_h.$$

The pointwise orthogonal decomposition (3.4) gives

$$\|\partial_u \Psi_k\|^2 = \frac{m_k'^2}{4m_k} + \|n_k\|^2.$$

Integrate and multiply by four. This proves both statements and the finiteness of $\mathcal{I}_{h,k}$. All masses and the factor four remain.

This is an application of the projection mechanism underlying Fisher-information convolution inequalities (Johnson–Barron), with the actual arithmetic amplitude, its phase, and its residual relative vector kept explicitly. It does not rely on log-concavity, a Gaussian approximation, or locations of the zeros beyond the stipulated reflection-stable packet and the source construction of F_h .

The pair $(\Pi_{0,u}\partial_u \Psi_k, n_k)$ retains the full derivative by addition. Its split lift can have an active zero in either component without converting the other component or its source to τ . Dropping n_k would replace equality (3.5) by an inequality and lose the quantified relative contribution; the actual projection and its complementary map remain available.

S12.1.4 4. The full relative matrix has a computed connection and normal variation

Take any finite linearly independent family of relative polynomials $\theta_\alpha(\mathbf{z})$, independent of S . For the original degree- M source use the actual coefficient functions

$$P(\mathbf{s}) = \sum_{\alpha} \theta_{\alpha}(\mathbf{z}) f_{\alpha}(S), \quad \deg f_{\alpha} \leq M - \deg \theta_{\alpha}. \quad (4.1)$$

Choose a homogeneous relative monomial basis, or the stated unscaled invariant basis on a symmetric summand, so that these degree conditions specify the exact original filtered image. This is the centered-coordinate version of the preceding matrix weight; substitution $s_i = S/k + z_i$ is triangular by total degree and invertible.

Define the actual fibre map, its derivative, and its three Gram matrices:

$$\begin{aligned} j_u : \mathbb{C}^r &\rightarrow \mathcal{H}_u, & j_u c &= \Psi_k(u, \mathbf{y}) \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}, \\ W(u) &= j_u^* j_u, & B(u) &= j_u^* j'_u, & T(u) &= (j'_u)^* j'_u. \end{aligned} \quad (4.2)$$

These are integrals in the same arithmetic measure. $W(u)$ is strictly positive because the relative polynomials are independent and Ψ_k is nonzero almost everywhere. Every entry is smooth by differentiated Schwartz estimates on compact u intervals. Its domain has its stated finite size r ; no assertion is made that r remains fixed as the source degree grows.

In the fixed-phase case (3.2),

$$B(u) = \frac{1}{2} W'(u). \quad (4.3)$$

Indeed $\overline{\Psi_k} \partial_u \Psi_k$ is real almost everywhere, hence B is Hermitian; differentiate W to get $W' = B + B^*$. The integrals remain valid at the zero set because the smooth product amplitude, not its logarithm, is used.

Let

$$\Pi_u = j_u W^{-1} j_u^*, \quad \Gamma_u = W^{-1} B = \frac{1}{2} W^{-1} W', \quad N_u = (1 - \Pi_u) j'_u. \quad (4.4)$$

The exact differential map is

$$\boxed{\partial_u(j_u c(u)) = j_u(c'(u) + \Gamma_u c(u)) + N_u c(u), \quad j_u^* N_u = 0.} \quad (4.5)$$

Thus $\nabla_u c = c' + \Gamma_u c$ is a metric-compatible connection on this finite coefficient bundle:

$$\Gamma_u^* W + W \Gamma_u = W'. \quad (4.6)$$

The complete normal form is

$$\boxed{\mathcal{N}(u) := N_u^* N_u = T - B^* W^{-1} B = T - \frac{1}{4} W' W^{-1} W' \succeq 0.} \quad (4.7)$$

In particular no commutation of W and W' is assumed. Multiplication proves the identity and positivity follows from $1 - \Pi_u$ being an orthogonal projection. The norm decomposition is

$$\boxed{\|\partial_u(j_u c)\|^2 = (\nabla_u c)^* W (\nabla_u c) + c^* \mathcal{N} c.} \quad (4.8)$$

For the one-column constant family, (4.8) is exactly (3.5) before integration. For the full relative family it retains every matrix entry in both terms. Scalar inequality (3.6) is not inserted as a bound

on this larger matrix: its precise connection is the inclusion of the constant column and the pair of projections $\Pi_{0,u}, \Pi_u$ with $\text{im } \Pi_{0,u} \subset \text{im } \Pi_u$.

S12.1.5 Change of relative frame

For an explicitly given invertible smooth matrix $C(u)$, replace the fibre map by $j_u^C = j_u C(u)$. The exact transformed connection and normal map are

$$W^C = C^* W C, \quad \Gamma^C = C^{-1} \Gamma C + C^{-1} C', \quad N^C = N C. \quad (4.12)$$

These follow by differentiating j^C and applying the same orthogonal projection. This is the additional derivative term needed when using the earlier pointwise triangular square-completion map, whose coefficients can depend on u . Multiplication by S still intertwines without such a term; the derivative's relationship to it is the explicit commutator (2.5). The exact filtered image of the original coefficient functions is retained after this change of frame.

S12.1.6 4.1 The differentiated filtered image is retained

For the actual source coefficient function $c_P(u) = (f_\alpha(k/2 + iu))_\alpha$, define the linear graph map

$$\mathcal{D}_M : P \mapsto (j_u \nabla_u c_P, N_u c_P). \quad (4.9)$$

Its values lie in the specified graph image $\mathcal{G}_M$, not in two freely chosen function spaces. Addition of the two components recovers

$$\text{Add } \mathcal{D}_M(P) = \partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P = i \mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} P. \quad (4.10)$$

The two summands are pointwise orthogonal, so this addition is an isometric injection on the graph into the derivative's actual image. This supplies a precise way to recover its arithmetic jet:

$$\mathcal{G}_M \xrightarrow{\text{Add}} i \mathcal{U}_k (\mathcal{B}^{\widehat{\otimes} k}) \xrightarrow{-i \mathcal{U}_k^{-1}} \mathcal{B}^{\widehat{\otimes} k} \xrightarrow{J^{(k)}} E_h^{\otimes k}. \quad (4.11)$$

No separate arithmetic-jet map on an arbitrary measurable normal section is asserted. The combined graph is exactly the domain on which the formula is defined. The original support label is retained in both components and in (4.11). The cohomological control still uses A_k, R_M, G_M, W_M and (1.1); (2.5), (4.10), and (4.11) are the actual comparison to this new analytic derivative.

S12.1.7 5. Differentiating a supported relation requires its first infinitesimal layer

The logarithmic observable in (2.5) has a calculated effect on the original theta relation, even in one variable:

$$\boxed{J_h(\log x \Theta \phi) = j_h(g') j_h(H_\phi)}. \quad (5.1)$$

Proof: logarithmic multiplication differentiates the actual Mellin transform, so

$$\mathcal{M}(\log x \Theta \phi) = g' H_\phi + g H'_\phi.$$

Taking full jets modulo h removes the second term, but not the first. For $\phi = P(D)\phi_*$, $H_\phi = P$; hence all E_h values of $j_h H_\phi$ are supplied by actual original source functions.

This is the map connecting the original supported boundary to the Jacobian trace contraction. In the original quotient $q\Theta = 0$, while $J_h(\log x \Theta)$ has the explicitly displayed image. The typed resolution of that effect is a first thickening of the *same* relation ideal.

Put

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)), \quad E = \mathcal{P}/I, \quad E^{[2]} = \mathcal{P}/I^2.$$

There is the actual exact sequence

$$0 \longrightarrow I/I^2 \longrightarrow E^{[2]} \xrightarrow{\pi} E \longrightarrow 0. \quad (5.2)$$

The coefficient conormal map is the isomorphism

$$\iota_N : E^k \xrightarrow{\sim} I/I^2, \quad (a_1, \dots, a_k) \mapsto \left[\sum_i h(s_i) \tilde{a}_i \right]. \quad (5.3)$$

A direct proof uses monic division in each variable. The polynomial ring is free over $\mathbb{C}[h(s_1), \dots, h(s_k)]$ with basis $\prod s_i^{a_i}$, $0 \leq a_i < d$. Modulo the square of the ideal of these latter variables, only coefficient degrees zero and one survive. This proves uniqueness and surjectivity in (5.3), retaining all nilpotents in E .

Define the sum derivative at fixed relative coordinates

$$\partial_S^{\text{rel}} = \frac{1}{k} \sum_i \partial_{s_i}. \quad (5.4)$$

It maps I^2 into I , so it induces a *complex-linear* map

$$\delta_S : E^{[2]} \longrightarrow E, \quad [P]_{I^2} \mapsto [\partial_S^{\text{rel}} P]_I. \quad (5.5)$$

It is a derivation along π :

$$\delta_S(ab) = \pi(a)\delta_S(b) + \pi(b)\delta_S(a), \quad \delta_S(S) = 1.$$

Its restriction to the retained conormal fibre is the E -linear row

$$\delta_S \iota_N(a_1, \dots, a_k) = \frac{1}{k} \sum_i h'(s_i) a_i. \quad (5.6)$$

The full conormal differential, before contraction by ∂_S^{rel} , is

$$I/I^2 \xrightarrow{d_I} \bigoplus_i E \, ds_i, \quad (a_i)_i \mapsto \sum_i h'(s_i) a_i \, ds_i. \quad (5.7)$$

This is the two-term cotangent presentation for the complete intersection, in cohomological degrees $-1, 0$. Its universal differential and (5.5) are related by contraction $\sum b_i ds_i \mapsto k^{-1} \sum b_i$. This is the established conormal/cotangent construction applied to this exact packet ideal (Stacks, Tags 00S0 and 08SH).

The original split diagram is

$$G(E^{[2]}) \xrightarrow{G(\pi)} G(E), \quad \delta_S^\tau : G(E^{[2]}) \longrightarrow G(E) \quad \text{as } G(\mathbb{C})\text{-semimodules.} \quad (5.8)$$

The first is a semiring morphism; the second is a split-linear map of coefficient modules, not a semiring morphism. Both send external absence to external absence and supported zero to supported zero. A supported conormal class has image e_E under the first map, while its image under the second is the supported scalar specified in (5.6). The larger domain retains the datum needed to differentiate that relation.

For one local block $h = (s - \rho)^m$ times a retained unit, (5.6) detects the z^{m-1} coefficient with its factor m . In an ordered product of local block lengths m_i , the image of (5.6) has dimension

$$\prod_i m_i - \prod_i (m_i - 1), \quad (5.9)$$

because it is the ideal generated by the unit multiples of $z_i^{m_i-1}$. Each unit is invertible in its actual local algebra. Monomials missing all top exponents form the complementary vector count; they are not discarded from E .

S12.1.8 5.1 Carry the arithmetic jet unit through the derivative

Let $U(\mathbf{s}) = \prod_i v_h(s_i)$, and retain its full Taylor class $\widehat{U} \in E^{[2]}$. This class is invertible. Define

$$\widehat{\mathcal{J}}P = \widehat{U}[P]_{I^2}, \quad \mathcal{J}P = U[P]_I, \quad \beta_h = U^{-1}\delta_S\widehat{U} \in E. \quad (5.10)$$

The finite Taylor realization here uses the entire functions on neighbourhoods of the finite zero set; polynomial Hermite interpolation supplies their classes. $E^{[2]}$ is *not* being called a new zeta-zero packet: it also retains first-order source relations not annihilated by g .

The exact commuting equation is

$$\boxed{J^{(k)}\mathcal{L}_k\mathcal{T}_h^{(k)}P = \delta_S\widehat{\mathcal{J}}P = \mathcal{J}(\partial_S^{\text{rel}}P) + \beta_h\mathcal{J}P.} \quad (5.11)$$

Proof: its Mellin transform is $\partial_S^{\text{rel}}(UP)$. Apply the product rule before taking jets. Thus the coefficient-connection term is $\beta_h = k^{-1}\sum_i j(v'_h/v_h)(s_i)$, with no discarded local unit.

For an original relation $P = \sum_i h(s_i)P_i$, the last term vanishes and

$$\boxed{J^{(k)}\mathcal{L}_k\mathcal{T}_h^{(k)}P = \frac{1}{k}U\sum_i h'(s_i)[P_i]_I.} \quad (5.12)$$

Modulo I , $v_h(s_i)h'(s_i) = g'(s_i)$, so the same equation can be written with $g'(s_i)$ and the other retained v_h factors. This is the precise relationship between the analytic sum derivative in (4.10) and the conormal row (5.6).

In one variable the source's residue pairing therefore obeys

$$\boxed{\mathcal{R}_Z(f, J_h(\log x \Theta(P(D)\phi_*))) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho).} \quad (5.13)$$

Indeed (5.1) makes the second argument $g'P$, and g'/g has residue m_ρ . The residue pairing, its Jacobian contraction, and the logarithmic derivative of an actual original theta relation have now been connected by a single formula. No new positivity assertion is inferred from that equality.

S12.1.9 6. All higher derivative layers and the old cochain signs remain available

For $r \geq 1$, the derivation gives actual maps

$$\partial_S^{\text{rel}} : \mathcal{P}/I^{r+1} \longrightarrow \mathcal{P}/I^r. \quad (6.1)$$

They commute with the quotient maps wherever their sources and targets match. The associated-graded algebra is explicitly

$$\text{gr}_I \mathcal{P} \cong E[\eta_1, \dots, \eta_k], \quad \eta_i \longleftrightarrow [h(s_i)]. \quad (6.2)$$

On it, the degree-minus-one derivative is

$$\boxed{\text{gr}(\partial_S^{\text{rel}}) = \frac{1}{k} \sum_i h'(s_i) \partial_{\eta_i}.} \quad (6.3)$$

The proof expands $\partial(h^\alpha P)$. The term differentiating P stays in $I^{|\alpha|}$ and disappears only in the explicitly specified quotient $I^{|\alpha|-1}/I^{|\alpha|}$. The other terms give (6.3) with every factor α_i retained. Consequently this layer has rank $\binom{r+k-1}{k-1}$ over E , not one new scalar support.

There is also a full mixed derivative formula retaining all coordinate directions:

$$\boxed{\left[\partial_{s_1} \cdots \partial_{s_k} \left(UP \prod_i h(s_i) \right) \right]_I = \left[UP \prod_i h'(s_i) \right]_I = \left[P \prod_i g'(s_i) \right]_I.} \quad (6.4)$$

Every other product-rule term contains an undifferentiated $h(s_i)$ and maps to the supported zero in E . At the test-function level these derivatives are multiplication by $\prod_i \log x_i$. At the residue level (6.4) is the product Jacobian in the existing tensor arithmetic trace. The ordered volume form $ds_1 \wedge \cdots \wedge ds_k$ and the previously specified tensor-duality sign are retained; no new identification of the iterated residue with an unsigned cochain supertrace is made.

The original primitive of $h(s_i)P_i$ still has the sign $(-1)^{i-1}$ from the prior product complex. Its cochain differential supplies the second $(-1)^{i-1}$. Conormal reduction records the resulting coefficient $[P_i]_I$. It does not replace that source primitive or the higher syzygies of its Koszul complex.

For a finite total-degree relation space, (6.1) is applied to the image of that exact space in $\mathcal{P}/I^{r+1}$. It does not enlarge the set of admissible representatives. The degree- M next-relation map from the attachment feeds (5.12) by its actual polynomial numerator. This supplies a differentiated observation of those very same e -relations.

S12.1.10 7. Exact calibrations and a bounded arithmetic seed evaluation

S12.1.11 7.1 Gaussian fibre: the relative connection can be flat without deleting it

Take the declared calibration $a(t) = e^{-t^2/4}$, $w(t) = e^{-t^2/2}$, $\mu = \sqrt{2\pi}$. This is a test of the formula, not the arithmetic input. For two factors,

$$\Psi_2(u, y) = e^{-u^2/8} e^{-y^2/2}, \quad m_2(u) = \sqrt{\pi} e^{-u^2/4}.$$

For every fixed relative polynomial frame, $j'_u = -(u/4)j_u$, so

$$\Gamma_u = -\frac{u}{4}I, \quad \mathcal{N}(u) = 0, \quad \mathcal{I}_2 = \frac{\mu \mathcal{I}_1}{2} = \pi. \quad (7.1)$$

All relative directions and their Gram entries remain; zero normal variation is a proved consequence of this particular product, not an instruction to erase the relative fibre in the arithmetic case.

S12.1.12 7.2 A zero-bearing calibration has nonzero retained normal variation

Take instead $a(t) = te^{-t^2/4}$ and $w(t) = t^2e^{-t^2/2}$. Then

$$\begin{aligned}\mu &= \sqrt{2\pi}, & \mathcal{I}_1 &= 3\sqrt{2\pi}, \\ \Psi_2(u, y) &= (u^2/4 - y^2)e^{-u^2/8 - y^2/2}, \\ m_2(u) &= \frac{\sqrt{\pi}}{16}e^{-u^2/4}(u^4 - 4u^2 + 12).\end{aligned}\tag{7.2}$$

These identities follow from the complete Gaussian even moments; the polynomial has positive minimum 8, so m_2 has no zero. For $u \neq 0$, $\partial_u \Psi_2$ is not a multiple of Ψ_2 as a polynomial in y , so $n_2(u) \neq 0$. The exact identity is

$$\mathcal{I}_2 + 4 \int \|n_2\|^2 = 3\pi, \quad \mathcal{I}_2 < 3\pi.\tag{7.3}$$

The checker calculates the two-column matrix connection and nonzero positive normal Gram at $u = 1$, retaining the relative basis and its mass factors. This also tests the handling of zeros without assuming logarithmic scores are bounded.

S12.1.13 7.3 The actual arithmetic seed, with no selected zero coordinates

For $h = 1$, $F_h = f_0 = \Theta\phi_*$. Its exact inversion is $f_0(x) = x^{-1}f_0(1/x)$. Hence

$$\mu_1 = 2 \int_1^\infty f_0(x)^2 dx, \quad \mathcal{I}_1 = 8 \int_1^\infty (\log x)^2 f_0(x)^2 dx.\tag{7.4}$$

On that half-line use the actual absolutely convergent sum

$$f_0(x) = 2 \sum_{n \geq 1} (4\pi^2 n^4 x^4 - 6\pi n^2 x^2) e^{-\pi n^2 x^2}.$$

The numerical evaluations at 45 digits with six integer terms and at 65 digits with eight integer terms agree on the displayed 35-digit values:

$$\mu_1 \approx 1.2790072478464851404795335922671933, \quad \mathcal{I}_1 \approx 0.12575246821348846545509759756793817.$$

The optional computation evaluates (7.4) at two stated decimal precisions and two integer cutoffs. Its output is labeled numerical, not an interval certificate. The theorem uses the exact integrals; no numerical value or cutoff is imported into a purity assertion.

S12.1.14 8. What the new estimate controls in the programme

The earlier arithmetic defect remains

$$W_M = A_k^* G_M + G_M A_k - k G_M, \quad V_M = A_k K_M + K_M A_k^* - k K_M, \quad W_M = G_M V_M G_M. \tag{8.1}$$

The new operations retain the path

$$\mathcal{T}_k(P) \xrightarrow{\mathcal{U}_k} \Psi_{kCP} \xrightarrow{\partial_u} \mathcal{G}_M \xrightarrow{\text{Add}, -i\mathcal{U}_k^{-1}} \mathcal{L}_k \mathcal{T}_k(P) \xrightarrow{J^{(k)}} \delta_S \hat{\mathcal{J}} P. \tag{8.2}$$

This is the explicit connection between the positive differential estimate, the matrix-valued relative fibre, and the first conormal layer of the original arithmetic quotient. It does not replace the scaling generator A_k with differentiation. Their relation is the commutator (2.5), and the source's original representative relation $D^{(k)}R_M - R_MA_k \in B_{M+1}$ continues to determine (8.1).

The result of this continuation is a genuine $1/k$ factor in (3.5), with the literal mass μ_h^{k-1} , and an exact decomposition of the full matrix derivative. Uniform relative interpolation control is still needed to transfer that integrated smoothing to the extremal finite-packet estimate (8.1). The required comparison cannot be inferred from the scalar density alone; (4.9)–(4.11) and (5.11) provide the actual objects on which to work. All normal terms and the conormal Jacobian remain in that comparison.

Deligne's source architecture requires a further estimate after the product-to-one-parameter operation, on the same arithmetic image and with its duality. This note constructs and estimates a differential of that new-base pushforward, and identifies its exact interaction with the trace-producing Jacobian. It does not assert that a Fisher inequality is already Deligne purity, or that the analytic matrix bundle is a finite-field lisse sheaf. The finite-point cohomology, spectral-sum restriction functor, kernel-layer calculation, and first infinitesimal quotient remain connected by the typed arrows above.

S12.1.15 References and reading record

1. Supplied *Tau Kernel Layer Integration*, NOTE.tex and RESEARCH_NOTE.md, read for the original relation-layer and antidual comparisons; the archive matched the earlier copy exactly and all manifest entries verified. The source's eighteen-method exact suite was rerun normally and with optimized Python. This is not a new formal certificate.
2. Supplied *Tau Spectral Sum Descent*, NOTE.tex, read in full, especially §§2, 5, 6, 8 and 10. The same sum coordinate, matrix density, relative polynomial image, full jets, and residue comparison are used.
3. Supplied *SplitZero: reconstruction, internal homology, and balanced finite jets*, §§1–4, for the existing original internal quotient and supported cohomology interfaces. No new scalar implementation is introduced.
4. Oliver Johnson and Andrew Barron, *Fisher Information inequalities and the Central Limit Theorem*, arXiv:math/0111020, published in Probability Theory and Related Fields 129 (2004), 391–409. Their projection method is established background; Theorem 3.1 is proved here directly with the unaltered arithmetic mass and phase. <https://arxiv.org/abs/math/0111020>.
5. The Stacks Project, Tags 00S0 and 08SH, the conormal presentation and cotangent complex of a local complete intersection. These are reusable algebraic constructions; the explicit arithmetic conormal row and the trace connection are computed in §§5–6. <https://stacks.math.columbia.edu/tag/00S0> and <https://stacks.math.columbia.edu/tag/08SH>.
6. Deligne, *La conjecture de Weil II*, IHES 52 (1980), 137–252. This continuation relies on the previously identified tensor/pushforward/dual-control architecture, not a fresh reading or independent audit of all pages. No finite-field theorem is asserted on the new base merely from terminology.

S13 Cyclic arithmetic sum control and the complete contracted relation tower

Source attribution. Complete TeX witness supplied in the Cyclic Sum Control archive. Its exact source and archive hashes identify the delivered text. The seven retained prior source files are identified separately; the earlier 48-entry validation claim concerns the separate Sum Connection archive. The new 24-method finite checker has a separate independent replay record. No Lean execution, remote CI or uniform arithmetic estimate is inferred from this source witness.

Exact revision. 2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c

Source file. Tau_Cyclic_Sum_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Source witness SHA-256; no remote Git revision is assigned to this delivery

Source SHA-256. 2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c

Original archive. Tau_Cyclic_Sum_Control_2026-09-12.zip

Archive SHA-256. e0e4d0ecda69e4738dcb4cdd3b750e623d863eac8b44f353737393e720a840a7

S13.1 Sum-generated arithmetic cohomology, scalar control, and exact conormal depth

Owner-directed continuation of the split-zero programme — 12 September 2026.

Status. Written deductions for mathematical review. This note uses the original theta source and its stated finite-packet realization from the preceding work. It does not claim a new Lean certificate, a global comparison-kernel theorem, or a uniform arithmetic purity estimate. The new constructions are (i) an explicitly injected sum-generated arithmetic submodule containing every amplified spectral value, (ii) its scalar convolution norm and rank-two control formula, (iii) raw-moment formulas for the complete differentiated polynomial image, and (iv) the exact contraction of every original conormal thickening to that sum coordinate. The full relative quotient, its action, and its trace are retained.

S13.1.1 1. Source intake and fixed objects

The two new transcript files were searched for changes to the arithmetic/control and formalization interfaces. They preserve the earlier full-jet kernel/layer calculations and the formalized internal quotient; they do not supply a later analytic proof of the remaining estimate. The previous Tau_Sum_Connection_Control archive was opened directly, its LaTeX and mathematical note were read, and all 48 entries of its supplied manifest verified. Its scalar contraction and matrix connection are inputs [S1], not attributed to the present continuation. The previous transcript's checked algebraic interfaces remain within their reported scope [S2–S4].

The original base is still b_τ . Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\begin{array}{ccc}
G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\
p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\
\mathbb{Z} & \xrightarrow{j} & \mathbb{C}.
\end{array} \tag{1.1}$$

The target of $p\mathbb{Z}$ is infinite. A ring homomorphism is lifted through G . A complex-linear map $T : V \rightarrow W$ has instead the explicitly defined $G(\mathbb{C})$ -linear lift $T^\tau(v^\bullet) = (Tv)^\bullet$, $T^\tau(\tau) = \tau$. In a support diagram the formula is $(\lambda, v) \mapsto (\varphi(\lambda), T_\lambda v)$, with its stated support map φ . These formulas commute with the respective amplitude observations.

Use exactly

$$\begin{aligned}
V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\
\mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\
C_+ &= [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \\
\phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad g(s) = 2\xi(s), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1.
\end{aligned} \tag{1.2}$$

Let $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$ be a **nonempty** finite packet of actual zeros of g , with their full orders. Assume it is stable under $\rho \mapsto 1 - \bar{\rho}$. Put

$$d = \deg h, \quad v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(v_h) \in E_h^\times. \tag{1.3}$$

The preceding source provides the actual test function F_h and map

$$\begin{aligned}
\mathcal{M}F_h &= v_h, \quad h(D)F_h = \Theta\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \\
\mathcal{T}_h(hP) &= \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h, \quad q\mathcal{T}_h(P) = \sigma_h(v_h[P]_h).
\end{aligned} \tag{1.4}$$

For $k \geq 2$, define

$$\begin{aligned}
\mathcal{P}_k &= \mathbb{C}[s_1, \dots, s_k], \quad I_k = (h(s_1), \dots, h(s_k)), \quad B_k = \mathcal{P}_k/I_k = E_h^{\otimes k}, \\
S &= s_1 + \dots + s_k, \quad A_k = M_S, \quad U_k = v_h^{\otimes k} \in B_k^\times.
\end{aligned} \tag{1.5}$$

The ordinary invariant submodule is $B_k^{\text{sym}} = B_k^{S_k}$. Its cochain realization is the existing signed chain projector $\frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi$: in top degree the Koszul sign in T_π cancels the displayed sign. The complementary cochain summand and both traces remain in the programme. No change to that projector is made here.

S13.1.2 2. The sum-generated submodule and its exact spectrum

Index the distinct roots by $\rho_1, \dots, \rho_a$, with orders $m_1, \dots, m_a$. For an occupation vector $\mathbf{n} = (n_1, \dots, n_a)$ with $\sum n_i = k$, set

$$\lambda_{\mathbf{n}} = \sum_i n_i \rho_i, \quad L_{\mathbf{n}} = 1 + \sum_i n_i (m_i - 1). \tag{2.1}$$

Let Λ_k be the set of distinct values of $\lambda_{\mathbf{n}}$, and put

$$\ell_\lambda = \max_{\mathbf{n}: \lambda_{\mathbf{n}} = \lambda} L_{\mathbf{n}}, \quad \chi_{h,k}(S) = \prod_{\lambda \in \Lambda_k} (S - \lambda)^{\ell_\lambda}. \tag{2.2}$$

Equal sums are grouped by their exact equality; no assumption of linear independence or of absent collisions is used. The distinct occupation idempotents stay in B_k^{sym} even when their sums agree.

Theorem 2.1. The polynomial (2.2) is the minimal polynomial of A_k both on B_k and on B_k^{sym} . There are an injective unital algebra map and a separate injective module map

$$\begin{aligned} \alpha_{h,k} : C_{h,k} &:= \mathbb{C}[S]/(\chi_{h,k}) \hookrightarrow B_k^{\text{sym}}, \quad [P] \mapsto P(A_k)1, \\ \eta_{h,k} = M_{U_k} \alpha_{h,k} : C_{h,k} &\hookrightarrow B_k^{\text{sym}}, \quad [P] \mapsto U_k P(A_k)1. \end{aligned} \quad (2.3)$$

The map M_{U_k} is an invertible B_k^{sym} -module map on its carrier. It is not asserted to be a unital algebra homomorphism; its inverse is $M_{U_k^{-1}}$. The full jet unit is retained by this pair of maps rather than assigned the value one.

Proof. At an ordered root tuple the local factor is

$$L_\rho = \mathbb{C}[z_1, \dots, z_k]/(z_1^{m_{i_1}}, \dots, z_k^{m_{i_k}}), \quad A_k = \lambda I + N, \quad N = M_{z_1 + \dots + z_k}.$$

Every monomial of degree greater than $\sum_j (m_{i_j} - 1)$ vanishes. In precisely that degree, $N^D 1$ contains the nonzero monomial $\prod z_j^{m_{i_j} - 1}$ with coefficient $D! / \prod (m_{i_j} - 1)!$. Thus its nilpotence index is $D + 1$. The minimal polynomial on the product takes the maximum index over factors at the same eigenvalue, which is (2.2). For the invariant algebra, the vector 1 is invariant and its annihilator under $\mathbb{C}[S]$ is already the same polynomial; indeed $P(A_k)1 = 0$ is equivalent to the algebra element $P(S)$ being zero. Thus restriction to invariants does not reduce this minimal polynomial. Multiplication by the actual unit U_k has zero kernel and commutes with A_k . This proves all assertions.

In particular, for every $\rho \in Z$, the value $k\rho$ occurs in Λ_k . The cyclic module has a nonzero eigenvector at every $\lambda \in \Lambda_k$: take the local class $(S - \lambda)^{\ell_\lambda - 1}$ in its Chinese-remainder factor, and zero in its other factors. Its image under $\eta_{h,k}$ is still nonzero and has eigenvalue λ .

This retains every exponent required for tensor amplification. It does not claim that a chosen repeated pure tensor is literally the same vector as this cyclic eigenvector when different occupations have the same sum. The original repeated-local-factor projection is the explicit map

$$C_{h,k} \xrightarrow{\eta_{h,k}} B_k^{\text{sym}} \longrightarrow L_{(\rho, \dots, \rho)}; \quad (2.4)$$

its complete nilpotent image is generated by the local unit times $1, N, \dots, N^{k(m_\rho - 1)}$. A cyclic eigenvector supported at a collided sum can have zero projection to a shorter local block. The spectral value is nevertheless present by the proved injection (2.3).

S13.1.3 The remainder of the arithmetic module remains as a quotient

Let

$$T_{h,k} = B_k^{\text{sym}} / \eta_{h,k} C_{h,k}.$$

Then

$$0 \rightarrow C_{h,k} \xrightarrow{\eta_{h,k}} B_k^{\text{sym}} \rightarrow T_{h,k} \rightarrow 0 \quad (2.5)$$

is an exact sequence of $\mathbb{C}[S]$ -modules. This is a module quotient, not a new algebra quotient of B_k . Its split lift is the original $G(\mathbb{C})$ -linear internal quotient. An element of the retained submodule maps to the supported zero; τ remains τ .

The sequence also has a constructed $\mathbb{C}[S]$ -linear retraction. On the generalized λ -part let $N_\lambda = A_k - \lambda I$, $w_\lambda = e_\lambda(A_k)U_k 1$, and $\ell = \ell_\lambda$. Choose the fixed-order root tuple attaining ℓ and let ϑ_λ extract

one of the top monomial coefficients of $N_\lambda^{\ell-1}w_\lambda$ there. That coefficient is its nonzero local arithmetic unit times the corresponding multinomial coefficient. In particular

$$\vartheta_\lambda(N_\lambda^{\ell-1}w_\lambda) \neq 0.$$

Define

$$\pi_\lambda^0(v) = \sum_{j=0}^{\ell-1} \vartheta_\lambda(N_\lambda^{\ell-1-j}v)X^j \pmod{X^\ell}, \quad p_\lambda(X) = \pi_\lambda^0(w_\lambda).$$

Its constant term is nonzero, so p_λ has a unique multiplicative inverse modulo X^ℓ . The actual retraction is

$$\boxed{\pi_\lambda(v) = p_\lambda(X)^{-1}\pi_\lambda^0(v)}. \quad (2.5a)$$

Indeed $\pi_\lambda^0(N_\lambda v) = X\pi_\lambda^0(v)$, and $\pi_\lambda(w_\lambda) = 1$. Combining these maps with the CRT idempotents gives $\pi : B_k^{\text{sym}} \rightarrow C_{h,k}$ with $\pi\eta_{h,k} = 1$. Thus the inverse pair is

$$\begin{aligned} C_{h,k} \oplus \ker \pi &\longrightarrow B_k^{\text{sym}}, & (c, t) &\mapsto \eta_{h,k}c + t, \\ v &\longmapsto (\pi v, v - \eta_{h,k}\pi v). \end{aligned} \quad (2.5b)$$

The quotient $T_{h,k}$ is linked to $\ker \pi$ by $[v] \mapsto v - \eta\pi v$, with inverse given by inclusion followed by the quotient. This retains every remaining nilpotent block; at each sum precisely one maximal-length block is assigned to the cyclic summand. The choice of top coefficient is recorded. This splitting is not asserted orthogonal: an arithmetic Gram H pulls back to the full block matrix containing $\eta^*H\eta$, $\eta^*H|_{\ker \pi}$, and its adjoint cross term, not just their diagonal blocks.

On a fixed sum λ , the dimension of the symmetric generalized eigenspace is

$$a_\lambda = \sum_{\mathbf{n}:\lambda_{\mathbf{n}}=\lambda} \prod_i \binom{n_i + m_i - 1}{m_i - 1}. \quad (2.6)$$

Thus $T_{h,k}$ has dimension $a_\lambda - \ell_\lambda$ there. For every entire f , trace additivity in (2.5) proves

$$\text{Tr}(f(A_k)|B_k^{\text{sym}}) = \sum_\lambda \ell_\lambda f(\lambda) + \sum_\lambda (a_\lambda - \ell_\lambda) f(\lambda). \quad (2.7)$$

The first term is the cyclic trace; the second is the retained quotient trace. All terms get the same cohomological factor $(-1)^k$ in top-degree supertrace. This calculation is valid with nilpotents; it does not replace their modules by their residue values.

For example, the algebraic fixture with simple roots $1/4, 1/2, 3/4$ and $k = 2$ has a six-dimensional symmetric module but a five-dimensional cyclic module. The two occupations with sum 1 give a one-dimensional quotient at that sum. The fixture with one root of order three and $k = 2$ has symmetric dimension six, cyclic length five, and quotient length one at 2ρ . These are fixtures of the formulas, not assertions of zeta zeros at those locations or multiplicities.

S13.1.4 3. An original source realization whose norm is the scalar sum density

Define the linear map

$$\boxed{\mathcal{V}_{h,k} : \mathbb{C}[S] \rightarrow \mathcal{B}^{\otimes k}, \quad P \mapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k})}. \quad (3.1)$$

Its Mellin transform is

$$\mathcal{M}^{(k)}\mathcal{V}_{h,k}(P) = \left(\prod_i v_h(s_i) \right) P(s_1 + \cdots + s_k).$$

It is injective: if this analytic function vanishes, it vanishes on an open set where all v_h are nonzero, forcing the polynomial P to vanish identically. Its actual arithmetic map is

$$\boxed{J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi,} \quad (3.2)$$

where $\pi_\chi(P) = [P]_\chi$. Thus its arithmetic kernel is exactly (χ) and not a selected subset of the original relations.

Let $q_k = \deg \chi_{h,k}$ and $N \geq q_k - 1$. At finite source degree,

$$\boxed{0 \rightarrow \mathbb{C}[S]_{\leq N-q_k} \xrightarrow{\times \chi} \mathbb{C}[S]_{\leq N} \xrightarrow{\pi_\chi} C_{h,k} \rightarrow 0.} \quad (3.3)$$

A negative bound means the zero space. Define the actual boundary space

$$\mathcal{D}_{h,k,N} = \mathcal{V}_{h,k}(\chi \mathbb{C}[S]_{\leq N-q_k}).$$

Because $\chi(s_1 + \cdots + s_k) \in I_k$, fixed-order monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}). \quad (3.4)$$

Each term is an original top cochain boundary: replace F_h in factor i by ϕ_* , apply the polynomial $Q_i(D_1, \dots, D_k)$, and give the primitive $(-1)^{i-1}$. The tensor differential gives the second $(-1)^{i-1}$. Their product is $+1$. This supplies the original primitive rather than defining a new vanishing relation by fiat.

The split quotient of (3.3) has equality criterion $P \sim P'$ precisely when $P - P' \in \chi \mathbb{C}[S]_{\leq N-q_k}$. A relation maps to the supported zero at label λ_N . Degree transport retains the preceding label and its killed-representative kernel.

S13.1.5 The exact norm

Retain from [S1]

$$a_h(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h(t)|^2, \quad \mu_h = \int w_h, \quad m_{h,k} = w_h^{*k}.$$

In centered sum coordinates $u = \sum t_i$, $y_i = t_i - u/k$, the amplitude is $\Psi_{h,k}(u, \mathbf{y}) = \prod_i a_h(u/k + y_i)$. Its relative integral is $m_{h,k}(u)$. Therefore

$$\boxed{\|\mathcal{V}_{h,k}P\|_{L^2(d^k x)}^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du.} \quad (3.5)$$

This is the restriction of the preceding full matrix-weight realization to the **specific source family** $P(S)$, through (3.1) and (3.2). It does not identify every relative coefficient with the constant relative coefficient.

The full relative module remains in (2.5). For comparison with the previous least-norm representative, at the same ambient degree N the maps obey

$$\mathcal{V}_{h,k}(\mathbb{C}[S]_{\leq N}) \hookrightarrow \mathcal{T}_h^{(k)}(\mathcal{P}_{k,N}), \quad \mathcal{D}_{h,k,N} \hookrightarrow \mathcal{B}_{k,N}.$$

Let $R_{k,N}^{\text{full}}$ be the old joint minimum, and $\mathcal{R}_N$ the cyclic minimum constructed next. Their exact relationship is

$$\Delta_N = \mathcal{R}_N - R_{k,N}^{\text{full}} \eta_{h,k} \in \text{Hom}(C_{h,k}, \mathcal{B}_{k,N}),$$

$$\boxed{\mathcal{G}_N = \eta_{h,k}^* G_{k,N}^{\text{full}} \eta_{h,k} + \Delta_N^* \Delta_N.} \quad (3.6)$$

Both maps represent the same full jets; their difference is in the original relation ideal, and the full minimum is orthogonal to it. This proves (3.6). The corresponding control difference is

$$\mathcal{W}_N - \eta_{h,k}^* W_{k,N}^{\text{full}} \eta_{h,k} = A_\chi^* \Delta_N^* \Delta_N + \Delta_N^* \Delta_N A_\chi - k \Delta_N^* \Delta_N. \quad (3.7)$$

This explicitly retains why the Gram inequality does not itself order the control forms.

The scalar source must reach degree $q_k - 1$ to realize all cyclic jets. This can exceed $k(d - 1)$. The reduction in relative rank has a degree cost which remains in every estimate.

S13.1.6 4. Canonical scalar representatives and rank-two arithmetic control

Apply Gram-Schmidt by subtracting lower-degree projections from $1, S, S^2, \dots$ for the exact measure in (3.5). Let $p_j^{[h,k]}$ be the resulting polynomial with leading coefficient one, and retain

$$\omega_j^{[h,k]} = \int |p_j^{[h,k]}(k/2 + iu)|^2 m_{h,k}(u) du > 0. \quad (4.1)$$

No polynomial is divided by its norm. In particular $\omega_0^{[h,k]} = \mu_h^k$. For this section suppress the indices h, k and write

$$b_j = [p_j]_\chi \in C_{h,k}, \quad \mathcal{K}_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad \mathcal{G}_N = \mathcal{K}_N^{-1}. \quad (4.2)$$

The coordinates on $C_{h,k}$ are the fixed remainder basis $1, S, \dots, S^{q_k-1}$. Its inclusion into the original full jets is $\eta_{h,k}$, carrying U_k ; b_j is not an assertion that $U_k = 1$.

The unique least-norm representative modulo the specified $\mathcal{D}_{h,k,N}$ is

$$\boxed{\mathcal{R}_N u = \mathcal{V}_{h,k} \left(\sum_{j=0}^N p_j \frac{b_j^* \mathcal{G}_N u}{\omega_j} \right).} \quad (4.3)$$

Its identities are

$$J^{(k)} \mathcal{R}_N = \eta_{h,k}, \quad q^{(k)} \mathcal{R}_N = \sigma_h^{\otimes k} \eta_{h,k}, \quad \mathcal{R}_N^* \mathcal{R}_N = \mathcal{G}_N. \quad (4.4)$$

Proof. The coefficient map in (4.3) is the weighted adjoint of the full remainder map followed by $(JJ^*)^{-1}$ in the displayed orthogonal coordinates. Multiplication verifies its right-inverse identity. If Z is another polynomial with the same remainder, their difference lies in the exact kernel (3.3); the cross term with (4.3) is zero by the adjoint formula. The norm is therefore the displayed minimum plus the squared norm of that difference. This proves uniqueness and all equations, before any completion of the arithmetic quotient.

Let $A_\chi = M_S$ on $C_{h,k}$. Multiplication by S has adjoint multiplication by $k - S$ on the integration line. Orthogonality and leading coefficients give

$$Sp_j = p_{j+1} + \beta_j p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}, \quad \Re \beta_j = k/2, \quad (4.5)$$

with the last term absent at $j = 0$. This is the ordinary orthogonal-polynomial recurrence in the retained complex coordinate; its general framework is recorded in DLMF §18.2 [E1].

Theorem 4.1 (two-row control on a spectrum-complete submodule).

$$\mathcal{Z}_N := A_\chi \mathcal{K}_N + \mathcal{K}_N A_\chi^* - k \mathcal{K}_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (4.6)$$

Consequently

$$\mathcal{W}_N := A_\chi^* \mathcal{G}_N + \mathcal{G}_N A_\chi - k \mathcal{G}_N = \mathcal{G}_N \mathcal{Z}_N \mathcal{G}_N, \quad \text{rank } \mathcal{W}_N \leq 2. \quad (4.7)$$

Proof. Substitute (4.5) into the finite sum (4.2). Each adjacent pair internal to $0, \dots, N$ cancels because the negative lower coefficient is exactly $-\omega_j/\omega_{j-1}$. Each diagonal pair is $k b_j b_j^*/\omega_j$. The only remaining adjacent pair is the one in (4.6). Congruence gives (4.7). This is a Christoffel–Darboux cancellation on the actual cyclic source, not a statement that the full relative module’s control has rank two.

S13.1.7 The original next relation is retained

Define

$$\mathfrak{b}_N = \mathcal{V}_{h,k}(p_{N+1}) - \mathcal{R}_N b_{N+1}.$$

Its full jets vanish and its polynomial degree is $N + 1$. It gives the explicit isomorphism

$$\mathbb{C} \xrightarrow{c \mapsto c \mathfrak{b}_N} \mathcal{D}_{h,k,N+1} \cap \mathcal{D}_{h,k,N}^\perp \xrightarrow{z \mapsto [z]} \mathcal{D}_{h,k,N+1} / \mathcal{D}_{h,k,N}. \quad (4.8)$$

The leading term proves injectivity; the one-dimensional degree increment proves surjectivity. Its norm and pairing are

$$\|\mathfrak{b}_N\|^2 = \omega_{N+1} + b_{N+1}^* \mathcal{G}_N b_{N+1}, \quad \mathfrak{b}_N^* \mathcal{R}_N = -b_{N+1}^* \mathcal{G}_N. \quad (4.9)$$

The class is nonzero at degree N and becomes the receiving supported zero at degree $N + 1$. Its theta primitive is obtained from the original division (3.4). The scalar support itself is not adjoined again.

The exact update is

$$\begin{aligned} \mathcal{K}_{N+1} &= \mathcal{K}_N + b_{N+1} b_{N+1}^* / \omega_{N+1}, \\ \mathcal{G}_{N+1} &= \mathcal{G}_N - \frac{\mathcal{G}_N b_{N+1} b_{N+1}^* \mathcal{G}_N}{\omega_{N+1} + b_{N+1}^* \mathcal{G}_N b_{N+1}}. \end{aligned} \quad (4.10)$$

S13.1.8 Same-metric reflection and exact excess

Let $P^{\dagger k}(S) = \overline{P(k - \bar{S})}$. The polynomial χ obeys $\chi^{\dagger k} = (-1)^{q_k} \chi$. The induced antilinear involution on the **weighted** cyclic module is

$$j_C[P] = (-1)^{kd} [P^{\dagger k}]. \quad (4.11)$$

This sign is forced by $U_k^\dagger = (-1)^{kd}U_k$ in the original embedding. It intertwines with the actual function reflection and preserves the source degree and norm. Hence the unique minimum obeys reflection on the same $\mathcal{G}_N$.

Uniqueness of the monic orthogonal polynomial gives $p_j^{\dagger k} = (-1)^j p_j$. In particular

$$c_N = b_N^* \mathcal{G}_N b_{N+1} \quad \text{satisfies} \quad \bar{c}_N = -c_N.$$

Write $a_N = b_N^* \mathcal{G}_N b_N$ and $d_N = b_{N+1}^* \mathcal{G}_N b_{N+1}$. The nonzero generalized control eigenvalues are the eigenvalues of

$$\frac{1}{\omega_N} \begin{pmatrix} \bar{c}_N & d_N \\ a_N & c_N \end{pmatrix}.$$

Therefore the exact least two-sided excess is

$$\epsilon_{h,k,N}^{\text{cyc}} = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}. \quad (4.12)$$

The determinant is nonnegative by the Gram inequality. Zero and rank-one degeneracies are covered by the same formula; in the reflection-stable case a nonzero rank-one signed defect cannot occur. No phase is dropped.

For the actual eigenvector at $k\rho$ constructed in Section 2,

$$k|2\Re\rho - 1| \leq \epsilon_{h,k,N}^{\text{cyc}}. \quad (4.13)$$

This is a proved diagnostic of this explicit family. A sublinear bound on the calculated right side as k grows is still to be established from the arithmetic input. The rank-two identity is not such a bound by itself.

S13.1.9 5. Keep packet changes through their actual source seeds

Let $h \mid H$ include complete multiplicities, and let $m = H/h$. Then

$$F_h = m(D)F_H, \quad \mathcal{V}_{h,k}P = \mathcal{T}_H^{(k)} \left(\prod_i m(s_i) P(S) \right). \quad (5.1)$$

The full CRT inclusion on jets is the $\mathbb{C}[S]$ -linear map which inserts zero jets at the added centres. Its old unit maps to the associated idempotent, so it is not described as a unital algebra inclusion into the entire larger packet.

The right side of (5.1) has the fixed seed $a_{hH}(\mathbf{s}) = \prod_i m(s_i)$. Its source norm is exactly the old norm, since $(g/H)m = g/h$ in every factor. It therefore defines an isometric map of the marked polynomial sources and their internal quotients, with the old cyclic annihilator $\chi_{h,k}$.

This is the correct packet-coherence comparison. The new seed is generally not a polynomial in S alone. For example, if $m(s) = s - 1/2$ and $k = 2$, then

$$m(s_1)m(s_2) = \frac{(S-1)^2 - (s_1 - s_2)^2}{4}. \quad (5.2)$$

The relative square and its coefficient are retained. The explicit source-seed map (5.1), followed by the larger packet's full relative source inclusion, replaces any unjustified identification with that larger packet's unmarked constant-seed cyclic submodule. No original packet, supported relation, or canonical scalar norm is changed by this marked comparison.

S13.1.10 6. The scalar Fisher estimate now acts on this source, and all polynomial derivative moments are calculable

From [S1],

$$\mathcal{I}_{h,k} + 4 \int \|n_{h,k}(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h, \quad \mathcal{I}_h = 4 \int |a'_h(t)|^2 dt. \quad (6.1)$$

The cyclic source (3.5) uses **that very density** $m_{h,k}$, not another scalar replacement for the full matrix norm. The full derivative still has the specified two components:

$$\partial_u(\Psi_{h,k}P(k/2 + iu)) = \Psi_{h,k} \left(iP'(k/2 + iu) + \frac{m'_{h,k}}{2m_{h,k}} P(k/2 + iu) \right) + n_{h,k}P(k/2 + iu). \quad (6.2)$$

They are pointwise orthogonal. The second component is retained even though the original coefficient polynomial depends only on the sum. Its arithmetic interpretation is the graph map and logarithmic observable from [S1]. The generator remains A_k ; the derivative corresponds to $i\mathcal{L}_k$, with $[D^{(k)}, \mathcal{L}_k] = -1$.

S13.1.11 Raw moments give the complete differentiated finite source without multidimensional quadrature

Define the finite moments

$$\mu_j = \int t^j |a_h(t)|^2 dt, \quad \nu_j = \int t^j |a'_h(t)|^2 dt, \quad \theta_j = \int t^j \overline{a_h(t)} a'_h(t) dt.$$

Fixed arithmetic phase and integration by parts give

$$\theta_0 = 0, \quad \theta_j = -\frac{j}{2} \mu_{j-1} \quad (j \geq 1). \quad (6.3)$$

Use the **formal** exponential generating series

$$\mathcal{M}(z) = \sum_{j \geq 0} \mu_j z^j / j!, \quad \mathcal{N}(z) = \sum_{j \geq 0} \nu_j z^j / j!.$$

No convergence of these series is needed for the finite coefficient identities. Their constant terms remain the actual mass and derivative energy.

Let

$$M_{j,k} = \int u^j m_{h,k}(u) du, \quad B_{j,k} = \int u^j \|\partial_u \Psi_{h,k}(u, \cdot)\|^2 du.$$

Theorem 6.1 (all polynomial derivative moments).

$$M_{j,k} = j! [z^j] \mathcal{M}(z)^k, \quad (6.4)$$

$$B_{j,k} = j! [z^j] \left[\frac{1}{k} \mathcal{N}(z) \mathcal{M}(z)^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}(z)^k \right]. \quad (6.5)$$

Proof. Expand $u^j = (t_1 + \cdots + t_k)^j$ and use Fubini; all integrals are absolutely finite. For (6.5), the diagonal terms of $|k^{-1} \sum_i a'_i \prod_{r \neq i} a_r|^2$ give $k^{-1} \mathcal{N} \mathcal{M}^{k-1}$. The distinct-index terms give

$(k-1)\Theta(z)^2\mathcal{M}^{k-2}/k$, where $\Theta(z) = \sum \theta_j z^j/j! = -z\mathcal{M}(z)/2$ by (6.3). Substitution gives the displayed expression. The second term is not discarded: the cross terms vanish only for the unweighted zeroth moment, not for arbitrary polynomial moments.

In particular,

$$\begin{aligned} M_{0,k} &= \mu_0^k, & M_{1,k} &= k\mu_1\mu_0^{k-1}, \\ M_{2,k} &= k\mu_2\mu_0^{k-1} + k(k-1)\mu_1^2\mu_0^{k-2}, \\ B_{0,k} &= \frac{1}{k}\nu_0\mu_0^{k-1} = \frac{\mu_h^{k-1}}{4k}\mathcal{I}_h. \end{aligned} \quad (6.6)$$

The first orthogonal norm and diagonal coefficient on the sum line are accordingly

$$\omega_0 = \mu_0^k, \quad \beta_0 = \frac{k}{2} + i\frac{k\mu_1}{\mu_0}, \quad \omega_1 = k\mu_0^{k-2}(\mu_0\mu_2 - \mu_1^2). \quad (6.7)$$

No symmetry assumption setting $\mu_1 = 0$ is made.

For any two polynomials P, Q , let $c_P(u) = P(k/2 + iu)$ and $c_Q(u) = Q(k/2 + iu)$. Their complete graph-energy form is

$$\begin{aligned} \mathfrak{D}_k(P, Q) &= \int \overline{c_P}c_Q \|\Psi'_k\|^2 du + \int \overline{c'_P}c'_Q m_k du \\ &\quad + \frac{1}{2} \int \left(\overline{c_P}c'_Q + \overline{c'_P}c_Q \right) m'_k du. \end{aligned} \quad (6.8)$$

All four terms remain. Equations (6.4)–(6.5) and $\frac{1}{2} \int u^j m'_k = -jM_{j-1,k}/2$ evaluate every matrix entry on a fixed polynomial degree using only μ_j, ν_j . This is the complete derivative graph norm from the existing sum-fibre construction, not merely the scalar score norm. It is positive definite on every nonzero polynomial source: a zero derivative would make $\Psi_k P$ constant in u for almost every relative coordinate, while its decay forces that constant and then P to be zero.

For a finite canonical representative, its graph-energy matrix is obtained by applying its actual coefficient matrix from (4.3) on both sides of (6.8). Thus there are now explicit moment matrices for both the original metric and the differentiated representative. No continuous jet functional on arbitrary L^2 classes is assumed.

S13.1.12 7. Exact sum-coordinate conormal depth at every original thickening

The derivative graph still needs the class **before** an original relation is sent to e . The required domains can now be computed exactly on the sum line.

For $r \geq 1$, retain the original thickened ring

$$B_{k,r} = \mathcal{P}_k/I_k^r.$$

Define, for an ordered root tuple $\rho = (\rho_{i_1}, \dots, \rho_{i_k})$,

$$\begin{aligned} L_{\rho,r} &= 1 + \sum_{j=1}^k (m_{i_j} - 1) + (r-1) \max_j m_{i_j}, \\ \ell_{\lambda,r} &= \max_{\rho: \sum \rho_{i_j} = \lambda} L_{\rho,r}, \end{aligned} \quad \boxed{\chi_{h,k,r}(S) = \prod_{\lambda \in \Lambda_k} (S - \lambda)^{\ell_{\lambda,r}}.} \quad (7.1)$$

Theorem 7.1. The exact contracted ideal is

$$\boxed{I_k^r \cap \mathbb{C}[S] = (\chi_{h,k,r}).} \quad (7.2)$$

Consequently there is an injective unital algebra map

$$\boxed{\alpha_r : C_{h,k,r} := \mathbb{C}[S]/(\chi_{h,k,r}) \hookrightarrow B_{k,r}^{S_k}, \quad [P] \mapsto [P(S)].} \quad (7.3)$$

Proof. At a root tuple, the units in $h(\rho_i + z_i) = u_i(z_i)z_i^{m_i}$ make the original local ideal equal to $(z_1^{m_1}, \dots, z_k^{m_k})$, with its r th power retained. A monomial z^α survives precisely when

$$\sum_i \left\lfloor \frac{\alpha_i}{m_i} \right\rfloor \leq r - 1. \quad (7.4)$$

The largest surviving total degree is $\sum_i (m_i - 1) + (r - 1) \max_i m_i$. It is attained by assigning all $r - 1$ full blocks to an index of maximal m_i , and taking remainders $m_i - 1$ in every index. The corresponding term of $(\sum z_i)^D$ has coefficient $D! / \prod \alpha_i! \neq 0$. All terms of degree $D + 1$ vanish. Thus the exact nilpotence index is (7.1). Take the maximum over collided sums and then the product over different sums. This gives precisely the annihilator of the algebra element S , proving (7.2). The same cyclic image is in the invariants because S and 1 are invariant. No local unit has been discarded: it enters the specified invertible change of generators of the ideal, not an alteration of the physical coordinate S .

S13.1.13 Quotient and derivative maps on the same tower

There are commuting quotient squares

$$\begin{array}{ccc} C_{h,k,r+1} & \xrightarrow{\alpha_{r+1}} & B_{k,r+1} \\ \downarrow & & \downarrow \\ C_{h,k,r} & \xrightarrow{\alpha_r} & B_{k,r}. \end{array} \quad (7.5)$$

Each quotient is a ring map, hence has the original split lift. Its kernel is retained. On the sum side,

$$0 \rightarrow \mathbb{C}[S]/(\chi_{r+1}/\chi_r) \xrightarrow{\times \chi_r} C_{r+1} \rightarrow C_r \rightarrow 0. \quad (7.6)$$

The first arrow is explicitly a module map. Under α_{r+1} its image lies in I_k^r/I_k^{r+1} ; this is the actual relation fibre sent to the supported zero by the right vertical quotient.

The exact exponent satisfies $\ell_{\lambda,r+1} \geq \ell_{\lambda,r} + 1$. Hence χ_r divides χ'_{r+1} , and the linear derivative map is well-defined:

$$\boxed{\partial_r : C_{r+1} \rightarrow C_r, \quad [P] \mapsto [P']}. \quad (7.7)$$

It commutes with the original relative-sum derivative

$$\delta_r : B_{k,r+1} \rightarrow B_{k,r}, \quad [P] \mapsto \left[\frac{1}{k} \sum_i \partial_{s_i} P \right],$$

$$\boxed{\delta_r \alpha_{r+1} = \alpha_r \partial_r.} \quad (7.8)$$

The derivatives are coefficient-linear maps and are lifted as such, not called ring homomorphisms. For example,

$$h(s) = (s - \rho)^2, \quad k = 2, \quad X = S - 2\rho \implies \chi_r = X^{2r+1}. \quad (7.9)$$

The class X^3 is nonzero modulo X^5 , maps to the supported zero modulo X^3 , and its derivative maps to $3X^2$, which is nonzero there.

This also calculates the relationship to a naive scalar power thickening:

$$\mathbb{C}[S]/(\chi_1^r) \twoheadrightarrow C_r \hookrightarrow B_{k,r}. \quad (7.10)$$

For the fixture in (7.9) at $r = 2$, the first map is $\mathbb{C}[X]/(X^6) \twoheadrightarrow \mathbb{C}[X]/(X^5)$, with kernel the retained class of X^5 . The original joint thickening therefore carries a different sum-coordinate depth, and the precise quotient above gives their relationship. This is a calculation about powers of the original relation ideal; it is not iteration of the scalar zero adjunction.

S13.1.14 The arithmetic Taylor unit remains through differentiation

Let $U_{k,r}$ be the full Taylor class of $\prod v_h(s_i)$ in $B_{k,r}$. It is invertible and compatible under the quotient tower. Define the coefficient-linear inclusion

$$\eta_r = M_{U_{k,r}} \alpha_r.$$

The full equation is

$$\boxed{\delta_r \eta_{r+1}(P) = U_{k,r} \alpha_r(P') + (\delta_r U_{k,r+1}) \alpha_r(P).} \quad (7.11)$$

The second term remains in the relative ring $B_{k,r}$. It is not presumed to be in the cyclic image. Its exact map is $M_{\delta_r U_{k,r+1}} \alpha_r$, and the complementary quotient $B_{k,r}/\eta_r C_r$ records its noncyclic part. This is the algebraic counterpart of the retained normal derivative in (6.2).

At $r = 1$, the quotient and derivative thus give two actual maps out of the same lifted thickening. A conormal relation can map to e under the quotient and to a nonzero supported value under the derivative. Differentiating the original theta relation, not the scalar e itself, is what produces that value.

S13.1.15 8. The conormal derivative reaches both cyclic and full arithmetic traces, with their difference retained

Choose an original division

$$\chi_1(S) = \sum_i h(s_i) B_i(\mathbf{s}).$$

Applying the actual derivative and then the original quotient yields

$$\boxed{\alpha_1(\chi_1') = \frac{1}{k} \sum_i h'(s_i) [B_i]_{I_k}.} \quad (8.1)$$

The right side is the source's conormal row. For the weighted relation $U \chi_1 P$, the term containing δU has an undifferentiated χ_1 and vanishes in B_k . Therefore

$$\boxed{\delta_1[U \chi_1 P]_{I_k^2} = U_k \alpha_1(\chi_1' P).} \quad (8.2)$$

It has the actual analytic representative

$$J^{(k)} \mathcal{L}_k \mathcal{V}_{h,k}(\chi_1 P) = U_k \alpha_1(\chi'_1 P), \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i. \quad (8.3)$$

Here the left source is an original theta boundary, supplied by (3.4). Its first differentiated arithmetic value is read before it is collapsed to its supported zero.

For $C_1 = \mathbb{C}[S]/(\chi_1)$, the explicitly chosen finite residue pairing is

$$R_\chi(f, P) = \sum_\lambda \text{Res}_{S=\lambda} \frac{f^{\dagger k}(S) P(S)}{\chi_1(S)} dS.$$

Its Jacobian contraction is

$$R_\chi(f, \chi'_1 P) = \sum_\lambda \ell_\lambda \overline{f(k - \bar{\lambda})} P(\lambda) = \text{Tr}(M_{f^{\dagger k} P} | C_1). \quad (8.4)$$

This is the cyclic trace. The original full symmetric trace has a_λ instead of ℓ_λ and is their sum with the quotient trace (2.7). Equivalently, let $\chi_{\text{full}}(S) = \prod_\lambda (S - \lambda)^{a_\lambda}$. Its difference is the explicit residue functional

$$\text{Tr}(f(A_k) | T_{h,k}) = \sum_\lambda \text{Res}_{S=\lambda} f(S) \left(\frac{\chi'_{\text{full}}}{\chi_{\text{full}}} - \frac{\chi'_1}{\chi_1} \right) dS. \quad (8.5)$$

There is a further explicit comparison for the original arithmetic vector pairing. Its pullback through the **weighted** inclusion is

$$\text{Tr}_{B_k^{\text{sym}}} M_{(\eta f)^{\dagger} \eta P} = \text{Tr}_{B_k^{\text{sym}}} M_{U_k^{\dagger} U_k \alpha_1(f^{\dagger k} P)}. \quad (8.6)$$

The factor $U_k^{\dagger} U_k$ remains. It need not lie in the sum-generated subalgebra. The trace decomposition (2.7) applies to the invariant operator $f(A_k)$; it is not an assertion that this weighted vector pairing is automatically the cyclic residue form (8.4).

Neither polynomial is substituted for the original $g = 2\xi$. Their finite trace comparison is (8.5), while $U_{k,r}$ and the map (8.3) retain their arithmetic source. The original product residue/Jacobian trace can still be evaluated on the full module and its retained quotient. No trace contribution is declared absent.

S13.1.16 9. Checks, counterfactual tests, and the next estimate

The exact checker implements finite polynomial, matrix, and Gaussian-amplitude fixtures. Its tests cover exact thickening dimensions and nilpotence; collision grouping; cyclic injections and trace complements; the non-naive thickening map; the derivative square; supported-zero versus absence; the original conormal row; full unit differentiation; scalar minimum and updates; rank-two control; reflection with a nonzero imaginary cross term; amplified off-line test exponents; raw convolution and complete polynomial derivative moments; and the marked source-seed transition. These are regression checks, not tests asserting that a fixture's roots are zeta zeros.

In particular an off-line reflected polynomial fixture remains off-line under the new construction: its exact eigenline identity is still $2\Re\lambda - k$. The scalar Fisher contraction does not annul this value. This guards against an inference from convolution smoothing alone to purity. The constructive object supplied instead is (4.12): two explicitly computed full-jet rows relative to the exact cyclic interpolation metric, with every scalar moment obtainable from the original one-factor moments by (6.4).

The degree cost $N \geq \deg \chi_{h,k} - 1$ and the reciprocal metric in (4.12) are retained as k grows. The derivative matrix (6.8) likewise keeps the polynomially weighted normal contribution. Controlling those quantities on the actual arithmetic inputs, rather than just the constant polynomial, is the remaining quantitative step of this branch.

The connection to the Deligne amplification programme is now a smaller, exactly included cohomological image with every amplified exponent represented; its source norm is the scalar sum-density norm for which (6.1) was proved. The original multi-direction module, its complementary trace, and all powers of its conormal ideal remain as explicit surrounding objects. This is not a transfer of finite-field purity by notation.

No remote repository is modified in this delivery. The user has assigned GitHub integration and Lean work to the parallel session. The authored note, checker, and a bounded handoff are ready for that session without changes to its existing files.

S13.1.17 Source record and references

- **[S1]** Owner programme, `Tau_Sum_Connection_Control/NOTE.tex` and `RESEARCH_NOTE.md`, archive `Tau_Sum_Connection_Control_2026-09-12.zip`. Selected analytic source, scalar information identity, full matrix connection, original first and higher conormal derivative. Exact hashes and source-manifest verification are in `checks/source_manifest.json` and `SOURCE_READING.json`.
- **[S2]** User upload `Untitled\1765(1).md`: retained progression from globalization to coherent interpolation, highest relation layer and spectral-sum descent. Read as a research transcript, not as an independent certificate for every embedded claim.
- **[S3]** User upload `Untitled\1766(1).md`: formalization progression, retained bounded scopes, and the kernel-layer integration. No new Lean run is claimed in this note.
- **[S4]** `SplitZero_Derived_Mathematics_2026-09-12(1).md` and retained source-derived interfaces: internal quotient, transported-class kernel and tensor base-change. Used within their stated hypotheses.
- **[E1]** NIST, *Digital Library of Mathematical Functions*, §18.2, particularly recurrence, Christoffel–Darboux, and moments. <https://dlmf.nist.gov/18.2>. General framework only; the arithmetic source maps and cyclic calculation are proved here.
- **[E2]** The Stacks Project, Tag 00S0, *The naive cotangent complex*. <https://stacks.math.columbia.edu/tag/00S0>. Background for the conormal map; (7.1)–(7.11) compute the exact present ideals and coordinate derivative.
- **[E3]** O. Johnson and A. Barron, *Fisher Information inequalities and the Central Limit Theorem*, arXiv:math/0111020. <https://arxiv.org/abs/math/0111020>. General convolution-projection mechanism; the mass-retaining arithmetic identity is the earlier source [S1].
- **[D]** P. Deligne, *La conjecture de Weil. II*, supplied TeX/source excerpts in the programme, especially §3.2.13 and §3.3.4–3.3.6. This turn uses the previously read amplification target; it does not claim to have reread or independently verified the whole paper.

S14 Exterior trace amplification with the original arithmetic Gram and full quartet multiplicities

Source attribution. Complete TeX witness supplied in the Exterior Trace Amplification archive. The original Gram, additive exterior action, factorial tensor inclusion, full generalized eigenspaces and quartet multiplicities are retained. The source supplies an amplification lower bound and explicitly leaves the actual subcubic arithmetic upper estimate unproved. Its finite fixtures and source-reported verification have their recorded scopes; the separate PR 21 formalization is identified independently.

Exact revision. c7050e1c2f76b31b72af1d237efd29f656fd34a92492e5a11f76d8bc55b2acd9

Source file. Tau_Exterior_Trace_Amplification/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Complete source witness SHA-256; cited repository main is a source baseline

Source SHA-256. c7050e1c2f76b31b72af1d237efd29f656fd34a92492e5a11f76d8bc55b2acd9

Original archive. Tau_Exterior_Trace_Amplification_2026-09-12.zip

Archive SHA-256. 168f977a9b54cf461f564e099fe55dc7b8feffbc09c563f541e0be2bb5746a94

S14.1 Result and scope

The previous cyclic-sum construction has a Hermitian control form of rank at most two. The continuation here uses that specific property, rather than changing its metric. Passing to an exterior power of the same arithmetic module does **not** multiply the control allowance by the exterior degree. It makes it possible to control a determinant line containing the sum of the positive real-part defects.

For the actual cyclic packet and its specified least-norm representative, the new inequality is

$$\sum_{\Re \lambda > k/2} \ell_{\lambda}(2\Re \lambda - k) \leq \epsilon_{h,k,N}^{\text{cyc}}.$$

For a complete nonreal off-line quartet with real displacement $\delta > 0$ and common multiplicity m , it gives the exact lower bound

$$\epsilon_{h,k,N}^{\text{cyc}} \geq 2\delta [1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad N \geq [1 + k(m-1)](k+1)^2 - 1.$$

For simple roots this grows as $\delta k^3/2$. The corresponding sufficient analytic target is therefore a **subcubic** upper bound on this same explicitly defined allowance, rather than the earlier sublinear target obtained from one repeated eigenvector. This note proves the lower bound and constructs the exterior/cohomological maps. It does not prove that analytic upper bound, and it does not claim RH.

The finite-dimensional extremal-trace ingredient is classical: it is a rank-two case of the Ky Fan principle. The contribution here is its application through the existing arithmetic representatives, their original theta boundaries, their split-support lifts, and the exact sum-set and multiplicity calculation. General finite-dimensional linear algebra is not presented as a new classical theorem.

S14.2 1. GitHub corrections integrated at a pinned revision

The repository read for this continuation is KokunoYumeto/zeta-function-research-reader, with pinned main revision

a4494fba4968db837a8af6fcb952c8688cd5d1da.

The inspection was bounded: the latest PR metadata, the main revision, the current sum-connection note, selected parts of the new Stieltjes/conormal continuation, and the formal frontier note. It was not an independent audit of the entire 292-page edition.

S14.2.1 1.1 The empty packet

The current PR #20 source explicitly corrects the scope of its monic-freeness argument. For a nonempty packet $d = \deg h \geq 1$, the original conormal calculation applies. For $h = 1$ and $k \geq 2$,

$$I = (h(s_1), \dots, h(s_k)) = \mathbb{C}[s_1, \dots, s_k], \quad \mathbb{C}[\mathbf{s}]/I^r = 0 \quad (r \geq 1).$$

The algebraic spectral module is then zero. Its split lift $G(0)$ still contains the external absence τ and the supported zero e . The analytic amplitude g and its mass do not vanish. No Gram inverse on a nonzero spectral space is inferred from that analytic amplitude.

Every inverse matrix below is on a **nonempty** actual spectral packet. The empty case has the unique maps between zero coefficient modules and the corresponding original split lifts.

The reviewed public note blob is 797d8bc298aec06b112b53a03654a9664323e748. Its historical validation file records an earlier note blob; that historical receipt is not relabelled as a new proof check.

S14.2.2 1.2 Metric restriction retains its hypotheses

The current formal frontier note requires, for rectangular inclusion and extraction matrices,

$$LI = 1, \quad P = IL = P^*, \quad PK = KP, \quad GK = 1$$

before concluding

$$(I^*GI)(LKL^*) = 1.$$

The spectral projection used later need not be orthogonal for the arithmetic metric. Accordingly this note uses the actual compressed Gram $C_+^*GC_+$ and its inverse. It does not replace that inverse by $C_+^*G^{-1}C_+$.

S14.2.3 1.3 Formalization and source boundaries

The source library already constructs the original internal quotient, cohomology maps, comparison kernels, support-changing morphisms, and homology of a cochain-idempotent image. These are inputs. This continuation neither reruns Lean nor extends the scope of their published certificates to the analytic theta construction.

The new tandem note also constructs the conormal algebra map into dual numbers and keeps the variable-frame connection term. Those results remain compatible inputs; they are not recounted as new results here.

S14.3 2. The original scalar and arithmetic source

Retain the original split scalar

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet.$$

Its original quotient and coefficient maps remain

$$\boxed{\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}}$$

Here j is the specified unital embedding. The target of $p\mathbb{Z}$ is infinite. Every lifted coefficient map sends e to e and τ to τ before the arithmetic observation.

Keep the source spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0 \right\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = H^1(C_+) = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Retain

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g,$$

$$\boxed{g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).}$$

These definitions and the continuous/finite source comparisons are inherited from the supplied cyclic-sum note. Their analytic proofs are not newly re-established in this note.

For a nonempty finite reflection-stable actual zero packet with complete orders, set

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h, \quad v_h = j_h(v_h) \in E_h^\times.$$

For $k \geq 1$, the full tensor arithmetic algebra, generator and unit are

$$B_k = E_h^{\otimes k}, \quad S = s_1 + \cdots + s_k, \quad A_k = M_S, \quad U_k = v_h^{\otimes k}.$$

At a sum λ , its maximal local nilpotent length is

$$\ell_\lambda = \max_{\rho_1 + \cdots + \rho_k = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) \right).$$

Use the original cyclic algebra and its arithmetic inclusion

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{\ell_\lambda}, \quad C = \mathbb{C}[S]/(\chi), \quad q_k = \deg \chi,$$

$$\alpha : C \hookrightarrow B_k^{S_k}, \quad [P] \mapsto P(A_k)1, \quad \eta = M_{U_k} \alpha : C \hookrightarrow B_k^{S_k}.$$

The first arrow is a unital algebra map. The map M_{U_k} and its inverse $M_{U_k}^{-1}$ are explicit module automorphisms. Thus η retains the arithmetic unit without declaring it to be a unital algebra inclusion.

The source already provides an equivariant retraction $\pi : B_k^{S_k} \rightarrow C$, $\pi\eta = 1$, and retains $\ker \pi$. Together with the full finite jet map and the original signed symmetrization this gives a left inverse to the cyclic inclusion in the top cohomology of $C_+^{\widehat{\otimes} k}$. No cyclic complement is discarded in this continuation.

S14.4 3. The existing canonical metric and its rank-two operator

The actual source map is

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = g/h.$$

It obeys

$$J^{(k)}\mathcal{V}_{h,k} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta\pi_\chi.$$

The norm remains

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du,$$

where

$$a_h(t) = \frac{(g/h)(1/2 + it)}{\sqrt{2\pi}}, \quad w_h = |a_h|^2, \quad \mu_h = \int_{\mathbb{R}} w_h, \quad m_{h,k} = w_h^{*k}.$$

The total mass is μ_h^k . Construct the monic orthogonal polynomials $p_j(S)$ by subtracting lower-degree projections from S^j ; retain their squared norms ω_j , including $\omega_0 = \mu_h^k$.

For $N \geq q_k - 1$, in the fixed remainder coordinates on C , put

$$b_j = [p_j]_\chi, \quad K = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad G = K^{-1},$$

$$Ru = \mathcal{V}_{h,k} \left(\sum_{j=0}^N p_j \frac{b_j^* G u}{\omega_j} \right).$$

These are the actual canonical representative and metric:

$$R^*R = G, \quad J^{(k)}R = \eta, \quad q^{(k)}R = \sigma_h^{\otimes k}\eta.$$

For $A = M_S$ on C , the source's exact boundary control is

$$W = A^*G + GA - kG = G \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N} G.$$

Introduce the metric raising map $G : C \rightarrow C^\#$, $v \mapsto (w \mapsto w^* G v)$, where $C^\#$ is the space of conjugate-linear functionals. Its inverse is K . Define the corresponding Gram adjoint and control endomorphism by the typed composites

$$A^{\sharp G} = G^{-1} A^* G : C \rightarrow C, \quad H = G^{-1} W = A^{\sharp G} + A - kI_C.$$

Then $H^{\sharp G} = H$ and $\text{rank } H \leq 2$. Reflection pairs λ with $k - \bar{\lambda}$, with equal full length. Hence

$$\text{Tr } H = 2\Re \text{Tr } A - kq_k = 0.$$

Thus the nonzero spectrum of H is $\{+\epsilon, -\epsilon\}$ for the actual nonnegative number

$$\epsilon = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}, \quad a_N = b_N^* G b_N, \quad d_N = b_{N+1}^* G b_{N+1}, \quad c_N = b_N^* G b_{N+1}.$$

The source reflection gives $\bar{c}_N = -c_N$. Equivalently $\epsilon^2 = \text{Tr}(H^2)/2$. The case $\epsilon = 0$ means $H = 0$ by its G -self-adjointness. No orthonormal replacement of the retained polynomial coordinates is made.

S14.5 4. Exterior control: the allowance does not multiply by exterior degree

Let $1 \leq p \leq q_k$. On $\bigwedge^p C$, define the **additive exterior generator**

$$A^{[p]}(v_1 \wedge \cdots \wedge v_p) = \sum_{j=1}^p v_1 \wedge \cdots \wedge A v_j \wedge \cdots \wedge v_p.$$

Its relation to the multiplicative exterior functor is

$$A^{[p]} = \left. \frac{d}{dt} \right|_{t=0} \bigwedge^p (e^{tA}), \quad \bigwedge^p (e^{tA}) = e^{tA^{[p]}}.$$

In the unscaled increasing-index wedge basis, the Gram matrix is

$$G_{I,J}^{[p]} = \det G_{I,J}.$$

It is positive definite and is the determinant inner product induced from the original G .

Theorem 4.1. The exact exterior control identity is

$$(A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} = G^{[p]} H^{[p]}.$$

For $1 \leq p \leq q_k - 1$,

$$-\epsilon G^{[p]} \preceq (A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} \preceq \epsilon G^{[p]}.$$

If $\epsilon > 0$, this is the exact optimal allowance. For $p = q_k$ the control form is zero.

Proof. The induced Gram adjoint satisfies

$$(A^{[p]})^{\sharp G^{[p]}} = (A^{\sharp G})^{[p]}.$$

This follows by applying the product rule to the determinant pairing of decomposable vectors. Since $(kI_C)^{[p]} = kpI$, add the two terms to obtain the identity.

For the order bound, keep the G -orthogonal eigenspace decomposition

$$C = L_+ \oplus Z_0 \oplus L_-, \quad H|_{L_+} = \epsilon, \quad H|_{Z_0} = 0, \quad H|_{L_-} = -\epsilon.$$

Both L_+ and L_- have dimension one, and $\dim Z_0 = q_k - 2$. The exterior product of each direct-summand inclusion gives the complete decomposition

$$\begin{aligned} \bigwedge^p C = \bigwedge^p Z_0 \oplus (L_+ \wedge \bigwedge^{p-1} Z_0) \\ \oplus (L_- \wedge \bigwedge^{p-1} Z_0) \oplus (L_+ \wedge L_- \wedge \bigwedge^{p-2} Z_0). \end{aligned}$$

The eigenvalues of $H^{[p]}$ on these four summands are respectively $0, \epsilon, -\epsilon, 0$. The multiplicity of each nonzero eigenvalue is $\binom{q_k-2}{p-1}$; summands with impossible degree are zero modules. This proves every assertion, including the top-degree case. When $\epsilon = 0$, the identity gives the conclusion directly. $\square$

This is the mechanism that makes a determinant-line estimate stronger than testing one eigenvector: a wedge cannot contain the same one-dimensional positive control direction twice.

S14.6 5. The exterior maps live on the original cochain source

Define the unscaled alternating map

$$\text{Alt}_p : \bigwedge^p C \hookrightarrow C^{\otimes p}, \quad v_1 \wedge \cdots \wedge v_p \mapsto \sum_{\pi \in S_p} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(p)}.$$

If $\text{wed} : C^{\otimes p} \rightarrow \bigwedge^p C$ is the exterior quotient, then

$$\text{wed} \text{Alt}_p = p! I.$$

The actual source representative is

$$R_p = R^{\otimes p} \text{Alt}_p : \bigwedge^p C \longrightarrow \mathcal{B}^{\widehat{\otimes} kp}.$$

It retains

$$R_p^* R_p = p! G^{[p]}, \quad J^{(kp)} R_p = \eta^{\otimes p} \text{Alt}_p,$$

$$q^{(kp)} R_p = (\sigma_h^{\otimes k} \eta)^{\otimes p} \text{Alt}_p.$$

The last map is injective: compose with the existing finite-jet/retraction map in each block, then with $\text{wed} / p!$. The factorial is part of this inverse composite; the representative itself is not divided by its norm.

The actual source control is therefore

$$W_p^{\text{source}} = p! \left((A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} \right).$$

Both its Gram and its control retain the same literal $p!$. Its relative allowance is the original ϵ .

S14.6.1 5.1 The cochain idempotent and both permutation signs

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$. Let T_π be the actual Koszul permutation on $\mathcal{C}_k^{\widehat{\otimes} p}$. Its top-degree action is $(\text{sgn } \pi)^k P_\pi$, with P_π ordinary block permutation. The chain idempotent for the exterior top-degree component is

$$A_{p,k} = \frac{1}{p!} \sum_{\pi \in S_p} (\text{sgn } \pi)^{k+1} T_\pi.$$

The two top-degree signs multiply to $\text{sgn } \pi$. Its square equals itself by the group-algebra calculation, with the factor $1/p!$ retained. The complementary idempotent $1 - A_{p,k}$ remains, with its own cohomology and trace. The existing projected-homology theorem supplies inclusion and retraction on the original cycle/boundary quotient.

These sign rules are the standard symmetric-monoidal structure on complexes, instantiated here on the retained arithmetic degree k ; they are not an assumption that degree-one vectors commute without sign.

S14.6.2 5.2 Propagate the actual theta primitive

The previous source construction supplies a map

$$K_N^{\text{prim}} : C \rightarrow \mathcal{C}_k^{k-1}, \quad D^{(k)} R - RA = dK_N^{\text{prim}}.$$

Define

$$K_{N,p}^{\text{prim}} = \sum_{j=1}^p (-1)^{k(j-1)} R^{\otimes(j-1)} \otimes K_N^{\text{prim}} \otimes R^{\otimes(p-j)} \text{Alt}_p.$$

The source and target are $\bigwedge^p C$ and $\mathcal{C}_k^{\widehat{\otimes} p, kp-1}$. Applying the tensor differential contributes the same $(-1)^{k(j-1)}$ on the displayed summand. Differentials on the top-degree R factors vanish. Consequently

$$D^{(kp)} R_p - R_p A^{[p]} = dK_{N,p}^{\text{prim}}.$$

This is the exact boundary producing the exterior control form on the original source.

S14.6.3 5.3 Original support and the alternative full-product minimum

Every coefficient map above has its original $G(\mathbb{C})$ -linear reconstructed lift. At a chosen active support label, a term with zero amplitude remains at that label's zero. The sums of differently located primitives use the joins of their actual tensor support masks; the character average uses the corresponding orbit join. External scalar τ maps every input to external absence, while e maps it to its fibre zero. In particular the boundary in the preceding equation is sent by the next internal quotient to the receiving supported zero, not to τ .

There is also an exact comparison with the earlier full-product minimum. Set

$$I_{k,p} = \eta^{\otimes p} \text{Alt}_p, \quad M \geq \max\{pN, kp(\deg h - 1)\}.$$

Let $R_{kp,M}^{\text{full}}$ be the already-constructed canonical full-jet representative. Both it and R_p lie in the stated degree- M source after applying $I_{k,p}$. Hence

$$\Delta = R_p - R_{kp,M}^{\text{full}} I_{k,p}$$

has zero full jets, and its numerator belongs to the original ideal $(h(s_1), \dots, h(s_{kp}))$. Thus it is an original theta boundary with the source's polynomial-division primitive. Orthogonality of the full minimum gives

$$p!G^{[p]} = I_{k,p}^* G_{kp,M}^{\text{full}} I_{k,p} + \Delta^* \Delta.$$

The complete control comparison is

$$\begin{aligned} W_p^{\text{source}} &= I_{k,p}^* W_{kp,M}^{\text{full}} I_{k,p} \\ &\quad + (A^{[p]})^* \Delta^* \Delta + \Delta^* \Delta A^{[p]} - kp \Delta^* \Delta. \end{aligned}$$

This connects the two representative choices without identifying their metrics or discarding their boundary difference.

S14.7 6. A determinant line measures the sum of the positive defects

Let

$$C_{>} = \bigoplus_{\Re \lambda > k/2} C_{\lambda} \subseteq C$$

be the sum of the **full generalized eigenspaces**, of total dimension p . Here $C_{\lambda} \cong \mathbb{C}[S]/(S - \lambda)^{\ell_{\lambda}}$. Write its inclusion as $I_{>} : C_{>} \hookrightarrow C$. Its determinant line has the actual injection

$$\det(C_{>}) = \bigwedge^p C_{>} \xrightarrow{\bigwedge^p I_{>}} \bigwedge^p C \xrightarrow{R_p} \mathcal{B}^{\widehat{\otimes} kp}.$$

The additive exterior generator acts on this line by

$$\zeta_{>} = \text{Tr}(A|_{C_{>}}) = \sum_{\Re \lambda > k/2} \ell_{\lambda} \lambda.$$

Indeed expansion of A on a wedge of a basis keeps only its diagonal coefficients: every off-diagonal replacement repeats one basis vector and has zero wedge. The full nilpotent modules are retained before this determinant map; their dimensions appear in the trace. No simplicity assertion follows: a critical-line Jordan block can have nonzero numerical control while contributing zero real-part defect.

Define

$$L_{h,k} := 2\Re \zeta_{>} - kp = \sum_{\Re \lambda > k/2} \ell_{\lambda} (2\Re \lambda - k).$$

Theorem 6.1. For every admitted N ,

$$0 \leq L_{h,k} \leq \epsilon_{h,k,N}^{\text{cyc}}.$$

Proof. If $p = 0$, the left side is zero. Otherwise reflection implies $p < q_k$. Let v be the nonzero determinant vector of an unscaled basis of $C_{>}$. On this actual eigenline,

$$\frac{v^* W_p^{\text{source}} v}{v^* (p!G^{[p]}) v} = 2\Re \zeta_{>} - kp = L_{h,k}.$$

Theorem 4.1 bounds the same quotient by ϵ . $\square$

The old eigenline argument gave the maximum individual defect. This theorem controls their entire positive sum, with their full lengths. Reflection also gives

$$L_{h,k} = \frac{1}{2} \sum_{\lambda} \ell_{\lambda} |2\Re \lambda - k| = \sum_{\lambda} \ell_{\lambda} |\Re \lambda - k/2|.$$

S14.7.1 6.2 The exact arithmetic volume equation

Let $B_{>} : \mathbb{C}^p \rightarrow C$ be the column map of a fixed unscaled basis of $C_{>}$, and set $G_{>} = B_{>}^* G B_{>}$. Its determinant vector $v_{>}$ has squared source norm

$$\|R_p v_{>}\|^2 = p! \det G_{>}.$$

For the original parameter $a > 0$, let $U_{>}(a) = a^{A|C_{>}}$. Then

$$\det(U_{>}(a)^* G_{>} U_{>}(a)) = a^{kp+L_{h,k}} \det G_{>}.$$

The entire exterior control integrates for $a \geq 1$ to

$$a^{kp-\epsilon} p! \det G_{>} \leq \|R_p \bigwedge^p U_{>}(a) v_{>}\|^2 \leq a^{kp+\epsilon} p! \det G_{>}.$$

Differentiating the norm along $a = e^t$ and applying the exact quadratic inequality proves the bound. Comparing with the exact determinant equality again gives $L_{h,k} \leq \epsilon$. The determinant, its norm, and its weight are all produced by maps from the original arithmetic module.

S14.8 7. Spectral and metric projections are linked explicitly

The determinant argument has an equivalent trace calculation useful for computation. Let $Q_{>} = e_{>}(A)$ be the actual CRT spectral idempotent, where $e_{>}$ is one modulo each positive-defect factor and zero modulo every other factor. It commutes with A .

Choose the original CRT column map $B_{>} : \mathbb{C}^p \rightarrow C$, including every jet. Its G -orthogonal projection is

$$P_{>} = B_{>} (B_{>}^* G B_{>})^{-1} B_{>}^* G.$$

Their exact relationship is

$$Q_{>} P_{>} = P_{>}, \quad P_{>} Q_{>} = Q_{>}, \quad P_{>}^{\sharp G} = P_{>}, \quad (P_{>} - Q_{>})^2 = 0.$$

The first two identities follow from their common range and identity restrictions there. Expanding the square gives the last identity. Invariance of that range gives $P_{>} A P_{>} = A P_{>}$ and

$$\mathrm{Tr}(P_{>} A) = \mathrm{Tr}(Q_{>} A) = \zeta_{>}.$$

Since $\mathrm{Tr}(P_{>} A^{\sharp G})$ is the conjugate of $\mathrm{Tr}(P_{>} A)$,

$$\mathrm{Tr}(P_{>} H) = L_{h,k}.$$

For $\epsilon > 0$, the two G -orthogonal rank-one control projections are the exact polynomials

$$F_+ = \frac{H^2 + \epsilon H}{2\epsilon^2}, \quad F_- = \frac{H^2 - \epsilon H}{2\epsilon^2}.$$

They give the full slack formula

$$L_{h,k} = \epsilon(1 - \eta_{h,k,N}), \quad \eta_{h,k,N} = 1 - \text{Tr}(P_{>}F_+) + \text{Tr}(P_{>}F_-) \geq 0.$$

With $\|T\|_{\text{HS},G}^2 = \text{Tr}(T^\sharp T)$, this is

$$\eta_{h,k,N} = \|(1 - P_{>})F_+\|_{\text{HS},G}^2 + \|P_{>}F_-\|_{\text{HS},G}^2.$$

Thus the gap has a specified projection meaning rather than an omitted term. This is the elementary rank-two instance of the classical extremal-trace principle.

The retained finite trace has its resolvent description as well:

$$\zeta_{>} = \frac{1}{2\pi i} \int_{\Gamma_{>}} S \text{Tr}((SI - A)^{-1}) dS,$$

where the positively oriented contour encloses exactly the indicated finite blocks. The complementary trace remains

$$\text{Tr } f(A) = \text{Tr}(Q_{>}f(A)) + \text{Tr}((1 - Q_{>})f(A)).$$

Multiplication by $(-1)^k$ gives the inherited cohomological sign. The determinant line occurs in degree kp , retaining $(-1)^{kp}$ for its trace. Split lifts of either projector send an existing class with zero projected amplitude to a supported zero, while the paired projector retains the complementary component.

S14.9 8. Exact sum counting for a nonreal quartet

Take a complete actual nontrivial quartet

$$\rho_{\varepsilon,\eta} = \frac{1}{2} + \varepsilon\delta + i\eta\gamma, \quad \varepsilon, \eta \in \{+1, -1\}, \quad \delta > 0, \quad \gamma > 0,$$

with common full multiplicity m . This is a specification for the following theorem, not an assertion that an off-line zero has been found. The two standard symmetries of $g = 2\xi$ give the full quartet and equal orders from any such actual zero.

The distinct k -fold sums are precisely

$$\lambda_{a,b} = \frac{k}{2} + \delta(2a - k) + i\gamma(2b - k), \quad 0 \leq a, b \leq k.$$

Proof of existence. Let a count positive real signs and b positive imaginary signs. Choose an integer t with

$$\max(0, a + b - k) \leq t \leq \min(a, b).$$

The four occupation counts

$$(t, a - t, b - t, k - a - b + t)$$

are nonnegative and sum to k , with the required two margins. Conversely every tuple has such margins. Since δ and γ are nonzero real numbers, equality of two sums forces equality of both margins. No hypothesis of algebraic independence of zero ordinates is used.

All ordered tuple blocks have the same maximal length

$$\ell_k = 1 + k(m - 1).$$

The nilpotency proof is the original one: the top monomial in $(z_1 + \cdots + z_k)^{k(m-1)}$ has coefficient $(k(m-1))!/((m-1)!)^k \neq 0$. It survives before the next power vanishes. Hence

$$\chi_{h,k}(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^{\ell_k}, \quad q_k = \ell_k(k+1)^2.$$

For the positive-defect subspace,

$$p = \ell_k(k+1) \left\lceil \frac{k}{2} \right\rceil.$$

Summing its real defects gives

$$\begin{aligned} L_{h,k} &= 2\delta\ell_k(k+1) \sum_{a=\lfloor k/2 \rfloor + 1}^k (2a - k) \\ &= 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \end{aligned}$$

For $k = 2n$, the last sum is $n(n+1)$; for $k = 2n+1$, it is $(n+1)^2$. The imaginary parts cancel in the determinant trace because $\sum_{b=0}^k (2b - k) = 0$.

Combining this calculation with Theorem 6.1 proves

$$2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}^{\text{cyc}}, \quad N \geq [1 + k(m-1)](k+1)^2 - 1.$$

Every multiplicity is retained. For simple roots the leading term is $\delta k^3/2$; for $m > 1$ it is $\delta(m-1)k^4/2$.

For the two-point reflected pair alone, the analogous exact lower bound is

$$\epsilon_{h,k,N}^{\text{cyc}} \geq 2\delta[1 + k(m-1)] \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

The quartet supplies the additional, rigorously counted factor $k+1$ through its distinct imaginary sums.

S14.10 9. The quantitative target after this construction

The current source formula is still

$$\epsilon_{h,k,N}^{\text{cyc}} = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}.$$

A contradiction to any actual off-line quartet now follows from a bound

$$\frac{\sqrt{a_N d_N - |c_N|^2}}{k^3 \omega_N} \rightarrow 0$$

along a permitted choice $N = N(k) \geq [1 + k(m-1)](k+1)^2 - 1$, proved for its actual arithmetic measure and complete jet map. For this contradiction the packet h is fixed: constants may depend

on h , and uniformity over all zero locations is not required. The expression itself has not been shown to tend to zero here. The result is a stronger amplification theorem and a weaker sufficient growth target for the next analytic estimate.

For a simple quartet, the source degree is at least quadratic in k . This cost is retained; a fixed-degree central-limit estimate cannot simply be inserted at that growing degree. The previously proved scalar Fisher contraction, with its factor μ_h^{k-1}/k , is not assigned to every polynomial moment.

The existing polynomial derivative formulas supply those actual moments:

$$\begin{aligned}\mathcal{M}_h(z) &= \sum_{j \geq 0} \mu_j \frac{z^j}{j!}, & \mathcal{N}_h(z) &= \sum_{j \geq 0} \nu_j \frac{z^j}{j!}, \\ \int u^j m_{h,k}(u) du &= j! [z^j] \mathcal{M}_h(z)^k, \\ \int u^j \|\partial_u \Psi_{h,k}\|^2 du &= j! [z^j] \left[\frac{1}{k} \mathcal{N}_h \mathcal{M}_h^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}_h^k \right].\end{aligned}$$

All masses, polynomial degrees and cross terms remain. The new determinant bound uses their already-defined Gram and control matrices rather than changing the source in order to obtain a favourable value.

The real interval $0 < s < 1$ causes no missing quartet case: the alternating zeta series there is positive when grouped into consecutive pairs, while $1 - 2^{1-s} < 0$, so $\zeta(s) < 0$. An actual non-trivial off-line zero is therefore nonreal and is covered by its complete quartet. This argument only identifies the case split; it supplies no upper estimate.

S14.11 10. Deligne architecture and the boundary retained in this step

The inherited Weil-II reading identifies an amplification pattern: carry the same arithmetic spectral information into a cohomological tensor image, obtain a further estimate there, and transport dual control without changing the representation. The finite-field geometric theorem is not being applied on a renamed base.

Here the new characteristic-zero operation is

$$\begin{aligned}& \text{original theta complex over } \mathfrak{b}_\tau \\ & \downarrow \text{existing } k\text{-fold cyclic sum image} \\ & (C, A, R_N, G_N, W_N) \\ & \downarrow \text{signed exterior cochain map} \\ & \left(\bigwedge^p C, A^{[p]}, R_p, p!G^{[p]} \right) \\ & \uparrow \det(C_{>}) \\ & \text{exact sum of positive arithmetic defects.}\end{aligned}$$

The exterior operation controls that determinant line with the **same** allowance as the cyclic image. The original e -relation primitives are carried by $K_{N,p}^{\text{prim}}$; the scalar τ remains external absence. The spectral and metric projectors, their difference, the cyclic complement, the exterior complement and their finite traces are all explicit.

No direct implication from the cardinality of the infinite quotient to positivity is used. Its role is to remain the structural arithmetic quotient throughout the construction, not to be replaced by a finite-field scalar or by the Boolean support observation.

S14.12 11. Verification and reproducibility

The supplied cyclic-sum archive was safely extracted and all 42 entries of its SHA-256 manifest verified. Its 24-method checker was rerun normally and under Python optimization, with matching success records. The new checker contains 22 exact finite test methods; its normal, optimized and deliberately failing results are recorded in checks/.

The Gaussian calculations in the checker are declared algebraic calibrations. Their literal measure is

$$\frac{7^k}{\sqrt{2\pi k}} e^{-u^2/(2k)} du.$$

They test companion matrices, full repeated-root modules, the canonical inverse-moment Gram, exterior factorials, the additive generator, the trace bound and the quartet counting. The mass is 7^k , not one. These examples do not claim that the calibration roots are zeta zeros.

The written arguments are general. Finite tests are regression evidence, not a Lean certificate, interval arithmetic for the actual spectral integrals, or independent certification of the whole analytic source corpus. No repository or other-session branch is changed by this continuation.

S14.13 References and provenance

1. Owner programme, **Tau Cyclic Sum Control**, delivered NOTE.tex, 12 September 2026. The original packet, norm, canonical section, rank-two identity, reflection and thickened relation maps are source inputs. Archive and member hashes are in checks/source_manifest.json.
2. Owner repository, pinned main a4494fba4968db837a8af6fcb952c8688cd5d1da. Reviewed workbenches/tau-sum-connection/RESEARCH_NOTE.md, current blob 797d8bc298aec06b112b53a03654a9664323e748; workbenches/tau-frontier-formal/RESEARCH_NOTE.md; selected sum_connection_stieltjes.tex and conormal_finite_weyl.tex. Exact reading scope is in SOURCE_REVIEW.md.
3. Original SplitZero derived/reconstruction note and the actual universal property of its internal quotient; retained in the owner's supplied formalization material.
4. Overton, M. L., and Womersley, R. S. (1992), **On the Sum of the Largest Eigenvalues of a Symmetric Matrix**, SIAM Journal on Matrix Analysis and Applications 13(1), 41–45, doi:10.1137/0613006. Used only as attribution for the classical extremal-trace ingredient. The rank-two proof used here is fully written above.
5. The Stacks Project, Tag **OGWN**, tensor products of complexes. Used for the standard tensor differential and Koszul permutation types; the specific arithmetic signs and primitive are calculated above.
6. NIST DLMF §25.4, equations 25.4.3–25.4.4, the functional equation and the literal definition of ξ . The original arithmetic function in this program remains 2ξ .
7. Deligne, P. (1980), **La conjecture de Weil II**, Publications Mathématiques de l'IHÉS 52, 137–252. The supplied TeX/reading record at §3.2.13 motivates the amplification architecture. This note does not claim a new full reading or a finite-field purity theorem on the τ -base.

S15 Source and relation volume dynamics for the original arithmetic control

Source attribution. Complete source note from the original Toda Volume Control archive. The source and relation Hankel determinants, tilt derivatives, imaginary cross term, first-degree endpoint, both loss squares and original theta input are retained. Its endpoint proof uses a vector symbol for a scalar norm; the explicit typed correction is proved in the integrated Toda bridge, while these original source bytes remain unchanged. The delivered floating evaluations are historical computations without interval or tail certification. The new local theta-tail proof and its separately scoped interval certificate supply that additional evidence for the analytic seed. Neither source asserts the growing-degree arithmetic estimate, and no local Lean execution is attributed to this edition.

Exact revision. 027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375

Source file. Tau_Toda_Volume_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Complete delivered source witness SHA-256; no new remote revision is assigned

Source SHA-256. 027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375

Original archive. Tau_Toda_Volume_Control_2026-09-12.zip

Archive SHA-256. 80b365233ba5ebaf1b52aeea369a7771d74bf774dbb4416f10a56503321b8f41

S15.1 Results and scope

The preceding exterior construction proves an aggregate lower bound on the canonical arithmetic control. This continuation constructs a scalar route to its upper estimate without selecting another metric on the arithmetic quotient.

The actual canonical quotient determinant is

$$V_N(\theta) = \det G_N(\theta) = \frac{\mathfrak{D}_{N+1}(\theta)}{\mathfrak{B}_{N-q+1}(\theta)}. \quad (1)$$

The numerator is the Gram determinant of the original polynomial source. The denominator is the Gram determinant of the original theta relations in that source. Both determinants come from one retained arithmetic Laplace transform and an explicitly given differential filter. Both obey the classical Hankel-Toda identity. No new general integrable-system theorem is claimed.

For $N \geq q$, put

$$v_N = \log \frac{V_{N-1}}{V_N}, \quad a_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \quad \ell_N = \log V_N, \quad \sigma = \frac{\text{Tr } A - (k/2)q}{i} \in \mathbb{R}.$$

The following is an exact formula, at every real tilt in the stated domain:

$$\epsilon_N^2 = a_{N+1}(1 - e^{-v_N})(e^{v_{N+1}} - 1) - (\sigma - \partial_\theta \ell_N)^2. \quad (2)$$

At $\theta = 0$, ϵ_N is the original allowance in the exterior-amplification note. The new formula retains the cross-term phase through a derivative of a scalar determinant. It is not an estimate that forgets that term.

In particular, with the actual two-step volume ratio

$$\mathcal{R}_N = \frac{V_{N-1}}{V_{N+1}},$$

one obtains the proved finite upper bound

$$\epsilon_N \leq \sqrt{a_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}. \quad (3)$$

Both nonnegative losses in this upper estimate are computed below. The previous aggregate spectral lower bound therefore forces an explicit minimum two-step contraction of the source/boundary volume ratio. This turns the remaining asymptotic question into a comparison between two specified scalar determinant flows. It does not prove that this comparison has the required subcubic bound.

S15.2 1. Integration of the new formalization report

The supplied report identifies verified source commit 2abc351424ba87aeda948a5cfb846e15ed9373d1 in draft PR #21. The current PR metadata and its complete mathematical note were read. The report's successful execution is used with its stated scope; this session did not run Lean or replay the workflow log.

The checked coefficient interfaces are the original quotient derivative

$$\delta_r : A_0/I^{r+1} \longrightarrow A_0/I^r,$$

its square with quotient transitions, the cyclic injection obtained from the **literal** pullback

$$\mathcal{K}_r = \{P : P(S) \in I^r\},$$

and the full multiplier identity

$$\delta_r([uP(S)]) = [uP'(S)] + [(\partial u)P(S)].$$

The report separately identifies the local quotient-basis and CRT interpretation as written proofs. That separation is maintained. Its relation-depth maximum includes attainment and permits the tuple maximizing a collided sum to change with depth.

Here the scalar relation determinant uses the first cyclic relation $\chi = \chi_{h,k,1}$. The weight $|\chi|^2$ is the **squared Hilbert norm of the multiplication map** M_χ ; it is not a declaration that the next contracted ideal is (χ^2) . The map used is displayed in (11). At higher original depth the relevant polynomial remains $\chi_{h,k,r}$, through the checked pullback $\mathcal{K}_r$, not an assumed power of χ .

The empty-packet correction is also kept. For $h = 1$, I is the whole coefficient ring and every positive-depth quotient is zero. The analytic seed g and its mass remain. The split lift $G(0)$ retains two elements. In that case the determinant quotient below is the determinant of the zero-dimensional module, namely 1, and no positive-dimensional inverse or spectral estimate is inferred.

S15.3 2. Original base, coefficient square, and arithmetic source

Keep

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the original square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (4)$$

The target of $p\mathbb{Z}$ remains infinite. The base remains $\mathfrak{b}_\tau$. Every lifted coefficient map fixes external absence and carries represented zero to represented zero.

The analytic source remains

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = \int \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \quad \forall b \in \mathbb{Z}, j \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Let h be a nonempty reflection-stable packet of actual zeros with complete orders. Set

$$E_h = \mathbb{C}[s]/h, \quad v_h = g/h, \quad v_h = j_h(v_h) \in E_h^\times, \quad \mathcal{M}F_h = v_h.$$

Fix tensor degree $k \geq 1$. Use the cyclic algebra already constructed:

$$C = \mathbb{C}[S]/\chi, \quad q = \deg \chi \geq 1, \quad c = k/2, \quad A = M_S,$$

$$\eta : C \hookrightarrow (E_h^{\otimes k})^{S_k}, \quad \eta[P] = v_h^{\otimes k} P \left(\sum_i M_{s_i} \right) 1.$$

The source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}).$$

It retains

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi. \quad (5)$$

Here $q^{(k)}$ denotes the original top cohomology quotient; $q = \deg \chi$ denotes a dimension. The full arithmetic unit is still in η .

The scalar norm is

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c+iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad w_h(t) = \left| \frac{v_h(1/2+it)}{\sqrt{2\pi}} \right|^2. \quad (6)$$

Its mass is μ_h^k , with $\mu_h = \int w_h$. No measure is divided by that mass.

S15.4 3. One arithmetic function generates both determinant flows

S15.4.1 3.1 Analytic domain

Define

$$M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad Z_{h,k}(\theta) = M_h(\theta)^k. \quad (7)$$

These are analytic on $|\Re \theta| < \pi/2$. Here and below positivity statements concern real θ in that interval.

A direct bound suffices. At $\Re s = 1/2$, bounded partial sums of the alternating series and summation by parts give $|\eta(s)| \leq C(1 + |s|)$. The denominator $|1 - 2^{1-s}|$ is bounded below by $\sqrt{2} - 1$. Stirling's vertical-strip estimate then gives, beyond a fixed compact set,

$$w_h(t) \leq C_h(1 + |t|)^{11/2 - 2 \deg h} e^{-\pi|t|/2}.$$

On the compact set, g/h is entire because the selected zero orders are complete. Thus every differentiated integral is locally uniformly dominated on the displayed strip. This is a sufficient domain, not a claim that it is the maximal analytic continuation of M_h .

Fubini and the exact sum coordinate give

$$Z_{h,k}(\theta) = \int e^{\theta u} m_k(u) du, \quad Z_{h,k}^{(j)}(\theta) = \int u^j e^{\theta u} m_k(u) du. \quad (8)$$

In particular $Z_{h,k}(0) = \mu_h^k$.

S15.4.2 3.2 The source and the relation filter

Let $\bar{\chi}$ mean coefficientwise conjugation, and define the differential operator and its output

$$\mathcal{F}_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta), \quad Z_\chi(\theta) = \mathcal{F}_\chi Z_{h,k}(\theta). \quad (9)$$

Differentiation under the integral proves

$$\boxed{Z_\chi(\theta) = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0.} \quad (10)$$

Both factors, their signs, and the coefficient $c = k/2$ remain. This filter is multiplication by a source polynomial under the Laplace transform. It is not the quotient-level derivation δ_r .

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$ and $N \geq q - 1$, the actual source sequence is

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad (11)$$

with $\mathcal{P}_{-1} = 0$. Applying $\mathcal{V}$ identifies its first image with the original admitted theta boundaries.

The Laplace observation gives the typed identity

$$\mathcal{L}_\theta(SP) = (c + i\partial_\theta)\mathcal{L}_\theta(P), \quad \mathcal{L}_\theta(P) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Equation (10) is the Gram observation of M_χ . A source relation has nonzero positive norm before the quotient sends it to a fibre zero.

S15.4.3 3.3 The two determinants

Put

$$\mathfrak{D}_n(\theta) = \det[Z_{h,k}^{(i+j)}(\theta)]_{i,j=0}^{n-1}, \quad \mathfrak{B}_n(\theta) = \det[Z_\chi^{(i+j)}(\theta)]_{i,j=0}^{n-1}, \quad (12)$$

with the actual empty determinants $\mathfrak{D}_0 = \mathfrak{B}_0 = 1$. Every positive-order determinant is positive. A zero norm polynomial would vanish on a full-measure subset of the real line.

The use of u -moments is linked to the original S -basis by

$$S^j = (c + iu)^j = \sum_{r=0}^j \binom{j}{r} c^{j-r} i^r u^r.$$

This triangular coordinate map has determinant $i^{n(n-1)/2}$ on dimension n . Its Gram determinant multiplier is exactly 1, because that determinant has modulus one. No polynomial coefficient or factorial has been dropped.

The determinant has the exact integral representation

$$\mathfrak{D}_n(\theta) = \frac{1}{n!} \int_{\mathbb{R}^n} \prod_{i < j} (u_i - u_j)^2 \prod_{j=1}^n e^{\theta u_j} m_k(u_j) d^n u. \quad (13)$$

For $\mathfrak{B}_n$, insert $\prod_j |\chi(c + iu_j)|^2$. Expanding both Vandermonde determinants and integrating proves the formula, including $1/n!$. The total mass in the first expression scales as $(\mu_h^k)^n$, not as one.

S15.5 4. The quotient determinant is the source/boundary volume ratio

At real θ equip the same polynomial source with

$$\langle P, Q \rangle_\theta = \int \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (14)$$

Let $H_\theta = L^2(\mathbb{R}, e^{\theta u} m_k(u) du)$. The map $f \mapsto e^{\theta u/2} f$ is an isometry $H_\theta \rightarrow H_0$ with inverse $f \mapsto e^{-\theta u/2} f$. The source in each space is exactly the image of $\mathcal{P}_N$ under evaluation. This is an observation of the original source, not an enlargement of its admitted polynomials.

Let J_N be its quotient matrix and $M_N(\theta)$ its source Gram. Define

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad \hat{R}_N = M_N^{-1} J_N^* G_N, \quad R_N = \mathcal{V} \hat{R}_N. \quad (15)$$

The adjoints in this calculation use (14). At $\theta = 0$ these are the preceding canonical arithmetic representatives and their original norm. At every θ ,

$$J^{(k)} R_N = \eta, \quad q^{(k)} R_N = \sigma_h^{\otimes k} \eta.$$

Use the source basis

$$1, S, \dots, S^{q-1}, \quad \chi, \chi S, \dots, \chi S^{N-q}.$$

Its change from $1, S, \dots, S^N$ is triangular with diagonal one because the original χ is monic. The Schur complement of its boundary Gram is G_N . This proves

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (16)$$

Equivalently, the determinant line of (11) is mapped by the wedge of the **specified** kernel columns and quotient lifts. Orthogonal quotient representatives supply the Schur complement. This is the exact determinant-line map behind the ratio.

The original support lift is

$$(\lambda_N, P) \mapsto (\lambda_N, [P]_\chi), \quad (\lambda_N, \chi Q) \mapsto (\lambda_N, 0), \quad \tau \mapsto \tau. \quad (17)$$

The determinant $\mathfrak{B}_{N-q+1}$ retains the norm-volume of those nonzero source relations. Replacing that source by its quotient value would incorrectly replace the positive denominator by information-free zeros. The construction evaluates its Gram before applying (17).

For $h = 1$, $\chi = 1$ and $C = 0$. Then $Z_\chi = Z_{h,k}$ and (16) is 1 identically. This is the empty determinant, not an inverse on a nonzero spectral module.

S15.6 5. The two classical Toda identities have their actual arithmetic input

The source determinants satisfy

$$\mathfrak{D}_n \mathfrak{D}_n'' - (\mathfrak{D}_n')^2 = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1}, \quad n \geq 1. \quad (18)$$

The relation determinants satisfy the same equation with $\mathfrak{D}$ replaced by $\mathfrak{B}$.

Here is a proof with the constants retained. For a real positive weight $e^{\theta u} d\mu(u)$, let Q_j be the monic orthogonal polynomial obtained by subtracting lower-degree projections, with norm h_j . Multiplication has the recurrence

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1}, \quad a_0 = 0,$$

The variable remains u . Differentiating orthogonality gives

$$\partial_\theta Q_j = -a_j Q_{j-1}, \quad \partial_\theta \log h_j = b_j, \quad b_j' = a_{j+1} - a_j.$$

For the last identity, differentiate $h_j b_j = \int u Q_j^2 e^{\theta u} d\mu$, substitute the derivative of Q_j , and use the recurrence. Since $\mathfrak{D}_n = \prod_{j < n} h_j$, the sum telescopes:

$$\partial_\theta^2 \log \mathfrak{D}_n = a_n = \frac{\mathfrak{D}_{n+1} \mathfrak{D}_{n-1}}{\mathfrak{D}_n^2}.$$

This proves (18). The relation weight is the same positive weight multiplied by $|\chi(c+iu)|^2$, which is independent of θ , so the proof applies without alteration to its determinants.

The original S -monic polynomial norms are exactly

$$\omega_j = \frac{\mathfrak{D}_{j+1}}{\mathfrak{D}_j}, \quad \nu_j = \frac{\mathfrak{B}_{j+1}}{\mathfrak{B}_j}. \quad (19)$$

The phases linking monic polynomials in S and u are i^j and i^{-j} ; their norms agree because the phase has modulus one.

For $m = N - q + 1 \geq 0$ this gives the exact quotient-volume curvature

$$\partial_\theta^2 \log V_N = \frac{\mathfrak{D}_{N+2} \mathfrak{D}_N}{\mathfrak{D}_{N+1}^2} - \frac{\mathfrak{B}_{m+1} \mathfrak{B}_{m-1}}{\mathfrak{B}_m^2}. \quad (20)$$

The second term is defined to be zero for $m = 0$, because $\log \mathfrak{B}_0 = 0$. It is not a quotient involving $\mathfrak{B}_{-1}$.

S15.6.1 5.1 The curvature comes from two actual source maps

Let X be multiplication by u on the polynomial realization in H_θ . Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ project to the original boundary subspace $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$. Write R for its least-norm polynomial lift.

Differentiating the quotient condition and its orthogonality proves

$$\partial_\theta R = -P_{\mathcal{D}_N} X R. \quad (21)$$

Thus its derivative is an original boundary, with a primitive obtained by literal monic division by χ and the existing theta division maps. Its quotient class and arithmetic jets stay fixed.

Define

$$O_N = (1 - P_N) X R : C \rightarrow \mathcal{P}_{N+1} \cap \mathcal{P}_N^\perp, \quad I_N = P_{\mathcal{D}_N} X R : C \rightarrow \mathcal{D}_N.$$

Direct differentiation, with the derivative of the measure included, yields

$$\boxed{G_N'' - G_N' G_N^{-1} G_N' = O_N^* O_N - I_N^* I_N.} \quad (22)$$

Indeed $G_N' = R^* X R$, $G_N'' = R^* X^2 R - 2R^* X P_{\mathcal{D}_N} X R$, and $R G_N^{-1} R^*$ is the projection to the orthogonal quotient representatives. Their orthogonal direct sum with $\mathcal{D}_N$ is $\mathcal{P}_N$, which proves (22).

Both maps have rank at most one. For O_N this follows from the highest polynomial degree. For I_N^* , multiplication sends every boundary of degree below its highest degree back into $\mathcal{D}_N$, orthogonal to R ; only the highest boundary direction can contribute.

Taking $\text{Tr}(G_N^{-1} \cdot)$ gives (20). The positive source contribution and the negative boundary contribution are retained. Positivity of each Gram separately is not assigned to their difference.

S15.7 6. The phase in the rank-two control is a scalar volume derivative

Put $T = (A - cI)/i$ on C , with $c = k/2$. This is the coefficient operator corresponding to multiplication by u under the explicit ring map $S \mapsto c + iu$. The actual scaling generator is still $A = cI + iT$.

Reflection makes the polynomial $i^{-q} \chi(c + iu)$ real. This assertion follows by substituting $S = c + iu$ into $\chi^\dagger = (-1)^q \chi$. The retained leading phase of $\chi(c + iu)$ is i^q . Its remainder ideal and coefficient extraction are transported through that scalar unit, rather than altering χ in the arithmetic source.

In those real coefficient coordinates let $Q_j(u)$ be the monic source polynomials and $d_j = [Q_j]$. Their norm is ω_j . The metric and representatives have the same quotient definition. Define

$$\ell_N = \log V_N, \quad \sigma = \text{Tr } T \in \mathbb{R}.$$

The exact identity is

$$\boxed{\frac{d_N^* G_N d_{N+1}}{\omega_N} = \sigma - \ell_N'.} \quad (23)$$

To prove it, let $\mathfrak{b}_N = Q_{N+1} - R_N d_{N+1}$ in the polynomial source. It is the actual next boundary and

$$R_N^* \mathfrak{b}_N = -G_N d_{N+1}.$$

Multiplying the least-norm polynomial representative by u , its highest coefficient is $d_N^* G_N / \omega_N$. Therefore, modulo old boundaries,

$$X R_N - R_N T = \mathfrak{b}_N \frac{d_N^* G_N}{\omega_N}.$$

Multiplication by R_N^* eliminates precisely those old boundary terms. Consequently

$$\mathrm{Tr}(G_N^{-1} R_N^* X R_N) = \mathrm{Tr} T - \frac{d_N^* G_N d_{N+1}}{\omega_N}.$$

The left side is ℓ'_N by (21) and differentiation of the norm. This proves (23).

In the original S -polynomial coordinates, $p_j(S)$ maps to $i^j Q_j(u)$. Hence the preceding source cross term satisfies

$$\boxed{\frac{b_N^* G_N b_{N+1}}{\omega_N} = i(\sigma - \ell'_N).} \quad (24)$$

Both the sign and the factor i are retained. In particular an auxiliary real tilt generally creates a nonzero imaginary cross term even from an initially even measure.

S15.8 7. The exact scalar formula and a two-step upper bound

For $N \geq q$, define

$$\delta_N = \frac{V_N}{V_{N-1}}, \quad v_N = -\log \delta_N \geq 0, \quad a_{N+1} = \frac{\omega_{N+1}}{\omega_N}.$$

The nonnegativity follows from the same minimum-norm comparison at nested source degrees. The rank-one kernel update and the matrix determinant lemma give

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - \delta_N, \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1. \quad (25)$$

Substitute (24)–(25) into the inherited exact rank-two radius. This proves

$$\boxed{\epsilon_N^2 = a_{N+1}(1 - e^{-v_N})(e^{v_{N+1}} - 1) - (\sigma - \ell'_N)^2.} \quad (26)$$

All scalar inputs come from (7), (9), and (12). In particular

$$\delta_N = \frac{\omega_N}{\nu_{N-q}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_{N-q+1}}.$$

This does not assert that Hankel determinant evaluation is well-conditioned; the exact formula still requires error enclosures for numerical certification.

S15.8.1 7.1 The first admitted degree

At $N = q - 1$, K_{N-1} is not invertible. The displayed formula is replaced by its direct endpoint calculation:

$$\boxed{\epsilon_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - (\sigma - \ell'_{q-1})^2.} \quad (27)$$

Here $\nu_0 = Z_\chi$ is the actual norm of the first source relation. To prove (27), the first q remainder vectors are a basis, so the final included column has leverage one; the next determinant update gives $d_N = \nu_0 - \omega_q$. Formula (24) supplies the phase. No V_{q-2}^{-1} is introduced.

S15.8.2 7.2 Both losses in the finite upper bound

For $N \geq q$, put

$$s_N = \frac{v_N + v_{N+1}}{2}, \quad t_N = \frac{v_{N+1} - v_N}{2}.$$

Direct scalar multiplication gives

$$(1 - e^{-v_N})(e^{v_{N+1}} - 1) = \sinh^2 s_N - (e^{t_N} - \cosh s_N)^2.$$

Thus

$$\boxed{\epsilon_N^2 = a_{N+1} \sinh^2 s_N - a_{N+1} (e^{t_N} - \cosh s_N)^2 - (\sigma - \ell'_N)^2.} \quad (28)$$

In particular, with $\mathcal{R}_N = e^{2s_N} = V_{N-1}/V_{N+1} \geq 1$,

$$\boxed{\epsilon_N^2 \leq a_{N+1} \frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N}.} \quad (29)$$

This upper estimate retains two computable losses: imbalance between consecutive volume contractions and the original cross-term phase. When the bound is too coarse, (26) or (28) is the exact replacement within the same construction.

S15.9 8. What the exterior lower bound forces on these arithmetic volumes

The preceding exterior theorem gives

$$L_{h,k} := \sum_{\Re \lambda > k/2} \ell_\lambda (2\Re \lambda - k) \leq \epsilon_{h,k,N}.$$

It uses the original representative on its determinant line, not a newly chosen scalar metric. Combining it with (29), at $\theta = 0$, proves

$$\boxed{\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L_{h,k}}{\sqrt{a_{N+1}}} \right), \quad N \geq q.} \quad (30)$$

The equivalent inequality without logarithms is

$$4\mathcal{R}_N L_{h,k}^2 \leq a_{N+1} (\mathcal{R}_N - 1)^2.$$

For disjoint successive two-degree steps the logarithms telescope. For every $N_0 \geq q$ and integer $r \geq 1$,

$$\boxed{\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L_{h,k}}{\sqrt{a_{N_0+2j+1}}} \right).} \quad (31)$$

This is a source-volume consequence for the actual nested relation quotients, not a claim that arithmetic classes disappear when their representative norms decrease. At every degree their jets and original classes remain fixed by (5) and (15).

For a full off-line quartet with displacement $\delta > 0$, height $\gamma > 0$, and common multiplicity m , the previous count is

$$q = [1 + k(m - 1)](k + 1)^2,$$

$$L_{h,k} = 2\delta[1 + k(m - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor. \quad (32)$$

The quotient (16) and its derivative therefore give an entirely scalar test of the exterior obstruction. For this full quartet m_k and $|\chi(c + iu)|^2$ are even, so $\sigma = 0$ and $\ell'_N(0) = 0$. The phase term vanishes at zero tilt by this **additional** symmetry; it has not been removed from the general formula.

The sufficient subcubic target can now be tested from the quantities

$$\frac{\sqrt{a_{N(k)+1}}}{k^3} \frac{\mathcal{R}_{N(k)} - 1}{2\sqrt{\mathcal{R}_{N(k)}}}, \quad N(k) \geq q, \quad (33)$$

or, more sharply, from the full expression (26). An upper limit of zero for (33) would contradict (32) for that fixed quartet. This continuation proves the formula, the finite upper inequality, and the forced volume lower inequalities. It does not prove that limit.

S15.10 9. The Laplace input can be evaluated from the original theta source

The inverse Mellin representation makes F_h holomorphic in $|\arg z| < \pi/4$. Uniform vertical-strip bounds for v_h permit differentiation and shifts on closed subsectors, and shifting the Mellin contour gives the required rapid endpoint bounds there.

The exact analytic-continuation map is

$$E_\theta F_h(y) = e^{(y+i\theta/2)/2} F_h(e^{y+i\theta/2}).$$

For $|\theta| < \pi/2$, its Fourier transform, with kernel e^{ity} , is

$$\mathcal{F}_+ E_\theta F_h(t) = e^{\theta t/2} v_h(1/2 + it).$$

Contour shift proves the equality; Plancherel yields

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (34)$$

The phase $e^{i\theta/4}$ in E_θ has modulus one and is retained before taking the norm.

Dagger stability gives the analytic identity

$$F_h(1/\bar{z}) = (-1)^{\deg h} \bar{z} \overline{F_h(z)}.$$

Substitute $z = e^{i\theta/2}x$ and change variables on $(0, 1)$. This proves

$$M_h(\theta) = 2 \int_1^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (35)$$

Evenness in θ requires conjugation stability as well; it is not inferred just from this endpoint identity.

For the **analytic seed** $h = 1$, keep

$$f_0(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}.$$

Let $c_1 = -6, c_2 = 4$ and

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}).$$

Substitution into (35), where $\Re C_{mn} > 0$, gives

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(r+s+1/2, C_{mn}(\theta))}{C_{mn}(\theta)^{r+s+1/2}}. \quad (36)$$

The complex power is the analytic branch on the open right half-plane agreeing with the positive real power on its positive axis. The sum converges locally uniformly for $|\theta| < \pi/2$, by the exterior Gaussian estimates. Thus all derivatives used by (12) can be taken there. The factor 4 includes both the original two-sided theta sum and the change from the integral to the incomplete gamma function.

The accompanying numerical script evaluated (36) at 50 and 70 digits, with integer cutoffs 6 and 8. Their printed values agree:

$$M_1(0) \approx 1.279007247846485140479533592267193274492,$$

$$M_1(0.1) \approx 1.345907178458249796968220592357629805161.$$

An independent source integral gives

$$M_1''(0) = \|(D - 1/2)f_0\|^2 \approx 13.05554930257055843539268465812319368243.$$

Here $M_1'(0) = 0$. Consequently the first source Toda coefficient for tensor degree k is exactly

$$a_1^{(k)} = k \frac{M_1''(0)}{M_1(0)} \approx 10.207564753485721688865761252237555175 k. \quad (37)$$

This evaluates initial analytic data of the determinant flow. It is not a finite-packet control estimate: for $h = 1$ the arithmetic cyclic quotient is zero. The numerical results have neither directed-rounding enclosures nor a numerical omitted-tail certificate.

S15.11 10. Exact finite checks, an essential phase example, and current endpoint

The checker independently constructs source Gram matrices, quotient maps, representatives, action matrices, and scalar determinants in rational Gaussian fixtures. It compares the scalar result against $\text{Tr}((G^{-1}W)^2)/2$.

For the declared coefficient fixture with weight $7e^{-u^2/2} du / \sqrt{2\pi}$, relation $u^2 + u + 1$, and degree $N = 3$, it obtains

$$\ell'_N(0) = -52/111, \quad \sigma = -1, \quad \epsilon_N^2 = 1156/333.$$

The phase subtraction is $(59/111)^2$, which is nonzero. Omitting it changes the answer. A tilted even-relation fixture gives the same warning: parity at zero tilt is not imposed at another tilt.

Twenty-four exact test methods passed in normal and optimized Python, with identical successful records. Both deliberate-failure controls failed. Repeated roots, source-coordinate phases, mass factors, the minimal-degree endpoint, both Toda equations, the curvature difference, the complete scalar radius, and the supported-zero relation are included. These are finite regression tests, not Lean certificates or verification of infinite arithmetic integrals.

All 34 members named in the preceding exterior package's manifest verified. Its 22-method checker passed a fresh normal run. A combined command attempting a subsequent optimized rerun timed out; no completed optimized rerun is claimed for that inherited suite. The new suite passed in both modes.

The present advance is an upper-estimate mechanism on the exact original family: its input is the one-factor arithmetic Laplace transform; its relation filter is the original cyclic annihilator; its output is the canonical quotient volume and complete control radius. The source-boundary curvature and the positive volume retained before an e -quotient are explicit maps. No uniform subcubic upper estimate, RH/GRH proof, additional Lean execution, or remote repository modification is claimed.

S15.12 References and source boundary

The original source, support, and cohomological conventions are those in the supplied `Tau_Exterior_Trace_Amplification` package and its predecessor `Tau_Cyclic_Sum_Control`. The new formalization report and pinned PR #21 note supply the checked coefficient interfaces, not the analytic estimates in this note.

The general determinant, norm, and Christoffel–Darboux identities are classical. NIST DLMF §18.2(iv), (v), and (ix) records them, including the caution that raw Hankel determinant formulas are often poorly conditioned numerically: <https://dlmf.nist.gov/18.2>. The proof in §5 derives the needed Toda identity directly for the actual exponential deformation, without an isomonodromic or rational-logarithmic-derivative hypothesis. Gamma vertical-strip estimates used in §3 are recorded in DLMF §5.11: <https://dlmf.nist.gov/5.11>.

The finite exterior trace bound remains the theorem proved in the supplied preceding note; this continuation changes its analytic upper-estimate realization, not its algebraic proof or the scope of the parallel formalization.

S16 Supplied conormal derivative, symmetric restriction and uncompiled tower draft

Source attribution. This appendix reproduces the complete supplied mathematical continuation. Its first-person reports about PR 16, PR 18, combined CI, Mathlib inspection and 161 Wolfram checks are historical source statements, not additional verification performed for this edition. Both Lean blocks are retained; the proposed continuation is explicitly uncompiled. The monic-division coordinates and displayed dimension formulas have positive packet degree and tensor degree at least one; the empty packet is handled separately in the integrated proof. Full arithmetic unit and reflection formulas retain the cited source packet hypotheses.

Exact revision. 18f86ad411d0dfafc003aabddf25bccae2d38dcc4006c4d27e5be8aece9347b8

Source file. sources/web_conormal_tower_paste/CONORMAL_TOWER_SOURCE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 18f86ad411d0dfafc003aabddf25bccae2d38dcc4006c4d27e5be8aece9347b8

Formal status. Both Lean blocks are source text; the draft has not been compiled for this edition.

Codex's integration has advanced: PR #16 is merged, and the combined frontier/Codex run has now passed. I checked the current repository state and the completed job: it rebuilt the inherited sources, checked the frontier files, imported both developments into one Lean environment, and completed their selected axiom audits. The “combined run pending” sentence in PR #18's description is therefore out of date.

I continued with the new sum-connection note rather than duplicating that work. **The additional result is a complete calculation of what its conormal derivative detects, what it misses, and how that calculation restricts to the symmetric tensor summand.** It also packages the quotient and derivative together as a genuine algebra homomorphism.

The new results below have complete mathematical proofs and exact finite algebra checks. The Lean continuation at the end is an **uncompiled draft**: the local execution service is timing out, and the currently exposed GitHub actions provide no write operation. No remote files were changed in this turn.

S16.0.1 1. The derivative now has a complete quotient construction

Retain the source's notation, with tensor degree $k \geq 1$:

$$P = \mathbb{C}[s_1, \dots, s_k], \quad h_i = h(s_i), \quad I = (h_1, \dots, h_k),$$

$$E = P/I, \quad E^{[2]} = P/I^2, \quad \partial_S = \frac{1}{k} \sum_{i=1}^k \partial_{s_i}.$$

Here h is the original monic packet polynomial, including all multiplicities. These are precisely the relation ideal and derivative in the new attachment.

For any derivation D of a commutative algebra and any ideal I ,

$$\boxed{D(I^{r+1}) \subseteq I^r \quad (r \geq 0).}$$

Proof. An element of I^{r+1} is a sum of multiples of products of $r+1$ elements of I . The product rule differentiates one factor at a time, leaving at least r factors in I . Differentiating an exterior coefficient leaves all $r+1$ factors. Additivity completes the proof.

Consequently the source's maps exist at every order:

$$\delta_r : P/I^{r+1} \longrightarrow P/I^r, \quad [F] \longmapsto [\partial_S F].$$

They commute with the quotient transitions. In particular,

$$\delta_1 : E^{[2]} \longrightarrow E$$

is a derivation **along** the quotient $\pi : E^{[2]} \rightarrow E$:

$$\boxed{\delta_1(xy) = \pi(x)\delta_1(y) + \pi(y)\delta_1(x).}$$

This is the appropriate product rule; δ_1 is not an endomorphism of the arithmetic algebra E .

Monic division constructs all the retained layers There is a unique finite expansion

$$F = \sum_{\alpha \in \mathbb{N}^k} h_1^{\alpha_1} \cdots h_k^{\alpha_k} r_\alpha, \quad \deg_{s_i} r_\alpha < d := \deg h.$$

Existence follows by repeated monic division in the separate variables. For uniqueness, the leading monomials of

$$h^\alpha s^\beta, \quad 0 \leq \beta_i < d,$$

have exponent vectors $d\alpha + \beta$, which are distinct. A finite dependence therefore has a unique largest leading monomial and cannot cancel.

This gives explicit coordinates—not just an existence statement—for

$$I^r/I^{r+1} \cong E[\eta_1, \dots, \eta_k]_r, \quad \eta_i \longleftrightarrow [h_i].$$

In particular,

$$I/I^2 \cong E^k, \quad (a_i) \longmapsto \left[\sum_i h_i \tilde{a}_i \right],$$

and

$$\dim_{\mathbb{C}} P/I^{r+1} = d^k \binom{k+r}{k}.$$

On the associated graded, the derivative is exactly

$$\boxed{\text{gr}(\partial_S) = \frac{1}{k} \sum_i [h'_i] \partial_{\eta_i}.$$

Indeed, differentiating a coefficient of $h^\alpha r_\alpha$ stays in $I^{|\alpha|}$, while differentiating one h_i -factor lowers the ideal order and contributes the literal exponent α_i . This proves the higher-layer formula in the attachment without deleting those exponent factors.

The standard conormal construction supplies the map

$$I/I^2 \longrightarrow \Omega_{P/\mathbb{C}} \otimes_P E.$$

The calculation here identifies its coordinates for this particular ideal and then contracts it by the specified sum direction; those are distinct operations (The Stacks Project Authors, n.d., Tag 00S0). (The Stacks Project)

S16.0.2 2. The exact information detected by the sum derivative

Let

$$A_h = \mathbb{C}[s]/(h), \quad E = A_h^{\otimes k},$$

and write

$$c_i = [h'(s_i)] \in E.$$

Under the preceding conormal coordinates, the restriction of δ_1 is the E -linear map

$$\partial_N : E^k \longrightarrow E, \quad (a_1, \dots, a_k) \longmapsto \frac{1}{k} \sum_i c_i a_i.$$

The attachment gives this row. The following kernel, cokernel, and rank calculations are additional deductions.

Let

$$b = \deg \gcd(h, h').$$

Over $\mathbb{C}$, if h has q distinct roots, then

$$b = d - q = \sum_{\rho} (m_{\rho} - 1).$$

The image and cokernel are

$$\mathrm{im} \partial_N = (c_1, \dots, c_k) \subseteq E,$$

$$\mathrm{coker} \partial_N \cong (\mathbb{C}[s]/(h, h'))^{\otimes k}.$$

Proof. The image is the ideal generated by the row entries because $1/k$ is invertible. Quotienting E by those entries imposes $h(s_i) = h'(s_i) = 0$ in each separate variable, giving the displayed tensor product.

Therefore

$$\mathrm{rank}_{\mathbb{C}} \partial_N = d^k - b^k, \quad \dim_{\mathbb{C}} \ker \partial_N = (k-1)d^k + b^k.$$

These formulas separate two sources of information lost by this particular observation.

When h is squarefree, $b = 0$, so the contracted derivative is onto. Nevertheless, for $k > 1$ its kernel still has dimension $(k-1)d^k$: **contracting to the sum direction does not observe every relative conormal direction, even without repeated roots.**

When roots are repeated, the additional term b^k records the failure of the Jacobian entries to generate the unit ideal.

For example, take the declared algebraic calibration

$$h(s) = s^2, \quad k = 2, \quad E = \mathbb{C}[x, y]/(x^2, y^2).$$

Then

$$\partial_N(a, b) = xa + yb.$$

Its image is

$$\text{span}\{x, y, xy\},$$

so its rank is three and its kernel has dimension five. The conormal vector $(x, 0)$ is nonzero but lies in that kernel. Its representative is x^3 modulo

$$I^2 = (x^4, x^2y^2, y^4),$$

so the retained relation is genuinely nonzero before observation.

These are ranks of the **conormal restriction**, not of the whole map $\delta_1 : E^{[2]} \rightarrow E$. The latter is onto: in coordinates

$$S = \sum_i s_i, \quad z_i = s_i - S/k,$$

one has $\partial_S = \partial/\partial S$, and a polynomial primitive is obtained by integrating its powers of S . The same argument makes every δ_r onto.

Package the quotient and derivative as one algebra map Introduce an ordinary formal square-zero parameter ϵ , distinct from the supported scalar e . Define

$$\boxed{\mathcal{J}_S^1 : E^{[2]} \longrightarrow E[\epsilon]/(\epsilon^2), \quad x \longmapsto \pi(x) + \epsilon \delta_1(x).}$$

This is a unital $\mathbb{C}$ -algebra homomorphism. Multiplication follows from the relative product rule:

$$(\pi x + \epsilon \delta x)(\pi y + \epsilon \delta y) = \pi(xy) + \epsilon \delta(xy).$$

Thus the source's two differently typed maps can be retained together in a single multiplicative observation. Its kernel is exactly

$$\ker \mathcal{J}_S^1 = \{n \in I/I^2 : \partial_N n = 0\},$$

and consequently

$$\boxed{\dim \ker \mathcal{J}_S^1 = (k-1)d^k + b^k, \quad \dim \text{im } \mathcal{J}_S^1 = 2d^k - b^k.}$$

The proof of the image dimension is the exact sequence

$$0 \longrightarrow \epsilon \text{ im } \partial_N \longrightarrow \text{im } \mathcal{J}_S^1 \longrightarrow E \longrightarrow 0.$$

This also describes the precise quotient of the first thickening seen by the sum derivative. It does not silently identify that quotient with the entire first thickening.

Retaining every first derivative is a different observation If all coordinate derivatives are retained, the map is

$$\mathcal{J}_{\text{all}}^1 : E^{[2]} \longrightarrow E \oplus E^k, \quad [F] \longmapsto ([F], ([\partial_i F])_i),$$

where E^k is a square-zero ideal in the target.

On $I/I^2 \cong E^k$, its derivative component is

$$(a_i) \longmapsto (c_i a_i)_i.$$

Each multiplication map M_{c_i} has kernel dimension $b d^{k-1}$. Hence

$$\dim \ker \mathcal{J}_{\text{all}}^1 = k b d^{k-1}.$$

In the squarefree case this full first-jet map is an algebra isomorphism onto the square-zero extension $E \oplus E^k$. With repeated roots it still loses the explicitly computed Jacobian-annihilator directions, which remain in the higher source layers.

The split lifts retain their original meaning throughout:

$$G(\pi)(n^\bullet) = e_E, \quad G(\mathcal{J}_S^1)(n^\bullet) = (\epsilon \partial_N n)^\bullet.$$

External absence maps to external absence. This is a calculation of a retained relation before quotienting, not a rule that differentiates e into a nonzero scalar. The attachment makes that distinction explicitly.

S16.0.3 3. This calculation restricts to the symmetric tensor construction

The preceding maps commute with simultaneous permutation of variables and conormal indices. Let

$$N = I/I^2 \cong E^k, \quad T = \mathbb{C}[s]/(h, h').$$

The cokernel sequence is equivariant:

$$N \xrightarrow{\partial_N} E \longrightarrow T^{\otimes k} \longrightarrow 0.$$

Taking S_k -invariants remains exact: given an invariant element in an image, average any preimage with the literal factor $1/k!$. Thus

$$\text{coker}(\partial_N : N^{S_k} \rightarrow E^{S_k}) \cong (T^{\otimes k})^{S_k}.$$

The invariant conormal domain has the explicit description

$$N^{S_k} \cong A_h \otimes (A_h^{\otimes(k-1)})^{S_{k-1}}.$$

An invariant tuple is determined by its first component, which is invariant under permutations fixing the first index; the other components are its prescribed translates. This constructs both directions of the identification.

Consequently,

$$\dim N^{S_k} = d \binom{d+k-2}{k-1},$$

$$\text{rank } \partial_N^{S_k} = \binom{d+k-1}{k} - \binom{b+k-1}{k},$$

and

$$\dim \ker \partial_N^{S_k} = d \binom{d+k-2}{k-1} - \binom{d+k-1}{k} + \binom{b+k-1}{k}.$$

For $b = 0$, the final binomial is zero.

For $h = s^2$ and $k = 2$, the symmetric conormal domain has dimension four, the symmetric arithmetic target has dimension three, and the restricted derivative has rank two. Its kernel still has dimension two.

These are ordinary variable-permutation invariants of the relation algebra. To place them in the original cochain construction, use the source's **signed chain idempotent**

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi.$$

Its two top-degree signs produce precisely the ordinary symmetric tensors used here. Replacing it by the unsigned chain average would change the selected summand.

This gives the smaller cohomological calculation its own exact derivative kernel; selecting the symmetric summand does not amount to presuming that the remaining relative information vanishes.

S16.0.4 4. The arithmetic unit and trace remain attached

The new note retains

$$U = \prod_i (g/h)(s_i)$$

and its Taylor class $\widehat{U} \in E^{[2]}$. The relative Leibniz rule proves

$$\delta_1(\widehat{U}x) = U \delta_1(x) + \delta_1(\widehat{U})\pi(x).$$

On a conormal class n , the second term vanishes:

$$\delta_1(\widehat{U}n) = U \partial_N n.$$

Because U is a unit, this arithmetic multiplication changes neither the kernel nor the ranks calculated above. It is nevertheless retained in the actual map, rather than assigned the value one. This is the source's specified unit correction.

There is also an important distinction between the **sum-Jacobian response** and the **product Jacobian used by the full product trace**.

Multiplication by

$$\prod_i h'(s_i)$$

on $E = A_h^{\otimes k}$ has rank

$$\boxed{q^k},$$

where q is the number of distinct roots of h . Its kernel has dimension

$$d^k - q^k$$

and is the nilradical of E .

Proof. At a root ρ of multiplicity m , write

$$h(\rho + z) = z^m u_\rho(z), \quad u_\rho(0) \neq 0.$$

Modulo z^m ,

$$h'(\rho + z) = m u_\rho(0) z^{m-1}.$$

Multiplication by this element has one-dimensional image and kernel (z) . At a k -tuple of roots, the product Jacobian similarly sends the constant coefficient to a nonzero multiple of

$$z_1^{m_1-1} \cdots z_k^{m_k-1}$$

and annihilates the maximal ideal. There is one image dimension per root tuple, giving q^k .

This is not the rank $d^k - b^k$ of the contracted conormal row. The two operators use the same derivative factors but perform different observations.

The one-variable trace identity remains exact. Let

$$\lambda_h([P]) = \sum_{\rho} \operatorname{Res}_{s=\rho} \frac{P(s)}{h(s)} ds.$$

Then

$$\boxed{\lambda_h(h'P) = \operatorname{Tr}(M_P \mid A_h).}$$

Locally, $h'/h = m_\rho/(s - \rho) + u'_\rho/u_\rho$, so its residue against P is $m_\rho P(\rho)$. Multiplication by P on the local algebra has that same trace because its nilpotent terms have zero diagonal.

Writing $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$, the actual arithmetic pairing gives

$$\begin{aligned} \mathcal{R}_Z(f, j_h(g')P) &= \lambda_h(\varepsilon_h f^\dagger v_h h'P) \\ &= \lambda_h(f^\dagger h'P) \\ &= \operatorname{Tr}(M_{f^\dagger P} \mid A_h). \end{aligned}$$

Both unit factors are present before cancellation. This is the algebraic form of the attachment's observation of

$$(\log x)\Theta(P(D)\phi_*),$$

with the original $g = 2\xi$, reflection, and multiplicities retained.

S16.0.5 5. Lean continuation against the integrated library

The merged Codex source already supplies

`SplitZero.RelationLayer.derivative`

as the quotient map induced by a linear map that carries the source relations into the target relations. I read its implementation and reused that interface below rather than introducing another quotient.

The following proposed `SplitZeroConormalTower.lean` proves the ideal-power containment from the derivation law, uses it to construct every descended derivative, proves compatibility between levels, and computes the derivative of an original relation. Its proof scripts are complete, but **compilation has not been checked**.

```
import SplitZeroRelationLayerIntegration
import Mathlib.RingTheory.Derivation.Basic
import Mathlib.RingTheory.Ideal.Operations

/-!
Derivative maps between the original ideal-power quotients.

This draft reuses RelationLayer.derivative. It does not assume that
the ideal is derivative-stable or that the packet is squarefree.
-/

noncomputable section

namespace SplitZero.ConormalTower

variable {R A : Type*}
[CommRing R] [CommRing A] [Algebra R A]

/-- The original ideal-power relation, viewed over the coefficient ring. -/
def level (I : Ideal A) (r : ℕ) : Submodule R A :=
  (I ^ r).restrictScalars R

/-- A derivation lowers ideal order by at most one. -/
theorem deriv_mem_pow
  (D : Derivation R A A) (I : Ideal A) (r : ℕ)
  {x : A} (hx : x ∈ I ^ (r + 1)) :
  D x ∈ I ^ r := by
  induction r generalizing x with
  | zero =>
    rw [Submodule.pow_zero, Ideal.one_eq_top]
    exact Submodule.mem_top
  | succ r ih =>
    rw [Submodule.pow_succ] at hx
    refine Submodule.mul_induction_on hx ?_ ?_
    · intro a ha b hb
      rw [D.leibniz]
      change a * D b + b * D a ∈ I ^ (r + 1)
      have hd : D a * b ∈ I ^ (r + 1) := by
        rw [Submodule.pow_succ]
        exact Ideal.mul_mem_mul (ih ha) hb
      exact (I ^ (r + 1)).add_mem
        ((I ^ (r + 1)).mul_mem_right (D b) ha)
        (by simp only [mul_comm] using hd)
    · intro a b ha hb
      rw [map_add]
      exact (I ^ (r + 1)).add_mem ha hb

/-- The derivative on A/I^(r+1), with values in A/I^r. -/
```



```

def descended
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) :
  (A / level (R := R) I (r + 1)) →1[R]
  (A / level (R := R) I r) :=
  SplitZero.RelationLayer.derivative
  (level I (r + 1)) (level I r) D.toLinearMap
  (fun _ hx => deriv_mem_pow D I r hx)

@[simp]
theorem descended_mk
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) (x : A) :
  descended D I r ((level I (r + 1)).mkQ x) =
  (level I r).mkQ (D x) :=
  rfl

/-- Forget one ideal-adic layer without differentiating. -/
def projection (I : Ideal A) (r : ℕ) :
  (A / level (R := R) I (r + 1)) →1[R]
  (A / level (R := R) I r) :=
  SplitZero.RelationLayer.transport
  (level I (r + 1)) (level I r)
  (fun _ hx => Ideal.pow_le_pow_right (Nat.le_succ r) hx)

@[simp]
theorem projection_mk (I : Ideal A) (r : ℕ) (x : A) :
  projection (R := R) I r ((level I (r + 1)).mkQ x) =
  (level I r).mkQ x :=
  rfl

/-- Derivative and quotient transition form the actual commuting square. -/
theorem tower_square
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) :
  (projection I r).comp (descended D I (r + 1)) =
  (descended D I r).comp (projection I (r + 1)) := by
  apply LinearMap.ext
  intro z
  obtain (x, rfl) :=
    Submodule.Quotient.mk_surjective (level I (r + 2)) z
  rfl

/-- On a retained relation  $z \cdot a$ , only  $a \cdot D(z)$  survives modulo  $I$ . -/
theorem conormal_product
  (D : Derivation R A A) (I : Ideal A)
  (z a : A) (hz : z ∈ I) :
  descended D I 1 ((level I 2).mkQ (z * a)) =
  (level I 1).mkQ (a * D z) := by
  rw [descended_mk, D.leibniz]
  change
    (level I 1).mkQ (z * D a + a * D z) =
    (level I 1).mkQ (a * D z)
  rw [map_add]
  have hzero : (level (R := R) I 1).mkQ (z * D a) = 0 := by
    apply (Submodule.Quotient.mk_eq_zero _).mpr
    change z * D a ∈ I ^ 1
    rw [Submodule.pow_one]
    exact I.mul_mem_right (D a) hz
  rw [hzero, zero_add]

end SplitZero.ConormalTower

```

The derivation and ideal-product APIs used here were checked against the repository's existing Mathlib revision, rather than assumed from a different release. That source inspection does not substitute for running Lean.

S16.0.6 Verification and the remaining quantitative connection

The fresh Wolfram calculations passed **161 exact checks**: conormal ranks and invariant restrictions, full first-jet ranks, ideal-square annihilation, residue traces, product-Jacobian ranks, higher derivative identities, and retained-unit trace identities. The fixtures include squarefree polynomials, repeated roots, mixed multiplicities, and tensor degrees through three. An initial variable-binding error in my checker was corrected before the successful runs. These are finite algebra checks, not a new Lean certificate or evaluations of arithmetic zero locations.

The new result does not identify the Hilbert-space normal map N_u with the algebraic conormal module I/I^2 . The attachment's filtered graph supplies the stated route from the full derivative to arithmetic jets. Its exact scalar identity

$$\mathcal{I}_{h,k} + 4 \int \|n_k(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h$$

retains the Hilbert normal contribution; the conormal calculations above now identify the algebraic kernel encountered by the subsequent sum-direction observation. Those two pieces must remain separately typed when estimating the original weight-control form.

The continuation now provides an exact first-jet algebra map, its full retained kernel, the corresponding symmetric restriction, and all-order quotient derivatives. Codex can build those on the merged quotient library without recreating the scalar, homology, or boundary-control implementations.

Reference The Stacks Project Authors. (n.d.). *The naive cotangent complex* (Tag 00S0). *The Stacks Project*. ([The Stacks Project](#))

S17 PR 21: conormal tower and cyclic depth formalization with its complete mathematical note

Source attribution. Complete mathematical note at PR 21 implementation commit 2abc351424ba87aeda948a5cfb846e15ed9373d1. The staged source provenance identifies all eight added Git blobs. The historical pasted draft, these repository modules and the observed remote CI evidence retain their distinct revisions and verification scopes.

Exact revision. 2abc351424ba87aeda948a5cfb846e15ed9373d1

Source file. `sources/web_conormal_cyclic_formal_delivery/repository/workbenches/tau-conormal-cyclic-formal/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 202d7e9111c0a88fdd30c15554f3dd1e0907b2fc4fb2e11da129c97b10a053fd

Formal status. The associated two Lean modules and observed remote CI evidence are archived separately. No local Lean execution is attributed to this edition.

S17.1 Original conormal and cyclic relation towers

S17.1.1 Scope and source reconciliation

This add-only continuation starts from main at a4494fba4968db837a8af6fcb952c8688cd5d1da. It retains the merged Codex quotient/homotopy/boundary implementation, the merged finite-frontier implementation, and the reviewed sum-connection note. It uses the owner's attached **Handoff: sum-generated arithmetic control**, especially targets 1, 2, 4 and 8. The current upload is an analytic/algebraic continuation with a separately reported 24-method finite checker; it is not a Lean certificate. The local execution service failed before its ZIP could be opened, so neither the ZIP manifest nor that checker has been independently rerun here.

Two actual GitHub corrections are incorporated, not merely acknowledged:

- Commit 600ab7a527f7a626c162bf0199d093d454b86c14 makes the nonempty hypothesis explicit in the monic-freeness argument and calculates $h=1$ separately. Every positive-depth relation ideal for $h=1$ is the whole polynomial ring. The corresponding underlying quotients are zero, while the split lift still retains tau and supported zero. The analytic theta seed and its mass do not disappear. `ConormalTower.top_level_zero` checks the full-ideal quotient at every level.
- Commit 7aec03a93b66d3d97fbf6f7042f58bb2c1019543 corrects the completed integration audit to 68 selected declarations and enables the existing frontier workflow on main and relevant pull requests. This branch inherits those files unchanged. The current 68-target audit, not the historical 66-target description, is rerun by the added workflow.

The original scalar G , e , tau, arithmetic projection to the infinite ring, reconstructed carriers, quotient transports and theta normalization $g=2$ xi are unchanged. No claim about a global arithmetic small-control estimate follows from the present algebraic results.

S17.1.2 1. An actual derivative between consecutive original quotients

Let R and A be commutative rings, A an R -algebra, I an ideal of A , and D an R -derivation of A . Then

$D(I^{(r+1)})$ is contained in I^r , $r \geq 0$.

Proof: use induction on r and the additive generation of an ideal product by products. For a in $I^{(r+1)}$, b in I , the Leibniz terms $a D(b)$ and $b D(a)$ both lie in $I^{(r+1)}$: the first by the ideal property, the second by the induction hypothesis. At $r=0$ the target ideal is the whole ring.

SplitZeroConormalTower.lean implements that proof as `deriv_mem_pow`. It then uses the merged `SplitZero.RelationLayer.derivative` to construct

$\text{delta}_r : A/I^{(r+1)} \rightarrow_R A/I^r, [x] \mapsto [D x]$.

It separately constructs the quotient projection p_r between the same types and proves

$p_r \text{delta}_{(r+1)} = \text{delta}_r p_{(r+1)}$.

The derivative is not an endomorphism of A/I . These are coefficient-linear maps on the actual original quotient types, not new quotient carriers. At a retained relation z in I ,

$\text{delta}_1([z a]) = [a D(z)]$.

For the full arithmetic multiplier u , the implemented formula is

$\text{delta}_r([u x]) = [u D(x)] + [x D(u)]$.

It applies in particular to the retained Taylor unit, without assigning the unit the value one or deleting its derivative. The definitions work for arbitrary commutative rings and do not assume characteristic zero or squarefreeness.

S17.1.3 2. The exact cyclic pullback, before a polynomial is used to describe it

Fix S in A . At each depth define the actual coefficient submodule

$K_r = \{P \text{ in } R[X] : P(S) \text{ belongs to } I^r\}$.

The implementation calls this `cyclicLevel`; it is literally the comap of the original relation submodule along `Polynomial.aeval\S`. Its constructed quotient map is

$\text{alpha}_r : R[X]/K_r \rightarrow_R A/I^r, [P] \mapsto [P(S)]$.

`evaluation_injective` proves injectivity by the two original quotient equality criteria. No assumed injectivity of polynomial evaluation is required. This implementation is a linear map; its usual algebra-homomorphism packaging is not claimed as a new bundled declaration.

If $D(S)=1$, Mathlib's formal polynomial chain rule gives

$D(P(S)) = P'(S)$.

Thus P in $K_{(r+1)}$ implies P' in K_r . The code constructs the cyclic derivative and proves the full square

$$\begin{array}{ccc} R[X]/K_{(r+1)} & \xrightarrow{-d/dX-} & R[X]/K_r \\ \downarrow \text{alpha}_{(r+1)} & & \downarrow \text{alpha}_r \\ A/I^{(r+1)} & \xrightarrow{\quad D \quad} & A/I^r. \end{array}$$

The actual unit comparison is also proved:

$\text{delta}_r([u P(S)]) = [u P'(S)] + [D(u) P(S)]$.

The final term need not be in the cyclic image. The equation retains its larger target. Most importantly, K_r is not replaced by K_1^r . The explicit annihilator calculation below explains the difference.

S17.1.4 3. Exact local maximum, including an attaining monomial

Here the base field is $\mathbb{C}$. Let $k \geq 1$, $d \geq 1$, $r \geq 1$ and write the complete packet polynomial as

$$h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}.$$

For a root tuple $\mathbf{\rho} = (\rho_1, \dots, \rho_k)$, put $m_i = m_{(\rho_i)}$ and $y_i = s_i - \rho_i$. Locally the ideal $(h(s_i))$ is $(y_i^{m_i})$, because each remaining factor is a unit. In the quotient by its r -th power the monomial basis is precisely

$$y^{\mathbf{a}} \text{ such that } \sum_i \lfloor a_i / m_i \rfloor < r.$$

Write $a_i = m_i q_i + t_i$ with $0 \leq t_i < m_i$. Its total degree satisfies

$$\sum_i a_i \leq \sum_i (m_i - 1) + (r - 1) \max_i m_i.$$

The proof uses $\sum_i q_i \leq r - 1$ and bounds each m_i by its maximum. This bound is attained: choose j with $m_j = \max_i m_i$, let every $t_i = m_i - 1$, let $q_j = r - 1$, and let every other $q_i = 0$.

SplitZeroCyclicDepth.lean formalizes the quotient/remainder predicate, the degree bound, that explicit exponent vector, and attainment. The floor-sum formulation is linked through Euclidean division. This is an all-index, all-depth result, not a finite calibration. The additional algebraic identification of these monomials with the polynomial quotient and the characteristic-zero multinomial step below have written proofs, not an additional Lean certificate in these two modules.

Let $N = y_1 + \dots + y_k$. Its exact nilpotency index is

$$L_r(\mathbf{\rho}) = 1 + \sum_i (m_i - 1) + (r - 1) \max_i m_i.$$

Indeed every monomial of total degree L_r is killed. The attaining monomial at degree $L_r - 1$ has coefficient

$$(L_r - 1)! / \prod_i a_i!$$

in $\mathbb{N}^{L_r - 1}$, a nonzero positive integer in characteristic zero. Distinct surviving monomials cannot cancel. This proves both vanishing and minimality.

The proof also counts the local dimension as

$$(\prod_i m_i) * \text{binomial}(k + r - 1, k).$$

The q_i with sum less than r give the binomial factor, and their remainders give the product factor.

S17.1.5 4. Collisions require a maximum at every depth

Set $S = \sum_i s_i$. At a collided sum λ , take the maximum over all root tuples with that sum:

$$\begin{aligned} \text{ell}(\lambda, r) &= \max_{\sum \rho_i = \lambda} L_r(\mathbf{\rho}). \\ \chi_r(X) &= \prod_{\lambda} (X - \lambda)^{(\text{ell}(\lambda, r))}. \end{aligned}$$

The Chinese-remainder decomposition and the exact local nilpotency prove

$$I^{\wedge r} \text{ intersect } \mathbb{C}[S] = (\chi_r(S)).$$

The same annihilator is obtained in the variable-permutation invariant subalgebra because every polynomial in S is already invariant. Thus α_r identifies the cyclic algebra $C[X]/(\chi_r)$ with its actual image, with no squarefree simplification.

The maximizing tuple can change with depth. In the declared calibration packet with roots 0,1,2 of multiplicities 5,4,1, respectively, the sum $\lambda=2$ in tensor degree two has

$$\text{ell}_r(2, r) = \max(5r, 4r+3).$$

At depths 1,...,5 the orders are 7,11,15,20,25. A tuple chosen only at depth one is insufficient for the whole tower. This is an algebraic test packet, not a statement about actual zeta zeros.

For every λ ,

$$\begin{aligned} \text{ell}_r(\lambda, r+1) &\geq \text{ell}_r(\lambda, r)+1, \\ \text{ell}_r(\lambda, r) &\leq r \text{ell}_r(\lambda, 1). \end{aligned}$$

The first follows by retaining a maximizing tuple from depth r ; its next slope is a positive multiplicity. For the second, note $\max_i m_i \leq 1 + \sum_i (m_i - 1)$ for every tuple. Hence

$$\begin{aligned} \chi_r &\text{ divides } \chi_{r+1}, \\ \chi_r &\text{ divides } \chi_{r+1}', \\ \chi_r &\text{ divides } \chi_1^r. \end{aligned}$$

The last divisibility supplies the stated surjection $C[X]/(\chi_1^r) \rightarrow C[X]/(\chi_r)$, not an equality. For $h(s)=(s-\rho)^2$ and $k=2$,

$$\begin{aligned} \chi_r(X) &= (X-2\rho)^{(2r+1)}, \\ \chi_1(X)^r &= (X-2\rho)^{(3r)}. \end{aligned}$$

They differ as soon as $r>1$. The derivative maps $C[X]/\chi_{r+1} \rightarrow C[X]/\chi_r$ are onto over C : integrate any polynomial representative, then take its source class. Their kernel dimensions are $\deg(\chi_{r+1}) - \deg(\chi_r)$.

For the empty packet $h=1$, handled separately as required by the GitHub correction, $I=P$, $K_r=C[X]$, and $\chi_r=1$ for all positive depths. The cyclic arithmetic quotient is zero. No maximum over an empty tuple set or inverse of a zero-dimensional Gram matrix is used to define it. The original nonzero analytic seed remains available as an analytic object.

S17.1.6 5. Explicit equivariant retraction of the cyclic summand (written proof)

This checks the handoff's retraction formula, including its inverse factor. On a λ -primary part M of the symmetric finite algebra, let $N=S-\lambda$, $N^{\text{ell}}=0$, and let w be the λ -component of the retained arithmetic unit U . Exact maximality gives $N^{(\text{ell}-1)}w \neq 0$. Choose the stated top-monomial coefficient functional θ with $\theta(N^{(\text{ell}-1)}w) \neq 0$. If it is realized first on an ordered root component, it can be restricted to the invariant subspace; the component of the symmetric vector is still detected.

Define

$$\pi_0(v) = \sum_{j=0}^{(\text{ell}-1)} \theta(N^{(\text{ell}-1-j)}v) X^j \bmod X^{\text{ell}}.$$

Coefficient comparison gives $\pi_0(Nv)=X \pi_0(v)$: the constant term on the left is $\theta(N^{\text{ell}}v)=0$ and the other coefficients shift exactly. Put $c=\pi_0(w)$. Its constant coefficient is nonzero, so it is invertible modulo X^{ell} . Explicitly, if $c=c_0(1+n)$, the inverse is $c_0^{-1} \sum_{j=0}^{(\text{ell}-1)} (-n)^j$.

Then $\pi=c^{-1}\pi_0$ satisfies

$$\pi(P(N)w) = P(X) \bmod X^{\text{ell}}.$$

Therefore the unit-weighted cyclic injection η has an S -equivariant left inverse. Summing over λ constructs the full retraction, and $\ker(\pi)$ is an invariant complement. This is a module splitting, not an assertion that π is an algebra map or that the summands are orthogonal for the canonical arithmetic Gram. All cross terms under that splitting remain.

This cyclic summand retains each amplified value $k \rho$; collided repeated pure tensors need not equal the particular cyclic eigenvector. The cyclic residue trace and the full symmetric trace have their distinct multiplicities. No part of the present calculation identifies those traces or drops the U -dagger U factor in the ambient arithmetic pairing.

S17.1.7 6. Formal versus analytic scope

The new formal outputs are the ideal-power derivative, actual quotient transitions, exact cyclic pull-back/injection, derivative square, unit rule, full-ideal case, and the all-degree attained monomial ceiling. Existing SplitZero support maps lift their coefficient maps at the original labels. No duplicate scalar construction or alternative homology core is introduced.

The exact χ_r annihilator, local basis/multinomial identification and the retraction in Sections 3-5 are supplied here as complete written arguments. Their finite calibrations are separate Python tests. The new upload's analytic cyclic convolution measure, canonical Gram, differentiated moments, rank-two control and full trace comparison remain its separately stated inputs and targets. This work does not independently certify that archive, an analytic moment enclosure, or a tensor-uniform small-control bound.

S17.1.8 Reproduction

Run the added SplitZero\conormal\cyclic\tower workflow on this branch. It installs unchanged Lean 4.31.0 and Mathlib fabf563a7c95a166b8d7b6efca11c8b4dc9d911f, rebuilds all inherited modules, checks every new source with `- \/-trust=0 \-DwarningAsError=true`, imports the combined sources, and rejects any selected transitive axiom outside `propext`, `Classical.choice` and `Quot.sound`. It reuses the previously tested axiom parser. The original core's byte pin is checked by the inherited prepare script.

The new finite checker directly multiplies the nilpotent sum in explicit local monomial quotients. It tests attainment and its literal multinomial coefficient, collided maxima, derivative divisibility, local dimensions and the $h=1$ branch, in normal and optimized Python with deliberate-failure controls. These finite tests do not replace the general Lean proofs or the written polynomial arguments. The final PR record identifies the exact completed run; no pending run is a certificate.

S18 The coordinating Zeta contribution: determinant losses and spectral coupling

Source attribution. Complete mathematical note supplied by the coordinating Zeta task on 13 September 2026. The direct sum of the two exact identities retains the original quotient metric, phase, spectral coupling and compression remainder. Its full formulas, domains, endpoint scope and Sylvester comparison are reproduced as supplied. The integrated bridge gives the same join with its complete derivation. No asymptotic estimate or novelty claim is inferred.

Exact revision. 71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370

Source file. sources/web_toda_parent_join/PARENT_EXACT_JOIN.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370

Formal status. Complete written join of Toda (28) and frozen exterior identity E8; no new Lean execution.

S18.1 Exact join: determinant losses and spectral coupling

This short continuation joins two already written identities on the same finite arithmetic packet and its original least-norm metric. It does not replace that metric or claim an asymptotic estimate. It is a direct algebraic consequence of the supplied Toda note (28) and the frozen companion reader's exterior equality calculation (E8); no novelty claim or Lean execution.

S18.1.1 Original objects and notation

Fix the existing nonempty packet, its tensor degree k , cyclic algebra $C = C[S]/\chi$, $q = \deg \chi$, $A = M_S$, and an admitted degree $N \geq q$. All quantities below are at the original observation $\theta = 0$. Retain G_N , the least-norm quotient Gram, and the adjoint $A^\sharp = G_N^{-1} A^* G_N$. Write $H_N = A^\sharp + A - kI$. Its eigenvalues are $+\epsilon_N$, $-\epsilon_N$, and zero, with the multiplicities from the rank-two calculation.

Let V_+ be the full generalized right-half-plane spectral subspace of A , and P_N its G_N -orthogonal projector. Its positive excess is

$$L = \sum_{\text{Re } \lambda > k/2} \ell_\lambda (2 \text{Re } \lambda - k).$$

Consider the nonempty V_+ case, so $L > 0$ and $\epsilon_N > 0$ by the proved exterior trace bound. Let u_+ , u_- be G_N -unit eigenvectors of H_N at $+\epsilon_N$ and $-\epsilon_N$. The coordinate injection $E_N : C^2 \rightarrow C$ sends (z_+, z_-) to $z_+ u_+ + z_- u_-$. Set $M_N = E_N^\sharp P_N E_N$. Thus $0 \leq M_N \leq I_2$. These eigen-coordinates describe a compression only; they do not replace the arithmetic metric.

Use $C_N = P_N A (1 - P_N) : V_+^\perp \rightarrow V_+$ for the actual off-diagonal coupling. The symbol C_N is used here to avoid confusing this coupling with the Toda note's operator $T = (A - kI/2)/i$. Every Hilbert-Schmidt norm below uses the original restricted G_N metrics.

For the scalar quantities retain

$$V_N = \det G_N = D_{(N+1)}/B_{(N-q+1)}, \quad v_N = \log(V_{(N-1)}/V_N), \quad s_N = (v_N + v_{(N+1)})/2, \quad t_N = (v_{(N+1)} - v_N)/2, \quad a_{(N+1)} = \omega_{(N+1)}/\omega_N, \quad \phi_N = \sigma - \partial_\theta \log V_N|_{\theta=0}, \quad \sigma = \text{Tr}((A - kI/2)/i).$$

Here D and B are exactly the source and relation Hankel determinant sequences in the supplied note, not alternative free sequences.

S18.1.2 Complete loss identity

The Toda identity gives

$$a_{(N+1)} \sinh(s_N)^2 - \epsilon_N^2 = a_{(N+1)} (\exp(t_N) - \cosh(s_N))^2 + \phi_N^2.$$

The companion's E8 gives, on the same original metric,

$$\epsilon_N^2 - L^2 = \|C_N\|_-(HS, G_N)^2 + \epsilon_N^2 (\det M_N + \det(I_2 - M_N)).$$

Adding these equalities cancels precisely ϵ_N^2 and proves

$$a_{(N+1)} \sinh(s_N)^2 - L^2 = a_{(N+1)} (\exp(t_N) - \cosh(s_N))^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2 + \epsilon_N^2 (\det M_N + \det(I_2 - M_N)).$$

All four terms on the right are nonnegative. The first two are the successive-volume imbalance and the retained phase. The last two are the original spectral-subspace coupling and the compression remainder. There is no cancellation between them.

The exact Sylvester morphism is also retained. In the G_N -orthogonal decomposition $C = V_+ \text{ plus } V_+^{\perp}$, write $A = [[B_N, C_N], [0, D_N^{\text{op}}]]$, and the CRT spectral idempotent as $Q = [[I, X_N], [0, 0]]$. The identity $AQ = QA$ is exactly

$$B_N X_N - X_N D_N^{\text{op}} = C_N.$$

Its inverse is $\int_0^\infty \exp(-t B_N) Y \exp(t D_N^{\text{op}}) dt$; the strict real spectral separation and the complete finite nilpotent polynomial factors give convergence. Consequently the coupling term controls $\|Q - P_N\|$ by the actual smallest singular value of this Sylvester map, as in E14-E18. No uniform bound for that singular value is inserted here.

S18.1.3 What this adds to the two-step inequality

Let $R_N = V_{(N-1)}/V_{(N+1)} = \exp(2s_N)$. Dropping only the imbalance square and the nonnegative compression remainder gives the stronger finite inequality

$$a_{(N+1)} (R_N - 1)^2 / (4 R_N) \geq L^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2.$$

Since $s_N \geq 0$, this is equivalently

$$\log R_N \geq 2 \operatorname{arsinh}(\sqrt{L^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2} / \sqrt{a_{(N+1)}}).$$

For a full quartet at zero tilt, the supplied symmetry proves $\phi_N = 0$; the coupling term is still present. If that coupling is nonzero, the original lower bound with L alone is strictly sharpened at that N . Nothing here supplies uniform strictness or large- k decay. The retained endpoint $N = q-1$ uses the source note's separate endpoint formula rather than these preceding-volume ratios.

The empty selected subspace has $L = 0$ and should use the Toda identity directly; no positive-eigenline construction is asserted when $\epsilon_N = 0$.

S18.1.4 Evidence and reading scope

The parent read the complete supplied NOTE.tex, including its sections 1-10, references and explicit first-degree calculation, directly from `Tau_Toda_Volume_Control_2026-09-12.zip` (SHA-256 80b365233ba5ebaf1b52aeea369a7771d74bf774dbb4416f10a56503321b8f41). The companion E8/E14-E18 source is pinned by GitHub commit 161089942cce70a09fbe441d08ba4d36c1fd810c, under `workbenches/splitzero-tandem/continuations/20260912-cyclic/tex/` `exterior_trace_equality_continuation.tex`. The displayed join was checked by literal addition of the two equalities. The independent delivery/checker review is separate in `toda_review/`.

Primary-source context checked in this turn: DLMF 5.11.9 for gamma vertical-strip decay, and DLMF 18.2(ix), especially 18.2.30 and the following numerical-conditioning warning. The supplied 1997 Toda paper was verified at publisher abstract/metadata scope; its full proof was not fetched. These references identify the classical ingredients, not a prior source for this arithmetic specialization.

S19 Gamma convolution descent and the original arithmetic coefficient correction

Source attribution. Complete original Gamma Convolution Descent Markdown note delivered on 13 September 2026. Its archive and alternate complete TeX note are retained byte-for-byte. The full local coefficient chapter proves the exact coordinate maps, finite derivative indices, cochain primitive and transform domains alongside this unmodified source witness. Source reports about remote revisions and computations retain the verification scope of their original receipts.

Exact revision. 886c82e2562d9b4846adb44640d28a34552e38eb2518a3cd47f90dbbedd95e5b

Source file. `sources/web_gamma_convolution_delivery/Tau_Gamma_Convolution_Descent/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 886c82e2562d9b4846adb44640d28a34552e38eb2518a3cd47f90dbbedd95e5b

Formal status. Complete source witness. Finite checker records are separately retained. No new local Lean execution is claimed.

S19.1 1. Scope and the inspected collaboration state

This is a mathematical continuation of **Tau Toda Volume Control** and **Tau Exterior Trace Amplification**. It retains their arithmetic source, quotient, scaling action, full packet multiplicities, and canonical minimum-norm metric. The new calculation links those objects to the gamma reference that is already in the repository. The gamma orthogonal polynomials and their general convolution identities are classical. The contribution here is the explicit arithmetic coefficient transform, its source and quotient maps, its quantitative bounds, and the resulting exact substitution into the existing Toda/exterior estimate.

The GitHub read was pinned to main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f. The separate formalization was read at PR #22 head 811210d24b80813a08972ca23f919db015383137. At that inspected revision the PR explicitly said strict verification was still in progress. No successful certificate for that revision is inferred here. Its invariant-filtration trace budget is retained as the other session's result, not presented as a new deduction of this note.

The main-branch gamma source read was `workbenches/splitzero-tandem/continuations/20260912-stieltjes/tex/theta_gamma_reference.tex`, equations TG.2–TG.14. It already constructs the exact one-factor gamma measure, its constants, its monic polynomials and its arithmetic multiplier. We extend that specified reference through the sum pushforward at every tensor degree.

There is no new RH proof or arithmetic asymptotic assertion here. Theorems below have written proofs. The checker supplies finite exact regression evidence, not analytic interval certification and not Lean execution. No inherited file or other-session branch is modified.

S19.2 2. The original arithmetic source and its split maps

Retain

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

with the original maps

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The target of the first quotient is infinite. Its structural base is the same absolute pointed base $\mathfrak{b}_\tau$. The supported scalar and the external absorber have not changed.

The source complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x,$$

where V consists of the original even Schwartz functions with both original moments zero, and $\mathcal{B}$ has the original rapid decay at both multiplicative ends with all D -derivatives. Keep

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Fix a nonempty finite packet $h(s) = \prod_\rho (s - \rho)^{m_\rho}$ of actual zeros, with every selected order complete. For reflection statements require the same dagger stability as the source. Write

$$v_h = g/h, \quad \mathcal{M}F_h = v_h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int w_h(t) dt. \quad (2)$$

The cyclic tensor source and its maps remain

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad C_{h,k} = \mathbb{C}[S]/\chi_{h,k}, \quad A = M_S, \quad (3)$$

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi,$$

where $S = s_1 + \cdots + s_k$ and $\eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1$, with $v_h = j_h(g/h)$. All local units remain in this map.

At polynomial degree N , the original relation map is

$$M_\chi : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad P \mapsto \chi P, \quad q = \deg \chi, \quad (4)$$

and its image under $\mathcal{V}_{h,k}$ consists of the already constructed theta boundaries. Every norm construction below concerns this same map and its quotient. Its squared-norm weight $|\chi|^2$ does not mean that the next conormal ideal is (χ^2) : that ideal remains the exact pullback of the original I^r .

For $h = 1$, the analytic source (2) still exists, but the finite arithmetic quotient is the zero module. Its split lift has e and τ . No nonzero packet or Gram inverse is inferred from that analytic branch.

S19.3 3. Carry the repository's gamma reference through every tensor power

For a fixed real $\lambda > 0$ define exactly

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad d\sigma_\lambda(t) = r_\lambda(t)dt, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda). \quad (5)$$

The repository proves, by the beta integral and Fourier inversion,

$$\int e^{iyt} d\sigma_\lambda(t) = c_\lambda (\cosh y)^{-2\lambda}. \quad (6)$$

Its factor 2 comes from $t = 2u$ in the beta integral. Thus c_λ , rather than 1, is the mass. Fourier uniqueness for finite measures gives the all-tensor identity

$$r_{\lambda,k} := r_\lambda^{*k} = \kappa_{\lambda,k} r_{k\lambda}, \quad \kappa_{\lambda,k} = \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (7)$$

In particular the mass of the sum measure is c_λ^k . For $\lambda = 1/4$ this reads

$$r_{1/4,k}(u) = \frac{2^{k-1} \pi^{k/2}}{\Gamma(k/2)} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad \int r_{1/4,k} = (2\pi)^{k/2}. \quad (8)$$

The literal coordinate map is $(t_1, \dots, t_k) \mapsto u = \sum_i t_i$; u is not replaced by an average or a rescaled variable. The original arithmetic scaling coordinate is still $S = k/2 + iu$.

For real $|\theta| < \pi/2$, analytic continuation of (6) gives

$$Z_{\lambda,k}^\Gamma(\theta) := \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}. \quad (9)$$

The exponential integrability follows either from the original beta-contour estimate or the gamma vertical-strip bound. This justifies all finite differentiated moment integrals used below.

S19.4 4. The sum projection and its complete polynomial complement

Define actual Hilbert spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du).$$

Their isometric inclusion is

$$U_\Sigma : \mathcal{H}_\Sigma \rightarrow \mathcal{H}_k, \quad (U_\Sigma f)(\mathbf{t}) = f(t_1 + \dots + t_k). \quad (10)$$

For $k \geq 2$, its adjoint has the explicit disintegration formula

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (11)$$

Its denominator is the actual fibre mass. Fubini proves $C_\Sigma U_\Sigma = 1$ and $C_\Sigma = U_\Sigma^*$. For $k = 1$ both maps are the identity. Put $P_\Sigma = U_\Sigma C_\Sigma$ and retain

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n. \quad (12)$$

These are inverse Hilbert isometries onto $\mathcal{H}_\Sigma \oplus \ker C_\Sigma$. The second component is not discarded. At each original support label ℓ , their reconstructed maps are $(\ell, F) \mapsto (\ell, C_\Sigma F)$ and $(\ell, F) \mapsto (\ell, (1 - P_\Sigma)F)$. A nonzero normal vector has first image $(\ell, 0)$, while its second image retains that vector. Both maps send external absence to external absence; their paired inverse is still (12). Norm and Gram observations have their actual quadratic types, not the type of a linear quotient: $\text{Gram} : \text{Hom}_\mathbb{C}(E, \mathcal{H}) \rightarrow \text{Herm}(E), R \mapsto R^* R$.

S19.4.1 4.1 Exact action on every product-polynomial degree

Let $b_n^{(\lambda)}(t)$ be the repository's monic gamma polynomial, with

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}.$$

Set $\alpha = 2\lambda$. Its complete norm and recurrence are

$$\|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \quad (13)$$

The polynomials are not divided by their norms. Multiplying the k generating functions proves the literal addition formula

$$b_n^{(k\lambda)}\left(\sum_i t_i\right) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \quad (14)$$

For a multi-index $\mathbf{n}$, put $n = \sum_i n_i$. The exact sum projection is

$$C_\Sigma \left(\prod_i b_{n_i}^{(\lambda)}(t_i) \right) = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (15)$$

Proof. Pair the product against (14), in the actual product measure. Orthogonality makes the answer zero unless the sum degree is n , and at degree n the answer is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$. The squared norm of the sum polynomial is $c_\lambda^k n! (k\alpha)_n$. Their quotient gives (15).

These tests identify the whole conditional projection, not merely its finite polynomial moments: the gamma sum measure has exponential moments, so its polynomials are dense in L^2 . To verify that last statement, an L^2 vector orthogonal to all polynomials defines a finite measure whose Fourier transform is analytic on a strip by Cauchy–Schwarz and exponential integrability. All derivatives at zero vanish; analytic uniqueness and uniqueness of Fourier transforms give the zero vector.

The retained normal polynomial is

$$n_{\mathbf{n}} = \prod_i b_{n_i}^{(\lambda)}(t_i) - \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}\left(\sum_i t_i\right).$$

Its exact squared norm is

$$\|n_{\mathbf{n}}\|^2 = c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right]. \quad (16)$$

Thus every degree, every factorial, and the relative component have an explicit map and norm. This is the classical Meixner–Pollaczek addition mechanism, applied with the source's exact scale and masses; it is not a claim to have discovered a new general family of orthogonal polynomials.

S19.5 5. Insert the complete arithmetic multiplier, not a substituted gamma measure

Define the complex amplitude and its squared modulus by

$$A_{h,\lambda}(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda}(t) = |A_{h,\lambda}(t)|^2, \quad w_h = B_{h,\lambda} r_\lambda. \quad (17)$$

The entire quotient g/h supplies values at the selected zeros. No pole is inserted into the multiplier at those locations. The full complex amplitude, not only its modulus, remains in the source comparison

$$P \mapsto \prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} A_{h,\lambda}(t_i) P(k/2 + i \sum_i t_i). \quad (18)$$

This is the original Mellin image of $\mathcal{V}_{h,k}P$.

The scalar multiplier on the sum line is

$$B_{h,\lambda;k} = C_\Sigma(B_{h,\lambda}^{\otimes k}).$$

The exact pushforward identity is

$$\boxed{m_{h,k}(u) = w_h^{*k}(u) = r_{\lambda,k}(u) B_{h,\lambda;k}(u)}. \quad (19)$$

This is a norm observation of the original source; it does not project the original finite arithmetic module or change its trace. The maps in (3), including the full unit in η , remain attached to every polynomial. The full field $B_{h,\lambda}^{\otimes k}$ is recoverable from (12), so its relative complement is retained too.

S19.5.1 5.1 A bounded multiplier for every full quartet

For $\lambda = 1/4$, the original formula is

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (20)$$

The minus sign and phase are still in (18). The elementary continuation formula

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx$$

gives, on this line,

$$|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|).$$

Let $d = \deg h \geq 3$ and take $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. For $|t| \geq T_h$, $|h(1/2 + it)| \geq (|t|/2)^d$, and direct substitution gives

$$B_{h,1/4}(t) \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d}.$$

Consequently the finite, specified constant

$$C_h = \max \left\{ \sup_{|t| \leq T_h} |A_{h,1/4}(t)|^2, \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} T_h^{6-2d} \right\}$$

satisfies $0 \leq B_{h,1/4} \leq C_h$. The compact supremum uses the analytic quotient, not the expression with a removable denominator left untreated. A full nonreal quartet has $d = 4m \geq 4$, so this applies to the entire packet used in the exterior amplification, with constants allowed to depend on that fixed packet.

The positive integral in (11) now proves

$$\boxed{0 < B_{h,1/4;k}(u) \leq C_h^k \quad (k \geq 2), \quad 0 \leq m_{h,k}(u) \leq C_h^k r_{1/4,k}(u).} \quad (21)$$

Strict positivity for $k \geq 2$ follows because the one-factor multiplier only vanishes at a discrete set; its excluded hyperplanes have zero fibre measure. For $k = 1$ retain its actual isolated zeros.

S19.5.2 5.2 The other packet sizes and the empty analytic source

For a chosen reference below, C_h always denotes an explicitly specified upper bound for that reference's $B_{h,\lambda}$. An explicit alternate reference is available without changing the arithmetic source. For integer $L \geq 0$,

$$A_{h,1/4+L}(t) = \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}. \quad (22)$$

Its product is a known nonvanishing gamma-shift factor on the integration line. Taking $L \geq \max(0, 3 - d)$ makes its squared modulus bounded by the same compact/tail argument. Equations (17)–(19) connect both references to the identical original w_h ; no coordinate or original mass is reassigned.

For the actual seed $h = 1$, choosing $L = 3$ gives the explicit all-real estimate

$$\boxed{B_{1,13/4}(t) \leq 1024/\sqrt{\pi}.} \quad (23)$$

Indeed the gamma-shift denominator is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, and $|\zeta(1/2 + it)| \leq 4\sqrt{t^2 + 1/4}$. The remaining fraction is bounded by one since $(t^2 + 1/4)^2 \leq (t^2 + 25/4)(t^2 + 81/4)$.

This is an analytic estimate for a nonzero seed. Its finite spectral quotient is still zero, as required by the merged correction.

S19.6 6. Exact arithmetic coefficients at every tensor degree

Use a reference for which $B = B_{h,\lambda}$ is bounded, as constructed above. Define its actual coefficients

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_\lambda j! (\alpha)_j}, \quad \mathcal{A}_h(z) = \sum_{j \geq 0} (\alpha)_j c_{h,j} z^j. \quad (24)$$

The series is used coefficientwise; no convergence radius for this formal series is stipulated. Each coefficient is an actual one-factor arithmetic integral. Define

$$d_{h,k,n} = [z^n] \mathcal{A}_h(z)^k.$$

Then the complete $L^2(r_{\lambda,k} du)$ expansion is

$$\boxed{B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u).} \quad (25)$$

Proof. Pair $B^{\otimes k}$ with the finite addition formula (14). Each one-factor pairing is $c_\lambda j! (\alpha)_j c_{h,j}$. The multinomial coefficient cancels the displayed factorials and gives

$$\int B_{h,\lambda;k} b_n^{(k\lambda)} r_{\lambda,k} = c_\lambda^n n! [z^n] \mathcal{A}_h(z)^k.$$

Divide by the exact norm $c_\lambda^k n! (k\alpha)_n$. Completeness gives (25). No exchange of an infinite product of formal series with an integral was needed: every coefficient used a finite polynomial identity.

At degree zero,

$$c_{h,0} = \mu_h / c_\lambda, \quad d_{h,k,0} = (\mu_h / c_\lambda)^k,$$

so (25) recovers exactly $\int m_{h,k} = \mu_h^k$.

The retained relative residual is

$$\mathcal{N}_{h,k} = B^{\otimes k} - U_\Sigma B_{h,\lambda;k}.$$

Its whole norm is calculated by Pythagoras:

$$\|\mathcal{N}_{h,k}\|^2 = \left(\int B(t)^2 d\sigma_\lambda(t) \right)^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (26)$$

The difference is nonnegative because it is the norm of the explicitly retained vector. It is not set to zero by calling the sum map a pushforward.

S19.6.1 6.1 Finite exactness and approximation errors

Let $B_L = \sum_{j=0}^L c_{h,j} b_j^{(\lambda)}$ and $B_{k,L} = C_\Sigma(B_L^{\otimes k})$. Projection contraction and the tensor telescoping identity give

$$\|B_{h,\lambda;k} - B_{k,L}\| \leq k \|B\|^{k-1} \|B - B_L\|. \quad (27)$$

These norms have the respective original reference measures. The proof uses $\|B_L\| \leq \|B\|$ and expands the tensor difference into k terms. Both sides retain the full mass factors.

More strongly, the moments of $B_{k,L} r_{\lambda,k}$ against every polynomial of degree at most L are **exactly** those of $m_{h,k}$, since coefficient $n \leq L$ in (25) uses only $c_{h,j}$ with $j \leq n$.

It follows that the entire degree- N source Gram, and its degree- N relation Gram, are determined exactly by $c_{h,0}, \dots, c_{h,2N}$. The relation integrands are also degree at most $2N$. Their tensor dependence is the explicit finite coefficient power in (24), not a fresh k -dimensional integral.

A truncated $B_{k,L}$ need not be nonnegative pointwise. Its relationship to the true positive metric is the exact finite-moment equality just proved when $L \geq 2N$, or the error estimate (27) and Cauchy-Schwarz for the specified polynomial tests when $L < 2N$. Positivity is not inferred from a plotted truncated density.

S19.6.2 6.2 Actual seed coefficients, and their tensor cross term

The supplied numerical driver evaluates the original theta seed on $x \geq 1$, using its inversion symmetry. With $L = D - 1/2$, the even spectral moments are $\mu_{2r} = 2 \int_1^\infty |L^r f_0(x)|^2 dx$. It constructs their source polynomials by $q_0(z) = 4z^2 - 6z$ and $q_{r+1} = 2z(q_r - q'_r) - q_r/2$, and integrates the integer double sums using upper incomplete gamma integrals. This is a numerical check, not a validated interval computation.

For $h = 1$ and the bounded-multiplier reference $\lambda = 13/4$, the 50-digit, cutoff-6 and 70-digit, cutoff-8 calculations agree in the displayed values:

Quantity	Numerical value
μ_0	1.279007247846485140479533592267
μ_2	13.055549302570558435392684658123
μ_4	383.274341717364242554407237479131
μ_6	17946.5273016052076235667646919834
$c_{h,0}$	0.201056688692785865809916163220
$c_{h,2}$	0.007645442999486401188133851481
$c_{h,4}$	-0.000030516920803762043746724840

The reference mass is $c_{13/4} = 6.361426004587219292629853548733 \dots$, so it has not been assigned the actual mass μ_0 or the value one. The negative fourth expansion coefficient is retained; a positive density need not have positive coefficients in this orthogonal basis.

For every even arithmetic multiplier, the first tensor terms from (25) are exactly

$$d_{h,k,2} = k c_{h,0}^{k-1}(\alpha)_2 c_{h,2},$$

$$d_{h,k,4} = k c_{h,0}^{k-1}(\alpha)_4 c_{h,4} + \binom{k}{2} c_{h,0}^{k-2}((\alpha)_2 c_{h,2})^2.$$

Both terms in the fourth coefficient remain. These are coefficients of the exact arithmetic source norm; the empty packet still supplies no nonzero finite spectral quotient. Agreement of two numerical truncations is not a bound on their shared truncation or rounding error.

S19.7 7. Feed the result into the two original Toda determinants

Put $\alpha_k = 2k\lambda$. The full gamma source determinants are explicit:

$$\mathfrak{D}_n^\Gamma(\theta) = c_\lambda^{kn} \prod_{j=0}^{n-1} j! (\alpha_k)_j (\cos \theta)^{-n(\alpha_k+n-1)}. \quad (28)$$

At $\theta = 0$ this is the product of the monic norms from (13) and (7). One direct proof for all real θ starts with $\mathfrak{D}_0^\Gamma = 1$ and $\mathfrak{D}_1^\Gamma = c_\lambda^k \cos^{-\alpha_k} \theta$. Apply the already-proved Hankel-Toda identity recursively; the second derivative of the logarithm of the right side is $n(\alpha_k + n - 1) \sec^2 \theta$. Substitution proves the next determinant with all constants. Strict positivity makes the recurrence division valid.

In particular

$$\omega_n^\Gamma(\theta) = c_\lambda^k n! (\alpha_k)_n (\cos \theta)^{-(\alpha_k+2n)}, \quad a_n^\Gamma(\theta) = n(n + \alpha_k - 1) \sec^2 \theta. \quad (29)$$

The arithmetic source determinant is the previous $\mathfrak{D}_n$. The reference relation determinant is

$$\mathfrak{B}_n^\Gamma(\theta) = \det \left[\partial_\theta^{i+j} \left\{ \bar{\chi}(k/2 - i\partial_\theta) \chi(k/2 + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \right\} \right]_{i,j < n}. \quad (30)$$

It is the Gram of the **same** multiplication map M_χ . This is an explicit finite expression in gamma constants, $\cos \theta$, $\tan \theta$, and the original coefficients of χ .

Define positive scalar corrections, including the empty-index value one,

$$X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_n = \mathfrak{B}_n / \mathfrak{B}_n^\Gamma.$$

The source-boundary determinant theorem now factors as

$$\boxed{V_N = V_N^\Gamma T_N, \quad V_N^\Gamma = \frac{\mathfrak{D}_{N+1}^\Gamma}{\mathfrak{B}_{N-q+1}^\Gamma}, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}.} \quad (31)$$

This retains the entire arithmetic correction in both numerator and denominator. It is not an assertion that the arithmetic quotient metric equals the reference metric.

S19.7.1 7.1 The representative map between the two metrics

Use the same finite polynomial source. Let B_χ be the injective matrix of its original relation columns, and let R_N^Γ be the least-norm lift for the gamma observation. Let M_h be the actual arithmetic source Gram. Then the actual lift is

$$\boxed{R_N = R_N^\Gamma - B_\chi (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_N^\Gamma.} \quad (32)$$

For an empty boundary block the correction is the zero map. The proof is direct: the correction is an old relation, so the quotient remains the identity; multiplication by $B_\chi^* M_h$ gives zero. These two properties characterize the arithmetic least-norm lift.

Applying $\mathcal{V}_{h,k}$ to (32) supplies the original theta primitive of the difference. In the reconstructed support carrier, the difference maps to the supported zero in its fibre. It does not disappear from the source metric or from its boundary primitive. The comparison keeps the complete unit in (3).

S19.7.2 7.2 A genuine source-side upper bound

From (21), for every admitted polynomial,

$$\|P\|_h^2 \leq C_h^k \|P\|_\Gamma^2.$$

Evaluate this inequality on $R_N^\Gamma u$ and then minimize on the arithmetic side. This proves

$$\boxed{G_N \preceq C_h^k G_N^\Gamma, \quad 0 < T_N \leq C_h^{kq}.} \quad (33)$$

The second assertion is the determinant inequality in dimension q . It has the quotient dimension, rather than the full polynomial-source dimension, in its exponent. The corresponding source estimate is $0 < X_n \leq C_h^{kn}$, and similarly $0 < Y_n \leq C_h^{kn}$.

These are one-sided estimates. They do not bound the consecutive ratio T_{N-1}/T_{N+1} , which is the quantity required by the Toda upper bound. Its exact map to that estimate is the following.

S19.8 8. The current upper-control expression, with all reference costs exposed

Define

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

At the original observation $\theta = 0$, (29)–(31) give

$$\boxed{a_{N+1} = (N+1)(N+\alpha_k)Q_N, \quad \mathcal{R}_N = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}.} \quad (34)$$

The preceding exact phase identity is also retained through

$$\ell'_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N.$$

Thus its finite upper bound becomes exactly

$$\epsilon_{h,k,N} \leq \sqrt{(N+1)(N+\alpha_k)Q_N} \frac{\mathcal{R}_N^\Gamma T_{N-1}/T_{N+1} - 1}{2\sqrt{\mathcal{R}_N^\Gamma T_{N-1}/T_{N+1}}}. \quad (35)$$

The nonnegative numerator uses the original monotonicity of the actual quotient minimum, not a presumed monotonicity of T_N . When this inequality is coarse, the inherited equality with the volume-imbalance square and the phase square remains the sharper calculation.

For the full quartet one can keep $\lambda = 1/4$, so $\alpha_k = k/2$. The gamma source recurrence is now explicit at every degree; the remaining arithmetic dependence is in Q_N and the relative source-boundary correction T_{N-1}/T_{N+1} . The reference relation term still contains the full spectral annihilator χ . It is not bounded by pretending its roots are on the integration line.

The exterior theorem remains

$$L_{h,k} \leq \epsilon_{h,k,N}, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

Neither (33) nor (35) proves a subcubic bound. Equation (35) identifies exactly which degree-to-degree arithmetic corrections must be controlled next. All entries of those finite determinants can now be generated by the all- k coefficient transform (25), with the source-boundary comparison map (32) retained.

S19.9 9. A proved all-degree tail estimate for the actual arithmetic moments

The multiplier bound gives more than a change of coordinates. For $a, b > 0$ with $a + b < \pi/2$, every integer $r \geq 0$ and $T \geq 0$,

$$\int_{|u|>T} |u|^r m_{h,k}(u) du \leq 2C_h^k c_\lambda^k r! a^{-r} e^{-bT} (\cos(a+b))^{-2k\lambda}. \quad (36)$$

Proof. For $x \geq 0$, $x^r \leq r! a^{-r} e^{ax}$. On the indicated tail, $1 \leq e^{b(x-T)}$. Apply (21) and integrate. Since the reference density is even,

$$\int e^{(a+b)|u|} r_{\lambda,k}(u) du \leq Z_{\lambda,k}^\Gamma(a+b) + Z_{\lambda,k}^\Gamma(-a-b).$$

Equation (9) gives (36). This estimate retains the original mass and the complete tensor cost. It is a theorem with specified constants, not a claim that a floating-point quadrature has validated them numerically.

For a source or relation moment whose integrand is a polynomial $P(u) = \sum_j p_j u^j$, sum $|p_j|$ times (36). In particular this controls every entry of a degree- N source and relation Gram, including the full $|\chi|^2$ factor. It is then combined with an independently certified compact-interval quadrature and the existing matrix inverse/error inequalities; it does not certify that quadrature by itself.

This is an explicit tail interface for the actual arithmetic input at all degrees and tensor powers. For the empty analytic seed, (23) gives a completely stated bound without locating any zeta zero. For a nonempty packet the compact constant in C_h must be enclosed using the actual analytic quotient.

S19.10 10. What has and has not been concluded

The constructed chain is

- original split-supported polynomial source
- full gamma product fibre with arithmetic amplitude
- sum density and retained relative component
- exact all-tensor arithmetic coefficients
- the same source–boundary Toda determinant ratio.

The finite coefficient projection, polynomial degree, all masses, all gamma factors, the arithmetic unit, and the original theta primitives are explicit. The same finite quotient is used throughout; an existing relation becomes e only under its original quotient map, while the source norm and all derivative data remain available before that map. No new scalar-zero adjunction is performed at a tensor or relation level.

There is now a proved bound for the full arithmetic sum density, explicit closed-form reference determinants at every tensor degree, an exact finite coefficient algorithm, and a rigorous moment-tail bound. There is not yet a uniform small bound for the consecutive arithmetic determinant correction. The upper-estimate problem is sharpened to that specific comparison rather than being claimed solved by gamma convolution or by positivity of individual source Grams.

S19.11 References and exact source roles

- Repository main at `fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f`, `tex/theta_gamma_reference.tex` in the 20260912-stieltjes continuation, TG.2–TG.14: original one-factor gamma reference, arithmetic amplitude, constants and polynomial norms. Only the stated TeX window was read in this review; no complete reader audit is claimed.
- PR #22 head `811210d24b80813a08972ca23f919db015383137`, `workbenches/tau-exterior-trace-formal/RESEARCH_NOTE.md`: current exterior and filtration formalization work, with its separate verification scope. Its reported in-progress status is not promoted to a completed certificate.
- Delivered `Tau_Toda_Volume_Control/NOTE.tex`: original source and relation determinants, quotient metric, phase derivative and finite upper inequality. Its supplied manifest was independently checked here.
- NIST Digital Library of Mathematical Functions, §§18.22 and 18.23: classical Meixner–Pollaczek recurrence and generating function. The formulas here retain the repository’s original $t/2$ scale and all factorials.
- Koelink, H. T., and Van der Jeugt, J. (1998). Convolutions for orthogonal polynomials from Lie and quantum algebra representations. *SIAM Journal on Mathematical Analysis*, 29(3), 794–822. arXiv:q-alg/9607010. Background for the classical addition/convolution mechanism, not a claim that this article treats the present arithmetic quotient.

The proofs in this note establish the stated specialized maps directly. The regression checker tests finite exact polynomial and matrix consequences; it does not replace the analytic proofs or establish the outstanding uniform estimate.

S20 Arithmetic endpoint bounds: complete original norm proof

Source role. Complete original delivered witness; historical statements retain their source revision.

Exact revision. 8fa97624a55c2f1baad54e465e4637c584329f24f6c198bb5d168a443811a280

Complete source. sources/web_arithmetic_endpoint_delivery/Tau_Arithmetic_Endpoint_Bounds/RESEARCH_NOTE.md

Source SHA-256. 8fa97624a55c2f1baad54e465e4637c584329f24f6c198bb5d168a443811a280

Source metadata: author (MetaInlines)

Owner-directed SplitZero analytic continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

A uniform polynomial-norm estimate; integration with the four-volume criterion

Source metadata: title (MetaInlines)

Arithmetic endpoint bounds on the original τ -base source

S20.1 1. Result and exact source boundary

The formalization session's endpoint criterion selects an actual member of the canonical arithmetic family. This continuation proves one of its previously unproved analytic inputs. For the **unchanged arithmetic measure**, a fixed finite packet h , and integers $n \geq k \geq 3$, there is a finite constant C_h , independent of n and k , such that

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n. \quad (1)$$

The norms are the original monic polynomial norms for $m_{h,k} = w_h^{*k}$; the total mass remains μ_h^k . No root-location hypothesis is used in this bound. The proof retains a compact-source mass, a lower convolution constant, and both exponential moments. These constants are defined explicitly below; they have not been numerically enclosed.

For the packet consisting exactly of a hypothetical nonreal off-line quartet, let

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (2)$$

The existing exterior theorem gives $L_{h,k} \leq \epsilon_{h,k,N}$ at every admitted degree. Combining (1) with the other session's endpoint theorem gives

$$\min_{q_k \leq N < 2q_k} \epsilon_{h,k,N} \leq C_h q_k \sinh\left(\frac{\mathcal{B}_{h,k}}{2q_k}\right), \quad \mathcal{B}_{h,k} = \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}}. \quad (3)$$

Consequently such a quartet forces

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh}\left(\frac{\delta k}{2C_h}\right), \quad \liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 2. \quad (4)$$

The upper bound for this four-volume quantity is not proved here. In particular, (1) does not establish RH. It completes the polynomial-norm input in (3) on a specific admissible window and leaves the source/relation-volume comparison with its exact maps. Section 9 expresses that remaining quantity as two positive block log-determinants of actual new relation layers.

The finite quotient, canonical source, residue action and rank-two control are inputs from the preceding analytic notes. The new proofs in Sections 3–7 use only their explicitly defined arithmetic measure. The endpoint theorem is the formalizer's written result, not attributed to this continuation. Its finite telescopes are reported kernel-checked; the analytic estimates below have written proofs, not new Lean certificates.

S20.2 2. The base, source, and both kinds of quotient

Retain the original marked coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (5)$$

Here $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$, $e_R = 0_R^\bullet$, $p_R(\tau) = 0_R$, and $p_R(r^\bullet) = r$. The target of $p_{\mathbb{Z}}$ is infinite. The construction remains over the original pointed base $\mathfrak{b}_\tau$; no new scalar zero is adjoined at a tensor degree or at a polynomial relation layer.

Keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $Q = \mathcal{B}/\Theta V$, and $D = -x\partial_x$. The source spaces and the Fourier transform remain those of the earlier theta construction. In particular,

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}\Theta\phi_* = g. \quad (6)$$

Let h be a fixed finite product of zeros of g , with the full order of each selected zero retained, and write $d = \deg h$. Define

$$v_h = g/h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_{h,k} = w_h^{*k}. \quad (7)$$

The quotient v_h is entire, including at the selected centres. It is not the meromorphic expression evaluated before removing its actual removable factors. For $h = 1$, the same analytic definitions are meaningful, but the finite arithmetic packet is zero. Only the nonempty case is used for the finite Gram inverses below.

The source map is

$$\mathcal{V}_{h,k} : \mathbb{C}[S] \longrightarrow \mathcal{B}^{\widehat{\otimes} k}, \quad P \longmapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h. \quad (8)$$

The original cyclic module is $E_{h,k} = \mathbb{C}[S]/(\chi_{h,k})$. Its arithmetic inclusion keeps the Taylor unit:

$$\eta_{h,k}[P] = v_h^{\otimes k} P(A_k)1, \quad v_h = j_h(g/h), \quad A_k = \sum_i M_{s_i}. \quad (9)$$

The supplied source identities are

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi. \quad (10)$$

The actual norm of (8) is

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du. \quad (11)$$

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$, the monic norm is the quotient norm of the leading-coefficient map

$$0 \longrightarrow \mathcal{P}_{n-1} \longrightarrow \mathcal{P}_n \xrightarrow{\text{lc}_n} \mathbb{C} \longrightarrow 0, \quad \omega_{h,k,n} = \inf_{\text{lc}_n} \|\mathcal{V}_{h,k}P\|^2. \quad (12)$$

The arithmetic quotient is the other specified map from the same source:

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_\chi} E_{h,k} \longrightarrow 0, \quad q = \deg \chi, \quad (13)$$

for $n \geq q - 1$, with the preceding relation space zero at $n = q - 1$. Multiplication by the monic χ carries $\mathcal{P}_{n-q}/\mathcal{P}_{n-q-1}$ isomorphically to $\mathcal{P}_n/\mathcal{P}_{n-1}$, with leading coefficient unchanged. The map π_χ kills that very relation. Thus (12), (13), and multiplication by χ specify the relation between the degree-line norm and the arithmetic quotient metric. One norm is not substituted for the other.

At each admitted original support label λ , the lifted quotient is $(\lambda, P) \mapsto (\lambda, [P]_\chi)$. An original relation has image $(\lambda, 0)$; external absence has image τ . Its norm before this projection remains available through (11).

S20.3 3. A quantitative local arithmetic mass lemma

S20.3.1 3.1 Uniform polynomial growth on a fixed strip

There is a constant $C_0 \geq 1$ such that, for $|T| \geq 10$, the function

$$f_T(z) = \zeta(1/2 + iT + iz) \quad (14)$$

is holomorphic on and inside the ellipse

$$z = \frac{1}{2}(w + w^{-1}), \quad |w| = 8, \quad (15)$$

and has modulus at most $M_T = C_0(1 + |T|)^6$ there.

The ellipse has real and imaginary semiaxes $65/16$ and $63/16$. Hence its image in the s -plane lies in $-55/16 \leq \Re s \leq 71/16$ and $|\Im s - T| \leq 65/16$. The pole of ζ is outside this set when $|T| \geq 10$.

For $\Re s \geq 1/4$, use the identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (16)$$

It follows by summing the steps of $[x]$ for $\Re s > 1$ and then by analytic continuation for $\Re s > 0$, $s \neq 1$. Its integral is bounded by $1/\Re s$, giving $O(1 + |T|)$ on the indicated part of the ellipse. For $\Re s \leq 1/4$, the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s) \quad (17)$$

and the vertical-strip gamma estimate give $O((1 + |T|)^{3/2 - \Re s})$. On the stated strip this exponent is below 5. The common power 6 in (14) is therefore a valid upper bound with one fixed constant. The gamma estimate and functional equation are the classical inputs recorded in DLMF §§5.11 and 25.4; none assumes RH.

Every disc of radius one about a point of $[-1, 1]$ lies inside the ellipse. The maximum principle and Cauchy's derivative formula therefore also give

$$\sup_{x \in [-1, 1]} |f'_T(x)| \leq M_T. \quad (18)$$

S20.3.2 3.2 Propagate a nonzero value to the critical-line interval

Put

$$r_* = (3 + \sqrt{13})/2 < 4, \quad z_* = -3i/2 = \frac{1}{2}(-ir_* + (-ir_*)^{-1}), \quad c_* = \zeta(2)^{-1}. \quad (19)$$

The absolutely convergent reciprocal Dirichlet series gives $|1/\zeta(2 + iT)| \leq \zeta(2)$, so $|f_T(z_*)| \geq c_* > 0$. Let $m_T = \max_{[-1, 1]} |f_T|$.

Apply the maximum principle on $1 \leq |w| \leq 8$ to $f_T((w + w^{-1})/2)$. Equivalently subtract from its logarithmic modulus the harmonic function linear in $\log |w|$ with boundary values $\log m_T$ and $\log M_T$. Zeros are handled by the usual subharmonic maximum principle. At $w = -ir_*$ this gives

$$c_* \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}. \quad (20)$$

Here $0 < c_* < 1$ and $M_T \geq 1$. Consequently

$$m_T \geq c_*^3 M_T^{-2}. \quad (21)$$

This is the three-circles argument with every coordinate map and radius fixed. No polynomial approximation to ζ or assumption about its zeros is used.

Choose a point attaining m_T . By (18), on a one-sided subinterval of length $m_T/(2M_T) \leq 1/2$ inside $[-1, 1]$, the modulus is at least $m_T/2$. Thus

$$\int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq \frac{m_T^3}{8M_T} \geq \frac{c_*^9}{8C_0^7} (1 + |T|)^{-42}. \quad (22)$$

For $T \in [-10, 10]$, the integral in (22) is continuous and strictly positive: otherwise ζ would vanish on an interval and hence identically on its holomorphic domain. Its minimum on that compact interval is positive. Combining it with (22) proves the all-real statement

$$\boxed{\exists c_\zeta > 0 \quad \int_{T-1}^{T+1} |\zeta(1/2 + it)|^2 dt \geq c_\zeta (1 + |T|)^{-42} \quad (T \in \mathbb{R}).} \quad (23)$$

The exponent 42 is deliberately an explicit sufficient exponent, not an optimal local mean-square estimate. The proof is a classical analytic propagation argument specialized here to the needed arithmetic interval.

S20.4 4. A lower envelope for the actual convolution, after three factors

S20.4.1 4.1 Local mass of the unchanged g/h

The gamma estimate, with a positive lower constant valid also on compact sets, implies

$$|\Gamma(1/4 + it/2)|^2 \geq c_\Gamma (1 + |t|)^{-1/2} e^{-\pi|t|/2}. \quad (24)$$

Let $H_h = \prod_{\rho \in Z} (1 + |\rho - 1/2|)^{m_\rho}$, so $|h(1/2 + it)| \leq H_h (1 + |t|)^d$. The entire identity $g = hv_h$ makes this bound valid at selected centres as well: $|g| \leq H_h (1 + |t|)^d |v_h|$. The factors in (6), including $|s(s-1)|^2 = (t^2 + 1/4)^2$, then give a positive constant $c_h^{(0)}$ with

$$w_h(t) \geq c_h^{(0)} e^{-\alpha_0|t|} (1 + |t|)^{-2d} |\zeta(1/2 + it)|^2, \quad \alpha_0 = \pi/2. \quad (25)$$

Equation (25) is a weaker bound than the available factor $(1 + |t|)^{7/2-2d}$; it follows by an inequality, not by changing the density. On $[T-1, T+1]$, its exponential and polynomial weights are bounded below by $e^{-\alpha_0} e^{-\alpha_0|T|}$ and $2^{-2d} (1 + |T|)^{-2d}$, respectively. Hence (23) proves

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\alpha_0|T|} (1 + |T|)^{-B_h}, \quad B_h = 42 + 2d. \quad (26)$$

S20.4.2 4.2 The positive compact convolution factor

The function w_h is continuous, bounded, integrable, and positive outside a discrete set. Thus $m_{h,2} = w_h * w_h$ is continuous and strictly positive at every real point. Indeed its integrand is positive almost everywhere; continuity follows from translation continuity in L^1 and boundedness of the other factor. Define the **actual** compact quantities

$$b_h = \min_{|v| \leq 1} m_{h,2}(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0. \quad (27)$$

Both are determined by the original density; neither is assigned the value one. Restricting a nonnegative convolution integral gives

$$m_{h,3}(u) \geq b_h \int_{u-1}^{u+1} w_h(t) dt \geq c_h e^{-\alpha_0|u|} (1 + |u|)^{-B_h}, \quad c_h = b_h c_h^{(1)} > 0. \quad (28)$$

The omitted part is the nonnegative integral over $|v| > 1$ and remains part of $m_{h,3}$. We use a lower bound, not a new compactly supported density.

For $k \geq 3$, restrict the remaining $k-3$ variables to $[-1, 1]$ and use (28). Their sum has absolute value at most $k-3$. This gives the central analytic estimate

$$\boxed{m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(|u|+k-3)} (1 + |u| + k - 3)^{-B_h} \quad (k \geq 3, u \in \mathbb{R}).} \quad (29)$$

All mass and tensor factors are explicit. For $k = 3$ the factor ϑ_h^0 is the empty product. No statement for $k = 1, 2$ is inferred by using a negative exponent. The amplification below only needs $k \geq 3$.

S20.5 5. Upper moments and exact lower monic norms

Fix $0 < b < \pi/2$ and retain both one-factor Laplace values

$$M_h(\pm b) = \int e^{\pm bt} w_h(t) dt > 0. \quad (30)$$

They are finite by the original gamma decay. Fubini and positivity give

$$\int e^{b|u|} m_{h,k}(u) du \leq M_h(b)^k + M_h(-b)^k. \quad (31)$$

The elementary exponential-series inequality $x^j \leq j! b^{-j} e^{bx}$ for $x \geq 0$ yields

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k). \quad (32)$$

The trial polynomial is $(S - k/2)^j$, whose value on $S = k/2 + iu$ is $i^j u^j$. Its phase has modulus one and remains in the source map; the generator $A = M_S$ has not been shifted to define a different spectral problem.

For every leading coefficient of modulus one, the exact least squared norm of a degree- j polynomial on $[-L, L]$ with Lebesgue measure is

$$\mathfrak{l}_j(L) = \frac{2L^{2j+1}}{2j+1} \left(\frac{2^j (j!)^2}{(2j)!} \right)^2. \quad (33)$$

To check all constants, Rodrigues' formula gives the Legendre polynomial $P_j(x)$ with leading coefficient $(2j)!/(2^j (j!)^2)$ and squared norm $2/(2j+1)$. Its scaled monic version on $[-L, L]$ is the unique minimizer: subtract it from any competing monic polynomial, and the difference has lower degree and is orthogonal to it. For the source coordinate $S = k/2 + iu$, multiply this minimizer by the retained leading phase i^j . This proves (33) at the original type. See also DLMF §§18.3 and 18.5.

Restricting (11) to $[-L, L]$ and using (29) proves

$$\omega_{h,k,j} \geq c_h \vartheta_h^{k-3} e^{-\alpha_0(L+k-3)} (1+L+k-3)^{-B_h} \mathfrak{l}_j(L) \quad (k \geq 3, L > 0). \quad (34)$$

The observation has the explicit isometric splitting into the inside and outside interval components. Addition recovers the full source function, and their squared norms add. In the split lift each zero component is the supported zero at its label. The outside component has not been declared absent.

S20.6 6. The norm-window estimate is proved

Set $j = n$ and $L = n$ in (34), and use $j = 2n$ in (32). The exact finite bound is

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq \left[\frac{(4n)! b^{-4n} (M_h(b)^k + M_h(-b)^k) e^{\alpha_0(n+k-3)} (1+n+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2n)}. \quad (35)$$

For $n \geq k \geq 3$, this is at most $C_h n$. Here is one explicit sufficient constant, with $M_* = \max\{1, M_h(b), M_h(-b)\}$:

$$C_h = 64b^{-2} \exp(\alpha_0) 2^{B_h/2} \exp\left(\frac{B_h}{2 \exp(1)}\right) \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}. \quad (36)$$

Proof of the uniform estimate: $\binom{2n}{n} \leq 4^n$ in (33) gives $\mathfrak{l}_n(n) \geq 2n^{2n+1}/((2n+1)4^n)$. Also $(4n)! \leq (4n)^{4n}$, $M_h(b)^k + M_h(-b)^k \leq 2M_*^k$, $n+k-3 \leq 2n$, and $1+n+k-3 \leq 2n$. After taking the $2n$ -th root the factorial and monic factors contribute at most $64n$. The mass factors contribute at most the square root

in (36); the exponential contributes at most $\exp(\alpha_0)$. Finally $n^{1/n} \leq \exp(1/\exp(1))$ gives $(2n)^{B_h/(2n)} \leq 2^{B_h/2} \exp(B_h/(2\exp(1)))$. This proves (1).

The constant is intentionally not optimized. Its dependence on the fixed packet is allowed; its independence of the polynomial and tensor degrees is what is needed. Defining it via positive compact minima does not claim to have provided a numerical interval enclosure for those minima.

For a general source window the same two inequalities give, without asymptotic notation,

$$\left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \leq \left[\frac{(2(n+r))! b^{-2(n+r)} (M_h(b))^k + M_h(-b)^k e^{\alpha_0(L+k-3)} (1+L+k-3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(L)} \right]^{1/(2r)}. \quad (37)$$

S20.7 7. Compose the proved bound with the formalizer's endpoint theorem

For a nonempty cyclic packet $E_{h,k}$, let G_N be its original canonical quotient metric, $V_N = \det G_N > 0$, and $\epsilon_{h,k,N}$ the original rank-two control allowance. The new endpoint theorem supplied by PR #23 is

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \sinh \left(\frac{1}{2r} \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}} \right), \quad n \geq q. \quad (38)$$

Its proof multiplies the local Toda bounds, applies concavity of $\log \sinh x$, and uses the two exact telescopes. The zero-contraction case is handled before taking logarithms. This result is credited to the parallel formalization/research session. The proof and the distinction between checked finite components and written Jensen assembly are retained in its source.

Choose $n \geq \max(q, k)$ and $r = n$. Equations (1) and (38) yield the actual canonical selection estimate

$$\boxed{\min_{n \leq N < 2n} \epsilon_{h,k,N} \leq C_h n \sinh \left(\frac{1}{2n} \log \frac{V_{n-1} V_n}{V_{2n-1} V_{2n}} \right).} \quad (39)$$

The chosen index is a member of the existing canonical family. The source representative, Taylor unit, kernel, action and trace are not chosen again after selecting this index.

For the packet consisting exactly of the quartet in (2), $q_k \geq k$ and $L_{h,k}/q_k \geq \delta k/2$. The earlier exterior bound holds at every admitted index, including the one selected by (39). Since $\sinh$ is increasing on $[0, \infty)$, (4) follows. Moreover $\operatorname{arsinh}(ak) = \log k + O_a(1)$ for every fixed $a > 0$, which proves the stated threshold 2.

A proved upper bound with $\limsup \mathcal{B}_{h,k}/(q_k \log k) < 2$ would contradict (4). Such an upper bound is **not** an output of this note. The previous sufficient target $\mathcal{B}_{h,k} = o(q_k \log k)$ remains sufficient, but (4) specifies the larger quantitative threshold against which a future upper estimate can be compared. No repeated cost-free exterior operation is used: the multiplicity correction in the new formalization is retained.

S20.8 8. The remaining four volumes retain their finite jet maps

The previous source/relation determinant identity remains

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (40)$$

In the parallel full-jet transfer coordinates (PR #24), let $a = N - q + 1$, let F_a be its $2q$ by $2q$ raw confluent-jet matrix, and let V be its raw-derivative Vandermonde matrix, with every factorial retained. Its established identity is

$$V_N = \left(\prod_{j=a}^{a+q-1} \omega_{h,k,j} \right) \frac{\det V}{\det F_a}. \quad (41)$$

For the window $n = r = q$, the four transfer indices are exactly $0, 1, q, q+1$. Put $\Omega_j = \prod_{i=j}^{j+q-1} \omega_{h,k,i}$. Then

$$\boxed{\exp \mathcal{B}_{h,k} = \frac{\Omega_0 \Omega_1}{\Omega_q \Omega_{q+1}} \frac{\det F_q \det F_{q+1}}{\det F_0 \det F_1}.} \quad (42)$$

The common Vandermonde cancels in this explicit ratio, not inside any underlying map. The transfer matrices and their inverses still need their actual arithmetic control; the fixed width alone gives no such bound.

The empty packet has $q = 0$, $V_N = 1$ as an empty determinant, and no positive-rank inverse or quartet estimate. The analytic inequalities (23)–(37) still apply to its theta seed, but are not statements about a nonexistent finite packet.

S20.9 9. The volume window is also two explicit retained-boundary costs

There is a second useful representation of the same remaining quantity, with no reference-density change. Fix $i < j$, both admissible. In the original monic polynomial basis put

$$F_{i,j} = [b_{i+1}, \dots, b_j], \quad \Omega_{i,j} = \text{diag}(\omega_{i+1}, \dots, \omega_j), \quad b_a = [p_a]_\chi. \quad (43)$$

The kernel update is the actual sum $K_j = K_i + F_{i,j}\Omega_{i,j}^{-1}F_{i,j}^*$, where $K_i = G_i^{-1}$. The matrix determinant lemma gives

$$\boxed{\frac{V_i}{V_j} = \det(I_{j-i} + \Omega_{i,j}^{-1}F_{i,j}^*G_iF_{i,j})}. \quad (44)$$

This determinant has positive real value. Its factors are $1 + \lambda_a$, where $\lambda_a \geq 0$ are the generalized eigenvalues of the Hermitian pair $(F_{i,j}^*G_iF_{i,j}, \Omega_{i,j})$. No non-Hermitian matrix is asserted positive in an unspecified Euclidean metric.

Let $T_{i,j}$ have columns $\mathcal{V}_{h,kp_{i+1}}, \dots, \mathcal{V}_{h,kp_j}$. The original boundary-layer isomorphism is

$$\mathbf{b}_{i,j} = T_{i,j} - R_i F_{i,j} : \mathbb{C}^{j-i} \xrightarrow{\sim} \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathbf{b}_{i,j}^* \mathbf{b}_{i,j} = \Omega_{i,j} + F_{i,j}^* G_i F_{i,j}. \quad (45)$$

Its jets vanish; its highest polynomial coefficients give injectivity. Any new boundary can be reduced by these highest coefficients to an old boundary, and orthogonality then gives surjectivity. These statements prove the displayed map with the same original relation spaces, not only a dimension count. The representative update is

$$R_j = R_i + \mathbf{b}_{i,j}(\Omega_{i,j} + F_{i,j}^* G_i F_{i,j})^{-1} F_{i,j}^* G_i. \quad (46)$$

Every correction in (46) is an original theta boundary. Its split quotient image is e at the receiving fibre, and the input relation remains available in (45). Its cochain primitives are the same fixed-order divisions and tensor signs as in the previous cyclic source construction.

Apply (44) to $(i, j) = (q-1, 2q-1)$ and $(q, 2q)$. This proves

$$\boxed{\mathcal{B}_{h,k} = \sum_a \log(1 + \lambda_a^{(0)}) + \sum_a \log(1 + \lambda_a^{(1)})}. \quad (47)$$

In particular

$$\mathcal{B}_{h,k} \leq \text{Tr}(\Omega_{q-1,2q-1}^{-1} F^* G_{q-1} F) + \text{Tr}(\Omega_{q,2q}^{-1} F^* G_q F), \quad (48)$$

where each F has its own indicated source and target from (43). When this trace bound is too large, (47) retains the full logarithmic cost rather than replacing it by its linear upper bound. The analytic norm estimate in (1) does not bound (47); it supplies the other factor in (39).

S20.10 10. Deligne reading and what has actually been transferred

The six supplied S20 parts were reassembled in this turn. Their individual chunk hashes matched, and the assembled archive has 216,580,466 bytes and SHA-256 e2005bf31e1fc362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8. This verifies the supplied bytes, not every transcription claim in them.

I read the TeX for Definition 1.3.5, Lemmas 3.2.10–3.2.12, the tensor argument 3.2.13, and Corollary 3.3.6 in the nested source files. I compared those exact passages with the supplied top-level French/English frozen page records at printed pages 158, 203, and 206. No PDF text extraction or OCR was used. The source review

records a genuine inherited transcription fault: old `fr_seg2.tex` writes a weight bound ≤ 2 at 3.2.10, whereas the controlling page record says < 2 ; only the latter, with integrality, implies ≤ 1 . The old file also corrupts the explicit squared-eigenvalue step in 3.2.13. Those old files remain unmodified; they are not used as authority for these comparisons. No claim of a full independent audit of Weil II is made.

Deligne’s determinant/rank operation and the upper/dual-lower architecture remain attached to the programme’s specified exterior and residue maps. Here the additional estimate is a characteristic-zero analytic estimate for the actual source, (1), followed by the endpoint selection (39). This is not a statement that the new base satisfies Deligne’s finite-field hypotheses. The remaining volume estimate has not been inferred from purity, positivity of a Gram, or the mere existence of the support.

The new result is therefore a completed analytic component of the current route: its two endpoint monic norms have the required $O_h(q_k)$ scale on the actual quartet window. The other session’s four-volume theorem now leaves the explicitly specified relation-volume cost as the remaining quantitative input on that window.

S20.11 11. Proof and computation status

Sections 3–7 contain the new written analytic argument. Sections 8–9 contain explicit finite comparisons and a handoff for checking them using the existing quotient and matrix library. The standard complex-analysis, gamma and Legendre facts used in the argument are proved to the extent needed or referenced below. No general novelty claim is made for these classical ingredients.

The accompanying checker tests finite exact identities: coordinate factors, Legendre extremal norms, the local algebra in the interval argument, convolution moments on declared Laplace models, source quotients, block updates, raw confluent-jet endpoint factors, the volume telescope, and retained support cases. These tests do not certify the analytic inequalities by sampling them. There is no new Lean certificate and no interval-certified value of C_h .

This continuation makes no remote changes. The integration files are add-only, and the parallel formalization branches and the reader’s existing sources are left untouched.

S20.12 References and pinned inputs

- Deligne, P. *La conjecture de Weil. II*. Publications Mathématiques de l’IHÉS **52** (1980), 137–252. Supplied S20 source and current page-local records; the selected reading and discrepancies are itemized in `SOURCE_REVIEW.md`.
- NIST Digital Library of Mathematical Functions: §§25.4 (zeta functional equation), 5.11 (gamma estimates), 18.3 and 18.5 (Legendre polynomials). Public source pages consulted: <https://dlmf.nist.gov/25.4>, <https://dlmf.nist.gov/5.11>, <https://dlmf.nist.gov/18.3>, <https://dlmf.nist.gov/18.5>.
- Original programme: the delivered *Tau Gamma Convolution Descent*, *Tau Toda Volume Control*, *Tau Cyclic Sum Control*, and *Tau Exterior Trace Amplification* notes. Their definitions of the source, action, unit and canonical quotient are retained; this is not a complete replay of all their upstream analytic claims.
- Parallel endpoint criterion: PR #23, `c720f40530eed2f5969dbabe94dfd3fddc0f507f`, `workbenches/ta u-toda-volume-formal/ENDPOINT_CRITERION.md`.
- Parallel confluent transfer: PR #24, `dfcbb5cbf7fec8c9301fe242c13e741d762013e`, `workbenches/ta u-confluent-transfer/RESEARCH_NOTE.md`.
- Inspected main: `b32ca2128e0deb0eec6eafb860776a5d6a28dcbb`, after its merge of the original-Gram exterior/trace contribution.

S21 Complete supplied endpoint window product continuation

Source role. Complete original mathematical paste, with every formula and calculation.

Exact revision. f34647902fd4f6a913b5adae421c3df20bb1773902a797ec934750b02ed90ddb

Complete source. sources/endpoint_user_sources/ARITHMETIC_ENDPOINT_USER_SOURCE_1_20260913.txt

Source SHA-256. f34647902fd4f6a913b5adae421c3df20bb1773902a797ec934750b02ed90ddb

The attached four-endpoint criterion can be sharpened using its own exact radius identity. The improvement comes from multiplying the consecutive-step identities *before* discarding their imbalance terms. This preserves the fact that each interior volume contraction occurs in two adjacent control estimates.

The resulting four-endpoint bound is never weaker than the attachment's bound. It also has a simpler corollary:

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} \right)^{1/(2r)}. \quad (1)$$

This corollary uses only two quotient volumes and two monic norms. It is useful for controlling large contractions, although it need not outperform the original bound when the contractions are small.

These are deductions below from the supplied exact-radius, positivity, and monotonicity hypotheses—not an assertion that the remaining arithmetic endpoint estimates have been established. The attachment itself distinguishes that analytic obligation from the existing finite formalizations.

S21.0.1 1. An exact product identity for a whole degree window

Fix the original packet h , tensor degree k , and cyclic dimension $q \geq 1$. Work at the original observation $\theta = 0$, or at a fixed admissible tilt for which the supplied identities hold. Suppress h, k, θ from the notation.

Retain

$$V_N = \det G_N > 0, \quad \omega_N > 0, \quad \delta_N = \frac{V_N}{V_{N-1}} \in (0, 1],$$

and write

$$\phi_N = \sigma - \partial_\theta \log V_N$$

for the original phase. The source identity is

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1), \quad N \geq q. \quad (2)$$

The phase is therefore retained on the left, not set to zero.

Take a consecutive window

$$N = n, \dots, n + r - 1, \quad n \geq q, \quad r \geq 1,$$

and define

$$d_j = \delta_{n+j} \quad (0 \leq j \leq r), \quad \Omega_{n,r} = \frac{\omega_{n+r}}{\omega_n}.$$

Multiplication of (2) gives

$$\begin{aligned} \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) &= \Omega_{n,r} \prod_{j=0}^{r-1} \frac{(1 - d_j)(1 - d_{j+1})}{d_{j+1}} \\ &= \Omega_{n,r} \frac{(1 - d_0)(1 - d_r)}{\prod_{j=1}^r d_j} \prod_{j=1}^{r-1} (1 - d_j)^2. \end{aligned}$$

Since

$$\prod_{j=1}^r d_j = \frac{V_{n+r}}{V_n},$$

the exact window formula is

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (3)$$

The squared interior factors are essential: each interior contraction belongs to two consecutive radius identities. The first and last contractions occur only once.

Put

$$E_{n,r} = \min_{0 \leq j < r} \epsilon_{n+j}.$$

Then

$$E_{n,r}^{2r} \leq \prod_{j=0}^{r-1} \epsilon_{n+j}^2 \leq \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2).$$

All factors $1 - d_j$ lie in $[0, 1)$. Dropping them from the right of (3) proves (1).

This proof does not require a supremum over recurrence coefficients, an estimate for every intermediate inverse matrix, or a choice of a different Gram.

S21.0.2 2. The sharper bound from the same four endpoint volumes

For $r \geq 2$, define the three endpoint ratios

$$\alpha = \frac{V_n}{V_{n-1}}, \quad \beta = \frac{V_{n+r}}{V_{n+r-1}}, \quad t = \frac{V_{n+r-1}}{V_n}.$$

Thus

$$d_0 = \alpha, \quad d_r = \beta, \quad \prod_{j=1}^{r-1} d_j = t,$$

with $0 < \alpha, \beta, t \leq 1$. Also retain the attachment's norm factor

$$\mathcal{A}_{n,r} = \left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)}.$$

The sharpened theorem is

$$E_{n,r} \leq \mathcal{U}_{n,r}, \quad (4)$$

where

$$\mathcal{U}_{n,r} = \mathcal{A}_{n,r} \left(\frac{(1 - \alpha)(1 - \beta)}{\beta t} \right)^{1/(2r)} \left(1 - t^{1/(r-1)} \right)^{(r-1)/r}. \quad (5)$$

Only

$$V_{n-1}, \quad V_n, \quad V_{n+r-1}, \quad V_{n+r}, \quad \omega_n, \quad \omega_{n+r}$$

are needed—the same endpoint data used in the attachment. Its existing criterion combines the four volumes into one logarithmic budget; (5) retains their separate contributions.

Proof Let $s = r - 1$. For positive contractions $d_1, \dots, d_s \leq 1$ with product t ,

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (6)$$

If a $d_j = 1$, the left side is zero, so the inequality is immediate. Otherwise set $v_j = -\log d_j > 0$. The function

$$f(v) = \log(1 - e^{-v})$$

has

$$f''(v) = -\frac{e^{-v}}{(1 - e^{-v})^2} < 0.$$

Jensen's inequality gives

$$\sum_{j=1}^s \log(1 - e^{-v_j}) \leq s \log \left(1 - e^{-(v_1 + \dots + v_s)/s} \right).$$

Exponentiating and using $e^{-\sum v_j} = t$ proves (6). This is an application of the ordinary concavity form of Jensen's inequality, not a new general inequality (National Institute of Standards and Technology [NIST], n.d.-b, §1.7(iv)). (DLMF)

Insert (6) into (3):

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) \leq \Omega_{n,r} \frac{(1 - \alpha)(1 - \beta)}{\beta t} (1 - t^{1/s})^{2s}.$$

Taking the nonnegative $2r$ -th root proves (4)–(5). $\square$

For $r = 1$, there is no interior product and no fractional exponent involving $r - 1$. The separate formula is simply

$$\epsilon_n \leq \sqrt{\frac{\omega_{n+1}}{\omega_n}} \sqrt{(1 - \alpha)(\beta^{-1} - 1)}. \quad (7)$$

It retains the exact contraction contribution at that single step.

Comparison with the attached criterion The attachment proves

$$E_{n,r} \leq \mathcal{A}_{n,r} \sinh \left(\frac{\mathcal{B}_{n,r}}{2r} \right), \quad \mathcal{B}_{n,r} = \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}.$$

Its proof first replaces each exact contraction product by a hyperbolic-sine square and then applies Jensen. The new endpoint expression satisfies

$$\mathcal{U}_{n,r} \leq \mathcal{A}_{n,r} \sinh \left(\frac{\mathcal{B}_{n,r}}{2r} \right). \quad (8)$$

To see this, fix the four endpoints. Equation (6) shows that the product of the exact local contraction terms is maximized when all interior ratios equal $t^{1/(r-1)}$. Apply the attachment's local upper inequality and its Jensen inequality to that maximizing scalar sequence. Their endpoint budget is still exactly $\mathcal{B}_{n,r}$, so their upper bound dominates the maximum used in (5). Degenerate cases follow directly or by continuity.

A useful explicit comparison occurs for geometric volume contraction:

$$V_{n+j} = V_{n-1} d^{j+1}, \quad d = e^{-v} \in (0, 1).$$

Then

$$\alpha = \beta = d, \quad t = d^{r-1},$$

and (5) becomes

$$\mathcal{U}_{n,r} = \mathcal{A}_{n,r} \frac{1-d}{\sqrt{d}} = 2\mathcal{A}_{n,r} \sinh(v/2). \quad (9)$$

The previous bound is $\mathcal{A}_{n,r} \sinh v$. Their ratio is

$$\frac{\text{previous bound}}{\mathcal{U}_{n,r}} = \cosh(v/2).$$

The improvement can therefore be substantial, rather than merely a rearrangement of the same estimate.

Exact scope of sharpness Equation (5) is sharp for the **relaxed scalar problem** consisting of positive norm sequences, nonincreasing positive volume sequences, and identity (2).

For strictly contracting endpoints, choose all interior $d_j = t^{1/(r-1)}$. Write

$$f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0.$$

Set

$$\frac{\omega_{n+j+1}}{\omega_{n+j}} = \frac{\mathcal{U}_{n,r}^2}{f_j}, \quad \phi_{n+j} = 0, \quad \epsilon_{n+j} = \mathcal{U}_{n,r}.$$

The product of the specified norm ratios is exactly $\Omega_{n,r}$, so the prescribed norm endpoints are respected and equality holds.

This is **not** an assertion that every such extremizing scalar sequence arises from the arithmetic convolution measure. The additional moment, recurrence, and interpolation constraints of that measure remain relevant.

Neither the phase nor the interior imbalance needs to be forgotten With all interior contractions strictly below one, define the nonnegative concavity loss

$$\mathcal{J}_{n,r} = (r-1) \log(1 - t^{1/(r-1)}) - \sum_{j=1}^{r-1} \log(1 - d_j).$$

Equation (3) becomes the exact identity

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \mathcal{U}_{n,r}^{2r} e^{-2\mathcal{J}_{n,r}}. \quad (10)$$

Thus the additional loss in passing to the endpoint bound is explicitly recorded.

For example, when the phases have been calculated, the sharper conclusion is

$$\prod_{j=0}^{r-1} (E_{n,r}^2 + \phi_{n+j}^2) \leq \mathcal{U}_{n,r}^{2r} e^{-2\mathcal{J}_{n,r}}. \quad (11)$$

The left side is an increasing polynomial in $E_{n,r}^2$. Its unique nonnegative threshold gives a phase-sensitive bound. No phase is presumed to vanish at a general auxiliary tilt.

S21.0.3 3. The simpler corollary reduces to a ratio of two determinant sequences

The quantity in (1) admits a cancellation that reduces the number of endpoint objects further. Define

$$\Lambda_N = \frac{\omega_N}{V_N}.$$

The supplied source formulas are

$$\omega_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{D}_N}, \quad V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}.$$

They use the original source and relation Gram determinants. The ordinary monic-norm and moment-determinant framework is also recorded in DLMF §18.2(iv), (ix); the arithmetic identification of the relation determinant is the supplied construction. (DLMF)

Consequently,

$$\Lambda_N = \frac{\mathfrak{B}_{N-q+1}}{\mathfrak{D}_N}. \quad (12)$$

The source determinant $\mathfrak{D}_{N+1}$ cancels literally. Equation (1) is therefore

$$E_{n,r} \leq \left(\frac{\Lambda_{n+r}}{\Lambda_n} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{n+r-q+1} \mathfrak{D}_n}{\mathfrak{D}_{n+r} \mathfrak{B}_{n-q+1}} \right)^{1/(2r)}. \quad (13)$$

This uses two source determinants and two relation determinants. It does not identify the relation space with its quotient value.

The mass bookkeeping remains exact. Rescaling the fixed source measure by $c > 0$ would give

$$\omega_N \mapsto c\omega_N, \quad V_N \mapsto c^q V_N, \quad \Lambda_N \mapsto c^{1-q} \Lambda_N.$$

Hence Λ_{n+r}/Λ_n is unchanged. No rescaling is performed; this calculation explains the cancellation while retaining the original mass.

Connection to the confluent-transfer interface The attachment supplies

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det V}{\det F_a}, \quad a = N - q + 1,$$

with the full confluent-jet conventions retained.

It follows that

$$\Lambda_N = \frac{\det F_a}{\det V \prod_{j=a}^{N-1} \omega_j}.$$

For $a = n - q + 1$, equation (13) becomes

$$E_{n,r} \leq \left[\frac{\det F_{a+r}}{\det F_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}. \quad (14)$$

At $q = 1$, the product is empty and equals one.

Thus the simpler criterion requires **two transfer determinants**, rather than four, together with the displayed norm bands. The common confluent Vandermonde factor cancels through an explicit ratio. No modulus is inserted to conceal a phase: positivity of the total expression follows from its equality with $\Lambda_{n+r}/\Lambda_n > 0$.

A quantitative target with an explicit threshold The supplied exterior result gives

$$L_{h,k} \leq \epsilon_{h,k,N}$$

at every admitted degree. For the packet consisting exactly of the hypothetical quartet,

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} \geq \frac{\delta}{2} k q_k.$$

The exact-quartet qualification is necessary.

Therefore

$$\boxed{L_{h,k}^{2r} \leq \frac{\Lambda_{n+r}}{\Lambda_n}.} \quad (15)$$

A sufficient asymptotic estimate is now

$$\boxed{\frac{1}{2r(k)} \log \frac{\Lambda_{n(k)+r(k)}}{\Lambda_{n(k)}} - \log(k q_k) \rightarrow -\infty.} \quad (16)$$

For a more concrete sufficient condition, suppose

$$\mathcal{A}_{n,r} \leq C_h q_k$$

and, for some fixed $0 < \eta < 2$,

$$\frac{1}{r} \log \frac{V_n}{V_{n+r}} \leq (2 - \eta) \log k.$$

Then (1) gives

$$E_{n,r} \leq C_h q_k k^{1-\eta/2} = o(k q_k). \quad (17)$$

These are still estimates to prove for the actual arithmetic family. The deduction specifies their required scale; it does not derive them from determinant positivity.

S21.0.4 4. Independent calculation on a canonical polynomial fixture

I checked the product identities and the numerical improvement using a declared Gaussian fixture—not an actual zeta-zero packet.

Take the measure

$$d\mu(u) = \frac{7}{\sqrt{2\pi}} e^{-u^2/2} du,$$

of literal mass seven, and the quotient

$$C = \mathbb{C}[u]/(u^2 + 1).$$

Its monic orthogonal polynomials satisfy

$$Q_0 = 1, \quad Q_1 = u, \quad Q_{j+1} = uQ_j - jQ_{j-1},$$

with

$$\omega_j = 7j!.$$

In the quotient basis $(1, u)$, let $b_j = [Q_j]$, and construct

$$K_N = \sum_{j=0}^N \frac{b_j b_j^T}{\omega_j}, \quad G_N = K_N^{-1}.$$

These are constrained-minimum Grams calculated from the source, not assigned matrices. The resulting values include

$$G_2 = \text{diag} \left(\frac{7}{3}, 7 \right),$$

$$G_3 = \text{diag} \left(\frac{7}{3}, \frac{21}{11} \right), \quad G_4 = \text{diag} \left(\frac{42}{43}, \frac{21}{11} \right).$$

Retain the coordinate generator

$$A = \frac{1}{2}I + iT, \quad T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

For $G = \text{diag}(g_0, g_1)$, direct calculation of

$$A^*G + GA - G = i(GT - T^T G)$$

gives

$$\epsilon^2 = \frac{(g_0 + g_1)^2}{g_0 g_1}.$$

Hence

$$(\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = \left(\frac{16}{3}, \frac{400}{99}, \frac{4225}{946} \right).$$

The five required volumes are

$$(V_1, V_2, V_3, V_4, V_5) = \left(49, \frac{49}{3}, \frac{49}{11}, \frac{882}{473}, \frac{980}{1333} \right).$$

For the window $n = 2, r = 3$,

$$\Omega_{2,3} = 60, \quad \alpha = \frac{1}{3}, \quad \beta = \frac{110}{279}, \quad t = \frac{54}{473}.$$

The comparison is

$$\begin{aligned} E_{2,3} &= 2.0100756305\dots, \\ (\epsilon_2 \epsilon_3 \epsilon_4)^{1/3} &= 2.1407199197\dots, \\ \mathcal{U}_{2,3} &= 2.1666731523\dots, \\ \text{attached endpoint bound} &= 2.5178436564\dots \end{aligned}$$

Thus

$$\boxed{2.0100756305 \leq 2.1407199197 \leq 2.1666731523 < 2.5178436564.} \quad (18)$$

A Wolfram kernel independently verified the rational Grams, all three exact radius identities, and symbolic versions of the window factorization for $r = 1, \dots, 8$. The general proof is the algebra above; these finite calculations are checks, not arithmetic interval certificates or additional Lean certificates.

S21.0.5 5. What remains attached to the selected degree

For the fixed arithmetic packet and window, let N_* be the first index attaining the finite minimum. Its representative remains the existing

$$R_{N_*} : C_{h,k} \longrightarrow \mathcal{B}^{\widehat{\otimes} k},$$

with

$$J^{(k)} R_{N_*} = \eta_{h,k}, \quad q^{(k)} R_{N_*} = \sigma_h^{\otimes k} \eta_{h,k}.$$

These are the supplied source and quotient maps. Selecting a degree does not alter their arithmetic unit, nilpotent jets, or cohomology classes.

At the corresponding support label, the lifted map is still

$$(\lambda_{N_*}, c) \longmapsto (\lambda_{N_*}, R_{N_*} c), \quad \tau \longmapsto \tau.$$

An original relation is sent by the quotient to the supported zero at that label. Its pre-quotient norm remains recorded by the relation determinant in (12)–(14).

There is also a computational qualification: **endpoint data certify the existence of a suitable degree; they do not generally identify which degree minimizes the allowance.** Locating N_* still requires evaluating candidate allowances or obtaining additional information about their distribution.

The completed continuation is the exact window product (3), its sharper four-endpoint consequence (5), and the reduced determinant/transfer criteria (13)–(14). No GitHub changes or new Lean verification were performed for these deductions. The remaining arithmetic task is now expressible as the specific growth estimate (16), with the full phase and interior-contraction loss available whenever the coarse endpoint observation is insufficient.

References

National Institute of Standards and Technology. (n.d.-a). General orthogonal polynomials (§18.2). In *NIST Digital Library of Mathematical Functions*. (DLMF)

National Institute of Standards and Technology. (n.d.-b). Inequalities (§1.7). In *NIST Digital Library of Mathematical Functions*. (DLMF)

S22 Complete supplied arithmetic norm continuation

Source role. Complete original mathematical paste, with every formula and calculation.

Exact revision. abdacca551cc7178170128c522beef8ac07d3948bf9ceea792177d0e0e8a21a8

Complete source. sources/endpoint_user_sources/ARITHMETIC_ENDPOINT_USER_SOURCE_2_20260913.txt

Source SHA-256. abdacca551cc7178170128c522beef8ac07d3948bf9ceea792177d0e0e8a21a8

One of the analytic estimates in the new endpoint criterion is now proved for the actual arithmetic measure. For a fixed packet h , retaining all its selected zero orders, I obtain

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n, \quad n \geq k \geq 3,$$

where $C_h < \infty$ depends on h , but **not** on the tensor degree k or polynomial degree n .

This establishes the $O_h(q_k)$ polynomial-norm input proposed in the formalization session's handoff, on the explicit window $q_k \leq N < 2q_k$. That handoff had correctly left this arithmetic estimate unproved.

Combining the bound with its endpoint-selection theorem gives a sharper statement about the remaining source-relation volumes: **an off-line quartet would force**

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 2.$$

The upper bound excluding that volume growth is not proved here. The norm factor in this particular criterion, however, is no longer an assumed estimate.

Full derivation in the HTML reader · LaTeX · Formalization handoff · Complete package

S22.0.1 1. The estimate concerns the original arithmetic source

I read the current main revision, which now includes the exterior/trace formalization, and the endpoint and confluent-transfer developments in PRs #23 and #24. Their source maps remain the inputs here; I have not replaced them with another positive metric.

The base remains $\mathfrak{b}_\tau$, carrying

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of $p\mathbb{Z}$ remains infinite. The maps preserve $e = 0^\bullet$ and external absence τ separately before the arithmetic observation.

Retain

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi.$$

For the fixed packet,

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad v_h = \frac{g}{h},$$

define the same density and convolution as before:

$$w_h(t) = \frac{|v_h(\frac{1}{2} + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \int_{\mathbb{R}} m_{h,k} = \mu_h^k.$$

The source map is

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = v_h.$$

Its norm is exactly

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du.$$

The arithmetic observation still satisfies

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_{\chi},$$

where

$$\eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h).$$

Thus the full arithmetic Taylor unit stays in the map.

The norm $\omega_{h,k,n}$ is the infimum over **monic degree- n polynomials in this source**. It belongs to the leading-coefficient quotient

$$\mathcal{P}_n/\mathcal{P}_{n-1}.$$

The arithmetic quotient is connected through the other exact sequence

$$0 \longrightarrow \mathcal{P}_{n-q} \xrightarrow{\times\chi} \mathcal{P}_n \xrightarrow{\pi_{\chi}} \mathbb{C}[S]/(\chi) \longrightarrow 0.$$

Multiplication by monic χ carries the leading-coefficient line at degree $n - q$ to the one at degree n . The arithmetic quotient sends that same relation to its supported zero. These maps connect the two norm calculations without identifying them. The original internal quotient retains labels and identifies precisely differences lying in the specified relation submodule.

S22.0.2 2. The new analytic input is a lower bound after three actual convolutions

The earlier gamma comparison supplied an upper envelope. For the endpoint norm ratio, I constructed a lower envelope for the **unchanged arithmetic density**.

The starting lemma is

$$\int_{T-1}^{T+1} |\zeta(\tfrac{1}{2} + it)|^2 dt \geq c_{\zeta}(1 + |T|)^{-42}, \quad T \in \mathbb{R},$$

for a fixed positive constant c_{ζ} . The exponent 42 is a sufficient explicit exponent; no optimal local mean-square estimate is claimed.

The local-mass proof For $|T| \geq 10$, put

$$f_T(z) = \zeta(\tfrac{1}{2} + iT + iz).$$

On the ellipse

$$z = \frac{1}{2}(w + w^{-1}), \quad |w| = 8,$$

the functional equation and the vertical-strip gamma estimate give

$$|f_T(z)| \leq M_T = C_0(1 + |T|)^6.$$

The relevant strip is fixed, and the zeta pole lies outside the ellipse. For the part with positive real component, the direct identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx$$

gives the needed polynomial growth; reflection handles the remaining part. These use the classical zeta functional equation and gamma estimates, without RH. (DLMF)

The absolutely convergent reciprocal Dirichlet series gives

$$|\zeta(2 + iT)| \geq \zeta(2)^{-1}.$$

Let

$$r_* = \frac{3 + \sqrt{13}}{2} < 4.$$

Then

$$\frac{1}{2} \left(-ir_* + \frac{1}{-ir_*} \right) = -\frac{3i}{2},$$

so f_T at this point is $\zeta(2 + iT)$.

Write

$$m_T = \max_{x \in [-1, 1]} |f_T(x)|.$$

The maximum principle on the annulus $1 \leq |w| \leq 8$, applied through the displayed coordinate map, gives

$$\zeta(2)^{-1} \leq m_T^\vartheta M_T^{1-\vartheta}, \quad \vartheta = 1 - \frac{\log r_*}{\log 8} > \frac{1}{3}.$$

Hence

$$m_T \geq \zeta(2)^{-3} M_T^{-2}.$$

Cauchy's estimate also gives

$$\sup_{[-1, 1]} |f'_T| \leq M_T.$$

Around a point attaining m_T , there is a one-sided interval of length $m_T/(2M_T)$ on which $|f_T| \geq m_T/2$. Therefore

$$\int_{T-1}^{T+1} |\zeta(\tfrac{1}{2} + it)|^2 dt \geq \frac{m_T^3}{8M_T} \geq \frac{\zeta(2)^{-9}}{8C_0^7} (1 + |T|)^{-42}.$$

On the remaining compact range of T , the integral is continuous and strictly positive. Its minimum supplies the remaining positive constant.

Carry the bound through g/h The original gamma factor and the elementary polynomial bound for h imply

$$w_h(t) \geq c_h^{(0)} e^{-\pi|t|/2} (1 + |t|)^{-2 \deg h} |\zeta(\tfrac{1}{2} + it)|^2.$$

This inequality is valid at selected centres through the entire identity $g = hv_h$; it does not divide by an unevaluated zero.

Consequently,

$$\int_{T-1}^{T+1} w_h(t) dt \geq c_h^{(1)} e^{-\pi|T|/2} (1 + |T|)^{-B_h}, \quad B_h = 42 + 2 \deg h.$$

Now w_h is continuous, integrable, and positive outside a discrete set. Thus $w_h * w_h$ is continuous and strictly positive everywhere. Define the actual compact quantities

$$b_h = \min_{|v| \leq 1} (w_h * w_h)(v) > 0, \quad \vartheta_h = \int_{-1}^1 w_h(v) dv > 0.$$

Restricting a nonnegative convolution integral gives

$$m_{h,3}(u) \geq b_h \int_{u-1}^{u+1} w_h(t) dt.$$

For the remaining $k - 3$ factors, restrict their integration variables to $[-1, 1]$. Their sum has absolute value at most $k - 3$. This proves

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\frac{\pi}{2}(|u|+k-3)} (1 + |u| + k - 3)^{-B_h}, \quad k \geq 3.$$

The original mass has not been changed. The complementary integration regions remain part of $m_{h,k}$; this is a lower bound obtained from one positive contribution to that same density.

S22.0.3 3. This proves the required endpoint norm bound

Fix $0 < b < \pi/2$, and retain the two original Laplace values

$$M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bt} w_h(t) dt.$$

The exponential-series inequality and convolution identity give

$$\omega_{h,k,j} \leq (2j)! b^{-2j} (M_h(b)^k + M_h(-b)^k).$$

The trial polynomial is $(S - k/2)^j$, whose value on the original integration line is $i^j u^j$. The operator remains multiplication by S .

For the lower estimate, the exact least squared norm of a monic polynomial of degree n on $[-L, L]$ is

$$\mathfrak{l}_n(L) = \frac{2L^{2n+1}}{2n+1} \left(\frac{2^n (n!)^2}{(2n)!} \right)^2.$$

This follows from the leading coefficient and orthogonality of the Legendre polynomial. Under $S = k/2 + iu$, the retained leading phase is i^n , whose squared modulus is one. (DLMF)

Applying the lower convolution envelope on $[-L, L]$ yields

$$\omega_{h,k,n} \geq c_h \vartheta_h^{k-3} e^{-\frac{\pi}{2}(L+k-3)} (1 + L + k - 3)^{-B_h} \mathfrak{l}_n(L).$$

Set $L = n$, and apply the upper estimate at degree $2n$. The exact finite bound is

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq \left[\frac{(4n)! b^{-4n} (M_h(b)^k + M_h(-b)^k) e^{\frac{\pi}{2}(n+k-3)} (1 + n + k - 3)^{B_h}}{c_h \vartheta_h^{k-3} \mathfrak{l}_n(n)} \right]^{1/(2n)}.$$

For $n \geq k \geq 3$, this is at most $C_h n$.

The degree dependence follows directly from

$$\binom{2n}{n} \leq 4^n, \quad (4n)! \leq (4n)^{4n}, \quad n + k - 3 \leq 2n.$$

The factorial and monic-norm factors contribute $O(n)$ after taking the $2n$ -th root. The retained mass factors contribute bounded factors of the form $\exp(O_h(k/n))$. The polynomial lower-envelope factor contributes

$$(2n)^{B_h/(2n)},$$

which is uniformly bounded.

Equation (36) of the note gives a literal sufficient C_h , retaining all these constants. It is deliberately not optimized or presented as a numerically certified constant.

This establishes a bound on the actual arithmetic norm sequence, uniformly in the required tensor and polynomial degrees.

S22.0.4 4. The endpoint criterion now leaves a precise four-volume threshold

The formalization session's theorem is

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \left(\frac{\omega_{h,k,n+r}}{\omega_{h,k,n}} \right)^{1/(2r)} \sinh \left(\frac{1}{2r} \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}} \right).$$

It selects a member of the existing canonical family. Its proof retains both telescopes and the zero-contraction case; the report distinguishes its checked finite components from the written Jensen assembly.

Taking $r = n \geq \max(q, k)$ now gives

$$\min_{n \leq N < 2n} \epsilon_{h,k,N} \leq C_h n \sinh \left(\frac{1}{2n} \log \frac{V_{n-1} V_n}{V_{2n-1} V_{2n}} \right).$$

For the packet **consisting exactly of** an off-line quartet

$$\frac{1}{2} \pm \delta \pm i\gamma, \quad \delta, \gamma > 0,$$

with common multiplicity m , retain

$$q_k = [1 + k(m-1)](k+1)^2$$

and the existing aggregate defect

$$L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

Here $q_k \geq k$, and

$$\frac{L_{h,k}}{q_k} \geq \frac{\delta k}{2}.$$

The exterior theorem bounds $L_{h,k}$ by every admitted allowance, including the degree selected above.

Define

$$\mathcal{B}_{h,k} = \log \frac{V_{q_k-1} V_{q_k}}{V_{2q_k-1} V_{2q_k}}.$$

Then

$$\frac{\delta k}{2} \leq C_h \sinh \left(\frac{\mathcal{B}_{h,k}}{2q_k} \right),$$

and therefore

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right).$$

Since h is fixed,

$$\operatorname{arsinh}\left(\frac{\delta k}{2C_h}\right) = \log k + O_h(1).$$

This proves the threshold

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 2.$$

The remaining upper estimate in this endpoint criterion is therefore sharply specified. The previously proposed

$$\mathcal{B}_{h,k} = o_h(q_k \log k)$$

would suffice; an upper limit strictly below 2 would already contradict the displayed lower threshold. **Neither upper estimate is being claimed here.**

No second cost-free exterior iteration is used. The formalization session's correction to that iteration's multiplicity cost remains intact.

S22.0.5 5. Those four volumes are two explicit costs of retained e -relations

I also expressed the remaining volume quantity directly through the original relation layers.

For admitted degrees $i < j$, put

$$F_{i,j} = [b_{i+1}, \dots, b_j], \quad \Omega_{i,j} = \operatorname{diag}(\omega_{i+1}, \dots, \omega_j),$$

where $b_a = [p_a]_\chi$ retains all quotient coordinates.

The existing full-jet kernel has the exact update

$$K_j = K_i + F_{i,j} \Omega_{i,j}^{-1} F_{i,j}^*, \quad K_i = G_i^{-1}.$$

Thus

$$\frac{V_i}{V_j} = \det(I + \Omega_{i,j}^{-1} F_{i,j}^* G_i F_{i,j}).$$

The source map behind this determinant is

$$\mathfrak{b}_{i,j} = T_{i,j} - R_i F_{i,j} : \mathbb{C}^{j-i} \xrightarrow{\sim} \mathcal{D}_j \cap \mathcal{D}_i^\perp,$$

where $T_{i,j}$ has the actual source columns $\mathcal{V}_{h,kp_{i+1}}, \dots, \mathcal{V}_{h,kp_j}$.

Its jets vanish, its highest polynomial coefficients give the inverse onto the new relation layer, and its Gram is

$$\mathfrak{b}_{i,j}^* \mathfrak{b}_{i,j} = \Omega_{i,j} + F_{i,j}^* G_i F_{i,j}.$$

The representative update remains

$$R_j = R_i + \mathfrak{b}_{i,j} (\Omega_{i,j} + F_{i,j}^* G_i F_{i,j})^{-1} F_{i,j}^* G_i.$$

Every correction is an original theta boundary. The complete transport is

$$\mathbb{C}^{j-i} \xrightarrow{\mathfrak{b}_{i,j}} \mathcal{D}_j / \mathcal{D}_i \hookrightarrow \mathcal{B}^{\widehat{\otimes} k} / \mathcal{D}_i \longrightarrow \mathcal{B}^{\widehat{\otimes} k} / \mathcal{D}_j.$$

Its final composite is zero. In the original split lift, it is the receiving fibre's **supported zero**, not τ .

Applying this to

$$(i, j) = (q_k - 1, 2q_k - 1), \quad (i, j) = (q_k, 2q_k)$$

gives

$$\mathcal{B}_{h,k} = \sum_a \log(1 + \lambda_a^{(0)}) + \sum_a \log(1 + \lambda_a^{(1)}),$$

where each $\lambda_a^{(\ell)} \geq 0$ is a generalized eigenvalue of the corresponding explicit pair

$$(F_{i,j}^* G_i F_{i,j}, \Omega_{i,j}).$$

This is the remaining volume cost in terms of the relations that become e -zero. Their source norms and cross-pairings are still present before that quotient.

The PR #24 transfer coordinates provide another exact calculation of the same four endpoints. Only its matrices at indices

$$0, \quad 1, \quad q_k, \quad q_k + 1$$

are needed, together with the specified products of ω_j . The common confluent Vandermonde cancels only in the displayed endpoint ratio; its raw-derivative factorials remain in the underlying maps. The note spells out that formula.

S22.0.6 6. The Deligne reading is now tied to this estimate

I reassembled the six supplied S20 parts and verified their chunk hashes and the full archive hash. I then read the TeX passages for Definition 1.3.5, Lemmas 3.2.10–3.2.12, the tensor step 3.2.13, and Corollary 3.3.6, comparing them with the archive’s current page-local French and English records.

There is a genuine inherited transcription trap: the older French TeX gives ≤ 2 in Lemma 3.2.10, while the current page record gives < 2 . The subsequent use of integral weights requires the strict inequality to conclude ≤ 1 . The older TeX also corrupts the explicit squared-eigenvalue step in 3.2.13. Both discrepancies are recorded; the old files were not overwritten.

The relevant operation remains the one identified in the other session’s reading: **an additional estimate before tensor amplification improves the weight bound**. The determinant/exterior map itself does not supply that estimate.

This continuation supplies a characteristic-zero estimate for the actual source—the norm-window bound above—and composes it with the endpoint criterion. It does not import finite-field purity hypotheses onto $\mathfrak{b}_\tau$.

Source reading and discrepancy record

Verification and handoff The new checker passed **18 exact finite test methods** normally and under python -0, with identical successful records; deliberately failing controls failed in both modes. All **37 entries** of the preceding gamma package’s manifest verified, and its **17-method suite** passed fresh normal and optimized runs.

The analytic theorem has a written proof. These finite checks are not a Lean certificate, a numerical enclosure of C_h , or a proof of the remaining volume estimate.

The [add-only integration patch](#) adds fifteen files and passed local application and byte-for-byte checks. No remote branch was changed.

The completed advance is the required $O_h(q_k)$ norm estimate on the actual quartet window. The remaining endpoint-volume problem is now a concrete two-block cost of the original supported relations, with a proved off-line threshold of $2q_k \log k$.

S23 PR 23: complete original endpoint and multiplicity criterion

Source role. Original pinned PR23 source; later results have their own complete proofs.

Exact revision. 61dd03044ad56b2baf3671fa217497b0f4749c1259f841cdc8491846abbd2e46

Complete source. sources/local_endpoint_continuations/endpoint_criterion_pr23_c720f405_20260913.md

Source SHA-256. 61dd03044ad56b2baf3671fa217497b0f4749c1259f841cdc8491846abbd2e46

S23.1 Sharper endpoint-only criterion and the scope of exterior iteration

This is a mathematical continuation of the supplied Toda and exterior notes. The general Jensen argument and exterior-multiplicity count below have written proofs; they are not additional Lean declarations. The norm-ratio product and overlapping-volume telescope are included in the finite formalization target list. No arithmetic asymptotic bound is claimed.

S23.1.1 1. Eliminate the supremum over recurrence coefficients

Retain a nonempty fixed packet and tensor degree k , cyclic dimension q , the actual positive monic norm sequence ω_N , the actual quotient volume $V_N = \det G_N$, and the source's exact control allowance ϵ_N . Choose consecutive admitted degrees $N = n, \dots, n+r-1$ with $n \geq q$ and integer $r \geq 1$. Put

$$B_{(n,r)} = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})),$$

$$A_{(n,r)} = (\omega_{(n+r)} / \omega_n)^{1/(2r)}.$$

These are four endpoint quotient volumes and two endpoint polynomial norms of the existing family. Since V is positive and nonincreasing, $B_{(n,r)} \geq 0$. The source's finite upper bound implies

$$\min_{(0 \leq j < r)} \epsilon_{(n+j)} \leq A_{(n,r)} \sinh(B_{(n,r)}/(2r)). \quad (E1)$$

Unlike the preceding maximum-recurrence criterion, E1 needs no pointwise upper bound for every recurrence coefficient in the window.

Complete proof For $N = n+j$ set

$$x_j = (1/2) \log(V_{(N-1)} / V_{(N+1)}) \geq 0,$$

$$A_j = \sqrt{\omega_{(N+1)} / \omega_N} > 0.$$

The existing local estimate is $\epsilon_{(n+j)} \leq A_j \sinh(x_j)$. If any $x_j = 0$, the nonnegative allowance at that step is zero, and E1 follows. Otherwise all $x_j > 0$. The function $f(x) = \log(\sinh x)$ has

$$f''(x) = -1/\sinh(x)^2 < 0.$$

Finite Jensen therefore gives

$$\text{product}_j \sinh(x_j) \leq \sinh((\sum_j x_j)/r)^r.$$

Let E be the minimum of the nonnegative allowances. Multiplication of the local bounds gives

$$E^r \leq \text{product}_j \epsilon_{(n+j)} \leq (\text{product}_j A_j) \text{product}_j \sinh(x_j).$$

The actual products telescope:

$$\text{product}_j A_j = (\omega_{(n+r)} / \omega_n)^{1/2},$$

$$2 \sum_j x_j = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})).$$

Taking the nonnegative r -th root proves E1. A finite minimum is attained, so E1 selects an actual original degree N inside the specified window. No favorable metric is selected independently, and its original theta representative, quotient and full jets remain attached.

The recurrence product is $\text{VolumeWindow.norm_ratio_product}$; the second telescope is $\text{VolumeWindow.overlapping_log_telescope}$, applied to $V(j)=V_{(n-1+j)}$. The Jensen and nonnegative-root assembly is this written proof rather than a claim about an unimplemented exterior/analytic declaration.

S23.1.2 2. Every input is an endpoint of the two original determinant sequences

The source's $\omega_j = D_{(j+1)}/D_j$ gives

$$A_{(n,r)} = [D_{(n+r+1)} D_n / (D_{(n+r)} D_{(n+1)})]^{1/(2r)}.$$

The four V -values in B are evaluated by $V_N = D_{(N+1)}/B_{(N-q+1)}$, with the source's literal relation determinant B . All indices are nonnegative because $n \geq q$. No negative-index determinant or inverse of a rank-deficient earlier kernel is used. The common mass factors cancel only in these displayed ratios; the source $\text{norm } \omega_0 = \mu_h^k$ has not been reset.

For the original quartet,

$$q_k = [1+k(m-1)](k+1)^2, \\ L_{(h,k)} \geq \delta k^2 q_k / 2.$$

The exterior theorem gives $L_{(h,k)} \leq \epsilon_N$ at every admitted N . Thus E1 yields

$$L_{(h,k)} \leq A_{(n,r)} \sinh(B_{(n,r)})/(2r). \quad (\text{E2})$$

A proof that the right side is $o(k^2 q_k)$ along admitted $n=n(k)$, $r=r(k)$ would exclude that fixed off-line quartet. One sufficient set of estimates is

$$r(k) \geq c_h q_k, \quad A_{(n,r)} = O_h(q_k), \\ B_{(n,r)} = o_h(q_k \log k),$$

for $c_h > 0$. Indeed $B/(2r) = o(\log k)$, so $\sinh(B/(2r)) \leq \exp(o(\log k)) = k^{o(1)}$, and the ratio to $k^2 q_k$ tends to zero. These are candidate arithmetic estimates, not consequences of the formal telescope, positivity, the Toda equation, or Deligne's finite-field hypotheses.

The weighted endpoint criterion is sharp as an abstract sequence inequality: equality can occur when all x_j are equal and every $A_j \sinh(x_j)$ equals the same allowance. For arbitrary recurrence sequences it is strictly preferable to replacing $A_{(n,r)}$ by the largest A_j .

S23.1.3 3. A useful boundary on further exterior amplification

The supplied single exterior theorem is correct. It must not be reused with a rank-two hypothesis that no longer holds after exterior powering.

Suppose the original self-adjoint control H has spectrum

$$(+\epsilon, 0 \text{ repeated } q-2 \text{ times}, -\epsilon), \quad \epsilon > 0.$$

On the actual p -th exterior power, $1 \leq p \leq q$, the additive control has $+\epsilon$ and $-\epsilon$ each with multiplicity

$$r_p = \text{binomial}(q-2, p-1),$$

and zero with multiplicity $\text{binomial}(q-2, p) + \text{binomial}(q-2, p-2)$. This follows by the unscaled wedge basis: select the positive direction but not the negative, the negative but not the positive, or both/neither. In particular its operator norm is still ϵ , but its rank is $2r_p$, not generally two.

Apply an s -th exterior power to that whole space, whose dimension is $D_p = \text{binomial}(q, p)$. Its largest absolute additive-control eigenvalue is exactly

$$\epsilon \min(s, r_p, D_p - s), \quad 0 \leq s \leq D_p.$$

Proof: select positive eigenvectors first, zero eigenvectors next, and negative eigenvectors only when the degree forces them. There are r_p available positive directions and r_p negative ones. This attains the stated piecewise maximum; reversing the choices attains its negative. At $s=D_p$ the sum is zero. For example $q=4, p=2$ gives spectrum $(+\epsilon, +\epsilon, 0, 0, -\epsilon, -\epsilon)$, and the next exterior square has allowance 2ϵ .

This does not weaken the original determinant-line amplification or remove its nilpotent arithmetic action. It identifies the precise cost of iterating the operation on the entire exterior space, and prevents a spurious second cost-free amplification. The source's chosen determinant line has its own one-dimensional action and must remain distinguished from the complete first exterior power.

S23.1.4 4. Collaboration boundary

The formal work carries the exact source/quotient determinant and the two loss terms into the original-Gram trace certificate. The analytic collaborator can now estimate the two endpoint quantities in E1, or retain the exact phase and imbalance when the coarse upper bound is too large. The six-part Deligne archive and uploaded ZIP test suites were not opened in this execution environment; the source statements used here came from the complete pasted notes. The current GitHub correction and original source maps are unchanged.

S24 The complete original endpoint and Gamma join

Source role. Complete original EG.1-5 mathematical witness.

Exact revision. ce4af7908adc2f8003b6fe7e7f62068a1ee125c45412048159fde67f2cfb1b50

Complete source. sources/local_endpoint_continuations/owner_endpoint_gamma_join/ENDPOINT_GAMMA_JOIN.md

Source SHA-256. ce4af7908adc2f8003b6fe7e7f62068a1ee125c45412048159fde67f2cfb1b50

S24.1 Endpoint degree selection in the actual gamma-corrected coordinates

This short calculation joins the complete written endpoint criterion in PR #23 (reported source revision c720f40530eed2f5969dbabe94dfd3fddc0f507f) to the gamma convolution descent supplied on 13 September 2026. It uses the original arithmetic source, the original polynomial relation, and the same canonical quotient metric. It is a written substitution and finite-dependence proof, not an additional Lean theorem or a uniform arithmetic estimate.

S24.1.1 Objects and the estimate being evaluated

Fix the original packet h and tensor degree k . Put $\alpha_k = 2k$ and keep the source and reference Hankel determinants D_j and D_j^Γ of GD.28. Their positive quotient is $X_j = D_j/D_j^\Gamma$. For the original relation χ , of degree $q \geq 1$, let $Y_j = B_j/B_j^\Gamma$ and

$$T_N = \frac{X_{N+1}}{Y_{N-q+1}}, \quad V_N = \det G_N = V_N^\Gamma T_N, \quad N \geq q-1.$$

Here T_N is a positive scalar correction ratio, not the coordinate operator also denoted T in the Toda calculation. Let $n \geq q$ and $r \geq 1$ be integers. The endpoint criterion uses the actual quantities

$$\mathcal{A}_{n,r} = \left(\frac{\omega_{n+r}}{\omega_n} \right)^{1/(2r)}, \quad \mathcal{B}_{n,r} = \log \frac{V_{n-1} V_n}{V_{n+r-1} V_{n+r}}.$$

Its complete Jensen and telescoping proof gives

$$\min_{0 \leq j < r} \epsilon_{h,k,n+j} \leq \mathcal{A}_{n,r} \sinh \left(\frac{\mathcal{B}_{n,r}}{2r} \right). \quad (\text{EG.1})$$

No metric is selected here: the finite minimum is over the pre-existing canonical representatives at degrees n through $n+r-1$.

S24.1.2 Exact reference and arithmetic factors at the endpoints

At the original observation $\theta=0$, GD.28–29 gives

$$\omega_j = c_\lambda^k j! (\alpha_k)_j \frac{X_{j+1}}{X_j}. \quad (\text{EG.2})$$

Consequently,

$$\mathcal{A}_{n,r} = \left((n+1)_r (n+\alpha_k)_r \right)^{1/(2r)} \left(\frac{X_{n+r+1} X_n}{X_{n+r} X_{n+1}} \right)^{1/(2r)}. \quad (\text{EG.3})$$

The mass c_λ^k is retained in EG.2 and cancels between the two specified norms. The arithmetic mass μ_h^k was never assigned the value one. Each rising factorial in EG.3 is the exact r -factor product, not a large-degree approximation.

Substitution of $V_N = V_N^\Gamma T_N$ gives a second exact identity:

$$\mathcal{B}_{n,r} = \log \frac{V_{n-1}^\Gamma V_n^\Gamma}{V_{n+r-1}^\Gamma V_{n+r}^\Gamma} + \log \frac{T_{n-1} T_n}{T_{n+r-1} T_{n+r}}. \quad (\text{EG.4})$$

The reference quotient volumes in this expression are formed with the same χ , using its two conjugate differential factors in GD.30. The roots, their orders and the centre $k/2$ have not been changed. All ratios are positive, so the real logarithms and the indicated positive roots exist.

Equations EG.3–4, inserted in EG.1, express the endpoint selection estimate through an explicit reference factor, four source corrections X , and four source-to-relation corrections T . A pointwise upper bound $T_N \leq C_h(kq)$ does not control the denominator in EG.4. Neither of these exact identities supplies that missing quantitative estimate.

S24.1.3 Exact finite input count for the whole endpoint expression

The maximum source degree occurring in EG.1–4 is $n+r$: ω_{n+r} is its highest monic polynomial norm, and V_{n+r} is its highest quotient metric. Each defining source or relation Gram entry is the integral of a polynomial of degree at most $2(n+r)$ against the original arithmetic sum measure. The gamma coefficient theorem therefore proves that the complete endpoint expression is determined by

$$c_{h,0}, c_{h,1}, \dots, c_{h,2(n+r)}, \quad k, \lambda, \chi. \quad (\text{EG.5})$$

The zeroth coefficient carries the original total mass. The higher coefficients may have either sign. The finite coefficient-power formula and the Schur complement give the actual endpoint values; an arbitrary positive moment matrix cannot be inserted in their place. EG.5 gives a finite route for each chosen window. It does not bound the conditioning of the matrices or permit truncating the growing input count uniformly.

S24.1.4 Source roles

- [PR #23 endpoint proof at its supplied revision](#): finite Jensen selection, four-volume telescope, two-norm telescope. The user supplied its complete proof, which was read and the telescopes checked. Current CI/source review is recorded separately.
- Tau Gamma Convolution Descent, GD.25 and GD.28–35: exact coefficient transform and source/reference determinant ratios. The full original note and independent finite/analytic review are retained in this edition.
- GJ.1–8 in the gamma chapter: the explicit analytic map from the same one-factor moment function to the coefficient series, and finite source/quotient derivative jets.

The new content here is their literal endpoint-coordinate substitution and the maximum-degree input count. No additional theorem about zeta-zero locations or a completed growing-window bound is claimed.

S25 PR 24: complete original confluent transfer proof

Source role. Original historical PR24 proof, retained separately from the correction.

Exact revision. d1046b75762e4248ded8f0b901c1ac12dc01065e4e684ca2d166b6a535425763

Complete source. sources/owner_endpoint_review/review_pr24/files/workbenches/tau-confluent-transfer/RESEARCH_NOTE.md

Source SHA-256. d1046b75762e4248ded8f0b901c1ac12dc01065e4e684ca2d166b6a535425763

S25.1 Confluent transfer for the original source-relation volume

S25.1.1 Scope and current-main integration

This add-only analytic continuation starts from main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f, after the merge of PR #21 and its workflow-trigger correction. The PR #22 mathematical note was read at 811210d24b80813a08972ca23f919db015383137; its strict verification was in progress at that reading. This contribution neither modifies that branch nor treats pending checks as a certificate.

The source is the delivered Toda-volume continuation, following the cyclic and exterior-trace notes. Its full LaTeX was read directly. The new result is a fixed-width, full-jet transfer for the SAME canonical source/relation quotient metric. It applies the classical Christoffel transformation; it is not a claim to a new general orthogonal-polynomial theorem or to RH.

Keep $G(\mathbb{Z}) \rightarrow \mathbb{Z}$, with its infinite target, and its coefficient square to $G(\mathbb{C}) \rightarrow \mathbb{C}$. The structural base remains b_{τ} . Every supported coefficient map below retains $e=0^{\bullet}$ and external τ separately. The original theta complex is $C_{+}=[V \rightarrow \Theta \rightarrow B]$, with $D=-x \, d/dx$, $Q=B/\Theta V$ and $g=2 \, \xi$. No scalar, source function or measure is assigned a new normalization.

For a nonempty finite packet h of actual zeros with full orders, retain F_h with Mellin transform g/h . At tensor degree k , keep the actual cyclic sum module

$$\begin{aligned} C &= C[S]/(\chi), \quad q = \deg \chi \geq 1, \quad c = k/2, \quad A = M_S, \\ \eta[P] &= \epsilonpsilon_h^{(\text{tensor } k)} P(\sum_i M(s_i)), \\ \epsilonpsilon_h &= j_h(g/h), \\ \text{Vcal } P &= P(D_1 + \dots + D_k)(F_h^{(\text{tensor } k)}). \end{aligned}$$

The original equations are

$$\begin{aligned} J^{(k)} \text{Vcal} &= \eta \, \pi_{\chi}, \\ q^{(k)} \text{Vcal} &= \sigma_h^{(\text{tensor } k)} \eta \, \pi_{\chi}. \end{aligned}$$

Its norm is the actual integral against $m_k = w_h^{\{k\}}$, where

$$w_h(t) = |(g/h)(1/2 + i \, t)|^{2/(2 \, \pi)}, \quad \text{integral } m_k = \mu_h^k.$$

The auxiliary measure is $d\mu_{\theta}(u) = \exp(\theta u) m_k(u) du$, $|\theta| < \pi/2$. Its mass is $M_h(\theta)^k$. θ in this expression is the real tilt parameter, not the theta-summation map. The exponential-moment bound from the source makes all finite derivatives below valid. At tilt zero the metric is the original arithmetic metric.

For $h=1$ the finite arithmetic quotient is zero. Its determinant is the empty determinant 1; the analytic seed and its mass remain, and $G(0)$ still has e and τ . The positive-dimensional inverses below require $q \geq 1$.

S25.1.2 1. Exact coordinate and raw-jet maps

Use the ring isomorphism

$$\Psi: C[S] \rightarrow C[u], \quad P \mapsto P(c + i \, u), \quad u \mapsto (S - c)/i.$$

Thus $d_u \Psi(P) = i \, \Psi(P')$, and multiplication by S remains $c + i \, M_u$. Define the original norm polynomial

$$\Pi(u) = \text{conj}(\chi)(c - i \, u) \, \chi(c + i \, u).$$

Its leading coefficient is literally $(-i)^q i^q = 1$. It is real and nonnegative on $\mathbb{R}$. Reflection gives $\chi^\dagger = (-1)^q \chi$, hence

$\psi(u) = i^{(-q)} \chi(c + i u)$ is real and monic,
 $\Pi = \psi^2$, $\Psi(\chi) = i^q \psi$.

The factor i^q remains in the source multiplication map. For distinct roots λ_a of χ , of complete lengths ℓ_a , put

$\zeta_a = (\lambda_a - c)/i$, $r_a = 2 \ell_a$, $r = \sum r_a = 2q$.

Define raw, NOT factorial-divided, derivative coordinates

$J_2 f = (f^{(d)}(\zeta_a))_{(a, 0 \leq d < r_a)}$.

Their kernel is (Π) . Their product uses the full binomial rule. Multiplication by u has matrix

$(X_{\Pi} v)_{(a,d)} = \zeta_a v_{(a,d)} + d v_{(a,d-1)}$, $v_{(a,-1)} = 0$.

All nilpotent orders and the coefficient d are retained.

Let Q_j be the real monic orthogonal source polynomials for $d\mu_\theta$, obtained by subtracting lower-degree projections, and write

$\omega_j = \int Q_j^2 d\mu_\theta > 0$,
 $a_j = \omega_j / \omega_{j-1}$, $a_0 = 0$,
 $b_j = \int u Q_j^2 d\mu_\theta / \omega_j$.

The source recurrence is

$u Q_j = Q_{(j+1)} + b_j Q_j + a_j Q_{(j-1)}$, $Q_{(-1)} = 0$.

No Q_j is divided by its norm. Set

$v_j = J_2 Q_j$,
 $F_n = [v_n, \dots, v_{(n+r-1)}]$,
 $V = [J_2 1, \dots, J_2 u^{(r-1)}]$.

Both are r by r . The raw confluent determinant is

$\det V = (\prod_{a=0}^{r-1} (d=0)^{(r_a-1)} d!) \prod_{a < b} (\zeta_b - \zeta_a)^{(r_a r_b)}$.

The fixed row order is part of this formula. $F_0 = V$ times the actual unit-triangular polynomial coefficient matrix; only their determinants are equal.

S25.1.3 2. Invertibility and the original relation polynomial

Invertibility proof. Suppose F lies in $\text{span}(Q_n, \dots, Q_{(n+r-1)})$ and $J_2 F = 0$. Then $F = \Pi H$ with $\deg H < n$. Orthogonality gives

$0 = \langle H, F \rangle = \int \Pi |H|^2 d\mu_\theta$.

The integrand is nonnegative and its weight is positive almost everywhere except finitely many points. Thus $H=0$, then $F=0$. For $n=0$ the degree argument suffices. Hence every F_n is invertible. This does NOT estimate its inverse uniformly.

Solve

$d_n = F_n^{(-1)} v_{(n+r)}$, $\vartheta_n = -(d_n)_0$.

The polynomial

$$\text{Qhat}_n = (Q_{(n+r)} - \sum_{(j=0)}^{(r-1)} (d_n)_j Q_{(n+j)}) / \text{Pi}$$

exists by exact row-jet divisibility and is monic of degree n . Pairing its numerator with every polynomial of degree below n proves that Qhat_n is precisely the monic orthogonal polynomial for the ORIGINAL relation weight Pi dmu_θ . Its norm is

$$\begin{aligned} \text{nu}_n &= \int \text{Qhat}_n^2 \text{Pi dmu}_\theta \\ &= -(d_n)_0 \omega_n = \omega_n \quad \text{vartheta}_n > 0. \end{aligned} \quad (1)$$

Indeed only the Q_n column pairs with the degree- n quotient; its coefficient is $-(d_n)_0$. This proves $\text{vartheta}_n > 0$ with arbitrary nonreal and repeated nodes.

Let $K_n = [e_1, \dots, e_{(r-1)}, d_n]$. Then

$$\begin{aligned} F_{(n+1)} &= F_n K_n, \\ \det K_n &= (-1)^{(r-1)} (d_n)_0 = \text{vartheta}_n. \end{aligned} \quad (2)$$

The sign uses the displayed even $r=2q$. On the same jet algebra,

$$\begin{aligned} T_n &= F_n^{(-1)} X_{\text{Pi}} F_n, \\ T_{(n+1)} &= K_n^{(-1)} T_n K_n, \\ d_n &= (T_n - b_{(n+r-1)} I) e_{(r-1)} - a_{(n+r-1)} e_{(r-2)}. \end{aligned} \quad (3)$$

Thus the full operator and its jets have an explicit discrete transport, rather than only a determinant observation.

S25.1.4 3. Fixed-width determinant and quotient-volume formulas

Let D_n be the n by n source Hankel determinant, and B_n the corresponding determinant for Pi dmu_θ . $D_0 = B_0 = 1$. Orthogonality gives

$$D_n = \text{product}_{(j < n)} \omega_j, \quad B_n = \text{product}_{(j < n)} \text{nu}_j.$$

Multiplying (1) and (2) proves the complete confluent Christoffel identity

$$B_n / D_n = \text{product}_{(j < n)} \text{vartheta}_j = \det F_n / \det V. \quad (4)$$

This is a proof from the source division and norm, not a distinct-root approximation. Raw derivative factors remain in $\det V$.

For $N \geq q-1$ set $n = N - q + 1$. The source sequence is still

$$0 \rightarrow P_{(N-q)} \xrightarrow{\chi} P_N \rightarrow C[S] / \chi \rightarrow 0.$$

Its canonical least-norm quotient Gram G_N satisfies the existing Schur identity $\det G_N = D_{(N+1)} / B_n$. Consequently

$$V_N = \det G_N = (\text{product}_{(j=n)}^{(n+q-1)} \omega_j) \det V / \det F_n. \quad (5)$$

The constant coefficient map $S = c + i u$ has determinant of modulus one on each remainder basis, so these are the original quotient determinants, with that coordinate map retained.

For $n \geq 1$,

$$\begin{aligned} \delta_N &= V_N / V_{(N-1)} = \omega_N / \text{nu}_{(n-1)} \\ &= \omega_N / (\omega_{(n-1)} \text{vartheta}_{(n-1)}), \end{aligned} \quad (6)$$

$$\begin{aligned} R_N &= V_{(N-1)} / V_{(N+1)} \\ &= \omega_{(n-1)} \omega_n \text{vartheta}_{(n-1)} \text{vartheta}_n \\ &\quad / (\omega_N \omega_{(N+1)}). \end{aligned} \quad (7)$$

The transfer width $2q$ does not grow with N . The source recurrence and high-degree polynomial evaluation are still required; no computational-complexity or conditioning claim is inferred from the width alone.

S25.1.5 4. One additional solve retains the full phase

Differentiating actual source orthogonality under $\exp(\theta u)$ gives

$$d_{\theta} Q_j = -a_j Q_{(j-1)}, \quad d_{\theta} \log \omega_j = b_j.$$

For the first identity, every coefficient below $j-1$ vanishes by differentiated orthogonality; pairing with $Q_{(j-1)}$ gives $-\omega_j$. Let L_n have entries $(L_n)(j-1,j) = -a(n+j)$ for $1 \leq j \leq r$, all other entries zero. Then

$$F_n' = F_n L_n - a_n v_{(n-1)} e_{\theta}^T. \quad (8)$$

At $n=0$ the last term is zero. Trace of L_n is zero. Hence

$$\begin{aligned} (\log \det F_n)' &= -a_n e_{\theta}^T F_n^{-1} v_{(n-1)}, \\ (\log V_N)' &= \sum_{j=n}^{\infty} (n+j-1) b_j \\ &\quad + a_n e_{\theta}^T F_n^{-1} v_{(n-1)}. \end{aligned} \quad (9)$$

Retain $T = (A - cI)/i$ and $\sigma = \text{Tr } T$, which is real. The previous source cross term is exactly

$$(b_N * G_N b_{(N+1)}) / \omega_N = i(\sigma - (\log V_N)').$$

Here b_N in that formula denotes the remainder vector of the S -monic source polynomial, not the scalar recurrence b_j . The factor i and the distinction of types remain. The phase is $\phi_N = \sigma - (\log V_N)'$. At zero tilt for a full quartet, evenness gives $\phi_N = 0$; it is not set to zero at a general tilt.

S25.1.6 5. The positive gaps are norms of actual source vectors

Put

$$Z_j = \psi \hat{Q}_j - Q_{(j+q)} \text{ in } P_{(j+q-1)}.$$

The leading terms cancel. Source orthogonality therefore proves

$$\Delta_j := \|Z_j\|^2 = \nu_j - \omega_{(j+q)} \geq 0. \quad (10)$$

The first term $\psi \hat{Q}_j$ is, through the explicit unit $i^{(-q)}$, an original χ relation. Its quotient is its supported zero. The difference Z_j instead has class $-[Q_{(j+q)}]$ in the original quotient. Its norm is not assigned to an absent element.

The SAME canonical rank-two control is $W_N = A * G_N + G_N A - k G_N$. Combining its previously proved radius identity with (6) gives, for $n \geq 1$,

$$\epsilon_N^2 = \Delta_{(n-1)} \Delta_n / (\nu_{(n-1)} \omega_N) - \phi_N^2. \quad (11)$$

Thus every factor in the upper bound

$$\epsilon_N \leq \sqrt{\Delta_{(n-1)} \Delta_n / (\nu_{(n-1)} \omega_N)}$$

comes from a specified positive source norm. Both gaps are computed by the transfer:

$$\begin{aligned} \Delta_{(n-1)} &= \omega_{(n-1)} \vartheta_{(n-1)} - \omega_N, \\ \Delta_n &= \omega_n \vartheta_n - \omega_{(N+1)}. \end{aligned}$$

The previous two-step bound is now

$$\epsilon_N \leq \sqrt{\omega_{(N+1)} / \omega_N} \cdot (R_{N-1}) / (2 \sqrt{R_N}), \quad (12)$$

with R_N from (7). At the first degree $N=q-1$, use the separate expression

$$\epsilon_{(q-1)}^2 = (\nu_0 - \omega_q) / \omega_{(q-1)} - \phi_{(q-1)}^2.$$

No inverse at the previous rank-deficient degree is introduced.

S25.1.7 6. Keep the actual conormal quotient, not a repeated first ideal

The helper algebra for the norm computation is $C[u]/\Pi$, isomorphic through $S=c+iu$ to $C[S]/\chi_1^2$. The actual depth-two object is $C[S]/\chi_2$, defined using the literal pullback of the original I^2 . The merged conormal work gives $\chi_2 \mid \chi_1^2$. The actual maps are

$$C[S]/\chi_1^2 \twoheadrightarrow C[S]/\chi_2 \twoheadrightarrow C[S]/\chi_1. \quad (13)$$

Their kernels are respectively $(\chi_2)/(\chi_1^2)$ and $(\chi_1)/(\chi_2)$. In raw jets they truncate at the indicated full orders and intertwine multiplication by S . Differentiation into depth one also commutes, since $\chi_1 \mid \chi_2$. The coordinate relation $d_u \Psi = i \Psi d_S$ is retained.

For $h=(s-\rho)^2$ and tensor degree two, $X=S-2\rho$ gives

$$C[X]/X^6 \twoheadrightarrow C[X]/X^5 \twoheadrightarrow C[X]/X^3.$$

The first kernel is the class of X^5 . The retained depth-two class X^3 has derivative $3X^2$ at depth one. These maps use one original scalar support. Their split lifts send quotient-killed classes to e at the receiving label and fix external τ .

The full multiplier $U=\text{product}_i(g/h)(s_i)$ retains the checked equation $\delta(U)P=U \delta(P)+(\delta U)P$. The second term is not discarded or presumed cyclic. The 2q-jet norm calculation is not counted as construction of new actual spectral multiplicities.

S25.1.8 7. Residual-controlled numerical interface

For an actual F and a specified candidate inverse Y , suppose a justified bound gives $\|I-YF\|_\infty \leq \eta < 1$. For candidate $\hat{d}$ and residual $r=v-F \hat{d}$, the exact Neumann argument proves

$$\|\hat{d}-d\|_\infty \leq \|Y\|_\infty \|r\|_\infty / (1-\eta). \quad (14)$$

This bounds the first coordinates used in (1) and (9). Arithmetic application requires certified input enclosures for the moments, source polynomials and zeros. Floating-point output alone is not a certificate.

Positive enclosures of ω and ϑ bound (7). Since $(R-1)^2/(4R)$ is increasing for $R \geq 1$, (12) yields a valid upper enclosure of ϵ^2 . The full phase can improve it through (11). Finite positivity/invertibility is not substituted for a uniform estimate of these residuals or inverse norms.

S25.1.9 8. Exact calibration and analytic target

For a declared Gaussian measure of literal mass seven and variance one, $\psi=u^2+1$, $q=2, n=2, N=3$, the calculation gives

```
(omega_0,...,omega_4)=(7,7,14,42,168),
vartheta_1=22, vartheta_2=86/3,
Qhat_1=u, Qhat_2=u^2-11/3,
Delta_1=112, Delta_2=700/3,
G_3=diag(7/3,21/11), epsilon_3^2=400/99.
```

An independent quotient-Gram calculation with $A=(1/2)I+i M_u$ gives the same result. The nonreal u -roots are algebraic fixtures, not zeros of ξ ; the nonzero allowance explicitly tests that the method does not manufacture purity. Repeated nodes, complex quartets, asymmetric atomic tilt derivatives and mass changes are covered by the expanded checks.

For the actual quartet retain

$$q_k=[1+k(m-1)](k+1)^2, \\ L_{\text{h,k}}=2 \delta [1+k(m-1)](k+1) \lfloor (k+1)^2/4 \rfloor.$$

The source exterior/filtration result gives $L_{\text{h,k}} \leq \epsilon_{\text{h,k,N}}$, and $L_{\text{h,k}}/(k q_k) \geq \delta/2$. Thus $\epsilon_{\text{h,k,N}} = o(k q_k)$, with fixed h and admitted $N \geq q_k$, is sufficient for the intended contradiction. The new representation attaches that exact target to TWO consecutive source-to-relation transfer coefficients and ONE phase solve. It has NOT proved that asymptotic bound.

No new Lean execution or arithmetic interval certificate is claimed. All new mathematical claims have the written proofs above at the retained source inputs. The original arithmetic function, unit, action, theta relations, residue pairing and trace remain unchanged.

S25.1.10 References and checks

Classical background: C. Krattenthaler (2021), ‘A determinant identity for moments of orthogonal polynomials that implies Uvarov’s formula for the orthogonal polynomials of rationally related densities’, arXiv:2103.03969; NIST DLMF Section 18.2. The proofs above supply the particular confluent and arithmetic interfaces rather than attributing them wholesale to those references.

The expanded conversation delivery includes the complete LaTeX/HTML reader and twenty exact finite regression methods; both normal and optimized runs passed with matching records, and both deliberate-failure controls failed. The preceding Toda archive’s 35 manifest entries verified and its 24-method checker passed fresh normal and optimized runs. These are stated verification scopes, not new Lean certificates. The public core checker in this directory independently tests the principal raw-jet, transfer, norm, determinant, volume and phase interfaces.

S26 Complete scalar volume comparison and arithmetic maps

Source role. Full VC.1-43 and VC.15a-b proof invoked by the AU continuation.

Exact revision. 47ac2b43904ec43f3ce19d7c74ba760a29ce4d7a3e411132fe81ec0b27af6104

Complete source. sources/local_endpoint_continuations/volume_comparator_review_20260913.md

Source SHA-256. 47ac2b43904ec43f3ce19d7c74ba760a29ce4d7a3e411132fe81ec0b27af6104

S26.1 An explicit scalar comparison for the original arithmetic quotient volumes

Date: 13 September 2026. This is the independent scalar comparison requested by the arithmetic volume upper-estimate lane. All inequalities below concern the original source and its original monic quotient. The displayed coordinate maps are invertible maps used in the proof; their Jacobians, phases, and the quotient units are retained.

S26.1.1 1. Original data and the finite answer

Let the fixed packet be a complete off-line quartet $1/2 \pm \delta \pm i\gamma$, with $\delta, \gamma > 0$ and common order $m \geq 1$. Put

$$c = k/2, \quad \ell = 1 + k(m-1), \quad q = \ell(k+1)^2, \quad R_0 = \sqrt{\delta^2 + \gamma^2}, \quad k \geq 3. \quad (\text{VC.1})$$

Keep the given monic annihilator $\chi \in \mathbb{C}[S]$ of degree q , with all its original local lengths. Its roots in the chart $S = c + iu$ have absolute value at most $\rho = kR_0$. The source is

$$w_h(t) = \frac{|(2\xi/h)(1/2 + it)|^2}{2\pi}, \quad m_k = w_h^*, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad M_h(a) = \int_{\mathbb{R}} e^{at} w_h(t) dt. \quad (\text{VC.2})$$

The proved arithmetic density estimate supplies its actual constants $c_h > 0$, $\vartheta_h > 0$, $B_h = 42 + 2 \deg h$, and $\alpha = \pi/2$:

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B_h}. \quad (\text{VC.3})$$

Choose any $0 < b < \alpha$. These are the established analytic inputs of the supplied Arithmetic Endpoint Bounds note, rather than added estimates on its unknown coefficients. Define the following literal finite numbers:

$$\begin{aligned} D &= \max\{1, 2/b, 2R_0\}, \quad T = Dq, \quad r = \rho/T, \quad N = 2q - 1, \\ \lambda_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B_h}, \\ U_k(T) &= \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} (M_h(b)^k + M_h(-b)^k), \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r} \right)^q, \\ C_{k,q} &= \frac{U_k(T) A_{2q-1}^2 B_q(r)^2}{T \lambda_k(T)}. \end{aligned} \quad (\text{VC.4})$$

Here $q \geq k$, so $0 \leq r \leq 1/2$, and all denominators are strictly positive. The formula for U_k uses the usual $0! = 1$ convention, although the quartet has $q \geq 16$.

For every source degree $j \geq q-1$, let H_j be the positive definite Gram matrix in the original ordered monomials $1, S, \dots, S^j$, let J_j be the coefficient matrix of remainder modulo the original χ , and set

$$K_j = J_j H_j^{-1} J_j^*, \quad G_j = K_j^{-1}, \quad V_j = \det G_j. \quad (\text{VC.5})$$

The finite Hermitian comparison is

$$K_q \preceq K_{2q-1} \preceq C_{k,q} K_q, \quad 0 \leq \log \frac{V_q}{V_{2q-1}} \leq q \log C_{k,q}. \quad (\text{VC.6})$$

The proof below establishes every algebraic and comparison step in (VC.6). In particular it proves $C_{k,q} \geq 1$, since the matrices in (VC.6) are positive definite.

S26.1.2 2. Exact source, quotient, and raw-jet maps

For each finite degree define the complex-linear isomorphism

$$\Psi_T : \mathbb{C}[S]_{\leq j} \longrightarrow \mathbb{C}[x]_{\leq j}, \quad (\Psi_T P)(x) = P(c + iTx). \quad (\text{VC.7})$$

Its inverse is $f \mapsto f((S - c)/(iT))$. A change of real variable gives the equality of the original norms

$$\int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du = \int_{\mathbb{R}} |(\Psi_T P)(x)|^2 T m_k(Tx) dx. \quad (\text{VC.8})$$

Thus the target measure in this equality has mass exactly μ_h^k . There is no mass divisor in the map. Define

$$\chi_T(x) = (iT)^{-q} \chi(c + iTx). \quad (\text{VC.9})$$

Its leading coefficient is one, its roots are the original u -chart roots divided by T , with the same multiplicities, and their absolute values are at most r . For every polynomial Q ,

$$\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q. \quad (\text{VC.10})$$

The nonzero scalar $(iT)^q$ is the exact unit connecting the two displayed generators. Consequently (VC.7) induces a quotient isomorphism

$$\bar{\Psi}_T : \mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[x]/(\chi_T), \quad [P] \longmapsto [P(c + iTx)]. \quad (\text{VC.11})$$

Division is compatible with this map. Indeed, if $P = \chi Q + R$ with $\deg R < q$, then (VC.10) gives $\Psi_T P = (iT)^q \chi_T \Psi_T Q + \Psi_T R$, and $\deg \Psi_T R < q$; uniqueness of division proves

$$\text{rem}_{\chi_T} \Psi_T P = \Psi_T \text{rem}_{\chi} P. \quad (\text{VC.12})$$

If λ is a root in the original S chart and $z = (\lambda - c)/(iT)$, the raw derivative of order d obeys

$$(\Psi_T P)^{(d)}(z) = (iT)^d P^{(d)}(\lambda), \quad 0 \leq d < \text{ord}_{\lambda} \chi. \quad (\text{VC.13})$$

This follows by differentiating the affine composition d times. In local Taylor coordinates $P^{(d)}(\lambda)/d!$, the multiplier is still $(iT)^d$, while the factorial stays $d!$. The derivative phase has therefore been specified for every retained local length.

For precision at the matrix level, let $\mathbf{A}_j$ be the coefficient matrix of Ψ_T in increasing monomial degrees. Its columns are the coefficient vectors of $(c + iTx)^a$, $0 \leq a \leq j$; its diagonal is $(iT)^a$. Thus it is invertible and

$$\det \mathbf{A}_j = (iT)^{j(j+1)/2}. \quad (\text{VC.14})$$

Let tildes denote the target-coordinate Gram and remainder matrices. Equations (VC.8) and (VC.12) give

$$H_j = \mathbf{A}_j^* \tilde{H}_j \mathbf{A}_j, \quad \tilde{J}_j \mathbf{A}_j = \mathbf{A}_{q-1} J_j, \quad \tilde{K}_j = \mathbf{A}_{q-1} K_j \mathbf{A}_{q-1}^*. \quad (\text{VC.15})$$

The third equality follows by inserting the first two into $\tilde{J}_j \tilde{H}_j^{-1} \tilde{J}_j^*$. Hence $\tilde{G}_j = \mathbf{A}_{q-1}^{-*} G_j \mathbf{A}_{q-1}^{-1}$ and $\det \tilde{G}_j = T^{-q(q-1)} V_j$. The exact determinant factor is independent of j and cancels in the ratio in (VC.6). All subsequent norm estimates can therefore be pulled back to the original matrices by this proved congruence.

To state the same comparison in the original raw-jet coordinates, list the distinct roots of χ as λ_{ν} with full lengths d_{ν} , $\sum_{\nu} d_{\nu} = q$. In the fixed ordering of pairs (ν, a) , $0 \leq a < d_{\nu}$, define the square matrix

$$E_{(\nu,a),j} = \begin{cases} \frac{j!}{(j-a)!} \lambda_{\nu}^{j-a}, & j \geq a, \\ 0, & j < a, \end{cases} \quad 0 \leq j < q. \quad (\text{VC.15a})$$

Its action is the raw-derivative map on polynomials of degree below q . The exact Taylor formula shows that a polynomial has all these derivatives zero if and only if it is divisible by each $(S - \lambda_\nu)^{d_\nu}$, hence by their product χ . For degree below q , such a polynomial is zero. Thus E is injective, and equality of its domain and codomain dimensions proves that it is invertible. Every polynomial and its remainder differ by a multiple of χ , so their listed raw derivatives agree. Therefore the degree- j raw-jet matrix is exactly $F_j = EJ_j$, and

$$K_j^{\text{raw}} = F_j H_j^{-1} F_j^* = EK_j E^*, \quad G_j^{\text{raw}} = E^{-*} G_j E^{-1}, \quad \det G_j^{\text{raw}} = |\det E|^{-2} V_j. \quad (\text{VC.15b})$$

The multiplier $|\det E|^{-2}$ is independent of source degree. Consequently (VC.6) holds with the same constant in raw-jet coordinates and has the same determinant ratio. Equation (VC.13) gives the exact phase for each component when those raw jets are expressed in the affine chart. This is the direct typed relation between the remainder-kernel estimate and the raw-jet determinant calculation.

S26.1.3 3. The Legendre coefficient bound, proved in full

For a polynomial $f(x) = \sum a_j x^j$, write $\|f\|_{\text{coeff},1} = \sum |a_j|$. We prove

$$\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1],dx)}, \quad \deg f \leq N. \quad (\text{VC.16})$$

Define

$$L_j(x) = \frac{1}{2^j j!} \frac{d^j}{dx^j} (x^2 - 1)^j. \quad (\text{VC.17})$$

The leading coefficient is $\kappa_j = (2j)!/(2^j(j!)^2)$. Repeated integration by parts, with every boundary term zero because $(x^2 - 1)^j$ has zeros of order j at both endpoints, proves that L_j is orthogonal to every polynomial of degree below j . The same integration by parts gives

$$\int_{-1}^1 L_j(x)^2 dx = \frac{\kappa_j}{2^j} \int_{-1}^1 (1 - x^2)^j dx = \frac{2}{2j+1}. \quad (\text{VC.18})$$

Here the intermediate integral is $2^{2j+1}(j!)^2/(2j+1)!$. To verify that expression without any special-function identity, substitute $x = 2t - 1$, obtaining $2 \cdot 4^j \int_0^1 t^j (1-t)^j dt$. Repeated integration by parts proves $\int_0^1 t^a (1-t)^b dt = a!b!/(a+b+1)!$ for nonnegative integers a, b : integration by parts with antiderivative $t^{a+1}/(a+1)$ reduces b by one and increases a by one, and the case $b = 0$ is $1/(a+1)$. This proves (VC.18).

Leibniz's rule applied to $(x-1)^j(x+1)^j$ proves the exact finite formula

$$L_j(x) = 2^{-j} \sum_{a=0}^j \binom{j}{a}^2 (x-1)^{j-a} (x+1)^a. \quad (\text{VC.19})$$

The coefficient norm of a product is at most the product of its coefficient norms, by the triangle inequality for each convolution coefficient. The binomial theorem gives $\|(x \pm 1)^a\|_{\text{coeff},1} = 2^a$. Therefore

$$\|L_j\|_{\text{coeff},1} \leq \sum_{a=0}^j \binom{j}{a}^2 = \binom{2j}{j} \leq 4^j. \quad (\text{VC.20})$$

The middle equality is the coefficient of z^j in $(1+z)^j(1+z)^j$; the last inequality bounds one coefficient by the sum of all positive coefficients. The degrees and nonzero leading coefficients show that $L_0, \dots, L_N$ are a basis. Write $f = \sum_{j=0}^N a_j L_j$. By orthogonality and Cauchy-Schwarz,

$$|a_j| = \frac{2j+1}{2} \left| \int_{-1}^1 L_j(x) f(x) dx \right| \leq \sqrt{\frac{2j+1}{2}} \|f\|_{L^2([-1,1])}. \quad (\text{VC.21})$$

This formula is valid for complex coefficients because L_j is real. Combining (VC.20)–(VC.21), bounding $2j+1$ by $2N+1$, and summing the geometric series $\sum_{j=0}^N 4^j = (4^{N+1} - 1)/3$ proves (VC.16).

S26.1.4 4. The complete monic-division bound

Write the roots of χ_T , with repetitions, as $z_1, \dots, z_q$, where $|z_a| \leq r < 1$. Define the elementary symmetric polynomials e_a and complete homogeneous polynomials h_j by the finite coefficient identities

$$\prod_{a=1}^q (1 - z_a t) = \sum_{a=0}^q (-1)^a e_a t^a, \quad \prod_{a=1}^q (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j. \quad (\text{VC.22})$$

The second identity is a formal power-series identity: h_j is the sum of all monomials $z_1^{a_1} \cdots z_q^{a_q}$ with non-negative a_i and $\sum a_i = j$. There are $\binom{q+j-1}{j}$ such monomials: placing $q-1$ separators among $j+q-1$ positions gives a bijection with the exponent lists. Consequently

$$|e_a| \leq \binom{q}{a} r^a, \quad |h_j| \leq \binom{q+j-1}{j} r^j. \quad (\text{VC.23})$$

For $0 \leq s \leq q-1$, set $Q_s(x) = \sum_{j=0}^s h_j x^{s-j}$. Multiplying the two series in (VC.22) shows that every coefficient of degrees $q+s-1, q+s-2, \dots, q$ in $\chi_T Q_s$ is zero, while its coefficient of degree $q+s$ is one. It follows that

$$\text{rem}_{\chi_T} x^{q+s} = x^{q+s} - \chi_T Q_s, \quad \deg(x^{q+s} - \chi_T Q_s) < q. \quad (\text{VC.24})$$

The leading coefficient of $\chi_T Q_s$ is exactly one, so removing its leading term reduces its coefficient norm by exactly one. All remaining terms are the negative of the remainder. Thus

$$\begin{aligned} \|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} &= \|\chi_T Q_s\|_{\text{coeff},1} - 1 \\ &\leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq \left(\frac{1+r}{1-r}\right)^q. \end{aligned} \quad (\text{VC.25})$$

For the last inequality, the convergent geometric-series product at real $r < 1$ gives $\sum_{j \geq 0} \binom{q+j-1}{j} r^j = (1-r)^{-q}$. For $0 \leq j < q$, the remainder of x^j is x^j and its coefficient norm is one, also at most $B_q(r)$. Linearity and the triangle inequality now give the exact finite operator estimate

$$\|\text{rem}_{\chi_T} f\|_{\text{coeff},1} \leq B_q(r) \|f\|_{\text{coeff},1}, \quad \deg f \leq 2q-1. \quad (\text{VC.26})$$

Multiplicity was retained throughout the root list; no distinct-root interpolation or inverse Vandermonde estimate occurs here. If desired, the smaller literal finite constant

$$\max \left\{ 1, (1+r)^q \sum_{j=0}^{q-1} \binom{q+j-1}{j} r^j - 1 \right\} \quad (\text{VC.27})$$

can replace $B_q(r)$ in (VC.4), by the preceding proof.

S26.1.5 5. Original moment control and the remainder's source norm

For every nonnegative integer a and real u , the nonnegative Taylor terms of $e^{b|u|}$ imply

$$|u|^{2a} \leq (2a)! b^{-2a} e^{b|u|} \leq (2a)! b^{-2a} (e^{bu} + e^{-bu}). \quad (\text{VC.28})$$

Tonelli's theorem applied to the nonnegative product density and the identity $e^{b(t_1 + \cdots + t_k)} = \prod e^{bt_i}$ gives

$$\int e^{\pm bu} m_k(u) du = M_h(\pm b)^k, \quad \int m_k(u) du = \mu_h^k. \quad (\text{VC.29})$$

Consequently the actual $2a$ -th moment is at most the right side of (VC.28) integrated using (VC.29). If v has degree at most $q-1$, each monomial has absolute value at most $\max\{1, |x|^{q-1}\}$. This gives

$$\begin{aligned} \int_{\mathbb{R}} |v(u/T)|^2 m_k(u) du &\leq \|v\|_{\text{coeff},1}^2 \int \max\{1, |u/T|^{2q-2}\} m_k(u) du \\ &\leq U_k(T) \|v\|_{\text{coeff},1}^2. \end{aligned} \quad (\text{VC.30})$$

The second inequality uses $\max\{1, t\} \leq 1 + t$ for $t \geq 0$, and the proved moment bound with $a = q - 1$. On the other hand, (VC.3) on $[-T, T]$ gives

$$\|P\|_{H_N}^2 \geq T \lambda_k(T) \|\Psi_T P\|_{L^2([-1,1])}^2. \quad (\text{VC.31})$$

Set $f = \Psi_T P$ and $v = \text{rem}_{\chi_T} f$. Equations (VC.8), (VC.12), (VC.16), (VC.26), (VC.30), and (VC.31), in that order, prove

$$\|\text{rem}_{\chi} P\|_{H_{q-1}}^2 \leq U_k(T) B_q(r)^2 A_N^2 \|\Psi_T P\|_{L^2([-1,1])}^2 \leq C_{k,q} \|P\|_{H_N}^2. \quad (\text{VC.32})$$

To specify the map's type, let $\iota_j : \mathbb{C}^{j+1} \rightarrow \mathcal{P}_j$ be $\iota_j(a) = \sum_{d=0}^j a_d S^d$, and let $\kappa : \mathbb{C}[S]/(\chi) \rightarrow \mathbb{C}^q$ send a class to the coefficient vector of its unique degree-below- q representative. Write $\mathcal{J}_j = J_j \iota_j^{-1} : \mathcal{P}_j \rightarrow \mathbb{C}^q$, and $s : \mathbb{C}^q \rightarrow \mathcal{P}_{q-1}$, $s(y) = \sum_{d=0}^{q-1} y_d S^d$. Then $\mathcal{J}_{q-1} s = I_{\mathbb{C}^q}$, and the original abstract quotient map π satisfies $\pi s = \kappa^{-1}$. The operator estimated in (VC.32) is exactly $s \mathcal{J}_N : \mathcal{P}_N \rightarrow \mathcal{P}_{q-1}$. Its restriction to $\mathcal{P}_{q-1}$ is the identity. In the formulas that follow, coefficient vectors and polynomials are identified only by the displayed ι_j .

S26.1.6 6. From the bounded remainder to the kernel inequality

The positive density in (VC.3) makes every polynomial Gram matrix positive definite. The remainder map J_j is surjective because its restriction to degrees below q is the identity. Thus K_j is positive definite. For completeness, define

$$R_j = H_j^{-1} J_j^* K_j^{-1}. \quad (\text{VC.33})$$

Then $J_j R_j = I$ by the definition of K_j . For $J_j z = 0$, $z^* H_j R_j = z^* J_j^* K_j^{-1} = 0$. Every lift of a quotient vector y is uniquely $R_j y + z$, $J_j z = 0$, so

$$\|R_j y + z\|_{H_j}^2 = y^* K_j^{-1} y + z^* H_j z. \quad (\text{VC.34})$$

The minimum lift norm is therefore exactly $y^* G_j y$. Degree inclusion enlarges the set of lifts and gives $G_{j+1} \preceq G_j$. Taking $P = R_N y$ in (VC.32), the degree- $(q-1)$ remainder is a lift of the same y , so

$$y^* G_q y \leq y^* G_{q-1} y \leq \|\text{rem}_{\chi} R_N y\|_{H_{q-1}}^2 \leq C_{k,q} y^* G_N y. \quad (\text{VC.35})$$

Together with $G_N \preceq G_q$, this gives $C_{k,q}^{-1} G_q \preceq G_N \preceq G_q$. Congruence by $G_q^{-1/2}$ bounds every eigenvalue of the middle matrix between $C_{k,q}^{-1}$ and one. Inversion reverses these positive eigenvalue inequalities; conjugating back proves the first inequality chain in (VC.6). Taking products of the same q eigenvalues proves its determinant statement. This proves the finite answer.

S26.1.7 7. The complete growth calculation and its exact gap

Let

$$X = \max\{1, M_h(b), M_h(-b)\}, \quad K = \max\{1, \vartheta_h^{-1}\}, \quad C_0 = \max\{1, 2/(3c_h D)\}. \quad (\text{VC.36})$$

These are bounding constants for the unchanged source. Since $bT \geq 2q$, $(2q-2)!/(bT)^{2q-2} \leq 1$. Also $M_h(b)^k + M_h(-b)^k = \int 2 \cosh(bu) m_k(u) du \geq 2\mu_h^k$. It follows that

$$U_k(T) \leq \frac{3}{2} (M_h(b)^k + M_h(-b)^k) \leq 3X^k. \quad (\text{VC.37})$$

The constants in (VC.4) further satisfy

$$A_{2q-1}^2 \leq \frac{2q}{9} 256^q, \quad \vartheta_h^{-(k-3)} \leq K^k. \quad (\text{VC.38})$$

On $[0, 1/2]$, the derivative of $\log((1+t)/(1-t))$ is $2/(1-t^2) \leq 8/3$. Integration from zero to r gives

$$\log B_q(r)^2 \leq \frac{16}{3} qr = \frac{16R_0}{3D} k. \quad (\text{VC.39})$$

Finally $1 + Dq + k - 3 \leq 1 + (D+1)q$, and $e^{-3\alpha} \leq 1$. Substituting (VC.37)–(VC.39) into the literal constant in (VC.4), and using $T = Dq$, yields

$$\begin{aligned} \log C_{k,q} &\leq (\alpha D + \log 256)q + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k \\ &\quad + B_h \log(1 + (D+1)q) + \log C_0. \end{aligned} \quad (\text{VC.40})$$

Thus an explicit upper bound on the requested original central volume is

$$\begin{aligned} 0 \leq \log \frac{V_q}{V_{2q-1}} &\leq (\alpha D + \log 256)q^2 \\ &\quad + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq \\ &\quad + B_h q \log(1 + (D+1)q) + q \log C_0. \end{aligned} \quad (\text{VC.41})$$

All constants in (VC.41) depend only on the fixed packet and the chosen b ; no constant depends on k or q . Since $q = [1 + k(m-1)](k+1)^2$, one has $k/q \rightarrow 0$ and $(\log q)/q \rightarrow 0$. In particular

$$\limsup_{k \rightarrow \infty} \frac{1}{q_k^2} \log \frac{V_{q_k}}{V_{2q_k-1}} \leq \alpha D + \log 256. \quad (\text{VC.42})$$

The proved central-window lower estimate in the companion factor-four note is

$$\log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \left(\log k + \log \frac{\delta}{2C_h^{\text{bal}}} \right). \quad (\text{VC.43})$$

The precise comparison between (VC.41) and (VC.43) is therefore an interval of admissible growth, not an RH contradiction. After division by $q \log k$, the leading term of the upper bound is $(\alpha D + \log 256)q / \log k$. This diverges like $(\alpha D + \log 256)k^2 / \log k$ when $m = 1$, and like $(\alpha D + \log 256)(m-1)k^3 / \log k$ when $m > 1$. The upper bound is larger than the lower scale by those explicit factors. This calculation locates exactly the loss in this scalar route: (VC.16) controls every coefficient of a degree- $(2q-1)$ polynomial, and (VC.31) replaces the full source by its least value on a length- $2Dq$ interval. The remainder step itself has only $O(k)$ logarithmic cost by (VC.39).

The sharper raw-jet matrix determinant comparison in the parent lane retains the directional dependence discarded by the scalar estimates (VC.16) and (VC.31). Equations (VC.7)–(VC.15) and (VC.33)–(VC.35) give the exact maps through which the present scalar estimate bounds that same original quotient. No claim about the sharper determinant's asymptotic size is made here.

S26.1.8 8. Provenance and verification boundary

The inherited analytic input is (VC.3) and the finiteness of $M_h(\pm b)$, proved in the complete supplied Tau_Arithmetic_Endpoint_Bounds/NOTE.tex. The central lower bound (VC.43) is proved in the owner's complete FOUR_VOLUME_THRESHOLD.md continuation. This note proves (VC.4)–(VC.42) directly from those established input theorems, with full finite calculations. It does not replace the arithmetic weight, discard the mass, erase a raw-jet phase, shorten a local length, postulate the desired upper estimate, or assert RH. Any finite symbolic test accompanying this note tests the algebraic maps and constants; the written argument proves the estimates for all the indicated degrees.

S27 Complete all-degree division and affine raw-jet estimates

Source role. Full RB.1-37 proof with all original derivative factors.

Exact revision. 56bee5eba5f951570428939169f69020a5c2b71c2ffc5ceaa18d26a2fe43049d

Complete source. sources/local_endpoint_continuations/volume_remainder_bound_review_20260913.md

Source SHA-256. 56bee5eba5f951570428939169f69020a5c2b71c2ffc5ceaa18d26a2fe43049d

S27.1 Exact finite remainder bound and affine raw-jet transport

This note proves the finite coefficient estimate requested for the volume-comparator calculation. All roots are counted with their original multiplicities. The degree, affine unit, complex phase, domains, and codomains remain explicit throughout.

S27.1.1 1. Polynomial spaces and the remainder map

Let $q \geq 1$ be an integer and write

$$\chi_x(x) = \prod_{a=1}^q (x - \lambda_a), \quad |\lambda_a| \leq R, \quad R \geq 0. \quad (\text{RB.1})$$

Repeated entries of $(\lambda_1, \dots, \lambda_q)$ record root multiplicity. Set

$$\mathcal{P}_d(x) = \left\{ \sum_{j=0}^d u_j x^j : u_j \in \mathbb{C} \right\}, \quad \|P\|_{1,x} = \sum_{j=0}^d |u_j|. \quad (\text{RB.2})$$

The exact linear map in this note is

$$\mathcal{R}_{\chi_x} : \mathcal{P}_{2q-1}(x) \longrightarrow \mathcal{P}_{q-1}(x), \quad P = \chi_x Q + \mathcal{R}_{\chi_x} P, \quad Q \in \mathcal{P}_{q-1}(x). \quad (\text{RB.3})$$

Existence follows by subtracting the leading coefficient times the corresponding monomial multiple of the monic polynomial χ_x , decreasing the degree at each step until it is below q . For uniqueness, the difference of two decompositions satisfies $\chi_x(Q_1 - Q_2) = r_2 - r_1$, where the right side has degree below q . A nonzero left side has degree at least q , so both sides vanish. Applying this uniqueness to linear combinations proves linearity. In particular $\mathcal{R}_{\chi_x} x^j = x^j$ for $0 \leq j < q$.

S27.1.2 2. The complete homogeneous division formula

Let e_a be the elementary symmetric polynomial of degree a in the listed roots, with $e_0 = 1$, and define the complete homogeneous polynomials by

$$h_\ell(\lambda_1, \dots, \lambda_q) = \sum_{\substack{\alpha_1, \dots, \alpha_q \geq 0 \\ \alpha_1 + \dots + \alpha_q = \ell}} \lambda_1^{\alpha_1} \cdots \lambda_q^{\alpha_q}, \quad h_0 = 1. \quad (\text{RB.4})$$

As an identity of formal power series,

$$\left(\sum_{a=0}^q (-1)^a e_a t^a \right) \left(\sum_{\ell=0}^{\infty} h_\ell t^\ell \right) = \prod_{a=1}^q (1 - \lambda_a t) \prod_{a=1}^q \left(\sum_{b=0}^{\infty} (\lambda_a t)^b \right) = 1. \quad (\text{RB.5})$$

There is no convergence requirement: every coefficient uses only finitely many factors and terms. Thus, for every integer $s \geq 1$,

$$\sum_{a=0}^{\min(q,s)} (-1)^a e_a h_{s-a} = 0. \quad (\text{RB.6})$$

For $0 \leq m \leq q-1$, put

$$Q_m(x) = \sum_{\ell=0}^m h_\ell x^{m-\ell}. \quad (\text{RB.7})$$

The coefficient of x^{q+m} in $\chi_x Q_m$ is one. For $1 \leq s \leq m$, the coefficient of x^{q+m-s} is the left side of (RB.6), hence zero. Therefore

$$x^{q+m} - \chi_x(x) Q_m(x) \in \mathcal{P}_{q-1}(x).$$

The uniqueness in (RB.3) now gives the exact division identity

$$\boxed{\mathcal{R}_{\chi_x} x^{q+m} = x^{q+m} - \chi_x Q_m.} \quad (\text{RB.8})$$

For completeness its individual coefficients are

$$[x^r] \mathcal{R}_{\chi_x} x^{q+m} = - \sum_{\substack{0 \leq a \leq q, 0 \leq \ell \leq m \\ a+\ell=q+m-r}} (-1)^a e_a h_\ell, \quad 0 \leq r < q. \quad (\text{RB.9})$$

This expression remains valid for repeated and zero roots without taking limits.

S27.1.3 3. Explicit coefficient-norm bound

The coefficient norm is submultiplicative. Indeed, if $U = \sum u_j x^j$ and $V = \sum v_k x^k$, then

$$\|UV\|_{1,x} = \sum_r \left| \sum_{j+k=r} u_j v_k \right| \leq \sum_{j,k} |u_j| |v_k| = \|U\|_{1,x} \|V\|_{1,x}. \quad (\text{RB.10})$$

Consequently

$$\|\chi_x\|_{1,x} \leq \prod_{a=1}^q (1 + |\lambda_a|) \leq (1 + R)^q. \quad (\text{RB.11})$$

The number of weak compositions of ℓ into q nonnegative parts is $\binom{q+\ell-1}{\ell}$: place $q-1$ separators among a row of ℓ markers and those separators. Thus (RB.4) yields

$$|h_\ell| \leq \binom{q+\ell-1}{\ell} R^\ell, \quad \|Q_m\|_{1,x} \leq \sum_{\ell=0}^m \binom{q+\ell-1}{\ell} R^\ell. \quad (\text{RB.12})$$

In (RB.8), the leading monomial x^{q+m} and the remainder have disjoint degree supports. In particular

$$\|\chi_x Q_m\|_{1,x} = 1 + \|\mathcal{R}_{\chi_x} x^{q+m}\|_{1,x}. \quad (\text{RB.13})$$

Combining (RB.10)–(RB.13) proves, with the original radius retained,

$$\boxed{\|\mathcal{R}_{\chi_x} x^{q+m}\|_{1,x} \leq (1+R)^q \sum_{\ell=0}^m \binom{q+\ell-1}{\ell} R^\ell - 1, \quad 0 \leq m \leq q-1.} \quad (\text{RB.14})$$

When $R \leq 1$, the finite sum of binomial coefficients is

$$\sum_{\ell=0}^m \binom{q+\ell-1}{\ell} = \binom{q+m}{m}. \quad (\text{RB.15})$$

To verify this identity, its $m = 0$ case is $1 = 1$, and its induction step is Pascal's identity $\binom{q+m-1}{m-1} + \binom{q+m-1}{m} = \binom{q+m}{m}$. The right side of (RB.15) increases with m , as is also immediate from its sum expression. Define the exact integer

$$C_q = 2^q \binom{2q-1}{q-1} - 1. \quad (\text{RB.16})$$

Then $C_q \geq 1$, with $C_1 = 1$, and every column of the remainder map satisfies

$$\max_{0 \leq j \leq 2q-1} \|\mathcal{R}_{\chi_x} x^j\|_{1,x} \leq C_q. \quad (\text{RB.17})$$

The two central coefficients $\binom{2q-1}{q-1}$ and $\binom{2q-1}{q}$ are equal. Their sum is at most the sum of all coefficients of $(1+t)^{2q-1}$ at $t=1$, which is 2^{2q-1} . Therefore

$$C_q \leq 2^{3q-2} - 1 < 8^q. \quad (\text{RB.18})$$

For an arbitrary $P = \sum_{j=0}^{2q-1} u_j x^j$, linearity and the triangle inequality give

$$\|\mathcal{R}_{\chi_x} P\|_{1,x} \leq C_q \|P\|_{1,x} < 8^q \|P\|_{1,x} \quad (P \neq 0). \quad (\text{RB.19})$$

The non-strict version with 8^q holds also for $P = 0$. In fact the induced operator norm between these coefficient ℓ^1 spaces equals the maximum column norm: the displayed triangle inequality proves the upper bound, and applying the operator to a monomial attaining the finite maximum proves the reverse inequality.

S27.1.4 4. Raw jets and the exact quotient section

Write the distinct roots as $\lambda_1^*, \dots, \lambda_s^*$, with multiplicities $m_1, \dots, m_s$, so $\sum_{\nu} m_{\nu} = q$. Define the raw-jet space and map by

$$\mathcal{J}_x = \bigoplus_{\nu=1}^s \mathbb{C}^{m_{\nu}}, \quad J_{\chi_x, d} : \mathcal{P}_d(x) \longrightarrow \mathcal{J}_x, \quad P \longmapsto \left(P^{(r)}(\lambda_{\nu}^*) \right)_{1 \leq \nu \leq s, 0 \leq r < m_{\nu}}. \quad (\text{RB.20})$$

The exact Taylor identity at a root is

$$P(x) = \sum_{r=0}^d \frac{P^{(r)}(\lambda_{\nu}^*)}{r!} (x - \lambda_{\nu}^*)^r.$$

It shows that the first m_{ν} raw derivatives vanish precisely when $(x - \lambda_{\nu}^*)^{m_{\nu}}$ divides P . Simultaneous divisibility by these distinct-root factors is equivalent to divisibility by their product χ_x : in the factorization of a nonzero polynomial over $\mathbb{C}$, the exponent at each distinct root must be at least the listed multiplicity. The zero polynomial is divisible by every factor. Hence, for $d \geq q$,

$$\ker J_{\chi_x, d} = \chi_x \mathcal{P}_{d-q}(x). \quad (\text{RB.21})$$

The restriction $J_{\chi_x, q-1}$ has zero kernel by degree, and both its domain and codomain have dimension q . It is therefore an isomorphism. Equation (RB.3) gives

$$J_{\chi_x, q-1} \mathcal{R}_{\chi_x} = J_{\chi_x, 2q-1}, \quad \mathcal{R}_{\chi_x} = J_{\chi_x, q-1}^{-1} J_{\chi_x, 2q-1}. \quad (\text{RB.22})$$

Thus this remainder map preserves every specified raw jet, including every derivative order required by multiplicity. Equivalently, the inclusion of $\mathcal{P}_{q-1}(x)$ followed by the quotient map is an isomorphism onto $\mathbb{C}[x]/(\chi_x)$, and (RB.3) is the unique representative of degree below q in that quotient class.

S27.1.5 5. Original S variable, affine unit, and phase

Let the original monic annihilator be

$$\chi(S) = \prod_{\nu=1}^s (S - \zeta_{\nu})^{m_{\nu}}, \quad c \in \mathbb{C}, \quad T > 0. \quad (\text{RB.23})$$

Use the requested transformed-polynomial notation with its full definition:

$$\bar{\chi}(x) = (iT)^{-q} \chi(c + iTx) = \prod_{\nu=1}^s \left(x - \frac{\zeta_\nu - c}{iT} \right)^{m_\nu}. \quad (\text{RB.24})$$

Here the bar symbol is defined by (RB.24); complex conjugation of a polynomial, if used elsewhere, has its separate coefficientwise definition. The displayed unit $(iT)^{-q}$ proves monicity and retains the full complex phase. The transformed root radius is exactly bounded by

$$\left| \frac{\zeta_\nu - c}{iT} \right| = \frac{|\zeta_\nu - c|}{T}; \quad \text{in particular } \max_\nu |\zeta_\nu - c| \leq T \implies R \leq 1. \quad (\text{RB.25})$$

For every $d \geq 0$, define the invertible linear map

$$\begin{aligned} A_{c,T,d} : \mathcal{P}_d(S) &\longrightarrow \mathcal{P}_d(x), & (A_{c,T,d}P)(x) &= P(c + iTx), \\ A_{c,T,d}^{-1}F(S) &= F\left(\frac{S - c}{iT}\right). \end{aligned} \quad (\text{RB.26})$$

Substitution also defines an algebra isomorphism on the whole polynomial rings. In particular it sends the principal ideal (χ) onto $(\bar{\chi})$, because $A_\chi = (iT)^q \bar{\chi}$ and the factor $(iT)^q$ is a nonzero scalar unit. Thus the induced quotient map is exactly

$$\hat{A}_{c,T} : \mathbb{C}[S]/(\chi) \longrightarrow \mathbb{C}[x]/(\bar{\chi}), \quad [P] \longmapsto [P(c + iTx)], \quad (\text{RB.27})$$

with inverse induced by the second formula in (RB.26).

Apply (RB.26) to the original division $P = \chi Q + \mathcal{R}_\chi P$. One obtains

$$A_{c,T,2q-1}P = \bar{\chi}((iT)^q A_{c,T,q-1}Q) + A_{c,T,q-1}\mathcal{R}_\chi P. \quad (\text{RB.28})$$

The last term has degree below q . Uniqueness of division gives the exact commuting identity

$$\boxed{\mathcal{R}_{\bar{\chi}} A_{c,T,2q-1} = A_{c,T,q-1} \mathcal{R}_\chi.} \quad (\text{RB.29})$$

For raw jets, set $\lambda_\nu^* = (\zeta_\nu - c)/(iT)$. Let $J_{\chi,d}$ be defined using derivatives in S at ζ_ν , and let $J_{\bar{\chi},d}$ use derivatives in x at λ_ν^* , with the same ordering and multiplicities. Repeated application of the ordinary chain rule gives, for every integer $r \geq 0$,

$$\frac{d^r}{dx^r} P(c + iTx) = (iT)^r P^{(r)}(c + iTx).$$

Define the invertible diagonal map

$$D_T : \bigoplus_\nu \mathbb{C}^{m_\nu} \longrightarrow \bigoplus_\nu \mathbb{C}^{m_\nu}, \quad (v_{\nu,r}) \longmapsto ((iT)^r v_{\nu,r}). \quad (\text{RB.30})$$

The complete raw-jet transport identity is therefore

$$\boxed{J_{\bar{\chi},d} A_{c,T,d} = D_T J_{\chi,d}.} \quad (\text{RB.31})$$

Every scalar $(iT)^r$, including its phase, appears in (RB.30)–(RB.31); the jet coordinates throughout are raw derivatives. Combining (RB.22), (RB.29), and (RB.31) proves preservation of the original raw jets as well as the transformed raw jets.

S27.1.6 6. Exact norm transfer back to the original coordinates

Define a norm on the original polynomial space by the displayed affine transport,

$$\|P\|_{c,T,1} := \|A_{c,T,d}P\|_{1,x}. \quad (\text{RB.32})$$

This definition is independent of the choice of $d \geq \deg P$, because adding zero coefficients does not change the sum. Invertibility of A proves that it is a norm. Equations (RB.19) and (RB.29), under the root-radius hypothesis in (RB.25), yield

$$\boxed{\|\mathcal{R}_\chi P\|_{c,T,1} \leq C_q \|P\|_{c,T,1}, \quad P \in \mathcal{P}_{2q-1}(S)}. \quad (\text{RB.33})$$

The same estimate expressed using the untransformed coefficient norm also has fully explicit constants. For $0 \leq j \leq d$, the binomial theorem gives

$$\|A_{c,T,d}S^j\|_{1,x} = (|c| + T)^j, \quad \|A_{c,T,d}^{-1}x^j\|_{1,S} = \left(\frac{1 + |c|}{T}\right)^j. \quad (\text{RB.34})$$

The maximum-column characterization of the coefficient ℓ^1 operator norm, proved after (RB.19), consequently gives the exact operator norms

$$\|A_{c,T,d}\|_{1 \rightarrow 1} = \max\{1, (|c| + T)^d\}, \quad \|A_{c,T,d}^{-1}\|_{1 \rightarrow 1} = \max\left\{1, \left(\frac{1 + |c|}{T}\right)^d\right\}. \quad (\text{RB.35})$$

Thus the original-coordinate estimate is

$$\boxed{\|\mathcal{R}_\chi P\|_{1,S} \leq C_q \max\left\{1, \left(\frac{1 + |c|}{T}\right)^{q-1}\right\} \max\{1, (|c| + T)^{2q-1}\} \|P\|_{1,S}}. \quad (\text{RB.36})$$

Finally, for any $x \in \mathbb{C}$ and any polynomial r of degree below q , the coefficient triangle inequality gives $|r(x)| \leq \|r\|_{1,x} \max\{1, |x|^{q-1}\}$. Applying this to the exact representative in (RB.29) yields

$$\boxed{|(\mathcal{R}_\chi P)(c + iTx)| \leq C_q \|P\|_{c,T,1} \max\{1, |x|^{q-1}\}}. \quad (\text{RB.37})$$

In particular the maximum on the segment $S = c + iTx$, $-1 \leq x \leq 1$, is at most $C_q \|P\|_{c,T,1}$. Equations (RB.24), (RB.27), (RB.29), and (RB.31) specify the exact relation to the original annihilator, quotient, representative, and raw-jet coordinates; no measure or quotient metric is replaced in this coefficient calculation.

S28 The complete corrected endpoint and multiplicity criterion

Source role. Complete corrected PR23 witness with the full multiplicity proof.

Exact revision. a9a8b080b3779f82388a8bd4338ada2564ceb2410a0734bd58700f65c7789c62

Complete source. sources/owner_endpoint_review/review_pr24/join_inputs/PR23_ENDPOINT_CORRECTED.md

Source SHA-256. a9a8b080b3779f82388a8bd4338ada2564ceb2410a0734bd58700f65c7789c62

S28.1 Sharper endpoint-only criterion and the scope of exterior iteration

This is a mathematical continuation of the supplied Toda and exterior notes. The general Jensen argument and exterior-multiplicity count below have written proofs; they are not additional Lean declarations. The norm-ratio product and overlapping-volume telescope are included in the finite formalization target list. No arithmetic asymptotic bound is claimed.

S28.1.1 1. Eliminate the supremum over recurrence coefficients

Retain a nonempty fixed packet and tensor degree k , cyclic dimension q , the actual positive monic norm sequence ω_N , the actual quotient volume $V_N = \det G_N$, and the source's exact control allowance ϵ_N . Choose consecutive admitted degrees $N=n, \dots, n+r-1$ with $n \geq q$ and integer $r \geq 1$. Put

$$B_{(n,r)} = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})),$$

$$A_{(n,r)} = (\omega_{(n+r)} / \omega_n)^{1/(2r)}.$$

These are four endpoint quotient volumes and two endpoint polynomial norms of the existing family. Since V is positive and nonincreasing, $B_{(n,r)} \geq 0$. The source's finite upper bound implies

$$\min_{0 \leq j < r} \epsilon_{(n+j)} \leq A_{(n,r)} \sinh(B_{(n,r)}/(2r)). \quad (E1)$$

Unlike the preceding maximum-recurrence criterion, E1 needs no pointwise upper bound for every recurrence coefficient in the window.

Complete proof For $N=n+j$ set

$$x_j = (1/2) \log(V_{(N-1)}/V_{(N+1)}) \geq 0,$$

$$A_j = \sqrt{\omega_{(N+1)}/\omega_N} > 0.$$

The existing local estimate is $\epsilon_{(n+j)} \leq A_j \sinh(x_j)$. If any $x_j = 0$, the nonnegative allowance at that step is zero, and E1 follows. Otherwise all $x_j > 0$. The function $f(x) = \log(\sinh x)$ has

$$f''(x) = -1/\sinh(x)^2 < 0.$$

Finite Jensen therefore gives

$$\text{product}_j \sinh(x_j) \leq \sinh((\sum_j x_j)/r)^r.$$

Let E be the minimum of the nonnegative allowances. Multiplication of the local bounds gives

$$E^r \leq \text{product}_j \epsilon_{(n+j)} \leq (\text{product}_j A_j) \text{product}_j \sinh(x_j).$$

The actual products telescope:

$$\text{product}_j A_j = (\omega_{(n+r)}/\omega_n)^{1/2},$$

$$2 \sum_j x_j = \log(V_{(n-1)} V_n / (V_{(n+r-1)} V_{(n+r)})).$$

Taking the nonnegative r -th root proves E1. A finite minimum is attained, so E1 selects an actual original degree N inside the specified window. No favorable metric is selected independently, and its original theta representative, quotient and full jets remain attached.

The recurrence product is $\text{VolumeWindow.norm_ratio_product}$; the second telescope is $\text{VolumeWindow.overlapping_log_telescope}$, applied to $V(j)=V_{(n-1+j)}$. The Jensen and nonnegative-root assembly is this written proof rather than a claim about an unimplemented exterior/analytic declaration.

S28.1.2 2. Every input is an endpoint of the two original determinant sequences

The source's $\omega_j = D_{(j+1)}/D_j$ gives

$$A_{(n,r)} = [D_{(n+r+1)} D_{(n)} / (D_{(n+r)} D_{(n+1)})]^{1/(2r)}.$$

The four V -values in B are evaluated by $V_N = D_{(N+1)}/B_{(N-q+1)}$, with the source's literal relation determinant B . All indices are nonnegative because $n \geq q$. No negative-index determinant or inverse of a rank-deficient earlier kernel is used. The common mass factors cancel only in these displayed ratios; the source $\text{norm } \omega_0 = \mu_h^k$ has not been reset.

For the original quartet,

$$q_k = [1 + k(m-1)](k+1)^2, \\ L_{(h,k)} \geq \delta k^2 q_k / 2.$$

The exterior theorem gives $L_{(h,k)} \leq \epsilon_N$ at every admitted N . Thus E1 yields

$$L_{(h,k)} \leq A_{(n,r)} \sinh(B_{(n,r)}) / (2r). \quad (E2)$$

A proof that the right side is $o(k^2 q_k)$ along admitted $n=n(k)$, $r=r(k)$ would exclude that fixed off-line quartet. One sufficient set of estimates is

$$r(k) \geq c_h q_k, \quad A_{(n,r)} = O_h(q_k), \\ B_{(n,r)} = O_h(q_k \log k),$$

for $c_h > 0$. Indeed $B/(2r) = o(\log k)$, so $\sinh(B/(2r)) \leq \exp(o(\log k)) = k^{o(1)}$, and the ratio to $k^2 q_k$ tends to zero. These are candidate arithmetic estimates, not consequences of the formal telescope, positivity, the Toda equation, or Deligne's finite-field hypotheses.

The weighted endpoint criterion is sharp as an abstract sequence inequality: equality can occur when all x_j are equal and every $A_j \sinh(x_j)$ equals the same allowance. For arbitrary recurrence sequences the bound is never larger, and potentially strictly smaller, than the bound obtained by replacing $A_{(n,r)}$ by the largest A_j .

S28.1.3 3. A useful boundary on further exterior amplification

The supplied single exterior theorem is correct. It must not be reused with a rank-two hypothesis that no longer holds after exterior powering.

Suppose the original self-adjoint control H has spectrum

$$(+\epsilon, 0 \text{ repeated } q-2 \text{ times}, -\epsilon), \quad \epsilon > 0.$$

On the actual p -th exterior power, $1 \leq p \leq q$, the additive control has $+\epsilon$ and $-\epsilon$ each with multiplicity

$$r_p = \text{binomial}(q-2, p-1),$$

and zero with multiplicity $\text{binomial}(q-2, p) + \text{binomial}(q-2, p-2)$. This follows by the unscaled wedge basis: select the positive direction but not the negative, the negative but not the positive, or both/neither. In particular its operator norm is at most ϵ : it equals ϵ for $1 \leq p \leq q-1$ and is zero for $p=q$. Its rank is $2r_p$, not generally two.

Apply an s -th exterior power to that whole space, whose dimension is $D_p = \text{binomial}(q, p)$. Its largest absolute additive-control eigenvalue is exactly

$\epsilon \cdot \min(s, r_p, D_p - s), \quad 0 \leq s \leq D_p.$

Proof: select positive eigenvectors first, zero eigenvectors next, and negative eigenvectors only when the degree forces them. There are r_p available positive directions and r_p negative ones. This attains the stated piecewise maximum; reversing the choices attains its negative. At $s=D_p$ the sum is zero. For example $q=4, p=2$ gives spectrum $(+\epsilon, +\epsilon, 0, 0, -\epsilon, -\epsilon)$, and the next exterior square has allowance 2ϵ .

This does not weaken the original determinant-line amplification or remove its nilpotent arithmetic action. It identifies the precise cost of iterating the operation on the entire exterior space, and prevents a spurious second cost-free amplification. The source's chosen determinant line has its own one-dimensional action and must remain distinguished from the complete first exterior power.

S28.1.4 4. Collaboration boundary

The formal work carries the exact source/quotient determinant and the two loss terms into the original-Gram trace certificate. The analytic collaborator can now estimate the two endpoint quantities in E1, or retain the exact phase and imbalance when the coarse upper bound is too large. The six-part Deligne archive and uploaded ZIP test suites were not opened in this execution environment; the source statements used here came from the complete pasted notes. The current GitHub correction and original source maps are unchanged.

S29 Complete original endpoint window product proof

Source role. Complete terminal owner WP.1-13 witness, all six sections and every mathematical source map.

Exact revision. c97c831a5e1c2ef3685910c648bce0f9fbc54230bc8a1bec7e521bd808a1209

Complete source. sources/owner_endpoint_terminal/published/workbenches/tau-arithmetic-endpoint-bounds/window/WINDOW_PRODUCT_PROOF.md

Source SHA-256. c97c831a5e1c2ef3685910c648bce0f9fbc54230bc8a1bec7e521bd808a1209

S29.1 Exact consecutive-window control and endpoint determinants

This note proves the consecutive product, its sharper four-endpoint consequence, and its two-transfer reduction for the original canonical quotient metrics. Composed with the independently reviewed arithmetic norm bound on a balanced window, it gives a necessary four-volume growth threshold of **4**, improving the older threshold of 2. It proves no opposing arithmetic upper estimate and no Riemann-hypothesis conclusion. It belongs to the next integration cut, not the frozen Toda-Gamma edition.

S29.1.1 1. Original identity and exact window product

Fix a nonempty reflection-stable packet, tensor degree, cyclic dimension $q \geq 1$, and one admissible observation. Write $V_N = \det G_N > 0$, retain the original monic squared norms $\omega_N > 0$, and set $\delta_N = V_N/V_{N-1} \in (0, 1]$. The original Toda source, TVB.13, TVB.15 and TVB.22–27, supplies

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N)(\delta_{N+1}^{-1} - 1), \quad \phi_N = \sigma - \partial_\theta \log V_N, \quad N \geq q. \quad (\text{WP1})$$

The real phase is not set to zero. At a general tilt all norms, volumes and derivatives in WP1 are evaluated at that same tilt. The arithmetic norm and asymptotic composition below use the original observation $\theta = 0$.

For integers $n \geq q, r \geq 1$, put $d_j = \delta_{n+j}$ for $0 \leq j \leq r$, $\Omega = \omega_{n+r}/\omega_n$, and $E = \min_{0 \leq j < r} \epsilon_{n+j}$. Multiplying WP1 first, the norm ratios telescope, every interior $1 - d_j$ occurs twice, and $\prod_{j=1}^r d_j = V_{n+r}/V_n$. Therefore

$$\prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = \frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (\text{WP2})$$

Since $E^{2r} \leq \prod_j \epsilon_{n+j}^2 \leq \prod_j (\epsilon_{n+j}^2 + \phi_{n+j}^2)$, dropping only the displayed nonnegative factors, each at most one, gives

$$E \leq \left(\frac{\omega_{n+r} V_n}{\omega_n V_{n+r}} \right)^{1/(2r)}. \quad (\text{WP3})$$

No inverse at degree $q - 2$, reassigned Gram, or recurrence supremum occurs. The empty quotient $q = 0$ is outside this positive-dimensional theorem; its determinant and control are separately the empty determinant and zero.

S29.1.2 2. Sharper four-endpoint bound and comparison

For $r \geq 2$, define

$$\alpha = V_n/V_{n-1}, \quad \beta = V_{n+r}/V_{n+r-1}, \quad t = V_{n+r-1}/V_n, \quad A = \Omega^{1/(2r)}.$$

These are positive and at most one, and the $r - 1$ interior contractions have product t . For $s = r - 1$,

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (\text{WP4})$$

If any interior $d_j = 1$, the left side is zero. Otherwise set $v_j = -\log d_j > 0$. The second derivative of $f(v) = \log(1 - e^{-v})$ is $-e^{-v}/(1 - e^{-v})^2 < 0$, so finite concavity proves WP4. This is ordinary Jensen concavity, obtainable from the integral form by a step function; see [DLMF §1.7\(iv\)](#). Inserting WP4 into WP2 gives

$$E \leq U := A \left(\frac{(1 - \alpha)(1 - \beta)}{\beta t} \right)^{1/(2r)} \left(1 - t^{1/(r-1)} \right)^{(r-1)/r}. \quad (\text{WP5})$$

For $r = 1$ use the separate expression

$$E = \epsilon_n \leq \sqrt{\frac{\omega_{n+1}}{\omega_n} (1 - \alpha)(\beta^{-1} - 1)}. \quad (\text{WP6})$$

No $1/(r - 1)$ is evaluated in WP6. For any r , a unit contraction $d_j = 1$ makes an adjacent local radius sum zero; hence the corresponding allowance and phase both vanish and $E = 0$. At an interior unit contraction both neighboring local sums vanish. If $t = 1$, all interior contractions are one and WP5 is zero.

The corrected PR23 endpoint note proves the older bound

$$E \leq A \sinh\left(\frac{\mathcal{B}}{2r}\right), \quad \mathcal{B} = \log \frac{V_{n-1}V_n}{V_{n+r-1}V_{n+r}} = -\log \alpha - \log \beta - 2 \log t.$$

In fact $U \leq A \sinh(\mathcal{B}/(2r))$. To prove it without assuming an arithmetic extremizer, replace just the interior scalar contractions by their common geometric mean. This maximizes the exact product by WP4 while leaving the four endpoints fixed. For that scalar sequence put $v_j = -\log d_j$. The original local completion of the square is

$$(1 - d_j)(d_{j+1}^{-1} - 1) = \sinh^2 \frac{v_j + v_{j+1}}{2} - \left(e^{(v_{j+1} - v_j)/2} - \cosh \frac{v_j + v_{j+1}}{2} \right)^2.$$

Drop this square and apply concavity of $\log \sinh x$, whose second derivative is $-1/\sinh^2 x$, to the r nonnegative half-sums. Their sum is $\mathcal{B}/2$, proving the comparison. A zero half-sum is handled before logarithms. WP5 also dominates the simpler WP3 because all factors it retains are at most one; WP3 alone need not dominate the older bound for small contractions.

For constant $d = e^{-v} < 1$, WP5 becomes $2A \sinh(v/2)$, whereas the older expression is $A \sinh v$; their ratio is $\cosh(v/2)$.

The sharpness claim is only for the relaxed scalar data, not the arithmetic moment family. In its strictly positive case require **all three** $\alpha < 1, \beta < 1, t < 1$. Set the interior ratios equal, put $f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0$, choose successive norm ratios U^2/f_j , phases zero and all allowances U . The norm-ratio product is exactly Ω , so the prescribed endpoints and equality hold. Zero-product cases give a zero bound directly. Strict contraction of only the two outside ratios would not justify claiming $f_j > 0$ when $t = 1$.

S29.1.3 3. Retained phase and interior loss

When every interior contraction is strictly below one, define

$$J = (r - 1) \log(1 - t^{1/(r-1)}) - \sum_{j=1}^{r-1} \log(1 - d_j) \geq 0.$$

WP2 is precisely

$$\prod_j (\epsilon_{n+j}^2 + \phi_{n+j}^2) = U^{2r} e^{-2J}, \quad \prod_j (E^2 + \phi_{n+j}^2) \leq U^{2r} e^{-2J}. \quad (\text{WP7})$$

The polynomial $P(x) = \prod_j (x + \phi_{n+j}^2)$ is strictly increasing for $x \geq 0$. For consistent input, the right side R is at least $P(0)$, so $P(x) = R$ has a unique nonnegative solution x_* and $E \leq \sqrt{x_*}$. If proposed data instead have $R < P(0)$, they are inconsistent with the original radius identities; a nonnegative threshold cannot then be asserted. Unit-contraction cases are handled directly, without using an infinite or undefined J .

S29.1.4 4. Literal determinant and full-jet transfer cancellation

To distinguish this scalar from the existing exterior map named Λ_N , write $\Lambda_N^{\text{sc}} = \omega_N / V_N$. With the unchanged source and relation determinants, TVB.13 and TVB.15 give

$$\omega_N = \mathfrak{D}_{N+1} / \mathfrak{D}_N, \quad V_N = \mathfrak{D}_{N+1} / \mathfrak{B}_{N-q+1}, \quad \boxed{\Lambda_N^{\text{sc}} = \mathfrak{B}_{N-q+1} / \mathfrak{D}_N}. \quad (\text{WP8})$$

The ordinary monic recurrence and moment-determinant background is [DLMF §18.2\(iv\)](#) and [§18.2\(ix\)](#); the specific arithmetic relation determinant is the supplied construction, not a conclusion attributed to DLMF. Thus WP3 equals

$$E \leq \left(\frac{\Lambda_{n+r}^{\text{sc}}}{\Lambda_n^{\text{sc}}} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{n+r-q+1} \mathfrak{D}_n}{\mathfrak{D}_{n+r} \mathfrak{B}_{n-q+1}} \right)^{1/(2r)}. \quad (\text{WP9})$$

PR24's complete transfer note, equations (4)–(5), defines F_a using raw derivatives of the real monic source polynomials at every doubled root multiplicity, with fixed row order. It proves

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det V}{\det F_a}, \quad a = N - q + 1.$$

Consequently, setting now $a = n - q + 1$,

$$E \leq \left[\frac{\det F_{a+r}}{\det F_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}. \quad (\text{WP10})$$

This uses two transfer determinants; at $q = 1$ the norm product is empty. The common raw confluent Vandermonde, including every derivative factorial and its coordinate phase, cancels through the displayed ratio. No absolute value is inserted: positivity follows from equality with the positive scalar ratio in WP9. The relation map still contains the unit i^q , and F_0 need not equal V , although their determinants agree.

Literal mass bookkeeping confirms the cancellation: rescaling the fixed measure by $c > 0$ would give $\omega_N \mapsto c\omega_N$, $V_N \mapsto c^q V_N$, $\Lambda_N^{\text{sc}} \mapsto c^{1-q} \Lambda_N^{\text{sc}}$. Only the endpoint ratio is invariant. No such rescaling is performed here.

S29.1.5 5. Immediate arithmetic consequence: threshold four

The complete Arithmetic Endpoint Bounds note proves its original lower convolution envelope and exact general norm-window bound, equations (29), (33), (34), (37). Its independently checked balanced extension yields

$$A_{h,k,n,r} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n,$$

where, with $b \in (0, \pi/2)$, $B_h = 42 + 2 \deg h$, $X_h = \max(1, M_h(b), M_h(-b))$, $K_h = \max(1, \vartheta_h^{-1})$,

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) X_h K_h \max(1, b^{-3}) 6^{B_h/3}. \quad (\text{WP11})$$

The constant is finite and depends only on the fixed packet and chosen b ; it is not a numerically enclosed constant. A direct check of the extension uses $\mathfrak{l}_n(n) \geq 2n^{2n+1}/((2n+1)4^n)$, $(2(n+r))! \leq (4n)^{2(n+r)}$, and the two moment masses before taking roots. The factorial/Legendre contribution becomes $4n 8^{n/r} ((2n+1)/(c_h n))^{1/(2r)} \leq 256n \max(1, 3/c_h)$. The remaining factors are bounded by $X_h K_h \max(1, b^{-3}) e^\pi 6^{B_h/3}$, using $2r \geq n$ and decreasing $\log(2n)/n$ for $n \geq 3$.

For a packet consisting exactly of a hypothetical full off-line quartet of common multiplicity m , the original exterior lower allowance satisfies

$$q = q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} \geq \delta k q / 2 > 0, \quad L_{h,k} \leq \epsilon_N.$$

Choose $n = q$, $r = q - 1$. Here $q \geq k \geq 3$, the allowance indices are precisely $q, \dots, 2q - 2$, and the last norm/volume endpoint is $2q - 1$. WP3 and WP11 imply

$$C_k := \log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \log \frac{\delta k}{2C_h^{\text{bal}}}. \quad (\text{WP12})$$

The old four-volume budget obeys the exact decomposition

$$\mathcal{B}_{h,k} = \log \frac{V_{q-1}V_q}{V_{2q-1}V_{2q}} = 2C_k + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}} \geq 2C_k.$$

Therefore

$$\boxed{\mathcal{B}_{h,k} \geq 4(q_k - 1) \log \frac{\delta k}{2C_h^{\text{bal}}}, \quad \liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4.} \quad (\text{WP13})$$

The diagonal window also gives $\liminf \log(V_q/V_{2q})/(q_k \log k) \geq 2$. These lower bounds follow from the actual arithmetic norm estimate and original radius identities, not from scalar-fixture sampling. The old threshold 2 remains a valid weaker result. A separately proved $\limsup \mathcal{B}_{h,k}/(q_k \log k) < 4$ would now contradict the hypothetical quartet; **no such upper bound is established**. Equivalently, WP9 identifies the sufficient target $(2r)^{-1} \log(\Lambda_{n+r}^{\text{sc}}/\Lambda_n^{\text{sc}}) - \log(kq_k) \rightarrow -\infty$, but does not prove that target for the arithmetic family.

S29.1.6 6. Exact fixture and verification scope

The independently recomputed fixture is $d\mu = 7e^{-u^2/2} du/\sqrt{2\pi}$, $\chi = u^2 + 1$, $Q_{j+1} = uQ_j - jQ_{j-1}$, $\omega_j = 7j!$, and $G_N = (\sum_{j \leq N} [Q_j][Q_j]^T/\omega_j)^{-1}$. It gives exactly

$$G_2 = \text{diag}(7/3, 7), \quad G_3 = \text{diag}(7/3, 21/11), \quad G_4 = \text{diag}(42/43, 21/11), \\ (\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = (16/3, 400/99, 4225/946), \quad (V_1, \dots, V_5) = (49, 49/3, 49/11, 882/473, 980/1333).$$

For $n = 2, r = 3, \Omega = 60, \alpha = 1/3, \beta = 110/279, t = 54/473$. The minimum, geometric mean, sharper bound and older bound are respectively approximately 2.0100756305184, 2.1407199197265, 2.1666731523012 and 2.5178436563897. Their exact underlying quantities, not these decimal strings, enter the identity checks.

A separate original exponential tilt $\theta = 1/2$ retains mass $7e^{1/8}$. At degree 2 it gives $\phi_2 = 7/129 \neq 0$ and $\epsilon_2^2 = 18769/4128$; further nonzero phases, the independent derivative $\sigma - \partial_\theta \log V_N$, and the whole window product are checked. The polynomial $u^2 - 3$ supplies an actual interior unit contraction. A one-dimensional quotient tests zero control with potentially nonzero phase and the empty norm band in WP10. Raw derivatives at doubled nonreal nodes independently verify WP8–WP10, while the literal moment matrices verify the source/relation determinants.

The receipt separates exact symbolic/rational/data checks from four approximate decimal-display comparisons. Symbolic product factorizations were checked for $r = 1, \dots, 8$; rational phases and degeneracies, exact powered endpoint comparisons, mass scaling, full raw-jet transfers and balanced-window index/algebra checks are retained. These finite checks do not replace the written general proofs, are not arithmetic zero packets, and are not interval or Lean certificates. No Wolfram session was run or its reported execution adopted as independent evidence.

The complete submitted calculation, original TVB source sections used above, corrected PR23 endpoint note, complete PR24 transfer note, complete Arithmetic Endpoint Bounds note, balanced-window proof, and parent four-volume synthesis were read. Their immutable hashes are in EXACT_VERIFICATION_RECEIPT.json; no private attachment path occurs in the public derivative or receipt. Source attachments remain unchanged. The parent's FOUR_VOLUME_THRESHOLD.md was independently approved at SHA-256 9ff55c84d74db85306e7a68c70fe7b08d8e642e8ac1a3e6137649f68e102fc3a.

Selecting the finite minimizing degree keeps the existing representative $R_N, J^{(k)}R_N = \eta_{h,k}$, and $q^{(k)}R_N = \sigma_h^{\otimes k} \eta_{h,k}$, with the same arithmetic unit, jets, cohomology classes and supported-zero/absence distinction. Endpoint data certify existence, not identification of the minimizing degree. No owner scalar-budget optimization, canonical source, frozen edition, or remote publication was changed by this review.

S30 Complete original endpoint restriction control

Source role. Complete delivered proof witness. Its historical claims retain their source revision; corrections and full current maps are proved in the corresponding integrated chapter.

Exact revision. 905ad0dcd20a2630eb6338488206a387265223590a527f867c1266b85bb27491

Complete source. sources/web_endpoint_restriction_delivery/Tau_Endpoint_Restriction_Control/RESEARCH_NOTE.md

Source SHA-256. 905ad0dcd20a2630eb6338488206a387265223590a527f867c1266b85bb27491

Source metadata: author (MetaInlines)

Owner-directed SplitZero analytic continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

The adjoint of the original support transition, exterior minor weights, and a logarithmic control certificate

Source metadata: title (MetaInlines)

Canonical restriction and the arithmetic endpoint volume

S30.1 1. Result, inherited input, and scope

The remaining quantity in the current endpoint criterion is a ratio of four canonical arithmetic quotient volumes. This note constructs its underlying restriction operator, proves composition and action identities for that operator, and gives two further ways to calculate and bound the same volume. No new arithmetic metric is selected.

The principal map is

$$T_{i,j} : (E, G_j) \longrightarrow (E, G_i), \quad T_{i,j} = K_i G_j, \quad K_n = G_n^{-1}, \quad i \leq j. \quad (1)$$

It is the **metric adjoint of the original arithmetic identity transport** from degree i to degree j . It is also the map read by restricting the actual degree- j canonical polynomial representative to degrees at most i :

$$R_i T_{i,j} = \text{pr}_{\leq i} R_j. \quad (2)$$

The source coordinate inclusions in (2) are made explicit below. The return maps compose in the reverse direction,

$$T_{i,j} T_{j,l} = T_{i,l}, \quad \det T_{i,j} = V_j / V_i, \quad V_n = \det G_n. \quad (3)$$

Their eigenvalues $g_{i,j,a}$ are in $(0, 1]$ and have an exact source meaning: they are the squared singular values of the original identity transport equipped with its two actual quotient metrics, and also of the polynomial restriction on the canonical image. Hence

$$\log(V_i / V_j) = - \sum_{a=1}^{\dim E} \log g_{i,j,a}. \quad (4)$$

No arithmetic vector is killed by $T_{i,j}$ at a finite admitted degree. Its small eigenvalues quantify the cost of recovering the original arithmetic coordinates from a lower-degree portion of their representatives.

The new finite upper bounds express (4) through positive traces of powers of $I - T_{i,j}$, with an explicitly certified lower bound on $g_{i,j,a}$. A Cauchy–Binet expansion gives another exact representation, by nonnegative full-jet minor weights. The expansion retains the literal mass of the arithmetic source and does not assign its total weight the value one.

The preceding analytic continuation proved, for a fixed full packet h and $n \geq k \geq 3$,

$$\left(\frac{\omega_{h,k,2n}}{\omega_{h,k,n}} \right)^{1/(2n)} \leq C_h n. \quad (5)$$

We use this as an inherited written analytic theorem, not as a newly kernel-checked result. Composing the new upper certificates with that theorem and the formalizer's endpoint criterion gives an explicit control certificate for the original family.

For a packet consisting exactly of an off-line quartet, the already-established endpoint lower bound additionally implies the following new representative-level consequence: among the two indicated degree restrictions, at least one squared singular value is at most

$$\exp \left[-\operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right) \right]. \quad (6)$$

This is a necessary consequence of an off-line quartet, not an assertion that small singular values imply one. The calibration below retains a nilpotent on-line block with nontrivial restriction loss.

The general matrix and exterior mechanisms are classical. The contribution is their explicit propagation through this programme's actual source, full-jet quotient, support transitions, and arithmetic action. No upper estimate uniform in k for the logarithmic loss is proved in this note.

S30.2 2. Preserve the original base and source

Retain the marked coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (7)$$

Here $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$, $e_R = 0_R^\bullet$, and $p_R(\tau) = 0_R$, $p_R(r^\bullet) = r$. The target of $p_{\mathbb{Z}}$ is infinite. The structural base remains $\mathfrak{b}_\tau$. Polynomial degree and relation depth do not adjoin another scalar zero.

Keep

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi. \quad (8)$$

For a fixed finite packet, retain every selected zero with its complete order:

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad v_h = g/h, \quad \mathcal{M}F_h = v_h. \quad (9)$$

The arithmetic integration density at tensor degree k is

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}, \quad \mu_h = \int w_h, \quad \int m_{h,k} = \mu_h^k. \quad (10)$$

On the existing cyclic sum module use

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k}, \quad A = M_S, \quad S = s_1 + \cdots + s_k, \quad c = k/2. \quad (11)$$

The canonical arithmetic source is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad (12)$$

and its norm is

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du. \quad (13)$$

The full arithmetic coefficient maps remain

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1, \quad v_h = j_h(g/h). \quad (14)$$

All derivatives and arithmetic units in these jets remain. The quotient coordinate $[P]_\chi$ and its arithmetic image $\eta[P]_\chi$ are connected by this specified injective module map.

Let $p_0, p_1, \dots$ be the original monic S -polynomials orthogonal for (13), with squared norms $\omega_n > 0$. In particular $\omega_0 = \mu_h^k$. No polynomial is divided by its norm.

For $N \geq q - 1$, let

$$\mathcal{P}_N = \mathbb{C}[S]_{\leq N}, \quad \mathcal{D}_N = \chi \mathcal{P}_{N-q}, \quad \mathcal{P}_{-1} = 0. \quad (15)$$

The exact source sequence is

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (16)$$

Monicity proves surjectivity and the displayed kernel, including at $N = q - 1$. Under $\mathcal{V}$, the relation χP is the same original theta boundary whose primitive was constructed by fixed-order division into the tensor factors. Its support label and both occurrences of every Koszul sign remain in that primitive.

All finite Gram inverses in the note require $q \geq 1$. For $h = 1$ the finite arithmetic packet is zero, all displayed determinant-line observations have empty determinant one, and no positive-rank inverse is introduced. Its analytic source and mass remain as in (10).

S30.3 3. Construct the canonical degree transport and its adjoint

Use the ordered, unscaled basis $(p_0, \dots, p_N)$ of $\mathcal{P}_N$. Define

$$O_N = \text{diag}(\omega_0, \dots, \omega_N), \quad B_N = [b_0, \dots, b_N], \quad b_n = [p_n]_\chi. \quad (17)$$

Here $B_N : \mathbb{C}^{N+1} \rightarrow E$ is the remainder map in the fixed quotient basis $1, S, \dots, S^{q-1}$. Put

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N : E \rightarrow \mathcal{P}_N. \quad (18)$$

The formula gives

$$B_N R_N = I_E, \quad R_N^* O_N R_N = G_N, \quad \text{im } R_N = \mathcal{D}_N^\perp. \quad (19)$$

Indeed B_N is onto, so K_N is positive definite. For $z \in \ker B_N$, $z^* O_N R_N = z^* B_N^* G_N = 0$. Every representative is $R_N u + z$; the two summands are orthogonal. Thus (18) is the unique minimum-norm representative for the original relation space, not an independently selected form.

The actual test-function representative is $\mathcal{V} R_N$. Every identity involving R_N below passes to that source by (12)–(14).

For $i \leq j$, let $L_{i,j} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$ be coefficient padding and let $P_{i,j} : \mathcal{P}_j \rightarrow \mathcal{P}_i$ keep exactly the first $i + 1$ monic coordinates. Since the full monic family is orthogonal, this is the orthogonal projection onto the original lower-degree source. One has

$$B_j L_{i,j} = B_i, \quad L_{i,j}^* O_j L_{i,j} = O_i. \quad (20)$$

Let $\mathcal{E}_N$ denote E with its actual quotient metric G_N . The original quotient transport is

$$I_{i,j} : \mathcal{E}_i \rightarrow \mathcal{E}_j, \quad u \mapsto u, \quad B_j L_{i,j} = I_{i,j} B_i. \quad (21)$$

It intertwines A exactly. Its Gram decreases, because the larger source admits the old representative and additional original relations:

$$G_j \preceq G_i. \quad (22)$$

Its metric adjoint is the map in (1). To prove it, compute

$$\langle I_{i,j} u, v \rangle_{G_j} = u^* G_j v = u^* G_i (K_i G_j v) = \langle u, T_{i,j} v \rangle_{G_i}. \quad (23)$$

This is an adjoint of the **existing** transport. It is not asserted to be a unital algebra homomorphism.

Direct substitution into (18) proves the representative restriction identity

$$\boxed{L_{i,j}R_iT_{i,j} = P_{i,j}R_j.} \quad (24)$$

For example, the first $i + 1$ coordinates of R_j are $O_i^{-1}B_i^*G_j$; the right coordinates of $R_iT_{i,j}$ are $O_i^{-1}B_i^*G_iK_iG_j$, which are identical.

The adjoints compose exactly:

$$T_{i,j}T_{j,l} = K_iG_jK_jG_l = K_iG_l = T_{i,l}. \quad (25)$$

This agrees with contravariance of metric adjoints applied to the original identity transports. There is no intervening choice of a metric.

S30.4 4. Record the arithmetic action and the actual boundary correction

Set

$$H_N = G_N^{-1}(A^*G_N + G_NA - kG_N), \quad W_N = A^*G_N + G_NA - kG_N, \quad V_N = AK_N + K_NA^* - kK_N. \quad (26)$$

The adjoint-action identity gives

$$\boxed{AT_{i,j} - T_{i,j}A = H_iT_{i,j} - T_{i,j}H_j = V_iG_j - K_iW_j.} \quad (27)$$

For a direct proof, expand the rightmost expression and cancel the two occurrences of $K_iA^*G_j$ and of kK_iG_j . This records the full equivariance defect of the return map. The forward transport (21) is equivariant. The return need not be equivariant for the arithmetic action.

If ϵ_i, ϵ_j are the actual operator norms of H_i, H_j in their respective metrics, (22)–(23) imply the finite bound

$$\|AT_{i,j} - T_{i,j}A\|_{\mathcal{E}_j \rightarrow \mathcal{E}_i} \leq \epsilon_i + \epsilon_j. \quad (28)$$

No spectral subspace of the return map is therefore declared invariant under A merely because it is a useful metric diagnostic. Equation (27) is the explicit comparison.

Define the degree-restriction loss endomorphism

$$H_{i,j} = I_E - T_{i,j} = (K_j - K_i)G_j. \quad (29)$$

The original relation correction is

$$\boxed{\mathfrak{d}_{i,j} = L_{i,j}R_i - R_j = L_{i,j}R_iH_{i,j} - (I - P_{i,j})R_j.} \quad (30)$$

Both terms on the right have arithmetic remainder $H_{i,j}$. Their difference has zero remainder. Consequently

$$B_j\mathfrak{d}_{i,j} = 0, \quad \text{im } \mathfrak{d}_{i,j} \subseteq \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathfrak{d}_{i,j}^*O_j\mathfrak{d}_{i,j} = G_i - G_j. \quad (31)$$

The Gram identity follows because R_j is orthogonal to every relation in $\mathcal{D}_j$. Thus the lower-degree canonical representative is the higher-degree representative plus the explicit original relation (30).

The split map of a single active fibre is

$$\begin{aligned} I_{i,j}^\tau : \mathcal{E}_i^\tau &\rightarrow \mathcal{E}_j^\tau, & u^\bullet &\mapsto u^\bullet, & \tau &\mapsto \tau, \\ T_{i,j}^\tau : \mathcal{E}_j^\tau &\rightarrow \mathcal{E}_i^\tau, & u^\bullet &\mapsto (T_{i,j}u)^\bullet, & \tau &\mapsto \tau. \end{aligned} \quad (32)$$

These are $G(\mathbb{C})$ -linear single-fibre maps. The forward maps embed into the original directed support diagram. The reverse maps are the metric-adjoint diagram; an additional naturality across the original forward transports is not presumed. Their actual reverse-composition identity is (25). Inclusion of a single-fibre lift into the reconstructed carrier sends $u^\bullet$ to (ℓ_N, u) and its external τ to the carrier's external zero.

Under the original degree- j internal quotient, a supported value of $\mathfrak{d}_{i,j}$ becomes $(\ell_j, 0)$; external absence remains external absence. The relation before that map remains available in (30)–(31), including its original theta primitive.

S30.5 5. Exact volume as a restriction spectrum

The endomorphism $T_{i,j}$ is positive self-adjoint in both G_i and G_j :

$$G_i T_{i,j} = G_j, \quad G_j T_{i,j} = G_j K_i G_j. \quad (33)$$

Also $0 \prec T_{i,j} \preceq I$ in either metric. For G_i , this is (22); for G_j , use $K_i \preceq K_j$. Let its q eigenvalues, with multiplicities, be $g_a \in (0, 1]$.

The singular values of $I_{i,j} : \mathcal{E}_i \rightarrow \mathcal{E}_j$ are exactly $\sqrt{g_a}$ because $I_{i,j}^\dagger I_{i,j} = T_{i,j}$. The same nonzero singular values belong to $T_{i,j}$ in the reverse Hilbert types and to the source restriction (24). This is the concrete principal-angle construction for the canonical image and the lower-degree source; it uses their actual inner products.

Taking determinants gives

$$\boxed{\det T_{i,j} = \frac{\det K_i}{\det K_j} = \frac{V_j}{V_i}, \quad \mathcal{L}_{i,j} := \log \frac{V_i}{V_j} = - \sum_a \log g_a.} \quad (34)$$

The top exterior map $\bigwedge^q T_{i,j}$ acts by this literal determinant in the fixed determinant coordinates. With the source's unscaled alternating tensor representative, the determinant vector has squared norm $q! V_N$ at degree N . Both factorials remain in the ratio. This is determinant-line accounting through the same source-relation sequence as in the previous Deligne-oriented construction, not an imported purity conclusion.

An equivalent exact count is

$$\boxed{\mathcal{L}_{i,j} = \int_0^1 \frac{\#\{a : g_a < t\}}{t} dt = \int_0^\infty \#\{a : g_a < e^{-s}\} ds.} \quad (35)$$

For one $g \in (0, 1]$, the first integral is $\int_g^1 dt/t$; summing proves the formula. The second uses $t = e^{-s}$. Values at equality do not affect the integrals.

For the quartet endpoint window, (34) is applied twice, to

$$(i_0, j_0) = (q-1, 2q-1), \quad (i_1, j_1) = (q, 2q). \quad (36)$$

Their losses add to the original four-volume quantity $\mathcal{B}_{h,k}$.

S30.6 6. An exterior expansion retains every full-jet basis weight

For a q -element subset $D = \{d_1 < \dots < d_q\} \subseteq \{0, \dots, j\}$ define

$$a_D = \frac{|\det[b_{d_1}, \dots, b_{d_q}]|^2}{\prod_{d \in D} \omega_d}. \quad (37)$$

Cauchy–Binet applied to (18) gives

$$\boxed{\det K_j = \sum_{|D|=q} a_D.} \quad (38)$$

This is an identity of positive finite scalar observations, with no assignment of the sum to one. The actual exterior map giving the numerator is

$$\bigwedge^q B_j : \bigwedge^q \mathcal{P}_j \rightarrow \det E, \quad p_{d_1} \wedge \dots \wedge p_{d_q} \mapsto \det[b_{d_1}, \dots, b_{d_q}] e_0 \wedge \dots \wedge e_{q-1}. \quad (39)$$

The squared norm of the source wedge is $\prod_{d \in D} \omega_d$. The determinant-coordinate line used to read its coefficient is specified; it is not the canonical arithmetic Gram reset to the identity.

Introduce the finite scalar polynomial

$$Z_{i,j}(z) = \det(K_i + z(K_j - K_i)). \quad (40)$$

Multilinearity and Cauchy–Binet prove the full coefficient identity

$$Z_{i,j}(z) = \sum_{|D|=q} a_D z^{\#(D \cap \{i+1, \dots, j\})}. \quad (41)$$

In particular $Z(0) = \det K_i > 0$, $Z(1) = \det K_j$, and

$$\frac{Z_{i,j}(z)}{\det K_j} = \det(T_{i,j} + zH_{i,j}) = \prod_a (g_a + z(1 - g_a)). \quad (42)$$

The scalar division is the displayed observation map from the retained pair $(Z, \det K_j)$; the original total and mass remain in (38)–(41). This determinant generating mechanism is classical in projection determinantal measures. Its present interpretation concerns the original degree-admission and full-jet maps, not a randomly selected alternative arithmetic source.

The first two occupation ratios are

$$\frac{Z'(1)}{Z(1)} = \text{Tr } H_{i,j}, \quad \frac{Z''(1) + Z'(1)}{Z(1)} - \left(\frac{Z'(1)}{Z(1)} \right)^2 = \text{Tr}(H_{i,j} - H_{i,j}^2). \quad (43)$$

These identities follow by differentiating (42); every denominator is the original positive scalar total. The weighted counts in (41) remain available before taking those ratios.

There are two explicit ways a term can be missing from a scalar total. An admitted subset with dependent full-jet columns has $a_D = 0$, a supported zero. A subset containing an index above the admitted degree is outside that finite indexing object, and its extension to a larger indexing object is assigned external absence. For example, in the calibration $\chi = (S - 1)^2$, b_0 and b_2 are dependent: the subset $\{0, 2\}$ is admitted and has supported zero weight. The subset $\{0, 4\}$ is absent from degree three. The algebraic specialization $z = 0$ in (41) sends each positive-degree monomial to a supported scalar zero; its support record is not retroactively deleted.

Full derivatives are retained under a change to raw confluent jet coordinates. For roots λ_a of orders m_a , let

$$C_{\chi(a,r),b} = \partial_S^r S^b|_{S=\lambda_a} = \begin{cases} \frac{b!}{(b-r)!} \lambda_a^{b-r}, & b \geq r, \\ 0, & b < r. \end{cases} \quad (44)$$

Its determinant, in the stated block order, is

$$\det C_\chi = \left(\prod_a \prod_{r=0}^{m_a-1} r! \right) \prod_{a < b} (\lambda_b - \lambda_a)^{m_a m_b}. \quad (45)$$

For any explicitly retained invertible arithmetic unit matrix M_u , the quotient-coordinate change is $C_\chi M_u$. Every weight (37) acquires the same factor $|\det(C_\chi M_u)|^2$, both totals acquire it, and (34), (42) are unchanged through that specified ratio. The action and return are conjugated by this matrix, and the canonical representative is precomposed with its inverse. No nilpotent derivative order is removed.

For the actual cyclic inclusion $\eta : E \hookrightarrow B_k^{S_k}$, the ambient matrix can be rectangular. Its correct exterior comparison is

$$\bigwedge^q (\eta B_D) = (\det B_D) \left(\bigwedge^q \eta \right) (e_0 \wedge \cdots \wedge e_{q-1}).$$

The fixed nonzero image vector is retained. There is no determinant of a rectangular matrix being inverted. A square coordinate change $C_\chi M_u$ applies only on a specified coordinate model of the finite image, or for an actual unit endomorphism of E . In the ambient symmetric module the original map remains η , including the complete factor $v_h^{\otimes k}$.

S30.7 7. A certified finite logarithmic upper bound

Let $H = H_{i,j}$ and $s_m = \text{Tr}(H^m)$. Positivity in the original metric implies $s_m \geq 0$, and the eigenvalues lie in $[0, 1)$. Therefore

$$\mathcal{L}_{i,j} = \sum_{m=1}^{\infty} \frac{s_m}{m}. \quad (46)$$

A valid rational or real lower gap $g_0 > 0$ is supplied by the **actual matrix inequality**

$$K_i \succeq g_0 K_j. \quad (47)$$

It gives $H \preceq (1 - g_0)I$ in G_j . Let $r = 1 - g_0$. For integer $p \geq 1$ and $0 < r < 1$, set

$$c_p(r) = \frac{-\log(1 - r) - \sum_{m=1}^p r^m/m}{r^{p+1}}. \quad (48)$$

The finite enclosure is

$$\sum_{m=1}^p \frac{s_m}{m} \leq \mathcal{L}_{i,j} \leq U_p(H, g_0) := \sum_{m=1}^p \frac{s_m}{m} + c_p(r)s_{p+1}. \quad (49)$$

For $r = 0$, (47) and $K_i \preceq K_j$ give $H = 0$ and both bounds are zero.

Proof. For $0 \leq x \leq r$, the tail of $-\log(1 - x)$ is

$$x^{p+1} \sum_{m=p+1}^{\infty} \frac{x^{m-p-1}}{m} \leq x^{p+1} \sum_{m=p+1}^{\infty} \frac{r^{m-p-1}}{m} = c_p(r)x^{p+1}.$$

Apply this scalar inequality to the positive self-adjoint H and take its trace. For every fixed admitted pair and fixed certified g_0 , the lower and upper bounds converge to the same loss as $p \rightarrow \infty$. This is fixed-pair convergence, not a uniform assertion in tensor degree.

Already $p = 1$ gives

$$\mathcal{L}_{i,j} \leq s_1 + \frac{-\log g_0 - (1 - g_0)}{(1 - g_0)^2} s_2. \quad (50)$$

It retains a second moment rather than replacing every loss by the maximum one. Higher p retain the successive moments of the same operator. No eigenvalue of A is inferred from an eigenvalue of H ; their actual commutator is (27).

For the two windows in (36), verified gaps $g_0^{(0)}, g_0^{(1)}$ give

$$\mathcal{B}_{h,k} \leq U_p(H^{(0)}, g_0^{(0)}) + U_p(H^{(1)}, g_0^{(1)}). \quad (51)$$

Combining (5), the formalizer's endpoint theorem, and monotonicity of $\sinh$ gives the finite canonical-family estimate

$$\min_{q \leq N < 2q} \epsilon_{h,k,N} \leq C_h q \sinh \left(\frac{U_p(H^{(0)}, g_0^{(0)}) + U_p(H^{(1)}, g_0^{(1)})}{2q} \right), \quad q \geq k \geq 3. \quad (52)$$

This theorem uses the explicit matrix certificate (47). Existence of some positive gap for each finite pair does not imply the uniform gap or moment control that would finish the arithmetic argument. Formula (52) states exactly the quantity a certified finite or uniform analytic estimate must bound.

S30.7.1 Error-aware certificate types

The checker supplies exact rational lower and upper bounds for $\log x$ on positive rational inputs. It uses $x = 2^e y$, $1 \leq y < 2$, and

$$\log y = 2 \sum_{a=0}^{m-1} \frac{t^{2a+1}}{2a+1} + \mathcal{E}_m, \quad t = \frac{y-1}{y+1}, \quad 0 \leq \mathcal{E}_m \leq \frac{2t^{2m+1}}{(2m+1)(1-t^2)}. \quad (53)$$

The same formula encloses $\log 2$; reciprocation handles $x < 1$. Finite rational LDL or principal-minor checks certify (47). No floating-point eigenvalue is accepted as proof of that inequality.

For actual analytic matrix enclosures, a sufficient input is $K_i^- \preceq K_i$ and $K_j \preceq K_j^+$ together with $K_i^- \succeq g_0 K_j^+$. Then (47) follows by transitivity. The trace moments in (49) also need certified input enclosures, or exact arithmetic; they are not automatically exact because a gap was enclosed. Independently,

$$\mathcal{L}_{i,j} \leq \log \det K_j^+ - \log \det K_i^- \quad (54)$$

when both bounding matrices are positive definite. The implemented test fixtures have exact rational entries. This note supplies no interval certificate for the actual arithmetic integrals or root coordinates.

S30.8 8. The off-line alternative forces a precise restriction phenomenon

For the packet consisting exactly of a quartet with displacement $\delta > 0$, height $\gamma > 0$, and common multiplicity m , retain

$$q_k = [1 + k(m-1)](k+1)^2. \quad (55)$$

The inherited endpoint theorem and its proved norm input give

$$\mathcal{B}_{h,k} \geq 2q_k \operatorname{arsinh} \left(\frac{\delta k}{2C_h} \right). \quad (56)$$

There are $2q_k$ positive eigenvalues g_a across the two return maps in (36). Equations (34) and (56) imply that at least one satisfies (6): otherwise every $-\log g_a$ would be smaller than the common lower average in (56), contradicting their sum.

An eigenvector v for that g_a is nonzero in the original arithmetic module. Equation (24) and the actual Gram calculation give

$$\boxed{\frac{\|P_{i,j} R_j v\|_{\mathcal{O}_j}^2}{\|R_j v\|_{\mathcal{O}_j}^2} = g_a, \quad \frac{\|(I - P_{i,j}) R_j v\|_{\mathcal{O}_j}^2}{\|R_j v\|_{\mathcal{O}_j}^2} = 1 - g_a.} \quad (57)$$

The old portion has arithmetic coordinate $g_a v$, not v . The exact original representative of v at degree i is recovered by applying $T_{i,j}^{-1}$ in (24). The difference from the full degree- j representative is the actual relation (30). Thus no arithmetic class is called absent, and no numerical near-zero is set equal to e .

The exact distributional statement is stronger than one selected direction:

$$\sum_{a=0}^1 \int_0^\infty \#\{b : g_b^{(a)} < e^{-s}\} ds = \mathcal{B}_{h,k}. \quad (58)$$

The growing-degree task can now be stated as control of this cumulative small-restriction spectrum, or as the positive trace certificates (51). A proved upper limit below 2 for $\mathcal{B}_{h,k}/(q_k \log k)$ would contradict (56). Such a uniform upper bound is not an output of the present finite identities.

S30.9 9. Exact calibration retaining nilpotence and mass

Use the explicitly declared algebraic fixture

$$E = \mathbb{C}[S]/((S-1)^2), \quad S = 1 + ix, \quad (59)$$

with the Gaussian source $7e^{-x^2/2}dx/\sqrt{2\pi}$, of literal mass seven and variance one. This fixture is not asserted to be a zeta packet. Its source monic polynomials and norms are

$$p_0 = 1, \quad p_1 = S - 1, \quad p_{n+1} = (S-1)p_n + np_{n-1}, \quad \omega_n = 7n!. \quad (60)$$

The coordinate basis $(1, S-1)$ is connected to $(1, S)$ by

$$C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad (1, S)C = (1, S - 1). \quad (61)$$

In the $(1, S - 1)$ coordinates, the kernel matrices are

$$\begin{aligned} K_1 &= \text{diag}(1/7, 1/7), & K_2 &= \text{diag}(3/14, 1/7), \\ K_3 &= \text{diag}(3/14, 5/14), & K_4 &= \text{diag}(15/56, 5/14). \end{aligned} \quad (62)$$

The action remains $A = I + N$ with $N \neq 0$ and $N^2 = 0$. It is not replaced by its reduced one-dimensional quotient.

The two return spectra are exactly

$$\text{Spec } T_{1,3} = \{2/3, 2/5\}, \quad \text{Spec } T_{2,4} = \{4/5, 2/5\}. \quad (63)$$

Consequently

$$\mathcal{B} = \log(15/4) + \log(25/8) = \log(375/32) \approx 2.46119012317068427. \quad (64)$$

For $H_{1,3}$ the first two trace moments are $14/15$ and $106/225$; for $H_{2,4}$ they are $4/5$ and $2/5$. Using $g_0 = 2/5$ and $p = 2$ in (49) gives the two rigorously enclosed upper bounds whose displayed decimals are

$$1.32854908781965490, \quad 1.14133853675838303. \quad (65)$$

Their sum is below 2.469887624578038 . The earlier direct bound by $\text{Tr}(K_i^{-1}(K_j - K_i))$ gives $2 + 7/4 = 3.75$ on these two windows. Thus the new finite certificate makes a strict, verified improvement on the same source and same metrics. The exact logarithmic enclosures, rather than the printed rounded values, certify the comparison.

The dependent subset $\{0, 2\}$ in (37) has a supported zero minor because $[p_2] = [p_0]$ in this quotient. Nevertheless the first derivative coordinate at $S = 1$ is retained and makes the quotient dimension two. A distinct-root replacement would not give the same matrices.

The test suite also contains off-line algebraic fixtures, nonconstant invertible jet units, and repeated confluent derivatives. It does not infer a zeta-zero statement from these fixtures.

S30.10 10. Integration with the Deligne-oriented programme

The same cohomological transition now carries an explicit positive adjoint and a determinant-line norm. Its action covariance is (27), its theta-boundary correction is (30), and its original support quotient is retained in (32). This is a metric calculation on the existing characteristic-zero realization over b_+ .

Deligne's use of determinant weights and the compact-to-ordinary image requires an actual estimate on the same image. This note supplies additional calculation and finite upper certificates for that estimate; it does not invoke his finite-field purity hypotheses for the theta object. The inherited source review's strict inequality in Lemma 3.2.10 and its squared-eigenvalue step in 3.2.13 are retained. This turn did not newly reassemble the Deligne archive or claim a complete rereading.

Current repository input was checked at main 7ea0a49945390eae14d3160a5730858899768b5f. The current endpoint criterion retains the corrected top-exterior case and the actual second-exterior multiplicity cost. No second cost-free exterior amplification is used here. The proof status of the original norm bound (5) remains the inherited written analytic status; the present finite tests do not promote it to a Lean certificate.

The principal next quantity is no longer only an unnamed ratio of large determinants. It is the explicit log-loss of the original identity transports, with the canonical representative restriction, all original relations, full jets, arithmetic action, and complementary polynomial degrees attached. The uniform bound still needed is precisely the one on that constructed family.

S30.11 References and provenance

1. Owner-directed *Arithmetic endpoint bounds on the original tau-base source*, supplied TeX and archive, 13 September 2026. Equations (1), (35)–(39), and (43)–(48) are the inherited analytic input, endpoint composition, and original relation-layer formulas.

2. Parallel formalization session, `workbenches/tau-toda-volume-formal/ENDPOINT_CRITERION.md`, read at main `7ea0a49945390eae14d3160a5730858899768b5f`; fetched Git blob `09d658b1c50245b9e340722817bc9487f1ce409a`. Endpoint selection is its written Jensen theorem; the finite telescope components have their separately reported formal scope.
3. Original SplitZero reconstruction/internal quotient and supplied full-jet kernel-layer note. The present construction reuses their quotient and support maps.
4. Lyons, R. (2003). *Determinantal probability measures*. Publications mathématiques de l’IHES 98, 167–212; arXiv:math/0204325. Background for the exterior/Cauchy–Binet generating mechanism, not a source for the arithmetic objects constructed here.
5. Drmac, Z. (2000). *On principal angles between subspaces of Euclidean space*. SIAM Journal on Matrix Analysis and Applications 22, 173–194, doi:10.1137/S0895479897320824. Background for principal angles and singular values. All weighted source maps needed here are proved explicitly.
6. Deligne, P. (1980). *La conjecture de Weil. II*. Publications mathématiques de l’IHES 52, 137–252. Used through the inherited precise reading record; no new finite-field purity application is claimed.

S31 Complete original two-trace relation control

Source role. Complete delivered proof witness. Its historical claims retain their source revision; corrections and full current maps are proved in the corresponding integrated chapter.

Exact revision. 17e14c11ca60906604f66ddf9e6a758412a16143d5742736f3a5708658270795

Complete source. sources/web_relation_moment_delivery/Tau_Relation_Moment_Control/RESEARCH_NOTE.md

Source SHA-256. 17e14c11ca60906604f66ddf9e6a758412a16143d5742736f3a5708658270795

Source metadata: author (MetaInlines)

Owner-directed SplitZero analytic continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

An upper determinant certificate without a separately supplied restriction gap

Source metadata: title (MetaInlines)

Two-moment control of the original arithmetic relation blocks

S31.0.1 1. Result and the current inherited theorem

This note continues the canonical endpoint-restriction construction. It does not choose a different positive metric or replace the arithmetic source by a comparison measure. Its new application is an upper bound for each original relation-block volume from two source-derived traces, together with a complete positive remainder and a source interpretation of the second trace as exterior area.

The general sharp determinant optimization with prescribed dimension, trace, and squared trace is classical; Grone, Johnson, Marques de Sá, and Wolkowicz (1984) study precisely that optimization. The proof below is included in full, using a scalar Hermite interpolation identity. No priority claim is made for the general inequality. The work here is its composition with the actual arithmetic representative, internal quotient, inverse restriction, exterior source maps, and consecutive-window theorem.

For one original block, the output is

$$\boxed{\frac{V_i}{V_j} \leq (1 + a_{i,j})(1 + b_{i,j})^{q-1}.} \quad (1)$$

Both $a_{i,j}$ and $b_{i,j}$ are calculated from the first two traces of a specified endomorphism. There is no assumption that the arithmetic action is normal, no assumption that a restriction eigenspace is invariant under it, and no independently supplied lower gap for the restriction.

Integration of the two new reports The attachment reporting PR #25 supplies the phase-retaining consecutive product, the separate first-admissible-degree step, a two-sided written arithmetic norm envelope, and its comparable-window consequence. Its resulting off-line-quartet lower budgets are

$$\log(V_q/V_{2q}) \geq 2q \log(D_h k), \quad \log(V_{q-1}/V_{2q-1}) \geq 2q \log(D_h k), \quad (2)$$

for a fixed positive D_h , hence the four-volume lower threshold is **four**, not merely the earlier sufficient lower threshold two. This is inherited from the parallel contributions, not claimed as a discovery of this note. The source explicitly distinguishes its checked finite implications from the written analytic estimates and the asymptotic conclusion.

Its regularity clarification is retained: the actual arithmetic density is in $L^1 \cap L^\infty$, and

$$\|(\tau_a w_h - w_h) * w_h\|_\infty \leq \|\tau_a w_h - w_h\|_1 \|w_h\|_\infty \longrightarrow 0.$$

Here τ_a denotes translation of a function, not the split scalar τ ; its definition is $(\tau_a f)(t) = f(t - a)$. This supplies continuity of the convolution used in the source's lower-envelope argument.

The second attachment is a working transcript reporting further Gamma, coefficient-error, endpoint, and reflected-algebra work. Its reported moment certificates and unpublished upper-bound details are not independently reproduced here or silently made premises. In particular, this note does not duplicate its claimed fixed-budget extremizer or claim to have executed its local Windows commands.

GitHub main was read at 952ef9fee1e1419b6858d920354de8fa99430b7d. PR #25 was read at its reported verified head 66fac5e7885a40cd4873e74b754907320f05b067, including its full RESEARCH_NOT E.md. Its CI status is a source report; this session did not execute Lean.

S31.0.2 2. Original arithmetic source and support

Retain the original scalar object and amplitude map

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The original coefficient square remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (3)$$

Its arithmetic target $\mathbb{Z}$ is infinite. The geometric base remains $\mathfrak{b}_\tau$. The square does not replace the internal support fibres by the ordinary amplitude observation.

Keep the original theta complex and generator

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi.$$

For a nonempty full packet h of actual zeros and tensor degree k , retain the cyclic module and its action

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad q = \deg \chi_{h,k}, \quad A = M_S, \quad S = s_1 + \cdots + s_k.$$

The source and arithmetic inclusion are the constructed maps

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = g/h,$$

$$\eta[P] = v_h^{\otimes k} P(A_k) 1, \quad v_h = j_h(g/h), \quad (4)$$

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k} \eta\pi_\chi.$$

The original norm is the integral against $m_{h,k} = w_h^{*k}$ on $S = k/2 + iu$, where

$$w_h(t) = \frac{|(g/h)(1/2 + it)|^2}{2\pi}, \quad \int m_{h,k} = \mu_h^k.$$

No mass is assigned a new value. All nilpotent jet orders and the unit in (4) remain in the arithmetic maps.

For $h = 1$ the finite packet is the zero coefficient module. Its determinant is the empty determinant one, its log-volume is zero, and $G(0)$ still contains the two split elements. The analytic theta seed remains nonzero. Positive-dimensional formulas below use $q \geq 1$; the divisions by $q - 1$ use $q \geq 2$.

Fixed polynomial coordinates Let $p_0, p_1, \dots$ be the original monic orthogonal polynomials and $\omega_n > 0$ their actual squared norms. Keep $\omega_0 = \mu_h^k$. In fixed remainder coordinates on E , set

$$b_n = [p_n]_\chi, \quad B_N = [b_0, \dots, b_N], \quad O_N = \text{diag}(\omega_0, \dots, \omega_N).$$

For every admitted $N \geq q - 1$,

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad V_N = \det G_N. \quad (5)$$

Thus $B_N R_N = I$ and $R_N^* O_N R_N = G_N$. The actual test-function representative is obtained by the specified polynomial-column realization $\mathcal{V}$, not by treating a coefficient vector as a function without that map.

S31.0.3 3. The positive cost operator comes from the original restriction

Fix $q - 1 \leq i \leq j$. Set $r = j - i$ and retain the new columns and their norms

$$F = F_{i,j} = [b_{i+1}, \dots, b_j] : \mathbb{C}^r \rightarrow E, \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j),$$

$$D_{i,j} = K_j - K_i = F \Omega^{-1} F^*. \quad (6)$$

The Hilbert spaces $E_i = (E, G_i)$ and $E_j = (E, G_j)$ are connected by the actual forward transport $I_{i,j} : E_i \rightarrow E_j$, identity on the fixed coefficient space. Its adjoint is

$$T_{i,j} = I_{i,j}^\dagger = K_i G_j : E_j \rightarrow E_i.$$

To form an endomorphism of one specified Hilbert object, compose first:

$$Q_{i,j} = T_{i,j} I_{i,j} : E_i \rightarrow E_i.$$

The coefficient matrix of $Q_{i,j}$ is $K_i G_j$. Define

$$Z_{i,j} = Q_{i,j}^{-1} - I_E = (K_j - K_i) G_i = D_{i,j} G_i. \quad (7)$$

It is positive self-adjoint on E_i , because

$$G_i Z_{i,j} = G_i D_{i,j} G_i \succeq 0.$$

Its eigenvalues λ_a are nonnegative. The restriction eigenvalues in the preceding note are precisely

$$g_a = \frac{1}{1 + \lambda_a}, \quad \frac{V_i}{V_j} = \det(I_E + Z_{i,j}), \quad \log(V_i/V_j) = \sum_{a=1}^q \log(1 + \lambda_a). \quad (8)$$

No inverse-square-root transformation changes the canonical arithmetic metric. Equation (7) is a typed inverse of the already-constructed restriction endomorphism.

Source and action maps remain attached The source map $F : (\mathbb{C}^r, \Omega) \rightarrow (E, G_i)$ has adjoint

$$F^\dagger = \Omega^{-1} F^* G_i, \quad Z_{i,j} = F F^\dagger. \quad (9)$$

Let $H_N = G_N^{-1}(A^* G_N + G_N A - k G_N)$ be the inherited control endomorphism. Put $\hat{H}_j = I_{i,j}^{-1} H_j I_{i,j}$ on E_i . The previous action defect and inversion give

$$[A, Z_{i,j}] = \hat{H}_j (I + Z_{i,j}) - (I + Z_{i,j}) H_i. \quad (10)$$

Indeed $[A, Q] = H_i Q - Q \hat{H}_j$ and $[A, Q^{-1}] = -Q^{-1} [A, Q] Q^{-1}$. This is the actual relation between the cost operator and the arithmetic generator. A cost eigenspace is not presumed invariant for A .

S31.0.4 4. The two input traces are norms of actual source maps

Write

$$t_1 = \text{Tr } Z_{i,j}, \quad t_2 = \text{Tr}(Z_{i,j}^2), \quad \mathcal{A}_2 = \frac{t_1^2 - t_2}{2}. \quad (11)$$

The scalar $\mathcal{A}_2$ is the second exterior coefficient of this cost operator. Its split observation is $\mathcal{A}_2^\bullet \in G(\mathbb{R})$; a vanishing value maps to the represented scalar e .

From (6) and cyclicity of the finite trace,

$$t_1 = \sum_{n=i+1}^j \frac{b_n^* G_i b_n}{\omega_n},$$

$$t_2 = \sum_{n,m=i+1}^j \frac{|b_n^* G_i b_m|^2}{\omega_n \omega_m}, \quad (12)$$

$$\mathcal{A}_2 = \sum_{i < n < m \leq j} \frac{(b_n^* G_i b_n)(b_m^* G_i b_m) - |b_n^* G_i b_m|^2}{\omega_n \omega_m}. \quad (13)$$

The off-diagonal pairings in (12)–(13) are essential.

Let $\Phi = R_i F$ with its actual degree- i source norm. Then $t_1 = \|\Phi\|_{\text{HS}}^2$. Equip unscaled exterior bases with determinant Grams. The precise second identity is

$$\mathcal{A}_2 = \left\| \bigwedge^2 \Phi \right\|_{\text{HS}}^2. \quad (14)$$

Proof: in the coordinate wedge indexed by $n < m$, the squared source norm is $\omega_n \omega_m$. Its image squared norm is the two-by-two Gram determinant in (13). Summing proves (14). An alternating-tensor realization instead has its literal factor $2!$ in each wedge Gram; this statement uses the specified exterior quotient, so no factorial is dropped.

Let T_{new} denote the actual newly admitted orthogonal polynomial columns in the degree- j source. The source relation is

$$\mathbf{b}_{i,j} = T_{\text{new}} - L_{i,j} R_i F, \quad B_j \mathbf{b}_{i,j} = 0,$$

$$\mathbf{b}_{i,j}^* O_j \mathbf{b}_{i,j} = \Omega + F^* G_i F. \quad (15)$$

Thus (12)–(14) measure the old representative components needed to cancel the new columns' jets. The relation itself maps to the receiving supported zero through the original coequalizer. Neither component is declared absent.

An admitted pair with dependent images has zero exterior area in (13), a represented zero. The pair and its original support label remain available. A pair outside the degree-indexing object is absent. No statement that these two cases have the same upstream datum is used.

S31.0.5 5. A sharp two-trace upper bound, with a complete proof

Suppose $q \geq 2$. Define

$$d = \sqrt{\frac{qt_2 - t_1^2}{q-1}}, \quad a = \frac{t_1 + (q-1)d}{q}, \quad b = \frac{t_1 - d}{q}. \quad (16)$$

Both discriminants and signs come from the actual positive spectrum:

$$\begin{aligned} \frac{t_1^2}{q} &\leq t_2 \leq t_1^2, & 0 &\leq b \leq a, \\ a + (q-1)b &= t_1, & a^2 + (q-1)b^2 &= t_2. \end{aligned} \quad (17)$$

The spectral cap is proved, not supplied For any eigenvalue x of Z , Cauchy–Schwarz on the other $q-1$ eigenvalues gives

$$t_2 \geq x^2 + \frac{(t_1 - x)^2}{q-1}.$$

Solving this quadratic yields $x \leq a$. Hence the same two traces also produce the actual restriction-gap bound

$$Q_{i,j} \succeq \frac{1}{1+a} I_E \quad \text{in the original metric } G_i. \quad (18)$$

This is a finite derived bound, not a uniform estimate in k .

The optimal determinant bound from these two traces The result is

$$\log \det(I + Z) \leq U_2(Z) := \log(1+a) + (q-1) \log(1+b). \quad (19)$$

For $a > b$, define the quadratic

$$P_{a,b}(x) = \log(1+b) + \frac{x-b}{1+b} + c_{a,b}(x-b)^2, \\ c_{a,b} = \frac{\log(1+a) - \log(1+b) - (a-b)/(1+b)}{(a-b)^2} < 0. \quad (20)$$

It matches $\log(1+x)$ and its derivative at b , and the value at a . For $0 \leq x \leq a$ the exact remainder is

$$P_{a,b}(x) - \log(1+x) = (a-x)(x-b)^2 \int_1^\infty \frac{ds}{(s+x)(s+b)^2(s+a)} \geq 0. \quad (21)$$

Proof of (21): use $\log(1+x) = \int_1^\infty (1/s - 1/(s+x)) ds$ and the exact third divided difference of $1/(s+x)$. Interpolation at the repeated point b and at a gives the displayed rational denominator. The integral converges as s^{-4} ; all factors have their shown positive signs. This also directly proves the Hermite remainder without an unspecified intermediate point.

Sum (21) over the actual eigenvalues. Since a quadratic trace depends only on q, t_1, t_2 , equations (17) give

$$\text{Tr } P_{a,b}(Z) = P_{a,b}(a) + (q-1)P_{a,b}(b) = U_2(Z).$$

This proves (19) and (1).

If $a = b$, equality in $qt_2 \geq t_1^2$ forces every eigenvalue to equal b , so (19) is equality without the quotient in (20). If $q = 1$, the exact answer is $\log(1+t_1)$. If $t_1 = 0$, positivity gives $Z = 0$ and the volume is zero. If $q = 0$, use the empty determinant, without forming (16).

For $a > b$, the integrand in (21) is strictly positive except at $x = a, b$. Equality in (19) therefore holds exactly when the spectrum consists of one a and $q-1$ copies of b . These values have the original two traces by (17), so the bound is best possible given only those traces and the dimension. This does not assert that every such extremizer arises from an arithmetic packet.

The general optimization is classical. The source application uses precisely $Z = (K_j - K_i)G_i$, not an independently chosen Hermitian matrix.

S31.0.6 6. Retain the complete remainder; a third trace improves the bound

Let

$$C_3 = \text{Tr}((aI - Z)(Z - bI)^2) = -t_3 + (a+2b)t_2 - (2ab + b^2)t_1 + qab^2,$$

where $t_3 = \text{Tr } Z^3$. Positivity on the spectrum proves $C_3 \geq 0$.

The full loss in the upper bound is

$$U_2(Z) - \log \det(I + Z) = \int_1^\infty \frac{\text{Tr}((aI - Z)(Z - bI)^2(sI + Z)^{-1})}{(s + b)^2(s + a)} ds. \quad (22)$$

All four operator factors commute because they are functions of this same G_i -self-adjoint Z . The expression is positive in that same metric. No claim about commutation with A replaces (10).

Since $0 \leq x, b \leq a$, the integrand denominator in (21) is at most $(s + a)^4$. Therefore

$$\log \det(I + Z) \leq U_2(Z) - \frac{C_3}{3(1 + a)^3}. \quad (23)$$

This retains a concrete part of the higher-spectrum correction. It requires one further trace of the same source cost, not a new spectral gap or arithmetic realization.

Equation (19) is always at least as sharp as the trace-only bound

$$U_2(Z) \leq q \log(1 + t_1/q), \quad (24)$$

by scalar concavity and (17). Equality in (24), apart from a zero-dimensional case, requires $a = b$. It is not asserted to dominate every higher-moment certificate from the previous note; those can be retained and their certified upper bounds compared.

S31.0.7 7. The simplest bound exposes independent pair growth

For $q \geq 2$ and $t_1 > 0$, (17) implies

$$\mathcal{A}_2 = (q - 1)b \left(t_1 - \frac{q}{2}b \right).$$

Because $0 \leq b \leq t_1/q$,

$$\frac{\mathcal{A}_2}{(q - 1)t_1} \leq b \leq \frac{2\mathcal{A}_2}{(q - 1)t_1}, \quad a \leq t_1.$$

Consequently there is a radical-free finite upper bound

$$\frac{V_i}{V_j} \leq (1 + t_1) \left(1 + \frac{2\mathcal{A}_2}{(q - 1)t_1} \right)^{q-1}. \quad (25)$$

The quantity $\mathcal{A}_2$ is exactly the sum of the original two-column exterior areas (13), including every cross-pairing. The first trace alone cannot distinguish nearly parallel old representative corrections from many independent ones; (13) does make that distinction quantitatively.

If $\mathcal{A}_2 = 0$ and $t_1 > 0$, the positive operator has rank one, and the sharper formula (19) is the exact value $\log(1 + t_1)$. This is a statement about the rank of this specified comparison map, not a claim about uniqueness of the original scalar support or about the number of higher support fibres.

S31.0.8 8. Compose with the new consecutive-window theorem

The inherited phase-retaining product implies for every original admitted window $i, \dots, j - 1$, $r = j - i$, including $i = q - 1$ through its separately proved first-degree step,

$$\min_{i \leq N < j} \epsilon_N \leq \left(\frac{\omega_j}{\omega_i} \right)^{1/(2r)} \left(\frac{V_i}{V_j} \right)^{1/(2r)}. \quad (26)$$

This is credited to PR #25 and its source continuation. It uses the original norm and volume sequence. It does not require constructing an inverse before the first admissible degree.

Combining (1) and (26) proves the finite upper certificate

$$\min_{i \leq N < j} \epsilon_N \leq \left(\frac{\omega_j}{\omega_i} \right)^{1/(2r)} ((1+a)(1+b)^{q-1})^{1/(2r)}. \quad (27)$$

For either quartet block $(i, j) = (q, 2q)$ or $(q-1, 2q-1)$, the reviewed comparable-window norm bound supplies an inherited constant C_h^{win} independent of k . Enlarge it once to account for $i = q-1$. Then

$$\min_{i \leq N < j} \epsilon_N \leq C_h^{\text{win}} q(1+a)^{1/(2q)} (1+b)^{(q-1)/(2q)}. \quad (28)$$

The selected member is still a canonical degree in the original window. Its source relation, arithmetic unit, and action are not chosen again after selecting the degree.

Quantitative consequence of the already established off-line lower bound For the packet consisting exactly of an off-line quartet with full common multiplicity, retain

$$q_k = [1 + k(m-1)](k+1)^2, \quad L_{h,k} \geq \delta k q_k / 2.$$

Let $D_h > 0$ be the fixed constant in the inherited per-block lower inequality (2). The new upper bound implies, on each original block,

$$(D_h k)^{2q} \leq (1+a)(1+b)^{q-1}. \quad (29)$$

In the source area quantities it implies

$$\frac{2\mathcal{A}_2}{(q-1)t_1} \geq \left(\frac{(D_h k)^{2q}}{1+t_1} \right)^{1/(q-1)} - 1, \quad (30)$$

whenever that hypothetical lower budget is positive, so $t_1 > 0$. When the right side is negative the inequality remains valid but has no positive force.

This strengthens the earlier statement that the mean relation cost must grow: unless the total cost has an extremely large individual scale, there must also be substantial *two-direction exterior area* in the original correction map. It is not a lower bound on every correction eigenvalue.

For example, along such a family, a proved estimate $\log(1+t_1) = o(q \log k)$ would turn (30) into the necessary lower exponent

$$\liminf \frac{\log(1 + 2\mathcal{A}_2/((q-1)t_1))}{\log k} \geq 2.$$

This last statement is a consequence with that displayed premise, not a claimed arithmetic estimate. The actual unresolved calculation is to bound the explicit sums (12)–(13) along the required growing-degree quartet family. Their uniform size has not been established in this note.

For the two-block total, (19) yields

$$\mathcal{B}_{h,k} \leq U_2(Z^{(0)}) + U_2(Z^{(1)}).$$

The inherited lower threshold remains four. No new purity conclusion is inferred from finite positivity or the optimality of a generic determinant inequality.

S31.0.9 9. A finite certificate robust to moment enclosures

The sharp formula (16) uses exact traces. One must not insert uncertified decimal traces into it and call the result an upper certificate.

There is a direct error-aware alternative. Suppose proved enclosures give $t_1 \in [t_1^-, t_1^+]$ and $t_2 \in [t_2^-, t_2^+]$. A rational cap $A_0 \geq t_1^+$ bounds every nonnegative eigenvalue. Choose a rational $0 \leq B_0 < A_0$ and use the explicit polynomial P_{A_0, B_0} in (20).

The same positive remainder proves

$$\log \det(I + Z) \leq q \log(1 + B_0) + \frac{t_1 - qB_0}{1 + B_0} + c_{A_0, B_0}(t_2 - 2B_0 t_1 + qB_0^2). \quad (31)$$

Every coefficient in (31) is a rational combination of two logarithms of positive rationals. Rational interval arithmetic, including all products, gives a valid upper enclosure from the two trace intervals. It needs no separately certified bound $K_i \geq g_0 K_j$.

The checker encloses $\log x$ by writing $x = 2^e y$, $1 \leq y < 2$, and retaining both factors in

$$\log y = 2 \sum_{n=0}^{M-1} \frac{z^{2n+1}}{2n+1} + R_M, \quad z = \frac{y-1}{y+1}, \quad 0 \leq R_M \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

The same formula encloses $\log 2$; reciprocation handles $x < 1$. These finite interval checks certify their supplied rational data only. An arithmetic application must supply certified moment/jet enclosures and hence valid trace intervals.

Arithmetic input size remains explicit For a direct construction through degree j , the source monic coefficients and norms are determined by the original convolution moments through degree $2j$, and the quotient columns retain the full coefficients of χ and the specified arithmetic inclusion. The sums (12)–(13) use G_i ; they do not require a second inverse G_j or an independent spectral-gap proof. This is not a claim that the required source moments are low-order or uniformly bounded. In the two length- q endpoint blocks, this direct route can require moments through $4q$. The parallel session's separate lower-input upper-volume comparison is not being replaced or credited as proved here.

Coordinate and mass propagation Under a fixed invertible quotient-coordinate map C , the actual data become

$$F \mapsto CF, \quad G_i \mapsto C^{-*} G_i C^{-1}, \quad Z \mapsto CZC^{-1}.$$

Thus both traces, $\mathcal{A}_2$, and the bound remain unchanged through that similarity. Raw confluent derivative coordinates keep their factorials in C ; the map is not silently made unitary.

A common source mass multiplier $c > 0$ gives

$$\Omega \mapsto c\Omega, \quad K_N \mapsto c^{-1} K_N, \quad G_N \mapsto cG_N,$$

so $Z, t_1, t_2, \mathcal{A}_2$ are unchanged by these displayed composites. The original mass has not been reset. An ambient rectangular arithmetic inclusion η is retained as an injection, not used as an invertible square coordinate map.

S31.0.10 10. Exact source calibrations

These are declared Gaussian polynomial fixtures, not asserted zeta packets. Use original coordinate $S = 1+ix$, literal Gaussian mass seven, variance one, and quotient $\chi = (S-1)^q$. The monic source recurrence and norms are

$$p_0 = 1, \quad p_1 = S - 1, \quad p_{n+1} = (S-1)p_n + np_{n-1}, \quad \omega_n = 7n!.$$

All quotient matrices below are computed in the original remainder basis $1, S, \dots, S^{q-1}$.

For $q = 2$, the blocks (1, 3) and (2, 4) give

$$(t_1, t_2) = \left(2, \frac{5}{2}\right), \quad (t_1, t_2) = \left(\frac{7}{4}, \frac{37}{16}\right).$$

Their exact two-trace factors are $(a, b) = (3/2, 1/2)$ and $(3/2, 1/4)$, so

$$\frac{V_1}{V_3} = \frac{15}{4}, \quad \frac{V_2}{V_4} = \frac{25}{8}, \quad B = \log(375/32).$$

The two-moment upper bound is exact in dimension two. This matches the preceding restriction example without supplying a spectral gap.

For $q = 3$, the original block $(3, 6)$ has

$$t_1 = 11, \quad t_2 = \frac{3365}{32}, \quad \mathcal{A}_2 = \frac{507}{64}, \quad \frac{V_3}{V_6} = \frac{5145}{256},$$

$$a = \frac{11}{3} + \frac{7\sqrt{127}}{12}, \quad b = \frac{11}{3} - \frac{7\sqrt{127}}{24}.$$

Numerical evaluations of these exact expressions are

$$\log(V_3/V_6) \approx 3.000603203789, \quad U_2 \approx 3.063328308618, \quad 3 \log(1 + t_1/3) \approx 4.621335122841.$$

The exact checks use rational enclosures and polynomial identities, not these decimals as proof. The original operator has its full nonzero nilpotent jets. No inference from a restriction eigenvalue to an off-line arithmetic zero is made.

S31.0.11 11. What is established and what is handed onward

Established here are the source cost map (7)–(10), the first and second exterior source norms (12)–(15), the sharp two-trace upper bound with a proved cap (16)–(21), its positive remainder and third-trace refinement, the area-only upper certificate, and their composition with the current first-degree-correct consecutive theorem. These are written finite theorems; exact algebraic regression checks support their implementation. They are not additional Lean certificates.

The same canonical arithmetic volume is now bounded without a separate restriction-gap premise. Estimating the growing arithmetic values of (12)–(13) remains a real analytic task. This note does not assert a uniform bound completing RH, does not erase a non-radical packet, and does not turn an e -valued quotient observation into absence.

No remote branch, publication cut, or formalization source was changed. The handoff lists bounded reusable targets for the parallel session and leaves its own interval-moment and fixed-budget work untouched.

S31.0.12 References and source locators

- Owner-supplied *Tau Endpoint Restriction Control*, NOTE.tex and original checker in the 13 September 2026 ZIP. Read as source; its finite map definitions are inputs here.
- Owner-supplied Pasted markdown(20260913-004825).md: PR #25 report, norm-review clarification, exact product, first-degree step, and threshold-four result. Source claims are distinguished from this session's executions.
- Owner-supplied Pasted markdown (2)(5).md: ongoing collaboration transcript; partial reported results are not substituted for their missing full proofs.
- GitHub KokunoYumeto/zeta-function-research-reader, main 952ef9fee1e1419b6858d920354de8fa99430b7d; PR #25 66fac5e7885a40cd4873e74b754907320f05b067, workbenches/tau-consecutive-window-formal/RESEARCH_NOTE.md.
- Grone, B., Johnson, C. R., Marques de Sá, E., and Wolkowicz, H. (1984). *Improving Hadamard's inequality*. Linear and Multilinear Algebra 16(1–4), 305–322. DOI: 10.1080/03081088408817634. The publisher abstract and author-hosted bibliography were inspected; the present proof is given explicitly rather than attributed to an unread theorem number.
- NIST Digital Library of Mathematical Functions, §3.3, especially the divided-difference and confluent interpolation formulas. <https://dlmf.nist.gov/3.3>

S32 Complete original polynomial-exponential cohomology comparison

Source role. Complete delivered proof witness. Its historical claims retain their source revision; corrections and full current maps are proved in the corresponding integrated chapter.

Exact revision. c2246c9f6046347bbd246890db88b1676b1a53487d66583a13b5883528bc0402

Complete source. sources/web_deligne_exponential_delivery/Tau_Deligne_Exponential_Comparison/RESEARCH_NOTE.md

Source SHA-256. c2246c9f6046347bbd246890db88b1676b1a53487d66583a13b5883528bc0402

Source metadata: author (MetaInlines)

Owner-directed SplitZero research continuation

Source metadata: date (MetaInlines)

13 September 2026

Source metadata: subtitle (MetaInlines)

A flat rank- q deformation, its exact operator connection, and the logarithmic image of the preceding comparison curve

Source metadata: title (MetaInlines)

Deligne's exponential-family step and the original arithmetic packet

S32.0.1 1. What is new, and what the source actually supplies

This continuation reads further into Deligne's *La conjecture de Weil II*, especially §§3.7.2–3.7.4, while completing the cohomological calculation of the preceding representative curve. There are two new applications, with different roles.

First, the logarithmic de Rham complex of that curve is calculated, rather than inferred from its underlying coherent bundle. Its arithmetic coefficients occur in degrees zero and two; its degree-one middle image is zero. This identifies it as a constant-coefficient comparison curve, not as a nonconstant constituent in Deligne's pencil argument. The full coefficients and the source-induced metric are retained by explicit maps.

Second, an actual polynomial-exponential family is built from the existing cyclic relation polynomial. Its characteristic-zero cohomology is a free module of exactly the original packet rank, including at repeated roots. Its special fibre is the original cyclic quotient, and the original sum operator is the exact polar coefficient of its parameter connection. Its finite-field realizations have precisely the form to which Deligne applies his exponential-sum theorem. The two realizations are exhibited as base changes of the same polynomial data, not declared isomorphic and not assigned the same operator.

The new algebraic statements below have proofs. The finite-field purity statement is an application of Deligne's theorem under the hypotheses verified in Section 9. No arithmetic norm comparison between these exponential periods and the original theta metric is proved here. In particular, the finite-field Frobenius has not been identified with the original sum operator.

S32.0.2 2. Reading Deligne, rather than using the word purity

The supplied S20 archive was reassembled in this turn. All six binary parts were checked against their declared lengths and SHA-256 hashes. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256 e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The retained French and English LaTeX were read at the selected passages recorded in SOURCE_REVIEW.md, alongside the current page-local French records. No PDF extraction or OCR was used. The historical TeX is retained as a witness, not automatically treated as an error-free edition.

On printed pages 202–203, Deligne's argument has three separate inputs: the exceptional-fibre exact sequence, the integer weights of nonconstant constituents, and the independent bound strictly below two. Constant constituents have zero first cohomology on the projective line. The tensor-square step then improves the bound on the same Frobenius eigenvalue. On page 206, compact-support and ordinary-cohomology bounds meet on the image of their actual comparison.

The additional reading is printed pages 215–216. For a degree- d polynomial with d prime to the

residue characteristic and nonsingular leading projective hypersurface, Deligne proves concentration of the exponential-sheaf cohomology in degree n , dimension $(d-1)^n$, and weight n . His next step is to put the polynomials into a parameter space, prove lissity by local acyclicity including at infinity, and use connectedness and the weight results to reduce to a particular polynomial. The purity propagation used in that argument is explicitly stated on page 178, Corollary 1.8.12: on a connected base, a lisse mixed sheaf pure at one point is pointwise pure. Its proof passes through the filtration into pure constituents, rather than inferring purity merely from constant rank. This supplies an actual geometric family to build from our relation polynomial. The supplied reference for the final monomial case has an internal numbering mismatch, recorded in SOURCE_REVIEW.md; the displayed exponential theorem is used with its stated scope, not represented as a newly verified entire historical proof.

The source does not say that the critical points of a polynomial have real part one half. The operator singled out by its purity theorem is finite-field Frobenius. The connection coefficient calculated below is a different output of the same potential data, linked through the displayed realization diagram rather than an assumed equality.

S32.0.3 3. Original data and their supported realization

Retain the scalar construction and its specified maps:

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r. \quad (1)$$

The original coefficient square remains

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (2)$$

The first target is infinite. Every lifted scalar specialization below fixes external absence and sends a represented scalar zero to the represented zero of the receiving ring.

Keep the original theta complex, source function, full packet, and sum coordinate:

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x, \quad g = 2\xi, \quad h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}. \quad (3)$$

At tensor degree k , write

$$E = \mathbb{C}[S]/(\chi), \quad \chi = \chi_{h,k}, \quad q = \deg \chi > 0, \quad A = M_S, \quad (4)$$

where $S = s_1 + \cdots + s_k$, not its average. The existing cyclic inclusion is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_k) 1, \quad v_h = j_h(g/h). \quad (5)$$

Its characteristic polynomial as a cyclic sum module need not be the characteristic polynomial of the entire tensor module. The actual injection (5), the previously constructed equivariant retraction, and its complementary module retain that distinction.

For an admitted degree N , retain the original canonical representative R_N and the source map $\mathcal{V}$:

$$J^{(k)} \mathcal{V} R_N = \eta, \quad q^{(k)} \mathcal{V} R_N = \sigma_h^{\otimes k} \eta, \quad R_N^* O_N R_N = G_N. \quad (6)$$

The monic norms in O_N have their original mass $\omega_0 = \mu_h^k$. No metric is chosen in this note to make an operator unitary. The empty-packet case has $E = 0$ and is handled outside all formulas requiring q positive.

S32.0.4 4. First compute the logarithmic cohomology of the preceding curve

For an original degree pair $i \leq j$, retain

$$\Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j, \quad R(z) = R_j + z\Delta, \quad G(z) = G_j + |z|^2 C. \quad (7)$$

The predecessor constructs the image lattice on $X = \mathbb{P}_{\mathbb{C}}^1$, with boundary $D_{\infty} = \{\infty\}$ and $w = 1/z$:

$$\mathcal{M}_{E_0} = E_0\mathcal{O} + wE\mathcal{O}, \quad E_0 = \ker \Delta. \quad (8)$$

Its smallest A-stable enlargement is $\mathcal{M}_F = F\mathcal{O} + wE\mathcal{O}$, where

$$F = E_0 + AE_0 + \cdots + A^{q-1}E_0. \quad (9)$$

All statements below concern this original target E. Let $Q_F = E/F$ denote this finite quotient, not the theta quotient Q.

4.1 The actual differential

For any positive integer m, define the lattice

$$\mathcal{M}_{F,m} = F\mathcal{O} + \mathcal{O}(-mD_{\infty})E \subset E\mathcal{O}, \quad (10)$$

and its two-term logarithmic complex, in degrees zero and one,

$$\mathcal{K}_{F,m} = [\mathcal{M}_{F,m} \xrightarrow{d} \mathcal{M}_{F,m} \otimes \Omega_X^1(\log D_{\infty})]. \quad (11)$$

The differential is the restriction of the constant coefficient connection: in the local frame w^me its coefficient is $w\partial_w + m$. Consequently it preserves (10) after tensoring with logarithmic forms. Because F is A-stable, A acts on the entire complex. A choice of complement $E = F \oplus J$ gives the coefficient description $\mathcal{M}_{F,m} = F\mathcal{O} \oplus J\mathcal{O}(-m)$; the action can have an off-diagonal block and is not asserted to split with that choice.

Computed cohomology. The canonical identifications as A-modules are

$$\boxed{\mathbb{H}^0(X, \mathcal{K}_{F,m}) = F, \quad \mathbb{H}^1(X, \mathcal{K}_{F,m}) = 0, \quad \mathbb{H}^2(X, \mathcal{K}_{F,m}) = Q_F.} \quad (12)$$

Here is a direct calculation retaining the coefficients. For the compactly supported component, use the two affine charts and the finite frame at z. The Cech groups are

$$H^1(\mathcal{O}(-mD_{\infty})) = \text{span}\{z^{-1}, \dots, z^{-(m-1)}\},$$

$$H^1(\mathcal{O}(-mD_{\infty}) \otimes \Omega^1(\log D_{\infty})) = \text{span}\{z^{-1}dz, \dots, z^{-m}dz\}.$$

The induced differential is exactly

$$z^{-a} \mapsto -a z^{-(a+1)}dz, \quad 1 \leq a < m. \quad (13)$$

It is injective, and its cokernel is the line generated by $z^{-1}dz$. For the ordinary logarithmic component the complex is $[\mathcal{O} \rightarrow \mathcal{O}(-1)]$, with only the constant H^0 . The exact sequence of complexes with subobject F and quotient Q_F then gives (12) canonically, without an A-invariant complement. In particular the degree-two identification is residue in the displayed coordinate.

If the original arithmetic cochain degree k is restored by the shift $[-k]$, the coefficient groups are in degrees k and k+2. For every polynomial f,

$$\boxed{\text{Str}(f(A) \mid R\Gamma\mathcal{K}_{F,m}[-k]) = (-1)^k (\text{Tr } f(A|F) + \text{Tr } f(\bar{A}|Q_F)) = (-1)^k \text{Tr } f(A|E).} \quad (14)$$

The quotient retains all induced Jordan data. This is trace additivity on the original exact sequence, not deletion of its second term.

4.2 Compact, middle, and ordinary comparisons

Put

$$\mathcal{K}_! = [E\mathcal{O}(-D_{\infty}) \rightarrow E\Omega_X^1], \quad \mathcal{K}_{\text{mid}} = [E\mathcal{O} \rightarrow E\Omega_X^1], \quad \mathcal{K}_* = [E\mathcal{O} \rightarrow E\Omega_X^1(\log D_{\infty})]. \quad (15)$$

Their natural inclusions compute the usual compact-support, smooth middle, and ordinary comparisons for the constant coefficient system on $\mathbb{A}^1$. At m=1 the mixed complex fits $\mathcal{K}_! \rightarrow \mathcal{K}_{F,1} \rightarrow \mathcal{K}_*$. The cohomology groups are

$$H^a(\mathcal{K}_!) = \begin{cases} E & a = 2, \\ 0 & a \neq 2, \end{cases} \quad H^a(\mathcal{K}_*) = \begin{cases} E & a = 0, \\ 0 & a \neq 0, \end{cases} \quad (16)$$

and $\mathcal{K}_{\text{mid}}$ has E in degrees zero and two. In particular,

$$\boxed{\text{im}(H_c^1(\mathbb{A}^1, E) \rightarrow H^1(\mathbb{A}^1, E)) = 0.} \quad (17)$$

This is an application of Deligne's constant-constituent test to the actual constructed curve. It locates its role: it is a metric comparison with retained boundary data, not yet the nonconstant cohomological image carrying E in degree one. The constructive continuation to such a cohomological family begins in Section 5.

The two boundary comparisons remain explicit. For $m=1$, the termwise quotient $\mathcal{K}_*/\mathcal{K}_{F,1}$ is the punctual complex $[Q_F \xrightarrow{0} Q_F]$ in degrees zero and one, while $\mathcal{K}_{F,1}/\mathcal{K}_! = [F \xrightarrow{0} F]$. For general m the first quotient has local coefficient differential

$$w^a \mapsto aw^a dw/w, \quad 0 \leq a < m. \quad (18)$$

Every positive a is paired with its exact inverse coefficient $1/a$; the constant class survives in both degrees. Thus the extra coherent length is kept before its logarithmic differential is taken. In the split construction those cycles, boundaries, and their quotients use the original fibrewise operations. External absence is never used as the image of a represented boundary.

4.3 Ramification and saturation retain the metric cost The ramified parameter map $f_m : \mathbb{P}^1 \rightarrow \mathbb{P}^1$, $z \mapsto z^m$, gives the actual representative $R_m(z) = R_j + z^m \Delta$. Its pulled-back determinant curvature has radial density

$$\kappa_m(t) = m^2 t^{m-1} \text{Tr}(Z(I + t^m Z)^{-2}), \quad Z = G_j^{-1} C. \quad (19)$$

Substitution $v = t^m$ proves

$$\boxed{\int_0^\infty \kappa_m(t) dt = m \text{rank } C, \quad \int_0^1 \log(1/t) \kappa_m(t) dt = \log(V_i/V_j).} \quad (20)$$

The pullback on the surviving degree-two de Rham class is $dz/z \mapsto m dz/z$. It is not an identity disguised by dividing the trace by m .

In the A -stable enlargement, let $s_A = \dim F - \dim E_0$. Its determinant metric in the infinity chart is the old one multiplied by $|w|^{-2ms_A}$. With the retained curvature definition $(i/2\pi)\partial\bar{\partial} \log \det G$, the current is

$$\boxed{c_1((\det \mathcal{M}_{F,m})^\vee, h^\vee) = \mu_m - ms_A \delta_\infty.} \quad (21)$$

Indeed $(i/2\pi)\partial\bar{\partial} \log |w|^2 = \delta_\infty$. Its total mass is $m(q - \dim F)$, the degree of the dual determinant. Full saturation subtracts $m \text{rank } C$ at infinity and has total degree zero, but leaves the endpoint logarithmic integral (20) unchanged. The negative atom is a geometric current; it is not a negative support label.

S32.0.5 5. An actual rank- q cohomological deformation of the cyclic packet

Write the original monic polynomial in its original coordinate:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a. \quad (22)$$

Introduce two new parameters u and t . They are independent of the scalar τ , the arithmetic coordinate S , the logarithmic comparison parameter z , and the tensor degree k . Put $R = \mathbb{C}[u, t]$ and define

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1}, \quad \Phi_t(S) = \Phi(S) - tS. \quad (23)$$

This is the specified primitive with $\Phi(0) = 0$, with every coefficient $1/(a+1)$ written. It changes no source coefficient in $g = 2\xi$. Define the two-term complex

$$\mathcal{C}_{u,t} = [R[S] \xrightarrow{\mathcal{D}_{u,t}} R[S]dS], \quad \mathcal{D}_{u,t}(P) = (uP'(S) + (\chi(S) - t)P(S))dS, \quad (24)$$

in degrees zero and one.

Flatness and exact rank. There is a direct sum of R-modules

$$R[S] = \mathcal{D}_{u,t}(R[S])/dS \oplus R[S]_{<q}. \quad (25)$$

In particular,

$$H^0(\mathcal{C}_{u,t}) = 0, \quad \mathcal{H} := H^1(\mathcal{C}_{u,t}) \cong R^q, \quad \mathcal{H}/(u,t)\mathcal{H} \cong E. \quad (26)$$

Proof. If P has degree a and leading coefficient b, then $uP' + (\chi - t)P$ has degree $a+q$ and leading coefficient b. It cannot be zero. For any polynomial F of degree at least q, subtract $\mathcal{D}_{u,t}(bS^{\deg F - q})/dS$ to remove exactly its highest term. Repeating terminates with a remainder of degree below q. Uniqueness follows from the leading-degree statement. The coefficients stay in R throughout. The same proof survives arbitrary base change of R, so (25) is a split exact presentation, not only a dimension count over its fraction field. Specializing $u=t=0$ turns (24) into multiplication by the literal χ , proving the final isomorphism.

Let $\text{red}_{u,t}$ denote the resulting coefficient-linear remainder map and $s_{<q}$ its specified polynomial section. The fibre map in (26) is $[P]_{\mathcal{D}} \mapsto [P|_{u=t=0}]_{\chi}$, with inverse induced by the chosen remainder basis.

There is no ring quotient asserted for u nonzero. Its exact relationship to multiplication is the commutator

$$[\mathcal{D}_{u,t}/dS, M_S] = uI. \quad (27)$$

For example, $\mathcal{D}(1)/dS = \chi - t$, while $\mathcal{D}(S)/dS = u + S(\chi - t)$; hence $[S(\chi - t)] = -u[1]$ in $\mathcal{H}$. The image of the differential is generally not an ideal. Formula (25) supplies the coefficient module and its section; the next paragraph supplies the operator that really descends.

S32.0.6 6. The parameter connection contains the original sum operator

On $\mathcal{H}[u^{-1}]$, define

$$\nabla_t[P] = [\partial_t P - SP/u]. \quad (28)$$

The exact relation

$$[\mathcal{D}_{u,t}/dS, \partial_t] = I, \quad [\mathcal{D}_{u,t}/dS, -M_S/u] = -I \quad (29)$$

proves that their sum commutes with the differential. Thus (28) descends without asserting that multiplication by S separately descends.

In the specified remainder basis $1, S, \dots, S^{q-1}$, the connection is

$$\nabla_t = \partial_t - \frac{A(t)}{u}, \quad A(t) = A + t\mathcal{R}, \quad \mathcal{R} = e_0 e_{q-1}^T. \quad (30)$$

Here $Ae_a = e_{a+1}$ for $a < q-1$ and

$$A(t)e_{q-1} = (-c_0 + t)e_0 - c_1 e_1 - \dots - c_{q-1} e_{q-1}. \quad (31)$$

Only a degree-q polynomial has to be reduced in deriving (30); $\mathcal{D}(1) = \chi - t$ gives exactly (31), including the sign of t.

The operator $-u\nabla_t = -u\partial_t + A(t)$ is regular in u and obeys the exact coefficient rule $(-u\nabla_t)(fv) = f(-u\nabla_t)v - u(\partial_t f)v$. It is a parameter-scaled connection in the t direction, not a derivative in u. Its reduction at $u=0$ is the honest endomorphism $A(t)$, and

$$\boxed{(-u\nabla_t) \bmod (u, t) = A = M_S \text{ on } E.} \quad (32)$$

This is the precise operator bridge. It contains a minus sign fixed by (24) and (28). The original arithmetic scaling is not renamed Frobenius.

Carry the parameter through the original theta representative Let $r_N = \mathcal{V}R_N : E \rightarrow \mathcal{B}^{\widehat{\otimes} k}$. On its specified test-function domain, use the earlier full jet map followed by the fixed equivariant cyclic retraction to obtain j_E with

$$j_E r_N = I_E, \quad j_E D^{(k)} = A j_E, \quad j_E \text{ kills the original theta boundaries.} \quad (33)$$

The construction of j_E uses the retained equivariant retraction, not an arbitrary quotient of the full tensor packet. Define the concrete finite-rank perturbation

$$\boxed{D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E.} \quad (34)$$

Then

$$j_E D_t^{(k)} = A(t) j_E, \quad \boxed{D_t^{(k)} r_N - r_N A(t) = D^{(k)} r_N - r_N A = dK_N.} \quad (35)$$

Both identities follow by substitution. The added term vanishes on every original boundary because j_E does. Thus the deformation has an explicit source realization preserving the original relation primitive. At $t=0$ it is exactly the original generator. At t nonzero it is the displayed finite-rank deformation, not a new assertion about the zero set of g .

The canonical arithmetic metric remains G_N . Its parameter control is explicitly

$$\boxed{W_N(t) = W_N(0) + t G_N \mathcal{R} + \bar{t} \mathcal{R}^* G_N.} \quad (36)$$

No positive metric has been selected from the exponential family to replace G_N .

6.1 The exponential periods are constructed and proved to span Fix a nonzero complex u , write $d=q+1$, and fix an argument of u . Set

$$\theta_0 = \frac{\arg u + \pi}{d}, \quad \theta_j = \theta_0 + \frac{2\pi j}{d} \quad (0 \leq j < d). \quad (36a)$$

Let ℓ_j be the oriented ray from zero to infinity with angle θ_j , and set $\Gamma_j = -\ell_0 + \ell_j$ for $1 \leq j \leq q$. The minus sign makes the initial endpoint contributions cancel. For a polynomial P define

$$\mathcal{P}_j(P) = \int_{\Gamma_j} e^{\Phi_t(S)/u} P(S) dS, \quad \Pi_{jb}(u, t) = \mathcal{P}_j(S^b), \quad 0 \leq b < q. \quad (36b)$$

These are absolutely convergent contour integrals. Along each ray the leading exponent is $-r^d/(d|u|)$. Lower-degree terms are dominated by that negative term outside a sufficiently large compact interval, uniformly when the lower coefficients and t range over a compact set. The same estimate after multiplying by any polynomial proves holomorphic dependence on those parameters and justifies their differentiation under the integral.

Integration of the actual total derivative gives

$$\mathcal{P}_j(uP' + (\chi - t)P) = u \int_{\Gamma_j} d(e^{\Phi_t/u} P) = 0. \quad (36c)$$

Both ends at infinity have zero boundary value, and the two values at zero cancel. Consequently the periods descend through the very differential (24), not through a quotient chosen afterwards.

The period matrix is invertible for every choice of the lower coefficients and every t , at this fixed nonzero u . Here is a direct proof, including the singular critical-point cases.

Temporarily regard the coefficients $c_0, \dots, c_{q-1}$ as independent parameters. The coefficient-direction connections are

$$\nabla_{c_a} = \partial_{c_a} + \frac{S^{a+1}}{(a+1)u}, \quad [\mathcal{D}/dS, \nabla_{c_a}] = 0. \quad (36d)$$

The two commutator terms are $-S^a$ and $+S^a$. Reducing $S^{a+b+1}/((a+1)u)$ by (25) gives the b-th column of a q-by-q matrix B_a . Every entry is a polynomial in the lower coefficients and t, with powers of the fixed u^{-1} . Thus

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u. \quad (36e)$$

At the monomial potential $\Phi = S^d/d$, $t=0$, direct substitution in the ray integral gives

$$\Pi_{jb}^{\text{mon}} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} \left(e^{2\pi i j(b+1)/d} - 1\right). \quad (36f)$$

All Gamma factors, all powers, and all phases remain. The final q-by-q matrix is nonsingular: a dependence among its columns would give a polynomial $\sum_{l=1}^q a_l(z^l - 1)$ of degree at most q vanishing at every d-th root of unity, including 1. It has $d=q+1$ distinct zeros, so it is the zero polynomial and every a_l vanishes.

For the phase matrix $V_{jl} = e^{2\pi i j l/d} - 1$, $1 \leq j, l < d$, finite root-of-unity summation also gives $V^*V = d(I_q + \mathbf{1}\mathbf{1}^T)$ and $\det(V^*V) = d^d$. This evaluates its nonzero size without absorbing any of the Gamma factors in (36f).

Join this monomial parameter to the actual parameter by a straight line in the coefficient space, keeping the rays fixed. Equations (36e) give $\Pi' = \Pi B$ with an everywhere continuous, explicitly reduced matrix B. Differentiation of the determinant, valid through the adjugate formula even at a putative singular matrix, yields

$$(\det \Pi)' = (\text{Tr } B) \det \Pi. \quad (36g)$$

Its initial value is nonzero by (36f). The scalar equation has solution given by that value times the exponential of the integral of $\text{Tr } B$, and never vanishes. This proves the stated invertibility without assuming simple critical points or a basis of unproved cycles.

It follows that the actual period map is the coefficient-linear isomorphism

$$\boxed{\mathcal{H}_{u,t} \xrightarrow{(\mathcal{P}_1, \dots, \mathcal{P}_q)} \mathbb{C}^q, \quad [P] \mapsto \Pi(u, t) \text{red}_{u,t}(P),} \quad (36h)$$

with inverse given by $\Pi(u, t)^{-1}$ followed by the specified degree-below-q section. Polynomial-exponential de Rham cohomology and rapid-decay period pairings are established machinery; (36a)–(36h) give an elementary proof of the particular one-variable comparison used here, with its contours and factors fixed.

6.2 Transport the period norm through the original source map Use the explicit coefficient identification

$$j_{u,t} : E \xrightarrow{\sim} \mathcal{H}_{u,t}, \quad [P]_{\chi} \mapsto [\text{rem}_{\chi} P]_{\mathcal{D}_{u,t}}. \quad (36i)$$

This is a complex-linear basis comparison. It is not a ring morphism when the target differential image is not an ideal. Equations (30)–(35) are the operator comparisons attached to it.

The original representative image now has an actual period isomorphism and inverse:

$$\boxed{r_N(E) \xrightarrow{\Pi(u,t)j_E} \mathbb{C}^q, \quad y \mapsto r_N \Pi(u, t)^{-1} y.} \quad (36j)$$

Their two composites are identities on the specified domains, by $j_E r_N = 1$. The same map applied to a retained theta boundary gives zero because j_E kills that boundary; its split lift gives the receiving supported zero, not external absence.

The operator comparison also holds at the level of actual period functions. For a t-independent F in the admitted theta test-function domain, (33), (35), and (36e) give

$$\boxed{-u \partial_t (\Pi(u, t) j_E F) = \Pi(u, t) j_E D_t^{(k)} F.} \quad (36j.1)$$

At $t=0$ the right side uses the original $D^{(k)}$. On the full test-function domain the kernel of this period observation is exactly $\ker j_E$, because Π is invertible. It remains an explicit retained kernel; the isomorphism (36j) concerns only the stated representative image $r_N(E)$.

Give the period-coordinate target its explicitly chosen ordinary coordinate Gram I_q and write

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (36k)$$

Then B_N^{per} is positive self-adjoint in the original G_N , and

$$\|\Pi v\|^2 = v^* G_N B_N^{\text{per}} v, \quad \boxed{\log \frac{V_i}{V_j} = \log \frac{\det B_j^{\text{per}}}{\det B_i^{\text{per}}}.} \quad (36l)$$

The latter follows from $\det B_N^{\text{per}} = \det H_\Pi / \det G_N$ at the same (u, t) . Thus the endpoint cost has been carried into two explicit period-to-source norm comparisons; it has not been set to zero by choosing period coordinates. The degree of the cyclic packet and the fixed contour-coordinate metric remain in the formula. Equation (36l) is a comparison identity, not a uniform bound on those determinants.

For $q=1$, $\chi(S) = S - \lambda$, real $u > 0$, and the contour $i\mathbb{R}$ oriented from $-i\infty$ to $+i\infty$, the integral is exactly

$$\boxed{\Pi(u, t) = i\sqrt{2\pi u} \exp\left(-\frac{(\lambda+t)^2}{2u}\right), \quad -u \partial_t \log \Pi = \lambda + t.} \quad (36m)$$

This uses the displayed orientation; it is the negative of the $q=1$ ray-difference cycle in (36a). The positive metric value for real λ, t is $2\pi u \exp(-(\lambda+t)^2/u)$. Its exponentially singular u -behaviour is retained. Thus even in rank one, passage through $u=0$ is a substantive norm-comparison question, not a consequence of finite-field purity of a different operator.

S32.0.7 7. A constant residue pairing gives the exact dual connection

On the remainder basis define the complex-bilinear form

$$\mathcal{S}_t(P, Q) = \sum_{\chi(\lambda)=t} \text{Res}_{S=\lambda} \frac{P(S)Q(S)}{\chi(S) - t} dS. \quad (37)$$

Multiplicity is handled by the residue, not by an assumption that the roots are simple. The form can equivalently be evaluated as the coefficient of S^{-1} in the Laurent expansion at infinity of the displayed rational function. This is minus the residue at infinity, so the sign agrees with the sum of the finite residues.

For degree-below- q P, Q ,

$$\boxed{\mathcal{S}_t(P, Q) = \mathcal{S}_0(P, Q).} \quad (38)$$

Indeed,

$$\frac{1}{\chi - t} - \frac{1}{\chi} = \frac{t}{\chi(\chi - t)},$$

and multiplication by PQ , whose degree is at most $2q - 2$, gives a Laurent expansion beginning at S^{-2} . Its S^{-1} coefficient is zero. Let S be the constant matrix. Its entries below the first nonzero antidiagonal vanish and that antidiagonal has entries one, so

$$\det S = (-1)^{q(q-1)/2}, \quad \boxed{A(t)^T S = S A(t).} \quad (39)$$

The latter equation follows because multiplication by S is self-adjoint for the residue pairing on the algebra $\mathbb{C}[S]/(\chi - t)$; (30) is precisely its multiplication matrix in this basis.

Let ι_u be the base involution $(u, t) \mapsto (-u, t)$. The actual dual map is

$$\boxed{\mathcal{H}[u^{-1}] \longrightarrow (\iota_u^* \mathcal{H}[u^{-1}])^\vee, \quad v \longmapsto (w \mapsto v^T S w).} \quad (40)$$

It is horizontal:

$$\partial_t(v^T S w) = (\nabla_t^u v)^T S w + v^T S (\nabla_t^{-u} w). \quad (41)$$

This is a bilinear duality; the original Hermitian arithmetic pairing is connected on the special fibre by its retained antilinear reflection $f \mapsto f^{\dagger k}$. In particular the Jacobian contraction is still

$$\boxed{\mathcal{S}_0(f^{\dagger k}, \chi' v) = \text{Tr}_E M_{f^{\dagger k} v}.} \quad (42)$$

The proof is the local equality $\chi'/\chi = \ell_\lambda/(S-\lambda) + \text{a holomorphic term}$. The trace on the full original tensor or symmetric packet is still obtained through η and its retained complement; (42) is not silently substituted for that larger trace. Likewise the original factor $v_h^{\otimes k}$ stays in (5) and (6).

S32.0.8 8. The split coefficient maps and the original derivative tower

The family is over $R = \mathbb{C}[u, t]$, with scalar map $G(\mathbb{C}) \rightarrow G(R)$. Its reconstruction at any original support label ℓ has classes (ℓ, v) , and

$$e_R(\ell, v) = (\ell, 0), \quad \tau(\ell, v) = \tau. \quad (43)$$

The specialization $R \rightarrow \mathbb{C}$, $u, t \mapsto 0$, lifts to a semiring map $G(R) \rightarrow G(\mathbb{C})$, fixing τ and carrying both $u^\bullet, t^\bullet$ to the receiving e . The induced coefficient map is

$$(\ell, [P]_{\mathcal{D}}) \longmapsto (\ell, [P(0, 0, S)]_{\chi}). \quad (44)$$

For every P , the exact family relation is

$$[\chi P] = t[P] - u[P'] \quad \text{in } \mathcal{H}. \quad (45)$$

At $u=t=0$, this becomes the original supported zero. This family identity must not be substituted for the already formalized derivative between powers of the original relation ideal. Their precise polynomial relationship is

$$\partial_S(\chi P) = \chi' P + \chi P', \quad [\partial_S(\chi P)]_{\chi} = [\chi' P]_{\chi}, \quad (46)$$

whereas the u -term of (45) contains $-P'$. Both are obtained from the same polynomial and the displayed maps. The conormal response $\chi' P$ remains exactly the Jacobian in (42). None of the new parameters is another scalar zero-adjunction, and no original ideal-power level is replaced by a power of a contracted first ideal.

S32.0.9 9. This potential has actual Deligne exponential-sheaf fibres

Let R_0 be the finitely generated subring of $\mathbb{C}$ generated by the coefficients c_a and $1/(q+1)!$. Keep its specified embedding into $\mathbb{C}$. Equations (23) and (24) are defined over $R_0[u, t]$.

For every specified ring map $R_0 \rightarrow \mathbb{F}_Q$ with residue characteristic $p > q+1$, and every $u_0 \in \mathbb{F}_Q^*$ and $t_0 \in \mathbb{F}_Q$, the same potential gives the explicit Artin-Schreier cover

$$\boxed{Y^p - Y = \frac{\Phi(S) - t_0 S}{u_0}.} \quad (47)$$

The infinity calculation in Deligne's lissity proof can be made explicitly on this family. Put $d = q+1$ and $w = 1/S$. Then

$$\frac{\Phi(S) - tS}{u} = w^{-d}b(w, u, t), \quad b(w, u, t) = \frac{1}{u} \left[\frac{1}{d} + \sum_{a=0}^{q-1} \frac{c_a}{a+1} w^{q-a} - tw^q \right]. \quad (47a)$$

At $w=0$ its value is the unit $1/(du)$. On the neighborhood where b is invertible, adjoin the unit v by $bv^d = 1$ and define $x_\infty = wv$. This is an explicit finite étale cover: the derivative in v is dbv^{d-1} , whose inverse on the cover is v/d . The new relative coordinate has invertible derivative at $w=0$. Direct substitution gives

$$\boxed{\frac{\Phi(S) - tS}{u} = x_\infty^{-d}, \quad x_\infty = wv, \quad bv^d = 1.} \quad (47b)$$

Every coefficient in b is retained, as is the cover's root-of-unity action $v \mapsto \zeta v, \zeta^d = 1$. This is the local, étale source map used to see a constant exponential model at infinity. Together with the locally constant sheaf on the smooth affine part, it is exactly the local-acyclicity mechanism used in Deligne's §3.7.3. The open u nonzero remains necessary; this map makes no claim at $u=0$.

Choose a nontrivial additive character $\psi : \mathbb{F}_p \rightarrow \overline{\mathbb{Q}}_\ell^*$, with $\ell \neq p$, and let $\mathcal{L}_\psi((\Phi - t_0S)/u_0)$ be the corresponding eigensheaf. The polynomial on the right has degree $q+1$, prime to p , and leading coefficient $1/((q+1)u_0)$. In one variable its leading projective zero locus in $\mathbb{P}^0$ is empty, and is smooth. Thus the precise hypotheses of Deligne §3.7.2 hold.

His theorem supplies

$$\boxed{H_c^a(\mathbb{A}_{\mathbb{F}_Q}^1, \mathcal{L}_\psi((\Phi - t_0S)/u_0)) = 0 \ (a \neq 1), \quad \dim H_c^1 = q, \quad H_c^1 \text{ is pure of weight } 1.} \quad (48)$$

The universal-polynomial argument in §§3.7.3–3.7.4 supplies a lisse family on the parameter open $u \neq 0$, after that finite-field specialization. This is a direct application of Deligne's result, not a new proof of it.

The exact realization diagram is a span of specified base changes:

$$(R_0, \Phi, u, t) \longrightarrow \begin{cases} (\mathbb{C}, \mathcal{C}_{u,t}, \nabla_t), \\ (\mathbb{F}_Q, \mathcal{L}_\psi((\Phi - tS)/u), R^1p_!). \end{cases} \quad (49)$$

Its characteristic-zero branch has the proved specialization (26) and operator formula (32). Its finite-field branch has Frobenius and (48). No field map $\mathbb{C} \rightarrow \mathbb{F}_Q$ is asserted; both arrows start at the actual ring R_0 . Nor is a comparison equating the two operators supplied. The arithmetic theta metric G_N is not part of Deligne's finite-field conclusion until an additional comparison theorem carries it there.

This places the cyclic relation in a genuine family of the kind Deligne studies, instead of attributing his nonconstant-constituent argument to the constant representative curve. It leaves an explicit operator/norm comparison to investigate.

A calibration of the operator distinction Take the declared algebraic polynomial $\chi(S) = S - 3/4$; this is not an asserted zeta packet. Then

$$\Phi(S) = S^2/2 - 3S/4, \quad A(t) = 3/4 + t, \quad \nabla_t = \partial_t - (3/4 + t)/u. \quad (50)$$

All the rank, specialization, and duality results above hold. At every finite-field realization of characteristic greater than two where the coefficient is defined, (48) still gives weight-one Frobenius. The characteristic-zero special coefficient remains $3/4$. This exact example shows why a purity theorem for the displayed Artin-Schreier family does not, by itself, locate the spectrum of (32). The additional comparison must use the specific original arithmetic source, not only the fact that a polynomial-exponential realization exists.

S32.0.10 10. What this changes in the programme

The predecessor's nonconcentration problem remains

$$\mathcal{B}_{h,k} = \sum_{\ell=0}^1 \int_{|z|<1} \log(1/|z|^2) d\mu_\ell. \quad (51)$$

The reviewed source-norm theorem and consecutive-window result still force the inherited off-line threshold of four. Neither log extension, finite ramification, nor saturation changes its original endpoint metric cost; (20) and (21) state precisely how those operations act.

What is added is a cohomological family with the exact original cyclic special fibre, an explicit connection containing A , an explicit dual connection, and finite-field exponential fibres covered by a theorem actually read in Deligne. The source-side realization (34)–(36) specifies the finite-rank deformation and keeps the theta boundary. The contour periods now give actual isomorphisms at u nonzero and the matrices $B_N^{\text{per}} = G_N^{-1} \Pi^* \Pi$ give their exact comparison to the fixed arithmetic metric. Equation (36l) carries the endpoint volume into their determinant ratio. The remaining estimate must control those specified matrices through singular $u=0$ and along the original tensor/degree windows. The existence of the period isomorphism or of scalar support is not itself that uniform estimate.

This continuation proves no new upper bound contradicting the endpoint threshold. It does provide the actual family and the exact maps on which a Deligne-style comparison can be attempted, after calculating and retaining the limitations of the earlier constant curve.

S32.0.11 References and verification scope

Deligne, P. (1980), *La conjecture de Weil II*, Publications Mathématiques de l’IHÉS 52, 137–252. Selected current source records: printed pages 201–203, 206, 215–216, 248. Retained French and English TeX windows are listed in SOURCE_REVIEW.md.

The Stacks Project, Section 50.15, Tag 0FMU, *Log poles along a divisor*. Its residue exact sequence fixes the complexes in (15); the actual calculations in (12), (13), and (18) are given here. The comparison is made in complexes of complex vector-space sheaves, since their connections are coefficient-linear, not $\mathcal{O}_X$ -linear.

The new exact tests are finite regression checks of the displayed polynomial, matrix, quotient, connection, residue, and Čech formulas. They do not certify infinite arithmetic integrals, compile new Lean modules, or prove a realization comparison absent from this note. The frozen predecessor is preserved. No remote branch is modified by this delivery.

Marco Hien (2009), *Periods for flat algebraic connections*, Inventiones Mathematicae 178, 1–22, DOI 10.1007/s00222-009-0185-7, is primary background for de Rham/rapid-decay duality. Only its bibliographic record and abstract were consulted here; no general theorem from it is needed in the direct contour proof (36a)–(36h).

S33 The original complete comparable-window norm proof

Source role. Complete original pinned PR25 source. Proposal and adopted status are recorded separately.

Exact revision. 10b89fa667c5fe88ada6631b67af81754773aeeb6efe377c3f3d6b26d4bc86c3

Complete source. sources/owner_pr25/original_66/workbenches/tau-consecutive-window-formal/NORM_REV IEW.md

Source SHA-256. 10b89fa667c5fe88ada6631b67af81754773aeeb6efe377c3f3d6b26d4bc86c3

S33.1 Arithmetic norm review and comparable-degree windows

The owner supplied the 2026-09-13 arithmetic endpoint note. Its principal input is the actual density $w_h = |2\chi/h|^2/(2\pi)$, its k -fold convolution, and the unscaled monic squared norms $\omega_{h,k,j}$. This is a review and extension of that written proof, not a Lean certificate for zeta estimates or arithmetic integrals. The locally linked delivery archive was not accessible in this runtime; its reported test and manifest results are not counted as fresh executions.

S33.1.1 1. Audit of the analytic steps

The Joukowski ellipse with $|w|=8$ has semiaxes $65/16$ and $63/16$. For $f_T(z) = \zeta(1/2 + iT + iz)$, $|T| \geq 10$ keeps the pole outside its closure. Its image lies in the fixed real strip $[-55/16, 71/16]$, with imaginary magnitude between $|T| - 65/16$ and $|T| + 65/16$. Split that strip at $\operatorname{Re}(s) = 1/4$. Above this value, the fractional-part integral gives polynomial growth $O((1+|T|))$; below it the functional equation and vertical-strip gamma estimate give $O((1+|T|)^{(1/2 - \operatorname{Re}(s) + 1)})$. The largest exponent is $79/16 < 6$. Thus the claimed envelope $M_T = C_0(1+|T|)^6$ is sufficient, uniformly on the ellipse. Increase C_0 so $M_T \geq 1$.

The point $z = -3i/2$ corresponds to $|w| = (3 + \sqrt{13})/2 < 4$ and hence to $\zeta(2 + iT)$. The reciprocal Dirichlet series gives modulus at least $\zeta(2)^{-1}$. The annular maximum principle gives $m_T \geq \zeta(2)^{-3} M_T^{-2}$. Every closed unit disk centred on $[-1, 1]$ lies strictly within the ellipse: its surrounding rectangle $[-2, 2] \times [-1, 1]$ already satisfies the ellipse inequality. The maximum principle and Cauchy estimate therefore give $\sup |f_T| \leq M_T$. At a maximizer use the longer one-sided interval in $[-1, 1]$, of length at least one; $m_T/(2M_T) \leq 1/2$. Integration gives $m_T^3/(8M_T)$ and the stated exponent 42. The compact remaining T -range is handled by continuity and analytic nonvanishing on an interval. No nonvanishing on the critical line is assumed.

For the passage to $|g/h|$, use $g = h(g/h)$ before taking absolute values. The polynomial upper bound on $|h|$ avoids division at a selected zero. Gamma asymptotics yield the claimed interval-mass bound. One regularity detail should be explicit: w_h is bounded as well as integrable, by its continuous entire quotient on compact sets and exponential tail. Thus $w_h * w_h$ is continuous, using L^1 translation continuity against the bounded factor. Its integrand is positive almost everywhere, so the convolution is strictly positive everywhere. This justifies its positive minimum on $[-1, 1]$. Restricting the remaining factors to that interval proves the note's lower envelope with the original factor $\vartheta_h(k-3)$.

The monic Legendre norm on $[-L, L]$ is exactly

$$1_n(L) = 2^{-L} \frac{(2n+1)!}{(2n)!} [2^n (n!)^2 / (2n)!]^{1/2}.$$

The trial upper bound at degree j is $(2j)! b^{(-2j)} (M_h(b)^k + M_h(-b)^k)$, for fixed $0 < b < \pi/2$. Both are used on the original shifted line $S = k/2 + iu$; multiplication by i^j has modulus one and does not change the leading coefficient norm. These arguments validate the supplied $O_h(n)$ doubling-window result as a written theorem, not as a floating-point or Lean assertion.

S33.1.2 2. Stronger, two-sided envelope

Write $a = \pi/2$ and retain the lower-envelope constants $c > 0$, $\vartheta > 0$, $B = 42 + 2 \deg(h)$. Put

$$\begin{aligned} t &= \min(1, \vartheta), \quad c_0 = \min(1, 2c/3), \\ \text{ell} &= (1/2) \exp(-a) \sqrt{c_0 t} 2^{(-B/2)}, \\ M &= \max(1, M_h(b), M_h(-b)), \quad U = 4 \sqrt{M}/b. \end{aligned}$$

Then, for every $n \geq k \geq 3$,

$$(e11\ n)^{(2n)} \leq \omega_{(h,k,n)} \leq (U\ n)^{(2n)}. \quad (A)$$

Proof of the lower bound: $l_n(n) \geq (2/3)n^{(2n)4^{(-n)}}$; $\vartheta^{(k-3)} \geq t^n$; $\exp(-a(n+k-3)) \geq \exp(-2an)$; $(1+n+k-3)^{(-B)} \geq (2n)^{(-B)} \geq 2^{(-nB)}$, since $2n \leq 2^n$ for integer $n \geq 1$; finally $2c/3 \geq c0^n$. Multiply. For the upper bound use $(2n)! \leq (2n)^{(2n)}$, $k \leq n$ and $M_h(b)^k + M_h(-b)^k \leq 2M^n$, then $2 \leq 4^n$. This proves (A), retaining all masses in constants rather than resetting them.

For $n \geq k \geq 3$ and $n/2 \leq r \leq 2n$, let $m = n + r$. From (A),

$$\begin{aligned} & (\omega_m/\omega_n)^{(1/(2r))} \\ & \leq n\ U^{(1+n/r)}\ e11^{(-n/r)}(1+r/n)^{(1+n/r)} \\ & \leq C_h\ n, \\ C_h = & 27\ \max(1,U)^3\ \max(1,e11^{(-1)})^2. \end{aligned} \quad (B)$$

This extends the doubling-window estimate to the overlapping windows needed by the volume theorem. Constants depend only on the fixed h and chosen b , not k, n, r . No monotonicity of the monic norms is assumed. The inequality is valid at $r = n$ and at $r = n + 1$ when $n \geq 1$.

References: NIST DLMF 25.4 (functional equation), 5.11.9 (uniform bounded-real-part gamma asymptotic), 18.3 (Legendre normalization). The general arguments above are written proofs; the present Lean contribution formalizes their finite norm-to-volume consequences, not their analytic hypotheses.

S34 The original complete consecutive-window proof

Source role. Complete original pinned PR25 source. Proposal and adopted status are recorded separately.

Exact revision. cc61ab74ee33eeb43f0bd968a9825be35b3eb3475fa2ff0678f833ba22daa332

Complete source. sources/owner_pr25/original_66/workbenches/tau-consecutive-window-formal/RESEARCH_NOTE.md

Source SHA-256. cc61ab74ee33eeb43f0bd968a9825be35b3eb3475fa2ff0678f833ba22daa332

S34.1 Consecutive losses and the arithmetic endpoint threshold

S34.1.1 Source, scope, and notation

This add-only continuation starts from main 7ea0a49945390eae14d3160a5730858899768b5f. The source is the owner's arithmetic endpoint note dated 2026-09-13 and the previously supplied exact Toda and exterior formulas. NORM_REVIEW.md audits its analytic proof and establishes the comparable-window extension used below. Those analytic results have written proofs, not a new Lean certificate. The Lean files prove the finite propagation independently of assuming RH, normality of the arithmetic action, or an estimate for the quotient volume.

Keep the original packet h , $g=2x_i$, theta source, full Taylor unit, canonical representatives R_N , and arithmetic cyclic module $C[S]/\chi(h,k)$. Write $q=\dim C$. The norms ω_N are the unscaled monic source norms; $V_N=\det G_N$ are its canonical quotient volumes. The phase is $\phi_N=\sigma\text{-partial_theta} \log V_N$. Λ_N below is ω_N/V_N , NOT the global theta retraction denoted Λ in earlier notes.

S34.1.2 1. Exact consecutive product

For $N \geq q$ the existing identity is

$$\begin{aligned} \epsilon_N^{2+\phi_N} &= (\omega_{N+1}/\omega_N)(1-\delta_N)(\delta_{N+1})^{(-1)-1}, \\ \delta_N &= V_N/V_{N-1} \text{ in } (0,1]. \end{aligned}$$

Rearrange it as

$$\begin{aligned} (\epsilon_N^{2+\phi_N}) \Lambda_N &= \Lambda_{N+1}(1-\delta_N)(1-\delta_{N+1}). \end{aligned} \quad (1)$$

For $n \geq q$ and $r \geq 1$, multiplication and cancellation of positive Λ -values gives

$$\begin{aligned} \text{product}_{(j < r)} (\epsilon_{n+j}^{2+\phi_{n+j}}) &= (\omega_{n+r} V_n) / (\omega_n V_{n+r}) \\ &\quad (1-\delta_n)(1-\delta_{n+r}) \\ &\quad \text{product}_{(0 < j < r)} (1-\delta_{n+j})^2. \end{aligned} \quad (2)$$

Every interior contraction occurs twice. This is an exact phase-retaining identity; the Lean theorem `exact_product` proves its general algebraic form. In particular,

$$\begin{aligned} \min_{(j < r)} \epsilon_{n+j} &\leq [(\omega_{n+r} V_n) / (\omega_n V_{n+r})]^{1/(2r)}. \end{aligned} \quad (3)$$

The finite minimum exists. Equation (3) follows by bounding each factor $1-\delta$ by one, comparing the product with the minimum's $2r$ -th power, and taking a nonnegative root. This proof does not require Jensen or the earlier four-endpoint hyperbolic-sine bound. It need not be sharper when contractions are small.

The local coarse bound $\epsilon_N^2 \Lambda_N \leq \Lambda_{N+1}$ also holds at $N=q-1$. Use the separate first-degree identity

$$\begin{aligned} \epsilon_{q-1}^{2+\phi_{q-1}} &= (\nu_0 - \omega_q) / \omega_{q-1}, \\ V_q/V_{q-1} &= \omega_q/\nu_0. \end{aligned}$$

It follows that $\epsilon_{(q-1)}^2 \Lambda_{(q-1)} \leq \Lambda_q$. No inverse at degree $q-2$ is introduced. Thus (3) and the lower-allowance consequence below apply to windows starting at $q-1$, although the exact two-contraction identity (2) is used only from q onward.

S34.1.3 2. The actual spectral lower allowance forces each single block

The existing original-Gram exterior/trace theorem gives $L_{(h,k)} \leq \epsilon_N$ for every admitted N . Iterating the coarse local inequality proves

$$L_{(h,k)}^{(2r)} \Lambda_n \leq \Lambda_{(n+r)}. \quad (4)$$

If the actual norm bound gives $\omega_{(n+r)}/\omega_n \leq C^{(2r)}$, then

$$(L_{(h,k)}/C)^{(2r)} V_{(n+r)} \leq V_n. \quad (5)$$

For $L > 0, C > 0$ this is

$$\log(V_n/V_{(n+r)}) \geq 2r \log(L_{(h,k)}/C). \quad (6)$$

The formal propagation, norm comparison, and log conversion have separate declarations, so the analytic premise is explicit.

For the exact hypothetical quartet, retain

$$\begin{aligned} q_k &= [1+k(m-1)](k+1)^2, \\ L_{(h,k)} &\geq \delta k^2 q_k/2, \quad \delta > 0. \end{aligned}$$

By NORM_REVIEW.md, both windows $(n,r)=(q_k,q_k)$ and (q_{k-1},q_k) have norm factors at most $C_h q_k$. Indeed $q_{k-1} \geq k$ for $k \geq 3$, and r is between $n/2$ and $2n$. The first window uses only regular local degrees; the second begins at the separately checked first degree. Let $D_h = \delta/(2C_h) > 0$. Then

$$\begin{aligned} \log(V_{(q_k)}/V_{(2q_k)}) &\geq 2q_k \log(D_h k), \\ \log(V_{(q_{k-1})}/V_{(2q_{k-1})}) &\geq 2q_k \log(D_h k). \end{aligned} \quad (7)$$

Therefore the source's SAME four-volume budget satisfies the stronger implication

$$\begin{aligned} B_{(h,k)} &= \log[V_{(q_{k-1})}V_{(q_k)}/(V_{(2q_{k-1})}V_{(2q_k)})] \\ &\geq 4q_k \log(D_h k), \\ \liminf B_{(h,k)}/(q_k \log k) &\geq 4. \end{aligned} \quad (8)$$

The source's threshold 2 remains correct but is not optimal after retaining the adjacency of the exact radius equations. This is a new deduction, not a correction falsely attributed to its original proof. No upper estimate below 4 is claimed. Either one-block upper limsup below 2 would already suffice for contradiction, while a total upper limsup below 4 would also suffice. Multiplicities remain in q_k .

S34.1.4 3. Relation-layer meaning

For the original block (i,j) , put $F=[b_{(i+1)}, \dots, b_j]$, $\Omega = \text{diag}(\omega_{(i+1)}, \dots, \omega_j)$. The supplied map $b = T \cdot R_i F$ is an isomorphism onto $D_j \cap D_i^\perp$, with Gram $\Omega + F^* G_i F$, and original quotient value zero. Hence

$$V_i/V_j = \det(I + \Omega^{-1} F^* G_i F).$$

Take the positive generalized eigenvalues λ_a of $(F^* G_i F, \Omega)$. The two blocks $(q-1, 2q-1)$ and $(q, 2q)$ each have q source coordinates. Equations (7) imply, for each block separately,

$$\begin{aligned} \text{product}_a (a=1..q) (1+\lambda_a) &\geq (D_h k)^{(2q)}, \\ \text{geometric_mean}(1+\lambda_a) &\geq (D_h k)^2, \\ (1/q) \sum_a \lambda_a &\geq (D_h k)^2 - 1. \end{aligned} \quad (9)$$

The arithmetic-geometric mean inequality proves the last line. This does NOT force every λ_a to grow. Their multiplicity and rank profile remain. On the split lift the actual final quotient sends each retained relation to the receiving label's supported zero, not to external τ . Its source Gram is measured before that map.

S34.1.5 4. Constant and source accounting

No measure is normalized. The original mass $\omega_0 = \mu_h^k$ remains, and its effect is in the explicit constants of NORM_REVIEW.md. The Taylor multiplier and its derivative are not omitted. The norm leading-coefficient quotient is not identified with the arithmetic quotient. No complementary arithmetic block, nilpotent direction, or cochain sign is removed.

The analytic proof and the asymptotic consequence (8) are written mathematics. The new Lean run, when successful, certifies the finite local/product/forcing interfaces named in AuditConsecutiveWindow.lean; it does not certify the complex-analytic zeta argument, the arithmetic integrals, or a new weight theorem. Consult the PR for the exact completed run rather than treating this source note as an execution receipt.

S35 Complete supplied specialization and curvature continuation

Source role. Complete supplied mathematical text, retained with its original formulas and source chronology. The present integrated chapters state and prove the subsequent corrections and exact maps.

Exact revision. ed20dd4c189eb741d666edbaa964a03f616279a90e88b171bb9f31bba8652106

Complete source. sources/endpoint_user_sources/USER_SPECIALIZATION_CURVATURE_PROOF_20260913.txt

Source SHA-256. ed20dd4c189eb741d666edbaa964a03f616279a90e88b171bb9f31bba8652106

I added an actual compactification, specialization calculation, and dual boundary to the existing arithmetic representatives. The main result is that the endpoint volume we have been trying to control is exactly a logarithmic curvature integral on that compactified curve:

$$\log \frac{V_i}{V_j} = \int_{|z|<1} \log \frac{1}{|z|^2} c_1((\det \mathcal{E}_{ij})^\vee).$$

The curve is constructed from the **original theta relation**

$$\Delta_{ij} = L_{ij}R_i - R_j.$$

Its exceptional fibre, the directions killed by specialization, and their duals are calculated. The previous two trace inputs become the first two curvature coefficients of this same object.

There is also an additional repair: when the original arithmetic action develops a pole at the exceptional fibre, I construct its **smallest action-stable enlargement** and calculate the metric contribution that enlargement retains.

[Full derivation in the HTML reader](#) · [LaTeX](#) · [Formalization handoff](#) · [Complete package](#)

S35.0.1 1. What the Deligne rereading adds

I reassembled and hash-checked the six supplied S20 parts, then read the relevant retained French and English LaTeX against the current page-local French records. The [reading record](#) identifies the exact files, ranges, and transcription discrepancies. This is a selected reading of *La conjecture de Weil II*, not a new audit of its entire proof. ([Numdam](#))

The decisive addition is **not just another tensor operation**.

In §§3.2.8–3.2.13, Deligne first controls what occurs at the exceptional fibres. The resulting calculation gives integral weights for the relevant nonconstant constituents. The independent bound is **strictly below two**, so those integral weights are at most one. Constant constituents contribute zero to the specified first cohomology of the projective line. Only after those steps do the Leray and tensor-square maps improve the bound on the same eigenvalue.

The older French transcription loses two essential details: it writes a non-strict inequality in 3.2.10 and loses the square on the tensor eigenvalue in 3.2.13. The current page record retains both.

The later purity statement also concerns a particular comparison:

$$H_c^i(X, \mathcal{F}) \twoheadrightarrow \text{im } c_i \hookrightarrow H^i(X, \mathcal{F}).$$

The two already-proved bounds meet on that image; the mixed source and target are not removed. The degree and dualizing twist are part of the argument.

What I added here is the corresponding concrete work on our side: construct the curve, calculate the exceptional comparison and its complete fibre, retain the action, and calculate the determinant metric on that object.

S35.0.2 2. Build the curve from the existing representatives

Keep the original coefficient square over the marked τ -base:

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The first arithmetic target remains infinite. The split coefficients retain $e = 0^\bullet$ and external absence τ before applying p .

For the current finite cyclic object,

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad A = M_S,$$

retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N.$$

Here B_N is the original full remainder matrix, O_N contains the actual monic source norms, and the arithmetic inclusion still contains $v_h^{\otimes k}$. The canonical metric is determined by that source and its admitted relations.

For $i \leq j$, pad the earlier source through

$$L_{ij} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$$

and define

$$\Delta = L_{ij} R_i - R_j : E \longrightarrow \mathcal{P}_j.$$

Both representatives have remainder identity, and the later representative is orthogonal to its relation space. Therefore

$$B_j \Delta = 0, \quad R_j^* O_j \Delta = 0,$$

and

$$C := \Delta^* O_j \Delta = G_i - G_j \succeq 0.$$

Now form the actual family

$$R(z) = R_j + z\Delta : E \longrightarrow \mathcal{P}_j.$$

Direct expansion gives

$$B_j R(z) = I_E, \quad G(z) := R(z)^* O_j R(z) = G_j + |z|^2 C.$$

Every finite z therefore retains the same arithmetic jets and cohomology class:

$$J^{(k)} \mathcal{V} R(z) = \eta, \quad q^{(k)} \mathcal{V} R(z) = \sigma_h^{\otimes k} \eta.$$

The marked endpoints are

$$R(0) = R_j, \quad R(1) = L_{ij} R_i.$$

The new coordinate z parametrizes this comparison. The original spectral coordinate remains S , with its original action A .

S35.0.3 3. Compactify it and compute the exceptional fibre

Set

$$E_0 = \ker \Delta, \quad E_1 = E_0^{\perp G_j}, \quad r = \dim E_1.$$

Let P_0, P_1 be the corresponding specified projections.

On $\mathbb{P}_{\mathbb{C}}^1$, with homogeneous coordinates $[u : v]$, define

$$\mathcal{E}_{ij} = (E_0 \otimes \mathcal{O}) \oplus (E_1 \otimes \mathcal{O}(-1)).$$

The map into the original source is

$$\begin{aligned} \Psi : \mathcal{E}_{ij} &\longrightarrow \mathcal{P}_j \otimes \mathcal{O}, \\ x_0 + x_1 \otimes (u, v) &\longmapsto R_j x_0 + u R_j x_1 + v \Delta x_1. \end{aligned}$$

This is injective on every fibre. For $u \neq 0$, applying B_j forces $x_0 = x_1 = 0$. At $u = 0$, it first forces $x_0 = 0$, and injectivity of $\Delta|_{E_1}$ forces $x_1 = 0$.

On the finite chart $z = v/u$, this is exactly $R(z)$. On the infinity chart $w = u/v$, it is

$$\Psi_{\infty}(w) = R_j P_0 + w R_j P_1 + \Delta P_1.$$

The original arithmetic observation becomes

$$\alpha = B_j \Psi : \mathcal{E}_{ij} \longrightarrow E \otimes \mathcal{O},$$

with local expression

$$\alpha_{\infty} = P_0 + w P_1.$$

Consequently,

$$0 \longrightarrow \mathcal{E}_{ij} \xrightarrow{\alpha} E \otimes \mathcal{O} \longrightarrow i_{\infty*} E_1 \longrightarrow 0.$$

This is the effective-divisor exact sequence on the modified columns, with its actual source map supplied above. ([The Stacks Project](#))

The lost fibre is retained in the derived comparison Specializing $w = 0$ gives P_0 . The derived fibre of the displayed cokernel is the actual complex

$$[E \xrightarrow{P_0} E] \quad \text{in degrees } -1, 0.$$

Its cohomology is

$$H^{-1} = E_1, \quad H^0 = E_1.$$

Intrinsically, the kernel carries

$$E_1 \otimes \mathcal{O}(-1)|_{\infty};$$

the chosen infinity frame gives the displayed coefficient identification.

In the split lift,

$$w^{\bullet} \longmapsto e$$

under specialization. A represented vector killed by P_0 maps to the receiving supported zero, **and remains present in the derived kernel**. External absence maps to τ . This is precisely the original internal-homology treatment of a class whose image becomes a boundary.

The global comparison is also calculated:

$$\boxed{H^0(\mathbb{P}^1, \mathcal{E}_{ij}) = E_0, \quad H^1(\mathbb{P}^1, \mathcal{E}_{ij}) = 0.}$$

Its source image is $E_0 \subseteq E$. The full E remains the target and the saturation of the image lattice; it is not silently assigned to the smaller global image.

S35.0.4 4. Duality and the arithmetic action now have explicit boundary maps

Dualizing the comparison gives

$$\boxed{0 \rightarrow E^* \otimes \mathcal{O} \rightarrow \mathcal{E}_{ij}^\vee \rightarrow i_{\infty*}(E_1^* \otimes \mathcal{O}(1)|_\infty) \rightarrow 0.}$$

The two boundary factors pair through

$$\boxed{(E_1 \otimes \mathcal{O}(-1)|_\infty) \otimes (E_1^* \otimes \mathcal{O}(1)|_\infty) \longrightarrow \mathbb{C}_\infty.}$$

Both line factors remain.

The connection transported through α is

$$\nabla_\infty = d + P_1 \frac{dw}{w}.$$

Its residue is P_1 , and its horizontal monodromy is

$$\exp(-2\pi i P_1) = I.$$

The original arithmetic operator transports through the same map:

$$\boxed{\hat{A}_\infty = (P_0 + w^{-1}P_1)A(P_0 + wP_1) = \begin{pmatrix} A_{00} & wA_{01} \\ w^{-1}A_{10} & A_{11} \end{pmatrix}.$$

Thus the precise possible pole is

$$w^{-1}P_1AP_0.$$

The kernel of a metric comparison has not been presumed A -invariant.

Construct the smallest enlargement that repairs that pole Take the actual invariant hull

$$\boxed{F_A = E_0 + AE_0 + \cdots + A^{q-1}E_0, \quad q = \dim E.}$$

Cayley–Hamilton proves that this is the smallest A -invariant subspace containing E_0 .

Let $\mathcal{M} = \alpha(\mathcal{E}_{ij})$. At infinity,

$$\mathcal{M}_\infty = E_0\mathcal{O} + wE\mathcal{O}.$$

Its smallest coherent A -stable enlargement inside the original target is

$$\boxed{\mathcal{M}_A = \sum_{a=0}^{q-1} A^a \mathcal{M}, \quad (\mathcal{M}_A)_\infty = F_A\mathcal{O} + wE\mathcal{O}.$$

The exact sequences are

$$0 \rightarrow \mathcal{M} \rightarrow \mathcal{M}_A \rightarrow i_{\infty*}(F_A/E_0) \rightarrow 0,$$

and

$$0 \rightarrow \mathcal{M}_A \rightarrow E \otimes \mathcal{O} \rightarrow i_{\infty*}(E/F_A) \rightarrow 0.$$

The second sequence now carries the regular original action throughout. It gives the equivariant global sequence

$$0 \rightarrow F_A \rightarrow E \rightarrow E/F_A \rightarrow 0$$

and retains the trace

$$\mathrm{Tr} f(A) = \mathrm{Tr}(f(A)|F_A) + \mathrm{Tr}(f(\bar{A})|E/F_A).$$

The enlargement has a calculated source cost. Put

$$s_A = \dim F_A - \dim E_0.$$

The inclusion of the old lattice into the enlarged one has determinant w^{s_A} . Pulling through the **original** source metric gives

$$\det G_A(w) = |w|^{-2s_A} \det G_{\mathcal{M}}(w).$$

Regularizing the arithmetic action transfers the retained boundary contribution into a specified metric pole. It does not make that contribution disappear.

S35.0.5 5. The endpoint cost is the determinant curvature of this curve

Let

$$\mathcal{L}_{ij} = (\det \mathcal{E}_{ij})^\vee.$$

Its metric in the finite chart is

$$(\det G(z))^{-1}.$$

Define the curvature with its complete coefficient:

$$c_1(\mathcal{L}_{ij}) = \frac{i}{2\pi} \partial \bar{\partial} \log \det G(z).$$

The operator in this formula is exactly the preceding relation-cost operator:

$$Z = G_j^{-1}(G_i - G_j) = (K_j - K_i)G_i.$$

It is positive self-adjoint for G_j .

Differentiation gives

$$c_1(\mathcal{L}_{ij}) = \frac{i}{2\pi} \mathrm{Tr} [Z(I + |z|^2 Z)^{-2}] dz \wedge d\bar{z}.$$

Write

$$\kappa_{ij}(t) = \mathrm{Tr} [Z(I + tZ)^{-2}].$$

The coordinate identity

$$i dz \wedge d\bar{z} = 2 dx \wedge dy$$

and polar integration give

$$\int_{\mathbb{P}^1(\mathbb{C})} c_1(\mathcal{L}_{ij}) = \int_0^\infty \kappa_{ij}(t) dt = r.$$

That agrees with the independently computed divisor:

$$\det \alpha = u^r, \quad \deg \mathcal{L}_{ij} = r.$$

Recover the exact endpoint volume Put

$$f(t) = \log \det(I + tZ), \quad M(t) = t f'(t).$$

Then

$$M'(t) = \kappa_{ij}(t), \quad M(0) = 0.$$

Fubini gives

$$\begin{aligned} \int_0^1 \log(1/t) \kappa_{ij}(t) dt &= \int_0^1 \frac{1}{s} \int_0^s \kappa_{ij}(t) dt ds \\ &= \int_0^1 f'(s) ds \\ &= \log \det(I + Z). \end{aligned}$$

Since $G(1) = G_i$ and $G(0) = G_j$,

$$\log \frac{V_i}{V_j} = \int_0^1 \log(1/t) \kappa_{ij}(t) dt.$$

This is the same endpoint ratio as before, now computed on a compactified, marked curve with its boundary sheaf and metric.

Its second fundamental map retains the arithmetic cancellation The actual source projection is

$$P_z = R(z)G(z)^{-1}R(z)^*O_j.$$

Define

$$\mathcal{N}_z = (I - P_z)\Delta.$$

Then

$$\mathcal{N}_z^* O_j \mathcal{N}_z = C - |z|^2 C G(z)^{-1} C,$$

and

$$\mathrm{Tr}[G(z)^{-1} \mathcal{N}_z^* O_j \mathcal{N}_z] = \kappa_{ij}(|z|^2).$$

The individual arithmetic observations are

$$B_j P_z \Delta = \bar{z} G(z)^{-1} C,$$

$$B_j \mathcal{N}_z = -\bar{z} G(z)^{-1} C.$$

Their sum is the original relation. **Neither projected part is declared a boundary separately.** Their cancellation is retained before its quotient value becomes e .

S35.0.6 6. The previous trace certificate is now a curvature-jet estimate

Differentiating the curvature density at the marked point gives

$$\kappa_{ij}^{(n)}(0) = (-1)^n (n+1)! \operatorname{Tr}(Z^{n+1}).$$

Therefore the last note's quantities satisfy

$$t_1 = \kappa_{ij}(0), \quad t_2 = -\frac{1}{2} \kappa'_{ij}(0).$$

Its independent-pair quantity becomes

$$\mathcal{A}_2 = \frac{1}{2} \left(\kappa_{ij}(0)^2 + \frac{1}{2} \kappa'_{ij}(0) \right).$$

Thus the already-proved two-trace bound is now a bound on the logarithmic curvature moment of this particular determinant line:

$$\int_{|z|<1} \log(1/|z|^2) c_1(\mathcal{L}_{ij}) \leq \log(1+a) + (q-1) \log(1+b).$$

The a, b on the right are precisely the earlier functions of q, t_1, t_2 . No new estimate is being attributed to Deligne here; the geometric object underlying our existing estimate has been constructed.

S35.0.7 7. The construction also follows the actual arithmetic constituents

Deligne's determinant-weight bookkeeping concerns constituents, so I carried this construction through an actual invariant inclusion

$$I_F : F \hookrightarrow E, \quad A I_F = I_F A_F.$$

Set

$$R_F(z) = R(z) I_F, \quad G_F(z) = I_F^* G(z) I_F, \quad \Delta_F = \Delta I_F.$$

The resulting constituent bundle maps into $\mathcal{E}_{ij}$. On its modified part, the map is explicitly

$$x \otimes (u, v) \longmapsto u P_0 I_F x + P_1 I_F x \otimes (u, v).$$

Its composite with Ψ is exactly the constituent source map, including at infinity.

The original control compresses through the same inclusion:

$$W_F(z) = I_F^* W(z) I_F = A_F^* G_F(z) + G_F(z) A_F - k G_F(z).$$

Its determinant action satisfies

$$2\Re \operatorname{Tr} A_F - k \dim F = \operatorname{Tr} [G_F(z)^{-1} W_F(z)].$$

The original theta primitive also varies explicitly:

$$K(z) = (1 - z)K_j + zK_i,$$

$$D^{(k)}\mathcal{V}R(z) - \mathcal{V}R(z)A = dK(z).$$

Consequently,

$$W(z) = W_j + |z|^2(W_i - W_j) = -[(\mathcal{V}R(z))^* dK(z) + (dK(z))^* \mathcal{V}R(z)].$$

This connects the constituent curve, its determinant metric, and its **actual spectral action**. The integer curve degree is not substituted for the trace on the left.

S35.0.8 8. What is now available for the remaining estimate

For the two original endpoint blocks, let their determinant-curvature measures be μ_0, μ_1 . Then

$$\mathcal{B}_{h,k} = \sum_{\ell=0}^1 \int_{|z|<1} \log(1/|z|^2) d\mu_\ell, \quad \int d\mu_\ell = r_\ell \leq q_k.$$

The inherited off-line-quartet implication remains

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4.$$

I checked the pinned GitHub proof and retained its stated scope.

So the remaining upper estimate is now a **nonconcentration estimate for two specified source-induced curvature measures**, with their exceptional fibres and arithmetic action still available.

There is an exact calibration showing why the total degree alone cannot supply it. The family

$$G_j = I_r, \quad C = LI_r, \quad L > 0$$

has total curvature r for every L , but its logarithmic moment is

$$r \log(1 + L).$$

Each component has density

$$\frac{L}{(1 + Lt)^2},$$

and half its mass lies before $t = 1/L$. This is a declared algebraic family, not an asserted arithmetic packet.

The arithmetic work is therefore to control that concentration **using the original theta source and its relation maps**. Its curvature jets, second fundamental map, action-stable enlargement, and metric-pole contribution are now explicit quantities for doing so.

Deligne's finite-field argument supplies a model for the architecture—exceptional-fibre calculation, constituent control, then amplification—but its arithmetic integrality and strict weight estimate have not been proved for this new coherent-sheaf realization. Likewise, the calculated global image E_0 or F_A is retained rather than being relabelled as the entire arithmetic E .

S35.0.9 Verification and handoff

The new checker passed **23 exact finite test methods** normally and under `python -0`, with identical successful records. Both deliberately false controls were rejected in both modes. The tests include the derived special-fibre kernel, dual line factors, curvature sign, constituent maps, the minimal action-stable enlargement, and its determinant-metric pole.

All **36 manifest entries** of the preceding relation-moment archive verified, and its unchanged **22-method suite** passed fresh normal and optimized runs. The HTML reader was rendered and inspected at desktop and mobile widths; its displayed equation labels and local equation scrolling are retained.

The package includes a [tested eighteen-file add-only integration patch](#). No remote branch was changed, no new Lean execution is claimed, and no uniform arithmetic nonconcentration estimate is claimed.

The added machinery is now geometric in a concrete sense: a curve of the original representatives, its complete specialization and dual boundary, the minimal repair of its arithmetic action, and a determinant curvature whose logarithmic moment is exactly the endpoint cost already under study.

S36 Complete supplied logarithmic constituent and coupling calculation

Source role. Complete supplied mathematical text, retained with its original formulas and source chronology. The present integrated chapters state and prove the subsequent corrections and exact maps.

Exact revision. 501decfd268d2e9a382196f61169e9cd3fb27484b5ff03119f4855955328c864

Complete source. sources/endpoint_user_sources/LOGARITHMIC_CONSTITUENT_USER_SOURCE.txt

Source SHA-256. 501decfd268d2e9a382196f61169e9cd3fb27484b5ff03119f4855955328c864

I have now carried the previous curve through its logarithmic de Rham complex and used the resulting subobject-quotient comparison to improve the upper-volume calculation. The new quantitative term is a **subtraction computed from the coupling between an invariant arithmetic subspace and its quotient**:

$$\log \frac{V_i}{V_j} \leq \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det(Q_0 + C_Q)}{\det Q_0} - \sum_{r=1}^p \frac{\text{Tr}(\Xi_F^r)}{r}.$$

Every matrix here comes from the **same original canonical representatives**. The final term is nonnegative before subtraction, and its full infinite sum recovers the exact volume. It is not a freely chosen correction or a consequence stipulated under the name “purity.”

The cohomological calculation also identifies exactly which degrees carry the arithmetic information on the auxiliary curve. That matters for using Deligne’s comparison correctly.

[Full derivation in the HTML reader](#) · [LaTeX](#) · [Formalization handoff](#) · [Complete package](#)

S36.0.1 1. The Deligne operation being applied

The operative passage remains the actual comparison

$$H_c^a(X, \mathcal{F}) \rightarrow \text{im } c_a \hookrightarrow H^a(X, \mathcal{F}).$$

Deligne’s upper and dual lower estimates meet on that **same image**, with the cohomological degree and dualizing twist retained. Neither the source nor its mixed boundary contribution is erased.

I used that structure in two concrete ways here: first, by constructing the compact-support and ordinary complexes of the connection already present in our curve; second, by carrying the original metric and arithmetic action through an invariant inclusion and its quotient.

The new argument starts with the existing coefficient square:

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of $p_{\mathbb{Z}}$ remains infinite. The supported scalar $e = 0^\bullet$ and external absence τ remain separately represented before applying p .

For the current cyclic arithmetic module,

$$E = \mathbb{C}[S]/(\chi_{h,k}), \quad A = M_S, \quad q = \dim E,$$

retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N.$$

For an admitted degree pair $i \leq j$, put

$$G_0 = G_j, \quad \Delta = L_{ij} R_i - R_j, \quad C = \Delta^* O_j \Delta = G_i - G_j.$$

The original relation and orthogonality equations are

$$B_j \Delta = 0, \quad R_j^* O_j \Delta = 0.$$

They give precisely the previous family

$$R(z) = R_j + z\Delta, \quad G(z) = G_0 + |z|^2 C.$$

Every finite z retains the same full arithmetic jets and cohomology class. These are inherited source identities, not a new choice of metric.

S36.0.2 2. The compact-support and ordinary complexes are now calculated

Let

$$X = \mathbb{P}_{\mathbb{C}}^1, \quad w = z^{-1},$$

and take a specified A -invariant subspace $F \subseteq E$. In particular, the previous action-stable repair supplies

$$F_A = \sum_{a=0}^{q-1} A^a \ker \Delta.$$

Define the actual lattice

$$\mathcal{M}_F = F\mathcal{O}_X + \mathcal{O}_X(-\infty)E \subseteq E\mathcal{O}_X$$

and its logarithmic complex

$$\mathcal{K}_F = \left[\mathcal{M}_F \xrightarrow{\nabla} \mathcal{M}_F \otimes \Omega_X^1(\log \infty) \right],$$

in degrees 0, 1.

The differential is the restriction of the trivial coefficient connection. It is a map of sheaves of $\mathbb{C}$ -vector spaces satisfying the Leibniz rule. In a fixed local complement to F , its two pieces are

$$d, \quad d + \frac{dw}{w}.$$

The logarithmic construction and its residue map are standard; here the lattice, differential, and comparison maps are specified explicitly. ([The Stacks Project](#))

There are actual inclusions of complexes

$$\mathcal{K}_0 \longrightarrow \mathcal{K}_F \longrightarrow \mathcal{K}_E,$$

where

$$\mathcal{K}_0 = [E\mathcal{O}(-1) \longrightarrow E\Omega_X^1]$$

is the compact-support model, and

$$\mathcal{K}_E = [E\mathcal{O} \longrightarrow E\Omega_X^1(\log \infty)]$$

is the ordinary model.

The global result Let

$$\mathbb{T} = H^1(X, \Omega_X^1)$$

with its specified trace

$$\mathrm{tr}[dz/z] = 1, \quad dz/z = -dw/w.$$

The full calculation gives

$$\mathbb{H}^a(X, \mathcal{K}_F) = \begin{cases} F, & a = 0, \\ 0, & a = 1, \\ (E/F) \otimes \mathbb{T}, & a = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. A chosen coefficient complement writes the underlying complex as

$$F[\mathcal{O} \rightarrow \mathcal{O}(-1)] \oplus (E/F)[\mathcal{O}(-1) \rightarrow \mathcal{O}(-2)].$$

The two-chart Laurent calculation gives

$$H^0(\mathcal{O}) = \mathbb{C}, \quad H^a(\mathcal{O}(-1)) = 0, \quad H^1(\mathcal{O}(-2)) = \mathbb{T}.$$

These yield the displayed groups. The identifications are compatible with the original action even when the chosen complement is not invariant.

The action is therefore

$$A_F \quad \text{on degree zero}, \quad A_{E/F} \otimes I_{\mathbb{T}} \quad \text{on degree two},$$

and

$$\mathrm{Str}(f(A) \mid R\Gamma(X, \mathcal{K}_F)) = \mathrm{Tr} f(A_F) + \mathrm{Tr} f(A_{E/F}) = \mathrm{Tr} f(A).$$

Placing the coefficient object in the existing arithmetic degree k puts these contributions in degrees k and $k + 2$, both with sign $(-1)^k$.

This adds a term the degree-zero bundle calculation alone did not contain: **the quotient E/F survives in degree two.**

The compact-to-ordinary maps are now explicitly

$$0 \longrightarrow F \longrightarrow E$$

in degree zero, and

$$E \otimes \mathbb{T} \longrightarrow (E/F) \otimes \mathbb{T} \longrightarrow 0$$

in degree two.

For this constant-coefficient parameter curve, the degree-one comparison image is zero. That is a statement about this particular auxiliary connection, not about the arithmetic theta quotient Q . The full arithmetic E remains in the outer comparison complexes and their original action.

The distinction also preserves extensions. For example,

$$A = \begin{pmatrix} \rho & 1 \\ 0 & \rho \end{pmatrix}, \quad F = \mathbb{C}e_1,$$

induces multiplication by ρ on both one-dimensional constituents. Their two induced actions alone would not reconstruct the nonzero map $e_2 \mapsto e_1$. That map remains in the original coefficient sequence and in the action comparison calculated below.

The dual boundary is retained For

$$F^{\text{ann}} = \{\lambda \in E^\vee : \lambda(F) = 0\},$$

evaluation gives the actual map

$$\mathcal{M}_F \otimes \mathcal{M}_{F^{\text{ann}}} \longrightarrow \mathcal{O}_X(-\infty).$$

Its constant term vanishes at infinity, so it induces

$$\boxed{\mathcal{K}_F(E) \otimes \mathcal{K}_{F^{\text{ann}}}(E^\vee) \longrightarrow \mathcal{K}_0(\mathbb{C}).}$$

The resulting degree- a , degree- $(2-a)$ pairings are the perfect pairings

$$F \otimes (E^\vee / F^{\text{ann}}) \rightarrow \mathbb{C}, \quad (E/F) \otimes F^{\text{ann}} \rightarrow \mathbb{C},$$

with $\mathbb{T}$ and its residue orientation retained.

The note also gives the explicit conjugation map from the linear dual to the metric antidual. Thus the original $kI - A^*$ antidual action is not substituted for a linear transpose without its coefficient map.

S36.0.3 3. Higher layers remain above the same split support

For every positive integer r , the same construction gives

$$\mathcal{M}_{F,r} = F\mathcal{O}_X + \mathcal{O}_X(-r\infty)E.$$

Its modified connection has residue r :

$$d + r \frac{dw}{w}.$$

The quotient comparing level r with level one has the actual coefficient basis

$$w, w^2, \dots, w^{r-1}$$

in both degrees, with differential

$$\boxed{\text{diag}(1, 2, \dots, r-1).}$$

Its inverse is the explicit contracting homotopy. The higher coefficient classes are present; their vanishing in cohomology follows from this differential.

This divisor-depth calculation is connected to the original conormal-depth calculation by

$$\boxed{\frac{\mathbb{C}[w] \otimes \mathcal{P}}{(w^r) + I^a(\mathbb{C}[w] \otimes \mathcal{P})} \xrightarrow{\sim} (\mathbb{C}[w]/w^r) \otimes (\mathcal{P}/I^a),}$$

where

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)).$$

The map sends $[f(w)P]$ to $[f] \otimes [P]$; its inverse multiplies the representatives.

Restriction and differentiation in the two directions therefore have their actual commuting maps. The two depth indices are retained independently, over the same scalar G -construction.

When a relation vanishes, its split lift lands at its receiving fibre's zero. This is the original internal-homology operation, which keeps both the source class and the comparison kernel available.

S36.0.4 4. Carry the original metric through an invariant arithmetic quotient

This is the new quantitative part.

Keep the actual invariant maps

$$I : F \hookrightarrow E, \quad AI = IA_F,$$

$$\pi : E \twoheadrightarrow E/F, \quad \pi A = A_Q \pi.$$

Write $t = |z|^2$, so

$$G(t) = G_0 + tC.$$

Choose any fixed coefficient lift $\overline{H} : E/F \rightarrow E$. Its **specified G_0 -orthogonal lift** is

$$H_0 = \overline{H} - I(I^*G_0I)^{-1}I^*G_0\overline{H}.$$

It satisfies

$$\pi H_0 = I, \quad I^*G_0H_0 = 0.$$

Define

$$G_{F,0} = I^*G_0I, \quad C_F = I^*CI,$$

$$D_F = I^*CH_0, \quad Q_0 = H_0^*G_0H_0, \quad C_Q = H_0^*CH_0.$$

In the actual coefficient coordinates $[I, H_0]$, the original metric is

$$\begin{pmatrix} G_{F,0} + tC_F & tD_F \\ tD_F^* & Q_0 + tC_Q \end{pmatrix}.$$

Now set

$$G_F(t) = G_{F,0} + tC_F,$$

$$H_t = H_0 - tIG_F(t)^{-1}D_F.$$

Then

$$\pi H_t = I, \quad I^*G(t)H_t = 0,$$

and the actual quotient metric is

$$Q(t) = H_t^*G(t)H_t = Q_0 + tC_Q - t^2D_F^*G_F(t)^{-1}D_F.$$

The correction $H_t - H_0$ lies in F . The full map $[I, H_t]$ remains invertible and retains that subspace. It is not a declaration that F is a theta boundary.

The coupling has an exact sign and size Define

$$\beta_t = G_F(t)^{-1} I^* C H_t : E/F \rightarrow F.$$

Differentiating the displayed maps gives

$$H'_t = -I\beta_t, \quad Q'(t) = H_t^* C H_t, \quad Q''(t) = -2\beta_t^* G_F(t) \beta_t.$$

For the first metric derivative, the terms containing H'_t vanish by orthogonality. Differentiating once more gives the last equation, with its minus sign and factor two.

Integrating with the original endpoints retained proves

$$D_F^* G_F(1)^{-1} D_F = 2 \int_0^1 (1-t) \beta_t^* G_F(t) \beta_t dt.$$

The subtraction in the quotient metric is therefore exactly an integrated squared coupling map. Its finite matrix formula defines it without requiring a new numerical quadrature.

S36.0.5 5. The same arithmetic action and theta relation reach this quotient

The actual lift defect is the map $a_t : E/F \rightarrow F$ defined by

$$A H_t - H_t A_Q = I a_t.$$

Differentiation gives

$$a'_t = \beta_t A_Q - A_F \beta_t.$$

Thus the metric coupling is related to the original arithmetic action through a concrete Sylvester map. On distinct primary blocks, the note constructs its finite inverse, retaining every power of the nilpotent parts. At coincident primary values, it retains the map with its kernel and image instead of using a nonexistent inverse.

For the original control form

$$W(t) = A^* G(t) + G(t) A - k G(t),$$

both comparisons are exact:

$$I^* W(t) I = A_F^* G_F(t) + G_F(t) A_F - k G_F(t),$$

$$H_t^* W(t) H_t = A_Q^* Q(t) + Q(t) A_Q - k Q(t).$$

The second equation uses $H_t^* G(t) I = 0$, not invariance of the orthogonal complement.

At the original test-function level,

$$D^{(k)} \mathcal{V} R(z) H_t - \mathcal{V} R(z) H_t A_Q = dK(z) H_t + \mathcal{V} R(z) I a_t.$$

The first term is the original theta boundary. The second belongs to the retained arithmetic subspace F , and becomes zero only under the further quotient by that included subspace. Both terms, and their different quotient maps, remain in the calculation.

This is the source-level connection between the constituent calculation and the original e -valued relation.

S36.0.6 6. The resulting upper bound keeps the extension contribution

At the endpoint, put

$$Q_+ = Q_0 + C_Q, \quad \Pi_F = D_F^* G_F(1)^{-1} D_F, \quad \boxed{\Xi_F = Q_+^{-1} \Pi_F.}$$

The source proves

$$Q_+ \succ 0, \quad Q(1) = Q_+ - \Pi_F \succ 0.$$

Hence Ξ_F is positive self-adjoint in Q_+ , with eigenvalues in $[0, 1)$.

The determinant of the coefficient transition $[I, H_0] \rightarrow [I, H_t]$ is independent of t . Taking determinants of the original metric therefore gives

$$\boxed{\log \frac{V_i}{V_j} = \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} + \log \det(I - \Xi_F).}$$

Now

$$-\log \det(I - \Xi_F) = \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r},$$

with nonnegative terms. Retaining the first p proves

$$\boxed{\log \frac{V_i}{V_j} \leq \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^p \frac{\text{Tr}(\Xi_F^r)}{r}.}$$

The upper bounds decrease to the exact volume. No separately assumed uniform restriction gap is needed to justify a finite subtraction.

The two positive terms can themselves be bounded by the previous two-trace certificate, applied to the actual compressed operators

$$Z_F = G_{F,0}^{-1} C_F, \quad Z_Q = Q_0^{-1} C_Q.$$

Writing that established majorant as U , the combined result is

$$\boxed{\log \frac{V_i}{V_j} \leq U(Z_F) + U(Z_Q) - \sum_{r=1}^p \frac{\text{Tr}(\Xi_F^r)}{r}.}$$

This uses the invariant subobject and quotient without estimating them as independent, uncoupled objects.

An exact improvement on the same source fixture For the preceding mass-seven Gaussian calibration,

$$\chi = (S-1)^3, \quad \omega_n = 7n!, \quad (i, j) = (3, 6),$$

take the actual invariant line generated by $(S-1)^2$.

The calculated matrices give

$$\frac{\det G_F(1)}{\det G_F(0)} = \frac{63}{8}, \quad \frac{\det Q_+}{\det Q_0} = \frac{245}{32},$$

and

$$\Xi_F = \begin{pmatrix} 2/3 & 2/3 \\ 0 & 0 \end{pmatrix}.$$

This matrix is self-adjoint in Q_+ , with eigenvalues $2/3, 0$.

The exact original loss is

$$\log(V_3/V_6) = \log(5145/256) = 3.000603203788587506 \dots$$

Four terms of the new subtraction give

$$\log(V_3/V_6) \leq \log(15435/256) - \frac{28}{27} = 3.062178455419660160 \dots$$

The preceding full two-trace bound was approximately 3.063328308618.

This is a strict improvement on that **same canonical source and metric**. It is a declared finite calibration, not an asserted zeta packet or an asymptotic estimate.

S36.0.7 7. The bound is composed with the existing endpoint theorem

Let $\mathcal{U}_{ij}$ be either new upper expression. The inherited consecutive-window theorem gives

$$\min_{i \leq N < j} \epsilon_N \leq \left(\frac{\omega_j}{\omega_i} \right)^{1/(2(j-i))} \exp \left(\frac{\mathcal{U}_{ij}}{2(j-i)} \right).$$

The selected degree belongs to the existing canonical family. No metric is selected again after estimating it.

For either original quartet block, the supplied arithmetic norm estimate bounds the first factor by $C_h^{\text{win}} q_k$. The other session's finite propagation and written analytic composition retain the lower cost

$$2q_k \log(D_h k)$$

for each block, and the combined limiting threshold four.

The present calculation therefore gives a constituent-level upper side of those **same inequalities**, including a negative coupling contribution that was previously unavailable in that form.

The regular extension at infinity is accounted for as well. Passing to the residue-zero extension gives, with the original metric retained,

$$c_1((\det E\mathcal{O}_X)^\vee) = \mu_{ij} - \text{rank}(C) \delta_\infty.$$

The positive curvature measure and its logarithmic moment remain. The boundary atom makes the total degree zero; it does not make that logarithmic moment small.

That is why this continuation uses the actual subquotient estimate rather than inferring arithmetic purity from the compactification's integer degree.

S36.0.8 Verification and current result

I checked GitHub main at 1f7e7c02343884a17df9873566e81a9083c70951; its current four-volume source retains the same threshold and proof scope.

The new checker passed **16 exact finite test methods**, normally and under python -0, with identical successful records. Both deliberately false controls failed in both modes. The parent specialization-curvature archive's **41 manifest entries** verified, and its unchanged **23-method suite** passed fresh normal and optimized runs.

The supplied Deligne archive was reassembled and hash-checked; the relevant LaTeX and page-local text passages were read. No PDF extraction or OCR was used. The HTML reader was rendered and inspected at desktop and mobile widths.

The package includes a **tested 23-file add-only integration patch**. No remote branch was changed, and no new Lean certificate is claimed.

The completed advance is the actual logarithmic comparison, its retained degree-two and dual boundary information, and an upper-volume estimate that uses the calculated coupling between arithmetic constituents. A uniform bound below the inherited quartet threshold is still not established. The quantitative work is now focused on the original compressed source matrices and that explicit subtraction—not on assigning purity to a curve merely because it has been compactified.

S37 Complete exact restriction fixture and its source maps

Source role. Full finite-calibration derivation with original masses, every map and the exact scope of its executed checks.

Exact revision. f5ef0746d22ba641aa2fd0d50c0f19954fe9c367d8cacd0206aa1e28f6702b29

Complete source. sources/local_endpoint_continuations/endpoint_restriction_join_fixture_20260913.md

Source SHA-256. f5ef0746d22ba641aa2fd0d50c0f19954fe9c367d8cacd0206aa1e28f6702b29

S37.1 Exact lower-source gap and two-pair restriction certificates

This is a complete finite calculation for the explicitly declared auxiliary source below. It joins the lower-source comparison in AU.11–29 to the restriction and trace construction in equations (24), (34), and (47)–(51) of the supplied restriction note. The original coordinate is $S=1+iu$, the relation is $\chi(S)=S^2-2S-1$, the quotient has dimension two, and the source mass is exactly $9/16$.

The result is an explicit lower gap $108/763$ obtained from a positive source comparison and an exact remainder operator norm. The actual sharp gap is $135/331$. Both apply to the two pairs (1,3) and (2,4). Their trace certificates are computed with exact rational logarithm enclosures. These are finite conclusions for this specified source; no estimate for a growing arithmetic quartet family is inferred.

S37.1.1 1. Source, lower source, and literal masses

For positive integer λ define

$$r_\lambda(u) = \frac{|\Gamma(\lambda + iu/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda).$$

The measures used here are

$$w(t) = \left(1 + \frac{t^2}{4}\right) r_1(t) = r_2(t), \quad m = w * w = \frac{1}{70} r_4, \quad v = r_1 * r_1 = \frac{1}{3} r_2, \quad d\nu = \frac{54}{35} v(u) du. \quad (\text{F.1})$$

Here is the measure identity with its constants. Euler's gamma integral gives $\Gamma(z+1)=z \Gamma(z)$ by integration by parts when $\Re z > 0$. Multiplying this equality by its conjugate proves the first equality in (F.1). The change from two positive gamma-integral variables to their sum and ratio, with Jacobian equal to the sum, gives

$$\Gamma(a)\Gamma(b) = \Gamma(a+b) \int_0^1 x^{a-1}(1-x)^{b-1} dx, \quad \Re a, \Re b > 0.$$

Set $a=\lambda+iu/2$, $b=\lambda-iu/2$, and then $x=e^t/(1+e^t)$. This gives

$$|\Gamma(\lambda + iu/2)|^2 = \Gamma(2\lambda) \int_{\mathbb{R}} (2 \cosh(t/2))^{-2\lambda} e^{iut/2} dt.$$

After $t=2y$, Fourier inversion, with convention $\hat{f}(t)=\int_{\mathbb{R}} f(u)e^{-itu}du$, proves

$$\hat{r}_\lambda(t) = 2\Gamma(2\lambda)(2 \cosh t)^{-2\lambda} = c_\lambda(\cosh t)^{-2\lambda}. \quad (\text{F.2})$$

All inversions and moment differentiations used here are legitimate. The integrand $(2 \cosh y)^{-2\lambda}$ and every derivative decay exponentially on the real axis. For each $0 < a < \pi/2$, it is holomorphic on the closed strip $|\Im y| < a$; shifting the Fourier integral up or down inside that strip bounds $r_\lambda(u)$ by a constant times $\exp(-a|u|)$. The vertical sides of a rectangular contour tend to zero because the integrand decays exponentially in $|\Re y|$, uniformly in that strip. Thus every polynomial moment exists, the Fourier transform extends to imaginary arguments near zero, and its moment-generating function is

$$\int e^{tu} r_\lambda(u) du = c_\lambda(\cos t)^{-2\lambda}, \quad |t| < \pi/2.$$

Multiplying (F.2) proves

$$r_a * r_b = \frac{c_a c_b}{c_{a+b}} r_{a+b}.$$

The constants are $c_1=1/2$, $c_2=3/4$, and $c_4=315/8$. Consequently the masses of w, m, v, ν are respectively

$$\left[\frac{3}{4}, \frac{9}{16}, \frac{1}{4}, \frac{27}{70} \right] \quad (\text{F.3})$$

Two applications of the gamma recurrence give the pointwise identity

$$m(u) = \left(\frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120} \right) v(u). \quad (\text{F.4})$$

Indeed $r_4/r_2=(4+u^2/4)(9+u^2/4)$ and $(r_4/70)/(r_2/3)$ multiplies that product by $3/70$. Every coefficient in (F.4) is retained. Since $v(u) > 0$, this proves $d\nu \leq m(u)du$ and hence the contraction

$$\mathbb{C}[S]_{\leq j} \longrightarrow \mathbb{C}[S]_{\leq j}, \quad P \longmapsto P, \quad \|P\|_\nu^2 \leq \|P\|_m^2. \quad (\text{F.5})$$

The forms in (F.5) integrate $|p(1+iu)|^2$. The polynomial map, quotient map, and original coordinate are unchanged.

The even moments of m through degree eight are

$$(9/16, 9/2, 117, 5472, 385272),$$

and those of v are

$$(1/4, 1, 14, 376, 16064). \quad (\text{F.6})$$

The ν moments are the latter multiplied by $54/35$; all odd moments are zero. These values follow by differentiating the functions in (F.2). The executable constructs them by the equivalent finite multiplication recurrence

$$ub_j = b_{j+1} + j(j + \alpha - 1)b_{j-1}, \quad \alpha = 2\lambda,$$

starting from $b_0=1$, retaining the coefficient of b_0 and multiplying by the literal mass. On the finite degrees needed here it also verifies them by two independent constructions: binomial convolution of $(1+t^2/4)r_1$ moments, and the full three-term density (F.4). The monic-polynomial orthogonality needed below is verified directly against every Gram entry, so no unchecked infinite recurrence is used as a finite matrix certificate.

S37.1.2 2. Complete original-coordinate matrices

The polynomial space has ordered basis $1, S, \dots, S^N$. Its conjugate-linear-first Gram is

$$(H_N)_{ab} = \sum_{x=0}^a \sum_{y=0}^b \binom{a}{x} \binom{b}{y} (-1)^x i^{x+y} m_{x+y}. \quad (\text{F.7})$$

The imaginary terms vanish for the stated even source and are separately accumulated and checked in the executable. The full largest matrices are

$$H_4 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 & 1449 \\ 9 & 81 & 81 & -1863 & -5751 \\ -63 & 81 & 2025 & 2025 & -89343 \\ -207 & -1863 & 2025 & 93393 & 93393 \\ 1449 & -5751 & -89343 & 93393 & 6526089 \end{pmatrix}, \quad (\text{F.8})$$

$$H_4^\vee = \frac{1}{70} \begin{pmatrix} 27 & 27 & -81 & -297 & 891 \\ 27 & 135 & 135 & -1485 & -4725 \\ -81 & 135 & 1755 & 1755 & -41985 \\ -297 & -1485 & 1755 & 45495 & 45495 \\ 891 & -4725 & -41985 & 45495 & 1906875 \end{pmatrix}. \quad (\text{F.9})$$

Every smaller source form is the corresponding leading principal submatrix. The polynomial relation yields

$$J_1 = I_2, \quad J_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{pmatrix}, \quad J_4 = \begin{pmatrix} 1 & 0 & 1 & 2 & 5 \\ 0 & 1 & 2 & 5 & 12 \end{pmatrix}. \quad (\text{F.10})$$

These are the maps to coefficients in $E = \mathbb{C}[S]/(\chi)$ with basis $(1, S)$. The recurrence for their columns is $J_{-1}(\cdot, a) = 2J_{-1}(\cdot, a-1) + J_{-1}(\cdot, a-2)$, exactly the relation $S^2 = 2S + 1$.

For each degree define

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N, \quad V_N = \det G_N. \quad (\text{F.11})$$

Positivity of the density proves $H_N > 0$. Surjectivity of J_N proves $K_N > 0$. Direct multiplication gives $J_N R_N = I$ and $R_N^* H_N R_N = G_N$. For any z with $J_N z = 0$, $z^* H_N R_N = z^* J_N^* G_N = 0$. Therefore every representative $R_N x + z$ has squared norm $x^* G_N x + z^* H_N z$, proving its minimum property on the original quotient fibre.

For transparent scalar calculations introduce the explicitly typed coordinate isomorphism

$$\mathbb{C}[x]/(x^2 - 2) \xrightarrow{\sim} E, \quad [x] \mapsto [S - 1], \quad C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad (1, S)C = (1, S - 1). \quad (\text{F.12})$$

Its inverse substitutes $S = x + 1$. Its determinant is one. The source observation of x is still iu , with its sign and phase retained. A quotient form transforms by $G_x = C^* G_S C$; its inverse transforms by $K_x = C^{-1} K_S C^{-*}$. No measure is changed by these identities.

The following finite polynomials and norms follow by substitution into (F.6)–(F.9). They are monic in S ; writing them in $x = S - 1$ is only (F.12).

Degree	Original source polynomial	Squared norm	Remainder in $(1, x)$
0	1	9/16	1
1	x	9/2	x
2	$x^2 + 8$	81	10
3	$x^3 + 26x$	2430	28x
4	$x^4 + 56x^2 + 240$	106920	356

Degree	Lower-source polynomial	Squared norm	Remainder in $(1, x)$
0	1	27/70	1
1	x	54/35	x
2	$x^2 + 4$	108/7	6
3	$x^3 + 14x$	1944/7	16x
4	$x^4 + 32x^2 + 72$	7776	140

Let P_N have these original S coefficient columns and let O_N be the diagonal of their squared norms. The checked identity is the full matrix equation $P_N^* H_N P_N = O_N$. Its inverse and (F.11) give

$$K_N = (J_N P_N) O_N^{-1} (J_N P_N)^*. \quad (\text{F.13})$$

Thus the inverse quotient form is the sum of each listed remainder outer product divided by its original norm. Explicitly,

$$\begin{aligned}
K_{1,x} &= \text{diag}(16/9, 2/9), & K_{2,x} &= \text{diag}(244/81, 2/9), \\
K_{3,x} &= \text{diag}(244/81, 662/1215), & K_{4,x} &= \text{diag}(56102/13365, 662/1215), \\
K_{3,x}^\nu &= \text{diag}(133/27, 763/486), & K_{4,x}^\nu &= \text{diag}(3619/486, 763/486).
\end{aligned} \tag{F.14}$$

The full original-basis matrices and least representatives at all four degrees are included as rational entries in the JSON. Equations (F.8)–(F.11) are their complete finite construction. Their quotient volumes are

$$V_1 = 81/32, \quad V_2 = 729/488, \quad V_3 = 98415/161528, \quad V_4 = 16238475/37139524. \tag{F.15}$$

S37.1.3 3. A proved gap from the exact remainder operator

Let $B=H_1$, the norm on the unique representative of degree below two. In (F.12),

$$B_x = C^*BC = \text{diag}(9/16, 9/2).$$

Consider the actual remainder operator

$$\mathcal{R}_j : (\mathbb{C}[S]_{\leq j}, H_j^\nu) \longrightarrow (\mathbb{C}[S]_{\leq 1}, B), \quad P \longmapsto \text{rem}_\chi P, \quad j = 3, 4. \tag{F.16}$$

Its coefficient matrix is exactly J_j . Its squared norm is the greatest eigenvalue of $B^{\wedge\{1/2\}} J_j (H_j^\nu)^{\wedge\{-1\}} J_j^* B^{\wedge\{1/2\}}$. To see this, the substitutions $a = (H_j^\nu)^{\wedge\{1/2\}} v$ and $b = B^{\wedge\{1/2\}} J_j v$ transport its norm ratio to the Euclidean operator $B^{\wedge\{1/2\}} J_j (H_j^\nu)^{\wedge\{-1\}}$. For a matrix L , its greatest squared singular value is the greatest eigenvalue of LL^* : the nonzero spectra of L^*L and LL^* coincide by the maps L and L^* , and maximization of the positive Hermitian quadratic form is diagonalization by an orthonormal eigenbasis.

By (F.14), these positive eigenvalues are exactly

j	first	second
3	133/48	763/108
4	3619/864	763/108.

Both first entries are smaller than the second: the differences are 1855/432 and 2485/864. Consequently

$$\boxed{\|\mathcal{R}_3\|^2 = \|\mathcal{R}_4\|^2 = \frac{763}{108}.} \tag{F.17}$$

This norm is attained. In the original S coefficient basis use

$$f_3(S) = -\frac{245}{36} + \frac{791}{108}S - \frac{7}{9}S^2 + \frac{7}{27}S^3, \quad f_4 = f_3 \in \mathbb{C}[S]_{\leq 4}. \tag{F.18}$$

This is $f_j = (H_j^\nu)^{\wedge\{-1\}} J_j^* B(-1, 1)^{\wedge\tau}$; the two expressions agree by direct multiplication of the matrices (F.8)–(F.10). The norm and remainder norm are

$$\|f_j\|_\nu^2 = \frac{763}{24}, \quad \|\text{rem}_\chi f_j\|_B^2 = \frac{582169}{2592},$$

whose ratio is 763/108. This supplies a concrete extremizing polynomial, in addition to the complete matrix certificate

$$\frac{763}{108} H_j^\nu - J_j^* B J_j \succeq 0. \tag{F.19}$$

The JSON records every principal minor of (F.19), including its zero minors. For a Hermitian matrix, non-negativity of all principal minors implies positive semidefiniteness: the coefficients of $\det(tI+A)$ are the sums of its principal minors of each size, so this polynomial is strictly positive for $t \succ 0$; a negative eigenvalue of A would give a positive root. This proves the exact PSD validation criterion used here without a floating-point eigenvalue.

The source contraction (F.5) now gives

$$\|\text{rem}_x P\|_B^2 \leq \frac{763}{108} \|P\|_\nu^2 \leq \frac{763}{108} \|P\|_m^2. \quad (\text{F.20})$$

Apply this to the actual least representative R_{jx} . Its remainder is x , so

$$G_j \succeq \frac{108}{763} B.$$

The degree-one remainder is an admitted representative for every $i \geq 1$, hence $G_{i \leq B}$ by the minimum property. Therefore for $(i,j)=(1,3),(2,4)$,

$$\boxed{G_j \succeq \frac{108}{763} G_i, \quad K_i \succeq \frac{108}{763} K_j.} \quad (\text{F.21})$$

The inverse inequality follows by conjugating the first by $G_{i^{-1/2}}$, inverting its positive eigenvalues, and transporting back. The gap $108/763$ is thus obtained from the explicitly proved source comparison and exact remainder norm. Its construction uses the low-degree actual form B and the declared comparison moments through degree eight.

S37.1.4 4. Restriction in the original polynomial coordinates

Define the typed restriction on the same quotient by

$$T_{i,j} : (E, G_j) \longrightarrow (E, G_i), \quad T_{i,j} = K_i G_j. \quad (\text{F.22})$$

For x, y in E , $x^* G_{jy} = x^* G_{iT_{i,j}y}$; this proves it is the adjoint of identity transport $(E, G_i) \rightarrow (E, G_j)$. In the original $(1, S)$ coordinates the two return maps are

$$T_{1,3} = \begin{pmatrix} 36/61 & 3681/20191 \\ 0 & 135/331 \end{pmatrix}, \quad T_{2,4} = \begin{pmatrix} 20130/28051 & 2876145/9284881 \\ 0 & 135/331 \end{pmatrix}. \quad (\text{F.23})$$

The coordinate matrices in (F.12) diagonalize these endomorphisms by $T_x = C^{-1} T_{SC}$, giving spectra

$$\{36/61, 135/331\}, \quad \{20130/28051, 135/331\}. \quad (\text{F.24})$$

Both eigenvalues in each pair lie in $(0,1)$. The smallest is $135/331$ for both pairs, and hence this is the exact best scalar in $\kappa_{i \geq g} K_j$ for these actual source forms. It is stronger than (F.21), through the strictly positive difference $135/331 - 108/763$.

Let $L_{\{i,j\}}$ pad original polynomial coefficients with zeros. Since $P_j^* H_j P_j = 0_j$, the orthogonal projection from degree j to degrees at most i is the endomorphism

$$Q_{i,j} = P_j \text{diag}(\underbrace{1, \dots, 1}_{i+1}, 0, \dots, 0) P_j^{-1}. \quad (\text{F.25})$$

Its range is $L_{\{i,j\}} P_i$; direct multiplication proves $Q^2 = Q$ and $Q^* H_j = H_j Q$. In the two original S coordinate spaces these are

$$Q_{1,3} = \begin{pmatrix} 1 & 0 & -9 & 0 \\ 0 & 1 & 2 & -23 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$Q_{2,4} = \begin{pmatrix} 1 & 0 & 0 & 27 & -189 \\ 0 & 1 & 0 & -29 & 0 \\ 0 & 0 & 1 & 3 & -50 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{F.26})$$

In monic polynomial coordinates the first $i+1$ rows of the degree- j least lift are $0_{i^{-1}} (J_i P_i)^* G_j$; multiplication of the degree- i lift by $T_{\{i,j\}}$ gives precisely that expression because $G_i K_i = I$. Transporting by the original coefficient matrices therefore proves

$$\boxed{L_{i,j}R_iT_{i,j} = Q_{i,j}R_j.} \quad (\text{F.27})$$

Its Gram is $G_jK_iG_j$. For an eigenvector v of T with eigenvalue g , this becomes

$$\frac{\|Q_{i,j}R_jv\|_{H_j}^2}{\|R_jv\|_{H_j}^2} = g, \quad \frac{\|(I - Q_{i,j})R_jv\|_{H_j}^2}{\|R_jv\|_{H_j}^2} = 1 - g. \quad (\text{F.28})$$

The second equation follows from orthogonal decomposition by (F.25). The same original representative determines both quantities.

Writing $D = L_{\{i,j\}}R_i - R_j$ and $H = I - T_{\{i,j\}}$, equation (F.27) gives the exact correction

$$D = L_{i,j}R_iH - (I - Q_{i,j})R_j, \quad J_jD = 0, \quad D^*H_jD = G_i - G_j. \quad (\text{F.29})$$

The remainder equation follows from $J_jL_{\{i,j\}}R_i = J_jR_j = I$. For the Gram identity, R_j is orthogonal to every relation in $\ker J_j$ by (F.11), so $L_{\{i,j\}}R_i = R_j + D$ is an orthogonal sum. The return maps compose by $K_iG_jK_jG_i = K_iG_i$.

The quotient multiplication action is retained as

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

For $W_N = A^*G_N + G_{NA} - 2G_N$ and $H_N = K_{NW}_N$, expansion gives

$$AT - TA = H_iT - TH_j. \quad (\text{F.30})$$

Indeed the two terms $K_iA^*G_j$ and the two terms $2K_iG_j$ cancel. This records the exact action defect of the return map. Each finite matrix equation (F.27), (F.29), and (F.30) is checked in the literal original coordinates.

S37.1.5 5. Full finite logarithmic certificates

The determinant identity in (F.22) is

$$\det T_{i,j} = V_j/V_i.$$

Since the two positive eigenvalues are (F.24), put $s_a = \text{Tr}(I - T)^a$. Expansion of $-\log g = \sum_{a=1}^{\infty} (1-g)^a/a$ proves

$$\mathcal{L}_{i,j} = \log(V_i/V_j) = \sum_{a \geq 1} \frac{s_a}{a}. \quad (\text{F.31})$$

For any of the proved lower gaps g_0 , let $r = 1 - g_0$. Each loss eigenvalue $x = 1 - g$ satisfies $0 < x < r < 1$. The exact tail comparison is

$$\sum_{a=p+1}^{\infty} \frac{x^a}{a} = x^{p+1} \sum_{a=p+1}^{\infty} \frac{x^{a-p-1}}{a} \leq x^{p+1} \frac{-\log g_0 - \sum_{a=1}^p r^a/a}{r^{p+1}}.$$

Thus

$$\sum_{a=1}^p \frac{s_a}{a} \leq \mathcal{L}_{i,j} \leq U_p(H, g_0) := \sum_{a=1}^p \frac{s_a}{a} + \frac{-\log g_0 - \sum_{a=1}^p r^a/a}{r^{p+1}} s_{p+1}. \quad (\text{F.32})$$

For these finite pairs, the executable computes every trace through power six directly by matrix multiplication and independently from the rational spectrum. Its exact first three values are:

Pair	s_1	s_2	s_3
(1,3)	20231/20191	211421561/407676481	2275697407691/8231395827871
(2,4)	8119847/9284881	37102062682217/86209015184161	184216076205697661987/800440447112127969841

The two determinant ratios and their product are

$$\frac{V_1}{V_3} = \frac{20191}{4860}, \quad \frac{V_2}{V_4} = \frac{9284881}{2717550}, \quad \boxed{\frac{V_1 V_2}{V_3 V_4} = \frac{3073295611}{216513000}}. \quad (\text{F.33})$$

All logarithms in (F.31)–(F.32) have rational arguments. The executable uses exact rational enclosures. For $1 < y < 2$, put $t = (y-1)/(y+1)$. Integrating the geometric series $2/(1-t^2) = 2 \sum_{a \geq 0} t^{2a}$ from zero gives

$$2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} \leq \log y \leq 2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} + \frac{2t^{2M+1}}{(2M+1)(1-t^2)}. \quad (\text{F.34})$$

The upper tail follows by replacing all positive denominators by the first denominator and summing the geometric series. For $z \geq 1$, finite multiplication writes $z = 2^e y$ with $1 < y < 2$, and $\log z = e \log 2 + \log y$ transports the two intervals. For $z < 1$, $\log z = -\log(1/z)$ reverses interval endpoints. These are exact scalar logarithm identities; the source measures remain (F.1). The script uses $M=90$, checks loss widths below 10^{-70} , and checks certificate widths below 10^{-65} . Every enclosure endpoint and exact tail coefficient is retained in the JSON.

For a compact rational display, all following intervals have denominator 10^{15} . Their endpoints were obtained by outward integer rounding of the saved exact fractions, retaining the correct inequalities:

Quantity	Lower numerator	Upper numerator
$\log(V_1 V_2 / (V_3 V_4))$	2652855161899890	2652855161899891
Sum of p=2 upper certificates using 108/763	2934134222167925	2934134222167926
Sum of p=2 upper certificates using 135/331	2666680039877747	2666680039877748

The exact old direct trace bound is

$$\text{Tr}(G_1(K_3 - K_1)) + \text{Tr}(G_2(K_4 - K_2)) = \frac{1159}{540} + \frac{334321}{181170} = \frac{1446331}{362340}. \quad (\text{F.35})$$

To prove it bounds the loss, diagonalize $G_i^{-1/2} K_j G_i^{-1/2}$, whose eigenvalues are $1/g$; the scalar inequality $\log x \leq x-1$ for $x \geq 1$ follows by integrating $1/t \leq 1$ from one. Summing gives (F.35). Exact rational comparison verifies that each new p=2 total upper bound is strictly below (F.35), and that the sharp-gap certificate is strictly below the lower-source-gap certificate.

The cost of using the explicit lower-source gap can also be inspected pair by pair. The intervals below again have denominator 10^{15} and enclose the excess $U_{2-\text{mathcal L}}$:

Pair and gap	Lower numerator	Upper numerator
(1,3), lower-source 108/763	155314375058502	155314375058503
(2,4), lower-source 108/763	125964685209532	125964685209533
(1,3), actual 135/331	9359847652264	9359847652265
(2,4), actual 135/331	4465030325591	4465030325592

The script additionally computes the exact (F.32) enclosures for $p=1,3,5$. Each is checked against the independently enclosed determinant logarithm. These finitely many calculations make no claim about a uniform gap in a tensor-degree limit.

S37.1.6 6. Complete executable replay and provenance

Only these three files were authored for this lane:

- `endpoint_restriction_join_fixture_20260913.py`: complete standard-library rational calculation and subprocess driver.
- `endpoint_restriction_join_fixture_20260913.json`: all matrices, principal minors, witnesses, trace coefficients, logarithm enclosures, check labels, full stdout/stderr, actual commands, timings, return codes, and hashes for every subprocess.
- This Markdown proof and local continuity record.

The complete positive run contains **303 distinct mathematical checks**. The driver ran the positive calculation and seven changed calculations in both ordinary Python and `python -O`: **16 actual complete subprocess jobs**. Every job reaches all 303 checks. Assertions are not used as tests, and no changed computation is rejected by a startup guard or deliberately raised exception. The changed computations fail at the following counts, identically in both modes:

Changed calculation	Failed checks
Set the actual source mass to one	20
Set the lower-source mass to one	30
Replace K_{iG_j} by the reverse K_{jG_i}	36
Truncate original monomial coefficients in place of the orthogonal projection	8
Use the smaller comparison eigenvalue as the remainder norm cap	24
Replace trace of a power by power of the trace	23
Use denominator r^p in place of $r^{(p+1)}$ in the logarithmic tail coefficient	24

Each unchanged run has zero failures and exit code zero; each changed run has the listed positive failure count and exit code one. The driver separately checks full counts, unique labels, outcomes, return codes, empty stderr, and equality of complete result objects between ordinary and optimized modes. These execution/provenance checks and repeated runs do not increase the count of distinct mathematical checks.

The mathematical inputs were read completely before construction:

- `arithmetic_volume_upper_route_20260913.tex`, SHA256 21aaedc3585887fa7d2cb3e37e68d14415829a4b213566f7a08f4619835ef38b.
- Supplied `Tau_Endpoint_Restriction_Control/NOTE.tex`, SHA256 ff4cc53cb4d3b10a9547a8a2365db7fd5d874bf1ecf7dd8542b8d1f05f433ef6.
- Earlier auxiliary-source proof `gamma_endpoint_window_bridge_fixture_20260913.md`, SHA256 05108b93cfd1e15e782f171f37569e3d77c8c4775b1bf8349f4021a0eb4315ed.

No existing executable was imported into this fixture. An independent mathematical reviewer derived the same source moments, remainder kernels, comparison eigenvalues, both original-coordinate return matrices, and exact trace values without importing this script or writing files.

The completed independent read-only audit found no defect. It read all 415 script lines and inspected the normal result and all 16 replay outcomes. Its separate calculation passed **170 audit predicates** without importing or running this fixture script. It reconstructed the original S Grams, all four kernels and least lifts, and the projections directly from $L_{H_i}^{-1} L^{*} H_j$, independently of the monic-basis implementation. It verified the extremizer (F.18), all six trace powers, and the four $p=2$ certificates. A separate 1,024-term rational logarithm series without range reduction produced intervals contained in every audited saved logarithm/certificate interval. Every stdout/stderr hash, Windows CRLF reconstruction, unique check label, complete job count, return code, and ordinary/optimized equality matched; all 88 driver provenance checks passed. These 170 audit predicates remain separate from the fixture's 303 mathematical checks. The reviewer wrote no files.

Final executable SHA256: 88bcee664a3812dd914e5d164de37d48f4388307e2359062ace360e988d9df14.

Final replay JSON SHA256: a8205b83417feb94baf903f53f9bdd0d2e38cf325fc28ceaac6a098530b4075d.

Assigned work, verbatim

New bounded independent task while I derive restriction-AU join: read full work/arithmetic_volume_upper_route_20260913.tex and new restriction full NOTE once intake path available (ask root/arithmetic_endpoint_intake). Construct meaningful exact rational fixtures validating two-pair restriction $T=KiGj$, projection, spectral trace finite upper certificate, explicit remainder-comparator g_0 for $(q-1, 2q-1), (q, 2q)$, retaining literal source mass. Prefer existing EW auxiliary source $S=1+iu$ $\chi=S^2-2S-1$, mass9/16, or actual new Gaussian nilpotent mass7. Do full derivation of finite g_0 from exact polynomial remainder norm and source lower form, and compare bound's gap and trace certificate. Send formulas early; only new work/endpoint_restriction_join_fixture_20260913.{py,md,json} files. Full exact output, actual complete mutations not assertions, independent ordinary+optimized if apt. No edits to frozen files, no Lean or publication. Do not assume a uniform gap.

The selected source is the first explicit option. The parent received early exact comparator and spectral formulas, followed by the replay counts and upper-bound comparison. Existing frozen files, inherited intake checks, Lean calculations, and publication are outside this independent lane and were not changed.